SYNOPSYS®

DesignWare® Cores Ethernet PCS

Databook

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Revision History

Revision history for DesignWare® Cores DWC_xpcs Databook.

Version Number	Date	Description
3.50a	April 2022	Added: New sections to describe programming sequence for multi-protocol 32G PHY. Updated: ■ "Initializing the DWC_xpcs Controller" on page 1650 ■ "Switching to 10G XAUI Mode or KX4 Mode" on page 1652 ■ "Safety Subordinate Feature" on page 1761
3.41a	December 2019	■ Updated the APB3 Interface section ■ Updated Programming Sequence for Multi-Protocol 25G PHY ■ Updated Table B-1, Description of Documents in Automotive Safety Package ■ Updated Reference Documentation Section ■ Updated Standard Compliance Section ■ Updated the Parameter, Register, and Signal chapters
3.40a	July 2019	■ Added details of Multi-port USXGMII ■ Added Automotive Safety Features ■ Added details of Multi-protocol 25G PHY
3.30a	December 2017	■ Added details of Enterprise Gen5 16G PHY ■ Added support for USXGMII in Backplane Ethernet Configurations ■ Added details of Multi-port USXGMII ■ Added details of Backplane 2.5G/5G Modes
3.20a	July 2016	■ Added the details of QSGMII feature ■ Added new section on Universal Serial XGMII ■ Added new section on Programming guidelines to switch to USXGMII speed mode ■ Added new section on Programming guidelines to switch to QSGMII speed mode

Version Number	Date	Description
3.11b	June 2015	■ Added details of Enterprise Gen5 12G PHY ■ Added new section on Programing Guidelines for 2.5G GMII Mode ■ Added Reference Clock Source configuration parameter ■ Added PD_CTRL register bit ■ The xpcs_pdown_o signal is not registered ■ The contents of the “SR PMA MMD Devices in Package Register 1” and “SR PMA MMD Devices in Package Register 2” have been interchanged ■ The contents of the “SR XS or PCS MMD Devices in Package Register 1” and “SR XS or PCS MMD Devices in Package Register 2” have been interchanged ■ Added new signal “xgxs_ssc_ref_clk_sel_o[8:0]” and modified the signal description for “xgxs_los{lane}_ctl_o[1:0]” ■ Removed the xgxs_mpll_half_rate_o signal ■ Added the following new registers: ❑ “VR XS,PMA or MII Synopsys Consumer Gen4 PHY Rx Loss-of-Signal Control 0 Register” ❑ “VR XS,PMA or MII Synopsys Consumer Gen4 PHY Rx Loss-of-Signal Control 1 Register” on page 397 ❑ “VR XS,PMA or MII Synopsys Consumer Gen4 PHY Miscellaneous Control 2 Register” ■ Updated the programming guidelines in the following sections to separate out Synopsys Consumer Gen2 PHY and Synopsys Consumer Gen4 PHY information: ❑ “Programming Sequence for Synopsys Consumer Gen2 or Gen4 PHY” while switching to 10G XAUI or KX4 mode ❑ “Programming Sequence for Synopsys Consumer Gen2 PHY or Gen4 PHY or Non-Synopsys PHY” while switching to 10G RXAUI mode ❑ “Programming Sequence for Synopsys Consumer Gen2 or Gen4 PHY” while switching to 1G or KX mode

Version Number	Date	Description
3.10a	August 2013	■ DWC_xpcs now supports Synopsys Consumer Gen4 and Enterprise Gen5 10G PHY. ■ The following PHY names have been modified: ❑ Synopsys 3G PHY is now referred to as Synopsys Consumer Gen1 PHY ❑ Synopsys 6G PHY is now referred to as Synopsys Consumer Gen2 PHY ■ Added options to choose Synopsys Consumer Gen4 and Enterprise Gen5 10G PHYs in the Synopsys PHY Type configuration parameter ■ Added the following configuration parameter: Add SGMII PHY Automate ■ Added the following signal to existing signal table: rpcs_lpitx_alert_o ■ Modified the signal tables to add Synopsys Consumer Gen4 PHY-related signals: ■ The following registers have been added: ❑ VR XS,PMA or MII Synopsys Consumer Gen4 PHY Tx General Control Register 2 ❑ VR XS,PMA or MII Synopsys Consumer Gen4 PHY Tx General Control Register 3 ❑ VR XS, PMA, or MII Synopsys Consumer Gen4 PHY Rx Equalization Control Register ❑ VR XS, PMA or MII Synopsys Enterprise Gen5 10G PHY MPLL Status Register ❑ VR XS, PMA, or MII Synopsys Enterprise Gen5 12G PHY Rx Power State Control Register ❑ VR XS,PMA or MII Synopsys Enterprise Gen5 10G PHY Tx Control Register ❑ VR XS,PMA or MII Synopsys Enterprise Gen5 10G PHY Tx Status Register ❑ VR XS,PMA or MII Synopsys Enterprise Gen5 10G PHY Rx Control Register ❑ VR XS, PMA or MII Synopsys Enterprise Gen5 10G PHY Rx Status Register ❑ VR XS, PMA or MII Synopsys Enterprise Gen5 10G PHY MPLL Control Register ❑ VR XS, PMA or MII Synopsys Enterprise Gen5 10G General Control Register ❑ VR XS, PMA or MII Synopsys Enterprise Gen5 10G PHY Tx Power State Control Register ❑ VR XS, PMA or MII Synopsys Enterprise Gen5 10G PHY Rx Power State Control Register ❑ VR XS, PMA or MII Synopsys Enterprise Gen5 10G PHY Miscellaneous Control Register ❑ VR XS, PMA or MII Synopsys Enterprise Gen5 10G PHY Miscellaneous Status Register (Contd. on next page)

Version Number	Date	Description
		■ The following sections in Register Descriptions are modified to add Synopsys Consumer Gen4 PHY-related registers: ❑ VR XS, PMA, or MII Synopsys Consumer Gen2 PHY or Synopsys Consumer Gen4 PHY Tx Boost Control Register 0 ❑ VR XS, PMA, or MII Synopsys Consumer Gen2 PHY or Synopsys Consumer Gen4 PHY Tx Boost Control Register 1 ❑ VR XS, PMA, or MII Synopsys Consumer Gen2 PHY or Synopsys Consumer Gen4 PHY Tx Level Control Register 0 ❑ VR XS, PMA, or MII Synopsys Consumer Gen2 PHY or Synopsys Consumer Gen4 PHY Tx Level Control Register 1 ❑ VR XS, PMA, or MII Synopsys Consumer Gen2 PHY or Synopsys Consumer Gen4 PHY Tx General Control Register 0 ❑ VR XS, PMA, or MII Synopsys Consumer Gen2 PHY or Synopsys Consumer Gen4 PHY Tx General Control Register 1 ❑ VR XS, PMA, or MII Synopsys Consumer Gen2 or Synopsys Consumer Gen4 PHY Rx General Control Register 0 ❑ VR XS, PMA, or MII Synopsys Consumer Gen2 PHY or Synopsys Consumer Gen4 PHY Rx General Control Register 1 ❑ VR XS, PMA, or MII Synopsys Consumer Gen2 PHY or Synopsys Consumer Gen4 PHY MPLL Control 0 Register ❑ VR XS, PMA, or MII Synopsys Consumer Gen2 PHY or Synopsys Consumer Gen4 PHY MPLL Control 1 Register ❑ VR XS, PMA, or MII Synopsys Consumer Gen2 PHY or Synopsys Consumer Gen4 PHY Level Control Register ❑ VR XS, PMA, or MII Synopsys Consumer Gen2 PHY or Synopsys Consumer Gen4 PHY Miscellaneous Control 0 Register ❑ “VR XS, PMA, or MII Synopsys Consumer Gen2 PHY or Synopsys Consumer Gen4 PHY Miscellaneous Control 1 Register” ■ Modified the following registers: ❑ VR XS, PMA, or MII Synopsys Consumer Gen1, Gen2, Gen4 PHY or Synopsys Enterprise Gen5 PHY CR Data Register ❑ VR XS, PMA, or MII Synopsys Consumer Gen1, Gen2, Gen4 PHY or Synopsys Enterprise Gen5 PHY CR Control Register ❑ VR XS, PMA, or MII Synopsys Consumer Gen1, Gen2, Gen4 PHY or Synopsys Enterprise Gen5 PHY CR Address Register ❑ VR XS,PMA or MII Synopsys Enterprise Gen5 10G PHY Tx Control Register ❑ VR XS,PMA or MII Synopsys Enterprise Gen5 10G PHY Tx Status Register ❑ VR XS,PMA or MII Synopsys Enterprise Gen5 10G PHY Rx Control Register ❑ VR XS, PMA or MII Synopsys Enterprise Gen5 10G PHY Rx Status Register ❑ VR XS, PMA or MII Synopsys Enterprise Gen5 10G PHY MPLL Control Register (new register bit MPLL_EN added) ❑ VR XS, PMA or MII Synopsys Enterprise Gen5 10G General Control Register (new register bits LANE_PWR_OFF_3_1 and LANE_P-WR_OFF_0 added, and PHY_RESET bit is now reserved bit) (Continued on next page)

Version Number	Date	Description
		(Continued from previous page) ■ Added the following new programming flows: ❑ Added initialization sequences: “Initializing the DWC_xpcs Core” on page 1279 and “Initialization Guidelines for Synopsys Consumer Gen4 PHY” ❑ Added programming sequence for Synopsys Enterprise Gen5 10G PHY in “Switching to 10G XAUI Mode or KX4 Mode” ❑ Added programming sequence for Synopsys Enterprise Gen5 10G PHY in “Switching to 10G RXAUI Mode” ❑ Added programming sequence for Synopsys Enterprise Gen5 10G PHY in “Switching to 1G or KX Mode and 10 or 100 Mbps SGMII Speed” ❑ “Switching to 10G KR Speed Mode”
3.00a	December 2011	Added: ■ The part number in the cover page ■ The note in EEE Receive Controller and the note related to Clause 37 and Clause 73 auto-negotiation flows in Clause 73 Auto-Negotiation Process ■ The note in Programming Guidelines for Clause 37 Auto-Negotiation and Programming Guidelines for Clause 73 Auto-Negotiation Updated: ■ The Value Range field for many parameters in “Configuration Parameters” ■ The signal name from rpcs_tx_data_o[9] to xgxs_tx_data_o[9] in PCS-X Transmit Bit Ordering ■ Clock Rate Compensation Block ■ XGMII or GMII Loopback ■ The note in PCS-R Data Path, EEE Controller, and FEC Decoder ■ Bit Error Rate Monitor ■ MCI Interface with Indirect Access and APB3 Interface with Indirect Access ■ Step 2 in DME Page Transmitter ■ Auto-Negotiation CSR Interface Description of the following signals: xpcs_mii_ctrl_o, clk_mii_tx_i, clk_mii_rx_i, xgxs_tx_data{lane}_o[9:0], xpcs_loopback_en_o{lane}, xgxs_tx{lane}_margin_o[1:0], xgxs_mpll_init_ctl_o[2:0], rpcs_ktx_ceu{filter}_ack_i, rpcs_ktx_ceu{filter}_sts_i [1:0], rpcs_krx_ceu{filter}_ack_o ■ Description of the following registers: VR XS, PMA, or MII Synopsys Consumer Gen2 PHY or Synopsys Consumer Gen4 PHY Tx Boost Control Register 1, VR XS, PMA, or MII Synopsys Consumer Gen2 PHY or Synopsys Consumer Gen4 PHY Tx Level Control Register 1 (Contd. on next page)

Version Number	Date	Description
3.00a	December 2011	Continued from previous page ■ PCS-R Data Path ■ EEE Controller and Clause 72 KR Training ■ APB3 Interface ■ FEC Error Correction Memory Interface Signals ■ FEC Error Forward Memory Interface Signals ■ Clause 72 KR Training Signals ■ PCS-R PHY Interface ■ Initiate the Vendor specific software reset by writing 1'b1 to VR_RST bit [15] of VR XS or PCS MMD Digital Control1 Register ■ SGMII Auto-Negotiation ■ Programming Guidelines for Clause 73 Auto-Negotiation ■ Programming Guidelines for Energy Efficient Ethernet Updated: ■ Link Types Supported Through PMA Layer and List of Features ■ Parameters Description" Chapter ■ Clause 73 Auto-Negotiation Block o ■ PCS-X Data Path" ■ Power Sequence and Reset Generation Block ■ Management Interface ■ Power, Area, and Latency Restructured: ■ Register Description ■ Programming Guidelines

Version Number	Date	Description
2.11a	August 2010	Added: ■ The Reference Clock Frequency option and 1000BASEX-Only PCS in the Mode of Operation option in Configuration Parameter Chapter. ■ The sections in “GMII Rx RAL Block” in Architecture Chapter ■ The section System Interface. ■ The signals “clk_mii_tx_i” and “clk_mii_rx_i”. ■ The bit MII_CTRL in Register Table. ■ The section “Clocking Scheme for the SGMII Interface”. Updated: ■ The List of Features. ■ The description of the following parameters in Parameter Description Chapter: ❑ SUPPORT_1G, CL37_AN, CLKCOMP, EXT_DSKW, and LEEE_EN. ❑ XGMII, RXAUI, and RXAUI_DEFAULT. ■ The description of the Package, PMA Device, and AN device related parameters in Register Parameters section. ■ The description of the gmii_rxd_o and gmii_rx_er signals in Signals Chapter. ■ The description of the following bits in Register Description Chapter: ❑ DTXLANED, CR_CJN, and DSKBYP. ❑ TP_MIS_EN, DSKW_EC_EN, and CHKEND_EC_EN. ❑ INV_XGMII_CHAR, INV_XGMII_T, and INV_XGMII_SOP. ❑ EC1, EC3, EC5, and EC5. ❑ TX_DONE[3:0], RX_DPLL_MODE3, MPLL_NCY, MPLL_NCY5, MPLL_PRESCALE, MPLL_MULTIPLIER. ❑ SGMII_LINK_STS, TX_CONFIG, PCS_MODE, and CLS37_AN_SGMII_STS. ■ The CSR address maps in Register Chapter and grouped them based on the MMDs. ■ The section “GMII Interface”. ■ The sections in Programming Guidelines Chapter. ■ The Tables A-1, A-2, and A-3 in Power Area, and Latency Chapter.

Version Number	Date	Description
2.10a	June 2010	Added: ■ The sections on XAUI and RXAUI ■ The new parameters in System Configuration section ■ The section Clause 37 Auto-Negotiation Block ■ The signals specific to Synopsys 6G SerDes, RXAUI, and Clause 37 auto-negotiation support in Signals chapter. ■ The registers specific to Synopsys 6G SerDes, RXAUI, Clause 37 auto-negotiation, and Energy Efficient Ethernet support in Register Description Chapter ■ The section “Clocking and Resetting with Synopsys 6G SerDes” and “Data and Clock Multiplexing Requirements for XAUI and RXAUI” ■ Programming Guideline Chapter ■ An appendix A on Area, Power, and Latency. This information was present in Chapter 1 in previous versions. Updated: ■ The section List of Features. ■ The Control and Status Memory Map ■ The sections “Timing Diagrams” on page 305, “Clocking and Resetting With Synopsys Consumer Gen1 SerDes” on page 312, and “Clocking and Resetting with Non-Synopsys SerDes” on page 315 ■ Workspace Directory and Files
2.00a	August 2008	Added ■ Bit-swapping support on 10-bit XAUI PHY interface ■ User programmable number of Lanes/Channels ■ inverter to XGXS Rx clk input to ease synthesis ■ Backplane Ethernet -negotiation support and 1G support Updated ■ Modifying the ASYNC FIFO used in clock rate compensation block ■ Correcting the width of Device and Package OUI number ■ VR-RST bit placed at incorrect register ■ Insufficient Read and Write pointer offset in Deskew module ■ Registering the data output at XGMII in case of DDR and DDW mode
1.00a	May 2006	Initial release

Preface

This document describes Synopsys' DesignWare® 10G Ethernet, which extends the operational distance of the 1 Gigabit or 10 Gigabit Media-Independent Interface while reducing the number of interface signals required, in conformity with the IEEE 802.3 specification.

The Ethernet PCS corresponds to DWC Ether PCS in the SolvNetPlus database.

This chapter has the following sections:

- "Reference Documentation" on page 22
- "Web Resources" on page 23
- "STAR on the Web (SotW)" on page 23
- "Synopsys Statement on Inclusivity and Diversity" on page 23
- "Customer Support" on page 23

Reference Documentation

- IEEE 802.3-2018, Clause 35, 36, 37, 45, 46, 47, 48, 49, 69, 72, 73, 74, and 78.
- Reuse Methodology Manual for System-on-a-Chip Designs by Michael Keating and Pierre Bricaud (Editors). Springer; 3rd edition, June 30, 2002.
- RXAUI - Reduced Pin XAUI specification, Dune Networks, January 2008.
- RXAUI Interface and RXAUI Adapter Specifications, Marvell, June 2008.
- DesignWare Cores® XAUI PHY for TSMC 40-nm LP 1.1/2.5 V Databook, November 2012 (Consumer multi-protocol 6G PHY).
- DesignWare® Cores Enterprise 12G PHY for TSMC 16FFLL Databook, February 2014, (Enterprise Gen5 12G PHY).
- DesignWare® Cores Enterprise 12G PHY for GF 14 LPP Databook, November 2014, (Enterprise Gen5 12G PHY).
- DesignWare® Cores Enterprise 12G PHY for TSMC 28 HPC Databook, February 2016. (Enterprise Gen5 12G PHY)
- DesignWare® Cores Enterprise 16G PHY for TSMC 16FFPGL, April 2016
- DesignWare Cores multi-protocol 32G PHY for TSMC 7FF Databook, June 2019
- Serial-GMII (SGMII) Specification, Revision 1.8, Cisco Systems
- QSGMII Specification, Revision 1.3, Cisco Systems
- AMBA 3 APB Protocol, version 1.0, ARM Limited
- Universal SXGMII PHY-MAC Interface for Single and Multiple Network Port: version 2.0, dated 08/21/2015
- Universal SXGMII PHY-MAC Interface for Multiple Network Port:version2.15,dated 07/30/2019
- Universal SXGMII Interface for a Single MultiGigabit CopperNetworkPort:version 2.4, dated 07/30/2019
- IEEE P802.3cb-20xx™/D3.0, 22 May 2017

Web Resources

- DesignWare IP product information: https://www.synopsys.com/designware-ip.html
- Your custom DesignWare IP page: https://www.synopsys.com/dw/mydesignware.php
- Documentation through SolvNetPlus: https://solvnetplus.synopsys.com (Synopsys password required)
- Synopsys Common Licensing (SCL): https://www.synopsys.com/support/licensing-installation-computeplatforms/licensing.html

STAR on the Web (SotW)

You must review all STARs on the Web (SotWs) associated with your product. SotWs are considered a part of the Synopsys documentation suite, and show critical information related to your product. To review product SotWs, refer to the DesignWare IP product information:

https://www.synopsys.com/designware-ip.html

Synopsys Statement on Inclusivity and Diversity

Synopsys is committed to creating an inclusive environment where every employee, customer, and partner feels welcomed. We are reviewing and removing exclusionary language from our products and supporting customer-facing collateral. Our effort also includes internal initiatives to remove biased language from our engineering and working environment, including terms that are embedded in our software and IPs. At the same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IPs implement industry-standard specifications that are currently under review to remove exclusionary language.

Customer Support

To obtain support for your product, prepare the required files and contact the support center using one of the methods described:

- Prepare the following debug information, if applicable:
  - For environment set-up problems or failures with configuration, simulation, or synthesis that occur within coreConsultant or coreAssembler, select the following menu:
    - **File > Build Debug Tar-file**

      Check all the boxes in the dialog box that apply to your issue. This option gathers all the Synopsys product data needed to begin debugging an issue and writes it to the <*core tool startup directory*>/debug.tar.gz file.
  - For simulation issues outside of coreConsultant or coreAssembler:
    - Create a waveform file (such as VPD or VCD).
    - Identify the hierarchy path to the DesignWare instance.
    - Identify the timestamp of any signals or locations in the waveforms that are not understood.

- For the fastest response, enter a case through SolvNetPlus:

  a. https://solvnetplus.synopsys.com

  > **Note** SolvNetPlus does not support Internet Explorer. Use a supported browser such as Microsoft Edge, Google Chrome, Mozilla Firefox, or Apple Safari.

  b. Click the **Cases** menu and then click **Create a New Case** (below the list of cases).

  c. Complete the mandatory fields that are marked with an asterisk and click **Save**.

  Make sure to include the following:

  - **Product L1:** DesignWare Cores
  - **Product L2:** 10G Ethernet

  d. After creating the case, attach any debug files you created.

  For more information about general usage information, refer to the following article in SolvNetPlus:

  https://solvnetplus.synopsys.com/s/article/SolvNetPlus-Usage-Help-Resources

- Or, send an e-mail message to mailto:support_center@synopsys.com (your email will be queued and then, on a first-come, first-served basis, manually routed to the correct support engineer):
  - Include the Product L1 and Product L2 names, and Version number in your e-mail so it can be routed correctly.
  - For simulation issues, include the timestamp of any signals or locations in waveforms that are not understood.
  - Attach any debug files you created.
- Or, telephone your local support center:
  - North America:

    Call 1-800-245-8005 from 7 AM to 5:30 PM Pacific time, Monday through Friday.
  - All other countries:

    https://www.synopsys.com/support/global-support-centers.html

1

Product Overview

This chapter introduces the Synopsys DesignWare Cores Ethernet PCS (DWC_xpcs) features and deliverables. It contains the following sections:

- "General Product Description" on page 26
- "PCS Layer Overview" on page 27
- "Link Types Supported Through PMA Layer" on page 35
- "List of Features" on page 38
- "Deliverables" on page 41
- "Standards Compliance" on page 42

1.1 General Product Description

The Physical Coding Sublayer (PCS) layer provides an interface between the Media Access Control (MAC) and Physical Medium Attachment Sublayer (PMA) through a Media independent interface. The following are these interfaces:

- GMII: 1000BASE-X PCS

  The GMII interface is defined for a single lane which operates at 125 MHz to support 1000BASE-X PMA.

- XGMII: 10GBASE-X PCS or 10GBASE-R PCS or 5GBASE-R PCS or 2.5GBASE-X PCS

  For 10GBASE-X PCS, the XGMII interface is defined to be four lanes. It interfaces with 10GBASE-X PMA with 4-lanes working at 10 Bits x 312.5 MHz. For 10GBASE-R/5GBASE-R PCS also, the XGMII interface is defined to be four lanes. However, it interfaces with 10GBASE-R/5GBASE-R PMA with single lane working at 16 Bits x 644.53 MHz/322.26 MHz.

To maintain transition density and DC balancing, 1000BASE-X, 2.5GBASE-X, and 10GBASE-X PCS use 8B/10B encoding/decoding and 10GBASE-R/5GBASE-R PCS use 64B/66B encoding or decoding and scrambling technique. The DWC_xpcs is designed to support the following derivatives of the PCS layer:

- XGXS PCS
- 10GBASE-X PCS
- 10GBASE-R/5GBASE-R PCS with optional USXGMII support
- 10G, 5G, 2.5G, or 1G PCS with Backplane Ethernet auto-negotiation support
- 1000BASE-X and SGMII /QSGMII with Clause 37 auto-negotiation support

1.2 PCS Layer Overview

This section contains the description of the PCS layer, including auto-negotiation and Backplane Ethernet.

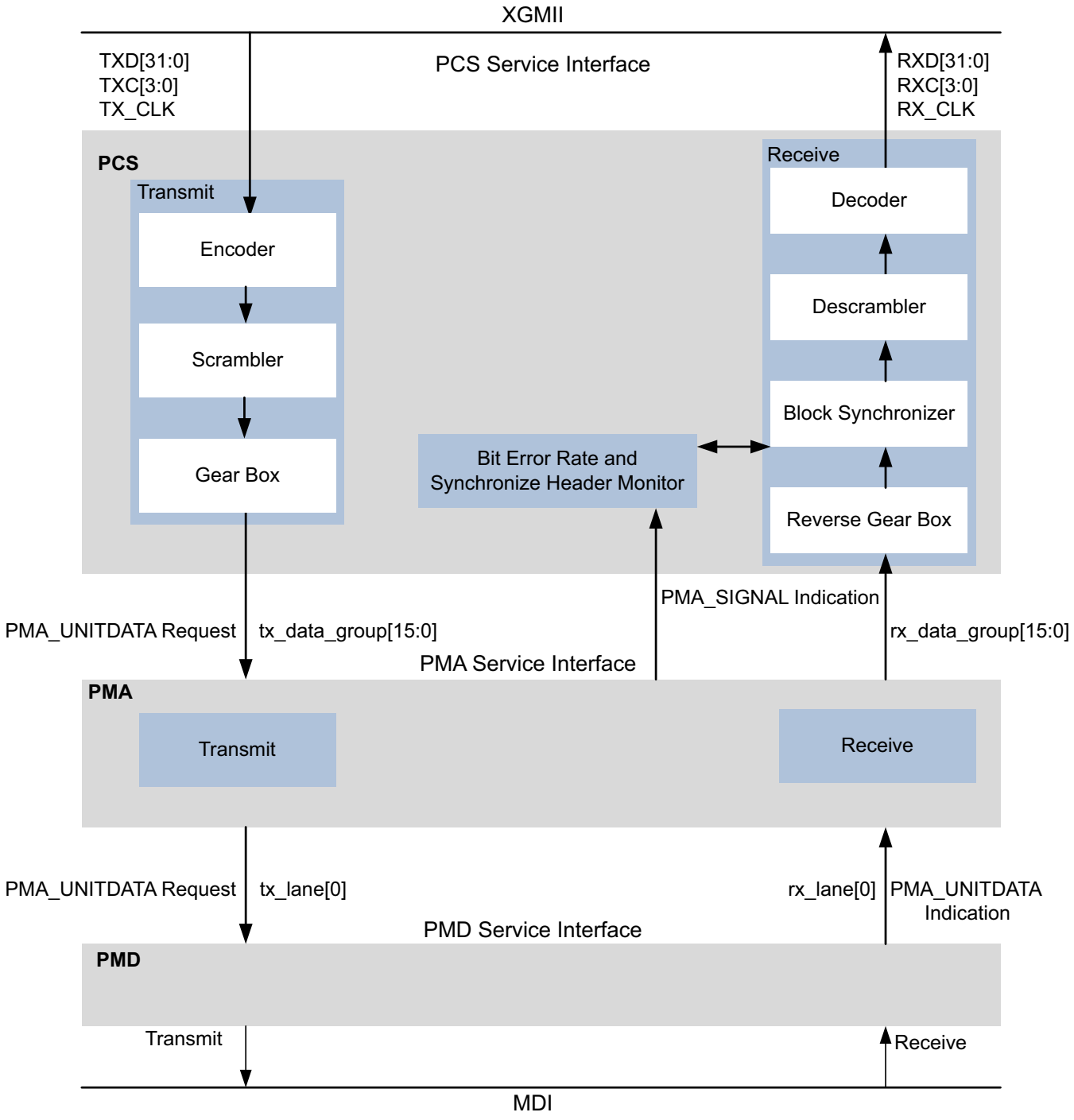
1.2.1 10GBASE-R or 5GBASE-R PCS Layer

The 10GBASE-R or 5GBASE-R PCS layer has the following main blocks:

- Transmit: The Transmit block interfaces with the XGMII 32-bit data and four control lines at the clock rate of 312.5 MHz. The Transmit block converts two XGMII data blocks into 66-bit blocks (64-bit data + 2-bit Sync Header) at the rate of 156.25 MHz (for 10GBASE-R) or 78.125 MHz (for 5GBASE-R). The data part of encoded 66-bit block is scrambled to achieve required transition density. The encoded and scrambled data is passed to the PMA (SerDes). The PMA serializes this data at 10.3125 Gbps (for 10GBASE-R) or 5.15625 Gbps (for 5GBASE-R) data rate.
- Receive: The Receive block receives the serialized self-clocked data from the PMA (SerDes). The clock and data recovery are done in the PMA and the unaligned data is presented to the PCS Receive block. In the PCS Receive block, 66-bit data boundary detection is done by successive validation of sync-header pattern of the 66-bit encoded data. The block aligned data is decoded through 64/66 decoder. The aligned and decoded data is forwarded over XGMII Interface (32-bit data and four control lines).

Figure 1-1 shows the 10GBASE-R or 5GBASE-R physical layer functional block diagram.

Figure 1-1 DWC_xpcs Functional Block Diagram: 10GBASE-R or 5GBASE-R PHY



1.2.2 10GBASE-X PCS Layer

The 10GBASE-X PCS layer has the following main blocks:

- Transmit: The Transmit block interfaces with the XGMII 32-bit data and four control lines at the clock rate of 312.5 MHz. The Transmit block converts the XGMII idle control characters to randomized sequence of code groups to enable the following:
  - Serial lane synchronization
  - Clock rate compensation
  - Code group alignment

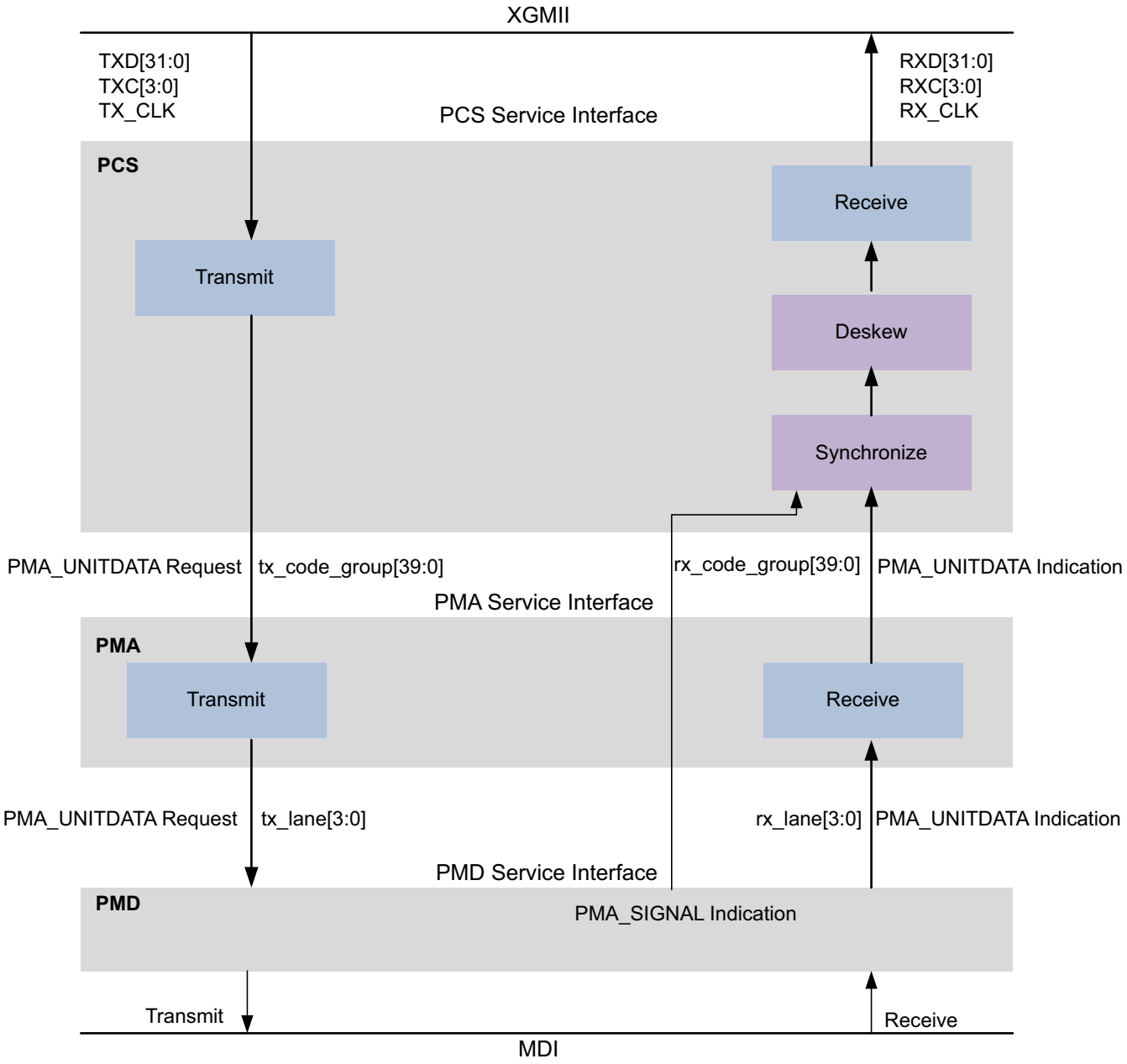
  The Transmit block performs 8B/10B encoding for each of the 8-bit data and passes each of the 10 bits per-lane to the PMA (SerDes). The PMA serializes the 10-bit data on each lane at the data rate of 3.125 Gbps.
- Receive: The Receive block receives the serialized self-clocked data from the PMA (SerDes). The clock and data recovery are done in the PMA and the raw 40-bit unaligned data is presented to the PCS Receive block.

  In the PCS Receive block, 10-bit data boundary detection is done through comma |K| pattern on each lane, and synchronization is done based on received |VALID| code groups to see if the underlying PMA is working reliably on individual lanes. The synchronized data is decoded through 10B/8B decoder on each lane. It is then passed on for lane-to-lane deskewing where non-coherency among lanes is removed and the aligned data is recovered. The synchronized, aligned, and decoded data is forwarded over the XGMII Interface (32-bit data and 4 control lines).

Note Auto-negotiation is not defined for 10GBASE-X physical layer. In addition, the half-duplex operation is not defined for 10GBASE-X. Therefore, the CRS and COL signals generation blocks are not required.

Figure 1-2 shows the 10GBASE-X physical layer functional block diagram.

Figure 1-2 DWC_xpcs Functional Block Diagram: 10GBASE-X PHY



1.2.3 1000BASE-X PCS Layer

The 1000BASE-X PCS layer has the following main blocks:

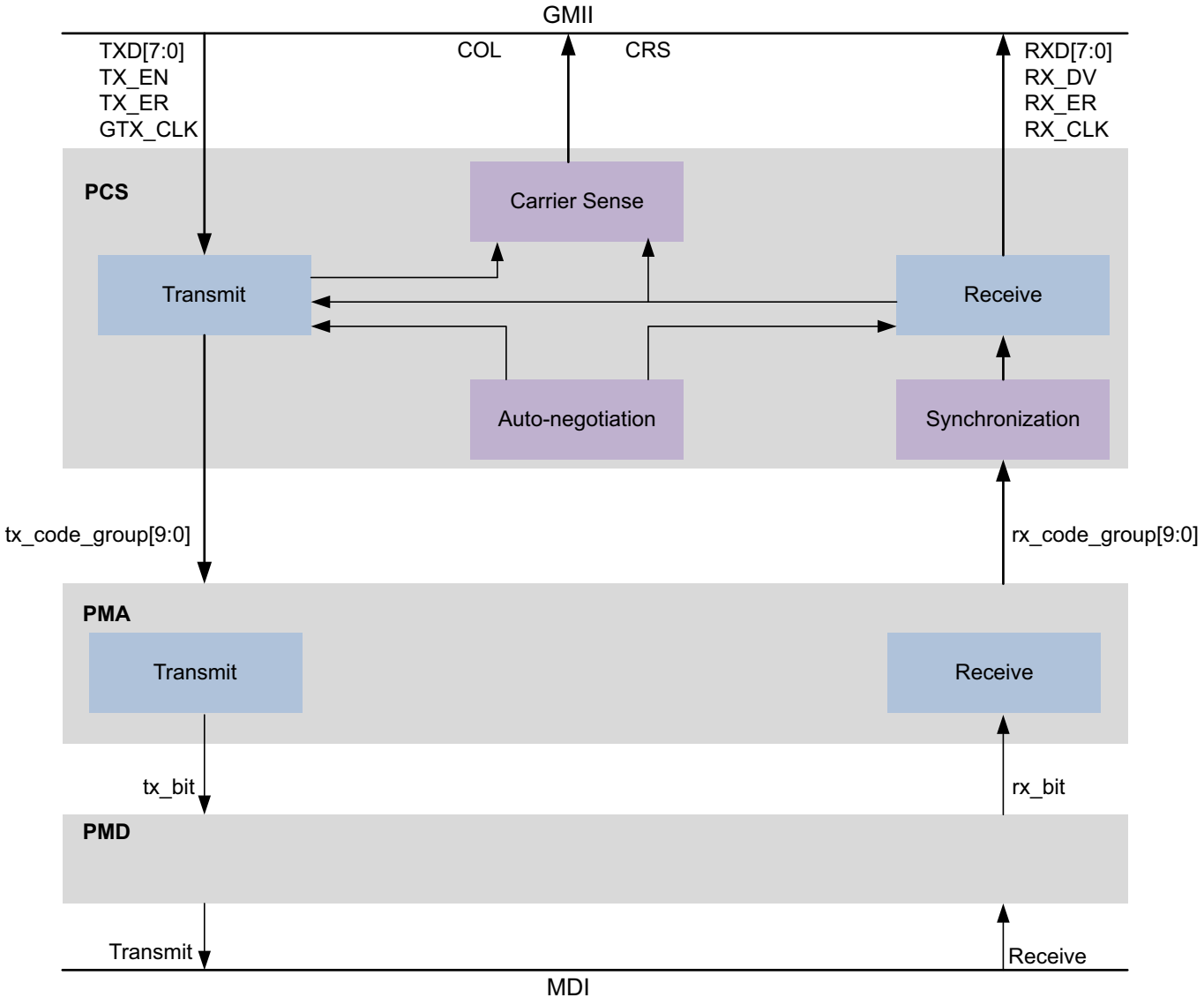
- The Transmit block interfaces with the GMII 8-bit data and two control lines at clock rate of 125 MHz. It performs 8B/10B encoding. Data encoding is controlled through GMII interface and Clause 37 auto-negotiation block. In addition to encoding the normal data, the Transmit block also encodes several control order sets (as defined IEEE Std 802.3, Table 36-3) such as |C|, |I|,|R|, |S|, |T|, and |V| order sets. The Transmit block sends the 10-bit encoded data to the PMA Serializer-Deserializer (SerDes) which performs the serialization at the rate of 1.25 Gbps.

- The Receive block receives the serialized self-clocked data from the PMA (SerDes). The clock recovery and deserialization are done in the PMA. The 10-bit unaligned or aligned data is given to the PCS Receive block. The PCS Receive block performs the following functions:
  - Comma-detection if the data is not aligned
  - Synchronization on the incoming data to determine whether the link is reliable or not
  - 10B/8B PCS decoding
  - Mapping of the decoded data to the GMII interface and the auto-negotiation block

Note	The auto-negotiation is achieved through /C/ order sets in the PCS layer itself. This auto-negotiation is defined to operate only with another 1000BASE-X-based PHY.

Figure 1-3 shows the 1000BASE-X physical layer functional block diagram.

Figure 1-3 DWC_xpcs Functional Block Diagram: 1000BASE-X PHY



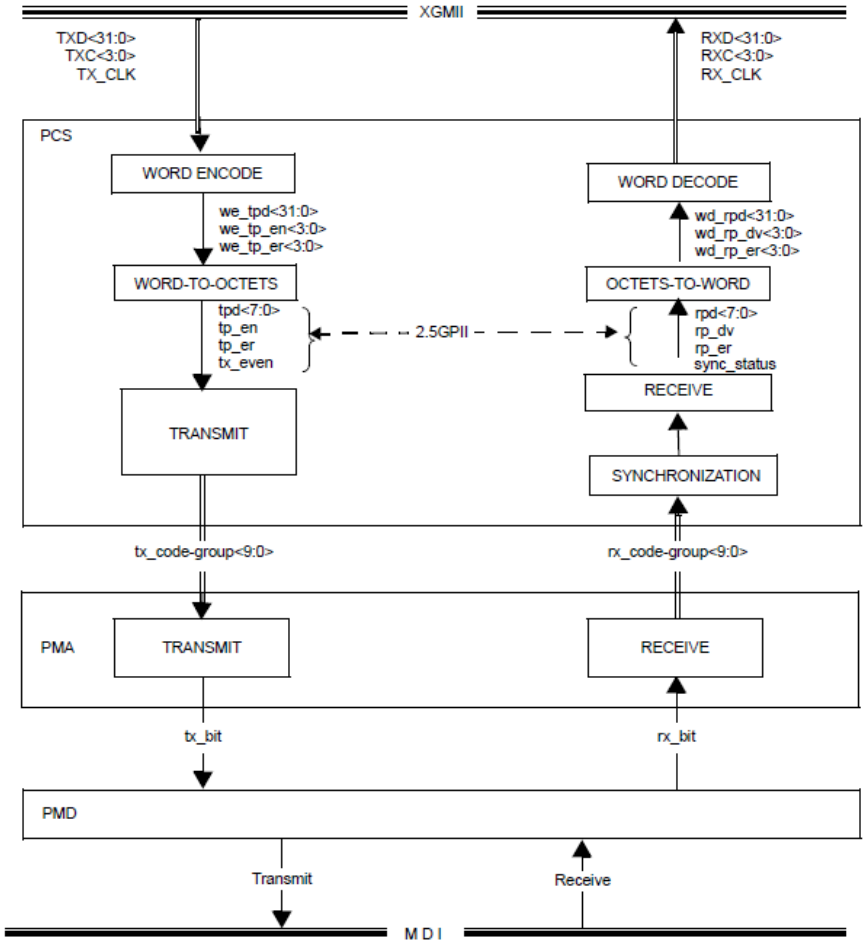
1.2.4 2.5GBASE-X PCS Layer

The 2.5GBASE-X PCS layer has the following main blocks:

- The Transmit block interfaces with the XGMII Tx interface at clock rate of 78.125 MHz. The XGMII data is first translated to 2.5GPII format (tpd<7:0>, tp_en, tp_er) using Word-Encode and Word-to-Octet functions. The Transmit block performs 8b/10b encoding of the 2.5GPII data. In addition to encoding the normal data, the transmit block also encodes several control order sets (as defined in IEEE Std 802.3-cb, Table 127-5) such as /I/, /R/, /S/, /T/, /Q/and /V/ order sets. The Transmit block sends the 10-bit encoded data to the PMA (SerDes) which performs the serialization at the rate of 3.125 Gbps.

- The Receive block receives the serialized self-clocked data from the PMA (SerDes). The clock recovery and deserialization are done in the PMA. The 10-bit unaligned or aligned data is given to the PCS Receive block. The PCS Receive block performs the following functions:
  - Comma-detection, if the data is not aligned
  - Synchronization on the incoming data to determine if the link is reliable or not
  - 10B/8B PCS decoding
  - Octet-to-Word Conversion: Converts the 2.5GPII Rx Octet Data to Word format
  - Word Decode: Converts the Word data to XGMII Rx data

Figure 1-4 DWC_xpcs Functional Block Diagram: 2.5GBASE-X Physical Layer



1.2.5 Auto-Negotiation Overview

Auto-negotiation enables a device to understand capabilities and status of the remote-end device. Auto-negotiation allows devices to advertise and share information, including speed, modes, fault signaling, and other control information.

- Auto-negotiation is defined in Clause 28 for any speed BASE-T (twisted-pair copper) PHY device and Clause 36 and Clause 37 for 1000BASE-X (based on PCS |C| order set). Auto-negotiation is not defined for 10GBASE-X and 10GBASE-R.

- Backplane Ethernet is defined to include the extension of 1000BASE-X, 2.5GBASE-X, 5GBASE-R, 10GBASE-X, and 10GBASE-R called 1000BASE-KX, 2.5GBASE-KX, 5GBASE-KR, 10GBASE-KX, and 10GBASE-KR, ap-2007/802.3cb-2017, Clause 73.

  During auto-negotiation, the information is exchanged through serial Differential Manchester Encoding (DME) data bits. The serial data pages, called Link Code Words, are sent through newly defined DME pages (as defined in IEEE Std 802.3ap/802.3cb, Clause 73).

1.2.6 PCS Layer with Backplane Ethernet Support

The Backplane Ethernet is defined in Clause 73 of the IEEE Std 802.3. This clause defines a new common auto-negotiation protocol which uses the signaling that is independent of the standard speed modes (10GBASE-KX4, 10GBASE-KR, 5GBASE-KR, 2.5GBASE-KX, or 1000BASE-KX). The links can communicate with each other to determine the link speeds during auto-negotiation. There is a provision in the auto-negotiation to detect the legacy links which support auto-negotiation. The following links are defined for the Backplane applications using Ethernet technology:

- 10GBASE-KX4 is compatible with the XAUI and 10GBASE-X links.
- 10GBASE-KR is compatible with the 10GBASE-R link.
- 1000BASE-KX is compatible with the SGMII and 1000BASE-X links.
- 5GBASE-KR is compatible with 5GBASE-R link
- 2.5GBASE-KX is compatible with 2.5GBASE-X link

The IEEE Std 802.3 also defines special equalization protocol called Clause 72 KR Training and a Forward Error Correction (FEC) block for 10GBASE-KR links to maximize the link reliability. This is because the 10GASE-KR links work at a very high speed (10.3125 Gbps) on an electrical media.

To support the Backplane Ethernet auto-negotiation without any additional functions in the PHY, the PCS layer implements the following additional functions:

- Clause 73 auto-negotiation (Rx, Tx, and Arbiter) state machine at the parallel PHY Interface
- DME page decoding and encoding at the parallel PHY Interface
- Auto-negotiation Manageable Device (MMD) for Management Interface register sets to handle auto-negotiation
- Switching between the 1000BASE-KX, 2.5GBASE-KX, KX4, 10GBASE-KR and 5GBASE-KR modes based on the auto-negotiation outcome when supporting multiple operating modes
- Parallel detection for legacy devices which do not support Clause 73 auto-negotiation

Note

In the Backplane Ethernet PCS configurations, the parallel detection for legacy devices has the following limitations:

- In KX or KR mode, the detection of the legacy KX4 device is not supported.
- In KX4 or KR mode, the detection of the legacy KX device is not supported.
- In KX4 or KX mode, the detection of the legacy KR device is not supported.

This is because the DWC_xpcs assumes that the PMA layer is common for all speed modes, and it supports a single speed mode at a time.

1.3 Link Types Supported Through PMA Layer

The DWC_xpcs supports the following link types through PMA layer:

- XAUI
- RXAUI
- 1000BASE-X or SGMII
- 10GBASE-X
- 10GBASE-R
- 10GBASE-KX4, 10GBASE-KR, 5GBASE-KR, 2.5GBASE-KX, 1000BASE-KX
- QSGMII
- USXGMII

1.3.1 XAUI

The 10 Gigabit Attachment Unit Interface (XAUI) interface is defined in IEEE Std 802.3, Clause 47. The XAUI interface is the solution to extend the XGMII interface between the 10G MAC and the PHY as a chip-to-chip interface. Because of its high signal density and speed, the XGMII interface is limited to short distances (less than 7 cm) on the boards. In addition, it is also difficult to route such wide buses with minimal skews between the chips. Therefore, you can use the XAUI interface to serialize the XGMII signals (four physical differential serial lines) and extend the connectivity distance (50 cm) between the 10G MAC and the PHY.

XAUI is a full-duplex interface with four physical differential lanes. These lanes work at the speed of 3.125 Gbps (with specific electrical requirements defined in the standard). These lanes are also referred to as 4x3.125 Gbps lanes. The effective data transfer rate, considering the overhead of 8B/10B encoding or decoding, is 4x2.5 Gbps (10G data transfer rate).

The DWC_xpcs supports the XAUI PHY interface that has the following:

- Four lanes (each of 10-bit data width)
- Clock rate of 312.5 MHz

1.3.2 RXAUI

The Reduced XAUI (RXAUI) is also a chip-to-chip interface between the 10G GMAC and the PHY. As compared to XAUI, the RXAUI interface uses reduced number of serial lines at higher speeds. The RXAUI interface, a non-IEEE standard, is defined separately by Dune Networks and Marvell. The RXAUI is a full-duplex interface with two physical differential lanes. These lanes work at the speed of 6.25 Gbps and are also referred to as 2x6.25 Gbps lanes. The effective data transfer rate, considering the overhead of 8B/10B encoding or decoding, is 2x5 Gbps (10G data transfer rate)

The DWC_xpcs supports the RXAUI PHY interface (defined by Dune Networks as well as Marvell) that has the following:

- Two lanes (each of 20-bit data width)
- Clock rate of 312.5 MHz

1.3.3 1000BASE-X or SGMII

The 1000BASE-X PCS functionality is defined in IEEE Std 802.3, Clause 36. The DWC_xpcs supports 1000BASE-X PCS functionality with single physical lane working at 1.25 Gbps. The effective data rate, considering the 8B/10B encoding or decoding, is 1 Gbps (1 GB data transfer rate)

The DWC_xpcs supports the 1000BASE-X or SGMII PHY interface that has the following:

- One lane working at 10-bit data width
- Clock rate of 125 MHz

1.3.4 10GBASE-X

The 10GBASE-X PCS layer is specified for 10 Gbps PHYs which transfer data across four serial lines. The PHY can be of following two types: Optical (10GBASE-LX4) or Copper (10GBASE-CX4). The 10GBASE-CX4 supports the link distance of up to 15 meters on twin-axial cables by adopting suitable electrical signaling.

The DWC_xpcs supports the 10GBASE-X PCS functionality and provides the XGMII interface to the 10G MAC and four lanes of 10-bit interface to the PHY. The DWC_xpcs supports the 10GBASE-X PHY interface that has the following:

- Four lanes (each of 10-bit data width)
- Clock rate of 312.5 MHz

1.3.5 10GBASE-R

The 10GBASE-R PCS layer is defined mainly for the optical fiber media. The 10GBASE-R link is deployed for 10 Gbps LAN or WAN applications. The DWC_xpcs provides the 10GBASE-R PCS functionality to interface with XGMAC (through XGMII interface) on one side and with the PHY (supporting 10GBASE-R serial link) on the other side. The 10GBASE-LR, 10GBASE-SR, 10GBASE-ER, and 10GBASE-LRM are various derivatives of 10GBASE-R using different wave lengths on the serial link.

The DWC_xpcs supports the 10GBASE-SR, 10GBASE-LR, 10GBASE-ER, and 10GBASE-LRM serial interfaces through appropriate PHY to achieve the serial data rate of 10.3125 Gbps. The DWC_xpcs can support any one of the following interfaces to the PHY:

- 16-bit data at 644.53 MHz clock rate (XSBI)
- 20-bit data at 515.625 MHz clock rate
- 32-bit data at 322.265 MHz clock rate
- 33-bit data at 312.5 MHz clock rate
- 64-bit data at 161.1328125 MHz clock rate
- 66-bit data at 156.25 MHz clock rate
- 40-bit data at 257.8125 MHz clock rate

1.3.6 10GBASE-KX4, 10GBASE-KR, 5GBASE-KR, 2.5GBASE-KX, 1000BASE-KX

The 10GBASE-KX4, 10GBASE-KR, 5GBASE-KR, 2.5GBASE-KX, and 1000BASE-KX links are defined for the Backplane applications which use the Ethernet technology. The 10GBASE-KX4 link is functionally compatible with the XAUI and 10GBASE-X links. The 10GBASE-KR link is compatible with the 10GBASE-R links and the 1000BASE-KX link is compatible with the SGMII and 1000BASE-X links. 5GBASE-KR link is similar to 10GBASE-KR link except that it operates at half the rate (baud-rate of 5.15625Gbps).

2.5GBASE-KX is a single-lane link which operates at a baud-rate of 3.125Gbps. The electrical characteristics of the Backplane Ethernet are different to support distance of more than 1 meter on a printed circuit board (including two connectors).

The IEEE Std 802.3, Clause 73 defines a new common auto-negotiation protocol which uses the DME signaling. The DME signaling is independent of any of the standard speed modes (10GBASE-KX4, 10GBASE-KR, 5GBASE-KR, 2.5GBASE-KX, and 1000BASE-KX). Therefore, the links can communicate with each other to determine the link speeds during auto-negotiation. There is also a provision in the auto-negotiation to detect the legacy links which do not support the Clause 73 auto-negotiation.

The IEEE Std 802.3 also defines a special equalization protocol called Clause 72 KR Training and Forward Error Correction block for 10GBASE-KR links to maximize the link reliability. This is because KR needs to work at a very high speed (10.3125 Gbps) on an electrical media.

1.3.7 QSGMII

The Quad Serial Gigabit Media Independent Interface (QSGMII) is an extension of SGMII. It is a chip-to-chip interface between a 4-port MAC (or 4 MAC s) and a 4-port PHY. Each port of the MAC can operate at a data rate of 10Mbps, 100Mbps or 1Gbps. The data from the 4 ports are multiplexed on to a single serial line operating at a baud rate of 5Gbps (4 x 1.25 Gbps). In this mode, DWC_xpcs interfaces with a 4-port MAC (or a 4-port PHY) on one-side (using 4 GMII interfaces) and with a SERDES on the other side (with a 20-bit interface operating at 250MHz).

1.3.8 USXGMII

The Universal Serial 10 Gigabit Media Independent Interface (USXGMII) is a chip-to-chip interface meant to transfer data between a 10M/100M/1G/10G MAC (one or many) and PHY (one or many) via a serial line operating at 10.3125 Gbps/5.15625 Gbps/2.578125 Gbps. This protocol leverages the BASE-R PCS layer defined by IEEE 802.3, Clause 49. DWC_xpcs supports the following variants of USXGMII:

- 10G-SXGMI
- 5G-SXGMII
- 2.5G-SXGMII
- 10G-DXGMII
- 5G-DXGMII
- 10G-QXGMII

1.4 List of Features

The following features are implemented in the DWC_xpcs:

- System level features
  - XGXS PCS:
    - XGXS DTE configuration with DTE_XS MMD
    - XGXS PHY configuration with PHY_XS MMD
    - XAUI and RXAUI interface support
    - Optional 1G mode support
    - Clause 37 auto-negotiation support in 1G mode
    - Optional SGMII/QSGMII interface support
  - Backplane Ethernet PCS:
    - KX4_KX configuration
    - KX4_Only configuration
    - KX_Only configuration
    - KR_Only configuration
    - KR_KX4 configuration
    - Optional support for 5GBASE-KR Mode
    - Optional support for 2.5GBASE-KX Mode
    - Forward Error Correction (FEC) in the 10GBASE-KR mode
    - Clause 72 KR Training support in the 10GBASE-KR mode
    - Clause 73 auto-negotiation support
    - Optional USXGMII support
    - Optional support for Clause 37 auto-negotiation in 1000BASE-KX mode
    - Auto-negotiation MMD
    - PMA and PMD MMD
    - PCS MMD
    - RXAUI interface support
    - (Optional) 1G mode support
    - Clause 37 auto-negotiation support in 1G mode
    - Optional SGMII/QSGMII interface support
    - Optional USXGMII support
  - 10GBASE-X:
    - PCS MMD
    - PMA MMD support when configured for Synopsys PHY
    - RXAUI interface support
    - Optional 1G mode support

        - Clause 37 auto-negotiation support in 1G mode
        - Optional SGMII/QSGMII interface support
    - Configurable Management Interface:
        - MDIO (As defined in IEEE Std 802.3, Clause 45)
        - MCI vendor-specific parallel interface
        - AMBA APB-3 Interface (as defined in AMBA 3 APB Protocol, version 1.0)
        - AMBA APB-3 (16-bit register)
        - AMBA APB-3 (32-bit register)
- Configurable MAC Interface:
    - For 10G (XGXS PCS, KX4, KR, 10GBASE-X PCS, 10GBASE-R PCS) XGMII Interface:
        - Single Data Rate (SDR): Single clock edge 312.5 MHz 32-bit data, 4 control line interface
        - Double Data Rate (DDR): Dual clock edge 156.25 MHz 32-bit data, 4 control line interface
        - Dual Data Width (DDW): Single clock edge 156.25 MHz 64-bit data, 8 control line interface
    - For 1G mode (KX_Only and 1000BASEX-Only):
        - GMII interface
    - For 10G or 1G mode:
        - GMII interface as a separate parallel interface along with XGMII interface
        - GMII interface shared with the XGMII interface signals
- Configurable Lane Control support for 10G modes (XGXS PCS, KX4, and 10GBASE-X PCS)
    - By default, four lanes exist, supporting the following combinations:
        - One lane support: 2.5 Gbps throughput
        - Two lanes support: 5 Gbps throughput
        - Three lanes support: 7.5 Gbps throughput
        - Four lanes support: 10 Gbps throughput
    - Dynamic switching between different speeds
- Configurable (Synopsys or Non-Synopsys) PHY support
    - Synopsys multi-protocol 6G PHY (see DesignWare Cores® XAUI PHY for TSMC 40-nm LP 1.1/2.5 V Databook)
    - Synopsys multi-protocol 12G PHY (see DesignWare® Cores Multi-Protocol 12G PHY for TSMC 16FFLL Databook or DesignWare® Cores Multi-Protocol 12G PHY for GF 14 LPP Databook, November 2014 or DesignWare® Cores Multi-Protocol 12G PHY for TSMC 28 HPC Databook, February 2016)
    - Synopsys multi-protocol 16G PHY (*see DesignWare®Cores Multi-Protocol 16G PHY for TSMC 16FFPGL Databook, April 2016*)
    - Synopsys multi-protocol 25G PHY (*see DesignWare®Cores Multi-Protocol 25G PHY for GF 14LPP Databook, February 2018*)
- Configurable Device and Package Identifier (OUI numbers)
- Vendor-specific Register sets (VR) to control the PCS functions and Synopsys PHY functionality

- Selectable lane-to-lane deskewing range (40UI/80UI) in the XGXS PCS, KX4, and 10GBASE-X PCS modes
- HFTP, LFTP, MFTP, CRTP, and CJTP test pattern generation in the XGXS PCS, KX4, and 10GBASE-X PCS modes
- HFTP, LFTP, and MFTP test patterns checking in the XGXS PCS, KX4 and 10GBASE-X PCS modes
- HFTP, LFTP and MFTP test pattern generation in 1000BASEX_Only PCS mode
- PRBS 31, PRBS 9, pseudo-random, and square wave test pattern generation in the 10GBASE-R and 10GBASE-KR modes
- PRBS-31, PRBS-9, and pseudo-random test pattern checking support in the 10GBASE-R and 10GBASE-KR modes
- Clock Rate compensation of 200ppm (+/- 100 ppm) on the Receive MAC interface
- Energy Efficient Ethernet Support

1.4.1 Monitoring, Test, and Debugging Support

DWC_xpcs provides the following features to support monitoring, testing, and debugging:

- Provides the following MDIO Manageable Devices (MMD): Debug PCS, PMA, PMD, XGXS, and auto-negotiation.
- Supports loopback control from the PHY Transmit to the PHY Receive lane.
- Supports internal loopback on XGMII for debugging through Receive to Transmit digital path.
- Supports various error status signals and statistics for monitoring and debugging.
- Provides programmable control to reduce the simulation time by reducing the timer values defined in the IEEE standard.
- Provides programmable Device or Package identifiers (OUI numbers as defined in IEEE Std 802.3).
- Provides programmable features to dynamically enable or disable MMD devices present in some configurations.

1.5 Deliverables

The DWC_xpcs is packaged as a .run file. The license file, required to use this Synopsys DesignWare IP, is delivered separately. You use the Synopsys coreConsultant tool to configure, synthesize, and simulate the DWC_xpcs controller. The DWC_xpcs image includes the following:

- Verilog RTL source code
- A Verilog test suite that integrates the DUT (which consists of the DWC_xpcs, integrated with a pin-compatible behavioral model of the Synopsys PHY) and the Ethernet VIP
- Synthesis scripts for Synopsys Design Compiler and Synplify Pro
- Regression scripts for the following simulators: Synopsys VCS, ModelSim, and NC-Verilog
- Spyglass checker rules used for linting and CDC analysis
- Support for creating gate-level Device Under Test (DUT) models using a GTECH library

1.5.1 Licensing Features

The licenses provided for DWC_xpcs are described in the *DesignWare Cores Ethernet PCS Installation Guide*.

1.6 Standards Compliance

The DWC_xpcs complies with the following IEEE Standards:

- IEEE Std 802.3-2018 Clause 35, 36, 37, 45, 46, 47, 48, 49, 69, 72, 73, 74, and 78 (Depending on your configuration)
- RXAUI - Reduced Pin XAUI specification, Dune Networks, January 2008.
- RXAUI Interface and RXAUI Adapter Specifications, Marvell, June 2008.
- Serial-GMII (SGMII) Specification, Revision 1.8, Cisco Systems.
- AMBA 3 APB Protocol, version 1.0, ARM Limited.
- QSGMII Specification, Revision 1.3, Cisco Systems.
- Universal SXGMII PHY-MAC Interface for Single and Multiple Network Port: version 2.0.
- Universal SXGMII PHY-MAC Interface for Multiple Network Port:version 2.15, dated 07/30/2019.
- Universal SXGMII Interface for a Single MultiGigabit CopperNetwork Port: version 2.4, dated 07/30/2019
- IEEE P802.3cb-20xx™/D3.0, 22 May 2017.

2

Architecture

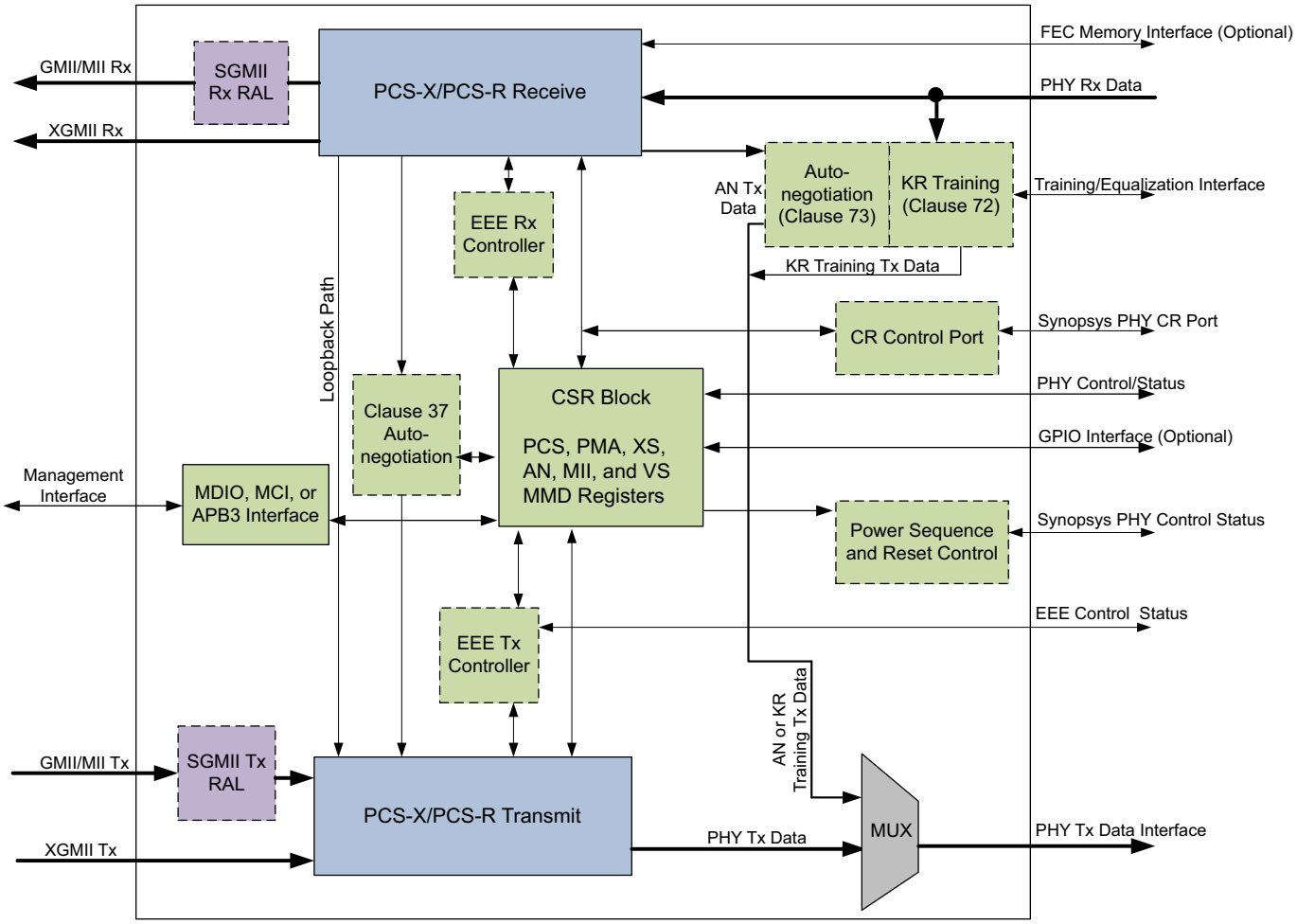
This chapter describes the DWC_xpcs architecture in functional blocks. It contains the following sections:

- "Overview of Architecture" on page 44
- "Modes of Operations" on page 45
- "PCS-X Data Path" on page 46
- "PCS-R Data Path" on page 65
- "Universal Serial XGMII (USXGMII)" on page 83
- "2.5G GMII Mode" on page 91
- "Command and Status Register Block" on page 93
- "Management Interface" on page 96
- "Clause 73 Auto-Negotiation" on page 106
- "Clause 72 KR Training" on page 122
- "Clause 37 Auto-Negotiation" on page 144
- "Serial Gigabit Media Independent Interface (SGMII)" on page 148
- "Quad SGMII (QSGMII)" on page 151
- "EEE Controller" on page 158
- "Power Sequence and Reset Generation" on page 171
- "Control Register Port Controller" on page 174
- "Loopback" on page 176
- "Timing Diagrams" on page 178

2.1 Overview of Architecture

Figure 2-1 illustrates the functional blocks of DWC_xpcs.

Figure 2-1 DWC_xpcs Block Diagram



2.2 Modes of Operations

The main modes of operations of DWC_xpcs are as follows:

- XGXS PCS
- 10GBASE-X PCS
- Backplane Ethernet PCS
- 1000BaseX-Only PCS
- 10GBASE-R PCS

These modes fall under one of the following categories of data path:

- PCS-X Data Path

  The PCS-X data path is applicable when the DWC_xpcs is configured for any of the following modes: XGXS PCS, 10GBASE-X PCS, KX4, KX, KX (1000BASE-KX and 2.5GBASE-KX), 1000BASEX-Only PCS, or SGMII.

- PCS-R Data Path

  This data path is applicable when the DWC_xpcs is enabled to support the 10GBASE-R PCS, or KR (10G or 5G) mode.

2.2.1 Configuring the Mode of Operation

To configure the mode of operation during the Specify Configuration Activity in coreConsultant, select the required mode of operation from the "Mode of Operation" drop-down parameter, in the System Configurations Settings window.

For details about this option, see System Configurations Settings Parameters in the "*Parameter Descriptions*" chapter.



See the "Enabled" field of the parameter descriptions to understand the dependencies.

2.3 PCS-X Data Path

This topic contains the following sections:

- "Overview of PCS-X Transmit Block" on page 46
- "Description of PCS-X Transmit Block" on page 47
- "Overview of PCS-X Receive Block" on page 55
- "Description of PCS-X Receive Block" on page 57
- "Error Counters" on page 63
- "Programming Guidelines for the PCS-X Data Path" on page 64

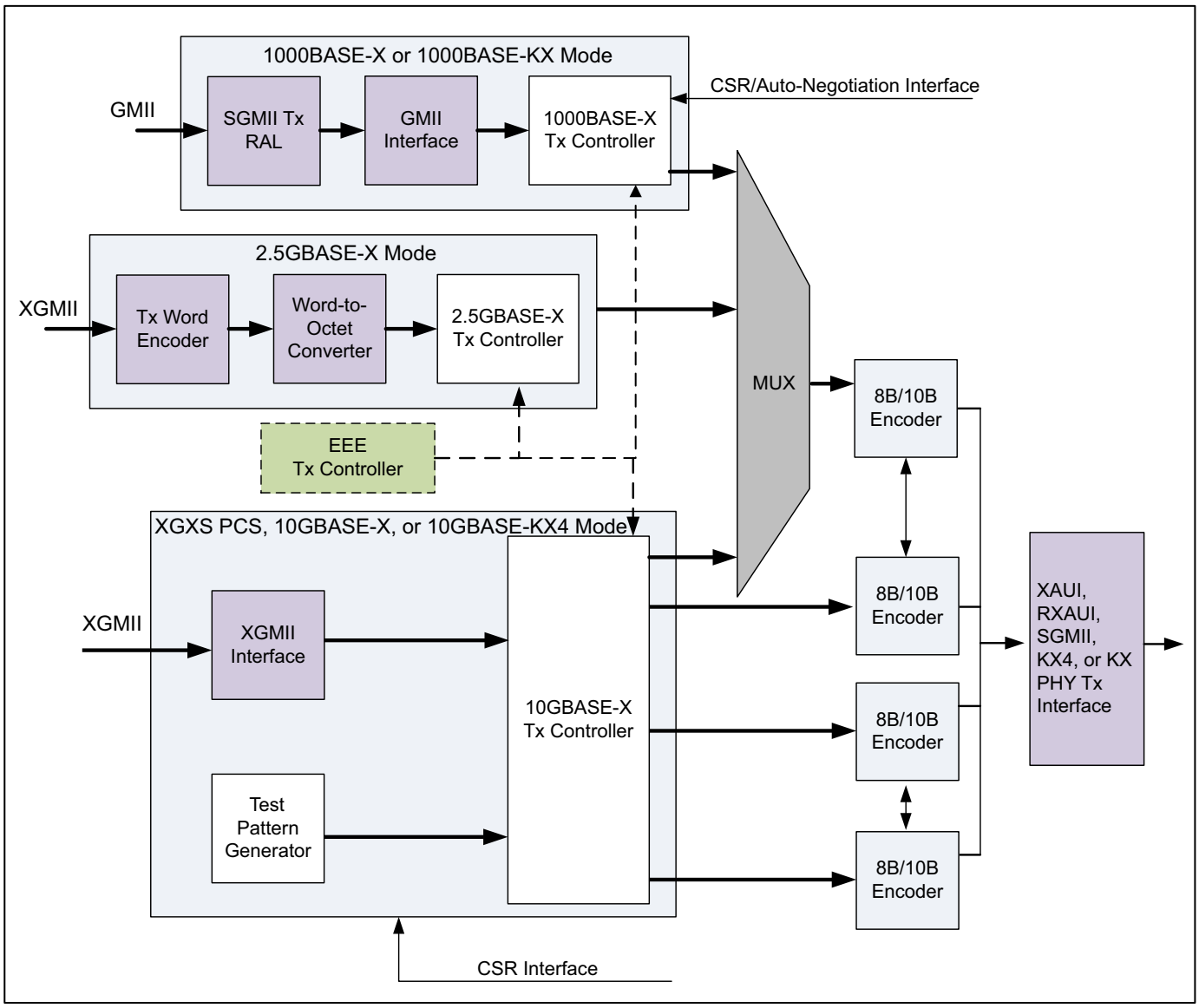
2.3.1 Overview of PCS-X Transmit Block

The PCS-X Transmit block implements the following functions:

- Supports the following interface protocols:
  - GMII interface in 1000BASE-X or 1000BASE-KX mode; XGMII interface in XGXS PCS, 10GBASE-X PCS, 2.5GBASE-KX, or 10GBASE-KX4 mode
  - XAUI or KX4, RXAUI, and SGMII/QSGMII or KX support at the PHY serial interface
  - SGMII Rate Adapter block to support 10 Mbps or 100 Mbps in SGMII/QSGMII mode
- Performs 8B/10B data encoding
- Generates Idle pattern (as defined in Clause 36 for 1000BASE-X or 1000BASE-KX mode and Clause 48 for XGXS PCS, 10GBASE-X PCS, or 10GBASE-KX4 mode)
- Generates the following test patterns defined in IEEE Std 802.3, Annexure 48A:
  - MFTP, LFTP, and HFTP
  - CRTP and CJTP in 10G mode
- Provides lane disable control for Lane 3, Lane 2, and Lane 1.

Figure 2-2 displays the components of the PCS-X Transmit block.

Figure 2-2 PCS-X Transmit Block Diagram



2.3.2 Description of PCS-X Transmit Block

2.3.2.1 GMII Tx RAL Block

The GMII Tx Rate Adapter Logic (RAL) block is an optional block. This block is present only when SGMII_EN = Enabled or QSGMII_EN = Enabled. This block converts the data (with different speeds and data widths) from the MAC clocking to a common internal speed and data width (125 MHz, 8-bit). For information about the supported SGMII/QSGMII speed modes, see Table 2-26.

2.3.2.2 GMII Interface Block

The GMII interface block interfaces with the MAC. The GMII interface has 8-bit data, two control bits, and it operates at 125 MHz.

2.3.2.3 Tx Word Encoder

This block is applicable for 2.5GBASE-X PCS mode. In this mode, the XGMII Tx data from 2.5G MAC is first word-encoded and then converted to an internal interface termed as '2.5G PCS Internal Interface' (GPII). This 2.5GPII interface consists of the following signals: tp_en, tp_er and tpd<7:0>. Table 2-1 shows the permissible encodings at 2.5GPII Tx.

Table 2-1 Encoding of 2.5GPII Tx

tp_en	tp_er	tpd<7:0>	Description	Abbreviation
0	0	0x00 to 0xFF	Normal Inter-frame	IDLE
0	1	0x00	Reserved	
0	1	0x1	LPI	LPI
0	1	0x02 to 0x9B	Reserved	
0	1	0x9C	Sequence	Seq
0	1	0x9D to 0xFF	Reserved	
1	1	0x00 to 0xFF	Data	Data X
1	0	0x00 to 0xFF	Transmit Error	Err

Tx Word Encoder maps the four XGMII lanes onto four 2.5GPII symbols as shown in Table 2-2. The mapping of sequence ordered set is dependent on the current state of the encoder (denoted by wencode_state variable). The state of the encode is updated once the mapping occurs as per the last column.

Table 2-2 Tx Word Encoder

XGMII						2.5GPII				
Lane 0	Lane 1	Lane 2	Lane 3	wencode_state(n)		2.5GPII <0>	2.5GPII <1>	2.5GPII <2>	2.5GPII <3>	wencode_state (n+1)
Data A or Err	Data B or Err	Data C or Err	Data D or Err	Don't care	=>	Data A or Err	Data B or Err	Data C or Err	Data D or Err	DATA
Idle	Idle	Idle	Idle	Don't care	=>	Idle	Idle	Idle	Idle	IDLE
LPI	LPI	LPI	LPI	Don't care	=>	LPI	LPI	LPI	LPI	DATA
start	Data A or Err	Data B or Err	Data C or Err	Don't care	=>	Data 0x55	Data A or Err	Data B or Err	Data C or Err	DATA
Terminate	Idle	Idle	Idle	Don't care	=>	Idle	Idle	Idle	Idle	IDLE

XGMII						2.5GPII				
Lane 0	**Lane 1**	**Lane 2**	**Lane 3**	**wencode_state(n)**		**2.5GPII <0>**	**2.5GPII <1>**	**2.5GPII <2>**	**2.5GPII <3>**	**wencode_state (n+1)**
Data A or Err	Terminate	Idle	Idle	Don't care	=>	Data A or Err	Idle	Idle	Idle	DATA
Data A or Err	Data B or Err	Terminate		Don't care	=>	Data A or Err	Data B or Err	Idle	Idle	DATA
Data A or Err	Data B or Err	Data C or Err	Terminate	Don't care	=>	Data A or Err	Data B or Err	Data C or Err	Idle	DATA
Sequence	Data X	Data Y	Data Z	IDLE	=>	Seq	Data S0	Seq	Data S1	SEQ
Sequence	Data X	Data Y	Data Z	SEQ	=>	Seq	Prev Data S2	Seq	Prev Data S3	IDLE
Sequence	Data X	Data Y	Data Z	DATA	=>	Idle	Idle	Idle	Idle	IDLE
else				Don't care	=>	Err	Err	Err	Err	DATA

The 24-bit Data X, Data Y, and Data Z from the sequence ordered set is mapped to Data S0, Data S1, Data S2, Data S3 as follows:

- S0<7> = S3<7> = 0, S1<7> = S2<7> = 1
- S0<5:0> = Data X<5:0>
- S1<5:0> = Data Y<3:0>, Data X<7:6>
- S2<5:0> = Data Z<1:0>, Data Y<7:4>
- S3<5:0> = Data Z<7:2>
- Sn<6> = Sn<7> if Sn<2> = 0
- Sn<6> = Sn<5> if Sn<2> = 1

Note Tx Word Encoder in DWC_xpcs does not support the signal ordered-set (|Fsig|).

2.3.2.4 Word-to-Octet Converter

This block takes the output of the Tx Word Encoder (four 2.5GPII symbols) as input and outputs them sequentially one-by-one in the following order: 2.5GPII<0>, 2.5GPII<1>, 2.5GPII<2> and finally 2.5GPII<3>.

This block comprises of a free-running FIFO which transfers the data from clk_xgmii_tx_i domain to clk_xgxs_tx_i domain.

2.3.2.5 2.5GBASE-KX Tx Control Block

The 2.5BASE-KX Tx control block is responsible for generating the PCS order sets as defined in IEEE Std 802.3cb, Clause 127. This block is controlled by the 2.5GPII interface. This block generates the required control or data signal to drive the encoder block. The 2.5GPII data is mapped to any of the order sets (|I1|, |I2|, |S|, |T|, |V|, |R|, or |Q|), as shown in Table 2-3. If EEE_EN = Enabled, this block also generates the LPI code groups.

Table 2-3 Clause 127 Order Sets

Code	Ordered Set	Number of Code Groups	Encoding
/I/	IDLE		Correcting /I1/, Preserving /I2/
/I1/	IDLE 1	2	/K28.5/D5.6/
/I2/	IDLE 2	2	/K28.5/D16.2/
/R/	Carrier_Extend	1	/K23.7/
/S/	Start_of_Packet	1	/K27.7/
/T/	End_of_Packet	1	/K29.7/
/V/	Error_Propagation	1	/K30.7/
/Q/	Sequence_ordered_set	8	/K28.5/W0/K28.5/W1/K28.5/W2/ K28.5/W3/
/LI/	LPI		Correcting /LI1/, Preserving /LI2/
/L1/	LPI 1	2	/K28.5/D6.5/
/L2/	LPI 2	2	/K28.5/D26.4/

2.3.2.6 XGMII Interface Block

The XGMII interface block interfaces with the MAC. As specified in the IEEE specification, the XGMII interface data operates on both (dual) edges of the 156.25 MHz clock. This block converts the data to a single-edged clock operating at 312.5 MHz.

The PCS-X also supports an optional configuration Dual Data Width (DDW), in which the XGMII data bus width is doubled (64 bits), operating with a 156.25 MHz single edge. If configured, this block converts the 64-bit data bus to a 32-bit data bus operating at 312.5 MHz.

Note The XGMII interface block is removed in configurations where XGMII operates at 312.5 MHz (SDR).

2.3.2.7 1000BASE-X or 1000BASE-KX Tx Control Block

The 1000BASE-X or 1000BASE-KX Tx control block is responsible for generating the PCS order sets as defined in IEEE Std 802.3, Clause 36. This block is controlled by the GMII interface and Clause 37

auto-negotiation. The |C1| and |C2| order sets are generated based on Clause 37 auto-negotiation. All other order sets are based on the GMII interface.

This block generates the required control or data signal to drive the encoder block. The GMII data is mapped to any of the order sets (|I1|, |I2|, |S|, |T|, |V|, |R|, or |D|), as shown in Table 2-4. If EEE_EN = Enabled, this block also generates the LPI code groups.

Table 2-4 Clause 36 Order Sets

Code	Ordered Set	Number of Code Groups	Encoding
|C|	Configuration		Alternating |C1| and |C2|
|C1|	Config1	4	|K28.5| or |D21.5| or Config_Reg†
|C2|	Config2	4	|K28.5| or |D2.2| or Config_Reg†
|I|	IDLE		Correcting |I1|, Preserving |I2|
|I1|	IDLE1	2	|K28.5| or |D5.6|
|I2|	IDLE2	2	|K28.5| or |D16.2|
|R|	Carrier_Extend	1	|K23.7|
|S|	Start_of_Packet	1	|K27.7|
|T|	End_of_Packet	1	|K29.7|
|V|	Error_Propagation	1	|K30.7|
|D|	Data	1	10-bit Encoded Data
|LI|	LPI		Correcting |LI1|, Preserving |LI2|
|LI1|	LPI 1	2	|K28.5| or |D6.5|
|LI2|	LPI 2	2	|K28.5| or |D26.4|
†Two data code groups representing the Config_Reg value for Clause 37 auto-negotiation.			

2.3.2.8 XGXS, 10GBASE-X, or 10GBASE-KX4 Tx Control Block

The XGXS, 10GBASE-X, or 10GBASE-KX4 Tx control block implements the Transmit control as defined in IEEE Std 802.3, Clause 48. It controls the generation of ||K||, ||R||, and ||A|| patterns during normal idle transmission, and||D||order set during data or frame transmission.

This block can also handle different vendor-specific order sets based on the XGMII interface signals. It generates the required control or data signal for the encoder block. If EEE_EN = Enabled, this block also encodes the LPI patterns from the XGMII interface. The XGMII data can be mapped to any of the order sets in Table 2-5.

Table 2-5 Clause 48 PCS Order Sets

XGMII TXC	XGMII TXD	PCS Code Group	Description
0	00h through FFh	Dxx.y	Normal data transmission
1	06h	K28.0, K28.3, K28.5, or D20.5	Low Power Idle /LPI/
1	07h	K28.0 or K28.3 or K28.5	Idle in /I/
1	07h	K28.5	Idle in /T/, |K|
1	9Ch	K28.4	Sequence
1	FBh	K27.7	Start
1	FDh	K29.7	Terminate
1	FEh	K30.7	Error
1	Other value in Table 36-2 of the IEEE Std 802.3.	See Table 36-2 of the IEEE Std 802.3.	Reserved XGMII character
1	Any other value	K30.7	Error XGMII character

2.3.2.9 PCS-X 8B/10B Encoder Block

The PCS-X 8B/10B encoder block implements the 8B/10B conversion according to the IEEE Std 802.3 requirements for given data and current running disparity. This block takes the data octet and special code group control signal from the Tx controller and generates the equivalent 10-bit output code group.

In 1000BASE-X or 1000BASE-KX or 2.5GBASE-KX mode, encoding is done only on Lane 0. In XGXS PCS, 10GBASE-X PCS, or 10GBASE-KX4 mode, encoding is required on all four lanes.

2.3.2.10 Test Pattern Generation Block

The test pattern generation block generates the following test patterns defined in IEEE Std 802.3, Annexure 48A:

- Mixed-frequency test pattern (MFTP), low-frequency test pattern (LFTP), and high-frequency test pattern (HFTP)
- Continuous random test pattern (CRTP) and continuous jitter test pattern (CJTP) in 10G mode

You can control the generation of these patterns through SR XS or PCS MMD 10GBASE-X Test Control Register.



Test Patterns are not applicable in QSGMII Mode.

2.3.2.11 RXAUI Support

The DWC_xpcs supports the following implementations of the RXAUI standard:

- Dune Networks implementation
- Marvell implementation

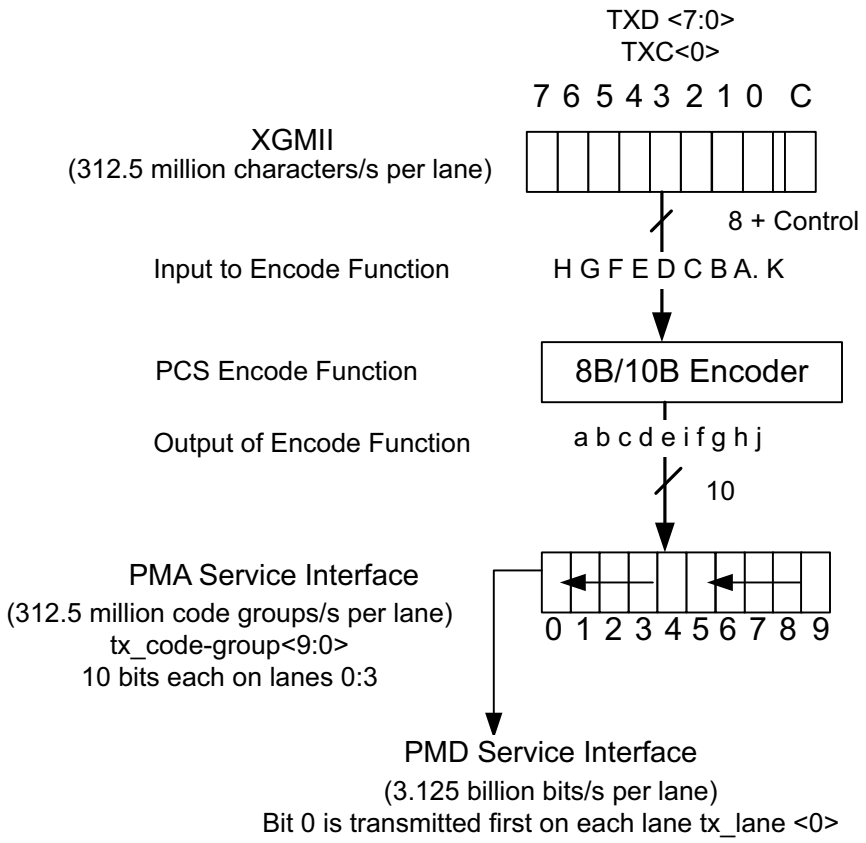
These implementations differ in the way the 8B/10B encoding is done for each lane (which corresponds to 20 bits). In Marvell implementation, the 8B/10B encoding or decoding is done independently on lower 10-bits and upper 10-bits of the same lane. In Dune Network implementation, the 8B/10B encoding or decoding is done such that the running disparity output of the lower 10-bits is used as current running disparity on upper 10-bits. The output of the running disparity of the upper 10-bits is used as current running disparity for next 10-bits to be sent on lower 10-bits.

You can enable, disable, or control the RXAUI mode through VR XS or PCS MMD XAUI Mode Control Register.

2.3.2.12 PCS-X Transmit Bit Ordering

Figure 2-3 shows the transmit bit order for the PCS-X for a single lane. In XGXS PCS, 10GBASE-X PCS, or KX4 mode, there are four such lanes. In 1000BASE-X or KX mode, there is only one lane operating at 125 MHz on the GMII interface and 1.24 million code-groups at the PMA Service Interface.

Figure 2-3 Transmit Bit Order in PCS-X



In XGMII interface, the TXD[7] corresponds to xgmii_txd_i[7] and TXC[0] corresponds to xgmii_txc_i[0] for Lane 0. The mapping is similar for other lanes.

In PMA Interface, the tx_data-group<0> is expected to be transmitted first. When TX_LSB_FIRST = Enabled, this bit corresponds to xgxs_tx_data_o[0]. When TX_LSB_FIRST = Disabled, this bit corresponds to xgxs_tx_data_o[9].

2.3.2.13 Transmit Lane Control

In 10G PCS-X mode, there are four lanes by default. The 10 Gbps speed mode is achieved through these four lanes. The DWC_xpcs provides an option to program the number of lanes to achieve variable speed modes. Along with the default 10 Gbps speed mode, you can enable 2.5 Gbps, 5.0 Gbps, and 7.5 Gbps speed modes by disabling some of the operating lanes in a vendor-specific way. You can control the Transmit lanes using Bits[15:13] of VR XS or PCS MMD Digital Control2 Register. Lane 0 is always present and must not be disabled.

Table 2-6 shows the valid combinations for 10G XAUI mode.

Table 2-6 Valid Lane Combinations for 10G XAUI Mode

Lane 3	Lane 2	Lane 1	Lane 0	Validity	Speed
Enable	Enable	Enable	Enable	Valid	10 Gbps
Disable	Enable	Enable	Enable	Valid	7.5 Gpbs
Disable	Disable	Enable	Enable	Valid	5 Gbps
Disable	Disable	Disable	Enable	Valid	2.5 Gpbs
Any other Combination				Not Valid	N/A

Table 2-7 shows the valid combinations for 10G RXAUI mode.

Table 2-7 Valid Lane Combinations for 10G RXAUI Mode

Lane 3	Lane 2	Lane 1	Lane 0	Validity	Speed
Enable	Enable	Enable	Enable	Valid	10 Gbps
Disable	Disable	Enable	Enable	Valid	5 Gpbs
Any Other Combination				Not Valid	N/A

In the Transmit path, the XGMII data of disabled lanes is ignored and the IDLE code groups are driven towards the PHY. In the Receive path, the PHY input data of disabled lanes is ignored and the IDLE characters are driven towards the XGMII.

Note The timing diagram shown in Figure 2-4 is specific to the XAUI mode. For RXAUI mode, the timing diagrams are similar except that there are only two physical lanes instead of four.

Figure 2-4 DWC_xpcs Behavior with Lane 2 and Lane 3 Disabled

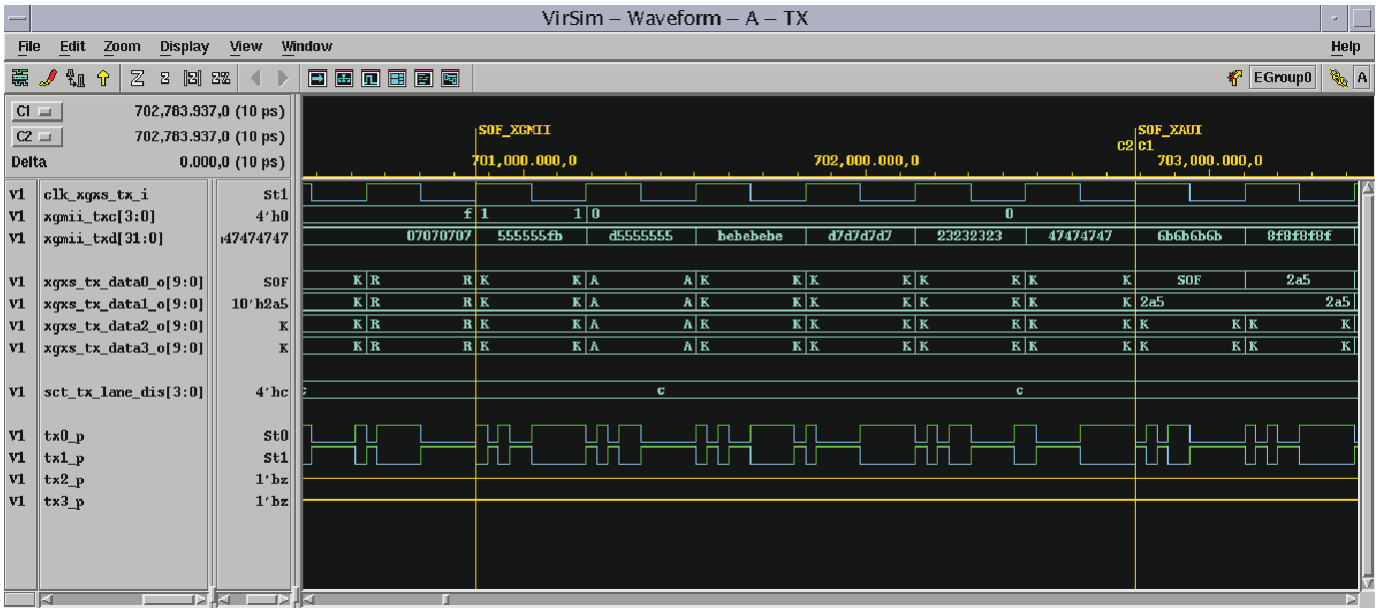


Figure 2-4 shows the DWC_xpcs behavior when Lane 2 and Lane 3 are disabled. The xgmii_txc_i[3:2] and xgmii_txd_i[31:16] signals are ignored, and the DWC_xpcs processes only xgmii_txc_i[1:0] and xgmii_txd_i[15:0] signals and drives them towards the PHY. Figure 2-4 shows the frame transmission and its encoded data to the PHY. On Lane 2 and Lane 3, IDLE (|K| pattern) characters are given as data to the PHY and on other two enabled lanes, proper data characters are given.

Note In Figure 2-4, the PHY differential data on Lane 2 and Lane 3 is tri-state because the corresponding PHY lanes are disabled. If the PHY lanes are still active, the IDLE patterns are driven out of these disabled lanes.

2.3.3 Overview of PCS-X Receive Block

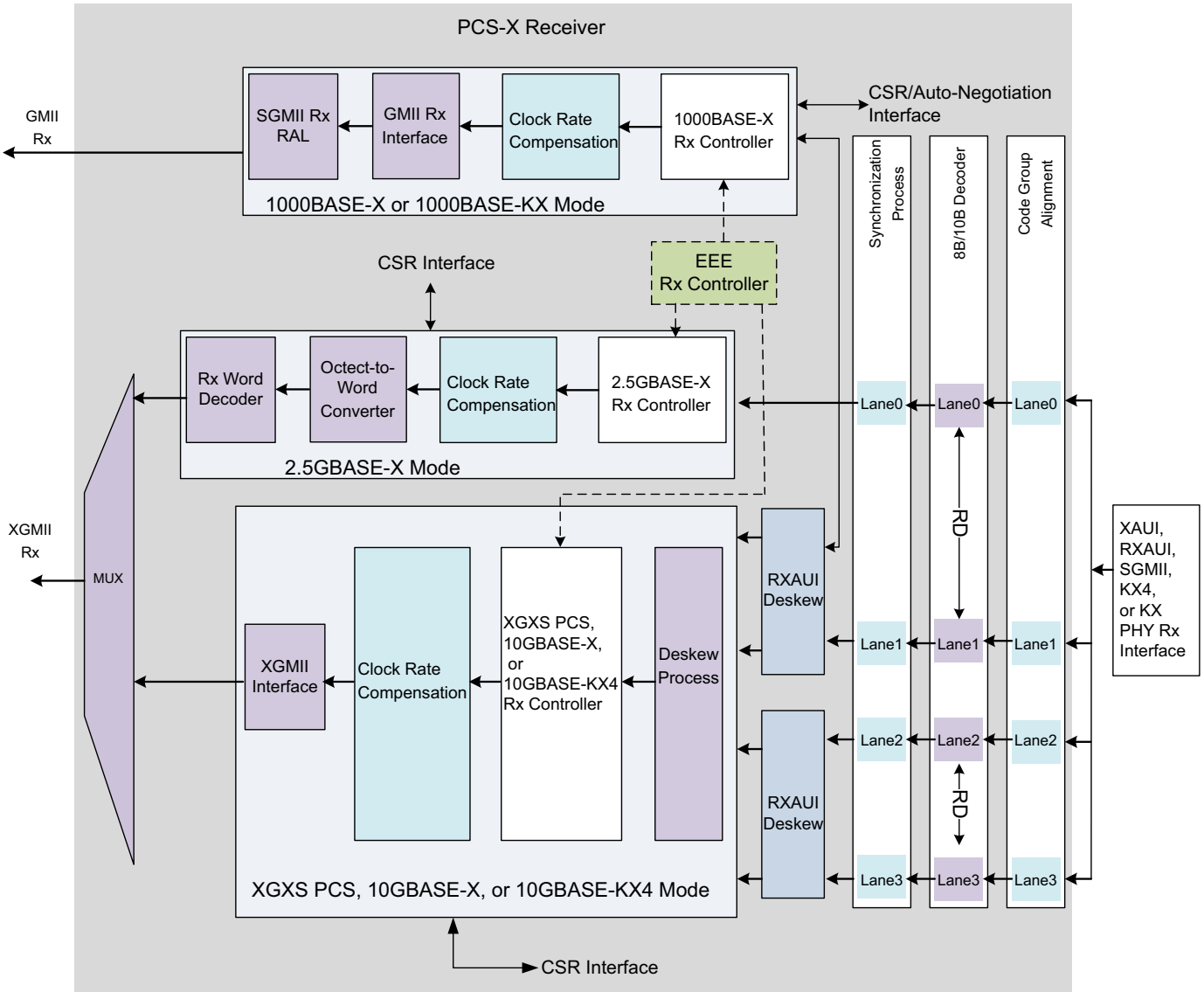
The PCS-X Receive block implements the following functions:

- Supports the following interface protocols:
  - GMII in the 1000BASE-X or 1000BASE-KX or 2.5GBASE-KX mode; XGMII in the XGXS PCS, 10GBASE-X PCS, or 10GBASE-KX4 mode
  - XAUI or KX4, RXAUI, and SGMII/QSGMII or KX (1000BASE-KX and 2.5GBASE-KX) support at PHY parallel interface
  - SGMII Rate Adapter block to support the 10 Mbps or 100 Mbps SGMII/QSGMII mode
- Performs code group alignment on each lane
- Performs 8B/10B data decoding on each lane
- Performs synchronization on each lane
- Performs deskew process among all four lanes when operating in the XGXS PCS, 10GBASE-X PCS, or 10GBASE-KX4 mode

- Performs check-end function (as defined in Clause 36 for the 1000BASE-X or 1000BASE-KX mode and Clause 48 for the XGXS PCS, 10GBASE-X PCS, or 10GBASE-KX4 mode)
- Detects Idle pattern (as defined in Clause 36 for the 1000BASE-X or 1000BASE-KX mode and Clause 48 for the XGXS PCS, 10GBASE-X PCS, or 10GBASE-KX4 mode)
- Provides lane disable control for Lane 3, Lane 2, and Lane 1

Figure 2-5 shows the components of the PCS-X Receive block.

Figure 2-5 PCS-X Receive Block Diagram



2.3.4 Description of PCS-X Receive Block

The following sections describe the various sub-blocks of the PCS-X receive block.

2.3.4.1 Code Group Alignment Block

The code group alignment block performs the code group boundary alignment based on |K| code group. This block is controlled through the synchronization block. The code group realignment is performed when the synchronization process is in the 'Not OK' state. When alignment is detected, this block locks to that particular alignment until the synchronization process subsequently goes back to the 'Not OK' state.

If the DWC_xpcs is configured for Synopsys PHY, the code group alignment functionality is bypassed. For Synopsys multi-protocol 6G PHY, you can enable the code group alignment using Bits[7:4] of VR_XS_PMA_MP_6G_RXGENCTRL0 Register.

2.3.4.2 8B/10B Decoder Block

The 8B/10B decoder block performs 8B/10B decoding on the 10-bit aligned code group. This is done only on single lane in the 1000BASE-X or 1000BASE-KX or 2.5GBASE-KX mode. Decoding is done on all four lanes in the XGXS PCS, 10GBASE-X PCS, or 10GBASE-KX4 mode. This block is also responsible for detecting the RD Errors and INVALID code groups.

2.3.4.3 Synchronization Block

The synchronization block performs the synchronization process as defined in IEEE Std 802.3, Clause 36 for the 1000BASE-X or 1000BASE-KX mode, Clause 127 for 2.5GBASE-KX, and Clause 48 for the XGXS PCS, 10GBASE-X PCS, or 10GBASE-KX4 mode. This block validates that the underlying PMA is operating reliably on per-lane basis. Success or failure is based on the number of |K| and |INVALID| code groups received on each lane.

In 10G mode, the synchronization process is considered successful when synchronization of all lanes is successful in the XGXS PCS, 10GBASE-X PCS, or 10GBASE-KX4 mode. If synchronization in any lane is not successful, the synchronization is considered as unsuccessful. If you have disabled any lane by programming Bits[11:9] of VR XS or PCS MMD Digital Control2 Register, the disabled lane is considered as synchronized.

The synchronization process for 1000BASE-X, 1000BASE-KX, or 2.5GBASE-KX mode differs from the XGXS PCS, 10GBASE-X PCS, or 10GBASE-KX4 mode. Lane 0 synchronization is capable of switching between two modes when the DWC_xpcs is configured for the XGXS PCS, 10GBASE-X PCS, or 10GBASE-KX4 and 1000BASE-X, 1000BASE-KX, or 2.5GBASE-KX modes.

2.3.4.4 RXAUI Deskew Block

The RXAUI deskew block is present only when the DWC_xpcs is configured to support RXAUI. This block determines which alternative received code group corresponds to the least significant code group of the single physical RXAUI data stream. The RXAUI deskew block determines this based on the first |A| code group that it receives. The first received |A| is declared as the least significant code group and is remapped to the logical Lane 0 or Lane 2 for XGXS PCS, 10GBASE-X PCS, or 10GBASE-KX4 deskew process. The

DWC_xpcs supports the Dune Networks implementation and Marvell implementation of RXAUI. You can control the RXAUI mode using VR XS or PCS MMD XAUI Mode Control Register.

2.3.4.5 Deskew Process Block

The deskew process block removes skew among the four lanes. This block performs lane-to-lane deskewing based on a special code group called |A|.

This block is operational only in the XGXS PCS, 10GBASE-X PCS, or 10GBASE-KX4 mode.

This block determines whether the Receive path is capable of providing coherent data to the upper layers (XGMAC). The deskew process starts when the synchronization process or deskew process status output has the FAIL value. The deskew process continues until synchronization is acquired in all lanes, achieving deskew as defined in IEEE Std 802.3, Clause 48. For debugging purpose, you can disable and bypass the deskew function by setting Bit 0 of VR XS or PCS MMD Digital Control1 Register.

2.3.4.6 1000BASE-X or 1000BASE-KX Rx Frame Process Block

The 1000BASE-X or 1000BASE-KX Rx frame process block takes the output from the synchronization process of Lane 0. This block then maps the incoming data stream to corresponding idle |I|, data |D|, error |V|, and configuration |C| characters.

This block is present only when the DWC_xpcs is configured for the 1000BASE-X or 1000BASE-KX mode.

2.3.4.7 2.5GBASE-KX Rx Controller Block

This block is valid for 2.5GBASE-KX mode of operation. This block takes the output of synchronization process of Lane 0 and maps it to corresponding idle |I|, data |D|, error |V| and sequence |Q| characters. The output from this block is passed on to the 2.5GPII Rx interface.

2.3.4.8 XGXS, 10GBASE-X, or 10GBASE-KX4 Rx Controller Block

The XGXS, 10GBASE-X, or 10GBASE-KX4 Rx controller block takes the data from the deskew process. This block maps the incoming data stream corresponding to idle ||I||, data ||D||, and error |E| characters and fault signals (such as LF, RF |Q| and check-end violations).

This block also implements error counters such as test pattern mismatch, deskew error counter, and INVALID code group counters (per lane). These error counters can be enabled or disabled through VR XS or PCS MMD Digital Error Count Select Register and the error count registers can be read through following registers:

- VR_XS_PCS_ICG_ERRCNT1 Register
- VR_XS_PCS_ICG_ERRCNT2 Register

- VR_XS_PCS_DSKW_ERRCTR Register
- VR_XS_PCS_TPM_ERRCTR Register

2.3.4.9 Clock Rate Compensation Block

The clock rate compensation block is used to compensate for frequency variations of the recovered clock (Rx Lane 0 clock) and the local clock (Tx clock) used for driving the data on the XGMII or GMII interface. To achieve the clock rate compensation, the DWC_xpcs inserts or deletes the Idle, LPI, or Sequence Order Sets as applicable during IFG or IDLE period. This is done only if the IFG or IDLE period meets the minimum IFG requirement on XGMII or GMII.

The clock rate compensation block supports the frequency variations of maximum +/-100ppm between the recovered clock and the local clock. In addition, this block is designed to support Jumbo frames with maximum size of 16K bytes. The clock rate compensation block is present even when the rate-compensation feature is not selected while configuring the DWC_xpcs. It is required for the loopback path from the Receive path to the Transmit path on the XGMII or GMII interface. Bits[6:5] of the VR XS or PCS MMD Digital Status Register provide the status of the clock rate compensation FIFO underflow or overflow.

2.3.4.10 GMII Interface Block

The GMII interface block takes the data and control signals from the 1000BASE-X or 1000BASE-KX Rx Controller and maps them to the GMII interface to drive the gmii_rx_er_o, gmii_rxd_o, and gmii_rx_dv_o signals.



This block is present only when the DWC_xpcs is configured for 1000BASE-X or 1000BASE-KX mode.

2.3.4.11 XGMII Interface Block

The XGMII interface block is configurable using XGMII Interface option. If XGMII interface is configured in the Single Data Rate (SDR) mode at 312.5 MHz, this block is not present and the XGMII data is directly driven from the clock rate compensation or the Rx frame process block.

To send the XGMII data in the Double Data Rate (DDR) mode, the data must be registered on both edges of the clock running at 156.25 MHz. Similarly, to send the data in the Dual Data Width (DDW) mode, the data bus width is double (64-bit with 8-control lines) at 156.25 MHz.

2.3.4.12 XGMII or GMII Loopback

This block is used to have loop-backs at the XGMII and GMII interface. The DWC_xpcs supports loopback of the received data to the Transmit data path at the XGMII interface. The loopback path does not include the conversion of data between 156.25 MHz and 312.5 MHz domains in the DDW or DDR mode.

To compensate for any clock frequency differences, the DWC_xpcs inserts or deletes the Idle characters during the Inter Frame Gap (IFG), provided it meets the minimum IFG requirement on the XGMII or GMII interface. This module supports frequency variations of maximum +/-100ppm between the Transmit clock and Receive clock. This block is designed to support Jumbo frames with maximum size of 9,022 bytes.

2.3.4.13 GMII Rx RAL Block

The GMII Rx RAL block is an optional block. This block is present only when you select the Add SGMII Support or Add QSGMII Support option. Depending on current SGMII/QSGMII speed mode, this block converts the data (at 125 MHz, 8-bit) from 1000BASE-X or 1000BASE-KX PCS controller to different speeds and data widths.

For information about the supported SGMII speed modes, see Table 2-26.

2.3.4.14 Octet-to-Word Converter

This block de-serializes a sequence of 2.5GPII Rx symbols (rp_dv, rp_er, rpd<7:0>) received from the 2.5GBASE-KX Rx controller to four 2.5GPII symbols (rp_dv<3:0>, rp_er<3:0>, rpd<3:0><7:0>).

The Octets-to-Word conversion process inserts 2.5GPII idle symbols, or deletes 2.5GPII symbols from the sequence of 2.5GPII symbols received to achieve the following conditions.

- A transition of a 2.5GPII idle symbol to a data or error symbol places the data or error symbol on index 0.
- A transition of a 2.5GPII idle symbol to a LPI symbol places the LPI symbol on either index 0 or 2.
- A transition of a 2.5GPII LPI symbol to an idle symbol places the idle symbol on either index 0 or 2.
- The start of |Q| ordered set always occurs on index 0.

This module comprises of a free-running FIFO to transfer data from clk_xgxs_rx0_i (or clk_xgxs_tx_i in case of CLKCOMP) domain to clk_xgmii_rx_i domain.

Figure 2-6 shows the permissible encodings of 2.5GPII Rx interface.

Figure 2-6 Permissible Encodings of 2.5GPII Rx Interface

rp_dv	rp_er	rpd<7:0>	Description	Mnemonic
0	0	0x00 to 0xFF	Normal Inter-frame	Idle
0	1	0x00	Reserved	
0	1	0x01	Assert LPI	LPI
0	1	0x02 to 0x0D	Reserved	
0	1	0x0E	False carrier indication	FCI
0	1	0x0F	Carrier Extend (odd byte packets)	CE
0	1	0x10 to 0x9B	Reserved	
0	1	0x9C	Sequence	Seq
0	1	0x9D to 0xFF	Reserved	
1	0	0x00 to 0xFF	Data	Data X
1	1	0x00 to 0xFF	Receive Error	Err

2.3.4.15 Rx Word Decoder

This block maps the four 2.5GPII Rx symbols onto the four XGMII lanes as shown in Table 2-8.

Table 2-8 Rx Word Decoder

2.5GPII							XGMII				
2.5GPII <0>	2.5GPII <1>	2.5GPII <2>	2.5GPII <3>	wencode_state(n)	next seq_s2s3		Lane 0	Lane 1	Lane 2	Lane 3	wencode_state (n+1)
Data A or Err	Data B or Err	Data C or Err	Data D or Err	!IDLE	Don't care	=>	Data A or Err	Data B or Err	Data C or Err	Data D or Err	DATA
Data *	Data A or Err	Data B or Err	Data C or Err	IDLE	Don't care	=>	SOP	Data A or Err	Data B or Err	Data C or Err	DATA
Idle	Idle	Idle	Idle	!DATA	Don't care	=>	Idle	Idle	Idle	Idle	IDLE
Idle	Idle	Idle	Idle	DATA	Don't care	=>	Terminate	Idle	Idle	Idle	IDLE
Data A or Err	Idle or CE	Idle	Idle	DATA	Don't care	=>	Data A or Err	Terminate	Idle	Idle	IDLE
Data A or Err	Data B or Err	Idle	Idle	DATA	Don't care	=>	Data A or Err	Data B or Err	Terminate	Idle	IDLE
Data A or Err	Data B or Err	Data C or Err	Idle or CE	DATA	Don't care	=>	Data A or Err	Data B or Err	Data C or Err	Terminate	IDLE
LPI	LPI	LPI	LPI	!DATA	Don't care	=>	LPI	LPI	LPI	LPI	IDLE
Idle	Idle	LPI	LPI	Don't care	Don't care	=>	LPI	LPI	LPI	LPI	IDLE
LPI	LPI	Idle	Idle	!DATA	Don't care	=>	Idle	Idle	Idle	Idle	IDLE
Seq	Data S0	Seq	Data S1	!DATA	TRUE	=>	Sequence	Data X	Data Y	Data Z	SEQ
Seq	Data S0	Seq	Data S1	!DATA	FALSE	=>	Idle	Idle	Idle	Idle	IDLE
Seq	Data S2	Seq	Data S3	SEQ	X	=>	Sequence	Data X	Data Y	Data Z	IDLE
else				Don't care		=>	Err	Err	Err	Err	ERR

This mapping is dependent on the current state of the decoder (denoted by wdecode_state variable) and whether the next set of four 2.5GPII symbols correspond to latter half of sequence ordered set (denoted by next_seq_s2_s3 variable). The state of the decoder is updated once the mapping occurs as per the last column. The 24-bit Data X, Data Y and Data Z of the sequence ordered set is reconstructed from Data S0, Data S1, Data S3 and Data S3 according to the following:

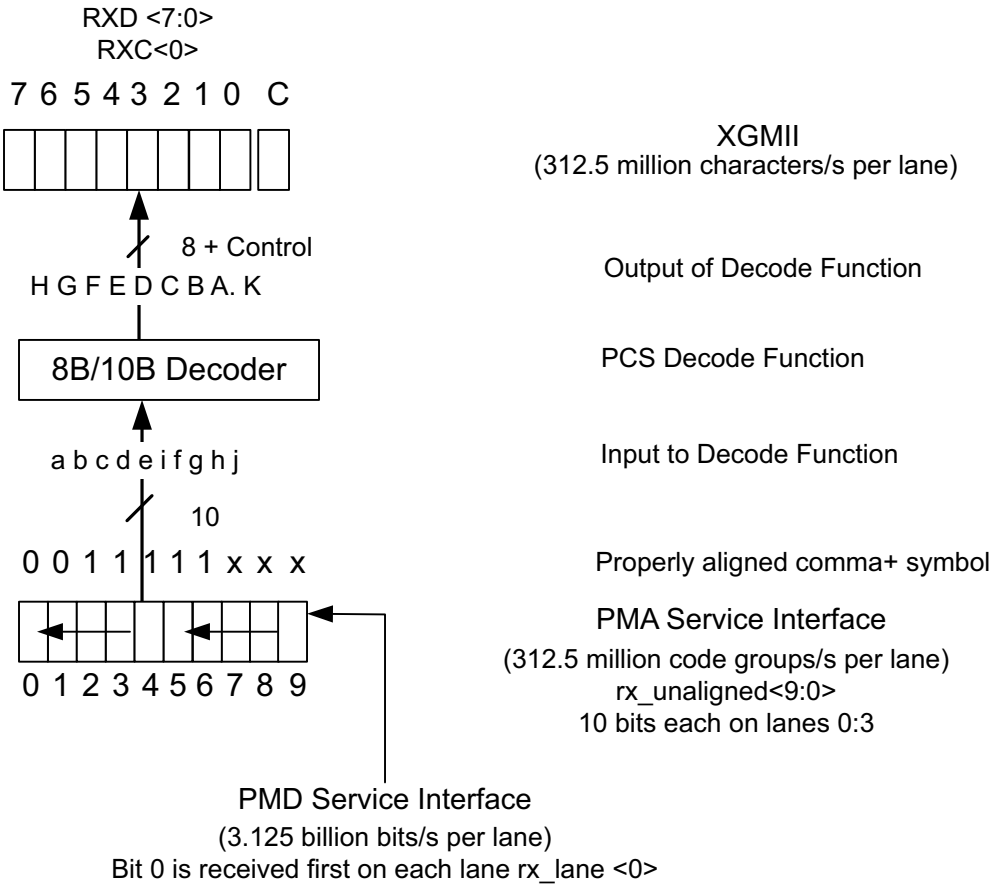
```
(S0<7>, S1<7>, S2<7>, S3<7> = 0110) then outputXGMII = Sequence, Data X, Data Y,
Data Z where    Data X<7:0> = S1<1:0>, S0<5:0>    Data Y<7:0> = S2<3:0>, S1<5:2>
Data Z<7:0> = S3<5:0>, S2<5:4 else XGMII = Idle, Idle, Idle, Idle
```

This block drives local-fault pattern (32'h0100009c) on XGMII Rx pattern till the receiver is synchronized to the incoming data-stream.

2.3.4.16 PCS-X Receive Bit Ordering

Figure 2-7 shows the receive bit order for PCS-X for a single lane. In XGXS PCS, 10GBASE-X PCS, or KX4 mode, there are four such lanes. In 1000BASE-X or KX mode, there is only one lane operating at 125 MHz on the GMII interface and 1.24 million code groups at the PMA Service Interface.

Figure 2-7 Receive Bit Order in PCS-X



In XGMII interface, the RXD[7] corresponds to xgmii_rxd_o[7] and RXC[0] corresponds to xgmii_rxc_o[0] for Lane 0. The mapping is similar for other lanes.

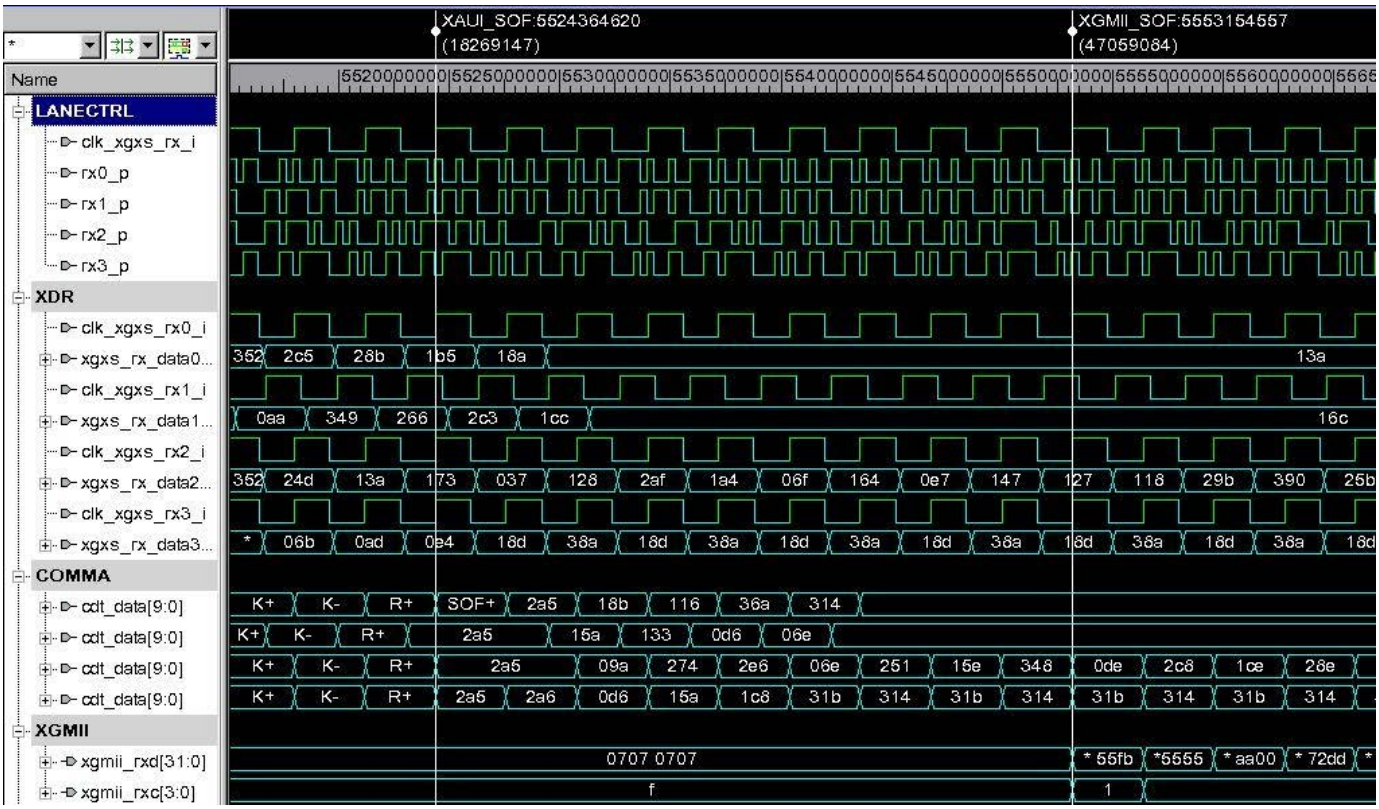
In PMA Interface, the rx_data-group<0> is expected to be received first. When RX_LSB_FIRST = Enabled, this bit corresponds to xgxs_rx_data_i[0]. When RX_LSB_FIRST = Disabled, this bit corresponds to xgxs_rx_data_i[9].

2.3.4.17 Receive Lane Control

In 10G PCS-X mode, there are four lanes by default. The DWC_xpcs provides an option to program the number of lanes to achieve variable speed modes. Along with the default 10 Gbps speed mode, you can enable 2.5 Gbps, 5.0 Gbps, and 7.5 Gbps speed modes by disabling some of the operating lanes in a vendor-specific way. You can control the Transmit lanes using Bits[11:9] of VR XS or PCS MMD Digital Control2 Register. Lane 0 is always present and must not be disabled.

Figure 2-8 shows the DWC_xpcs behavior when Lane 2 and Lane 3 are disabled in the Receive path. The inputs from the PHY are unaligned data on all four lanes with four different recovered clocks. This data is aligned on each lane based on the comma patterns (cdt_data[9:0]). The skew between Lane 0 and Lane 1 is removed, and the deskewed data is driven out on xgmii_rxc_o[1:0] and xgmii_rxd_o[15:0]. The disabled Lane 2 and Lane 3 show IDLE patterns on XGMII even though some inputs are received from the PHY.

Figure 2-8 DWC_xpcs Behavior With Lane 2 and Lane 3 Disabled



Note In Figure 2-8, Lane 2 and Lane 3 of the PHY differential lines are active. To minimize power consumption, you can disable unused lanes in the PHY when the corresponding lanes are disabled in the DWC_xpcs.

2.3.5 Error Counters

The DWC_xpcs implements 8-bit error counters for counting the following:

- Invalid code group count in each lane

- Check-end error count
- Deskew alignment error count
- Test pattern mismatch count (only for HFTP, MFTP, and LFTP test patterns)

Each counter group has an individual enable to start the counter, as described in following:

- VR_XS_PCS_ICG_ERRCNT1 Register
- VR_XS_PCS_ICG_ERRCNT2 Register
- VR_XS_PCS_DSKW_ERRCTR Register
- VR_XS_PCS_TPM_ERRCTR Register

All counters saturate with all-ones, and are cleared on a read access. When Bit 0 of VR XS or PCS MMD Invalid Code Group Error Count1 Register is set to 1, the counters are cleared every time a particular counter is read.

2.3.6 Programming Guidelines for the PCS-X Data Path

- "Switching to 10G RXAUI Mode" on page 1661
- "Switching to 1G or KX Mode and 10 or 100 Mbps SGMII Speed" on page 1669
- "Programming Guidelines for QSGMII Mode" on page 1684
- "Programming Guidelines for 2.5G GMII Mode" on page 1692
- "Programming Guidelines for 2.5GBASE-KX Mode (over XGMII)" on page 1696

2.4 PCS-R Data Path

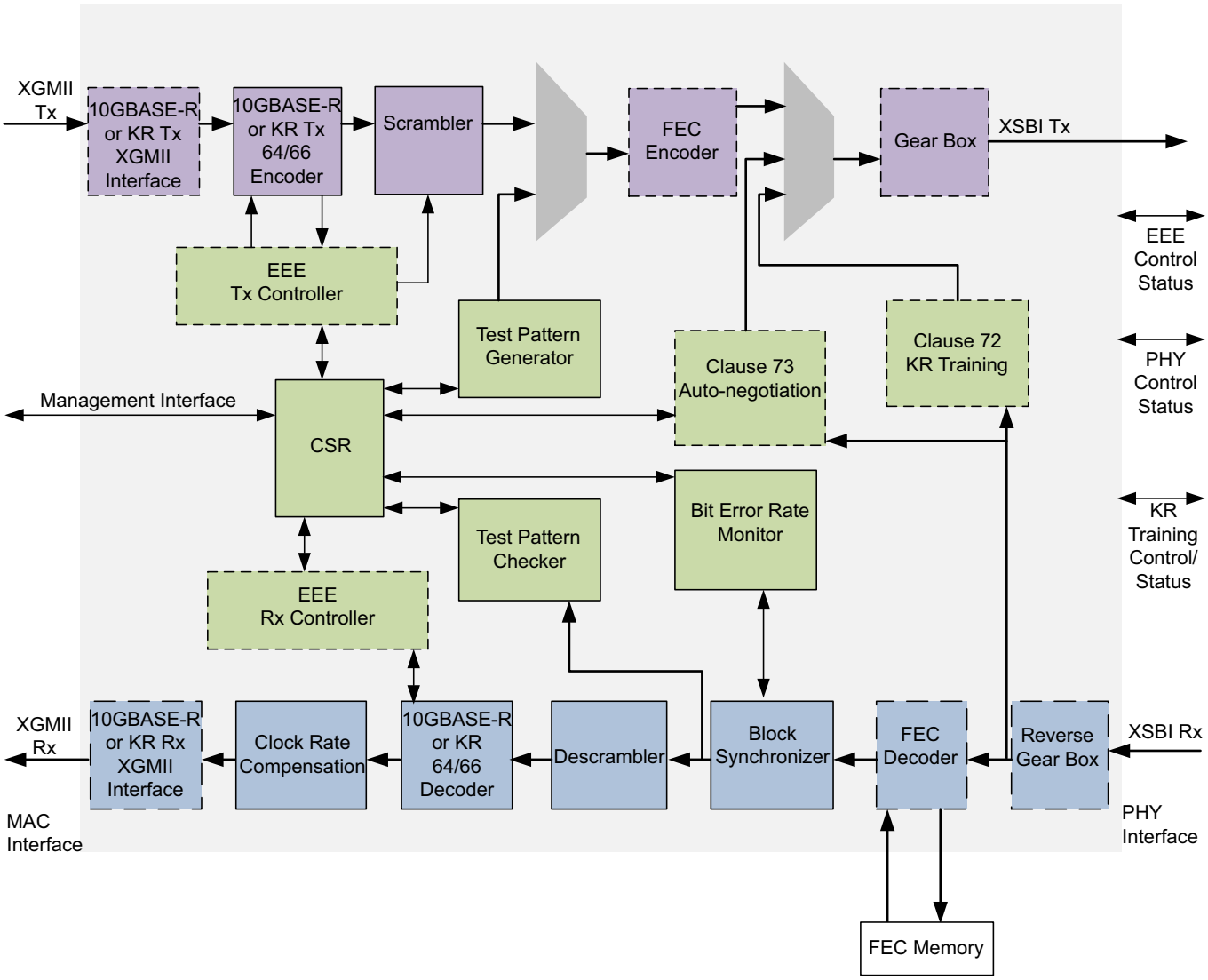
This topic contains the following sections:

- "Overview of PCS-R Data Path" on page 65
- "Overview of PCS-R Transmit Block" on page 66
- "Description of PCS-R Transmit Block" on page 67
- "Overview of PCS-R Receive Block" on page 74
- "Description of PCS-R Receive Block" on page 75
- "10GBASE-R or 10G/5GBASE-KR Test Pattern Generator and Checker Block" on page 81
- "Programming Guidelines for the PCS-R" on page 82

2.4.1 Overview of PCS-R Data Path

Figure 2-11 displays the components of the PCS-R data path.This section is applicable for 10GBASE-R PCS and 5GBASE-R PCS Modes of operation. 5GBASE-R PCS mode of operation is almost identical to 10GBASE-R PCS except that the data-rate and clock frequencies are reduced by half. Also, there is a difference with regard to HIBER generation.

Figure 2-9 PCS-R Data Path Components



Note A dual-port/two-port RAM (or register array) of 32 depth and 66-bit width (32x66) is required for FEC decoding. If FEC error forwarding feature is enabled, two such blocks are required.

2.4.2 Overview of PCS-R Transmit Block

The PCS-R Transmit block implements the following functions:

- Implements the following interfaces:
  - XGMII interface towards the MAC
  - 10GBASE-R or 10GBASE-KR or 5GBASE-KR parallel interface towards the PHY
- Performs 64B/66B encoding for block boundary determination
- Performs optional Forward Error Correction (FEC) encoding

- Performs scrambling to ensure transition density
- Generates the following test patterns:
  - Pseudo Random
  - PRBS 31
  - PRBS 9
  - Square wave

2.4.3 Description of PCS-R Transmit Block

2.4.3.1 Tx 10GBASE-R or 10G/5G BASE-KR XGMII Interface

The Tx 10GBASE-R or 10G/5G BASE-KR XGMII block interfaces with the MAC. This block converts the XGMII Transmit data interface, operating in the SDR mode (32-bit data and 4-bit control at 312.5 MHz for 10GBASE-R or 156.25 MHz for 5GBASE-R, to the DDW mode (64-bit data and 8-bit control) required for 10GBASE-R/5GBASE-R. This block also converts the XGMII interface operating in the DDR mode (32-bit data and 4-bit control at 156.25 MHz running on the dual-edge clock) to the DDW mode.

Note This block is not present when XGMII = DDW.

2.4.3.2 64/66 Encoder

The 64/66 encoder block encodes the incoming data in DDW format (64-bit data and 8-bit control at 156.25 MHz) to encoded 66-bit data. This is done by adding the 2-bit Sync Header and following the encoding rules defined in the IEEE Std 802.3, Clause 49.

If the XGMII input has all the data (of format D0 D1 D2 D3 D4 D5 D6 D7), the encoding is done by just adding the Sync Header of 2'b01 to form 66-bits.

Input XGMII Data	Encoded Output Data	
	Sync Header	Data Encoding
D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7	2'b01	D_0 D_1 D_2 D_3 D_4 D_5 D_6 D_7

where, D_n = XGMII Data Character in n^{th} byte on xgmii_txd_i[] with xgmii_txc_i[n] being set to 0.

If the XGMII input has the non-data (that is, having format other than D0 D1 D2 D3 D4 D5 D6 D7), encoding is done in the following way:

1. By adding 2-bit Sync Header with the value 2'b10
2. By adding 8-bit Block Type field depending on the following factors:
   - Start of Frame control character (|S|) and location of |S| on 64-bit data
   - Terminate control character (|T|) and location of |T| on 64-bit data

- Order set |O| (Link Message) and location of |O| on 64-bit data
- Error control character |E|
- Idle control character |I|

Figure 2-10 shows how the Encoder maps input XGMII control characters to output.

Figure 2-10 Encoded Output Data for Input XGMII Control Characters

$C_0 C_1 C_2 C_3/C_4 C_5 C_6 C_7$	10	0x1e	C_0	C_1	C_2	C_3	C_4	C_5	C_6	C_7
$C_0 C_1 C_2 C_3/O_4 D_5 D_6 D_7$	10	0x2d	C_0	C_1	C_2	C_3	O_4	D_5	D_6	D_7
$C_0 C_1 C_2 C_3/S_4 D_5 D_6 D_7$	10	0x33	C_0	C_1	C_2	C_3		D_5	D_6	D_7
$O_0 D_1 D_2 D_3/S_4 D_5 D_6 D_7$	10	0x66	D_1	D_2	D_3	O_0		D_5	D_6	D_7
$O_0 D_1 D_2 D_3/O_4 D_5 D_6 D_7$	10	0x55	D_1	D_2	D_3	O_0	O_4	D_5	D_6	D_7
$S_0 D_1 D_2 D_3/D_4 D_5 D_6 D_7$	10	0x78	D_1	D_2	D_3	D_4		D_5	D_6	D_7
$O_0 D_1 D_2 D_3/C_4 C_5 C_6 C_7$	10	0x4b	D_1	D_2	D_3	O_0	C_4	C_5	C_6	C_7
$T_0 C_1 C_2 C_3/C_4 C_5 C_6 C_7$	10	0x87		C_1	C_2	C_3	C_4	C_5	C_6	C_7
$D_0 T_1 C_2 C_3/C_4 C_5 C_6 C_7$	10	0x99	D_0		C_2	C_3	C_4	C_5	C_6	C_7
$D_0 D_1 T_2 C_3/C_4 C_5 C_6 C_7$	10	0xaa	D_0	D_1		C_3	C_4	C_5	C_6	C_7
$D_0 D_1 D_2 T_3/C_4 C_5 C_6 C_7$	10	0xb4	D_0	D_1	D_2		C_4	C_5	C_6	C_7
$D_0 D_1 D_2 D_3/T_4 C_5 C_6 C_7$	10	0xcc	D_0	D_1	D_2	D_3		C_5	C_6	C_7
$D_0 D_1 D_2 D_3/D_4 T_5 C_6 C_7$	10	0xd2	D_0	D_1	D_2	D_3	D_4		C_6	C_7
$D_0 D_1 D_2 D_3/D_4 D_5 T_6 C_7$	10	0xe1	D_0	D_1	D_2	D_3	D_4	D_5		C_7
$D_0 D_1 D_2 D_3/D_4 D_5 D_6 T_7$	10	0xff	D_0	D_1	D_2	D_3	D_4	D_5	D_6	

where in XGMII Input Data Notations:

- Cn = XGMII Control Character in nth byte (xgmii_txc_i[n] being set to 1)
- Dn = XGMII Data Character in nth byte on xgmii_txd_i[] with xgmii_txc_i[n] being set to 0
- S0/4 = XGMII Start Control Character in 0/4 byte (XGMII_DATA[] = 8'hFB and xgmii_txc_i [0/4] = 1)
- Tn = XGMII Terminate Character in nth byte (XGMII_DATA[] = 8'hFD and xgmii_txc_i[n] = 1)
- O0/4 = XGMII Link Message Control Character in 0/4 byte (xgmii_txd_i[0/4] = 8'h9C and xgmii_txc_i [0/4] = 1)

and in Output Notations:

- Cn = Control Code in nth byte (7-bit field with 7'h00 value for Idle and 7'h06 value for |LPI|)
- O0/4 = Message Control Code in 0th or 4th byte (4-bit field with 4'h0 value)
- Dn = XGMII Data Character in nth byte on xgmii_txd_i[]
- || = 1-bit data with value 0 (1-bit field with 1'b0 value)

The output encoded control codes (Cn, O0/4, |S|, |T|, |E|, and other reserved control characters) are shown in Table 2-9. If EEE_EN = Enabled, the DWC_xpcs also encodes the XGMII |LI| (LPI) characters as shown in Table 2-9.

Table 2-9 XGMII Output Control Characters

Control Character	Notation	XGMII Control Code	10GBASE-R Control Code	10GBASE-R O Code
Idle	/I/	0x07	0x00	-
Start	/S/	0xfb	Encoded by the Block Type field	-
Terminate	/T/	0xfd	Encoded by the Block Type field	-
LPI	/LI/	0x06	0x06	-
Error	/E/	0xfe	0x1e	-
Sequence ordered_set	/Q/	0x9c	Encoded by the Block Type field plus O code	0x0
reserved0	/R/	0x1c	0x2d	-
reserved1	-	0x3c	0x33	-
reserved2	/A/	0x7c	0x4b	-
reserved3	/K/	0xbc	0x55	-
reserved4	-	0xdc	0x66	-
reserved5	-	0xf7	0x78	-
Signal ordered_set	/Fsig/	0x5c	Encoded by the Block Type field	0xF

2.4.3.3 Scrambler

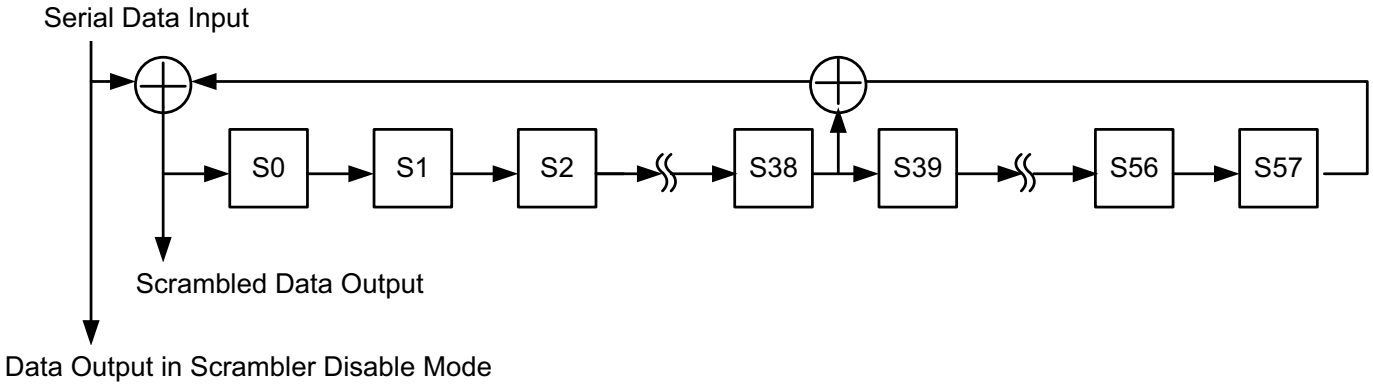
The encoded data is passed through the scrambler. The scrambler takes 64-bit data and generates the scrambled data high in transition density. The scrambler does not alter the 2-bit Sync header because it is used for Block alignment at the receiver and it should not be scrambled.

The scrambling is done using the following polynomial.

$G(x) = 1+ x^{39} + x^{58}$

The initial seed for this scrambler can be any value. Figure 2-11 shows the serial implementation of the scrambler. The scrambler is bypassed depending on Bit 8 of VR PCS 10GBASE-R Control Register. If EEE_EN = Enabled and FEC_EN = Enabled, the scrambler is bypassed during low-power mode of the EEE protocol if Bit 5 of VR XS or PCS MMD EEE Mode Control Register set.

Figure 2-11 Scrambler Implementation



2.4.3.4 FEC Encoder

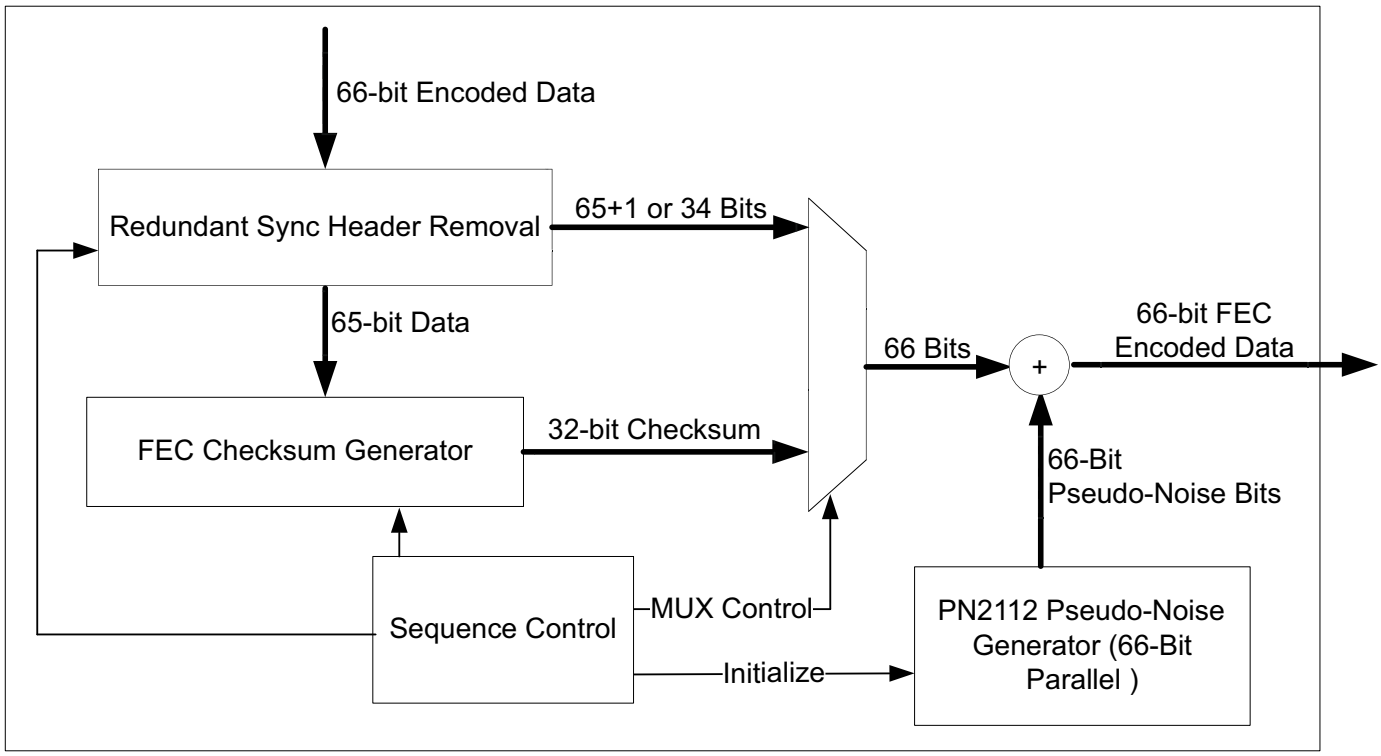
The Forward Error Correction (FEC) encoder block is an optional block. This block is present only when the Add Forward Error Correction Support option is selected. This block enables the error correction mechanism on the Receive path. You can select this option in the KR mode to reduce the effective bit error rate (BER).

This block is needed for erroneous channels, especially when 10GBASE-R type serial data (10.3125 Gbps) is transmitted or received on copper (Backplane Ethernet applications called 10GBASE-KR). This block uses one of the Cyclic Codes which has both Error Detection and Error Correcting capabilities. It can detect and correct up to 11-bit burst errors.

Note The FEC capabilities are exchanged between the local device and the link partner through Clause 73 auto-negotiation. If both local device and link partner support FEC, the FEC feature can be enabled in the local device. You can obtain the FEC ability status through SR_PMA_KR_FEC_ABL Register and you can control the FEC features through SR_PMA_KR_FEC_CTRL Register.

Figure 2-12 shows the FEC encoding process.

Figure 2-12 FEC Generation Process



If FEC is enabled in your configuration and the FEC feature needs to be auto-negotiated, you need to enable the FEC by setting Bit 0 of SR PMA MMD 10GBASE-KR FEC Control Register. The FEC is considered to be successfully auto-negotiated if the following settings are negotiated during base page exchange:

- Bit 14 of SR_AN_ADV3 Register
- Bit 15 of SR_AN_ADV3 Register is set to 1 indicating that the local device is requesting the FEC mode
- Bit 14 of the SR_AN_LP_ABL3 Register
- Bit 15 of the SR_AN_LP_ABL3 Register

In addition, if CL72_EN = Enabled, you need to program as follows:

1. The host should enable the Clause 72 start-up protocol by writing to Bit 1 of SR_PMA_KR_PMD_CTRL Register.
2. The host should restart the Clause 72 start-up protocol by writing to Bit 0 of SR_PMA_KR_PMD_CTRL Register.

FEC Checksum Generator

This block removes one of the redundant bits of Sync Header of 32-blocks (each block is of 66-bit) data and adds 32-bit checksum at the end. It uses the redundant bit (of Sync Header portion) of 64/66 encoded data to add the checksum so that the effective data rate does not change.

The polynomial used for FEC encoder is:

$g(x) = x^{32} + x^{23} + x^{21} + x^{11} + x^{2} + 1$

If $m(x)$ is the message polynomial and $g(x)$ is the generator polynomial, the checksum $p(x)$ is calculated by dividing the message polynomial with the generated polynomial. The remainder is the checksum.

$p(x)= x^{32}m(x) \bmod g(x)$

Note The CRC polynomial is of Cyclic Code (42,987, 42,955) which means the code length is 42,987 bits. This Cyclic Code has error correcting capability of up to 11-bit burst error. In DWC_xpcs, it is used in the Shortened form of Code (2112, 2080) to detect up to 11-bit burst for each 2112 (65x33) block size. This makes the error correcting capability of 11-bit burst error for every 65x33 data block size.

The FEC block `c(x)` is generated by appending or adding the 32-blocks (65 bits) `m(x)` with checksum P(x):

$c(x)= x^{32}m(x)+ p(x)$

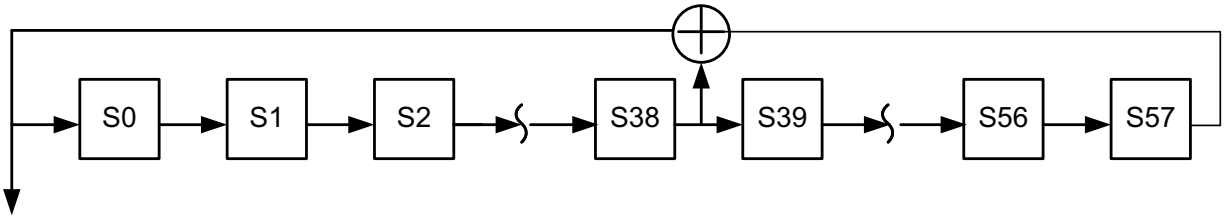
PN2112 Pseudo-Noise Generator

The output of the FEC Encoder block is mixed with the PN2112 pseudo-noise generator to ensure transition density for the FEC encapsulated block. The polynomial used for pseudo-noise generator is:

$r(x) = 1+x^{39}+x^{58}$

Figure 2-13 shows the serial implementation of the PN2112 pseudo-noise generator.

Figure 2-13 Tx Pseudo-Noise Generator



The following is the initial state of the PN2112 pseudo-noise generator:

```
S[57:0] = {29{2'b10}};
```

Note The pseudo-noise generator is initialized with this value before each FEC block (32-block of 66-bit encoded input).

2.4.3.5 Gear Box

The gear box block is an optional block. This block is not present when RPCS PHY Interface Type is 66 x 156.25 MHz. The scrambled or FEC encoded data goes to the gear box which converts the data, from 66-bit data clocking at 156.25 MHz, to one of the following data formats:

- 16-bit data clocking at 644.53125 MHz (Standard XSBI interface format)
- 20-bit data clocking at 515.625MHz (20-bit XSBI interface format)
- 32-bit data clocking at 322.265625 MHz (32-bit XSBI interface format)
- 33-bit data clocking at 312.5 MHz
- 64-bit data clocking at 161.1328125 MHz

- 40-bit data clocking at 257.8125 MHz

The 33-bit data format conversion does not require the rate adaptation because the effective data rate of input and output is same. However, the gear box handles the phase difference between two clocks. The operating frequency and width of the output data depends on the PCS-R PHY interface type.

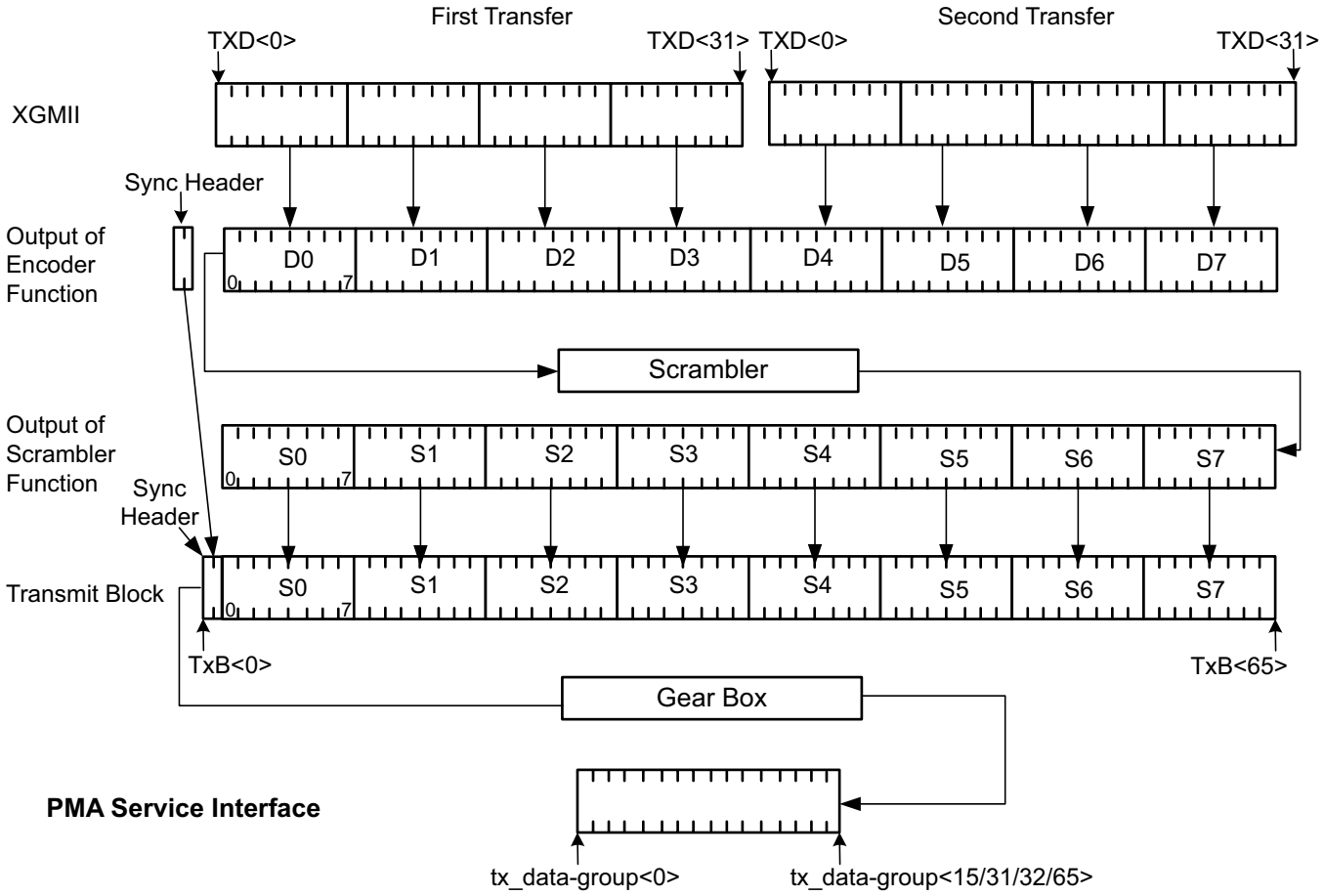
Note The frequencies mentioned in this section apply to 10GBASE-R PCS mode. For 5GBASE-R PCS, all the clock frequencies get reduced by half.

2.4.3.6 PCS-R Transmit Bit Ordering

In 64/66 PCS Layer, the transmit bit ordering is implemented as defined in IEEE Std 802.3, Clause 49. The Sync Header data is transmitted first, followed by the Block Type and Data or Control sub-blocks, as shown in Figure 2-14.

In Figure 2-14, the tx_data-group<0> is expected to be transmitted first. If TX_LSB_FIRST = Enabled, this bit corresponds to rpcs_tx_data_o[0]. If TX_LSB_FIRST = Disabled, this bit corresponds to rpcs_tx_data_o[15/19/31/32/39/63/65].

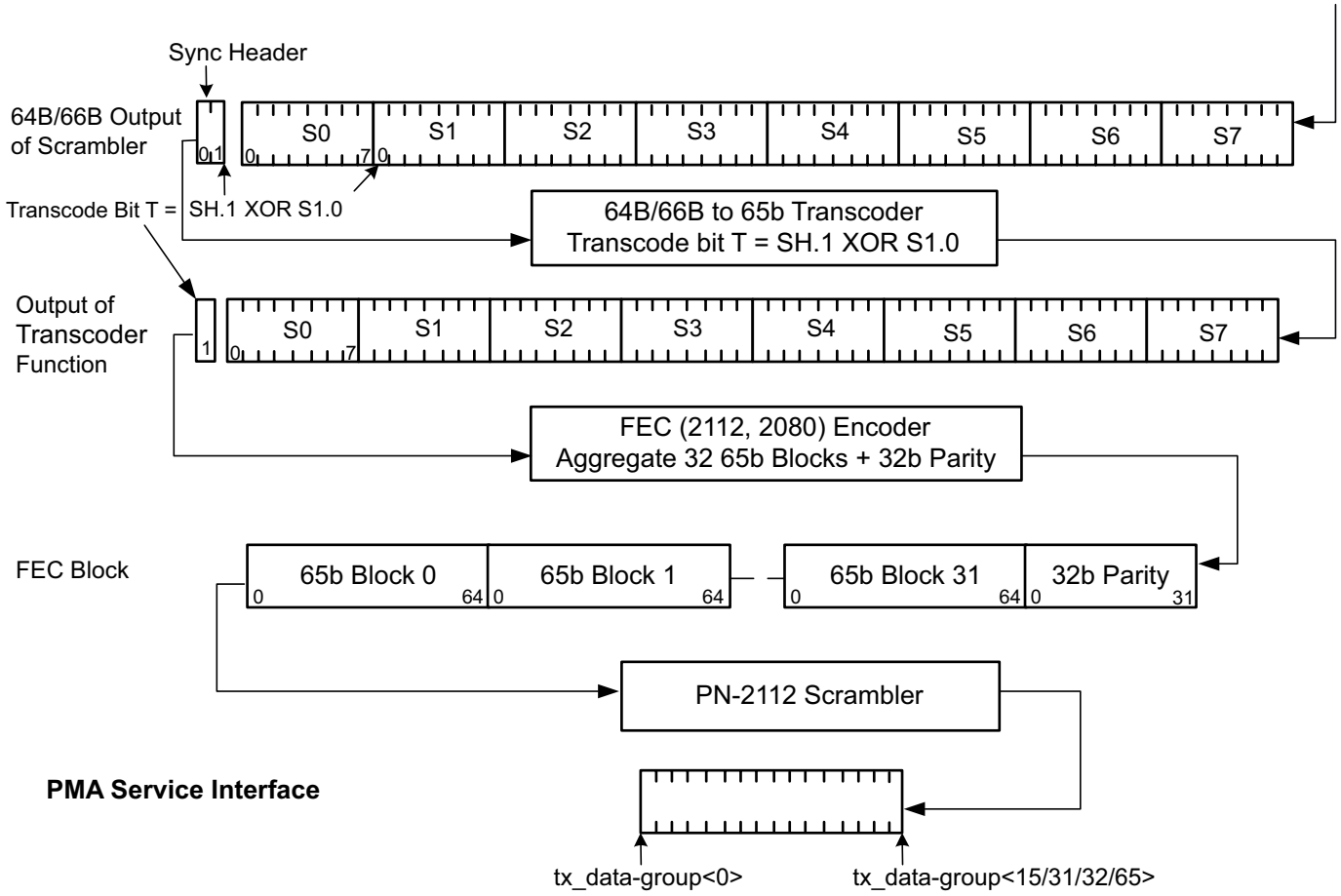
Figure 2-14 Transmit Bit Ordering in 64/66 PCS Layer



In the FEC block, the transmit bit ordering is implemented as defined in IEEE Std 802.3, Clause 74. Figure 2-15 shows the transmit bit ordering in the FEC block.

In Figure 2-15, the tx_data-group<0> is expected to be transmitted first. If TX_LSB_FIRST = Enabled, this bit corresponds to rpcs_tx_data_o[0]. If TX_LSB_FIRST = Disabled, this bit corresponds to rpcs_tx_data_o[15/19/31/32/39/63/65].

Figure 2-15 Transmit Bit Ordering in the FEC Block



2.4.4 Overview of PCS-R Receive Block

The following are the PCS-R Receive block features:

- Implements the following interfaces:
  - XGMII interface towards the MAC
  - 10GBASE-R and 10GBASE-KR or 5GBASE-KR parallel interfaces from the PHY
- Performs 66-bit block alignment
- Performs optional FEC decoding
- Descrambles data to recover the original unscrambled data
- Performs 64B/66B decoding

- Checks the following test patterns:
  - Pseudo Random
  - PRBS 31
  - PRBS 9
- Performs Clock Rate compensation

2.4.5 Description of PCS-R Receive Block

2.4.5.1 Reverse Gear Box

The reverse gear box converts the data from 16-bit, 20-bit, 32-bit, 33-bit, 40-bit, or 64-bit data for 10GBASE-R (clocking at the rate of 644.53 MHz, 515.625 MHz, 322.265 MHz, 312.5 MHz, 257.8125 MHz, or 161.1328125 MHz (XSBI format) to 66-bit data clocking at 156.25 MHz. This block performs the reverse process of the "Gear Box" on page 72.

Note The frequencies mentioned in this section apply to 10GBASE-R PCS mode. For 5GBASE-R PCS, all the clock frequencies are reduced by half.

2.4.5.2 FEC Decoder

The Receive Forward Error Correction (FEC) Decoder block is an optional block. It is present only when the Add Forward Error Correction Support option is selected. This option enables the error correction mechanism on the Receive path. This block performs the following functions on the 66-bit data output from the reverse gear box:

- FEC block synchronization
- FEC decoding (One FEC Decoder for one PCS lane)
- Error correction or error forwarding

If FEC_ERROR_FWD = Enabled, the FEC decoder block provides additional information about uncorrectable errors to the PCS-X block. You can enable the FEC Error Indication control bit by setting Bit 1 of the SR PMA MMD 10GBASE-KR FEC Control Register. You can find the FEC correctable error and uncorrectable error count information through following registers:

- SR_PMA_KR_FEC_CORR_BLK1
- SR_PMA_KR_FEC_CORR_BLK2
- SR_PMA_KR_FEC_UCORR_BLK1
- SR_PMA_KR_FEC_UCORR_BLK2

You can get the FEC ability status through SR_PMA_KR_FEC_ABL Register. You can control the FEC features through SR PMA MMD 10GBASE-KR FEC Control Register.

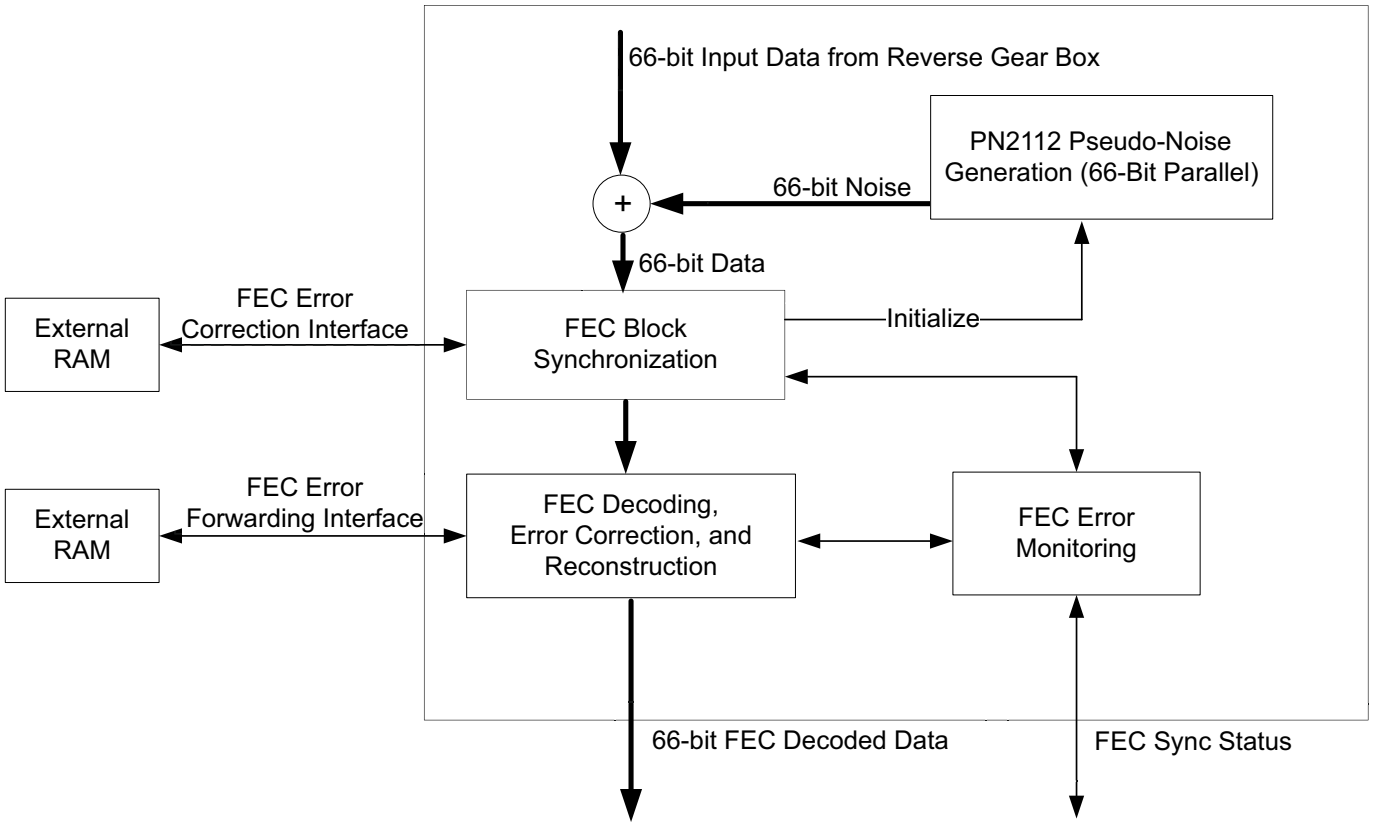
Note An external dual-port/two-port RAM (or register array) of 32 depth and 66-bit width (32x66) is required for FEC decoding. If FEC error forwarding feature is enabled, two such blocks are required.

Note FEC is applicable only for 10GBASE-R PCS. It is not valid in 5GBASE-R PCS mode.

FEC Decoding Process

Figure 2-16 shows the FEC decoding process.

Figure 2-16 FEC Decoding Process



The FEC block synchronization is done in the following way:

1. Test a potential candidate block-start position by following these steps:
   a. Descramble the block by using the PN2112 pseudo-noise generator to derive the raw data.
   b. Evaluate CRC check for the potential block and do one of the following:
      - If the CRC check matches, go to step 2.
      - If the CRC check does not match (that is, the received CRC does not match the computed CRC), shift start-candidate by 1 bit position and repeat step 1.
2. Validate that the potential block start position has a good CRC for n consecutive blocks. Do one of the following:
   - If n (where n = 4) consecutive blocks are received with good CRC, the FEC block synchronization is established.

- If m (where m = 8) consecutive blocks are received with bad CRC, drop the FEC block synchronization and again repeat step 1.

This procedure can be repeated maximum 2,111 times for all bits positions in the 2,112 (32 x 66) block. An FEC decoding error is indicated to the CSR and the 64/66 decoder blocks of the Receive path. After successful decoding, the FEC block synchronizer performs the following:

1. Adds the additional redundant bit to form the original 66-bit Sync Header
2. Removes the CRC bits so that the effective data rate does not change

The input to the FEC block is 32*65 Bit Data + 32 Bit checksum. The data is reconstructed by removing the 32-bit checksum and adding the redundant bit of the Sync Header to all 32-blocks (without affecting the total number of bits).

The DWC_xpcs indicates the decoding errors (error forwarding option) through Sync Header.

1. If decoding is successful (because the CRC matches or the FEC block is correctable) and the descrambled received transcode bit (T) is 1, the sync bits take a value of {SH.0,SH.1} = 01. If the descrambled received transcode bit (T) is 0, the sync bits take a value of {SH.0,SH.1} = 10.
2. If the FEC Error Forwarding is enabled (SR_PMA_KR_FEC_CTRL Register Bit 1 = 1) to indicate errors to the descrambler and the received FEC block has uncorrectable errors, the 1st, 9th, 17th, 25th, and 32nd Sync Header bits for 32 decoded blocks take a value of {SH.0,SH.1} = 11. The Sync Header bits for all remaining blocks take values as described in step 1.

   -or-

   If the FEC Error Forwarding is disabled (SR_PMA_KR_FEC_CTRL Register Bit 1 = 0) and the received FEC block has uncorrectable errors, the sync bits take a value as described in step 1.

Block Alignment

The FEC block synchronizer uses the following method to determine the block boundary of 66-bit encoded data based on the 2-bit Sync Header:

1. The block synchronizer looks for valid Sync Header (2'b10 or 2'b01) on 66-bit data.
2. The block synchronizer monitors the subsequent blocks to validate further consecutive 66-bit data based on the Sync Header (2'b10 or 2'b01).

   If the invalid block count is NORMAL, it locks to the boundary chosen in step 1 and the block alignment is declared as successful. If the invalid block count is 'Not NORMAL', it goes back to step 1 and tries to lock it to a new boundary on 64-bit data which has a valid Sync Header.

PN2122 Pseudo-Noise Generation

The PN2122 pseudo-noise generator block generates the pseudo noise which is mixed with the received data to recover the raw FEC encoded data. The pseudo-noise generator uses the following polynomial:

$r(x) = 1+x^{39}+x^{58}$

The linear feedback shift register (LFSR) implementation of this polynomial and the initial seed used in this block are similar to the one used in the transmitter.

FEC Memory Interfaces

Figure 2-17 shows the timing diagram for FEC write interface. When the rpcs_ec_wr_en_o signal goes high and the rpcs_ec_wr_csn_o signal goes low, the data on the rpcs_ec_wr_data_o signal is written to the memory location indicated by the rpcs_ec_wr_addr_o signal. For example, as shown in Figure 2-17,

0x3_157e_6ebe_87d0_0d3eOx2_10fc_7981_fc8c_ba95 data is written to 0x0 memory location and 0x2_dc2d_2e18_2294_0fa5 is written to 0x1 memory location.

Figure 2-17 FEC Write Interface

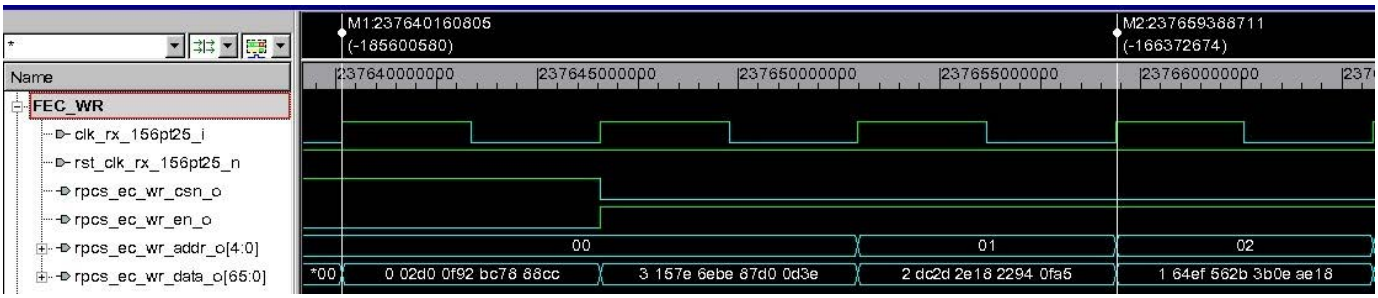
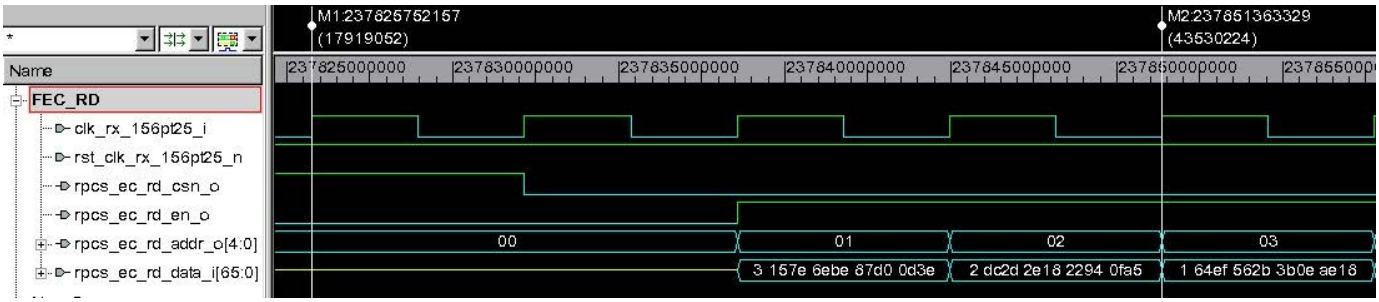


Figure 2-18 shows the timing diagram for FEC read interface. The rpcs_ec_rd_csn_o signal goes low to indicate the start of the read operation. In the next clock edge, the rpcs_ec_rd_en_o signal goes high and the data (rpcs_ec_rd_data_i) from the memory location, indicated by value of rpcs_ec_rd_addr_o at previous clock edge (when rpcs_ec_rd_csn_o goes low), is available.

The address indicated by the rpcs_ec_rd_addr_o signal is valid as long as the rpcs_ec_rd_csn_o signal is low. For example, as shown in Figure 2-18, Ox3_157e_6ebe_87d0_0d3e 0x2_10fc_7981_fc8c_ba95 data is read from 0x0 memory location and 0x2_dc2d_2e18_2294_0fa5 is read from 0x1 memory location.

Figure 2-18 FEC Read Interface



2.4.5.3 Bit Error Rate Monitor

To determine if the locked input data is showing high bit error rate (BER), the BER monitor continuously monitors the input data and validates whether it has a valid Sync Header. If less than 16 erroneous blocks are detected in 125 µs (for 10GBASE-R) or 250 µs (for 5GBASE-R) duration, it is considered as LOW BER. If 16 or more erroneous blocks are detected in 125 µs (for 10GBASE-R) or 250 µs (for 5GBASE-R) duration, it is considered as HIGH BER.

Note For declaring the Receive link to be up (indicated by Bit 2 of SR_XS_PCS_STS1 Register), the block alignment must be successful and HIGH BER must be low.

2.4.5.4 Descrambler

This block descrambles the boundary aligned data by using the polynomial defined in “Scrambler” on page 69. This block performs the reverse process of the scrambler and recovers the original data. If Bit 9 of

VR_PCS_KR_CTRL Register is set, the descrambler is bypassed. This feature can be used for debugging purpose.

2.4.5.5 64/66 Decoder

This block performs the decoding of the 66-bit descrambled data to 64-bit data and 8-bit control as defined in IEEE Std 802.3, Clause 49. This block performs exactly inverse operation of the "64/66 Encoder" on page 67. When it receives any invalid 66-bit data, it maps the data to |E| (error character) in the XGMII format.

2.4.5.6 Clock Rate Compensation

This block performs the data transfer from the Rx clock domain to the Tx clock domain. It removes any clock frequency jitter of +/-100ppm between the Tx and Rx clocks for the maximum frame size of 10,000 bytes. The rate is matched by inserting or deleting the IDLE, LPI, or Sequence Order Sets as applicable during IFG or IDLE period. This is done only if the IFG or IDLE period meets the minimum IFG requirement on XGMII or GMII.

If CLKCOMP = Disabled, this block is not present in the Rx data path but this block is present in the loopback path.

Note This block can handle packets of size up to 16k bytes, only if the IPG between the incoming packets is large. If the incoming 16k bytes packet stream has minimum IPG, this block might corrupt the received data.

2.4.5.7 10GBASE-R or 10GBASE-KR Rx XGMII Interface

This block performs the data format conversion from the DDW format (64-bit data with 8-bit control clocking at 156.25 MHz for 10GBASE-R or 78.125 MHz for 5GBASE-R) to the following formats:

- SDR: 32-bit data with 4-bit control clocking at 312.5 MHz for 10GBASE-R or 156.25 MHz for 5GBASE-R when XGMII = SDR
- DDR: 32-bit data with 4-bit control clocking at 156.25 MHz on both clock edges when XGMII = DDR

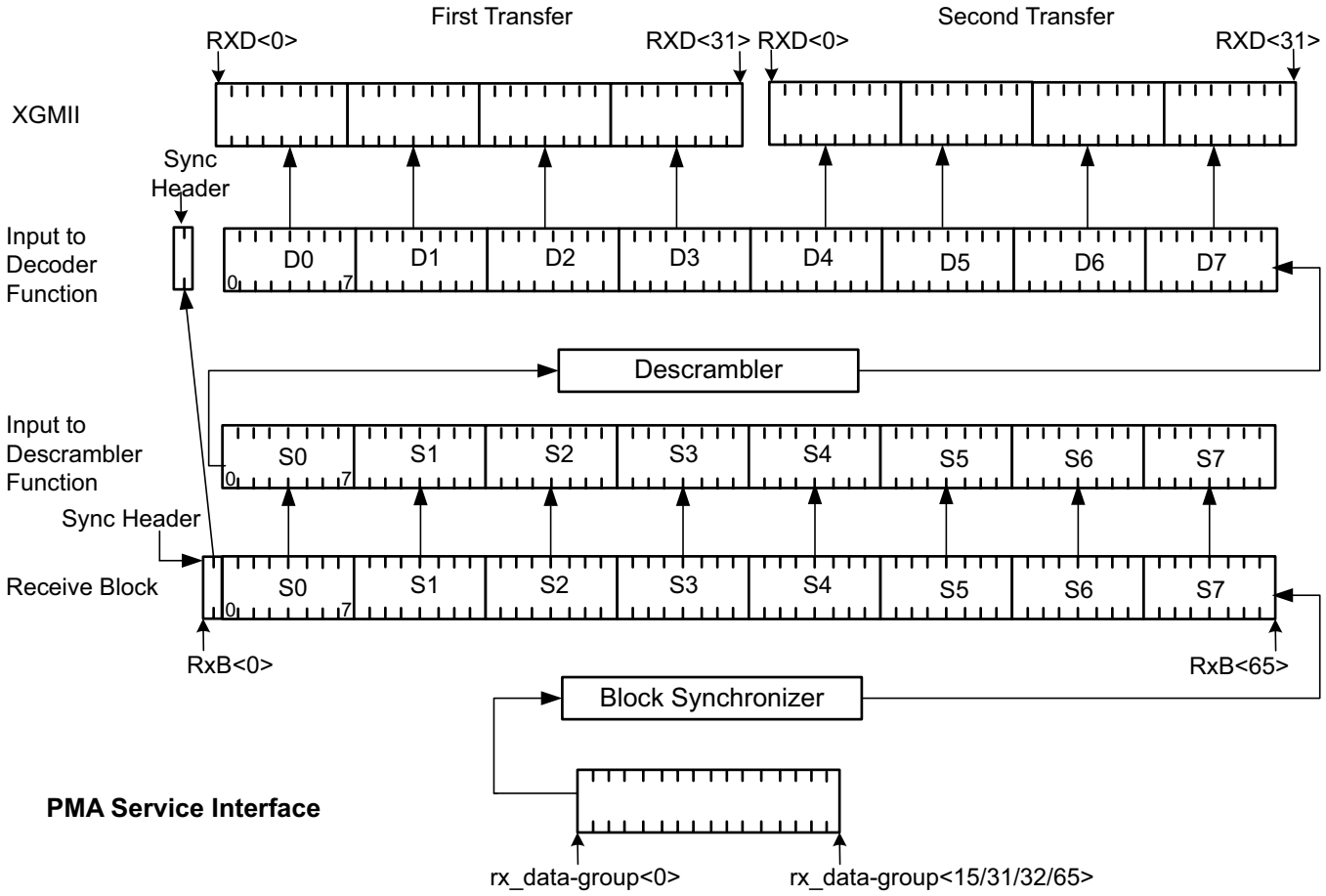
Note This block is not present when XGMII = DDW

2.4.5.8 PCS-R Receive Bit Ordering

In 64/66 PCS Layer, the receive bit ordering is implemented as defined in IEEE Std 802.3, Clause 49. Figure 2-19 shows the receive bit ordering in the 64/66 PCS Layer.

In Figure 2-19, the rx_data-group<0> is expected to be received first. If RX_LSB_FIRST = Enabled, this bit corresponds to rpcs_rx_data_i[0]. If RX_LSB_FIRST = Disabled, this bit corresponds to rpcs_rx_data_o[15/31/32/65].

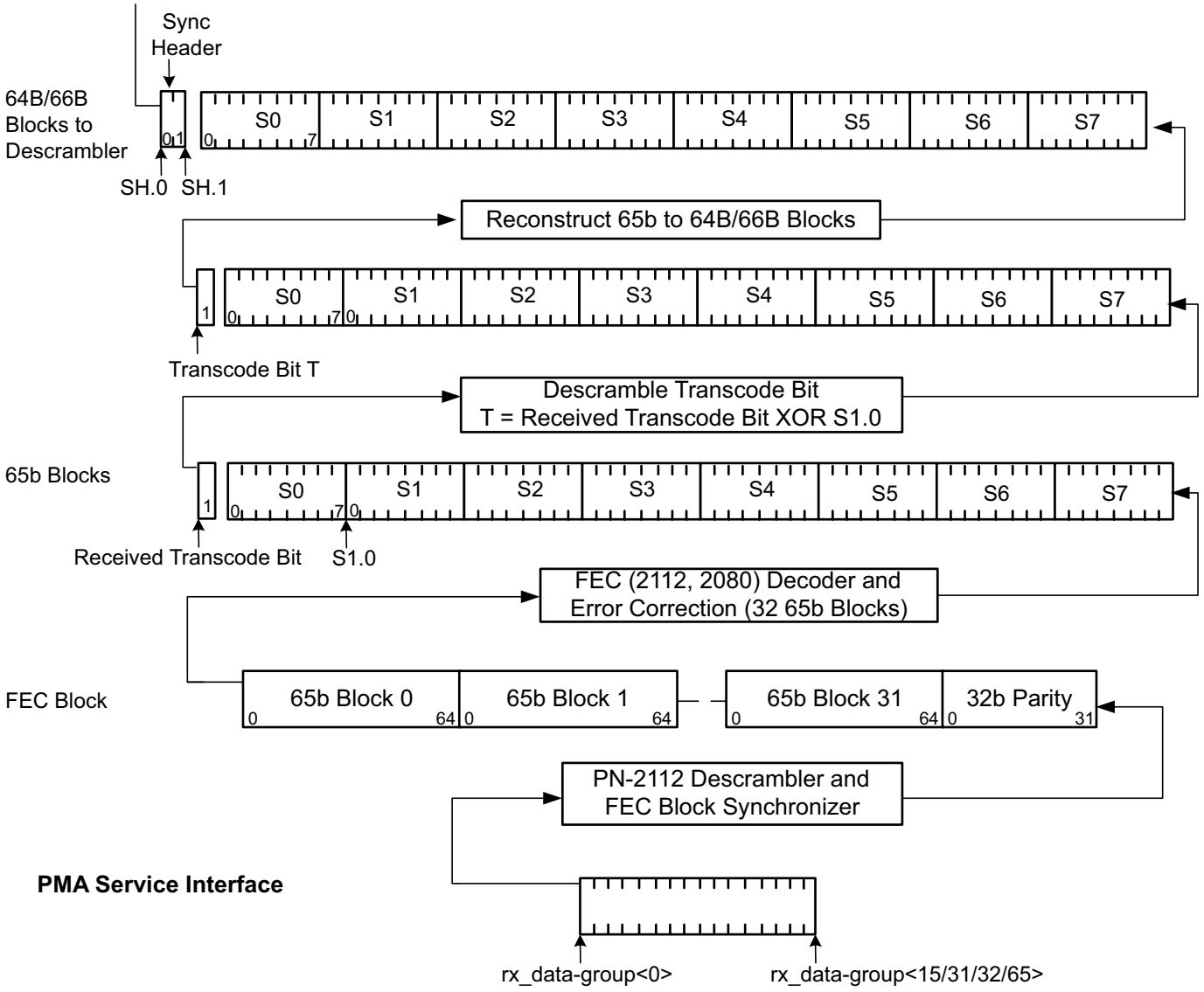
Figure 2-19 Receive Bit Ordering in 64/66 PCS Layer



In the FEC block, the Receive bit ordering is implemented as defined in the IEEE Std 802.3, Clause 74. Figure 2-20 shows the receive bit ordering in the FEC block.

In Figure 2-20, the rx_data-group<0> is expected to be received first. If RX_LSB_FIRST = Enabled, this bit corresponds to rpcs_rx_data_i[0]. If RX_LSB_FIRST = Disabled, this bit corresponds to rpcs_rx_data_o[15/31/32/65].

Figure 2-20 Receive Bit Ordering in the FEC Block



2.4.6 10GBASE-R or 10G/5GBASE-KR Test Pattern Generator and Checker Block

This block generates or checks the Test pattern defined in the IEEE Std 802.3, Clause 49. This block can generate or check the following Test patterns:

- Square Wave

  In the Transmit path, the DWC_xpcs sends the repeating pattern of n number of ones followed by n number of zeros (on serial line) where n can be any number between 4 and 11. You can program the value of n in Bits [6:4] of VR_PCS_KR_CTRL

  The DWC_xpcs does not support detection of this pattern in the Receive path.

- Pseudo Random

  In the Transmit path, the scrambler generates the test pattern for a given set of Seeds and Data inputs. The management can program the seeds and the data. The scrambler is loaded with a seed or its invert at the start of every 128th block. The seeds are loaded in the following pattern:

  a. Seed A
  b. Seed A Invert
  c. Seed B
  d. Seed B Invert

  You can select the data patterns for these seeds by using the SR PCS 10GBASE-R Test Pattern Seed A and B registers (described in the "*Register Descriptions*" chapter). After loading Seed A or Seed B, the scrambler input is driven with the data pattern. After loading the Seed A invert or Seed B Invert, the scrambler input is driven with the inverse of the data pattern. In pseudo-random pattern, the Sync Header is the control Sync Header (10). Therefore, the pseudo-random test pattern is a series of blocks with a control Sync Header and a pseudo-random payload.

  In the Receive path, the DWC_xpcs performs the test pattern checking by monitoring the received descrambled data every 128 clocks.

- PRBS31 or PRBS9

  In the Transmit path, the output of this block directly drives the gear box. In PRBS31 test pattern, the DWC_xpcs generates the pseudo-random sequence corresponding to the following polynomial:

  $G(x) = 1+x^{28}+x^{31}$

  In PRBS9 test pattern, the DWC_xpcs generates the pseudo-random sequence corresponding to the following polynomial:

  $G(x) = 1+x^{5}+x^{9}$

In the Receive path, the DWC_xpcs descrambles the PRBS31 or PRBS9 test pattern. If there are no errors, the descrambled output is expected to be 0.

If there is any error, the test pattern mismatch counter is incremented in the SR_XS_PCS_TP_ERRCTR Register. You can control the test pattern generation or checking through SR_XS_PCS_TP_CTR and VR_PCS_KR_CTRL Registers. When Test Pattern checking is enabled in the Receive path, you can use the error counters to monitor or test the link reliability and perform jitter testing.



Note The error count register indicates the number of 66-bit blocks received with error. It does not indicate the number of bit errors.

2.4.7 Programming Guidelines for the PCS-R

For more information, see, "Switching to 10G/5G KR Speed Mode" on page 1676.

2.5 Universal Serial XGMII (USXGMII)

This topic contains the following sections:

- "Overview of Universal Serial XGMII (USXGMII)" on page 83
- "Operating Speed Modes" on page 85
- "GMII/MII to XGMII Converter" on page 87
- "Data Replication" on page 88
- "Auto-Negotiation" on page 88
- "Enabling the USXGMII Mode" on page 90
- "Signals Related to USXGMII Mode" on page 90
- "Registers Related to USXGMII Mode" on page 90
- "Programming Guidelines to Switch to USXGMII Mode" on page 90

2.5.1 Overview of Universal Serial XGMII (USXGMII)

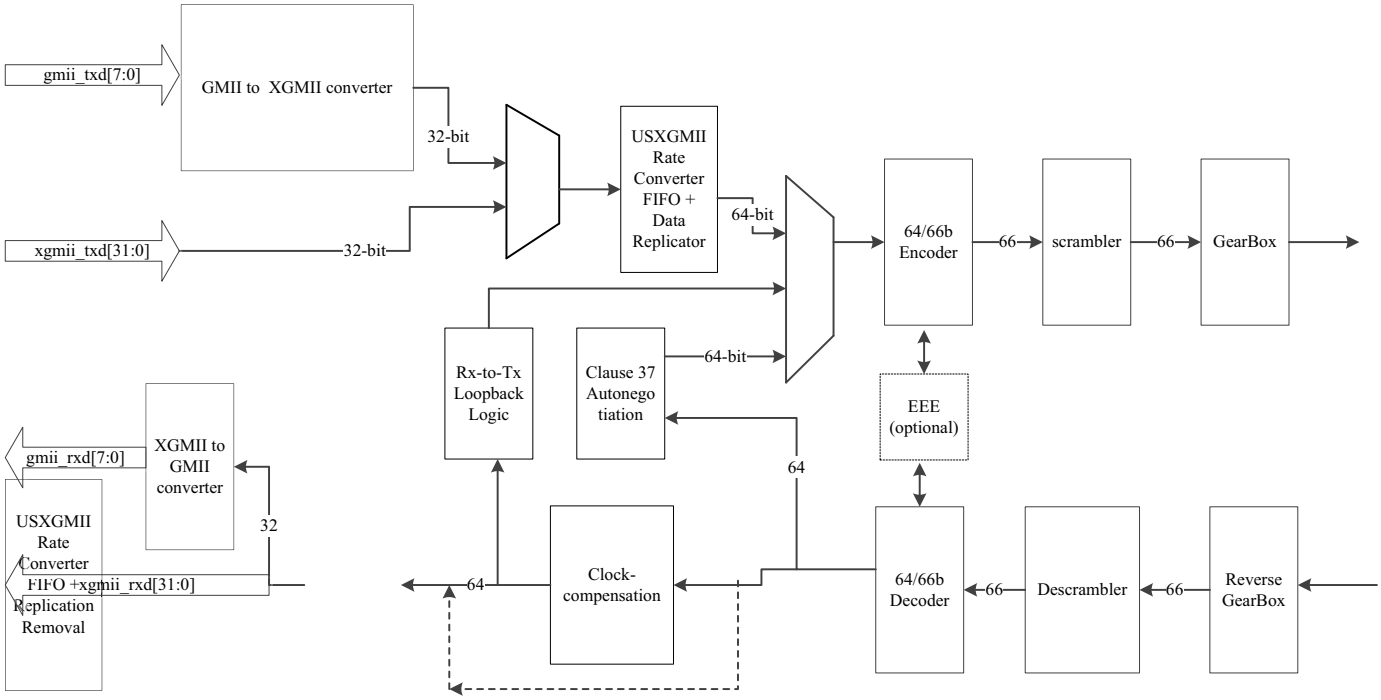
USXGMII protocol is used to convey 10/100/1000Mbps GMII/MII data or 10G/5G/2.5G XGMII data, between MAC and PHY, over a serial line operating at 10.3125Gbps/5.15625 Gbps / 2.578125 Gbps. DWC_xpcs supports the following variants of USXGMII:

- Single Port: 10G-SXGMII, 5G-SXGMII, 2.5G-SXGMII
- Dual Port: 10G-DXGMII, 5G-DXGMII
- Quad Port: 10G-XGMII

The main highlights of 10G-USXGMII are:

- USXGMII architecture of DWC_xpcs supports one/two/four GMII port and one/two/four XGMII port
- GMII port is used to convey data from/to a 10/100/1000Mbps MAC
- XGMII port is used to convey data from/to a 2.5/5/10G MAC
- One serial lane operating at 10.3125Gbps/5.15625 Gbps / 2.578125 Gbps
- GMII/XGMII ports operate at clk_xgmii_tx_i/clk_xgmii_rx_i /clk_xgmii_tx_p{1,2,3}_i/clk_xgmii_rx_p{1,2,3}_i clocks

Figure 2-21 Single-port USXGMII block Diagram Support



Single-port Mode:

The XGMII transmit data (after replication/converting GMII to XGMII) is encoded using a 64/66b encoder (as described in clause 49 of IEEE 802.3-2012), which is then scrambled and fed to a Gear-Box to convert the data to a width suitable for the SERDES.

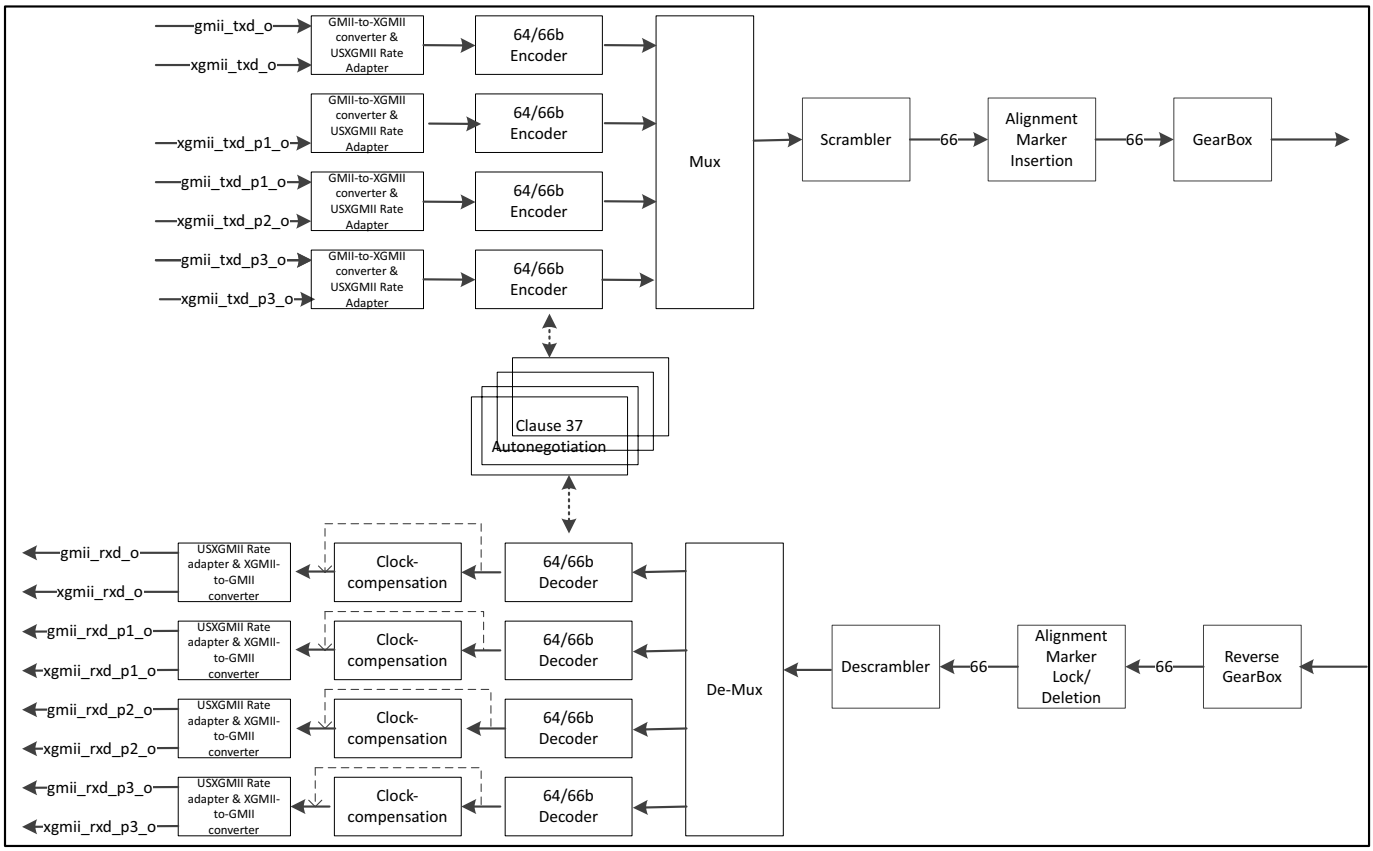
In the receive path, data received data from SERDES is first fed to a reverse gear-box to convert it into 66-bit format. The data is then descrambled and 66/64b decoded. The data-replication (for speeds other than 10G) is removed and then suitably fed to XGMII or GMII port based on the operating speed mode.

Multi-port Mode:

The XGMII transmit data from each of the active ports (one, two or four) is first 64/66b encoded individually. The data outputs from the encoders are time-division multiplexed and then fed to a common scrambler. Alignment markers (set of four) are inserted periodically at an interval of 16400 blocks in the output stream of the scrambler and then finally fed to the Gear-Box.

In the receive path, data received from SERDES is fed to a reverse gear-box to convert it into 66-bit format. The output of the gear-box is fed to an alignment marker detection module to delete alignment markers from the incoming stream. The data stream is then descrambled and de-multiplexed to separate data-streams (two or four). Each data-stream is then separately decoded by a 64/66b decoder, data-replication removed and suitably fed to the corresponding XGMII or GMII Rx port.

Figure 2-22 Multi-Port USXGMII Block Diagram



2.5.2 Operating Speed Modes

DWC_xpcs supports 10G, 5G, 2.5G, 1G, 100M, and 10M speeds in USXGMII mode.

2.5.2.1 10G Speed Mode

This is the default speed mode of DWC_xpcs. In this mode, MAC is expected to send data at 10 Gbps rate through the XGMII interface. The XGMII interface operates at 32-bit @ 312.5 MHz. The 32-bit XGMII data is converted to 64-bit at 156.25MHz internally. This data is processed as per Clause 49 of IEEE 802.3 specification and finally sent of DWC_xpcs towards the PHY at a rate of 10.3125Gbps.

2.5.2.2 5G Speed Mode

In this mode, MAC sends data at 5 Gbps rate through the XGMII interface. The XGMI interface operates at 32-bit @ 156.25 MHz.

For 10G-SXGMII, the 32-bit XGMII data is replicated twice to form 64-bit data at 156.25 MHz. This data is then processed as per Clause 49 and finally sent towards the PHY at a rate of 10.3125Gbps.

For 5G-SXGMII, the 32-bit XGMII data at 156.25 MHz is converted to 64-bit data at 78.125 MHz. This data is then processed as per Clause 49 and finally sent towards the PHY at a rate of 5.15625Gbps.

For 10G-DXGMII, the 32-bit XGMII data (from each of the two ports) at 156.25 MHz is first converted to 64-bit data at alternate 156.25 MHz. This data is then processed as per Clause 49, multiplexed with data from other port, alignment markers inserted and finally sent towards the PHY at a rate of 10.3125Gpbs.

2.5.2.3 2.5G Speed Mode

In this mode, MAC sends data at 2.5 Gbps rate through the XGMII interface. The XGMI interface operates at 32-bit @ 78.125 MHz.

For 2.5G-SXGMII, the 32-bit XGMII data at 78.125 MHz is converted to 64-bit data at 39.0625 MHz. This data is then processed as per Clause 49 and finally sent towards the PHY at a rate of 2.578125 Gbps.

For 5G-SXGMII, the 32-bit XGMII data at 78.125 MHz is replicated twice to form 64-bit data at 78.125 MHz. This data is then processed as per Clause 49 and finally sent towards the PHY at a rate of 5.51625 Gbps.

For 10G-SXGMII, the 32-bit XGMII data is replicated four-times and converted into 64-bit data at 156.25 MHz domain. This data is then processed as per Clause 49 and finally sent towards the PHY at a rate of 10.3125Gbps.

For 10G-QXGMII, the 32-bit XGMII data (from each of the 4 ports) at 78.125 MHz is first converted to 64-bit data at once in 4 clocks of 156.25 MHz. This data is then processed as per Clause 49, multiplexed with data from other port, alignment markers inserted and finally sent towards the PHY at a rate of 10.3125Gbps.

For 10G-DXGMII, the 32-bit XGMII data (from each of the 2 ports) at 78.125 MHz is replicated twice and is converted to 64-bit data at once in 2 clocks of 156.25 MHz. This data is then processed as per Clause 49, multiplexed with data from other port, alignment markers inserted and finally sent towards the PHY at a rate of 10.3125Gbps.

For 5G-DXGMII, the 32-bit XGMII data (from each of the 2 ports) at 78.125 MHz is converted to 64-bit data at once in 2 clocks of 78.125 MHz. This data is then processed as per Clause 49, multiplexed with data from other port, alignment markers inserted and finally sent towards the PHY at a rate of 5.15625Gbps.



USXG_2PT5G_GMII field of VR_XS_PCS_KR_CTRL register can be set when operating in single-port USXGMII modes to make use of GMII interface (instead of XGMII) at 2.5G speed mode. However, this feature is not available when operating in multi-port modes.

2.5.2.4 1G Speed Mode

In this mode, MAC sends data at 1 Gbps rate through the GMII interface. The GMI interface operates at 8-bit @ 125 MHz. The 8-bit GMII data is first converted to a 32-bit XGMII format. Thereafter, this 32-bit XGMII data is replicated 10/5/2.5 times (based on the USXGMII variant) and finally it is converted into 64-bit data at 156.25/78.125/39.0625 MHz format. This data is then processed as per Clause 49 and finally sent towards the PHY at a rate of 10.3125/5.15625/2.578125 Gbps.

2.5.2.5 100Mbps Speed Mode

In this mode, MAC sends data at 100 Mbps rate through the MII interface (MII interface is physically same as that the GMII interface). The MI interface can operate either at 8-bit @ 12.5 MHz or 4-bit @ 25 MHz (selected based on MII_CTRL bit of VR MII MMD AN Control Register). This MII data is first converted to a 32-bit XGMII data. Thereafter, this 32-bit XGMII data is replicated 100/50/25 times (based on USXGMII

variant) and finally it is converted into 64-bit data at 156.25 MHz /78.125/39.0625 format. This data is then processed as per Clause 49 and finally sent towards the PHY at a rate of 10.3125/5.15625/2.578125 Gbps.

2.5.2.6 10Mbps Speed Mode

In this mode, MAC sends data at 10 Mbps rate through the MII interface (MII interface is physically same as that the GMII interface). The MI interface can operate either at 8-bit @ 1.25 MHz or 4-bit @ 2.5 MHz (selected based on MII_CTRL bit of VR MII MMD AN Control Register). This MII data is first converted to a 32-bit XGMII data. Thereafter, this 32-bit XGMII data is replicated 1000/500/250 times (based on USXGMII variant) and finally it is converted into 64-bit data at 156.25/78.125/39.0625 MHz format. This data is then processed as per Clause 49 and finally sent towards the PHY at a rate of 10.3125/5.15625/2.578125 Gbps.

2.5.3 GMII/MII to XGMII Converter

When operating in 1G Mode, the 8-bit GMII Tx data is first converted to 32-bit XGMII format. Similarly, when operating in 10/100Mbps mode, the 4/8-bit MII Tx data is first converted to 32-bit XGMII format. In order to do this conversion, 4 consecutive bytes of GMII/MII data are clubbed together.

Table 2-10 GMII-to-XGMII Mapping

GMII Input	Corresponding 32-Bit XGMII
TX_EN=0, TX_ER=0, TXD=<anything> (except for TX_EN: 1 -> 0 transition)	IDLE: txd = 32'h07070707 txc = 4'hf
TX_EN=1, TX_ER= 0, TXD=<anything> (except for TX_EN: 0 -> 1 transition)	DATA: txd = 32'hD3D2D1D0 txc = 4'h0
TX_EN= 1, TX_ER=0, TXD = <anything> (during TX_EN: 0 -> 1 transition)	SOP: txd = 32'hD3D2D1fb txc = 4'h1
TX_EN= 0, TX_ER= 0, TXD = <anything> (during TX_EN: 1-> 0 transition)	EOP: Any of the following: a. 32'h070707fd b. 32'h0707fdD0 c. 32'h07fdD1D0 d. 32'hfdD2D1D0
TX_EN= 1 and TX_ER =1 (in the middle of a frame)	Treated as txc[i]=1'b1 and txd[i]=8'hfe Consequently, 64-bit XGMII data is interpreted as 'Error' Block by 64/66b encoder.
TX_EN = 0 , TX_ER =1, TXD = 8'h01	LPI IDLE: txd = 32'h06060606 txc = 4'hf
Carrier Extension /Carrier Extension Error	Not supported.

Whenever 4 bytes of GMII/MII Data are grouped as XGMII, care is taken to ensure that SOP character (8'hfb) gets aligned to bits [7:0] of XGMII. This is done by adding/deleting IDLE bytes suitably. After the start of a packet, a maximum of 3 bytes of IDLE may be added. After the end of the packet, the same number of IDLE bytes is removed.

In the receive path, the 32-bit XGMII data is converted to 8-bit/4-bit GMII data when operating in 1G/100M/100M speed modes.

2.5.4 Data Replication

In the transmit path, data is replicated (in terms of 32-bit) 2, 4, 10,100 or 1000 times when operating in 5G,2.5G,1G, 100M, 10M speed modes respectively.

- SOP word (4-bytes) is transmitted only once and in the remainder of the 1 (5G), 3(2.5G), 9 (1G), 99 (100M) or 999 (10M) 4-byte words, 0xfb is replaced by 0xaa.
- Each word (4-bytes) of the preamble and payload is replicated 2/4/10/100/1000 times for 5G/2.5G/1G/100M/10M respectively.
- EOP is transmitted once and the remaining 1/3/9/99/999 bytes are idles
- Each idle byte is replicated 2/4/10/100/1000 times.
- All other sequence ordered sets and control sets are replicated 2/4/10/100/1000 times.

In the receive path, the data-replication is removed from the incoming data stream before passing on to XGMII/GMII Rx interface.

Note The replication numbers mentioned in this section corresponds to 10G-SXGMII mode. For other USXGMII modes, the replication numbers are suitably modified to match the serial baud-rate.

2.5.5 Auto-Negotiation

USXGMII supports Clause 37 auto-negotiation feature to enable the exchange of capabilities between the MAC and PHY. The auto-negotiation Configuration Word (UsxgmiiChannelInfo) mapping is shown in Table 2-11.

Table 2-11 Auto-Negotiation Configuration Register

Bit	Tx_Config_Reg[15:0] Sent from PHY to MAC	Tx_Config_Reg[15:0] Sent from MAC to PHY
15	Link: 1= Link up, 0 = Link Down	DWC_xpcs always sends this field as 0
14	Auto-negotiation acknowledge. This field is derived from SGMII_LINK_STS bit of VR MII MMD AN Control Register.	1
13	0: Reserved for future use	0: Reserved for future use
12	Duplex Mode: 1= full_duplex, 0 = half_duplex DWC_xpcs always sends 1, as USXGMII does not support half-duplex links.	Duplex Mode: 1= full_duplex, 0 = half_duplex DWC_xpcs always sends 1, as USXGMII does not support half-duplex links

Bit	Tx_Config_Reg[15:0] Sent from PHY to MAC	Tx_Config_Reg[15:0] Sent from MAC to PHY
11:9	Speed: 000 = 10Mbps 001 = 100 Mbps 010 = 1000 Mbps 011 = 10 Gbps 100 = 2.5 Gbps 101 = 5 Gbps 110 = Reserved 111 = Reserved When DWC_xpcs is configured as USXGMII PHY, this field is derived from SS5,SS6 and SS13 bits of SR MII MMD Control Register.	Speed: DWC_xpcs (configured as MAC) sends this value as 011(10Gbps). The link operates at the speed conveyed by the PHY to the MAC.
8	EEE capability:1= supported, 0 = not supported If EEE_EN=Enabled, this field is transmitted as 1. Otherwise, it is 0.	EEE capability:1= supported, 0 = not supported If EEE_EN=Enabled, this field is transmitted as 1. Otherwise, it is 0.
7	EEE clock stop capability: 1= supported, 0 = not supported If EEE_EN=Enabled, this field is transmitted as 1. Otherwise, it is 0.	EEE clock stop capability: 1= supported, 0 = not supported If EEE_EN=Enabled, this field is derived from CS_EN bit of SR XS or PCS MMD Control 1 Register. Otherwise, it is 0.
6:1	0: Reserved for future use	0: Reserved for future use
0	1	1

Note

The content of the configuration word sent from MAC to PHY (speed and duplex mode) is based on the following information:

- Status of the link/link-speed should be ideally conveyed by PHY to the MAC.
- MAC does not convey these values (by itself) to the PHY. One possible option is to get this information (link status/link speed) from the config_reg received from the PHY and thereafter MAC can echo-back these values in its transmitted config_reg. However, this strategy would result in 'consistency_match' evaluating to a FALSE condition (ability match config_reg and acknowledge_match config_reg would be different), which would in turn result in an unnecessary restart of the auto-negotiation. Therefore, DWC_xpcs does not adopt the strategy of echoing back the link status/link-speed values. Instead, it always sends link-status as 0 and link-speed as 3'b011 in its transmitted config_reg.

The auto-negotiation configuration word is mapped into an XGMII ordered set as: 32'h03_Config[7:0]_Config[15:8]_9c

This XGMII ordered set is 64/66b encoded as per Clause 49, scrambled and then sent to the link partner during auto-negotiation.

Auto-negotiation process follows figure 37-6 (Auto-Negotiation state diagram) of IEEE 802.3 with few modifications as USXGMII auto-negotiation exchanges data using 64/66b 10GBASE-R PCS.

2.5.6 Enabling the USXGMII Mode

To enable the USXGMII mode during the Specify Configuration Activity in coreConsultant, select the "Add Single-port USXGMII Support" parameter, in the System Configurations Settings window.

For details about this option, see System Configurations Settings Parameters in the "*Parameter Descriptions*" chapter.

Note See the "Enabled" field of the parameter descriptions to understand the dependencies.

2.5.7 Signals Related to USXGMII Mode

See the "*Signal Descriptions*" chapter.

2.5.8 Registers Related to USXGMII Mode

- VR_XS_PCS_DIG_CTRL1
- VR_MII_AN_CTRL
- SR_MII_CTRL

2.5.9 Programming Guidelines to Switch to USXGMII Mode

See "Switching to USXGMII Mode" on page 1682

2.6 2.5G GMII Mode

This topic contains the following sections:

- "Overview of 2.5G GMII Mode" on page 91
- "Enabling 2.5G GMII Mode Support" on page 92
- "Signals Related to 2.5G GMII Support" on page 92
- "Registers Related to 2.5G GMII Support" on page 92
- "Programming Guidelines for 2.5G GMII Support" on page 92

2.6.1 Overview of 2.5G GMII Mode

DWC_xpcs supports 2.5G mode of operation over GMII interface, when the selected configuration has 1000BASE-X/KX mode support. The 2.5G mode of operation is functionally the same as 1000BASE-X/KX mode, except that the clock-rate is 2.5 times the original rate. In this mode, the Serdes/PHY operates at a serial baud-rate of 3.125 Gbps and DWC_xpcs data-path and the GMII interface to MAC operates at 312.5 MHz (instead of 125 MHz).

The 2.5G speed mode can be enabled in DWC_xpcs by setting it to 1000BASE-X/KX Mode (by programming SS13 bit of SR_XS_PCS_CTRL1 to 0) and providing a 312.5 MHz clocks to the TX and RX paths of the DWC_xpcs. In addition, the host can set the EN_2_5G_MODE (bit [2]) of the VR_MII_DIG_CTRL1 (for 1000BASE-X-Only PCS configurations) or VR_XS_PCS_DIG_CTRL1 (for all other configurations) to 1. Correspondingly, the 2.5G speed mode is indicated on the 'xpcs_2pt5g_mode_o' output port. This port can be used to select the appropriate clock from the PHY. For more details, see the Clocking Diagrams in the DesignWare Cores Ethernet PCS User Guide.

Note

When operating in the 2.5G speed mode:

- There is no automatic way for DWC_xpcs to detect or indicate that the link is operating in 2.5G rate or 1G rate. So, both sides of the link should be configured by the software to operate in 2.5G mode. The link fails to establish connection if one side is in 1G rate and the other is in 2.5G rate.
- Clause 73 auto-negotiation is not supported in 2.5G mode. So, when the node is operating in 2.5G mode, the Clause 73 Auto-negotiation protocol using DME encoding might not work.
- Clause 37 (& SGMII) auto-negotiation is supported in 2.5G mode if the link partner is also operating in the equivalent 2.5G mode.

When DWC_xpcs is operating in 2.5G SGMII mode, the speed mode bits (SS13 and SS6 of SR MII MMD Control Register) give the speeds for the standard encoding as mentioned in Table 2-12. In other words, the DWC_xpcs operates either in 10/100/1000Mbps rate or 25/250/2500Mbps rates respectively with SGMII auto-negotiation. The DWC_xpcs cannot support switching or negotiation between 1G and 2.5G rates using auto-negotiation.

Table 2-12 SGMII 2.5G Speed Modes

SS6	SS13	Speed
1	1	Reserved
1	0	2500 Mbps

SS6	SS13	Speed
0	1	250 Mbps
0	0	25 Mbps

During the Clause 37/SGMII as the auto-negotiation link timer operates with a faster clock in the 2.5G mode, the timeout duration reduces by a factor of 2.5. To restore the standard specified timeout period, program the registers as follows:

- SGMII Mode: To restore the 1.6ms link timer
  - Program the VR MII MMD Link Timer Register to 16'h7A1
  - Program bit[3] (CL37_TMR_OVR_RIDE) of VR MII MMD Digital Control 1 Register to 1
- 1000BASE-X mode: To restore the 10ms link timer
  - Program VR MII MMD Link Timer Register to 16'h2FAF
  - Program bit[3] (CL37_TMR_OVR_RIDE) of VR MII MMD Digital Control 1 Register to 1

The DWC_xpcs also supports Energy Efficient Ethernet in 2.5G mode to enable low-power operations. It follows the same steps/sequences prescribed by IEEE specification for 1000BASE-KX mode with same timer periods. No specific or alternative programming steps are required for this operation.

2.6.2 Enabling 2.5G GMII Mode Support

2.5G GMII mode is present by default if the selected configuration has 1G/KX Mode support.

2.6.3 Signals Related to 2.5G GMII Support

The main signal corresponding to this mode is

- xpcs_2pt5g_mode_o

2.6.4 Registers Related to 2.5G GMII Support

- VR_MII_DIG_CTRL1

2.6.5 Programming Guidelines for 2.5G GMII Support

See "Programming Guidelines for 2.5G GMII Mode" on page 1692.

2.7 Command and Status Register Block

This topic contains the following sections:

- "Overview of CSR Block" on page 93
- "CSR Addressing" on page 94

2.7.1 Overview of CSR Block

The Command and Status Registers (CSR) block performs the following functions:

- Maintains the following MMDs register sets (depending on the configuration and current operating mode):
  - PCS
  - PMA or PMD
  - XGXS (DTE or PHY configurable)
  - Auto-negotiation
  - Vendor-specific MII MMD
  - Vendor-specific Control MMD
- Interacts with all blocks to provide the control and status information

The CSR register sets are used as Command and Status registers. A management agent can issue commands to set attributes supported by the DWC_xpcs, or to read the status of the device or command. This block implements all register sets required for implementing the controller and complying with IEEE std, Clause 45.

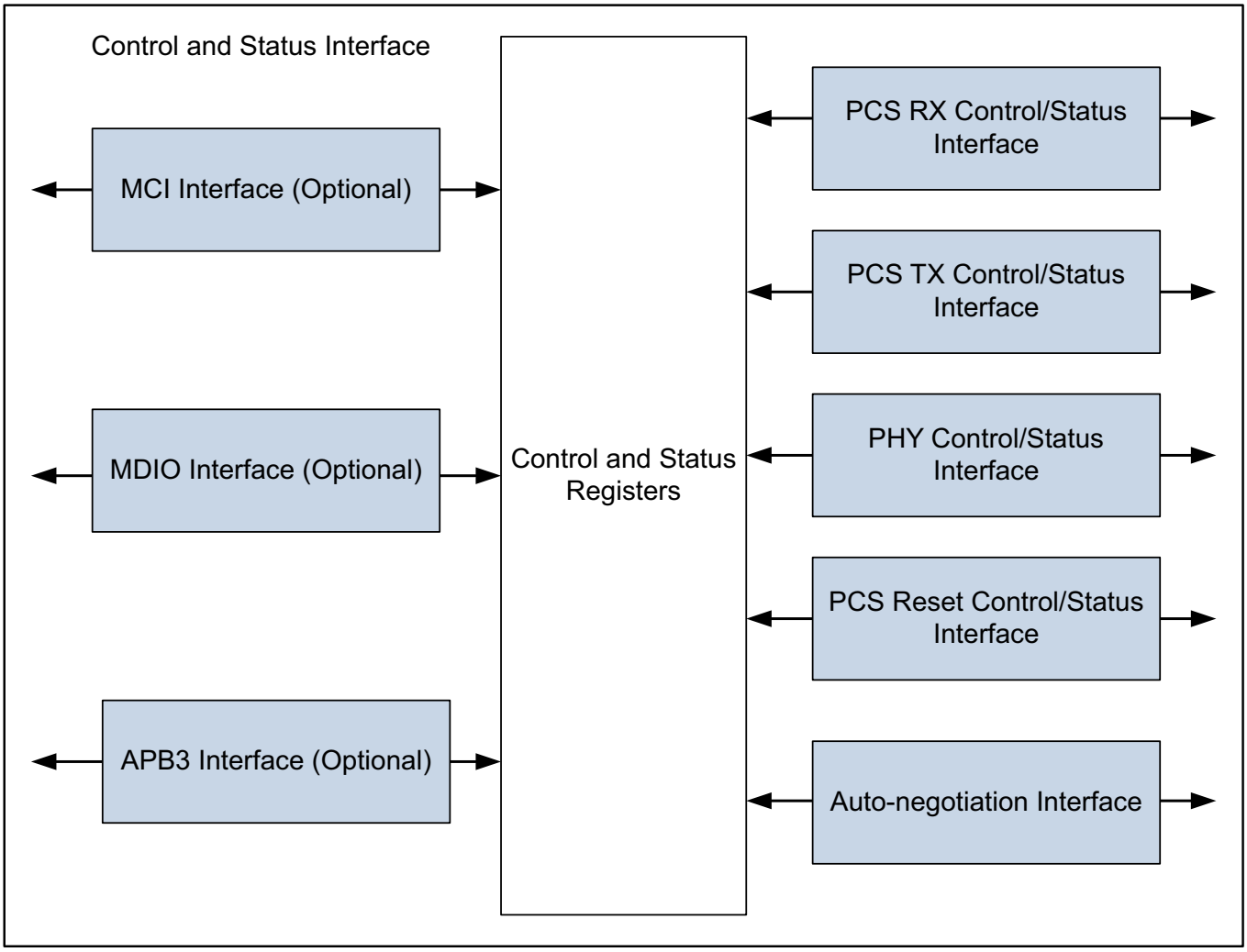
Along with the standard register sets, this block implements a vendor-specific register set to deploy additional features. Vendor-specific register sets provide flexibility to the management agent to use features and attributes supported by the DWC_xpcs and not defined in the Standard.

The host can access the CSR block through the following interfaces:

- Generic parallel MCI interface
- Standard serial MDIO interface
- AMBA APB 3 Interface (3 variants)

You can select one of these interfaces through the Management Interface option in coreConsultant. Figure 2-23 displays the CSR block.

Figure 2-23 CSR Interface Block Diagram



2.7.2 CSR Addressing

The CSR register set addressing includes the following factors:

- MMD Device Address
- Register Address
- Port ID

Note The Port ID is not needed for the MCI or APB3 Interface. The Port ID is needed only when CSR_INTERFACE = MDIO.

2.7.2.1 MMD Device Address

The 5-bit MMD device address specifies which MMD device is being addressed by the host. The DWC_xpcs supports various MMDs depending on the configuration.If MAIN_MODE = XGXS PCS, only PHY XS or DTE XS MMDs are present. If MAIN_MODE = Backplane Ethernet PCS, the PCS, PMA, PMD, and AN MMDs are present.

The DWC_xpcs supports the MMDs based on the configuration and current mode of operation. The "*Register Descriptions*" chapter describes the Device Address map for various MMDs.

2.7.2.2 Register Address

The 16-bit register address indicates which register in selected MMD device is being addressed. If the 16th bit is zero, it indicates that the lower bits are addressing the Standard Register set. If the 16th bit is one, it indicates that the lower bits are addressing a vendor-specific register set.

2.7.2.3 Port ID

The port_id_i[4:0] input signal is one of the primary inputs for the DWC_xpcs. For MDIO interface, the DWC_xpcs responds to read and write transactions only when the PORT_ID of the MDIO frame matches this primary input signal. If there is a mismatch, the DWC_xpcs ignores the transaction. This input signal is not present for the MCI or APB3 interface.

2.8 Management Interface

The MCI, MDIO, or APB3 block functions as the management interface implementing the MCI, MDIO, or APB3 protocols, respectively. All CSR register sets can be accessed through these interfaces.

This topic contains the following sections:

- "Configuring the Management Interface" on page 96
- "MDIO Interface" on page 96
- "MCI Interface" on page 100
- "APB3 Interface" on page 102

2.8.1 Configuring the Management Interface

To configure the Management interface during the Specify Configuration Activity in coreConsultant, select the required interface from the "Management Interface" drop-down parameter, in the System Configurations Settings window.

For details about this option, see System Configurations Settings Parameters in the "*Parameter Descriptions*" chapter.


Note See the "Enabled" field of the parameter descriptions to understand the dependencies.

2.8.2 MDIO Interface

This topic contains the following sections:

- "Overview of MDIO Interface" on page 96
- "MDIO Register Access" on page 97
- "MDIO Timing Specification" on page 97
- "Clause 45 MDIO Frame Structure" on page 98
- "Signals Related to MDIO Interface" on page 100

2.8.2.1 Overview of MDIO Interface

The IEEE Std 802.3 defines the MDIO Management Interface which serves to access the Management register of 802.3 compliant devices. This is a two-line interface including MDC (clock) and MDIO (bi-directional data). The standard defines two protocols associated with the same Management Interface.

- Clause 22 defines access for the MII or GMII register sets
- Clause 45 defines access for the MDIO Manageable device (MMD) devices register sets


Note The DWC_xpcs supports only Clause 45 MDIO interfaces.

2.8.2.2 MDIO Register Access

The MDIO register access happens in the following two phases:

- Address Write Phase: In this phase, the MAC provides the following:
  - Device address of 5-bits in the control field, indicating the MMD devices address
  - Register address of 16-bits in the data field, indicating the Register implemented in a MMD
- Data Access Phase: In this phase, the MAC accesses the data from the device.
  - Read Operation:
    - The MAC drives the device address in the Control field.
    - The DWC_xpcs drives the data of the addressed resister in the Data field.
  - Write Operation:
    - The MAC drives the device address in the Control field.
    - The MAC drives the data of the addressed resister in the Data field.

Note DWC_xpcs discards the first MDIO transaction after the power-on reset. However, this does not impact the functionality. Because, as per DWC_xpcs initialization sequence, after power-on-reset, software must poll bit[15] of the SR_XS_PCS_CTRL1/SR_MII_CTRL register until it becomes 0, before proceeding to any other programming task. So, this behavior of MDIO is not visible to software.

2.8.2.3 MDIO Timing Specification

The DWC_xpcs assumes that the MDIO timing is compatible with IEEE 802.3 Std, Clause 45. When the DWC_xpcs is sampling the MDIO line, it assumes that the data is available 10 ns before the MDC positive edge (Setup time) and 10 ns after the MDC positive edge (Hold time) as described in Figure 2-24. Similarly, when the DWC_xpcs is driving the data on MDIO, it meets the hold time requirements as given in the Figure 2-25. Additionally, the frame structure and access methods are different. Figure 2-24 and Figure 2-25 show the timing relationship between MDIO and MDC.

Note If the MDIO timing is not compatible with the standard in your system, you need to change input or output delay constraints for the mdi_i and mdo_o signals during synthesis.

Figure 2-24 MDIO Driven by STA (Management)

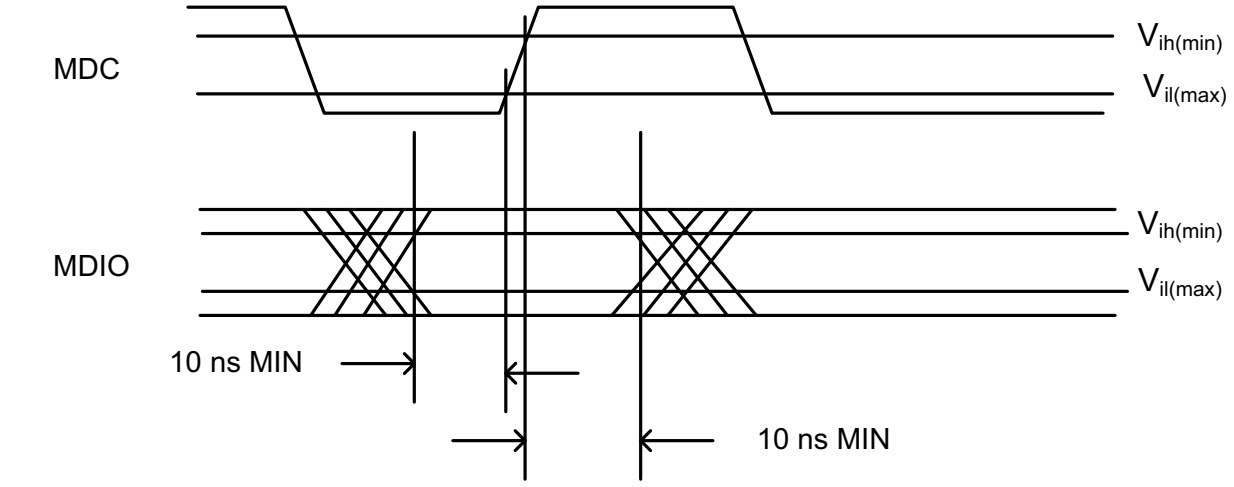
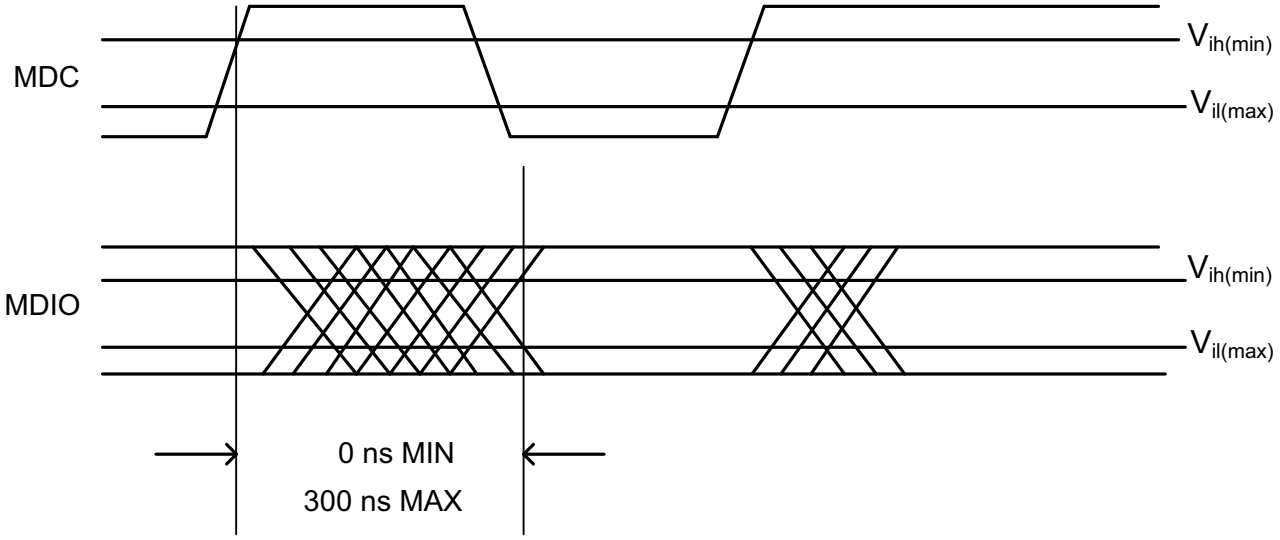


Figure 2-25 MDIO Driven by Device



2.8.2.4 Clause 45 MDIO Frame Structure

Table 2-13 describes the MDIO Frame structure.

Table 2-13 Clause 45 MDIO Frame Structure

	Management FrameFields							
	PREAMBLE	SOF	OPcode	Port Address	Device Address	Turnaround Time	Address/Data	IDLE
Address	1... 1	00	00	PPPPP	EEEEE	10	AAAAAAAAAAAAAAAA	Z
Write	1... 1	00	01	PPPPP	EEEEE	10	DDDDDDDDDDDDDDDD	Z
Read	1... 1	00	11	PPPPP	EEEEE	Z0	DDDDDDDDDDDDDDDD	Z

	Management FrameFields							
	PREAMBLE	SOF	OPcode	Port Address	Device Address	Turnaround Time	Address/Data	IDLE
Post-read increment address	1... 1	00	10	PPPPP	EEEEE	Z0	DDDDDDDDDDDDDDDD	Z

Table 2-14 describes the MDIO Frame fields.

Note All fields are ordered with MSb first.

Table 2-14 MDIO Frame Field Descriptions

Field	Description
Idle	This is when MDIO Line is high impedance and no body is driving the MDIO line. The pull-up resistor keeps the MDIO line in high state.
Preamble	Consists of 32-bit contiguous ones.
SOF	Start of Frame delimiter. This has a pattern 00, indicating start of the frame.
OPcode	Operation Code. Indicates a read, write, or address phase. ■ 00: Address phase ■ 01: Data Write phase ■ 10: Data Read phase with automatic address increment ■ 11: Data Read phase
DEVAD	Devices address. Allows addressing a specific MMD such as PCS, XS, PMA, and auto-negotiation.
Port Address	PHY Address field. This 5-bit address corresponds to port_id_i[4:0]. If this field is not equal to port_id_i[4:0], the DWC_xpcs does not respond to any read operation and write operation is ignored.
Turnaround Time	This is 2-bit field. ■ Read operation: ❑ The first bit is not driven. ❑ The DWC_xpcs drives the next immediate bit with 0. ■ Write operation: ❑ The Management drives the first bit with 1. ❑ The Management drives the second bit with 0.

Field	Description
Address/Data	This field specifies the address or data. ■ OPcode field = 00 specifies the register address management wants to read or write. ■ Any other value specifies the register read or write contents. ❑ Read: The DWC_xpcs drives the data. ❑ Write: The Management drives the data.

2.8.2.5 Signals Related to MDIO Interface

See the "*Signal Descriptions*" chapter.

2.8.3 MCI Interface

This topic contains the following sections:

- "Overview of MCI Interface" on page 100
- "MCI Interface without Indirect Access" on page 100
- "MCI Interface with Indirect Access" on page 101
- "Signals Related to MCI Interface" on page 102

2.8.3.1 Overview of MCI Interface

The MCI interface is a vendor-specific parallel interface to access the CSR registers in the DWC_xpcs. The method in which the MCI interface accesses the registers depends on the INDIRECT_ACCESS parameter.

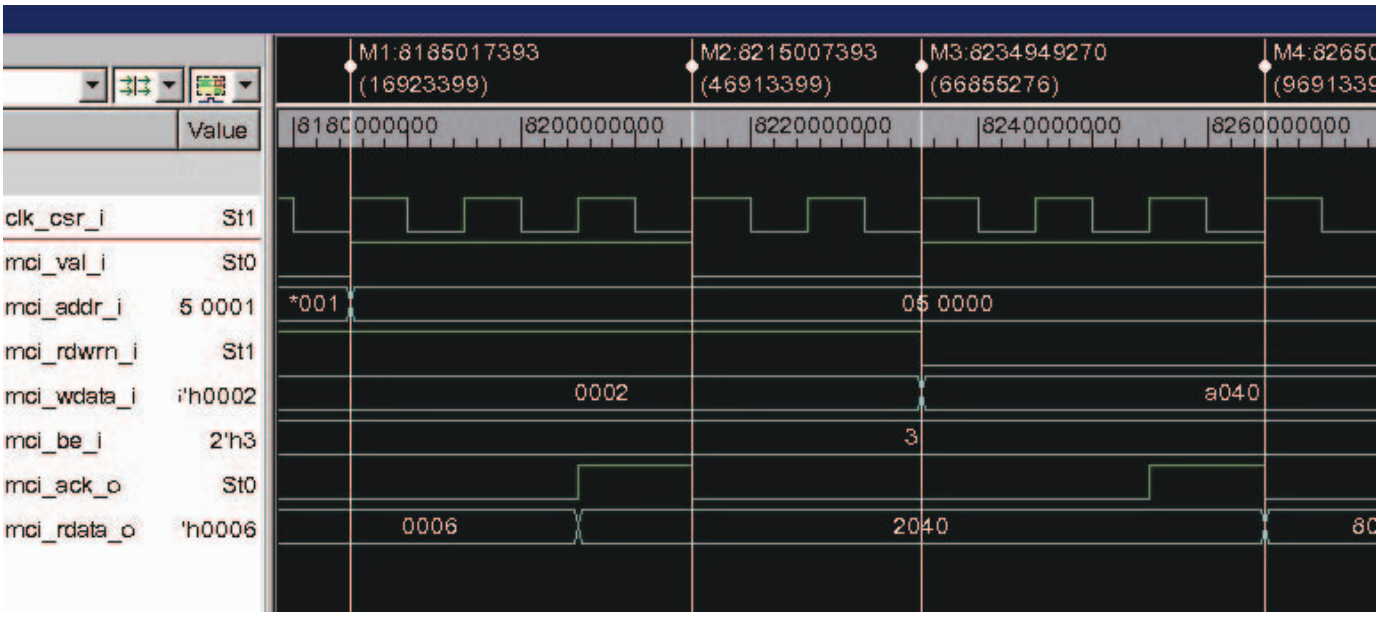
2.8.3.2 MCI Interface without Indirect Access

If INDIRECT_ACCESS = Disabled, the MCI interface accesses the registers in the DWC_xpcs in one phase. The width of the address bus without indirect option is 21-bits (mci_addr_i[20:0]). The following list describes the flow of register access:

- The assertion of mci_val_i signal indicates the start of the register access. When the mci_val_i signal is asserted, the mci_rdwrn_i, mci_be_i[1:0], and mci_addr_i[20:0] signals are valid. The assertion of mci_val_i signal also indicates that the mci_wdata_i[15:0] signal is valid during write operation. When the mci_rdwrn_i signal is high, it indicates the Read access; otherwise, it indicates the Write access. The mci_addr_i[20:16] signal indicates the MMD Device address and the mci_addr_i[15:0] signal indicates the Register address.
- The DWC_xpcs drives the mci_rdata_o[15:0] signal which contains the data of the register being addressed during Read access.
- The DWC_xpcs drives the mci_ack_o signal high which indicates the completion of the register read or write access. In addition, it indicates that the mci_rdata_o[15:0] has valid data during read access. The host should de-assert the mci_val_i signal when the DWC_xpcs asserts the mci_ack_o signal. If the mci_val_i signal continues to be asserted, it is taken as the next read or write access. If the host accesses a register with a reserved or unmapped address, the DWC_xpcs returns "all-zero" data during reads and ignores the data during write transfers.

Figure 2-26 shows an example of register read and write access.

Figure 2-26 MCI Interface Register Access (without Indirect Access)



In Figure 2-26, the marker M1 indicates the start of the read access at address 21'b005000. The marker M2 indicates the end of the read access. The data read is 16'h2040. Similarly, the marker M3 indicates the start of the write access at address 21'h05000 with write data 16'ha040. The marker M4 indicates the end of the write access.

2.8.3.3 MCI Interface with Indirect Access

If INDIRECT_ACCESS = Enabled, the MCI interface accesses the registers in the DWC_xpcs in two phases. With indirect access, the memory range in the DWC_xpcs is reduced to only 256 register sets. Therefore, the width of address bus is only 8-bits (mci_addr_i[7:0]). The MCI interface accesses the CSR registers in the following two phases:

- Address Phase

  In this phase, the host performs the write operation at register address 8'hFF of any of the supported MMDs in the DWC_xpcs. The write data should contain the base address of the 21-bit address. The base address corresponds to address bits[20:8] of 21-bit full address. The base address is driven on the mci_wdata_i[12:0] bus.

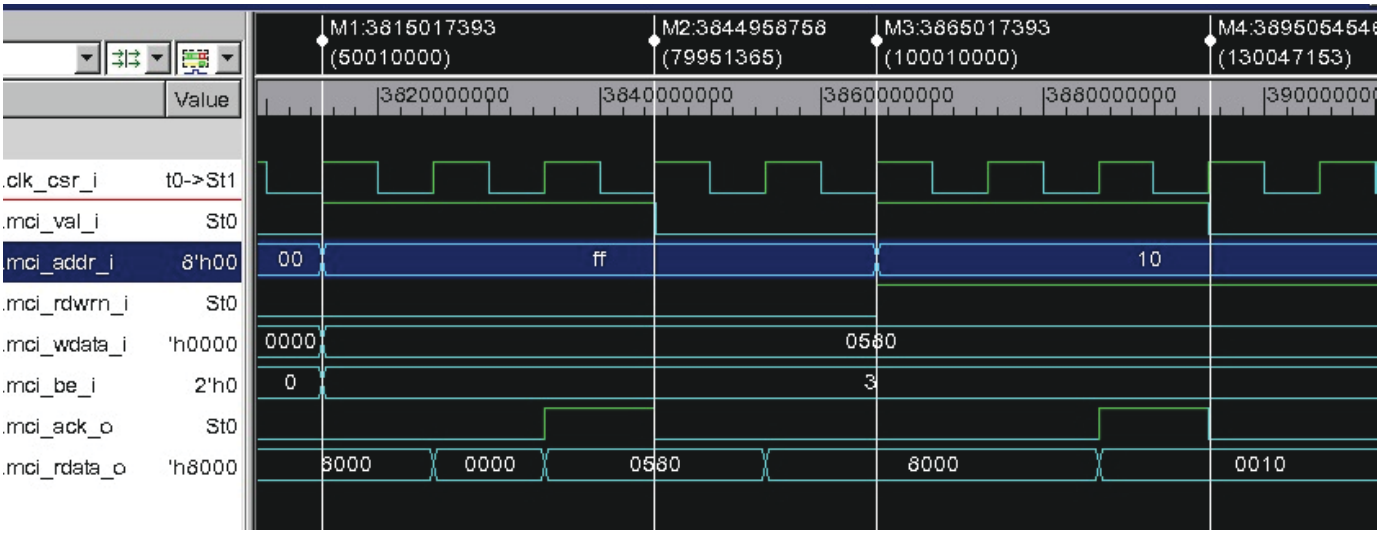
- Data Phase

  In this phase, the management provides the offset address of the 21-bit address. The offset address corresponds to address bits [7:0] of the 21-bit full address and it is driven on the mci_addr_i[7:0] bus.

  The DWC_xpcs derives the 21-bit address from the base address of Address phase and the offset address of the Data phase and performs the read or write access in the Data phase. You can perform multiple or consecutive Data phase transfers if the base address remains same.

Figure 2-27 shows an example of indirect register read access.

Figure 2-27 MCI Interface Register Access (with Indirect Access)



In Figure 2-27, the marker M1 indicates the start of the Address phase. It is a write access at address 8'hFF with base address 13'h0580, which corresponds to base address [20:8] of the full address. The marker M2 indicates the end of the Address phase.

The marker M3 indicates the start of the Data phase. It is a read access to the offset address of 8'h10 which corresponds to offset address [7:0] of the full address. Therefore, the full register address is 21'h058010. The DWC_xpcs returns the read data (for this full address) with value 16'h0010. The marker M4 indicates the end of Data phase.

2.8.3.4 Signals Related to MCI Interface

See "Micro Controller Interface Signals" in the "*Signal Descriptions*" chapter.

2.8.4 APB3 Interface

This topic contains the following sections:

- "Overview of APB3 Interface" on page 102
- "APB3 Interface without Indirect Access" on page 103
- "APB3 Interface with Indirect Access" on page 104
- "Signals Related to APB3 Interface" on page 105

2.8.4.1 Overview of APB3 Interface

The AMBA APB 3 interface is compliant with AMBA 3 APB Protocol Specification, v1.0. It is a parallel interface to access the CSR registers in the DWC_xpcs. The method in which the APB3 interface accesses the registers depends on the INDIRECT_ACCESS parameter.

Note Error response (PSLVERR) is not supported with the APB3 interface because the DWC_xpcs does not generate any error response.

DWC_xpcs also supports the following additional variants of APB3 interface, in addition to the legacy APB3 interface:

- APB3_16

  In this configuration, though the registers are 16-bit wide, addresses of registers are byte-aligned, which means, address value is in multiples of 2.
- APB3_32

  In this configuration, registers are 32-bit wide, addresses of registers are multiples of 4.

In the legacy AB3 configuration, DWC_xpcs registers are 16-bit wide and addresses of registers are not byte-aligned. So, addresses are multiples of 1.

2.8.4.2 APB3 Interface without Indirect Access

If INDIRECT_ACCESS = Disabled, the APB3 interface accesses the registers in the DWC_xpcs in one phase. The width of the address bus without indirect option is:

- 22 bits for APB3_16
- 23 bits for APB3_32
- 21 bits for legacy APB3

The following list describes the flow of register access:

- The assertion of psel_i signal indicates the start of the register access. When psel_i signal is asserted, the pwrite_i and paddr_i[20/21/22:0] signals are valid. The assertion of psel_i signal also indicates that the pwdata_i[15:0] signal is valid during write operation. When the pwrite_i signal is high, it indicates the write access; otherwise, it indicates the read access. Table shows the signals used to indicate the MMD Device address and the Register address, in the APB3 variants.

Table 2-15 Signals Indicating the Device and Register Addresses in the APB3 Variants

APB3 Variant	MMD Device Address	Register Address
APB3_32	paddr_i[22:18]	paddr_i[17:0]
APB3_16	paddr_i[21:17]	paddr_i[16:0]
Legacy APB3	paddr_i[20:16]	paddr_i[15:0]

- Based on the type of the APB3 interface, DWC_xpcs drives one of the following signals which contains the data of the register being addressed, during Read access.

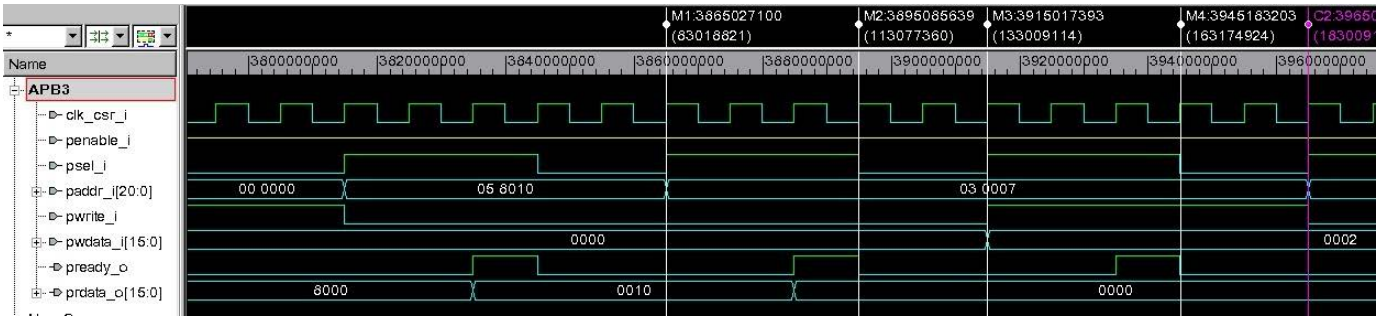
Table 2-16 Signal Driven by DWC_xpcs for APB3 Variants

APB3 Variant	Signal Driven by DWC_xpcs
APB3_32	prdata_o[31:0] The prdata_o[31:16] always indicate 0. Thought the registers are 32-bit wide, upper 16-bits are always reserved and read-only.
APB3_16	prdata_o[15:0]
Legacy APB3	

- The DWC_xpcs drives the pready_o signal high which indicates the completion of the register access. In addition, it indicates that the prdata_o[15/31:0] has valid data during read access. The host should de-assert the psel_i signal when the DWC_xpcs asserts the pready_o signal. If the psel_i signal continues to be asserted, it is taken as next read or write access. If the host accesses a register with a reserved or unmapped address, the DWC_xpcs returns "all-zero" data during reads and ignores the data during write transfers.

Figure 2-28 shows an example of register read or write access.

Figure 2-28 APB3 Interface Register Access (without Indirect Access)



In Figure 2-28, the marker M1 indicates the start of the read access at address 21'b030007. The marker M2 indicates the end of the read access. The data read is 16'h0000. Similarly, the marker M3 indicates the start of the write access at address 21'h030007 with write data 16'h0002. The marker M4 indicates the end of the write access.

2.8.4.3 APB3 Interface with Indirect Access

If INDIRECT_ACCESS = Enabled, the APB3 interface accesses the registers in the DWC_xpcs in two phases. With indirect access, the memory range in the DWC_xpcs is reduced to,

- 256 register sets for legacy APB3
- 512 register sets for APB3_16
- 1024 register sets for APB_32

Therefore, the width of address bus is 8 bits for legacy APB3, 9 bits for APB3_16, or 10 bits for APB3_32.

The APB3 interface accesses the CSR registers in the following two phases:

- Address Phase

  In this phase, the host performs the write operation at the following register address of any of the supported MMDs in the DWC_xpcs.

  - 8'hFF for legacy APB3
  - 9'h1FE for APB3_16
  - 10'h3FC for APB3_32

  The write data should contain the base address of the 21-bit/22-bit/23-bit address. The base address corresponds to the following address bits of 21-bit/22-bit/23-bit full address. The base address is driven on the pwdata_i[12:0] bus.

  - bits[20:8] for legacy APB3
  - bits[21:9] for APB3_16

- bits[22:10] for APB3_32

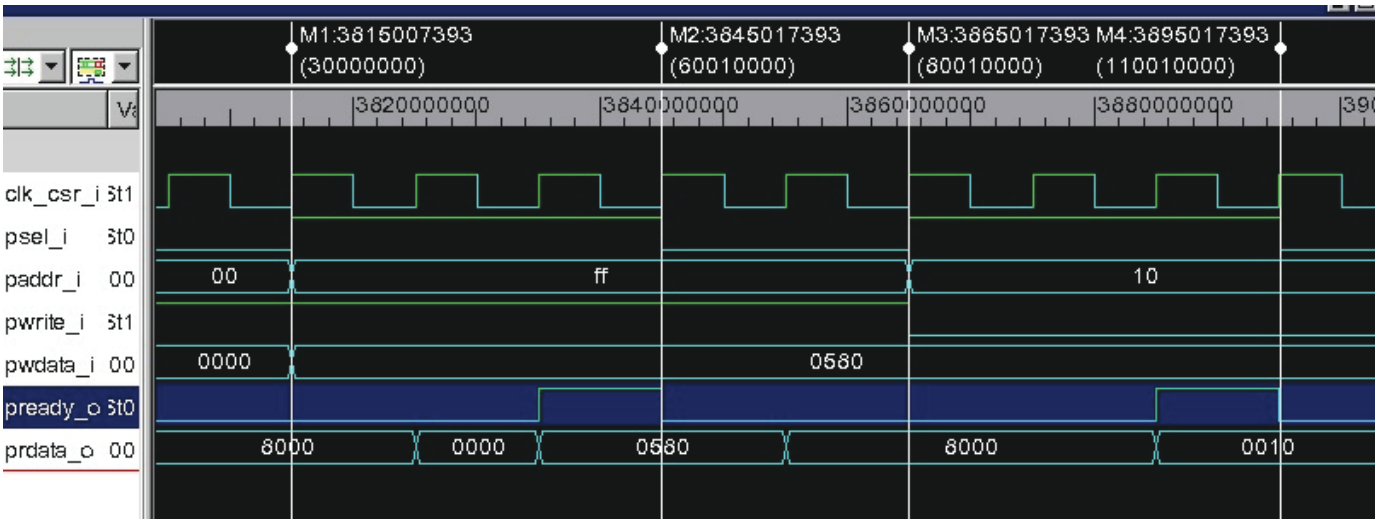
- Data Phase

  In this phase, the management provides the offset address of the 21-bit/22-bit/23-bit address. The offset address corresponds to address bits[7/8/9:0] of the 21-bit/22-bit/23-bit full address and it is driven on the paddr_i[7/8/9:0] bus.

  The DWC_xpcs derives the 21-bit/22-bit/23-bit address, from the base address of Address phase and the offset address of the Data phase, and performs the read or write access in the Data phase. You can perform multiple or consecutive Data phase transfers if the base address remains the same.

Figure 2-29 shows an example of indirect register read access.

Figure 2-29 APB3 Interface Register Access (with Indirect Access)



In Figure 2-29, the marker M1 indicates the start of the Address phase. It is a write access at address 8'hFF with base address 13'h0580, which corresponds to base address[20:8] of the full address. The marker M2 indicates the end of the Address phase.

The marker M3 indicates the start of the Data phase. It is a read access to the offset address of 8'h10 which corresponds to offset address[7:0] of the full address. Therefore, the full register address is 21'h058010. The DWC_xpcs returns the read data (for this full address) with value 16'h0010. The marker M4 indicates the end of Data phase.

2.8.4.4 Signals Related to APB3 Interface

See the "*Signal Descriptions*" chapter.

2.9 Clause 73 Auto-Negotiation

This topic contains the following sections:

- "Overview of Clause 73 Auto-Negotiation" on page 106
- "Auto-Negotiation Format: Differential Manchester Encoding (DME)" on page 107
- "Base Link Code Word Format" on page 109
- "Next Page Link Code Word Format" on page 110
- "Auto-Negotiation Block: Functional Components" on page 111
- "Auto-Negotiation CSR Interface" on page 120
- "Enabling Clause 73 Auto-Negotiation" on page 121
- "Registers Related to Clause 73 Auto-Negotiation" on page 121
- "Programming Guidelines for Clause 73 Auto-Negotiation" on page 121

2.9.1 Overview of Clause 73 Auto-Negotiation

The Clause 73 auto-negotiation is an optional feature. This feature is available only when the mode of operation (MAIN_MODE) is Backplane Ethernet PCS. The Backplane Ethernet is an extension of 1000BASE-X, 10GBASE-X, 2. GBASE-X, 5GBASE-R, and 10GBASE-R and is referred to as 1000BASE-KX, 10GBASE-KX4, 2.5GBASE-KX, 5GBASE-KR, and 10GBASE-KR respectively.

The auto-negotiation block implements the following functions:

- Recognizing the Differential Manchester Encoded (DME) Base page and Next pages on the Rx path and generating the DME pages on the Tx path during the auto-negotiation process.
- Connecting and disconnecting the Tx and Rx lanes (Lane 1, Lane 2, and Lane 3) based on outcome of auto-negotiation process.
- Implementing Parallel detection to detect legacy 1000BASE-X, 10GBASE-X, 10GBASE-R, 2.5BASE-X, 5GBASE-R, or Backplane devices (1000BASE-KX, 2.5GBASE-KX, KX4, 10GBASE-KR or 5GBASE-KR) without Clause 73 auto-negotiation.

Auto-negotiation allows devices to advertise and share information for detecting capabilities and status of remote devices such as:

- Speed
- Modes
- Fault Signaling
- Legacy device detection
- PAUSE abilities
- Duplex information
- Next Page ability
- Technology ability
- Forward Error Correction (FEC)

2.9.2 Auto-Negotiation Format: Differential Manchester Encoding (DME)

A common auto-negotiation process is independent of any specific technology (KX, KX4, KR). The auto-negotiation information is exchanged through serial DME at 312.5 MHz. The serial data, called Link Code Words, is sent through encoded pages called Differential Manchester Encoding (DME) pages.

2.9.2.1 DME Pseudo-Random Number Generator

This block generates the pseudo-random numbers. For generating DME page random numbers, the following polynomial is used:

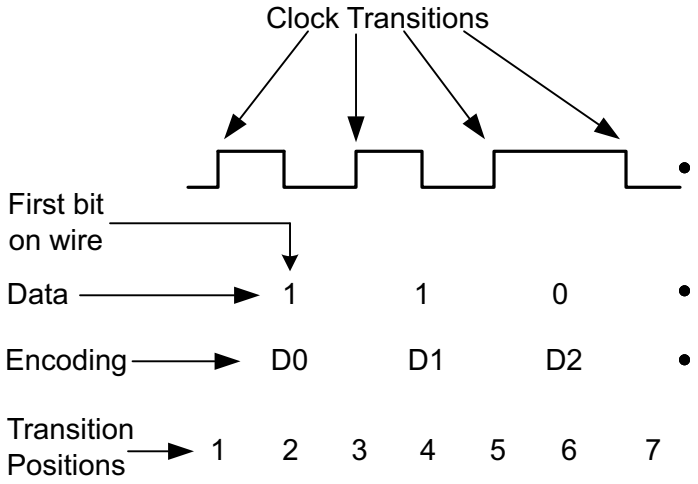
`X7+X3+1`

The pseudo-random number is used on the DME page as a last data bit. The same polynomial is used in the arbiter to generate Transmit Nonce field of the Base Page.

2.9.2.2 DME Page Timing Specification

Figure 2-30 shows the example encoding.

Figure 2-30 Data Bit Encoding with DME Pages



2.9.2.3 DME Page Transition Timing

Figure 2-31 and Figure 2-32 show the DME page transition timing.

Figure 2-31 DME Page Timing

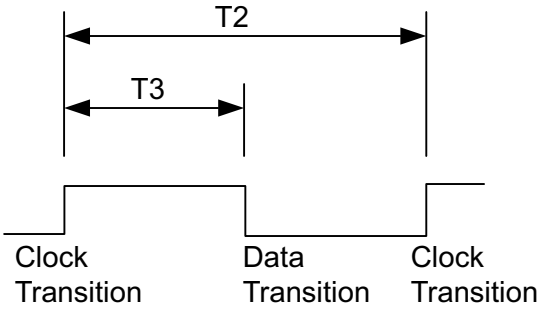


Figure 2-32 DME Page Transition

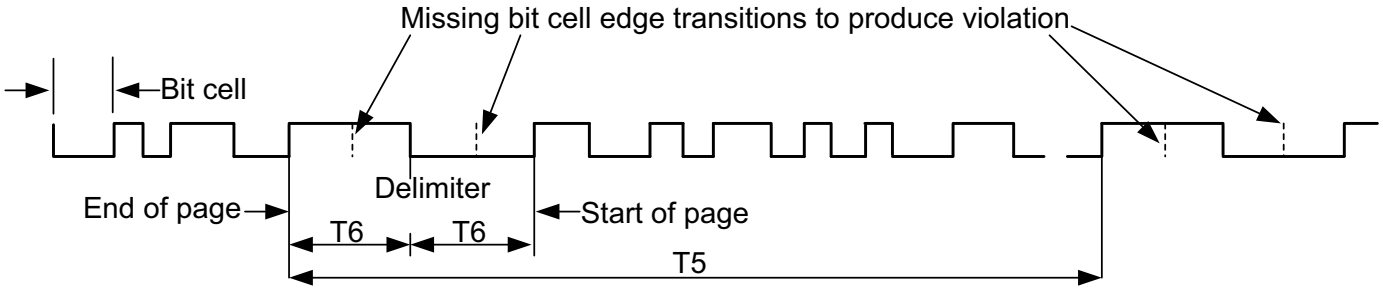


Table 2-17 describes the DME page transition timing.

Table 2-17 DME Page Transmit Timing as Specified by IEEE Std 802.3, Clause 73

#	Parameter	Minimum	Typical	Maximum	Unit
T1	Transition position spacing	3.2 +/- 0.01%	3.2	3.2 +/- 0.01%	ns
T2	Clock Transition to Clock Transition	6.2	6.4	6.6	ns
T3	Clock Transition to Data Transition	3.0	3.2	3.4	ns
T4	Transitions on a DME page	51	-	100	-
T5	DME Page Width	338.8	339.2	339.6	ns
T6	DME MVD Width	12.6	12.8	13.0	ns

2.9.2.4 DME Page Structure

A DME page contains 106 evenly-spaced transition positions, contained in 53 bits of 3.2 ns bit time. These 53 bits include the following:

- Manchester's Violation Delimiter (MVD): The first eight transition positions (4 bits) contain MVD, which marks the beginning of the page. The Manchester violation contains the transition at position 1 and 5, and no transition in the remaining transition position. Two consecutive MVDs are used for page synchronization.

- The Auto-Negotiation page: The auto-negotiation page is 96 transition positions (48 bits) corresponding to the Link Code word (described in section "Base Link Code Word Format" on page 109) encoded in the DME format. The following list describes the encoding rules:
  - Odd numbered transition positions contain clock information. Odd-numbered transition positions always contain a transition.
  - Even-numbered transition positions contain data information with the following transition rules:
    - Transitions present represent the data corresponding to logical 1.
    - Transitions absent represent the data corresponding to logical 0.
- A pseudo-random bit: The 53rd bit of the DME page contains a pseudo-random data bit.

2.9.3 Base Link Code Word Format

Figure 2-33 and Table 2-18 define the format of the Link Code Word used with AN Advertisement Registers 1-3.

Figure 2-33 Base Page: Link Code Word Form

D 0	D 1	D 2	D 3	D 4	D 5	D 6	D 7	D 8	D 9	D 10	D 11	D 12	D 13	D 14	D 15
S 0	S 1	S 2	S 3	S 4	E 0	E 1	E 2	E 3	E 4	C 0	C 1	C 2	RF	ACK	NP

F

D 16	D 17	D 18	D 19	D 20	D 21	D 22	D 23	D 24	D 25	D 26	D 27	D 28	D 45	D 46	D 47
T 0	T 1	T 2	T 3	T 4	A 0	A 1	A 2	A 3	A 4	A 5	A 6	A 7	A 24	F 0	F 1

Table 2-18 Base Link Code Word Format Data

Item	Definition
NP	Next page to be sent or received
ACK	Code received successfully from the link partner
RF	Remote fault
C0	SYNC_PAUSE_ABILITY
C1	ASYNC_PAUSE_ABILITY
C2	Reserved
E[4:0] Echoed Nonce Field	Received Nonce field containing Transmitted Nonce field of remote device
S[4:0] Selector Field	IEEE standard with which this device is compliant
T[4:0]	Transmitted Nonce field.

Item	Definition
A[24:0]	Technology Ability field Indicates the supported technologies: ■ A0: When this bit is set to 1, the 1000BASE-KX technology is supported. Otherwise, the 1000BASE-KX technology is not supported. ■ A1: When this bit is set to 1, the 10GBASE-KX4 technology is supported. Otherwise, the 10GBASE-KX4 technology is not supported. ■ A2: When this bit is set to 1, the 10GBASE-KR technology is supported. Otherwise, the 10GBASE-KR technology is not supported. ■ A3-A10: Not applicable for DWC_xpcs ■ A11: When this bit is set to 1, the 2.5GBASE-KX technology is supported. Otherwise, 2.5GBASE-KX technology is not supported ■ A12: When this bit is set to 1, the 5GBASE-KR technology is supported. Otherwise, 5GBASE-KR technology is not supported ■ A13-A24: Reserved for future technology
F[1:0]	Forward Error Correction (FEC) Capable

2.9.4 Next Page Link Code Word Format

Figure 2-34 and Figure 2-35, along with Table 2-19, define the format of the Next Page Link Code Word used with SR_AN_XNP_TX1–TX3.

Figure 2-34 Message Next Page

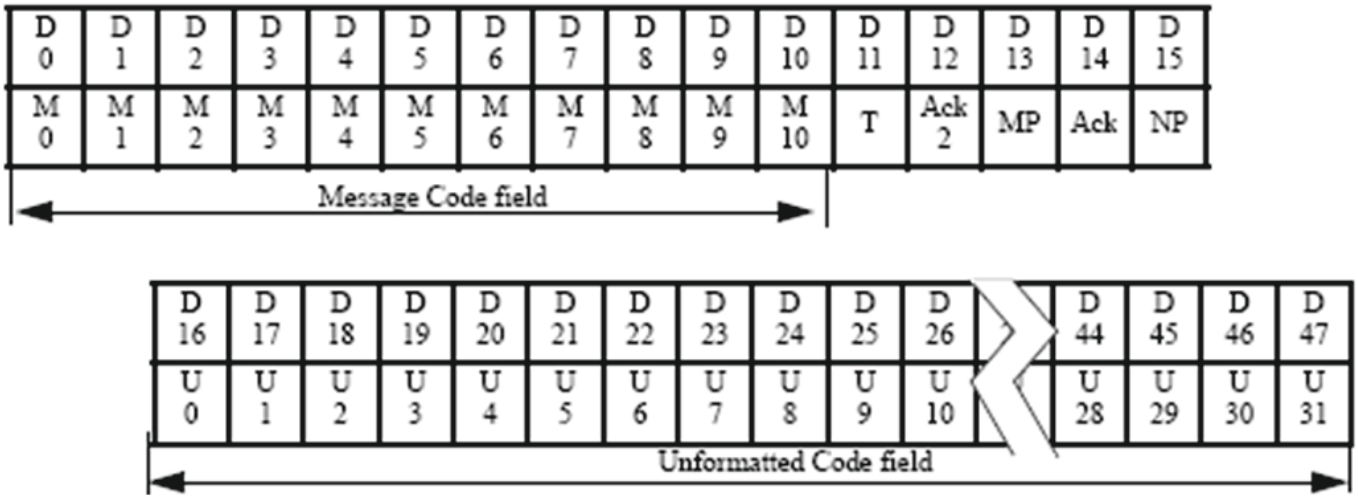


Figure 2-35 Unformatted Next Page

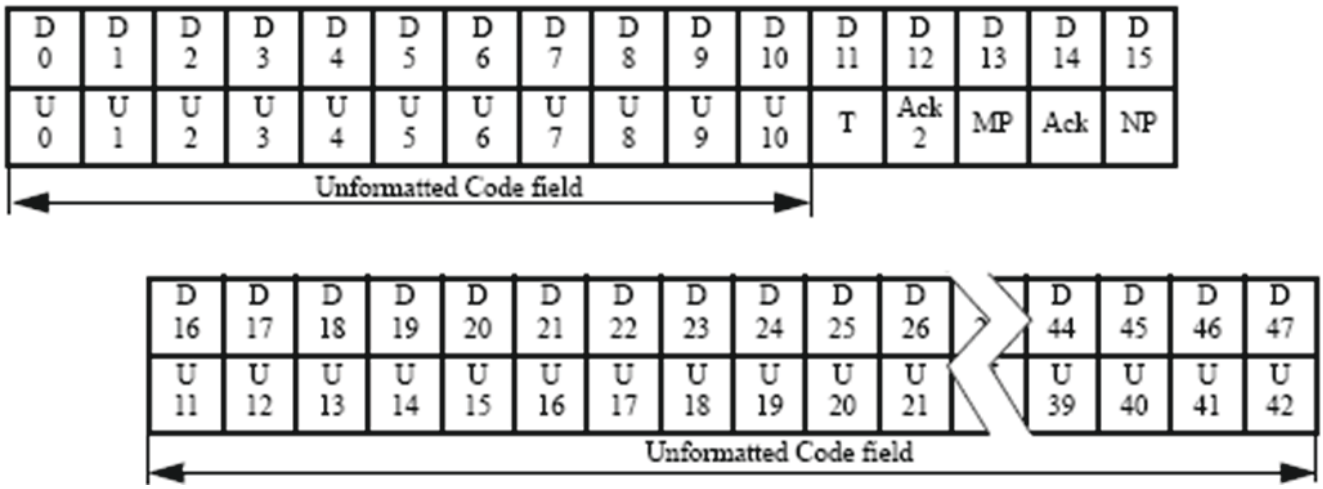


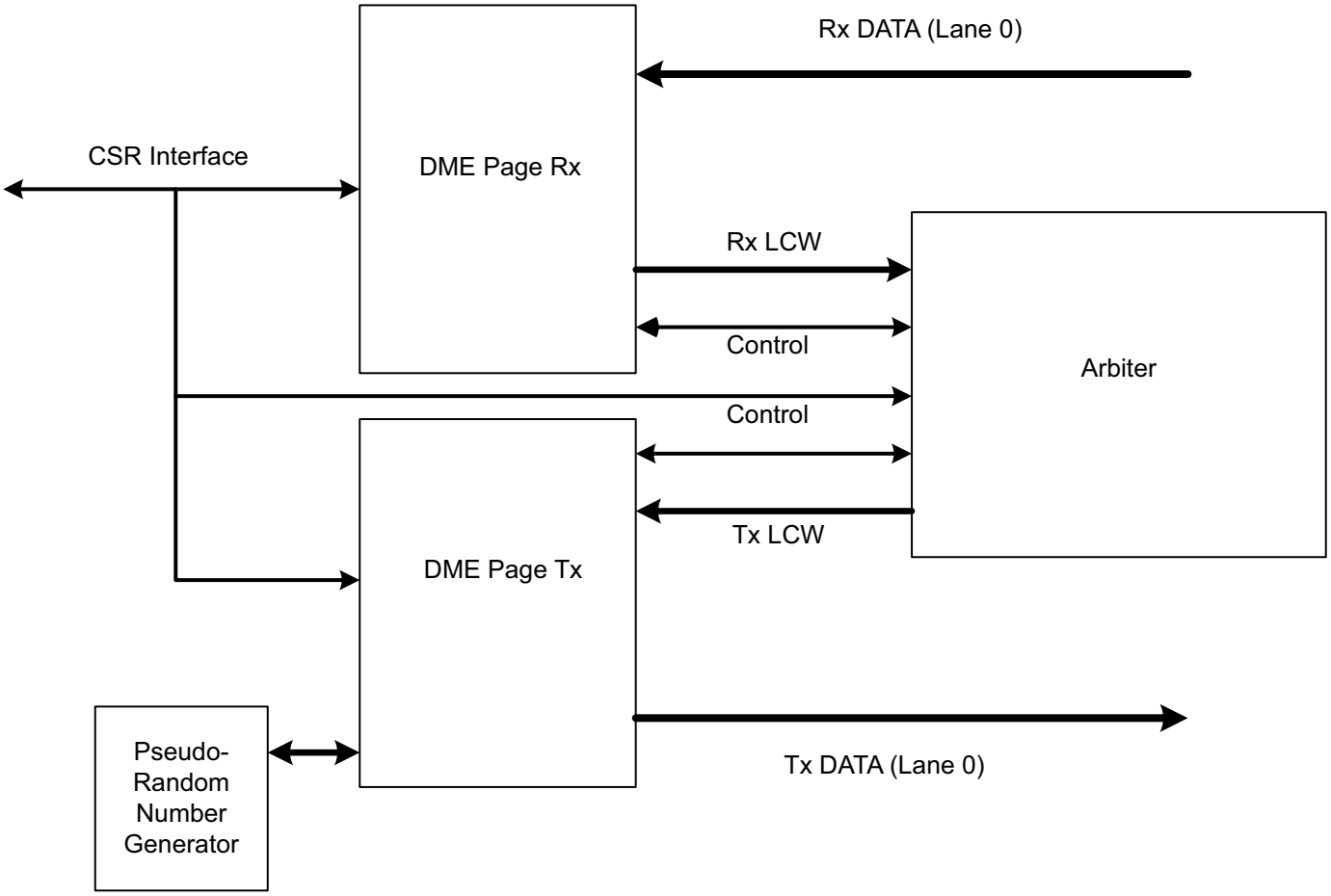
Table 2-19 Next Page Link Code Word Format Data

Item	Definition
NP	Next page to be sent or received
ACK	Code received successfully from link partner
MP	1: Formatted Message Page 0: Unformatted Message Page
ACK2	Code received successfully and act upon the information defined on the received message
T	Toggle bit
M[10:0]	Message field as defined in the Annex 73A of IEEE Std 802.3ap
U[31:0]	Unformatted code field on the formatted message page
U[42:0]	Unformatted code received on the Unformatted message page

2.9.5 Auto-Negotiation Block: Functional Components

Figure 2-36 displays the components of the auto-negotiation block.

Figure 2-36 Backplane-PCS Auto-Negotiation Block Detail



2.9.5.1 Arbiter

The arbiter component controls the DME page transmitter and DME page receiver. The arbiter implements the following features:

- Tx lane disabling or enabling
- Rx lane disabling or enabling and scan for carrier operation
- Parallel detection
- Next Page transmission or reception
- Link code word decoding

The arbiter contains various timers as defined in the IEEE Std 802.3ap, Clause 73. This block is controlled by the Management requests to start, abort, and restart auto-negotiation. The arbiter also implements the state machine as defined in the IEEE Std 802.3ap, Clause 73, Figure 73-11.

Parallel Detection in the Arbiter

The parallel detection is done during the Clause 73 auto-negotiation to detect the legacy devices which do not support Clause 73 auto-negotiation. When the auto-negotiation is going on, the received data from the

PHY is given to both Clause 73 auto-negotiation and KX(1000Base-KX and 2.5GBase-KX), KX4, or KR(10GBASE-KR and 5GBASE-KR), receive logic. While expecting the Clause 73 compliance DME page, if the arbiter state machine sees the Link Up status from the receive logic, it means one of the following:

- If the current speed mode is KX4, the remote link is 10GBASE-X.
- If the current speed mode is 10GBASE-KR, the remote link is 10GBASE-R.
- If the current speed mode is 1000BASE-KX, the remote link is 1000BASE-X.
- If the current speed is 2.5GBASE-KX, the remote link is 2.5GBASE-X.
- If the current speed is 5GBASE-KR, the remote link is 5GBASE-R.

Note

- The DWC_xpcs assumes that the PHY is configured to operate in the same speed mode as the DWC_xpcs. Therefore, the parallel detection is done only in the active speed mode.
- IEEE does not define parallel detection for 10GBASE-KR or 5GBASE-KR modes. Support for these in DWC_xpcs is a vendor specific feature.

If DWC_xpcs is configured to have multiple speed modes (KX4_KX, KR_KX4_KX, KR_KX4, or KR_KX) and the auto-negotiation fails (AN Incompatible Link received) in the current speed mode, the management should switch to some other supported speed mode to detect the compatible legacy device.

Table 2-20 Compatible Legacy Speed Mode Device Detectable Through Parallel Detection

Mode	Compatible Legacy Speed Mode
10GBASE-KR	10GBASE-R
10GBASE-KX4 (KX4)	10GBASE-X
1000BASE-KX	1000BASE-X
2.5GBASE-KX	2.5GBASE-X
5GBASE-KR	5GBASE-R

2.9.5.2 DME Page Transmitter

When requested by the arbiter, the DME page transmitter implements the logic related to the DME page encoding of the Link Code Word (LCW) to be transmitted. The logic implemented for DME page encoding varies based on the current speed mode. This is because although the DME timings are defined for a serial interface which is common across all speed modes, the bit time for each speed mode (KX4, KX, or KR) is different.

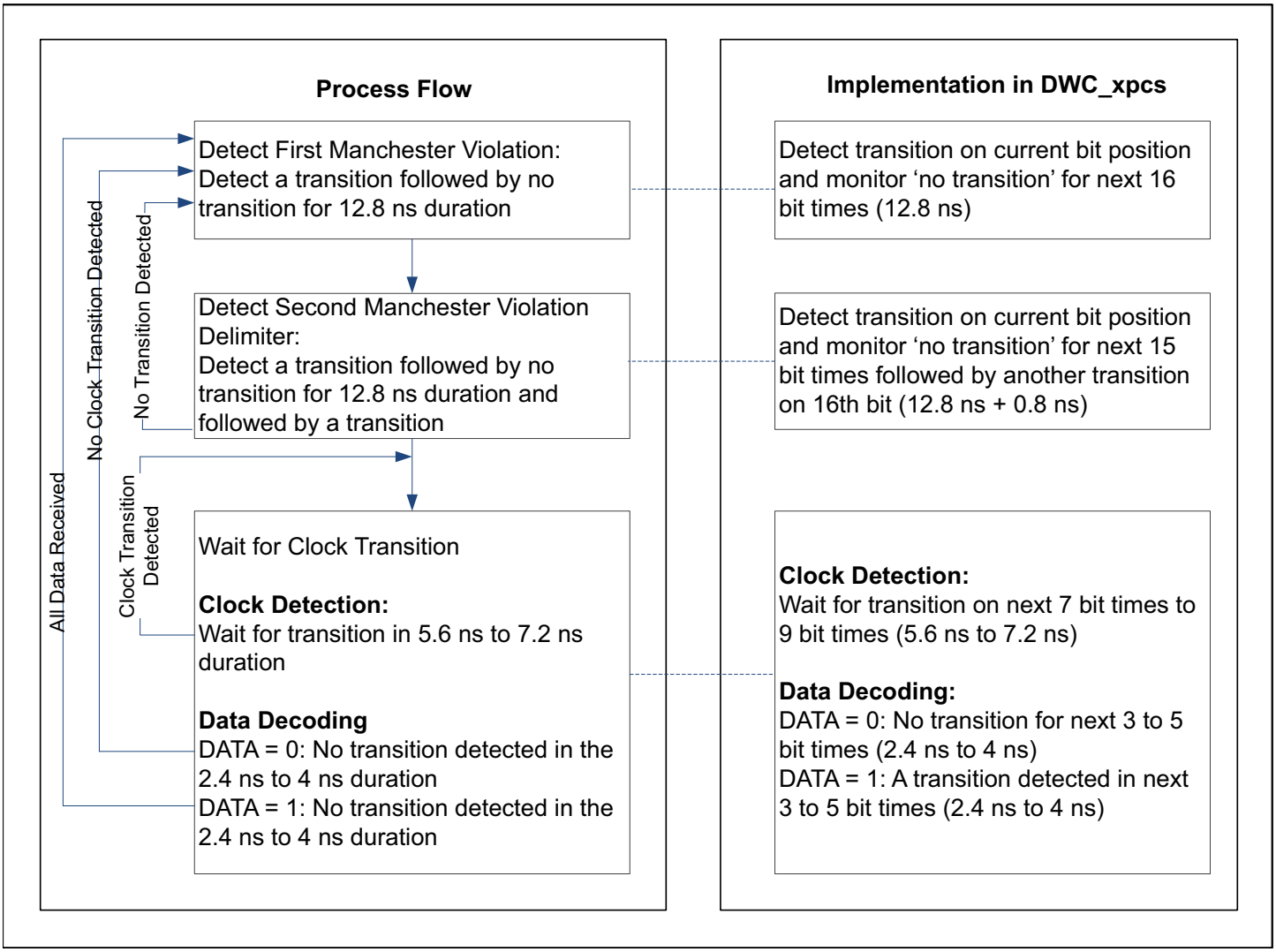
The DME page encoding has the following stages:

1. MV Delimiter: Two transitions separated by T6 (12.8 ns) time period
2. Clock: Occurs at every T2 (6.4 ns) time period
3. Data: Occurs at every T3 (3.2 ns) time period from the previous clock transition

DME Page Encoding for KX4 Mode

In the KX4 mode, 1 bit time corresponds to 0.32 ns. Figure 2-37 shows the process for DME page encoding for the KX4 mode and also how this process is implemented in DWC_xpcs.

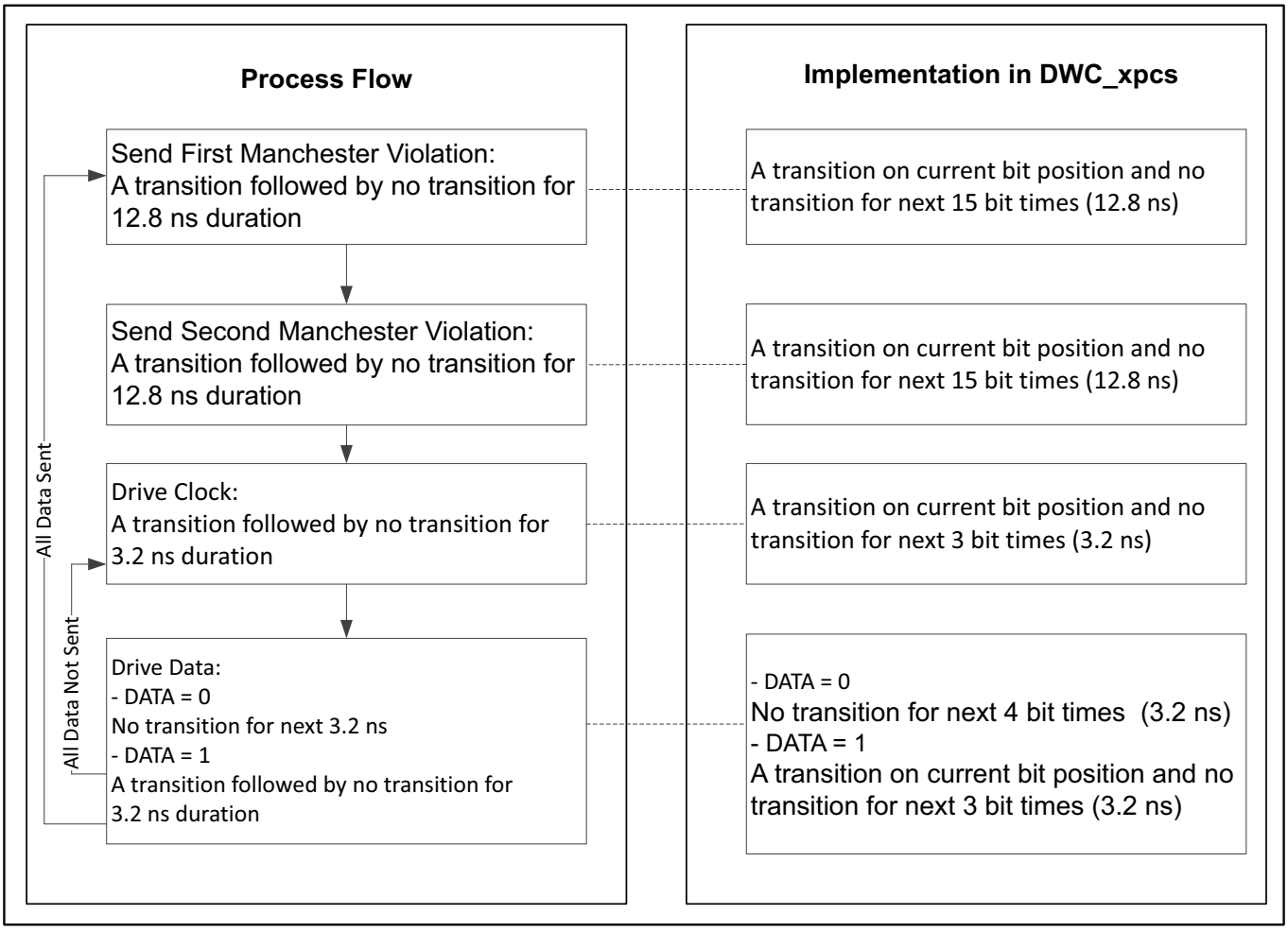
Figure 2-37 DME Page Encoding Process for KX4 Mode



DME Page Encoding for 1000BASE-KX Mode

In the KX mode, 1 bit time corresponds to 0.8 ns. Figure 2-38 shows the process for DME page encoding for KX mode and also how this process is implemented in the DWC_xpcs.

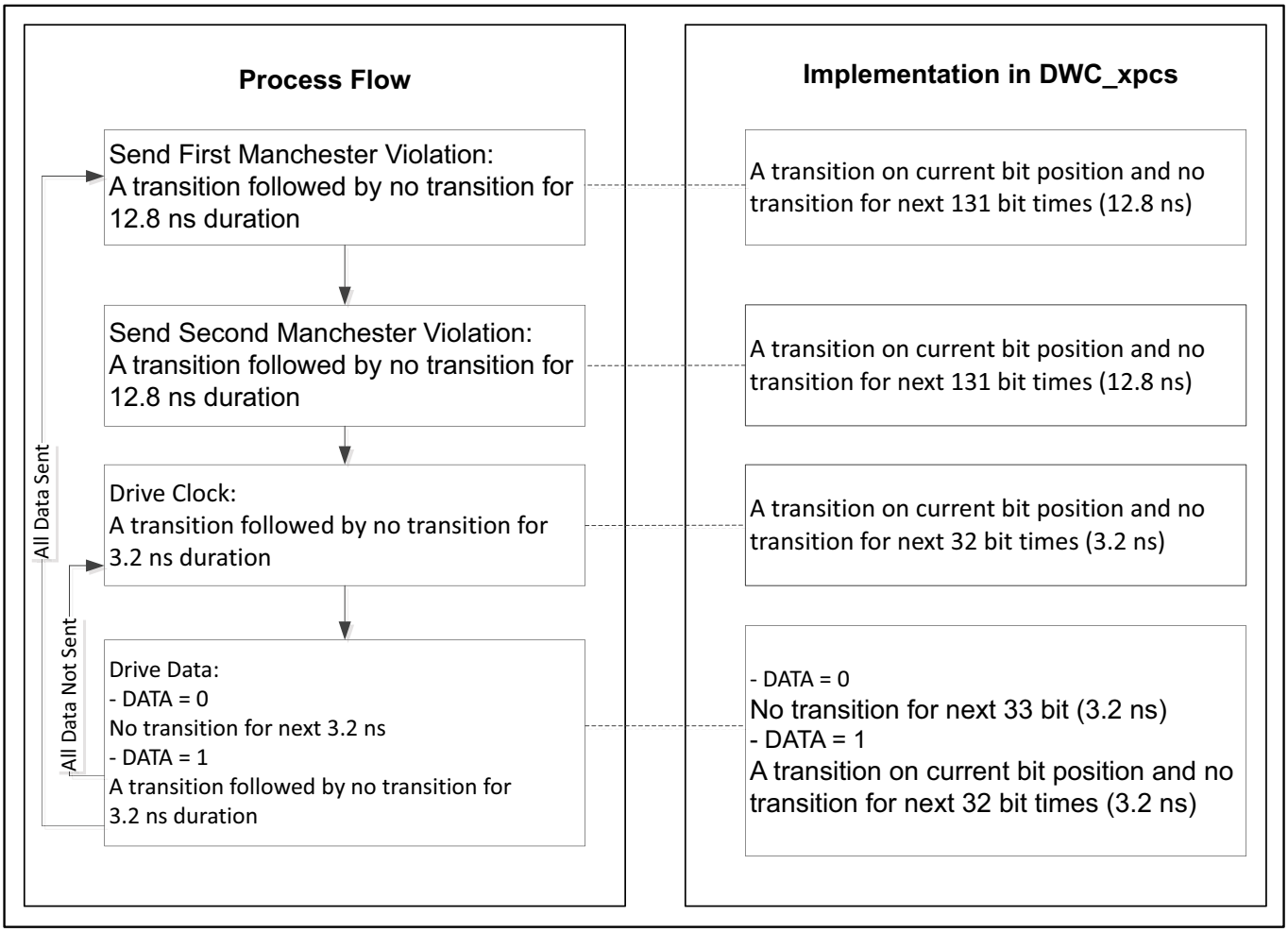
Figure 2-38 DME Page Encoding Process for 1000BASE-KX Mode



DME Page Encoding for 10GBASE-KR Mode

In the 10GBASE-KR mode, 1 bit time corresponds to 0.09696ns. Figure 2-39 shows the process for DME page encoding for KR mode and also how this process is implemented in DWC_xpcs.

Figure 2-39 DME Page Encoding Process for 10GBASE-KR Mode



2.9.5.3 DME Page Receiver

The DME page receiver implements the logic related to the DME page decoding of the Link Code Word. The logic implemented for DME page encoding varies based on the current speed mode. This is because although the DME timings are defined for a serial interface which is common across all speed modes, the bit time for each speed mode (KX4, KX, or KR) is different.

The DME page decoding timing is different from the encoding timing. The decoding is implemented based on the timing range as described in Table 2-21.

Note In KR speed mode, the DWC_xpcs implements the DME page decoding as per KR with Reduced Granularity timing (described in the last column in Table 2-21).

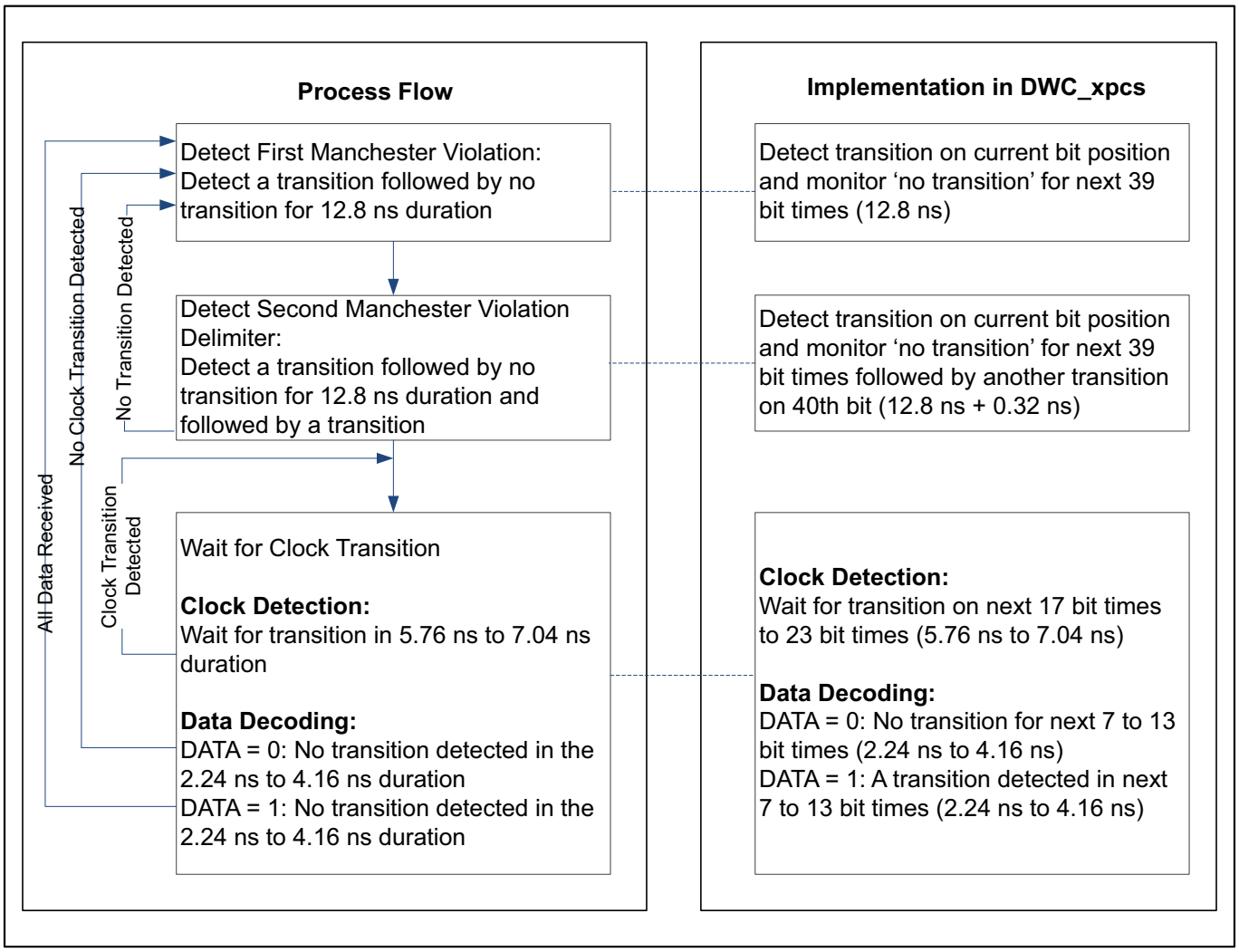
Table 2-21 DME Page Decoding Timing

Parameter	IEEE Value Range	1000BASE-KX (0.8 ns Bit Period)	2.5GBASE-KX/ 10GBASE-KX4 (0.32 ns Bit Period)	10GBASE-KR (0.096 ns Bit Period)	10GBASE-KR (Reduced Granularity) *	5GBASE-KR (0.192 ns Bit Period)
Clock detect min Timer	4.8 ns to 6.2 ns	Value = 5.6 ns Number of bits = 7	Value = 5.76 ns Number of bits = 18	Value = 5.8 ns Number of bits = 60	Value = 5.8 ns Number of bits = 20	Value = 5.6 ns Number of bits = 29
Clock detect max Timer	6.6 ns to 8.0 ns	Value= 7.2 ns Number of bits = 9	Value = 7.04 ns Number of bits = 22	Value = 7.2 ns Number of bits = 75	Value = 7.2 ns Number of bits = 25	Value = 7.2 ns Number of bits = 38
Data detect Min Timer	1.6ns to 3.0 ns	Value = 2.4 ns Number of bits = 3	Value = 2.56 ns Number of bits = 8	Value = 2.424 ns Number of bits = 25	Value = 2.6 ns Number of bits = 9	Value = 2.4 ns Number of bits = 12
Data detect Max Timer	3.4ns to 4.8 ns	Value = 4.0 ns Number of bits = 5	Value = 4.16 ns Number of bits = 13	Value = 4.0 ns Number of bits = 42	Value = 4.0 ns Number of bits = 14	Value = 4.0 ns Number of bits = 21
Page Test Min Timer	305 ns to 330 ns	Value = 312 ns Number of bits = 390	Value = 310 ns Number of bits = 970	Value = 305 ns Number of bits = 3196	Value = 305 ns Number of bits = 1065	Value = 307.2 ns Number of bits = 3168
Page Test Max Timer	350 ns to 375 ns	Value = 368 ns Number of bits = 460	Value = 371 ns Number of bits = 1160	Value = 370ns Number of bits = 3814	Value = 370 ns Number of bits = 1272	Value = 358.4 ns Number of bits = 3828

DME Page Decoding for KX4 Mode

In the KX4 mode, 1 bit time corresponds to 0.32 ns. Figure 2-40 shows the process for DME page decoding for the KX4 mode and also how this process is implemented in the DWC_xpcs.

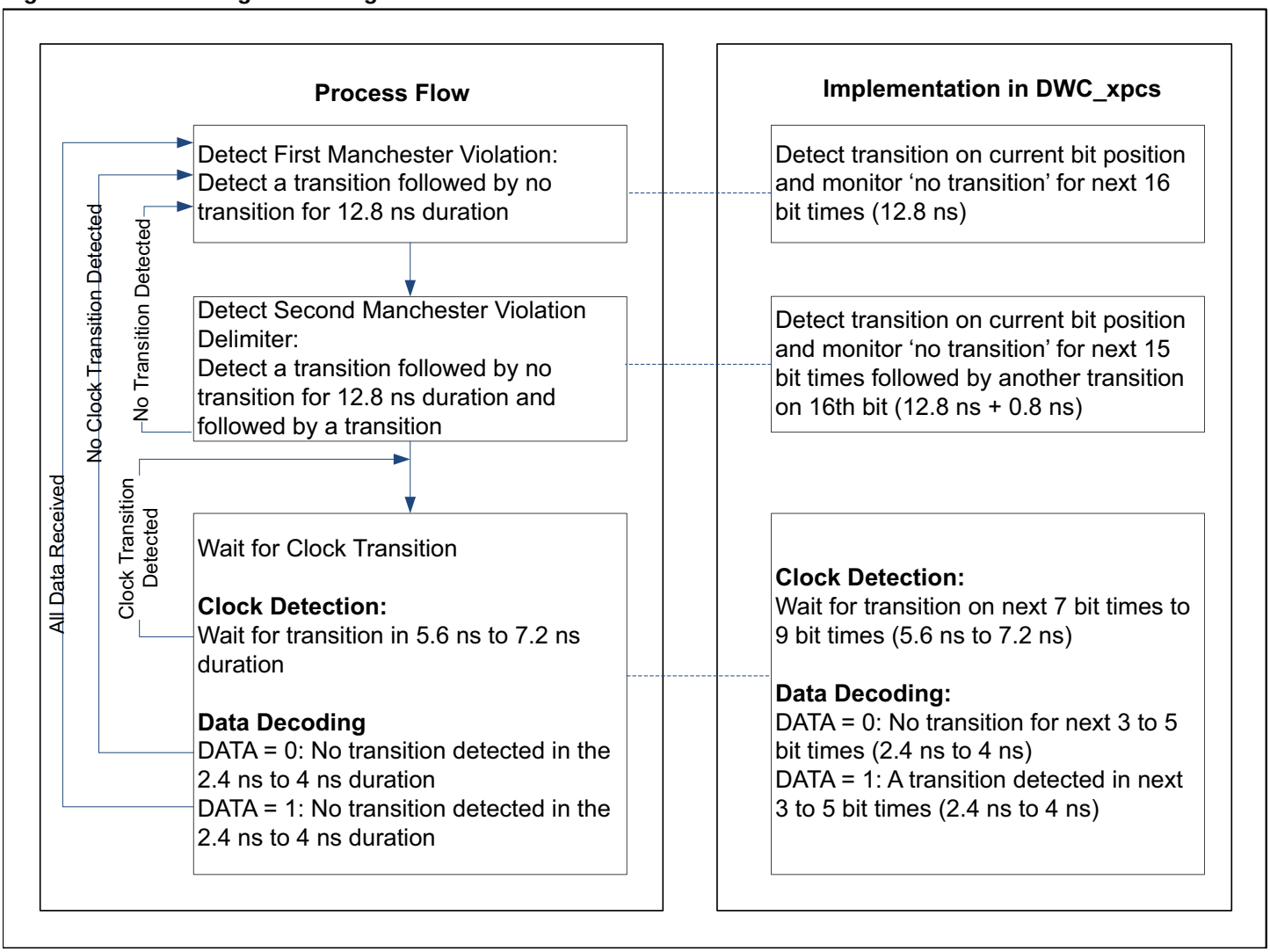
Figure 2-40 DME Page Decoding Process for KX4 Mode



DME Page Decoding for 1000BASE-KX Mode

In the 1000BASE-KX mode, 1 bit time corresponds to 0.8 ns. Figure 2-41 shows the process for DME page decoding for the 1000BASE-KX mode and also how this process is implemented in DWC_xpcs.

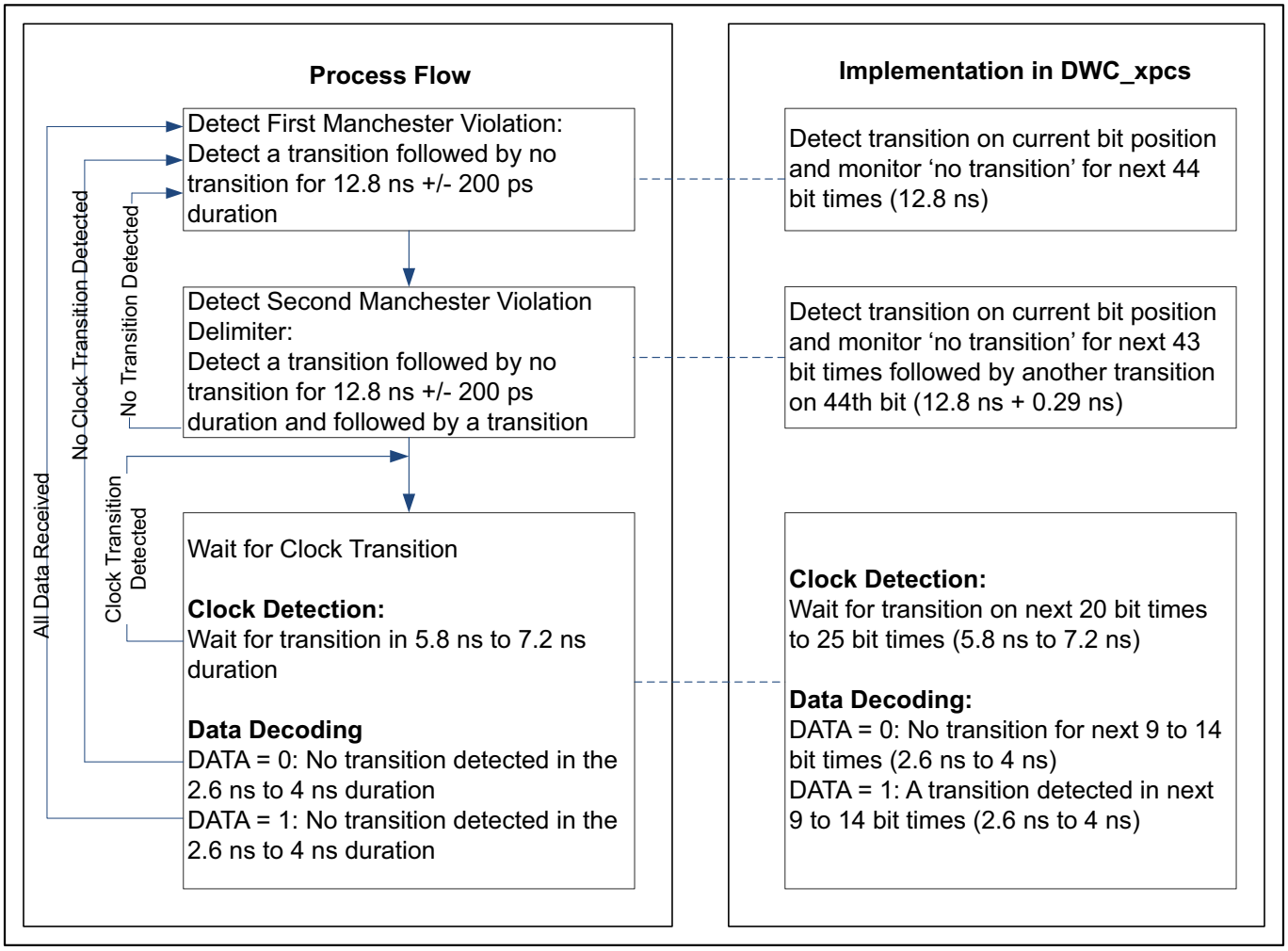
Figure 2-41 DME Page Decoding Process for 1000BASE-KX Mode



DME Page Decoding for 10GBASE-KR Mode

In the 10GBASE-KR DWC_xpcs, the implementation is done on reduced granularity by combining 3 bits of 10GBASE-KR speed into 1 bit. Therefore, the granularity is 0.29 ns. Figure 2-42 shows the process for DME page decoding for the 10GBASE-KR mode and also how this process is implemented in the DWC_xpcs. The +200 ps or – 200 ps resolution is implemented by considering 0.09696 ns bit granularity.

Figure 2-42 DME Page Decoding Process for 10GBASE-KR Mode



2.9.6 Auto-Negotiation CSR Interface

The auto-negotiation CSR interface block interfaces with the CSR block to send or receive various types of information, such as the following:

- Enabling or disabling auto-negotiation
- Auto-negotiation status (PASS or FAIL)
- Lane enabling or disabling
- DME page Error indication
- Link code word to be transmitted or received
- Parallel detection Link Status
- Priority Resolution among 1000BASE-KX, 2.5GBASE-KX, 10GBASE-KX4, 5GBASE-KR and 10GBASE-KR
- Test and Debug support

- Programmable auto-negotiation timers

The Clause 73 auto-negotiation specific registers are listed in the "*Register Descriptions*" chapter.

2.9.7 Enabling Clause 73 Auto-Negotiation

Clause 73 Auto-Negotiation feature is present by default when the MAIN_MODE is selected to be Backplane Ethernet PCS.

2.9.8 Registers Related to Clause 73 Auto-Negotiation

Following are the main registers related to Clause 73 Auto-Negotiation support:

- SR_AN_ADV3
- SR_AN_ADV2
- SR_AN_ADV1
- SR_AN_XNP_TX3
- SR_AN_XNP_TX2
- SR_AN_XNP_TX1
- VR_AN_INTR
- SR_AN_LP_ABL3
- SR_AN_LP_ABL2
- SR_AN_LP_ABL1

2.9.9 Programming Guidelines for Clause 73 Auto-Negotiation

See the "Programming Guidelines for Clause 73 Auto-Negotiation" on page 1712.

2.10 Clause 72 KR Training

This topic contains the following sections:

- "Overview of Clause 72 KR Training" on page 122
- "Description of Clause 72 KR Training" on page 122
- "Training Frame Structure" on page 123
- "Training Pattern" on page 124
- "Transmit Filter Structure" on page 124
- "Start-Up Protocol Flow" on page 125
- "Enabling Clause 72 KR Training" on page 142
- "Signals Related to Clause 72 KR Training" on page 142
- "Registers Related to Clause 72 KR Training" on page 142
- "Programming Guidelines for Clause 72 KR Training" on page 143

Note KR training is applicable only for 10GBASE-KR mode. It is not valid in 5GBASE-KR mode.

2.10.1 Overview of Clause 72 KR Training

The Clause 72 KR training is an optional feature in the DWC_xpcs. You can enable the Clause 72 KR training feature by selecting the Add Clause 72 KR Training Support option while configuring the DWC_xpcs. The IEEE Std 802.3, Clause 72 defines a start-up protocol between the local device and its link partner to facilitate the timing recovery and equalization tuning. It provides a mechanism through which a Local Receiver can tune the filter coefficients of the Remote Transmitter to optimize the signal quality over Backplane Interconnect. This is achieved by exchanging specialized frames called Training frames between the local device and its link partner.

2.10.2 Description of Clause 72 KR Training

The IEEE Std 802.3, Clause 72 start-up protocol is initiated after exchanging all pages of Clause 73 auto-negotiation. The start-up protocol should be enabled only when the auto-negotiation priority resolution indicates that the remote device is of KR type and the local device is also configured in the KR mode.

A training frame consists of a Preamble, a Control field, and a Training field as shown in Figure 2-43. The Control field is encoded using Differential Manchester Encoding (DME) at lower speed, that is, one fourth of 10.3125 Gbps. For information about the DME encoding, see "Auto-Negotiation Format: Differential Manchester Encoding (DME)" on page 107. The Training field has the PRBS11 random sequence using standard PRBS11 sequence generator at 10.3125 Gbps speed. With this scheme, the Control field is always exchanged successfully without any need for equalization because it is transferred at a lower baud rate.

The Control field of a training frame has requests and responses. During the start-up protocol, the local device and the remote device simultaneously send or receive these requests or responses, respectively. The following list describes this:

- The local device sends requests to tune the Transmit filter coefficients of the remote device.
- The remote device sends requests to tune the Transmit filter coefficients of the local device.

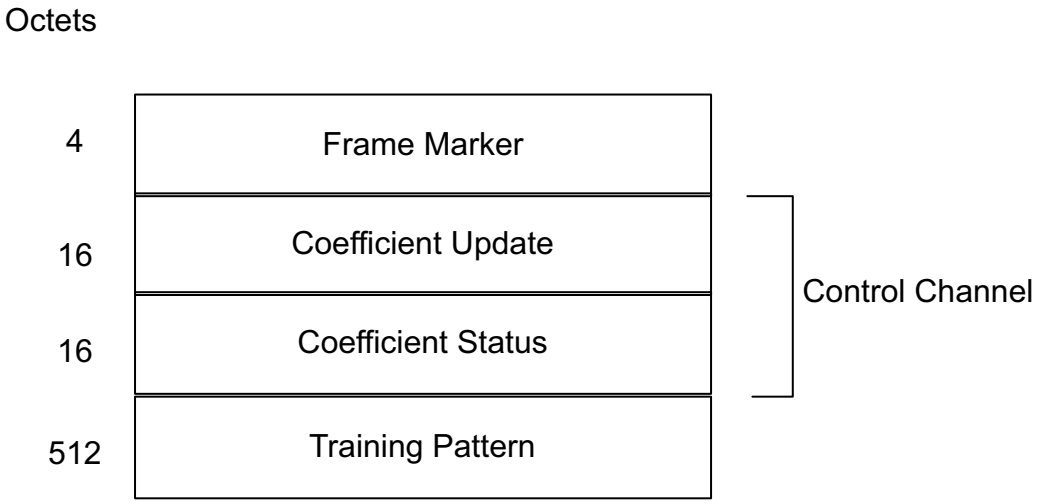
- The local device receives status for requests given to the remote device.
- The remote device receives status for requests given to the local device.

Note The information in the following subsections is provided from the perspective of a local device. This information is also applicable to a remote device.

2.10.3 Training Frame Structure

Figure 2-43 shows the frame structure specified in IEEE Std 802.3, Clause 72 for KR training.

Figure 2-43 Clause 72 KR Training Frame Structure



2.10.3.1 Frame Marker

The frame marker acts as the training frame delimiter. It has a fixed 32-bit pattern of 32'hFFFF0000. This 32-bit pattern does not appear in any other field (control channel or the training pattern).

2.10.3.2 Coefficient Update and Status (Control Channel)

The Coefficient Update and Status fields (control channel) are encoded with slower data rate by using the Differential Manchester Encoding (DME). Each DME Cell is of eight 10GBASE-KR Unit Intervals (UI), that is, 8-bit duration. The middle bit (4th bit) indicates whether data is 1 (when there is an additional transition) or 0 (when there is no transition).

On a transmitted training frame, the Coefficient Update field has the requests from the local device to tune the remote transmitter. The Coefficient Status field has the status of the requests received from the remote device. On a received training frame, the Coefficient Update field has the requests from the remote device to tune the local transmitter. The Coefficient Status field has the status of the requests transmitted to remote device.

For detailed information about these fields, see "Coefficient Update and Coefficient Status Fields during Tx and Rx Link Equalization" on page 127.

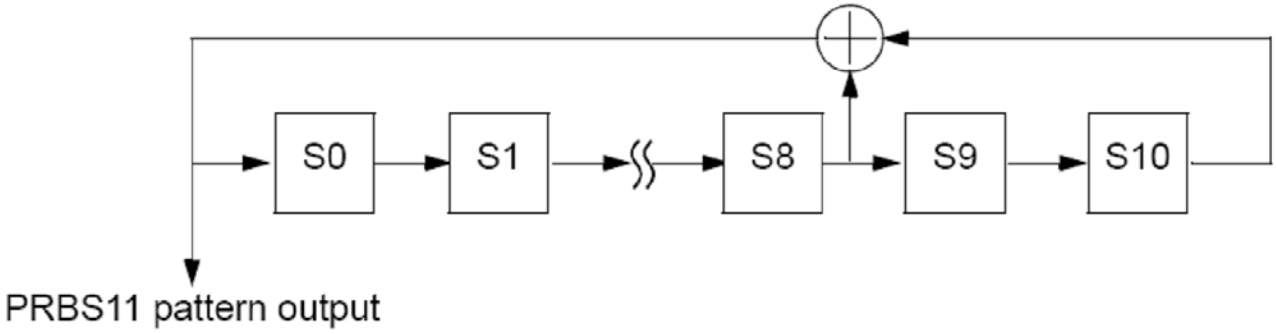
2.10.4 Training Pattern

The training pattern is 512 bytes of pseudo-random bit sequence generated through PRBS11 generator with initial random seed. It is not encoded by using the DME format. The following is the PRBS11 generator polynomial:

$G(x) = 1 + X^9 + X^{11}$

The PRBS11 pattern is sent or received at 10.3125 Gbps rate. During reception, this pattern can used to analyze the Rx link reliability or quality.

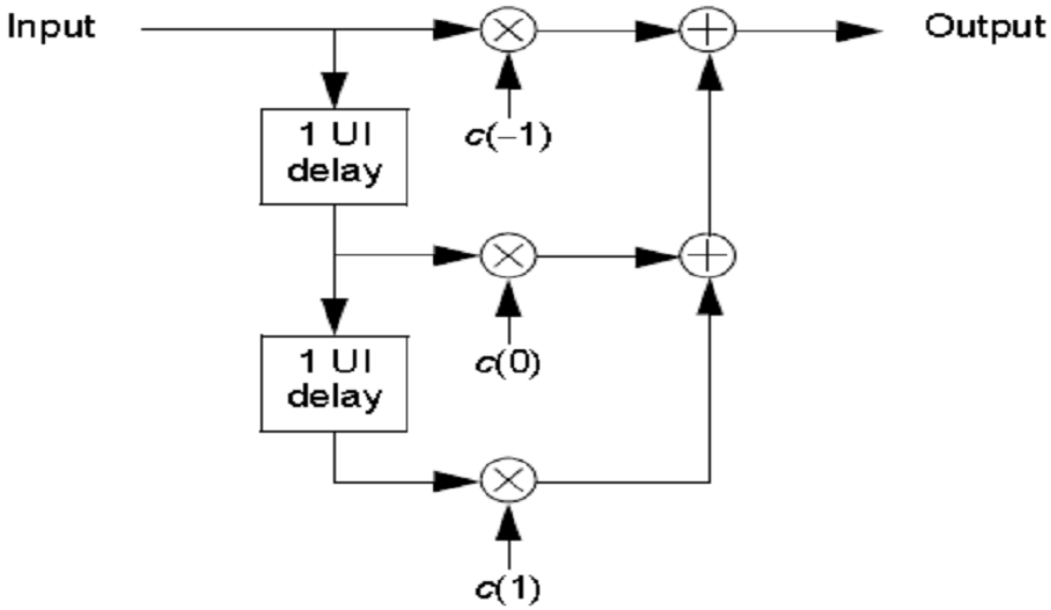
Figure 2-44 PRBS11 Pattern Output



2.10.5 Transmit Filter Structure

Figure 2-45 shows the filter structure that a transmitter can have at the serial outputs as defined in IEEE Std 802.3, Clause 72. There are three Transmit filter coefficients c(-1), c(0), and c(1). The equalization is achieved by tuning these Transmit filter coefficients. The tuning information and the corresponding status are exchanged through control channel of the training frame.

Figure 2-45 Transmit Filter Structure



2.10.6 Start-Up Protocol Flow

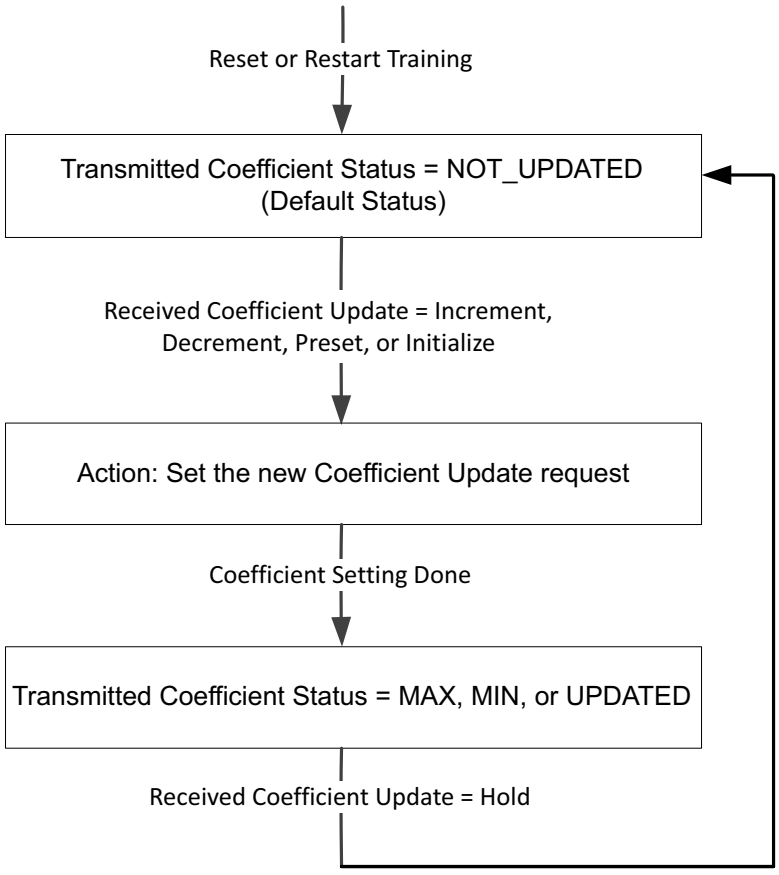
This section describes how the Tx and Rx link equalization is achieved. For programming guidelines for start-up protocol, see "Programming Guidelines for Clause 72 Start-Up Protocol" on page 1708.

2.10.6.1 Tx Link Equalization Flow

The Tx link equalization is achieved by tuning the local transmitter as per the requests in the Control field of the received training frame. During start-up protocol, the remote device first initializes the local Transmit filter coefficients and then issues the increment or decrement request to tune these Transmit filter coefficients.

Figure 2-46 shows the Tx link equalization flow. This flow is applicable for filter coefficients c(1), c(0), and c(-1). All three filter coefficients can be tuned simultaneously or independently.

Figure 2-46 Transmit Link Equalization Flow



The following list describes the flow shown in Figure 2-46:

1. The local device sends the status as Not_Updated (default response) on the transmitted training frame. The local device waits for the increment, decrement, preset, or initialize request in the incoming frame.
2. On receiving the request, the local device initiates the requested action by setting its transmitter filter coefficients.

3. After doing the requested settings, the local device responds with Max, Min, or Updated status in the transmitted training frame. Any of these responses acts as an acknowledgment to the remote device.
4. On receiving the acknowledgment, the remote device starts sending the hold request. At the same time, the remote device also analyzes its Rx link reliability and decides on further tuning of the local transmitter filters.
5. On receiving the hold request, the local device sends the Not_Updated response to indicate that it is ready for processing the next request.

This cycle continues until the remote receiver is satisfied with the quality of the received data (in such case, it continues to send the hold requests) or a timer as specified by the standard runs out.

The actual Tx equalization is to tune the filters of the local PHY transmitter. The DWC_xpcs provides the following interface signals to achieve Tx link equalization in the PHY:

- Local Tx Filter Coefficient Update Interface

  This output interface of the DWC_xpcs provides the filter tuning requests received from the remote device to the PHY. The requests are issued in the form of increment, decrement, preset, or initialize requests. The hold requests are not given out to the PHY because it is not required.

- Local Tx Filter Coefficient Status Interface

  This input interface of the DWC_xpcs is used to get the status of the tuning requests that are issued through Local Tx Filter Coefficient Update Interface to the PHY. The DWC_xpcs reports this status back to the remote device through training frames generated in the DWC_xpcs.

In the "*Signal Descriptions*" chapter, all top-level interface signals with the prefix `rpcs_ktx_*` are related to Tx Link equalization.

2.10.6.2 Rx Link Equalization Flow

The Rx link equalization is achieved by requesting the remote transmitter to tune its Transmit filters based on the Rx Link reliability analysis. The flow for Rx link equalization is similar to the one explained in "Tx Link Equalization Flow" on page 125. The only difference is that the roles of the local device and the remote device are swapped.

To achieve the Rx link equalization in the PHY, the DWC_xpcs provides the following interfaces to the PHY:

- Rx Filter Coefficient Update Interface

  This is the input interface to the DWC_xpcs. The Rx Equalization algorithm in the PHY uses this interface to issue requests to tune Transmit filters of the remote device. The requests can be issued in the form of preset, initialize, increment, or decrement.

- Rx Filter Coefficient Status Interface

  This output interface of the DWC_xpcs provides the received status of the requests sent to the remote device through Rx Filter Coefficient Update interface. The status is provided in the form of updated, maximum, or minimum as per the received training frame from the remote device. The PHY also uses this status as acknowledgment of its requests.

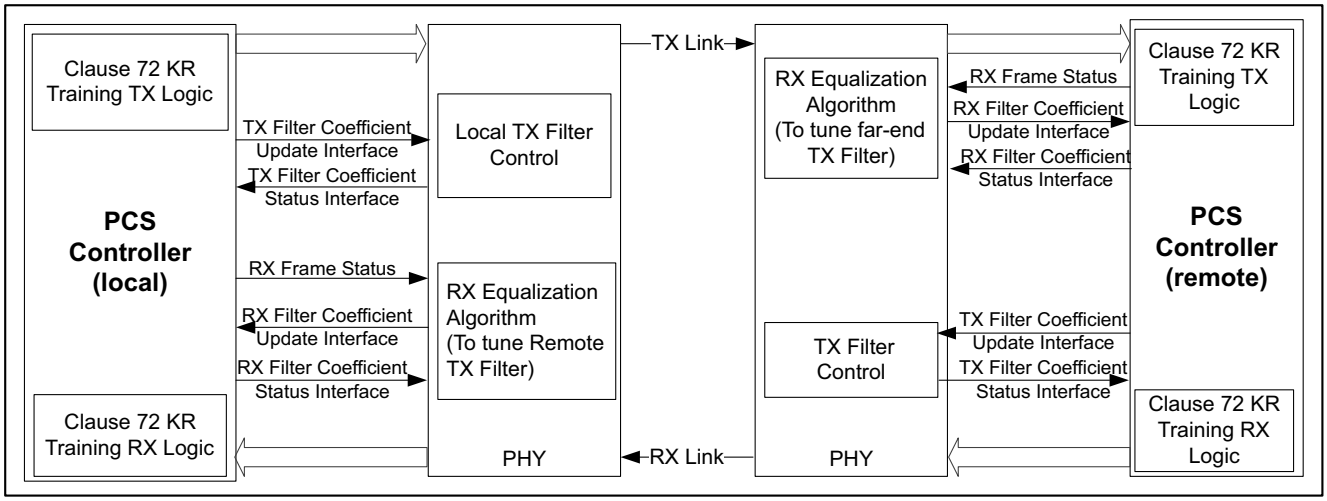
- Rx Frame Status Interface

  This output interface of the DWC_xpcs provides the status or quality of the received Training frame such as PRBS11 decoding status, frame lock status, or frame received event. The Rx Equalization algorithm in the PHY can use this status to decide when to terminate or initiate further training.

In the "*Signal Descriptions*" chapter, all top-level interface signals with prefix `rpcs_krx_*` are related to Rx Link equalization.

Figure 2-47 shows all interfaces provided by the DWC_xpcs to the local PHY and equivalent implementation that exists in the remote device to achieve both Tx and Rx link equalization.

Figure 2-47 Clause 72 KR Training Equalization Interfaces



2.10.6.3 Coefficient Update and Coefficient Status Fields during Tx and Rx Link Equalization

Table 2-22 provides information about the coefficient update field during Tx and Rx link equalization.

Table 2-22 Coefficient Update Field during Tx and Rx Link Equalization

Field	Description	Generation	Reception
15:14	Reserved	N/A	N/A

Field	Description	Generation	Reception
13	Preset ■ 1: Preset ■ 0: No Preset	The Rx equalization process in the PHY generates this request. The PHY can generate this request as a first request, that is, the first request after the rpcs_ktr_restart_o output pulse is asserted. The PHY can also generate this request when it receives NOT_UPDATED as status from the remote device for all the filters. When the PHY issues the preset request, the DWC_xpcs continues to send the preset request till it receives updated or maximum as status from the remote device. When the DWC_xpcs receives the status, it updates this received status to the PHY and starts sending the hold request. Note: This request is usually sent in the beginning of the start-up protocol to analyze the link behavior when equalization is turned off.	When the DWC_xpcs receives this request, it requests the local PHY to perform preset. The PHY should preset its Tx filter coefficients such that c(-1) and c(1) are set to zero and c(0) is set to the maximum value. After these settings are done, the PHY should update the status as updated or maximum to the DWC_xpcs. The DWC_xpcs then sends this status to the remote device.
12	Initialize ■ 1: Initialize ■ 0: No Initialize	The Rx equalization process in the PHY generates this request. The PHY can generate this request as a first request, that is, the first request after the rpcs_ktr_restart_o output pulse is asserted. The PHY can also generate this request when it receives NOT_UPDATED as status from the remote device for all the filters. When the initialize request is issued, the DWC_xpcs continues to send the initialize request till it receives updated, maximum, or minimum as status from the remote device. When the DWC_xpcs receives the status, it updates this received status to the PHY and starts sending the hold request. Note: This request is usually sent in the beginning of the start-up protocol to set the Transmit filters to predefined values before issuing the increment or decrement requests.	When the DWC_xpcs receives this request, it requests the local PHY to perform initialize.
11:6	Reserved	N/A	N/A

Field	Description	Generation	Reception
5:4	Coefficient update for c(1) filter ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold	The Rx equalization process in the PHY generates the increment or decrement request. This field is for c(1) filter. The DWC_xpcs automatically generates the hold request as part of the handshake and also when no request is issued by the PHY. The PHY can generate this request as a first request, that is, the first request after the rpcs_ktr_restart_o output pulse is asserted. The PHY can also generate this request when it receives NOT_UPDATED as status from the remote device for c(1) filter. When the PHY issue the increment or decrement request, the DWC_xpcs continues to send the increment or decrement request till it receives updated, maximum, or minimum as status from the remote device. When the DWC_xpcs receives the status, it updates this received status to the PHY and starts sending the hold request until the next request is received. Note: The generation of increment or decrement is based on the Rx equalization algorithm which is based on the Rx link reliability analysis.	When the DWC_xpcs receives the increment or decrement request, it sends the request to the local PHY for c(1) filter. The PHY should increment or decrement its c(1) Tx filter coefficients as defined in the IEEE Std 802.3, Clause 72. The PHY should update the status as updated, maximum, or minimum to the DWC_xpcs. The DWC_xpcs then sends this status to the remote device.
3:2	Coefficient update for c(0) filter	This field is for c(0) filter. The information for this field is similar to the information given in Bits[5:4] of this table for c(1) filter.	This field is for c(0) filter. The information for this field is similar to the information given in Bits[5:4] of this table for c(1) filter.
1:0	Coefficient update for c(-1) filter	This field is for c(-1) filter. The information for this field is similar to the information given in Bits[5:4] of this table for c(1) filter.	This field is for c(-1) filter. The information for this field is similar to the information given in Bits[5:4] of this table for c(1) filter.

Table 2-23 provides information about the coefficient update field during Tx and Rx link equalization.

Table 2-23 Coefficient Status Format

Field	Description	Generation	Reception
15	Receiver Ready	The DWC_xpcs sets this bit to 1 when the local device Rx equalization process determines that the Rx link is ready for KR speed traffic. The Rx equalization process generates this status by asserting the rpcs_rx_trained_i input signal any time after the rpcs_ktr_restart_o output pulse is asserted. Otherwise, this bit is set to 0. Note: The Rx equalization process sets this input by monitoring the reliability of the Rx link based on successful decoding of training patterns (PRBS11) of the received training frame.	When the DWC_xpcs receives this status, it indicates that the remote device does not generate any more requests to tune the local transmitter filters. Note: This status is received in the DWC_xpcs. This status is not given to the PHY.
14:6	Reserved	N/A	N/A
5:4	Coefficient Status of c(1) ■ 11: Max ■ 10: Min ■ 01: Updated ■ 00: Not_Updated	The local PHY should generate this status for c(1) filter as a response to the preset, initialize, increment, or decrement request received for c(1) filter. Note: The DWC_xpcs automatically generates the Not_Updated status during handshaking. The DWC_xpcs transmits this status back to remote device. Therefore, the PHY does not need to give the Not_Updated status.	When the DWC_xpcs receives the max, min, or updated status, it sends the status to the PHY Rx equalization process. The DWC_xpcs receives this status as a response to the preset, initialize, increment, or decrement request sent to the remote device. When the DWC_xpcs receives this status, the Rx equalization can again evaluate the Rx link and issue further requests to tune the remote Transmit filters.
3:2	Coefficient Status of c(0)	This field is for c(0) filter. The information for this field is similar to the information given in Bits[5:4] of this table for c(1) filter.	This field is for c(0) filter. The information for this field is similar to the information given in Bits[5:4] of this table for c(1) filter.
1:0	Coefficient Status of c (-1)	This field is for c(-1) filter. The information for this field is similar to the information given in Bits[5:4] of this table for c(1) filter.	This field is for c(-1) filter. The information for this field is similar to the information given in Bits[5:4] of this table for c(1) filter.

2.10.6.4 CL72 Training Control Options

DWC_xpcs supports some custom features to control the mode of operation of CL72 KR Training.

PRBS31 Support

DWC_xpcs, by default, uses PRBS11 as the training pattern in the transmitted KRTR frames, as required by IEEE specification. However, DWC_xpcs has the capability to send and monitor PRBS31 pattern (instead of PRBS11) in the KRTR frames. This feature can be enabled by programming bit [1] (PRBS31_EN) of VR_PMA_KRTR_PRBS_CTRL0. If this mode is enabled, the 'Training Pattern' field of the transmitted CL72 Frame has PRBS31 instead of PRBS11. In addition, the PRBS checker in the receive path of DWC_xpcs monitors PRBS31 errors instead of PRBS11 errors.

PRBS Mode to Determine Successful Rx Equalization

DWC_xpcs has a control (bit [0] of VR_PMA_KRTR_PRBS_CTRL0) to enable PRBS mode to determine successful receive equalization. By default, this mode is not enabled, and DWC_xpcs determines successful Rx Equalization based on the status of the primary input 'rpcs_rx_trained_i' (for non-Synopsys PHY configurations) or based on the expiry of an internal Rx-Train-Time timer (for Synopsys PHY configurations).

If PRBS mode is enabled, DWC_xpcs starts monitoring the received CL72 Frames (after the assertion of 'rpcs_rx_trained_i' or after the expiry of Rx-Train-Time timer) for any PRBS errors, for an interval determined by the value programmed to VR_PMA_KRTR_PRBS_CTRL1. If the number of PRBS errors detected is less than or equal to the tolerance limit programmed to VR_PMA_KRTR_PRBS_CTRL2, then DWC_xpcs concludes that the receive equalization has been successful; otherwise this is treated as a 'TRAINING FAILURE'.

Sequential/Parallel Mode of Coefficient Update

During Receive equalization, DWC_xpcs makes Rx-link evaluation request to the PHY. In turn, PHY provides INC/DEC commands for the pre (C(-1)), post (C(1)) or main(C(0)) coefficients. In parallel mode (default mode), DWC_xpcs transmits the received commands for all the three coefficients (if all three have been provided by the PHY) to the remote-end device in the same frame.

In sequential mode, DWC_xpcs transmits the three coefficient update commands sequentially one after the other. DWC_xpcs starts with sending C(-1) update , then transmits C(0) update and ends with C(1) update. Prior to sending a new coefficient update, DWC_xpcs waits for an acknowledgment from the remote-end device for the previously transmitted coefficient update. This mode is useful if the remote-end device is not capable of handling multiple coefficient update request at the same time.

This feature is available only if the selected configuration has Synopsys PHY. This mode is controlled through bit[8] (SEQ_EQ_EN) of VR_XS_PMA_MP_12G_16G_25G_RX_EQ_CTRL4 register.

Ping-Pong Mode of Rx Equalization

If Ping-Pong mode is enabled, DWC_xpcs sends exactly one of C(-1) or C(1) requests to the remote-end device , even if PHY has given INC/DEC commands for both C(-1) and C(1) in a particular iteration of Rx equalization. DWC_xpcs alternates between C(-1) and C(1 between iterations, i.e., if C(-1) has been serviced in nth iteration, then DWC_xpcs would serve C(1) (if any) in the (n+1)th iteration and vice versa. This mode is enabled by programming bit[9] (PING_PONG_EN) of VR_XS_PMA_MP_12G_16G_25G_RX_EQ_CTRL4 register. This feature is available only if the selected configuration has Synopsys PHY.

Ping-Pong mode works well when automatic generation of main coefficient is enabled. Auto-generation of main coefficient can be enabled by programming bit[10] (SELF_MAIN_EN) of VR_XS_PMA_MP_12G_16G_25G_RX_EQ_CTRL4 register. If auto-generation of main is enabled,

DWC_xpcs would ignore the update command received from PHY for C(0) (if any) and would transmit INC/DEC based on the update command received for C(-1) or C(1).

If DWC_xpcs has decided to serve C(-1) in a particular iteration and if C(-1) is an INC command, then DWC_xpcs would transmit INC command for both C(-1) and C(0). On the other hand, if C(-1) is a DEC command, then DWC_xpcs would transmit DEC command for both C(-1) and C(0). If sequential mode is also enabled (along with Ping-Pong mode), then DWC_xpcs would transmit DEC C(0) first and then transmit DEC(-1) sequentially; otherwise both DEC C(0) and DEC C(-1) are transmitted simultaneously.

If DWC_xpcs has decided to serve C(1) in a particular iteration and if C(1) is an INC command, then DWC_xpcs would transmit INC command for both C(1) and C(0). On the other hand, if C(1) is a DEC command, then DWC_xpcs would transmit DEC command for both C(1) and C(0). If sequential mode is also enabled (along with Ping-Pong mode), then DWC_xpcs would transmit DEC C(0) first and then transmit DEC(1) sequentially; otherwise both DEC C(0) and DEC C(1) are transmitted simultaneously.

During ping-pong mode, DWC_xpcs first checks for the validity of the coefficient update commands provided by the PHY. DWC_xpcs would process and send the command to the link partner only if it is valid.

A INC command for C(-1) is considered valid, only if the status returned by link partner for c(-1) and C(0) coefficients are both not MAX for the previous iteration. Similarly, a INC command for C(+1) is considered valid, only if the status returned by link partner for c(+1) and C(0) coefficients are both not MAX for the previous iteration.

A DEC command for C(-1) is considered valid, only if the status returned by link partner for c(-1) and C(0) coefficients are both not MIN for the previous iteration. Similarly, a DEC command for C(+1) is considered valid, only if the status returned by link partner for c(+1) and C(0) coefficients are both not MIN for the previous iteration.

Management mode of Rx Equalization

Management mode of Rx Equalization can be enabled by setting bit [15] (RX_EQ_MM) of VR_PMA_KRTR_RX_EQ_CTRL register to 1. This bit should be set prior to enabling KR training, that is, before setting TR_EN bit of SR_PMA_KR_PMD_CTRL register to 1.

During Rx equalization process (of KR training), DWC_xpcs sends training frames (to link partner) with Coefficient Update commands to tune the filter coefficients of the link-partner. These commands are reflected in SR_PMA_KR_LD_CEU Register. These coefficient update commands are generated by the local PHY receiver and is provided to DWC_xpcs through its input ports (rpcs_krx_ceu{filter}_vld_i for Non-Synopsys configurations; rpcs_krx_{pre,main,post}_dir_i for Synopsys PHY configurations).

During software (management) mode, the management can program desired coefficient update commands to VR_PMA_KRTR_RX_EQ_CTRL register. The values programmed to this register are transmitted (instead of the commands given by the local PHY) to the link partner using training frames. The values programmed by management to this register is also reflected in SR_PMA_KR_LD_CEU register. The link-partner, in turn, sends frames with the 'Status' information to DWC_xpcs. This 'Status' information is available in SR_PMA_KR_LP_STS register. During hardware mode, this status information is passed on to the local PHY receiver through the 'rpcs_krx_ceu{filter}_sts_o' ports of DWC_xpcs (for Non-Synopsys PHY configurations). However, in software mode, management can choose to provide any desired status to the local PHY by programming suitable values to 'VR_PMA_KRTR_RX_EQ_STS_CTRL' register. Though the coefficient update commands given by the local PHY are ignored by DWC_xpcs (in software mode), this information is captured in VR_PMA_PHY_RX_EQ_CEU Register.

In addition, software can program the 'RR_RDY' field of VR_PMA_KRTR_RX_EQ_CTRL register to 1, once it decides to send no more coefficient update commands to the link-partner. Consequently, the link partner infers that the local receiver has trained successfully. During this mode, DWC_xpcs ignores the value of the 'rpcs_rx_trained_i' port of DWC_xpcs.

Software Mode of Rx equalization has two independent processes under the hood as illustrated in Figure 2-48, Figure 2-49, and Figure 2-50.

Figure 2-48 Software Mode of Rx Equalization - Process 1

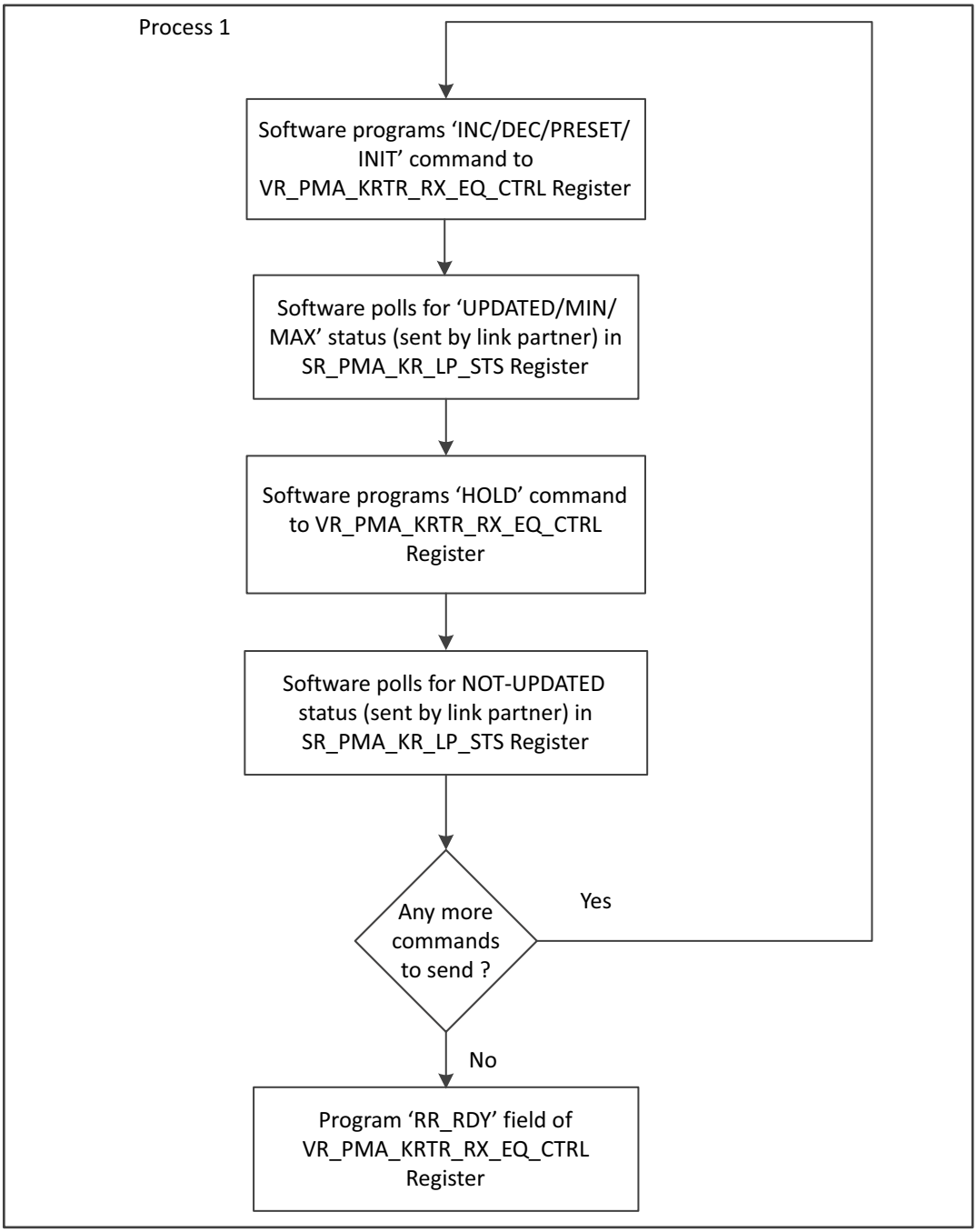


Figure 2-49 Software Mode of Rx Equalization - Process 2: Non-Synopsys PHY

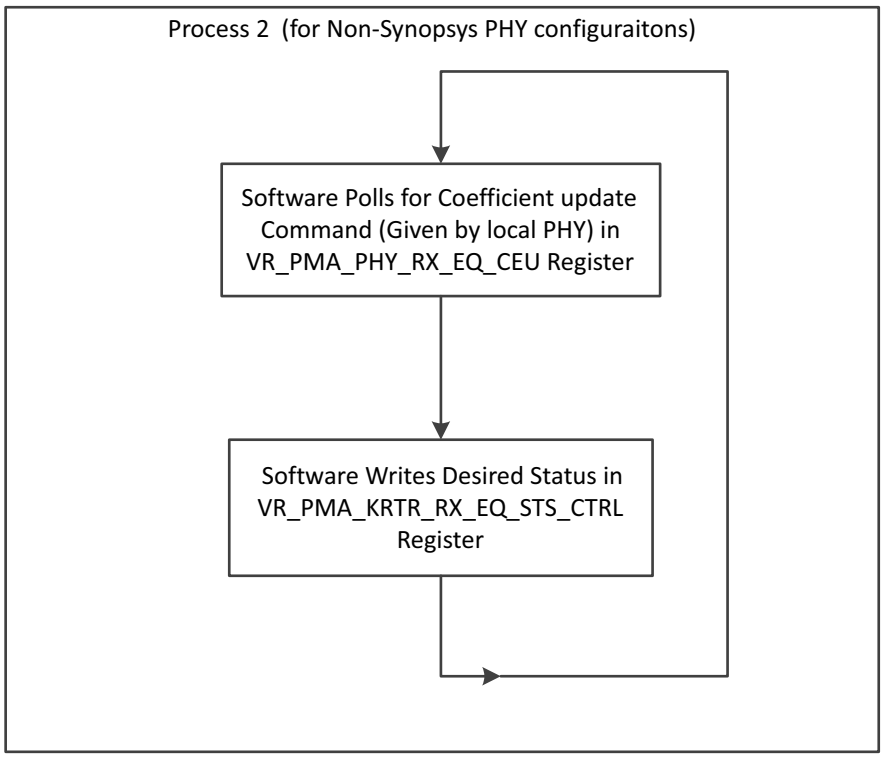
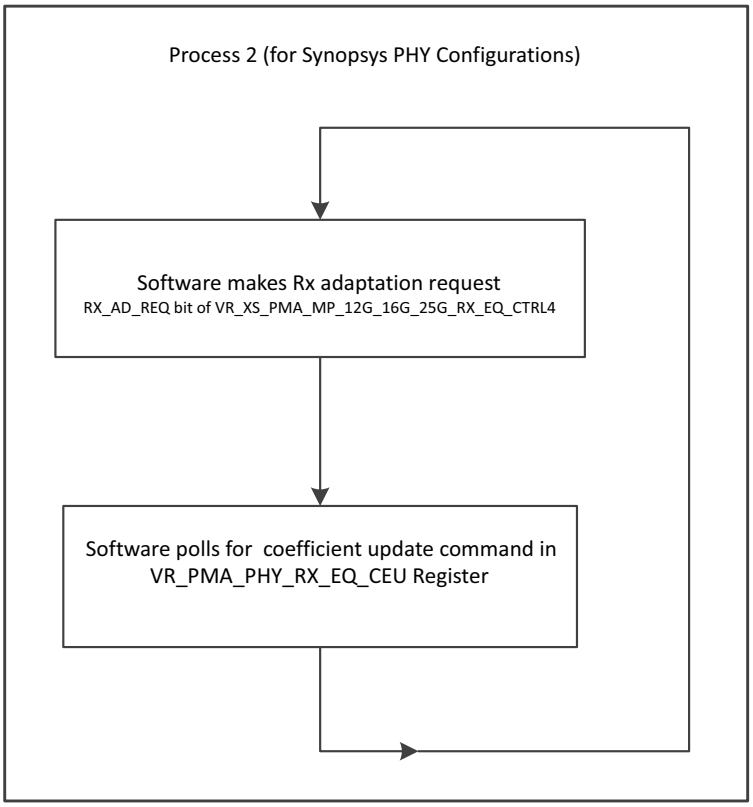
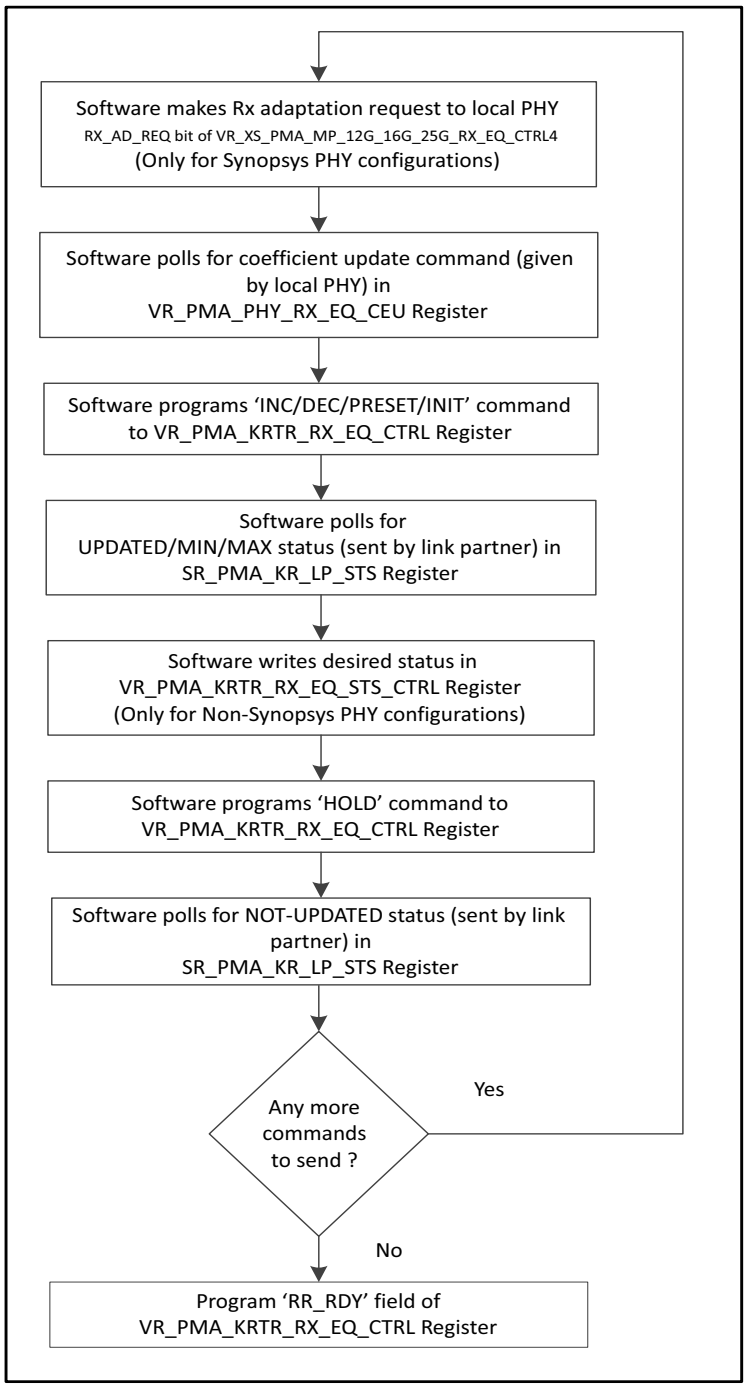


Figure 2-50 Software Mode of Rx Equalization - Process 2: Synopsys PHY



Though these two processes can be performed independent of each other (while training is enabled), software can optionally decide to perform Rx Equalization in the sequence shown in Figure 2-51.

Figure 2-51 Rx Equalization Sequence



Management Mode of Tx Equalization

Management Mode of Tx Equalization can be enabled by setting bit [15] (TX_EQ_MM) of VR_PMA_KRTR_TX_EQ_STS_CTRL register to 1. This bit should be set prior to enabling KR training, i.e., before setting TR_EN bit of SR_PMA_KR_PMD_CTRL register to 1.

During Tx equalization process (of KR training), DWC_xpcs receives training frames (from link partner) with Coefficient Update commands for the local PHY transmitter. These commands are reflected in SR_PMA_KR_LP_CEU Register. In Hardware Mode, these coefficient update commands get driven to the local PHY TX filter inputs. The PHY responds to these commands and provides the corresponding 'Status' information. This 'status' information is then sent back to the link-partner. The 'Status' information sent to the link-partner is reflected in SR_PMA_KR_LD_STS Register.

During software (management) mode, the TX coefficient update commands driven to the PHY TX filter inputs no longer depend on the commands received from the link-partner. Software can program the desired coefficient update commands to 'VR_PMA_KRTR_TX_EQ_CFF_CTRL' register. In addition, the 'Status' that is returned to the link-partner is also determined by the management. The 'status' values programmed to VR_PMA_KRTR_TX_EQ_STS_CTRL register are packed into the transmitted training frames. Additionally, the values programmed to this register gets reflected in SR_PMA_KR_LD_STS register, as it indicates the 'status' information in the most recently transmitted frame. Though the 'Status' returned by the local PHY is ignored by DWC_xpcs, it is captured in VR_PMA_PHY_TX_EQ_STS register.

Software Mode of TX equalization has two independent processes under the hood as explained by Figure 2-52 and Figure 2-53:

Figure 2-52 Software Mode of Tx Equalization - Process 1

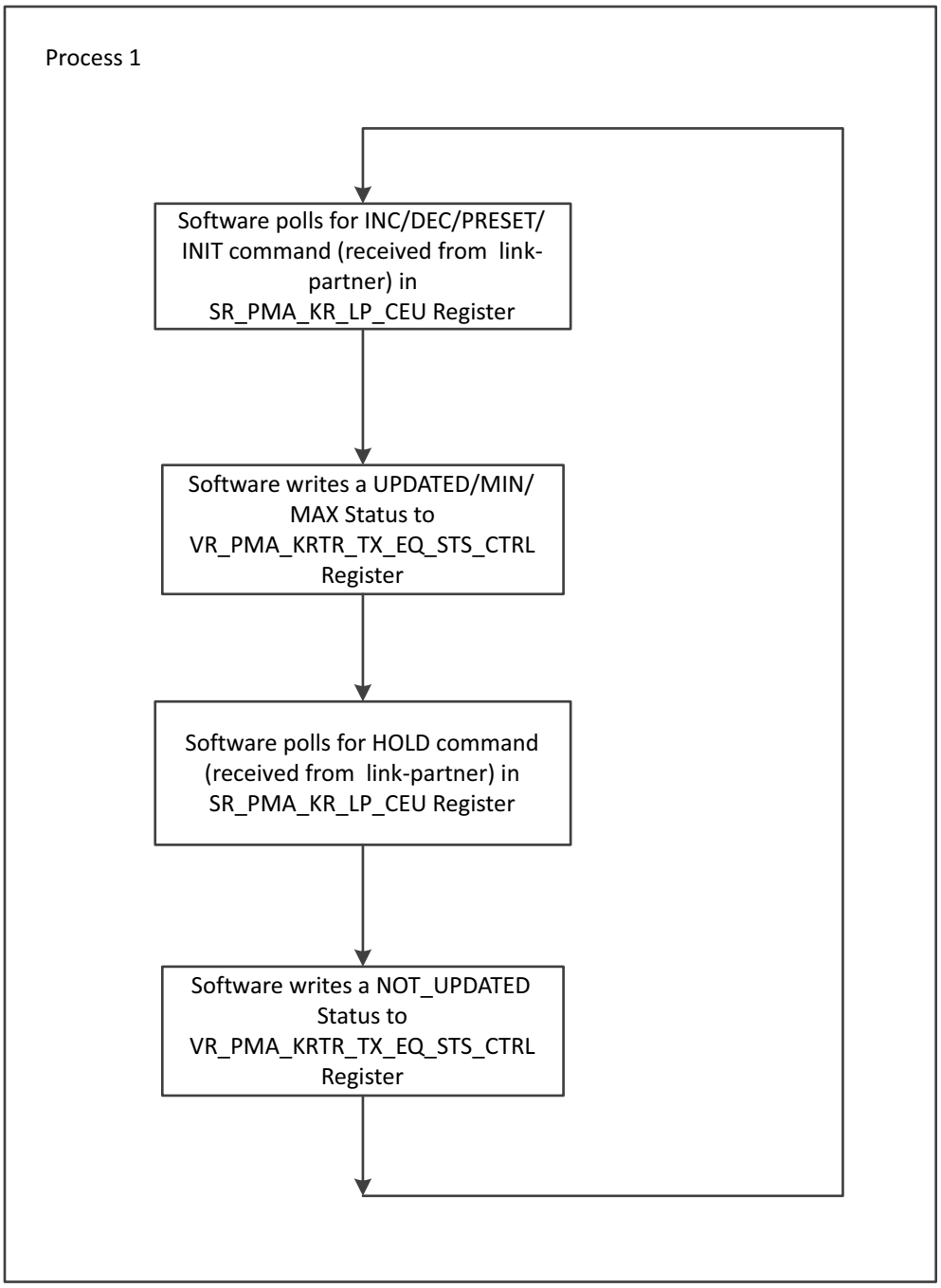
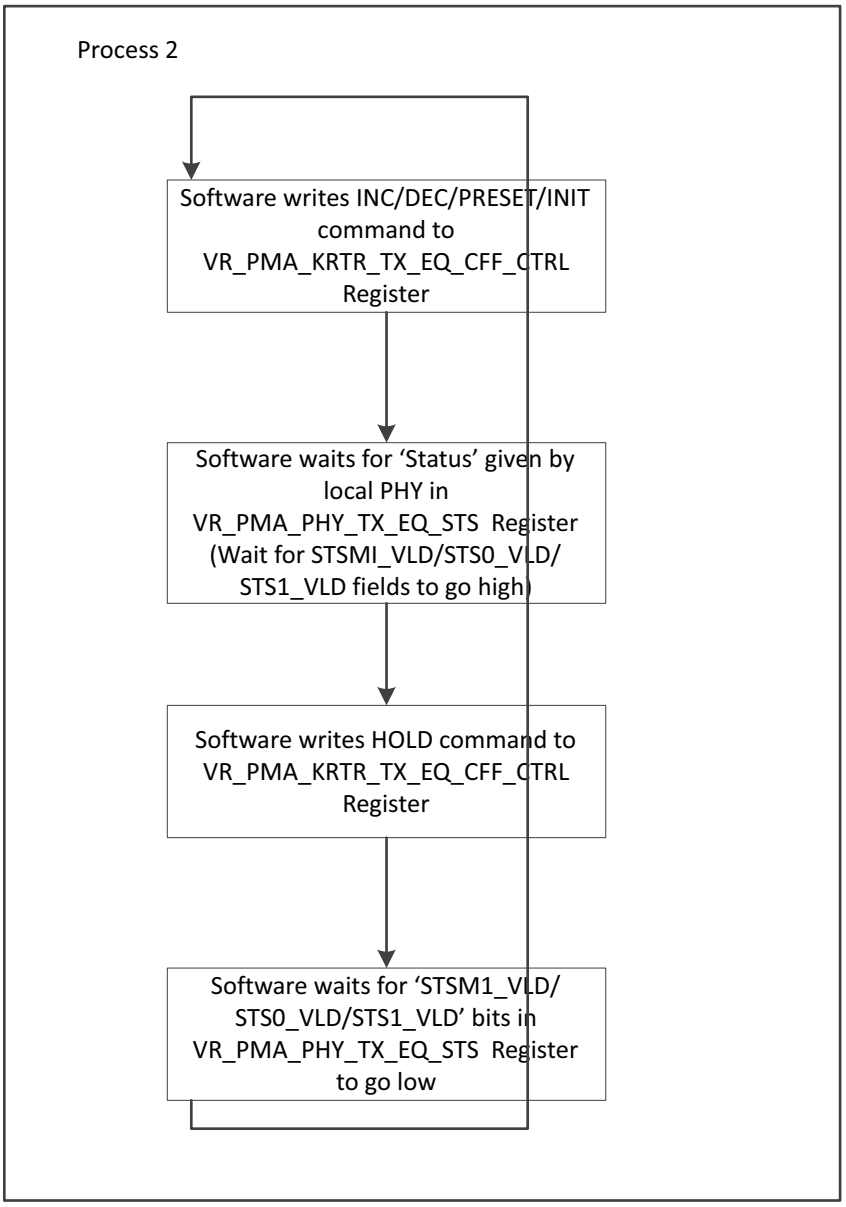
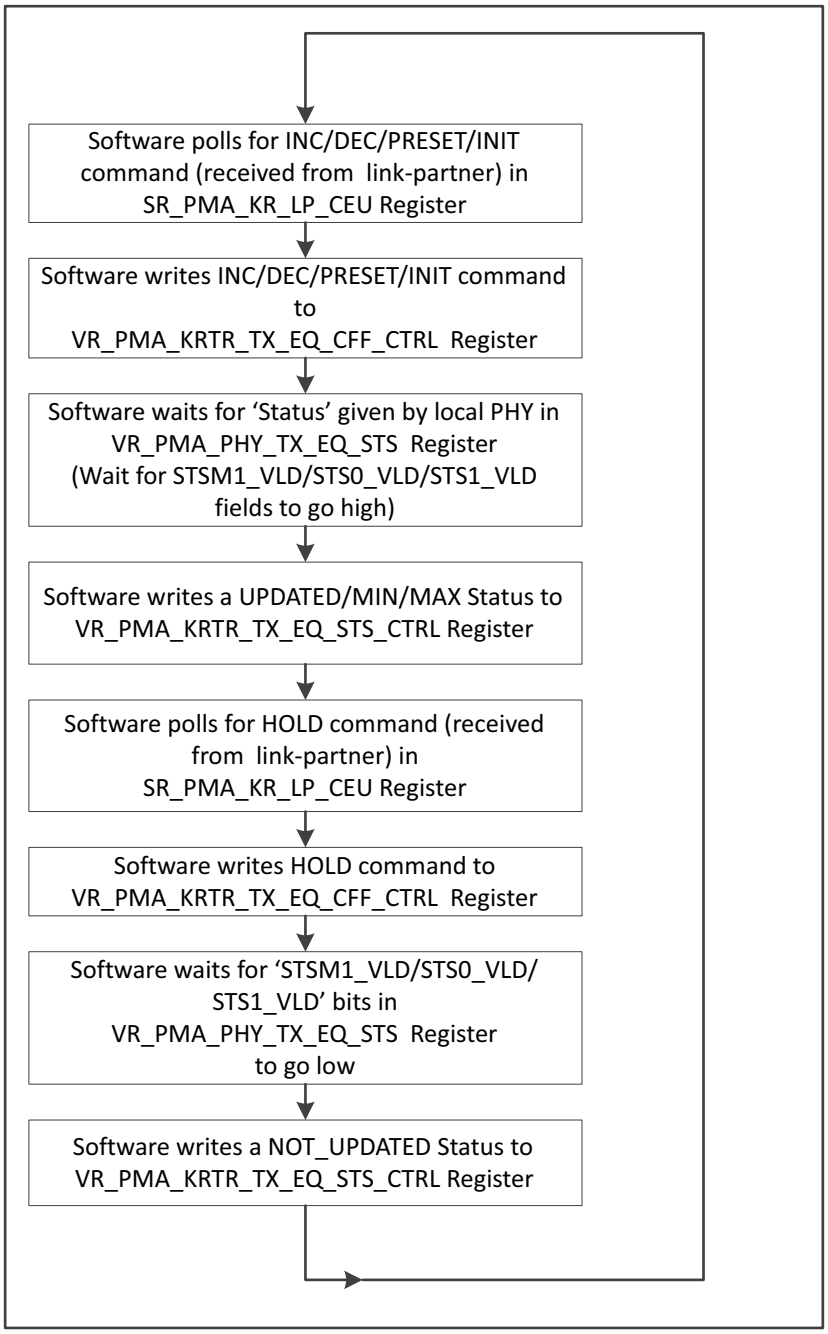


Figure 2-53 Software Mode of Tx Equalization - Process 2



Though these two processes can be performed independent of each other (while training is enabled), software can optionally decide to perform TX Equalization in the sequence (which is more logical) as shown in Figure 2-54:

Figure 2-54 Tx Equalization Sequence



Programmable Timers

Clause 72 defines two timers, max_wait_timer and wait_timer. As per the specification, the value of max_wait_timer should be 500ms and the value of wait_timer should be between 100 and 300 training frames.

DWC_xpcs offers the flexibility to program the values of these timers. Application can program the following registers to update the values of these timers:

- VR_PMA_KRTR_TIMER_CTRL0
- VR_PMA_KRTR_TIMER_CTRL1

The default values of these registers are such that the timers run for the duration specified by IEEE specification.

For configurations with Synopsys PHY, an additional timer determines the duration for receive equalization. This timer is programmable through VR_PMA_KRTR_TIMER_CTRL2.

Any change to these programmable timer registers should be done before enabling the KR training operation.

Constant Amplitude Tx Equalization

During KR training, DWC_xpcs translates the INC/DEC commands (received from the far-end device) to appropriate transmit filter coefficients (rpcs_ktx_pre_o, rpcs_ktx_main_o and rpcs_ktx_post_o). The default Tx equalization algorithm changes these filter coefficients in such a manner that might result in the reduction of signal amplitude. This might deteriorate the channel quality. Hence, DWC_xpcs provides another Tx Equalization algorithm that preserves the signal amplitude on receiving INC/DEC commands for PRE or POST coefficients. This mode can be enabled by setting bit [8] (CA_TX_EQ) of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL1 Register. This feature is applicable only for Synopsys multi-protocol 12G, 16G, or 25G PHY configurations.

When this mode is enabled in the Tx side, software must ensure that Ping-Pong mode with automatic main generation (SELF_MAIN_EN) is not enabled on the Rx side.

2.10.7 Enabling Clause 72 KR Training

To enable Clause 72 KR Training during the Specify Configuration Activity in coreConsultant, select the "Add Clause 72 KR Training Support" check box, in the System Configurations Settings window.

For details about this option, see the "*Parameter Descriptions*" chapter.

2.10.8 Signals Related to Clause 72 KR Training

Based on the type of PHY used, see the appropriate section in the "*Signal Descriptions*" chapter:

- Clause 72 KR Training Signals for Synopsys PHY
- Clause 72 KR Training Signals for Non-Synopsys PHY

2.10.9 Registers Related to Clause 72 KR Training

Following are the registers related to Clause 72 KR Training support:

- SR_PMA_KR_PMD_CTRL
- SR_PMA_KR_LP_CEU
- SR_PMA_KR_LP_CESTS
- SR_PMA_KR_LD_CEU
- SR_PMA_KR_LD_CESTS

2.10.10 Programming Guidelines for Clause 72 KR Training

See the "Programming Guidelines for Clause 72 Start-Up Protocol" on page 1708.

2.11 Clause 37 Auto-Negotiation

This topic contains the following sections:

- "Overview of Clause 37 Auto-Negotiation" on page 144
- "Description of Clause 37 Auto-Negotiation" on page 144
- "Enabling Clause 37 Auto-Negotiation" on page 146
- "Registers Related to Clause 37 Auto-Negotiation" on page 146
- "Programming Guidelines for Clause 37 Auto-Negotiation" on page 147

2.11.1 Overview of Clause 37 Auto-Negotiation

The DWC_xpcs supports auto-negotiation in the 1000BASE-X and SGMII/QSGMII modes. Auto-negotiation enables the DWC_xpcs to do the following:

- Advertise its speed, duplex mode (full-duplex or half-duplex), and flow control capabilities.
- Acquire capabilities of its remote partner and link status to converge on a common operating mode and flow-control mechanism.

This block performs the following functions:

- Recognizes or generates the |C| order set
- Supports the SGMII and 1000BASE-X Clause 37 auto-negotiation protocol

2.11.2 Description of Clause 37 Auto-Negotiation

The Clause 37 auto-negotiation is enabled by default when Mode of Operation (MAIN_MODE) is 1000BASEX-Only PCS. The DWC_xpcs supports the IEEE-defined MII and auto-negotiation registers as defined in Clause 22 and Clause 37. For information about these registers, see the "*Register Descriptions*" chapter. The Clause 37 auto-negotiation is based on the PCS /C/ order set shown in Table 2-24.

Table 2-24 Configuration Order Sets

/C/	Configuration	Number of Code Groups	Alternating /C1/ and /C2/
/C1/	Configuration 1	4	/K28.5/D21.5/Config_Reg
/C2/	Configuration 2	4	/K28.5/D2.2/Config_Reg

The /C/ order set is sent or received during auto-negotiation. The link partners exchange information in a 16-bit word known as configuration register (Config_Reg) which is a part of the /C/ order set. The link partners first exchange the Config_Reg Base Page, and then may exchange the Config_Reg Next Page. Table 2-25 describes the bits of the Config_Reg in the SGMII mode and the 1000BASE-X mode. In the 1000BASE-X mode, the values of the Config_Reg bits are taken from the SR_MII_AN_ADV.

Table 2-25 Config_Reg Bits in the 1000BASE-X and SGMII Mode

Config_Reg Bit	1000BASE-X Mode	SGMII ModeValue When TX_CONFIG=0	Value When TX_CONFIG = 1
15	Next page support (NP)	0	Link UP or Link Down. If bit[0] of VR_MII_DIG_CTRL1 is 0, this bit is derived from Bit 4 (SGMII_LINK_STS) of the VR_MII_AN_CTRL. If bit[0] of VR_MII_DIG_CTRL1 is 1, this bit is derived from the input port ‘xpcs_sgmii_link_sts_i’
14	ACK	1	1
13	RF[1]	0	0
12	RF[0]	0	FULL_DUPLEX If bit[0] of VR_MII_DIG_CTRL1 is 0, this bit is derived from Bit 5 (FD) of the SR_MII_AN_ADV. If bit[0] of VR_MII_DIG_CTRL1 is 1, this bit is derived from the input port ‘xpcs_sgmii_full_duplex_i’
11:10	Reserved	0	If bit[0] of VR_MII_DIG_CTRL1 is 0,the value of these bits is derived from Bit 13 (SS13) and Bit 6 (SS6) of the SR_MII_CTRL. If bit[0] of VR_MII_DIG_CTRL1 is 1, this bit is derived from the input port ‘xpcs_sgmii_link_speed_i[1:0]’ ■ 10: 1000 Mbps ■ 01: 100 Mbps ■ 00: 10 Mbps
9	0 (Reserved)	0	0
8	PAUSE[1]	0	0
7	PAUSE[0]	0	0
6	HALF_DUPLEX	0	0
5	FULL_DUPLEX	0	0
4	0 (Reserved)	0	0
3	0 (Reserved)	0	0
2	0 (Reserved)	0	0
1	0 (Reserved)	0	0
0	0 (Reserved)	1	1

Note	If TX_CONFIG=1 and Bit[0] (SGMII_PHY_MODE_CTRL) of VR_MII_DIG_CTRL1 = 0, program the SR_MII_AN_ADV only after programming 'SGMII_LINK_STS' bit (of VR_MII_AN_CTRL) and SS13 and SS6 bits (of SR_MII_CTRL)

2.11.3 Enabling Clause 37 Auto-Negotiation

The Clause 37 auto-negotiation feature is supported only if the DWC_xpcs is configured for 1000BASE-X or 1000BASE-KX speed mode.

The Clause 37 auto-negotiation is available in the following configurations:

- Configurations with SUPPORT_1G = Enabled
- Configurations with MAIN_MODE = Backplane Ethernet PCS and BACKPLANE_ETH_CONFIG =KX_Only, KX4_KX, KR_KX, or KR_KX4_KX
- Configurations with MAIN_MODE = 1000BaseXOnly-PCS

If you select "Mode of Operation" as 1000BASEX-Only PCS during the Specify Configuration activity in coreConsultant, Clause 37 auto-negotiation feature is enabled by default. This is indicated by the Enable Clause 37 auto-negotiation check box.

If you select any of the following as the "Mode of Operation":

- XGXS PCS
- 10GBASE-X PCS
- 10GBASE-R PCS

Then

1. Enable "Add 1G Support (1000BASE-X)".
2. Select "Enable Clause 37 auto-negotiation" check box.

For details about this option, see the "*Parameter Descriptions*" chapter.

Note	See the "Enabled" field of the parameter descriptions to understand the dependencies.

2.11.4 Registers Related to Clause 37 Auto-Negotiation

Following are the registers related to Clause 37 Auto-Negotiation:

- SR_MII_AN_ADV
- VR_MII_AN_CTRL
- VR_MII_AN_INTR_STS
- SR_MII_LP_BABL
- SR_MII_CTRL
- MII_AN_INTR_EN
- VR_MII_DIG_CTRL1
- VR_MII_LINK_TIMER_CTRL

2.11.5 Programming Guidelines for Clause 37 Auto-Negotiation

See the "Programming Guidelines for Clause 37 Auto-Negotiation" on page 1704.

2.12 Serial Gigabit Media Independent Interface (SGMII)

This topic contains the following sections:

- "Overview of Serial Gigabit Media Independent Interface" on page 148
- "SGMII Auto-Reconfiguration" on page 149
- "Enabling Serial Gigabit Media Independent Interface" on page 149
- "Signals Related to Serial Gigabit Media Independent Interface" on page 149
- "Registers Related to Serial Gigabit Media Independent Interface" on page 150
- "Programming Guidelines for the Serial Gigabit Media Independent Interface" on page 150

2.12.1 Overview of Serial Gigabit Media Independent Interface

The Serial Gigabit Media Independent Interface (SGMII) converts the GMII or MII interface, between the Gigabit MAC and the Gigabit PHY, to a serial interface. Using the SGMII interface lowers the pin count required for the GMII interface. However, the reduced pin count may increase the power consumption because the SGMII interface maintains a constant 1250 Mbps data rate, irrespective of the operating speed of the MAC.

The SGMII block has the following features:

- Supports 10 Mbps, 100 Mbps, and 1000 Mbps speed modes.

  The DWC_xpcs uses the RAL blocks (Table 2-26 GMII Tx RAL Block and GMII Rx RAL Block) to convert the data (with different speed and data width) from the MAC to the internal speed of 125 MHz, 8-bit required for the SGMII link speed (1.25 Gbps). Table 2-26 describes the supported speed and data width.

Table 2-26 Supported Speeds for SGMII

Link Speed	GMII or MII Data Width	GMII or MII Clock Rate
10 Mbps	8 MII	1.25 MHz
10 Mbps	4 MII	2.5 MHz
100 Mbps	8 MII	12.5 MHz
100 Mbps	4 MII	25 MHz
1000 Mbps	8 GMII	125 MHz

- Provides status signals (link speed, duplex mode, and link status) from the PHY to the MAC.

  The SGMII interface uses the Clause 37 auto-negotiation to know the speed, duplex mode, and link status negotiated by the PHY. To enable the Clause 37 auto-negotiation in the SGMII mode, you need to program Bits[2:1] of the VR_MII_AN_CTRL to 2'b10. For information about the |C| order set data format used during Clause-37 auto-negotiation in the SGMII mode, see Table 2-25.

Note

- The SGMII interface requires a separate lower-speed clock at the GMII or MII interface along with the 125 MHz internal clock. For information about the clock connections with SGMII interface, see *DesignWare Cores Ethernet PCS User Guide*.
- When operating in 4-bit MII mode, gmii_txd_i[7:4] should be driven to 0.

2.12.2 SGMII Auto-Reconfiguration

DWC_xpcs supports automatic reconfiguration of SGMII speed/duplex mode based on the outcome of auto-negotiation. This feature can be enabled by programming bit[9] (MAC_AUTO_SW) of VR_MII_DIG_CTRL1 when operating in SGMII MAC mode. DWC_xpcs is initially configured in the speed/duplex mode as programmed in the SR_MII_CTRL. If MAC_AUTO_SW bit is enabled, DWC_xpcs automatically switches to the negotiated speed mode after the completion of CL37 Auto-negotiation. This eliminates the software over-head of reading CL37 AN SGMII Status from VR_MII_AN_INTR_STS and then programming SS13 and SS6 speed-selection bits of SR_MII_CTRL appropriately.

When DWC_xpcs is configured in PHY Mode and if bit[0] (PHY_MODE_CTRL) of VR_MII_DIG_CTRL1 is set to 1, DWC_xpcs configures itself to the speed/duplex mode as indicated by xpcs_sgmii_link_speed_i[1:0] and xpcs_sgmii_full_duplex_i inputs (which are provided by the underlying copper PHY). In addition, when there is a change in any of these inputs DWC_xpcs initiates/restarts SGMII auto-negotiation.

The SGMII Link speed, duplex mode, and link status are indicated on the xpcs_link_speed_o, xpcs_sgmii_full_duplex_o, and xpcs_sgmii_link_sts_o output ports respectively. These ports reflect the values as shown in Table 2-27 based on the selected configuration/register settings.

Table 2-27 SGMII Ports and Their Values Based on the Selected Configuration

<table>
<tr><th></th><th>MAC_AUTO_SW</th><th>PHY_MODE_CTRL</th><th>xpcs_link_speed_o</th><th>xpcs_sgmii_link_sts_o</th><th>xpcs_sgmii_full_duplex_o</th></tr>
<tr><td rowspan="2">MAC-SGMII mode (TX_CONFIG=0)</td><td>1</td><td>0</td><td>CL37_ANSGM_STS[2:1]</td><td>CL37_ANSGM_STS[3]</td><td>CL37_ANSGM_STS[0]</td></tr>
<tr><td>0</td><td>0</td><td rowspan="2">ss6, ss13</td><td rowspan="2">SGMII_LINK_STS</td><td rowspan="2">DUPLEX_MODE when SGMII ANEG is disabled else CL37_ANSGM_STS[0]</td></tr>
<tr><td>PHY-SGMII mode (TX_CONFIG=1)</td><td>0</td><td>0</td></tr>
<tr><td></td><td>0</td><td>1</td><td>xpcs_link_speed_i</td><td>xpcs_sgmii_link_sts_i</td><td>xpcs_sgmii_full_duplex_i</td></tr>
</table>

2.12.3 Enabling Serial Gigabit Media Independent Interface

To enable the SGMII mode, during the Specify Configuration activity in coreConsultant:

1. Select 1000BASEX-Only PCS as the "Mode of Operation".
2. Click the "Add SGMII Support" check box.

For details about this option, see the "*Parameter Descriptions*" chapter.

Note See the "Enabled" field of the parameter descriptions to understand the dependencies.

2.12.4 Signals Related to Serial Gigabit Media Independent Interface

- SGMII/QSGMII control and clock interface signals in the "*Signal Descriptions*" chapter.
- SGMII/QSGMII/USXGMII control interface signals in the "*Signal Descriptions*" chapter.

2.12.5 Registers Related to Serial Gigabit Media Independent Interface

Following are the main registers related to SGMII:

- VR_MII_AN_CTRL
- VR_MII_DIG_CTRL1
- SR_MII_CTRL

2.12.6 Programming Guidelines for the Serial Gigabit Media Independent Interface

See "SGMII Auto-Negotiation" on page 1705.

2.13 Quad SGMII (QSGMII)

This topic contains the following sections:

- "Overview of Quad SGMII" on page 151
- "Config_Reg Bits in QSGMII Mode" on page 152
- "Enabling QSGMII Support" on page 156
- "Signals Related to QSGMII Support" on page 157
- "Registers Related to QSGMII Support" on page 157
- "Programming Guidelines for the QSGMII Support" on page 157

2.13.1 Overview of Quad SGMII

Quad Serial Gigabit Media Independent Interface is an extension of SGMII. QSGMII interface transfers data between a 4-port Gigabit MAC (or 4 separate MACs) and a 4-port Gigabit PHY (or 4 separate PHYs) over a serial line operating at 5Gbps. QSGMII eliminates the need to have 4 separate GMII or MII interfaces between the MAC and PHY and reduces the pin count considerably. Each port can operate at a data rate of 10/100/100Mbps.

QSGMII Mode can be enabled in DWC_xpcs by programming bits[2:1] of VR MII MMD AN Control Register to 2'b11. Figure 2-55 the components of QSGMII Transmitter.

Figure 2-55 QSGMII Transmitter Architecture

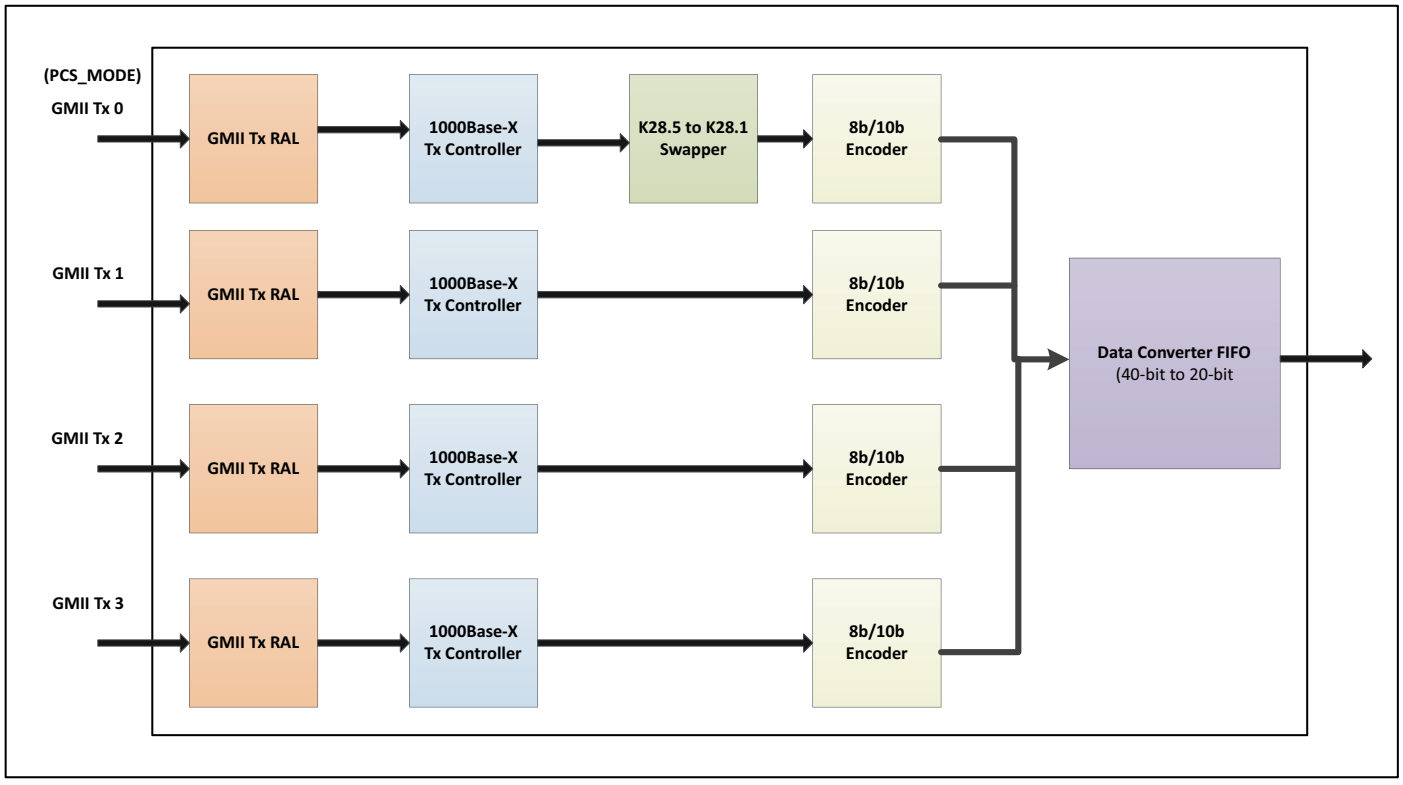
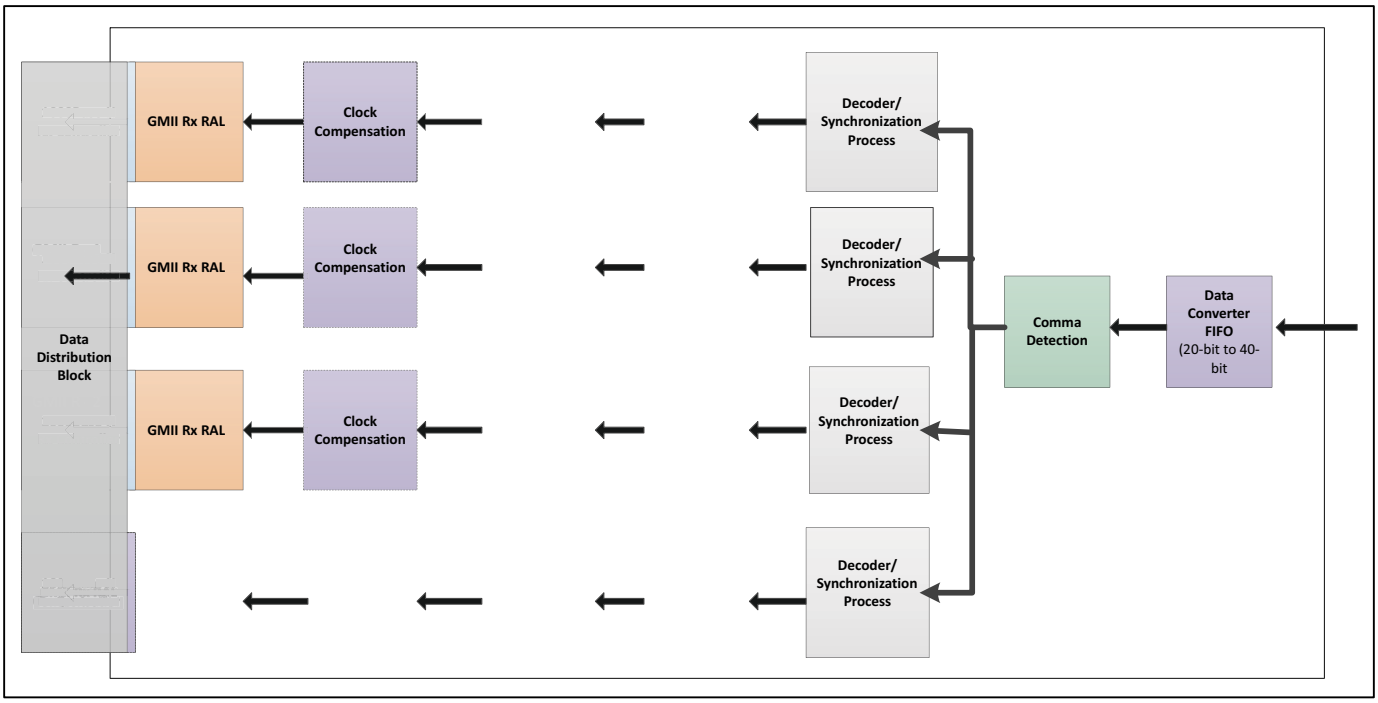


Figure 2-56 displays the components of QSGMII Receiver.

Figure 2-56 QSGMII Receiver Architecture



DWC_xpcs supports the following features in QSGMII mode:

- Supports 10 Mbps, 100 Mbps and 1000 Mbps along each port

  DWC_xpcs uses the RAL blocks at each of the 4 GMII/MII interface to convert the data (with different speed and data width) from the MACs to 8-bit data operating at 125MHz. In the transmit path, the data from the four MAC ports are time-division multiplexed, processed suitably (as defined by the specification) and finally converted to 20-bit data at 250MHz to interface with the SERDES. As per QSGMII specification, K28.5 pattern is replaced with K28.1 pattern in the data sent from port 0 MAC. In the receive path, DWC_xpcs receives 20-bit data at 250MHz from the SERDES. The received data is processed suitably, de-multiplexed and separated-out into 4 separate data traffics. The K28.1 character plays a key role in ensuring that the received data traffic is routed correctly to the intended GMII/MII port.

- Supports Clause 37 auto-negotiation for each port

  Each of the 4 ports can make use of QSGMII auto-negotiation (based on IEEE 802.3- Clause 37) to exchange the speed, duplex mode and the link status with the link partner. Auto-negotiation operation along one port is independent of the other ports. DWC_xpcs can be used either at MAC end of QSGMII link (TX_CONFIG=0) or at the PHY end. The auto-negotiation operation is similar to that of SGMII and involves the exchange of |C| order sets.

2.13.2 Config_Reg Bits in QSGMII Mode

The contents of the Configuration Register (Config_Reg), which is a part of |C| ordered set is selected as per Table 2-28.

Table 2-28 Config_Reg Bits in QSGMII Mode

Config_Reg Bit	Value When TX_CONFIG=0(QSGMII MAC)	Value When TX_CONFIG=1(QSGMII PHY)
15	0	Link Up or Link Down Port 0: If bit [0] of VR MII MMD Digital Control 1 Register is 0, this bit is derived from Bit [4] (SGMII_LINK_STS) of the VR MII MMD AN Control Register. If bit [0] of VR MII MMD Digital Control 1 Register is 1, this bit is derived from the input port 'xpcs_sgmii_link_sts_i'. Port 1: If bit [0] of VR MII_1 MMD Digital Control 1 Register is 0, this bit is derived from bit [4] (QSGMII_LINK_STS) of the VR MII_1 MMD AN Control Register. If bit [0] of VR MII_1 MMD Digital Control 1 Register is 1, this bit is derived from the input port 'xpcs_qsgmii_link_sts_p1_i'. Port 2: If bit [0] of VR MII_2 MMD Digital Control 1 Register is 0, this bit is derived from bit [4] (QSGMII_LINK_STS) of the VR MII_2 MMD AN Control Register. If bit [0] of VR MII_2 MMD Digital Control 1 Register is 1, this bit is derived from the input port 'xpcs_qsgmii_link_sts_p2_i'. Port 3: If bit [0] of VR MII_3 MMD Digital Control 1 Register is 0, this bit is derived from bit [4] (QSGMII_LINK_STS) of the VR MII MMD AN Control Register. If bit [0] of VR MII_3 MMD Digital Control 1 Register is 1, this bit is derived from the input port 'xpcs_qsgmii_link_sts_p3_i'.
14	1(ACK)	1(ACK)
13	0	0

Config_Reg Bit	Value When TX_CONFIG=0(QSGMII MAC)	Value When TX_CONFIG=1(QSGMII PHY)
12	0	Duplex Mode Port 0: If bit [0] of VR MII MMD Digital Control 1 Register is 0, this bit is derived from bit [5] (FD) of the SR MII MMD AN Advertisement Register. If bit [0] of VR MII MMD Digital Control 1 Register is 1, this bit is derived from the input port 'xpcs_sgmii_full_duplex_i'. Port 1: If bit [0] of VR MII_1 MMD Digital Control 1 Register is 0, this bit is derived from bit [5] (FD) of the SR MII_1 MMD AN Advertisement Register. If bit [0] of VR MII_1 MMD Digital Control 1 Register is 1, this bit is derived from the input port 'xpcs_qsgmii_full_duplex_p1_i'. Port 2: If bit [0] of VR MII_2 MMD Digital Control 1 Register is 0, this bit is derived from bit [5] (FD) of the SR MII_2 MMD AN Advertisement Register. If bit [0] of VR MII_2 MMD Digital Control 1 Register is 1, this bit is derived from the input port 'xpcs_qsgmii_full_duplex_p2_i'. Port 3: If bit [0] of VR MII_3 MMD Digital Control 1 Register is 0, this bit is derived from bit [5] (FD) of the SR MII_3 MMD AN Advertisement Register. If bit [0] of VR MII_3 MMD Digital Control 1 Register is 1, this bit is derived from the input port 'xpcs_qsgmii_full_duplex_p3_i'.

Config_Reg Bit	Value When TX_CONFIG=0(QSGMII MAC)	Value When TX_CONFIG=1(QSGMII PHY)
11:10	0	Speed The speed encoding is as follows: ■ 10: 1000Mbps ■ 01: 100Mbps ■ 00: 10Mbps Port 0: If bit [0] of VR MII MMD Digital Control 1 Register is 0, the value of these bits is derived from bit [13] (SS13) and bit [6] (SS6) of the SR MII MMD Control Register. If bit [0] of VR MII MMD Digital Control 1 Register is 1, this bit is derived from the input port 'xpcs_sgmii_link_speed_i[1:0]' Port 1: If bit [0] of VR MII_1 MMD Digital Control 1 Register is 0, the value of these bits is derived from bit [13] (SS13) and bit [6] (SS6) of the SR MII_1 MMD Control Register. If bit [0] of VR MII_1 MMD Digital Control 1 Register is 1, this bit is derived from the input port 'xpcs_qsgmii_link_speed_p1_i[1:0]'. Port 2: If bit [0] of VR MII_2 MMD Digital Control 1 Register is 0, the value of these bits is derived from bit [13] (SS13) and bit [6] (SS6) of the SR MII_2 MMD Control Register. If bit [0] of VR MII_2 MMD Digital Control 1 Register is 1, this bit is derived from the input port 'xpcs_qsgmii_link_speed_p2_i[1:0]'. Port 3: If bit [0] of VR MII_3 MMD Digital Control 1 Register is 0, the value of these bits is derived from bit [13] (SS13) and bit [6] (SS6) of the SR MII_3 MMD Control Register. If bit [0] of VR MII_3 MMD Digital Control 1 Register is 1, this bit is derived from the input port 'xpcs_qsgmii_link_speed_p3_i[1:0]'.
9	0	EEE capability If EEE_EN= Enabled, this bit is 1 else 0.

Config_Reg Bit	Value When TX_CONFIG=0(QSGMII MAC)	Value When TX_CONFIG=1(QSGMII PHY)
8	EEE clock stop enable Port 0: This bit is derived from bit [10] (CS_EN) of VR MII MMD Digital Control 1 Register (for 1000BaseX-Only PCS configurations) or SR XS or PCS MMD Control 1 Register (for other configurations). Port 1: This bit is derived from bit [10] (CS_EN) of VR MII_1 MMD Digital Control 1 Register. Port 2: This bit is derived from bit [10] (CS_EN) of VR MII_2 MMD Digital Control 1 Register. Port 3: This bit is derived from bit [10] (CS_EN) of VR MII_3 MMD Digital Control 1 Register.	EEE clock stop capability If EEE_EN= Enabled, this bit is 1 else 0.
7:1	0	0
0	1	1

2.13.3 Enabling QSGMII Support

To enable QSGMII during the Specify Configuration Activity:

1. Check the Add QSGMII Support check box.

Note

The QSGMII Support check box can be checked in one of the following settings

- Select the Mode of Operation 1000BaseX-Only PCS and do not check the Add SGMII Support check box.
- Select the Mode of Operation Backplane Ethernet PCS and check the Enable Clause 37 Auto-Negotiation check box

For details about this option, see the "*Parameter descriptions*" chapter.

Note See the "Enabled" field of the parameter descriptions to understand the dependencies.

2.13.4 Signals Related to QSGMII Support

- SGMII/QSGMII control and clock interface signals in the "*Signal descriptions*" chapter.
- SGMII/QSGMII/USXGMII control interface signals in the "*Signal descriptions*" chapter.
- QSGMII Control and Clock Interface Signals in the "*Signal descriptions*" chapter.

2.13.5 Registers Related to QSGMII Support

Following are the main registers related to QSGMII support:

- SR_MII_CTRL/SR_MII__CTRL/SR_MII_2_CTRL/SR_MII_3_CTRL
- VR_MII_AN_CTRL/VR_MII_1_AN_CTRL/VR_MII_2_AN_CTRL/VR_MII_3_AN_CTRL
- VR_MII_DIG_CTRL1/VR_MII_1_DIG_CTRL1/VR_MII_2_DIG_CTRL1/VR_MII_3_DIG_CTRL1

2.13.6 Programming Guidelines for the QSGMII Support

See the "Programming Guidelines for QSGMII Mode" on page 1684.

2.14 EEE Controller

This topic contains the following sections:

- "Overview of EEE Controller" on page 158
- "EEE Transmit Controller" on page 158
- "EEE Receive Controller" on page 162
- "Reference EEE Flow" on page 167
- "Enabling EEE Support" on page 169
- "Signals Related to EEE" on page 169
- "Registers Related to EEE" on page 169
- "Programming Guidelines for EEE" on page 170

2.14.1 Overview of EEE Controller

Energy Efficient Ethernet, specified in IEEE Std 802.3az-2010, helps to save energy or reduce power consumption in the PHY when no data transmission or reception is required for a very long duration. The basic principle of this feature is that when the MAC transmitter indicates that it does not have any data to transmit, the local transmitter (which includes the PCS and the PHY) and the remote-end PHY receiver can be put in the power-save or sleep mode to save energy.

2.14.2 EEE Transmit Controller

2.14.2.1 Overview of EEE Transmit Controller

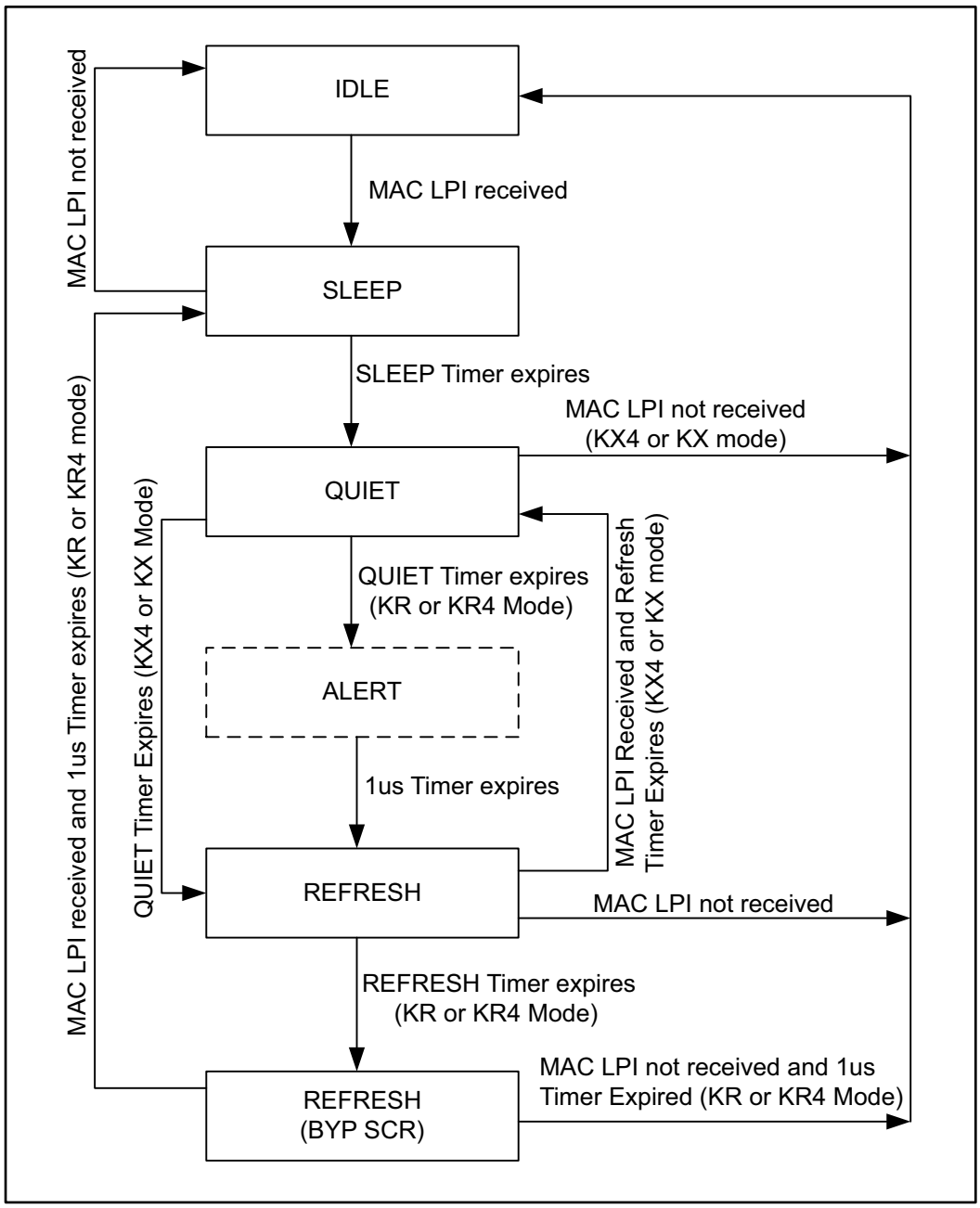
The DWC_xpcs Transmit path implements the following functions when the EEE support is selected:

- Monitor the XGMII or GMII interface for Low Power Idle (LPI) pattern
- Maintain the following EEE states in the Transmit path:
  - Sleep (LPI request monitoring state)
  - Quiet (Power-save state)
  - Alert (Pre-wakeup state) in the PCS-R mode
  - Wakeup (Refresh state)
  - Bypass Scrambler state (in the KR mode to enable FEC Fast Synchronization)

Figure 2-57 shows the simplified version of the Transmit LPI state machine. The Transmit LPI flow is triggered when the MAC drives the LPI signals on the XGMII or GMII interface. The DWC_xpcs maintains various states in which it periodically includes refresh signaling and freezing the transmitter in the quiet state. By maintaining this predefined sequence, the protocol ensures that the link is alive and at the same time power is saved. When the MAC stops sending the LPI pattern (and starts sending the IDLE pattern), the EEE Transmit controller also comes out of this sequence and reverts to the IDLE (or normal transmit) state.

The timers used in the EEE Transmit Controller have the default values given in the IEEE standard. You can change these default values (within a range) according to your requirement.

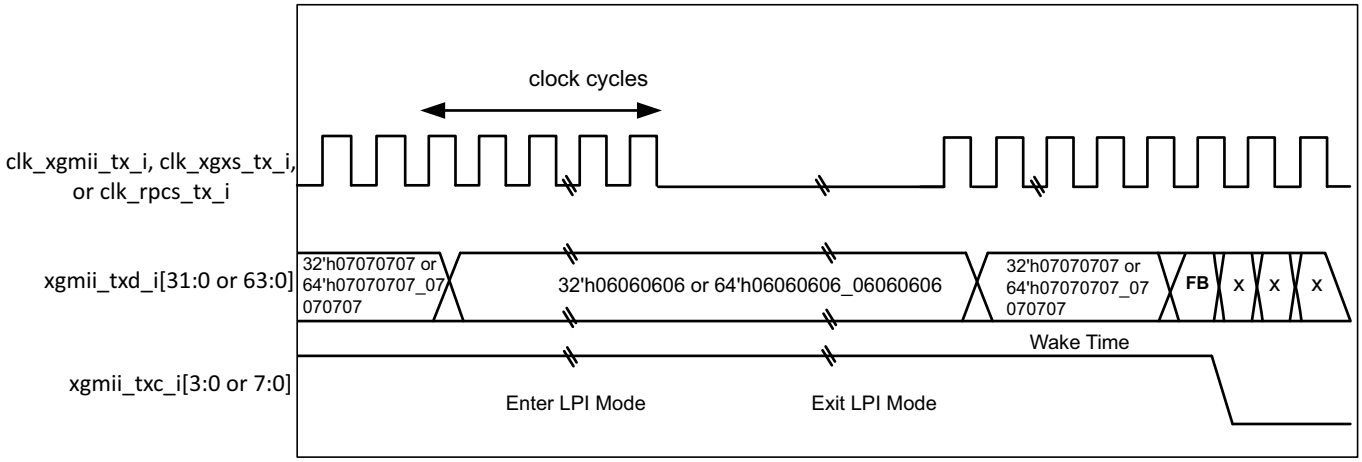
Figure 2-57 Transmit LPI Flow



2.14.2.2 10G MAC Transmit LPI

Figure 2-58 describes the Transmit LPI flow on the XGMII Interface. The MAC triggers the LPI mode in the DWC_xpcs by driving the xgmii_txd_i[31:0 or 63:0] = 32'h06060606 or 64'h06060606_06060606 and xgmii_txc_i[3:0 or 7:0] = 4'hF or 8'hFF.

Figure 2-58 10G MAC Transmit LPI



In configurations in which the clk_xgmii_tx_i clock is used to sample the xgmii_txd_i[] or xDWC_xpcs, the MAC can also gate the clock (clk_xgmii_tx_i) after 128 clocks of asserting the LPI Idle to save additional power in the PCS layer. In such configurations, the XGMII interface is also directly sampled with the reference EEE controller clock (clk_eee_i) to monitor the LPI pattern in the wake-up cycle.In configurations in which the clk_xgxs_i or clk_rpcs_tx_i clocks are used to sample the xgmii_txd_i[] or xgmii_txc_i[] signals, the gating of the MAC transmitter clock does not affect the DWC_xpcs.

The XGMAC can come out of the LPI mode at any time by driving Normal Idle instead of LPI Idle with values xgmii_txd_i[31:0 or 63:0] = 32'h07070707 or 64'h07070707_07070707 and xgmii_txc_i[3:0 or 715:0] = 4'hF or 8'hFF.

After coming out of the LPI mode, the MAC waits for the wake-time period before it starts sending the frames. The wake-time limit depends on various factors such as the following:

- Local transmitter PCS or PHY layer latency in coming to normal IDLE state
- Pre-wakeup timing (ALERT in the case of 10GBASE-R/5GBASE-R)
- Remote Link PHY Latency
- Remote PCS or MAC wake-up requirements

The IEEE standard provides an option to exchange the wake-up time requirements, by using DDLP link layer protocol with the link partner, before any of the nodes goes into the power-save mode.

Note LPI Transmit behavior for 5G MAC and 2.5G MAC (over XGMII) is same as that of 10G MAC.

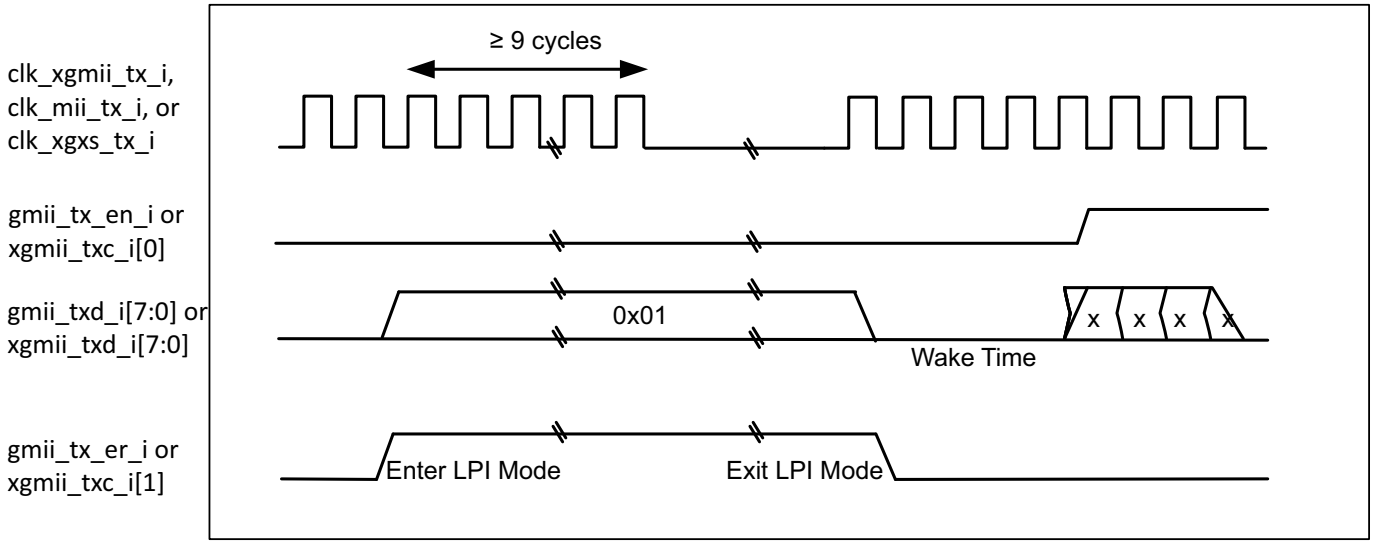
2.14.2.3 1G MAC Transmit LPI

Figure 2-59 describes the Transmit LPI flow on the GMII Interface. The MAC triggers the LPI mode in the DWC_xpcs by driving the following:

- gmii_txd_i[7:0] = 8'h01, gmii_en_i = 0, and gmii_tx_er_i = 1 when GMII port is present in your configuration
- xgmii_txd_i[7:0] = 8'h01, xgmii_txc_i[0] = 0, and xgmii_txc_i[1] = 1 when the GMII interface is shared with the XGMII pins in your configuration

If the clk_xgmii_tx_i or clk_mii_tx_i clock (SGMII mode) is used in the DWC_xpcs to sample GMII, the MAC can gate the clk_xgmii_tx_i or clk_mii_tx_i clock after 9 clocks of asserting the LPI Idle. If the clk_xgxs_i or clk_rpcs_tx_i clock is used to sample GMII, clock gating based on the MAC is not applicable.

Figure 2-59 1G MAC Transmit LPI



Note LPI Transmit behavior for 2.5G MAC (over XGMII) is same as that of 1G MAC.

2.14.2.4 PHY Transmit LPI Control

The DWC_xpcs generates the xpcs_lpitx_quiet_o signal when the Transmit LPI state machine is in the Quiet state. When this signal is asserted, the PHY can save power on the Transmit path. The power-saving strategy in the PHY is implementation dependent. When the Transmit LPI state machine is in the Quiet state, the PHY should freeze the Transmit serial lines or drive these to high-impedance. The PHY can switch off the power (or gate off the clocks) to some of its internal components only if these components can be restored quickly when the xpcs_lpitx_quiet_o signal is de-asserted. For example, you can gate off the PHY Tx Interface clock inputs by using the xpcs_lpitx_quiet_o signal. For more information about these options and their dependencies, see the DesignWare Cores Ethernet PCS User Guide.

When DWC_xpcs is in the Tx Quiet state, it carries out the actions specified in Table 2-29.

Table 2-29 Actions Taken in Tx Quiet State

PHY	Actions Taken in Tx Quiet State
Synopsys Multi-Protocol 6G PHY	Drive xgxs_tx{lane}_en_o[2:0] and xgxs_tx_data_en_o{lane} to zero
Synopsys Multi-Protocol 12G/16G/25G/32G PHY	Drive xgxs_tx_data_en_o{lane} to zero
Non-Synopsys PHY	Drive xpcs_tx_en_o{lane} to zero

Note

- DWC_xpcs has recovery time after it comes out of the LPI mode and before it is ready to send the frames. The worst-case recovery time is equal to the following:

  `4 Clocks of clk_eee_i period + 3 clocks of clk_xgmii_tx_i, clk_xgxs_tx_i, or clk_rpcs_tx_i` (whichever is slower among these three clock)

Note

- DWC_xpcs is ready to transmit the packets once it comes out of the LPI mode. However, the reliable packet transmission over the link occurs only after the WAKE TIMER (maintained in MAC layer as defined in IEEE802.3az standard) expires.
- EEE is not supported when operating in 10Mbps speed mode.
- In USXGMII Mode, DWC_xpcs does not support gating-off of clk_xgmii_tx_i clock when operating in 1Gbps or 100Mbps speed mode.

2.14.2.5 Transparent LPI Mode

In addition to the standard EEE Mode, DWC_xpcs supports the transparent LPI Mode in its transmit path. In this mode, EEE operation involves mere encoding of LPI patterns (received on XGMII/GMII Tx interface) and sending it to the SERDES. On detecting Lower-Power Idle on XGMII/GMII Tx interface, DWC_xpcs goes to the TX_SLEEP state and remains in this state till MAC stops sending LPI. It does not transition to TX_QUIET state. This mode does not involve gating-off of any clocks to DWC_xpcs. In addition, the serdes transmitter is not disabled (xpcs_tx_data_en_o signal remains high). MAC should ensure that it does not gate-off XGMII/GMII Tx clock to DWC_xpcs during this mode of operation.

This mode can be enabled by programming 'TRN_LPI' bit of VR XS or PCS MMD EEE Mode Control 1 Register (or VR MII MMD PCS MMD EEE Mode Control 1 Register for 1000BaseX-Only PCS configurations) in addition to 'LTX_EN' bit of VR XS or PCS MMD EEE Mode Control 0 Register.

2.14.3 EEE Receive Controller

2.14.3.1 Overview of EEE Receive Controller

Note

DWC_xpcs is ready to transmit the packets once it comes out of the LPI mode. However, the reliable packet transmission over the link occurs only after the WAKE TIMER (maintained in MAC layer as defined in IEEE802.3az standard) expires.

The PCS Receive path implements the following functions when the EEE support is selected:

- Decode the received LPI code group from the link partner.
- Maintain the following EEE states in the Receive path:
  - Sleep (LPI request monitoring state)
  - Quiet (Power-Save state)
  - Wake (Refresh state)

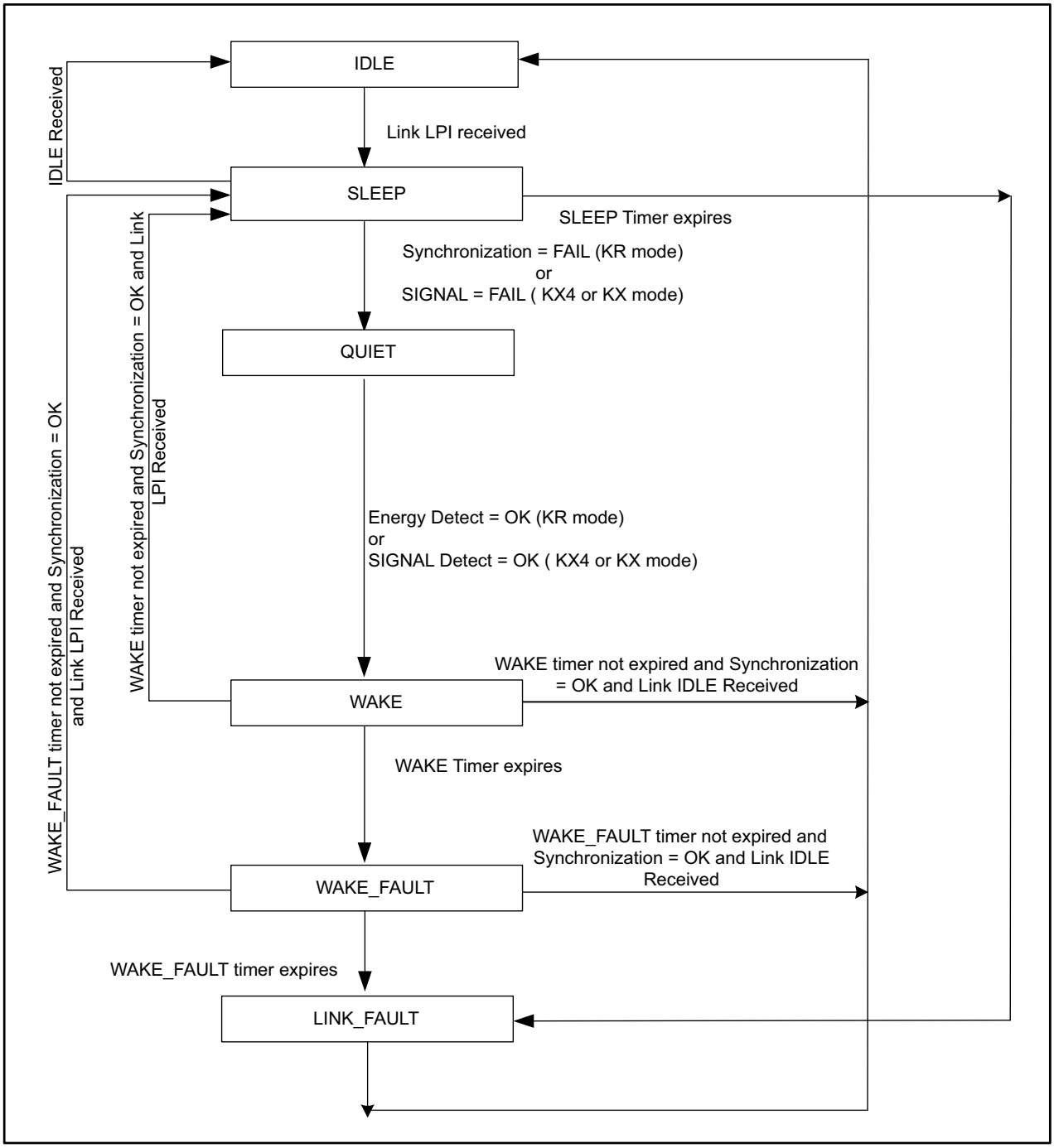
- Wake-up Fault (Error Monitoring)

Figure 2-60 describes the simplified version of the LPI Receive state machine. The sequence in the state machine helps in saving the power in the PHY or PCS because some of the functional blocks can be switched off in the Quiet state. In addition, during Refresh state, the Rx path is re-synchronized so that it can monitor the link status. If FEC is enabled in KR mode, the Rx path also implements the Rapid Block Synchronization to minimize the FEC block synchronization time. The timers used in the EEE Rx Controller have the default values as given in the IEEE standard. You can change these default values (within a range) according to your requirement.

Note

- Loss-of-Signal from Synopsys PHYs might be unreliable in RXAUI mode. Therefore, EEE Rx FSM of DWC_xpcs might misbehave. In such cases, program the EEE_SUPRESS_LOS bit (of VR_XS_PCS_DEBUG_CTRL Register), to prevent the entry of Rx FSM to QUIET state. However, this mode provides only minimal power saving.
- EEE is not supported when operating in 10Mbps speed mode
- In USXGMII Mode, DWC_xpcs does not support gating-off of clk_xgmii_rx_i clock when operating in 1Gbps or 100Mbps speed mode. Hence, CS_EN bit of SR_XS_PCS_CTRL1 Register should be set to 0.
- When operating in 10G/5G/2.5G USXGMII speed modes, xpcs_lrx_clk_gat_o signal (used to gate-off clk_xgmii_rx_i clock) gets asserted on detecting LPI, after a duration approximately equal to 128 clocks of 156.25MHz (if CS_EN bit of SR_XS_PCS_CTRL1 Register is set to 1).

Figure 2-60 Receive LPI Flow



2.14.3.2 PHY LPI Control

The Rx low power mode is triggered when the DWC_xpcs receives the LPI Idle code group on the link. It exits the LPI mode when the link partner stops transmitting the LPI Idle code groups.

The DWC_xpcs enters the Quiet state after successfully detecting LPI from the link partner followed by SIGNAL detect logic which indicates 'No Signal' on the Rx serial line. In normal flow, the DWC_xpcs is in the Quiet state as long as the SIGNAL is not detected on the serial line. The DWC_xpcs exits the Quiet state when SIGNAL is detected (different signal inputs for KX4, KX, and KR modes) and waits for the PCS synchronization before decoding the incoming code groups. If the received code groups are LPIs, the cycle is repeated and the DWC_xpcs again enters the Quiet state. If the incoming code groups are normal Idle, the DWC_xpcs exits the LPI mode.

The DWC_xpcs asserts the xpcs_lpirx_quiet_o signal when the Rx LPI state machine is in the QUIET state. When this signal is asserted, the PHY can save power on its Receive path. The power-saving strategy in the PHY is implementation dependent. For example, the PHY Rx interface clocks can be gated by using the xpcs_lpirx_quiet_o signal. For more information about these options and their dependencies, see the *DesignWare Cores Ethernet PCS User Guide*.

When the Rx LPI is in the Quiet state, the PHY can switch off some of the Rx functional blocks except the SIGNAL detect logic. However, the PHY should do it only if the switched-off blocks can be restored quickly when the xpcs_lpirx_quiet_o signal is de-asserted. Otherwise, the PCS receiver may not get sufficient time to synchronize successfully.

For configurations without Synopsys 12G/16G/25G/32G PHY, when the DWC_xpcs is in the Rx quiet state, it de-asserts the xpcs_rx{lane}_en_o output signal. When the DWC_xpcs comes out of the Rx quiet state, it enables back this signal. For Synopsys Multi-Protocol 6G PHY configurations, DWC_xpcs de-asserts 'xgxs_rx_pll_en_o' output signal also during Rx quiet state. For configurations with Synopsys multi-protocol 12G/16G/25G/32G PHY, DWC_xpcs drives xpcs_rx{lane}_pstate_0 to P0s power state when it is in Rx quiet state. When DWC_xpcs comes out of Rx quiet state, xpcs_rx{lane}_pstate_o is driven to whatever value is programmed in VR_XS_MP_12G_16G_25G_RX_POWER_STATE_CTRL or VR_MII_MP_12G_16G_25G_RX_POWER_STATE_CTRL.

2.14.3.3 10G MAC Receive LPI

Note

DWC_xpcs needs some recovery time after it comes out of the LPI mode and before it is ready to receive the frames.

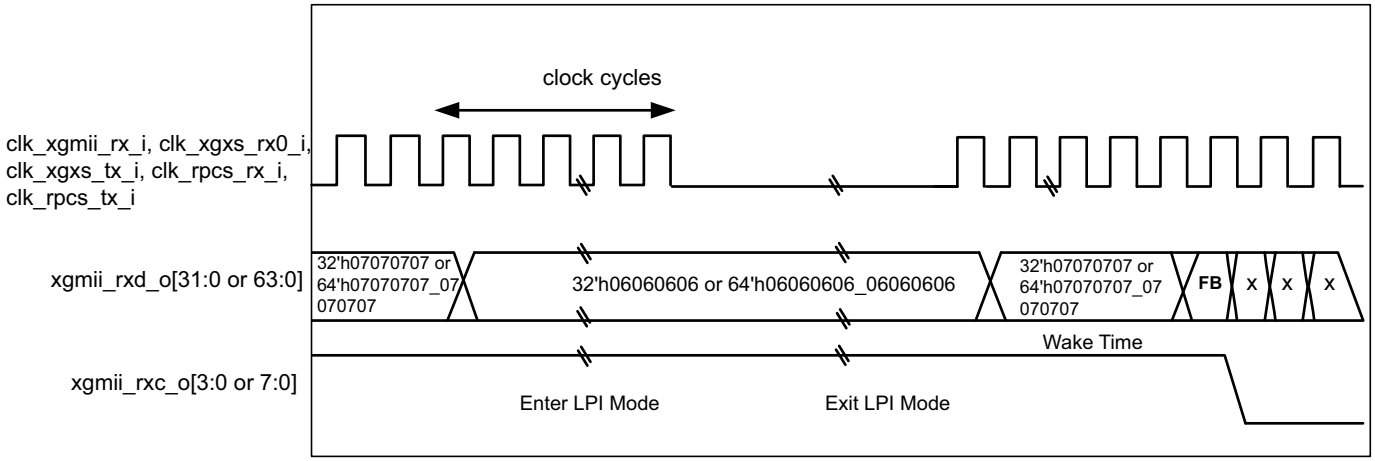
The worst-case recovery time is equal to the following:

11 Clocks of clk_eee_i period + 3 clocks of clk_xgmii_rx_i, clk_xgxs_rx_i, or clk_rpcs_rx_i (whichever is slower among these three clocks).

Figure 2-61 describes the Rx LPI flow on the XGMII Interface. The Rx LPI mode is triggered when the DWC_xpcs receives the LPI from the link partner. After successful decoding, the controller enters the Sleep state and drives the LPI patterns on the Rx XGMII signals as shown in the Figure 2-61 (xgmii_rxd_i[31:0 or 63:0] = 32'h06060606 or 64'h06060606_06060606 and xgmii_rxc_i[3:0 or 715:0] = 4'hF or 8'hFF).

The xpcs_lrx_clk_gat_o signal is asserted after 128 clocks of driving the LPI pattern on the XGMII interface. The xpcs_lrx_clk_gat_o signal can be used to gate the receiver clock towards the MAC and also the clk_xgmii_rx_i clock input of the DWC_xpcs if it is used to drive the XGMII interface in your configuration. The xpcs_lrx_clk_gat_o signal is de-asserted when the DWC_xpcs fails to detect the LPI Idle during the Wake state or when the Sleep timer or the Wake timer expires and the controller enters the Idle or Fault states.

Figure 2-61 10G MAC Receive LPI



Note The 128 clock duration is the default setting. The host can increase this duration by programming Bits[15:12] of VR_XS_PCS_EEE_MCTRL0 Register.

Note LPI Receive behavior for 5G MAC and 2.5G MAC(overXGMII)is same as that of 10G MAC.

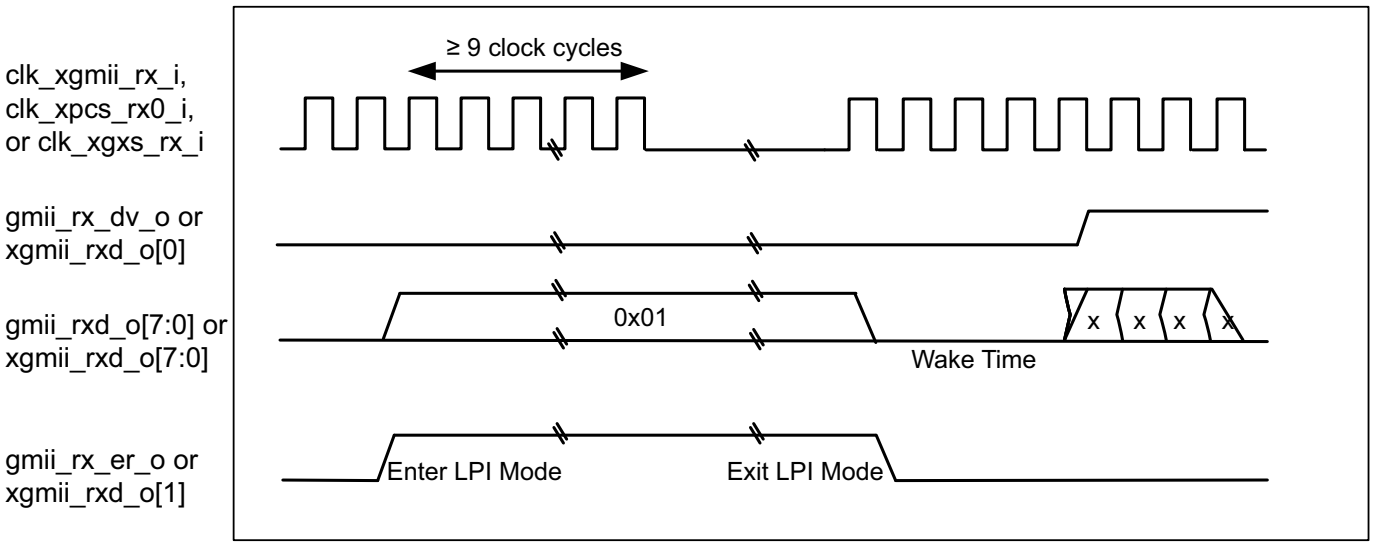
2.14.3.4 1G MAC Receive LPI

Figure 2-62 describes the Rx LPI flow on the GMII Interface. The flow remains the same as explained in "10G MAC Receive LPI" on page 165. The only difference is that the following LPI pattern is driven to the MAC on GMII:

- gmii_rxd_o[7:0] or xgmii_rxd_o[7:0] = 8'h06
- gmii_rx_dv_o or xgmii_rxd_o[0] = 0
- gmii_rx_er_o or xgmii_rxd_o[1] = 1

In this mode, the xpcs_lrx_clk_gat_o signal is asserted after the 9 clocks of driving the LPI pattern on the GMII interface. It can be used to gate off the clocks on the GMII Rx interface on the MAC and the DWC_xpcs. The 9 clock duration can be increased by programming Bits[15:12] of VR_XS_PCS_EEE_MCTRL0 Register VR PCS MMD EEE Mode Control Register.

Figure 2-62 1G MAC Receive LPI



Note LPI Receive behavior for 2.5G MAC (over GMII) is same as that of 1G MAC.

2.14.4 Reference EEE Flow

Figure 2-63 provides a summary of the sequence of events (in time) that occur in the EEE mode in the local PCS Transmitter and remote Receiver (or vice versa). You can use this figure to determine the time margins available in the system for recovery during refresh periods. The details of these latencies (in DWC_xpcs) are given in Table 2-30. The DWC_xpcs Tx sends the LPI IDLE code groups during the Sleep and Refresh periods (except during Tx Alert period in KR mode). During the Quiet period, the PHY transmitter drives the serial lines to HiZ.

Figure 2-63 EEE Flow

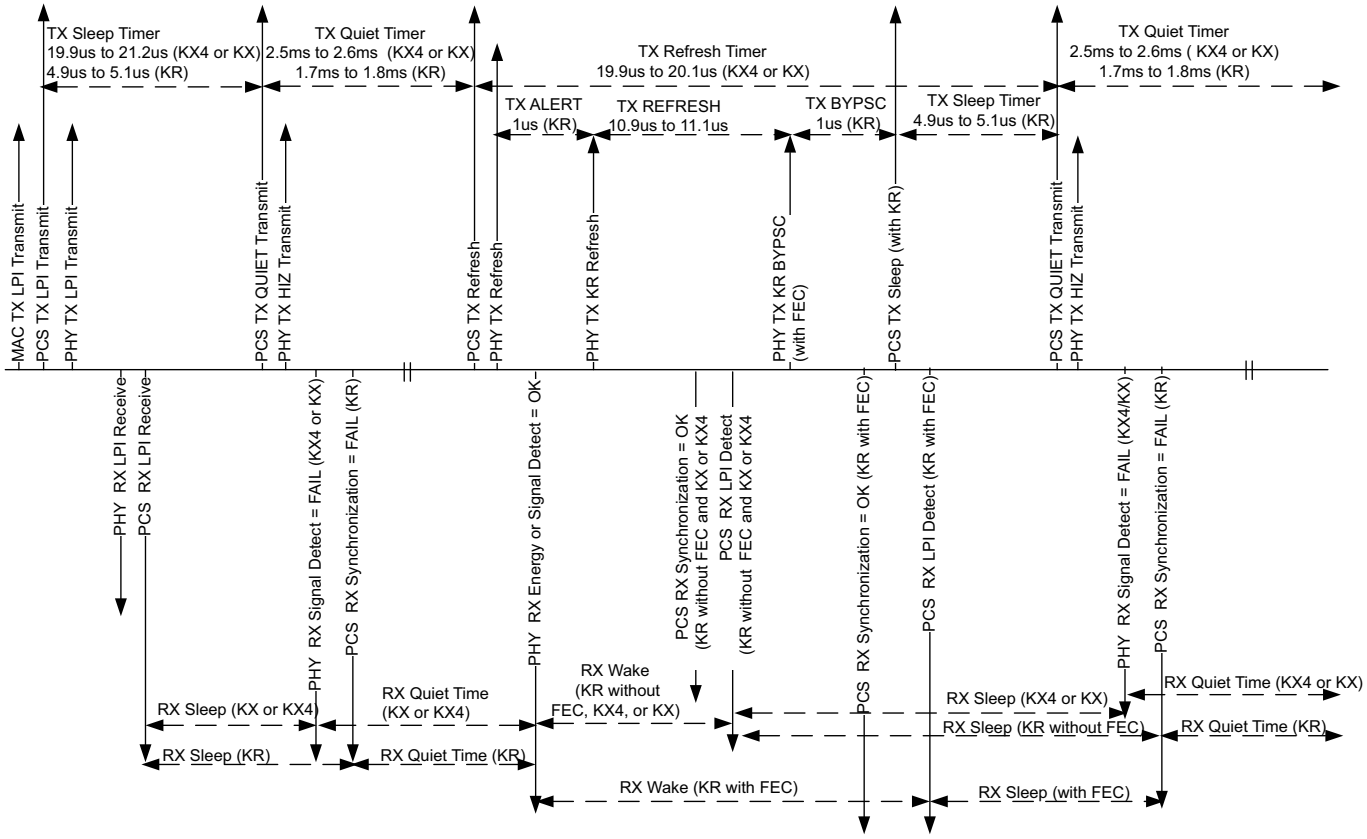


Table 2-30 gives the DWC_xpcs Rx timing. The DWC_xpcs meets all transmitter timings requirements specified in IEEE Std 802.3.

Table 2-30 DWC_xpcs EEE Rx Timing

Parameter	IEEE 802.3 Specification for Transmitter	IEEE 802.3 Specification for Receiver	DWC_xpcs Rx Timing
Sleep Timer	■ KX or KX4 mode: 19.9 µs to 21.1µs ■ KR mode: 4.9 µs to 5.1 µs	■ KX or KX4 mode: 3 ms to 4 ms ■ KR mode: 2 ms to 3 ms Note: This is the maximum time for which the Rx tries to detect the Signal Detect or Synchronization FAIL.	Meets the Rx requirements

Parameter	IEEE 802.3 Specification for Transmitter	IEEE 802.3 Specification for Receiver	DWC_xpcs Rx Timing
Quiet Timer (Power-save mode)	■ KX or KX4 mode: 2.5 ms to 2.6 ms ■ KR mode: 1.7 to 1.8 ms	N/A Rx QUIET time = Tx QUIET Time – (Signal Detect FAIL delay + PCS Synchronization FAIL delay) + Signal Detect OK delay These Rx delays should be as minimum as possible to Save maximum power in Rx path.	Maximum PCS Synchronization FAIL delay assuming incoming data is INVALID. ■ KX mode: 120 ns ■ KX4 mode: 54.4 ns ■ KR without FEC: 230.4 ns ■ KR with FEC: 2124 ns ■ KR with FEC and EF: 2329 ns
Refresh (Refresh period during which the Rx Link should be synchronized)	■ KX or KX4 mode: 19.1 µs to 20.1 µs ■ KR mode: 1 µs ALERT followed by 10.9 µs to 11.1 µs followed by 1µs BYPASS Scrambler	■ KX mode: 11 µs ■ KX4 Mode: 9 µs ■ KR without FEC: 11.5 µs ■ KR with FEC: 13.7 µs The Signal Detect or Energy Detect FAIL delay + PCS Synchronization OK delay should meet this value.	The minimum PCS Synchronization OK delay assuming that the DWC_xpcs received reliable data ■ KX Mode: 56 ns ■ KX4 Mode: 35.2 ns ■ KR without FEC: 454.2 ns ■ KR with FEC: 1100 ns ■ KR with FEC and EF: 1510 ns

2.14.5 Enabling EEE Support

To enable the EEE support during the Specify Configuration Activity in coreConsultant, select the "Add EEE Support" parameter, in the System Configurations Settings window.

For details about this option, see the "*Parameter Descriptions*" chapter.

Note The KX, KX4, and KR modes in this section are also applicable for 1000BASE-X/SGMII, 10GBASE-X/XGXS, and 10GBASE-R configurations, respectively.
In KX4, 10GBASE-X, or XGXS mode, Energy Efficient Ethernet is not supported if any lane is disabled by programming the VR_XS_PCS_DIG_CTRL2 Register.

Note See the "Enabled" field of the parameter descriptions to understand the dependencies.

2.14.6 Signals Related to EEE

See Energy Efficient Ethernet PHY Interface Signals in the "*Signal Descriptions*" chapter.

2.14.7 Registers Related to EEE

- VR_XS_PCS_EEE_MCTRL0 or VR_MII_EEE_MCTRL0
- VR_XS_PCS_EEE_MCTRL1 or VR_MII_EEE_MCTRL1

- VR_XS_PCS_EEE_TXTIMER or VR_MII_EEE_TXTIMER
- VR_XS_PCS_EEE_RXTIMER or VR_MII_EEE_RXTIMER
- SR_XS_PCS_EEE_ABL
- SR_XS_PCS_EEE_WKERR

2.14.8 Programming Guidelines for EEE

See Programming Guidelines for Energy Efficient Ethernet "Programming Guidelines for Energy Efficient Ethernet" on page 1719.

2.15 Power Sequence and Reset Generation

The power sequence and reset generation block performs the following functions:

- Controls and tracks the power-up sequence of the Synopsys PHY.
- Generates the reset to various blocks of the DWC_xpcs.

2.15.1 Power Sequence

The power sequence and reset generation block controls the power-on and power-off sequence of the Synopsys PHY. The PCS logic starts functioning only when the power-up sequence state is in the normal state. This ensures stable Transmit clock propagation and proper data transmission. The DWC_xpcs supports the following power-off modes. You can program both modes by using the CSR Registers. In both the modes, all the internal blocks of the DWC_xpcs are in the reset state, except the management interface block and the CSR block. The difference is only in the control signals driven to the PHY.

- Power Save Mode

  The MPLL clock remains active and the PHY transmitter and receiver are turned off. Powering up the transmit and receive paths from this state can be accomplished without waiting for the MPLL clock to lock.

  DWC_xpcs asserts the

  - xpcs_tx_lpd_o, xpcs_rx_lpd_o output signals for the multi-protocol 12G/16G/25G/32G PHY configurations.
  - xgxs_tx_reset_o/xgxs_rx_reset_o signals for the multi-protocol 6G PHY configurations.

  You can enable the power-save mode by programming Bit 11 of any of the following registers:

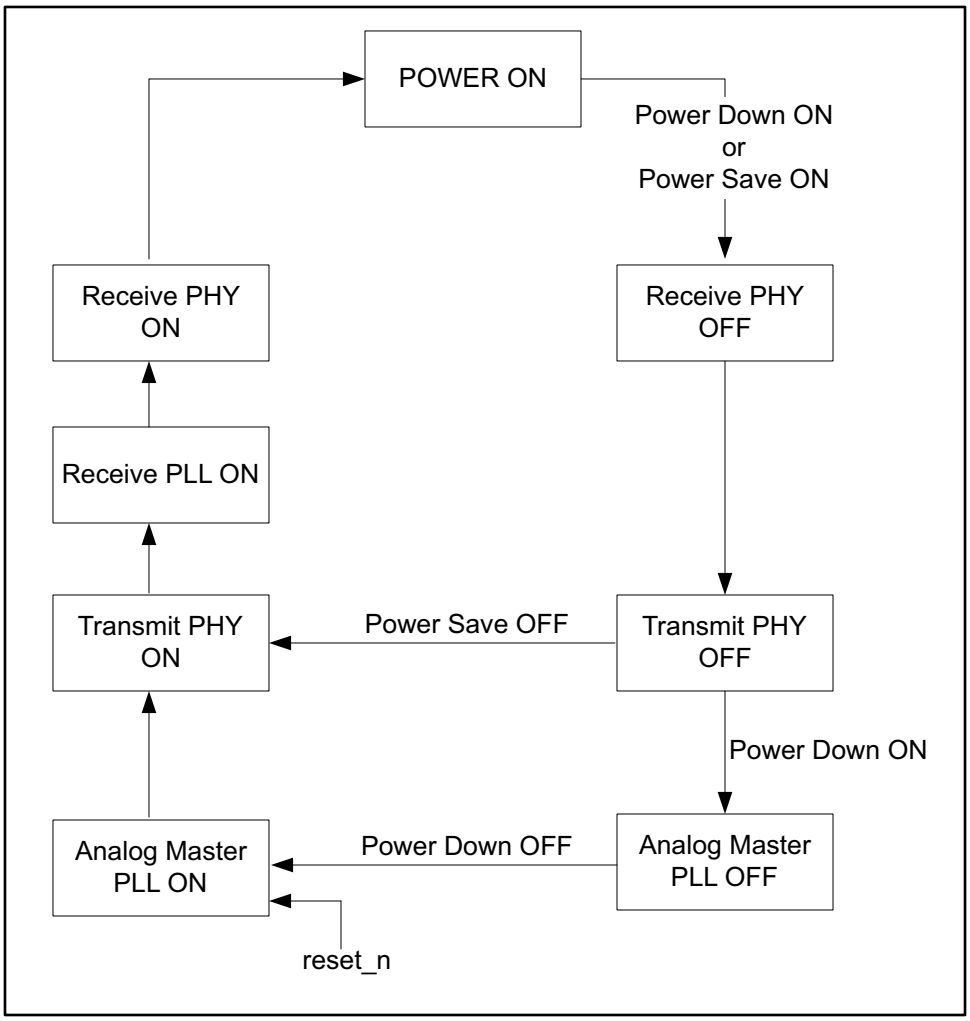
  - VR_PMA_DIG_CTRL1 Register
  - VR_XS_PCS_DIG_CTRL1 Register
  - VR_AN_DIG_CTRL Register
  - VR_MII_DIG_CTRL1 Register
- Power Down Mode:
  - DWC_xpcs asserts the xpcs_tx_reset_o, xpcs_rx_reset_o output signals for the multi-protocol 12G/16G/25G/32G PHY configurations.
  - For the multi-protocol 6G PHY configurations, DWC_xpcs
    - De-asserts the xgxs_mpll_pwr_on_o signal
    - Asserts the xgxs_tx_reset_o and xgxs_rx_reset_o output signals

  You can enable the power-down mode by programming Bit 11 of any of the following registers:

  - SR_PMA_CTRL1
  - SR_XS_PCS_CTRL1
  - SR_AN_CTRL
  - SR_MII_CTRL

Figure 2-64 shows the power-on and power-off sequence.

Figure 2-64 Power ON and OFF Sequence



2.15.2 Reset Generation

The reset generation block generates the reset for other blocks of the DWC_xpcs. The DWC_xpcs supports the following types of reset:

- Hardware Reset

  You can issue the hardware reset by asserting the pwr_on_rst_n signal. When this reset is asserted, all internal blocks of the DWC_xpcs are reset. In addition, the xpcs_reset_n_o signal is asserted to reset the PHY.

- Standard Soft Reset

  You can issue the standard soft reset by setting Bit 15 of any of the following registers:

  - SR_PMA_CTRL1 Register
  - SR_XS_PCS_CTRL1 Register
  - SR_AN_CTRL Register

- SR_MII_CTRL Register

The DWC_xpcs clears this bit after

- 32 clocks cycles of the clk_csr_i clock for MCI or APB3 interface in the non-Synopsys PHY configuration.
- 1 clock cycle of the clk_mdc_i clock for MDIO interface configurations in the non-Synopsys PHY configuration.
- PHY powers up, for the Synopsys PHY configuration.

When this reset bit is set, all internal blocks of the DWC_xpcs, except the Management Interface block, get reset. In addition, the xpcs_reset_n_o signal is asserted to reset the PHY.

- Vendor- Specific Soft Reset

You can issue the vendor-specific soft reset by setting Bit 15 of any of the following registers:

- VR_PMA_DIG_CTRL1 Register
- VR_XS_PCS_DIG_CTRL1 Register
- VR_AN_DIG_CTRL1 Register

The DWC_xpcs clears this bit after

- 32 clocks cycles of the clk_csr_i clock for MCI or APB3 interface in the non-Synopsys PHY configuration.
- 1 clock cycle of the clk_mdc_i clock for MDIO interface configurations in the non-Synopsys PHY configuration.
- PHY powers up, for Synopsys PHY configuration.

When this reset bit is set, all internal blocks of the DWC_xpcs are reset, except the Management Interface block and the CSR block. In addition, the xpcs_reset_n_o signal is asserted to reset the PHY.

- Power-Down Mode

You can issue the power-down mode by setting Bit 11 of any of the following registers:

- SR_PMA_CTRL1 register
- SR_XS_PCS_CTRL1 register
- SR_AN_CTRL register
- SR_MII_CTRL register

When this bit is set, the host needs to clear this bit when it wants to power up the DWC_xpcs. When this bit is set, all internal blocks of the DWC_xpcs are reset, except the Management Interface block and the CSR block. The xpcs_reset_n_o output is not asserted during the power-down mode.

2.16 Control Register Port Controller

This topic has the following sections:

- "Overview of Control Register Port Controller" on page 174
- "Enabling Control Register Port Controller" on page 175
- "Signals Related to Control Register Port Controller" on page 175
- "Registers Related to Control Register Port Controller" on page 175
- "Programming Guidelines for the Control Register Port Controller" on page 175

2.16.1 Overview of Control Register Port Controller

The Control Register (CR) Port Controller block implements the CR Port Interface protocol to access the internal register sets of the Synopsys PHY. This block interfaces with the CSR block of the DWC_xpcs through data, address, read or write, and miscellaneous control signals. Upon request from the CSR block, the CR block generates a read or write transaction on the asynchronous port of the Synopsys PHY.

This block is activated when the Station Management wants to access the internal registers set of the Synopsys PHY through the following registers:

- VR_XS_PMA_SNPS_CR_CTRL or VR_MII_SNPS_CR_CTRL Register
- VR_XS_PMA_SNPS_CR_ADDR or VR_MII_SNPS_CR_ADDR Register
- VR_XS_PMA_SNPS_CR_DATA or VR_MII_SNPS_CR_DATA Register

The CR access protocols are defined in the following databooks:

- Synopsys DesignWare Cores® XAUI PHY for TSMC 40-nm LP 1.1/2.5 V Databook
- DesignWare® Cores Enterprise 12G PHY for TSMC 16FFLL Databook, February 2014
- DesignWare® Cores Enterprise 12G PHY for GF LPP Databook, November 2014
- DesignWare® Cores Enterprise 12G PHY for TSMC 28 HPC Databook, February 2016
- DesignWare Cores Enterprise 16G PHY for TSMC 16FFPGL Databook, April 2016
- DesignWare Cores multi-protocol 25G PHY for GF 14LPP Databook, February 2018
- DesignWare Cores multi-protocol 32G PHY for TSMC 7FF Databook, June 2019

The programming guidelines for CR port read and write operations are covered in "Programming Guidelines for CR Port Read or Write Operation" on page 1702. The following list provides a brief overview of the read or write transfer protocol specific to Synopsys PHYs on the CR port:

Write Protocol for Synopsys Multi-Protocol 6G PHY

1. Supply the address on xgxs_cr_data_o.
2. Assert xgxs_cr_cap_addr _o until xgxs_cr_ack_i is asserted.
3. Wait for xgxs_cr_ack_i to de-assert.
4. Supply the data on xgxs_cr_data_o.
5. Assert xgxs_cr_cap_data_o until xgxs_cr_ack_i is asserted.
6. Wait for xgxs_cr_ack_i to de-assert.
7. Assert xgxs_cr_write_o until xgxs_cr_ack_i is asserted.

8. Wait for xgxs_cr_ack_i to de-assert.

Read Protocol for Synopsys Multi-Protocol 6G PHY

1. Supply the address on xgxs_cr_data_o.
2. Assert xgxs_cr_cap_addr_o until xgxs_cr_ack_i is asserted.
3. Wait for xgxs_cr_ack_i to de-assert.
4. Assert xgxs_cr_read_o.
5. When xgxs_cr_ack_i is asserted, sample the data from xgxs_cr_data_i.
6. De-assert xgxs_cr_read_o and wait for xgxs_cr_ack_i to de-assert.

Write protocol for Synopsys Multi-Protocol 12G/16G/25G/32G PHY

1. Supply the register address on xgxs_cr_reg_address_o
2. Supply the data on xgxs_cr_data_o
3. Assert xgxs_cr_write_o (for one clock cycle) to initiate a write transfer to the PHY register.
4. Wait for xgxs_cr_ack_i to assert. This indicates the completion of write operation.

Read protocol for Synopsys Multi-Protocol 12G/16G/25G/32G PHY

1. Supply the register address on xgxs_cr_reg_address_o
2. Assert xgxs_cr_read_o (for one clock cycle) to initiate a read transfer to the PHY register.
3. Sample the data on 'xgxs_cr_data_i' when xgxs_cr_ack_i is asserted.

The format of the CR data for all registers is compatible with the Synopsys PHY specification.

2.16.2 Enabling Control Register Port Controller

If you select any of the Synopsys PHY, this interface gets added.

2.16.3 Signals Related to Control Register Port Controller

The main signals of the Control Register Port Controller are listed in the section “Synopsys PHY CR Port Interface Signals” in the “*Signal Descriptions*” chapter.

2.16.4 Registers Related to Control Register Port Controller

- VR_XS_PMA_SNPS_CR_CTRL or VR_MII_SNPS_CR_CTRL
- VR_XS_PMA_SNPS_CR_ADDR or VR_MII_SNPS_CR_ADDR
- VR_XS_PMA_SNPS_CR_DATA or VR_MII_SNPS_CR_DATA

2.16.5 Programming Guidelines for the Control Register Port Controller

See“Programming Guidelines for CR Port Read or Write Operation” on page 1702.

2.17 Loopback

DWC_xpcs supports two kinds of loopback operations:

- Tx-to-Rx Serial Loopback in the PHY

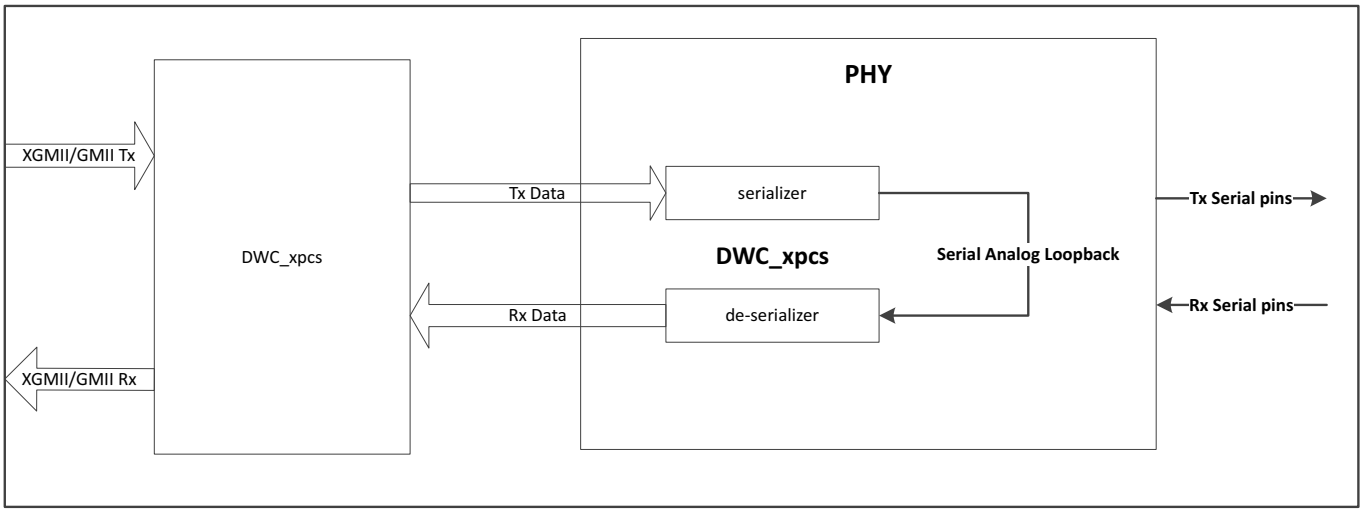
  This loopback operation takes place inside the PHY (external to DWC_xpcs). During this mode, PHY loops back the data from its analog transmitter to its receiver. This is a serial loopback operation.

  This mode can be enabled by programming any of the following register bits:

  - Bit [0] of SR PMA MMD Control 1 Register
  - Bit [14] of SR XS or PCS MMD Control 1 Register
  - Bit [14] of SR MII MMD Control Register

  DWC_xpcs asserts the 'xpcs_loopback_en_o' output port in this mode. Use this signal to enable the loopback operation in the PHY.

Figure 2-65 Tx-to-Rx Serial Loopback in the PHY



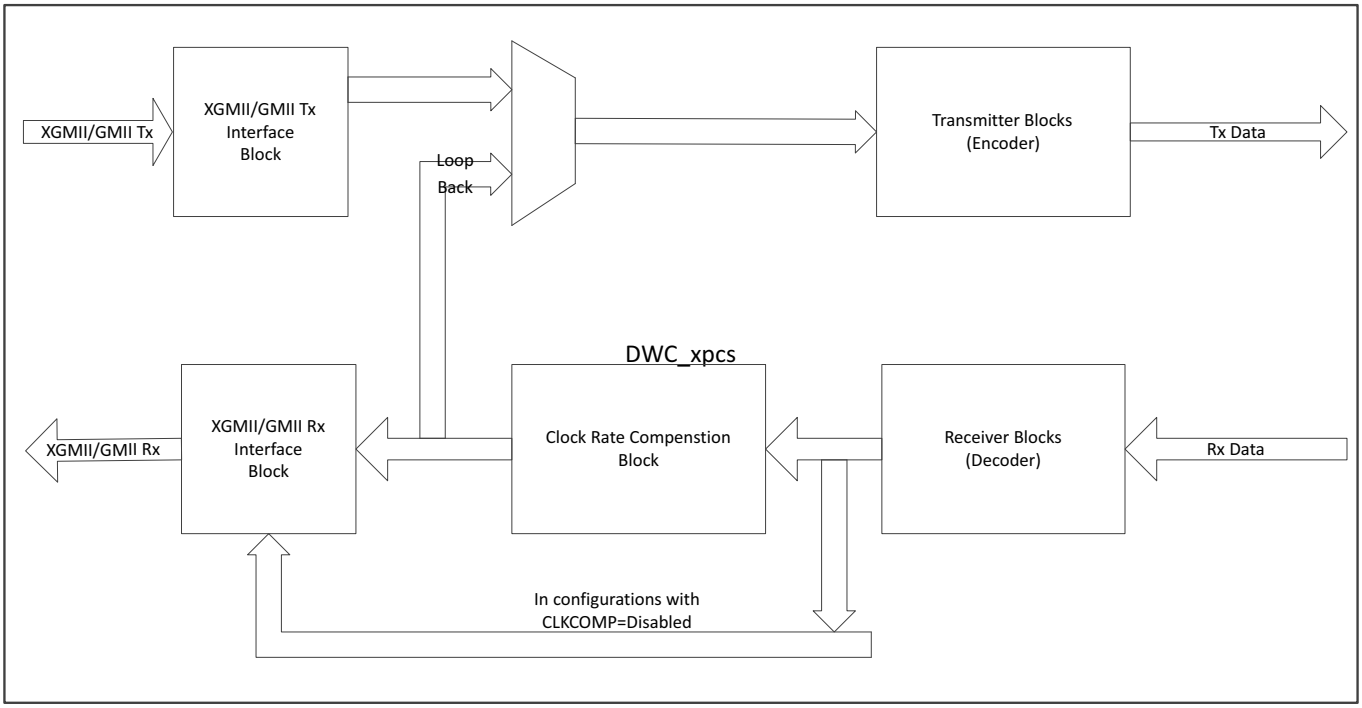
- Rx-to-Tx parallel Loopback in DWC_xpcs

  This loopback operation takes place internal to DWC_xpcs at the XGMII/GMII interface Level.

  During this mode, DWC_xpcs receiver processes the data received from link-partner and converts it into XGMII/GMII format. This data is loop-backed to the DWC_xpcs transmitter internally.

  This mode can be enabled by programming 'R2TLBE' bit of VR XS or PCS MMD Digital Control 1 Register or VR MII MMD Digital Control 1 Register. The clock-compensation FIFO gets used in this loopback path as the transmitter and receiver are operating at different clocks.

Figure 2-66 Rx-to-Tx Parallel Loopback in DWC_xpcs



2.18 Timing Diagrams

This section contains timing information for XGMII in SDR, DDR, and DDW configurations, as well as GMII timing through XGMII signals.

2.18.1 XGMII SDR

The XGMII waveform in SDR configuration is shown in Figure 2-67. It has 32-bit data and 4-control bits and clock frequency is 312.5 MHz.

Figure 2-67 illustrates start of the normal frame transmission. The XGMII IDLE (with xgmii_txc_i[] = 1 and xgmii_txd_i[] = 8'h07) followed by SOF (xgmii_txc_i[0] = 1 with xgmii_txd_i[7:0] = 8'hFB), and then the frame DATA (with xgmii_txc_i[] = 0 and xgmii_txd_i[] = Data).

Figure 2-67 Start of Normal Frame Transmission

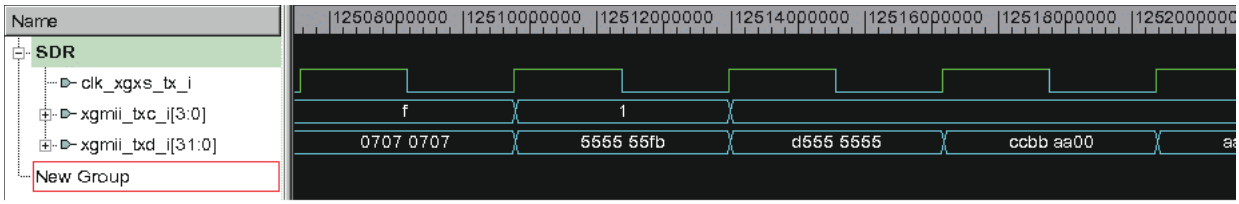


Figure 2-68 illustrates end of the normal frame transmission. The XGMII frame DATA (with xgmii_txc_i[] = 0 and xgmii_txd_i[] = Data), followed by XGMII EOF(xgmii_txc_i[2] = 1 and xgmii_txd_i[23:16] = 8'hFD), followed in turn by XGMII IDLE (with xgmii_txc_i[] = 1 and xgmii_txd_i[] = 8'h07). The EOF (End of Frame) can be inserted on any of the four lanes. In Figure 2-68, it is inserted on lane 2.

Figure 2-68 End of Normal Frame Transmission

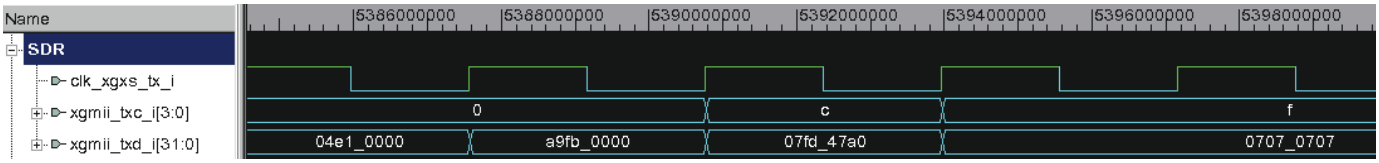
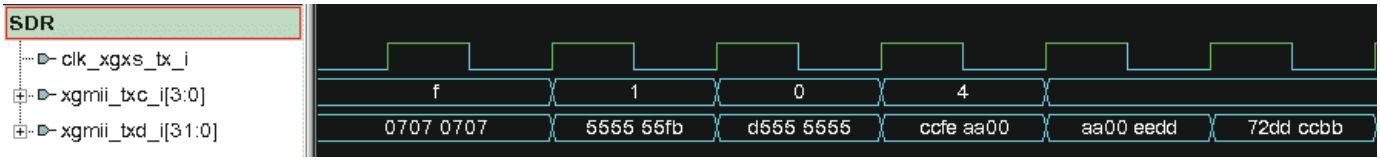


Figure 2-69 shows how an ERROR character can be inserted into a frame that is being sent. The ERROR character can be inserted by setting xgmii_txc_i[] = 1 with xgmii_txd_i[7:0] = 8'hFE. Similarly, any of the other XGMII control characters can be inserted in between Frame/Idle transmission.The reception is similar to that of transmission on XGMII Rx interface (xgmii_rxc_o[3:0] and xgmii_rxd_o[31:0]).

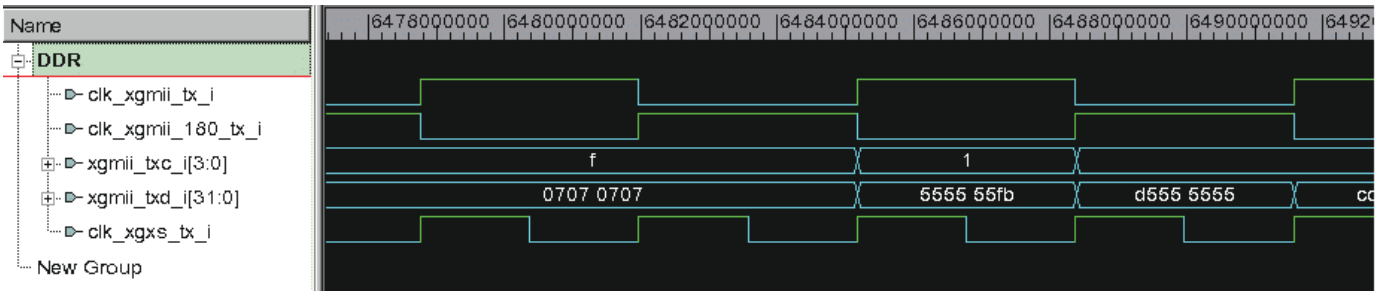
Figure 2-69 ERROR Character Insertion During Frame Transmission



2.18.2 XGMII DDR

The XGMII waveform in DDR configuration is shown in Figure 2-70. It has 32-bit data and 4-control bits and the clock frequency is 156.25 MHz but sampled or driven with two clocks which are 180 degrees phase shifted to each other (Dual Data Rate).

Figure 2-70 XGMII Waveform in DDR Configuration



The data is sampled at the rising edge of both clocks (clk_xgmii_tx_i and clk_xgmii_180_tx_i) and is processed in the same order in which it is sampled. In Figure 2-70, the data xgmii_txd_i[31:0] = 32'h555555FB and xgmii_txc_i[3:0] = 4'h1 is sampled at the rising edge of clk_xgmii_tx_i and data xgmii_txd_i[31:0] = 32'h555555d5 and xgmii_txc_i[3:0] = 4'h0 are sampled at the rising edge of clk_xgmii_180_tx_i. These are processed inside the DWC_xpcs in the same order.

The reception is similar to transmission on XGMII Rx interface (xgmii_rxc_o[3:0] and xgmii_rxd_o[31:0]) driven using two clocks (clk_xgmii_rx_i and clk_xgmii_180_rx_i).

2.18.3 XGMII DDW

Figure 2-71 shows the XGMII waveform for DDW configuration. It has 64-bit data, 8 control bits, and 156.25 MHz clock frequency.

Note Frame transmission or reception can be started with any of the upper or lower 32-bit boundaries. The following diagrams show how the frame transmission or reception can start.

Figure 2-71 XGMII Waveform in DDW Configuration

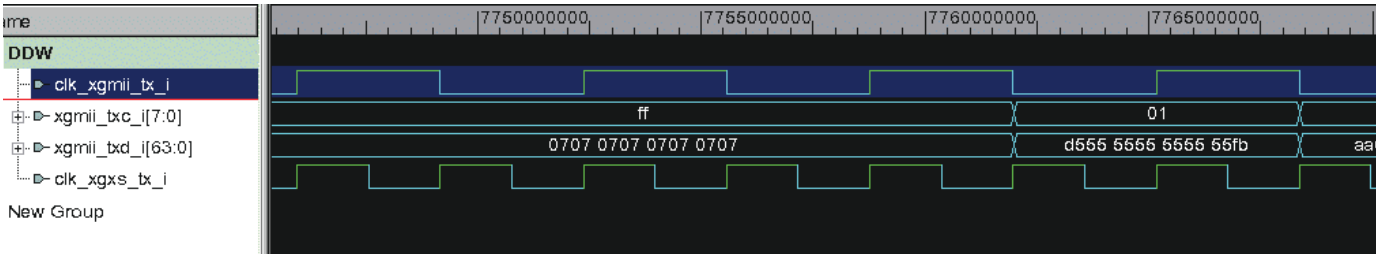
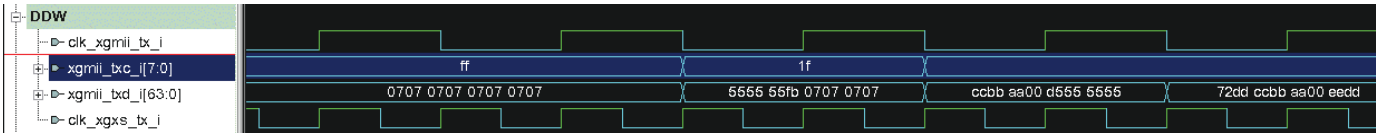


Figure 2-72 shows that the frame transmission started with the lower 32 bits of the xgmii_txd_i[63:0]. The data is sampled using the positive edge of the clk_xgmii_tx_i clock, and converted into two 32-bit patterns and processed internally with the lower 32 bits as the first pattern, and the upper 32 bits as the second.

Figure 2-72 shows how the frame transmission can be started with upper 32 bits of the XGMII interface.

Note The reception is similar to that of transmission. The reception of a frame can also start with either lower or upper 32-bit of the XGMII Rx interface.

Figure 2-72 Frame Transmission Starting with the Upper 32 XGMII Bits



2.18.4 GMII Interface Through XGMII Signals

When the DWC_xpcs is configured for KX4_KX with the ADD_GMII option not selected, the same XGMII interface is used in KX (1G) mode of operation for transmission/reception as is used in a GMII. Note that the XGMII clock is 125 MHz in KX (1G) mode.

Figure 2-73 XGMII Used as a GMII

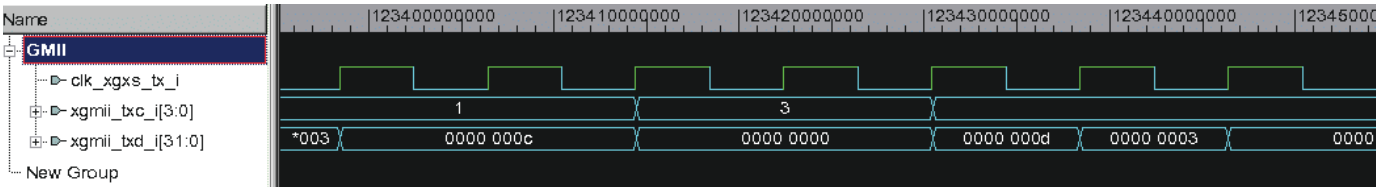


Figure 2-73 shows how the XGMII interface is used as a GMII interface. The XGMII or GMII interface mapping is done as follows:

- xgmii_txd_i[[7:0] = gmii_txd[7:0]
- xgmii_txc_i[0] = gmii_tx_en_i
- xgmii_txc_i[1] = gmii_tx_er_i
- xgmii_txc_i[3:2] = Not used in 1G mode

Figure 2-73 shows transmission of IDLE followed by frame SFD and frame data. Figure 2-74 shows how errors can be inserted in the frame by asserting xgmii_txc_i[1] (which is equivalent to gmii_tx_er_i).

Figure 2-74 Error Insertion in Frame



Reception is similar to transmission, with the following XGMII or GMII mapping:

- xgmii_rxd_o[7:0] = gmii_rxd_o[7:0]
- xgmii_rxc_o[0] = gmii_rx_dv_o
- xgmii_rxc_o[1] = gmii_rx_er_o

2.18.5 GMII Interface

The GMII interface is present in the following configurations:The GMII interface is used when the DWC_xpcs is programmed to work in KX (1G) mode.Figure 2-75 shows frame transmission on GMII interface. The first sequence of data (8'h55) is a preamble, and is followed by SFD(8'hD5) and frame data.

- KX_Only
- 1000BASEX-Only PCS
- KX4_KX and 1G Support configurations with a separate GMII port (ADD_GMII = Enabled).

Figure 2-75 Frame Transmission on GMII

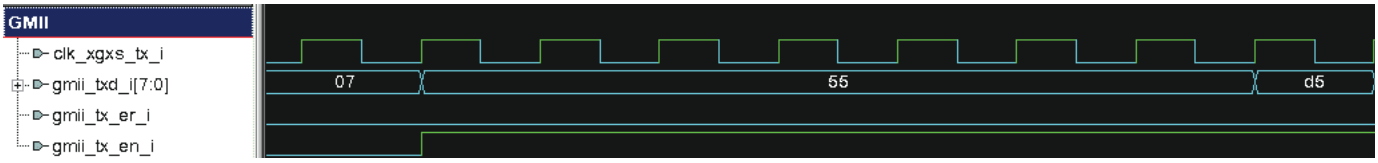
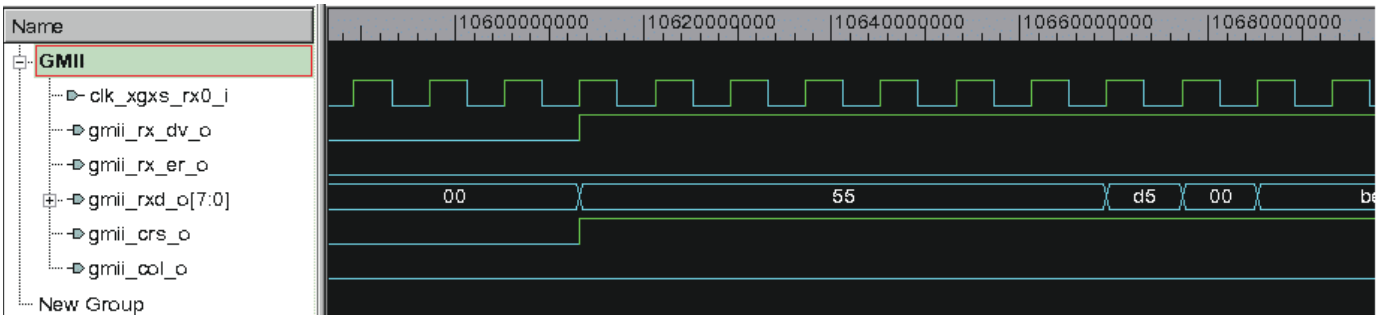


Figure 2-76 shows frame reception on GMII interface. The first sequence of data (8'h55) is preamble, followed by SFD (8'hD5) and frame data.

Figure 2-76 GMII Frame Reception



3

Registers Overview

This section describes the register access types and the writing conventions used in field descriptions. The DWC_xpcs implements the standard register sets for the different MDIO Manageable device as defined in the IEEE Std 802.3, Clause 45. The DWC_xpcs additionally includes vendor-specific register sets.:

- "Special Case Register Reset" on page 184
- "Read or Write During Software Reset" on page 184
- "Conventions" on page 186
- "Device Address Map for MMDs" on page 187

3.1 Special Case Register Reset

This section describes what happens when a register is accessed during a software-triggered reset, or when an address is not implemented or supported.

3.1.1 Read or Write During Software Reset

The software reset period is the duration between the following:

- The time at which the host resets any of the soft reset bits. The following bits can trigger a soft reset:
  - Bit 15 (RST) of the SR_PMA_CTRL1, SR_XS_PCS_CTRL1, SR_AN_CTRL, or SR_MII_CTRL Register.
  - Bit 15 (VR_RST) of the VR_PMA_DIG_CTRL1, VR_XS_PCS_DIG_CTRL1, VR_AN_VR_DIG_CTRL, or VR_MII_DIG_CTRL1 Register.
- The time at which the soft reset bit gets self-cleared after completion of reset in all clock domains.

Note When the host sets any soft reset bit to 1 to initiate reset, the host must wait till the DWC_xpcs clears the bit that was set (resets bit to 0).

During a software reset period, the following is the behavior:

- Writes to any register are ignored.
- Reads to any register return the following:
  - If SNPS_XAUI_PHY = Enabled, the following values are returned:
    - Bits[4:2] of any register indicate Bits[4:2] of VR_XS_PCS_DIG_STS Register.
    - Bit 15 of any register returns 1.
  - If SNPS_XAUI_PHY = Disabled, reads to any register return 8000H.

Preferably, do not program the RST or VR_RST bits while LPI patterns are exchanged.

3.1.2 Read or Write During Power-Down or Power-Save Mode

To trigger

- Power-down mode, program Bit 11 (LPM) of the SR_PMA_CTRL1, SR_XS_PCS_CTRL1, SR_AN_CTRL, or SR_MII_CTRL register to 1.
- Power-save mode, program Bit 11 (PWRSV) of the VR_PMA_DIG_CTRL1, VR_XS_PCS_DIG_CTRL1, VR_AN_VR_DIG_CTRL, or VR_MII_DIG_CTRL1 register to 1.

Note To initiate the power-down or power-save mode, the host must set one of these register bits to 1. When the host needs to power up the DWC_xpcs, it must clear the bit that was set (reset bit to 0).

In the power-down or power-save mode, the behavior is as follows:

- Write operations to any register are ignored.
- Read operations to any register return the following:

- If SNPS_XAUI_PHY = Enabled, the following values are returned:
  - Bits[4:2] of any register indicate Bits[4:2] of VR_XS_PCS_DIG_STS Register.
  - Bit 15 of any register returns 1.
- If SNPS_XAUI_PHY = Disabled, reads to any register return an undefined value.

Preferably, do not program the LPM or PWRSV bits while LPI patterns are exchanged.

3.2 Read or Write of Unsupported Addresses

If an address is not supported in the current configuration, the following happens:

- Writes are ignored
- Reads return 16'h0000

3.3 Special Case Register Access for Multi-Protocol 12G/16G/25G/32G PHY

DWC_xpcs generally does not allow read or write to its registers till the reset/power-up sequence (during hard-reset/soft-reset/power-down/power-save) is complete. However, for configurations with Synopsys Multi-Protocol 12G/16G/25G/32G PHY, the following registers are allowed to be accessed even if the reset/power-up sequence is in progress:

- VR_XS_PMA_SNPS_CR_CTRL or VR_MII_SNPS_CR_CTRL
- VR_XS_PMA_SNPS_CR_ADDR or VR_MII_SNPS_CR_ADDR
- VR_XS_PMA_SNPS_CR_DATA or VR_MII_SNPS_CR_DATA
- VR_XS_PMA_MP_12G_16G_25G_SRAM or VR_MII_MP_12G_16G_25G_SRAM

3.4 Other Special Case Register Accesses

Following are the other special case register accesses. Do not access these registers or register bits, until Rx clocks are available, specifically when Rx EEE mode is active.

- Error-count registers.
- INIT and USRA_RST bits of the VR_XS_PCS_DIG_CTRL register.

3.5 Conventions

This section describes the conventions used in register and bit-field descriptions.

3.5.1 Register Types

Registers are grouped into the categories shown in Table 3-1

Table 3-1 Register Types

Prefix	Description
SR	Standard register set for XS, PCS, PMS, or PMDMMD.
VR	Vendor-specific register set
AN	Standard AN MMD register set
VR AN	Vendor-specific AN MMD register set

3.5.2 Register Access

Table 3-2 displays the access types used in the description of register fields.

Table 3-2 Access Types

Access	Description
LH	Latched High bit. The default value is 0. The DWC_xpcs sets the bit to 1 based on the described event. When this bit is set, it is not changed until the host reads the register.After the host reads the register, this bit indicates the current status of the event.
LL	Latched Low bit. The default value is 1. The DWC_xpcs clears the bit to 0 based on the described event. When this bit is cleared, it does not change until the host reads the register. After the host reads the register, this bit indicates the current status of the event.
RO	Read Only bit
RW	Read/Write bit
SC	Self Clearing bit. The host sets this bit.TheDWC_xpcsclearsthis bit on its own.
SS	Self Setting bit. The DWC_xpcs sets this bit.In some cases, the DWC_xpcs clears it based on the functional requirements.
WS	Write Set. The host can set this bit only duringwriteoperations.Writeof 0 is ignored.
COR	Clear on Read bit. This bit is cleared when the host reads the register containing this bit.



- In KX_Only and 1000BASEX-Only PCS mode, only the register fields corresponding to Lane 0 are present. In addition, the register fields corresponding to Lane 1, 2, and 3 are reserved.
- Synopsys PHY-specific registers are present in MII MMD only for 1000BaseX-Only PCS configurations.

Table 3-3 Device Address Map for MMDs

Device Address	MMD	Implemented Configuration
00	Reserved	N/A
01	PMA or PMD	■ KX4_KX, KX4_Only, KX_Only, KR_KX4_KX, KR_KX4, KR_KX, and KR_Only ■ 10GBASE-X PCS with Synopsys PHY ■ 10GBASE-R PCS with Synopsys PHY
02	WIS	N/A
03	PCS	■ KX4_KX, KX4_Only, KX_Only, KR_KX4_KX, KR_KX4, KR_KX, and KR_Only ■ 10GBASE-X PCS ■ 10GBASE-R PCS
04	PHY XS	MAIN_MODE = XGXS PCS and DTE_PHY_SEL = XGXS in PHY
05	DTE XS	MAIN_MODE = XGXS PCS and DTE_PHY_SEL = XGXS in DTE
06	TC	N/A
07	Auto-negotiation (AN)	KX4_KX, KX4_Only, KX_Only, KR_KX4_KX, KR_KX4, KR_KX, and KR_Only
08-10	Reserved	N/A
1A	Vendor-specific MII_1 MMD	QSGMII or multi-protocol USXGMII configurations Note: This MMD is accessible only after switching to 1G or USXGMII mode
1B	Vendor-specific MII_2 MMD	QSGMII or Quad-port USXGMII configurations Note: This MMD is accessible only after switching to 1G or USXGMII mode
1C	Vendor-specific MII_3 MMD	QSGMII or Quad-port USXGMII configurations Note: This MMD is accessible only after switching to 1G or USXGMII mode
1D	Clause 22 extension	N/A
1E	Vendor-Specific 1 (Control MMD)	All configurations Note: This Control MMD is used to control other MMDs present in your configuration. This MMD is also used to enable fast simulation during gate-level simulation. For more information, see Bit 4 of SR_VSMMD_CTRL Register.
1F	Vendor-Specific MII MMD (VR MII MMD)	CL37_AN = Enabled or MAIN_MODE = 1000BASEX-Only PCS Note: This MMD is accessible in 1G or USXGMII mode.

4

Parameter Descriptions

This chapter details all the configuration parameters. **You can use the coreConsultant GUI configuration reports to determine the complete configuration state of the controller.** Some expressions might refer to TCL functions or procedures (sometimes identified as **<functionof>**) that coreConsultant uses to make calculations. The exact formula used by these TCL functions is not provided in this chapter. However, when you configure the controller in coreConsultant, all TCL functions and parameters are evaluated completely; and the resulting values are displayed where appropriate in the coreConsultant GUI reports.

The parameter descriptions in this chapter include the **Enabled:** attribute which indicates the values required to be set on other parameters before you can change the value of this parameter.

These tables define all of the configuration options for this component.

- "Features / System Configurations Settings Parameters" on page 190
- "Features / XGMII Interface Parameters" on page 195
- "Features / PHY Interface Parameters" on page 196
- "Registers / IEEE PORT ID Defaults in CSR Parameters" on page 200
- "Registers / IEEE PMA Delay Defaults in CSR Parameters" on page 203
- "SafetySubordinateCoreConfiguration Parameters" on page 205

4.1 Features / System Configurations Settings Parameters

Table 4-1 Features / System Configurations Settings Parameters

Label	Description
Mode of Operation	Configures the mode of operation of the DWC_xpcs. **Values:** ■ XGXS PCS (0) ■ 10GBASE-X PCS (1) ■ Backplane Ethernet PCS (2) ■ 1000BaseX-Only PCS (3) ■ 10GBASE-R PCS (4) **Default Value:** XGXS PCS **Enabled:** Always **Parameter Name:** MAIN_MODE
Add 1G Support (1000BASE-X)	Adds the 1000BASE-X PCS functionality. **Values:** 0, 1 **Default Value:** 0 **Enabled:** MAIN_MODE == 0 || MAIN_MODE == 1 || MAIN_MODE == 4 **Parameter Name:** SUPPORT_1G
XGXS Mode	Configures the DWC_xpcs to operate on the DTE or the PHY side. **Values:** ■ XGXS in DTE (0) ■ XGXS in PHY (1) **Default Value:** XGXS in DTE **Enabled:** MAIN_MODE == 0 **Parameter Name:** DTE_PHY_SEL
Backplane Ethernet PCS Speed	Specifies the speed mode for Backplane Ethernet PCS configurations. **Values:** ■ KX4_KX (KX4 and KX Links) (0) ■ KX4_Only (4x3.125Gbps link) (1) ■ KX_Only (1x1.25Gbps link) (2) ■ KR_KX4_KX(KR, KX4 and KX Links) (3) ■ KR_KX4(KR and KX4 links) (4) ■ KR_KX(KR and KX Links) (5) ■ KR_Only (1x10.3125Gbps link) (6) **Default Value:** KX4_KX (KX4 and KX Links) **Enabled:** MAIN_MODE == 2 **Parameter Name:** BACKPLANE_ETH_CONFIG

Label	Description
Add 2.5GBASE-X Support	Adds XGMII-interface based 2.5GBASE-X functionality as per IEEE 802.3cb, to the controller. Overclocked GMII-based 2.5GBASE-X functionality is automatically supported when selected configuration has 1000BASE-X feature. **Values:** 0, 1 **Default Value:** 0 **Enabled:** KX4_KX||KR_KX4_KX||KR_KX||((MAIN_MODE==1||MAIN_MODE==4)&&SUPPORT_1G) **Parameter Name:** ADD_2PT5G
Add 5GBASE-R Support	Adds 5GBASE-R functionality to the controller, as per the IEEE 802.3cb standard. **Values:** 0, 1 **Default Value:** 0 **Enabled:** KR_ONLY||KR_KX||KR_KX4||KR_KX4_KX||(MAIN_MODE==4) **Parameter Name:** ADD_5G
Add Quad-port USXGMII Support	Adds Support for 10G-QXGMII mode in the controller. If this option is selected, Single-port and Dual-port support is automatically added. **Values:** 0, 1 **Default Value:** 0 **Enabled:** (MAIN_MODE == 4||KR_PRESENT)&&<DWC-ETHERNET-XPCS-MULTIPORT feature authorized> **Parameter Name:** USXG_QPORT
Add Dual-port USXGMII Support	Adds support for 10G-DXGMII and 5G-DXGMII support in the controller. If this option is selected, Single-port support is automatically added. **Values:** 0, 1 **Default Value:** USXG_QPORT **Enabled:** (MAIN_MODE == 4||KR_PRESENT)&&!USXG_QPORT&&<DWC-ETHERNET-XPCS-MULTIPORT feature authorized> **Parameter Name:** USXG_DPORT
Add Single-port USXGMII Support	Adds Support for Single-port USXGMII modes (10G-SXGMII,5G-SXGMII,2.5G-SXGMII modes). **Values:** 0, 1 **Default Value:** USXG_QPORT||USXG_DPORT **Enabled:** (MAIN_MODE == 4||KR_PRESENT)&&!USXG_DPORT **Parameter Name:** USXG_SPORT

Label	Description
Management Interface	Specifies the parallel micro-controller interface (MCI), the serial Management Data Input/Output (MDIO) interface, or APB3 interface (3 types) with the Control and Status registers. For 'APB3 Interface (legacy)' interface, DWC_xpcs registers are 16-bit wide and addresses of registers are multiples of 1. This is the legacy APB3 interface supported till 3.30a. For 'APB3 Interface (16-bit register)' interface, DWC_xpcs registers are 16-bit wide but addresses of registers are multiples of 2. For 'APB3 Interface (32-bit register)' interface, DWC_xpcs registers are 32-bit wide but addresses of registers are multiples of 4. **Values:** ■ MDIO Interface (0) ■ MicroController Interface (1) ■ APB3 Interface (legacy) (2) ■ APB3 Interface (16-bit register) (3) ■ APB3 Interface (32-bit register) (4) **Default Value:** MDIO Interface **Enabled:** Always **Parameter Name:** CSR_INTERFACE
Enable Indirect Register Access	Enables indirect access to registers in the DWC_xpcs. By default, each DWC_xpcs register is accessible through 21-bit/22-bit(APB3_16)/23-bit (APB3_32) wide address. By enabling this mode, the upper 13-bit of the address can be written to a pre-defined register so that only 8-bit or 9-bit (APB3_16) or 10-bit (APB3_32) address bus is used for all register accesses in the DWC_xpcs. **Values:** 0, 1 **Default Value:** 0 **Enabled:** CSR_INTERFACE!=0 **Parameter Name:** INDIRECT_ACCESS
Add Forward Error Correction Support	Enables the support for the Forward Error Correction (FEC) sub layer embedded in the 10GBASE-KR PCS. This feature is useful for erroneous channels, especially for the backplane application with 10GBASE-KR, when 10GBASE-R type serial data (10.31232 Gbps) is transmitted or received on copper. **Values:** 0, 1 **Default Value:** 0 **Enabled:** KR_ONLY||KR_KX4||KR_KX||KR_KX4_KX **Parameter Name:** FEC_EN
Add Error Forwarding Support	Enables the FEC block to forward error information to the Block Synchronization module. This option adds more design latency and hardware in the Receive path. **Values:** 0, 1 **Default Value:** 0 **Enabled:** FEC_EN **Parameter Name:** FEC_ERROR_FWD

Label	Description
Add Clause 72 KR Training Support	Enables the support for Clause 72 KR training protocol in the PCS layer. **Values:** 0, 1 **Default Value:** KR_ONLY||KR_KX4||KR_KX||KR_KX4_KX)&&(KR_PHY||(E12&&!SNPS_RSVDPARAM_39&&!SNPS_RSVDPARAM_22&&!SNPS_RSVDPARAM_33)||E16||E25||E32 **Enabled:** (KR_ONLY||KR_KX4||KR_KX||KR_KX4_KX)&&!SNPS_XAUI_PHY **Parameter Name:** CL72_EN
Enable Clause 37 Auto-Negotiation	Configures the DWC_xpcs to support the Clause 37 auto-negotiation feature. **Values:** 0, 1 **Default Value:** BASE1000X_ONLY||USXGMII_EN **Enabled:** (SUPPORT_1G == 1||KX_ONLY||KX4_KX||KR_KX||KR_KX4_KX)&&!USXGMII_EN **Parameter Name:** CL37_AN
Add SGMII or QSGMII Support	Add Support for SGMII or QSGMII functionality into the controller. **Values:** ■ None (0) ■ SGMII Only (1) ■ SGMII and QSGMII (2) **Default Value:** None **Enabled:** (CL37_AN&&(SUPPORT_1G||KX_PRESENT))||BASE1000X_ONLY **Parameter Name:** SGM_OR_QSGM_SEL
Default SGMII PHY	Configures default operating mode of DWC_xpcs as SGMII-PHY.This option is available when Mode of operation is "1000BaseX-Only PCS" and 'Add SGMII or QSGMII Support' option is chosen as "SGMII Only". **Values:** 0, 1 **Default Value:** 0 **Enabled:** SGMII_EN&&BASE1000X_ONLY **Parameter Name:** SGMII_PHY_DFLT
Add SGMII PHY Automate	Support to automate the clause 37 SGMII auto-negotiation restart in PHY mode. When this option is enabled, additional input ports to indicate the speed (xpcs_sgmii_link_speed_i), duplex mode (xpcs_sgmii_full_duplex_i) and link-status (xpcs_sgmii_link_sts_i) of the network link gets added to DWC_xpcs controller. In this mode, any change to these input ports results in automatic re-triggering of SGMII auto-negotiation to convey the new speed/duplex mode/link-status to the MAC-side of the SGMII link. You can enable this option to use DWC_xpcs at PHY-side of the SGMII link and indicate the network link speed/status to DWC_xpcs using input ports (instead of Register programmed values). **Values:** 0, 1 **Default Value:** 0 **Enabled:** SGMII_EN **Parameter Name:** SGMII_PHY_AN_AUTO_RESTART

Label	Description
Add QSGMII PHY Automate	Support to automate the clause 37 QSGMII auto-negotiation restart in PHY mode.When this option is enabled, additional input ports (4 sets of ports) to indicate the speed, duplex mode and link-status of the network link gets added to DWC_xpcs controller. In this mode, any change to these input ports results in automatic re-triggering of corresponding QSGMII auto-negotiation to convey the new speed/duplex mode/link-status to the corresponding MAC at the other side of the QSGMII link. You can enable this option to use DWC_xpcs at PHY-side of the QSGMII link and indicate the network link speed/status to DWC_xpcs using input ports (instead of Register programmed values). **Values:** 0, 1 **Default Value:** 0 **Enabled:** QSGMII_EN **Parameter Name:** QSGMII_PHY_AN_AUTO_RESTART
Add USXGMII PHY Automate	Support to automate the clause 37 USXGMII auto-negotiation restart in PHY mode.When this option is enabled, additional input ports to indicate the speed, duplex mode, link-status, EEE capability and EEE clock-stop capability of the network link gets added to DWC_xpcs controller. In this mode, any change to these input ports results in automatic re-triggering of corresponding USXGMII auto-negotiation to convey the new speed/duplex mode/link-status/EEE cap/EEE clock-stop cap to the corresponding MAC at the other side of the USXGMII link. You can enable this option to use DWC_xpcs at PHY-side of the USXGMII link and indicate the network link speed/status/EEE_cap/EEE_clk_stop_cap to DWC_xpcs using input ports (instead of Register programmed values). **Values:** 0, 1 **Default Value:** 0 **Enabled:** USXGMII_EN **Parameter Name:** USXG_PHY_AN_AUTO_RESTART
Add EEE Support	Enables support for Energy Efficient Ethernet (EEE). **Values:** 0, 1 **Default Value:** 0 **Enabled:** Always **Parameter Name:** EEE_EN
Add Automotive Safety Support	Enables support for Automotive Safety (ISO26262) features like FSM/Data-path parity protection, FSM time-outs and so on. **Values:** 0, 1 **Default Value:** 0 **Enabled:** SNPS_XAUI_PHY||!MDIO_INTF **Parameter Name:** DWC_XPCS_ASP
Add Safety Subordinate Support	Enables support for Safety Subordinate features. **Values:** 0, 1 **Default Value:** 0 **Enabled:** DWC_XPCS_ASP **Parameter Name:** DWC_XPCS_SS

4.2 Features / XGMII Interface Parameters

Table 4-2 Features / XGMII Interface Parameters

Label	Description
Add Separate GMII Port	Configures the DWC_xpcs to have a parallel GMII port, along with the XGMII Port, for the 1G mode. This port is used when the DWC_xpcs is operating in the 1G mode. **Values:** 0, 1 **Default Value:** ((SGMII_EN||QSGMII_EN)&&!(BASE1000X_ONLY||KX_ONLY))||KRKX_OR_RX||USXGMII_EN **Enabled:** (KX4_KX||KR_KX4_KX||SUPPORT_10GX_1GX)&&!(SGMII_EN||QSGMII_EN||USXGMII_EN) **Parameter Name:** ADD_GMII
XGMII Interface	Configures the XGMII interface to operate in the Single Data Rate (SDR), Dual Data Rate (DDR), or Double Data Rate (DDW) mode. **Values:** ■ Single Data Rate (SDR) (0) ■ Double Data Rate (DDR) (1) ■ Double Data Width (DDW) (2) **Default Value:** USXGMII_EN ? 0: RPCS_ONLY ? 2 : 0 **Enabled:** !(KX_ONLY||BASE1000X_ONLY)&&!MP_USXGMII **Parameter Name:** XGMII
Enable Clock Compensation	Enables the clock compensation module in the Receive path. It enables the transfer of XGMII or GMII receive data to the XGMII or GMII Transmit clock domain and thus, having a common clock domain for XGMII or GMII Transmit and Receive interfaces. **Values:** 0, 1 **Default Value:** KR_PHY&&RPCS_PRESENT **Enabled:** !(KR_PHY&&RPCS_PRESENT) **Parameter Name:** CLKCOMP

4.3 Features / PHY Interface Parameters

Table 4-3 Features / PHY Interface Parameters

Label	Description
PHY Receives LSb First	Indicates that the PHY first receives the least significant bit of the xgxs_rx_data{lane}_i[9:0] or rpcs_rx_data_i[rpcs_phw:0]. **Values:** 0, 1 **Default Value:** 0 **Enabled:** !SNPS_XAUI_PHY **Parameter Name:** RX_LSB_FIRST
PHY Transmits LSb First	Indicates that the PHY first transmits the least significant bit of xgxs_tx_data{lane}_o[9:0] or rpcs_tx_data_o[rpcs_phw:0]. **Values:** 0, 1 **Default Value:** 0 **Enabled:** !SNPS_XAUI_PHY **Parameter Name:** TX_LSB_FIRST
Enable Synopsys PHY Interface	Enables the logic and input and output signals to interface with the Synopsys PHY. **Values:** 0, 1 **Default Value:** 1 **Enabled:** Always **Parameter Name:** SNPS_XAUI_PHY
Synopsys PHY Type	Select the Synopsys PHY Type **Values:** ■ Multi-protocol 6G (0) ■ Multi-protocol 12G (1) ■ Multi-protocol 16G (2) ■ Multi-protocol 25G (3) ■ Multi-protocol 32G (4) **Default Value:** (RPCS_PRESENT&&(SNPS_XAUI_PHY==1)) ? 1 : 0 **Enabled:** SNPS_XAUI_PHY==1 **Parameter Name:** SNPS_PHY_TYPE

Label	Description
Reference Clock Frequency	Indicates the reference clock frequency for Synopsys PHY to program the default MPLL parameters of PHY during power up. **Values:** ■ 156.25 MHz (0) ■ 125 MHz (1) ■ 78.125 MHz (2) ■ 62.5 MHz (3) ■ 50 MHz (4) ■ 25 MHz (5) ■ 312.5 MHz (6) **Default Value:** 156.25 MHz **Enabled:** SNPS_XAUI_PHY **Parameter Name:** REF_CLK_FREQ
Reference Clock Source	Selects the source of reference clock for Synopsys PHY. **Values:** ■ Pad Clk (0) ■ Alt/Core Clk (1) ■ Alt 1 Clk (2) **Default Value:** Pad Clk **Enabled:** SNPS_XAUI_PHY **Parameter Name:** REF_CLK_SRC
Tx Context Select Value	Default value of Tx context select value that is driven to PHY. Set this to appropriate value for the default operating speed mode of DWC_xpcs as per the the PHY ROM/SRAM image used **Values:** 0x0, ..., 0xff **Default Value:** SNPS_RSVDPARAM_37 ? 0x2E : E32 ? 0x4 : 0x0 **Enabled:** E32||SNPS_RSVDPARAM_37 **Parameter Name:** TX_CNTX_SEL
Common Context Select Value	Default value of Common context select value that is driven to PHY. Set this to appropriate value for the default operating speed mode of DWC_xpcs as per the the PHY ROM/SRAM image used **Values:** 0x0, ..., 0xff **Default Value:** E32 ? 0x04 : 0x00 **Enabled:** E32 **Parameter Name:** CM_CNTX_SEL
Rx Context Select Value	Default value of Rx context select value that is driven to PHY. Set this to appropriate value for the default operating speed mode of DWC_xpcs as per the the PHY ROM/SRAM image used **Values:** 0x0, ..., 0xff **Default Value:** SNPS_RSVDPARAM_37 ? 0x2E : E32 ? 0x4 : 0x0 **Enabled:** E32||SNPS_RSVDPARAM_37 **Parameter Name:** RX_CNTX_SEL

Label	Description
Add side-band signal for Non-Synopsys PHY 'Ready' Indication	If this option is selected, a primary input 'xpcs_phy_rdy_i' gets added to DWC_xpcs controller. This input is used to indicate if non-Synopsys PHY is 'ready' or not. This input should be driven to 1 when PHY is stable and all clocks from the PHY are at desired frequencies. DWC_xpcs keeps its internal data-path modules at reset, when this signal is 0. **Values:** 0, 1 **Default Value:** 0 **Enabled:** !SNPS_XAUI_PHY **Parameter Name:** ADD_PHY_RDY
RPCS PHY Interface Type	Specifies the data width and the operating frequency of the data interface (rpcs_tx_data_o[rpcs_phw:0] and rpcs_rx_data_i[rpcs_phw:0]) to the 10GBASE-R or KR PHY. **Values:** ■ 16x644.53MHz (0) ■ 32x322.265MHz (1) ■ 33x312.5MHz (2) ■ 66x156.25MHz (3) ■ 20x515.625MHz (4) ■ 64x161.1328125MHz (5) ■ 40x257.8125MHz (6) **Default Value:** 16x644.53MHz **Enabled:** RPCS_PRESENT **Parameter Name:** R_SRDS_INTF
Enable Lane Control	Enables control over the number of lanes in the XGXS PCS, 10GBASE-X PCS, and KX4 speed modes. Using this option, you can enable 2.5 Gbps, 5.0 Gbps, and 7.5 Gbps speed modes, along with the default 10 Gbps speed mode, by disabling some of the operating lanes in a vendor-specific way. **Values:** 0, 1 **Default Value:** !(KX_ONLY||BASE1000X_ONLY||(MAIN_MODE==4)||KR_ONLY||KR_KX) **Enabled:** !(KX_ONLY||BASE1000X_ONLY||(MAIN_MODE==4)||KR_ONLY||KR_KX) **Parameter Name:** LANE_CONTROL
Enable Extended Lane-to-Lane De-Skew	Enables the extended lane-to-lane deskew of 80UI. The default lane-to-lane deskew is 40UI. **Values:** 0, 1 **Default Value:** 0 **Enabled:** !SINGLE_LANE_CFG **Parameter Name:** EXT_DSKW

Label	Description
Enable RXAUI Support(2x6.25 Gbps Link)	Enables the RXAUI functionality. The DWC_xpcs supports the implementations of the RXAUI. A. Dune Networks implementation B.Marvell implementation. When this feature is enabled, the DWC_xpcs supports RXAUI with two lanes (each of 20-bit data width), operating at 312.5 MHz clock, to get the speed of 2x6.25 Gbps. When this feature is disabled, the DWC_xpcs does not support RXAUI. **Values:** 0, 1 **Default Value:** 0 **Enabled:** !(KX_ONLY||BASE1000X_ONLY||(MAIN_MODE==4)||KR_ONLY||KR_KX) **Parameter Name:** RXAUI
Enable RXAUI Mode by Default	Enables the RXAUI mode by default in the DWC_xpcs. You can switch to the XAUI mode by programming registers. **Values:** 0, 1 **Default Value:** RXAUI&&!BACKPLANE_ETHERNET_PCS **Enabled:** RXAUI&&!BACKPLANE_ETHERNET_PCS **Parameter Name:** RXAUI_DEFAULT
Enable GPIO	Enables the General Purpose Inputs/Outputs (GPIO) for Synopsys PHY. You can use these IOs as the Control or Status interface to a PHY or from a PHY. **Values:** 0, 1 **Default Value:** 0 **Enabled:** Always **Parameter Name:** GPIO_EN

4.4 Registers / IEEE PORT ID Defaults in CSR Parameters

Table 4-4 Registers / IEEE PORT ID Defaults in CSR Parameters

Label	Description
Support to Modify Default Values IEEE Registers	Makes the IEEE Device or Package Identifier registers programmable through Vendor-Specific Control MMD. **Values:** 0, 1 **Default Value:** 0 **Enabled:** Always **Parameter Name:** IEEE_REG_WR_SUPPORT
XS/PCS Device OUI (24-bit)	Configures the default value of the 24-bit IEEE XGXS/PCS Device Organizationally Unique Identifier. **Values:** 0x0, ..., 0xffffff **Default Value:** 0xcda679 **Enabled:** Always **Parameter Name:** IEEE_DEV_OUI
XS/PCS Device Model Number (6-bit)	Configures the default value of the 6-bit IEEE XGXS/PCS Device Model Number. **Values:** 0x0, ..., 0x2f **Default Value:** 0x2d **Enabled:** Always **Parameter Name:** IEEE_DEV_MODEL_NO
XS/PCS Device Revision Number (4-bit)	Configures the default value of the 4-bit IEEE XGXS/PCS Device Revision Number. **Values:** 0x0, ..., 0xf **Default Value:** 0x0 **Enabled:** Always **Parameter Name:** IEEE_DEV_REV_NO
Package OUI (24-bit)	Configures the default value of the 24-bit IEEE Package Organizationally Unique Identifier. **Values:** 0x0, ..., 0xffffff **Default Value:** 0x0 **Enabled:** BASE1000X_ONLY == 0 **Parameter Name:** IEEE_PKG_OUI
Package Model Number (6-bit)	Configures the default value of the 6-bit IEEE Package Model Number. **Values:** 0x0, ..., 0x2f **Default Value:** 0x0 **Enabled:** BASE1000X_ONLY == 0 **Parameter Name:** IEEE_PKG_MODEL_NO
Package Revision Number (4-bit)	Configures the default value of the 4-bit IEEE Package Revision Number. **Values:** 0x0, ..., 0xf **Default Value:** 0x0 **Enabled:** BASE1000X_ONLY == 0 **Parameter Name:** IEEE_PKG_REV_NO

Label	Description
PMA Device OUI (24-bit)	Configures the default value of the 24-bit IEEE PMA Device Organizationally Unique Identifier. **Values:** 0x0, ..., 0xffffff **Default Value:** 0x0 **Enabled:** PMA_MMD == 1 **Parameter Name:** IEEE_PMA_DEV_OUI
PMA Device Model Number (6-bit)	Configures the default value of the 6-bit IEEE PMA Device Model Number. **Values:** 0x0, ..., 0x2f **Default Value:** 0x0 **Enabled:** PMA_MMD == 1 **Parameter Name:** IEEE_PMA_DEV_MODEL_NO
PMA Device Revision Number (4-bit)	Configures the default value of the 4-bit IEEE PMA Device Revision Number. **Values:** 0x0, ..., 0xf **Default Value:** 0x0 **Enabled:** PMA_MMD == 1 **Parameter Name:** IEEE_PMA_DEV_REV_NO
AN Device OUI (24-bit)	Configures the default value of 24-bit IEEE AN Device Organizationally Unique Identifier (OUI). **Values:** 0x0, ..., 0xffffff **Default Value:** 0x0 **Enabled:** BACKPLANE_ETHERNET_PCS == 1 **Parameter Name:** IEEE_AN_DEV_OUI
AN Device Model Number (6-bit)	Configures the default value of 6-bit IEEE AN Device Model Number. **Values:** 0x0, ..., 0x2f **Default Value:** 0x0 **Enabled:** BACKPLANE_ETHERNET_PCS == 1 **Parameter Name:** IEEE_AN_DEV_MODEL_NO
AN Device Revision Number (4-bit)	Configures the default value of 4-bit IEEE AN Device Revision Number. **Values:** 0x0, ..., 0xf **Default Value:** 0x0 **Enabled:** BACKPLANE_ETHERNET_PCS == 1 **Parameter Name:** IEEE_AN_DEV_REV_NO
Vendor-Specific MMD1 Device OUI (24-bit)	Configures the default value of 24-bit IEEE Vendor-Specific MMD1 Device Organizationally Unique Identifier (OUI). **Values:** 0x0, ..., 0xffffff **Default Value:** 0x0 **Enabled:** Always **Parameter Name:** IEEE_VS_DEV_OUI

Label	Description
Vendor-Specific MMD1 Device Model Number (6-bit)	Configures the default value of 6-bit IEEE Vendor-Specific MMD1 Device Model Number. **Values:** 0x0, ..., 0x2f **Default Value:** 0x0 **Enabled:** Always **Parameter Name:** IEEE_VS_DEV_MODEL_NO
Vendor Specific MMD1 Device Revision Number (4-bit)	Configures the default value of 4-bit IEEE Vendor-Specific MMD1 Device Revision Number. **Values:** 0x0, ..., 0xf **Default Value:** 0x0 **Enabled:** Always **Parameter Name:** IEEE_VS_DEV_REV_NO

4.5 Registers / IEEE PMA Delay Defaults in CSR Parameters

Table 4-5 Registers / IEEE PMA Delay Defaults in CSR Parameters

Label	Description
PMA Tx Max Delay-lower (16-bit)	Configures the default value of the 16-bit IEEE Time Sync PMA/PMD Transmit Path Maximum Data Delay (lower 16-bit). **Values:** 0x0, ..., 0xffff **Default Value:** 0x0 **Enabled:** PMA_MMD == 1 **Parameter Name:** PMA_TX_MAX_DLY_LWR
PMA Tx Max Delay-upper (16-bit)	Configures the default value of the 16-bit IEEE Time Sync PMA/PMD Transmit Path Maximum Data Delay (upper 16-bit). **Values:** 0x0, ..., 0xffff **Default Value:** 0x0 **Enabled:** PMA_MMD == 1 **Parameter Name:** PMA_TX_MAX_DLY_UPR
PMA Tx Min Delay-lower (16-bit)	Configures the default value of the 16-bit IEEE Time Sync PMA/PMD Transmit Path Minimum Data Delay (lower 16-bit). **Values:** 0x0, ..., 0xffff **Default Value:** 0x0 **Enabled:** PMA_MMD == 1 **Parameter Name:** PMA_TX_MIN_DLY_LWR
PMA Tx Min Delay-upper (16-bit)	Configures the default value of the 16-bit IEEE Time Sync PMA/PMD Transmit Path Minimum Data Delay (upper 16-bit). **Values:** 0x0, ..., 0xffff **Default Value:** 0x0 **Enabled:** PMA_MMD == 1 **Parameter Name:** PMA_TX_MIN_DLY_UPR
PMA Rx Max Delay-lower (16-bit)	Configures the default value of the 16-bit IEEE Time Sync PMA/PMD Receive Path Maximum Data Delay (lower 16-bit). **Values:** 0x0, ..., 0xffff **Default Value:** 0x0 **Enabled:** PMA_MMD == 1 **Parameter Name:** PMA_RX_MAX_DLY_LWR
PMA Rx Max Delay-upper (16-bit)	Configures the default value of the 16-bit IEEE Time Sync PMA/PMD Receive Path Maximum Data Delay (upper 16-bit). **Values:** 0x0, ..., 0xffff **Default Value:** 0x0 **Enabled:** PMA_MMD == 1 **Parameter Name:** PMA_RX_MAX_DLY_UPR

Label	Description
PMA Rx Min Delay-lower (16-bit)	Configures the default value of the 16-bit IEEE Time Sync PMA/PMD Receive Path Minimum Data Delay (lower 16-bit). **Values:** 0x0, ..., 0xffff **Default Value:** 0x0 **Enabled:** PMA_MMD == 1 **Parameter Name:** PMA_RX_MIN_DLY_LWR
PMA Rx Min Delay-upper (16-bit)	Configures the default value of the 16-bit IEEE Time Sync PMA/PMD Receive Path Minimum Data Delay (upper 16-bit). **Values:** 0x0, ..., 0xffff **Default Value:** 0x0 **Enabled:** PMA_MMD == 1 **Parameter Name:** PMA_RX_MIN_DLY_UPR

4.6 SafetySubordinateCoreConfiguration Parameters

Table 4-6 SafetySubordinateCoreConfiguration Parameters

Label	Description
Core register start offset	Indicates the Core register start offset. This parameter is used to select the starting offset of the core diagnostic register. Minimum Value = 0x20, Maximum = (based on the SSM_BUS_ADDR_SIZE) **Values:** 32, ..., 65516 **Default Value:** 32 **Enabled:** DWC_XPCS_SS **Parameter Name:** SS_CORE_REG_START_OFFSET
Core register depth	Indicates the Core register depth. Minimum Depth = 4(Here there are no diagnostic registers), Maximum Depth = 68 (4 core registrs+ 64 diagnostic registers) **Values:** 4, ..., 68 **Default Value:** 10 **Enabled:** DWC_XPCS_SS **Parameter Name:** SS_CORE_REG_DEPTH

5

Register Descriptions

This chapter details all possible registers in the controller. They are arranged hierarchically into maps and blocks (banks). For configurable IP titles, your actual configuration might not contain all of these registers.

Attention: For configurable IP titles, do not use this document to determine the exact attributes of your register map. It is for reference purposes only.

When you configure the controller in coreConsultant, you must access the register attributes for your actual configuration at `workspace/report/ComponentRegisters.html` or `workspace/report/ComponentRegisters.xml` after you have completed the report creation activity. That report comes from the exact same source as this chapter but removes all the registers that are not in your actual configuration. This does not apply to non-configurable IP titles. In addition, all parameter expressions are evaluated to actual values. Therefore, the Offset and Memory Access values might change depending on your actual configuration.

Some expressions might refer to TCL functions or procedures (sometimes identified as **<functionof>**) that coreConsultant uses to make calculations. The exact formula used by these TCL functions is not provided in this chapter. However, when you configure the controller in coreConsultant, all TCL functions and parameters are evaluated completely; and the resulting values are displayed where appropriate in the coreConsultant GUI reports.

Exists Expressions

These expressions indicate the combination of configuration parameters required for a register, field, or block to exist in the memory map. The expression is only valid in the local context and does not indicate the conditions for existence of the parent. For example, the expression for a bit field in a register assumes that the register exists and does not include the conditions for existence of the register.

Offset

The term *Offset* is synonymous with *Address.*

Memory Access Attributes

The Memory Access attribute is defined as `<ReadBehavior>/<WriteBehavior>` which are defined in the following table.

Table 5-1 Possible Read and Write Behaviors

Read (or Write) Behavior	Description
RC	A read clears this register field.
RS	A read sets this register field.
RM	A read modifies the contents of this register field.
Wo	You can only write once to this register field.
W1C	A write of 1 clears this register field.
W1S	A write of 1 sets this register field.
W1T	A write of 1 toggles this register field.
W0C	A write of 0 clears this register field.
W0S	A write of 0 sets this register field.
W0T	A write of 0 toggles this register field.
WC	Any write clears this register field.
WS	Any write sets this register field.
WM	Any write toggles this register field.
no Read Behavior attribute	You cannot read this register. It is Write-Only.
no Write Behavior attribute	You cannot write to this register. It is Read-Only.

Table 5-2 Memory Access Examples

Memory Access	Description
R	Read-only register field.
W	Write-only register field.
R/W	Read/write register field.
R/W1C	You can read this register field. Writing 1 clears it.
RC/W1C	Reading this register field clears it. Writing 1 clears it.
R/Wo	You can read this register field. You can only write to it once.

Special Optional Attributes

Some register fields might use the following optional attributes.

Table 5-3 Optional Attributes

Attribute	Description
Volatile	As defined by the IP-XACT specification. If true, indicates in the case of a write followed by read, or in the case of two consecutive reads, there is no guarantee as to what is returned by the read on the second transaction or that this return value is consistent with the write or read of the first transaction. The element implies there is some additional mechanism by which this field can acquire new values other than by reads/writes/resets and other access methods known to IP-XACT. For example, when the controller updates the register field contents.
Testable	As defined by the IP-XACT specification. Possible values are unconstrained, untestable, readOnly, writeAsRead, restore. Untestable means that this field is untestable by a simple automated register test. For example, the read-write access of the register is controlled by a pin or another register. readOnly means that you should not write to this register; only read from it. This might apply for a register that modifies the contents of another register.
Reset Mask	As defined by the IP-XACT specification. Indicates that this register field has an unknown reset value. For example, the reset value is set by another register or an input pin; or the register is implemented using RAM.
* Varies	Indicates that the memory access (or reset) attribute (read, write behavior) is not fixed. For example, the read-write access of the register is controlled by a pin or another register. Or when the access depends on some configuration parameter; in this case the post-configuration report in coreConsultant gives the actual access value.

5.1 Component Memory Maps

The following table documents the memory maps visible in this component.

Table 5-4 DWC_xpcs Memory Maps

Memory Map	Description
“Registers for the DWC_xpcs_map Memory Map” on page 211	The memory map DWC_xpcs_map contains Several Address Blocks **Exists:** Always
“Registers for the DWC_xpcs_safetyslave_map Memory Map” on page 1378	The memory map DWC_xpcs_safetyslave_map contains Several Address Blocks **Exists:** (DWC_XPCS_SS==1)

5.2 DWC_xpcs_map Memory Map Registers

Register definitions for each component memory map.

Table 5-5 Registers for the DWC_xpcs_map Memory Map

Register	Offset	Description
PMA MMD Standard (SR) and Vendor Specific Register (VR) (except SERDES Registers) address block PMA_MMD Exists: PMA_MMD		
"SR_PMA_CTRL1" on page 245	* Varies	SR PMA MMD Control1 Register
"SR_PMA_STATUS1" on page 250	* Varies	SR PMA MMD Status1 Register
"SR_PMA_DEV_ID_1" on page 252	* Varies	SR PMA MMD Device Identifier 1 Register
"SR_PMA_DEV_ID_2" on page 253	* Varies	SR PMA MMD Device Identifier 2 Register
"SR_PMA_SPD_ABL" on page 255	* Varies	SR PMA MMD Speed Ability Register
"SR_PMA_DEV_PKG1" on page 256	* Varies	SR PMA MMD Devices in Package 1 Register
"SR_PMA_DEV_PKG2" on page 258	* Varies	SR PMA MMD Devices in Package 2 Register
"SR_PMA_CTRL2" on page 259	* Varies	SR PMA MMD Control2 Register
"SR_PMA_STATUS2" on page 260	* Varies	SR PMA MMD Status2 Register
"SR_PMA_TX_DIS" on page 263	* Varies	SR PMA MMD Transmit Disable Register Bits[4:1] of this register are shared with the following...
"SR_PMA_RX_SIG_DET" on page 265	* Varies	SR PMA MMD Receive Signal Detect Register
"SR_PMA_EXT_ABL" on page 267	* Varies	SR PMA or PMD Extended Ability Register
"SR_PMA_PKG1" on page 269	* Varies	SR PMA MMD Package Identifier 1 Register
"SR_PMA_PKG2" on page 270	* Varies	SR PMA MMD Package Identifier 2 Register
"SR_PMA_2PT5G_5G_EXT_ABL" on page 272	* Varies	SR PMA MMD 2.5G/5G Extended Ability Register
"SR_PMA_KR_PMD_CTRL" on page 274	* Varies	SR PMA MMD 10GBASE-KR PMD Control Register
"SR_PMA_KR_PMD_STS" on page 275	* Varies	SR PMA MMD 10GBASE-KR Status Register
"SR_PMA_KR_LP_CEU" on page 277	* Varies	SR PMA MMD 10GBASE-KR LP Coefficient Update Register. This register is present when the Clause...
"SR_PMA_KR_LP_CESTS" on page 280	* Varies	SR PMA MMD 10GBASE-KR LP Coefficient Status Register This register is present when the Clause 72...
"SR_PMA_KR_LD_CEU" on page 282	* Varies	SR PMA MMD 10GBASE-KR LD Coefficient Update This register is present when the Clause 72 KR Training...
"SR_PMA_KR_LD_CESTS" on page 285	* Varies	SR PMA MMD 10GBASE-KR LD Coefficient Status This register is present only for the PMA MMD when...

Register	Offset	Description
“SR_PMA_KX_CTRL” on page 287	* Varies	SR PMA or PMD KX Control Register This register is present only in the configurations with KX_Present...
“SR_PMA_KX_STS” on page 288	* Varies	SR PMA or PMD KX Status Register This register is present only in the configurations with KX_Present...
“SR_PMA_KR_FEC_ABL” on page 290	* Varies	SR PMA MMD 10GBASE-KR FEC Ability Register This register is present only if FEC_EN = Enabled. Otherwise,...
“SR_PMA_KR_FEC_CTRL” on page 291	* Varies	SR PMA MMD 10GBASE-KR FEC Control Register This register is present only if FEC_EN = Enabled. Otherwise,...
“SR_PMA_KR_FEC_CORR_BLK1” on page 292	* Varies	SR PMA MMD 10GBASE-KR Lane0 FEC Correctable Error Block Counter 1 Register This register is present...
“SR_PMA_KR_FEC_CORR_BLK2” on page 293	* Varies	SR PMA MMD 10GBASE-KR Lane0 FEC Correctable Error Block Counter 2 Register This register is present...
“SR_PMA_KR_FEC_UCORR_BLK1” on page 294	* Varies	SR PMA MMD 10GBASE-KR Lane0 FEC Uncorrectable Error Block Counter 1 Register This register is present...
“SR_PMA_KR_FEC_UCORR_BLK2” on page 295	* Varies	SR PMA MMD 10GBASE-KR Lane0 FEC Uncorrectable Error Block Counter 2 Register This register is present...
“SR_PMA_TIME_SYNC_PMA_ABL” on page 296	* Varies	SR PMA MMD Time Sync Capability Register
“SR_PMA_TIME_SYNC_TX_MAX_DLY_LWR” on page 297	* Varies	SR PMA MMD Time Sync Tx Max Delay Lower Register
“SR_PMA_TIME_SYNC_TX_MAX_DLY_UPR” on page 298	* Varies	SR PMA MMD Time Sync Tx Max Delay Upper Register
“SR_PMA_TIME_SYNC_TX_MIN_DLY_LWR” on page 299	* Varies	SR PMA MMD Time Sync Tx Min Delay Lower Register
“SR_PMA_TIME_SYNC_TX_MIN_DLY_UPR” on page 300	* Varies	SR PMA MMD Time Sync Tx Min Delay Upper Register
“SR_PMA_TIME_SYNC_RX_MAX_DLY_LWR” on page 301	* Varies	SR PMA MMD Time Sync Rx Max Delay Lower Register
“SR_PMA_TIME_SYNC_RX_MAX_DLY_UPR” on page 302	* Varies	SR PMA MMD Time Sync Rx Max Delay Upper Register
“SR_PMA_TIME_SYNC_RX_MIN_DLY_LWR” on page 303	* Varies	SR PMA MMD Time Sync Rx Min Delay Lower Register
“SR_PMA_TIME_SYNC_RX_MIN_DLY_UPR” on page 304	* Varies	SR PMA MMD Time Sync Rx Min Delay Upper Register
“VR_PMA_DIG_CTRL1” on page 305	* Varies	VR PMA MMD Digital Control1 Register
“VR_PMA_KRTR_PRBS_CTRL0” on page 310	* Varies	VR PMA MMD KRTR PRBS Control0 Register

Register	Offset	Description
"VR_PMA_KRTR_PRBS_CTRL1" on page 311	* Varies	VR PMA MMD KRTR PRBS Control1 Register
"VR_PMA_KRTR_PRBS_CTRL2" on page 312	* Varies	VR PMA MMD KRTR PRBS Control2 Register
"VR_PMA_KRTR_TIMER_CTRL0" on page 313	* Varies	VR PMA MMD KRTR Timer Control 0 Register
"VR_PMA_KRTR_TIMER_CTRL1" on page 314	* Varies	VR PMA MMD KRTR Timer Control 1 Register
"VR_PMA_KRTR_TIMER_CTRL2" on page 315	* Varies	VR PMA MMD KRTR Timer Control 2 Register
"VR_PMA_KRTR_RX_EQ_CTRL" on page 316	* Varies	VR PMA MMD KRTR Receive Equalization Control Register This register is present only if CL72_EN...
"VR_PMA_KRTR_TX_EQ_STS_CTRL" on page 319	* Varies	VR PMA MMD KRTR Transmit Equalization Status Control Register This register is present only if...
"VR_PMA_KRTR_TX_EQ_CFF_CTRL" on page 322	* Varies	VR PMA MMD KRTR Transmit Equalization Coefficient Control Register. This register is present...
"VR_PMA_PHY_TX_EQ_STS" on page 324	* Varies	VR PMA MMD PHY Transmit Equalization Status Register This register is present only if CL72_EN...
"VR_PMA_PHY_RX_EQ_CEU" on page 326	* Varies	VR PMA MMD PHY Receive Equalization Coefficient Update This register is present when the Clause...
"VR_PMA_KRTR_RX_EQ_STS_CTRL" on page 329	* Varies	VR PMA MMD KRTR Receive Equalization Status Control Register This register is present only if...
"VR_PMA_DIG_STS" on page 332	* Varies	VR PMA MMD Digital Status Register
"VR_PMA_GPIO" on page 334	* Varies	VR PMA MMD GPIO Register This register is present only if the DWC_xpcs is configured to have the...
PCS MMD /DTE XS MMD /PHY XS MMD Standard (SR) & Vendor Specific (VR) Registers address block Based on the configuration, this address block is either PCS MMD or XS (DTE/PHY) MMD. If Mode of Operation = XGXS and XGXS Mode = XGXS in DTE, then this is DTE XS MMD (MMD Address = 5'h05). If Mode of Operation = XGXS and XGXS Mode = XGXS in PHY, then this is PHY XS MMD (MMD Address = 5'h04). If Mode of Operation = 10GBASE-R PCS or Backplane Ethernet PCS, then this is PCS MMD (MMD Address = 5'h03) XS_PCS_MMD Exists: ((PCS_MMD||XGXS_PCS))		
"SR_XS_PCS_CTRL1" on page 335	* Varies	SR XS or PCS MMD Control1 Register
"SR_XS_PCS_STS1" on page 340	* Varies	SR XS or PCS MMD Status1 Register
"SR_XS_PCS_DEV_ID1" on page 344	* Varies	SR XS or PCS MMD Device Identifier 1 Register
"SR_XS_PCS_DEV_ID2" on page 345	* Varies	SR XS or PCS MMD Device Identifier 2 Register
"SR_XS_PCS_SPD_ABL" on page 347	* Varies	SR XS or PCS MMD Speed Ability Register
"SR_XS_PCS_DEV_PKG1" on page 348	* Varies	SR XS or PCS MMD Devices in Package 1 Register

Register	Offset	Description
“SR_XS_PCS_DEV_PKG2” on page 350	* Varies	SR XS or PCS MMD Devices in Package 2 Register
“SR_XS_PCS_CTRL2” on page 351	* Varies	SR PCS Control2 Register This register is applicable only for the PCS MMD.
“SR_XS_PCS_STS2” on page 353	* Varies	SR XS or PCS MMD Status2 Register
“SR_XS_PCS_STS3” on page 355	* Varies	SR PCS MMD Status3 Register This register is applicable only for the PCS MMD
“SR_XS_PCS_PKG1” on page 356	* Varies	SR XS or PCS MMD Package Identifier 1 Register
“SR_XS_PCS_PKG2” on page 357	* Varies	SR XS or PCS MMD Package Identifier 2 Register
“SR_XS_PCS_EEE_ABL” on page 359	* Varies	SR XS or PCS MMD EEE Capability Register
“SR_XS_PCS_EEE_ABL2” on page 362	* Varies	SR PCS MMD EEE Control and Capability2 Register
“SR_XS_PCS_EEE_WKERR” on page 364	* Varies	SR XS or PCS MMD EEE Wake Error Counter Register This register is present only when EEE_EN =...
“SR_XS_PCS_LSTS” on page 365	* Varies	SR XS or PCS MMD 10GBASE-X Lane Status Register This register is applicable only for the XS and...
“SR_XS_PCS_TCTRL” on page 367	* Varies	SR XS or PCS MMD 10GBASE-X Test Control Register This register is present only with the XS and...
“SR_XS_PCS_KR_STS1” on page 369	* Varies	SR PCS MMD 10GBASE-R Status1 Register This register is present only with the PCS MMD in the following...
“SR_XS_PCS_KR_STS2” on page 371	* Varies	SR PCS MMD 10GBASE-R Status2 Register This register is present only with the PCS MMD in the following...
“SR_XS_PCS_TP_A0” on page 373	* Varies	SR PCS MMD 5G/10GBASE-R Pseudo-Random Test Pattern Seed A0 Register This register contains Bits[15:0]...
“SR_XS_PCS_TP_A1” on page 374	* Varies	SR PCS MMD 5G/10GBASE-R Pseudo-Random Test Pattern Seed A1 Register
“SR_XS_PCS_TP_A2” on page 375	* Varies	SR PCS MMD 5G/10GBASE-R Pseudo-Random Test Pattern Seed A2 Register
“SR_XS_PCS_TP_A3” on page 376	* Varies	SR PCS MMD 5G/10GBASE-R Pseudo-Random Test Pattern Seed A3 Register
“SR_XS_PCS_TP_B0” on page 377	* Varies	SR PCS MMD 5G/10GBASE-R Pseudo-Random Test Pattern Seed B0 Register This register contains Bits[15:0]...
“SR_XS_PCS_TP_B1” on page 378	* Varies	SR PCS MMD 5G/10GBASE-R Pseudo-Random Test Pattern Seed B1 Register
“SR_XS_PCS_TP_B2” on page 379	* Varies	SR PCS MMD 5G/10GBASE-R Pseudo-Random Test Pattern Seed B2 Register

Register	Offset	Description
“SR_XS_PCS_TP_B3” on page 380	* Varies	SR PCS MMD 5G/10GBASE-R Pseudo-Random Test Pattern Seed B3 Register
“SR_XS_PCS_TP_CTRL” on page 381	* Varies	SR PCS BASE-R Test Pattern Control Register This register is present only with the PCS MMD in the...
“SR_XS_PCS_TP_ERRCTR” on page 384	* Varies	SR PCS BASE-R Test Pattern Error Counter Register This register is present only with the PCS MMD...
“SR_PCS_TIME_SYNC_PCS_ABL” on page 385	* Varies	SR PCS MMD Time Sync Capability Register
“SR_PCS_TIME_SYNC_TX_MAX_DLY_LWR” on page 388	* Varies	SR PCS MMD Time Sync Tx Max Delay Lower Register
“SR_PCS_TIME_SYNC_TX_MAX_DLY_UPR” on page 389	* Varies	SR PCS MMD Time Sync Tx Max Delay Upper Register
“SR_PCS_TIME_SYNC_TX_MIN_DLY_LWR” on page 390	* Varies	SR PCS MMD Time Sync Tx Min Delay Lower Register
“SR_PCS_TIME_SYNC_TX_MIN_DLY_UPR” on page 391	* Varies	SR PCS MMD Time Sync Tx Min Delay Upper Register
“SR_PCS_TIME_SYNC_RX_MAX_DLY_LWR” on page 392	* Varies	SR PCS MMD Time Sync Rx Max Delay Lower Register
“SR_PCS_TIME_SYNC_RX_MAX_DLY_UPR” on page 394	* Varies	SR PCS MMD Time Sync Rx Max Delay Upper Register
“SR_PCS_TIME_SYNC_RX_MIN_DLY_LWR” on page 395	* Varies	SR PCS MMD Time Sync Rx Min Delay Lower Register
“SR_PCS_TIME_SYNC_RX_MIN_DLY_UPR” on page 397	* Varies	SR PCS MMD Time Sync Rx Min Delay Upper Register
“SR_PCS_TIME_SYNC_TX_MAX_FINE_DLY” on page 398	* Varies	SR PCS MMD Time Sync Tx Max Fine Delay Register
“SR_PCS_TIME_SYNC_TX_MIN_FINE_DLY” on page 399	* Varies	SR PCS MMD Time Sync Tx Min Fine Delay Register
“SR_PCS_TIME_SYNC_RX_MAX_FINE_DLY” on page 400	* Varies	SR PCS MMD Time Sync Rx Max Fine Delay Register
“SR_PCS_TIME_SYNC_RX_MIN_FINE_DLY” on page 401	* Varies	SR PCS MMD Time Sync Rx Min Fine Delay Register
“SR_PCS_TIME_SYNC_CFG” on page 402	* Varies	SR PCS MMD Time Sync Configuration Register
“VR_XS_PCS_DIG_CTRL1” on page 403	* Varies	VR XS or PCS MMD Digital Control1 Register
“VR_XS_PCS_DIG_CTRL2” on page 410	* Varies	VR XS or PCS MMD Digital Control2 Register

Register	Offset	Description
“VR_XS_PCS_DIG_ERRCNT_SEL” on page 414	* Varies	VR XS or PCS MMD Digital Error Count Select Register. This register is not present in the following...
“VR_XS_PCS_XAUI_CTRL” on page 417	* Varies	VR XS or PCS MMD XAUI Mode Control Register This register is present only when the DWC_xpcs is...
“VR_XS_PCS_DEBUG_CTRL” on page 419	* Varies	VR XS or PCS MMD Debug Control Register
“VR_XS_PCS_EEE_MCTRL0” on page 425	* Varies	VR XS or PCS MMD EEE Mode Control Register This register enables or disables the Energy Efficient...
“VR_XS_PCS_KR_CTRL” on page 429	* Varies	VR PCS 10GBASE-R Control Register This register is present only with the PCS MMD in the following...
“VR_XS_PCS_EEE_TXTIMER” on page 433	* Varies	VR XS or PCS MMD EEE Tx Timer Register This register is present only if EEE_EN = Enabled.
“VR_XS_PCS_EEE_RXTIMER” on page 437	* Varies	VR XS or PCS MMD EEE Rx Timer Register This register is present only if EEE_EN = Enabled.
“VR_XS_PCS_AM_CTRL” on page 440	* Varies	VR PCS MMD Alignment Marker Control Register
“VR_XS_PCS_EEE_MCTRL1” on page 442	* Varies	VR XS or PCS MMD EEE Mode Control 1 Register This register also controls the Energy Efficient Ethernet...
“VR_XS_PCS_DIG_STS” on page 446	* Varies	VR XS or PCS MMD Digital Status Register
“VR_XS_PCS_ICG_ERRCNT1” on page 450	* Varies	VR XS or PCS MMD Invalid Code Group Error Count1 Register The seven error counters EC0-EC6 do not...
“VR_XS_PCS_ICG_ERRCNT2” on page 451	* Varies	VR XS or PCS MMD Invalid Code Group Error Count2 Register The seven error counters EC0-EC6 do not...
“VR_XS_PCS_DSKW_ERRCTR” on page 452	* Varies	VR XS or PCS MMD Check End or Deskew Error Count Register The seven error counters EC0-EC6 do not...
“VR_XS_PCS_TPM_ERRCTR” on page 453	* Varies	VR XS or PCS MMD Test Pattern Mismatch Error Count Register The seven error counters EC0-EC6 do...
“VR_XS_PCS_GPIO” on page 454	* Varies	VR XS or PCS MMD GPIO Register This register is present only if the DWC_xpcs is configured to have...
“VR_XS_PCS_CDT_STS” on page 455	* Varies	VR XS or PCS MMD Comma-Detect Status Register
“VR_XS_PCS_MISC_STS” on page 457	* Varies	VR XS or PCS MMD Miscellaneous Status Register
“VR_XS_PCS_NSNPS_EEE_CTRL” on page 458	* Varies	VR XS or PCS MMD Non-Synopsys PHY EEE Control Register This register is present only in configurations...
“VR_XS_PCS_SFTY_UE_INTR0” on page 460	* Varies	VR XS or PCS MMD Uncorrectable Safety Error Interrupt Register
“VR_XS_PCS_SFTY_ERR_INJ” on page 465	* Varies	VR XS or PCS MMD Safety Error Injection Register

Register	Offset	Description
“VR_XS_PCS_SFTY_CE_INTR” on page 469	* Varies	VR XS or PCS MMD Correctable Safety Error Interrupt Register This register is present only in PCS...
“VR_XS_PCS_SFTY_DBG_CTRL” on page 471	* Varies	VR XS or PCS MMD Safety Debug Control Register This register provides the flexibility to control...
“VR_XS_PCS_SFTY_DISABLE” on page 474	* Varies	VR XS or PCS MMD Safety Disable Register
“VR_XS_PCS_SFTY_TMR_CTRL” on page 475	* Varies	VR XS or PCS MMD Safety Timer Control Register
“VR_XS_PCS_EC_MEM_ECC_STS” on page 477	* Varies	VR XS or PCS MMD EC Memory ECC Status Register. This register is present only in PCS MMD.
“VR_XS_PCS_EF_MEM_ECC_STS” on page 479	* Varies	VR XS or PCS MMD EF Memory ECC Status Register. This register is present only in PCS MMD.
“VR_XS_PCS_SFTY_UE_INTR1” on page 481	* Varies	VR XS or PCS MMD Uncorrectable Safety Error Interrupt Register 1
SERDES Registers Address Block. This register block contains the registers specific (mainly) to Synopsys PHYs. Based on the configuration, these registers are present in XS MMD or PMA MMD. If Mode of Operation = XGXS and XGXS Mode = XGXS in DTE, then these registers are present in DTE XS MMD (MMD Address = 5'h05). If Mode of Operation = XGXS and XGXS Mode = XGXS in PHY, then these registers are present in PHY XS MMD (MMD Address = 5'h04). For other configurations, these registers are present in PMA MMD (MMD Address = 5'h01). XS_PMA_MMD Exists: (PMA_MMD||XGXS_PCS)		
“VR_XS_PMA_RX_LSTS” on page 482	* Varies	VR XS or PMA PHY Rx Lane Status Register In KX_Only, KR_Only, KR_KX, 10GBASE-R PCS, and 1000BASEX-Only...
“VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL0” on page 485	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Tx General 0 Register
“VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1” on page 488	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Tx General 1 Register
“VR_XS_PMA_MP_12G_16G_TX_GENCTRL2” on page 491	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G PHY Tx General 2 Register
“VR_XS_PMA_MP_25G_TX_GENCTRL2” on page 495	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY Tx General 2 Register
“VR_XS_PMA_MP_12G_16G_25G_TX_BOOST_CTRL” on page 498	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Tx Boost Control Register
“VR_XS_PMA_MP_32G_TX_GEN_CTRL3” on page 500	* Varies	VR XS or PMA Synopsys Multi-protocol 32G PHY Tx General Control 3 Register

Register	Offset	Description
“VR_XS_PMA_MP_12G_16G_25G_TX_RATE_CTRL” on page 502	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Tx Rate Control Register
“VR_XS_PMA_MP_32G_TX_GEN_CTRL4” on page 505	* Varies	VR XS or PMA Synopsys Multi-protocol 32G PHY Tx General Control 4 Register
“VR_XS_PMA_MP_12G_16G_25G_TX_POWER_STATE_CTRL” on page 508	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Tx Power State Register
“VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL0” on page 510	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Tx Equalization Control 0 Register
“VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL1” on page 512	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Tx Equalization Control 1 Register
“VR_XS_PMA_MP_6G_TX_BSTCTRL0” on page 515	* Varies	VR XS or PMA Synopsys Multi-protocol 6G PHY Tx Boost Control Register 0
“VR_XS_PMA_MP_6G_TX_BSTCTRL1” on page 516	* Varies	VR XS or PMA Synopsys Multi-protocol 6G PHY Tx Boost Control 1 Register This register is present...
“VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL3” on page 518	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Tx Equalization Control 3 Register
“VR_XS_PMA_MP_6G_TX_LVLCTRL0” on page 520	* Varies	VR XS or PMA Synopsys Multi-protocol 6G PHY Tx Level Control 0 Register
“VR_XS_PMA_MP_12G_16G_25G_EQ_INIT_CTRL0” on page 521	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY INIT Control 0 Register
“VR_XS_PMA_MP_6G_TX_LVLCTRL1” on page 523	* Varies	VR XS or PMA Synopsys Multi-protocol 6G PHY Tx Level Control 1 Register
“VR_XS_PMA_MP_12G_16G_25G_EQ_INIT_CTRL1” on page 525	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY INIT Control 1 Register
“VR_XS_PMA_MP_6G_TXGENCTRL0” on page 526	* Varies	VR XS or PMA Synopsys Multi-protocol 6G PHY Tx General Control 0 Register
“VR_XS_PMA_MP_16G_25G_TX_GENCTRL3” on page 529	* Varies	VR XS or PMA Synopsys Multi-protocol 16G/25G PHY TX General Control 3 Register
“VR_XS_PMA_MP_32G_TX_CM_CNTX_SEL0” on page 530	* Varies	VR XS or PMA Synopsys Multi-protocol 32G Tx Common Context Select 0 Register
“VR_XS_PMA_MP_6G_TXGENCTRL1” on page 531	* Varies	VR XS or PMA Synopsys Multi-protocol 6G PHY Tx General Control 1 Register
“VR_XS_PMA_MP_16G_25G_TX_GENCTRL4” on page 534	* Varies	VR XS or PMA Synopsys Multi-protocol 16G/25G PHY TX General Control 4 Register
“VR_XS_PMA_MP_32G_TX_CM_CNTX_SEL1” on page 535	* Varies	VR XS or PMA Synopsys Multi-protocol 32G Tx Common Context Select 1 Register

Register	Offset	Description
“VR_XS_PMA_MP_6G_TXGENCTRL2” on page 536	* Varies	VR XS or PMA Synopsys Multi-protocol 6G PHY Tx General Control 2 Register
“VR_XS_PMA_MP_32G_TX_CNTX_CTRL0” on page 538	* Varies	VR XS or PMA Synopsys Multi protocol E32 PHY TX Context Control 0 Register
“VR_XS_PMA_MP_6G_TXGENCTRL3” on page 539	* Varies	VR XS or PMA Synopsys Multi-protocol 6G PHY Tx General Control 3 Register
“VR_XS_PMA_MP_16G_25G_TX_MISC_CTRL1” on page 541	* Varies	VR XS or PMA Synopsys Multi-protocol 16G/25G PHY TX Miscellaneous Control 1 Register
“VR_XS_PMA_MP_32G_TX_CNTX_CTRL1” on page 542	* Varies	VR XS or PMA Synopsys Multi Protocol E32 PHY TX Context Control 1 Register
“VR_XS_PMA_MP_12G_16G_25G_TX_STS” on page 543	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Tx Status Register
“VR_XS_PMA_MP_25G_TX_TERM_CTRL” on page 545	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY Transmit Termination Control Register
“VR_XS_PMA_MP_25G_TX_DCC_CTRL” on page 547	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY Tx DCC Control Register
“VR_XS_PMA_MP_25G_TX_WIDTH_CTRL” on page 549	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY Tx Width Control Register
“VR_XS_PMA_MP_6G_TX_STS” on page 551	* Varies	VR XS or PMA Synopsys Multi-protocol 6G PHY Tx Status Register
“VR_XS_PMA_MP_PHY_EQ_PRESET_CTRL0” on page 554	* Varies	VR XS or PMA Synopsys Multi-protocol PHY PRESET Control 0 Register
“VR_XS_PMA_MP_PHY_EQ_PRESET_CTRL1” on page 555	* Varies	VR XS or PMA Synopsys Multi-protocol PHY PRESET Control 1 Register
“VR_XS_PMA_MP_12G_16G_25G_RX_GENCTRL0” on page 556	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Rx General Control 0 Register
“VR_XS_PMA_MP_12G_16G_25G_RX_GENCTRL1” on page 559	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Rx General Control 1 Register
“VR_XS_PMA_MP_12G_16G_RX_GENCTRL2” on page 562	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G PHY Rx General Control 2 Register
“VR_XS_PMA_MP_25G_RX_GENCTRL2” on page 566	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY Rx General Control 2 Register
“VR_XS_PMA_MP_12G_16G_RX_GENCTRL3” on page 568	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G PHY Rx General Control 3 Register
“VR_XS_PMA_MP_25G_RX_GENCTRL3” on page 571	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY Rx General Control 3 Register

Register	Offset	Description
"VR_XS_PMA_Consumer_10G_RX_GENCTRL3" on page 573	* Varies	VR XS or PMA Synopsys Consumer 10G PHY Rx General Control 3 Register
"VR_XS_PMA_MP_12G_16G_25G_RX_RATE_CTRL" on page 575	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Rx Rate Control Register
"VR_XS_PMA_MP_32G_RX_MRGN_CTRL4" on page 578	* Varies	VR XS or PMA Synopsys Multi protocol E32 PHY Rx Margin Control 4 Register
"VR_XS_PMA_MP_12G_16G_25G_RX_POWER_STATE_CTRL" on page 580	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Rx Power State Register
"VR_XS_PMA_MP_12G_16G_25G_RX_CDR_CTRL" on page 583	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Rx CDR Control Register
"VR_XS_PMA_MP_12G_16G_25G_RX_ATTN_CTRL" on page 586	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Rx Attenuation Control Register
"VR_XS_PMA_MP_6G_RXGENCTRL0" on page 588	* Varies	VR XS or PMA Synopsys Multi-protocol 6G PHY Rx General Control 0 Register
"VR_XS_PMA_MP_12G_RX_EQ_CTRL0" on page 591	* Varies	VR XS or PMA Synopsys Multi-protocol 12G PHY Rx Equalization Control 0 Register
"VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL0" on page 593	* Varies	VR XS or PMA Synopsys Multi-protocol 16G/25G PHY Rx Equalization Control 0 Register
"VR_XS_PMA_MP_32G_RX_MRGN_CTRL0" on page 595	* Varies	VR XS or PMA Synopsys Multi protocol E32 PHY Rx Margin Control 0 Register
"VR_XS_PMA_MP_6G_RXGENCTRL1" on page 596	* Varies	VR XS or PMA Synopsys Multi-protocol 6G PHY Rx General Control 1 Register
"VR_XS_PMA_MP_12G_RX_EQ_CTRL1" on page 599	* Varies	VR XS or PMA Synopsys Multi-protocol 12G PHY Rx Equalization Control 1 Register
"VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL1" on page 601	* Varies	VR XS or PMA Synopsys Multi-protocol 16G/25G PHY Rx Equalization Control 1 Register
"VR_XS_PMA_MP_32G_RX_MRGN_CTRL1" on page 603	* Varies	VR XS or PMA Synopsys Multi protocol E32 PHY Rx Margin Control 1 Register
"VR_XS_PMA_MP_12G_RX_EQ_CTRL2" on page 604	* Varies	VR XS or PMA Synopsys Multi-protocol 12G PHY Rx Equalization Control 2 Register
"VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL2" on page 606	* Varies	VR XS or PMA Synopsys Multi-protocol 16G/25G PHY Rx Equalization Control 2 Register
"VR_XS_PMA_MP_32G_RX_MRGN_CTRL2" on page 608	* Varies	VR XS or PMA Synopsys Multi protocol E32 PHY Rx Margin Control 2 Register
"VR_XS_PMA_MP_6G_RXEQ_CTRL" on page 609	* Varies	VR XS or PMA Synopsys Multi-protocol 6G PHY Rx Equalization Control Register All four Rx Equalization...

Register	Offset	Description
“VR_XS_PMA_MP_12G_RX_EQ_CTRL3” on page 611	* Varies	VR XS or PMA Synopsys Multi-protocol 12G PHY Rx Equalization Control 3 Register
“VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL3” on page 613	* Varies	VR XS or PMA Synopsys Multi-protocol 16G/25G PHY Rx Equalization Control 3 Register
“VR_XS_PMA_MP_32G_RX_MRGN_CTRL3” on page 615	* Varies	VR XS or PMA Synopsys Multi protocol E32 PHY Rx Margin Control 3 Register
“VR_XS_PMA_MP_12G_16G_25G_RX_EQ_CTRL4” on page 616	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Rx Equalization Control 4 Register
“VR_XS_PMA_MP_32G_RX_EQ_CTRL4” on page 621	* Varies	VR XS or PMA Synopsys Multi protocol E32 PHY Rx Equalization Control 4 Register
“VR_XS_PMA_MP_6G_RXLOS_CTRL0” on page 624	* Varies	VR XS or PMA Synopsys Multi-protocol 6G PHY Rx Loss-of-Signal Control 0 Register
“VR_XS_PMA_MP_12G_AFE_DFE_EN_CTRL” on page 626	* Varies	VR XS or PMA Synopsys Multi-protocol 12G PHY AFE-DFE Enable Register
“VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL5” on page 628	* Varies	VR XS or PMA Synopsys Multi-protocol 16G/25G PHY Rx Equalization Control 5 Register
“VR_XS_PMA_MP_32G_RX_ADPT_CTRL” on page 631	* Varies	VR XS PMA Synopsys Multi-protocol 32G PHY Rx Adapt Control Register
“VR_XS_PMA_MP_6G_RXLOS_CTRL1” on page 633	* Varies	VR XS or PMA Synopsys Multi-protocol 6G PHY Rx Loss-of-Signal Control 1 Register
“VR_XS_PMA_MP_12G_16G_25G_DFE_TAP_CTRL0” on page 634	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY DFE Tap Control 0 Register
“VR_XS_PMA_MP_12G_16G_25G_DFE_TAP_CTRL1” on page 635	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY DFE Tap Control 1 Register
“VR_XS_PMA_MP_12G_16G_25G_RX_STS” on page 637	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Rx Status Register
“VR_XS_PMA_MP_16G_25G_RX_PPM_STS0” on page 640	* Varies	VR XS or PMA Synopsys Multi-protocol 16G/25G PHY Rx PPM Status 0 Register
“VR_XS_PMA_MP_16G_25G_RX_PPM_STS1” on page 642	* Varies	VR XS or PMA Synopsys Multi-protocol 16G/25G PHY Rx PPM Status 1 Register
“VR_XS_PMA_MP_25G_RX_TERM_CTRL” on page 644	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY Receive Termination Control Register
“VR_XS_PMA_MP_16G_RX_CDR_CTRL1” on page 646	* Varies	VR XS or PMA Synopsys Multi-protocol 16G PHY Rx CDR Control1 Register
“VR_XS_PMA_MP_16G_25G_RX_PPM_CTRL0” on page 649	* Varies	VR XS or PMA Synopsys Multi-protocol 16G/25G PHY Rx PPM Control 0 Register

Register	Offset	Description
“VR_XS_PMA_MP_16G_25G_RX_PPM_CTRL1” on page 651	* Varies	VR XS or PMA Synopsys Multi-protocol 16G/25G PHY Rx PPM Control 1 Register
“VR_XS_PMA_MP_16G_25G_RX_GENCTRL4” on page 653	* Varies	VR XS or PMA Synopsys Multi-protocol 16G/25G PHY Rx General Control 4 Register
“VR_XS_PMA_Consumer_10G_RX_GENCTRL4” on page 655	* Varies	VR XS or PMA Consumer 10G PHY Rx General Control 4 Register
“VR_XS_PMA_MP_16G_25G_RX_MISC_CTRL0” on page 656	* Varies	VR XS or PMA Synopsys Multi-protocol 16G/25G PHY RX Miscellaneous Control 0 Register
“VR_XS_PMA_MP_32G_RX_MRGN_STS0” on page 657	* Varies	VR XS or PMA Synopsys Multi Protocol E32 PHY RX Margin Status 0 Register
“VR_XS_PMA_MP_16G_25G_RX_MISC_CTRL1” on page 658	* Varies	VR XS or PMA Synopsys Multi-protocol 16G/25G PHY RX Miscellaneous Control 1 Register
“VR_XS_PMA_MP_32G_RX_MRGN_STS1” on page 659	* Varies	VR XS or PMA Synopsys Multi Protocol E32 RX Margin Status 1 Register
“VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL0” on page 660	* Varies	VR XS or PMA Synopsys Multi-protocol 16G/25G PHY RX IQ Control 0 Register
“VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL1” on page 662	* Varies	VR XS or PMA Synopsys Multi-protocol 16G/25G PHY RX IQ Control 1 Register
“VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL2” on page 664	* Varies	VR XS or PMA Synopsys Multi-protocol 16G/25G PHY RX IQ Control 2 Register
“VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL3” on page 666	* Varies	VR XS or PMA Synopsys Multi-protocol 16G/25G PHY RX IQ Control 3 Register
“VR_XS_PMA_MP_12G_16G_25G_MPLL_CMN_CTRL” on page 668	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY MPLL Common Control Register
“VR_XS_PMA_MP_12G_16G_MPLLA_CTRL0” on page 669	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G PHY MPLLA Control 0 Register
“VR_XS_PMA_MP_25G_MPLLA_CTRL0” on page 672	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLA Control 0 Register
“VR_XS_PMA_MP_12G_MPLLA_CTRL1” on page 674	* Varies	VR XS or PMA Synopsys Multi-protocol 12G PHY MPLLA Control 1 Register
“VR_XS_PMA_MP_16G_MPLLA_CTRL1” on page 676	* Varies	VR XS or PMA Synopsys Multi-protocol 16G PHY MPLLA Control 1 Register
“VR_XS_PMA_MP_25G_MPLLA_CTRL1” on page 677	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLA Control 1 Register
“VR_XS_PMA_MP_32G_MPLLA_CTRL1” on page 679	* Varies	VR XS or PMA Synopsys Multi Protocol 32G PHY MPLLA Control 1 Register

Register	Offset	Description
“VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2” on page 681	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G PHY MPLLA Control 2 Register
“VR_XS_PMA_MP_25G_MPLLA_CTRL2” on page 684	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLA Control 2 Register
“VR_XS_PMA_MP_12G_16G_MPLLB_CTRL0” on page 686	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G PHY MPLLB Control 0 Register
“VR_XS_PMA_MP_25G_MPLLB_CTRL0” on page 688	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLB Control 0 Register
“VR_XS_PMA_MP_12G_MPLLB_CTRL1” on page 690	* Varies	VR XS or PMA Synopsys Multi-protocol 12G PHY MPLLB Control 1 Register
“VR_XS_PMA_MP_16G_MPLLB_CTRL1” on page 692	* Varies	VR XS or PMA Synopsys Multi-protocol 16G PHY MPLLB Control 1 Register
“VR_XS_PMA_MP_25G_MPLLB_CTRL1” on page 694	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLB Control 1 Register
“VR_XS_PMA_MP_32G_MPLLB_CTRL1” on page 697	* Varies	VR XS or PMA Synopsys Multi Protocol E32 PHY MPLLB Control 1 Register
“VR_XS_PMA_MP_12G_16G_MPLLB_CTRL2” on page 699	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G PHY MPLLB Control 2 Register
“VR_XS_PMA_MP_25G_MPLLB_CTRL2” on page 702	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLB Control 2 Register
“VR_XS_PMA_MP_12G_MPLLA_CTRL3” on page 704	* Varies	VR XS or PMA Synopsys Multi-protocol 12G PHY MPLLA Control 3 Register
“VR_XS_PMA_MP_25G_MPLLA_CTRL3” on page 705	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLA Control 3 Register
“VR_XS_PMA_MP_16G_MPLLA_CTRL3” on page 706	* Varies	VR XS or PMA Synopsys Multi-protocol 16G PHY MPLLA Control 3 Register
“VR_XS_PMA_MP_6G_MPLL_CTRL0” on page 707	* Varies	VR XS or PMA Synopsys Multi-protocol 6G PHY MPLL Control 0 Register
“VR_XS_PMA_MP_12G_MPLLB_CTRL3” on page 709	* Varies	VR XS or PMA Synopsys Multi-protocol 12G PHY MPLLB Control 3 Register
“VR_XS_PMA_MP_25G_MPLLB_CTRL3” on page 710	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLB Control 3 Register
“VR_XS_PMA_MP_16G_MPLLB_CTRL3” on page 711	* Varies	VR XS or PMA Synopsys Multi-protocol 16G PHY MPLLB Control 3 Register
“VR_XS_PMA_MP_6G_MPLL_CTRL1” on page 712	* Varies	VR XS or PMA Synopsys Multi-protocol 6G PHY MPLL Control 1 Register

Register	Offset	Description
“VR_XS_PMA_MP_32G_PLL_CMN_CTRL” on page 714	* Varies	VR XS or PMA Synopsys Multi-protocol 32G PLL Common Control Register
“VR_XS_PMA_MP_16G_MPLLA_CTRL4” on page 716	* Varies	VR XS or PMA Synopsys Multi-protocol 16G PHY MPLLA Control 4 Register
“VR_XS_PMA_MP_25G_MPLLA_CTRL4” on page 717	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLA Control 4 Register
“VR_XS_PMA_MP_25G_MPLLB_CTRL4” on page 718	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLB Control 4 Register
“VR_XS_PMA_MP_16G_MPLLA_CTRL5” on page 719	* Varies	VR XS or PMA Synopsys Multi-protocol 16G PHY MPLLA Control 5 Register
“VR_XS_PMA_MP_16G_MPLLB_CTRL4” on page 720	* Varies	VR XS or PMA Synopsys Multi-protocol 16G PHY MPLLB Control 4 Register
“VR_XS_PMA_MP_25G_MPLLA_CTRL5” on page 721	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLA Control 5 Register
“VR_XS_PMA_MP_16G_MPLLB_CTRL5” on page 722	* Varies	VR XS or PMA Synopsys Multi-protocol 16G PHY MPLLB Control 5 Register
“VR_XS_PMA_MP_25G_MPLLB_CTRL5” on page 723	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLB Control 5 Register
“VR_XS_PMA_MP_8G_MPLLA_CTRL6” on page 724	* Varies	VR XS or PMA Synopsys Multi-protocol 8G PHY MPLLA Control 6 Register
“VR_XS_PMA_MP_25G_MPLLA_BW_HI_CTRL” on page 725	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLA Bandwidth High Control Register
“VR_XS_PMA_MP_25G_MPLLA_BW_LO_CTRL” on page 726	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLA Bandwidth Low Control Register
“VR_XS_PMA_MP_8G_MPLLA_CTRL7” on page 727	* Varies	VR XS or PMA Synopsys Multi-protocol 8G PHY MPLLA Control 7 Register
“VR_XS_PMA_MP_25G_MPLLB_BW_HI_CTRL” on page 728	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLB Bandwidth High Control Register
“VR_XS_PMA_MP_25G_MPLLB_BW_LO_CTRL” on page 729	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLB Bandwidth Low Control Register
“VR_XS_PMA_MP_6G_MPLL_STS” on page 730	* Varies	VR XS or PMA Synopsys Multi-protocol 6G PHY MPLL Status Register
“VR_XS_PMA_MP_12G_16G_25G_MISC_CTRL0” on page 731	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Miscellaneous Control 0 Register
“VR_XS_PMA_MP_12G_16G_25G_REF_CLK_CTRL” on page 734	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Reference Control Register

Register	Offset	Description
“VR_XS_PMA_MP_6G_EEE_CTRL” on page 738	* Varies	VR XS or PMA Synopsys Multi-protocol 6G PHY EEE Control Register
“VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD0” on page 739	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY VCO Calibration Load 0 Register
“VR_XS_PMA_MP_32G_RX_CNTX_CTRL0” on page 741	* Varies	VR XS or PMA Synopsys Multi protocol E32 PHY RX Context Control 0 Register
“VR_XS_PMA_MP_6G_MISC_CTRL2” on page 742	* Varies	VR XS or PMA Synopsys Multi-protocol 6G PHY Miscellaneous Control 2 Register
“VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD1” on page 743	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY VCO Calibration Load 1 Register
“VR_XS_PMA_MP_32G_RX_CNTX_CTRL1” on page 744	* Varies	VR XS or PMA Synopsys Multi protocol E32 PHY RX Contex Control 1 Register
“VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD2” on page 745	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G PHY VCO Calibration Load 2 Register
“VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD3” on page 746	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY VCO Calibration Load 3 Register
“VR_XS_PMA_MP_32G_REF_CLK_CTRL0” on page 747	* Varies	VR XS or PMA Synopsys Multi-protocol 32G Reference clock control 0 Register
“VR_XS_PMA_MP_12G_VCO_CAL_REF0” on page 749	* Varies	VR XS or PMA Synopsys Multi-protocol 12G PHY VCO Calibration Reference 0 Register
“VR_XS_PMA_MP_16G_25G_VCO_CAL_REF0” on page 751	* Varies	VR XS or PMA Synopsys Multi-protocol 16G/25G PHY VCO Calibration Reference 0 Register
“VR_XS_PMA_MP_32G_REF_CLK_CTRL1” on page 753	* Varies	VR XS or PMA Synopsys Multi-protocol 32G PHY Reference Clock Control 1 Register
“VR_XS_PMA_MP_12G_VCO_CAL_REF1” on page 755	* Varies	VR XS or PMA Synopsys Multi-protocol 12G PHY VCO Calibration Reference 1 Register
“VR_XS_PMA_MP_16G_25G_VCO_CAL_REF1” on page 757	* Varies	VR XS or PMA Synopsys Multi-protocol 16G/25G PHY VCO Calibration Reference 1 Register
“VR_XS_PMA_MP_32G_LN_LINK_CTRL” on page 759	* Varies	VR XS or PMA Synopsys Multi protocol E32 PHY Lane Link Control Register
“VR_XS_PMA_MP_6G_LVL_CTRL” on page 761	* Varies	VR XS or PMA Synopsys Multi-protocol 6G PHY Level Control Register
“VR_XS_PMA_MP_12G_16G_25G_MISC_STS” on page 763	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Miscellaneous Status Register
“VR_XS_PMA_MP_6G_MISC_CTRL0” on page 765	* Varies	VR XS or PMA Synopsys Multi-protocol 6G PHY Miscellaneous Control 0 Register

Register	Offset	Description
"VR_XS_PMA_MP_12G_16G_25G_MISC_CTRL1" on page 767	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Miscellaneous Control 1 Register
"VR_XS_PMA_MP_6G_MISC_CTRL1" on page 768	* Varies	VR XS or PMA Synopsys Multi-protocol 6G PHY Miscellaneous Control 1 Register
"VR_XS_PMA_MP_12G_16G_25G_EEE_CTRL" on page 772	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY EEE Control Register This register is present...
"VR_XS_PMA_MP_12G_16G_25G_SRAM" on page 774	* Varies	VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY SRAM Register. Note: This is a special register...
"VR_XS_PMA_MP_16G_25G_MISC_CTRL2" on page 776	* Varies	VR XS or PMA Synopsys Multi-protocol 16G/25G PHY Miscellaneous Control 2 Register
"VR_XS_PMA_MP_32G_CNTX_CTRL" on page 778	* Varies	VR XS or PMA Synopsys Multi-protocol 32G PHY Context Control Register
"VR_XS_PMA_MP_32G_MISC_STS1" on page 779	* Varies	VR XS or PMA Synopsys Multi-protocol 32G PHY Miscellaneous Status 1 Register
"VR_XS_PMA_SNPS_CR_CTRL" on page 781	* Varies	VR XS or PMA Synopsys PHY CR Control Register
"VR_XS_PMA_SNPS_CR_ADDR" on page 783	* Varies	VR XS or PMA Synopsys PHY CR Address Register
"VR_XS_PMA_SNPS_CR_DATA" on page 784	* Varies	VR XS or PMA Synopsys CR Data Register
"VR_XS_PMA_MP_25G_RX_WIDTH_CTRL" on page 785	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY Rx Width Control Register
"VR_XS_PMA_MP_25G_RX_AFE_RATE_CTRL" on page 787	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY Rx AFE Rate Control Register
"VR_XS_PMA_MP_25G_RX_RECAL_CTRL" on page 789	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY Rx Re-calibration Control Register
"VR_XS_PMA_MP_25G_RX_DCC_CTRL" on page 791	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY Rx DCC Control Register
"VR_XS_PMA_MP_25G_RX_VCO_CFG0" on page 793	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY Rx CDR VCO Configuration 0 Register
"VR_XS_PMA_MP_25G_RX_VCO_CFG1" on page 794	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY Rx CDR VCO Configuration 1 Register
"VR_XS_PMA_MP_25G_RX_VCO_CFG2" on page 795	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY Rx CDR VCO Configuration 2 Register
"VR_XS_PMA_MP_25G_RX_VCO_CFG3" on page 796	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY Rx CDR VCO Configuration 3 Register
"VR_XS_PMA_MP_25G_RX_LF_TRSHLD" on page 797	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY RX Low Frequency Threshold Register

Register	Offset	Description
"VR_XS_PMA_MP_25G_RX_HF_TRSHLD" on page 799	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY RX High Frequency Threshold Register
"VR_XS_PMA_C16G_RX_EQ_AFE_CTRL0" on page 801	* Varies	VR XS or PMA Synopsys Consumer 16G PHY RX Equalization AFE Control 0 Register
"VR_XS_PMA_C16G_RX_EQ_AFE_CTRL1" on page 803	* Varies	VR XS or PMA Synopsys Consumer 16G PHY RX Equalization AFE Control 1 Register
Auto-Negotiation (AN) MMD Standard (SR) & Vendor Specific (VR) address block AN_MMD Exists: (BACKPLANE_ETHERNET_PCS)		
"SR_AN_CTRL" on page 805	* Varies	SR AN MMD Control Register
"SR_AN_STS" on page 808	* Varies	SR AN MMD Status Register
"SR_AN_DEV_ID1" on page 811	* Varies	SR AN MMD Device Identifier Register 1
"SR_AN_DEV_ID2" on page 812	* Varies	SR AN MMD Device Identifier Register 2
"SR_AN_DEV_PKG1" on page 814	* Varies	SR AN MMD Devices in Package Register 1
"SR_AN_DEV_PKG2" on page 816	* Varies	SR AN MMD Devices in Package Register 2
"SR_AN_PKG1" on page 817	* Varies	SR AN MMD Package Identifier Register 1
"SR_AN_PKG2" on page 818	* Varies	SR AN MMD Package Identifier Register 2
"SR_AN_ADV1" on page 820	* Varies	SR AN MMD Advertisement Register 1 The AN Advertisement Register 1, 2, and 3 contain the Base Page...
"SR_AN_ADV2" on page 822	* Varies	SR AN MMD Advertisement Register 2
"SR_AN_ADV3" on page 823	* Varies	SR AN MMD Advertisement Register 3
"SR_AN_LP_ABL1" on page 824	* Varies	SR AN MMD LP Base Page Ability Register 1 The management uses the SR AN LP Base Page Ability Registers1,...
"SR_AN_LP_ABL2" on page 826	* Varies	SR AN MMD LP Base Page Ability Register 2
"SR_AN_LP_ABL3" on page 827	* Varies	SR AN MMD LP Base Page Ability Register 3
"SR_AN_XNP_TX1" on page 828	* Varies	SR AN MMD XNP Transmit Register 1 This register holds the next page Link Code Word to be transmitted...
"SR_AN_XNP_TX2" on page 830	* Varies	SR AN MMD XNP Transmit Register 2
"SR_AN_XNP_TX3" on page 831	* Varies	SR AN MMD XNP Transmit Register 3
"SR_AN_LP_XNP_ABL1" on page 832	* Varies	SR AN MMD LP XNP Ability Register 1 The management uses the AN LP XNP Ability Register 1, 2, or...
"SR_AN_LP_XNP_ABL2" on page 834	* Varies	SR AN MMD LP XNP Ability Register 2

Register	Offset	Description
“SR_AN_LP_XNP_ABL3” on page 835	* Varies	SR AN MMD LP XNP Ability Register 3
“SR_AN_COMP_STS” on page 836	* Varies	SR AN MMD Completion Status Register
“SR_AN_EEE_ABL1” on page 838	* Varies	SR AN MMD EEE Capability 1 Register
“SR_AN_EEE_LP_ABL1” on page 840	* Varies	SR AN MMD EEE Link Partner Capability Register
“SR_AN_EEE_ABL2” on page 842	* Varies	SR AN MMD EEE Capability 2 Register
“SR_AN_EEE_LP_ABL2” on page 844	* Varies	SR AN MMD EEE Link Partner Capability 2 Register
“VR_AN_DIG_CTRL1” on page 846	* Varies	VR AN MMD Digital Control1 Register
“VR_AN_INTR_MSK” on page 850	* Varies	VR AN MMD Interrupt Mask Register
“VR_AN_INTR” on page 851	* Varies	VR AN MMD Interrupt Register Note: The interrupt output port 'sbd_intr_o' goes high when any bit...
“VR_AN_KR_MODE_CTRL” on page 853	* Varies	VR AN MMD KR Mode Control Register
“VR_AN_TIMER_CTRL0” on page 854	* Varies	VR AN MMD Timer Control 0 Register
“VR_AN_TIMER_CTRL1” on page 855	* Varies	VR AN MMD Timer Control 1 Register
“VR_AN_DIG_STS” on page 857	* Varies	VR AN MMD Digital Status Register
Vendor Specific MII_1 MMD Standard (SR) & Vendor Specific (VR) Registers address block This MMD is present if QSGMII support or Multi-port USXGMII support is selected during the controller configuration. VS_MII_1_MMD Exists: QSGMII_EN||MP_USXGMII		
“SR_MII_1_CTRL” on page 860	* Varies	SR MII_1 MMD Control Register The host can use this register to control (enable or disable) some...
“SR_MII_1_STS” on page 865	* Varies	SR MII_1 MMD Status Register
“SR_MII_1_DEV_ID1” on page 869	* Varies	SR MII_1 MMD Device Identifier Register 1 This register returns the configurable Organizationally...
“SR_MII_1_DEV_ID2” on page 870	* Varies	SR MII_1 MMD Device Identifier Register 2 This register returns the configurable Organizationally...
“SR_MII_1_AN_ADV” on page 872	* Varies	SR MII_1 MMD AN Advertisement Register Except for bit[5], rest of the fields of this register reflects...
“SR_MII_1_LP_BABL” on page 874	* Varies	SR MII_1 MMD AN Link Partner Base Ability Register This register merely reflects the value of SR_MII_LP_BABL...
“SR_MII_1_AN_EXPN” on page 876	* Varies	SR MII_1 MMD AN Expansion Register
“SR_MII_1_EXT_STS” on page 877	* Varies	SR MII_1 MMD Extended Status Register This register reflects the same value as that of SR_MII_EX_STS...
“SR_MII_1_TIME_SYNC_ABL” on page 878	* Varies	SR MII 1 MMD Time Sync Capability Register

Register	Offset	Description
“SR_MII_1_TIME_SYNC_TX_MAX_DLY_LWR” on page 881	* Varies	SR MII 1 MMD Time Sync Tx Max Delay Lower Register
“SR_MII_1_TIME_SYNC_TX_MAX_DLY_UPR” on page 882	* Varies	SR MII 1 MMD Time Sync Tx Max Delay Upper Register
“SR_MII_1_TIME_SYNC_TX_MIN_DLY_LWR” on page 883	* Varies	SR MII 1 MMD Time Sync Tx Min Delay Lower Register
“SR_MII_1_TIME_SYNC_TX_MIN_DLY_UPR” on page 884	* Varies	SR MII 1 MMD Time Sync Tx Min Delay Upper Register
“SR_MII_1_TIME_SYNC_RX_MAX_DLY_LWR” on page 885	* Varies	SR MII 1 MMD Time Sync Rx Max Delay Lower Register
“SR_MII_1_TIME_SYNC_RX_MAX_DLY_UPR” on page 886	* Varies	SR MII 1 MMD Time Sync Rx Max Delay Upper Register
“SR_MII_1_TIME_SYNC_RX_MIN_DLY_LWR” on page 887	* Varies	SR MII 1 MMD Time Sync Rx Min Delay Lower Register
“SR_MII_1_TIME_SYNC_RX_MIN_DLY_UPR” on page 888	* Varies	SR MII 1 MMD Time Sync Rx Min Delay Upper Register
“SR_MII_1_TIME_SYNC_TX_MAX_FINE_DLY” on page 889	* Varies	SR MII 1 MMD Time Sync Tx Max Fine Delay Register
“SR_MII_1_TIME_SYNC_TX_MIN_FINE_DLY” on page 890	* Varies	SR MII 1 MMD Time Sync Tx Min Fine Delay Register
“SR_MII_1_TIME_SYNC_RX_MAX_FINE_DLY” on page 891	* Varies	SR MII 1 MMD Time Sync Rx Max Fine Delay Register
“SR_MII_1_TIME_SYNC_RX_MIN_FINE_DLY” on page 892	* Varies	SR MII 1 MMD Time Sync Rx Min Fine Delay Register
“SR_MII_1_TIME_SYNC_CFG” on page 893	* Varies	SR MII 1 MMD Time Sync Configuration Register
“VR_MII_1_DIG_CTRL1” on page 894	* Varies	VR MII_1 MMD Digital Control1 Register
“VR_MII_1_AN_CTRL” on page 901	* Varies	VR MII_1 MMD AN Control Register
“VR_MII_1_AN_INTR_STS” on page 903	* Varies	VR MII_1 MMD AN Interrupt and Status Register
“VR_MII_1_TC” on page 906	* Varies	VR MII_1 MMD Test Control Register. Note: This register is present only in 1000BaseX-Only PCS configurations...
“VR_MII_1_DIG_CTRL3” on page 907	* Varies	VR MII_1 MMD Digital Control 3 Register
“VR_MII_1_DBG_CTRL” on page 910	* Varies	VR MII_1 MMD Debug Control Register The contents of this register is valid only in QSGMII...
“VR_MII_1_EEE_MCTRL0” on page 912	* Varies	VR MII_1 MMD EEE Mode Control Register This register is present only for configurations with MAIN_MODE...

Register	Offset	Description
"VR_MII_1_EEE_TXTIMER" on page 914	* Varies	VR MII_1 MMD EEE Tx Timer Register This register is present only for configurations with MAIN_MODE...
"VR_MII_1_EEE_RXTIMER" on page 916	* Varies	VR MII_1 MMD EEE Rx Timer Register This register is present only for configurations with MAIN_MODE...
"VR_MII_1_LINK_TIMER_CTRL" on page 917	* Varies	VR MII_1 MMD Link Timer Control Register This register merely reflects the value of VR_MII_LINK_TIMER_CTRL...
"VR_MII_1_EEE_MCTRL1" on page 918	* Varies	VR MII_1 MMD EEE Mode Control 1 Register This register is present only for configurations with...
"VR_MII_1_DIG_STS" on page 919	* Varies	VR MII_1 MMD Digital Status Register.
"VR_MII_1_ICG_ERRCNT1" on page 923	* Varies	VR MII_1 MMD Invalid Code Group Error Count1 Register
"VR_MII_1_GPIO" on page 924	* Varies	VR MII_2 MMD GPIO Register This register is present only if the DWC_xpcs is configured to have...
"VR_MII_1_DIG_CTRL2" on page 925	* Varies	VR MII_1 MMD Digital Control2 Register Note: This register is present only in 1000BaseX-Only PCS...
"VR_MII_1_DIG_ERRCNT_SEL" on page 926	* Varies	VR MII_1 MMD Digital Error Count Select Register This register is applicable only in QSGMII...
Vendor Specific MII_2 MMD Standard (SR) & Vendor Specific (VR) Registers address block This MMD is present if QSGMII_EN support or Quad-Port USXGMII support is selected during the controller configuration. VS_MII_2_MMD Exists: QSGMII_EN||USXG_QPORT		
"SR_MII_2_CTRL" on page 928	* Varies	SR MII_2 MMD Control Register The host can use this register to control (enable or disable) some...
"SR_MII_2_STS" on page 933	* Varies	SR MII_2 MMD Status Register
"SR_MII_2_DEV_ID1" on page 937	* Varies	SR MII_2 MMD Device Identifier Register 1 This register returns the configurable Organizationally...
"SR_MII_2_DEV_ID2" on page 938	* Varies	SR MII_2 MMD Device Identifier Register 2 This register returns the configurable Organizationally...
"SR_MII_2_AN_ADV" on page 940	* Varies	SR MII_2 MMD AN Advertisement Register Except for bit[5], rest of the fields of this register reflects...
"SR_MII_2_LP_BABL" on page 942	* Varies	SR MII_2 MMD AN Link Partner Base Ability Register This register merely reflects the value of SR_MII_LP_BABL...
"SR_MII_2_AN_EXPN" on page 944	* Varies	SR MII_2 MMD AN Expansion Register
"SR_MII_2_EXT_STS" on page 945	* Varies	SR MII_2 MMD Extended Status Register This register reflects the same value as that of SR_MII_EX_STS...
"SR_MII_2_TIME_SYNC_PCS_ABL" on page 946	* Varies	SR MII 2 MMD Time Sync Capability Register

Register	Offset	Description
“SR_MII_2_TIME_SYNC_TX_MAX_DLY_LWR” on page 949	* Varies	SR MII 2 MMD Time Sync Tx Max Delay Lower Register
“SR_MII_2_TIME_SYNC_TX_MAX_DLY_UPR” on page 950	* Varies	SR MII 2 MMD Time Sync Tx Max Delay Upper Register
“SR_MII_2_TIME_SYNC_TX_MIN_DLY_LWR” on page 951	* Varies	SR MII 2 MMD Time Sync Tx Min Delay Lower Register
“SR_MII_2_TIME_SYNC_TX_MIN_DLY_UPR” on page 952	* Varies	SR MII 2 MMD Time Sync Tx Min Delay Upper Register
“SR_MII_2_TIME_SYNC_RX_MAX_DLY_LWR” on page 953	* Varies	SR MII 2 MMD Time Sync Rx Max Delay Lower Register
“SR_MII_2_TIME_SYNC_RX_MAX_DLY_UPR” on page 954	* Varies	SR MII 2 MMD Time Sync Rx Max Delay Upper Register
“SR_MII_2_TIME_SYNC_RX_MIN_DLY_LWR” on page 955	* Varies	SR MII 2 MMD Time Sync Rx Min Delay Lower Register
“SR_MII_2_TIME_SYNC_RX_MIN_DLY_UPR” on page 956	* Varies	SR MII 2 MMD Time Sync Rx Min Delay Upper Register
“SR_MII_2_TIME_SYNC_TX_MAX_FINE_DLY” on page 957	* Varies	SR MII 2 MMD Time Sync Tx Max Fine Delay Register
“SR_MII_2_TIME_SYNC_TX_MIN_FINE_DLY” on page 958	* Varies	SR MII 2 MMD Time Sync Tx Min Fine Delay Register
“SR_MII_2_TIME_SYNC_RX_MAX_FINE_DLY” on page 959	* Varies	SR MII 2 MMD Time Sync Rx Max Fine Delay Register
“SR_MII_2_TIME_SYNC_RX_MIN_FINE_DLY” on page 960	* Varies	SR MII 2 MMD Time Sync Rx Min Fine Delay Register
“SR_MII_2_TIME_SYNC_CFG” on page 961	* Varies	SR MII 2 MMD Time Sync Configuration Register
“VR_MII_2_DIG_CTRL1” on page 962	* Varies	VR MII_2 MMD Digital Control1 Register
“VR_MII_2_AN_CTRL” on page 969	* Varies	VR MII_2 MMD AN Control Register
“VR_MII_2_AN_INTR_STS” on page 971	* Varies	VR MII_2 MMD AN Interrupt and Status Register
“VR_MII_2_TC” on page 974	* Varies	VR MII_2 MMD Test Control Register. Note: This register is present only in 1000BaseX-Only PCS configurations...
“VR_MII_2_DIG_CTRL3” on page 975	* Varies	VR MII_2 MMD Digital Control 3 Register
“VR_MII_2_DBG_CTRL” on page 978	* Varies	VR MII_2 MMD Debug Control Register
“VR_MII_2_EEE_MCTRL0” on page 980	* Varies	VR MII_2 MMD EEE Mode Control Register This register is present only for configurations with MAIN_MODE...

Register	Offset	Description
"VR_MII_2_EEE_TXTIMER" on page 982	* Varies	VR MII_2 MMD EEE Tx Timer Register This register is present only for configurations with MAIN_MODE...
"VR_MII_2_EEE_RXTIMER" on page 984	* Varies	VR MII_2 MMD EEE Rx Timer Register This register is present only for configurations with MAIN_MODE...
"VR_MII_2_LINK_TIMER_CTRL" on page 985	* Varies	VR MII_2 MMD Link Timer Control Register This register merely reflects the value of VR_MII_LINK_TIMER_CTRL...
"VR_MII_2_EEE_MCTRL1" on page 986	* Varies	VR MII_2 MMD EEE Mode Control 1 Register This register is present only for configurations with...
"VR_MII_2_DIG_STS" on page 987	* Varies	VR MII_2 MMD Digital Status Register.
"VR_MII_2_ICG_ERRCNT1" on page 991	* Varies	VR MII_2 MMD Invalid Code Group Error Count1 Register
"VR_MII_2_GPIO" on page 992	* Varies	VR MII_2 MMD GPIO Register This register is present only if the DWC_xpcs is configured to have...
"VR_MII_2_DIG_CTRL2" on page 993	* Varies	VR MII_2 MMD Digital Control2 Register Note: This register is present only in 1000BaseX-Only PCS...
"VR_MII_2_DIG_ERRCNT_SEL" on page 994	* Varies	VR MII_2 MMD Digital Error Count Select Register This register is applicable only in QSGMII...
Vendor Specific MII_3 MMD Standard (SR) & Vendor Specific (VR) Registers address block This MMD is present if QSGMII_EN support or Quad-port USXGMII support is selected during the configuration. VS_MII_3_MMD Exists: QSGMII_EN||USXG_QPORT		
"SR_MII_3_CTRL" on page 996	* Varies	SR MII_3 MMD Control Register The host can use this register to control (enable or disable) some...
"SR_MII_3_STS" on page 1001	* Varies	SR MII_3 MMD Status Register
"SR_MII_3_DEV_ID1" on page 1005	* Varies	SR MII_3 MMD Device Identifier Register 1 This register returns the configurable Organizationally...
"SR_MII_3_DEV_ID2" on page 1006	* Varies	SR MII_3 MMD Device Identifier Register 2 This register returns the configurable Organizationally...
"SR_MII_3_AN_ADV" on page 1008	* Varies	SR MII_3 MMD AN Advertisement Register Except for bit[5], rest of the fields of this register reflects...
"SR_MII_3_LP_BABL" on page 1010	* Varies	SR MII_3 MMD AN Link Partner Base Ability Register This register merely reflects the value of SR_MII_LP_BABL...
"SR_MII_3_AN_EXPN" on page 1012	* Varies	SR MII_3 MMD AN Expansion Register
"SR_MII_3_EXT_STS" on page 1013	* Varies	SR MII_3 MMD Extended Status Register This register reflects the same value as that of SR_MII_EX_STS...
"SR_MII_3_TIME_SYNC_PCS_ABL" on page 1014	* Varies	SR MII 3 MMD Time Sync Capability Register

Register	Offset	Description
“SR_MII_3_TIME_SYNC_TX_MAX_DLY_LWR” on page 1017	* Varies	SR MII 3 MMD Time Sync Tx Max Delay Lower Register
“SR_MII_3_TIME_SYNC_TX_MAX_DLY_UPR” on page 1018	* Varies	SR MII 3 MMD Time Sync Tx Max Delay Upper Register
“SR_MII_3_TIME_SYNC_TX_MIN_DLY_LWR” on page 1019	* Varies	SR MII 3 MMD Time Sync Tx Min Delay Lower Register
“SR_MII_3_TIME_SYNC_TX_MIN_DLY_UPR” on page 1020	* Varies	SR MII 3 MMD Time Sync Tx Min Delay Upper Register
“SR_MII_3_TIME_SYNC_RX_MAX_DLY_LWR” on page 1021	* Varies	SR MII 3 MMD Time Sync Rx Max Delay Lower Register
“SR_MII_3_TIME_SYNC_RX_MAX_DLY_UPR” on page 1022	* Varies	SR MII 3 MMD Time Sync Rx Max Delay Upper Register
“SR_MII_3_TIME_SYNC_RX_MIN_DLY_LWR” on page 1023	* Varies	SR MII 3 MMD Time Sync Rx Min Delay Lower Register
“SR_MII_3_TIME_SYNC_RX_MIN_DLY_UPR” on page 1024	* Varies	SR MII 3 MMD Time Sync Rx Min Delay Upper Register
“SR_MII_3_TIME_SYNC_TX_MAX_FINE_DLY” on page 1025	* Varies	SR MII 3 MMD Time Sync Tx Max Fine Delay Register
“SR_MII_3_TIME_SYNC_TX_MIN_FINE_DLY” on page 1026	* Varies	SR MII 3 MMD Time Sync Tx Min Fine Delay Register
“SR_MII_3_TIME_SYNC_RX_MAX_FINE_DLY” on page 1027	* Varies	SR MII 3 MMD Time Sync Rx Max Fine Delay Register
“SR_MII_3_TIME_SYNC_RX_MIN_FINE_DLY” on page 1028	* Varies	SR MII 3 MMD Time Sync Rx Min Fine Delay Register
“SR_MII_3_TIME_SYNC_CFG” on page 1029	* Varies	SR MII 3 MMD Time Sync Configuration Register
“VR_MII_3_DIG_CTRL1” on page 1030	* Varies	VR MII_3 MMD Digital Control1 Register
“VR_MII_3_AN_CTRL” on page 1037	* Varies	VR MII_3 MMD AN Control Register
“VR_MII_3_AN_INTR_STS” on page 1039	* Varies	VR MII_3 MMD AN Interrupt and Status Register
“VR_MII_3_TC” on page 1043	* Varies	VR MII_3 MMD Test Control Register. Note: This register is present only in 1000BaseX-Only PCS...
“VR_MII_3_DIG_CTRL3” on page 1044	* Varies	VR MII_3 MMD Digital Control 3 Register
“VR_MII_3_DBG_CTRL” on page 1047	* Varies	VR MII_3 MMD Debug Control Register
“VR_MII_3_EEE_MCTRL0” on page 1049	* Varies	VR MII_3 MMD EEE Mode Control Register This register is present only for configurations with MAIN_MODE...

Register	Offset	Description
"VR_MII_3_EEE_TXTIMER" on page 1051	* Varies	VR MII_3 MMD EEE Tx Timer Register This register is present only for configurations with MAIN_MODE...
"VR_MII_3_EEE_RXTIMER" on page 1053	* Varies	VR MII_3 MMD EEE Rx Timer Register This register is present only for configurations with MAIN_MODE...
"VR_MII_3_LINK_TIMER_CTRL" on page 1054	* Varies	VR MII_3 MMD Link Timer Control Register This register merely reflects the value of VR_MII_LINK_TIMER_CTRL...
"VR_MII_3_EEE_MCTRL1" on page 1055	* Varies	VR MII_3 MMD EEE Mode Control 1 Register This register is present only for configurations with...
"VR_MII_3_DIG_STS" on page 1056	* Varies	VR MII_3 MMD Digital Status Register.
"VR_MII_3_ICG_ERRCNT1" on page 1060	* Varies	VR MII_3 MMD Invalid Code Group Error Count1 Register
"VR_MII_3_GPIO" on page 1061	* Varies	VR MII_3 MMD GPIO Register This register is present only if the DWC_xpcs is configured to have...
"VR_MII_3_DIG_CTRL2" on page 1062	* Varies	VR MII_2 MMD Digital Control2 Register Note: This register is present only in 1000BaseX-Only PCS...
"VR_MII_3_DIG_ERRCNT_SEL" on page 1063	* Varies	VR MII_3 MMD Digital Error Count Select Register This register is applicable only in QSGMII...
Vendor Specific MMD1 Registers address block VS_MMD1 Exists: Always		
"SR_VSMMD_PMA_ID1" on page 1065	* Varies	SR Control MMD PMA Device Identifier Register 1
"SR_VSMMD_PMA_ID2" on page 1066	* Varies	SR Control MMD PMA Device Identifier Register 2
"SR_VSMMD_DEV_ID1" on page 1068	* Varies	SR Control MMD Device Identifier Register 1
"SR_VSMMD_DEV_ID2" on page 1069	* Varies	SR Control MMD Device Identifier Register 2
"SR_VSMMD_PCS_ID1" on page 1071	* Varies	SR Control MMD PCS Device Identifier Register 1
"SR_VSMMD_PCS_ID2" on page 1072	* Varies	SR Control MMD PCS Device Identifier Register 2
"SR_VSMMD_AN_ID1" on page 1074	* Varies	SR Control MMD AN Device Identifier Register 1
"SR_VSMMD_AN_ID2" on page 1075	* Varies	SR Control MMD AN Device Identifier Register 2
"SR_VSMMD_STS" on page 1077	* Varies	SR Control MMD Status Register
"SR_VSMMD_CTRL" on page 1078	* Varies	SR Control MMD Control Register
"SR_VSMMD_PKGID1" on page 1081	* Varies	SR Control MMD Package Identifier Register 1
"SR_VSMMD_PKGID2" on page 1082	* Varies	SR Control MMD Package Identifier Register 2 Note: The values written in the fields of this register...

Register	Offset	Description
Vendor Specific MII MMD Standard (SR) & Vendor Specific (VR) Registers address block This MMD is present if CL37_AN = Enabled.Synopsys PHY Registers are present in MII MMD only when MAIN_MODE=1000BaseX-Only PCS. VS_MII_MMD Exists: (CL37_AN||BASE1000X_ONLY)		
"SR_MII_CTRL" on page 1084	* Varies	SR MII MMD Control Register The host can use this register to control (enable or disable) some...
"SR_MII_STS" on page 1089	* Varies	SR MII MMD Status Register The host uses this register to know the features supported by the DWC_xpcs...
"SR_MII_DEV_ID1" on page 1093	* Varies	SR MII MMD Device Identifier Register 1 This register returns the configurable Organizationally...
"SR_MII_DEV_ID2" on page 1094	* Varies	SR MII MMD Device Identifier Register 2 This register returns the configurable Organizationally...
"SR_MII_AN_ADV" on page 1096	* Varies	SR MII MMD AN Advertisement Register The host uses this register to advertise the abilities and...
"SR_MII_LP_BABL" on page 1098	* Varies	SR MII MMD AN Link Partner Base Ability Register The host uses this page to know the link partner's...
"SR_MII_AN_EXPN" on page 1100	* Varies	SR MII MMD AN Expansion Register
"SR_MII_EXT_STS" on page 1101	* Varies	SR MII MMD Extended Status Register This register is present only for configurations with 1G/KX...
"SR_MII_TIME_SYNC_ABL" on page 1103	* Varies	SR MII MMD Time Sync Capability Register. This register is present only in 1000BaseX-Only PCS...
"SR_MII_TIME_SYNC_TX_MAX_DLY_LWR" on page 1106	* Varies	SR MII MMD Time Sync Tx Max Delay Lower Register. This register is present only in 1000BaseX-Only...
"SR_MII_TIME_SYNC_TX_MAX_DLY_UPR" on page 1107	* Varies	SR MII MMD Time Sync Tx Max Delay Upper Register. This register is present only in 1000BaseX-Only...
"SR_MII_TIME_SYNC_TX_MIN_DLY_LWR"on page 1108	* Varies	SR MII MMD Time Sync Tx Min Delay Lower Register. This register is present only in 1000BaseX-Only...
"SR_MII_TIME_SYNC_TX_MIN_DLY_UPR"on page 1109	* Varies	SR MII MMD Time Sync Tx Min Delay Upper Register. This register is present only in 1000BaseX-Only...
"SR_MII_TIME_SYNC_RX_MAX_DLY_LWR" on page 1110	* Varies	SR MII MMD Time Sync Rx Max Delay Lower Register. This register is present only in 1000BaseX-Only...
"SR_MII_TIME_SYNC_RX_MAX_DLY_UPR" on page 1111	* Varies	SR MII MMD Time Sync Rx Max Delay Upper Register. This register is present only in 1000BaseX-Only...
"SR_MII_TIME_SYNC_RX_MIN_DLY_LWR" on page 1112	* Varies	SR MII MMD Time Sync Rx Min Delay Lower Register. This register is present only in 1000BaseX-Only...
"SR_MII_TIME_SYNC_RX_MIN_DLY_UPR"on page 1113	* Varies	SR MII MMD Time Sync Rx Min Delay Upper Register This register is present only in 1000BaseX-Only...

Register	Offset	Description
“SR_MII_TIME_SYNC_CFG” on page 1114	* Varies	SR MII MMD Time Sync Configuration Register
“VR_MII_DIG_CTRL1” on page 1115	* Varies	VR MII MMD Digital Control1 Register
“VR_MII_AN_CTRL” on page 1123	* Varies	VR MII MMD AN Control Register
“VR_MII_AN_INTR_STS” on page 1126	* Varies	VR MII MMD AN Interrupt and Status Register
“VR_MII_TC” on page 1129	* Varies	VR MII MMD Test Control Register. Note: This register is present only in 1000BaseX-Only PCS...
“VR_MII_DIG_CTRL3” on page 1131	* Varies	VR MII MMD Digital Control 3 Register
“VR_MII_DBG_CTRL” on page 1134	* Varies	VR MII MMD Debug Control Register Note: This register is present only in 1000BaseX-Only PCS...
“VR_MII_EEE_MCTRL0” on page 1137	* Varies	VR MII MMD EEE Mode Control Register This register enables or disables the Energy Efficient Ethernet...
“VR_MII_EEE_TXTIMER” on page 1141	* Varies	VR MII MMD EEE Tx Timer Register This register is present only for configurations with MAIN_MODE...
“VR_MII_EEE_RXTIMER” on page 1143	* Varies	VR MII MMD EEE Rx Timer Register This register is present only for configurations with MAIN_MODE...
“VR_MII_LINK_TIMER_CTRL” on page 1146	* Varies	VR MII MMD Link Timer Control Register
“VR_MII_EEE_MCTRL1” on page 1148	* Varies	VR MII MMD EEE Mode Control 1 Register This register also controls the Energy Efficient Ethernet...
“VR_MII_DIG_STS” on page 1151	* Varies	VR MII MMD Digital Status Register. Note: This register is present only in 1000BaseX-Only PCS...
“VR_MII_ICG_ERRCNT1” on page 1154	* Varies	VR MII MMD Invalid Code Group Error Count1 Register. Note: This register is present only for 1000BaseX-Only...
“VR_MII_GPIO” on page 1155	* Varies	VR MII MMD GPIO Register This register is present only if the DWC_xpcs is configured to have the...
“VR_MII_MISC_STS” on page 1156	* Varies	VR MII MMD Miscellaneous Status Register. This register is present only when the DWC_xpcs is configured...
“VR_MII_NSNPS_EEE_CTRL” on page 1157	* Varies	VR MII MMD Non-Synopsys PHY EEE Control Register This register is present only for configurations...
“VR_MII_RX_LSTS” on page 1158	* Varies	VR MII PHY Rx Lane Status Register In KX_Only, KR_Only, KR_KX, 10GBASE-R PCS, and 1000BASEX-Only...
“VR_MII_MP_12G_16G_25G_TX_GENCTRL0” on page 1161	* Varies	VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Tx General Register 0
“VR_MII_MP_12G_16G_25G_TX_GENCTRL1” on page 1163	* Varies	VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Tx General Register 1

Register	Offset	Description
“VR_MII_MP_12G_16G_TX_GENCTRL2” on page 1165	* Varies	VR MII Synopsys Multi-protocol 12G/16G PHY Tx General Register 2
“VR_MII_MP_25G_TX_GENCTRL2” on page 1167	* Varies	VR MII Synopsys Multi-Protocol 25G PHY Tx General 2 Register
“VR_MII_MP_12G_16G_25G_TX_BOOST_CTRL” on page 1169	* Varies	VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Tx Boost Control Register
“VR_MII_MP_32G_TX_GEN_CTRL3” on page 1170	* Varies	VR MII Synopsys Multi-Protocol 32G PHY Tx General Control 3 Register
“VR_MII_MP_12G_16G_25G_TX_RATE_CTRL” on page 1171	* Varies	VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Tx Rate Control Register
“VR_MII_MP_32G_TX_GEN_CTRL4” on page 1172	* Varies	VR XS or PMA Synopsys Multi-Protocol 32G PHY Tx General Control 4 Register
“VR_MII_MP_12G_16G_25G_TX_POWER_STATE_CTRL” on page 1174	* Varies	VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Tx Power State Register
“VR_MII_MP_12G_16G_25G_TX_EQ_CTRL0” on page 1176	* Varies	VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Tx Equalization Control 0 Register
“VR_MII_MP_12G_16G_25G_TX_EQ_CTRL1” on page 1178	* Varies	VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Tx Equalization Control 1 Register
“VR_MII_MP_6G_TX_BSTCTRL0” on page 1179	* Varies	VR MII Synopsys Multi-protocol 6G Tx Boost Control Register 0
“VR_MII_MP_6G_TX_LVLCTRL0” on page 1180	* Varies	VR MII Synopsys Multi-protocol 6G Tx Level Control Register 0
“VR_MII_MP_6G_TXGENCTRL0” on page 1181	* Varies	VR MII Synopsys Multi-protocol 6G Tx General Control Register 0
“VR_MII_MP_32G_TX_CM_CNTX_SEL0” on page 1183	* Varies	VR XS or PMA Synopsys Multi-Protocol 32G Tx Common Context Select 0 Register
“VR_MII_MP_16G_25G_TX_GENCTRL3” on page 1184	* Varies	VR MII Synopsys Multi-Protocol 16G/25G PHY TX General Control 3 Register
“VR_MII_Consumer_20G_TX_GENCTRL3” on page 1185	* Varies	VR MII Synopsys Consumer 20G PHY TX General Control 3 Register
“VR_MII_MP_6G_TXGENCTRL1” on page 1187	* Varies	VR MII Synopsys Multi-protocol 6G Tx General Control Register 1
“VR_MII_MP_16G_25G_TX_GENCTRL4” on page 1188	* Varies	VR MII Synopsys Multi-Protocol 16G/25G PHY TX General Control 4 Register
“VR_MII_MP_6G_TXGENCTRL2” on page 1189	* Varies	VR MII Synopsys Multi-protocol 6G PHY Tx General Control Register 2

Register	Offset	Description
"VR_MII_MP_16G_25G_TX_MISC_CTRL0" on page 1190	* Varies	VR MII Synopsys Multi-Protocol 16G/25G PHY TX Miscellaneous Control 0 Register
"VR_MII_MP_32G_TX_CNTX_CTRL0" on page 1191	* Varies	VR MII Synopsys Multi Protocol E32 PHY TX Context Control 0 Register
"VR_MII_MP_12G_16G_25G_TX_STS" on page 1192	* Varies	VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Tx Status Register
"VR_MII_MP_25G_TX_TERM_CTRL" on page 1194	* Varies	VR MII Synopsys Multi-Protocol 25G PHY Transmit Termination Control Register
"VR_MII_MP_25G_TX_DCC_CTRL" on page 1195	* Varies	VR MII Synopsys Multi-Protocol 25G PHY Tx DCC Control Register
"VR_MII_MP_25G_TX_WIDTH_CTRL" on page 1196	* Varies	VR MII Synopsys Multi-Protocol 25G PHY Tx Width Control Register
"VR_MII_MP_6G_TX_STS" on page 1197	* Varies	VR MII Synopsys Multi-protocol 6G PHY Tx Status Register
"VR_MII_MP_12G_16G_25G_RX_GENCTRL0" on page 1199	* Varies	VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Rx General Control 0 Register
"VR_MII_MP_12G_16G_25G_RX_GENCTRL1" on page 1201	* Varies	VR MII Synopsys Multi-protocol 12G/16G/25G PHY Rx General Control 1 Register
"VR_MII_MP_12G_16G_RX_GENCTRL2" on page 1203	* Varies	VR MII Synopsys Multi-protocol 12G/16G PHY Rx General Control 2 Register
"VR_MII_MP_25G_RX_GENCTRL2" on page 1205	* Varies	VR XS or PMA Synopsys Multi-protocol 25G PHY Rx General Control 2 Register
"VR_MII_MP_12G_16G_RX_GENCTRL3" on page 1207	* Varies	VR MII Synopsys Multi-protocol 12G/16G PHY Rx General Control 3 Register
"VR_MII_MP_25G_RX_GENCTRL3" on page 1209	* Varies	VR MII Synopsys Multi-protocol 25G PHY Rx General Control 3 Register
"VR_MII_Consumer_10G_RX_GENCTRL3" on page 1210	* Varies	VR MII Synopsys Consumer 10G PHY Rx General Control 3 Register
"VR_MII_MP_12G_16G_25G_RX_RATE_CTRL" on page 1211	* Varies	VR MII Synopsys Multi-protocol 12G/16G/25G PHY Rx Rate Control Register
"VR_MII_MP_32G_RX_MRGN_CTRL4" on page 1212	* Varies	VR MII Synopsys Multi protocol E32 PHY Rx Margin Control 4 Register
"VR_MII_MP_12G_16G_25G_RX_POWER_STATE_CTRL" on page 1214	* Varies	VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Rx Power State Register
"VR_MII_MP_12G_16G_25G_RX_CDR_CTRL" on page 1216	* Varies	VR MII Synopsys Multi-protocol 12G/16G PHY/25G Rx CDR Control Register
"VR_MII_MP_12G_16G_25G_RX_ATTN_CTRL" on page 1218	* Varies	VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Rx Attenuation Control Register

Register	Offset	Description
"VR_MII_MP_6G_RXGENCTRL0" on page 1219	* Varies	VR MII Synopsys Multi-protocol 6G PHY Rx General Control Register 0
"VR_MII_MP_12G_RX_EQ_CTRL0" on page 1221	* Varies	VR MII Synopsys Multi-Protocol 12G PHY Rx Equalization Control 0 Register
"VR_MII_MP_16G_25G_RX_EQ_CTRL0" on page 1223	* Varies	VR MII Synopsys Multi-Protocol 16G/25G PHY Rx Equalization Control 0 Register
"VR_MII_MP_32G_RX_MRGN_CTRL0" on page 1225	* Varies	VR MII Synopsys Multi-Protocol E32 PHY Rx Margin Control 0 Register
"VR_MII_MP_6G_RXGENCTRL1" on page 1226	* Varies	VR MII Synopsys Multi-protocol 6G Rx General Control Register 1
"VR_MII_MP_6G_RXEQ_CTRL" on page 1228	* Varies	VR MII Synopsys Multi-Protocol 6G PHY Rx Equalization Control Register All four Rx Equalization...
"VR_MII_MP_12G_16G_25G_RX_EQ_CTRL4" on page 1229	* Varies	VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Rx Equalization Control 4 Register
"VR_MII_MP_32G_RX_EQ_CTRL4" on page 1231	* Varies	VR MII Synopsys Multi-Protocol E32 PHY Rx Equalization Control 4 Register
"VR_MII_MP_6G_RXLOS_CTRL0" on page 1232	* Varies	VR MII Synopsys Multi-Protocol 6G PHY Rx Loss-of-Signal Control 0 Register
"VR_MII_MP_12G_AFE_DFE_EN_CTRL" on page 1233	* Varies	VR MII Synopsys Multi-Protocol 12G PHY AFE-DFE Enable Register
"VR_MII_MP_16G_25G_RX_EQ_CTRL5" on page 1235	* Varies	VR MII Synopsys Multi-Protocol 16G/25G PHY Receive Equalization Control 5 Register
"VR_MII_MP_32G_RX_ADPT_CTRL" on page 1237	* Varies	VR XS PMA Synopsys Multi-Protocol 32G PHY Rx Adapt Control Register
"VR_MII_MP_12G_16G_25G_DFE_TAP_CTRL0" on page 1239	* Varies	VR MII Synopsys Multi-Protocol 12G/16G/25G PHY DFE Tap Control 0 Register
"VR_MII_MP_12G_16G_25G_RX_STS" on page 1240	* Varies	VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Rx Status Register
"VR_MII_MP_16G_25G_RX_PPM_STS0" on page 1242	* Varies	VR MII Synopsys Multi-Protocol 16G/25G PHY Rx PPM Status 0 Register
"VR_MII_MP_25G_RX_TERM_CTRL" on page 1244	* Varies	VR MII Synopsys Multi-Protocol 25G PHY Receive Termination Control Register
"VR_MII_MP_16G_RX_CDR_CTRL1" on page 1245	* Varies	VR MII Synopsys Multi-Protocol 16G PHY Rx CDR Control1 Register
"VR_MII_MP_16G_25G_RX_PPM_CTRL0" on page 1247	* Varies	VR MII Synopsys Multi-Protocol 16G/25G PHY Rx PPM Control 0 Register

Register	Offset	Description
"VR_MII_MP_16G_25G_RX_GENCTRL4" on page 1248	* Varies	VR MII Synopsys Multi-Protocol 16G/25G PHY Rx General Control 4 Register
"VR_MII_MP_16G_25G_RX_MISC_CTRL0" on page 1250	* Varies	VR MII Synopsys Multi-Protocol 16G/25G PHY RX Miscellaneous Control 0 Register
"VR_MII_MP_32G_RX_MRGN_STS0" on page 1251	* Varies	VR MII Synopsys Multi-Protocol E32 PHY RX Margin Status 0 Register
"VR_MII_MP_16G_25G_RX_IQ_CTRL0" on page 1252	* Varies	VR MII Synopsys Multi-Protocol 16G/25G PHY RX IQ Control 0 Register
"VR_MII_MP_12G_16G_25G_MPLL_CMN_CTRL" on page 1254	* Varies	VR MII Synopsys Multi-Protocol 12G/16G/25G PHY MPLL Common Control Register
"VR_MII_MP_12G_16G_MPLLA_CTRL0" on page 1256	* Varies	VR MII Synopsys Multi-Protocol 12G/16G PHY MPLLA Control 0 Register
"VR_MII_MP_25G_MPLLA_CTRL0" on page 1258	* Varies	VR MII Synopsys Multi-protocol 25G PHY MPLLA Control 0 Register
"VR_MII_MP_12G_MPLLA_CTRL1" on page 1260	* Varies	VR MII Synopsys Multi-protocol 12G PHY MPLLA Control 1 Register
"VR_MII_MP_16G_MPLLA_CTRL1" on page 1262	* Varies	VR MII Synopsys Multi-protocol 16G PHY MPLLA Control 1 Register
"VR_MII_MP_25G_MPLLA_CTRL1" on page 1263	* Varies	VR MII Synopsys Multi-protocol 25G PHY MPLLA Control 1 Register
"VR_MII_MP_32G_MPLLA_CTRL1" on page 1265	* Varies	VR MII Synopsys Multi Protocol 32G PHY MPLLA Control 1 Register
"VR_MII_MP_12G_16G_MPLLA_CTRL2" on page 1267	* Varies	VR MII Synopsys Multi-protocol 12G/16G PHY MPLLA Control 2 Register
"VR_MII_MP_25G_MPLLA_CTRL2" on page 1270	* Varies	VR MII Synopsys Multi-protocol 25G PHY MPLLA Control 2 Register
"VR_MII_MP_12G_16G_MPLLB_CTRL0" on page 1272	* Varies	VR MII Synopsys Multi-protocol 12G/16G PHY MPLLB Control 0 Register
"VR_MII_MP_25G_MPLLB_CTRL0" on page 1274	* Varies	VR MII Synopsys Multi-protocol 25G PHY MPLLB Control 0 Register
"VR_MII_MP_12G_MPLLB_CTRL1" on page 1276	* Varies	VR MII Synopsys Multi-protocol 12G PHY MPLLB Control 1 Register
"VR_MII_MP_16G_MPLLB_CTRL1" on page 1278	* Varies	VR MII Synopsys Multi-protocol 16G PHY MPLLB Control 1 Register
"VR_MII_MP_25G_MPLLB_CTRL1" on page 1279	* Varies	VR MII Synopsys Multi-protocol 25G PHY MPLLB Control 1 Register

Register	Offset	Description
“VR_MII_MP_32G_MPLLB_CTRL1” on page 1282	* Varies	VR MII Synopsys Multi-protocol E32 PHY MPLLB Control 1 Register
“VR_MII_MP_12G_16G_MPLLB_CTRL2” on page 1284	* Varies	VR MII Synopsys Multi-protocol 12G/16G PHY MPLLB Control 2 Register
“VR_MII_MP_25G_MPLLB_CTRL2” on page 1287	* Varies	VR MII Synopsys Multi-protocol 25G PHY MPLLB Control 2 Register
“VR_MII_MP_12G_MPLLA_CTRL3” on page 1289	* Varies	VR MII Synopsys Multi-protocol 12G PHY MPLLA Control 3 Register
“VR_MII_MP_16G_MPLLA_CTRL3” on page 1290	* Varies	VR MII Synopsys Multi-protocol 16G PHY MPLLA Control 3 Register
“VR_MII_MP_25G_MPLLA_CTRL3” on page 1291	* Varies	VR MII Synopsys Multi-protocol 25G PHY MPLLA Control 3 Register
“VR_MII_MP_6G_MPLL_CTRL0” on page 1292	* Varies	VR MII Synopsys Multi-protocol 6G MPLL Control 0 Register
“VR_MII_MP_12G_MPLLB_CTRL3” on page 1294	* Varies	VR MII Synopsys Multi-protocol 12G PHY MPLLB Control 3 Register
“VR_MII_MP_16G_MPLLB_CTRL3” on page 1295	* Varies	VR MII Synopsys Multi-protocol 16G PHY MPLLB Control 3 Register
“VR_MII_MP_25G_MPLLB_CTRL3” on page 1296	* Varies	VR MII Synopsys Multi-protocol 25G PHY MPLLB Control 3 Register
“VR_MII_MP_6G_MPLL_CTRL1” on page 1297	* Varies	VR MII Synopsys Multi-protocol 6G MPLL Control 1 Register
“VR_MII_MP_32G_PLL_CMN_CTRL” on page 1299	* Varies	VR MII Synopsys Multi-protocol 32G PLL Common Control Register
“VR_MII_MP_16G_MPLLA_CTRL4” on page 1301	* Varies	VR MII Synopsys Multi-protocol 16G PHY MPLLA Control 4 Register
“VR_MII_MP_25G_MPLLA_CTRL4” on page 1302	* Varies	VR MII Synopsys Multi-protocol 25G PHY MPLLA Control 4 Register
“VR_MII_MP_25G_MPLLB_CTRL4” on page 1303	* Varies	VR MII Synopsys Multi-protocol 25G PHY MPLLB Control 4 Register
“VR_MII_MP_16G_MPLLA_CTRL5” on page 1304	* Varies	VR MII Synopsys Multi-protocol 16G PHY MPLLA Control 5 Register
“VR_MII_MP_16G_MPLLB_CTRL4” on page 1305	* Varies	VR MII Synopsys Multi-protocol 16G PHY MPLLB Control 4 Register
“VR_MII_MP_25G_MPLLA_CTRL5” on page 1306	* Varies	VR MII Synopsys Multi-protocol 25G PHY MPLLA Control 5 Register

Register	Offset	Description
"VR_MII_MP_16G_MPLLB_CTRL5" on page 1307	* Varies	VR MII Synopsys Multi-protocol 16G PHY MPLLB Control 5 Register
"VR_MII_MP_25G_MPLLB_CTRL5" on page 1308	* Varies	VR MII Synopsys Multi-protocol 25G PHY MPLLB Control 5 Register
"VR_MII_MP_25G_MPLLA_BW_HI_CTRL" on page 1309	* Varies	VR MII Synopsys Multi-protocol 25G PHY MPLLA Bandwidth High Control Register
"VR_MII_MP_25G_MPLLA_BW_LO_CTRL" on page 1310	* Varies	VR MII Synopsys Multi-protocol 25G PHY MPLLA Bandwidth Low Control Register
"VR_MII_MP_25G_MPLLB_BW_HI_CTRL" on page 1311	* Varies	VR MII Synopsys Multi-protocol 25G PHY MPLLB Bandwidth High Control Register
"VR_MII_MP_25G_MPLLB_BW_LO_CTRL" on page 1312	* Varies	VR MII Synopsys Multi-protocol 25G PHY MPLLB Bandwidth Low Control Register
"VR_MII_MP_6G_MPLL_STS" on page 1313	* Varies	VR MII Synopsys Multi-protocol 6G PHY MPLL Status Register
"VR_MII_MP_12G_16G_25G_MISC_CTRL0" on page 1314	* Varies	VR MII Synopsys Multi-protocol 12G/16G/25G PHY Miscellaneous Control 0 Register
"VR_MII_MP_12G_16G_25G_REF_CLK_CTRL" on page 1316	* Varies	VR MII Synopsys Multi-protocol 12G/16G/25G PHY Reference Control Register
"VR_MII_MP_6G_EEE_CTRL" on page 1320	* Varies	VR MII Synopsys Multi-Protocol 6G PHY EEE Control Register
"VR_MII_MP_12G_16G_25G_VCO_CAL_LD0" on page 1321	* Varies	VR MII Synopsys Multi-protocol 12G/16G/25G PHY VCO Calibration Load 0 Register
"VR_MII_MP_32G_RX_CNTX_CTRL0" on page 1322	* Varies	VR MII Synopsys Multi protocol E32 PHY RX Contex Control 0 Register
"VR_MII_MP_6G_MISC_CTRL2" on page 1323	* Varies	VR MII Synopsys Multi-Protocol 6G PHY Miscellaneous Control 2 Register.
"VR_MII_MP_32G_REF_CLK_CTRL0" on page 1324	* Varies	VR MII Synopsys Multi-protocol 32G Reference clock control 0 Register
"VR_MII_MP_32G_REF_CLK_CTRL1" on page 1326	* Varies	VR MII Synopsys Multi-Protocol 32G PHY Reference Clock Control 1 Register
"VR_MII_MP_12G_VCO_CAL_REF0" on page 1328	* Varies	VR MII Synopsys Multi-protocol 12G PHY VCO Calibration Reference Register 0
"VR_MII_MP_16G_25G_VCO_CAL_REF0" on page 1330	* Varies	VR MII Synopsys Multi-protocol 16G/25G PHY VCO Calibration Reference Register 0
"VR_MII_MP_32G_LN_LINK_CTRL" on page 1331	* Varies	VR MII Synopsys Multi protocol E32 PHY Lane Link Control Register
"VR_MII_MP_6G_LVL_CTRL" on page 1332	* Varies	VR MII Synopsys Multi-protocol 6G Level Control Register

Register	Offset	Description
"VR_MII_MP_12G_16G_25G_MISC_STS" on page 1334	* Varies	VR MII Synopsys Multi-protocol 12G/16G/25G PHY Miscellaneous Status Register
"VR_MII_MP_6G_MISC_CTRL0" on page 1336	* Varies	VR MII Synopsys Multi-Protocol 6G Miscellaneous Control 0 Register.
"VR_MII_MP_12G_16G_25G_MISC_CTRL1" on page 1337	* Varies	VR MII Synopsys Multi-protocol 12G/16G/25G PHY Miscellaneous Control 1 Register
"VR_MII_MP_6G_MISC_CTRL1" on page 1338	* Varies	VR MII Synopsys Multi-protocol 6G Miscellaneous Control 1 Register
"VR_MII_MP_12G_16G_25G_EEE_CTRL" on page 1339	* Varies	VR MII Synopsys Multi-protocol 12G/16G/25G PHY EEE Control Register This register is present only...
"VR_MII_MP_12G_16G_25G_SRAM" on page 1340	* Varies	VR MII Synopsys Multi-protocol 12G/16G/25G PHY SRAM Register. Note: This is a special register...
"VR_MII_MP_16G_25G_MISC_CTRL2" on page 1342	* Varies	VR MII Synopsys Multi-protocol 16G/25G PHY Miscellaneous Control 2 Register
"VR_MII_MP_32G_CNTX_CTRL" on page 1344	* Varies	VR XS or PMA Synopsys Multi-protocol 32G PHY Context Control Register
"VR_MII_MP_32G_MISC_STS1" on page 1345	* Varies	VR XS or PMA Synopsys Multi-protocol 32G PHY Miscellaneous Status 1 Register
"VR_MII_SNPS_CR_CTRL" on page 1347	* Varies	VR MII Synopsys PHY CR Control Register
"VR_MII_SNPS_CR_ADDR" on page 1349	* Varies	VR MII Synopsys PHY CR Address Register
"VR_MII_SNPS_CR_DATA" on page 1350	* Varies	VR MII Synopsys CR Data Register
"VR_MII_MP_25G_RX_WIDTH_CTRL" on page 1351	* Varies	VR MII Synopsys Multi-protocol 25G PHY Rx Width Control Register
"VR_MII_MP_25G_RX_AFE_RATE_CTRL" on page 1352	* Varies	VR MII Synopsys Multi-protocol 25G PHY Rx AFE Rate Control Register
"VR_MII_MP_25G_RX_RECAL_CTRL" on page 1353	* Varies	VR MII Synopsys Multi-protocol 25G PHY Rx Re-calibration Control Register
"VR_MII_MP_25G_RX_DCC_CTRL" on page 1355	* Varies	VR MII Synopsys Multi-protocol 25G PHY Rx DCC Control Register
"VR_MII_MP_25G_RX_VCO_CFG0" on page 1356	* Varies	VR MII Synopsys Multi-protocol 25G PHY Rx CDR VCO Configuration 0 Register
"VR_MII_MP_25G_RX_LF_TRSHLD" on page 1357	* Varies	VR MII Synopsys Multi-protocol 25G PHY RX Low Frequency Threshold Register
"VR_MII_MP_25G_RX_HF_TRSHLD" on page 1358	* Varies	VR MII Synopsys Multi-protocol 25G PHY RX High Frequency Threshold Register

Register	Offset	Description
“VR_MII_C16G_RX_EQ_AFE_CTRL0” on page 1359	* Varies	VR MII Synopsys Consumer 16G PHY RX Equalization AFE Control0 Register
“VR_MII_C16G_RX_EQ_AFE_CTRL1” on page 1360	* Varies	VR MII Synopsys Consumer 16G PHY RX Equalization AFE Control1 Register
“VR_MII_DIG_CTRL2” on page 1361	* Varies	VR MII MMD Digital Control2 Register This register is present only when the DWC_xpcs is configured...
“VR_MII_DIG_ERRCNT_SEL” on page 1362	* Varies	VR MII MMD Digital Error Count Select Register This register is present only when the DWC_xpcs...
“VR_MII_SFTY_UE_INTR0” on page 1364	* Varies	VR MII MMD Uncorrectable Safety Error Interrupt Register
“VR_MII_SFTY_ERR_INJ” on page 1368	* Varies	VR MII MMD Safety Error Injection Register
“VR_MII_SFTY_DBG_CTRL” on page 1372	* Varies	VR MII MMD Safety Debug Control Register This register provides the flexibility to control the...
“VR_MII_SFTY_DISABLE” on page 1374	* Varies	VR MII MMD Safety Disable Register
“VR_MII_SFTY_TMR_CTRL” on page 1375	* Varies	VR MII MMD Safety Timer Control Register
“VR_MII_SFTY_UE_INTR1” on page 1377	* Varies	VR MII MMD Uncorrectable Safety Error Interrupt Register 1

5.2.1 DWC_xpcs_map/PMA_MMD Registers

5.2.1.1 SR_PMA_CTRL1

- **Description:** SR PMA MMD Control1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD

15	14	13	12	11	10:7	6	5:2	1	0
RST	Reserved_14	SS13	Reserved_12	LPM	Reserved_10_7	SS6	SS_5_2	Reserved_1	LB

Table 5-6 Fields for Register: SR_PMA_CTRL1

Bits	Name	Memory Access	Description
15	RST	R/W	Soft Reset (RW,SC Type) **For Synopsys PHY configurations:** When the host sets this bit, the CSR block triggers the software reset process in which all internal blocks are reset, except the Management Interface block. The registers are reset to their default values. When this bit is set, it also resets the PHY. This bit is self-cleared when PSEQ_STATE in VR_XS_PCS_DIG_STS Register is equal to 3'b100 , that is, Tx/Rx clocks are stable and in Power_Good state. **For Non-Synopsys PHY configurations:** When the host sets this bit, the CSR block triggers the software reset process in which all internal blocks are reset, except the Management Interface block. The registers are reset to their default values. This bit is self-cleared after the following: ■ 32 clk_csr_i clocks for the MCI interface ■ 1 MDC clock period for the MDIO interface **Note:** For information about read or write-access for registers during power-down mode, see "Special Case Register Access" section. **Value After Reset:** 0x0 **Exists:** Always
14	Reserved_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
13	SS13	* Varies	Speed Selection This bit indicates the operational mode of the PCS-X block of the DWC_xpcs. ■ 0: The DWC_xpcs is in the KX mode (1G speed). ■ 1: The DWC_xpcs is in the 10GBASE-X PCS or KX4 mode (10G speed). Setting or resetting this bit causes the xpcs_kx_kx4n_o output port (if present) to be asserted or de-asserted. **Default Value:** ■ In a KX_Only, 1000BASEX-Only PCS or KR_KX configuration or '10GBASE-R configuration with SUPPORT_1G= Enabled', this bit is always reset to 0 to indicate 1G speed mode. ■ In a KX4_Only, or 10GBASE-X PCS configuration with SUPPORT_1G = Disabled, this bit is always set to 1 to indicate 10G speed mode. ■ In KX4_Only, 10GBASE-X PCS, KR_KX4, KX4_KX or KR_KX4_KX configuration, this bit has a default value of 1. **Access Type:** ■ RW in KR_KX4_KX and KX4_KX configurations ■ RW in 10GBASE-X PCS configurations with SUPPORT_1G = Enabled and SNPS_XAUI_PHY = Enabled ■ RO for all other configurations **Value After Reset:** "(KX4_KX_PRESENT||SUPPORT_10GX_1GX) ? 0x1 : ((ONLY_1G_CFG||KR_KX||SUPPORT_10GR_1GX) ? 0x0 : 0x1)" **Exists:** Always **Memory Access:** "(KX4_KX_PRESENT||SUPPORT_10GX_1GX) ? \"read-write\" : \"read-only\"""
12	Reserved_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
11	LPM	R/W	Low-Power Enable This bit controls the power-down mode of the DWC_xpcs. ■ 0: Normal Operation ■ 1: The DWC_xpcs enters power-down mode along with the PHY. For non-Synopsys PHY, DWC_xpcs asserts xpcs_pdown_o port. For Synopsys PHY, DWC_xpcs does the following to trigger the power-down mode: ■ Turns off the PHY Receiver and Transmitter ■ Switches off all the clocks When the host clears this bit, DWC_xpcs resumes normal operation. In a configuration with Synopsys PHY, after clearing this bit, the host must wait until Bits[4:2] of VR_XS_PCS_DIG_STS Register indicate that DWC_xpcs is in normal state. **Note:** For information about read or write-access for registers during power-down mode, see "Special Case Register Access" section. **Value After Reset:** 0x0 **Exists:** Always
10:7	Reserved_10_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6	SS6	R	Speed Selection Bit 6 This bit always returns 1. **Value After Reset:** 0x1 **Exists:** Always
5:2	SS_5_2	R	Speed Selection Bits [5:2] This field always returns 4'b0000. This field indicates that the device is capable of working at 10 Gbps when both SS13 and SS6 are set. **Value After Reset:** 0x0 **Exists:** Always
1	Reserved_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	LB	R/W	Loopback Enable This bit enables the loopback control of the PHY. ■ 0: The DWC_xpcs de-asserts the xgxs_loopback_en_o{lane} signal. ■ 1: The DWC_xpcs asserts the xgxs_loopback_en_o{lane} signal. **Value After Reset:** 0x0 **Exists:** Always

5.2.1.2 SR_PMA_STATUS1

- **Description:** SR PMA MMD Status1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1 : (APB3_16) ? (0x1*0x2) : (0x1*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD)

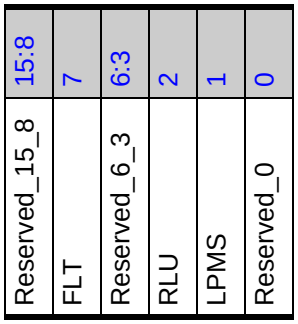


Table 5-7 Fields for Register: SR_PMA_STATUS1

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7	FLT	R	Fault Condition Detected This bit indicates that a fault condition is detected in the PHY. This bit is set when the Rx path of the PHY encounters any of the following fault conditions: ■ Loss-of-signal detection (xpcs_los_i{lane} signal is high). ■ xgxs_rx_valid_i{lane} signal is low. **Note:** For some Synopsys PHYs, xgxs_rx_valid_i remains low if rxX_align input of the PHY is 0. Therefore, do not consider assertion of this bit as a real fault condition in the PHY. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
6:3	Reserved_6_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
2	RLU	R	Rx Link Up (RO,LL Type) ■ 0: This bit goes low when the CDR status of the PHY goes low (indicated by xgxs_rx_valid_i{lane}). ■ 1: This bit is set when the CDR status of the PHY goes high (indicated by xgxs_rx_valid_i{lane}). **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
1	LPMS	R	Low-Power Mode Support When set, this bit indicates that DWC_xpcs supports low-power mode. When reset, this bit indicates that DWC_xpcs does not support the low-power mode. **Value After Reset:** 0x1 **Exists:** Always
0	Reserved_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.1.3 SR_PMA_DEV_ID_1

- **Description:** SR PMA MMD Device Identifier 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x2 : (APB3_16) ? (0x2*0x2) : (0x2*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD



Table 5-8 Fields for Register: SR_PMA_DEV_ID_1

Bits	Name	Memory Access	Description
15:0	PMA_DEV_OUI_3_18	R	Organizationally Unique Identifier[3:18] The DWC_xpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. The value is configured in coreConsultant for each of the MMDs. This read-only field contains Bits[18:3] of 24-bit OUI of device manufacturer. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:0] of SR_VSMMD_PMA_ID1 Register modifies the content of this register **Value After Reset:** "(PMA_MMD) ? IEEE_PMA_DEV_OUI_3_18 : 0x0" **Exists:** Always

5.2.1.4 SR_PMA_DEV_ID_2

- **Description:** SR PMA MMD Device Identifier 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x3 : (APB3_16) ? (0x3*0x2) : (0x3*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD

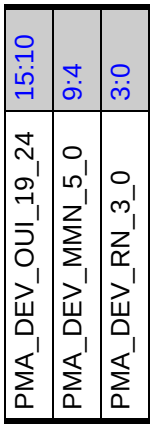


Table 5-9 Fields for Register: SR_PMA_DEV_ID_2

Bits	Name	Memory Access	Description
15:10	PMA_DEV_OUI_19_24	R	Organizationally Unique Identifier[19:24] The DWC_xpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. You can configure the value of each MMD using coreConsultant. This read-only field contains Bits[24:19] of 24-bit OUI of device manufacturer. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:10] of SR_VSMMD_PMA_ID2 Register modifies the content of this register. **Value After Reset:** "(PMA_MMD) ? IEEE_PMA_DEV_OUI_19_24 : 0x0" **Exists:** Always
9:4	PMA_DEV_MMN_5_0	R	Model Number This read-only field contains the 6-bit Model number of the device. You can configure the value of each MMD using coreConsultant. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[9:4] of SR_VSMMD_PMA_ID2 Register modifies the content of this register. **Value After Reset:** "(PMA_MMD) ? IEEE_PMA_DEV_MODEL_NO_0_5 : 0x0" **Exists:** Always

Bits	Name	Memory Access	Description
3:0	PMA_DEV_RN_3_0	R	Revision Number This read-only field contains the 4-bit Revision number of the device. You can configure the value of each MMD using coreConsultant. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[3:0] of SR_VSMMD_PMA_ID2 Register modifies the content of this register **Value After Reset:** "(PMA_MMD) ? IEEE_PMA_DEV_REV_NO_0_3 : 0x0" **Exists:** Always

5.2.1.5 SR_PMA_SPD_ABL

- **Description:** SR PMA MMD Speed Ability Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x4 : (APB3_16) ? (0x4*0x2) : (0x4*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD

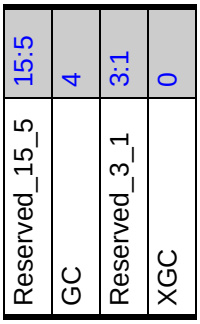
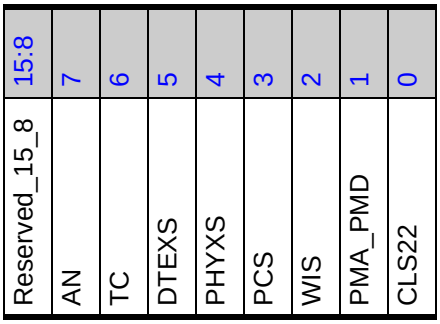


Table 5-10 Fields for Register: SR_PMA_SPD_ABL

Bits	Name	Memory Access	Description
15:5	Reserved_15_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	GC	R	1G Capable This bit indicates that the DWC_xpcs can support 1G mode. **Value After Reset:** "(KX_PRESENT) ? 0x1 : 0x0" **Exists:** Always
3:1	Reserved_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	XGC	R	10G Capable This bit indicates that the DWC_xpcs can support 10G mode. **Value After Reset:** "(ONLY_1G_CFG) ? 0x0 : 0x1" **Exists:** Always

5.2.1.6 SR_PMA_DEV_PKG1

- **Description:** SR PMA MMD Devices in Package 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x5 : (APB3_16) ? (0x5*0x2) : (0x5*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD



15:8	7	6	5	4	3	2	1	0
Reserved_15_8	AN	TC	DTEXS	PHYXS	PCS	WIS	PMA_PMD	CLS22

Table 5-11 Fields for Register: SR_PMA_DEV_PKG1

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7	AN	R	Auto-Negotiation MMD This bit indicates that an AN MMD is present in the package. **Value After Reset:** "(BACKPLANE_ETHERNET_PCS) ? 0x1 : 0x0" **Exists:** Always
6	TC	R	TC MMD This bit indicates that the device has the TC MMDs in the package. This bit is always 0. **Value After Reset:** 0x0 **Exists:** Always
5	DTEXS	R	DTE XGXS MMD This bit indicates that the device has the DTE XS MMD in the package. **Value After Reset:** "(XGXS_PCS&&(DTE_PHY_SEL==0)) ? 0x1 : 0x0" **Exists:** Always

Bits	Name	Memory Access	Description
4	PHYXS	R	PHY XGXS MMD This bit indicates that the device has the PHY XS MMD in the package. **Value After Reset:** "(XGXS_PCS&&(DTE_PHY_SEL==1)) ? 0x1 : 0x0" **Exists:** Always
3	PCS	R	PCS MMD This bit indicates that the device has PCS MMD in the package. **Default Value:** If MAIN_MODE = Backplane Ethernet PCS, 10GBASE-X PCS, or 10GBASE-R PCS, this bit returns 1. For all other configurations, this bit returns 0. **Value After Reset:** "(PCS_MMD) ? 0x1 : 0x0" **Exists:** Always
2	WIS	R	WIS MMD This bit indicates that the device has the WIS MMD in the package. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
1	PMA_PMD	R	PMA or PMD MMD This bit indicates that the device has the PMA or PMD MMD in the package. **Default Value:** ■ This bit is set to 1 for the following configurations: Backplane Ethernet PCS, 10GBASE-X PCS for Synopsys PHY, 10GBASE-R PCS for Synopsys PHY. ■ This bit returns 0 for all other configurations. **Value After Reset:** "(PMA_MMD) ? 0x1 : 0x0" **Exists:** Always
0	CLS22	R	Clause 22 Register Support This bit indicates that the Clause 22 (1G) functional register sets are implemented in this package. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always

5.2.1.7 SR_PMA_DEV_PKG2

- **Description:** SR PMA MMD Devices in Package 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x6 : (APB3_16) ? (0x6*0x2) : (0x6*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD

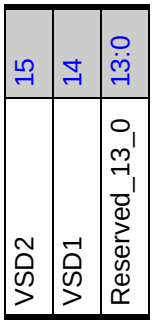


Table 5-12 Fields for Register: SR_PMA_DEV_PKG2

Bits	Name	Memory Access	Description
15	VSD2	R	Vendor-Specific Device 2 This bit indicates that a vendor-specific device 2 is present in the package. **Value After Reset:** "(BASE1000X_ONLY||CL37_AN) ? 0x1 : 0x0" **Exists:** Always
14	VSD1	R	Vendor-Specific Device 1 This bit indicates that the vendor-specific device 1 is present in the package. This bit always returns 1. **Value After Reset:** 0x1 **Exists:** Always
13:0	Reserved_13_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.1.8 SR_PMA_CTRL2

- **Description:** SR PMA MMD Control2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x7 : (APB3_16) ? (0x7*0x2) : (0x7*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD

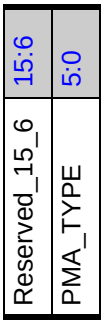
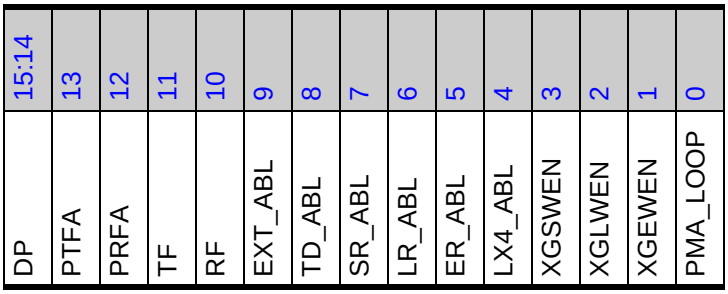


Table 5-13 Fields for Register: SR_PMA_CTRL2

Bits	Name	Memory Access	Description
15:6	Reserved_15_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:0	PMA_TYPE	R/W	PMA Type Select During initialization, the host must program an appropriate value depending upon the active PHY with which DWC_xpcs is interfacing. ■ 001101: 1000BASE-KX PMA or PMD type ■ 001011: 10GBASE-KR PMA or PMD type ■ 001010: 10GBASE-KX4 PMA or PMD type ■ 001000: 10GBASE-LRM PMA or PMD type ■ 000000: 10GBASE-CX4 PMA or PMD type ■ 111100: 5BASE-KR PMA or PMD type ■ 111011: 2.5GBASE-KX PMA or PMD type ■ Other Values: Reserved PMA/PMD supported by DWC_xpcs depends on the RTL configuration chosen. **Value After Reset:** 0x0 **Exists:** Always

5.2.1.9 SR_PMA_STATUS2

- **Description:** SR PMA MMD Status2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8 : (APB3_16) ? (0x8*0x2) : (0x8*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD



15:14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DP	PTFA	PRFA	TF	RF	EXT_ABL	TD_ABL	SR_ABL	LR_ABL	ER_ABL	LX4_ABL	XGSWEN	XGLWEN	XGEWEN	PMA_LOOP

Table 5-14 Fields for Register: SR_PMA_STATUS2

Bits	Name	Memory Access	Description
15:14	DP	R	Device Present Status The value 2'b10 indicates that the PMA MMD is present and responding to this register address. Any other value indicates that the PMA MMD is not present or not functioning properly. This field is always set to 2'b10. **Value After Reset:** 0x2 **Exists:** Always
13	PTFA	R	PMA Tx Fault Ability This bit indicates that the PMA MMD is able to detect a Tx fault. DWC_xpcs does not detect the PMA faults on the Tx path. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
12	PRFA	R	PMA Rx Fault Ability This bit indicates that the PMA MMD is able to detect an Rx fault. DWC_xpcs supports PMA fault detection on the Rx path based on xpcs_los_i{lane}. **Value After Reset:** 0x1 **Exists:** Always
11	TF	R	Transmitter Fault This bit indicates that DWC_xpcs encountered a remote fault condition on the PMA Tx path. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
10	RF	R	Receiver Fault (LH Type) This bit indicates that the device encountered a fault condition on the Rx path. This bit is set when any of the xpcs_los_i{lane} inputs goes high. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
9	EXT_ABL	R	**For configurations other than KX_Only:** This bit indicates that the PMA or PMD has extended abilities listed in the SR_PMA_EXT_ABL Register. **For KX_Only configurations:** Reserved For KX_Only configurations, this bit returns 0. For all other configurations, this bit returns 1. **Value After Reset:** "(!KX_ONLY) ? 0x1 : 0x0" **Exists:** Always
8	TD_ABL	R	PMA Tx Disable Ability This bit indicates that the PMD device is able to disable the Tx path. This bit always returns 1. **Value After Reset:** 0x1 **Exists:** Always
7	SR_ABL	R	10GBASE-SR Capable This bit indicates if the PMA or PMD can support the 10GBASE-SR operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
6	LR_ABL	R	10GBASE-LR Capable This bit indicates if the PMA or PMD can support the 10GBASE-LR operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
5	ER_ABL	R	10GBASE-ER Capable This bit indicates if the PMA or PMD can support the 10GBASE-ER operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
4	LX4_ABL	R	10GBASE-LX4 Capable This bit indicates if the PMA or PMD can support the 10GBASE-LX4 operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
3	XGSWEN	R	10GBASE-SW Capable This bit indicates if the PMA can support the 10GBASE-SW operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
2	XGLWEN	R	10GBASE-LW Capable This bit indicates if the PMA can support the 10GBASE-LW operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
1	XGEWEN	R	10GBASE-EW Capable This bit indicates if the PMA can support the 10GBASE-EW operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
0	PMA_LOOP	R	PMA Loopback Ability This bit indicates if the PMA has loopback ability. This bit always returns 1. **Value After Reset:** 0x1 **Exists:** Always

5.2.1.10 SR_PMA_TX_DIS

- **Description:** SR PMA MMD Transmit Disable Register
  Bits[4:1] of this register are shared with the following register bits:
  - bits [7:4] of VR_PMA_DIG_CTRL1 Register
  - bits [7:4] of VR_XS_PCS_DIG_CTRL1 Register
  - bit [4] of VR_MII_DIG_CTRL1 Register.

  Bit [0] of this register is shared with bit [0] of SR_PMA_KX_CTRL Register.
  Note: DWC_xpcs automatically updates the Bits[4:1] to 4'b1110 when the host changes the speed from 10G (KX4) to 1G (KX) in the following configurations:
  - KX4_KX or KR_KX4_KX
  - XGXS PCS or 10GBASE-X PCS configurations with SUPPORT_1G = Enabled

  Similarly, the DWC_xpcs automatically updates Bits[4:1] to 4'b0000 when the host changes the speed from 1G (KX) to 10G (KX4) in these configurations. You can use Bit 13 of the SR_PMA_CTRL1 to switch between 10G (KX4) and 1G (KX) speeds
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x9 : (APB3_16) ? (0x9*0x2) : (0x9*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD

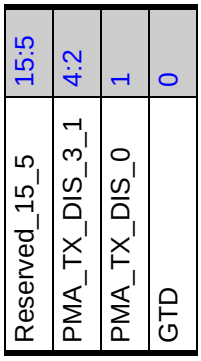


Table 5-15 Fields for Register: SR_PMA_TX_DIS

Bits	Name	Memory Access	Description
15:5	Reserved_15_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4:2	PMA_TX_DIS_3_1	* Varies	**For Multi-lane configurations:** Transmit Disable on Lanes[3:1] This field disables Tx on Lane 3, 2, and 1. This field is present only for the following configurations (with multiple lanes): ■ KX4_Present ■ 10GBASE-X PCS **For single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
1	PMA_TX_DIS_0	R/W	Transmit Disable on Lane 0 This bit disables Tx on Lane 0. **Value After Reset:** 0x0 **Exists:** Always
0	GTD	R/W	Global Transmit Disable This bit disables all Tx lanes. When this bit is set, Bits[4:1] of this register are set to 1. **Value After Reset:** 0x0 **Exists:** Always

5.2.1.11 SR_PMA_RX_SIG_DET

- **Description:** SR PMA MMD Receive Signal Detect Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0A : (APB3_16) ? (0x0A*0x2) : (0x0A*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD

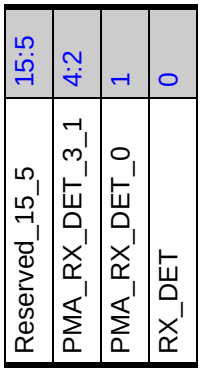


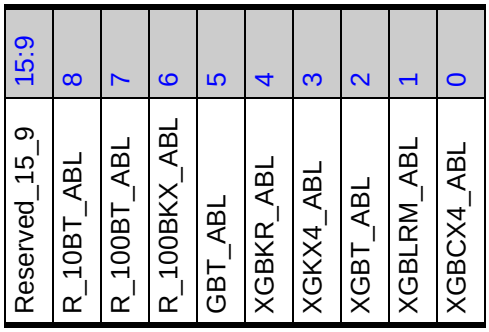
Table 5-16 Fields for Register: SR_PMA_RX_SIG_DET

Bits	Name	Memory Access	Description
15:5	Reserved_15_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4:2	PMA_RX_DET_3_1	R	**For Multi-lane configurations:** Receive Signal Detect on Lanes[3:1] This field shows if Rx signal has been detected on Lane 3, 2, and 1. This field is present only for the following configurations (with multiple lanes): ■ KX4_Present ■ 10GBASE-X PCS **For single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
1	PMA_RX_DET_0	R	Receive Signal Detect on Lane 0 This bit is high if Rx signal is detected on Lane 0. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

Bits	Name	Memory Access	Description
0	RX_DET	R	Global Receive Signal Detect This is a combined signal detect for all lanes. This bit is set only when all lanes are high. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

5.2.1.12 SR_PMA_EXT_ABL

- **Description:** SR PMA or PMD Extended Ability Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0B : (APB3_16) ? (0x0B*0x2) : (0x0B*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD



15:9	8	7	6	5	4	3	2	1	0
Reserved_15_9	R_10BT_ABL	R_100BT_ABL	R_100BKX_ABL	GBT_ABL	XGBKR_ABL	XGKX4_ABL	XGBT_ABL	XGBLRM_ABL	XGBCX4_ABL

Table 5-17 Fields for Register: SR_PMA_EXT_ABL

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	R_10BT_ABL	R	10BASE-T Capable This bit indicates if the PMA or PMD can support the 10BASE-T operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
7	R_100BT_ABL	R	100BASE-T Capable This bit indicates if the PMA or PMD can support the 100BASE-T operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
6	R_100BKX_ABL	R	1000BASE-KX Capable This bit indicates if the PMA or PMD can support the 1000BASE-KX operation. **Value After Reset:** "(KX_PRESENT) ? 0x1 : 0x0" **Exists:** Always

Bits	Name	Memory Access	Description
5	GBT_ABL	R	1000BASE-T Capable This bit indicates if the PMA or PMD can support the 1000BASE-T operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
4	XGBKR_ABL	R	10GBASE-KR Capable This bit indicates if the PMA or PMD can support the 10GBASE-KR operation. **Value After Reset:** "KR_PRESENT ? 0x1 : 0x0" **Exists:** Always
3	XGKX4_ABL	R	10GBASE-KX4 Capable This bit indicates if the PMA or PMD can support the 10GBASE-KX4 operation. **Value After Reset:** "(!SINGLE_LANE_CFG) ? 0x1 : 0x0" **Exists:** Always
2	XGBT_ABL	R	10GBASE-T Capable This bit indicates if the PMA or PMD can support the 10GBASE-T operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
1	XGBLRM_ABL	R	10GBASE-LRM Capable This bit indicates if the PMA or PMD can support the 10GBASE-LRM operation. **Default Value:** For 10GBASE-R PCS configuration, this bit returns 1. For all other configurations, this bit returns 0. **Value After Reset:** "(RPCS_EN&&!BACKPLANE_ETHERNET_PCS) ? 0x1 : 0x0" **Exists:** Always
0	XGBCX4_ABL	R	10GBASE-CX4 Capable This bit indicates if the PMA or PMD can support the 10GBASE-CX4 operation. **Value After Reset:** "(!SINGLE_LANE_CFG) ? 0x1 : 0x0" **Exists:** Always

5.2.1.13 SR_PMA_PKG1

- **Description:** SR PMA MMD Package Identifier 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0xe : (APB3_16) ? (0xe*0x2) : (0xe*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD



Table 5-18 Fields for Register: SR_PMA_PKG1

Bits	Name	Memory Access	Description
15:0	PMA_PKG_OUI_3_18	R	Organizationally Unique Identifier The DWC_xpcs offers 24 configurable bits[24:1], known as OUI, for identifying the device manufacturer. The value is configured within coreConsultant. This read-only field contains Bits[18:3] of 24-bit OUI of device manufacturer. This is a unique identifier for all the MMDs in the device. It is common for all MMDs (PMA, PCS, XS, AN, and MII) present in the package. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:0] of SR_VSMMD_PKGID1 Register modifies the content of this register. **Value After Reset:** "(!BASE1000X_ONLY) ? IEEE_PKG_OUI_3_18 : 0x0" **Exists:** Always

5.2.1.14 SR_PMA_PKG2

- **Description:** SR PMA MMD Package Identifier 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0xf : (APB3_16) ? (0xf*0x2) : (0xf*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD

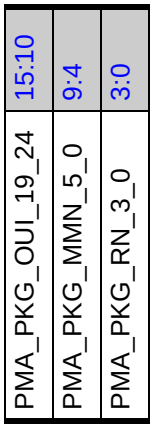


Table 5-19 Fields for Register: SR_PMA_PKG2

Bits	Name	Memory Access	Description
15:10	PMA_PKG_OUI_19_24	R	Organizationally Unique Identifier DWC_xpcs offers 24 configurable bits[24:1], known as OUI, for identifying the device manufacturer. This value is configured by the host. This read-only field contains Bits[24:19] of 24-bit OUI of device manufacturer. This is a unique identifier for all the MMDs in the device. It is common for all the MMDs (PMA, PCS, XS, AN, and MII) present in the package. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:10] of SR_VSMMD_PKGID2 Register modifies the content of this register. **Value After Reset:** "(!BASE1000X_ONLY) ? IEEE_PKG_OUI_19_24 : 0x0" **Exists:** Always
9:4	PMA_PKG_MMN_5_0	R	Model Number This read-only field contains the 5-bit Model Number of the Package in which the MMDs are installed. This value is configured by the host. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[9:4] of SR_VSMMD_PKGID2 Register modifies the content of this register. **Value After Reset:** "(!BASE1000X_ONLY) ? IEEE_PKG_MODEL_NO_0_5 : 0x0" **Exists:** Always

Bits	Name	Memory Access	Description
3:0	PMA_PKG_RN_3_0	R	Revision Number This read-only field contains the 4-bit Revision Number of the Package. This value is configured by the host. It is common for all the MMDs in the device. It is common for all MMDs (PMA, PCS, XS, AN, and MII) present in the package. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[3:0] of SR_VSMMD_PKGID2 Register modifies the content of this register. **Value After Reset:** "(!BASE1000X_ONLY) ? IEEE_PKG_REV_NO_0_3 : 0x0" **Exists:** Always

5.2.1.15 SR_PMA_2PT5G_5G_EXT_ABL

- **Description:** SR PMA MMD 2.5G/5G Extended Ability Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x15 : (APB3_16) ? (0x15*0x2) : (0x15*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD

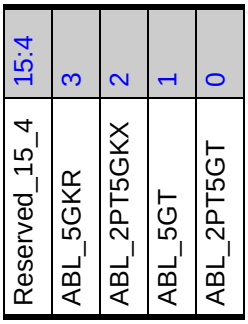


Table 5-20 Fields for Register: SR_PMA_2PT5G_5G_EXT_ABL

Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3	ABL_5GKR	R	5GBASE-KR Capable This bit indicates if the PMA or PMD can support the 5GBASE-KR operation. **Value After Reset:** "ADD_5G ? 0x1 : 0x0" **Exists:** Always
2	ABL_2PT5GKX	R	2.5GBASE-KX Capable This bit indicates if the PMA or PMD can support the 2.5GBASE-KX operation. **Value After Reset:** "ADD_2PT5G ? 0x1 : 0x0" **Exists:** Always
1	ABL_5GT	R	5GBASE-T Capable This bit indicates if the PMA or PMD can support the 5GBASE-T operation. For all configurations, this bit returns 0. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	ABL_2PT5GT	R	2.5GBASE-T Capable This bit indicates if the PMA or PMD can support the 2.5GBASE-T operation. For all configurations, this bit returns 0. **Value After Reset:** 0x0 **Exists:** Always

5.2.1.16 SR_PMA_KR_PMD_CTRL

- **Description:** SR PMA MMD 10GBASE-KR PMD Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x96 : (APB3_16) ? (0x96*0x2) : (0x96*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD&&CL72_EN)

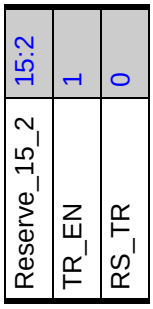


Table 5-21 Fields for Register: SR_PMA_KR_PMD_CTRL

Bits	Name	Memory Access	Description
15:2	Reserve_15_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1	TR_EN	R/W	Training Enable When this bit is set, DWC_xpcs enables the Clause 72 KR Training protocol (for optimizing the link reliability through Transmit equalization settings). ■ 1: Enable the 10GBASE-KR start-up protocol ■ 0: Disable the 10GBASE-KR start-up protocol If this bit is set before or during Clause 73 auto-negotiation, the training starts after all pages have been exchanged during Clause 73 auto-negotiation. **Value After Reset:** 0x0 **Exists:** Always
0	RS_TR	R/W	Restart Training (RW,SC Type) When this bit is set, DWC_xpcs restarts the Clause 72 Training protocol if Bit 1 of this register is set. ■ 1: Restart the 10GBASE-KR start-up protocol ■ 0: Normal operation This bit is self-cleared to 0 after restarting the 10GBASE-KR startup protocol. **Value After Reset:** 0x0 **Exists:** Always

5.2.1.17 SR_PMA_KR_PMD_STS

- **Description:** SR PMA MMD 10GBASE-KR Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x97 : (APB3_16) ? (0x97*0x2) : (0x97*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD&&CL72_EN)

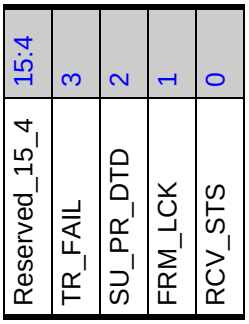


Table 5-22 Fields for Register: SR_PMA_KR_PMD_STS

Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3	TR_FAIL	R	Training Failure ■ 1: Training is completed with failure. ■ 0: Training is not yet complete. When this bit is set to 1, it indicates that the host enabled the Clause 72 KR training protocol but the training is not completed successfully. This can happen because of the following reasons: 1. The Clause 72 KR Training is not supported or enabled in the link partner. 2. A reliable link could not be established through Clause 72 KR Training protocol within 500 ms. When the host gets the Training Failure status, it can do one of the following: ■ Disable the Clause 72 KR Training and rely on the default Transmit equalization settings for the link. ■ Restart the Clause 72 KR Training to try again. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
2	SU_PR_DTD	R	Start-Up Protocol Status When this bit is set to 1, it indicates that the Clause 72 KR Training is either in progress or completed with failure. ■ 1: Start-up protocol in progress ■ 0: Start-up protocol complete **Value After Reset:** 0x0 **Exists:** Always
1	FRM_LCK	R	Frame Lock When this bit is set to 1, it indicates that the DWC_xpcs receiver successfully and consistently decoded the incoming Clause 72 KR training frame from the link partner. ■ 1: Training frame delineation successfully detected ■ 0: Training frame delineation not detected **Value After Reset:** 0x0 **Exists:** Always
0	RCV_STS	R	Receiver Status When this bit is set to 1, it indicates that the Rx link is tuned for KR speed mode. The external equalization process indicates this based on the Rx link reliability status. ■ 1: Receiver trained and ready to receive data ■ 0: Receiver training **Value After Reset:** 0x0 **Exists:** Always

5.2.1.18 SR_PMA_KR_LP_CEU

- **Description:** SR PMA MMD 10GBASE-KR LP Coefficient Update Register.
  This register is present when the Clause 72 KR Training support is enabled. This register indicates the most recently received training control field from the link partner (LP) when KR training is enabled. If KR training is disabled, the management can write to this register to tune the transmit filter coefficients of local device.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x98 : (APB3_16) ? (0x98*0x2) : (0x98*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD&&CL72_EN)

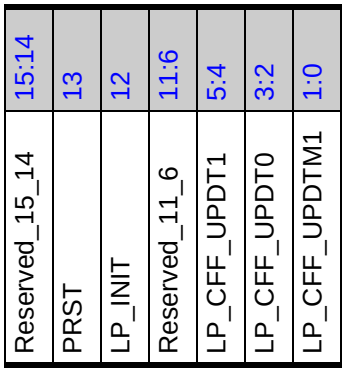


Table 5-23 Fields for Register: SR_PMA_KR_LP_CEU

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13	PRST	R/W	Preset Coefficient When this bit is set to 1, it indicates that the link partner requested the transmitter filter coefficients of the local device to be set to a state where transmit equalization is turned off. This means that the filter coefficient value, corresponding to filter (-1) and filter (+1), must be 0 and the filter coefficient filter, corresponding to filter (0), must be maximum. ■ 1: Pre-initialize coefficients ■ 0: Normal operation **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
12	LP_INIT	R/W	Initialize Coefficient When this bit is set to 1, it indicates that the link partner requested the transmitter filter coefficients of the local device to be initialized. The initialize requirements of the filter coefficient is defined in IEEE Std 802.3, Clause 72, section 72.6.10.4.2. This request is usually received at the beginning of the start-up protocol. ■ 1: Initialize the local device transmit filter coefficient ■ 0: Normal operation **Value After Reset:** 0x0 **Exists:** Always
11:6	Reserved_11_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	LP_CFF_UPDT1	R/W	Co-efficient Update (+1) When this bit is set to 1, it indicates that the link partner requested the local device to increment, decrement, or hold the transmit filter (+1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always
3:2	LP_CFF_UPDT0	R/W	Co-efficient Update (0) When this bit is set to 1, it indicates that the link partner requested the local device to increment, decrement, or hold the transmit filter (0) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
1:0	LP_CFF_UPDTM1	R/W	Coefficient Update (-1) When this bit is set to 1, it indicates that the link partner requested the local device to increment, decrement, or hold the transmit filter (-1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

5.2.1.19 SR_PMA_KR_LP_CESTS

- **Description:** SR PMA MMD 10GBASE-KR LP Coefficient Status Register
  This register is present when the Clause 72 KR Training support is enabled. This register indicates the most recently received training status field from the link partner (LP).
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x99 : (APB3_16) ? (0x99*0x2) : (0x99*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD&&CL72_EN)

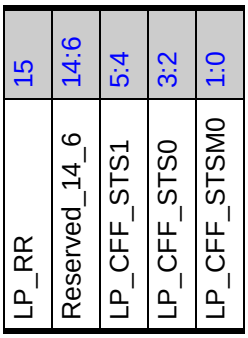


Table 5-24 Fields for Register: SR_PMA_KR_LP_CESTS

Bits	Name	Memory Access	Description
15	LP_RR	R	Receiver Ready When this bit is set to 1, it indicates that the transmit link is reliable to transmit the KR speed data. The link partner sends this information based on successful and consistent reception of the PRBS data field in the KR Training frame (transmitted by the local device). ■ 1: The LP receiver has determined that the training is complete, and it is ready to receive data. ■ 0: The LP receiver is requesting to continue the training. **Value After Reset:** 0x0 **Exists:** Always
14:6	Reserved_14_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
5:4	LP_CFF_STS1	R	Coefficient Status (+1) This field indicates the status from the link partner for the (increment, decrement, or hold) request sent by the local device for transmit filter (+1). ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always
3:2	LP_CFF_STS0	R	Coefficient Status (0) This field indicates the status from the link partner for the (increment, decrement, or hold) request sent by the local device for transmit filter (0). ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always
1:0	LP_CFF_STSM0	R	Coefficient Status (-1) This field indicates the status from the link partner for the (increment, decrement, or hold) request sent by the local device for transmit filter (-1). ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always

5.2.1.20 SR_PMA_KR_LD_CEU

- **Description:** SR PMA MMD 10GBASE-KR LD Coefficient Update
  This register is present when the Clause 72 KR Training support is enabled. This register indicates the most-recently transmitted training control field from the local device (LD) to the link partner.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x9A : (APB3_16) ? (0x9A*0x2) : (0x9A*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD&&CL72_EN)

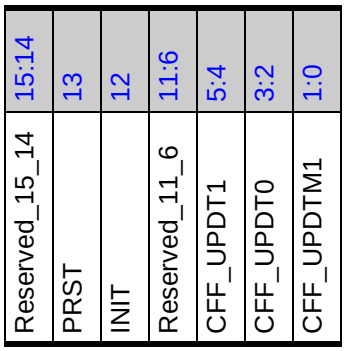


Table 5-25 Fields for Register: SR_PMA_KR_LD_CEU

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13	PRST	R	Pre-initialize the Transmit Filter Coefficient of the Link Partner When this bit is set to 1, it indicates that the local device is requesting the transmitter filter coefficients of link partner to be set to a state where transmit equalization is turned off. This means the filter coefficient value, corresponding to filter (-1) and filter (+1), must be zero and the filter coefficient value, corresponding to filter (0), must be maximum. ■ 1: Preset coefficients ■ 0: Normal operation **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
12	INIT	R	Initialize the Transmit Filter Coefficient of the Link Partner When this bit is set to 1, it indicates that the local device requested the transmitter filter coefficients of link partner to be initialized. The initialize requirements of the filter coefficient is defined in IEEE Std 802.3, Clause 72, section 72.6.10.4.2. This request is usually received at the beginning of the start-up protocol. ■ 1: Initialize the local device transmit filter coefficient ■ 0: Normal operation **Value After Reset:** 0x0 **Exists:** Always
11:6	Reserved_11_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	CFF_UPDT1	R	Filter Coefficient Update (+1) Request to the Link Partner Transmitter When this bit is set to 1, it indicates that the local device requested the link partner to increment, decrement, or hold the transmit filter (+1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_UPDT0	R	Filter Coefficient Update (0) Request to the Link Partner Transmitter When this bit is set to 1, it indicates that the local device requested the link partner to increment, decrement, or hold the transmit filter (0) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
1:0	CFF_UPDTM1	R	Filter Coefficient Update (-1) Request to the Link Partner Transmitter When this bit is set to 1, it indicates that the local device requested the link partner to increment, decrement, or hold the transmit filter (-1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

5.2.1.21 SR_PMA_KR_LD_CESTS

- **Description:** SR PMA MMD 10GBASE-KR LD Coefficient Status
  This register is present only for the PMA MMD when the Clause 72 KR Training support is enabled. This register indicates the most-recently transmitted training status field from the local device (LD) to the link partner.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x9B : (APB3_16) ? (0x9B*0x2) : (0x9B*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD&&CL72_EN)

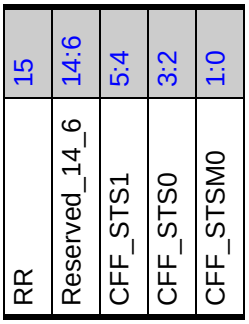


Table 5-26 Fields for Register: SR_PMA_KR_LD_CESTS

Bits	Name	Memory Access	Description
15	RR	R	Receiver Ready When this bit is set to 1, it indicates that the receive link is reliable to receive the KR speed data. The local device Rx adaptation process sets this bit based on successful and consistent reception of the PRBS data field in received training frame at KR speed. ■ 1: The receiver of local device has determined that the training is complete, and it is ready to receive data. ■ 0: The receiver of local device is requesting to continue the training. **Value After Reset:** 0x0 **Exists:** Always
14:6	Reserved_14_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
5:4	CFF_STS1	R	Filter Coefficient Status (+1) to Link Partner Transmitter from Local Device This field indicates the status from the local device for the (increment, decrement, or hold) request received from the link partner for transmit filter (+1). ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_STS0	R	Filter Coefficient Status (0) to Link Partner Transmitter from Local Device This field indicates the status from the local device for the (increment, decrement, or hold) request received from the link partner for transmit filter (0). ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always
1:0	CFF_STSM0	R	Filter Coefficient Status (-1) to Link Partner Transmitter from Local Device This field indicates the status from the local device for the (increment, decrement, or hold) request received from the link partner for transmit filter (-1). ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always

5.2.1.22 SR_PMA_KX_CTRL

- **Description:** SR PMA or PMD KX Control Register
  This register is present only in the configurations with KX_Present mode.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0xA0 : (APB3_16) ? (0xA0*0x2) : (0xA0*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD

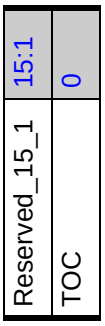


Table 5-27 Fields for Register: SR_PMA_KX_CTRL

Bits	Name	Memory Access	Description
15:1	Reserved_15_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	TOC	* Varies	**For KX_Present configurations:** Tx Output Control ■ 0: Enable Tx Output (Lane 0 in KX4_KX mode) ■ 1: Disable Tx Output (Lane 0 in KX4_KX mode) This bit is shared with the following bits: ■ Bit 4 of VR_PMA_DIG_CTRL1 Register, VR_X-S_PCS_DIG_CTRL1 Register, and VR_MII_DIG_CTRL1 Register ■ Bit 1 of SR_PMA_TX_DIS Register **For configurations without KX:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(KX_PRESENT) ? \"read-write\" : \"read-only\""

5.2.1.23 SR_PMA_KX_STS

- **Description:** SR PMA or PMD KX Status Register
  This register is present only in the configurations with KX_Present mode. This register can be read only if SS13 bit is low
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0xA1 : (APB3_16) ? (0xA1*0x2) : (0xA1*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD

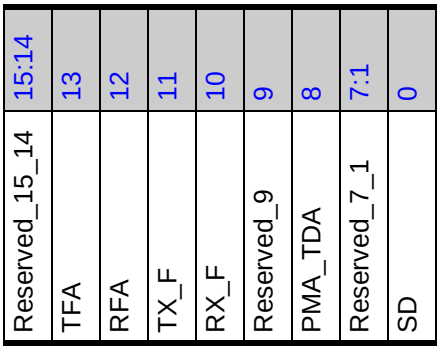


Table 5-28 Fields for Register: SR_PMA_KX_STS

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13	TFA	R	Tx Fault Ability This bit indicates that the PMA or PMD is able to detect the Tx fault. This bit always returns 0 because the Tx path fault detection is not supported for the PMA MMD. **Value After Reset:** 0x0 **Exists:** Always
12	RFA	R	Rx Fault Ability This bit indicates that the PMA or PMD is able to detect the Rx fault. This bit always returns 1 because DWC_xpcs supports Rx fault detection for PMA MMD. **Value After Reset:** "(KX_PRESENT) ? 0x1 : 0x0" **Exists:** Always **Testable:** untestable

Bits	Name	Memory Access	Description
11	TX_F	R	Tx Fault This bit indicates that fault condition is detected in the Tx path for PMA MMD. This bit always returns 0 because this feature is not supported. **Value After Reset:** 0x0 **Exists:** Always
10	RX_F	R	Rx Fault This bit indicates that fault condition is detected in the Rx path for PMA MMD. This bit is set when the xgxs_los_i[0] input goes high. This bit is cleared when the xgxs_los_i[0] input goes low. **Value After Reset:** 0x0 **Exists:** Always
9	Reserved_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	PMA_TDA	R	PMD Tx Disable Ability This bit indicates that the PMA has the ability to disable the transmission. This bit always returns 1 in KX mode. **Value After Reset:** "(KX_ONLY||KR_KX) ? 0x1 : 0x0" **Exists:** Always **Testable:** untestable
7:1	Reserved_7_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	SD	R	Signal Detect This bit indicates that the PMD detected the signal detect state. This bit is inverse of the xgxs_los_i[0] input. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

5.2.1.24 SR_PMA_KR_FEC_ABL

- **Description:** SR PMA MMD 10GBASE-KR FEC Ability Register
  This register is present only if FEC_EN = Enabled. Otherwise, this register is reserved.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0xAA : (APB3_16) ? (0xAA*0x2) : (0xAA*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD&&FEC_EN)

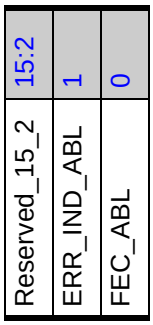


Table 5-29 Fields for Register: SR_PMA_KR_FEC_ABL

Bits	Name	Memory Access	Description
15:2	Reserved_15_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1	ERR_IND_ABL	R	10GBASE-KR FEC Error Indication Ability This bit indicates that the 10GBASE-KR PHY is able to report FEC decoding errors to the PCS layer. **Value After Reset:** "(FEC_ERROR_FWD) ? 0x1 : 0x0" **Exists:** Always
0	FEC_ABL	R	10GBASE-KR FEC Ability This bit indicates that the 10GBASE-KR PHY supports FEC. This bit always returns 1 indicating the FEC ability. **Value After Reset:** 0x1 **Exists:** Always

5.2.1.25 SR_PMA_KR_FEC_CTRL

- **Description:** SR PMA MMD 10GBASE-KR FEC Control Register
  This register is present only if FEC_EN = Enabled. Otherwise, this register is reserved.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0xAB : (APB3_16) ? (0xAB*0x2) : (0xAB*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD&&FEC_EN)

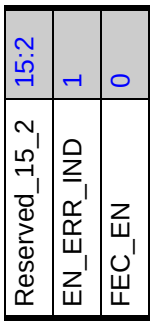


Table 5-30 Fields for Register: SR_PMA_KR_FEC_CTRL

Bits	Name	Memory Access	Description
15:2	Reserved_15_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1	EN_ERR_IND	* Varies	FEC Enable Error Indication When set, this bit configures the FEC decoder to indicate uncorrectable errors to the PCS layer. When reset, the FEC decoder does not pass the uncorrectable error information to the PCS layer. **Access Type:** If FEC_ERROR_FWD = Enabled, this bit is RW. Otherwise, this bit is RO. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(FEC_ERROR_FWD) ? \"read-write\" : \"read-only\""
0	FEC_EN	R/W	FEC Enable When set, this bit enables 10GBASE-KR FEC. When reset, this bit disables 10GBASE-KR FEC. The FEC ability of the local device and the link partner is negotiated through Clause 73 auto-negotiation. The host must enable this bit only if during auto-negotiation, both link partner and local device indicate that they have the FEC ability. **Value After Reset:** 0x0 **Exists:** Always

5.2.1.26 SR_PMA_KR_FEC_CORR_BLK1

- **Description:** SR PMA MMD 10GBASE-KR Lane0 FEC Correctable Error Block Counter 1 Register This register is present only if FEC_EN = Enabled. Otherwise, this register is reserved. This register indicates the number of FEC blocks with correctable errors after the FEC block lock is achieved. The total width of the error counter is 32-bit. This register holds the lower 16-bits of the 32-bits counter.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0xAC : (APB3_16) ? (0xAC*0x2) : (0xAC*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD&&FEC_EN)



Table 5-31 Fields for Register: SR_PMA_KR_FEC_CORR_BLK1

Bits	Name	Memory Access	Description
15:0	CORR_B_CNT1	R	FEC Corrected Blocks Lower (RO,COR type) This field indicates the FEC correctable error block counter. These are the lower 16-bits of the counter. This is a non-roll over counter. This field is cleared when this register is read. **Value After Reset:** 0x0 **Exists:** Always

5.2.1.27 SR_PMA_KR_FEC_CORR_BLK2

- **Description:** SR PMA MMD 10GBASE-KR Lane0 FEC Correctable Error Block Counter 2 Register This register is present only if FEC_EN = Enabled. Otherwise, this register is reserved. This register indicates the number of FEC blocks with correctable errors after the FEC block lock is achieved. The total width of the error counter is 32-bit. This register holds the higher 16-bits of the 32-bits counter.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0xAD : (APB3_16) ? (0xAD*0x2) : (0xAD*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD&&FEC_EN)



Table 5-32 Fields for Register: SR_PMA_KR_FEC_CORR_BLK2

Bits	Name	Memory Access	Description
15:0	CORR_B_CNT2	R	FEC Corrected Blocks Higher (RO,COR Type) This field indicates the FEC correctable error block counter. These are the higher 16-bits of the counter. This is a non-roll over counter. This field is cleared when SR_PMA_KR_FEC_CORR_BLK1 register is read. **Value After Reset:** 0x0 **Exists:** Always

5.2.1.28 SR_PMA_KR_FEC_UCORR_BLK1

- **Description:** SR PMA MMD 10GBASE-KR Lane0 FEC Uncorrectable Error Block Counter 1 Register This register is present only if FEC_EN = Enabled. Otherwise, this register is reserved. This register indicates the number of FEC blocks with uncorrectable errors after the FEC block lock is achieved. The total width of the Error counter is 32-bit. This register holds the lower 16-bits of the 32-bits counter.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0xAE : (APB3_16) ? (0xAE*0x2) : (0xAE*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD&&FEC_EN)



Table 5-33 Fields for Register: SR_PMA_KR_FEC_UCORR_BLK1

Bits	Name	Memory Access	Description
15:0	UCORR_B_CNT1	R	FEC Uncorrected Blocks Lower (RO,COR Type) This field indicates the FEC uncorrectable error block counter. These are the lower 16-bits of the counter. This is a non-roll over counter. This register is cleared when it is read. **Value After Reset:** 0x0 **Exists:** Always

5.2.1.29 SR_PMA_KR_FEC_UCORR_BLK2

- **Description:** SR PMA MMD 10GBASE-KR Lane0 FEC Uncorrectable Error Block Counter 2 Register This register is present only if FEC_EN = Enabled. Otherwise, this register is reserved. This register indicates the number of FEC blocks with uncorrectable errors after the FEC block lock is achieved. The total width of the Error counter is 32-bit. This register holds the higher 16-bits of the 32-bits counter.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0xAF : (APB3_16) ? (0xAF*0x2) : (0xAF*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD&&FEC_EN)



Table 5-34 Fields for Register: SR_PMA_KR_FEC_UCORR_BLK2

Bits	Name	Memory Access	Description
15:0	UCORR_B_CNT2	R	FEC Uncorrected Blocks Higher (RO,COR Type) This field indicates the FEC uncorrectable error block counter. These are the higher 16-bits of the counter. This is a non-roll over counter. This field is cleared when SR_PMA_KR_FEC_UCORR_BLK1 register is read. **Value After Reset:** 0x0 **Exists:** Always

5.2.1.30 SR_PMA_TIME_SYNC_PMA_ABL

- **Description:** SR PMA MMD Time Sync Capability Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0708 : (APB3_16) ? (0x0708*0x2) : (0x0708*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD



Table 5-35 Fields for Register: SR_PMA_TIME_SYNC_PMA_ABL

Bits	Name	Memory Access	Description
15:2	Reserved_15_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1	PMA_TX_DLY_ABL	R	PMA Transmit Path Data Delay Information Available. If this bit is set, it indicates that the information regarding the maximum and minimum transmit data delay of PMA is available in IEEE registers 1.1801 through 1.1804 **Value After Reset:** 0x1 **Exists:** Always
0	PMA_RX_DLY_ABL	R	PMA Receive Path Data Delay Information Available. If this bit is set, it indicates that the information regarding the maximum and minimum receive data delay of PMA is available in IEEE registers 1.1805 through 1.1808 **Value After Reset:** 0x1 **Exists:** Always

5.2.1.31 SR_PMA_TIME_SYNC_TX_MAX_DLY_LWR

- **Description:** SR PMA MMD Time Sync Tx Max Delay Lower Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0709 : (APB3_16) ? (0x0709*0x2) : (0x0709*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD

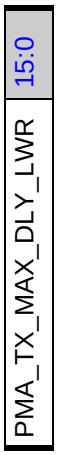


Table 5-36 Fields for Register: SR_PMA_TIME_SYNC_TX_MAX_DLY_LWR

Bits	Name	Memory Access	Description
15:0	PMA_TX_MAX_DLY_LWR	R	This field indicates the lower 16-bit of the 32-bit value which indicates the Maximum Data Delay in the Transmit Path (in nanoseconds) of PMA. This value is configurable in coreConsultant. **Value After Reset:** PMA_TX_MAX_DLY_LWR **Exists:** Always

5.2.1.32 SR_PMA_TIME_SYNC_TX_MAX_DLY_UPR

- **Description:** SR PMA MMD Time Sync Tx Max Delay Upper Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070A : (APB3_16) ? (0x070A*0x2) : (0x070A*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD



Table 5-37 Fields for Register: SR_PMA_TIME_SYNC_TX_MAX_DLY_UPR

Bits	Name	Memory Access	Description
15:0	PMA_TX_MAX_DLY_UPR	R	This field indicates the upper 16-bit of the 32-bit value which indicates the Maximum Data Delay in the Transmit Path (in nanoseconds) of PMA.This value is configurable in coreConsultant. **Value After Reset:** PMA_TX_MAX_DLY_UPR **Exists:** Always

5.2.1.33 SR_PMA_TIME_SYNC_TX_MIN_DLY_LWR

- **Description:** SR PMA MMD Time Sync Tx Min Delay Lower Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070B : (APB3_16) ? (0x070B*0x2) : (0x070B*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD



Table 5-38 Fields for Register: SR_PMA_TIME_SYNC_TX_MIN_DLY_LWR

Bits	Name	Memory Access	Description
15:0	PMA_TX_MIN_DLY_LWR	R	This field indicates the lower 16-bit of the 32-bit value which indicates the Minimum Data Delay in the Transmit Path (in nanoseconds) of PMA.This value is configurable in coreConsultant. **Value After Reset:** PMA_TX_MIN_DLY_LWR **Exists:** Always

5.2.1.34 SR_PMA_TIME_SYNC_TX_MIN_DLY_UPR

- **Description:** SR PMA MMD Time Sync Tx Min Delay Upper Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070C : (APB3_16) ? (0x070C*0x2) : (0x070C*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD



Table 5-39 Fields for Register: SR_PMA_TIME_SYNC_TX_MIN_DLY_UPR

Bits	Name	Memory Access	Description
15:0	PMA_TX_MIN_DLY_UPR	R	This field indicates the upper 16-bit of the 32-bit value which indicates the Minimum Data Delay in the Transmit Path (in nanoseconds) of PMA.This value is configurable in coreConsultant. **Value After Reset:** PMA_TX_MIN_DLY_UPR **Exists:** Always

5.2.1.35 SR_PMA_TIME_SYNC_RX_MAX_DLY_LWR

- **Description:** SR PMA MMD Time Sync Rx Max Delay Lower Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF| |MCI_INTF| |MDIO_INTF) ? 0x070D : (APB3_16) ? (0x070D*0x2) : (0x070D*0x4)" + "(APB3_INTF| |MCI_INTF| |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD



Table 5-40 Fields for Register: SR_PMA_TIME_SYNC_RX_MAX_DLY_LWR

Bits	Name	Memory Access	Description
15:0	PMA_RX_MAX_DLY_LWR	R	This field indicates the lower 16-bit of the 32-bit value which indicates the Maximum Data Delay in the Receive Path (in nanoseconds) of PMA.This value is configurable in coreConsultant **Value After Reset:** PMA_RX_MAX_DLY_LWR **Exists:** Always

5.2.1.36 SR_PMA_TIME_SYNC_RX_MAX_DLY_UPR

- **Description:** SR PMA MMD Time Sync Rx Max Delay Upper Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070E : (APB3_16) ? (0x070E*0x2) : (0x070E*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD



Table 5-41 Fields for Register: SR_PMA_TIME_SYNC_RX_MAX_DLY_UPR

Bits	Name	Memory Access	Description
15:0	PMA_RX_MAX_DLY_UPR	R	This field indicates the upper 16-bit of the 32-bit value which indicates the Maximum Data Delay in the Receive Path (in nanoseconds) of PMA.This value is configurable in coreConsultant **Value After Reset:** PMA_RX_MAX_DLY_UPR **Exists:** Always

5.2.1.37 SR_PMA_TIME_SYNC_RX_MIN_DLY_LWR

- **Description:** SR PMA MMD Time Sync Rx Min Delay Lower Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070F : (APB3_16) ? (0x070F*0x2) : (0x070F*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD



Table 5-42 Fields for Register: SR_PMA_TIME_SYNC_RX_MIN_DLY_LWR

Bits	Name	Memory Access	Description
15:0	PMA_RX_MIN_DLY_LWR	R	This field indicates the lower 16-bit of the 32-bit value which indicates the Minimum Data Delay in the Receive Path (in nanoseconds) of PMA.This value is configurable in coreConsultant **Value After Reset:** PMA_RX_MIN_DLY_LWR **Exists:** Always

5.2.1.38 SR_PMA_TIME_SYNC_RX_MIN_DLY_UPR

- **Description:** SR PMA MMD Time Sync Rx Min Delay Upper Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0710 : (APB3_16) ? (0x0710*0x2) : (0x0710*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD



Table 5-43 Fields for Register: SR_PMA_TIME_SYNC_RX_MIN_DLY_UPR

Bits	Name	Memory Access	Description
15:0	PMA_RX_MIN_DLY_UPR	R	This field indicates the upper 16-bit of the 32-bit value which indicates the Minimum Data Delay in the Receive Path (in nanoseconds) of PMA.This value is configurable in coreConsultant **Value After Reset:** PMA_RX_MIN_DLY_UPR **Exists:** Always

5.2.1.39 VR_PMA_DIG_CTRL1

- **Description:** VR PMA MMD Digital Control1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8000 : (APB3_16) ? (0x8000*0x2) : (0x8000*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD

15	14:12	11	10:8	7:5	4	3:2	1	0
VR_RST	Reserved_14_12	PWRSV	Reserved_10_8	DTXLANED_3_1	DTXLANED_0	Reserved_3_2	BYP_PWRUP	CL72_AUTO

Table 5-44 Fields for Register: VR_PMA_DIG_CTRL1

Bits	Name	Memory Access	Description
15	VR_RST	R/W	Vendor-specific Soft Reset (RW,SC Type) When the host sets this bit to 1, the DWC_xpcs initiates the reset for all the internal blocks except the Management Interface block and the CSR block. When this bit is set, it also resets the PHY. This bit is self cleared under the following conditions: ■ For Synopsys PHY: This bit is self cleared when Bits[4:2] in VR_X-S_PCS_DIG_STS Register is equal to 3'b100, that is, Tx/Rx clocks are stable and in Power_Good state. ■ For Non-Synopsys PHY: This bit is self cleared after the following: 32 clk_csr_i clocks for the MCI interface; 1 MDC clock period for the MDIO interface. **Note:** For information about the read or write access for any register during the reset process, see "Special Case Register Access" section. **Value After Reset:** 0x0 **Exists:** Always
14:12	Reserved_14_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
11	PWRSV	* Varies	**For Synopsys PHY configurations:** Power Save This bit is present only when the DWC_xpcs is configured for Synopsys PHY. ■ 0: Normal operation. ■ 1: DWC_xpcs and the PHY enter the power-save mode. When this bit is set, the DWC_xpcs triggers the power-down mode by turning off the PHY Receiver and Transmitter, and without turning off MPLL. For Synopsys Multi-protocol 12G/16G/25G PHY, DWC_xpcs asserts xpcs_tx_lpd_o{lane} and xpcs_rx_lpd_o{lane} when this bit is set. When the host clears this bit, DWC_xpcs resumes normal operation. After clearing this bit, the host must wait until Bits[4:2] of the VR_XS_PCS_DIG_STS Register indicate that the DWC_xpcs is in a normal state. For a description of read or write access for any register while this bit is set, see "Special Case Register Access" section. **For non-Synopsys PHY configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_XAUI_PHY) ? \"read-write\" : \"read-only\""
10:8	Reserved_10_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
7:5	DTXLANED_3_1	* Varies	**Single-lane configurations:** Reserved **Multi-lane configurations:** Tx Disable on Lanes[3:1] When specific bits of this field is 1, the DWC_xpcs disables the corresponding Tx lanes (3, 2, or 1) of the PHY. When specific bits of this field is 0, the DWC_xpcs enables the corresponding Tx lanes of the PHY. ■ This field is shared with the following fields: ❑ Bits[7:5] of VR_XS_PCS_DIG_CTRL1 Register ❑ Bits[4:2] of SR_PMA_TX_DIS Register **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SINGLE_LANE_CFG) ? \"read-only\" : \"read-write\""
4	DTXLANED_0	R/W	Tx Disable on Lane 0 When this bit is set, the DWC_xpcs disables the Tx Lane 0 of the PHY. When reset, the DWC_xpcs enables the Tx Lane 0 of the PHY. **Note:**This bit is shared with the following bits: ■ Bit 4 of VR_XS_PCS_DIG_CTRL1 and VR_MII_DIG_CTRL1 Register ■ Bit 1 of SR_PMA_TX_DIS Register ■ Bit 0 of SR_PMA_KX_CTRL Register **Value After Reset:** 0x0 **Exists:** Always
3:2	Reserved_3_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
1	BYP_PWRUP	* Varies	**For Synopsys PHY configurations:** Bypass Power-Up Sequence This bit is present only when the DWC_xpcs is configured for the Synopsys PHY. In all other configurations, this bit is reserved. When this bit is set to 1, the DWC_xpcs bypasses the normal flow of the power-up sequence and reaches the Power_Good state to enable transmission or reception. When this bit is set, the DWC_xpcs does not wait for the MPLL and Transmit or Receive PLL status from the Synopsys PHY. You can use this feature in the following scenarios: ■ When the DWC_xpcs is configured to interface with a specific Synopsys PHY. ■ When the data path needs to be interfaced to some other PHY. When this bit is set to 0, the DWC_xpcs waits for the MPLL, Tx, or Rx PLL status from the Synopsys PHY before resuming the normal transmission and reception. **Note:** This bit is shared with Bit 1 of VR_XS_PCS_DIG_CTRL1 Register, VR_AN_DIG_CTRL1 and VR_MII_DIG_CTRL1 Register. **Access Type:** If SNPS_XAUI_PHY= Enabled, this bit is RW. Otherwise, this bit is RO. **For non-Synopsys PHY configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_XAUI_PHY) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
0	CL72_AUTO	* Varies	Automatic Start/Restart of Clause 72 KR Training If this bit is set, DWC_xpcs automatically starts/restarts the Clause 72 KR training on reaching AN_GOOD_CHECK state of auto-negotiation if the following conditions are met : ■ Far-end advertises 10G-KR capability and ■ DWC_xpcs is configured in 10G-KR mode As per the programming sequence described in Figure 7-1 of Databook, software is expected to start KR training by programming 2'b11 to SR_PMA_KR_PMD_CTRL register (after completion of auto-negotiation page exchange). However, if this CL72_AUTO bit had been set upfront, software can skip that step. Advantage of this new mechanism is that, it saves the time-delay involved in turning-on KR training (by software). This time-delay is critical as only 500 milli-seconds is available in AN_GOOD_CHECK state for completion of KR training and for attainment of block-lock. If this bit is set, DWC_xpcs immediately starts sending KR training pattern, after the completion of AN page exchange.If this bit is not set, DWC_xpcs sends 10G-KR data, after the completion of AN page exchange, until software manually turns-on KR training by programming SR_PMA_KR_PMD_CTRL register. If you program this bit back to 0, KR training is turned-off Note: ■ This register bit acts as master control over TR_EN bit of SR_PMA_KR_PMD_CTRL register ■ This register bit is valid only if you plan to run Clause 72 training, in conjunction with Clause 73 auto-negotiation. ■ If you plan to run Clause 72 training, without Clause 73 auto-negotiation, you need to control KR training from SR_PMA_KR_PMD_CTRL register ■ KR training restarts itself everytime auto-negotiation state-machine reaches AN_GOOD_CHECK state (if this bit is 1) This bit is present only for configurations with CL72_EN = Enabled **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "CL72_EN ? \"read-write\" : \"read-only\""

5.2.1.40 VR_PMA_KRTR_PRBS_CTRL0

- **Description:** VR PMA MMD KRTR PRBS Control0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8003 : (APB3_16) ? (0x8003*0x2) : (0x8003*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD&&CL72_EN)

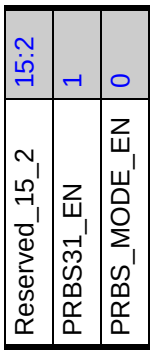


Table 5-45 Fields for Register: VR_PMA_KRTR_PRBS_CTRL0

Bits	Name	Memory Access	Description
15:2	Reserved_15_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1	PRBS31_EN	R/W	Enable PRBS31 as the KR Training Pattern If this bit is set, PCS sends PRBS31 as the training pattern during KR Training. It also checks for PRBS31 data (instead of PRBS11) in the received training frames. **Value After Reset:** 0x0 **Exists:** Always
0	PRBS_MODE_EN	R/W	Enable PRBS Mode to determine KR Training Status. If this bit is set, PCS monitors the received KRTR frames for any PRBS errors for a specified interval. (This interval starts after 300ms of KR training in case of SNPS PHYs OR after 'rpcs_rx_trained_i' is asserted for non-SNPS PHY configurations). If the number of PRBS errors are less than or equal to the programmed threshold, then Rx link is declared as Good, otherwise it is considered as a training failure. If this bit is not set, Rx link is assumed to be successfully trained after 300ms of KR Training or after 'rpcs_trained_i' input is asserted. **Value After Reset:** 0x0 **Exists:** Always

5.2.1.41 VR_PMA_KRTR_PRBS_CTRL1

- **Description:** VR PMA MMD KRTR PRBS Control1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8004 : (APB3_16) ? (0x8004*0x2) : (0x8004*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD&&CL72_EN)



Table 5-46 Fields for Register: VR_PMA_KRTR_PRBS_CTRL1

Bits	Name	Memory Access	Description
15:0	PRBS_TIM_LMT	R/W	PRBS Timer Duration Control If PRBS Mode (bit[0] of VR_PMA_KRTR_PRBS_CTRL0 Register) is enabled to determine successful Rx Equalization, DWC_xpcs monitors PRBS errors for a certain duration at the end of Rx Equalization. There is a 20-bit timer to control this monitoring interval and this timer runs at a clock frequency of 156.25 MHz. This field (PRBS_TIM_LMT) forms the upper 16-bit of the value to be loaded to the timer.The lower 4-bits are hard-coded as 4'h0. This field has a default value such that this timer runs for duration of 3ms. Application must program a non-zero value to this field if PRBS Mode is enabled. **Value After Reset:** 0x7270 **Exists:** Always

5.2.1.42 VR_PMA_KRTR_PRBS_CTRL2

- **Description:** VR PMA MMD KRTR PRBS Control2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8005 : (APB3_16) ? (0x8005*0x2) : (0x8005*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD&&CL72_EN)

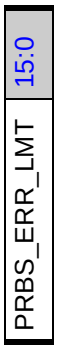


Table 5-47 Fields for Register: VR_PMA_KRTR_PRBS_CTRL2

Bits	Name	Memory Access	Description
15:0	PRBS_ERR_LMT	R/W	PRBS Tolerable Error Limit This field determines the maximum allowed PRBS errors in the received KR training frames during the interval specified by 'PRBS_TIM_LMT' field of VR_PMA_KRTR_PRBS_CTRL1 Register. If the number of PRBS errors is greater than the value programmed in this field, training is considered as a failure. **Value After Reset:** 0x0 **Exists:** Always

5.2.1.43 VR_PMA_KRTR_TIMER_CTRL0

- **Description:** VR PMA MMD KRTR Timer Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8006 : (APB3_16) ? (0x8006*0x2) : (0x8006*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD&&CL72_EN)



Table 5-48 Fields for Register: VR_PMA_KRTR_TIMER_CTRL0

Bits	Name	Memory Access	Description
15:0	MAX_WAIT_TIME	R/W	Maximum Wait Time for KR Training. This field determines the maximum wait time (max_wait_timer as per IEEE clause 72.6.10.3.2) allowed for successful KR Training. There is a 27-bit timer running at a clock frequency of 156.25 MHz to serve this purpose. This field forms the upper 16-bits of the value to be loaded to this timer. The lower 11-bits are hard-coded as 11'h7c8. The default value corresponds to 500ms as mandated by IEEE spec. **Value After Reset:** 0x9502 **Exists:** Always

5.2.1.44 VR_PMA_KRTR_TIMER_CTRL1

- **Description:** VR PMA MMD KRTR Timer Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8007 : (APB3_16) ? (0x8007*0x2) : (0x8007*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD&&CL72_EN)



Table 5-49 Fields for Register: VR_PMA_KRTR_TIMER_CTRL1

Bits	Name	Memory Access	Description
15:0	WAIT_TIME	R/W	Wait Time during KR Training. This field determines the wait time (wait_timer as per IEEE clause 72.6.10.3.2) in the LINK_READY state of KR Training. There is a 16-bit timer running at a clock frequency of 156.25 MHz to serve this purpose. This field forms the value to be loaded (at beginning) to this timer . The default value corresponds to ~64us. **Value After Reset:** 0x26eb **Exists:** Always

5.2.1.45 VR_PMA_KRTR_TIMER_CTRL2

- **Description:** VR PMA MMD KRTR Timer Control 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8008 : (APB3_16) ? (0x8008*0x2) : (0x8008*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD&&CL72_EN&&SNPS_XAUI_PHY)

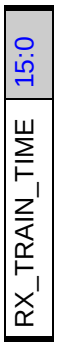


Table 5-50 Fields for Register: VR_PMA_KRTR_TIMER_CTRL2

Bits	Name	Memory Access	Description
15:0	RX_TRAIN_TIME	R/W	Timer interval for Rx Training. This field determines the duration for which Rx Equalization must be carried out for Synopsys PHY configurations. After this time interval, 'rx_trained' signal (as per IEEE clause 72.6.10.31.1) is made high by DWC_xpcs. The default value corresponds to 300ms. DWC_xpcs implements a 27-bit timer for this purpose and this register field forms the upper 16-bit of the value that is loaded to this timer. The lower 11-bits are hard-coded as 11'h178. DWC_xpcs sends coefficient update requests to remote-end partner during this time interval. There after it does not send coefficient update requests. **Value After Reset:** 0x5968 **Exists:** Always

5.2.1.46 VR_PMA_KRTR_RX_EQ_CTRL

- **Description:** VR PMA MMD KRTR Receive Equalization Control Register
  This register is present only if CL72_EN = Enabled.
  This register can be used by management to send coefficient update requests to the link partner when KR Training is enabled. This mode can be enabled by programming bit [15] of this register to 1 (This bit must be programmed to the desired value prior to enabling KR training). During this mode, the coefficient updates (and rx_trained information) given by the PHY are ignored. For Synopsys PHY configurations, DWC_xpcs does not make the receiver requests to the PHY, i.e, rpcs_krx_-adapt_req_o/rpcs_krx_eq_eval_o' ports are not asserted. The values programmed to the RR_RDY,PRST, INIT, CFF_UPDT1, CFF_UPDT0 and CFF_UPDTM1 fields of this register are sent by DWC_xpcs to the link partner.
  After programming a new coefficient update value (INC/DEC/PRST/INIT) to this register, application must wait till the response for the corresponding command is received (from link partner) in SR_PMA_KR_LP_CESTS Register. After receiving the 'UPDATED/MIN/MAX' status, application must send 'HOLD' command to the link partner by programming appropriate values to this register. Then, application must wait for 'NOT_UPDATED' status to be reflected in SR_PMA_KR_LP_CESTS Status Register.
  After a suitable number of training iterations, application can set the 'RR_RDY' field of this register to 1. At this point, DWC_xpcs infers that the local receiver is trained. Also, it passes this information to the link-partner by setting 'Receiver Ready' cell of the 'Status report' field of the KR training frame. Note that DWC_xpcs ignores the 'rpcs_rx_trained_i' input port during this mode.

  Additionally, during this software mode of receive equalization, the 'status' information received from the link partner are not driven on 'rpcs_krx_ceu{lane}_sts_o' port of DWC_xpcs. Instead, the 'status' programmed to various fields of VR_PMA_KRTR_RX_EQ_STS_CTRL Registers are driven towards the PHY.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8009 : (APB3_16) ? (0x8009*0x2) : (0x8009*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD&&CL72_EN)

15	14:9	8	7	6	5:4	3:2	1:0
RX_EQ_MM	Reserved_14_9	RR_RDY	PRST	INIT	CFF_UPDT1	CFF_UPDT0	CFF_UPDTM1

Table 5-51 Fields for Register: VR_PMA_KRTR_RX_EQ_CTRL

Bits	Name	Memory Access	Description
15	RX_EQ_MM	R/W	Enable Rx Equalization by Management This bit can be set to 1, to bypass/ignore Rx coefficient updates given by PHY. During this mode, the values programmed by management to the RR_DY,PRST, INIT, CFF_UPDT1, CFF_UPDT0 and CFF_UPDTM1 fields of this register are transmitted to the link partner. In addition, the 'status' values programmed to VR_PMA_KRTR_RX_EQ_STS_CTRL Register are driven on the 'rpcs_krx_ceu{filter}_sts_o' port of DWC_xpcs (towards the local PHY). **Value After Reset:** 0x0 **Exists:** Always
14:9	Reserved_14_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	RR_RDY	R/W	Local Receiver Ready This bit can be set to 1, to indicate to the link partner that the local receiver is trained and is ready to accept data. **Value After Reset:** 0x0 **Exists:** Always
7	PRST	R/W	Pre-initialize the Transmit Filter Coefficient of the Link Partner This bit can be set to 1, to request the link partner to set its transmitter filter coefficients to a state where transmit equalization is turned off. **Value After Reset:** 0x0 **Exists:** Always
6	INIT	R/W	Initialize the Transmit Filter Coefficient of the Link Partner This bit can be set to 1, to request the link partner to initialize its transmitter filter coefficients. **Value After Reset:** 0x0 **Exists:** Always
5:4	CFF_UPDT1	R/W	Filter Coefficient Update (+1) Request to the Link Partner Transmitter This field can be programmed suitably to request the link partner to increment, decrement, or hold the transmit filter (+1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
3:2	CFF_UPDT0	R/W	Filter Coefficient Update (0) Request to the Link Partner Transmitter This field can be programmed suitably to request the link partner to increment, decrement, or hold the transmit filter (0) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always
1:0	CFF_UPDTM1	R/W	Filter Coefficient Update (-1) Request to the Link Partner Transmitter This field can be programmed suitably to request the link partner to increment, decrement, or hold the transmit filter (-1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

5.2.1.47 VR_PMA_KRTR_TX_EQ_STS_CTRL

- **Description:** VR PMA MMD KRTR Transmit Equalization Status Control Register
  This register is present only if CL72_EN = Enabled.
  This register provides a mechanism to allow the software mode of Transmit Equalization. This mode can be enabled by programming bit [15] of this register to 1 (This bit must be programmed to the desired value prior to enabling KR training). During this mode the Tx coefficient update commands (driven towards local PHY) and status (sent to link-partner) are totally controlled by the software.

  The coefficient update commands received from the link partner are available in SR_PMA_KR_LP_CEU Register. Irrespective of the command received from the link-partner,software can drive any desired Tx coefficient Update command towards the local PHY by writing suitable values to VR_PMA_KRTR_TX_EQ_CFF_CTRL register. In addition, the 'status' to be returned to the link-partner is controlled by the values programmed to VR_PMA_KRTR_TX_EQ_STS_CTRL register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x800B : (APB3_16) ? (0x800B*0x2) : (0x800B*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD&&CL72_EN)

15	14:6	5:4	3:2	1:0
TX_EQ_MM	Reserved_14_6	CFF_STS1	CFF_STS0	CFF_STSM1

Table 5-52 Fields for Register: VR_PMA_KRTR_TX_EQ_STS_CTRL

Bits	Name	Memory Access	Description
15	TX_EQ_MM	R/W	Management Mode of Transmit Equalization. This bit can be set to 1 to allow the software to decide the transmit equalization coefficient/status. During this mode, the coefficient update commands in the received training frames are not passed on to the local PHY.Instead the values programmed to VR_PMA_KRTR_TX_EQ_CFF_CTRL register are driven towards the PHY. In addition, the status values (to be sent to the link partner) are also controlled by the software.DWC_xpcs ignores the coefficient status given by the local PHY (or computed internally by DWC_xpcs' training algorithm) during KR training operation. The values programmed by management to the CFF_STS1, CFF_STS0 and CFF_STSM1 fields of this register are transmitted to the link partner. **Value After Reset:** 0x0 **Exists:** Always
14:6	Reserved_14_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	CFF_STS1	R/W	Filter Coefficient (+1) Status to the Link Partner Transmitter This field can be programmed suitably to send min/max/updated/not-updated status information corresponding to the transmit filter (+1) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-updated **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_STS0	R/W	Filter Coefficient (0) Status to the Link Partner Transmitter This field can be programmed suitably to send min/max/updated/not-updated status information corresponding to the transmit filter (0) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-updated **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
1:0	CFF_STSM1	R/W	Filter Coefficient (-1) Status to the Link Partner Transmitter This field can be programmed suitably to send min/max/updated/not-updated status information corresponding to the transmit filter (-1) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-updated **Value After Reset:** 0x0 **Exists:** Always

5.2.1.48 VR_PMA_KRTR_TX_EQ_CFF_CTRL

- **Description:** VR PMA MMD KRTR Transmit Equalization Coefficient Control Register.

  This register is present only if CL72_EN = Enabled. This register can be used by management (software) to control the coefficient updates commands driven to the local PHY. This mode can be enabled by programming bit[15] of VR_PMA_KRTR_TX_EQ_STS_CTRL register to 1. During this mode, DWC_xpcs ignores the Tx coefficient update commands received from the link partner (reflected in SR_PMA_KR_LP_CEU Register). Instead, the values programmed by the management to the various fields of this register are considered as the Tx coefficient update commands that need to be driven to the local PHY.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x800C : (APB3_16) ? (0x800C*0x2) : (0x800C*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD&&CL72_EN)

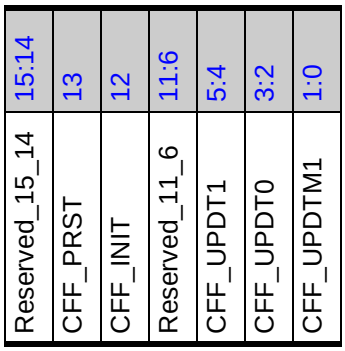


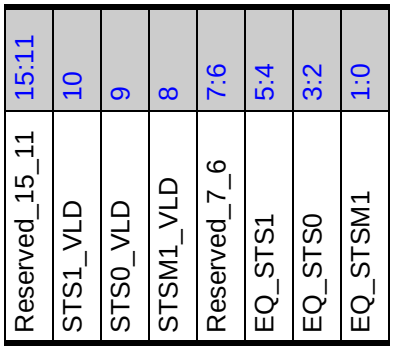
Table 5-53 Fields for Register: VR_PMA_KRTR_TX_EQ_CFF_CTRL

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13	CFF_PRST	R/W	Preset Coefficient Management can set this bit to 1 to request the local PHY to set its transmitter filter coefficients to a state where transmit equalization is turned off. **Note:**If this bit is set, then the values programmed to CFF_UPDTM1, CFF_UPDT0 and CFF_UPDT1 fields of this register are meaningless. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
12	CFF_INIT	R/W	Initialize Coefficient Management can set this bit to 1, to request the local PHY to initialize its transmitter filter coefficients **Note:** If this bit is set, then the values programmed to CFF_UPDTM1, CFF_UPDT0 and CFF_UPDT1 fields of this register are meaningless. **Value After Reset:** 0x0 **Exists:** Always
11:6	Reserved_11_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	CFF_UPDT1	R/W	Co-efficient Update (+1) Management can set this bit to 1, to request the local PHY to increment, decrement, or hold the transmit filter (+1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always
3:2	CFF_UPDT0	R/W	Co-efficient Update (0) Management can set this bit to 1, to request the local PHY to increment, decrement, or hold the transmit filter (0) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always
1:0	CFF_UPDTM1	R/W	Coefficient Update (-1) Management can set this bit to 1, to request the local PHY to increment, decrement, or hold the transmit filter (-1) coefficient. The coefficient is updated as follows: ■ 11: Reserved ■ 01: Increment ■ 10: Decrement ■ 00: Hold **Value After Reset:** 0x0 **Exists:** Always

5.2.1.49 VR_PMA_PHY_TX_EQ_STS

- **Description:** VR PMA MMD PHY Transmit Equalization Status Register
  This register is present only if CL72_EN = Enabled.This register indicates the Tx equalization status provided by the local PHY (on rpcs_ktx_ceu{filter}_sts_i ports). For Synopsys PHY configurations, this register indicates the Tx equalization status computed internally by the DWC_xpcs (based on the selected PHY algorithm).
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x800D : (APB3_16) ? (0x800D*0x2) : (0x800D*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD&&CL72_EN)



15:11	10	9	8	7:6	5:4	3:2	1:0
Reserved_15_11	STS1_VLD	STS0_VLD	STSM1_VLD	Reserved_7_6	EQ_STS1	EQ_STS0	EQ_STSM1

Table 5-54 Fields for Register: VR_PMA_PHY_TX_EQ_STS

Bits	Name	Memory Access	Description
15:11	Reserved_15_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
10	STS1_VLD	R	Filter Coefficient (1) Status is Valid. If this bit is 1, it indicates that the Status indicated in EQ_STS1 field is valid. **Value After Reset:** 0x0 **Exists:** Always
9	STS0_VLD	R	Filter Coefficient (0) Status is Valid. If this bit is 1, it indicates that the Status indicated in EQ_STS0 field is valid. **Value After Reset:** 0x0 **Exists:** Always
8	STSM1_VLD	R	Filter Coefficient (-1) Status is Valid. If this bit is 1, it indicates that the Status indicated in EQ_STSM1 field is valid. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	EQ_STS1	R	Filter Coefficient (1) Status given by local PHY. The status indicated in this field is valid when STS1_VLD is high. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always
3:2	EQ_STS0	R	Filter Coefficient (0) Status given by local PHY. The status indicated in this field is valid when STS0_VLD is high. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always
1:0	EQ_STSM1	R	Filter Coefficient (-1) Status given by local PHY. The status indicated in this field is valid when STSM1_VLD is high. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not updated **Value After Reset:** 0x0 **Exists:** Always

5.2.1.50 VR_PMA_PHY_RX_EQ_CEU

- **Description:** VR PMA MMD PHY Receive Equalization Coefficient Update
  This register is present when the Clause 72 KR Training support is enabled. This register indicates the command update commands given by the local PHY.
  During normal (hardware) mode of Rx equalization, these coefficient update commands are transmitted to the link partner. Therefore, the values appearing in this register reflects in the SR_PMA_KR_LD_CEU Register.
  If management mode of Rx equalization is enabled, the coefficient update commands given by local PHY are not transmitted to the link-partner; instead the values programmed to VR_PMA_KRTR_RX_EQ_CTRL Register is transmitted o the link partner. This register provides a means for the software to know the coefficient update commands given by the local PHY.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x800E : (APB3_16) ? (0x800E*0x2) : (0x800E*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD&&CL72_EN)

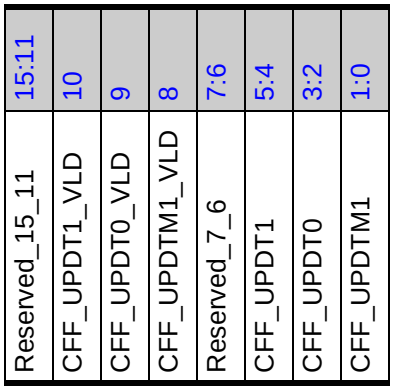


Table 5-55 Fields for Register: VR_PMA_PHY_RX_EQ_CEU

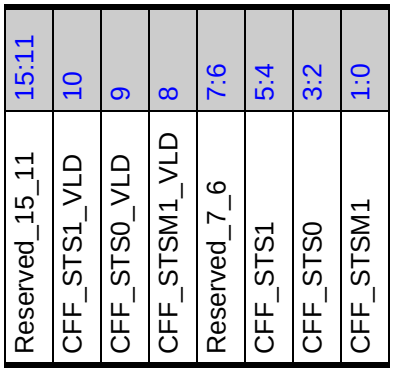
Bits	Name	Memory Access	Description
15:11	Reserved_15_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
10	CFF_UPDT1_VLD	R	Filter Coefficient Update (1) Request is Valid. When this bit is 1, it indicates that the coefficient update request in 'CFF_UPDT1' field of this registers is valid. ■ For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu2_vld_i' port. ■ For Synopsys Multi-protocol 12G/16G PHY configurations, this field reflects the value of 'rpcs_krx_adapt_ack_i' port. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
9	CFF_UPDT0_VLD	R	Filter Coefficient Update (0) Request is Valid. When this bit is 1, it indicates that the coefficient update request in 'CFF_UPDT0' field of this registers is valid. ■ For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu1_vld_i' port. ■ For Synopsys Multi-protocol 12G/16G PHY configurations, this field reflects the value of 'rpcs_krx_adapt_ack_i' port. **Value After Reset:** 0x0 **Exists:** Always
8	CFF_UPDTM1_VLD	R	Filter Coefficient Update (-1) Request is Valid. When this bit is 1, it indicates that the coefficient update request in 'CFF_UPDTM1' field of this register is valid. ■ For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu0_vld_i' port. ■ For Synopsys Multi-protocol 12G/16G PHY configurations, this field reflects the value of 'rpcs_krx_adapt_ack_i' port. **Value After Reset:** 0x0 **Exists:** Always
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	CFF_UPDT1	R	Filter Coefficient Update (+1) Request by local PHY receiver When this bit is set to 1, it indicates that the local device is requesting the link partner to increment, decrement, or hold the transmit filter (1) coefficient. The coefficient is updated as follows: ■ 11: Initialize (valid only for Non-Synopsys PHY configurations) ■ 01: Increment ■ 10: Decrement ■ 00: Preset (valid only for Non-Synopsys PHY configurations) For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu2_i' port. For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_main_di2_i' port. The validity of this field is indicated by bit[10] of this register. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
3:2	CFF_UPDT0	R	Filter Coefficient Update (0) Request by local PHY receiver When this bit is set to 1, it indicates that the local device is requesting the link partner to increment, decrement, or hold the transmit filter (0) coefficient. The coefficient is updated as follows: ■ 11: Initialize (valid only for Non-Synopsys PHY configurations) ■ 01: Increment ■ 10: Decrement ■ 00: Preset (valid only for Non-Synopsys PHY configurations) For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu1_i' port. For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_main_dir_i' port. The validity of this field is indicated by bit[9] of this register. **Value After Reset:** 0x0 **Exists:** Always
1:0	CFF_UPDTM1	R	Filter Coefficient Update (-1) Request by local PHY receiver When this bit is set to 1, it indicates that the local device is requesting the link partner to increment, decrement, or hold the transmit filter (-1) coefficient. The coefficient is updated as follows: ■ 11: Initialize (valid only for Non-Synopsys PHY configurations) ■ 01: Increment ■ 10: Decrement ■ 00: Preset (valid only for Non-Synopsys PHY configurations) For non-Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_ceu0_i' port. For Synopsys PHY configurations, this field reflects the value of 'rpcs_krx_pre_dir_i' port. The validity of this field is indicated by bit[8] of this register. **Value After Reset:** 0x0 **Exists:** Always

5.2.1.51 VR_PMA_KRTR_RX_EQ_STS_CTRL

- **Description:** VR PMA MMD KRTR Receive Equalization Status Control Register
  This register is present only if CL72_EN = Enabled and if SNPS_XAUI_PHY = Disabled.
  This register allows the software to control the Rx equalization status driven towards the local PHY whenever RX_EQ_MM bit (of VR_PMA_KRTR_RX_EQ_CTRL Register) is set to 1.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x800F : (APB3_16) ? (0x800F*0x2) : (0x800F*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD&&CL72_EN&&!SNPS_XAUI_PHY)



15:11	10	9	8	7:6	5:4	3:2	1:0
Reserved_15_11	CFF_STS1_VLD	CFF_STS0_VLD	CFF_STSM1_VLD	Reserved_7_6	CFF_STS1	CFF_STS0	CFF_STSM1

Table 5-56 Fields for Register: VR_PMA_KRTR_RX_EQ_STS_CTRL

Bits	Name	Memory Access	Description
15:11	Reserved_15_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
10	CFF_STS1_VLD	R/W	Filter Coefficient (1) Status driven to local PHY is Valid This field can be set to 1 to indicate (to the local PHY) that the value programmed to CFF_STS1 field is valid. The value programmed to this field is driven on 'rpcs_krx_ceu2_ack_o' port. Software must set this bit as 1, after programming either 01 or 10 or 11 to CFF_STS1 field. It is recommended that software does not assert this bit, if CFF_STS1 is programmed to 00 (Not-Updated). **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
9	CFF_STS0_VLD	R/W	Filter Coefficient (0) Status driven to local PHY is Valid This field can be set to 1 to indicate (to the local PHY) that the value programmed to CFF_STS0 field is valid. The value programmed to this field is driven on 'rpcs_krx_ceu1_ack_o' port. Software must set this bit as 1, after programming either 01 or 10 or 11 to CFF_STS0 field. It is recommended that software does not assert this bit, if CFF_STS0 is programmed to 00 (Not-Updated). **Value After Reset:** 0x0 **Exists:** Always
8	CFF_STSM1_VLD	R/W	Filter Coefficient (-1) Status driven to local PHY is Valid This field can be set to 1 to indicate (to the local PHY) that the value programmed to CFF_STSM1 field is valid. The value programmed to this field is driven on 'rpcs_krx_ceu0_ack_o' port. Software must set this bit as 1, after programming either 01 or 10 or 11 to CFF_STSM1 field. It is recommended that software does not assert this bit, if CFF_STSM1 is programmed to 00 (Not-Updated). **Value After Reset:** 0x0 **Exists:** Always
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	CFF_STS1	R/W	Filter Coefficient (+1) Status driven to local PHY This field can be programmed suitably to send MAX/MIN/UPDATED status information corresponding to the filter (0) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-Updated The value programmed to this field is reflected on 'rpcs_krx_ceu2_sts_o' port. The value programmed to this field is considered valid when bit[10] of this register is set to 1. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
3:2	CFF_STS0	R/W	Filter Coefficient (0) Status driven to local PHY This field can be programmed suitably to send MAX/MIN/UPDATED status information corresponding to the filter (0) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-Updated The value programmed to this field is reflected on 'rpcs_krx_ceu1_sts_o' port. The value programmed to this field is considered valid when bit[9] of this register is set to 1. **Value After Reset:** 0x0 **Exists:** Always
1:0	CFF_STSM1	R/W	Filter Coefficient (-1) Status driven to local PHY This field can be programmed suitably to send MAX/MIN/UPDATED status information corresponding to the filter (-1) coefficient. The status information is encoded as follows: ■ 11: Maximum ■ 10: Minimum ■ 01: Updated ■ 00: Not-updated The value programmed to this field is reflected on 'rpcs_krx_ceu0_sts_o' port. The value programmed to this field is considered valid when bit[8] of this register is set to 1. **Value After Reset:** 0x0 **Exists:** Always

5.2.1.52 VR_PMA_DIG_STS

- **Description:** VR PMA MMD Digital Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8010 : (APB3_16) ? (0x8010*0x2) : (0x8010*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** PMA_MMD

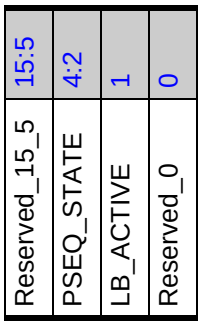


Table 5-57 Fields for Register: VR_PMA_DIG_STS

Bits	Name	Memory Access	Description
15:5	Reserved_15_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4:2	PSEQ_STATE	R	**For Synopsys PHY configurations:** Power Up Sequence State This field indicates the state variable value of the power-up sequence module. For Synopsys Multi-protocol 6G PHY: ■ 3'b000: Wait for MPLL ON ■ 3'b001: Wait for TX up ■ 3'b010: Wait for RX up ■ 3'b011/3'b100: Tx/Rx Stable (Power_Good state) ■ 3'b101: Wait for Rx down (MPLL still ON) ■ 3'b110: MPLL OFF For Synopsys Multi-protocol 12G/16G/25G/32G PHY ■ 3'b000: Wait for ACK High 0 ■ 3'b001: Wait for ACK Low 0 ■ 3'b010: Wait for ACK High 1 ■ 3'b011: Wait for ACK Low 1 ■ 3'b100: Tx/Rx stable (Power_Good state) ■ 3'b101: Power Save state ■ 3'b110: Power Down state ■ 3'b111: Other intermediate states (valid only for 32G PHY configurations) **For non-Synopsys PHY configurations:** Reserved **Value After Reset:** 0x4 **Exists:** Always
1	LB_ACTIVE	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	Reserved_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.1.53 VR_PMA_GPIO

- **Description:** VR PMA MMD GPIO Register
  This register is present only if the DWC_xpcs is configured to have the GPIO interface and if USXGMII_EN = Disabled.
  For configurations with USXGMII support,VR_XS_PCS_GPIO register is applicable.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8015 : (APB3_16) ? (0x8015*0x2) : (0x8015*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x010000 : (APB3_16) ? (0x010000*0x2) : (0x010000*0x4)"
- **Exists:** (PMA_MMD&&GPIO_EN&&!USXGMII_EN)

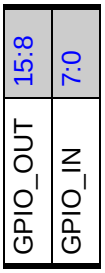


Table 5-58 Fields for Register: VR_PMA_GPIO

Bits	Name	Memory Access	Description
15:8	GPIO_OUT	R/W	GPIO Output The content written in this field is driven to the xpcs_gpo_o[7:0] output port. **Note:** This field is shared with Bits[15:8] of VR_XS_PCS_GPIO Register and VR_MII_GPIO Register. **Value After Reset:** 0x0 **Exists:** Always
7:0	GPIO_IN	R	GPIO Input This field indicates the content of the xpcs_gpo_i[7:0] port. **Value After Reset:** 0x0 **Exists:** Always

5.2.2 DWC_xpcs_map/XS_PCS_MMD Registers

5.2.2.1 SR_XS_PCS_CTRL1

- **Description:** SR XS or PCS MMD Control1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(PCS_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD | |XGXS_PCS))

15	14	13	12	11	10	9	8:7	6	5:2	1:0
RST	LBE	SS13	Reserved_12	LPM	CS_EN	XAUI_STOP	Reserved_8_7	SS6	SS_5_2	Reserved_1_0

Table 5-59 Fields for Register: SR_XS_PCS_CTRL1

Bits	Name	Memory Access	Description
15	RST	R/W	Soft Reset (RW,SC Type) **For Synopsys PHY configurations:** When the host sets this bit, the CSR block triggers the software reset process in which all internal blocks are reset, except the Management Interface block. The registers are reset to their default values. When this bit is set, it also resets the PHY. This bit is self-cleared when PSEQ_STATE in VR_XS_PCS_DIG_STS Register is equal to 3'b100, that is, Tx/Rx clocks are stable and in Power_Good state. **For Non-Synopsys PHY configurations:** When the host sets this bit, the CSR block triggers the software reset process in which all internal blocks are reset, except the Management Interface block. The registers are reset to their default values.When this bit is set, it also resets the PHY. This bit is self-cleared after the following: ■ 32 clk_csr_i clocks for the MCI interface ■ 1 MDC clock period for the MDIO interface **Note:** For a description of read or write access for registers during the reset process, see "Special Case Register Access" section. **Value After Reset:** 0x0 **Exists:** Always
14	LBE	* Varies	**When XS MMD:** Loopback Enable When set, this bit loops the PHY Tx lanes back to the PHY Rx lanes. ■ 0: DWC_xpcs de-asserts the xpcs_loopback_en_o signal. ■ 1: DWC_xpcs asserts xpcs_loopback_en_o signal. **When PCS MMD:**Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(XGXS_PCS) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
13	SS13	* Varies	Speed Selection This bit indicates the operational mode of the PCS-X block of the DWC_xpcs. ■ 0: DWC_xpcs is in the KX mode (1G speed) ■ 1: DWC_xpcs is in the KX4 mode (10G speed) Setting or resetting this bit causes the xpcs_kx_kx4n_o output port (if present) to be asserted or de-asserted. **Value After Reset:** "(KX4_KX_PRESENT||SUPPORT_10GX_1GX) ? 0x1 : ((ONLY_1G_CFG||KR_KX||SUPPORT_10GR_1GX) ? 0x0 : 0x1)" **Exists:** Always **Memory Access:** "(KX4_KX_PRESENT||SUPPORT_10GX_1GX) ? \"read-write\" : \"read-only\""
12	Reserved_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11	LPM	R/W	Low-Power Enable This bit controls the power-down mode of DWC_xpcs. ■ 0: Normal operation ■ 1: DWC_xpcs goes to power-down mode along with the PHY. For non-Synopsys PHY configurations, xpcs_pdown_o port is asserted during low-power mode. For Synopsys PHY configurations, DWC_xpcs does the following to trigger power-down mode: ■ Turns off the PHY Receiver and Transmitter ■ Switches off all the clocks When the host clears this bit, DWC_xpcs resumes normal operation. In a Synopsys PHY configuration, after clearing this bit, the host must wait until Bits[4:2] of VR_XS_PCS_DIG_STS Register indicate that DWC_xpcs is in normal state. **Note:** For a description of read or write access for registers during the reset process, see "Special Case Register Access" section. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
10	CS_EN	* Varies	**For EEE configurations:** Clock Stop Enable This bit controls stopping the clock during LPI mode on the GMII or XGMII Rx interface: ■ 1: The PHY might stop the clock during LPI mode. ■ 0: The clock cannot be stopped during LPI mode. You must program this bit based on the capability of the XGMAC during Rx LPI mode. **Access Type:** ■ For configurations with EEE_EN = Enabled, this bit is RW. ■ For all other configurations, this bit is reserved and RO. **For other configurations:** Reserved **Value After Reset:** "(EEE_EN) ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\""
9	XAUI_STOP	* Varies	**For XGXS configurations with EEE_EN = Enabled:** XAUI Stop This bit indicates that the PHY might freeze the transmit serial lines during LPI. ■ 1: The PHY might freeze the serial lines. ■ 0: The PHY must not freeze the serial lines. You must program this bit based on the system capability. This bit is present only for the XS MMD. For PCS MMD, this bit is reserved. **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(EEE_EN&&XGXS_PCS) ? \"read-write\" : \"read-only\""
8:7	Reserved_8_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6	SS6	R	Speed Selection Bit 6 This bit always returns 1. **Value After Reset:** 0x1 **Exists:** Always
5:2	SS_5_2	R	Speed Selection Bits [5:2] This field always returns 4'b0000. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
1:0	Reserved_1_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.2.2 SR_XS_PCS_STS1

- **Description:** SR XS or PCS MMD Status1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1 : (APB3_16) ? (0x1*0x2) : (0x1*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD||XGXS_PCS))

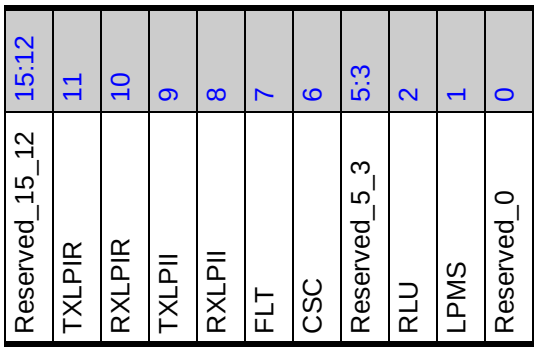


Table 5-60 Fields for Register: SR_XS_PCS_STS1

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11	TXLPIR	R	**For Configurations with EEE:** Tx LPI Received (RO,LH Type) This bit indicates if the DWC_xpcs Tx received LPI in the Transmit path from the GMII or XGMII interface of the MAC or XGMAC: ■ 1: LPI received ■ 0: LPI not received This bit is present only if EEE_EN = Enabled. Otherwise, this bit is reserved. **For all configurations without EEE:** Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
10	RXLPIR	R	**For Configurations with EEE:** Rx LPI Received (RO,LH Type) This bit indicates if the DWC_xpcs Rx received LPI in the Receive path from the link partner: ■ 1: LPI received ■ 0: LPI not received This bit is present only if EEE_EN = Enabled. Otherwise, this bit is reserved. **For all configurations without EEE:** Reserved **Value After Reset:** 0x0 **Exists:** Always
9	TXLPII	R	**For Configurations with EEE:** Tx LPI Indication This bit indicates if DWC_xpcs Tx is currently receiving LPI in the Transmit path from the GMII or XGMII interface of the MAC or XGMAC: ■ 1: DWC_xpcs Tx is currently receiving LPI ■ 0: DWC_xpcs Tx is currently not receiving LPI This bit is present only if EEE_EN = Enabled. Otherwise, this bit is reserved. **For all configurations without EEE:** Reserved **Value After Reset:** 0x0 **Exists:** Always
8	RXLPII	R	**For Configurations with EEE:** Rx LPI Indication This bit indicates if DWC_xpcs Rx is currently receiving LPI in the Receive path from the link partner: ■ 1: DWC_xpcs Rx is currently receiving LPI ■ 0: DWC_xpcs Rx is currently not receiving LPI This bit is present only if EEE_EN = Enabled. Otherwise, this bit is reserved. **For all configurations without EEE:** Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
7	FLT	R	Fault Condition Detected This bit indicates that a fault condition is detected in the DWC_xpcs. This bit is set when the Tx or Rx path of the DWC_xpcs encounters any of the following fault conditions: ■ Rx path did not align or synchronize ■ Tx fault encountered the remote fault signaling **Note:** This field is valid only in 10G Mode. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
6	CSC	R	Clock Stop Capable This bit indicates if the MAC can stop the clock during the LPI mode in the transmit path: ■ 1: The MAC can stop the clock during the LPI mode. ■ 0: The MAC cannot stop the clock during the LPI mode. This bit is present only if EEE_EN = Enabled. Otherwise, this bit is reserved. **Value After Reset:** "(PCS_XS_MMD&&EEE_EN) ? 0x1 : 0x0" **Exists:** Always
5:3	Reserved_5_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2	RLU	R	Rx Link Up (RO,LL Type) XS MMD and PCS MMD in 10G mode: ■ 0: When code group alignment and block lock are not successful. ■ 1: When code group alignment and block lock are successful. PCS MMD in 1G mode: ■ 0: When code group alignment on Lane 0 is not successful. ■ 1: When code group alignment on Lane 0 is successful. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
1	LPMS	R	Low-Power Mode Support This bit indicates that the DWC_xpcs supports the low-power mode. **Value After Reset:** 0x1 **Exists:** Always

Bits	Name	Memory Access	Description
0	Reserved_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.2.3 SR_XS_PCS_DEV_ID1

- **Description:** SR XS or PCS MMD Device Identifier 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x2 : (APB3_16) ? (0x2*0x2) : (0x2*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD||XGXS_PCS))



Table 5-61 Fields for Register: SR_XS_PCS_DEV_ID1

Bits	Name	Memory Access	Description
15:0	PCS_DEV_OUI_3_18	R	Organizationally Unique Identifier[3:18] The DWC_xpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. This value is configured in coreConsultant. This read-only field contains Bits[18:3] of the 24-bit OUI of device manufacturer. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:0] of SR_VSMMD_PCS_ID1 Register modifies the content of this register. **Value After Reset:** IEEE_DEV_OUI_3_18 **Exists:** Always

5.2.2.4 SR_XS_PCS_DEV_ID2

- **Description:** SR XS or PCS MMD Device Identifier 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x3 : (APB3_16) ? (0x3*0x2) : (0x3*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD||XGXS_PCS))

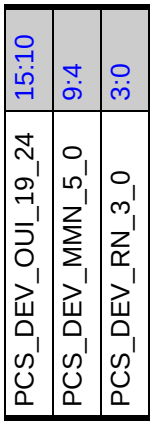


Table 5-62 Fields for Register: SR_XS_PCS_DEV_ID2

Bits	Name	Memory Access	Description
15:10	PCS_DEV_OUI_19_24	R	Organizationally Unique Identifier[19:24] The DWC_xpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. You can configure the value of each MMD using coreConsultant. This read-only field contains Bits[24:19] of 24-bit OUI of the device manufacturer. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:10] of SR_VSMMD_PCS_ID2 Register modifies the content of this register. **Value After Reset:** IEEE_DEV_OUI_19_24 **Exists:** Always
9:4	PCS_DEV_MMN_5_0	R	Model Number This read-only field contains the 6-bit Model number of the device. You can configure the value of each MMD using coreConsultant. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[9:4] of SR_VSMMD_PCS_ID2 Register modifies the content of this register. **Value After Reset:** IEEE_DEV_MODEL_NO_0_5 **Exists:** Always

Bits	Name	Memory Access	Description
3:0	PCS_DEV_RN_3_0	R	Revision Number This read-only field contains the 4-bit Revision number of the device. You can configure the value of each MMD using coreConsultant. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[3:0] of SR_VSMMD_PCS_ID2 Register modifies the content of this register. **Value After Reset:** IEEE_DEV_REV_NO_0_3 **Exists:** Always

5.2.2.5 SR_XS_PCS_SPD_ABL

- **Description:** SR XS or PCS MMD Speed Ability Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x4 : (APB3_16) ? (0x4*0x2) : (0x4*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD||XGXS_PCS))

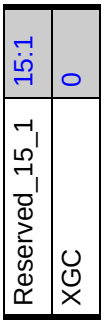


Table 5-63 Fields for Register: SR_XS_PCS_SPD_ABL

Bits	Name	Memory Access	Description
15:1	Reserved_15_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	XGC	R	10G Capable This bit indicates that DWC_xpcs can support 10G mode. **Value After Reset:** "(KX_ONLY) ? 0x0 : 0x1" **Exists:** Always

5.2.2.6 SR_XS_PCS_DEV_PKG1

- **Description:** SR XS or PCS MMD Devices in Package 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x5 : (APB3_16) ? (0x5*0x2) : (0x5*0x4)" + "(PCS_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD | |XGXS_PCS))

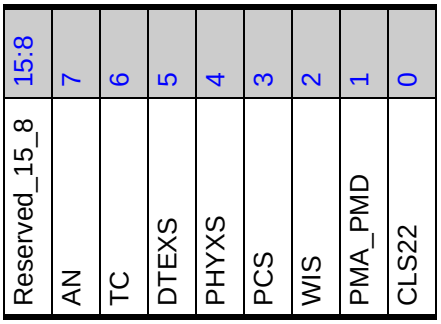


Table 5-64 Fields for Register: SR_XS_PCS_DEV_PKG1

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7	AN	R	Auto-Negotiation MMD This bit indicates that an AN MMD is present in the package. **Value After Reset:** "(BACKPLANE_ETHERNET_PCS) ? 0x1 : 0x0" **Exists:** Always
6	TC	R	TC MMD This bit indicates that the device has the TC MMDs in the package. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
5	DTEXS	R	DTE XGXS MMD This bit indicates that the device has the DTE XS MMD in the package. **Value After Reset:** "(XGXS_PCS&&(DTE_PHY_SEL==0)) ? 0x1 : 0x0" **Exists:** Always

Bits	Name	Memory Access	Description
4	PHYXS	R	PHY XGXS MMD This bit indicates that the device has the PHY XS MMD in the package. **Value After Reset:** "(XGXS_PCS&&(DTE_PHY_SEL==1)) ? 0x1 : 0x0" **Exists:** Always
3	PCS	R	PCS MMD This bit indicates that the device has the PCS MMD in the package. Default Value: If MAIN_MODE = Backplane Ethernet PCS, 10GBASE-X PCS, or 10GBASE-R PCS, this bit returns 1. For all other configurations, this bit returns 0. **Value After Reset:** "(PCS_MMD) ? 0x1 : 0x0" **Exists:** Always
2	WIS	R	WIS MMD This bit indicates that the device has the WIS MMDs in the package. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
1	PMA_PMD	R	PMA or PMD MMD This bit indicates that the device has PMA or PMD MMD in the package. Default Value: This bit is set to 1 for the following configurations: ■ Backplane Ethernet PCS ■ 10GBASE-X PCS with Synopsys PHY ■ 10GBASE-R PCS with Synopsys PHY This bit returns 0 for all other DWC_xpcs configurations. **Value After Reset:** "(PMA_MMD&&!BASE1000X_ONLY) ? 0x1 : 0x0" **Exists:** Always
0	CLS22	R	Clause 22 Register Support This bit indicates that the Clause 22 (1G) functional register sets are implemented in this package. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.7 SR_XS_PCS_DEV_PKG2

- **Description:** SR XS or PCS MMD Devices in Package 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x6 : (APB3_16) ? (0x6*0x2) : (0x6*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD||XGXS_PCS))

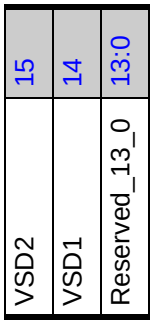


Table 5-65 Fields for Register: SR_XS_PCS_DEV_PKG2

Bits	Name	Memory Access	Description
15	VSD2	R	Vendor-Specific Device 2 This bit indicates that a vendor-specific device 2 (MII MMD) is present in the package. **Value After Reset:** "(BASE1000X_ONLY||CL37_AN) ? 0x1 : 0x0" **Exists:** Always
14	VSD1	R	Vendor-specific Device 1 This bit indicates that the vendor-specific device 1 is present in the package. This bit always returns 1. **Value After Reset:** 0x1 **Exists:** Always
13:0	Reserved_13_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.2.8 SR_XS_PCS_CTRL2

- **Description:** SR PCS Control2 Register
  This register is applicable only for the PCS MMD.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x7 : (APB3_16) ? (0x7*0x2) : (0x7*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD)

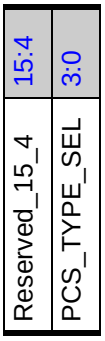


Table 5-66 Fields for Register: SR_XS_PCS_CTRL2

Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
3:0	PCS_TYPE_SEL	* Varies	PCS Type Select This field is used to select PCS operating mode in DWC_xpcs when multiple speed modes are present. ■ 0001: Select 10GBASE-X PCS Type ■ 0000: Select 10GBASE-R PCS Type ■ 1111: Select 5GBASE-R PCS Type (as per IEEE802.3cb) ■ 1110: Select 2.5GBASE-X PCS Type (as per IEEE802.3cb) ■ Other Values: Reserved When this field is written with 4'b0000, the 10GBASE-KR speed mode is selected.When this field is written to 4'b1111, 5GBASE-KR speed mode is selected. When this field is written to 4'b1110, 2.5GBASE-KX speed mode is selected. When this field is written with any other value, DWC_xpcs operates in KX4(10GBASE-X) or KX(1000Base-X) mode depending on Bit 13 of SR_PMA_CTRL1 Register or SR_XS_PCS_CTRL1 Register. **Note:** For USXGMII mode of operation, this field must be set as 4'b0000. This field is present only in the PCS MMD. **Value After Reset:** "(!RPCS_EN) ? 0x1 : 0x0" **Exists:** Always **Testable:** untestable **Memory Access:** "((RPCS_EN&&!RPCS_ONLY)||ADD_5G||ADD_2PT5G) ? \"read-write\" : \"read-only\""

5.2.2.9 SR_XS_PCS_STS2

- **Description:** SR XS or PCS MMD Status2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8 : (APB3_16) ? (0x8*0x2) : (0x8*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD||XGXS_PCS))

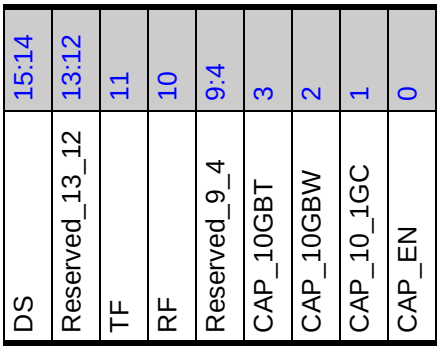


Table 5-67 Fields for Register: SR_XS_PCS_STS2

Bits	Name	Memory Access	Description
15:14	DS	R	Device Present Status ■ 2'b10: Indicates that the MMD is present and responding to this register address. ■ Other values: Indicates the MMD is not present or not functioning properly. These bits are always set to 2'b10. **Value After Reset:** 0x2 **Exists:** Always
13:12	Reserved_13_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11	TF	R	Transmitter Fault (LH Type) This bit indicates that DWC_xpcs encountered a remote fault condition being transmitted on the Tx path. This bit is not valid in 1G/KX modes. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

Bits	Name	Memory Access	Description
10	RF	R	Receiver Fault (LH Type) This bit indicates that the device encountered a fault condition on the Rx path. This bit is set when the code group alignment or lane-to-lane deskew status is not successful in PCS-X mode and Block Lock is not successful in the PCS-R mode. This bit is also set when the Rx path receives the remote-fault message. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
9:4	Reserved_9_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3	CAP_10GBT	R	10GBASE-T Capable This bit indicates if DWC_xpcs is capable of 10GBASE-T operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
2	CAP_10GBW	R	10GBASE-W Capable This bit indicates if DWC_xpcs is capable of the 10GBASE-W operation. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
1	CAP_10_1GC	R	10GBASE-X Capable This bit indicates if DWC_xpcs is capable of the 10GBASE-X operation. **Value After Reset:** "(!SINGLE_LANE_CFG) ? 0x1 : 0x0" **Exists:** Always
0	CAP_EN	R	10GBASE-R Capable This bit indicates if DWC_xpcs is capable of 10GBASE-R operation. ■ 1: DWC_xpcs can support 10GBASE-R PCS type ■ 0: DWC_xpcs cannot support 10GBASE-R PCS type **Value After Reset:** "(RPCS_EN) ? 0x1 : 0x0" **Exists:** Always

5.2.2.10 SR_XS_PCS_STS3

- **Description:** SR PCS MMD Status3 Register
  This register is applicable only for the PCS MMD
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x9 : (APB3_16) ? (0x9*0x2) : (0x9*0x4)" + "(PCS_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** PCS_MMD

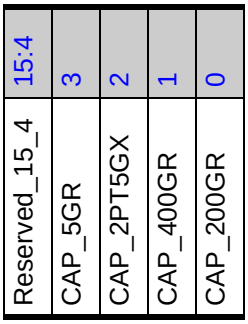


Table 5-68 Fields for Register: SR_XS_PCS_STS3

Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3	CAP_5GR	R	5GBASE-R Capable This bit if DWC_xpcs supports 5GBASE-R mode. **Value After Reset:** "ADD_5G ? 0x1 : 0x0" **Exists:** Always
2	CAP_2PT5GX	R	2.5GBASE-X Capable This bit if DWC_xpcs supports 2.5GBASE-X mode. **Value After Reset:** "ADD_2PT5G ? 0x1 : 0x0" **Exists:** Always
1	CAP_400GR	R	400GBASE-R Capable This bit always return 0. **Value After Reset:** 0x0 **Exists:** Always
0	CAP_200GR	R	200GBASE-R Capable This bit always return 0. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.11 SR_XS_PCS_PKG1

- **Description:** SR XS or PCS MMD Package Identifier 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0xe : (APB3_16) ? (0xe*0x2) : (0xe*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD||XGXS_PCS))



Table 5-69 Fields for Register: SR_XS_PCS_PKG1

Bits	Name	Memory Access	Description
15:0	PCS_PKG_OUI_3_18	R	Organizationally Unique Identifier DWC_xpcs offers 24 configurable bits[24:1], known as OUI, for identifying the device manufacturer. This value is configured in coreConsultant. This read-only field contains Bits[18:3] of 24-bit OUI of device manufacturer. This is a unique identifier for all the MMDs in the device. It is common for all MMDs (PMA, PCS, XS, AN, and MII) present in the package. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:0] of SR_VSMMD_PKGID1 Register modifies the content of this register. **Value After Reset:** "(!BASE1000X_ONLY) ? IEEE_PKG_OUI_3_18 : 0x0" **Exists:** Always

5.2.2.12 SR_XS_PCS_PKG2

- **Description:** SR XS or PCS MMD Package Identifier 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0xf : (APB3_16) ? (0xf*0x2) : (0xf*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD||XGXS_PCS))

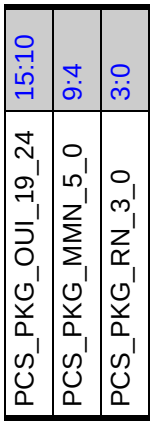


Table 5-70 Fields for Register: SR_XS_PCS_PKG2

Bits	Name	Memory Access	Description
15:10	PCS_PKG_OUI_19_24	R	Organizationally Unique Identifier DWC_xpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. This value is configured by the host. This read-only field contains Bits[24:19] of 24-bit OUI of device manufacturer. This is a unique identifier for all the MMDs in the device. It is common for all MMDs (PMA, PCS, XS, AN, and MII) present in the package. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:10] SR_VSMMD_PKGID2 Register modifies the contents of this register. **Value After Reset:** "(!BASE1000X_ONLY) ? IEEE_PKG_OUI_19_24 : 0x0" **Exists:** Always

Bits	Name	Memory Access	Description
9:4	PCS_PKG_MMN_5_0	R	Model Number This read-only field contains the 5-bit model number of the package that is common for all the MMDs in the Device. This value is configured by the host. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[9:4] of SR_VSMMD_PKGID2 Register modifies the content of this register. **Value After Reset:** "(!BASE1000X_ONLY) ? IEEE_PKG_MODEL_NO_0_5 : 0x0" **Exists:** Always
3:0	PCS_PKG_RN_3_0	R	Revision Number This read-only field contains the 4-bit revision number of the package in which the MMDs are installed. The value is configured by the host. It is common for all the MMDs in the device. It is common for all MMDs (PMA, PCS, XS, AN, and MII) present in the package. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[3:0] of SR_VSMMD_PKGID2 Register modifies the content of this register. Dependencies: The value is configured by the host. This value is common for all the MMDs in the device. **Value After Reset:** "(!BASE1000X_ONLY) ? IEEE_PKG_REV_NO_0_3 : 0x0" **Exists:** Always

5.2.2.13 SR_XS_PCS_EEE_ABL

- **Description:** SR XS or PCS MMD EEE Capability Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x14 : (APB3_16) ? (0x14*0x2) : (0x14*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD||XGXS_PCS)&&EEE_EN)

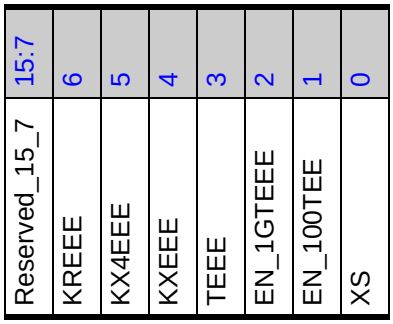


Table 5-71 Fields for Register: SR_XS_PCS_EEE_ABL

Bits	Name	Memory Access	Description
15:7	Reserved_15_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6	KREEE	R	**When PCS MMD:**10GBASE-KR Mode EEE Support This bit indicates if DWC_xpcs supports Energy Efficient Ethernet (EEE) in the 10GBASE-R PCS or 10GBASE-KR mode: ■ 0: EEE is not supported in 10GBASE-R PCS or 10GBASE-KR mode ■ 1: EEE is supported in 10GBASE-R PCS or 10GBASE-KR mode **When XS MMD:** Reserved **Value After Reset:** "(PCS_MMD&&LP_10GBR) ? 0x1 : 0x0" **Exists:** Always

Bits	Name	Memory Access	Description
5	KX4EEE	R	**When PCS MMD:**10GBASE-X PCS or 10GBASE-KX4 Mode EEE Support This bit indicates if DWC_xpcs supports EEE in the 10GBASEKX4 or 10GBASE-X PCS mode: ■ 0: EEE is not supported in 10GBASE-KX4 or 10GBASE-X PCS mode ■ 1: EEE is supported in 10GBASE-KX4 or 10GBASE-X PCS mode **When XS MMD:** Reserved **Value After Reset:** "(PCS_MMD&&LP_10GBX) ? 0x1 : 0x0" **Exists:** Always
4	KXEEE	R	XGXS, KX, or 1G EEE Support This bit indicates EEE support for XGXS PCS, KX, or 1G mode. When set to 1, this bit indicates that EEE is supported in the XGXS, KX, or 1G mode. When set to 0, this bit indicates that EEE is not supported in the XGXS PCS, KX, or 1G mode. This bit is valid for the following configurations: ■ Configurations with KX_Present mode ■ Configurations with MAIN_MODE = 10GBASE-X PCS and SUPPORT_1G = Enabled ■ Configurations with MAIN_MODE = 10GBASE-R PCS and SUPPORT_1G = Enabled ■ Configurations with MAIN_MODE = XGXS PCS **Value After Reset:** "(PCS_MMD&&LP_1GBX) ? 0x1 : ((XGXS_PCS&&EEE_EN) ? 0x1 : 0x0)" **Exists:** Always
3	TEEE	R	**When PCS MMD:**10GBASE-T Mode EEE Support This bit indicates the EEE support in the 10GBASE-T mode: ■ 0: EEE is not supported in the 10GBASE-T mode ■ 1: EEE is supported in the 10GBASE-T mode This bit is present only in the PCS MMD. In XS MMD, this bit is reserved and it always returns 0. **When XS MMD:** Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
2	EN_1GTEEE	R	**When PCS MMD:**1000BASE-T Mode EEE Support This bit indicates the EEE support in the 1000BASE-T mode: ■ 0: EEE is not supported in the 1000BASE-T mode ■ 1: EEE is supported in the 1000BASE-T mode This bit always returns 0. This bit is present only in the PCS MMD. In XS MMD, this bit is reserved and it always returns 0. **When XS MMD:** Reserved **Value After Reset:** 0x0 **Exists:** Always
1	EN_100TEE	R	**When PCS MMD:**100BASE-T Mode EEE Support This bit indicates the EEE support in the 100BASE-T mode: ■ 0: EEE is not supported in the 100BASE-T mode ■ 1: EEE is supported in the 100BASE-T mode This bit always returns 0. This bit is present only in the PCS MMD. In XS MMD, this bit is reserved and it always returns 0. **When XS MMD:** Reserved **Value After Reset:** 0x0 **Exists:** Always
0	XS	R	**For XGXS Configurations:** XGXS XAUI Stop Capable This bit indicates that the XGXS can support signal freezing on the XAUI serial lines during QUIET mode. ■ 0: XGXS does not support signal freezing on the XAUI interface ■ 1: XGXS supports signal freezing on the XAUI interface In XGXS PCS configurations, this bit always returns 1 to indicate the support for XAUI signal freezing. **For Other configurations:** Reserved **Value After Reset:** "(XGXS_PCS) ? 0x1 : 0x0" **Exists:** Always

5.2.2.14 SR_XS_PCS_EEE_ABL2

- **Description:** SR PCS MMD EEE Control and Capability2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x15 : (APB3_16) ? (0x15*0x2) : (0x15*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** PCS_MMD&&EEE_EN

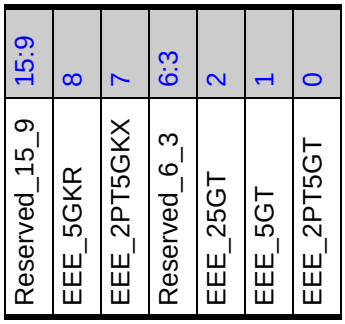


Table 5-72 Fields for Register: SR_XS_PCS_EEE_ABL2

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	EEE_5GKR	R	5GBASE-KR EEE Mode support. This bit indicates if EEE is supported in 5GBASE-KR mode. **Value After Reset:** "ADD_5G ? 0x1 : 0x0 " **Exists:** Always
7	EEE_2PT5GKX	R	2.5GBASE-KX EEE Mode support. This bit indicates if EEE is supported in 2.5GBASE-KX mode. **Value After Reset:** "ADD_2PT5G ? 0x1 : 0x0 " **Exists:** Always
6:3	Reserved_6_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2	EEE_25GT	R	25GBASE-T EEE Mode support. This bit indicates if EEE is supported in 25GBASE-T mode. This bit always returns zero. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
1	EEE_5GT	R	5GBASE-T EEE Mode support. This bit indicates if EEE is supported in 5GBASE-T mode. This bit always returns zero. **Value After Reset:** 0x0 **Exists:** Always
0	EEE_2PT5GT	R	2.5GBASE-T EEE Mode support. This bit indicates if EEE is supported in 2.5GBASE-T mode. This bit always returns zero. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.15 SR_XS_PCS_EEE_WKERR

- **Description:** SR XS or PCS MMD EEE Wake Error Counter Register
  This register is present only when EEE_EN = Enabled.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x16 : (APB3_16) ? (0x16*0x2) : (0x16*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD||XGXS_PCS)&&EEE_EN)



Table 5-73 Fields for Register: SR_XS_PCS_EEE_WKERR

Bits	Name	Memory Access	Description
15:0	EEE_WEC	R	EEE Wake Error Counter This field indicates the Rx wake up error counter in the EEE mode. This counter is incremented if the Refresh signaling is not received in time when DWC_xpcs is in low-power mode. When this counter reaches 16'hFFFF, it stops incrementing. This counter is cleared by reading this register. This counter is present only when EEE support is enabled. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.16 SR_XS_PCS_LSTS

- **Description:** SR XS or PCS MMD 10GBASE-X Lane Status Register
  This register is applicable only for the XS and PCS MMDs when the DWC_xpcs is not configured in the following modes:
  - 10GBASE-R PCS
  - KR_Only
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x18 : (APB3_16) ? (0x18*0x2) : (0x18*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD||XGXS_PCS))

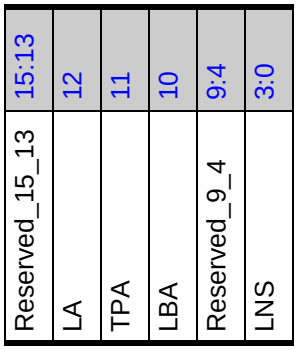


Table 5-74 Fields for Register: SR_XS_PCS_LSTS

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	LA	R	Receiver Lanes are Aligned This bit is set to 1 when Rx lanes are successfully deskewed. ■ 0: Rx lanes are not deskewed. ■ 1: Rx lanes are deskewed. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
11	TPA	R	Test Pattern Ability This bit indicates that DWC_xpcs is capable of generating test patterns. **Value After Reset:** "(!RPCS_ONLY) ? 0x1 : 0x0" **Exists:** Always

Bits	Name	Memory Access	Description
10	LBA	R	**When XS MMD:**Loopback Ability This bit is applicable for the XS MMD. This bit indicates that the device is capable of supporting the loopback function. **When XS MMD:**Reserved **Value After Reset:** "(XGXS_PCS) ? 0x1 : 0x0" **Exists:** Always
9:4	Reserved_9_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:0	LNS	R	Lane Synchronization LNS{lane} indicates that code group alignment and synchronization are successful in the corresponding lane. These bits are set to 1 when the corresponding lane acquires synchronization. When operating in QSGMII Mode, only bit [0] is valid. This bit is set if lane-synchronization is successful for all the 4 QSGMII data streams. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

5.2.2.17 SR_XS_PCS_TCTRL

- **Description:** SR XS or PCS MMD 10GBASE-X Test Control Register
  This register is present only with the XS and PCS MMDs when DWC_xpcs is not configured in the following modes:
  - 10GBASE-R PCS
  - KR_Only
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x19 : (APB3_16) ? (0x19*0x2) : (0x19*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD||XGXS_PCS)&&!RPCS_ONLY)

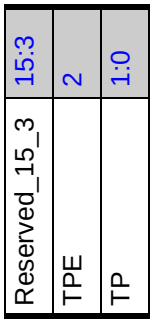


Table 5-75 Fields for Register: SR_XS_PCS_TCTRL

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2	TPE	R/W	Test Pattern Enable Lanes This bit indicates that a test pattern can be enabled in the Tx path after the current normal frame transmission is complete. ■ 0: Test pattern disabled ■ 1: Test pattern enabled **Dependencies:** The specific test pattern that is generated is based on the Bits[1:0] of this register. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
1:0	TP	R/W	Test Pattern Select This field indicates the pattern type that is enabled with Bit 2 (TPE). The following are the supported test patterns: ■ 2'b00: High Frequency Test Pattern ■ 2'b01: Low Frequency Test Pattern ■ 2'b10: Mixed Frequency Test Pattern ■ 2'b11: Continuous Test Pattern (CJPAT or CRPAT) Bit 3 of the VR_XS_PCS_DIG_CTRL1 Register determines the CJPAT or CRPAT pattern. This pattern is reserved in the KX_Only or 1000BASEX-Only PCS configurations. These test patterns are specified in IEEE Std 802.3, Annex 48A. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.18 SR_XS_PCS_KR_STS1

- **Description:** SR PCS MMD 10GBASE-R Status1 Register
  This register is present only with the PCS MMD in the following configurations:
  - 10GBASE-R PCS
  - KR_Only
  - KR_KX4_KX
  - KR_KX4
  - KR_KX
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x20 : (APB3_16) ? (0x20*0x2) : (0x20*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD&&RPCS_EN)

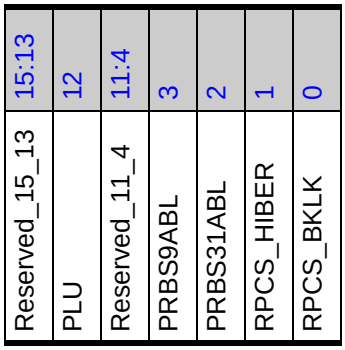


Table 5-76 Fields for Register: SR_XS_PCS_KR_STS1

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	PLU	R	RPCS Link Up When this bit is high, it indicates that the PCS-R 66-bit code group alignment is successful and the high bit rate condition is not encountered. If FEC is enabled, this bit also indicates that the FEC block lock is successful. ■ 1: 10GBASE-R or 10GBASE-KR PCS receive link is up ■ 0: 10GBASE-R or 10GBASE-KR PCS receive link is down **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

Bits	Name	Memory Access	Description
11:4	Reserved_11_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3	PRBS9ABL	R	PRBS9 Testing Ability ■ 1: The DWC_xpcs supports the PRBS9 pattern testing ■ 0: The DWC_xpcs does not support the PRBS9 pattern testing This bit always returns 1 because the DWC_xpcs supports the PRBS9 testing ability. **Value After Reset:** 0x1 **Exists:** Always
2	PRBS31ABL	R	PRBS31 Testing Ability ■ 1: The DWC_xpcs supports the PRBS31 pattern testing ■ 0: The DWC_xpcs does not support the PRBS31 pattern testing This bit always returns 1 because the DWC_xpcs supports the PRBS31 testing ability. **Value After Reset:** 0x1 **Exists:** Always
1	RPCS_HIBER	R	10GBASE-R PCS High Bit Error Rate ■ 1: 10GBASE-R or 10GBASE-KR PCS reporting a high BER ■ 0: 10GBASE-R or 10GBASE-KR PCS not reporting a high BER **Value After Reset:** 0x0 **Exists:** Always
0	RPCS_BKLK	R	10GBASE-R Block Lock ■ 1: 10GBASE-R is locked (Block Alignment Success) to received blocks ■ 0: 10GBASE-R is not locked to received blocks **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

5.2.2.19 SR_XS_PCS_KR_STS2

- **Description:** SR PCS MMD 10GBASE-R Status2 Register
  This register is present only with the PCS MMD in the following configurations:
  - 10GBASE-R PCS
  - KR_Only
  - KR_KX4_KX
  - KR_KX4
  - KR_KX
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x21 : (APB3_16) ? (0x21*0x2) : (0x21*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD&&RPCS_EN)

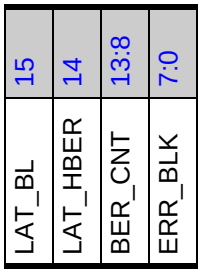


Table 5-77 Fields for Register: SR_XS_PCS_KR_STS2

Bits	Name	Memory Access	Description
15	LAT_BL	R	Latched Block Lock (RO,LL Type) ■ 1: 10GBASE-R or 10GBASE-KR PCS achieved the 66-bit code group alignment ■ 0: 10GBASE-R or 10GBASE-KR PCS did not achieve the 66-bit code group alignment **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

Bits	Name	Memory Access	Description
14	LAT_HBER	R	Latched High Bit Error Rate (RO,LH Type) High bit error is when more than 15 invalid 66-bit blocks are received in 125us duration. ■ 1: 10GBASE-R or 10GBASE-KR PCS is reporting a high BER ■ 0: 10GBASE-R or 10GBASE-KR PCS is not reporting a high BER **Value After Reset:** 0x0 **Exists:** Always
13:8	BER_CNT	R	Bit Error Rate Counter Value (RO,COR Type) This field indicates the high bit error count on the receive path. It stops incrementing when it reaches the value 6'h3F. This counter is cleared when read. **Value After Reset:** 0x0 **Exists:** Always
7:0	ERR_BLK	R	Error Block Counter Value (RO,COR Type) This field indicates invalid 66-bit blocks in the receive path. It stops incrementing when it reaches the value 8'hFF. This counter is cleared when read. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.20 SR_XS_PCS_TP_A0

- **Description:** SR PCS MMD 5G/10GBASE-R Pseudo-Random Test Pattern Seed A0 Register
  This register contains Bits[15:0] of Test pattern seed A. The content of this register is used as an initial seed to load scrambler for generating the pseudo-random test patterns. This register is present only with the PCS MMD in the following configurations:
  - 10GBASE-R PCS
  - KR_Only
  - KR_KX4_KX
  - KR_KX4
  - KR_KX
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x22 : (APB3_16) ? (0x22*0x2) : (0x22*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD&&RPCS_EN)



Table 5-78 Fields for Register: SR_XS_PCS_TP_A0

Bits	Name	Memory Access	Description
15:0	TP_SA0	R/W	Test Pattern Seed A0 This field contains Bits[15:0] of Test pattern seed A. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.21 SR_XS_PCS_TP_A1

- **Description:** SR PCS MMD 5G/10GBASE-R Pseudo-Random Test Pattern Seed A1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x23 : (APB3_16) ? (0x23*0x2) : (0x23*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD&&RPCS_EN)



Table 5-79 Fields for Register: SR_XS_PCS_TP_A1

Bits	Name	Memory Access	Description
15:0	TP_SA1	R/W	Test Pattern Seed A1 This field contains Bits[31:16] of Test pattern seed A. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.22 SR_XS_PCS_TP_A2

- **Description:** SR PCS MMD 5G/10GBASE-R Pseudo-Random Test Pattern Seed A2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x24 : (APB3_16) ? (0x24*0x2) : (0x24*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD&&RPCS_EN)



Table 5-80 Fields for Register: SR_XS_PCS_TP_A2

Bits	Name	Memory Access	Description
15:0	TP_SA2	R/W	Test Pattern Seed A2 This field contains Bits[47:32] of Test pattern seed A. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.23 SR_XS_PCS_TP_A3

- **Description:** SR PCS MMD 5G/10GBASE-R Pseudo-Random Test Pattern Seed A3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x25 : (APB3_16) ? (0x25*0x2) : (0x25*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD&&RPCS_EN)

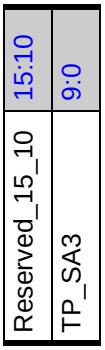


Table 5-81 Fields for Register: SR_XS_PCS_TP_A3

Bits	Name	Memory Access	Description
15:10	Reserved_15_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9:0	TP_SA3	R/W	Test Pattern Seed A3 This field contains Bits[57:48] of Test pattern seed A. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.24 SR_XS_PCS_TP_B0

- **Description:** SR PCS MMD 5G/10GBASE-R Pseudo-Random Test Pattern Seed B0 Register
  This register contains Bits[15:0] of Test pattern seed B. The content of this register is used as an initial seed to load scrambler for generating the pseudo-random test patterns. This register is present only with the PCS MMD in the following configurations:
  - 10GBASE-R PCS
  - KR_Only
  - KR_KX4_KX
  - KR_KX4
  - KR_KX
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x26 : (APB3_16) ? (0x26*0x2) : (0x26*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD&&RPCS_EN)



Table 5-82 Fields for Register: SR_XS_PCS_TP_B0

Bits	Name	Memory Access	Description
15:0	TP_SB0	R/W	Test Pattern Seed B0 This field contains Bits[15:0] of Test pattern seed B. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.25 SR_XS_PCS_TP_B1

- **Description:** SR PCS MMD 5G/10GBASE-R Pseudo-Random Test Pattern Seed B1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x27 : (APB3_16) ? (0x27*0x2) : (0x27*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD&&RPCS_EN)



Table 5-83 Fields for Register: SR_XS_PCS_TP_B1

Bits	Name	Memory Access	Description
15:0	TP_SB1	R/W	Test Pattern Seed B1 This field contains Bits[31:16] of Test pattern seed B. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.26 SR_XS_PCS_TP_B2

- **Description:** SR PCS MMD 5G/10GBASE-R Pseudo-Random Test Pattern Seed B2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x28 : (APB3_16) ? (0x28*0x2) : (0x28*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD&&RPCS_EN)

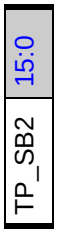


Table 5-84 Fields for Register: SR_XS_PCS_TP_B2

Bits	Name	Memory Access	Description
15:0	TP_SB2	R/W	Test Pattern Seed B2 This field contains Bits[47:32] of Test pattern seed B. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.27 SR_XS_PCS_TP_B3

- **Description:** SR PCS MMD 5G/10GBASE-R Pseudo-Random Test Pattern Seed B3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x29 : (APB3_16) ? (0x29*0x2) : (0x29*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD&&RPCS_EN)

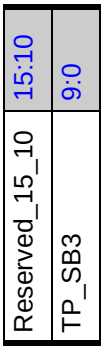
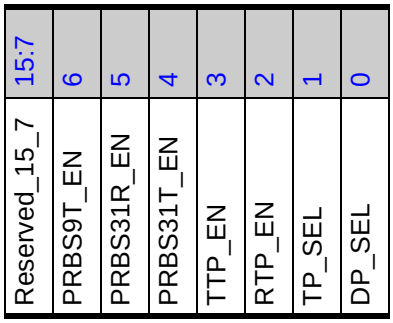


Table 5-85 Fields for Register: SR_XS_PCS_TP_B3

Bits	Name	Memory Access	Description
15:10	Reserved_15_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9:0	TP_SB3	R/W	Test Pattern Seed B3 This field contains Bits[57:48] of Test pattern seed B. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.28 SR_XS_PCS_TP_CTRL

- **Description:** SR PCS BASE-R Test Pattern Control Register
  This register is present only with the PCS MMD in the following configurations:
  - 10GBASE-R PCS
  - KR_Only
  - KR_KX4_KX
  - KR_KX4
  - KR_KX
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x2A : (APB3_16) ? (0x2A*0x2) : (0x2A*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD&&RPCS_EN)



15:7	6	5	4	3	2	1	0
Reserved_15_7	PRBS9T_EN	PRBS31R_EN	PRBS31T_EN	TTP_EN	RTP_EN	TP_SEL	DP_SEL

Table 5-86 Fields for Register: SR_XS_PCS_TP_CTRL

Bits	Name	Memory Access	Description
15:7	Reserved_15_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6	PRBS9T_EN	R/W	PRBS9 Transmit Test Pattern Enable ■ 1: Enable PRBS9 test pattern on the Transmit path when Bit 3 (TTP_EN), Bit 1 (TP_SEL), and Bit 4 (PRBS31T_EN) are set to 0 ■ 0: Disable PRBS9 test-pattern mode on the Transmit path **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
5	PRBS31R_EN	R/W	PRBS31 Receive Test Pattern Enable ■ 1: Enable PRBS31 test pattern checking on the Receive path when Bit 2 (RTP_EN) is set to 0 ■ 0: Disable PRBS31 test pattern checking on the Receive path **Value After Reset:** 0x0 **Exists:** Always
4	PRBS31T_EN	R/W	PRBS31 Transmit Pattern Enable ■ 1: Enable PRBS31 test pattern on the Transmit path when Bit 3 (TTP_EN) and Bit 1 (TP_SEL) of this register is set to 0 ■ 0: Disable PRBS31 test-pattern mode on the Transmit path **Value After Reset:** 0x0 **Exists:** Always
3	TTP_EN	R/W	Transmit Pseudo-Random or Square Wave Test Pattern Enable ■ 1: Enables pseudo-random or square wave test pattern depending on Bit 1 (TP_SEL) ■ 0: Disables pseudo-random or square wave test patterns This bit must be set to 0 to enable PRBS31 or PRBS9 test pattern. You can enable the PRBS31 or PRBS9 test pattern by setting Bit 4 (PRBS31T_EN) or Bit 6 (PRBS9T_EN) to 1, respectively. **Value After Reset:** 0x0 **Exists:** Always
2	RTP_EN	R/W	Receive Test Pattern Enable ■ 1: Enables receive pseudo-random test pattern when Bit 1 (TP_SEL) is set to 0 ■ 0: Disables receive pseudo-random test pattern This field must be set to 0 to enable PRBS31 or PRBS9 test pattern in the Receive path. You can enable the PRBS31 or PRBS9 test pattern by setting Bit 5 (PRBS31R_EN) of this register or Bit 7 (PRBS9RXEN) of VR_XS_PCS_KR_CTRL Register to 1, respectively. **Value After Reset:** 0x0 **Exists:** Always
1	TP_SEL	R/W	Transmit Test Pattern Select When set, DWC_xpcs transmits the square wave test pattern. Bits [6:4] of VR_XS_PCS_KR_CTRL Register specify the period of the square wave. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	DP_SEL	R/W	Data Pattern Select for Pseudo-Random Testing This bit selects the data given to the scrambler when pseudo-random test pattern is enabled (Bit 3 (TTP_EN) is set). ■ 1: All-zero ■ 0: Encoded Local-fault pattern (LF) You can change the data pattern by programming Bits [3:1] of VR_XS_PCS_KR_CTRL Register. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.29 SR_XS_PCS_TP_ERRCTR

- **Description:** SR PCS BASE-R Test Pattern Error Counter Register
  This register is present only with the PCS MMD in the following configurations:
  - 10GBASE-R PCS
  - KR_Only
  - KR_KX4_KX
  - KR_KX4
  - KR_KX
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x2B : (APB3_16) ? (0x2B*0x2) : (0x2B*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD&&RPCS_EN)



Table 5-87 Fields for Register: SR_XS_PCS_TP_ERRCTR

Bits	Name	Memory Access	Description
15:0	TP_ERR_CNT	R	Test Pattern Error Counter (RO,COR Type) If checking for any test pattern is enabled in the Rx path, this field indicates the test pattern mismatch error count. This counter stops when it becomes all-ones. It gets cleared when this register is read. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.30 SR_PCS_TIME_SYNC_PCS_ABL

- **Description:** SR PCS MMD Time Sync Capability Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0708 : (APB3_16) ? (0x0708*0x2) : (0x0708*0x4)" + "(PCS_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD)

15:14	13	12	11	10	9:4	3	2	1	0
Reserved_15_14	TREF_SFD	TREF_FS	MLS	NUC	Reserved_9_4	FINE_TX_DLY	FINE_RX_DLY	TX_DLY_ABL	RX_DLY_ABL

Table 5-88 Fields for Register: SR_PCS_TIME_SYNC_PCS_ABL

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13	TREF_SFD	R	Timestamp reference at SFD ■ 0: DWC_xpcs does not support data delay measurement point at the beginning of SFD ■ 1: DWC_xpcs supports data delay measurement point at the beginning of SFD **Value After Reset:** "SNPS_RSVDPARAM_11 ? 0x1 : 0x0" **Exists:** Always
12	TREF_FS	R	Timestamp reference at first symbol after SFD ■ 0: DWC_xpcs does not support data delay measurement point at the beginning of the first symbol after SFD ■ 1: DWC_xpcs supports data delay measurement point at the beginning of the first symbol after SFD **Value After Reset:** "SNPS_RSVDPARAM_11 ? 0x1 : 0x0" **Exists:** Always

Bits	Name	Memory Access	Description
11	MLS	R	Multi-lane Support. ■ 0: DWC_xpcs does not support reporting of multi-PCS lane path data delay ■ 1: DWC_xpcs supports reporting of multi-PCS lane path data delay **Value After Reset:** "SNPS_RSVDPARAM_11&&!SINGLE_LANE_CFG ? 0x1 : 0x0" **Exists:** Always
10	NUC	R	Num_Unit_Change Support Available. If this bit is set, it indicates that DWC_xpcs supports Transmit/Receive Num_Unit_Change indication capability **Value After Reset:** "SNPS_RSVDPARAM_11 ? 0x1 : 0x0" **Exists:** Always
9:4	Reserved_9_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3	FINE_TX_DLY	R	Fine Resolution Transmit Path Data Delay Available. If this bit is set, it indicates that DWC_xpcs provides information on transmit data path delay with sub-ns resolution in SR_PCS_TIME_SYNC_TX_MAX_FINE_DLY and SR_PCS_TIME_SYNC_TX_MIN_FINE_DLY registers. **Value After Reset:** "SNPS_RSVDPARAM_11 ? 0x1 : 0x0" **Exists:** Always
2	FINE_RX_DLY	R	Fine Resolution Receive Path Data Delay Available. If this bit is set, it indicates that DWC_xpcs provides information on receive data path delay with sub-ns resolution in SR_PCS_TIME_SYNC_RX_MAX_FINE_DLY and SR_PCS_TIME_SYNC_RX_MIN_FINE_DLY registers. **Value After Reset:** "SNPS_RSVDPARAM_11 ? 0x1 : 0x0" **Exists:** Always
1	TX_DLY_ABL	R	PCS Transmit Path Data Delay Information Available. If this bit is set, it indicates that the information regarding the maximum and minimum transmit data delay of PCS is available in ns-resolution in registers 3.1801 through 3.1804 (Register number, see the IEEE numbers) **Value After Reset:** 0x1 **Exists:** Always

Bits	Name	Memory Access	Description
0	RX_DLY_ABL	R	PCS Receive Path Data Delay Information Available. If this bit is set, it indicates that the information regarding the maximum and minimum receive data delay of PCS is available in ns-resolution in registers 3.1805 through 3.1808 (Register number, see the IEEE numbers) **Value After Reset:** 0x1 **Exists:** Always

5.2.2.31 SR_PCS_TIME_SYNC_TX_MAX_DLY_LWR

- **Description:** SR PCS MMD Time Sync Tx Max Delay Lower Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0709 : (APB3_16) ? (0x0709*0x2) : (0x0709*0x4)" + "(PCS_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD)

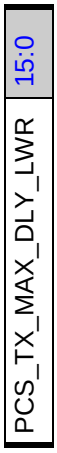


Table 5-89 Fields for Register: SR_PCS_TIME_SYNC_TX_MAX_DLY_LWR

Bits	Name	Memory Access	Description
15:0	PCS_TX_MAX_DLY_LWR	R	PCS Transmit Max Delay Lower This field indicates the lower 16-bit of the 32-bit value which indicates integer nano-seconds portion of the Maximum Data Delay in the Transmit Path of DWC_xpcs. The value returned by this field depends on the current operating mode. If TS_MODE bit is 0, this register indicates the end-to-end (MAC interface to PCS interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the transmit path from the DDMP point until the final Tx data output port. **Value After Reset:** "RPCS_PRESENT ? ((MP_USXGMII&&(R_SRDS_INTF==3)) ? 0x40 : (USXGMII_EN&&(R_SRDS_INTF==3)) ? 0x3A : MP_USXGMII ? 0x4D : USXGMII_EN ? 0x46 :((R_SRDS_INTF==3) ? (XGMII_SDR ? 0x33 : 0x20): (R_SRDS_INTF==2) ? (XGMII_SDR ? 0x46 : 0x33) : (XGMII_SDR ? 0x40 : 0x2D))) : ONLY_1G_CFG ? 0x40 : (XGMII_SDR ? 0x17 : 0x2A)" **Exists:** Always

5.2.2.32 SR_PCS_TIME_SYNC_TX_MAX_DLY_UPR

- **Description:** SR PCS MMD Time Sync Tx Max Delay Upper Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070A : (APB3_16) ? (0x070A*0x2) : (0x070A*0x4)" + "(PCS_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD)



Table 5-90 Fields for Register: SR_PCS_TIME_SYNC_TX_MAX_DLY_UPR

Bits	Name	Memory Access	Description
15:0	PCS_TX_MAX_DLY_UPR	R	PCS Transmit Max Delay Upper This field indicates the upper 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Maximum Data Delay in the Transmit Path of DWC_xpcs. If TS_MODE bit is 0, this register indicates the end-to-end (MAC interface to PCS interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the transmit path from the DDMP point until the final Tx data output port **Value After Reset:** 0x0 **Exists:** Always

5.2.2.33 SR_PCS_TIME_SYNC_TX_MIN_DLY_LWR

- **Description:** SR PCS MMD Time Sync Tx Min Delay Lower Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070B : (APB3_16) ? (0x070B*0x2) : (0x070B*0x4)" + "(PCS_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD)



Table 5-91 Fields for Register: SR_PCS_TIME_SYNC_TX_MIN_DLY_LWR

Bits	Name	Memory Access	Description
15:0	PCS_TX_MIN_DLY_LWR	R	PCS Transmit Min Delay Lower This field indicates the lower 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Minimum Data Delay in the Transmit Path of DWC_xpcs. The value returned by this register field depends on the current operating mode. If TS_MODE bit is 0, this register indicates the end-to-end (MAC interface to PCS interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the transmit path from the DDMP point until the final Tx data output port. **Value After Reset:** "RPCS_PRESENT ? (USXGMII_EN&&(R_SRDS_INTF==3) ? 0x26 : USXGMII_EN ? 0x33 :((R_SRDS_INTF==3) ? (XGMII_SDR ? 0x2D : 0x1A): (R_SRDS_INTF==2) ? (XGMII_SDR ? 0x40 : 0x2D) : (XGMII_SDR ? 0x3A : 0x26))) : ONLY_1G_CFG ? 0x38 : (XGMII_SDR ? 0x15 : 0x27)" **Exists:** Always

5.2.2.34 SR_PCS_TIME_SYNC_TX_MIN_DLY_UPR

- **Description:** SR PCS MMD Time Sync Tx Min Delay Upper Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070C : (APB3_16) ? (0x070C*0x2) : (0x070C*0x4)" + "(PCS_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD)



Table 5-92 Fields for Register: SR_PCS_TIME_SYNC_TX_MIN_DLY_UPR

Bits	Name	Memory Access	Description
15:0	PCS_TX_MIN_DLY_UPR	R	PCS Transmit Min Delay Upper This field indicates the upper 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Minimum Data Delay in the Transmit Path of DWC_xpcs. If TS_MODE bit is 0, this register indicates the end-to-end (MAC interface to PCS interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the transmit path from the DDMP point until the final Tx data output port **Value After Reset:** 0x0 **Exists:** Always

5.2.2.35 SR_PCS_TIME_SYNC_RX_MAX_DLY_LWR

- **Description:** SR PCS MMD Time Sync Rx Max Delay Lower Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070D : (APB3_16) ? (0x070D*0x2) : (0x070D*0x4)" + "(PCS_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD)



Table 5-93 Fields for Register: SR_PCS_TIME_SYNC_RX_MAX_DLY_LWR

Bits	Name	Memory Access	Description
15:0	PCS_RX_MAX_DLY_LWR	R	PCS Receive Max Delay Lower This field indicates the lower 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Maximum Data Delay in the Receive Path of DWC_xpcs.The value returned by this register field depends on the current operating mode and is valid after Rx link-up is attained. If TS_MODE bit is 0, this register indicates the end-to-end (PHY interface to MAC interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the receive path from Rx data input port until the DDMP point. **Value After Reset:** "RPCS_PRESENT ? ((USXGMII_EN&&!MP_USXGMII) ? (R_SRDS_INTF==3 ? (CLKCOMP ? 0x6A : 0x43) : (CLKCOMP ? 0x76 : 0x50)) : MP_USXGMII ? (R_SRDS_INTF==3 ? (CLKCOMP ? 0x70 : 0x4A) : (CLKCOMP ? 0x7D : 0x56)) : ((R_SRDS_INTF==3) ? (!CLKCOMP ? (XGMII_SDR ? 0x46 : 0x2D): (XGMII_SDR ? 0x60 : 0x40)) :(R_SRDS_INTF==2) ? (!CLKCOMP ? (XGMII_SDR ? 0x53 : 0x3A) : (XGMII_SDR ? 0x6D : 0x4D)): (!CLKCOMP ? (XGMII_SDR ? 0x4D : 0x33) : (XGMII_SDR ? 0x67 : 0x46)))) : ONLY_1G_CFG ? (!CLKCOMP ? 0x58 : 0xB8) : (!CLKCOMP ? (XGMII_SDR ? 0x30 : 0x44) : (XGMII_SDR ? 0x4A : 0x5D))" **Exists:** Always

5.2.2.36 SR_PCS_TIME_SYNC_RX_MAX_DLY_UPR

- **Description:** SR PCS MMD Time Sync Rx Max Delay Upper Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070E : (APB3_16) ? (0x070E*0x2) : (0x070E*0x4)" + "(PCS_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD)



Table 5-94 Fields for Register: SR_PCS_TIME_SYNC_RX_MAX_DLY_UPR

Bits	Name	Memory Access	Description
15:0	PCS_RX_MAX_DLY_UPR	R	PCS Receive Max Delay Upper This field indicates the upper 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Maximum Data Delay in the Receive Path of DWC_xpcs. If TS_MODE bit is 0, this register indicates the end-to-end (PHY interface to MAC interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the receive path from Rx data input port until the DDMP point. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.37 SR_PCS_TIME_SYNC_RX_MIN_DLY_LWR

- **Description:** SR PCS MMD Time Sync Rx Min Delay Lower Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070F : (APB3_16) ? (0x070F*0x2) : (0x070F*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD)



Table 5-95 Fields for Register: SR_PCS_TIME_SYNC_RX_MIN_DLY_LWR

Bits	Name	Memory Access	Description
15:0	PCS_RX_MIN_DLY_LWR	R	PCS Receive Min Delay Lower This field indicates the lower 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Minimum Data Delay in the Receive Path of DWC_xpcs.The value returned by this register field depends on the current operating mode and is valid after Rx link-up is attained. If TS_MODE bit is 0, this register indicates the end-to-end (PHY interface to MAC interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the receive path from Rx data input port until the DDMP point. **Value After Reset:** "RPCS_PRESENT ? (USXGMII_EN ? (R_SRDS_INTF==3 ? (CLKCOMP ? 0x3C : 0x30) : (CLKCOMP ? 0x4A : 0x3D)) : ((R_SRDS_INTF==3) ? (!CLKCOMP ? (XGMII_SDR ? 0x3A : 0x20): (XGMII_SDR ? 0x53 : 0x33)) :(R_SRDS_INTF==2) ? (!CLKCOMP ? (XGMII_SDR ? 0x46 : 0x2D) : (XGMII_SDR ? 0x60 : 0x40)): (!CLKCOMP ? (XGMII_SDR ? 0x40 : 0x26) : (XGMII_SDR ? 0x5A : 0x3A)))) : ONLY_1G_CFG ? (!CLKCOMP ? 0x50 : 0x88) : (!CLKCOMP ? (XGMII_SDR ? 0x2A : 0x3D) : (XGMII_SDR ? 0x43 : 0x56))" **Exists:** Always

5.2.2.38 SR_PCS_TIME_SYNC_RX_MIN_DLY_UPR

- **Description:** SR PCS MMD Time Sync Rx Min Delay Upper Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0710 : (APB3_16) ? (0x0710*0x2) : (0x0710*0x4)" + "(PCS_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD)



Table 5-96 Fields for Register: SR_PCS_TIME_SYNC_RX_MIN_DLY_UPR

Bits	Name	Memory Access	Description
15:0	PCS_RX_MIN_DLY_UPR	R	PCS Receive Min Delay Upper This field indicates the upper 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Minimum Data Delay in the Receive Path of DWC_xpcs. If TS_MODE bit is 0, this register indicates the end-to-end (PHY interface to MAC interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the receive path from Rx data input port until the DDMP point. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.39 SR_PCS_TIME_SYNC_TX_MAX_FINE_DLY

- **Description:** SR PCS MMD Time Sync Tx Max Fine Delay Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0711 : (APB3_16) ? (0x0711*0x2) : (0x0711*0x4)" + "(PCS_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD&&SNPS_RSVDPARAM_11)

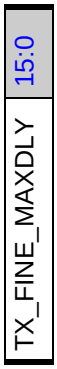


Table 5-97 Fields for Register: SR_PCS_TIME_SYNC_TX_MAX_FINE_DLY

Bits	Name	Memory Access	Description
15:0	TX_FINE_MAXDLY	R	Transmit Maximum Fine Delay This register indicates the sub-nano-seconds portion of the maximum delay inside the DWC_xpcs transmit path from the DDMP point until the final transmit output port. The delay is indicated in units of 2^-16 ns. Contents of this register is valid only when TS_MODE bit is programmed to 1. Default value of this register is 0, as reset value of TS_MODE bit is 0 **Value After Reset:** 0x0 **Exists:** Always

5.2.2.40 SR_PCS_TIME_SYNC_TX_MIN_FINE_DLY

- **Description:** SR PCS MMD Time Sync Tx Min Fine Delay Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0712 : (APB3_16) ? (0x0712*0x2) : (0x0712*0x4)" + "(PCS_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD&&SNPS_RSVDPARAM_11)



Table 5-98 Fields for Register: SR_PCS_TIME_SYNC_TX_MIN_FINE_DLY

Bits	Name	Memory Access	Description
15:0	TX_FINE_MINDLY	R	Transmit Minimum Fine Delay This register indicates the sub-nano-seconds portion of the minimum delay inside the DWC_xpcs transmit path from the DDMP point until the final transmit output port. The delay is indicated in units of 2^-16 ns. Contents of this register is valid only when TS_MODE bit is programmed to 1. Default value of this register is 0, as reset value of TS_MODE bit is 0 **Value After Reset:** 0x0 **Exists:** Always

5.2.2.41 SR_PCS_TIME_SYNC_RX_MAX_FINE_DLY

- **Description:** SR PCS MMD Time Sync Rx Max Fine Delay Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0713 : (APB3_16) ? (0x0713*0x2) : (0x0713*0x4)" + "(PCS_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD&&SNPS_RSVDPARAM_11)



Table 5-99 Fields for Register: SR_PCS_TIME_SYNC_RX_MAX_FINE_DLY

Bits	Name	Memory Access	Description
15:0	RX_FINE_MAXDLY	R	Receive Maximum Fine Delay This register indicates the sub-nano-seconds portion of the maximum delay inside the DWC_xpcs receive path from Rx data input port until the DDMP point. The delay is indicated in units of 2^-16 ns. Contents of this register is valid only when TS_MODE bit is programmed to 1. Default value of this register is 0, as reset value of TS_MODE bit is 0. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.42 SR_PCS_TIME_SYNC_RX_MIN_FINE_DLY

- **Description:** SR PCS MMD Time Sync Rx Min Fine Delay Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0714 : (APB3_16) ? (0x0714*0x2) : (0x0714*0x4)" + "(PCS_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD&&SNPS_RSVDPARAM_11)



Table 5-100 Fields for Register: SR_PCS_TIME_SYNC_RX_MIN_FINE_DLY

Bits	Name	Memory Access	Description
15:0	RX_FINE_MINDLY	R	Receive Minimum Fine Delay This register indicates the sub-nano-seconds portion of the minimum delay inside the DWC_xpcs receive path from Rx data input port until the DDMP point. The delay is indicated in units of 2^-16 ns. Contents of this register is valid only when TS_MODE bit is programmed to 1. Default value of this register is 0, as reset value of TS_MODE bit is 0. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.43 SR_PCS_TIME_SYNC_CFG

- **Description:** SR PCS MMD Time Sync Configuration Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0715 : (APB3_16) ? (0x0715*0x2) : (0x0715*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD&&SNPS_RSVDPARAM_11)

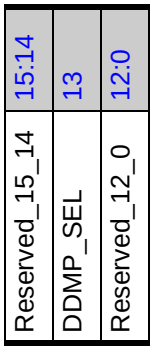


Table 5-101 Fields for Register: SR_PCS_TIME_SYNC_CFG

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13	DDMP_SEL	R/W	Data Delay Measurement Point Selection ■ 0: DWC_xpcs is configured to use the data delay measurement point at the beginning of SFD ■ 1: DWC_xpcs is configured to use the data delay measurement point at the beginning of the first symbol after SFD **Value After Reset:** 0x0 **Exists:** Always
12:0	Reserved_12_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.2.44 VR_XS_PCS_DIG_CTRL1

- **Description:** VR XS or PCS MMD Digital Control1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8000 : (APB3_16) ? (0x8000*0x2) : (0x8000*0x4)" + "(PCS_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD | |XGXS_PCS))

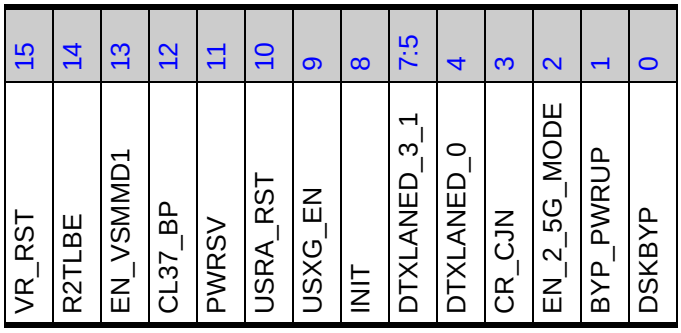


Table 5-102 Fields for Register: VR_XS_PCS_DIG_CTRL1

Bits	Name	Memory Access	Description
15	VR_RST	R/W	Vendor-Specific Soft Reset (RW,SC Type) When the host sets this bit, the CSR block triggers the software reset process in which all internal blocks are reset, except the Management Interface block and CSR block. When this bit is set, it also resets the PHY. This bit is self cleared under the following conditions: For Synopsys PHY, this bit is self cleared when Bits[4:2] in VR_XS_PCS_DIG_STS Register is equal to 3'b100, that is, Tx/Rx clocks are stable and in Power_Good state. For Non-Synopsys PHY, this bit is self cleared after the following: ■ 32 clk_csr_i clocks for the MCI interface ■ 1 MDC clock period for the MDIO interface **Note:** For information about the read or write access for any register during the reset process, see 'Special Case Register Access' section in the Databook. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
14	R2TLBE	R/W	Rx to Tx Loopback Enable This bit controls the loopback path from the XGMII or GMII Rx to the XGMII or GMII Tx at the XGMII interface. ■ 0: Loopback path is disabled. ■ 1: Loopback path is enabled. Whenever this bit is changed, RXFIFO_OVF/RXFIFO_UNF bits of VR_XS_PCS_DIG_STS Register might get set incorrectly. Therefore, read these register bits so that they are cleared. Thereafter, RXFIFO_OVF/RXFIFO_UNF bits would be reliable. You can use this bit for debugging purpose. **Note:** For performing loopback in 2.5GBASE-X mode, you must also set the TX_PMBL_CTL bit of VR_XS_PCS_DEBUG_CTRL register. **Value After Reset:** 0x0 **Exists:** Always
13	EN_VSMMD1	R/W	Enable Vendor-Specific MMD1 When this bit is set to 1, the Vendor-Specific MMD1 (VSMMD1) is enabled. When this bit is set to zero, VSMMD1 is disabled. **Value After Reset:** 0x1 **Exists:** Always
12	CL37_BP	* Varies	Enable Clause 37 AN in BackPlane (BP) Configurations When this bit is set to 1, Clause 37 auto-negotiation is enabled in Backplane Ethernet configurations. This bit is present when MAIN_MODE = Backplane Ethernet PCS and CL37_AN = Enabled. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(BACKPLANE_ETHERNET_PCS&&CL37_AN) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
11	PWRSV	* Varies	**For Synopsys PHY configurations:** Power Save This bit is present only when the DWC_xpcs is configured for Synopsys PHY. ■ 0: Normal operation. ■ 1: DWC_xpcs and the PHY enter the power-save mode. When this bit is set, the DWC_xpcs triggers the power-down mode by turning off the PHY Receiver and Transmitter, and without turning off MPLL. For Synopsys Multi-protocol 12G/16G PHY, DWC_xpcs asserts xpcs_tx_lpd_o{lane} and xpcs_rx_lpd_o{lane} when this bit is set. When the host clears this bit, DWC_xpcs resumes normal operation. After clearing this bit, the host must wait until Bits[4:2] of the VR_XS_PCS_DIG_STS Register indicate that the DWC_xpcs is in a normal state. **Note:** For information about the read or write access for any register during the reset process, see 'Special Case Register Access' section in the Databook. **For non-Synopsys PHY configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_XAUI_PHY) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
10	USRA_RST	* Varies	**For USXGMII configurations:** USXGMII Rate Adapter Reset (Port 0) This bit can be set to 1 to reset/initialize the USXGMII Rate Adapter Logic (for port 0) inside DWC_xpcs. The Rate Adapter Logic maps GMII data to XGMII (and vice-versa in Rx path) data and perform data replication for lower USXGMII speeds. This is a self-clear bit (RW,SC) and clears itself after the reset of Tx and Rx Rate Adapter Logic is complete. After setting this bit to 1, application must poll this bit till it self-clears. **Note:** If CLKCOMP = Enabled, this bit also initializes the clock-compensation FIFO (corresponding to port 0). When this bit is programmed to 1, RXFIFO_OVF/RXFIFO_UNF bits of VR_XS_PCS_DIG_STS Register might get set incorrectly. Therefore, read these register bits (RXFIFO_OVF and RXFIFO_UNF) so that they get cleared. Thereafter, RXFIOF_UNF and RXFIFO_UNF bits are reliable. **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "USXGMII_EN ? \"read-write\" : \"read-only\""
9	USXG_EN	* Varies	**For USXGMII configurations:** USXGMII Mode Enable When this bit is set, DWC_xpcs operates in 10G-USXGMII mode. Otherwise, DWC_xpcs operates in 10GBASE-R Mode. MII MMD Registers can be accessed only after this register bit is set to 1 **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "USXGMII_EN ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
8	INIT	R/W	Datapath Initialization Control This bit can be set to flush/initialize the various FIFOs implemented inside DWC_xpcs. This is a Self-Clear bit. After writing 1 to this bit, application must continuously poll this bit until it reads a 0. Application should proceed to any other operation, only after reading 0 from this bit. When this bit is programmed to 1, RXFIFO_OVF/RXFIFO_UNF bits of VR_XS_PCS_DIG_STS Register might get set incorrectly. Therefore, read these register bits (RXFIFO_OVF and RXFIFO_UNF) so that they get cleared. Thereafter, RXFIOF_UNF and RXFIFO_UNF bits are reliable. **Value After Reset:** 0x0 **Exists:** Always
7:5	DTXLANED_3_1	* Varies	**For Multi-lane configurations:** Tx Disable on Lanes[3:1] When any bits of this field are set, the DWC_xpcs disables the corresponding Tx lanes (3, 2, or 1) of the PHY. When reset, the DWC_xpcs enables the corresponding Tx lanes of the PHY. **Note:** For single-lane configurations, this field is reserved and RO. For all other configurations, this field is RW. This field is shared with the following fields: ■ Bits[7:5] of VR_PMA_DIG_CTRL1 Register ■ Bits[4:2] of SR_PMA_TX_DIS Register **For Single-lane configurations:**Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SINGLE_LANE_CFG) ? \"read-only\" : \"read-write\""
4	DTXLANED_0	R/W	Tx Disable on Lane 0 When this bit is 1, the DWC_xpcs disables the Tx Lane 0 of the PHY. When this bit is 0, the DWC_xpcs enables the Tx Lane 0 of the PHY. **Note:** This bit is shared with the following bits (if present): ■ Bit 4 of VR_PMA_DIG_CTRL1 Register and VR_MII_DIG_CTRL1 Register ■ Bit 1 of SR_PMA_TX_DIS Register ■ Bit 0 of SR_PMA_KX_CTRL Register **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
3	CR_CJN	* Varies	**For Multi-lane configurations:** Continuous Random or Continuous Jitter Select ■ 0: Continuous jitter test pattern is inserted in the Tx data path. ■ 1: Continuous random test pattern is inserted in the Tx data path. These patterns are defined in the IEEE Std 802.3, Annexure 48A. **Note:** This bit is not present in the KX_Only, 1000BASEX-Only PCS, 10GBASE-R PCS, KR_Only, and KR_KX configurations. **Access Type:** ■ For KX_Only, 1000BASEX-Only PCS, 10GBASE-R PCS, KR_Only, and KR_KX configurations, this bit is RO. ■ For all other configurations (that is, multi-lane configurations), this bit is RW. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SINGLE_LANE_CFG) ? \"read-only\" : \"read-write\""
2	EN_2_5G_MODE	* Varies	**For configurations with 1G/KX Speed Mode Support:**Enable 2.5G GMII Mode. Set this bit to 1 to enable 2.5G GMII Mode (GMII port operating at 312.5 MHz instead of 125 MHz) of operation. This bit drives the output port 'xpcs_2pt5g_mode_o'. This bit is shared with bit[2] of VR_MII_DIG_CTRL1 Register (if present). This bit is present only only in configurations with 'SUPPORT_1G = Enabled' or KX_Present configurations. **For configurations without 1G/KX Speed Mode Support:**Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SUPPORT_1G||BASE1000X_ONLY||KX_PRESENT) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
1	BYP_PWRUP	* Varies	**For Synopsys PHY configurations:** Bypass Power-Up Sequence This bit is present only when the DWC_xpcs is configured for the Synopsys PHY. In all other configurations, this bit is reserved. When this bit is set to 1, the DWC_xpcs bypasses the normal flow of the power-up sequence and reaches the Power_Good state to enable transmission or reception. When this bit is set, the DWC_xpcs does not wait for the MPLL and Transmit or Receive PLL status from the Synopsys PHY. You can use this feature in the following scenarios: ■ When the DWC_xpcs is configured to interface with a specific Synopsys PHY. ■ When the data path needs to be interfaced to some other PHY. When this bit is set to 0, the DWC_xpcs waits for the MPLL, Tx, or Rx PLL status from the Synopsys PHY before resuming the normal transmission and reception. **Note:** This bit is shared with Bit 1 of following registers (if present): ■ VR_XS_PCS_DIG_CTRL1 Register ■ VR_AN_DIG_CTRL1 Register ■ VR_MII_DIG_CTRL1 Register **Access Type:** If SNPS_XAUI_PHY= Enabled, this bit is RW. Otherwise, this bit is RO. **For non-Synopsys PHY configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_XAUI_PHY) ? \"read-write\" : \"read-only\""
0	DSKBYP	* Varies	**For Multi-lane configurations:** Deskew Bypass When this bit is set, the lane-to-lane deskew process is bypassed on the receiver in the PCS-X 10G mode. ■ 0: Normal operation ■ 1: Deskew bypassed If deskew bypass is enabled, Bit 12 of SR XS or PCS MMD 10GBASE-X Lane Status Register is set to 1 (status is always aligned) and the receiver never declares local fault because of loss-of-alignment. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SINGLE_LANE_CFG) ? \"read-only\" : \"read-write\""

5.2.2.45 VR_XS_PCS_DIG_CTRL2

- **Description:** VR XS or PCS MMD Digital Control2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8001 : (APB3_16) ? (0x8001*0x2) : (0x8001*0x4)" + "(PCS_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD | |XGXS_PCS)

15:13	12	11:9	8	7:5	4	3:1	0
PTX_LN_DIS_3_1	Reserved_12	PRX_LN_DIS_3_1	Reserved_8	TX_POL_INV_3_1	TX_POL_INV_0	RX_POL_INV_3_1	RX_POL_INV_0

Table 5-103 Fields for Register: VR_XS_PCS_DIG_CTRL2

Bits	Name	Memory Access	Description
15:13	PTX_LN_DIS_3_1	* Varies	**When LANE_CONTROL=Enabed:** Tx Lane Disable This field indicates PCS Tx Lane disable on the corresponding Lane 1, Lane 2, or Lane 3. This bit is present for all MMDs. When any of these bits are set, the corresponding lanes of the XGMII are ignored and the IDLE patterns are inserted in the PHY Transmit lanes. **Note:** XGMII Fault-Signaling protocol does not work, when any of these bits are set. The following are the valid combinations: ■ 000: Lanes 0 to 3 are enabled (10G Ethernet). ■ 100: Lane 3 is disabled. Lanes 0 to 2 are operational (7.5G Ethernet). ■ 110: Lanes 2 and 3 are disabled. Lanes 0 and 1 are operational (5.0G Ethernet application). ■ 111: Lanes 1, 2, and 3 are disabled. Lane 0 is operational (2.5G Ethernet application). ■ Other values: Reserved. This field is present only when LANE_CONTROL = Enabled. **Access Type:** ■ In configurations with LANE_CONTROL = Enabled, this field is RW. ■ For all other configurations, this field is RO. **When LANE_CONTROL=Disabled:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(LANE_CONTROL) ? \"read-write\" : \"read-only\""
12	Reserved_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
11:9	PRX_LN_DIS_3_1	* Varies	**When LANE_CONTROL = Enabed:** Rx Lane Disable This field indicates PCS Rx lane disable on the corresponding lane (Lane 1, 2, or 3). When this bit is set, the PHY data of the corresponding lanes are ignored. Synchronization and deskew are performed only among the enabled lanes. **Note:** XGMII Fault-Signaling protocol does not work, when any of these bits are set. The following are the valid combinations: ■ 000: Lanes 0, 1, 2, and 3 are enabled (10G Ethernet). ■ 100: Lane 3 is disabled. Lanes 0, 1, and 2 are operational (7.5G Ethernet). ■ 110: Lanes 2 and 3 are disabled. Lanes 0 and 1 are operational (5.0G Ethernet application). ■ 111: Lanes 1, 2, and 3 are disabled. Lane 0 is operational (2.5G Ethernet application). ■ Other values: Reserved. This field is present only when LANE_CONTROL = Enabled. **Access Type:** ■ In configurations with LANE_CONTROL = Enabled, this field is RW. ■ For all other configurations, this field is RO. **When LANE_CONTROL = Disabled:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(LANE_CONTROL) ? \"read-write\" : \"read-only\""
8	Reserved_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
7:5	TX_POL_INV_3_1	* Varies	**For Multi-lane configurations:** Tx Polarity Invert on Lanes [3:1] When set to 1, this field reverses the data polarity on the Tx differential lines of the corresponding lanes. ■ 3'b**1: Lane 1 Tx data inverted ■ 3'b*1*: Lane 2 Tx data inverted ■ 3'b1**: Lane 3 Tx data inverted **Access Type:** ■ In KX_Only, KR_Only, 10GBASE-R PCS, and KR_KX configurations, this field is reserved and RO. ■ In all other configurations, this field is RW. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SINGLE_LANE_CFG) ? \"read-only\" : \"read-write\""
4	TX_POL_INV_0	R/W	Tx Polarity Invert on Lane 0 When set to 1, this bit reverses the data polarity on the Tx differential lines of Lane 0. **Value After Reset:** 0x0 **Exists:** Always
3:1	RX_POL_INV_3_1	* Varies	**For Multi-lane configurations:** Rx Polarity Invert on Lanes [3:1] When set to 1, this field reverses the polarity on the data received from the PHY. ■ 3'b**1: Lane 1 Rx data inverted ■ 3'b*1*: Lane 2 Rx data inverted ■ 3'b1**: Lane 3 Rx data inverted **Access Type:** ■ In KX_Only, KR_Only, 10GBASE-R PCS, and KR_KX configurations, this field is reserved and RO. ■ In all other configurations, this field is RW. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SINGLE_LANE_CFG) ? \"read-only\" : \"read-write\""
0	RX_POL_INV_0	R/W	Rx Polarity Invert on Lane 0. ■ 1: This bit reverses the polarity on the data received from the PHY. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.46 VR_XS_PCS_DIG_ERRCNT_SEL

- **Description:** VR XS or PCS MMD Digital Error Count Select Register.
  This register is not present in the following configurations:
  - KR_Only configurations
  - 10GBASE-R PCS with SUPPORT_1G=Disabled

  In KR_KX, KR_KX4, or KR_KX4_KX mode or 10GBASE-R PCS (with SUPPORT_1G = Enabled) configurations, these register bits are applicable only when the DWC_xpcs is working in the KX4 or KX or 1G speed mode.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8002 : (APB3_16) ? (0x8002*0x2) : (0x8002*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD||XGXS_PCS)&&!RPCS_ONLY)

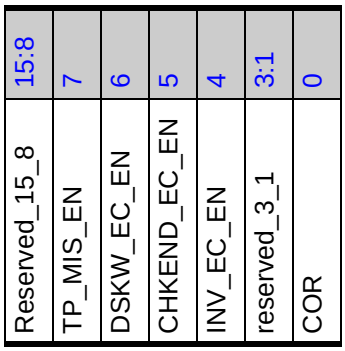


Table 5-104 Fields for Register: VR_XS_PCS_DIG_ERRCNT_SEL

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
7	TP_MIS_EN	* Varies	**For Single-lane Configurations:** Reserved **For Multi-lane configurations:** Test Pattern Mismatch Counter Enable When this bit is set, test pattern mismatch counters are enabled. Only the HF (D21.5 pattern), MF (K28.5 pattern), and LF (K28.7 pattern) test patterns are checked for mismatch. ■ 0: The counting pattern mismatch is disabled. ■ 1: The counting pattern mismatch is enabled. **Dependencies:** ■ Bit 2 of SR_XS_PCS_TCTRL Register must be set to 1. ■ Bits[1:0] of SR_XS_PCS_TCTRL Register must be set to 2'b11. ■ This bit is not present in the KX_Only or 1000BASEX-Only PCS configurations. For information about the Error count information, see VR_XS_PCS_TPM_ERRCTR Register. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SINGLE_LANE_CFG) ? \"read-only\" : \"read-write\""
6	DSKW_EC_EN	* Varies	**For Mutli-lane configurations:** Deskew Error Counter Enable When this bit is set, counting with deskew error counters is enabled. ■ 0: The counting of errors is disabled. ■ 1: The counting of errors is enabled. For information about the number of errors counted, see Bits[15:8] of VR_XS_PCS_DSKW_ERRCTR register. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SINGLE_LANE_CFG) ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
5	CHKEND_EC_EN	* Varies	**For Single-lane Configurations:** Reserved **For Multi-lane configurations:** Check End Error Counter Enable When this bit is set, the counting of check end error counters is enabled. ■ 0: The counting of errors is disabled. ■ 1: The counting of errors is enabled. For information about the number of errors counted, see Bits[7:0] of VR_XS_PCS_DSKW_ERRCTR Register. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SINGLE_LANE_CFG) ? \"read-only\" : \"read-write\""
4	INV_EC_EN	R/W	Invalid Code Group Error Counter Enable When this bit is set, the counting of invalid code group errors is enabled. ■ 0: The counting of errors is disabled ■ 1: The counting of errors is enabled For information about the fields containing the number of errors counted, see VR_XS_PCS_ICG_ERRCNT2 Register. **Value After Reset:** 0x0 **Exists:** Always
3:1	reserved_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	COR	R/W	Clear on Read When this bit is set and the host reads any error counter, that counter is cleared after the read cycle. ■ 0: Normal operation ■ 1: Clear any error counter that is read **Value After Reset:** 0x0 **Exists:** Always

5.2.2.47 VR_XS_PCS_XAUI_CTRL

- **Description:** VR XS or PCS MMD XAUI Mode Control Register
  This register is present only when the DWC_xpcs is configured for RXAUI.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8004 : (APB3_16) ? (0x8004*0x2) : (0x8004*0x4)" + "(PCS_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD | |XGXS_PCS)

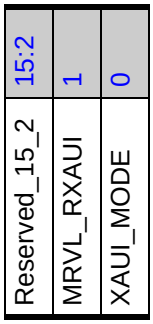


Table 5-105 Fields for Register: VR_XS_PCS_XAUI_CTRL

Bits	Name	Memory Access	Description
15:2	Reserved_15_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1	MRVL_RXAUI	* Varies	**When RXAUI Support is enabled:** Marvell RXAUI Mode When Bit 0 (XAUI_MODE) is set to 1, this bit indicates if the Marvell mode of RXAUI is enabled: ■ 0: Marvell mode RXAUI is not enabled. ■ 1: Marvell mode RXAUI enabled. **In all other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(RXAUI) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
0	XAUI_MODE	* Varies	**When RXAUI Support is enabled:** XAUI Mode This bit indicates the mode of the DWC_xpcs. ■ 0: XAUI Mode ■ 1: RXAUI Mode **In all other configurations:** Reserved **Value After Reset:** "(RXAUI_DEFAULT) ? 0x1 : 0x0" **Exists:** Always **Testable:** untestable **Memory Access:** "(RXAUI) ? \"read-write\" : \"read-only\""

5.2.2.48 VR_XS_PCS_DEBUG_CTRL

- **Description:** VR XS or PCS MMD Debug Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x8005 : (APB3_16) ? (0x8005*0x2) : (0x8005*0x4)" + "(PCS_MMD) ? ((APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD | | XGXS_PCS)

15:13	12	11	10	9	8	7	6	5	4	3:1	0
Reserved_15_13	PCH_MODE	TS_MODE	SS_CORE_EN	RX_PMBL_CTL	TX_PMBL_CTL	RX_SYNC_CTL	RX_DT_EN_CTL	SUPRESS_EEE_LOS_DET	SUPRESS_LOS_DET	RESTAR_SYNC_3_1	RESTAR_SYNC_0

Table 5-106 Fields for Register: VR_XS_PCS_DEBUG_CTRL

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
12	PCH_MODE	* Varies	PCH MODE This bit shall be set to 1, when PCH is used ethernet frames when operating in SGMII/QSGMII modes. During this mode, it is assumed that preamble is always 7 bytes long and xpcs_tx_ddmp_o/xpcs_rx_ddmp_o pulses are generated with this assumption. When PCH is not used in ethernet frames set this bit to 0. Value of this bit has no impact when operating in USXGMII mode. Irrespective of the value of this bit, in all BASE-R modes, including USXGMII, it is assumed that preamble is always 7 bytes long. **Note:**Whenever this bit is set to 1, you are expected to set TX_PMBL_CTL bit also to 1 to avoid possible truncation of premable in BASE-X transmit logic of DWC_xpcs. **Value After Reset:** 0x0 **Exists:** SNPS_RSVDPARAM_11 **Memory Access:** "SNPS_RSVDPARAM_11 ? \"read-write\" : \"read-only\""
11	TS_MODE	* Varies	Timestamp Mode When this bit is 0, DWC_xpcs indicates End-to-End Tx/Rx min/max delay values in the time-delay registers. Also, DWC_xpcs drives a value of 0 on xpcs_tx/rx_num_unit_chg_o and xpcs_tx/rx_ddmp_o ports. When this bit is 1, DWC_xpcs generates DDMP indication pulses on xpcs_tx/rx_ddmp_o ports and indicates dynamic delay variation on xpcs_tx/rx_num_unit_chg_o ports. Also, in this mode, the tx/rx delay in time-delay registers (ns and sub-ns registers) indicate the delay inside the DWC_xpcs with respect to the DDMP generation points. **Note:** When operating in Rx-to-Tx loopback mode, the behavior of xpcs_tx/rx_ddmp_o and xpcs_tx/rx_num_unit_chg_o ports are ambiguous and shall be ignored. **Value After Reset:** 0x0 **Exists:** SNPS_RSVDPARAM_11 **Memory Access:** "SNPS_RSVDPARAM_11 ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
10	SS_CORE_EN	* Varies	**When Safety Subordinate feature is enabled** Safety Subordinate Core enable Set this bit to 1 to enable the safety subordinate to monitor errors in DWC_xpcs. **Other Configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWC_XPCS_SS ? \"read-write\" : \"read-only\""
9	RX_PMBL_CTL	* Varies	Receive Preamble Control This bit can be set to 1 when DWC_xpcs is operating in 2.5GBASE-KX mode and far-end device is operating in 1000BASE-KX mode over-clocked by 2.5 times. As per IEE specification, it is permissible for a compliant 1000BASE-X PCS transmit process to truncate the first byte of preamble to align the start of the packet on the EVEN boundary. Therefore, when DWC_xpcs outputs the received data on its XGMII Rx interface, it might have one byte of preamble less. If this bit is set to 1, DWC_xpcs adds one additional byte of preamble to the incoming data before outputting it on XGMII Rx interface (if incoming data has truncated preamble). This bit need not be set, if the 2.5G MAC is able to accept packets with shortened preamble. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "ADD_2PT5G ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
8	TX_PMBL_CTL	* Varies	Transmit Preamble Control This bit can be set to 1 to prevent possible preamble truncation in the DWC_xpcs transmitter when operating in 1000BASE-X (or 2.5G Mode over GMII) Mode. As per IEEE spec, it is permissible for a compliant 1000BASE-X PCS transmit process to truncate the first byte of preamble to align the start of the packet on the EVEN boundary. If this bit is set to 1, DWC_xpcs does not delete any preamble bytes (received from GMII Tx interface) and passes on the same number of preamble bytes to its output. DWC_xpcs carries out this operation by adjusting the IPG. This mode is a deviation from Clause 36 of IEEE802.3 specification. But it can prove useful when operating at 2.5G Mode (over clocked 1000BASE-X Mode) and the far-end device is compliant to IEEE802.3bz/cb specification. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "KX_PRESENT||BASE1000X_ONLY||SUPPORT_1G ? \"read-write\" : \"read-only\""
7	RX_SYNC_CTL	* Varies	Receive Synchronization Control DWC_xpcs, by default, does not monitor the 'xgxs_rx_valid_i' input signal to determine Receive Synchronization Status. Receive Synchronization (sync_status) is attained by merely monitoring and detecting COMMA pattern in the input data, irrespective of the value of 'xgxs_rx_valid_i' signal. If this bit is set to 1, DWC_xpcs starts looking for COMMA in the input data stream only after PHY has asserted 'xgxs_rx_valid_i'. This bit is valid only when operating in PCS-X Mode (1G,XAUI,RXAUI and QSGMII modes). **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SNPS_XAUI_PHY&&!RPCS_ONLY ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
6	RX_DT_EN_CTL	* Varies	**For Multi-protocol 12G/16G PHY configurations:** Rx Data Enable Control. This bit controls the behavior of xpcs_rx_data_en_o{lane} output from DWC_xpcs. During normal operation DWC_xpcs de-asserts xpcs_rx_data_en_o on detecting of Loss-of-Signal (xpcs_los_i). However, if this bit is set, DWC_xpcs does not de-assert xpcs_rx_data_en_o on detection of Loss-of-Signal.Instead, the value of RX_DATA_EN bits from VR_XS_PMA_MP_12G_16G_25G_RX_GENCTRL0 Register is driven on xpcs_rx_data_en_o output port. **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "E12||E16||E25||E32 ? \"read-write\" : \"read-only\""
5	SUPRESS_EEE_LOS_DET	* Varies	**For PCS-X Configurations with EEE = Enabled:** Suppress EEE Loss of Signal Detection. When this field is set to 1, Loss of Signal indicated by the PHY (based on input port xpcs_los_i) is ignored by DWC_xpcs while evaluating the Receive link when operating in EEE mode. Receive link is evaluated based the on data received by DWC_xpcs from PHY. When this field is set to 0, Loss of signal indicated by the PHY is considered by the DWC_xpcs while evaluating the Receive link status in EEE mode. This field is present for all configurations (except 10GBASE-R and KR_ONLY) with EEE= Enabled. **For all other Configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(LP_1GBX||LP_10GBX) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
4	SUPRESS_LOS_DET	R/W	Suppress Loss of Signal Detection When this field is set to 1, Loss of Signal indicated by the PHY (based on input port xpcs_los_i) is ignored by DWC_xpcs while evaluating the Receive link. Receive link is evaluated based the Rx data received on the 'xgxs_rx_data{lane}_i' or 'rpcs_rx_data_i' port of DWC_xpcs. When this field is set to 0, Loss of signal indicated by the PHY is considered by the DWC_xpcs while evaluating the Receive link status. **Note**: This register bit has an impact on the EEE Rx behavior of DWC_xpcs when operating in QSGMII Mode for Multi-protocol 12G/16G/25G PHY configurations. if this register bit is set to 1, then entry to RX_QUIET state is based on 'code_sync_status' (generated by PCS Synchronization State Diagram) instead of 'signal_detect'. **Value After Reset:** 0x0 **Exists:** Always
3:1	RESTAR_SYNC_3_1	* Varies	**For Multi-lane configurations:** Restart Synchronization on Lanes[3:1] When set, the bits within this field restart the Rx synchronization state machine of PCS-X block on corresponding lane. The host must clear this field before setting it to 1 next time. **Access Type:** ■ For single-lane configurations, this field is reserved and RO. ■ For all other configurations, this field is RW. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SINGLE_LANE_CFG) ? \"read-only\" : \"read-write\""
0	RESTAR_SYNC_0	* Varies	Restart Synchronization on Lane 0 When set to 1, this bit restarts the Rx synchronization state machine of PCS-X block on Lane 0. The host must clear this bit to 0 before setting it to 1 next time. You can use this bit for debugging purpose. **Note:** This field is not present for the following configurations: ■ KR_ONLY configurations ■ 10GBASE-R PCS with SUPPORT_1G=Disabled **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(RPCS_ONLY) ? \"read-only\" : \"read-write\""

5.2.2.49 VR_XS_PCS_EEE_MCTRL0

- **Description:** VR XS or PCS MMD EEE Mode Control Register
  This register enables or disables the Energy Efficient Ethernet (EEE) support. This register is present only when the DWC_xpcs is configured to have the EEE support.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8006 : (APB3_16) ? (0x8006*0x2) : (0x8006*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD||XGXS_PCS)&&EEE_EN)

15:12	11:8	7	6	5	4	3	2	1	0
CLKSTOP	MULT_FACT_100NS	RX_EN_CTRL	SIGN_BIT	EEE_SLR_BYP	TX_EN_CTRL	RX_QUIET_EN	TX_QUIET_EN	LRX_EN	LTX_EN

Table 5-107 Fields for Register: VR_XS_PCS_EEE_MCTRL0

Bits	Name	Memory Access	Description
15:12	CLKSTOP	* Varies	Clock Stop This field controls the count value after which the Rx clock to the MAC can be stopped. The default value of this field is 8. The value programmed to this field gets added to the internal hard-coded value (120 in case of XGMII or 1 in case of GMII) to determine the number of clock cycles after which Rx clock to MAC can be stopped. The host must program this value depending on the capability of the MAC. This field is used to control the assertion of the xpcs_lrx_clk_gat_o signal (to gate the Rx clock of the MAC). By default, the xpcs_lrx_clk_gat_o signal is asserted after the PCS Rx module detects the LPI pattern followed by the following: Number of clk_xgxs_rx0_i or clk_rpcs_rx_i clocks (specified in this field + internal hard-coded value) + clk_eee_i synchronization delay + FIFO initialization delay **Value After Reset:** 0x8 **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\""
11:8	MULT_FACT_100NS	* Varies	100 ns Clock Tic Multiplying Factor This bit is the multiplying factor to the clk_eee_i clock time period to make it closer to 100 ns. For example, if the clk_eee_i clock time period (clk_eee_i_time_period) is 10 ns, the value of this field is 9, that is, 1 less than 10. Program this with a value such that the clk_eee_i_time_period * (MULT_FACT_100NS + 1) is within 80 ns to 120 ns. The default value of this register is 9, assuming the default clk_eee_i time period to be 10 ns. **Value After Reset:** 0x9 **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\""
7	RX_EN_CTRL	* Varies	Rx Control Enable This bit controls the generation of the xpcs_rx_en_o{lane} signal. When this bit is set to 1, the xpcs_rx_en_o signal is de-asserted when the EEE receive controller reaches the Quiet state. When this bit set to 0, the xpcs_rx_en_o signal is not de-asserted when the EEE transmit controller reaches the Quiet state. **Value After Reset:** "(EEE_EN) ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
6	SIGN_BIT	* Varies	Effective 100 ns Tic Value The host must use this bit to fine tune the EEE timing requirement. The host must set this bit to 0 when the clk_eee_i_time_period * (MULT_FACT_100NS + 1) value is more than or equal to 100 ns. The host must set this bit to 1 when the clk_eee_i clock period * (MULT_FACT_100NS + 1) value is less than 100 ns. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\""
5	EEE_SLR_BYP	* Varies	**For 10GBASE-R/KR configurations with EEE:** Scrambler Bypass This bit is used to bypass the transmit scrambler to enable the Rapid Block synchronization. When this bit is set to 1, the scrambler is bypassed in PCS-R during EEE Tx mode. When this bit is set to 0, the scrambler is not bypassed. This bit is enabled only when FEC_EN = Enabled. **Access Type:** For 10GBASE-R PCS or KR_Present mode configurations, this bit is RW. **For all other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(LP_10GBR) ? \"read-write\" : \"read-only\""
4	TX_EN_CTRL	* Varies	Tx Control Enable This bit controls the generation of the following signals: ■ xgxs_tx{lane}_en_o (for Synopsys PHY) ■ xpcs_tx_en_o{lane} (for non-Synopsys PHY) When this bit is set to 1, the xgxs_tx{lane}_en_o or xpcs_tx_en_o{lane} signal is de-asserted when the EEE transmit controller reaches the Quiet state. When this bit set to 0, the xgxs_tx{lane}_en_o or xpcs_tx_en_o{lane} signal is not de-asserted when the EEE transmit controller reaches the Quiet state. **Value After Reset:** "(EEE_EN) ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
3	RX_QUIET_EN	* Varies	Rx Quiet Enable This bit controls the generation of the xpcs_lpirx_quiet_o output. When this bit is set to 1, the xpcs_lpirx_quiet_o output is set to 1 when the EEE receive controller reaches the Quiet state. When this bit set to 0, the xpcs_lpirx_quiet_o output is not set to 1. **Value After Reset:** "(EEE_EN) ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\"""
2	TX_QUIET_EN	* Varies	Tx Quiet Enable This bit controls the generation of the xpcs_lpitx_quiet_o output. When this bit is set to 1, the xpcs_lpitx_quiet_o output is set to 1 when the EEE transmit controller reaches the Quiet state. When this bit set to 0, the xpcs_lpitx_quiet_o output is not set to 1. **Value After Reset:** "(EEE_EN) ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\"""
1	LRX_EN	* Varies	LPI Rx Enable When set to 1, this bit enables the Energy Efficient Ethernet support in the DWC_xpcs Receive path. When set to 0, it disables the support for Energy Efficient Ethernet in the DWC_xpcs Receive path. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\"""
0	LTX_EN	* Varies	LPI Tx Enable When set to 1, this bit enables the Energy Efficient Ethernet support in the DWC_xpcs Transmit path. When set to 0, it disables the support for Energy Efficient Ethernet in the DWC_xpcs Transmit path. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\"""

5.2.2.50 VR_XS_PCS_KR_CTRL

- **Description:** VR PCS 10GBASE-R Control Register
  This register is present only with the PCS MMD in the following configurations:
  - 10GBASE-R PCS
  - KR_Only
  - KR_KX4_KX
  - KR_KX4
  - KR_KX
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8007 : (APB3_16) ? (0x8007*0x2) : (0x8007*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD&&RPCS_EN)

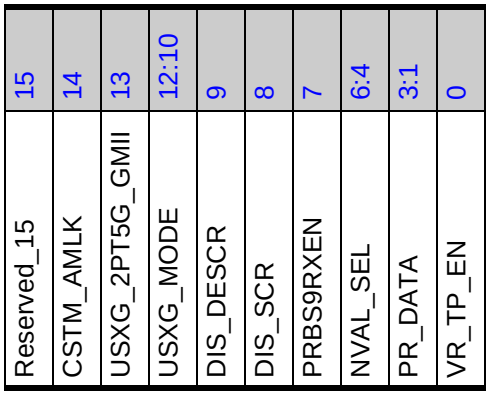


Table 5-108 Fields for Register: VR_XS_PCS_KR_CTRL

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
14	CSTM_AMLK	* Varies	Customer Alignment Marker Lock Mode Enable As per USXGMII specification, version 2.15, the only allowed values for port-number and inverse fields of the Alignment Markers are 0x00 and 0xFF respectively. The data immediately following the AM set (one AM set consists of 4 AMs) must correspond to port 0. Whenever the port-number and its inverse fields of the received Alignment marker do not match these values, it is considered as a bad alignment marker. If this CSTM_AMLK bit is set to 1, it enables a vendor-specific mode, where the port-number field (and its inverse field) can take even non-zero values and in such cases the data immediately following the received AM set is assumed to correspond to the port whose port-number is specified in the port-number field. Accordingly, the received data is de-multiplexed to the appropriate GMII/XGMII Rx ports. However, note that, when operating in 4-port mode, the only allowed values of port-number field are 0,1,2, and 3 and when operating in 2-port modes, the allowed values of port-number field are 0 and 1. **Note:** This field is applicable only for configurations with multi-port USXGMII support. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "MP_USXGMII ? \"read-write\": \"read-only\""
13	USXG_2PT5G_GMII	* Varies	MAC/PCS interface select in USXGMII 2.5G Mode As per Cisco USXGMII specification, when MAC operates in 2.5G mode, it must interface with DWC_xpcs through XGMII interface (32-bit at 78.125 MHz). However, this bit provides flexibility to operate through GMII interface (8-bit at 312.5MHz). ■ 0: Use XGMII interface ■ 1: Use GMII interface **Note:** This field is applicable only for configurations with USXGMII support. This bit is valid only when operating at single-port USXGMII mode. Therefore,this bit must not be set when operating in multi-port modes. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "USXGMII_EN ? \"read-write\": \"read-only\""

Bits	Name	Memory Access	Description
12:10	USXG_MODE	* Varies	USXGMII Mode Select This field chooses the current USXGMII mode in which DWC_xpcs must operate. The encoding of the mode is as follows: ■ 000: 10G-SXGMII ■ 001: 5G-SXGMII ■ 010: 2.5G-SXGMII ■ 011: 10G-DXGMII ■ 100: 5G-DXGMII ■ 101: 10G-QXGMII ■ 110/111: Reserved **Note:** Application must choose an appropriate and judicious value based on the USXGMII features selected during controller configuration. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(USXG_SPORT_FULL) ? \"read-write\" : \"read-only\""
9	DIS_DESCR	* Varies	Disable Descrambler When set to 1, this bit disables the 10GBASE-R PCS descrambling function. You can use this bit for debugging purpose. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(RPCS_EN) ? \"read-write\" : \"read-only\""
8	DIS_SCR	* Varies	Disable Scrambler When set to 1, this bit disables the 10GBASE-R PCS scrambling function. You can use this bit for debugging purpose. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(RPCS_EN) ? \"read-write\" : \"read-only\""
7	PRBS9RXEN	* Varies	Enable PRBS9 Testing in Receive Path When this bit is set to 1 and Bit 5 (PRBS31R_EN) and Bit 2 (RTP_EN) of SR PCS 10GBASE-R Test Pattern Control Register are set to 0, this bit enables the PRBS9 test pattern checking in 10GBASE-R PCS Receive path. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(RPCS_EN) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
6:4	NVAL_SEL	* Varies	Square Wave Control When square wave test pattern generation is enabled in SR PCS 10GBASE-R Test Pattern Control register, this field indicates the number of bit times for which Ones and Zeros must be driven on the serial lines (square wave period). Encoding: ■ 3'b000: NVAL_SEL = 4; ■ 3'b001: NVAL_SEL = 5; ■ 3'b010: NVAL_SEL = 6; ■ 3'b011: NVAL_SEL = 7; ■ 3'b100: NVAL_SEL = 8; ■ 3'b101: NVAL_SEL = 9; ■ 3'b110: NVAL_SEL = 10; ■ 3'b111: NVAL_SEL = 11; **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(RPCS_EN) ? \"read-write\" : \"read-only\""
3:1	PR_DATA	* Varies	Pseudo-Random Test Pattern Data When pseudo-random test pattern is enabled in Bit 3 (TTP_EN) of SR PCS 10GBASE-R Test Pattern Control Register and Bit 0 (VR_TP_EN) of this register is set to 1, this field selects the data input for the scrambler. Encoding: ■ 3'b000: Data = All zero (0x00); IEEE specified ■ 3'b001: Data = XGMII local fault (encoded version); IEEE specified ■ 3'b010: Data = All five (0x55); HFTP ■ 3'b011: Data = All A (0xAA); HFTP ■ 3'b100: Data = All ones (0xFF); LFTP ■ 3'b101: Data = All three (0x33); MFTP ■ 3'b110: Data = All C (0xCC); MFTP ■ 3'b111: Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(RPCS_EN) ? \"read-write\" : \"read-only\""
0	VR_TP_EN	* Varies	PCS-R Vendor-Specific Test Pattern Enable When this bit set to 1, the data for the pseudo-random test pattern is selected as per the PR_DATA selection. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(RPCS_EN) ? \"read-write\" : \"read-only\""

5.2.2.51 VR_XS_PCS_EEE_TXTIMER

- **Description:** VR XS or PCS MMD EEE Tx Timer Register
  This register is present only if EEE_EN = Enabled.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8008 : (APB3_16) ? (0x8008*0x2) : (0x8008*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD||XGXS_PCS)&&EEE_EN)

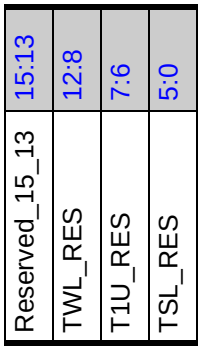


Table 5-109 Fields for Register: VR_XS_PCS_EEE_TXTIMER

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
12:8	TWL_RES	* Varies	**For Configurations With PCS-R and EEE:** TWL Resolution This field stores the resolution value for the TWL timer. When the generated 100 ns tic timer is not exactly 100 ns, this field is programmed with the required value. The programmed value is equivalent to the number of more counts (SIGN_BIT is low) or less counts (SIGN_BIT is high) than the ideal count to meet the range of the TWL timer. The DWC_xpcs maintains the default value of the TWL timer as 109, assuming clk_eee_i_time_period * (MULT_FACT_100NS + 1)is equal to 100 ns to produce 10900 ns (10.9 us) as per the TWL requirement in the IEEE standard. Example: For 10GBASE-R or KR_Present configuration, if clk_eee_i_time_period * (MULT_FACT_100NS + 1) is 90, the default TWL timer produces 9810 ns (9.81 us) which is lesser than the standard 10.9 us. To make it 10.9 us, you must program this register such that (109 +/- TWL_RES) * ((MULT_FACT_100NS + 1) * clk_eee_i_time_period) is greater than 10.9 us and less than 11 us (max limit), that is, (109 + TWL_RES)*90 = 10900 ns. TWL_RES = 13 meets the requirement which produces 10.98 us. The SIGN_BIT should be programmed as 1. **Access Type:** For 10GBASE-R PCS or KR_Present mode configurations, this bit is RW. For all other configurations, this bit is RO. **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(LP_10GBR) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
7:6	T1U_RES	* Varies	T1u Resolution This field stores the resolution value for the T1u timer. When the generated 100 ns tic timer is not exactly 100 ns, this field is programmed with the required value. The programmed value is equivalent to the number of more (SIGN_BIT is low) or less (SIGN_BIT is high) counts than the ideal count to meet the range of the T1u timer. The DWC_xpcs maintains the default value of the T1u timer as 11, assuming clk_eee_i_time_period * (MULT_FACT_100NS + 1) is equal to 100 ns to produce 1100 ns (1.1 us) as per the T1u requirement in the IEEE standard. Example: For 10GBASE-R or KR_Present configuration, if clk_eee_i_time_period * (MULT_FACT_100NS + 1)is 90, the default T1u timer produces 990 ns (0.99 us) which is lesser than the standard 1.1 us. To make it 1.1 us, you should program this register such that (11 +/- T1U_RES) * ((MULT_FACT_100NS + 1) * clk_eee_i_time_period) is greater than 1.1 us and less than 1.3 us (max limit), that is, (11+ T1U_RES)*90 = 1.1 us. T1U_RES = 2 meets the requirement which produces 1.17 us. Program the SIGN_BIT as 1. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(LP_10GBR) ? \"read-write\" : \"read-only\"""

Bits	Name	Memory Access	Description
5:0	TSL_RES	* Varies	TSL Resolution This field stores the resolution value for the TSL timer. When the generated 100 ns tic timer is not exactly 100 ns, this field is programmed with the required value. The programmed value is equivalent to the number of more (SIGN_BIT is low) or less (SIGN_BIT is high) counts than the ideal count to meet the range of the TSL timer. The DWC_xpcs maintains the following default value of the TSL timer: ■ 199 for all configurations ■ 49 for 10GBASE-R or KR_present configurations This value is maintained assuming clk_eee_i_time_period * (MULT_FACT_100NS + 1) is equal to 100 ns to produce the following as per the TSL requirement in the IEEE standard: ■ 19900 ns (19.9 us) for all configurations ■ 4900 ns (4.9 us) for 10GBASE-R or KR_Present configurations Example: For 1000BASEX-Only PCS configuration, if clk_eee_i_time_period * (MULT_FACT_100NS + 1) is 90, the default TSL timer produces 17910 ns (17.91 us) which is lesser than the standard value 19.9 us. To make it 19.9 us, you should program this register such that (199 +/- TSL_RES) * ((MULT_FACT_100NS + 1) * clk_eee_i_time_period) is greater than 19.9 us and less than 20.1 us (max limit), that is, (199 + TSL_RES)*90 = 19.9 us. TSL_RES = 23 meets the requirement which produces 19.98 us. Program the SIGN_BIT as 1. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\""

5.2.2.52 VR_XS_PCS_EEE_RXTIMER

- **Description:** VR XS or PCS MMD EEE Rx Timer Register
  This register is present only if EEE_EN = Enabled.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8009 : (APB3_16) ? (0x8009*0x2) : (0x8009*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD||XGXS_PCS)&&EEE_EN)

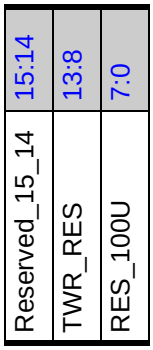


Table 5-110 Fields for Register: VR_XS_PCS_EEE_RXTIMER

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
13:8	TWR_RES	* Varies	TWR Resolution This field stores the resolution value for the TWR timer. When the generated 100 ns tic timer is not exactly 100 ns, this field is programmed with the required value. The programmed value is equivalent to the number of more (SIGN_BIT is low) or less (SIGN_BIT is high) counts than the ideal count to meet the range of the TWR timer. The following are the default value: ■ 9 us for KX4, XGXS PCS, and 10GBASE-X modes ■ 11 us for KX and 1000BASEX-Only PCS modes ■ 11.5 us for KR mode when scrambler is not bypassed ■ 13.7 us for KR mode when scrambler is bypassed These values are as per the requirements specified in the standard assuming that clk_eee_i_time_period * (MULT_FACT_100NS + 1) produces 100 ns tick. Example: For KX4 mode, the time requirement is 9 us. The DWC_xpcs maintains the default value of the TWR timer as 90, assuming clk_eee_i_time_period * (MULT_FACT_100NS + 1) is equal to 100 ns to produce 9000 ns (9 us). If clk_eee_i_time_period * (MULT_FACT_100NS + 1) is 110, the default TWR timer produces 9900 ns (9.9 us) which is more than 9 us. To make it 9 us, you must program this register such that (90 +/- TWR_RES) * ((MULT_FACT_100NS + 1) * clk_eee_i_time_period) = 9 us. Therefore, use the following equation: (90 - TWR_RES) * 110 <= 9000 ns TWR_RES = 9 meets the requirement which produces 8.91 us. Program the SIGN_BIT as 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
7:0	RES_100U	* Varies	100 us Resolution This field stores the resolution value for the 100 us timer. If the generated 100 ns tic timer is not exactly 100 ns, this field is programmed with the required value. The programmed value is equivalent to the number of more (SIGN_BIT is low) or less (SIGN_BIT is high) counts than the ideal count to meet the range of the 100 us timer. This field is used to control the generation of 100 us time tick using the clk_eee_i clock and MULT_FACT_100NS field. By default, the 100 us timer is generated assuming clk_eee_i_time_period * (MULT_FACT_100NS + 1) is equal to 100 ns. Therefore, the DWC_xpcs maintains the default value of the 100 us timer as 1000 to achieve 100000 ns (100 us). You must use this field if clk_eee_i_time_period * (MULT_FACT_100NS + 1) is not equal to 100 ns. To program this field, use the following equation: (1000 +/- RES_100US)* ((MULT_FACT_100NS + 1) * clk_eee_i_time_period) = 100 us Program the SIGN_BIT as 1 for (+) and 0 for (-). **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\""

5.2.2.53 VR_XS_PCS_AM_CTRL

- **Description:** VR PCS MMD Alignment Marker Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x800A : (APB3_16) ? (0x800A*0x2) : (0x800A*0x4)" + "(PCS_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD&&MP_USXGMII)

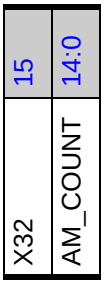


Table 5-111 Fields for Register: VR_XS_PCS_AM_CTRL

Bits	Name	Memory Access	Description
15	X32	R/W	Multiply AM_COUNT by 32 ■ 1: Alignment marker interval used by DWC_xpcs is 32 times the value programmed in the AM_COUNT field. ■ 0: Alignment marker interval corresponds to AM_COUNT itself. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
14:0	AM_COUNT	R/W	Alignment Marker Interval This field controls the Alignment Marker interval when operating in multi-port USXGMII Mode. In the transmit data-stream, Alignment Markers are inserted periodically at the interval (in terms of 66-bit blocks) defined by this field. Similarly, DWC_xpcs checks for Alignment Markers in the receive data-stream, at this interval. The default value of this field corresponds to the standard value of 16'd16400 as defined by the specification. However, application can modify this value based on its choice. Whenever X32 bit is 0, the programmed value must be a multiple of 4. Behavior of DWC_xpcs can be ambiguous, if the programmed value is not a multiple of 4. Also note that programming too small values can result in overflow or underflow of USXGMII related FIFOs. To support jumbo packets with minimum IPG, it is recommended to program larger (than default) values to this field. It is recommended to issue a VR_RST (vendor-specific soft-reset) after programming this field to a new value. **Value After Reset:** 0x4010 **Exists:** Always

5.2.2.54 VR_XS_PCS_EEE_MCTRL1

- **Description:** VR XS or PCS MMD EEE Mode Control 1 Register
  This register also controls the Energy Efficient Ethernet (EEE) related settings of DWC_xpcs . This register is present only when the DWC_xpcs is configured to have the EEE support.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x800B : (APB3_16) ? (0x800B*0x2) : (0x800B*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD||XGXS_PCS)&&EEE_EN)

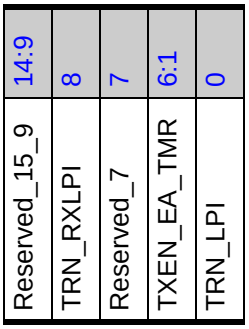


Table 5-112 Fields for Register: VR_XS_PCS_EEE_MCTRL1

Bits	Name	Memory Access	Description
14:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
8	TRN_RXLPI	R/W	Transparent Rx LPI Mode Enable. When set to 1 (along with LRX_EN=1), transparent LPI mode gets activated in the DWC_xpcs Receive path. On receiving Lower-Power Idles from link-partner, DWC_xpcs merely decodes it and passes the LPI pattern to MAC XGMII/GMII Rx interface. It does not initiate any low power actions (like switching PHY to low power state or gating-off clocks to DWC_xpcs). **1G/2.5G/QSGMII modes:** ■ The Receive LPI FSM goes to LRX_SLEEP/LRX_LPI_K state on receiving LPI pattern from link-partner. Thereafter, FSM transitions back to RX_ACTIVE state on receiving normal IDLE patterns from link-partner. To prevent DWC_xpcs from asserting xpcs_lrx_clk_gat_o during the transparent Rx LPI mode, program the CS_EN bit of the SR_XS_PCS_CTRL1 register to 0. **Other modes:** ■ The receive LPI FSM remains in RX_ACTIVE state itself during transparent Rx LPI mode. **Value After Reset:** 0x0 **Exists:** Always
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
6:1	TXEN_EA_TMR	R/W	Transmit Enable Early Assert Time Duration During normal Tx EEE operation, DWC_xpcs drives xpcs_tx_en_o/xgxs_tx_data_en_o low for the entire duration for which Tx EEE FSM remains in TX_QUIET state. On exiting TX_QUIET state, it asserts back xpcs_tx_en_o/xgxs_tx_data_en_o to 1. This field can be set to a non-zero value (ranging from 1 to 63), to allow early assertion of xpcs_tx_en_o/xgxs_tx_data_en_o signal prior to Tx FSM exiting TX_QUIET state. This field controls how many micro-seconds earlier to the exit of TX_QUIET state, must DWC_xpcs be driving xpcs_tx_en_o/xgxs_tx_data_en_o back to 1. This feature can be used to enable back the PHY transmitter, a little earlier than desired, so that it can be fully ready to transmit the desired patterns when Tx FSM reaches TX_ALERT/TX_WAKE states. Whenever clk_eee_i period is other than 10ns, the acutal early exit duration in micro-seconds does not exactly match the number programmed to this field. The exact granularity depends on the value of (clk_eee_i_time_period*(MULT_FACT_100NS+1)). A value of 1 programmed to this field corresponds to an early exit duration of (clk_eee_i_time_period*(MULT_FACT_100NS+1)*10 nano-seconds. If the value of 0 is programmed, DWC_xpcs retains the original legacy behavior,i.e, no early exit For Synopsys-PHY configurations, DWC_xpcs additionally de-asserts xpcs_lpitx_quiet_o signal earlier if a non-zero value is programmed. This is done to ensure that Tx clocks are available to DWC_xpcs and hence xgxs_tx_data_en_o can be asserted synchronously(to Tx clock) as required by the Synopsys PHY. **Note:**This feature is not supported in FASTSIM Mode. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	TRN_LPI	R/W	Transparent Tx LPI Mode Enable. When set to 1 (along with LTX_EN=1), transparent LPI mode gets activated in the DWC_xpcs Transmit path. In this mode, the transmit LPI state-machine does not move to TX_QUIET state. On detecting Lower-Power Idle on XGMII/GMII Tx interface, DWC_xpcs goes to the TX_SLEEP state and remains in this state till MAC stops sending LPI. In this mode, DWC_xpcs sends the encoded LPI pattern to the serdes. This mode does not involve gating-off of any clocks to DWC_xpcs. In addition, the serdes transmitter is not disabled (xpcs_tx_data_en_o signal remains high). MAC must ensure that it does not gate-off XGMII/GMII Tx clock to DWC_xpcs during this mode of operation. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.55 VR_XS_PCS_DIG_STS

- **Description:** VR XS or PCS MMD Digital Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8010 : (APB3_16) ? (0x8010*0x2) : (0x8010*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD||XGXS_PCS))

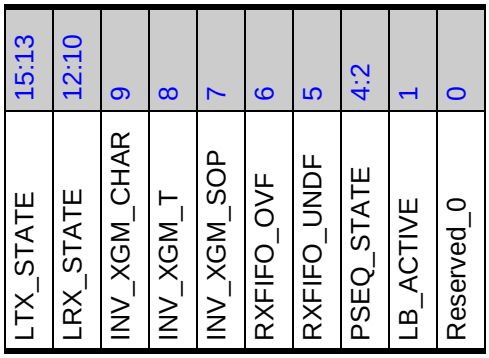


Table 5-113 Fields for Register: VR_XS_PCS_DIG_STS

Bits	Name	Memory Access	Description
15:13	LTX_STATE	R	**For EEE configurations:** LPI Transmit State This field indicates the current state of the LPI Transmit state Machine. This is a 2-bit field if the selected configuration does not have KR/10GBASE-R support.Otherwise, it is a 3-bit field. ■ 000: LTX_ACTIVE ■ 001: LTX_SLEEP ■ 010: LTX_QUIET ■ 011: LTX_REF_WAKE ■ 100: LTX_ALERT (valid only if PCS-R is present) ■ 101: LTX_SCR_BYP (valid only if PCS-R is present) **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
12:10	LRX_STATE	R	**For EEE configurations:** LPI Receive State This field indicates the current state of the LPI Receive State Machine ■ 000: LRX_ACTIVE ■ 001: LRX_SLEEP ■ 010: LRX_QUIET ■ 011: LRX_WAKE ■ 100: LRX_WTF ■ 101: LRX_LINK_FAIL ■ 110: LRX_LPI_K (present only in configurations with 1G speed) **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always
9	INV_XGM_CHAR	R	Invalid XGMII Character (RO,LH Type) This bit indicates an invalid XGMII character on the Transmit path. ■ 0: No Invalid XGMII character ■ 1: Invalid character This bit is set when one of the following conditions is true on the XGMII interface: ■ |Q| is detected on a lane other than Lane 0. ■ |Q| is detected on Lane 0 and also any of the following conditions is true: ❑ No 8'h00 data in Lane 1. ❑ No 8'h00 data in Lane 2. ❑ No 8'h01 or 8'h02 data on Lane 3. ■ |E| or |T| is encountered in non-data phase and |Q| is encountered in data phase. ■ |D| is detected during the Idle phase (after Terminate and before valid SOF). ■ |I| is detected during the data phase (after SOF and before Terminate). This bit is not present in KX_Only or 1000BaseX-Only PCS configurations. In configurations with both 1G and 10G speed support, this bit is valid only in 10G mode (that is when XGMII interface is active). **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
8	INV_XGM_T	R	Invalid XGMII T Character (RO,LH Type) This bit indicates that the transmit data received on XGMII has an invalid Terminate character. ■ 0: Normal operation ■ 1: Invalid Terminate character The following can cause an invalid Terminate character: ■ In Terminate column, before lane corresponds to the |T| character, there is a |I| or |Q| character. ■ In Terminate column, after lane corresponds to the |T| character, there is a non-|I| character This bit is not present in KX_Only or 1000BaseX-Only PCS configurations. In configurations with both 1G and 10G speed support, this bit is valid only in 10G mode (that is when XGMII interface is active). **Value After Reset:** 0x0 **Exists:** Always
7	INV_XGM_SOP	R	Invalid XGMII Start Character (RO,LH Type) This bit indicates that the XGMII transmit frame contains invalid SOP character. ■ 0: Normal operation ■ 1: Invalid SOP character An invalid SOP event occurs when a |S| character is detected in a lane other than Lane 0. This bit is not present in KX_Only or 1000BaseX-Only PCS configurations. In configurations with both 1G and 10G speed support, this bit is valid only in 10G mode (that is when XGMII interface is active). **Value After Reset:** 0x0 **Exists:** Always
6	RXFIFO_OVF	R	Rx FIFO Overflow (RO,LH Type) This bit indicates the clock rate compensation FIFO overflow. ■ 0: Normal operation ■ 1: FIFO overflow **Value After Reset:** 0x0 **Exists:** Always
5	RXFIFO_UNDF	R	Rx FIFO Underflow (RO,LH Type) This bit indicates the clock rate compensation FIFO underflow. ■ 0: Normal operation ■ 1: FIFO underflow **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4:2	PSEQ_STATE	R	**For Synopsys PHY configurations:** Power Up Sequence State This field indicates the state variable value of the power-up sequence module. For Synopsys Multi-protocol 6G PHY: ■ 3'b000: Wait for MPLL ON ■ 3'b001: Wait for TX up ■ 3'b010: Wait for RX up ■ 3'b011/3'b100: Tx/Rx Stable (Power_Good state) ■ 3'b101: Wait for Rx down (MPLL still ON) ■ 3'b110: MPLL OFF For Other Synopsys Multi-protocol PHYs ■ 3'b000: Wait for ACK High 0 ■ 3'b001: Wait for ACK Low 0 ■ 3'b010: Wait for ACK High 1 ■ 3'b011: Wait for ACK Low 1 ■ 3'b100: Tx/Rx stable (Power_Good state) ■ 3'b101: Power Save state ■ 3'b110: Power Down state ■ 3'b111: Other intermediate states (valid only for 32G PHY configurations) **For non-Synopsys PHY configurations:** Reserved **Value After Reset:** 0x4 **Exists:** Always
1	LB_ACTIVE	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	Reserved_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.2.56 VR_XS_PCS_ICG_ERRCNT1

- **Description:** VR XS or PCS MMD Invalid Code Group Error Count1 Register
  The seven error counters EC0-EC6 do not overflow. Each counter remains at all-ones on saturation. A counter is automatically cleared when it is read, with Bit 4 of VR_XS_PCS_DIG_ERRCNT_SEL Register set to 1.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8011 : (APB3_16) ? (0x8011*0x2) : (0x8011*0x4)" + "(PCS_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD | |XGXS_PCS)&&!(RPCS_ONLY))

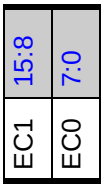


Table 5-114 Fields for Register: VR_XS_PCS_ICG_ERRCNT1

Bits	Name	Memory Access	Description
15:8	EC1	R	**For Multi-lane configurations:** Error Count1: Invalid Code Group Count on Lane 1 This field gives the Invalid Code Group count in Lane 1 when Bit 4 of VR_XS_PCS_DIG_ERRCNT_SEL Register is set to 1 (RO,COR Type). It is recommended to wait at the least for a time duration equal to 5 clocks of clk_xgxs_rx0_i/clk_rpcs_rx_i before initiating back-to-back register access. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always
7:0	EC0	R	Error Count0: Invalid Code Group Count on Lane 0 (RO,COR Type) This field is similar to the EC1 field and is applicable to Lane 0. See description of the EC1 field. This field is not valid for single-lane configurations. Wait at the least for a time duration equal to 5 clocks of clk_xgxs_rx0_i/clk_rpcs_rx_i before initiating back-to-back register access. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.57 VR_XS_PCS_ICG_ERRCNT2

- **Description:** VR XS or PCS MMD Invalid Code Group Error Count2 Register
  The seven error counters EC0-EC6 do not overflow. Each counter remains at all-ones on saturation. A counter is automatically cleared when it is read, with Bit 4 of VR_XS_PCS_DIG_ERRCNT_SEL Register set to 1.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8012 : (APB3_16) ? (0x8012*0x2) : (0x8012*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD||XGXS_PCS)&&!SINGLE_LANE_CFG)

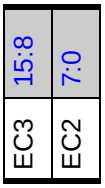


Table 5-115 Fields for Register: VR_XS_PCS_ICG_ERRCNT2

Bits	Name	Memory Access	Description
15:8	EC3	R	Error Count3: Invalid Code Group Count on Lane 3 (RO,COR Type) This field gives the Invalid Code Group count in Lane 3 when Bit 4 of VR_XS_PCS_DIG_ERRCNT_SEL Register is set to 1. Dependency: This field is not present in single-lane configurations. **Value After Reset:** 0x0 **Exists:** Always
7:0	EC2	R	Error Count2: Invalid Code Group Count on Lane 2 (RO,COR Type) This field is similar to the EC3 field and is applicable for Lane 2. See description of the EC3 field. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.58 VR_XS_PCS_DSKW_ERRCTR

- **Description:** VR XS or PCS MMD Check End or Deskew Error Count Register
  The seven error counters EC0-EC6 do not overflow. Each counter remains at all-ones on saturation. A counter is automatically cleared when it is read, with Bit 6 of VR_XS_PCS_DIG_ERRCNT_SEL Register set to 1.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8013 : (APB3_16) ? (0x8013*0x2) : (0x8013*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD||XGXS_PCS)&&!SINGLE_LANE_CFG)



Table 5-116 Fields for Register: VR_XS_PCS_DSKW_ERRCTR

Bits	Name	Memory Access	Description
15:8	EC5	R	Deskew Error Count (RO,COR Type) This field counts the deskew error count. This counter is enabled when Bit 6 of VR_XS_PCS_DIG_ERRCNT_SEL Register is set to 1. Dependency: This field is not present in the KX_Only, 10GBASE-R PCS, KR_Only, and KR_KX configurations. **Value After Reset:** 0x0 **Exists:** Always
7:0	EC4	R	Check-End Failure Error Counter (RO,COR Type) This field indicates the check-end failure count. This counter is enabled when Bit 5 of VR_XS_PCS_DIG_ERRCNT_SEL Register is set to 1. Dependency: This field is not present in the KX_Only, 10GBASE-R PCS, KR_Only, and KR_KX configurations. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.59 VR_XS_PCS_TPM_ERRCTR

- **Description:** VR XS or PCS MMD Test Pattern Mismatch Error Count Register
  The seven error counters EC0-EC6 do not overflow. Each counter remains at all-ones on saturation. A counter is automatically cleared when it is read, with Bit 7 of VR_XS_PCS_DIG_ERRCNT_SEL Register set to 1.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8014 : (APB3_16) ? (0x8014*0x2) : (0x8014*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD||XGXS_PCS)&&!SINGLE_LANE_CFG)

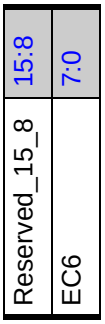


Table 5-117 Fields for Register: VR_XS_PCS_TPM_ERRCTR

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:0	EC6	R	Error Count6 Test Pattern Mismatch Error Count (RO,COR Type) This read-only field indicates the number of Test Pattern Mismatch Errors. Bit 5 of VR_XS_PCS_DIG_ERRCNT_SEL Register must be set to 1. Dependency: This field is not present in the KX_Only, 10GBASE-R PCS, KR_Only, and KR_KX configurations. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.60 VR_XS_PCS_GPIO

- **Description:** VR XS or PCS MMD GPIO Register
  This register is present only if the DWC_xpcs is configured to have the GPIO interface.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8015 : (APB3_16) ? (0x8015*0x2) : (0x8015*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** ((PCS_MMD||XGXS_PCS)&&GPIO_EN)

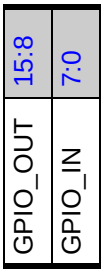


Table 5-118 Fields for Register: VR_XS_PCS_GPIO

Bits	Name	Memory Access	Description
15:8	GPIO_OUT	R/W	GPIO Output The content written on this field is driven to the xpcs_gpo_o[7:0] output port. Dependency: This field is valid only when GPIO_EN = Enabled. **Value After Reset:** 0x0 **Exists:** Always
7:0	GPIO_IN	R	GPIO Input This field indicates the content of the xpcs_gpo_i[7:0] port. Dependency: This field is valid only when GPIO_EN = Enabled. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.61 VR_XS_PCS_CDT_STS

- **Description:** VR XS or PCS MMD Comma-Detect Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8018 : (APB3_16) ? (0x8018*0x2) : (0x8018*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD||XGXS_PCS)

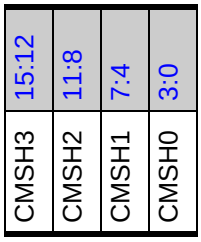


Table 5-119 Fields for Register: VR_XS_PCS_CDT_STS

Bits	Name	Memory Access	Description
15:12	CMSH3	R	Comma-detect Bit shift-lane 3 This field captures the bit-shift introduced in receive data due to comma-detection logic on lane 3. This field is present and valid only in multi-lane modes. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
11:8	CMSH2	R	Comma-detect Bit shift-lane 2 This field captures the bit-shift introduced in receive data due to comma-detection logic on lane 2. This field is present and valid only in multi-lane modes. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
7:4	CMSH1	R	Comma-detect Bit shift-lane 1 This field captures the bit-shift introduced in receive data due to comma-detection logic on lane 1. This field is present and valid only in multi-lane modes. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

Bits	Name	Memory Access	Description
3:0	CMSH0	R	Comma-detect Bit shift-lane 0 This field captures the bit-shift introduced in receive data due to comma-detection logic on lane 0. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

5.2.2.62 VR_XS_PCS_MISC_STS

- **Description:** VR XS or PCS MMD Miscellaneous Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8019 : (APB3_16) ? (0x8019*0x2) : (0x8019*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD||XGXS_PCS)

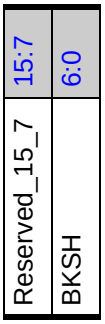


Table 5-120 Fields for Register: VR_XS_PCS_MISC_STS

Bits	Name	Memory Access	Description
15:7	Reserved_15_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:0	BKSH	R	Block-lock bit-shift This field captures the bit-shift introduced in receive data-path due to Clause 49 block-lock logic or Clause 74 FEC lock logic.This field is valid only when operating in PCS-R mode. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

5.2.2.63 VR_XS_PCS_NSNPS_EEE_CTRL

- **Description:** VR XS or PCS MMD Non-Synopsys PHY EEE Control Register
  This register is present only in configurations with PCS-R/QSGMII support and with EEE enabled.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x801F : (APB3_16) ? (0x801F*0x2) : (0x801F*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD||XGXS_PCS)&&!SNPS_XAUI_PHY&&(LP_10GBR||QSGMII_EN)&&EEE_EN

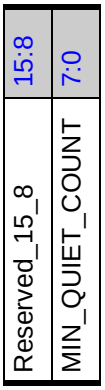


Table 5-121 Fields for Register: VR_XS_PCS_NSNPS_EEE_CTRL

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
7:0	MIN_QUIET_COUNT	R/W	Minimum Rx Quiet Duration This field determines the minimum duration for which Rx LPI FSM remains in the RX_QUIET state.DWC_xpcs implements a vendor-specific 'MIN QUIET counter' in the RX_QUIET state of LPI state machine. This counter (running at clk_rpcs_rx_i clock in PCS-R Mode or running at clk_xgxs_rx0_i clock in QSGMII Mode) starts as soon as RX_QUIET state is reached and it stops when it reaches a value equal to 8 times the value mentioned as 'MIN QUIET COUNT' field. DWC_xpcs start monitoring for 'energy_detect' (in PCS-R MOde) or 'signal_detect' (in QSGMII Mode) only after this counter has reached its maximum value (8 times MIN_QUIET_COUNT). The field becomes useful in preventing premature exit of RX_QUIET state in those cases where the PHY takes considerable amount of time to de-assert 'energy_detect' or 'signal_detect' signal on detecting electrical IDLE on its receive serial lines. **Note:** In QSGMII Mode, this timer is activated only if SUPRESS_LOS_DET bit of VR_XS_PCS_DEBUG_CTRL Register is set to 1. If this bit is 0, behavior of Rx EEE FSM is as per IEEE specification. **Value After Reset:** 0xa **Exists:** Always

5.2.2.64 VR_XS_PCS_SFTY_UE_INTR0

- **Description:** VR XS or PCS MMD Uncorrectable Safety Error Interrupt Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x80F0 : (APB3_16) ? (0x80F0*0x2) : (0x80F0*0x4)" + "(PCS_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD | |XGXS_PCS)&&DWC_XPCS_ASP

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PSEQ_ERR	EF_UE_INT	EC_UE_INT	DPP_INT	PHY_IFT_INT	APB3_IFT_INT	CKOFT_INT	EEEFT_INT	CSRFT_INT	RXFT_INT	TXFT_INT	CKOFP_INT	EEEFP_INT	CSRFP_INT	RXFP_INT	TXFP_INT

Table 5-122 Fields for Register: VR_XS_PCS_SFTY_UE_INTR0

Bits	Name	Memory Access	Description
15	PSEQ_ERR	R	Error in power-up sequence FSM This bit is read as 1 when the power-up sequence FSM is stuck in a state other than 'POWER_GOOD'. This field is present only for Synopsys PHY configurations. **Note:** When, power-up sequence FSM is any state other than 'POWER_GOOD', read to any register returns bit[15] as 1 and bits[4:2] indicate power-up sequence FSM state (PSEQ_STAE field of VR_XS_PCS_DIG_STS register). **Value After Reset:** 0x0 **Exists:** Always
14	EF_UE_INT	* Varies	Uncorrectable ECC Error Interrupt from 'FEC Error-Forwarding Memory' This bit gets set automatically when the safety feature in DWC_xpcs detects an uncorrectable ECC Error in 'FEC Error-Forwarding Memory'. Once this bit is set, it remains high till software clears it by writing 0. This field is present only for configurations with FEC_ERROR_FWD = Enabled. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "FEC_ERROR_FWD ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
13	EC_UE_INT	* Varies	Uncorrectable ECC Error Interrupt from 'FEC Error-Correction Memory' This bit gets set automatically when the safety feature in DWC_xpcs detects an uncorrectable ECC Error in 'FEC Error-Correction Memory'. Once this bit is set, it remains high till software clears it by writing 0. >This field is present only for configurations with FEC_EN = Enabled. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "FEC_EN ? \"read-write\" : \"read-only\""
12	DPP_INT	R/W	Datapath Parity Error Interrupt This bit gets set automatically when the safety feature in DWC_xpcs detects a parity error in the receive data-path. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always
11	PHY_IFT_INT	* Varies	PHY Interface Time-out Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects a time-out Error at the PHY interface. Once this bit is set, it remains high till software clears it by writing 0. This bit is valid only for configurations with Synopsys PHYs. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SNPS_XAUI_PHY ? \"read-write\" : \"read-only\""
10	APB3_IFT_INT	* Varies	APB3/MCI Interface Time-out Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects a time-out Error at the APB3/MCI interface. Once this bit is set, it remains high till software clears it by writing 0. This bit is not valid for MDIO configurations. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "MDIO_INTF ? \"read-only\" :\"read-write\""

Bits	Name	Memory Access	Description
9	CKOFT_INT	* Varies	CKO Alive FSM Time-out Error Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects a parity Error in FSMs operating in clk_cko_alive_i domain. Once this bit is set, it remains high till software clears it by writing 0. This bit is valid only for configurations with Synopsys multi-protocol 6G PHY. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SSP ? \"read-write\" : \"read-only\""
8	EEEFT_INT	* Varies	EEE FSM Time-out Error Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects a time-out Error in FSMs operating in clk_eee_i domain. Once this bit is set, it remains high till software clears it by writing 0. This bit is valid only for configurations with EEE_EN = Enabled. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "EEE_EN ? \"read-write\" : \"read-only\""
7	CSRFT_INT	R/W	CSR FSM Time-out Error Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects a parity Error in FSMs operating in CSR clock domain. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always
6	RXFT_INT	R/W	Rx FSM Time-out Error Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects a time-out Error in FSMs operating in Rx clock domain. Once this bit is set, it remains high till software clears it by writing 0 **Value After Reset:** 0x0 **Exists:** Always
5	TXFT_INT	R/W	Tx FSM Time-out Error Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects a Time-out Error in FSMs operating in Tx clock domain. Once this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4	CKOFP_INT	* Varies	CKO Alive FSM Parity Error Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects a parity Error in FSMs operating in clk_cko_alive_i domain. When this bit is set, it remains high till software clears it by writing 0. When this bit is cleared by software, it remains 0 even if the parity error continues to exist. This bit can get set again only if a new parity error is seen after the current parity error subsides. This bit is valid only for configurations with Synopsys multi-protocol 6G PHY. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SSP ? \"read-write\" : \"read-only\""
3	EEEFP_INT	* Varies	EEE FSM Parity Error Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects a parity Error in FSMs operating in clk_eee_i domain. When this bit is set, it remains high till software clears it by writing 0. When this bit is cleared by software, it remains 0 even if the parity error continues to exist. This bit can get set again only if a new parity error is seen after the current parity error subsides. This bit is valid only for configurations with EEE_EN = Enabled. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "EEE_EN ? \"read-write\" : \"read-only\""
2	CSRFP_INT	R/W	CSR FSM Parity Error Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects a parity Error in FSMs operating in CSR clock domain. When this bit is set, it remains high till software clears it by writing 0. When this bit is cleared by software, it remains 0 even if the parity error continues to exist. This bit can get set again only if a new parity error is seen after the current parity error subsides. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
1	RXFP_INT	R/W	Rx FSM Parity Error Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects a parity Error in FSMs operating in Rx clock domain. When this bit is set, it remains high till software clears it by writing 0. When this bit is cleared by software, it remain 0 even if the parity error continues to exist. This bit can get set again only if a new parity error is seen after the current parity error subsides. **Value After Reset:** 0x0 **Exists:** Always
0	TXFP_INT	R/W	Tx FSM Parity Error Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects a parity Error in FSMs operating in Tx clock domain. When this bit is set, it remains high till software clears it by writing 0. When this bit is cleared by software, it remains 0 even if the parity error continues to exist. This bit can get set again only if a new parity error is seen after the current parity error subsides. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.65 VR_XS_PCS_SFTY_ERR_INJ

- **Description:** VR XS or PCS MMD Safety Error Injection Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80F1 : (APB3_16) ? (0x80F1*0x2) : (0x80F1*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD||XGXS_PCS)&&DWC_XPCS_ASP

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
EF_ECCEI	EC_ECCEI	CSREI	DPPEI	PHYIFTEI	APB3IFTEI	CKOFTEI	EEEFTEI	CSRFTEI	RXFTEI	TXFTEI	CKOFPEI	EEEFPEI	CSRFPEI	RXFPEI	TXFPEI

Table 5-123 Fields for Register: VR_XS_PCS_SFTY_ERR_INJ

Bits	Name	Memory Access	Description
15	EF_ECCEI	* Varies	FEC Error Forward Memory ECC Error injection enable Whenever this bit is set to 1, DWC_xpcs injects error into the data that is read from FEC Error Forward (EF) Memory. Type of error that needs to be injected can be controlled by programming VR_XS_PCS_SFTY_DBG_CTRL register. Note that error is injected only for one clock cycle and this bit self-clears after the error interrupt is generated. This field is present only for configurations with FEC_ERROR_FWD = Enabled. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "FEC_ERROR_FWD ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
14	EC_ECCEI	* Varies	FEC Error Correction Memory ECC Error injection enable Whenever this bit is set to 1, DWC_xpcs injects error into the data that is read from FEC Error Correction (EC) Memory. Type of error that needs to be injected can be controlled by programming VR_XS_PCS_SFTY_DBG_CTRL register. Note that error is injected only for one clock cycle and this bit self-clears after the error interrupt is generated. This field is present only for configurations with FEC_EN = Enabled. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "FEC_EN ? \"read-write\" : \"read-only\""
13	CSREI	R/W	CSR Error Injection When this bit is set to 1, DWC_xpcs injects error into the CSR protection feature. Consequent CSR_INT interrupt gets set. **Note:** This bit self-clears after the error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always
12	DPPEI	R/W	Datapath Parity Error Inject When this bit is set to 1, DWC_xpcs injects parity error in its data-path. **Note:** This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always
11	PHYIFTEI	* Varies	PHY Interface Time-out Error Injection Enable ■ 1: Inject time-out error at PHY (CR port) interface. This bit is valid only for configurations with Synopsys PHYs. **Note:** This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SNPS_XAUI_PHY ? \"read-write\" : \"read-only\""
10	APB3IFTEI	* Varies	APB3/MCI Interface Time-out Error Injection Enable This bit can set to 1 to inject time-out error at APB3/MCI interface. This bit is not valid for MDIO configurations. **Note:** This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "MDIO_INTF ? \"read-only\" :\"read-write\""

Bits	Name	Memory Access	Description
9	CKOFTEI	* Varies	CKO Alive FSM Time-out Error Injection Enable ■ 1: Inject time-out error in all the FSMs operating with clk_cko_alive_i. This bit is valid only for configurations with Synopsys multi-protocol 6G PHY. **Note:** This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SSP ? \"read-write\" : \"read-only\""
8	EEEFTEI	* Varies	EEE FSM Time-out Error Injection Enable This bit can be set to 1 to inject time-out error in all the FSMs operating with clk_eee_i. This bit is valid only for configurations with EEE_EN = Enabled. **Note:** This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "EEE_EN ? \"read-write\" : \"read-only\""
7	CSRFTEI	R/W	CSR FSM Time-out Error Injection Enable This bit can be set to 1 to inject time-out error in all the FSMs operating with clk_csr_i. **Note:** This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always
6	RXFTEI	R/W	Rx FSM Time-out Error Injection Enable This bit can be set to 1 to inject time-out error in all the FSMs operating in receive clock domain(s). **Note:** This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always
5	TXFTEI	R/W	Tx FSM Time-out Error Injection Enable This bit can be set to 1 to inject time-out error in all the FSMs operating in transmit clock domain(s). **Note:** This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4	CKOFPEI	* Varies	CKO Alive FSM Parity Error Injection Enable This bit can be set to 1 to inject parity error in all the FSMs operating with clk_cko_alive_i. This bit is valid only for configurations with Synopsys multi-protocol 6G PHY. **Note:** This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SSP ? \"read-write\" : \"read-only\""
3	EEEFPEI	* Varies	EEE FSM Parity Error Injection Enable This bit can be set to 1 to inject parity error in all the FSMs operating with clk_eee_i. This bit is valid only for configurations with EEE_EN = Enabled. **Note:** This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "EEE_EN ? \"read-write\" : \"read-only\""
2	CSRFPEI	R/W	CSR FSM Parity Error Injection Enable This bit can be set to 1 to inject parity error in all the FSMs operating with clk_csr_i. **Note:** This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always
1	RXFPEI	R/W	Rx FSM Parity Error Injection Enable This bit can be set to 1 to inject parity error in all the FSMs operating in receive clock domain(s). **Note:**This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always
0	TXFPEI	R/W	Tx FSM Parity Error Injection Enable This bit can be set to 1 to inject parity error in all the FSMs operating in transmit clock domain(s). **Note:**This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.66 VR_XS_PCS_SFTY_CE_INTR

- **Description:** VR XS or PCS MMD Correctable Safety Error Interrupt Register
  This register is present only in PCS MMD.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x80F2 : (APB3_16) ? (0x80F2*0x2) : (0x80F2*0x4)" + "(PCS_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** PCS_MMD&&DWC_XPCS_ASP

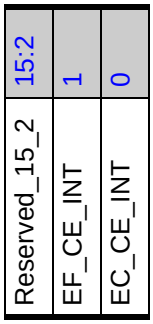


Table 5-124 Fields for Register: VR_XS_PCS_SFTY_CE_INTR

Bits	Name	Memory Access	Description
15:2	Reserved_15_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1	EF_CE_INT	* Varies	Correctable ECC Error Interrupt from Error-Forwarding Memory. This bit gets set automatically when the safety feature in DWC_xpcs detects a correctable ECC error in FEC Error Forwarding Memory. Once this bit is set, it remains high till software clears it by writing 0. This field is present only for configurations with FEC_ERROR_FWD = Enabled. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "FEC_ERROR_FWD ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
0	EC_CE_INT	* Varies	Correctable ECC Error Interrupt from Error-Correction Memory. This bit gets set automatically when the safety feature in DWC_xpcs detects a correctable ECC error in FEC Error Correction Memory. Once this bit is set, it remains high till software clears it by writing 0. This field is present only for configurations with FEC_EN = Enabled. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "FEC_EN ? \"read-write\" : \"read-only\""

5.2.2.67 VR_XS_PCS_SFTY_DBG_CTRL

- **Description:** VR XS or PCS MMD Safety Debug Control Register
  This register provides the flexibility to control the safety error injection mode. Program the fields of this register to desired values prior to enabling the error injection enable bits in VR_XS_PCS_SFTY_ERR_INJ register. If the fields of this register are changed after enabling the error injection enable bits, it can lead to ambiguous results.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80F3 : (APB3_16) ? (0x80F3*0x2) : (0x80F3*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD||XGXS_PCS)&&DWC_XPCS_ASP

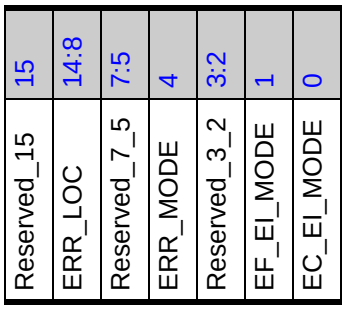


Table 5-125 Fields for Register: VR_XS_PCS_SFTY_DBG_CTRL

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
14:8	ERR_LOC	R/W	Error Location This field controls the bit-location at which Data Path Parity (DPP) error or ECC error is introduced. This is field is applicable for both DPP error injection and ECC error injection features. Software must program judicious values to this field depending on the width of the data and parity bus applicable to the feature selected and also depending on the current operating mode. If an invalid value is programmed to this field, then error injection behavior is undefined. The legal values for this field are as follows: ■ For ECC error injection: ❑ 7'd0 to 7'd65: Introduce error in FEC data bits (when ERR_MODE=0) ❑ 7'd66 to 7'd70: Introduce error in FEC address bits (when ERR_MODE=0) ❑ 7'd0 to 7'd7: Introduce error in FEC parity bits (when ERR_MODE=1) ■ For DPP error injection, when operating in 1000BASE-X/2.5GBASE-X/(Q)SGMII mode: ❑ 7'd0 to 7'd7: Introduce error in data bits (when ERR_-MODE=0) ❑ 7'd0: Introduce error in parity bit (when ERR_-MODE=1) ■ For DPP error injection, when operting in 10G/5G BASE-R Mode: ❑ 7'd0 to 7'd31 (for SDR) or 7'd63 (for DDW): Introduce error in data bits (when ERR_MODE=0) ❑ 7'd0 to 7'd3 (for SDR) or 7'd7 (for DDW): Introduce error in parity bits (when ERR_MODE=1) ■ For DPP error injection, when operating in 10GBASE-X Mode: ❑ 7'd0 to 7'd63 (for DDW): Introduce error in data bits (when ERR_MODE=0) ❑ 7'd0 to 7'd7 (for DDW): Introduce error in parity bits (when ERR_MODE=1) **Value After Reset:** 0x0 **Exists:** Always
7:5	Reserved_7_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4	ERR_MODE	R/W	Error Mode This bit determines whether DDP error or ECC error must be introduced in data bits or parity bits.This field is applicable for both DPP error injection and ECC error injection features. Encoding of this field is as follows: ■ 0: Introduce error in data bits ■ 1: Introduce error in parity bits **Value After Reset:** 0x0 **Exists:** Always
3:2	Reserved_3_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1	EF_EI_MODE	* Varies	Error injection Mode for FEC Error Forwarding Memory. This bit introduces whether single-bit or two-bit error must be introduced to the ECC logic catering to FEC Error Forwarding Memory. Encoding of this field is as follows: - 0: 1-bit ECC data/parity bit corruption - 1: 2-bit ECC data/parity bit corruption For 1-bit error, the bit-location of error injection is controlled by ERR_LOC field of this register. For 2-bit error mode, bit[0] is always corrupted. The 2nd location of error injection is controlled by ERR_LOC field. Therefore, software must program a non-zero value to ERR_LOC field. This field is present only for configurations with FEC_ERROR_FWD = Enabled. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "FEC_ERROR_FWD ? \"read-write\" : \"read-only\""
0	EC_EI_MODE	* Varies	Error injection Mode for FEC Error Correcting Memory. This bit introduces whether single-bit or two-bit error must be introduced to the ECC logic catering to FEC Error Correcting Memory. Encoding of this field is as follows: - 0: 1-bit ECC data/parity bit corruption - 1: 2-bit ECC data/parity bit corruption For 1-bit error, the bit-location of error injection is controlled by ERR_LOC field of this register. For 2-bit error mode, bit[0] is always corrupted. The 2nd location of error injection is controlled by ERR_LOC field. Therefore, software must program a non-zero value to ERR_LOC field. This field is present only for configurations with FEC_EN = Enabled. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "FEC_EN ? \"read-write\" : \"read-only\""

5.2.2.68 VR_XS_PCS_SFTY_DISABLE

- **Description:** VR XS or PCS MMD Safety Disable Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80F4 : (APB3_16) ? (0x80F4*0x2) : (0x80F4*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD||XGXS_PCS)&&DWC_XPCS_ASP

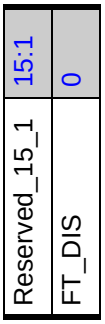


Table 5-126 Fields for Register: VR_XS_PCS_SFTY_DISABLE

Bits	Name	Memory Access	Description
15:1	Reserved_15_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	FT_DIS	R/W	FSM Time-out Disable. When this bit is set to 1, the FSM time-out safety mechanism inside DWC_xpcs gets disabled. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.69 VR_XS_PCS_SFTY_TMR_CTRL

- **Description:** VR XS or PCS MMD Safety Timer Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80F5 : (APB3_16) ? (0x80F5*0x2) : (0x80F5*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD||XGXS_PCS)&&DWC_XPCS_ASP

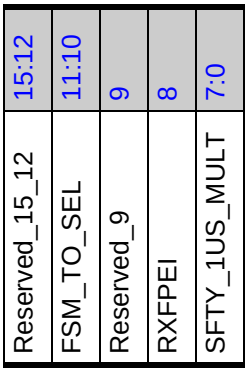


Table 5-127 Fields for Register: VR_XS_PCS_SFTY_TMR_CTRL

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:10	FSM_TO_SEL	R/W	FSM Time-out Select DWC_xpcs allocates appropriate time-out values to various FSMs inside it. These pre-allocated time-out values range from 10 micro-seconds to 1 second. This field provides an option to scale the time-out values if required. ■ 2'b00: Timers get scaled by 0.1 ■ 2'b01: Timers run for standard duration ■ 2'b10: Timers get scaled by 10 ■ 2'b11: Reserved This field has no effect if FASTSIM bit is set to 1. **Value After Reset:** 0x1 **Exists:** Always
9	Reserved_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
8	RXFPEI	R/W	Interface Time-out Select. This field can set to appropriate value to configure the time-out value of Interface time-outs. ■ 0: Interface time-out is 10 us ■ 1: Interface time-out is 50 us **Value After Reset:** 0x0 **Exists:** Always
7:0	SFTY_1US_MULT	R/W	Safety 1us Clock Tic Multiplying Factor This bit is the multiplying factor to the clk_csr_i clock time period to make it equal to 1us. Program this field such that the (clk_csr_i time_period * 2 * SFTY_1US_MULT) is 1us or as close as possible to 1 us. For example, if the clk_csr_i clock time period is 10 ns, the value of this field needs to be set to 50. The default value of this register is set as 50 corresponding to a 100 MHz clk_csr_i clock. **Value After Reset:** 0x32 **Exists:** Always

5.2.2.70 VR_XS_PCS_EC_MEM_ECC_STS

- **Description:** VR XS or PCS MMD EC Memory ECC Status Register.
  This register is present only in PCS MMD.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x80F6 : (APB3_16) ? (0x80F6*0x2) : (0x80F6*0x4)" + "(PCS_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** PCS_MMD&&FEC_EN&&DWC_XPCS_ASP

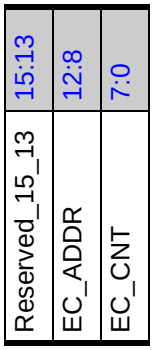


Table 5-128 Fields for Register: VR_XS_PCS_EC_MEM_ECC_STS

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:8	EC_ADDR	R	This field denotes the address of the 'FEC Error Correction Memory' data that has been corrected by ECC logic of DWC_xpcs. In case ECC has corrected data from multiple memory locations, this field indicates the address of the most recent location that has been corrected. This field is of 'Read-only, Clear-on-Read' type. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
7:0	EC_CNT	R	ECC Correctable Error Count from 'FEC Error Correction Memory' (RO,COR Type). This register field indicates the number of ECC code-words that have been corrected by the DWC_xpcs safety feature. The value of this register saturates on reaching 8'hFF. This field is of 'Read-only, Clear-on-Read' type. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.71 VR_XS_PCS_EF_MEM_ECC_STS

- **Description:** VR XS or PCS MMD EF Memory ECC Status Register.
  This register is present only in PCS MMD.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x80F7 : (APB3_16) ? (0x80F7*0x2) : (0x80F7*0x4)" + "(PCS_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** PCS_MMD&&FEC_ERROR_FWD&&DWC_XPCS_ASP

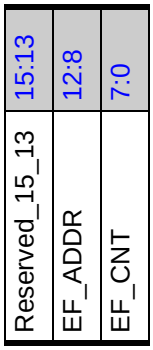


Table 5-129 Fields for Register: VR_XS_PCS_EF_MEM_ECC_STS

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:8	EF_ADDR	R	This field denotes the address of the 'FEC Error Forwarding Memory' data that has been corrected by ECC logic of DWC_xpcs. In case ECC has corrected data from multiple memory locations, this field indicates the address of the most recent location that has been corrected. This field is of 'Read-only, Clear-on-Read' type. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
7:0	EF_CNT	R	ECC Correctable Error Count from 'FEC Error Forwarding Memory' (RO,COR Type). This register field indicates the number of ECC code-words that have been corrected by the DWC_xpcs safety feature. The value of this register saturates on reaching 8'hFF. This field is of 'Read-only, Clear-on-Read' type. **Value After Reset:** 0x0 **Exists:** Always

5.2.2.72 VR_XS_PCS_SFTY_UE_INTR1

- **Description:** VR XS or PCS MMD Uncorrectable Safety Error Interrupt Register 1
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80F8 : (APB3_16) ? (0x80F8*0x2) : (0x80F8*0x4)" + "(PCS_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x030000 : (APB3_16) ? (0x030000*0x2) : (0x030000*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x050000 : (APB3_16) ? (0x050000*0x2) : (0x050000*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x040000 : (APB3_16) ? (0x040000*0x2) : (0x040000*0x4)))"
- **Exists:** (PCS_MMD||XGXS_PCS)&&DWC_XPCS_ASP

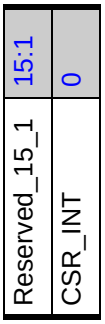


Table 5-130 Fields for Register: VR_XS_PCS_SFTY_UE_INTR1

Bits	Name	Memory Access	Description
15:1	Reserved_15_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	CSR_INT	R/W	Control & Status register Error Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects error in Control or Status register. When this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always

5.2.3 DWC_xpcs_map/XS_PMA_MMD Registers

5.2.3.1 VR_XS_PMA_RX_LSTS

- **Description:** VR XS or PMA PHY Rx Lane Status Register
  In KX_Only, KR_Only, KR_KX, 10GBASE-R PCS, and 1000BASEX-Only PCS configurations, only Lane 0 control and status information is present. In configurations with both 1G (KX) and 10G (XGXS PCS, 10GBASE-X PCS, or KX4) modes, only Lane 0 control and status information is used when the DWC_xpcs is operating in the 1G mode.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)

15:13	12	11:9	8	7:5	4	3:0
RX_VALID_3_1	RX_VALID_0	RX_PLL_STATE_3_1	RX_PLL_STATE_0	SIG_DET_3_1	SIG_DET_0	Reserved_3_0

Table 5-131 Fields for Register: VR_XS_PMA_RX_LSTS

Bits	Name	Memory Access	Description
15:13	RX_VALID_3_1	R	DPLL Lock Status for Lanes[3:1] This field indicates that the DPLL in the PHY is locked on the serial data in the corresponding lane. ■ 3'b**1: Lane 1 DPLL bit locked ■ 3'b*1*: Lane 2 DPLL bit locked ■ 3'b1**: Lane 3 DPLL bit locked **Dependency:** This field is present only the following configurations: ■ Synopsys Multi-protocol 12G/16G/25G PHY configurations ■ Backplane Ethernet PCS configurations with non-Synopsys PHY **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
12	RX_VALID_0	R	DPLL Lock Status for Lane 0 This field indicates that the DPLL in the PHY is locked on the serial data in Lane 0. The value 1'b1 indicates that Lane 0 DPLL bit is locked. **Dependency:** This field is present only the following configurations: ■ Synopsys Multi-protocol 12G/16G/25G PHY configurations ■ Backplane Ethernet PCS configurations with non-Synopsys PHY **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
11:9	RX_PLL_STATE_3_1	R	**For Multi-lane Synopsys Multi-protocol 6G PHY configurations:** DPLL State for Lanes[3:1] This field indicates that the Rx has successfully reached the PLL Power_Good during the power up sequence on the corresponding lane. ■ 3'b**1: Lane 1 Rx successful ■ 3'b*1*: Lane 2 Rx successful ■ 3'b1**: Lane 3 Rx successful **For other configurations:**Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

Bits	Name	Memory Access	Description
8	RX_PLL_STATE_0	R	**For Synopsys Multi-protocol 6G PHY configurations:**Rx DPLL State for Lane 0 This field indicates that the Rx has successfully reached the PLL Power_Good state during the power up sequence on Lane 0. The value 1'b1 indicates that Lane 0 Rx is successful. **For other configurations:**Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
7:5	SIG_DET_3_1	R	**For Multi-lane configurations:**Rx Signal Detect for Lanes[3:1] This field indicates that the Rx has detected the signal on the corresponding lanes. The bits in this field are the complement value of the signals input from PHY (xpcs_los_i{lane}) on the corresponding lane. ■ 3'b**1: Lane 1 signal detected ■ 3'b*1*: Lane 2 signal detected ■ 3'b1**: Lane 3 signal detected **For Single-lane configurations:**Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
4	SIG_DET_0	R	Rx Signal Detect for Lane 0 This bit indicates that the Rx detected the signal on Lane 0. This bit is the complement value of the signals input from PHY (xgxs_los_i[0]) on Lane 0. The value 1'b1 indicates that Lane 0 signal is detected. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
3:0	Reserved_3_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.3.2 VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL0

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Tx General 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0010 : (APB3_16) ? (0x0010*0x2) : (0x0010*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25||E32))

15:13	12	11:9	8	7:5	4	3:1	0
TX_DT_EN_3_1	TX_DT_EN_0	TX_RST_3_1	TX_RST_0	TX_INV_3_1	TX_INV_0	TXBCN_EN_3_1	TXBCN_EN_0

Table 5-132 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL0

Bits	Name	Memory Access	Description
15:13	TX_DT_EN_3_1	* Varies	**For Multi-lane configurations:** Transmit Data Enable on Lanes[3:1] This field controls the output port 'xgxs_tx_data_en_o[3:1]'. Clear this field when PHY Tx is not in P0 power state. **For Single-lane configurations:** Reserved **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : 0x7" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
12	TX_DT_EN_0	R/W	Tx Data Enable on lane 0 of PHY When this bit is set, Transmit Output Driver in the PHY is enabled. This bit controls the output port 'xgxs_tx_data_en_o[0]'. Clear this field when PHY Tx is not in P0 power state. **Value After Reset:** 0x1 **Exists:** Always

Bits	Name	Memory Access	Description
11:9	TX_RST_3_1	* Varies	**For Multi-lane configurations:** Transmit Reset on Lanes[3:1] This signal drives the output port 'xgxs_tx_reset_o[3:1]' when DWC_xpcs is in POWER_GOOD state. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
8	TX_RST_0	R/W	Tx Reset on lane 0 of PHY When this bit is set, PHY transmitter is reset, including common-mode adjustment and receiver detection state machines. This signal drives the output port 'xgxs_tx_reset_o[0]' when DWC_xpcs is in POWER_GOOD state. **Value After Reset:** 0x0 **Exists:** Always
7:5	TX_INV_3_1	* Varies	**For Multi-lane configurations:** Transmit Data Invert on Lanes[3:1] This field drives the output port 'xgxs_tx_invert_o[3:1]'. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
4	TX_INV_0	R/W	Tx Invert on lane 0 of PHY When this bit is set, the data on PHY Tx serial lines are logically inverted. This signal drives the output port 'xgxs_tx_invert_o[0]'. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
3:1	TXBCN_EN_3_1	* Varies	**For Multi-lane configurations:** Transmit Beacon Enable Control on Lanes[3:1] This field drives the output port 'xgxs_tx_beacon_en_o[3:1]'. When set to 1, this field enables the Tx beaconing. ■ TXBCN_EN_3_1[1]: Beacon enable control on Lane 1 ■ TXBCN_EN_3_1[2]: Beacon enable control on Lane 2 ■ TXBCN_EN_3_1[3]: Beacon enable control on Lane 3 **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
0	TXBCN_EN_0	R/W	Tx Beaconing Enable on lane 0 of PHY When this bit is set, PHY enables transmitter beaconing (LFPS). The period for transmit pulses is between 20-50ns. This field drives the output port 'xgxs_tx_beacon_en_o[0]'. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.3 VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Tx General 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0011 : (APB3_16) ? (0x0011*0x2) : (0x0011*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25||E32))

15:13	12	11	10:8	7:5	4	3:1	0
TX_CLK_RDY_3_1	TX_CLK_RDY_0	Reserved_11	VBOOST_LVL	VBOOST_EN_3_1	VBOOST_EN_0	DET_RX_REQ_3_1	DET_RX_REQ_0

Table 5-133 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1

Bits	Name	Memory Access	Description
15:13	TX_CLK_RDY_3_1	* Varies	**For Multi-lane configurations:** Transmitter Input clock ready on Lanes[3:1] This field drives the output xpcs_tx_clk_rdy_o[3:1]. This field must remain high, as long as PHY Tx Clock is active. **For Single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0 : 0x7" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
12	TX_CLK_RDY_0	R/W	Transmitter Input clock ready on lane 0 This field drives the output xpcs_tx_clk_rdy_o[0]. This field must remain high, as long as PHY Tx Clock is active **Value After Reset:** 0x1 **Exists:** Always
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
10:8	VBOOST_LVL	R/W	Tx Voltage Boost Maximum Level This field controls the maximum achievable Tx swing in the PHY This field drives the output port 'xpcs_tx_vboost_lvl_o[2:0]'. **Value After Reset:** "(SNPS_RSVDPARAM_39) ? 0x4 : (SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_33) ? 0x3 : (E25&&SNPS_RSVDPARAM_16&&!SNPS_RSVDPARAM_36) ? 0x7 : 0x5" **Exists:** !(SNPS_RSVDPARAM_37||E32)
7:5	VBOOST_EN_3_1	* Varies	**For Multi-lane configurations:** Transmitter Voltage Boost Enable on Lanes[3:1] This field drives the output port xpcs_tx_vboost_en_o[3:1] **For Single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG||SNPS_RSVDPARAM_22) ? 0x0 : 0x7" **Exists:** !(SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_37||E32) **Memory Access:** "(!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_22) ? \"read-write\" : \"read-only\"""
4	VBOOST_EN_0	* Varies	Tx voltage Boost Enable on lane 0 of PHY When this bit is set, the current mode Tx Swing boost in the PHY is enabled. This bit drives the output port 'xpcs_tx_vboost_en_o[0]'. **Value After Reset:** "(KX_ONLY)&&(E25||SNPS_RSVDPARAM_39) ? 0x0 : 0x1" **Exists:** !(SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_37||E32) **Memory Access:** "SNPS_RSVDPARAM_22 ? \"read-only\" : \"read-write\""

Bits	Name	Memory Access	Description
3:1	DET_RX_REQ_3_1	* Varies	**For Multi-lane configurations:** Transmitter Rx-Detection Request on Lanes[3:1] This field drives the output port 'xpcs_tx_detrx_req_o[3:1]'. When this field is set, a receiver detection request is made towards the PHY on lanes 3 to 1. The result of the Rx Detection operation is available at field 'DETRX_RSLT_3_1' whenever 'TX_ACK_3_1' is read as high. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
0	DET_RX_REQ_0	R/W	Transmitter Rx-Detection request on lane 0 of PHY. This field drives the output port 'xpcs_tx_detrx_req_o[0]'. When this bit is set, a receiver detection request is made towards the PHY on lane 0. When this bit is set, it must remain high till 'TX_ACK_0' bit is read as high. When 'TX_ACK_0' is read as high, the result of the Rx Detection operation is available at field 'DETRX_RSLT_0'. After obtaining the result of 'Rx-Detect operation', clear this bit. Rx detection operation is valid only when transmitter is in P1 power state. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.4 VR_XS_PMA_MP_12G_16G_TX_GENCTRL2

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G PHY Tx General 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0012 : (APB3_16) ? (0x0012*0x2) : (0x0012*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16))

15:14	13:12	11:10	9:8	7:5	4	3:1	0
TX3_WIDTH	TX2_WIDTH	TX1_WIDTH	TX0_WIDTH	TX_LPD_3_1	TX_LPD_0	TX_REQ_3_1	TX_REQ_0

Table 5-134 Fields for Register: VR_XS_PMA_MP_12G_16G_TX_GENCTRL2

Bits	Name	Memory Access	Description
15:14	TX3_WIDTH	* Varies	**For Multi-lane configurations:** Tx Datapath Width on lane 3 of the PHY This field drives the output port xpcs_tx3_width_o[1:0] . **For Single-lane configurations:** Reserved **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : (RXAUI_DEFAULT||(RPCS_PRESENT&&(R_SRDS_INTF==4))) ? 0x3 : (RPCS_PRESENT&&(R_SRDS_INTF==0)) ? 0x2 : 0x1" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
13:12	TX2_WIDTH	* Varies	**For Multi-lane configurations:** Tx Datapath Width on lane 2 of the PHY This field drives the output port xpcs_tx2_width_o[1:0]. **For Single-lane configurations:** Reserved **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : (RXAUI_DEFAULT||(RPCS_PRESENT&&(R_SRDS_INTF==4))) ? 0x3 : (RPCS_PRESENT&&(R_SRDS_INTF==0)) ? 0x2 : 0x1" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
11:10	TX1_WIDTH	* Varies	**For Multi-lane configurations:** Tx Datapath Width on lane 1 of the PHY This field drives the output port xpcs_tx1_width_o[1:0]. **For Single-lane configurations:** Reserved **Value After Reset:** "SINGLE_LANE_CFG ? \"0x0\" : (RXAUI_DEFAULT||(RPCS_PRESENT&&(R_SRDS_INTF==4))) ? \"0x3\" : (RPCS_PRESENT&&(R_SRDS_INTF==0)) ? \"0x2\" : \"0x1\"" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
9:8	TX0_WIDTH	R/W	Tx Datapath Width on lane 0 of the PHY This field controls the width of input transmit data on lane 0. The encoding of the width is as follows: ■ 2'b00: 8-bit ■ 2'b01: 10-bit ■ 2'b10: 16-bit ■ 2'b11: 20-bit This field drives the output port xpcs_tx0_width_o[1:0]. **Value After Reset:** "(RPCS_PRESENT&&(R_SRDS_INTF==0)) ? 0x2 : (RPCS_PRESENT||RXAUI_DEFAULT) ? 0x3 : 0x1" **Exists:** Always

Bits	Name	Memory Access	Description
7:5	TX_LPD_3_1	* Varies	**For Multi-lane configurations:** Transmitter Lane Power Down on Lanes[3:1] This field drives the output 'xpcs_tx_lpd_o[3:1]'. This field can be asserted to put the PHY transmitter to a power state equivalent to that of P1. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
4	TX_LPD_0	R/W	Transmitter Lane Power Down on lane 0 of PHY This field drives the output 'xpcs_tx_lpd_o[3:1]'. This field can be asserted to put the PHY transmitter to a power state equivalent to that of P1. **Value After Reset:** 0x0 **Exists:** Always
3:1	TX_REQ_3_1	* Varies	**For Multi-lane configurations:** Transmitter Request on Lanes[3:1] (RW,SC Type) This field drives the output port 'xpcs_tx_req_o[3:1]'. These bits can be set by application to initiate a Tx-Related request to the PHY and they are self-cleared when acknowledgment is received from the PHY. Functionality of this field is similar to that of 'TX_REQ_0' field. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
0	TX_REQ_0	R/W	Transmitter operation request on lane 0 of E12G/E16G PHY (RW,SC Type) This bit drives the output port 'xpcs_tx_req_o[0]. This bit can be set to initiate a new transmitter setting request. Set this bit only if TX_ACK_0 field is low. This bit gets self-cleared after the transmitter request operation has been successfully completed. This bit should be asserted to make sure that the PHY accepts any change to the following signals: ■ xpcs_tx0_pstate_o ■ xpcs_tx0_lpd_o ■ xpcs_tx0_rate_o ■ xpcs_tx0_width_o ■ xpcs_tx0_mpllb_sel. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.5 VR_XS_PMA_MP_25G_TX_GENCTRL2

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY Tx General 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0012 : (APB3_16) ? (0x0012*0x2) : (0x0012*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&(E25||E32)

15:12	11:9	8	7:5	4	3:1	0
Reserved_15_12	TX_XFER_3_1	TX_XFER_0	TX_LPD_3_1	TX_LPD_0	TX_REQ_3_1	TX_REQ_0

Table 5-135 Fields for Register: VR_XS_PMA_MP_25G_TX_GENCTRL2

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** !SNPS_RSVDPARAM_37
11:9	TX_XFER_3_1	* Varies	**For Multi-lane configurations:** Transmitter Alignment Wide Transfer Enable on Lanes[3:1] This field drives the output xpcs_tx_align_wide_xfer_en_o[3:1]'. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** (!(SNPS_RSVDPARAM_37||E32)) **Memory Access:** "(!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37&&!E32) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
8	TX_XFER_0	* Varies	Transmitter Alignment Wide Transfer Enable on lane 0 When this field is set, the internal data transfer (in PHY) from txX_clk domain to the internal clock domain is performed using a width of 32 bits. This field drives the xpcs_tx_align_wide_xfer_en_o[0] port. **Value After Reset:** 0x0 **Exists:** (!(SNPS_RSVDPARAM_37||E32)) **Memory Access:** "(!(SNPS_RSVDPARAM_37||E32)) ? \"read-write\" : \"read-only\""
7:5	TX_LPD_3_1	* Varies	**For Multi-lane configurations:** Transmitter Lane Power Down on Lanes[3:1] This field drives the output 'xpcs_tx_lpd_o[3:1]'. This field can be asserted to put the PHY transmitter to a power state equivalent to that of P1. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
4	TX_LPD_0	R/W	Transmitter Lane Power Down on lane 0 of PHY This field drives the output 'xpcs_tx_lpd_o[3:1]'. This field can be asserted to put the PHY transmitter to a power state equivalent to that of P1. **Value After Reset:** 0x0 **Exists:** Always
3:1	TX_REQ_3_1	* Varies	**For Multi-lane configurations:** Transmitter Request on Lanes[3:1] (RW,SC Type) This field drives the output port 'xpcs_tx_req_o[3:1]'. These bits can be set by application to initiate a Tx-Related request to the PHY and they are self-cleared when acknowledgment is received from the PHY. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
0	TX_REQ_0	R/W	Transmitter operation request on lane 0 of PHY (RW,SC Type) This bit drives the output port 'xpcs_tx_req_o[0]. This bit can be set to initiate a new transmitter setting request. Set this bit only if TX_ACK_0 field is low. This bit gets self-cleared after the transmitter request operation has been successfully completed. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.6 VR_XS_PMA_MP_12G_16G_25G_TX_BOOST_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Tx Boost Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0013 : (APB3_16) ? (0x0013*0x2) : (0x0013*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25)&&!(SNPS_RSVD-PARAM_22||SNPS_RSVDPARAM_37))

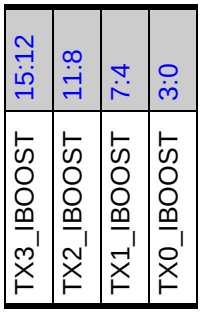


Table 5-136 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_TX_BOOST_CTRL

Bits	Name	Memory Access	Description
15:12	TX3_IBOOST	* Varies	**For Multi-lane configurations:** Tx current boost on lane 3 of the PHY This bit drives the output port 'xpcs_tx3_iboost_lvl_o[3:0]'. For details on how to set this field, see the PHY Databook. **For Single-lane configurations:** Reserved **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : 0xf" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
11:8	TX2_IBOOST	* Varies	**For Multi-lane configurations:** Tx current boost on lane 2 of the PHY This bit drives the output port 'xpcs_tx2_iboost_lvl_o[3:0]'. For details on how to set this field, see the PHY Databook. **For Single-lane configurations:** Reserved **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : 0xf" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
7:4	TX1_IBOOST	* Varies	**For Multi-lane configurations:** Tx current boost on lane 1 of the PHY This bit drives the output port 'xpcs_tx1_iboost_lvl_o[3:0]'. For details on how to set this field, see the PHY Databook. **For Single-lane configurations:** Reserved **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : 0xf" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
3:0	TX0_IBOOST	R/W	Tx current boost level on lane 0 of the PHY This bit drives the output port xpcs_tx0_iboost_lvl_o[3:0]. For details on how to set this field, see the PHY Databook. **Value After Reset:** "SNPS_RSVDPARAM_39 ? 0xC : 0xF" **Exists:** Always

5.2.3.7 VR_XS_PMA_MP_32G_TX_GEN_CTRL3

- **Description:** VR XS or PMA Synopsys Multi-protocol 32G PHY Tx General Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0013 : (APB3_16) ? (0x0013*0x2) : (0x0013*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E32))

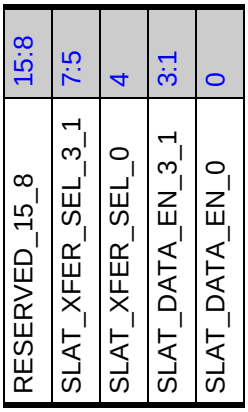


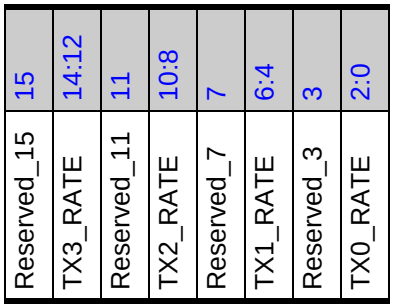
Table 5-137 Fields for Register: VR_XS_PMA_MP_32G_TX_GEN_CTRL3

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	SLAT_XFER_SEL_3_1	* Varies	**For Multi-lane configurations:** Transmitter Short Latency Transfer Select - for Lane 3:1. This field drives the output port xpcs_tx_slat_xfer_sel_o[3:1]. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
4	SLAT_XFER_SEL_0	R/W	Transmitter Short Latency Transfer Select - for Lane0. This field drives the output port xpcs_tx_slat_xfer_sel_o[0]. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
3:1	SLAT_DATA_EN_3_1	* Varies	**For Multi-lane configurations:** Transmitter Short Latency Data Enable - for Lane3:1. This field drives the output port xpcs_tx_slat_data_en_o[3:1]. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
0	SLAT_DATA_EN_0	R/W	Transmitter Short Latency Data Enable - for Lane0. This field drives the output port xpcs_tx_slat_data_en_o[0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.8 VR_XS_PMA_MP_12G_16G_25G_TX_RATE_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Tx Rate Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0014 : (APB3_16) ? (0x0014*0x2) : (0x0014*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25)&&!SNPS_RSVDPARAM_37



15	14:12	11	10:8	7	6:4	3	2:0
Reserved_15	TX3_RATE	Reserved_11	TX2_RATE	Reserved_7	TX1_RATE	Reserved_3	TX0_RATE

Table 5-138 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_TX_RATE_CTRL

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:12	TX3_RATE	* Varies	**For Multi-lane configurations:** Tx date rate on lane 3 of the PHY **For Single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0: (RXAUI_DEFAULT) ? (E25 ? 0x0 : 0x1) :(RPCS_PRESENT) ? (E25 ? 0x1 : 0x0) : ((E25||SNPS_RSVDPARAM_39) ? 0x1 : 0x2) " **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
10:8	TX2_RATE	* Varies	**For Multi-lane configurations:** Tx date rate on lane 2 of the PHY **For Single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0: (RXAUI_DEFAULT) ? (E25 ? 0x0 : 0x1) :(RPCS_PRESENT) ? (E25 ? 0x1 : 0x0) : ((E25||SNPS_RSVDPARAM_39) ? 0x1 : 0x2) " **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\"""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:4	TX1_RATE	* Varies	**For Multi-lane configurations:** Tx date rate on lane 1 of the PHY **For Single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0: (RXAUI_DEFAULT) ? (E25 ? 0x0 : 0x1) :(RPCS_PRESENT) ? (E25 ? 0x1 : 0x0) : ((E25||SNPS_RSVDPARAM_39) ? 0x1 : 0x2) " **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\"""
3	Reserved_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
2:0	TX0_RATE	R/W	Tx date rate on lane 0 of the PHY Data rate encoding is as follows: ■ 3'b000: baud ■ 3'b001: baud/2 ■ 3'b010: baud/4 ■ 3'b011: baud/8 ■ 3'b111: baud/10 ■ 3'b100-3'b110: Not supported **Value After Reset:** "(RPCS_PRESENT) ? ((E25||SNPS_RSVDPARAM_39) ? 0x1 : 0x0) : (RXAUI_DEFAULT) ? (E25 ? 0x0 : 0x1) : (BASE1000X_ONLY||KX_ONLY) ? ((E25||SNPS_RSVDPARAM_39) ? 0x2 : SNPS_RSVDPARAM_22 ? 0x0: SNPS_RSVDPARAM_35 ? 0x2 : 0x3) : ((E25||SNPS_RSVDPARAM_39) ? 0x1 : 0x2)" **Exists:** Always

5.2.3.9 VR_XS_PMA_MP_32G_TX_GEN_CTRL4

- **Description:** VR XS or PMA Synopsys Multi-protocol 32G PHY Tx General Control 4 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0014 : (APB3_16) ? (0x0014*0x2) : (0x0014*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E32))

15:14	13:12	11:10	9:8	7:5	4	3:1	0
TX_RCL_BANK_SEL_3	TX_RCL_BANK_SEL_2	TX_RCL_BANK_SEL_1	TX_RCL_BANK_SEL_0	TX_RCL_SKIP_EN_3_1	TX_RCL_SKIP_EN_0	TX_RCL_FRC_EN_3_1	TX_RCL_FRC_EN_0

Table 5-139 Fields for Register: VR_XS_PMA_MP_32G_TX_GEN_CTRL4

Bits	Name	Memory Access	Description
15:14	TX_RCL_BANK_SEL_3	* Varies	**For Multi-lane configurations:** Tx Recalibration Bank select for Lane 3 Selects the bank to store the Tx Duty cycle correction (DCC) values for Lane 3 **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
13:12	TX_RCL_BANK_SEL_2	* Varies	**For Multi-lane configurations:** Tx Recalibration Bank select for Lane 2 Selects the bank to store the Tx Duty cycle correction (DCC) values for Lane 2 **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
11:10	TX_RCL_BANK_SEL_1	* Varies	**For Multi-lane configurations:** **For Single-lane configurations:** Tx Recalibration Bank select for Lane 1 Selects the bank to store the Tx Duty cycle correction (DCC) values for Lane 1. Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
9:8	TX_RCL_BANK_SEL_0	R/W	Tx Recalibration Bank select for Lane 0 Selects the bank to store the Tx Duty cycle correction (DCC) values for Lane 0. **Value After Reset:** 0x0 **Exists:** Always
7:5	TX_RCL_SKIP_EN_3_1	* Varies	**For Multi-lane configurations:** Tx Recalibration Skip Enable on Lanes 3:1 Enables the skipping of Tx recalibration on lanes [3:1]. This field drives the output port 'xpcs_tx_recal_skip_en_o'. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
4	TX_RCL_SKIP_EN_0	R/W	Tx Recalibration Skip Enable on Lane 0 Enables the skipping of Tx recalibration on lane0. This field drives the output port 'xpcs_tx_recal_skip_en_o'. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
3:1	TX_RCL_FRC_EN_3_1	* Varies	**For Multi-lane configurations:** Tx Force Recalibration Enable on Lanes 3:1 Forces Recalibration on lanes [3:1].This field drives the output port 'xpcs_tx_recal_force_en_o'. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
0	TX_RCL_FRC_EN_0	R/W	Tx Force Recalibration Enable on Lane 0 Forces Recalibration on lane 0. This field drives the output port 'xpcs_tx_recal_force_en_o' **Value After Reset:** 0x0 **Exists:** Always

5.2.3.10 VR_XS_PMA_MP_12G_16G_25G_TX_POWER_STATE_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Tx Power State Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0015 : (APB3_16) ? (0x0015*0x2) : (0x0015*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25||E32))

15:12	11:9	8	7:6	5:4	3:2	1:0
Reserved_15_12	TX_DISABLE_3_1	TX_DISABLE_0	TX3_PSTATE	TX2_PSTATE	TX1_PSTATE	TX0_PSTATE

Table 5-140 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_TX_POWER_STATE_CTRL

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:9	TX_DISABLE_3_1	* Varies	**For Multi-lane configurations:** Transmitter Disable on lanes [3:1] This field drives the output port 'xpcs_tx_disable_o[3:1]'. These bits can be set in P1 power state to put transmitter (lanes 1-3) in a low-power mode. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\" "

Bits	Name	Memory Access	Description
8	TX_DISABLE_0	R/W	Transmitter Disable on lane 0 This field drives the output port 'xpcs_tx_disable_o[0]'. This bit can be set in P1 power state to put transmitter (lane 0) in a low-power mode. **Value After Reset:** 0x0 **Exists:** Always
7:6	TX3_PSTATE	* Varies	**For Multi-lane configurations:** Tx power state control for lane 3 of PHY **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\" "
5:4	TX2_PSTATE	* Varies	**For Multi-lane configurations:** Tx power state control for lane 2 of PHY **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\" "
3:2	TX1_PSTATE	* Varies	**For Multi-lane configurations:** Tx power state control for lane 1 of PHY **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\" "
1:0	TX0_PSTATE	R/W	Tx power state control for lane 0 of PHY Power state encoding is as follows: ■ 2'b00: P0 ■ 2'b01: P0s ■ 2'b10: P1 ■ 2'b11: P2 **Value After Reset:** 0x0 **Exists:** Always

5.2.3.11 VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL0

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Tx Equalization Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0016 : (APB3_16) ? (0x0016*0x2) : (0x0016*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25||E32))

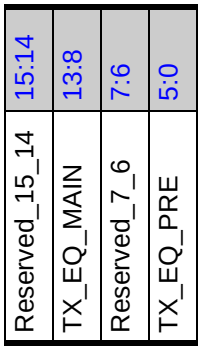


Table 5-141 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL0

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:8	TX_EQ_MAIN	R/W	Control for setting Tx driver output amplitude This field drives the output port 'rpcs_ktx_main_o' in the following conditions: ■ When CL72_EN = Enabled: if 'TX_EQ_OVR_RIDE' bit of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL1 Register is set ■ When CL72_EN = Disabled: always **Value After Reset:** "SNPS_RSVDPARAM_37 ? (RXAUI_DEFAULT ? 0x24 : 0x30) : SNPS_RSVDPARAM_22 ? 0x3E : SNPS_RSVDPARAM_33 ? 0x12 : (SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_39) ? 0x18 : E32 ? (RPCS_PRESENT ? 0x17 : 0x18) : RXAUI_DEFAULT ? (E25 ? 0x12 : 0x1E) : (E25 ? 0x18 : 0x28) " **Exists:** Always

Bits	Name	Memory Access	Description
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:0	TX_EQ_PRE	R/W	Tx Pre-Emphasis level adjustment Control This field controls the transmitter driver output pre-emphasis (pre-shoot coefficient). This field drives the output port 'rpcs_ktx_pre_o' in the following conditions: ■ When CL72_EN = Enabled: if 'TX_EQ_OVR_RIDE' bit of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL1 Register is set ■ When CL72_EN = Disabled: always **Value After Reset:** "(E32&&RPCS_PRESENT) ? 0x4 : 0x00" **Exists:** Always

5.2.3.12 VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Tx Equalization Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0017 : (APB3_16) ? (0x0017*0x2) : (0x0017*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25||E32))

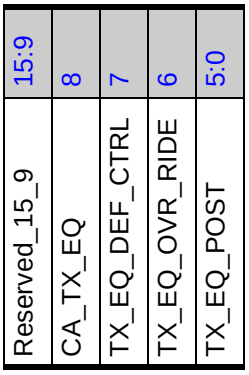


Table 5-142 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL1

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
8	CA_TX_EQ	* Varies	**For configurations with CL72_EN = Enabled:** Enable 'Constant Amplitude' Mode of Transmit Equalization. When this mode is enabled, on receiving INC/DEC commands for C(+1) or C(-1), transmit equalization algorithm updates the Tx filter coefficients such that the amplitude of the signal transmitted by the PHY is held constant. This method is optimal for Synopsys PHY. However, it does not conform to the KR voltage specifications mandated by IEEE. When this bit is low, the default transmit equalization conforming to IEEE specification gets selected. **Note:** Do not set this bit to 1 for Synopsys 32G PHY configurations. **For configurations with CL72_EN = Disabled:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "CL72_EN ? \"read-write\" : \"read-only\""
7	TX_EQ_DEF_CTRL	* Varies	**For configurations with CL72_EN = Enabled:** Transmit Equalization Default Settings Control This field controls the default values that are assigned to the Transmit Equalization coefficients (rpcs_ktx_pre_o, rpcs_ktx_main_o, rpcs_ktx_post_o), whenever KR Training is restarted. The KR adaptation algorithm implemented in DWC_xpcs makes use of these default values to compute the new values of the coefficients, when it receives a coefficient update request from the link partner.**Value :** ■ 0: The default values corresponds to CL72 INIT ■ 1: The default values corresponds to CL72 PRESET **For configurations with CL72_EN = Disabled:**Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "CL72_EN ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
6	TX_EQ_OVR_RIDE	* Varies	**For configurations with CL72_EN = Enabled:** Over-ride Tx Eq settings This bit can be set to override the Transmit Equalization settings generated by the Clause 72 Training module within DWC_xpcs. In the override mode, the values programmed to 'TX_EQ_POST' field of this register drives 'rpcs_ktx_post_o' and 'TX_EQ_MAIN' and 'TX_EQ_PRE' fields of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL0 Register drive 'rpcs_ktx_main_o' and 'rpcs_ktx_pre_o' outputs respectively. **For configurations with CL72_EN = Disabled:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "CL72_EN ? \"read-write\" : \"read-only\""
5:0	TX_EQ_POST	R/W	Tx Post-Emphasis level adjustment Control This field controls the transmitter driver output pre-emphasis (pre-shoot coefficient). This field drives the output port 'rpcs_ktx_post_o' in the following conditions: ■ When CL72_EN = Enabled: if 'TX_EQ_OVR_RIDE' bit is set ■ When CL72_EN = Disabled: always **Value After Reset:** "SNPS_RSVDPARAM_37 ? (RXAUI_DEFAULT ? 0xC : 0x0) : (RXAUI_DEFAULT) ? (E25 ? 0x18 : 0x28) : (SNPS_RSVDPARAM_33 ? 0x18: 0x0)" **Exists:** Always

5.2.3.13 VR_XS_PMA_MP_6G_TX_BSTCTRL0

- **Description:** VR XS or PMA Synopsys Multi-protocol 6G PHY Tx Boost Control Register 0
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0018 : (APB3_16) ? (0x0018*0x2) : (0x0018*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&SNPS_XAUI_PHY&&SSP

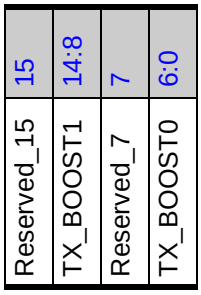


Table 5-143 Fields for Register: VR_XS_PMA_MP_6G_TX_BSTCTRL0

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:8	TX_BOOST1	* Varies	**For Multi-lane configurations:** Transmit Pre-Emphasis Level Adjustment Control for Lane 1 Control for setting the amplitude of the transmitter driver output on Lane 1. **For Single-lane configurations:**Reserved **Value After Reset:** "(!SINGLE_LANE_CFG) ? 0x0A : 0x0" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:0	TX_BOOST0	R/W	Transmit Pre-Emphasis Level Adjustment Control for Lane 0 This field is similar to the TX_BOOST1 field. It sets the amplitude of the transmitter driver output on Lane 0. **Value After Reset:** 0xa **Exists:** Always

5.2.3.14 VR_XS_PMA_MP_6G_TX_BSTCTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 6G PHY Tx Boost Control 1 Register This register is present only in multi-lane configurations.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0019 : (APB3_16) ? (0x0019*0x2) : (0x0019*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&SNPS_XAUI_PHY&&SSP

15	14:8	7	6:0
Reserved_15	TX_BOOST3	Reserved_7	TX_BOOST2

Table 5-144 Fields for Register: VR_XS_PMA_MP_6G_TX_BSTCTRL1

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:8	TX_BOOST3	* Varies	**For Multi-lane configurations:** Transmit Pre-Emphasis Level Adjustment Control for Lane 3 **For Single-lane configurations:** Reserved **Value After Reset:** "(!SINGLE_LANE_CFG) ? 0x0A : 0x0" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
6:0	TX_BOOST2	* Varies	**For Multi-lane configurations:** Transmit Pre-Emphasis Level Adjustment Control for Lane 2 This field is similar to the TX_BOOST3 field. It sets the amplitude of the transmitter driver output on Lane 2. **For Single-lane configurations:** Reserved **Value After Reset:** "(!SINGLE_LANE_CFG) ? 0x0A : 0x0" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""

5.2.3.15 VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL3

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Tx Equalization Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0019 : (APB3_16) ? (0x0019*0x2) : (0x0019*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||(E25&&!SNPS_RSVD-PARAM_37)||E32)&&CL72_EN)

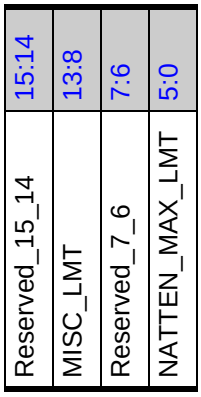


Table 5-145 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL3

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
13:8	MISC_LMT	R/W	Transmit Equalization Miscellaneous Limit Transmit Equalization process is carried out by DWC_xpcs subject to various rules, so that the computed values of Tx coefficients do not violate the IEEE standard requirements. The adaptation algorithm uses the following variables: ■ Lmain[7:0] ■ Lpost[5:0] ■ Lpre[5:0] This field is used in the following rule check: Lmain[7:0] - Lpost[5:0] - Lpre[5:0] >= (MISC_LMT) *4 The desired value of MISC_LMT is 4. However, this field provides flexibility to the application to change the limit. (if desired). **Value After Reset:** "(E25||E32) ? 0x2 : 0x4" **Exists:** Always
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:0	NATTEN_MAX_LMT	R/W	Maximum Value of Natten[5:0] The Transmit Equalization process makes use of a variable called Natten[5:0]. This field can be used to configure the maximum allowed value of Natten. The desired maximum limit of this variable is 6'h12. However, this field provides flexibility to the application to change the limit. (If desired). **Value After Reset:** "(E25||E32) ? 0xA : 0x12" **Exists:** Always

5.2.3.16 VR_XS_PMA_MP_6G_TX_LVLCTRL0

- **Description:** VR XS or PMA Synopsys Multi-protocol 6G PHY Tx Level Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x001A : (APB3_16) ? (0x001A*0x2) : (0x001A*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&SSP

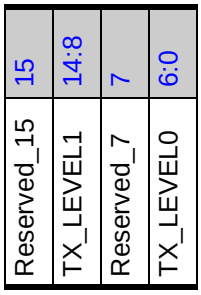


Table 5-146 Fields for Register: VR_XS_PMA_MP_6G_TX_LVLCTRL0

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:8	TX_LEVEL1	* Varies	**For Multi-lane configurations:** Transmit Amplitude Adjustment Control for Lane 1. This is the control field for adjusting the amplitude of the transmit driver output. **For Single-lane configurations:** Reserved **Value After Reset:** "(!SINGLE_LANE_CFG) ? 0x7F : 0x0" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:0	TX_LEVEL0	R/W	Transmit Amplitude Adjustment Control for Lane 0 This is the control field for adjusting the amplitude of the transmit driver output. **Value After Reset:** 0x7f **Exists:** Always

5.2.3.17 VR_XS_PMA_MP_12G_16G_25G_EQ_INIT_CTRL0

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY INIT Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x001A : (APB3_16) ? (0x001A*0x2) : (0x001A*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25||E32)&&CL72_EN)

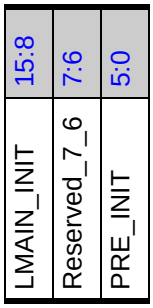


Table 5-147 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_EQ_INIT_CTRL0

Bits	Name	Memory Access	Description
15:8	LMAIN_INIT	R/W	Value of LMAIN corresponding to KR INIT. DWC_xpcs considers the value programmed to this field as the 'LMAIN' value corresponding to 'INIT' command. **Value After Reset:** "(E25||E32) ? 0x40 : 0x6C" **Exists:** !SNPS_RSVDPARAM_37
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
5:0	PRE_INIT	R/W	Value of PRE corresponding to KR INIT. PRE, LMAIN and POST are three parameters/coefficients which control the Multi-protocol 12G/16G/25G PHY Transmit Filter Values. As per the information provided in the 12G/16G PHY Databook, the values of these parameters corresponding to INIT are : PRE=8, LMAIN=108 and POST=44. As per the information provided in the 25G PHY Databook, the values of these parameters corresponding to INIT are : PRE=5, LMAIN=64 and POST=26. DWC_xpcs considers the value programmed to this field as the 'PRE' value corresponding to 'INIT' command. **Value After Reset:** "SNPS_RSVDPARAM_37 ? 0x1 : E25 ? 0x5 : E32 ? 0x4 : 0x8" **Exists:** Always

5.2.3.18 VR_XS_PMA_MP_6G_TX_LVLCTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 6G PHY Tx Level Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x001B : (APB3_16) ? (0x001B*0x2) : (0x001B*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&SSP)

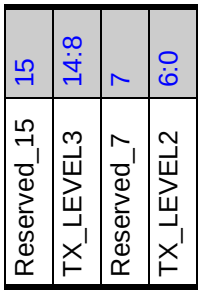


Table 5-148 Fields for Register: VR_XS_PMA_MP_6G_TX_LVLCTRL1

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:8	TX_LEVEL3	* Varies	**For Multi-lane configurations:** Transmit Amplitude Adjustment Control for Lane 3 This is the control field for adjusting the amplitude of the transmit driver output. **For Single-lane configurations:** Reserved **Value After Reset:** "(!SINGLE_LANE_CFG) ? 0x7F : 0x0" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
6:0	TX_LEVEL2	* Varies	**For Multi-lane configurations:** Transmit Amplitude Adjustment Control for Lane 2 This field is similar to the TX_LEVEL3 field. See description of TX_LEVEL3. **For Single-lane configurations:** Reserved **Value After Reset:** "(!SINGLE_LANE_CFG) ? 0x7F : 0x0" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""

5.2.3.19 VR_XS_PMA_MP_12G_16G_25G_EQ_INIT_CTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY INIT Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x001B : (APB3_16) ? (0x001B*0x2) : (0x001B*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25||E32)&&CL72_EN)

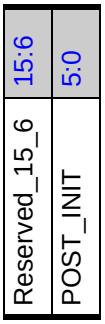


Table 5-149 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_EQ_INIT_CTRL1

Bits	Name	Memory Access	Description
15:6	Reserved_15_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:0	POST_INIT	R/W	Value of POST corresponding to KR INIT. PRE, LMAIN and POST are three parameters/coefficients which control the Multi-protocol 12G/16G/25G PHY Transmit Filter Values. As per the information provided in the 12G/16G PHY Databook, the values of these parameters corresponding to INIT are : PRE=8, LMAIN=108 and POST=44. As per the information provided in the 25G PHY Databook, the values of these parameters corresponding to INIT are : PRE=5, LMAIN=64 and POST=26. DWC_xpcs considers the value programmed to this field as the 'POST' value corresponding to 'INIT' command. **Value After Reset:** "SNPS_RSVDPARAM_37 ? 0x7 : E25 ? 0x1A : E32 ? 0x1C : 0x2C" **Exists:** Always

5.2.3.20 VR_XS_PMA_MP_6G_TXGENCTRL0

- **Description:** VR XS or PMA Synopsys Multi-protocol 6G PHY Tx General Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x001C : (APB3_16) ? (0x001C*0x2) : (0x001C*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&SNPS_XAUI_PHY&&SSP

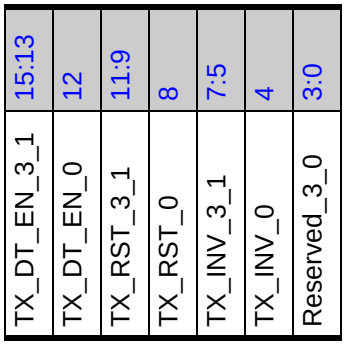


Table 5-150 Fields for Register: VR_XS_PMA_MP_6G_TXGENCTRL0

Bits	Name	Memory Access	Description
15:13	TX_DT_EN_3_1	* Varies	**For Multi-lane configurations:** Output Driver Enable on Lanes[3:1] This bit drives the xgxs_tx_data_en_o[3:1] port of DWC_xpcs. When set to 1, this field internally enables the TX_EN[] and TX_CM_EN[] signals inside the PHY (if not already set to 1 in its IO). ■ TX_DT_EN_3_1[1]: Enables output driver on Lane 1 ■ TX_DT_EN_3_1[2]: Enables output driver on Lane 2 ■ TX_DT_EN_3_1[3]: Enables output driver on Lane 3 **For Single-lane configurations:** Reserved **Value After Reset:** "(!SINGLE_LANE_CFG) ? \"0x7\" : \"0x0\"" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
12	TX_DT_EN_0	R/W	Output Driver Enable on Lane 0 This bit drives the xgxs_tx_data_en_o[0] port of DWC_xpcs. When set to 1, this bit internally enables the TX_EN[] and TX_CM_EN[] signals inside the Synopsys Multi-protocol 6G PHY (if not already set to 1 in its IO). This bit enables the output driver on Lane 0. **Value After Reset:** 0x1 **Exists:** Always
11:9	TX_RST_3_1	* Varies	**For Multi-lane configurations:** Transmit Reset on Lanes[3:1] This field disables the corresponding TX_EN[] and TX_DATA_EN[] signals. It overrides the IO settings and resets all transmit state machines, including the common-mode settings. ■ TX_RST_3_1[1]: Transmit reset on Lane 1 ■ TX_RST_3_1[2]: Transmit reset on Lane 2 ■ TX_RST_3_1[3]: Transmit reset on Lane 3 **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
8	TX_RST_0	R/W	Transmit Reset on Lane 0 This field disables the corresponding TX_EN[] and TX_DATA_EN[] signals. It overrides the IO settings and resets all transmit state machines, including the common-mode settings. This bit resets transmit on Lane 0. **Value After Reset:** 0x0 **Exists:** Always
7:5	TX_INV_3_1	* Varies	**For Multi-lane configurations:** Transmit Invert on Lanes[3:1] This field inverts the logic levels of tx{lane}_p and tx{lane}_m of the PHY output. ■ TX_INV_3_1[1]: Inverts the logic level of tx1_p and tx1_m ■ TX_INV_3_1[2]: Inverts the logic level of tx2_p and tx2_m ■ TX_INV_3_1[3]: Inverts the logic level of tx3_p and tx3_m **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
4	TX_INV_0	R/W	Transmit Invert on Lane 0 This bit inverts the logic levels of tx0_p and tx0_m of the PHY output. **Value After Reset:** 0x0 **Exists:** Always
3:0	Reserved_3_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.3.21 VR_XS_PMA_MP_16G_25G_TX_GENCTRL3

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G/25G PHY TX General Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x001C : (APB3_16) ? (0x001C*0x2) : (0x001C*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37

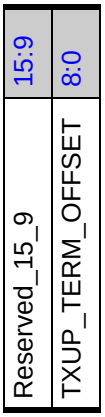


Table 5-151 Fields for Register: VR_XS_PMA_MP_16G_25G_TX_GENCTRL3

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8:0	TXUP_TERM_OFFSET	R/W	Offset for TX Up Termination. This field drives the output port xpcs_txup_term_offset_o[8:0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.22 VR_XS_PMA_MP_32G_TX_CM_CNTX_SEL0

- **Description:** VR XS or PMA Synopsys Multi-protocol 32G Tx Common Context Select 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x001C : (APB3_16) ? (0x001C*0x2) : (0x001C*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&E32

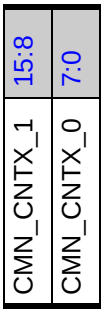


Table 5-152 Fields for Register: VR_XS_PMA_MP_32G_TX_CM_CNTX_SEL0

Bits	Name	Memory Access	Description
15:8	CMN_CNTX_1	* Varies	**For multi lane configurations:** TX Common Context Select Lane1. This field drives the output port 'xpcs_tx_cm1_cntx_sel_o'. **For single lane configurations:** Reserved **Value After Reset:** "!SINGLE_LANE_CFG ? CM_CNTX_SEL : 0x0" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
7:0	CMN_CNTX_0	R/W	TX Common Context Select Lane0. This field drives the output port 'xpcs_tx_cm0_cntx_sel_o'. **Value After Reset:** CM_CNTX_SEL **Exists:** Always

5.2.3.23 VR_XS_PMA_MP_6G_TXGENCTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 6G PHY Tx General Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x001D : (APB3_16) ? (0x001D*0x2) : (0x001D*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD | |XGXS_PCS)&&SNPS_XAUI_PHY&&SSP)

15:12	11:9	8	7:4	3:1	0
Reserved_15_12	RX_DET_REQ_3_1	RX_DET_REQ_0	Reserved_7_4	TX_CM_EN_3_1	TX_CM_EN_0

Table 5-153 Fields for Register: VR_XS_PMA_MP_6G_TXGENCTRL1

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
11:9	RX_DET_REQ_3_1	* Varies	**For Multi-lane configurations:** Receiver Detection Request on Lanes[3:1] When asserted, this field initiates the receiver detection operation for the Synopsys Multi-protocol 6G PHY. Set these bits to 1 only when the corresponding bits (RX_DET_ACK_ 3_1) in the VR_XS_PMA_MP_6G_TX_STS register are set to 0. When these bits are set to 1, do not change these bits to 0 until the corresponding bits (RX_DET_ACK_3_1) are set to 1. ■ RX_DET_REQ_3_1[1]: Receiver detection request on Lane 1 ■ RX_DET_REQ_3_1[2]: Receiver detection request on Lane 2 ■ RX_DET_REQ_3_1[3]: Receiver detection request on Lane 3 **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
8	RX_DET_REQ_0	R/W	Receiver Detection Request on Lane 0 When asserted, this bit initiates the receiver detection operation for the Synopsys Multi-protocol 6G PHY on Lane 0. Set this bit to 1 only when the corresponding bit (RX_DET_ACK_ 0) in the VR_XS_PMA_MP_6G_TX_STS Register is set to 0. When this bit is set to 1, do not change this bit to 0 until the RX_DET_ACK_0 is set to 1. **Value After Reset:** 0x0 **Exists:** Always
7:4	Reserved_7_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
3:1	TX_CM_EN_3_1	* Varies	**For Multi-lane configurations:** Transmit Common Mode Control Enable on Lanes[3:1] This field enables the transmit common mode to be held at the half value of the vptx. Otherwise, the common mode is weakly held to ground through a high impedance connection. ■ TX_CM_EN_3_1[1]: Transmit common mode control on Lane 1 ■ TX_CM_EN_3_1[2]: Transmit common mode control on Lane 2 ■ TX_CM_EN_3_1[3]: Transmit common mode control on Lane 3 **For Single-lane configurations:** Reserved **Value After Reset:** "(!SINGLE_LANE_CFG) ? 0x7 : 0x0" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
0	TX_CM_EN_0	R/W	Transmit Common Mode Control Enable on Lane 0 This field enables the transmit common mode to be held at the half value of the vptx on Lane 0. Otherwise, the common mode is weakly held to ground through a high impedance connection. **Value After Reset:** 0x1 **Exists:** Always

5.2.3.24 VR_XS_PMA_MP_16G_25G_TX_GENCTRL4

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G/25G PHY TX General Control 4 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x001D : (APB3_16) ? (0x001D*0x2) : (0x001D*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37

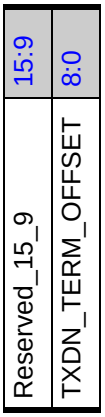


Table 5-154 Fields for Register: VR_XS_PMA_MP_16G_25G_TX_GENCTRL4

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8:0	TXDN_TERM_OFFSET	R/W	Offset for TX Down Termination. This field drives the output port xpcs_txdn_term_offset_o[8:0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.25 VR_XS_PMA_MP_32G_TX_CM_CNTX_SEL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 32G Tx Common Context Select 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x001D : (APB3_16) ? (0x001D*0x2) : (0x001D*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&E32&&!SINGLE_LANE_CFG

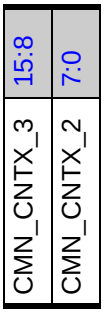


Table 5-155 Fields for Register: VR_XS_PMA_MP_32G_TX_CM_CNTX_SEL1

Bits	Name	Memory Access	Description
15:8	CMN_CNTX_3	* Varies	TX Common Context Select Lane3. This field drives the output port 'xpcs_tx_cm3_cntx_sel_o'. **Value After Reset:** CM_CNTX_SEL **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
7:0	CMN_CNTX_2	* Varies	TX Common Context Select Lane2. This field drives the output port 'xpcs_tx_cm2_cntx_sel_o'. **Value After Reset:** CM_CNTX_SEL **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""

5.2.3.26 VR_XS_PMA_MP_6G_TXGENCTRL2

- **Description:** VR XS or PMA Synopsys Multi-protocol 6G PHY Tx General Control 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x001E : (APB3_16) ? (0x001E*0x2) : (0x001E*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&SSP)

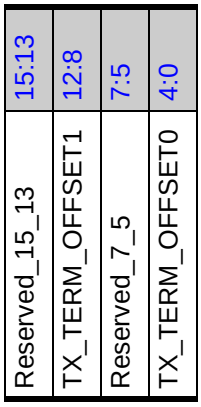


Table 5-156 Fields for Register: VR_XS_PMA_MP_6G_TXGENCTRL2

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:8	TX_TERM_OFFSET1	* Varies	**For multi-lane configurations:** Tx Termination Offset Lane 1 Applies an offset to the resistor calibration value. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
7:5	Reserved_7_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4:0	TX_TERM_OFFSET0	R/W	Tx Termination Offset Lane 0 This field applies an offset to the resistor calibration value on lane 0. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.27 VR_XS_PMA_MP_32G_TX_CNTX_CTRL0

- **Description:** VR XS or PMA Synopsys Multi protocol E32 PHY TX Context Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x001E : (APB3_16) ? (0x001E*0x2) : (0x001E*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(SNPS_RSVDPARAM_37||E32)

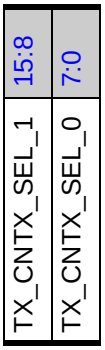


Table 5-157 Fields for Register: VR_XS_PMA_MP_32G_TX_CNTX_CTRL0

Bits	Name	Memory Access	Description
15:8	TX_CNTX_SEL_1	* Varies	**For Multi-lane configurations:** Transmit context select along lane 1 of the PHY specifies the context that needs to be restored for the TX block on lane 1. This field drives the output port 'xpcs_tx1_cntx_sel_o[7:0]'. **For single-lane configurations:** Reserved **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : E32 ? TX_CNTX_SEL : 0x2E" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" :\"read-only\""
7:0	TX_CNTX_SEL_0	R/W	Transmit context select along lane 0 of the PHY specifies the context that needs to be restored for the TX block on lane 0. This field drives the output port 'xpcs_tx0_cntx_sel_o[7:0]'. **Value After Reset:** TX_CNTX_SEL **Exists:** Always

5.2.3.28 VR_XS_PMA_MP_6G_TXGENCTRL3

- **Description:** VR XS or PMA Synopsys Multi-protocol 6G PHY Tx General Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x001F : (APB3_16) ? (0x001F*0x2) : (0x001F*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&SSP)

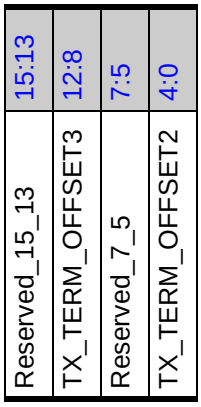


Table 5-158 Fields for Register: VR_XS_PMA_MP_6G_TXGENCTRL3

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:8	TX_TERM_OFFSET3	* Varies	**For multi-lane configurations:** Tx Termination Offset Lane 3 Applies an offset to the resistor calibration value. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
7:5	Reserved_7_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4:0	TX_TERM_OFFSET2	* Varies	**For multi-lane configurations:** Tx Termination Offset Lane 2 Applies an offset to the resistor calibration value. **For Single-lane configurations :**Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""

5.2.3.29 VR_XS_PMA_MP_16G_25G_TX_MISC_CTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G/25G PHY TX Miscellaneous Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x001F : (APB3_16) ? (0x001F*0x2) : (0x001F*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||E25||SNPS_RSVDPARAM_39)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37

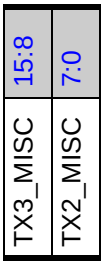


Table 5-159 Fields for Register: VR_XS_PMA_MP_16G_25G_TX_MISC_CTRL1

Bits	Name	Memory Access	Description
15:8	TX3_MISC	R/W	TX Miscellaneous control for lane3. This field drives the output port xpcs_tx3_misc_o[7:0]. **Value After Reset:** 0x0 **Exists:** Always
7:0	TX2_MISC	R/W	TX Miscellaneous control for lane2. This field drives the output port xpcs_tx2_misc_o[7:0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.30 VR_XS_PMA_MP_32G_TX_CNTX_CTRL1

- **Description:** VR XS or PMA Synopsys Multi Protocol E32 PHY TX Context Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x001F : (APB3_16) ? (0x001F*0x2) : (0x001F*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(SNPS_RSVDPARAM_37||E32)&&!SINGLE_LANE_CFG

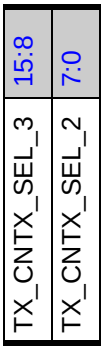


Table 5-160 Fields for Register: VR_XS_PMA_MP_32G_TX_CNTX_CTRL1

Bits	Name	Memory Access	Description
15:8	TX_CNTX_SEL_3	R/W	Transmit context select along lane 3 of the PHY specifies the context that needs to be restored for the TX block on lane 3. This field drives the output port 'xpcs_tx3_cntx_sel_o[7:0]'. **Value After Reset:** TX_CNTX_SEL **Exists:** Always
7:0	TX_CNTX_SEL_2	R/W	Transmit context select along lane 2 of the PHY specifies the context that needs to be restored for the TX block on lane 2. This field drives the output port 'xpcs_tx2_cntx_sel_o[7:0]'. **Value After Reset:** TX_CNTX_SEL **Exists:** Always

5.2.3.31 VR_XS_PMA_MP_12G_16G_25G_TX_STS

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Tx Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0020 : (APB3_16) ? (0x0020*0x2) : (0x0020*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25||E32))

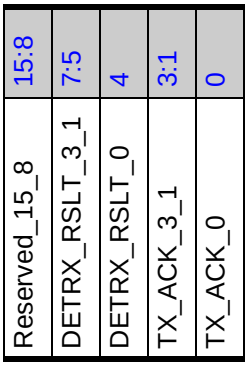


Table 5-161 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_TX_STS

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	DETRX_RSLT_3_1	R	**For Multi-lane configurations:** Receiver Detection Result on Lanes[3:1] This field captures the value of the input port 'xpcs_tx_detrx_result_i[3:1]'. The value of this field is valid when 'TX_ACK_3_1' is high. ■ 1'b0: Receiver not detected ■ 1'b1: Receiver detected **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4	DETRX_RSLT_0	R	Receiver Detection Result on lane 0 of PHY This field captures the value of the input port 'xpcs_tx_detrx_result_i[0]'. The value of this field is valid when 'TX_ACK_0' is high. ■ 1'b0: Receiver not detected ■ 1'b1: Receiver detected **Value After Reset:** 0x0 **Exists:** Always
3:1	TX_ACK_3_1	R	**For Multi-lane configurations:** Transmit Acknowledge on Lanes[3:1] This bit captures the value of the input port 'xpcs_tx_ack_i[3:1]'. In single-lane configurations, this field is reserved. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always
0	TX_ACK_0	R	Tx Acknowledge on lane 0 of PHY This bit captures the value of the input port 'xpcs_tx_ack_i[0]'. When this bit is read as high, it indicates that the requested transmitter setting is complete or the requested RX-detection operation is complete. After this bit is read high, it remains high till bit DET_RX_REQ_0 TX_REQ_0 or is self-cleared **Value After Reset:** 0x0 **Exists:** Always

5.2.3.32 VR_XS_PMA_MP_25G_TX_TERM_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY Transmit Termination Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0024 : (APB3_16) ? (0x0024*0x2) : (0x0024*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&E25&&!SNPS_RSVDPARAM_37

15	14:12	11	10:8	7	6:4	3	2:0
Reserved_15	TX3_TERM	Reserved_11	TX2_TERM	Reserved_7	TX1_TERM	Reserved_3	TX0_TERM

Table 5-162 Fields for Register: VR_XS_PMA_MP_25G_TX_TERM_CTRL

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:12	TX3_TERM	* Varies	Transmit Termination Control for lane 3 **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : SNPS_RSVDPARAM_36 ? 0x7 : (SNPS_RSVDPARAM_16) ? 0x4 : 0x2" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
10:8	TX2_TERM	* Varies	Transmit Termination Control for lane 2 **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : SNPS_RSVDPARAM_36 ? 0x7 : (SNPS_RSVDPARAM_16) ? 0x4 : 0x2" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:4	TX1_TERM	* Varies	Transmit Termination Control for lane 1 **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : SNPS_RSVDPARAM_36 ? 0x7 : (SNPS_RSVDPARAM_16) ? 0x4 : 0x2" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
3	Reserved_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	TX0_TERM	R/W	Transmit Termination Control for lane 0 **Value After Reset:** "SNPS_RSVDPARAM_36 ? 0x7 : (SNPS_RSVDPARAM_16) ? 0x4 : 0x2" **Exists:** Always

5.2.3.33 VR_XS_PMA_MP_25G_TX_DCC_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY Tx DCC Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0025 : (APB3_16) ? (0x0025*0x2) : (0x0025*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (XGXS_PCS||PMA_MMD)&&E25&&!SNPS_RSVDPARAM_37

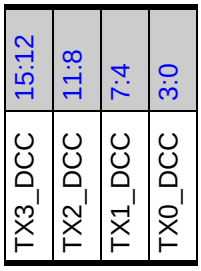


Table 5-163 Fields for Register: VR_XS_PMA_MP_25G_TX_DCC_CTRL

Bits	Name	Memory Access	Description
15:12	TX3_DCC	* Varies	Tx Internal Clock Duty Cycle Correction Control Range for lane 3. **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : SNPS_RSVDPARAM_36 ? 0xC : RPCS_PRESENT ? (SNPS_RSVDPARAM_16 ? 0xC : 0xB) : 0xA" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
11:8	TX2_DCC	* Varies	Tx Internal Clock Duty Cycle Correction Control Range for lane 2. **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : SNPS_RSVDPARAM_36 ? 0xC : RPCS_PRESENT ? (SNPS_RSVDPARAM_16 ? 0xC : 0xB) : 0xA" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
7:4	TX1_DCC	* Varies	Tx Internal Clock Duty Cycle Correction Control Range for lane 1. **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : SNPS_RSVDPARAM_36 ? 0xC : RPCS_PRESENT ? (SNPS_RSVDPARAM_16 ? 0xC : 0xB) : 0xA" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
3:0	TX0_DCC	R/W	Tx Internal Clock Duty Cycle Correction Control Range for lane 0. **Value After Reset:** "SNPS_RSVDPARAM_36 ? 0xC : RPCS_PRESENT ? (SNPS_RSVDPARAM_16 ? 0xC : 0xB) : 0xA" **Exists:** Always

5.2.3.34 VR_XS_PMA_MP_25G_TX_WIDTH_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY Tx Width Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0026 : (APB3_16) ? (0x0026*0x2) : (0x0026*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (XGXS_PCS||PMA_MMD)&&((E25&&!SNPS_RSVDPARAM_37)||E32)

15	14:12	11	10:8	7	6:4	3	2:0
Reserved_15	TX3_WIDTH	Reserved_11	TX2_WIDTH	Reserved_7	TX1_WIDTH	Reserved_3	TX0_WIDTH

Table 5-164 Fields for Register: VR_XS_PMA_MP_25G_TX_WIDTH_CTRL

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:12	TX3_WIDTH	* Varies	Tx PHY Width -lane 3. This field drives the xpcs_tx3_width_o output port and controls the width of PHY Tx lane 3. This field is valid only for multi-lane configurations. **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : RPCS_PRESENT ? (R_SRDS_INTF==0 ? 0x2 : R_SRDS_INTF==1 ? 0x4 : R_SRDS_INTF==4 ? 0x3 : 0x5) : RXAUI_DEFAULT ? 0x3 : 0x1" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
10:8	TX2_WIDTH	* Varies	Tx PHY Width -lane 2. This field drives the xpcs_tx2_width_o output port and controls the width of PHY Tx lane 2. This field is valid only for multi-lane configurations. **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : RPCS_PRESENT ? (R_SRDS_INTF==0 ? 0x2 : R_SRDS_INTF==1 ? 0x4 : R_SRDS_INTF==4 ? 0x3 : 0x5) : RXAUI_DEFAULT ? 0x3 : 0x1" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:4	TX1_WIDTH	* Varies	Tx PHY Width -lane 1. This field drives the xpcs_tx1_width_o output port and controls the width of PHY Tx lane 1. This field is valid only for multi-lane configurations. **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : RPCS_PRESENT ? (R_SRDS_INTF==0 ? 0x2 : R_SRDS_INTF==1 ? 0x4 : R_SRDS_INTF==4 ? 0x3 : 0x5) : RXAUI_DEFAULT ? 0x3 : 0x1" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
3	Reserved_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	TX0_WIDTH	R/W	Tx PHY Width -lane 0. This field drives the xpcs_tx0_width_o output port and controls the width of PHY Tx lane 0. **Value After Reset:** "RPCS_PRESENT ? (R_SRDS_INTF==0 ? 0x2 : R_SRDS_INTF==1 ? 0x4 : R_SRDS_INTF==4 ? 0x3 : 0x5) : RXAUI_DEFAULT ? 0x3 : 0x1" **Exists:** Always

5.2.3.35 VR_XS_PMA_MP_6G_TX_STS

- **Description:** VR XS or PMA Synopsys Multi-protocol 6G PHY Tx Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0028 : (APB3_16) ? (0x0028*0x2) : (0x0028*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&SSP)

15:13	12	11:9	8	7:5	4	3:1	0
RX_DET_RESLT_3_1	RX_DET_RESLT_0	RX_DET_ACK_3_1	RX_DET_ACK_0	TX_CMNST_3_1	TX_CMNST_0	TX_ST_3_1	TX_ST_0

Table 5-165 Fields for Register: VR_XS_PMA_MP_6G_TX_STS

Bits	Name	Memory Access	Description
15:13	RX_DET_RESLT_3_1	R	**For Multi-lane configurations:** Transmit Receiver Detect Result on Lanes[3:1] When set to 1, this field indicates that a receiver is detected. This signal is valid when the corresponding RX_DET_ACK[] signal is read as 1. ■ RX_DET_RESLT_3_1[1]: Transmit receiver detect result on Lane 1 ■ RX_DET_RESLT_3_1[2]: Transmit receiver detect result on Lane 2 ■ RX_DET_RESLT_3_1[3]: Transmit receiver detect result on Lane 3 **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
12	RX_DET_RESLT_0	R	Transmit Receiver Detect Result on Lane 0 When set to 1, this field indicates that a receiver is detected on Lane 0. This signal is valid when the corresponding TX_DET_RX_ACK[] signal is read as 1. **Value After Reset:** 0x0 **Exists:** Always
11:9	RX_DET_ACK_3_1	R	**For Multi-lane configurations:** Transmit Receiver Detect Acknowledgment Signal When set to 1, this field indicates that the receiver detection operation, requested by setting the TX_DET_RX_REQ[] signal, is complete. This is de-asserted only when the corresponding TX_DET_RX_REQ[] signal is set to 0. ■ RX_DET_ACK_3_1[1]: Tx receiver detect acknowledgment on Lane 1 ■ RX_DET_ACK_3_1[2]: Tx receiver detect acknowledgment on Lane 2 ■ RX_DET_ACK_3_1[3]: Tx receiver detect acknowledgment on Lane 3 **For Single-lane configurations:**Reserved **Value After Reset:** 0x0 **Exists:** Always
8	RX_DET_ACK_0	R	Transmit Receiver Detect Acknowledgment Signal When set to 1, this field indicates that the receiver detection operation, requested by setting the TX_DET_RX_REQ[] signal, is complete on Lane 0. This is de-asserted only when the corresponding TX_DET_RX_REQ[] signal is set to 0. **Value After Reset:** 0x0 **Exists:** Always
7:5	TX_CMNST_3_1	R	**For Multi-lane configurations:** Transmit Common Mode Control State on Lanes[3:1] When set to 1, this field indicates that the Tx common mode has reached the intended level. ■ TX_CMNST_3_1[1]: Tx common mode state on Lane 1 ■ TX_CMNST_3_1[2]: Tx common mode state on Lane 2 ■ TX_CMNST_3_1[3]: Tx common mode state on Lane 3 **For single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

Bits	Name	Memory Access	Description
4	TX_CMNST_0	R	Transmit Common Mode Control State on Lane 0 When set to 1, this field indicates that the Tx common mode has reached the intended level on Lane 0. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
3:1	TX_ST_3_1	R	**For multi-lane configurations:** Transmit Active Status of the PHY on Lanes[3:1] When set to 1, this field indicates that the PHY Tx has reached the intended level on the corresponding lane and is ready to transmit the data. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
0	TX_ST_0	R	Transmit Active Status of the Synopsys Multi-protocol 6G PHY on Lane 0 When set to 1, this field indicates that the PHY Tx has reached the intended level on Lane 0 and is ready to transmit the data. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

5.2.3.36 VR_XS_PMA_MP_PHY_EQ_PRESET_CTRL0

- **Description:** VR XS or PMA Synopsys Multi-protocol PHY PRESET Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x002E : (APB3_16) ? (0x002E*0x2) : (0x002E*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25||E32)&&CL72_EN)

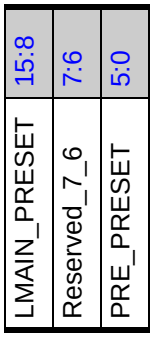


Table 5-166 Fields for Register: VR_XS_PMA_MP_PHY_EQ_PRESET_CTRL0

Bits	Name	Memory Access	Description
15:8	LMAIN_PRESET	R/W	Value of LMAIN corresponding to KR PRESET. DWC_xpcs considers the value programmed to this field as the 'LMAIN' value corresponding to 'PRESET' command. **Value After Reset:** "(E12||E16) ? 0xA0 : E25 ? 0x60 : 0x5C" **Exists:** !SNPS_RSVDPARAM_37
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:0	PRE_PRESET	R/W	Value of PRE corresponding to KR PRESET. PRE, LMAIN and POST are three parameters/coefficients which control the Multi-protocol PHY Transmit Filter Values. DWC_xpcs considers the value programmed to this field as the 'PRE' value corresponding to 'PRESET' command. **Value After Reset:** "E32 ? 0x4 : 0x0" **Exists:** Always

5.2.3.37 VR_XS_PMA_MP_PHY_EQ_PRESET_CTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol PHY PRESET Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x002F : (APB3_16) ? (0x002F*0x2) : (0x002F*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25||E32)&&CL72_EN)

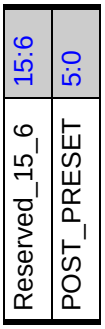


Table 5-167 Fields for Register: VR_XS_PMA_MP_PHY_EQ_PRESET_CTRL1

Bits	Name	Memory Access	Description
15:6	Reserved_15_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:0	POST_PRESET	R/W	Value of POST corresponding to KR PRESET. PRE, LMAIN and POST are three parameters/coefficients which control the Multi-protocol PHY Transmit Filter Values. DWC_xpcs considers the value programmed to this field as the 'POST' value corresponding to 'PRESET' command. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.38 VR_XS_PMA_MP_12G_16G_25G_RX_GENCTRL0

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Rx General Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0030 : (APB3_16) ? (0x0030*0x2) : (0x0030*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25||E32))

15:13	12	11:9	8	7:5	4	3:1	0
RX_CLKSFT_3_1	RX_CLKSFT_0	RX_DT_EN_3_1	RX_DT_EN_0	RX_ALIGN_EN_3_1	RX_ALIGN_EN_0	RX_TERM_EN_3_1	RX_TERM_EN_0

Table 5-168 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_RX_GENCTRL0

Bits	Name	Memory Access	Description
15:13	RX_CLKSFT_3_1	* Varies	Rx clock shift on Lanes[3:1] of Multi-protocol 12G PHY When these bits are set, a 1-bit shift of receive data happens relate to receive clock. This operation works only if alignment enable is disabled. This field drives the output port 'xpcs_rx_clk_shift_o[3:1]'. This field is reserved for Multi-protocol 16G/25G PHY configurations.In addition, this field is valid only in multi-lane configurations. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!(SINGLE_LANE_CFG||E16||E25||E32) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
12	RX_CLKSFT_0	* Varies	Rx clock shift on lane 0 of Multi-protocol 12G PHY When this bit is set, a 1-bit shift of receive data happens relate to receive clock. This operation works only if alignment enable is disabled. This bit drives the output port 'xpcs_rx_clk_shift_o[0]'. This field is reserved for Multi-protocol 16G/25G PHY configurations. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "E12 ? \"read-write\" : \"read-only\""
11:9	RX_DT_EN_3_1	* Varies	**For Multi-lane configurations:** Rx Data Enable on Lanes[3:1] Set this bit to enable the PHY receiver data output on lanes 3 to 1. This field drives the output port 'xpcs_rx_data_en_o[3:1]'. **For Single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0 : 0x7" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
8	RX_DT_EN_0	R/W	Rx Data Enable on lane 0 of PHY Set this bit to enable the PHY receiver data output on lane 0. This bit drives the output port 'xpcs_rx_data_en_o[0]'. **Value After Reset:** 0x1 **Exists:** Always
7:5	RX_ALIGN_EN_3_1	* Varies	Rx Data Alignment Enable on Lanes[3:1] of Multi-protocol 12G PHY These bits can be set to enable word alignment (based on k28.5 character) in the PHY on lanes 3 to 1. This field drives the output port 'xgxs_rx_align_en_o[3:1]'. This field is reserved for Multi-protocol 16G/25G PHY configurations.In addition, this field is valid only in multi-lane configurations. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!(SINGLE_LANE_CFG||E16||E25||E32) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
4	RX_ALIGN_EN_0	* Varies	Rx Data Alignment Enable on lane 0 of Multi-protocol 12G PHY This bit can be set to enable word alignment (based on k28.5 character) in the PHY. This field drives the output port 'xgxs_rx_align_en_o[0]'. This field is reserved for Multi-protocol 16G/25G PHY configurations. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "E12 ? \"read-write\" : \"read-only\""
3:1	RX_TERM_EN_3_1	* Varies	**For Multi-lane configurations:** Rx Termination Enable on Lanes[3:1] When these bits are set, PHY Rx is terminated with a nominal 50 ohm resistance. Otherwise, the termination is in high impedance. This field drives the output port 'xpcs_rx_term_en_o[3:1]'. This field is present only for multi-lane configurations. **For Single-lane configurations:** Reserved **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : 0x7" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
0	RX_TERM_EN_0	R/W	Rx Termination Enable on lane 0 of PHY When this bit is set, PHY Rx is terminated with a nominal 50 ohm resistance. Otherwise, the termination is in high impedance. This field drives the output port 'xpcs_rx_term_en_o[0]'. **Value After Reset:** 0x1 **Exists:** Always

5.2.3.39 VR_XS_PMA_MP_12G_16G_25G_RX_GENCTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Rx General Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0031 : (APB3_16) ? (0x0031*0x2) : (0x0031*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25||E32)

15:13	12	11:9	8	7:5	4	3:1	0
RX_DIV16P5_CLK_EN_3_1	RX_DIV16P5_CLK_EN_0	RX_TERM_ACDC_3_1	RX_TERM_ACDC_0	RX_RST_3_1	RX_RST_0	RX_INV_3_1	RX_INV_0

Table 5-169 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_RX_GENCTRL1

Bits	Name	Memory Access	Description
15:13	RX_DIV16P5_CLK_EN_3_1	* Varies	**For Multi-lane configurations:** Receiver Divide by 16.5 output clock on lanes [3:1] This field enables rxX_div16p5_clk on lanes 3 to 1. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** !(SNPS_RSVDPARAM_35||(SNPS_RSVDPARAM_22&&SNPS_RSVDPARAM_2)||SNPS_RSVDPARAM_37||E32) **Memory Access:** "(!(SINGLE_LANE_CFG||SNPS_RSVDPARAM_37)) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
12	RX_DIV16P5_CLK_EN_0	R/W	Receiver Divide by 16.5 output clock on lane 0 When this bit is set, it enables the divide by 16.5 Rx VCO recovered clock of the PHY. This clock is available at rx0_div16p5_clk output port of PHY. This bit must be set when operating in PCS-R mode. **Value After Reset:** "RPCS_PRESENT ? 0x1 : 0x0" **Exists:** !(SNPS_RSVDPARAM_35||(SNPS_RSVDPARAM_22&&SNPS_RSVDPARAM_2)||SNPS_RSVDPARAM_37||E32)
11:9	RX_TERM_ACDC_3_1	* Varies	For Multi-lane configurations: Rx Termination control on Lanes[3:1] This field controls the termination of PHY receiver ■ 0: Floating Rx Termination (DC coupled PHY) ■ 1: Grounded Rx Termination (AC coupled PHY) This field drives the output port 'xpcs_rx_term_acdc_o[3:1]'. **For Single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG||RXAUI_DEFAULT||E32) ? 0x0 : 0x7" **Exists:** Always **Memory Access:** "(!(SINGLE_LANE_CFG||E32)) ? \"read-write\" : \"read-only\""
8	RX_TERM_ACDC_0	* Varies	**For E32 Configurations:** Reserved **For non E32 Configurations:** Rx Termination control on lane 0 of PHY. ■ 0: DC Termination (Floating Rx) ■ 1: AC Termination (Grounded Rx) This field drives the output port xpcs_rx_term_acdc_o[0]. **Value After Reset:** "RXAUI_DEFAULT||E32 ? 0x0 : 0x1" **Exists:** Always **Memory Access:** "E32 ? \"read-only\" : \"read-write\" "

Bits	Name	Memory Access	Description
7:5	RX_RST_3_1	* Varies	**For Multi-lane configurations:** Rx reset on Lanes[3:1] When this field is set, RX data path, all the receiver settings and state machines of the PHY (on lanes 3 to 1) are reset This field drives the output port xgxs_rx_reset_o[3:1] when DWC_xpcs is in POWER_GOOD state. In single-lane configurations, this field is reserved. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
4	RX_RST_0	R/W	Rx reset on lane 0 of PHY When this bit is set, RX data path, all the receiver settings and state machines of the PHY are reset This field drives the output port xgxs_rx_reset_o[0] when DWC_xpcs is in POWER_GOOD state. **Value After Reset:** 0x0 **Exists:** Always
3:1	RX_INV_3_1	* Varies	**For Multi-lane configurations:** Rx Data Invert on Lanes[3:1] This field drives the output port xgxs_rx_invert_o[3:1] This field is present only in multi-lane configurations. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
0	RX_INV_0	R/W	Rx Data Invert on lane 0 of PHY When this bit is set, the data on PHY Rx serial lines are logically inverted. This signal drives the output port xgxs_rx_invert_o[0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.40 VR_XS_PMA_MP_12G_16G_RX_GENCTRL2

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G PHY Rx General Control 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0032 : (APB3_16) ? (0x0032*0x2) : (0x0032*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16))

15:14	13:12	11:10	9:8	7:5	4	3:1	0
RX3_WIDTH	RX2_WIDTH	RX1_WIDTH	RX0_WIDTH	RX_LPD_3_1	RX_LPD_0	RX_REQ_3_1	RX_REQ_0

Table 5-170 Fields for Register: VR_XS_PMA_MP_12G_16G_RX_GENCTRL2

Bits	Name	Memory Access	Description
15:14	RX3_WIDTH	* Varies	**For Multi-lane configurations:** Rx Datapath Width on lane 3 of the PHY This field drives the output port xpcs_rx3_width[1:0] This field is present only in multi-lane configurations. **For Single-lane configurations:** Reserved **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : (RXAUI_DEFAULT||(RPCS_PRESENT&&(R_SRDS_INTF==4))) ? \"0x3\" : (RPCS_PRESENT&&(R_SRDS_INTF==0)) ? 0x2 : 0x1" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
13:12	RX2_WIDTH	* Varies	**For Multi-lane configurations:** Rx Datapath Width on lane 2 of the PHY This field drives the output port xpcs_rx2_width[1:0] This field is present only in multi-lane configurations. **For Single-lane configurations:** Reserved **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : (RXAUI_DEFAULT||(RPCS_PRESENT&&(R_SRDS_INTF==4))) ? 0x3 : (RPCS_PRESENT&&(R_SRDS_INTF==0)) ? 0x2 : 0x1" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
11:10	RX1_WIDTH	* Varies	**For Multi-lane configurations:** Rx Datapath Width on lane 1 of the PHY This field drives the output port xpcs_rx1_width[1:0] This field is present only in multi-lane configurations. **For Single-lane configurations:** Reserved **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : (RXAUI_DEFAULT||(RPCS_PRESENT&&(R_SRDS_INTF==4))) ? 0x3 : (RPCS_PRESENT&&(R_SRDS_INTF==0)) ? 0x2 : 0x1" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
9:8	RX0_WIDTH	R/W	Rx Datapath Width on lane 0 of the PHY This field controls the width of output receive data from PHY on lane 3. The encoding of the width is as follows: ■ 2'b00: 8-bit ■ 2'b01: 10-bit ■ 2'b10: 16-bit ■ 2'b11: 20-bit This field drives the output port xpcs_rx0_width[1:0]. **Value After Reset:** "(RPCS_PRESENT&&(R_SRDS_INTF==0)) ? 0x2 : (RPCS_PRESENT||RXAUI_DEFAULT) ? 0x3 : 0x1" **Exists:** Always

Bits	Name	Memory Access	Description
7:5	RX_LPD_3_1	* Varies	**For Multi-lane configurations:** Receiver Lane Power Down on Lanes[3:1] This field drives the output port xpcs_rx_lpd_o[3:1]. These bits can be set to power down the receiver to a power state equivalent to that of P1.In single-lane configurations, this field is reserved. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
4	RX_LPD_0	R/W	Receiver Lane Power Down on lane 0 of PHY This bit can be set to power down the receiver to a power state equivalent to that of P1. This bit drives the output port 'xpcs_rx_lpd_o[0]'. **Value After Reset:** 0x0 **Exists:** Always
3:1	RX_REQ_3_1	* Varies	**For Multi-lane configurations:** Receiver Request on Lanes[3:1] (RW,SC Type) These bits can be set by application to make a Rx operation request.This bit drives the output port xpcs_rx_req_o[3:1]. Completion of the request is indicated by fields'RX_ACK_3_1'. These bits are self-cleared when the acknowledgment is asserted. For more details see the 'RX_REQ_0' field. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
0	RX_REQ_0	R/W	Receiver operation request on lane 0 of PHY (RW,SC Type). When this bit is set, a new receiver setting request is made towards the PHY.This bit drives the output port xpcs_rx_req_o[0]. When this bit is set, PHY captures its following input signals: ■ rx0_pstate[1:0] ■ rx0_lpd ■ rx0_rate[1:0] ■ rx0_width[1:0] ■ rx0_data_en ■ rx0_adapt_afe_en ■ rx0_adapt_dfe_en ■ rx0_eq_att_lvl[2:0] ■ rx0_eq_vga1_gain[3:0] ■ rx0_eq_vga2_gain[3:0] ■ rx0_eq_ctle_pole[2:0] ■ rx0_eq_ctle_boost[4:0] ■ rx0_eq_dfe_tap1[7:0] A successful completion of receiver request operation on lane 0 is indicated by bit RX_ACK_0. This bit is self-cleared when 'xpcs_tx_ack_i[0]' is asserted. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.41 VR_XS_PMA_MP_25G_RX_GENCTRL2

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY Rx General Control 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0032 : (APB3_16) ? (0x0032*0x2) : (0x0032*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&(E25 | |E32)

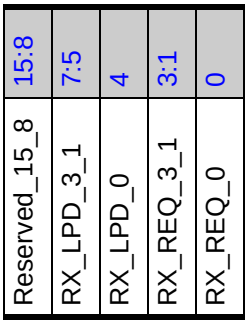


Table 5-171 Fields for Register: VR_XS_PMA_MP_25G_RX_GENCTRL2

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	RX_LPD_3_1	* Varies	**For Multi-lane configurations:** Receiver Lane Power Down on Lanes[3:1] This field drives the output port xpcs_rx_lpd_o[3:1]. These bits can be set to power down the receiver to a power state equivalent to that of P1.In single-lane configurations, this field is reserved. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
4	RX_LPD_0	R/W	Receiver Lane Power Down on lane 0 of PHY This bit can be set to power down the receiver to a power state equivalent to that of P1. This bit drives the output port 'xpcs_rx_lpd_o[0]'. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
3:1	RX_REQ_3_1	* Varies	**For Multi-lane configurations:** Receiver Request on Lanes[3:1] (RW,SC Type) These bits can be set by application to make a Rx operation request.This bit drives the output port xpcs_rx_req_o[3:1]. Completion of the request is indicated by fields'RX_ACK_3_1'. These bits are self-cleared when the acknowledgment is asserted. For more details see the 'RX_REQ_0' field. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
0	RX_REQ_0	R/W	Receiver operation request on lane 0 of PHY (RW,SC Type). When this bit is set, a new receiver setting request is made towards the PHY.This bit drives the output port xpcs_rx_req_o[0]. A successful completion of receiver request operation on lane 0 is indicated by bit RX_ACK_0. This bit is self-cleared when 'xpcs_tx_ack_i[0]' is asserted. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.42 VR_XS_PMA_MP_12G_16G_RX_GENCTRL3

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G PHY Rx General Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0033 : (APB3_16) ? (0x0033*0x2) : (0x0033*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16)&&!SNPS_RSVDPARAM_22)

15:13	12	11:9	8:6	5:3	2:0
LOS_LFPS_EN_3_1	LOS_LFPS_EN_0	LOS_TRSHLD_3	LOS_TRSHLD_2	LOS_TRSHLD_1	LOS_TRSHLD_0

Table 5-172 Fields for Register: VR_XS_PMA_MP_12G_16G_RX_GENCTRL3

Bits	Name	Memory Access	Description
15:13	LOS_LFPS_EN_3_1	* Varies	**For Multi-lane configurations:** Rx LOS LFPS Enable on lane [3:1] of the PHY This field drives the output port xpcs_rx_los_lfps_en_o[3:1] to enable the LFPS filter on lanes 3 to 1 of the PHY. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
12	LOS_LFPS_EN_0	R/W	Rx LOS LFPS Enable on lane 0 of the PHY This field drives the output port xpcs_rx_los_lfps_en_o[0] to enable the LFPS filter on lane 0 of the PHY. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
11:9	LOS_TRSHLD_3	* Varies	**For Multi-lane configurations:** Loss of signal threshold on Lane 3 This field drives the output port xpcs_rx3_los_threshold_o[2:0]. In single-lane configurations, this field is reserved. **For Single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0 : E16 ? 0x1 : SNPS_RSVDPARAM_39 ? 0x5 : 0x3" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
8:6	LOS_TRSHLD_2	* Varies	**For Multi-lane configurations:** Loss of signal threshold on Lane 2 This field drives the output port xpcs_rx2_los_threshold_o[2:0]. In single-lane configurations, this field is reserved. **For Single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0 : E16 ? 0x1 : SNPS_RSVDPARAM_39 ? 0x5 : 0x3" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
5:3	LOS_TRSHLD_1	* Varies	**For Multi-lane configurations:** Loss of signal threshold on Lane 1 This field drives the output port xpcs_rx1_los_threshold_o[2:0]. In single-lane configurations, this field is reserved. **For Single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0 : E16 ? 0x1 : SNPS_RSVDPARAM_39 ? 0x5 : 0x3" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
2:0	LOS_TRSHLD_0	R/W	Loss of signal threshold on lane 0 of PHY. This field drives the output port xpcs_rx0_los_threshold_o[2:0]. Threshold voltages for various values are as follows: ■ 3'b000 : Reserved ■ 3'b001 : 90 mVpp ■ 3'b010 : 120 mVpp ■ 3'b011 : 150 mVpp ■ 3'b100 : 180 mVpp ■ 3'b101 : 210 mVpp ■ 3'b110 : 240 mVpp ■ 3'b111 : 270 mVpp **Value After Reset:** "E16 ? 0x1 : SNPS_RSVDPARAM_39 ? 0x5 : 0x3 " **Exists:** Always

5.2.3.43 VR_XS_PMA_MP_25G_RX_GENCTRL3

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY Rx General Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0033 : (APB3_16) ? (0x0033*0x2) : (0x0033*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (XGXS_PCS||PMA_MMD)&&E25&&!SNPS_RSVDPARAM_37

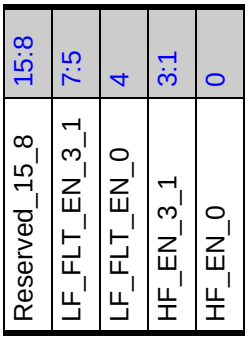


Table 5-173 Fields for Register: VR_XS_PMA_MP_25G_RX_GENCTRL3

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	LF_FLT_EN_3_1	* Varies	Receiver LFPS Signal Detection Filter Enable on lanes 3-1 This field drives the xpcs_rx_sigdet_lf_flt_en_o[3:1] port. This field is valid only for multi-lane configurations. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SINGLE_LANE_CFG ? \"read-only\" : \"read-write\""
4	LF_FLT_EN_0	R/W	Receiver LFPS Signal Detection Filter Enable on lane 0 This field drives the xpcs_rx_sigdet_lf_flt_en_o[0] port. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
3:1	HF_EN_3_1	* Varies	High frequency Signal Detect Enable on lanes 3-1 This field drives the xpcs_rx_sigdet_hf_en_o[3:1] port. This field is valid only for multi-lane configurations. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SINGLE_LANE_CFG ? \"read-only\" : \"read-write\""
0	HF_EN_0	R/W	High frequency Signal Detect Enable on lane 0 This field drives the xpcs_rx_sigdet_hf_en_o[0] port. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.44 VR_XS_PMA_Consumer_10G_RX_GENCTRL3

- **Description:** VR XS or PMA Synopsys Consumer 10G PHY Rx General Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0033 : (APB3_16) ? (0x0033*0x2) : (0x0033*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(SNPS_RSVDPARAM_22))

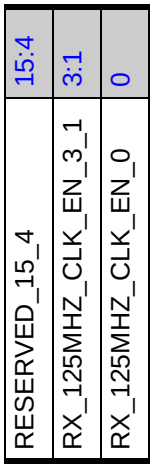


Table 5-174 Fields for Register: VR_XS_PMA_Consumer_10G_RX_GENCTRL3

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
3:1	RX_125MHZ_CLK_EN_3_1	* Varies	**For SNPS_RSVDPARAM_35 Gen2 PHY without USXGMII support** Reserved **For SNPS_RSVDPARAM_35 Gen 2 PHY with USXGMII support** Rx 125Mhz Clock Generation Enable on Lanes 3-1 When this bit is set 125Mhz recovered clock is generated by PHY.This field drives the output port xpcs_rx_125mhz_clk_en_o. For single lane configurations, this field is reserved. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!SNPS_RSVDPARAM_2&&!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
0	RX_125MHZ_CLK_EN_0	* Varies	**For SNPS_RSVDPARAM_35 Gen2 PHY without USXGMII support** Reserved **For SNPS_RSVDPARAM_35 Gen 2 PHY with USXGMII support** Rx 125Mhz Clock Generation Enable on Lane 1. When this bit is set 125Mhz recovered clock is generated by PHY. This bit drives the output port xpcs_rx_125mhz_clk_en_o. **Value After Reset:** "!SNPS_RSVDPARAM_2&&USXGMII_EN ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "!SNPS_RSVDPARAM_2 ? \"read-write\" : \"read-only\""

5.2.3.45 VR_XS_PMA_MP_12G_16G_25G_RX_RATE_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Rx Rate Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0034 : (APB3_16) ? (0x0034*0x2) : (0x0034*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25)&&!SNPS_RSVD-PARAM_37)

15	x:12	11	x:8	7	x:4	3	x:0
Reserved_15	RX3_RATE	Reserved_11	RX2_RATE	Reserved_7	RX1_RATE	Reserved_3	RX0_RATE

Table 5-175 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_RX_RATE_CTRL

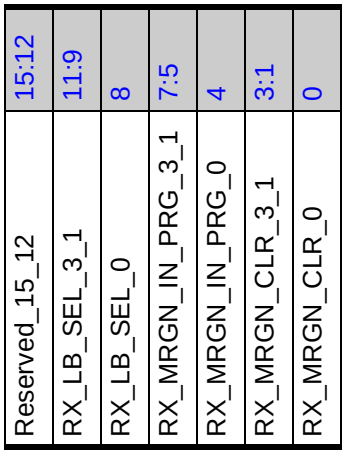
Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
x:12	RX3_RATE	* Varies	**For Multi-lane configurations:** Rx date rate on lane 3 of the PHY **For Single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0 : RXAUI_DEFAULT ? 0x1 : RPCS_PRESENT ? (E25 ? 0x1 : 0x0) : SNPS_RSVDPARAM_39 ? 0x1 : 0x2" **Exists:** Always **Range Variable[x]:** "E25 ? 3 : 2" + 11 **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
x:8	RX2_RATE	* Varies	**For Multi-lane configurations:** Rx date rate on lane 2 of the PHY **For Single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0 : RXAUI_DEFAULT ? 0x1 : RPCS_PRESENT ? (E25 ? 0x1 : 0x0) : SNPS_RSVDPARAM_39 ? 0x1 : 0x2" **Exists:** Always **Range Variable[x]:** "E25 ? 3 : 2" + 7 **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
x:4	RX1_RATE	* Varies	**For Multi-lane configurations:** Rx date rate on lane 1 of the PHY **For Single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0 : RXAUI_DEFAULT ? 0x1 : RPCS_PRESENT ? (E25 ? 0x1 : 0x0) : SNPS_RSVDPARAM_39 ? 0x1 : 0x2" **Exists:** Always **Range Variable[x]:** "E25 ? 3 : 2" + 3 **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
3	Reserved_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
x:0	RX0_RATE	R/W	Rx date rate on lane 0 of the PHY Data Rate Encoding for multi-protocol 25G PHY is as follows: ■ 3'b000: baud ■ 3'b001: baud/2 ■ 3'b010: baud/4 ■ 3'b011: baud/8 ■ 3'b100: baud/16 Data Rate Encoding for multi-protocol 12G/16G PHY is as follows: ■ 2'b00: baud ■ 2'b01: baud/2 ■ 2'b10: baud/4 ■ 2'b11: baud/8 **Value After Reset:** "RPCS_PRESENT ? (E25 ? 0x1 : 0x0) : RXAUI_DEFAULT ? 0x1 : KX_ONLY ? (E25 ? 0x4 : (SNPS_RSVDPARAM_39) ? 0x2 : 0x3) : 0x2" **Exists:** Always **Range Variable[x]:** "E25 ? 3 : 2" - 1

5.2.3.46 VR_XS_PMA_MP_32G_RX_MRGN_CTRL4

- **Description:** VR XS or PMA Synopsys Multi protocol E32 PHY Rx Margin Control 4 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0034 : (APB3_16) ? (0x0034*0x2) : (0x0034*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(SNPS_RSVDPARAM_37||E32))



15:12	11:9	8	7:5	4	3:1	0
Reserved_15_12	RX_LB_SEL_3_1	RX_LB_SEL_0	RX_MRGN_IN_PRG_3_1	RX_MRGN_IN_PRG_0	RX_MRGN_CLR_3_1	RX_MRGN_CLR_0

Table 5-176 Fields for Register: VR_XS_PMA_MP_32G_RX_MRGN_CTRL4

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:9	RX_LB_SEL_3_1	* Varies	**For SNPS_RSVDPARAM_37 Multi-lane configurations:** RX Loopback path select on lane 1-3 of the PHY This signal is connected to xpcs_rx_lb_sel_o[3:1] output port. **For E32 or SNPS_RSVDPARAM_37 Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!(SINGLE_LANE_CFG||E32)) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
8	RX_LB_SEL_0	* Varies	**SNPS_RSVDPARAM_37 configurations:** RX Loopback path select on lane 0 of the PHY This signal is connected to xpcs_rx_lb_sel_o[0] output port. **E32 configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SNPS_RSVDPARAM_37 ? \"read-write\" : \"read-only\""
7:5	RX_MRGN_IN_PRG_3_1	* Varies	**For Multi-lane configurations:** Receiver margining in progress on lane 1-3 of the PHY This signal is connected to xpcs_rx_margin_in_prg_o[3:1] output port. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
4	RX_MRGN_IN_PRG_0	R/W	Receiver margining in progress on lane 0 of the PHY This signal is connected to xpcs_rx_margin_in_prg_o[0] output port. **Value After Reset:** 0x0 **Exists:** Always
3:1	RX_MRGN_CLR_3_1	* Varies	**For Multi-lane configurations:** Clear signal for RX margining on lane 1-3 of the PHY This signal is connected to xpcs_rx_margin_clr_o[3:1] output port. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
0	RX_MRGN_CLR_0	R/W	Clear signal for RX margining on lane 0 of the PHY This signal is connected to xpcs_rx_margin_clr_o[0] output port. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.47 VR_XS_PMA_MP_12G_16G_25G_RX_POWER_STATE_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Rx Power State Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0035 : (APB3_16) ? (0x0035*0x2) : (0x0035*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25||E32))

15:13	12	11:9	8	7:6	5:4	3:2	1:0
Reserved_15_12	EEE_OVR_RIDE	RX_DISABLE_3_1	RX_DISABLE_0	RX3_PSTATE	RX2_PSTATE	RX1_PSTATE	RX0_PSTATE

Table 5-177 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_RX_POWER_STATE_CTRL

Bits	Name	Memory Access	Description
15:13	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	EEE_OVR_RIDE	* Varies	**For configurations with EEE:** Rx Power state override control during EEE DWC_xpcs switches Rx power state to P0s by itself during Rx QUIET state of EEE to save power in the PHY. This bit can be set to 1, to override this behavior of DWC_xpcs. If this bit is set, Rx power state remains unchanged during Rx EEE and retain the values as programmed in 'RX{lane}_PSTATE' fields of this register. **For configurations without EEE:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\" "

Bits	Name	Memory Access	Description
11:9	RX_DISABLE_3_1	* Varies	**For Multi-lane configurations:** Receiver Disable on lanes [3:1] These bits can be set in P1 power state to put the receiver in a low-power mode. This field drives the output port 'xpcs_rx_disable_o[3:1]'. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\" "
8	RX_DISABLE_0	R/W	Receiver Disable on lane 0 This bit can be set in P1 power state to put the receiver in a low-power mode. This field drives the output port 'xpcs_rx_disable_o[0]'. **Value After Reset:** 0x0 **Exists:** Always
7:6	RX3_PSTATE	* Varies	**For Multi-lane configurations:** Rx power state control for lane 3 of PHY **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\" "
5:4	RX2_PSTATE	* Varies	**For Multi-lane configurations:** Rx power state control for lane 2 of PHY **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\" "
3:2	RX1_PSTATE	* Varies	**For Multi-lane configurations:** Rx power state control for lane 1 of PHY **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\" "

Bits	Name	Memory Access	Description
1:0	RX0_PSTATE	R/W	Rx power state control for lane 0 of PHY Power state encoding is as follows: ■ 2'b00: P0 ■ 2'b01: P0s ■ 2'b10: P1 ■ 2'b11: P2 **Value After Reset:** 0x0 **Exists:** Always

5.2.3.48 VR_XS_PMA_MP_12G_16G_25G_RX_CDR_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Rx CDR Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0036 : (APB3_16) ? (0x0036*0x2) : (0x0036*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25||E32))

15:12	11:9	8	7:5	4	3:1	0
Reserved_15_12	VCO_LOW_FREQ_3_1	VCO_LOW_FREQ_0	CDR_SSC_EN_3_1	CDR_SSC_EN_0	CDR_TRACK_EN_3_1	CDR_TRACK_EN_0

Table 5-178 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_RX_CDR_CTRL

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:9	VCO_LOW_FREQ_3_1	* Varies	**For Multi-protocol 12G PHY Multi-lane configurations:** Rx VCO lower frequency band mode on lanes [3:1] of the PHY This field controls the frequency of the Rx VCO to a lower-frequency operating band. This field drives the output port xpcs_rx_cdr_vco_lowfreq_o[3:1]. This field is present only for multi-lane configurations. **For other configurations:** Reserved **Value After Reset:** "!(SINGLE_LANE_CFG||E16||E25||SNPS_RSVDPARAM_39||E32) ? 0x7 : 0x0" **Exists:** Always **Memory Access:** "!(SINGLE_LANE_CFG||E16||E25||E32) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
8	VCO_LOW_FREQ_0	* Varies	**For Multi-protocol 12G PHY configurations:** Rx VCO lower frequency band mode on lane 0 of the PHY This field controls the frequency of the Rx VCO to a lower-frequency operating band. This field drives the output port xpcs_rx_cdr_vco_lowfreq_o[0]. **For Other configurations:** Reserved **Value After Reset:** "(E16||SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_22||E25||SNPS_RSVDPARAM_39||SNPS_RSVDPARAM_33||E32) ? 0x0 : 0x1" **Exists:** Always **Memory Access:** "E12 ? \"read-write\" : \"read-only\""
7:5	CDR_SSC_EN_3_1	* Varies	**For Multi-lane configurations:** Rx CDR SSC-Mode Enable on lanes [3:1] of the PHY. This field controls the CDR tracking gains and duration. Set this bit to 3'b111 when receive data (on lanes 3 to 1) has a spread spectrum clock. Clear this bit if receive data does not have SSC. This bit drives the output port 'xpcs_rx_cdr_ssc_en_o[3:1]'. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
4	CDR_SSC_EN_0	R/W	Rx CDR SSC Mode Enable on lane 0 of the PHY This field controls the CDR tracking gains and duration. Set this bit to 1 when receive data has a spread spectrum clock.Clear this bit if receive data does not have SSC. This bit drives the output port 'xpcs_rx_cdr_ssc_en_o[0]'. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
3:1	CDR_TRACK_EN_3_1	* Varies	**For Multi-protocol 12G PHY Multi-lane configurations:** Rx CDR Tracking Enable on lanes [3:1] of the PHY Set this field to 3'b111 to enable CDR tracking of receive data on lanes 3 to 1 of the PHY. This bit drives the output port 'xpcs_rx_cdr_track_en_o[3:1]'. This field is present only for multi-lane configurations. **For other configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG||E16||E25||E32) ? 0x0 : 0x7" **Exists:** Always **Memory Access:** "!(SINGLE_LANE_CFG||E16||E25||E32) ? \"read-write\" : \"read-only\""
0	CDR_TRACK_EN_0	* Varies	**For Multi-protocol 12G PHY configurations:** Rx CDR Tracking Enable on lane 0 of the PHY Set this bit to enable CDR tracking of receive data on lane 0 of the PHY. This bit drives the output port 'xpcs_rx_cdr_track_en_o[0]'. **For other configurations:** Reserved **Value After Reset:** "E12 ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "E12 ? \"read-write\" : \"read-only\""

5.2.3.49 VR_XS_PMA_MP_12G_16G_25G_RX_ATTN_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Rx Attenuation Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0037 : (APB3_16) ? (0x0037*0x2) : (0x0037*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25)&&!SNPS_RSVD-PARAM_37)

15	14:12	11	10:8	7	6:4	3	2:0
Reserved_15	RX3_EQ_ATT_LVL	Reserved_11	RX2_EQ_ATT_LVL	Reserved_7	RX1_EQ_ATT_LVL	Reserved_3	RX0_EQ_ATT_LVL

Table 5-179 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_RX_ATTN_CTRL

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:12	RX3_EQ_ATT_LVL	* Varies	**For Multi-lane configurations:** Rx Equalization Attenuation level for lane 3 of the PHY. This field drives the output port xpcs_rx3_eq_att_lvl_o[2:0]. This field controls the AFE attenuation level of the PHY. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
10:8	RX2_EQ_ATT_LVL	* Varies	**For Multi-lane configurations:** Rx Equalization Attenuation level for lane 2 of the PHY. This field drives the output port xpcs_rx2_eq_att_lvl_o[2:0]. This field controls the AFE attenuation level of the PHY. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:4	RX1_EQ_ATT_LVL	* Varies	**For Multi-lane configurations:** Rx Equalization Attenuation level for lane 1 of the PHY This field drives the output port xpcs_rx1_eq_att_lvl_o[2:0]. This field controls the AFE attenuation level of the PHY. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
3	Reserved_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	RX0_EQ_ATT_LVL	R/W	Rx Equalization Attenuation level for lane 0 of the PHY This field drives the output port xpcs_rx0_eq_att_lvl_o[2:0]. This field controls the AFE attenuation level of the PHY. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.50 VR_XS_PMA_MP_6G_RXGENCTRL0

- **Description:** VR XS or PMA Synopsys Multi-protocol 6G PHY Rx General Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0038 : (APB3_16) ? (0x0038*0x2) : (0x0038*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&SSP)

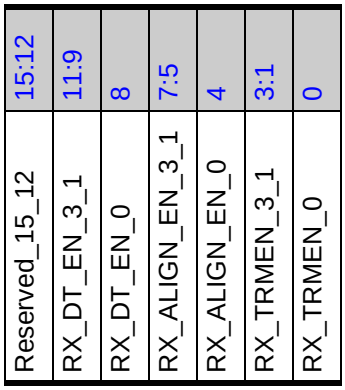


Table 5-180 Fields for Register: VR_XS_PMA_MP_6G_RXGENCTRL0

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:9	RX_DT_EN_3_1	* Varies	**For Multi-lane configurations:** Receiver Clock and Data Output Enable on Lanes[3:1] This signal drives the xpcs_rx_en_o[3:1] output port of DWC_xpcs. When set to 1, this field sets the corresponding RX_PLL_EN[] and RX_TERM_EN[] signals inside the PHY. ■ RX_DT_EN_3_1[1]: Enables receiver clock and data output on Lane 1 ■ RX_DT_EN_3_1[2]: Enables receiver clock and data output on Lane 2 ■ RX_DT_EN_3_1[3]: Enables receiver clock and data output on Lane 3 **For Single-lane configurations:** Reserved **Value After Reset:** "(!SINGLE_LANE_CFG) ? 0x7 : 0x0" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
8	RX_DT_EN_0	R/W	Receiver Clock and Data Output Enable on Lane 0 This bit enables receiver clock and data output on Lane 0. **Value After Reset:** 0x1 **Exists:** Always
7:5	RX_ALIGN_EN_3_1	* Varies	**For Multi-lane configurations:** Receiver Alignment Enable on Lanes[3:1] When set to 1, this field enables the 10-bit word alignment based on the COMMA character. You can set this field only when the corresponding RX_DATA_EN[] signal is 0. ■ RX_ALIGN_EN_3_1[1]: Enables receiver alignment on Lane 1 ■ RX_ALIGN_EN_3_1[2]: Enables receiver alignment on Lane 2 ■ RX_ALIGN_EN_3_1[3]: Enables receiver alignment on Lane 3 **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
4	RX_ALIGN_EN_0	R/W	Receiver Alignment Enable on Lane 0 When set to 1, this field enables the 10-bit word alignment on Lane 0 based on the COMMA character. You can set this field only when the corresponding RX_DATA_EN[] signal is 0. **Value After Reset:** 0x0 **Exists:** Always
3:1	RX_TRMEN_3_1	* Varies	**For Multi-lane configurations:** Receiver Termination Enable on Lanes[3:1] When set to 1, this field enables the 100 Ohm termination on rx{lane}_p and rx{lane}_m. ■ RX_TRMEN_3_1[1]: Enables receiver termination enable on Lane 1 ■ RX_TRMEN_3_1[2]: Enables receiver termination enable on Lane 2 ■ RX_TRMEN_3_1[3]: Enables receiver termination enable on Lane 3 **For Single-lane configurations:** Reserved **Value After Reset:** "(!SINGLE_LANE_CFG) ? 0x7 : 0x0" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
0	RX_TRMEN_0	R/W	Receiver Termination Enable on Lane 0 When set to 1, this bit enables the 100 Ohm termination on rx0_p and rx0_m. **Value After Reset:** 0x1 **Exists:** Always

5.2.3.51 VR_XS_PMA_MP_12G_RX_EQ_CTRL0

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G PHY Rx Equalization Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0038 : (APB3_16) ? (0x0038*0x2) : (0x0038*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&E12)

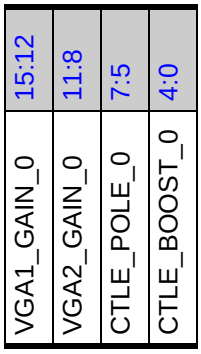


Table 5-181 Fields for Register: VR_XS_PMA_MP_12G_RX_EQ_CTRL0

Bits	Name	Memory Access	Description
15:12	VGA1_GAIN_0	R/W	Rx Equalization VGA1 Gain on lane 0 of the PHY This field drives the output port xpcs_rx0_eq_vga1_gain_o[3:0]. This field controls the AFE first stage Variable Gain amplifier gain. **Value After Reset:** "SNPS_RSVDPARAM_39 ? 0x3 : 0x7" **Exists:** !(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_33)
11:8	VGA2_GAIN_0	R/W	Rx Equalization VGA2 Gain on lane 0 of the PHY This field drives the output port xpcs_rx0_eq_vga2_gain_o[3:0].This field controls the AFE second stage Variable Gain amplifier gain. **Value After Reset:** "SNPS_RSVDPARAM_39 ? 0x3 : 0x7" **Exists:** !(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_33)

Bits	Name	Memory Access	Description
7:5	CTLE_POLE_0	R/W	Rx Equalization CTLE Pole Value on lane 0 of the PHY This field drives the output port xpcs_rx0_eq_ctle_pole_o[2:0].This field controls the continuous time linear equalizer boost pole location of the PHY. **Value After Reset:** "(SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_39) ? 0x0 : RPCS_PRESENT ? 0x6 : 0x5" **Exists:** !(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_33)
4:0	CTLE_BOOST_0	R/W	Rx Equalization CTLE Boost value on lane 0 of the PHY This field drives the output port xpcs_rx0_eq_ctle_boost_o[4:0].This field controls the CTLE boost level. **Value After Reset:** "(SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_33) ? 0x7 : SNPS_RSVDPARAM_39 ? 0x6 : RPCS_PRESENT ? 0xa : 0x6" **Exists:** Always

5.2.3.52 VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL0

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G/25G PHY Rx Equalization Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0038 : (APB3_16) ? (0x0038*0x2) : (0x0038*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||E25)&&!SNPS_RSVDPARAM_37

15	14:12	11	10:8	7	6:5	4:0
Reserved_15	VGA1_GAIN_0	Reserved_11	VGA2_GAIN_0	Reserved_7	CTLE_POLE_0	CTLE_BOOST_0

Table 5-182 Fields for Register: VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL0

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:12	VGA1_GAIN_0	R/W	Rx Equalization VGA1 Gain on lane 0 of the PHY This field drives the output port xpcs_rx0_eq_vga1_gain_o[3:0]. This field controls the AFE first stage Variable Gain amplifier gain. **Value After Reset:** "SNPS_RSVDPARAM_36 ? (RPCS_PRESENT ? 0x6 : RXAUI_DEFAULT ? 0x7 : (KX_ONLY||BASE1000X_ONLY) ? 0x4 : 0x5) : E25&&RPCS_PRESENT ? ((SNPS_RSVDPARAM_16||SNPS_RSVDPARAM_36) ? 0x6 : 0x5) : E16&&RPCS_PRESENT ? 0x5 : 0x4" **Exists:** Always
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
10:8	VGA2_GAIN_0	R/W	Rx Equalization VGA2 Gain on lane 0 of the PHY This field drives the output port xpcs_rx0_eq_vga2_gain_o[3:0].This field controls the AFE second stage Variable Gain amplifier gain. **Value After Reset:** "SNPS_RSVDPARAM_36 ? (RPCS_PRESENT ? 0x6 : RXAUI_DEFAULT ? 0x7 : (KX_ONLY||BASE1000X_ONLY) ? 0x4 : 0x5) : E25&&RPCS_PRESENT ? (SNPS_RSVDPARAM_16 ? 0x6 : 0x5) : E16&&RPCS_PRESENT ? 0x5 : 0x4" **Exists:** Always
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:5	CTLE_POLE_0	R/W	Rx Equalization CTLE Pole Value on lane 0 of the PHY This field drives the output port xpcs_rx0_eq_ctle_pole_o[1:0].This field controls the continuous time linear equalizer boost pole location of the PHY. **Value After Reset:** "((E25&&!SNPS_RSVDPARAM_16&&!SNPS_RSVDPARAM_36)||(E16&&SNPS_RSVDPARAM_15||((E16||SNPS_RSVDPARAM_36)&&!RPCS_PRESENT))) ? 0x0 : 0x2" **Exists:** Always
4:0	CTLE_BOOST_0	R/W	Rx Equalization CTLE Boost value on lane 0 of the PHY This field drives the output port xpcs_rx0_eq_ctle_boost_o[4:0].This field controls the CTLE boost level. **Value After Reset:** "SNPS_RSVDPARAM_36 ? (RPCS_PRESENT ? 0xE : RXAUI_DEFAULT ? 0x14 : (KX_ONLY||BASE1000X_ONLY) ? 0xF : 0x1F) : SNPS_RSVDPARAM_16 ? 0xe : (E25||E16)&&RPCS_PRESENT ? 0x10 : E25&&RXAUI_DEFAULT ? 0x8 : (E25||(E16&&SNPS_RSVDPARAM_15)) ? 0x6 : E16&&RXAUI_DEFAULT ? 0xE : 0x12" **Exists:** Always

5.2.3.53 VR_XS_PMA_MP_32G_RX_MRGN_CTRL0

- **Description:** VR XS or PMA Synopsys Multi protocol E32 PHY Rx Margin Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0038 : (APB3_16) ? (0x0038*0x2) : (0x0038*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(SNPS_RSVDPARAM_37||E32)

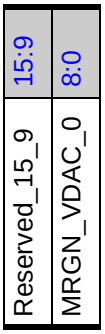


Table 5-183 Fields for Register: VR_XS_PMA_MP_32G_RX_MRGN_CTRL0

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8:0	MRGN_VDAC_0	R/W	Value for RX voltage margining along lane 0 of the PHY This field drives the output port xpcs_rx0_margin_vdac_o[8:0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.54 VR_XS_PMA_MP_6G_RXGENCTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 6G PHY Rx General Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0039 : (APB3_16) ? (0x0039*0x2) : (0x0039*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&SSP)

15:13	12	11:9	8	7:5	4	3:1	0
RX_LOS_EN_3_1	RX_LOS_EN_0	RX_PLL_EN_3_1	RX_PLL_EN_0	RX_RST_3_1	RX_RST_0	RX_INV_3_1	RX_INV_0

Table 5-184 Fields for Register: VR_XS_PMA_MP_6G_RXGENCTRL1

Bits	Name	Memory Access	Description
15:13	RX_LOS_EN_3_1	* Varies	**For Multi-lane configurations:** Receiver Loss-of-Signal Enable on Lanes[3:1] When set to 1, this field enables the receiver loss-of-signal detection. ■ RX_LOS_EN_3_1[1]: Enables receiver loss-of-signal detection on Lane 1 ■ RX_LOS_EN_3_1[2]: Enables receiver loss-of-signal detection on Lane 2 ■ RX_LOS_EN_3_1[3]: Enables receiver loss-of-signal detection on Lane 3 **For Single-lane configurations:** Reserved **Value After Reset:** "(!SINGLE_LANE_CFG) ? 0x7 : 0x0" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
12	RX_LOS_EN_0	R/W	Receiver Loss-of-Signal Enable on Lane 0 When set to 1, this bit enables the receiver loss-of-signal detection on Lane 0. **Value After Reset:** 0x1 **Exists:** Always
11:9	RX_PLL_EN_3_1	* Varies	**For Multi-lane configurations:** Receiver DPLL Enable on Lanes[3:1] When set to 1, this field enables the receiver DPLL on corresponding lane. Otherwise, it disables the receiver DPLL. ■ RX_PLL_EN_3_1[1]: Enables the receiver DPLL on Lane 1 ■ RX_PLL_EN_3_1[2]: Enables the receiver DPLL on Lane 2 ■ RX_PLL_EN_3_1[3]: Enables the receiver DPLL on Lane 3 **For Single-lane configurations:** Reserved **Value After Reset:** "(!SINGLE_LANE_CFG) ? 0x7 : 0x0" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
8	RX_PLL_EN_0	R/W	Receiver DPLL Enable on Lane 0 When set to 1, this field enables the receiver DPLL on Lane 0. Otherwise, it disables the receiver DPLL. **Value After Reset:** 0x1 **Exists:** Always
7:5	RX_RST_3_1	* Varies	**For Multi-lane configurations:** Receiver Reset on Lanes[3:1] When set to 1, this field resets the PHY receiver functionality, including RX_DATA_EN[] and RX_PLL_EN[] signals inside the PHY. ■ RX_RST_3_1[1]: Resets receiver on Lane 1 ■ RX_RST_3_1[2]: Resets receiver on Lane 2 ■ RX_RST_3_1[3]: Resets receiver on Lane 3 **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
4	RX_RST_0	R/W	Receiver Reset on Lane 0 When set to 1, this field resets the PHY receiver functionality including RX_DATA_EN[] and RX_PLL_EN[] signals inside the PHY. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
3:1	RX_INV_3_1	* Varies	**For Multi-lane configurations:** Receiver Invert on Lanes[3:1] When set to 1, this field inverts the logic levels of rx{lane}_p and rx{lane}_m. ■ RX_INV_3_1[1]: Inverts the logic level of rx1_p and rx1_m ■ RX_INV_3_1[2]: Inverts the logic level of rx2_p and rx2_m ■ RX_INV_3_1[3]: Inverts the logic level of rx3_p and rx3_m **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
0	RX_INV_0	R/W	Receiver Invert on Lane 0 When set to 1, this bit inverts the logic levels of rx0_p and rx0_m. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.55 VR_XS_PMA_MP_12G_RX_EQ_CTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G PHY Rx Equalization Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0039 : (APB3_16) ? (0x0039*0x2) : (0x0039*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&E12&&!SINGLE_LANE_CFG)

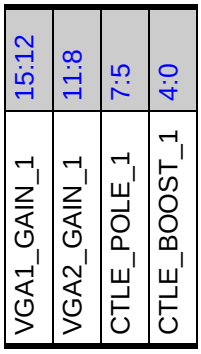


Table 5-185 Fields for Register: VR_XS_PMA_MP_12G_RX_EQ_CTRL1

Bits	Name	Memory Access	Description
15:12	VGA1_GAIN_1	R/W	Rx Equalization VGA1 Gain on lane 1 of the PHY This field drives the output port xpcs_rx1_eq_vga1_gain_o[3:0]. This field controls the AFE first stage Variable Gain amplifier gain. **Value After Reset:** "SNPS_RSVDPARAM_39 ? 0x3 : 0x7" **Exists:** !(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33)
11:8	VGA2_GAIN_1	R/W	Rx Equalization VGA2 Gain on lane 1 of the PHY This field drives the output port xpcs_rx1_eq_vga2_gain_o[3:0]. This field controls the AFE second stage Variable Gain amplifier gain. **Value After Reset:** "SNPS_RSVDPARAM_39 ? 0x3 : 0x7" **Exists:** !(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33)
7:5	CTLE_POLE_1	R/W	Rx Equalization CTLE Pole Value on lane 1 of the PHY This field drives the output port xpcs_rx1_eq_ctle_pole_o[2:0]. This field controls the continuous time linear equalizer boost pole location of the PHY **Value After Reset:** "(SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_35) ? 0x0 : RPCS_PRESENT ? 0x6 : 0x5" **Exists:** !(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33)

Bits	Name	Memory Access	Description
4:0	CTLE_BOOST_1	R/W	Rx Equalization CTLE Boost value on lane 1 of the PHY This field drives the output port xpcs_rx1_eq_ctle_boost_o[4:0]. This field controls the CTLE boost level. **Value After Reset:** "(SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_33) ? 0x7 : SNPS_RSVDPARAM_39 ? 0x6 : RPCS_PRESENT ? 0xa : 0x6" **Exists:** Always

5.2.3.56 VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G/25G PHY Rx Equalization Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0039 : (APB3_16) ? (0x0039*0x2) : (0x0039*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||E25)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37

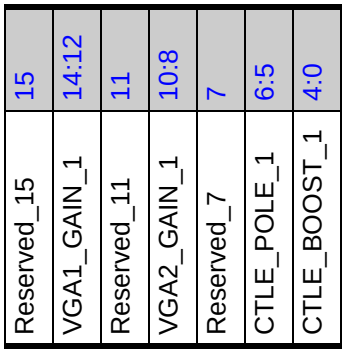


Table 5-186 Fields for Register: VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL1

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:12	VGA1_GAIN_1	R/W	Rx Equalization VGA1 Gain on lane 1 of the PHY This field drives the output port xpcs_rx1_eq_vga1_gain_o[3:0]. This field controls the AFE first stage Variable Gain amplifier gain. **Value After Reset:** "SNPS_RSVDPARAM_36 ? (RPCS_PRESENT ? 0x6 : RXAUI_DEFAULT ? 0x7 : 0x5) : (RPCS_PRESENT) ? 0x5 : 0x4" **Exists:** Always
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
10:8	VGA2_GAIN_1	R/W	Rx Equalization VGA2 Gain on lane 1 of the PHY This field drives the output port xpcs_rx1_eq_vga2_gain_o[3:0]. This field controls the AFE second stage Variable Gain amplifier gain. **Value After Reset:** "SNPS_RSVDPARAM_36 ? (RPCS_PRESENT ? 0x6 : RXAUI_DEFAULT ? 0x7 : 0x5) : (RPCS_PRESENT) ? 0x5 : 0x4" **Exists:** Always
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:5	CTLE_POLE_1	R/W	Rx Equalization CTLE Pole Value on lane 1 of the PHY This field drives the output port xpcs_rx1_eq_ctle_pole_o[2:0]. This field controls the continuous time linear equalizer boost pole location of the PHY **Value After Reset:** "(E16&&RPCS_PRESENT||SNPS_RSVDPARAM_36) ? 0x2 : 0x0" **Exists:** Always
4:0	CTLE_BOOST_1	R/W	Rx Equalization CTLE Boost value on lane 1 of the PHY This field drives the output port xpcs_rx1_eq_ctle_boost_o[4:0]. This field controls the CTLE boost level. **Value After Reset:** "SNPS_RSVDPARAM_36 ? (RPCS_PRESENT ? 0xE : RXAUI_DEFAULT ? 0x14 : 0x1F) : (E25||E16)&&RPCS_PRESENT ? 0x10 : E25&&RXAUI_DEFAULT ? 0x8 : (E25||E16&&SNPS_RSVDPARAM_15) ? 0x6 : E16&&RXAUI_DEFAULT ? 0xE : 0x12" **Exists:** Always

5.2.3.57 VR_XS_PMA_MP_32G_RX_MRGN_CTRL1

- **Description:** VR XS or PMA Synopsys Multi protocol E32 PHY Rx Margin Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0039 : (APB3_16) ? (0x0039*0x2) : (0x0039*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(SNPS_RSVDPAR-AM_37||E32)&&!SINGLE_LANE_CFG

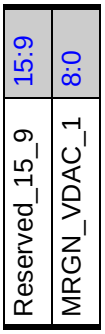


Table 5-187 Fields for Register: VR_XS_PMA_MP_32G_RX_MRGN_CTRL1

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8:0	MRGN_VDAC_1	R/W	Value for RX voltage margining along lane 1 of the PHY This field drives the output port xpcs_rx1_margin_vdac_o[8:0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.58 VR_XS_PMA_MP_12G_RX_EQ_CTRL2

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G PHY Rx Equalization Control 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x003A : (APB3_16) ? (0x003A*0x2) : (0x003A*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&E12&&!SINGLE_LANE_CFG)

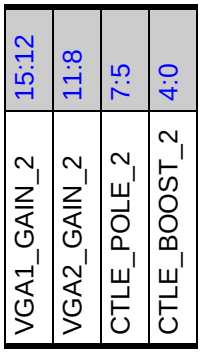


Table 5-188 Fields for Register: VR_XS_PMA_MP_12G_RX_EQ_CTRL2

Bits	Name	Memory Access	Description
15:12	VGA1_GAIN_2	R/W	Rx Equalization VGA1 Gain on lane 2 of the PHY This field drives the output port xpcs_rx2_eq_vga1_gain_o[3:0]. This field controls the AFE first stage Variable Gain amplifier gain. **Value After Reset:** "SNPS_RSVDPARAM_39 ? 0x3 : 0x7" **Exists:** !(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33)
11:8	VGA2_GAIN_2	R/W	Rx Equalization VGA2 Gain on lane 2 of the PHY This field drives the output port xpcs_rx2_eq_vga2_gain_o[3:0]. This field controls the AFE second stage Variable Gain amplifier gain **Value After Reset:** "SNPS_RSVDPARAM_39 ? 0x3 : 0x7" **Exists:** !(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33)
7:5	CTLE_POLE_2	R/W	Rx Equalization CTLE Pole Value on lane 2 of the PHY This field drives the output port 'xpcs_rx2_eq_ctle_pole_o[2:0]' This field controls the continuous time linear equalizer boost pole location of the PHY **Value After Reset:** "(SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_35) ? 0x0 : RPCS_PRESENT ? 0x6 : 0x5" **Exists:** !(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33)

Bits	Name	Memory Access	Description
4:0	CTLE_BOOST_2	R/W	Rx Equalization CTLE Boost value on lane 2 of the PHY This field drives the output port xpcs_rx2_eq_ctle_boost_o[4:0]. This field controls the CTLE boost level. **Value After Reset:** "(SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_33) ? 0x7 : SNPS_RSVDPARAM_39 ? 0x6 : RPCS_PRESENT ? 0xa : 0x6" **Exists:** Always

5.2.3.59 VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL2

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G/25G PHY Rx Equalization Control 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x003A : (APB3_16) ? (0x003A*0x2) : (0x003A*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||E25)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37

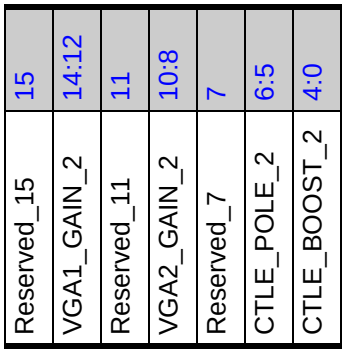


Table 5-189 Fields for Register: VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL2

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:12	VGA1_GAIN_2	R/W	Rx Equalization VGA1 Gain on lane 2 of the PHY This field drives the output port xpcs_rx2_eq_vga1_gain_o[3:0]. This field controls the AFE first stage Variable Gain amplifier gain. **Value After Reset:** "SNPS_RSVDPARAM_36 ? (RPCS_PRESENT ? 0x6 : RXAUI_DEFAULT ? 0x7 : 0x5) : (RPCS_PRESENT) ? 0x5 : 0x4" **Exists:** Always
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
10:8	VGA2_GAIN_2	R/W	Rx Equalization VGA2 Gain on lane 2 of the PHY This field drives the output port xpcs_rx2_eq_vga2_gain_o[3:0]. This field controls the AFE second stage Variable Gain amplifier gain **Value After Reset:** "SNPS_RSVDPARAM_36 ? (RPCS_PRESENT ? 0x6 : RXAUI_DEFAULT ? 0x7 : 0x5) : (RPCS_PRESENT) ? 0x5 : 0x4" **Exists:** Always
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:5	CTLE_POLE_2	R/W	Rx Equalization CTLE Pole Value on lane 2 of the PHY This field drives the output port 'xpcs_rx2_eq_ctle_pole_o[2:0]' This field controls the continuous time linear equalizer boost pole location of the PHY **Value After Reset:** "(E16&&RPCS_PRESENT||SNPS_RSVDPARAM_36) ? 0x2 : 0x0" **Exists:** Always
4:0	CTLE_BOOST_2	R/W	Rx Equalization CTLE Boost value on lane 2 of the PHY This field drives the output port xpcs_rx2_eq_ctle_boost_o[4:0]. This field controls the CTLE boost level. **Value After Reset:** "SNPS_RSVDPARAM_36 ? (RPCS_PRESENT ? 0xE : RXAUI_DEFAULT ? 0x14 : 0x1F) : (E25||E16)&&RPCS_PRESENT ? 0x10 : E25&&RXAUI_DEFAULT ? 0x8 : (E25||E16&&SNPS_RSVDPARAM_15) ? 0x6 : E16&&RXAUI_DEFAULT ? 0xE : 0x12" **Exists:** Always

5.2.3.60 VR_XS_PMA_MP_32G_RX_MRGN_CTRL2

- **Description:** VR XS or PMA Synopsys Multi protocol E32 PHY Rx Margin Control 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x003A : (APB3_16) ? (0x003A*0x2) : (0x003A*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(SNPS_RSVDPAR-AM_37||E32)&&!SINGLE_LANE_CFG

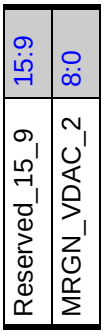


Table 5-190 Fields for Register: VR_XS_PMA_MP_32G_RX_MRGN_CTRL2

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8:0	MRGN_VDAC_2	R/W	Value for RX voltage margining along lane 2 of the PHY This field drives the output port xpcs_rx2_margin_vdac_o[8:0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.61 VR_XS_PMA_MP_6G_RXEQ_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 6G PHY Rx Equalization Control Register All four Rx Equalization fields use the following equation: Internal equalizer boost = ~(rx_eq_val+1)*0.5 dB
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x003B : (APB3_16) ? (0x003B*0x2) : (0x003B*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&SSP)

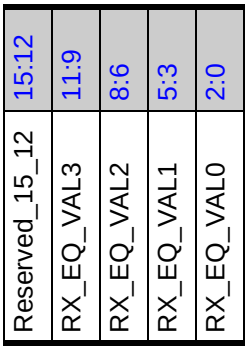


Table 5-191 Fields for Register: VR_XS_PMA_MP_6G_RXEQ_CTRL

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:9	RX_EQ_VAL3	* Varies	**For Multi-lane configurations:**Rx Equalization Control Lane 3 This field controls the internal linear equalizer boost on Lane 3. **For Single-lane configurations:**Reserved **Value After Reset:** "(!SINGLE_LANE_CFG) ? 0x2 : 0x0" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
8:6	RX_EQ_VAL2	* Varies	**For Multi-lane configurations:**Rx Equalization Control Lane 2 This field controls the internal linear equalizer boost on Lane 2. **For Single-lane configurations:**Reserved **Value After Reset:** "(!SINGLE_LANE_CFG) ? 0x2 : 0x0" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\"""
5:3	RX_EQ_VAL1	* Varies	**For Multi-lane configurations:**Rx Equalization Control Lane 1 This field controls the internal linear equalizer boost on Lane 1. **For Single-lane configurations:**Reserved **Value After Reset:** "(!SINGLE_LANE_CFG) ? 0x2 : 0x0" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\"""
2:0	RX_EQ_VAL0	R/W	Rx Equalization Control Lane 0 This field controls the internal linear equalizer boost on Lane 0. **Value After Reset:** 0x2 **Exists:** Always

5.2.3.62 VR_XS_PMA_MP_12G_RX_EQ_CTRL3

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G PHY Rx Equalization Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x003B : (APB3_16) ? (0x003B*0x2) : (0x003B*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&E12&&!SINGLE_LANE_CFG)

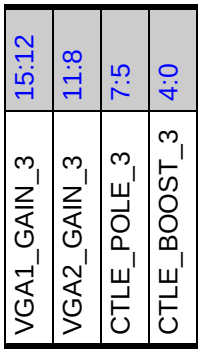


Table 5-192 Fields for Register: VR_XS_PMA_MP_12G_RX_EQ_CTRL3

Bits	Name	Memory Access	Description
15:12	VGA1_GAIN_3	R/W	Rx Equalization VGA1 Gain on lane 3 of the PHY. This field drives the output port xpcs_rx3_eq_vga1_gain_o[3:0]. This field controls the AFE first stage Variable Gain amplifier gain. **Value After Reset:** "SNPS_RSVDPARAM_39 ? 0x3 : 0x7" **Exists:** !(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33)
11:8	VGA2_GAIN_3	R/W	Rx Equalization VGA2 Gain on lane 3 of the PHY This field drives the output port xpcs_rx3_eq_vga2_gain_o[3:0]. This field controls the AFE second stage Variable Gain amplifier gain. **Value After Reset:** "SNPS_RSVDPARAM_39 ? 0x3 : 0x7" **Exists:** !(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33)
7:5	CTLE_POLE_3	R/W	Rx Equalization CTLE Pole Value on lane 3 of the PHY This field drives the output port xpcs_rx3_eq_ctle_pole_o[2:0] This field controls the continuous time linear equalizer boost pole location of the PHY **Value After Reset:** "(SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_35) ? 0x0 : RPCS_PRESENT ? 0x6 : 0x5" **Exists:** !(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33)

Bits	Name	Memory Access	Description
4:0	CTLE_BOOST_3	R/W	Rx Equalization CTLE Boost value on lane 3 of the PHY This field drives the output port xpcs_rx3_eq_ctle_boost_o[4:0]. This field controls the CTLE boost level. **Value After Reset:** "(SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_33) ? 0x7 : SNPS_RSVDPARAM_39 ? 0x6 : RPCS_PRESENT ? 0xa : 0x6" **Exists:** Always

5.2.3.63 VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL3

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G/25G PHY Rx Equalization Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x003B : (APB3_16) ? (0x003B*0x2) : (0x003B*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||E25)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37

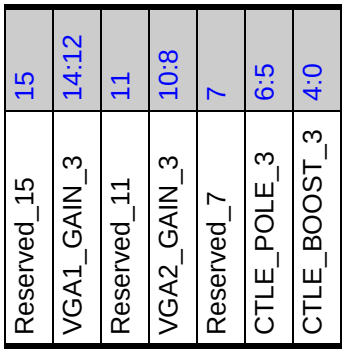


Table 5-193 Fields for Register: VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL3

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:12	VGA1_GAIN_3	R/W	Rx Equalization VGA1 Gain on lane 3 of the PHY. This field drives the output port xpcs_rx3_eq_vga1_gain_o[3:0]. This field controls the AFE first stage Variable Gain amplifier gain. **Value After Reset:** "SNPS_RSVDPARAM_36 ? (RPCS_PRESENT ? 0x6 : RXAUI_DEFAULT ? 0x7 : 0x5) : (RPCS_PRESENT) ? 0x5 : 0x4" **Exists:** Always
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
10:8	VGA2_GAIN_3	R/W	Rx Equalization VGA2 Gain on lane 3 of the PHY This field drives the output port xpcs_rx3_eq_vga2_gain_o[3:0]. This field controls the AFE second stage Variable Gain amplifier gain. **Value After Reset:** "SNPS_RSVDPARAM_36 ? (RPCS_PRESENT ? 0x6 : RXAUI_DEFAULT ? 0x7 : 0x5) : (RPCS_PRESENT) ? 0x5 : 0x4" **Exists:** Always
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:5	CTLE_POLE_3	R/W	Rx Equalization CTLE Pole Value on lane 3 of the PHY. This field drives the output port xpcs_rx3_eq_ctle_pole_o[2:0] This field controls the continuous time linear equalizer boost pole location of the PHY **Value After Reset:** "(E16&&RPCS_PRESENT||SNPS_RSVDPARAM_36) ? 0x2 : 0x0" **Exists:** Always
4:0	CTLE_BOOST_3	R/W	Rx Equalization CTLE Boost value on lane 3 of the PHY. This field drives the output port xpcs_rx3_eq_ctle_boost_o[4:0]. This field controls the CTLE boost level. **Value After Reset:** "SNPS_RSVDPARAM_36 ? (RPCS_PRESENT ? 0xE : RXAUI_DEFAULT ? 0x14 : 0x1F) : (E25||E16)&&RPCS_PRESENT ? 0x10 : E25&&RXAUI_DEFAULT ? 0x8 : (E25||E16&&SNPS_RSVDPARAM_15) ? 0x6 : E16&&RXAUI_DEFAULT ? 0xE : 0x12" **Exists:** Always

5.2.3.64 VR_XS_PMA_MP_32G_RX_MRGN_CTRL3

- **Description:** VR XS or PMA Synopsys Multi protocol E32 PHY Rx Margin Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x003B : (APB3_16) ? (0x003B*0x2) : (0x003B*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(SNPS_RSVDPAR-AM_37||E32)&&!SINGLE_LANE_CFG

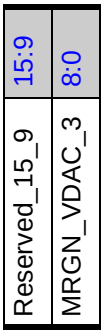


Table 5-194 Fields for Register: VR_XS_PMA_MP_32G_RX_MRGN_CTRL3

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8:0	MRGN_VDAC_3	R/W	Value for RX voltage margining along lane 3 of the PHY This field drives the output port xpcs_rx3_margin_vdac_o[8:0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.65 VR_XS_PMA_MP_12G_16G_25G_RX_EQ_CTRL4

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Rx Equalization Control 4 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x003C : (APB3_16) ? (0x003C*0x2) : (0x003C*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25)&&!SNPS_RSVD-PARAM_37)

15:13	12	11	10	9	8	7:5	4	3:1	0
Reserved_15_13	RX_AD_REQ	RX_EQ_STRT_CTRL	SELF_MAIN_EN	PING_PONG_EN	SEQ_EQ_EN	CONT_OFF_CAN_3_1	CONT_OFF_CAN_0	CONT_ADAPT_3_1	CONT_ADAPT_0

Table 5-195 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_RX_EQ_CTRL4

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
12	RX_AD_REQ	R/W	Receive Adaptation Request This bit drives the 'rpcs_krx_adapt_req_o' port. For configurations with CL72_EN = Enabled, this bit drives the 'rpcs_krx_adapt_req_o' port only when KR training is disabled (TR_EN bit of SR_PMA_KR_PMD_CTRL Register is 0). Software can set this bit to make an Receiver Adaptation Request to the PHY.The acknowledgment from the PHY (for the adaptation request made) is reflected in bit[8]/bit[9]/bit[10] of VR_PMA_PHY_RX_EQ_CEU Register (for configurations with CL72_EN = Enabled) or bit[12] (RX_ADPT_ACK) of VR_XS_PMA_MP_12G_16G_25G_MISC_STS Register (for configurations with CL72_EN = Disabled). After receiving the acknowledgment, software must clear this register field. **Value After Reset:** 0x0 **Exists:** Always
11	RX_EQ_STRT_CTRL	* Varies	**For configurations with CL72_EN = Enabled:** Receive Equalization Start Control This bits controls the behavior of DWC_xpcs at the start of receive equalization process. This bit decides whether DWC_xpcs sends PRESET or INIT as the first coefficient update request to the link-partner when KR Training is enabled. There after, DWC_xpcs sends INC/DEC as provided by the PHY. **Value:** ■ 0: DWC_xpcs sends INIT first ■ 1: DWC_xpcs sends PRESET first This bit is present only in configurations with CL72_EN Enabled. **For configurations with CL72_EN = Disabled:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "CL72_EN ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
10	SELF_MAIN_EN	* Varies	**For configurations with CL72_EN = Enabled:** Enable Automatic Generation of Main -C(0) If this bit is set, DWC_xpcs generates C(0) (Main) coefficient update requests by itself during Rx equalization process and ignores the update request given by PHY for C(0). This bit can be set to 1,when Ping Pong Method is selected for Rx equalization. If PRE/POST is a INC request, DWC_xpcs generates a INC request for MAIN. If PRE/POST is a DEC request, DWC_xpcs generates a INC request for MAIN. This bit is present only in configurations with CL72_EN Enabled. **For configurations with CL72_EN = Disabled:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "CL72_EN ? \"read-write\" : \"read-only\""
9	PING_PONG_EN	* Varies	**For configurations with CL72_EN = Enabled:** Ping-Pong Mode Enable for Rx equalization In ping-pong method, only one of PRE/POST update request is sent to remote-end device during one cycle of Rx equalization and the other coefficient update is ignored. DWC_xpcs alternates (ping-pongs) between PRE and POST requests between iterations. This method works well 'SEQ_EQ_EN' and 'SELF_MAIN_EN' bits are set to 1. This bit is present only in configurations with CL72_EN Enabled. **For configurations with CL72_EN = Disabled:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "CL72_EN ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
8	SEQ_EQ_EN	* Varies	**For configurations with CL72_EN = Enabled:** Sequential Mode Enable for Rx equalization This bit controls the mode for Rx Equalization. - 1: Sequential Mode - 0: Parallel Mode In parallel mode, DWC_xpcs sends all the coefficient update requests (C(-1), C(0), and C(1)) received from PHY (in a particular evaluation iteration) to the remote-end device simultaneously (in the same frame). In sequential mode DWC_xpcs sends the coefficient update commands (received in a particular iteration) one by one. DWC_xpcs starts with sending C(-1). Then it waits for UPDATED/MAX/MIN status for C(-1). Then it sends c(0) and waits for UPDATED/MAX/MIN status for C(0). Then it sends C(1) and wait for UPDATED/MAX/MIN status for C(1). This method can be used when the remote-end partner is not capable of receiving simultaneous coefficient update commands for all the three coefficients. **Note:** Set this bit to 1, if PING_PONG_EN = 1 and SELF_MAIN_EN = 0. This bit is present only in configurations with CL72_EN Enabled. **For configurations with CL72_EN = Disabled:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "CL72_EN ? \"read-write\" : \"read-only\""
7:5	CONT_OFF_CAN_3_1	* Varies	**For Multi-lane configurations:** Receiver offset cancellation continuous operation on lanes 3-1. This field drives the output port 'xpcs_rx_offcan_cont_o[3:1]'. **For single-lane configurations:** Reserved **Value After Reset:** "RXAUI_DEFAULT? 0x1 : SINGLE_LANE_CFG ? 0x0 : 0x7" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
4	CONT_OFF_CAN_0	R/W	Receiver offset cancellation continuous operation on lane 0 This bit can be set if continuous receiver offset cancellation is required. If this bit is 0, offset cancellation runs when receiver exits P2 power state. This bit drives the output port 'xpcs_rx_offcan_cont_o[0]'. **Value After Reset:** 0x1 **Exists:** Always

Bits	Name	Memory Access	Description
3:1	CONT_ADAPT_3_1	* Varies	**For Multi-lane configurations:** Receiver Adaptation Continuous Operation on lanes 3-1. These bit can be set to enable continuous receiver adaptation in the PHY. This field drives the output port 'xpcs_rx_offcan_cont_o[3:1]'. This field is present only in multi-lane configurations. **For single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
0	CONT_ADAPT_0	R/W	Receiver Adaptation Continuous Operation on lane 0 This bit can be set to enable continuous receiver adaptation in the PHY. This bit drives the output port 'xpcs_rx_offcan_cont_o'. **Value After Reset:** "RPCS_PRESENT ? 0x1 : 0x0" **Exists:** Always

5.2.3.66 VR_XS_PMA_MP_32G_RX_EQ_CTRL4

- **Description:** VR XS or PMA Synopsys Multi protocol E32 PHY Rx Equalization Control 4 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x003C : (APB3_16) ? (0x003C*0x2) : (0x003C*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD | |XGXS_PCS)&&SNPS_XAUI_PHY&&(SNPS_RSVDPARAM_37 | |E32))

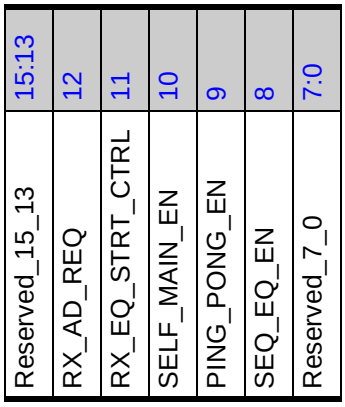


Table 5-196 Fields for Register: VR_XS_PMA_MP_32G_RX_EQ_CTRL4

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	RX_AD_REQ	R/W	Receive Adaptation Request This bit drives the 'rpcs_krx_adapt_req_o' port. For configurations with CL72_EN = Enabled, this bit drives the 'rpcs_krx_adapt_req_o' port only when KR training is disabled (TR_EN bit of SR_PMA_KR_PMD_CTRL Register is 0). Software can set this bit to make an Receiver Adaptation Request to the PHY.The acknowledgment from the PHY (for the adaptation request made) is reflected in bit[8]/bit[9]/bit[10] of VR_PMA_PHY_RX_EQ_CEU Register (for configurations with CL72_EN = Enabled) or bit[12] (RX_ADPT_ACK) of VR_XS_PMA_MP_12G_16G_25G_MISC_STS Register (for configurations with CL72_EN=Disabled). After receiving the acknowledgment, software must clear this register field. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
11	RX_EQ_STRT_CTRL	* Varies	**For configurations with CL72_EN = Enabled:** Receive Equalization Start Control This bits controls the behavior of DWC_xpcs at the start of receive equalization process. This bit decides whether DWC_xpcs sends PRESET or INIT as the first coefficient update request to the link-partner when KR Training is enabled. There after, DWC_xpcs sends INC/DEC as provided by the PHY. **Value:** ■ 0: DWC_xpcs sends INIT first ■ 1: DWC_xpcs sends PRESET first This bit is present only in configurations with CL72_EN Enabled. **For configurations with CL72_EN=Disabled:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "CL72_EN ? \"read-write\" : \"read-only\""
10	SELF_MAIN_EN	* Varies	**For configurations with CL72_EN = Enabled:** Enable Automatic Generation of Main -C(0) If this bit is set, DWC_xpcs generates C(0) (Main) coefficient update requests by itself during Rx equalization process and ignores the update request given by PHY for C(0). This bit can be set to 1,when Ping Pong Method is selected for Rx equalization. If PRE/POST is a INC request, DWC_xpcs generates a INC request for MAIN. If PRE/POST is a DEC request, DWC_xpcs generates a INC request for MAIN. This bit is present only in configurations with CL72_EN Enabled. **For configurations with CL72_EN=Disabled:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "CL72_EN ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
9	PING_PONG_EN	* Varies	**For configurations with CL72_EN = Enabled:** Ping-Pong Mode Enable for Rx equalization In ping-pong method, only one of PRE/POST update request is sent to remote-end device during one cycle of Rx equalization and the other coefficient update is ignored. DWC_xpcs alternates (ping-pongs) between PRE and POST requests between iterations. This method works well 'SEQ_EQ_EN' and 'SELF_MAIN_EN' bits are set to 1. This bit is present only in configurations with CL72_EN Enabled. **For configurations with CL72_EN=Disabled:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "CL72_EN ? \"read-write\" : \"read-only\""
8	SEQ_EQ_EN	* Varies	**For configurations with CL72_EN = Enabled:** Sequential Mode Enable for Rx equalization This bit controls the mode for Rx Equalization. - 1: Sequential Mode - 0: Parallel Mode In parallel mode, DWC_xpcs sends all the coefficient update requests (C(-1), C(0), and C(1)) received from PHY (in a particular evaluation iteration) to the remote-end device simultaneously (in the same frame). In sequential mode DWC_xpcs sends the coefficient update commands (received in a particular iteration) one by one. DWC_xpcs starts with sending C(-1). Then it waits for UPDATED/MAX/MIN status for C(-1). Then it sends c(0) and waits for UPDATED/MAX/MIN status for C(0). Then it sends C(1) and wait for UPDATED/MAX/MIN status for C(1). This method can be used when the remote-end partner is not capable of receiving simultaneous coefficient update commands for all the three coefficients. **Note:**Do not set this bit to 1, if PING_PONG_EN = 1 and SELF_MAIN_EN = 0. This bit is present only in configurations with CL72_EN Enabled. **For configurations with CL72_EN = Disabled:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "CL72_EN ? \"read-write\" : \"read-only\""
7:0	Reserved_7_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.3.67 VR_XS_PMA_MP_6G_RXLOS_CTRL0

- **Description:** VR XS or PMA Synopsys Multi-protocol 6G PHY Rx Loss-of-Signal Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x003D : (APB3_16) ? (0x003D*0x2) : (0x003D*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&SSP)

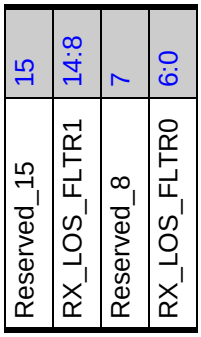


Table 5-197 Fields for Register: VR_XS_PMA_MP_6G_RXLOS_CTRL0

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:8	RX_LOS_FLTR1	* Varies	**For Multi-lane configurations:** Loss-of-Signal Filter Value for Lane 1 This field controls the loss-of-signal filter mode on Lane 1 by driving the output port 'xgxs_los1_ctl_o' of DWC_xpcs. **For Single-lane configurations:** Reserved **Value After Reset:** "(!SINGLE_LANE_CFG) ? 0xE :0x0" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" :\"read-only\""
7	Reserved_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
6:0	RX_LOS_FLTR0	R/W	Loss-of-Signal Filter Value for Lane 0 This field controls the loss-of-signal filter mode on Lane 0 by driving the output port 'xgxs_los0_ctl_o' of DWC_xpcs. **Value After Reset:** 0xe **Exists:** Always

5.2.3.68 VR_XS_PMA_MP_12G_AFE_DFE_EN_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G PHY AFE-DFE Enable Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x003D : (APB3_16) ? (0x003D*0x2) : (0x003D*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&E12)

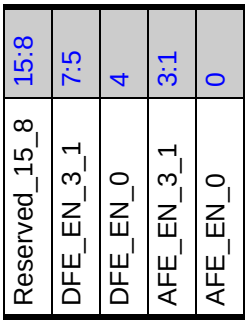


Table 5-198 Fields for Register: VR_XS_PMA_MP_12G_AFE_DFE_EN_CTRL

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	DFE_EN_3_1	* Varies	**For Multi-lane configurations:** Rx DFE Enable on lanes [3:1] of the PHY This bit drives the output port xpcs_rx_adapt_dfe_en_o[3:1]. **For Single-lane configurations:** Reserved **Value After Reset:** "RXAUI_DEFAULT ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
4	DFE_EN_0	R/W	Rx DFE Enable on lane 0 of the PHY This bit drives the output port xpcs_rx_adapt_dfe_en_o[0]. This bit can be set to enable Rx adaption and decision feedback equalization (DFE) circuitry and applies the input setting of DFE Tap1: ■ rx0_eq_dfe_tap1[7:0]. If this bit is not set, the change on the input DFE settings can be applied using a 'RX_REQ_0/RX_ACK_0' handshake. If this bit is set, DFE settings are initialized to the inputs and RX DFE adaption routines are executed to achieve optimal DFE settings. **Value After Reset:** "(SNPS_RSVDPARAM_22&&!SNPS_RSVDPARAM_2) ? 0x0 : (RPCS_PRESENT||RXAUI_DEFAULT) ? 0x1 : 0x0" **Exists:** Always
3:1	AFE_EN_3_1	* Varies	**For Multi-lane configurations:** Rx Adaptation AFE Enable on lanes [3:1] of the PHY This field drives the output port xpcs_rx_adapt_afe_en_o[3:1]. This field is present only in multi-lane configurations. **For Single-lane configurations:** Reserved **Value After Reset:** "RXAUI_DEFAULT ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
0	AFE_EN_0	R/W	Rx Adaptation AFE Enable on lane 0 of the PHY This bit drives the output port xpcs_rx_adapt_afe_en_o[0]. This bit can be set to enable Rx adaption circuitry and applies the following input receiver equalization settings to the PHY: - rx0_eq_att_lvl[2:0] - rx0_eq_vga1_gain[3:0] - rx0_eq_vga2_gain[3:0] - rx0_eq_ctle_pole[2:0] - rx0_eq_ctle_boost[4:0] If this bit is not set, change on input AFE settings can be applied using RX_REQ_0/RX_ACK_0 handshake. If this bit is set, AFE equalization settings are initialized to the PHY inputs, but any further change on these inputs are ignored by PHY and RX AFE adaption routines are executed to achieve optimal AFE settings. **Value After Reset:** "(SNPS_RSVDPARAM_22&&!SNPS_RSVDPARAM_2) ? 0x0 : (RPCS_PRESENT||RXAUI_DEFAULT) ? 0x1 : 0x0" **Exists:** Always

5.2.3.69 VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL5

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G/25G PHY Rx Equalization Control 5 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x003D : (APB3_16) ? (0x003D*0x2) : (0x003D*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||E25)&&!SNPS_RSVDPARAM_37

15:13	12	11:10	9:8	7:6	5:4	3:1	0
RX_AD_PROG_3_1	RX_ADPT_PROG_0	RX3_ADPT_MODE	RX2_ADPT_MODE	RX1_ADPT_MODE	RX0_ADPT_MODE	RX_ADPT_SEL_3_1	RX_ADPT_SEL_0

Table 5-199 Fields for Register: VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL5

Bits	Name	Memory Access	Description
15:13	RX_AD_PROG_3_1	* Varies	Receiver Adaptation in Progress -lanes 3 to 1 This field drives the output port xpcs_rx_adpt_in_prog_o[3:1]. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
12	RX_ADPT_PROG_0	R/W	Receiver Adaptation in Progress -lane 0 This field drives the output port xpcs_rx_adpt_in_prog_o[0]. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
11:10	RX3_ADPT_MODE	* Varies	**For Multi-lane configurations:** RX Adaptation for lane3 This field drives the output port xpcs_rx3_adpt_mode_o[1:0]. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
9:8	RX2_ADPT_MODE	* Varies	**For Multi-lane configurations:** RX Adaptation for lane2 This field drives the output port xpcs_rx2_adpt_mode_o[1:0]. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
7:6	RX1_ADPT_MODE	* Varies	**For Multi-lane configurations:** RX Adaptation for lane1 This field drives the output port xpcs_rx1_adpt_mode_o[1:0]. **For Single-lane configurations:** Reserved **Value After Reset:** "RXAUI_DEFAULT ? 0x3 : 0x0" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
5:4	RX0_ADPT_MODE	R/W	RX Adaptation for lane0 This field drives the output port xpcs_rx0_adpt_mode_o[1:0]. **Value After Reset:** "(RPCS_PRESENT|| RXAUI_DEFAULT) ? 0x3 : 0x0" **Exists:** Always

Bits	Name	Memory Access	Description
3:1	RX_ADPT_SEL_3_1	* Varies	**For Multi-lane configurations:** Select Storage Bank for RX adaptation on Lane 3 to 1 This field drives the output port xpcs_rx_adpt_sel_o[3:1]. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
0	RX_ADPT_SEL_0	R/W	Select Storage Bank for RX adaptation on Lane0 This field drives the output port xpcs_rx_adpt_sel_o[0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.70 VR_XS_PMA_MP_32G_RX_ADPT_CTRL

- **Description:** VR XS PMA Synopsys Multi-protocol 32G PHY Rx Adapt Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x003D : (APB3_16) ? (0x003D*0x2) : (0x003D*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&SNPS_XAUI_PHY&&(E32)

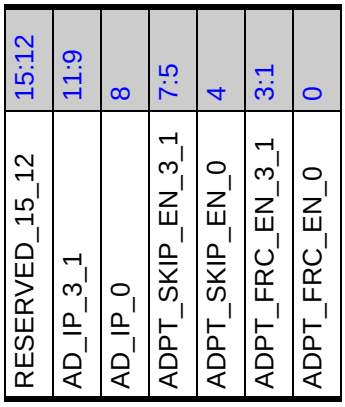


Table 5-200 Fields for Register: VR_XS_PMA_MP_32G_RX_ADPT_CTRL

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:9	AD_IP_3_1	* Varies	Rx Adaptation in Progress on lanes 3:1 This field drives the output port xpcs_rx_adpt_in_prog_o[3:1]. Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
8	AD_IP_0	R/W	Rx Adaptation in Progress on lane 0 This field drives the output port xpcs_rx_adpt_in_prog_o[0]. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
7:5	ADPT_SKIP_EN_3_1	* Varies	Rx Coarse adaptation skip enable on lanes 3:1 Forces Coarse adaptation on lanes[3:1]. This field drives the output port xpcs_rx_crse_adpt_skip_en_o. Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
4	ADPT_SKIP_EN_0	R/W	Rx Coarse adaptation skip enable on lane 0 Enables the skipping of Rx coarse adaptation on lane0. This field drives the output port xpcs_rx_crse_adpt_skip_en_o. **Value After Reset:** 0x0 **Exists:** Always
3:1	ADPT_FRC_EN_3_1	* Varies	Rx Force Coarse adaptation enable on lanes 3:1 Forces Coarse adaptation on lanes[3:1]. This field drives the output port xpcs_rx_crse_adpt_frc_en_o. Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
0	ADPT_FRC_EN_0	R/W	Rx Force Coarse adaptation enable on lane 0 Forces Coarse adaptation on lane0. This field drives the output port xpcs_rx_crse_adpt_frc_en_o. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.71 VR_XS_PMA_MP_6G_RXLOS_CTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 6G PHY Rx Loss-of-Signal Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x003E : (APB3_16) ? (0x003E*0x2) : (0x003E*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&SSP&&!SINGLE_LANE_CFG

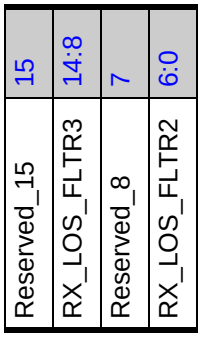


Table 5-201 Fields for Register: VR_XS_PMA_MP_6G_RXLOS_CTRL1

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:8	RX_LOS_FLTR3	R/W	Loss-of-Signal Filter Value for Lane 3 This field controls the loss-of-signal filter mode on Lane 3 by driving the output port 'xgxs_los03_ctl_o' of DWC_xpcs. **Value After Reset:** 0xe **Exists:** Always
7	Reserved_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:0	RX_LOS_FLTR2	R/W	Loss-of-Signal Filter Value for Lane 2 This field controls the loss-of-signal filter mode on Lane 2 by driving the output port 'xgxs_los2_ctl_o' of DWC_xpcs. **Value After Reset:** 0xe **Exists:** Always

5.2.3.72 VR_XS_PMA_MP_12G_16G_25G_DFE_TAP_CTRL0

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY DFE Tap Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x003E : (APB3_16) ? (0x003E*0x2) : (0x003E*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25)&&!SNPS_RSVD-PARAM_37)

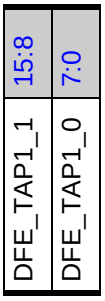


Table 5-202 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_DFE_TAP_CTRL0

Bits	Name	Memory Access	Description
15:8	DFE_TAP1_1	* Varies	**For Multi-lane configurations:** Rx Equalization DFE Tap1 value on lane 1 of the PHY This field drives the output port xpcs_rx1_eq_dfe_tap1_o[7:0] **For Single-lane configurations:** Reserved **Value After Reset:** "(!SINGLE_LANE_CFG&&SNPS_RSVDPARAM_36) ? 0xF: (E25&&RPCS_PRESENT&&!SINGLE_LANE_CFG) ? 0xA : 0x0" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
7:0	DFE_TAP1_0	R/W	Rx Equalization DFE Tap1 value on lane 0 of the PHY This field drives the output port xpcs_rx0_eq_dfe_tap1_o[7:0] **Value After Reset:** "(SNPS_RSVDPARAM_16||SNPS_RSVDPARAM_36) ? 0xF: (E25&&RPCS_PRESENT) ? 0xA : 0x0" **Exists:** Always

5.2.3.73 VR_XS_PMA_MP_12G_16G_25G_DFE_TAP_CTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY DFE Tap Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x003F : (APB3_16) ? (0x003F*0x2) : (0x003F*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25)&&!SINGLE_LANE_CFG)

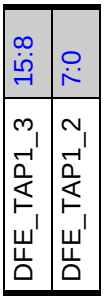


Table 5-203 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_DFE_TAP_CTRL1

Bits	Name	Memory Access	Description
15:8	DFE_TAP1_3	* Varies	**For Multi-lane configurations:** Rx Equalization DFE Tap1 value on lane 3 of the PHY This field drives the output port xpcs_rx3_eq_dfe_tap1_o[7:0]. **For Single-lane configurations:** Reserved **Value After Reset:** "SNPS_RSVDPARAM_36 ? 0xF : (E25&&RPCS_PRESENT&&!SINGLE_LANE_CFG) ? 0xA : 0x0" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
7:0	DFE_TAP1_2	* Varies	**For Multi-lane configurations:** Rx Equalization DFE Tap1 value on lane 2 of the PHY. This field drives the output port xpcs_rx2_eq_dfe_tap1_o[7:0] **For Single-lane configurations:** Reserved **Value After Reset:** "SNPS_RSVDPARAM_36 ? 0xF : (E25&&RPCS_PRESENT&&!SINGLE_LANE_CFG) ? 0xA : 0x0" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""

5.2.3.74 VR_XS_PMA_MP_12G_16G_25G_RX_STS

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Rx Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0040 : (APB3_16) ? (0x0040*0x2) : (0x0040*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25||E32))

15:12	11:9	8	7:5	4	3:1	0
Reserved_15_12	HF_SD_3_1	HF_SD_0	LF_SD_3_1	LF_SD_0	RX_ACK_3_1	RX_ACK_0

Table 5-204 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_RX_STS

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:9	HF_SD_3_1	R	**For Multi-protocol 25G PHY or Multi protocol 32G PHY or For Multi-protocol 10G Gen2 PHY configurations:** High-frequency Signal Detect on lanes 1-3. This field captures the value of the input port 'xpcs_rx_sigdet_hf_i[3:1]'.This field is valid only for multi-lane configurations. **For Other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
8	HF_SD_0	R	**Multi-protocol 25G PHY or Multi-protocol 32G PHY or For Multi-protocol 10G Gen2 PHY configurations:** High-frequency Signal Detect on lane 0. This field captures the value of the input port 'xpcs_rx_sigdet_hf_i[0]'. **For Other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	LF_SD_3_1	R	**For Multi-protocol 25G PHY or Multi-protocol 32G PHY or Multi-protocol 10G Gen2 PHY configurations:** Low-frequency Signal Detect on lanes 1-3. This field captures the value of the input port 'xpcs_rx_sigdet_lf_i[3:1]'.This field is valid only for multi-lane configurations. **For Other configurations:** Reserved **Value After Reset:** "(E25&&RXAUI_DEFAULT) ? 0x1 : ((SNPS_RSVDPARAM_22||E25||E32)&&!SINGLE_LANE_CFG) ? 0x7 :0x0" **Exists:** Always **Testable:** "((E25||E32)&&(RPCS_EN||RXAUI_DEFAULT)&&!SINGLE_LANE_CFG) ? \"untestable\" : \"unconstrained\"""
4	LF_SD_0	R	**For Multi-protocol 25G PHY or Multi-protocol 32G PHY or Multi-protocol 10G Gen2 PHY configurations:** Low-frequency Signal Detect on lane 0. This field captures the value of the input port 'xpcs_rx_sigdet_lf_i[0]'. **FOr other configurations:** Reserved **Value After Reset:** "(SNPS_RSVDPARAM_22||E25||E32) ? 0x1 : 0x0" **Exists:** Always **Testable:** "((E25||E32)&&RPCS_EN) ? \"untestable\" : \"unconstrained\"""
3:1	RX_ACK_3_1	R	**For Multi-lane configurations:** Receive Acknowledge on Lanes[3:1] This bit captures the value of the input port xpcs_rx_ack_i[3:1]. If this field is set, it indicates that the requested receiver setting is complete. This field forms a hand-shake with 'RX_REQ_3_1' field. When this field is set, 'RX_REQ_3_1' field is self-cleared. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	RX_ACK_0	R	Rx Acknowledge on lane 0 of PHY This bit captures the value of the input port xpcs_rx_ack_i[0]. If this bit is set, it indicates that the requested receiver setting is complete. This bit forms a hand-shake with 'RX_REQ_0' bit. When this bit is set, RX_REQ_0 bit is self-cleared. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.75 VR_XS_PMA_MP_16G_25G_RX_PPM_STS0

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G/25G PHY Rx PPM Status 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0041 : (APB3_16) ? (0x0041*0x2) : (0x0041*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||E25||SNPS_RSVDPARAM_39||E32)

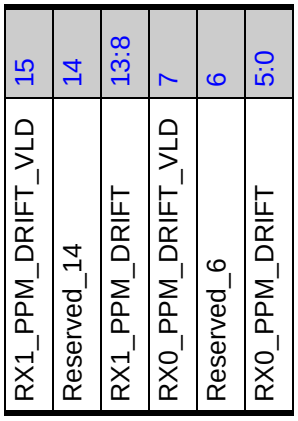


Table 5-205 Fields for Register: VR_XS_PMA_MP_16G_25G_RX_PPM_STS0

Bits	Name	Memory Access	Description
15	RX1_PPM_DRIFT_VLD	R	**For Multi-lane configurations:** RX CDR PPM Drift Valid on lane1 This field is driven by the input port xpcs_rx1_ppm_drift_vld_i. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always
14	Reserved_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
13:8	RX1_PPM_DRIFT	R	**For Multi-lane configurations:** RX CDR PPM Drift on lane1 This field is driven by the input port xpcs_rx1_ppm_drift_i[5:0]. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always
7	RX0_PPM_DRIFT_VLD	R	RX CDR PPM Drift Valid on lane0 This field is driven by the input port xpcs_rx0_ppm_drift_vld_i. **Value After Reset:** 0x0 **Exists:** Always
6	Reserved_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:0	RX0_PPM_DRIFT	R	RX CDR PPM Drift on lane0 This field is driven by the input port xpcs_rx0_ppm_drift_i[5:0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.76 VR_XS_PMA_MP_16G_25G_RX_PPM_STS1

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G/25G PHY Rx PPM Status 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0042 : (APB3_16) ? (0x0042*0x2) : (0x0042*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||E25||SNPS_RSVDPARAM_39||E32)&&!SINGLE_LANE_CFG

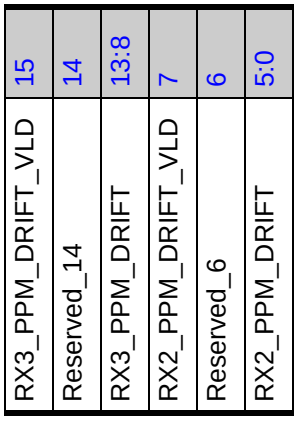


Table 5-206 Fields for Register: VR_XS_PMA_MP_16G_25G_RX_PPM_STS1

Bits	Name	Memory Access	Description
15	RX3_PPM_DRIFT_VLD	R	RX CDR PPM Drift Valid on lane3 This field is driven by the input port xpcs_rx1_ppm_drift_vld_i. **Value After Reset:** 0x0 **Exists:** Always
14	Reserved_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:8	RX3_PPM_DRIFT	R	RX CDR PPM Drift on lane3 This field is driven by the input port xpcs_rx3_ppm_drift_i[5:0]. **Value After Reset:** 0x0 **Exists:** Always
7	RX2_PPM_DRIFT_VLD	R	RX CDR PPM Drift Valid on lane2 This field is driven by the input port xpcs_rx2_ppm_drift_vld_i. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
6	Reserved_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:0	RX2_PPM_DRIFT	R	RX CDR PPM Drift on lane2 This field is driven by the input port xpcs_rx2_ppm_drift_i[5:0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.77 VR_XS_PMA_MP_25G_RX_TERM_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY Receive Termination Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0044 : (APB3_16) ? (0x0044*0x2) : (0x0044*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&E25&&!SNPS_RSVDPARAM_37

15	14:12	11	10:8	7	6:4	3	2:0
Reserved_15	RX3_TERM	Reserved_11	RX2_TERM	Reserved_7	RX1_TERM	Reserved_3	RX0_TERM

Table 5-207 Fields for Register: VR_XS_PMA_MP_25G_RX_TERM_CTRL

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:12	RX3_TERM	* Varies	Receive Termination Control for lane 3 **Value After Reset:** "!SINGLE_LANE_CFG ? 0x2 : 0x0" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
10:8	RX2_TERM	* Varies	Receive Termination Control for lane 2 **Value After Reset:** "!SINGLE_LANE_CFG ? 0x2 : 0x0" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
6:4	RX1_TERM	* Varies	Receive Termination Control for lane 1 **Value After Reset:** "!SINGLE_LANE_CFG ? 0x2 : 0x0" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
3	Reserved_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	RX0_TERM	R/W	Receive Termination Control for lane 0 **Value After Reset:** 0x2 **Exists:** Always

5.2.3.78 VR_XS_PMA_MP_16G_RX_CDR_CTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G PHY Rx CDR Control1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0044 : (APB3_16) ? (0x0044*0x2) : (0x0044*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||SNPS_RSVDPARAM_39))

15:14	13:12	11:10	9:8	7:5	4	3:1	0
VCO_FRQBAND_3	VCO_FRQBAND_2	VCO_FRQBAND_1	VCO_FRQBAND_0	VCO_STEP_CTRL_3_1	VCO_STEP_CTRL_0	VCO_TEMP_COMP_EN_3_1	VCO_TEMP_COMP_EN_0

Table 5-208 Fields for Register: VR_XS_PMA_MP_16G_RX_CDR_CTRL1

Bits	Name	Memory Access	Description
15:14	VCO_FRQBAND_3	* Varies	**For Multi-lane configurations:** RX CDR VCO Frequency Band lane3 This field drives the output port xpcs_vco_freqband3_o[1:0]. **For Single-lane configurations:** Reserved **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : (RPCS_PRESENT) ? 0x1 : SNPS_RSVDPARAM_39 ? 0x0 : 0x2" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
13:12	VCO_FRQBAND_2	* Varies	**For Multi-lane configurations:** RX CDR VCO Frequency Band lane2 This field drives the output port xpcs_vco_freqband2_o[1:0]. **For Single-lane configurations:** Reserved **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : (RPCS_PRESENT) ? 0x1 : SNPS_RSVDPARAM_39 ? 0x0 : 0x2" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
11:10	VCO_FRQBAND_1	* Varies	**For Multi-lane configurations:** RX CDR VCO Frequency Band lane1 This field drives the output port xpcs_vco_freqband1_o[1:0]. **For Single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0 : (RPCS_PRESENT) ? 0x1 : SNPS_RSVDPARAM_39 ? 0x0 : 0x2" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
9:8	VCO_FRQBAND_0	R/W	RX CDR VCO Frequency Band lane0 This field drives the output port xpcs_vco_freqband0_o[1:0]. **Value After Reset:** "SNPS_RSVDPARAM_39 ? 0x0 : (RPCS_PRESENT||BASE1000X_ONLY||KX_ONLY) ? 0x1 : 0x2" **Exists:** Always
7:5	VCO_STEP_CTRL_3_1	* Varies	**For Multi-lane configurations:** RX_CDR VCO Step Control Lane 3 to 1 This field drives the output port xpcs_vco_step_ctrl_o[3:1]. **For Single-lane configurations:** Reserved **Value After Reset:** "(!SINGLE_LANE_CFG)&&(E16) ? 0x7 : 0x0" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG)&&E16 ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
4	VCO_STEP_CTRL_0	* Varies	RX_CDR VCO Step Control Lane 0 This field drives the output port xpcs_vco_step_ctrl_o[0]. **Value After Reset:** "E16 ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(E16) ? \"read-write\" : \"read-only\""
3:1	VCO_TEMP_COMP_EN_3_1	* Varies	**For Multi-lane configurations:** RX_CDR VCO Temperature Compensation Enable on Lane 3 to 1 This field drives the output port xpcs_vco_temp_comp_en_o[3:1]. **For Single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG||SNPS_RSVDPARAM_39)? 0x0 :(RPCS_PRESENT) ? 0x7 : 0x0" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG)&&E16 ? \"read-write\" : \"read-only\""
0	VCO_TEMP_COMP_EN_0	* Varies	RX_CDR VCO Temperature Compensation Enable Lane 0 This field drives the output port xpcs_vco_temp_comp_en_o[0]. **Value After Reset:** "(RPCS_PRESENT||BASE1000X_ONLY||KX_ONLY)&&E16 ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(E16) ? \"read-write\" : \"read-only\""

5.2.3.79 VR_XS_PMA_MP_16G_25G_RX_PPM_CTRL0

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G/25G PHY Rx PPM Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0045 : (APB3_16) ? (0x0045*0x2) : (0x0045*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37

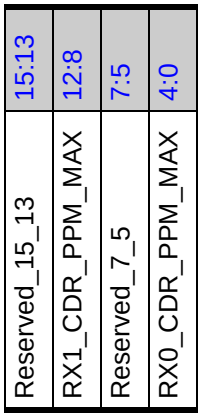


Table 5-209 Fields for Register: VR_XS_PMA_MP_16G_25G_RX_PPM_CTRL0

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:8	RX1_CDR_PPM_MAX	* Varies	**For Multi-lane configurations:** Maximum Allowed PPM on the RX1 CDR Clock This field drives the output port xpcs_rx1_cdr_ppm_max_o[4:0]. **For Single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG)? 0x0 : SNPS_RSVDPARAM_39 ? 0xD : (E25&&SNPS_RSVDPARAM_16) ? 0x12 : (E25&&REF_CLK_78&&(RPCS_PRESENT)) ? 0x13 : 0x12" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
7:5	Reserved_7_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4:0	RX0_CDR_PPM_MAX	R/W	Maximum Allowed PPM on the RX0 CDR Clock This field drives the output port xpcs_rx0_cdr_ppm_max_o[4:0]. **Value After Reset:** "SNPS_RSVDPARAM_39 ? 0xD : (E25&&SNPS_RSVDPARAM_16) ? 0x12 : ((E25&&REF_CLK_78&&(RPCS_PRESENT||KX_ONLY||BASE1000X_ONLY))|| (SNPS_RSVDPARAM_36&&KX_ONLY&&SNPS_RSVDPARAM_13)) ? 0x13 : 0x12" **Exists:** Always

5.2.3.80 VR_XS_PMA_MP_16G_25G_RX_PPM_CTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G/25G PHY Rx PPM Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0046 : (APB3_16) ? (0x0046*0x2) : (0x0046*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG

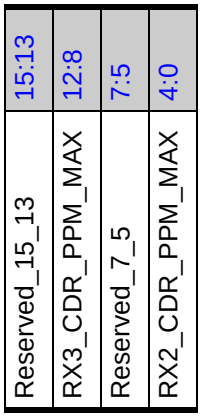


Table 5-210 Fields for Register: VR_XS_PMA_MP_16G_25G_RX_PPM_CTRL1

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:8	RX3_CDR_PPM_MAX	R/W	Maximum Allowed PPM on the RX3 CDR Clock This field drives the output port xpcs_rx3_cdr_ppm_max_o[4:0]. **Value After Reset:** "SNPS_RSVDPARAM_39 ? 0xD : (E25&&REF_CLK_78&&(RPCS_PRESENT)) ? 0x13 : 0x12" **Exists:** Always
7:5	Reserved_7_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4:0	RX2_CDR_PPM_MAX	R/W	Maximum Allowed PPM on the RX2 CDR Clock This field drives the output port xpcs_rx2_cdr_ppm_max_o[4:0]. **Value After Reset:** "SNPS_RSVDPARAM_39 ? 0xD : (E25&&REF_CLK_78&&(RPCS_PRESENT)) ? 0x13 : 0x12" **Exists:** Always

5.2.3.81 VR_XS_PMA_MP_16G_25G_RX_GENCTRL4

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G/25G PHY Rx General Control 4 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0048 : (APB3_16) ? (0x0048*0x2) : (0x0048*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37

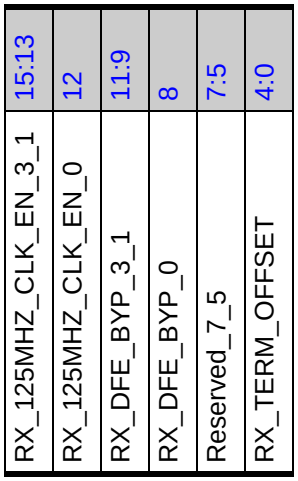


Table 5-211 Fields for Register: VR_XS_PMA_MP_16G_25G_RX_GENCTRL4

Bits	Name	Memory Access	Description
15:13	RX_125MHZ_CLK_EN_3_1	* Varies	RX 125MHz clock generation Enable on Lane 3 to 1 This field drives the output port xpcs_rx_125mhz_clk_en_o[3:1]. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_RSVDPARAM_39&&!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
12	RX_125MHZ_CLK_EN_0	* Varies	RX 125MHz clock generation Enable on Lane 0 This field drives the output port xpcs_rx_125mhz_clk_en_o[0]. **Value After Reset:** "(RPCS_PRESENT&&SNPS_RSVDPARAM_39) ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(SNPS_RSVDPARAM_39) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
11:9	RX_DFE_BYP_3_1	* Varies	**For Multi-lane configurations:** RX DFE Bypass Enable on Lane 3 to 1 This field drives the output port xpcs_rx_dfe_bypass_o[3:1]. **For Single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0 : RXAUI_DEFAULT ? 0x6 : 0x7" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_39) ? \"read-write\" : \"read-only\""
8	RX_DFE_BYP_0	* Varies	RX DFE Bypass Enable on Lane 0 This field drives the output port xpcs_rx_dfe_bypass_o[0]. **Value After Reset:** "(RPCS_PRESENT|| RXAUI_DEFAULT||SNPS_RSVDPARAM_39) ? 0x0 : 0x1" **Exists:** Always **Memory Access:** "(!SNPS_RSVDPARAM_39) ? \"read-write\" : \"read-only\""
7:5	Reserved_7_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4:0	RX_TERM_OFFSET	R/W	Offset for RX Termination This field drives the output port xpcs_rx_term_offset_o[4:0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.82 VR_XS_PMA_Consumer_10G_RX_GENCTRL4

- **Description:** VR XS or PMA Consumer 10G PHY Rx General Control 4 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0048 : (APB3_16) ? (0x0048*0x2) : (0x0048*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&SNPS_RSVDPARAM_33

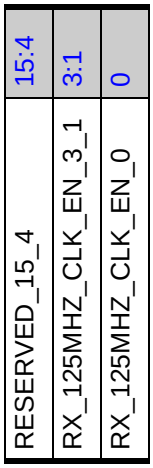


Table 5-212 Fields for Register: VR_XS_PMA_Consumer_10G_RX_GENCTRL4

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:1	RX_125MHZ_CLK_EN_3_1	* Varies	RX 125MHz clock generation Enable on Lanes [3:1] This field drives the output port xpcs_rx_125mhz_clk_en_o[3:1]. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
0	RX_125MHZ_CLK_EN_0	R/W	RX 125MHz clock generation Enable on Lane 0 This field drives the output port xpcs_rx_125mhz_clk_en_o[0]. **Value After Reset:** "(USXGMII_EN) ? 0x1 : 0x0" **Exists:** Always

5.2.3.83 VR_XS_PMA_MP_16G_25G_RX_MISC_CTRL0

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G/25G PHY RX Miscellaneous Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0049 : (APB3_16) ? (0x0049*0x2) : (0x0049*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37

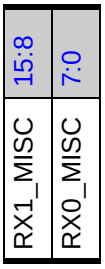


Table 5-213 Fields for Register: VR_XS_PMA_MP_16G_25G_RX_MISC_CTRL0

Bits	Name	Memory Access	Description
15:8	RX1_MISC	* Varies	**For Multi-lane configurations:** RX Miscellaneous control for lane1. This field drives the output port xpcs_rx1_misc_o[7:0]. **For Single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG||E25) ? 0x0 : SNPS_RSVDPARAM_39 ? (RPCS_PRESENT ? 0x3 : 0x0) : RPCS_PRESENT ? 0x12 : RXAUI_DEFAULT ? 0x15 : 0x17" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\" "
7:0	RX0_MISC	R/W	RX Miscellaneous control for lane0. This field drives the output port xpcs_rx0_misc_o[7:0]. **Value After Reset:** "E25 ? 0x0 : SNPS_RSVDPARAM_39 ? (RPCS_PRESENT ? 0x3 : 0x0) : (RPCS_PRESENT) ? 0x12: (BASE1000X_ONLY||KX_ONLY)?SNPS_RSVDPARAM_15?0x8 : 0x16 : RXAUI_DEFAULT ? 0x15 : 0x17" **Exists:** Always

5.2.3.84 VR_XS_PMA_MP_32G_RX_MRGN_STS0

- **Description:** VR XS or PMA Synopsys Multi Protocol E32 PHY RX Margin Status 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0049 : (APB3_16) ? (0x0049*0x2) : (0x0049*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&SNPS_XAUI_PHY&&(SNPS_RSVDPARAM_37 | |E32)

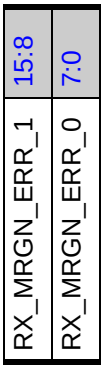


Table 5-214 Fields for Register: VR_XS_PMA_MP_32G_RX_MRGN_STS0

Bits	Name	Memory Access	Description
15:8	RX_MRGN_ERR_1	R	**For Multi-lane configurations:** Error counter for RX margining along Lane 1. This field is driven by xpcs_rx1_margin_err_i[7:0]. **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always
7:0	RX_MRGN_ERR_0	R	Error counter for RX margining along Lane 0. This field is driven by xpcs_rx0_margin_err_i[7:0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.85 VR_XS_PMA_MP_16G_25G_RX_MISC_CTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G/25G PHY RX Miscellaneous Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x004A : (APB3_16) ? (0x004A*0x2) : (0x004A*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||E25||SNPS_RSVDPARAM_39)&&(!SINGLE_LANE_CFG)&&!SNPS_RSVDPARAM_37)

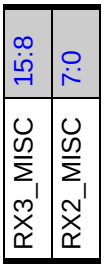


Table 5-215 Fields for Register: VR_XS_PMA_MP_16G_25G_RX_MISC_CTRL1

Bits	Name	Memory Access	Description
15:8	RX3_MISC	R/W	RX Miscellaneous control for lane3. This field drives the output port xpcs_rx3_misc_o[7:0]. **Value After Reset:** "E25 ? 0x0 : SNPS_RSVDPARAM_39 ? (RPCS_PRESENT ? 0x3 : 0x0) : (RPCS_PRESENT)? 0x12 : RXAUI_DEFAULT ? 0x15 : 0x17" **Exists:** Always
7:0	RX2_MISC	R/W	RX Miscellaneous control for lane2. This field drives the output port xpcs_rx2_misc_o[7:0]. **Value After Reset:** "E25 ? 0x0 : SNPS_RSVDPARAM_39 ? (RPCS_PRESENT ? 0x3 : 0x0) : (RPCS_PRESENT)? 0x12 : RXAUI_DEFAULT ? 0x15 : 0x17" **Exists:** Always

5.2.3.86 VR_XS_PMA_MP_32G_RX_MRGN_STS1

- **Description:** VR XS or PMA Synopsys Multi Protocol E32 RX Margin Status 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x004A : (APB3_16) ? (0x004A*0x2) : (0x004A*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(SNPS_RSVDPARAM_37||E32)&&!SINGLE_LANE_CFG

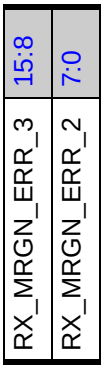


Table 5-216 Fields for Register: VR_XS_PMA_MP_32G_RX_MRGN_STS1

Bits	Name	Memory Access	Description
15:8	RX_MRGN_ERR_3	R	Error counter for RX margining along Lane 3. This field is driven by xpcs_rx3_margin_err_i[7:0]. **Value After Reset:** 0x0 **Exists:** Always
7:0	RX_MRGN_ERR_2	R	Error counter for RX margining along Lane 2. This field is driven by xpcs_rx2_margin_err_i[7:0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.87 VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL0

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G/25G PHY RX IQ Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x004B : (APB3_16) ? (0x004B*0x2) : (0x004B*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||E25||SNPS_RSVDPARAM_39||E32)

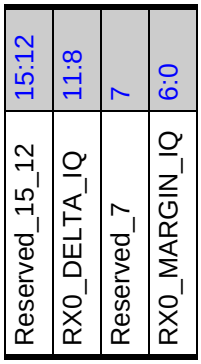


Table 5-217 Fields for Register: VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL0

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:8	RX0_DELTA_IQ	* Varies	**Non SNPS_RSVDPARAM_37/E32 configurations:** RX IQ Offset Value for lane0. This field drives the output port xpcs_rx0_delta_iq_o[3:0]. **For SNPS_RSVDPARAM_37/E32 configurations:** Reserved **Value After Reset:** "SNPS_RSVDPARAM_37||E32 ? 0x0 : (SNPS_RSVDPARAM_16&&!SNPS_RSVDPARAM_36) ? 0x6 : (RPCS_PRESENT&&SNPS_RSVDPARAM_36) ? 0x6 : (RPCS_PRESENT&&E25) ? 0x3 : 0x0" **Exists:** Always **Memory Access:** "(!(SNPS_RSVDPARAM_37||E32)) ? \"read-write\" :\"read-only\""

Bits	Name	Memory Access	Description
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:0	RX0_MARGIN_IQ	* Varies	Value of RX IQ Margining on lane0. This field drives the output port xpcs_rx0_margin_iq_o[6:0]. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!SNPS_RSVDPARAM_39 ? \"read-write\" : \"read-only\""

5.2.3.88 VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G/25G PHY RX IQ Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x004C : (APB3_16) ? (0x004C*0x2) : (0x004C*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||E25||SNPS_RSVDPARAM_39||E32)&&!SINGLE_LANE_CFG

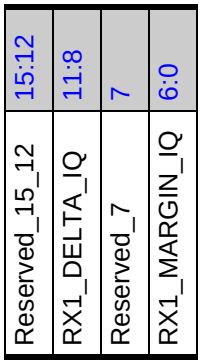


Table 5-218 Fields for Register: VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL1

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:8	RX1_DELTA_IQ	* Varies	**Non SNPS_RSVDPARAM_37/E32 configurations:** RX IQ Offset Value for lane1. This field drives the output port xpcs_rx1_delta_iq_o[3:0]. **For SNPS_RSVDPARAM_37/E32 configurations:** Reserved **Value After Reset:** "SNPS_RSVDPARAM_36&&RPCS_PRESENT ? 0x4 : (RPCS_PRESENT&&E25)&&!SNPS_RSVDPARAM_37 ? 0x3 : 0x0" **Exists:** Always **Memory Access:** "(!(SNPS_RSVDPARAM_37||E32)) ? \"read-write\" :\"read-only\""

Bits	Name	Memory Access	Description
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:0	RX1_MARGIN_IQ	* Varies	Value of RX IQ Margining on lane1. This field drives the output port xpcs_rx1_margin_iq_o[6:0]. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!SNPS_RSVDPARAM_39 ? \"read-write\" : \"read-only\""

5.2.3.89 VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL2

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G/25G PHY RX IQ Control 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x004D : (APB3_16) ? (0x004D*0x2) : (0x004D*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||E25||SNPS_RSVDPARAM_39||E32)&&!SINGLE_LANE_CFG

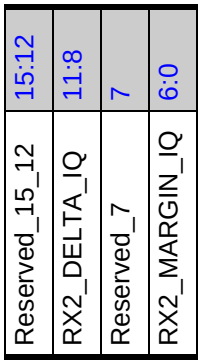


Table 5-219 Fields for Register: VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL2

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:8	RX2_DELTA_IQ	* Varies	**Non SNPS_RSVDPARAM_37/E32 configurations:** RX IQ Offset Value for lane2. This field drives the output port xpcs_rx2_delta_iq_o[3:0]. **For SNPS_RSVDPARAM_37/E32 configurations:** Reserved **Value After Reset:** "SNPS_RSVDPARAM_36&&RPCS_PRESENT ? 0x4 : (RPCS_PRESENT&&E25)&&!SNPS_RSVDPARAM_37 ? 0x3 : 0x0" **Exists:** Always **Memory Access:** "(!(SNPS_RSVDPARAM_37||E32)) ? \"read-write\" :\"read-only\""

Bits	Name	Memory Access	Description
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:0	RX2_MARGIN_IQ	* Varies	Value of RX IQ Margining on lane2. This field drives the output port xpcs_rx2_margin_iq_o[6:0]. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!SNPS_RSVDPARAM_39 ? \"read-write\" : \"read-only\""

5.2.3.90 VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL3

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G/25G PHY RX IQ Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x004E : (APB3_16) ? (0x004E*0x2) : (0x004E*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||E25||SNPS_RSVDPARAM_39||E32)&&!SINGLE_LANE_CFG

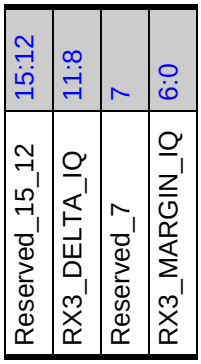


Table 5-220 Fields for Register: VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL3

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:8	RX3_DELTA_IQ	* Varies	**Non SNPS_RSVDPARAM_37/E32 configurations:** RX IQ Offset Value for lane3. This field drives the output port xpcs_rx3_delta_iq_o[3:0]. **For SNPS_RSVDPARAM_37/E32 configurations:** Reserved **Value After Reset:** "SNPS_RSVDPARAM_36&&RPCS_PRESENT ? 0x4 : (RPCS_PRESENT&&E25)&&!SNPS_RSVDPARAM_37 ? 0x3 : 0x0" **Exists:** Always **Memory Access:** "(!(SNPS_RSVDPARAM_37||E32)) ? \"read-write\" :\"read-only\""

Bits	Name	Memory Access	Description
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:0	RX3_MARGIN_IQ	* Varies	Value of RX IQ Margining on lane3. This field drives the output port xpcs_rx3_margin_iq_o[6:0]. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!SNPS_RSVDPARAM_39 ? \"read-write\" : \"read-only\""

5.2.3.91 VR_XS_PMA_MP_12G_16G_25G_MPLL_CMN_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY MPLL Common Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0050 : (APB3_16) ? (0x0050*0x2) : (0x0050*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25))

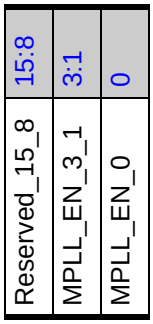


Table 5-221 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_MPLL_CMN_CTRL

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:1	MPLL_EN_3_1	* Varies	**For Multi-lane configurations:** Tx MPLL Enable-lanes[3:1] Set this bit to 1 to power-up the MPLL. This bit must be 1 for normal operation. **For single-lane configurations:** Reserved **Value After Reset:** "(!SINGLE_LANE_CFG) ? 0x7 : 0x0" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
0	MPLL_EN_0	R/W	Tx MPLL Enable-lane 0 Set this bit to power-up the MPLL. This bit must be 1, for normal operation. **Value After Reset:** 0x1 **Exists:** Always

5.2.3.92 VR_XS_PMA_MP_12G_16G_MPLLA_CTRL0

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G PHY MPLLA Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0051 : (APB3_16) ? (0x0051*0x2) : (0x0051*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16))

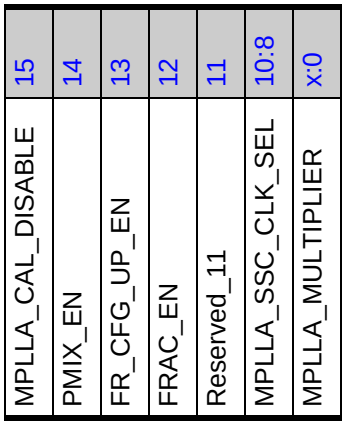


Table 5-222 Fields for Register: VR_XS_PMA_MP_12G_16G_MPLLA_CTRL0

Bits	Name	Memory Access	Description
15	MPLLA_CAL_DISABLE	R/W	MPLLA Calibration Disable This field can be programmed to 1, to disable calibration of MPLLA by PHY firmware. **Value After Reset:** 0x0 **Exists:** Always
14	PMIX_EN	* Varies	**For Multi-protocol 10G Gen2 configurations:** MPLLA PMIX Enabled This field drives the output port 'xpcs_mplla_pmix_en_o'. **For Other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SNPS_RSVDPARAM_22 ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
13	FR_CFG_UP_EN	* Varies	**For Multi-protocol 10G Gen2 configurations:** MPLLA Fractional Configuration Latch Signal This field drives the xpcs_mplla_fracn_cfg_updt_en_o port. **For Other configurations:** Reserved **Value After Reset:** "SNPS_RSVDPARAM_22 ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "SNPS_RSVDPARAM_22 ? \"read-write\" : \"read-only\""
12	FRAC_EN	* Varies	**For Multi-protocol 10G Gen2 configurations:** MPLLA Fractional Mode Enable This field drives the xpcs_mplla_fracn_en_o port. **For Other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SNPS_RSVDPARAM_22 ? \"read-write\" : \"read-only\""
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** !SNPS_RSVDPARAM_22
10:8	MPLLA_SSC_CLK_SEL	* Varies	This field controls MPLLA spread-spectrum clock select of Multi-protocol 10G Gen1 PHY. For other configurations, this is a reserved field. **Value After Reset:** 0x0 **Exists:** !SNPS_RSVDPARAM_22 **Memory Access:** "(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
x:0	MPLLA_MULTIPLIER	R/W	MPLLA frequency Multiplier Control This field controls the multiplication of reference clock to a frequency suitable for operating speed. Any change in this field must be followed by a vendor-specific soft reset to ensure that PHY is properly powered-up in the desired mode. **Value After Reset:** "(BASE1000X_ONLY||KX_ONLY)&&!SNPS_RSVDPARAM_35 ? (SNPS_RSVDPARAM_39 ? 0x40 : 0x20) : SNPS_RSVDPARAM_35&&!SNPS_RSVDPARAM_13 ? 0x50 : SNPS_RSVDPARAM_35&&SNPS_RSVDPARAM_13 ? 0x19 : (RPCS_PRESENT) ? (SNPS_RSVDPARAM_22 ? 0x64 : SNPS_RSVDPARAM_39 ? 0x67 : SNPS_RSVDPARAM_33 ? 0x42 : 0x21) : (SNPS_RSVDPARAM_39 ? 0x0 : 0x28)" **Exists:** Always **Range Variable[x]:** "SNPS_RSVDPARAM_22 ? 12 : 8" - 1

5.2.3.93 VR_XS_PMA_MP_25G_MPLLA_CTRL0

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLA Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0051 : (APB3_16) ? (0x0051*0x2) : (0x0051*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&E25&&!SNPS_RSVDPARAM_37

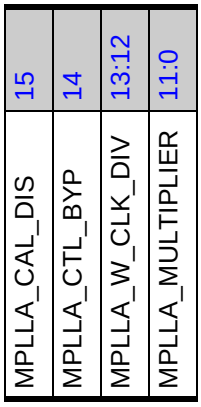


Table 5-223 Fields for Register: VR_XS_PMA_MP_25G_MPLLA_CTRL0

Bits	Name	Memory Access	Description
15	MPLLA_CAL_DIS	R/W	MPLLA Calibration Disable This field drives the xpcs_mplla_init_cal_disable_o port. **Value After Reset:** 0x0 **Exists:** Always
14	MPLLA_CTL_BYP	R/W	MPLLA Control Buffer Bypass This field drives the xpcs_mplla_ctl_buf_byp_o port. **Value After Reset:** 0x0 **Exists:** Always
13:12	MPLLA_W_CLK_DIV	R/W	MPLLA Word Clock Divider This field controls the MPLLA word clock dividers for generating mplla_word_clk.This field drives the xpcs_mplla_word_clk_div_o port. **Value After Reset:** "RPCS_PRESENT&&(R_SRDS_INTF==0||R_SRDS_INTF==1) ? 0x2 : ((R_SRDS_INTF==4||R_SRDS_INTF==6)||(SNPS_RSVDPARAM_36&&KX_ONLY)) ? 0x3 : 0x1" **Exists:** Always

Bits	Name	Memory Access	Description
11:0	MPLLA_MULTIPLIER	R/W	MPLLA frequency Multiplier Control This field controls the multiplication of reference clock to a frequency suitable for operating speed. Any change in this field must be followed by a vendor-specific Soft Reset to ensure that PHY is properly powered-up in the desired mode. **Value After Reset:** "(BASE1000X_ONLY||KX_ONLY) ? ((SNPS_RSVDPARAM_13&&SNPS_RSVDPARAM_36) ? 0x64 : 0x80) : RPCS_PRESENT ? (SNPS_RSVDPARAM_16 ? 0x10C : 0x84) : 0x78" **Exists:** Always

5.2.3.94 VR_XS_PMA_MP_12G_MPLLA_CTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G PHY MPLLA Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0052 : (APB3_16) ? (0x0052*0x2) : (0x0052*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD | |XGXS_PCS)&&SNPS_XAUI_PHY&&E12&&!SNPS_RSVDPAR-AM_22&&!SNPS_RSVDPARAM_39)

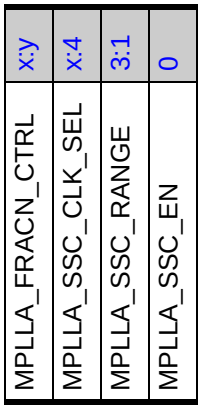


Table 5-224 Fields for Register: VR_XS_PMA_MP_12G_MPLLA_CTRL1

Bits	Name	Memory Access	Description
x:y	MPLLA_FRACN_CTRL	R/W	MPLLA Fractional Control This field drives the output port xpcs_mplla_fracn_ctrl_o. **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33) ? 11 : 9" + "(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33) ? 5 : 7" - 1 **Range Variable[y]:** "(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33) ? 5 : 7"

Bits	Name	Memory Access	Description
x:4	MPLLA_SSC_CLK_SEL	* Varies	**For Multi-protocol 12G configurations:** MPLLA SSC Clock Select This field drives the output port xpcs_mplla_ssc_clk_sel_o. **For Multi-protocol 10G configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "!(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33) ? 3 : 1" + 3 **Memory Access:** "!(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33) ? \"read-write\" : \"read-only\""
3:1	MPLLA_SSC_RANGE	R/W	MPLLA SSC Range This field selects the modulation to insert. This field applies a fixed PPM offset to phase accumulator. **mplla_ssc_rangePPM Offset** 3'b000 -4980 3'b001 -4492 3'b010 -4003 3'b011 -3027 3'b100 4980 3'b101 4492 3'b110 4003 3'b111 3027 **Value After Reset:** 0x0 **Exists:** Always
0	MPLLA_SSC_EN	R/W	MPLLA SSC Enable This field can be set to enable spread-spectrum generation on mplla_div_clk output. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.95 VR_XS_PMA_MP_16G_MPLLA_CTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G PHY MPLLA Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0052 : (APB3_16) ? (0x0052*0x2) : (0x0052*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||SNPS_RSVDPARAM_39))

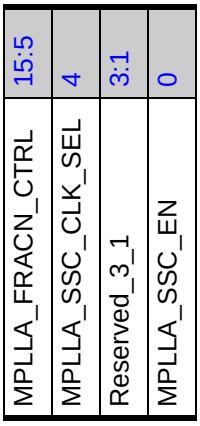


Table 5-225 Fields for Register: VR_XS_PMA_MP_16G_MPLLA_CTRL1

Bits	Name	Memory Access	Description
15:5	MPLLA_FRACN_CTRL	R/W	MPLLA Fractional Control This field drives the output port xpcs_mplla_fracn_ctrl_o. **Value After Reset:** "SNPS_RSVDPARAM_39 ? 0x186 : 0x0" **Exists:** Always
4	MPLLA_SSC_CLK_SEL	R/W	MPLLA SSC Clock Select This field drives the output port xpcs_mplla_ssc_clk_sel_o. **Value After Reset:** 0x0 **Exists:** Always
3:1	Reserved_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	MPLLA_SSC_EN	R/W	MPLLA SSC Enable This field can be set to enable spread-spectrum generation on mplla_div_clk output. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.96 VR_XS_PMA_MP_25G_MPLLA_CTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLA Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0052 : (APB3_16) ? (0x0052*0x2) : (0x0052*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&E25&&!SNPS_RSVDPARAM_37

15	14	13:12	11	10	9	8	7:4	3:2	1	0
SSC_SPD	SSC_EN	BW_TH	Reserved_11	FB_DIV4_EN	SH_LK_EN	FRAC_EN	Reserved_7_4	RCAL_BK_SEL	RCAL_SKP	RCAL_FRC

Table 5-226 Fields for Register: VR_XS_PMA_MP_25G_MPLLA_CTRL1

Bits	Name	Memory Access	Description
15	SSC_SPD	R/W	MPLLA Up Spread Enable This field drives the xpcs_mplla_scc_sp d_en_o port. **Value After Reset:** 0x0 **Exists:** Always
14	SSC_EN	R/W	MPLLA Spread Spectrum Enable This field drives the xpcs_mplla_ssc_en_o port. **Value After Reset:** 0x0 **Exists:** Always
13:12	BW_TH	R/W	MPLLA Bandwidth Threshold This field drives the xpcs_mplla_bw_th_o port. **Value After Reset:** 0x3 **Exists:** Always
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
10	FB_DIV4_EN	R/W	MPLLA DIV clock output frequency divider control This field drives the xpcs_mplla_fb_clk_div4_en_o port. **Value After Reset:** 0x1 **Exists:** Always
9	SH_LK_EN	R/W	MPLLA Short Lock Enable This field drives the xpcs_mplla_short_lock_en_o port. **Value After Reset:** 0x0 **Exists:** Always
8	FRAC_EN	R/W	MPLLA Fractional Enable This field drives the xpcs_mplla_fracn_en_o port. **Value After Reset:** "SNPS_RSVDPARAM_16 ? 0x1 : 0x0" **Exists:** Always
7:4	Reserved_7_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:2	RCAL_BK_SEL	R/W	MPLLA Recalibration Bank Select This field drives the xpcs_mplla_recal_bank_sel_o port. **Value After Reset:** 0x0 **Exists:** Always
1	RCAL_SKP	R/W	MPLLA Recalibration Skip Enable This field drives the xpcs_mplla_recal_skip_en_o port. **Value After Reset:** 0x0 **Exists:** Always
0	RCAL_FRC	R/W	MPLLA Recalibration Force Enable This drives the xpcs_mplla_recal_force_en_o port. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.97 VR_XS_PMA_MP_32G_MPLLA_CTRL1

- **Description:** VR XS or PMA Synopsys Multi Protocol 32G PHY MPLLA Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0052 : (APB3_16) ? (0x0052*0x2) : (0x0052*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&(SNPS_RSVDPARAM_37 | |E32)

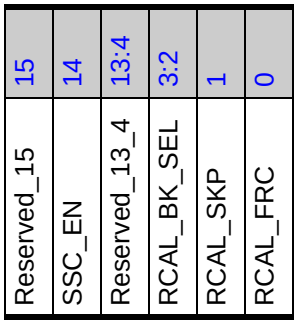


Table 5-227 Fields for Register: VR_XS_PMA_MP_32G_MPLLA_CTRL1

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14	SSC_EN	R/W	MPLLA Spread Spectrum Enable This field drives the xpcs_mplla_ssc_en_o port. **Value After Reset:** 0x0 **Exists:** Always
13:4	Reserved_13_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:2	RCAL_BK_SEL	R/W	MPLLA Recalibration Bank Select This field drives the xpcs_mplla_recal_bank_sel_o port. **Value After Reset:** 0x0 **Exists:** Always
1	RCAL_SKP	R/W	MPLLA Recalibration Skip Enable This field drives the xpcs_mplla_recal_skip_en_o port. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	RCAL_FRC	R/W	MPLLA Recalibration Force Enable This drives the xpcs_mplla_recal_force_en_o port. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.98 VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G PHY MPLLA Control 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0053 : (APB3_16) ? (0x0053*0x2) : (0x0053*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16)&&!SNPS_RSVDPARAM_22&&!SNPS_RSVDPARAM_33)

15	x:y	x:11	10	9	8	x:y	x:0
MPLLA_WRD_DIV2_EN	MPLLA_RECAL_BANK_SEL	MPLLA_TX_CLK_DIV	MPLLA_DIV16P5_CLK_EN	MPLLA_DIV10_CLK_EN	MPLLA_DIV8_CLK_EN	MPLLA_DIV_CLK_EN	MPLLA_DIV_MULT

Table 5-228 Fields for Register: VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2

Bits	Name	Memory Access	Description
15	MPLLA_WRD_DIV2_EN	* Varies	MPLLA Word Clock Divide by 2 This field controls the xpcs_mplla_word_div2_en_o port. **Note:**This field is valid only for Multi-protocol 8G PHY configurations. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SNPS_RSVDPARAM_39 ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
x:y	MPLLA_RECAL_BANK_SEL	* Varies	**For Multi-protocol 16G configurations:** MPLLA Re-calibration Bank Select This field drives the output port 'xpcs_mplla_recal_bank_sel_o[1:0]'. **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "!(SNPS_RSVDPARAM_39||SNPS_RSVDPARAM_35) ? 2 : 1" + "!(SNPS_RSVDPARAM_39||SNPS_RSVDPARAM_35) ? 13 : 14" - 1 **Range Variable[y]:** "!(SNPS_RSVDPARAM_39||SNPS_RSVDPARAM_35) ? 13 : 14" **Memory Access:** "(E16) ? \"read-write\" :\"read-only\""
x:11	MPLLA_TX_CLK_DIV	* Varies	MPLLA Tx Clock Divider This field drives the output port 'xpcs_mplla_tx_clk_div_o[1:0]'. This field is present only for configurations with Multi-protocol 16G PHY/Multi-protocol 10G PHY. **Value After Reset:** "(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33) ? 0x1 : SNPS_RSVDPARAM_39 ? 0x3 : 0x0" **Exists:** Always **Range Variable[x]:** "SNPS_RSVDPARAM_39||SNPS_RSVDPARAM_35 ? 3 : 2" + 10 **Memory Access:** "(E16||SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_39||SNPS_RSVDPARAM_33) ? \"read-write\" :\"read-only\""
10	MPLLA_DIV16P5_CLK_EN	R/W	MPLLA Divide by 16.5 Enable This bit can be set to enable output clocks derived from MPLLA based on 16.5,33 and 66 division ratios. **Value After Reset:** "RPCS_PRESENT ? 0x1 : 0x0" **Exists:** !(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33)
9	MPLLA_DIV10_CLK_EN	R/W	MPLLA Divide by 10 Enable When this bit is set, the frequency of the mplla_word_clk output clock from PHY is MPLLA frequency divided by 10. **Value After Reset:** "RPCS_PRESENT&&(R_SRDS_INTF==0) ? 0x0 : 0x1" **Exists:** Always

Bits	Name	Memory Access	Description
8	MPLLA_DIV8_CLK_EN	R/W	MPLLA Divide by 8 Enable When this bit is set, the frequency of the mplla_word_clk output clock from PHY is MPLLA frequency divided by 8. **Value After Reset:** "RPCS_PRESENT&&(R_SRDS_INTF==0) ? 0x1 : 0x0" **Exists:** Always
x:y	MPLLA_DIV_CLK_EN	R/W	Enable mplla_div_clk from PHY. When asserted, the frequency of mplla_div_clk from PHY is the MPLLA frequency divided by 'mplla_div_multiplier' **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33) ? 10 **Range Variable[y]:** 7"
x:0	MPLLA_DIV_MULT	R/W	MPLLA Output Frequency Multiplier Control This field controls the frequency multiplication factor used to generate MPLLA clock output from the reference clock input as seen by the MPLL. **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33) ? 8 : 7" - 1

5.2.3.99 VR_XS_PMA_MP_25G_MPLLA_CTRL2

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLA Control 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0053 : (APB3_16) ? (0x0053*0x2) : (0x0053*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&E25&&!SNPS_RSVDPARAM_37

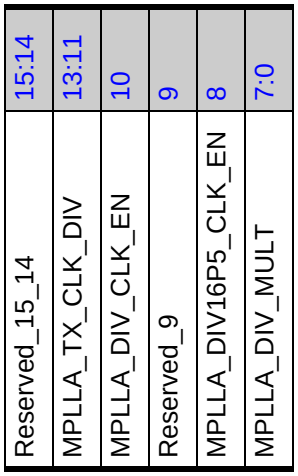


Table 5-229 Fields for Register: VR_XS_PMA_MP_25G_MPLLA_CTRL2

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:11	MPLLA_TX_CLK_DIV	R/W	MPLLA Tx Clock Divider This field drives the output port 'xpcs_mplla_tx_clk_div_o[1:0]'. **Value After Reset:** "RPCS_PRESENT ? 0x1 : KX_ONLY ? 0x4 : 0x3" **Exists:** Always
10	MPLLA_DIV_CLK_EN	R/W	Enable mplla_div_clk from PHY. When asserted, the frequency of mplla_div_clk from PHY is the MPLLA frequency divided by 'mplla_div_multiplier' **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
9	Reserved_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	MPLLA_DIV16P5_CLK_EN	R/W	MPLLA Divide by 16.5 Enable This bit can be set to enable output clocks derived from MPLLA based on 16.5,33 and 66 division ratios. **Value After Reset:** "RPCS_PRESENT ? 0x1 : 0x0" **Exists:** Always
7:0	MPLLA_DIV_MULT	R/W	MPLLA Output Frequency Multiplier Control This field controls the frequency multiplication factor used to generate MPLLA clock output from the reference clock input as seen by the MPLL.This field drives the xpcs_mplla_div_mult_o port. **Value After Reset:** "SNPS_RSVDPARAM_36 ? (RPCS_PRESENT ? 0x8 : RXAUI_DEFAULT ? 0xF : KX_ONLY ? 0x28 : 0x1E) : 0x0" **Exists:** Always

5.2.3.100 VR_XS_PMA_MP_12G_16G_MPLLB_CTRL0

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G PHY MPLLB Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0054 : (APB3_16) ? (0x0054*0x2) : (0x0054*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16)&&!SNPS_RSVDPARAM_39

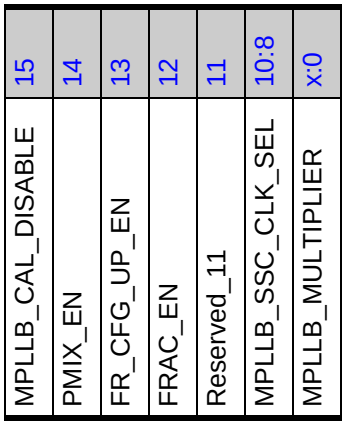


Table 5-230 Fields for Register: VR_XS_PMA_MP_12G_16G_MPLLB_CTRL0

Bits	Name	Memory Access	Description
15	MPLLB_CAL_DISABLE	R/W	MPLLB Calibration Disable This field can be programmed to 1, to disable calibration of MPLLB by PHY firmware. **Value After Reset:** 0x1 **Exists:** Always
14	PMIX_EN	* Varies	**For Multi-protocol 10G Gen2 configurations:** MPLLA PMIX Enabled This field drives the output port 'xpcs_mpllb_pmix_en_o'. **For Other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SNPS_RSVDPARAM_22 ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
13	FR_CFG_UP_EN	* Varies	**For Multi-protocol 10G Gen2 configurations:** MPLLB Fractional Configuration Latch Signal This field drives the xpcs_mpllb_fracn_cfg_updt_en_o port. **For Other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SNPS_RSVDPARAM_22 ? \"read-write\" : \"read-only\""
12	FRAC_EN	* Varies	**For Multi-protocol 10G Gen2 configurations:** MPLLB Fractional Mode Enable This field drives the xpcs_mpllb_fracn_en_o port. **For Other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SNPS_RSVDPARAM_22 ? \"read-write\" : \"read-only\""
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** !SNPS_RSVDPARAM_22
10:8	MPLLB_SSC_CLK_SEL	* Varies	This field controls MPLLB spread-spectrum clock select of Multi-protocol 10G Gen1 PHY. For other configurations, this is a reserved field. **Value After Reset:** 0x0 **Exists:** !SNPS_RSVDPARAM_22 **Memory Access:** "(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33) ? \"read-write\" : \"read-only\""
x:0	MPLLB_MULTIPLIER	R/W	MPLLB frequency Multiplier Control This field controls the multiplication of reference clock to a frequency suitable for operating speed. Any change in this field must be followed by a vendor-specific Soft Reset to ensure that PHY is properly powered-up in the desired mode. **Value After Reset:** "SNPS_RSVDPARAM_35&&!SNPS_RSVDPARAM_13?0x7d : SNPS_RSVDPARAM_35 ? 0x9c : 0x0" **Exists:** Always **Range Variable[x]:** "SNPS_RSVDPARAM_22 ? 12 : 8" - 1

5.2.3.101 VR_XS_PMA_MP_25G_MPLLB_CTRL0

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLB Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0054 : (APB3_16) ? (0x0054*0x2) : (0x0054*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&E25&&!SNPS_RSVDPARAM_37

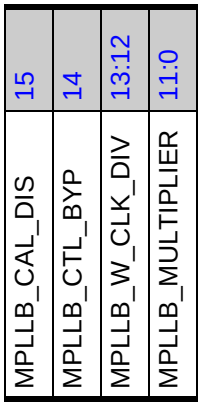


Table 5-231 Fields for Register: VR_XS_PMA_MP_25G_MPLLB_CTRL0

Bits	Name	Memory Access	Description
15	MPLLB_CAL_DIS	R/W	MPLLB Calibration Disable This field drives the xpcs_mpllb_init_cal_disable_o port. **Value After Reset:** 0x1 **Exists:** Always
14	MPLLB_CTL_BYP	* Varies	MPLLB Control Buffer Bypass This field drives the xpcs_mpllb_ctl_buf_byp_o port. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_RSVDPARAM_36) ? \"read-only\" :\"read-write\""
13:12	MPLLB_W_CLK_DIV	R/W	MPLLB Word Clock Divider This field controls the MPLLB word clock dividers for generating mpllb_word_clk.This field drives the xpcs_mpllb_word_clk_div_o port. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
11:0	MPLLB_MULTIPLIER	R/W	MPLLB frequency Multiplier Control This field controls the multiplication of reference clock to a frequency suitable for operating speed. Any change in this field must be followed by a vendor-specific Soft Reset to ensure that PHY is properly powered-up in the desired mode. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.102 VR_XS_PMA_MP_12G_MPLLB_CTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G PHY MPLLB Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0055 : (APB3_16) ? (0x0055*0x2) : (0x0055*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD | |XGXS_PCS)&&SNPS_XAUI_PHY&&E12&&!SNPS_RSVDPAR-AM_22&&!SNPS_RSVDPARAM_39)

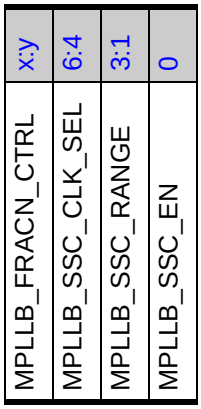


Table 5-232 Fields for Register: VR_XS_PMA_MP_12G_MPLLB_CTRL1

Bits	Name	Memory Access	Description
x:y	MPLLB_FRACN_CTRL	R/W	MPLLB Fractional Control This field drives the output port 'xpcs_mpllb_fracn_ctrl_o'. **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33) ? 11 : 9" + "(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33) ? 5 : 7" - 1 **Range Variable[y]:** "(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33) ? 5 : 7"
6:4	MPLLB_SSC_CLK_SEL	R/W	MPLLB Spread Spectrum Clock Select This field drives the output port 'xpcs_mpllb_ssc_clk_sel_o' **Value After Reset:** 0x0 **Exists:** !(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33)

Bits	Name	Memory Access	Description
3:1	MPLLB_SSC_RANGE	R/W	MPLLB Spread Spectrum Range This field selects the modulation to insert. This field applies a fixed PPM offset to phase accumulator. **mplla_ssc_range PPM Offset** 3'b000 -4980 3'b001 -4492 3'b010 -4003 3'b011 -3027 3'b100 4980 3'b101 4492 3'b110 4003 3'b111 3027 This field drives the output port 'xpcs_mpllb_ssc_range_o'. **Value After Reset:** 0x0 **Exists:** Always
0	MPLLB_SSC_EN	R/W	MPLLB Spread Spectrum Enable This field can be set to enable spread-spectrum generation on mplla_div_clk output.This bit drives the output port 'xpcs_mpllb_ssc_en_o'. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.103 VR_XS_PMA_MP_16G_MPLLB_CTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G PHY MPLLB Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0055 : (APB3_16) ? (0x0055*0x2) : (0x0055*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&SNPS_XAUI_PHY&&E16

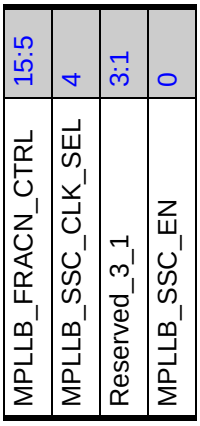


Table 5-233 Fields for Register: VR_XS_PMA_MP_16G_MPLLB_CTRL1

Bits	Name	Memory Access	Description
15:5	MPLLB_FRACN_CTRL	R/W	MPLLB Fractional Control This field drives the output port 'xpcs_mpllb_fracn_ctrl_o'. **Value After Reset:** 0x0 **Exists:** Always
4	MPLLB_SSC_CLK_SEL	R/W	MPLLB Spread Spectrum Clock Select This field drives the output port 'xpcs_mpllb_ssc_clk_sel_o' **Value After Reset:** 0x0 **Exists:** Always
3:1	Reserved_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	MPLLB_SSC_EN	R/W	MPLLB Spread Spectrum Enable This field can be set to enable spread-spectrum generation on mpllb_div_clk output.This bit drives the output port 'xpcs_mpllb_ssc_en_o'. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.104 VR_XS_PMA_MP_25G_MPLLB_CTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLB Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0055 : (APB3_16) ? (0x0055*0x2) : (0x0055*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&E25&&!SNPS_RSVDPARAM_37

15	14	13:12	11	10	9	8	7:6	5:4	3:2	1	0
SSC_SPD	SSC_EN	BW_TH	Reserved_11	FB_DIV4_EN	SH_LK_EN	FRAC_EN	MPLLB_V2I	MPLLB_FREQ_VCO	RCAL_BK_SEL	RCAL_SKP	RCAL_FRC

Table 5-234 Fields for Register: VR_XS_PMA_MP_25G_MPLLB_CTRL1

Bits	Name	Memory Access	Description
15	SSC_SPD	R/W	MPLLB Up Spread Enable This field drives the xpcs_mpllb_ssc_spd_en_o port. **Value After Reset:** 0x0 **Exists:** Always
14	SSC_EN	R/W	MPLLB Spread Spectrum Enable This field drives the xpcs_mpllb_ssc_en_o port. **Value After Reset:** 0x0 **Exists:** Always
13:12	BW_TH	* Varies	MPLLB Bandwidth Threshold This field drives the xpcs_mpllb_bw_th_o port. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_RSVDPARAM_36) ? \"read-only\" : \"read-write\""
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
10	FB_DIV4_EN	R/W	MPLLB DIV clock output frequency divider control This field drives the xpcs_mpllb_fb_clk_div4_en_o port. **Value After Reset:** 0x0 **Exists:** Always
9	SH_LK_EN	* Varies	MPLLB Short Lock Enable This field drives the xpcs_mpllb_short_lock_en_o port. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_RSVDPARAM_36) ? \"read-only\" : \"read-write\""
8	FRAC_EN	R/W	MPLLB Fractional Enable This field drives the xpcs_mpllb_fracn_en_o port. **Value After Reset:** 0x0 **Exists:** Always
7:6	MPLLB_V2I	* Varies	**For configurations with SNPS_RSVDPARAM_36:** MPLLB V2I Operating range This field drives the xpcs_mpllb_v2i_o port and allows the configuration of V2I operating range. **For non SNPS_RSVDPARAM_36 configurations:** Reserved For configurations without SNPS_RSVDPARAM_36 , this bit is Reserved **Value After Reset:** "SNPS_RSVDPARAM_36 ? 0x2 : 0x0" **Exists:** Always **Memory Access:** "(SNPS_RSVDPARAM_36) ? \"read-write\" : \"read-only\""
5:4	MPLLB_FREQ_VCO	* Varies	**For configurations with SNPS_RSVDPARAM_36:** MPLLB VCO Range Control This field drives the xpcs_mpllb_freq_vco_o port and allows to control the desired operating VCO range. **For non SNPS_RSVDPARAM_36 configurations:** Reserved For configurations without SNPS_RSVDPARAM_36 , this bit is Reserved **Value After Reset:** "SNPS_RSVDPARAM_36 ? 0x3 : 0x0" **Exists:** Always **Memory Access:** "(SNPS_RSVDPARAM_36) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
3:2	RCAL_BK_SEL	R/W	MPLLB Recalibration Bank Select This field drives the xpcs_mpllb_recal_bank_sel_o port. **Value After Reset:** 0x0 **Exists:** Always
1	RCAL_SKP	R/W	MPLLB Recalibration Skip Enable This field drives the xpcs_mpllb_recal_skip_en_o port. **Value After Reset:** 0x0 **Exists:** Always
0	RCAL_FRC	R/W	MPLLB Recalibration Force Enable This drives the xpcs_mpllb_recal_force_en_o port. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.105 VR_XS_PMA_MP_32G_MPLLB_CTRL1

- **Description:** VR XS or PMA Synopsys Multi Protocol E32 PHY MPLLB Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0055 : (APB3_16) ? (0x0055*0x2) : (0x0055*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&(SNPS_RSVDPARAM_37 | |E32)

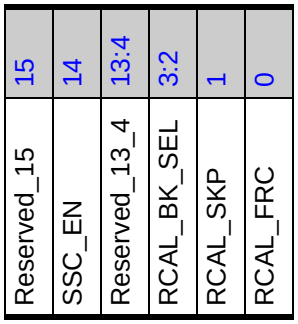


Table 5-235 Fields for Register: VR_XS_PMA_MP_32G_MPLLB_CTRL1

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14	SSC_EN	R/W	MPLLB Spread Spectrum Enable This field drives the xpcs_mplla_ssc_en_o port. **Value After Reset:** 0x0 **Exists:** Always
13:4	Reserved_13_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:2	RCAL_BK_SEL	R/W	MPLLB Recalibration Bank Select This field drives the xpcs_mplla_recal_bank_sel_o port. **Value After Reset:** 0x0 **Exists:** Always
1	RCAL_SKP	R/W	MPLLB Recalibration Skip Enable This field drives the xpcs_mplla_recal_skip_en_o port. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	RCAL_FRC	R/W	MPLLB Recalibration Force Enable This drives the xpcs_mplla_recal_force_en_o port. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.106 VR_XS_PMA_MP_12G_16G_MPLLB_CTRL2

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G PHY MPLLB Control 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0056 : (APB3_16) ? (0x0056*0x2) : (0x0056*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16)&&!SNPS_RSVDPARAM_22&&!SNPS_RSVDPARAM_39&&!SNPS_RSVDPARAM_33)

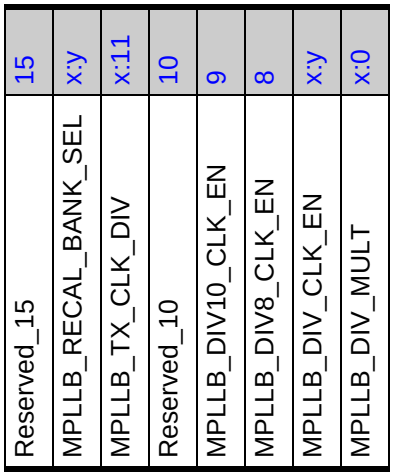


Table 5-236 Fields for Register: VR_XS_PMA_MP_12G_16G_MPLLB_CTRL2

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
x:y	MPLLB_RECAL_BANK_SEL	* Varies	**For Multi-protocol 16G configurations:** MPLLB Re-calibration Bank Select This field drives the output port 'xpcs_mpllb_recal_bank_sel_o[1:0]'. **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "!(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33) ? 2 : 1" + "!(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33) ? 13 : 14" - 1 **Range Variable[y]:** "!(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33) ? 13 : 14" **Memory Access:** "(E16) ? \"read-write\" :\"read-only\""
x:11	MPLLB_TX_CLK_DIV	* Varies	MPLLB Tx Clock Divider This field drives the output port 'xpcs_mpllb_tx_clk_div_o[1:0]' This field is valid only for configurations with Multi-protocol 16G PHY/Multi-protocol 10G PHY. **Value After Reset:** "(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33) ? 0x2 : 0x0" **Exists:** Always **Range Variable[x]:** "(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33) ? 3 : 2" + 10 **Memory Access:** "(E16||SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33) ? \"read-write\" :\"read-only\""
10	Reserved_10	R	Reserved **Value After Reset:** 0x0 **Exists:** !(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33)
9	MPLLB_DIV10_CLK_EN	R/W	MPLLB Divide by 10 Enable When this bit is set, the frequency of the mpllb_word_clk output clock from PHY is MPLLB frequency divided by 10. **Value After Reset:** 0x0 **Exists:** Always
8	MPLLB_DIV8_CLK_EN	R/W	MPLLB Divide by 8 Enable When this bit is set, the frequency of the mpllb_word_clk output clock from PHY is MPLLB frequency divided by 8. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
x:y	MPLLB_DIV_CLK_EN	R/W	Enable mpllb_div_clk from PHY When asserted, the frequency of mpllb_div_clk output from PHY is MPLLB frequency divided by 'mpllb_div_multiplier' **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33) ? 10 **Range Variable[y]:** 7"
x:0	MPLLB_DIV_MULT	R/W	MPLLB Output Frequency Multiplier Control This field controls the frequency multiplication factor used to generate MPLLB clock output from the reference clock input as seen by the MPLL. **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "SNPS_RSVDPARAM_35 ? 8 : 7" - 1

5.2.3.107 VR_XS_PMA_MP_25G_MPLLB_CTRL2

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLB Control 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0056 : (APB3_16) ? (0x0056*0x2) : (0x0056*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&E25&&!SNPS_RSVDPARAM_37

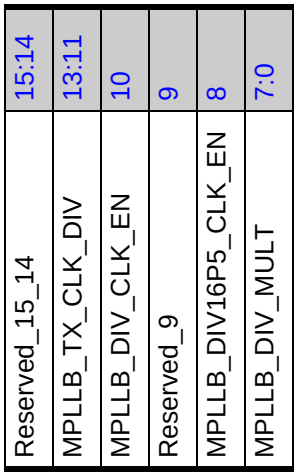


Table 5-237 Fields for Register: VR_XS_PMA_MP_25G_MPLLB_CTRL2

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:11	MPLLB_TX_CLK_DIV	R/W	MPLLB Tx Clock Divider This field drives the output port 'xpcs_mpllb_tx_clk_div_o[1:0]'. **Value After Reset:** 0x0 **Exists:** Always
10	MPLLB_DIV_CLK_EN	R/W	Enable mpllb_div_clk from PHY. When asserted, the frequency of mpllb_div_clk from PHY is the MPLLB frequency divided by 'mpllb_div_multiplier' **Value After Reset:** 0x0 **Exists:** Always
9	Reserved_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
8	MPLLB_DIV16P5_CLK_EN	* Varies	MPLLB Divide by 16.5 Enable This bit can be set to enable output clocks derived from MPLLB based on 16.5,33 and 66 division ratios. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_RSVDPARAM_36) ? \"read-only\" :\"read-write\""
7:0	MPLLB_DIV_MULT	R/W	MPLLB Output Frequency Multiplier Control This field controls the frequency multiplication factor used to generate MPLLB clock output from the reference clock input as seen by the MPLL.This field drives the xpcs_mpllb_div_mult_o port. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.108 VR_XS_PMA_MP_12G_MPLLA_CTRL3

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G PHY MPLLA Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0057 : (APB3_16) ? (0x0057*0x2) : (0x0057*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&E12&&!SNPS_RSVDPARAM_22&&!SNPS_RSVDPARAM_39)

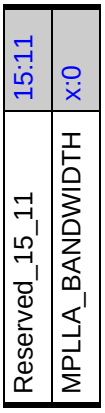


Table 5-238 Fields for Register: VR_XS_PMA_MP_12G_MPLLA_CTRL3

Bits	Name	Memory Access	Description
15:11	Reserved_15_11	R	Reserved **Value After Reset:** 0x0 **Exists:** !(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33)
x:0	MPLLA_BANDWIDTH	R/W	MPLLA Bandwidth Control This field controls the bandwidth of MPLLA present in Multi-protocol 12G PHY. This field drives the output port 'xpcs_mplla_bandwidth_o'. **Value After Reset:** "SNPS_RSVDPARAM_35&&SNPS_RSVDPARAM_13 ? 0x165 : SNPS_RSVDPARAM_33 ? 0x783 : SNPS_RSVDPARAM_35 ? 0x2B: RPCS_PRESENT ? 0x790 : (KX_ONLY||BASE1000X_ONLY) ? 0x798 : 0x783" **Exists:** Always **Range Variable[x]:** "(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33) ? 16 : 11" - 1

5.2.3.109 VR_XS_PMA_MP_25G_MPLLA_CTRL3

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLA Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0057 : (APB3_16) ? (0x0057*0x2) : (0x0057*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&E25&&!SNPS_RSVDPARAM_37



Table 5-239 Fields for Register: VR_XS_PMA_MP_25G_MPLLA_CTRL3

Bits	Name	Memory Access	Description
15:0	MPLLA_FRACN_QUOT	R/W	MPLLA fractional Quotient Control This field drives the xpcs_mplla_fracn_quot_o port. **Value After Reset:** "SNPS_RSVDPARAM_16 ? 0x4700 : 0x0" **Exists:** Always

5.2.3.110 VR_XS_PMA_MP_16G_MPLLA_CTRL3

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G PHY MPLLA Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0057 : (APB3_16) ? (0x0057*0x2) : (0x0057*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&E16)



Table 5-240 Fields for Register: VR_XS_PMA_MP_16G_MPLLA_CTRL3

Bits	Name	Memory Access	Description
15:0	MPLLA_BANDWIDTH	R/W	MPLLA Bandwidth Control This field controls the bandwidth of MPLLA present in Multi-protocol 16G PHY. This field drives the output port 'xpcs_mplla_bandwidth_o'. **Value After Reset:** "(RPCS_PRESENT) ? 0xA016 : (KX_ONLY) ? 0xA035 : 0xA017" **Exists:** Always

5.2.3.111 VR_XS_PMA_MP_6G_MPLL_CTRL0

- **Description:** VR XS or PMA Synopsys Multi-protocol 6G PHY MPLL Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0058 : (APB3_16) ? (0x0058*0x2) : (0x0058*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD | |XGXS_PCS)&&SNPS_XAUI_PHY&&SSP)

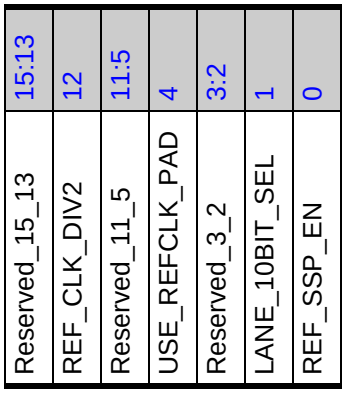


Table 5-241 Fields for Register: VR_XS_PMA_MP_6G_MPLL_CTRL0

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	REF_CLK_DIV2	R/W	Divides the reference clock by 2 before passing it to MPLL **Default Value:** ■ For 156.25 MHz, 125 MHz, 50 MHz Reference Clock Frequency: 1'b1 ■ For 62.5 MHz and 25 MHz Reference Clock Frequency: 1'b0 **Value After Reset:** "((REF_CLK_62==1||REF_CLK_25==1) ? 0x0 : 0x1)" **Exists:** Always
11:5	Reserved_11_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4	USE_REFCLK_PAD	R/W	Use Normal Reference Clock When set to 1, the Synopsys Multi-protocol 6G PHY selects the clock source connected to ref_pad_clk_p/m as its reference clock. When this bit is set to 0, clock source connected to ref_alt_clk_p/m is selected. **Value After Reset:** "(REF_CLK_SRC==0) ? 0x1 : 0x0" **Exists:** Always
3:2	Reserved_3_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1	LANE_10BIT_SEL	R/W	Lane Width Control When set to 1, the lane width is 10 bit and when set to 0, the lane width is 20 bit **Value After Reset:** "(RXAUI_DEFAULT)? 0x0 : 0x1" **Exists:** Always
0	REF_SSP_EN	R/W	Enables Reference Clock When set to 1, the reference clock to PHY is enabled. **Value After Reset:** 0x1 **Exists:** Always

5.2.3.112 VR_XS_PMA_MP_12G_MPLLB_CTRL3

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G PHY MPLLB Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0058 : (APB3_16) ? (0x0058*0x2) : (0x0058*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&E12&&!SNPS_RSVDPARAM_22&&!SNPS_RSVDPARAM_39)

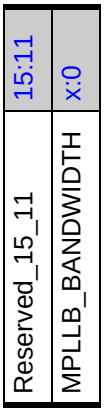


Table 5-242 Fields for Register: VR_XS_PMA_MP_12G_MPLLB_CTRL3

Bits	Name	Memory Access	Description
15:11	Reserved_15_11	R	Reserved **Value After Reset:** 0x0 **Exists:** !(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33)
x:0	MPLLB_BANDWIDTH	R/W	MPLLB Bandwidth Control This field controls the bandwidth of MPLLB present in Multi-protocol 12G PHY. This field drives the output port 'xpcs_mpllb_bandwidth_o'. **Value After Reset:** "(SNPS_RSVDPARAM_35) ? 0x44 : 0x0" **Exists:** Always **Range Variable[x]:** "(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33) ? 16 : 11" - 1

5.2.3.113 VR_XS_PMA_MP_25G_MPLLB_CTRL3

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLB Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0058 : (APB3_16) ? (0x0058*0x2) : (0x0058*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&E25&&!SNPS_RSVDPARAM_37



Table 5-243 Fields for Register: VR_XS_PMA_MP_25G_MPLLB_CTRL3

Bits	Name	Memory Access	Description
15:0	MPLLB_FRACN_QUOT	R/W	MPLLB fractional Quotient Control This field drives the xpcs_mpllb_fracn_quot_o port. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.114 VR_XS_PMA_MP_16G_MPLLB_CTRL3

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G PHY MPLLB Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0058 : (APB3_16) ? (0x0058*0x2) : (0x0058*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&E16)



Table 5-244 Fields for Register: VR_XS_PMA_MP_16G_MPLLB_CTRL3

Bits	Name	Memory Access	Description
15:0	MPLLB_BANDWIDTH	R/W	MPLLB Bandwidth Control This field controls the bandwidth of MPLLB present in Multi-protocol 16G PHY. This field drives the output port 'xpcs_mpllb_bandwidth_o'. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.115 VR_XS_PMA_MP_6G_MPLL_CTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 6G PHY MPLL Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0059 : (APB3_16) ? (0x0059*0x2) : (0x0059*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&SSP)

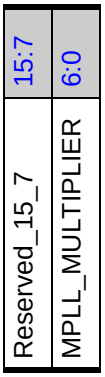


Table 5-245 Fields for Register: VR_XS_PMA_MP_6G_MPLL_CTRL1

Bits	Name	Memory Access	Description
15:7	Reserved_15_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
6:0	MPLL_MULTIPLIER	R/W	MPLL Frequency Multiplier Control **Default Value:** The default value of this bit depends on the reference clock frequency and the configuration: 156.25 MHz Reference Clock Frequency ■ 7'h20 for KX_Only or 1000BASEX-Only PCS configurations ■ 7'h28 for all other configurations. 125 MHz and 62.5 MHz Reference Clock Frequency ■ 7'h28 for KX_Only or 1000BASEX-Only PCS configurations ■ 7'h32 for all other configurations 50 MHz and 25 MHz Reference Clock Frequency ■ 7'h64 KX_Only or 1000BASEX-Only PCS configurations ■ 7'h7D for all other configurations **Value After Reset:** "(REF_CLK_156==1) ? ((SINGLE_LANE_CFG) ? 0x20 :0x28): ((REF_CLK_125==1||REF_CLK_62==1) ? ((SINGLE_LANE_CFG) ? 0x28 :0x32) : ((SINGLE_LANE_CFG) ? 0x64 :0x7d))" **Exists:** Always

5.2.3.116 VR_XS_PMA_MP_32G_PLL_CMN_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 32G PLL Common Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0059 : (APB3_16) ? (0x0059*0x2) : (0x0059*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&E32)

15:12	11:9	8	7:6	5	4	3:1	0
RESERVED_15_12	TX_RPLL_WRD_CLK_EN_3_1	TX_RPLL_WRD_CLK_EN_0	RESERVED_7_6	MPLLB_WRD_CLK_EN	MPLLA_WRD_CLK_EN	TX_PLL_EN_3_1	TX_PLL_EN_0

Table 5-246 Fields for Register: VR_XS_PMA_MP_32G_PLL_CMN_CTRL

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:9	TX_RPLL_WRD_CLK_EN_3_1	* Varies	**For multi lane configurations:** Tx ROPLL Word clock output enable Lane 3:1 The field drives the output port 'xpcs_tx_ropll_wrd_clk_en_o'. **For single lane configurations:** Reserved **Value After Reset:** "!SINGLE_LANE_CFG ? 0x7 : 0x0" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\" "

Bits	Name	Memory Access	Description
8	TX_RPLL_WRD_CLK_EN_0	R/W	Tx ROPLL Word clock output enable Lane 0 The field drives the output port 'xpcs_tx_ropll_wrd_clk_en_o'. **Value After Reset:** 0x1 **Exists:** Always
7:6	RESERVED_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5	MPLLB_WRD_CLK_EN	R/W	MPLLB word clock output enable. This field drives the output port 'xpcs_mpllb_wrd_clk_en_o'. **Value After Reset:** 0x0 **Exists:** Always
4	MPLLA_WRD_CLK_EN	R/W	MPLLA word clock output enable. This field drives the output port 'xpcs_mplla_wrd_clk_en_o'. **Value After Reset:** 0x0 **Exists:** Always
3:1	TX_PLL_EN_3_1	* Varies	**For multi lane configurations:** TX PLL Enable Lanes 3:1. Powers up or powers down the ROPLL or MPLLA or MPLLB. This field drives the output port 'xpcs_tx_pll_en_o'. **For single lane configurations:** Reserved **Value After Reset:** "!SINGLE_LANE_CFG ? 0x7 : 0x0" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\" "
0	TX_PLL_EN_0	R/W	TX PLL Enable Lane0. Powers up or powers down the ROPLL or MPLLA or MPLLB. This field drives the output port 'xpcs_tx_pll_en_o'. **Value After Reset:** 0x1 **Exists:** Always

5.2.3.117 VR_XS_PMA_MP_16G_MPLLA_CTRL4

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G PHY MPLLA Control 4 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0059 : (APB3_16) ? (0x0059*0x2) : (0x0059*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||SNPS_RSVDPARAM_39))



Table 5-247 Fields for Register: VR_XS_PMA_MP_16G_MPLLA_CTRL4

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:0	MPLLA_SSC_FRQ_CNT_INT	R/W	MPLLA SSC Frequency Counter Initialization. This field drives the output port xpcs_mplla_ssc_freq_cnt_init_o[11:0]. **Value After Reset:** "(SNPS_RSVDPARAM_39&RPCS_PRESENT) ? 0x27 : 0x0" **Exists:** Always

5.2.3.118 VR_XS_PMA_MP_25G_MPLLA_CTRL4

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLA Control 4 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0059 : (APB3_16) ? (0x0059*0x2) : (0x0059*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&E25&&!SNPS_RSVDPARAM_37



Table 5-248 Fields for Register: VR_XS_PMA_MP_25G_MPLLA_CTRL4

Bits	Name	Memory Access	Description
15:0	MPLLA_FRACN_REM	R/W	MPLLA fractional Remainder Control This field drives the xpcs_mplla_fracn_rem_o port. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.119 VR_XS_PMA_MP_25G_MPLLB_CTRL4

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLB Control 4 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x005A : (APB3_16) ? (0x005A*0x2) : (0x005A*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&E25&&!SNPS_RSVDPARAM_37



Table 5-249 Fields for Register: VR_XS_PMA_MP_25G_MPLLB_CTRL4

Bits	Name	Memory Access	Description
15:0	MPLLB_FRACN_REM	R/W	MPLLB fractional Remainder Control This field drives the xpcs_mpllb_fracn_rem_o port. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.120 VR_XS_PMA_MP_16G_MPLLA_CTRL5

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G PHY MPLLA Control 5 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x005A : (APB3_16) ? (0x005A*0x2) : (0x005A*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||SNPS_RSVDPARAM_39)

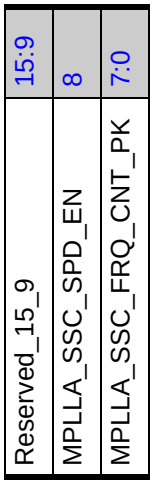


Table 5-250 Fields for Register: VR_XS_PMA_MP_16G_MPLLA_CTRL5

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	MPLLA_SSC_SPD_EN	R/W	MPLLA SSC Up Spread Enable. This field drives the output port xpcs_mplla_ssc_spd_en_o. **Value After Reset:** 0x0 **Exists:** Always
7:0	MPLLA_SSC_FRQ_CNT_PK	R/W	MPLLA SSC Frequency Counter Peak. This field drives the output port xpcs_mplla_ssc_freq_cnt_pk_o[7:0]. **Value After Reset:** "(SNPS_RSVDPARAM_39&&RPCS_PRESENT) ? 0x66 : 0x0" **Exists:** Always

5.2.3.121 VR_XS_PMA_MP_16G_MPLLB_CTRL4

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G PHY MPLLB Control 4 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x005B : (APB3_16) ? (0x005B*0x2) : (0x005B*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&E16



Table 5-251 Fields for Register: VR_XS_PMA_MP_16G_MPLLB_CTRL4

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:0	MPLLB_SSC_FRQ_CNT_INT	R/W	MPLLB SSC Frequency Counter Initialization. This field drives the output port xpcs_mpllb_ssc_freq_cnt_init_o[11:0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.122 VR_XS_PMA_MP_25G_MPLLA_CTRL5

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLA Control 5 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x005B : (APB3_16) ? (0x005B*0x2) : (0x005B*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&E25&&!SNPS_RSVDPARAM_37



Table 5-252 Fields for Register: VR_XS_PMA_MP_25G_MPLLA_CTRL5

Bits	Name	Memory Access	Description
15:0	MPLLA_FRACN_DEN	R/W	MPLLA fractional Denominator Control This field drives the xpcs_mplla_fracn_den_o port. **Value After Reset:** "SNPS_RSVDPARAM_16 ? 0x1 : 0x0" **Exists:** Always

5.2.3.123 VR_XS_PMA_MP_16G_MPLLB_CTRL5

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G PHY MPLLB Control 5 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x005C : (APB3_16) ? (0x005C*0x2) : (0x005C*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD | |XGXS_PCS)&&SNPS_XAUI_PHY&&E16)

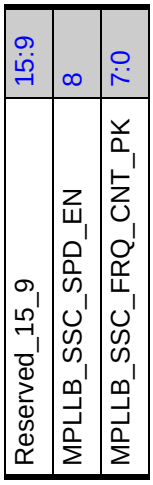


Table 5-253 Fields for Register: VR_XS_PMA_MP_16G_MPLLB_CTRL5

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	MPLLB_SSC_SPD_EN	R/W	MPLLB SSC Up Spread Enable. This field drives the output port xpcs_mpllb_ssc_spd_en_o. **Value After Reset:** 0x0 **Exists:** Always
7:0	MPLLB_SSC_FRQ_CNT_PK	R/W	MPLLB SSC Frequency Counter Peak. This field drives the output port xpcs_mpllb_ssc_freq_cnt_pk_o[7:0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.124 VR_XS_PMA_MP_25G_MPLLB_CTRL5

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLB Control 5 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x005C : (APB3_16) ? (0x005C*0x2) : (0x005C*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&E25&&!SNPS_RSVDPARAM_37



Table 5-254 Fields for Register: VR_XS_PMA_MP_25G_MPLLB_CTRL5

Bits	Name	Memory Access	Description
15:0	MPLLB_FRACN_DEN	R/W	MPLLB fractional Denominator Control This field drives the xpcs_mpllb_fracn_den_o port. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.125 VR_XS_PMA_MP_8G_MPLLA_CTRL6

- **Description:** VR XS or PMA Synopsys Multi-protocol 8G PHY MPLLA Control 6 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x005D : (APB3_16) ? (0x005D*0x2) : (0x005D*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_RSVDPARAM_39

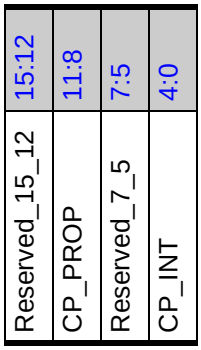


Table 5-255 Fields for Register: VR_XS_PMA_MP_8G_MPLLA_CTRL6

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:8	CP_PROP	R/W	MPLLA Charge Pump Proportional Control This field drives the xpcs_mplla_cp_prop_o port. **Value After Reset:** "RPCS_PRESENT ? 0x5 :KX_ONLY ? 0x6 : 0xA" **Exists:** Always
7:5	Reserved_7_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4:0	CP_INT	R/W	MPLLA Charge Pump Integral Control This field drives the xpcs_mplla_cp_int_o port. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.126 VR_XS_PMA_MP_25G_MPLLA_BW_HI_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLA Bandwidth High Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x005D : (APB3_16) ? (0x005D*0x2) : (0x005D*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&E25&&!SNPS_RSVDPARAM_37

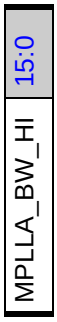


Table 5-256 Fields for Register: VR_XS_PMA_MP_25G_MPLLA_BW_HI_CTRL

Bits	Name	Memory Access	Description
15:0	MPLLA_BW_HI	R/W	MPLLA Bandwidth Control -High This field controls the bandwidth of MPLLA present in Multi-protocol 25G PHY. This field drives the output port 'xpcs_mplla_bw_high_o'. **Value After Reset:** "SNPS_RSVDPARAM_36 ? (RPCS_PRESENT ? 0x16 : 0x26) : SNPS_RSVDPARAM_16 ? 0xf : 0x7" **Exists:** Always

5.2.3.127 VR_XS_PMA_MP_25G_MPLLA_BW_LO_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLA Bandwidth Low Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x005E : (APB3_16) ? (0x005E*0x2) : (0x005E*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&E25&&!SNPS_RSVDPARAM_37



Table 5-257 Fields for Register: VR_XS_PMA_MP_25G_MPLLA_BW_LO_CTRL

Bits	Name	Memory Access	Description
15:0	MPLLA_BW_LO	R/W	MPLLA Bandwidth Control -Low This field controls the bandwidth of MPLLA present in Multi-protocol 25G PHY. This field drives the output port 'xpcs_mplla_bw_low_o'. **Value After Reset:** "SNPS_RSVDPARAM_36 ? (RPCS_PRESENT ? 0x16 : 0x26) : SNPS_RSVDPARAM_16 ? 0xf : 0x7" **Exists:** Always

5.2.3.128 VR_XS_PMA_MP_8G_MPLLA_CTRL7

- **Description:** VR XS or PMA Synopsys Multi-protocol 8G PHY MPLLA Control 7 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x005F : (APB3_16) ? (0x005F*0x2) : (0x005F*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_RSVDPARAM_39

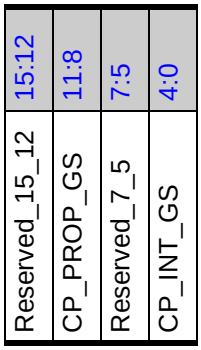


Table 5-258 Fields for Register: VR_XS_PMA_MP_8G_MPLLA_CTRL7

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:8	CP_PROP_GS	R/W	MPLLA Charge Pump Proportional Control in Gear-Shift Mode This field drives the xpcs_mplla_cp_prop_gs_o port. **Value After Reset:** "RPCS_PRESENT ? 0xA : KX_ONLY ? 0xC : 0xF" **Exists:** Always
7:5	Reserved_7_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4:0	CP_INT_GS	R/W	MPLLA Charge Pump Integral Control in Gear-Shift Mode This field drives the xpcs_mplla_cp_int_gs_o port. **Value After Reset:** 0x1 **Exists:** Always

5.2.3.129 VR_XS_PMA_MP_25G_MPLLB_BW_HI_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLB Bandwidth High Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x005F : (APB3_16) ? (0x005F*0x2) : (0x005F*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&E25&&!SNPS_RSVDPARAM_36&&!SNPS_RSVDPARAM_37

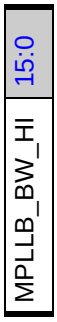


Table 5-259 Fields for Register: VR_XS_PMA_MP_25G_MPLLB_BW_HI_CTRL

Bits	Name	Memory Access	Description
15:0	MPLLB_BW_HI	R/W	MPLLB Bandwidth Control -High This field controls the bandwidth of MPLLB present in Multi-protocol 25G PHY. This field drives the output port 'xpcs_mpllb_bw_high_o'. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.130 VR_XS_PMA_MP_25G_MPLLB_BW_LO_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY MPLLB Bandwidth Low Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0061 : (APB3_16) ? (0x0061*0x2) : (0x0061*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&E25&&!SNPS_RSVDPARAM_36&&!SNPS_RSVDPARAM_37



Table 5-260 Fields for Register: VR_XS_PMA_MP_25G_MPLLB_BW_LO_CTRL

Bits	Name	Memory Access	Description
15:0	MPLLB_BW_LO	R/W	MPLLB Bandwidth Control -Low This field controls the bandwidth of MPLLB present in Multi-protocol 25G PHY. This field drives the output port 'xpcs_mpllb_bw_low_o'. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.131 VR_XS_PMA_MP_6G_MPLL_STS

- **Description:** VR XS or PMA Synopsys Multi-protocol 6G PHY MPLL Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0068 : (APB3_16) ? (0x0068*0x2) : (0x0068*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD | |XGXS_PCS)&&SNPS_XAUI_PHY&&SSP)

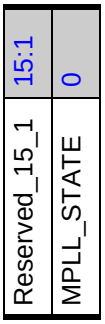


Table 5-261 Fields for Register: VR_XS_PMA_MP_6G_MPLL_STS

Bits	Name	Memory Access	Description
15:1	Reserved_15_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	MPLL_STATE	R	MPLL State Indicator This bit indicates the state of MPLL. When set to 1, it indicates that the MPLL is powered up and phase-locked. **Value After Reset:** 0x1 **Exists:** Always

5.2.3.132 VR_XS_PMA_MP_12G_16G_25G_MISC_CTRL0

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Miscellaneous Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0070 : (APB3_16) ? (0x0070*0x2) : (0x0070*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25||E32))

15	14	13	x:8	7:5	4	3:1	0
PLL_CTRL	CR_PARA_SEL	RTUNE_REQ	RX_VREF_CTRL	RX2TX_LB_EN_3_1	RX2TX_LB_EN_0	TX2RX_LB_EN_3_1	TX2RX_LB_EN_0

Table 5-262 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_MISC_CTRL0

Bits	Name	Memory Access	Description
15	PLL_CTRL	R/W	PLL Re-initialization Control If the PHY PLL does not lock for a long time, DWC_xpcs would initiate a Rx power-state change from P0 to P1 and then back to P0, if this bit is set. This process is done to re-initialize the PLL. The time duration to wait before reinitializing the PLL is determined by VR_XS_PMA_MP_12G_16G_25G_MISC_CTRL1 Register. **Value After Reset:** 0x0 **Exists:** Always
14	CR_PARA_SEL	R/W	Select CR Para/APB3 Port This bit select the interface for accessing PHY registers: ■ 0: JTAG ■ 1: CR parallel port Change the value of this field, only after disabling 'jtag_tck' to PHY. **Value After Reset:** 0x1 **Exists:** Always

Bits	Name	Memory Access	Description
13	RTUNE_REQ	R/W	Resistor Tuning Request This bit can be set to trigger a resistor tune request to the PHY. This bit controls the 'xgxs_rtune_req_o' output port. **Value After Reset:** 0x0 **Exists:** Always
x:8	RX_VREF_CTRL	* Varies	**Non SNPS_RSVDPARAM_37/Non E32 configurations:** Rx Biasing Current Control This field drives the output port 'xpcs_rx_vref_ctrl_o[4:0]'. This field sets the Rx biasing current for Rx analog front end. **For SNPS_RSVDPARAM_37/E32 configurations:** Reserved **Value After Reset:** "(SNPS_RSVDPARAM_37||E32) ? 0x0 : ((SNPS_RSVDPARAM_22||E25) ? 0x5 : 0x11)" **Exists:** Always **Range Variable[x]:** "SNPS_RSVDPARAM_22 ? 3 : 5" + 7 **Memory Access:** "(!(SNPS_RSVDPARAM_37||E32)) ? \"read-write\" :\"read-only\""
7:5	RX2TX_LB_EN_3_1	* Varies	**For Multi-lane configurations:** Enable Parallel Rx-to-Tx Loopback on lanes[3:1] This field is present only in multi-lane configurations. **For single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
4	RX2TX_LB_EN_0	R/W	Enable Parallel Rx-to-Tx Loopback on lane 0 When this bit is set, recovered parallel data from PHY receiver is looped back to the transmit serializer. This loop-back takes place internal to the PHY (not within DWC_xpcs). **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
3:1	TX2RX_LB_EN_3_1	* Varies	**For Multi-lane configurations:** Enable Analog Tx-to-Rx Serial Loopback on lanes[3:1] This field enable serial loopback in the PHY from Tx pre-driver to Rx analog front-end on lanes 3 to 1. This field is present only in multi-lane configurations. **For single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
0	TX2RX_LB_EN_0	R/W	Enable Analog Tx-to-Rx Serial Loopback on lane 0 This bit can be set to enable serial loopback in the PHY from Tx pre-driver to Rx analog front-end. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.133 VR_XS_PMA_MP_12G_16G_25G_REF_CLK_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Reference Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0071 : (APB3_16) ? (0x0071*0x2) : (0x0071*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25)&&!SNPS_RSVD-PARAM_37)

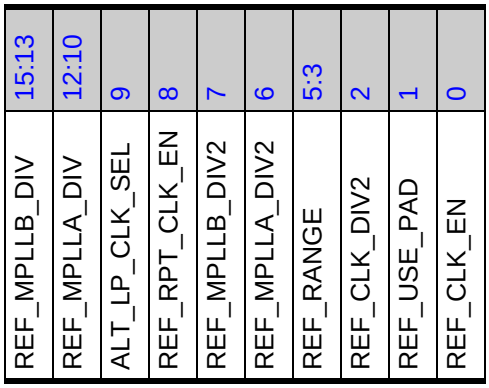


Table 5-263 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_REF_CLK_CTRL

Bits	Name	Memory Access	Description
15:13	REF_MPLLB_DIV	* Varies	**For Multi-protocol 25G PHY or Multi-protocol 10G Gen2 PHY configurations:** This field drives the xpcs_ref_clk_mpllb_div_o port. **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_RSVDPARAM_22||E25) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
12:10	REF_MPLLA_DIV	* Varies	**For Multi-protocol 25G PHY or Multi-protocol 10G Gen2 PHY configurations:** This field drives the xpcs_ref_clk_mplla_div_o port. **For other configurations:** Reserved **Value After Reset:** "SNPS_RSVDPARAM_22 ? (REF_CLK_78 ? 0x0 : 0x1) : E25 ? (REF_CLK_312 ? 0x2 : (SNPS_RSVDPARAM_16||SNPS_RSVDPARAM_13) ? 0x0 : REF_CLK_156 ? 0x1 : 0x0) : 0x0" **Exists:** Always **Memory Access:** "(SNPS_RSVDPARAM_22||E25) ? \"read-write\" : \"read-only\"""
9	ALT_LP_CLK_SEL	* Varies	**For Multi-protocol 10G Gen2 PHY configurations:** Alternate Low Precision Clock Select. This field drives the xpcs_alt_clk_lp_sel_o port. **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SNPS_RSVDPARAM_22 ? \"read-write\" : \"read-only\"""
8	REF_RPT_CLK_EN	R/W	Repeat Reference Clock Enable If this bit is set, 'ref_repeat_clk_{p,m}' clock from PHY is enabled. **Value After Reset:** 0x0 **Exists:** Always
7	REF_MPLLB_DIV2	R/W	MPLLB Reference Clock Divider Control The reference clock used for MPLLB calibration and locking can be divided by 2 by setting this bit to 1. This division is applied after ref_clk_div2_en. Therefore, the total division ratio (from the input reference clock) can be 1, 2 or 4 **Value After Reset:** 0x0 **Exists:** !SNPS_RSVDPARAM_22&&!E25&&!SNPS_RSVDPARAM_39

Bits	Name	Memory Access	Description
6	REF_MPLLA_DIV2	R/W	MPLLA Reference Clock Divider Control The reference clock used for MPLLA calibration and locking can be divided by 2 by setting this bit to 1. This division is applied after ref_clk_div2_en. Therefore, the total division ratio (from the input reference clock) can be 1, 2 or 4. **Value After Reset:** "(REF_CLK_FREQ==2)||(SNPS_RSVDPARAM_35&&SNPS_RSVDPARAM_13)||SNPS_RSVDPARAM_16 ? 0x0 : (SNPS_RSVDPARAM_39&&SNPS_RSVDPARAM_13) ? 0x1 : (SNPS_RSVDPARAM_39&&REF_CLK_25) ? 0x0 : 0x1" **Exists:** !SNPS_RSVDPARAM_22&&!E25
5:3	REF_RANGE	R/W	Input Reference Clock Range This field specifies the frequency range of the input reference clock (post ref_clk_div2_en division if any). The code mapping of REF_RANGE (for Multi-protocol 12G/16G PHY) is as follows: ■ 3'b000: 20 - 26 MHz ■ 3'b001: 26.1 - 52 MHz ■ 3'b010: 52.1 - 78 MHz ■ 3'b011: 78.1 - 104 MHz ■ 3'b100: 104.1 - 130 MHz ■ 3'b101: 130.1 - 156 MHz ■ 3'b110: 156.1 - 182 MHz ■ 3'b111: 182.1 - 200 MHz Set this field to appropriate values based on your reference clock frequency. **Value After Reset:** "SNPS_RSVDPARAM_22 ? (SNPS_RSVDPARAM_2 ? 0x5 : (REF_CLK_78 ? 0x3 : 0x6)) : ((SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_36)&&SNPS_RSVDPARAM_13) ? 0x3 : SNPS_RSVDPARAM_35 ? 0x2 : SNPS_RSVDPARAM_33 ? (REF_CLK_78 ? 0x1 : 0x2) : (SNPS_RSVDPARAM_39&&SNPS_RSVDPARAM_13) ? 0x1 : (SNPS_RSVDPARAM_39&&REF_CLK_25) ? 0x0 : SNPS_RSVDPARAM_16 ? 0x1 : (REF_CLK_FREQ==2) ? 0x3 : 0x6" **Exists:** Always

Bits	Name	Memory Access	Description
2	REF_CLK_DIV2	R/W	Reference Clock divide by 2 If this bit is set, reference clock provided to PHY gets divided by 2 internally in the PHY. **Value After Reset:** "(SNPS_RSVDPARAM_16||(SNPS_RSVDPARAM_13&&SNPS_RSVDPARAM_36)) ? 0x0 : ((REF_CLK_FREQ==6)||(SNPS_RSVDPARAM_35&&!SNPS_RSVDPARAM_13)||(SNPS_RSVDPARAM_33&&REF_CLK_78)||(SNPS_RSVDPARAM_39&&SNPS_RSVDPARAM_13)) ? 0x1 : (SNPS_RSVDPARAM_39&&REF_CLK_25) ? 0x0 : (SNPS_RSVDPARAM_33&&REF_CLK_156) ? 0x1 : 0x0" **Exists:** !SNPS_RSVDPARAM_22
1	REF_USE_PAD	R/W	Use Pad Clk as Reference Clock If this bit is set, PHY selects the clock connected to ref_pad_clk_p/m as its reference clock; otherwise it selects the clock connected to ref_alt_clk_p/m. **Value After Reset:** "(REF_CLK_SRC==0) ? 0x1 : 0x0" **Exists:** Always
0	REF_CLK_EN	R/W	Reference Clock Enable Set this bit to 1, to enable reference clock to the PHY. This bit controls the xpcs_ref_clk_en_o output port. The value programmed to this bit is driven on xpcs_ref_clk_en_o when LPM (power-down) bit is programmed to 1. When the LPM bit is programmed to 0, xpcs_ref_clk_en_o remains high. **Value After Reset:** 0x1 **Exists:** Always

5.2.3.134 VR_XS_PMA_MP_6G_EEE_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 6G PHY EEE Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0072 : (APB3_16) ? (0x0072*0x2) : (0x0072*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&SSP&&QSGMII_EN&&EEE_EN

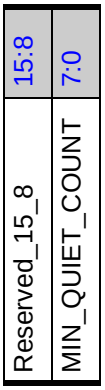


Table 5-264 Fields for Register: VR_XS_PMA_MP_6G_EEE_CTRL

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:0	MIN_QUIET_COUNT	R/W	This field allows application to determine the timeout of MIN QUIET Counter. MIN QUIET counter is implemented in the RX_QUIET state of LPI state machine. This counter starts when DWC_xpcs reaches RX_QUIET state (if SUPRESS_LOS_DET bit is set) and it stops when it reaches a value equals to 8 times MIN_QUIET_COUNT. DWC_xpcs starts monitoring for 'signal_detect' in RX_QUIET state only after this timer reaches its maximum value (8 * MIN_QUIE_COUNT). **Note:** This is a vendor-specific feature applicable only in QSGMII Mode.It is useful if xpcs_los_i input is unreliable at QSGMII rate. **Value After Reset:** 0xa **Exists:** Always

5.2.3.135 VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD0

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY VCO Calibration Load 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0072 : (APB3_16) ? (0x0072*0x2) : (0x0072*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25)&&!SNPS_RSVDPARAM_37

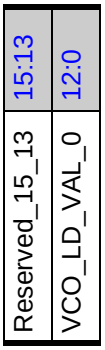


Table 5-265 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD0

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
12:0	VCO_LD_VAL_0	R/W	Rx VCO calibration load value on lane 0 of the PHY This field is used to load internal calibration registers in the PHY to perform Rx VCO calibration on lane 0. This field drives the output port 'xpcs_rx0_vco_ld_val_o[5:0]'. **Value After Reset:** "(SNPS_RSVDPARAM_13&&SNPS_RSVDPARAM_36&&KX_ONLY) ? 0x578 : SNPS_RSVDPARAM_16 ? 0x53F: (SNPS_RSVDPARAM_39&&(SNPS_RSVDPARAM_13||REF_CLK_25)) ? 0x407 : (BASE1000X_ONLY||KX_ONLY)&&!SNPS_RSVDPARAM_35 ? ((E25&&REF_CLK_78) ? 0x580 : (SNPS_RSVDPARAM_39&&REF_CLK_78) ? 0x400 : 0x540) : (RPCS_PRESENT&&REF_CLK_78) ? ((E25||SNPS_RSVDPARAM_33) ? 0x5AC : SNPS_RSVDPARAM_39 ? 0x420 : 0x56A) : (RPCS_PRESENT) ? ((E25||SNPS_RSVDPARAM_33) ? 0x56A : (SNPS_RSVDPARAM_39 ? 0x3FF : 0x549)) : SNPS_RSVDPARAM_13 ? 0x546 : 0x550" **Exists:** Always

5.2.3.136 VR_XS_PMA_MP_32G_RX_CNTX_CTRL0

- **Description:** VR XS or PMA Synopsys Multi protocol E32 PHY RX Context Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0072 : (APB3_16) ? (0x0072*0x2) : (0x0072*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&SNPS_XAUI_PHY&&(SNPS_RSVDPARAM_37 | |E32)

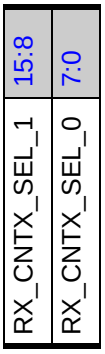


Table 5-266 Fields for Register: VR_XS_PMA_MP_32G_RX_CNTX_CTRL0

Bits	Name	Memory Access	Description
15:8	RX_CNTX_SEL_1	* Varies	**For Multi-lane configurations:** Receive context select along lane 0 of the PHY specifies the context that needs to be restored for the RX block on lane 1. This field drives the output port 'xpcs_rx1_cntx_sel_o[7:0]'. **For single-lane configurations:** Reserved **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : E32 ? RX_CNTX_SEL : 0x2E" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" :\"read-only\""
7:0	RX_CNTX_SEL_0	R/W	Receive context select along lane 0 of the PHY specifies the context that needs to be restored for the RX block on lane 0. This field drives the output port 'xpcs_rx0_cntx_sel_o[7:0]'. **Value After Reset:** RX_CNTX_SEL **Exists:** Always

5.2.3.137 VR_XS_PMA_MP_6G_MISC_CTRL2

- **Description:** VR XS or PMA Synopsys Multi-protocol 6G PHY Miscellaneous Control 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0073 : (APB3_16) ? (0x0073*0x2) : (0x0073*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&SSP)

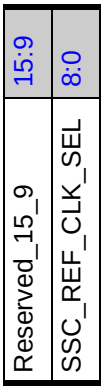


Table 5-267 Fields for Register: VR_XS_PMA_MP_6G_MISC_CTRL2

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8:0	SSC_REF_CLK_SEL	R/W	Spread Spectrum Reference Clock Shifting Control This register field drives the output port 'xgxs_ssc_ref_clk_sel_o' of DWC_xpcs, which in turn is connected to 'ssc_ref_clk_sel' input port of the PHY. This helps in enabling non-standard oscillator frequencies to generate targeted MPLL output rates. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.138 VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD1

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY VCO Calibration Load 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0073 : (APB3_16) ? (0x0073*0x2) : (0x0073*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37

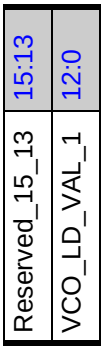


Table 5-268 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD1

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:0	VCO_LD_VAL_1	* Varies	Rx VCO calibration load value on lane 1 of the PHY This field is used to load internal calibration registers in the PHY to perform Rx VCO calibration on lane 1. This field drives the output port 'xpcs_rx1_vco_ld_val_o[5:0]'. **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0 : (RPCS_PRESENT&&REF_CLK_78) ? ((E25||SNPS_RSVDPARAM_33) ? 0x5AC : SNPS_RSVDPARAM_39 ? 0x420 : 0x56A) : (RPCS_PRESENT) ? ((E25||SNPS_RSVDPARAM_33) ? 0x56A : SNPS_RSVDPARAM_39 ? 0x3FF : 0x549) : SNPS_RSVDPARAM_39 ? 0x3E8 : 0x550" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" :\"read-only\""

5.2.3.139 VR_XS_PMA_MP_32G_RX_CNTX_CTRL1

- **Description:** VR XS or PMA Synopsys Multi protocol E32 PHY RX Contex Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0073 : (APB3_16) ? (0x0073*0x2) : (0x0073*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(SNPS_RSVDPARAM_37||E32)&&!SINGLE_LANE_CFG

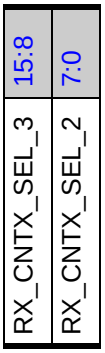


Table 5-269 Fields for Register: VR_XS_PMA_MP_32G_RX_CNTX_CTRL1

Bits	Name	Memory Access	Description
15:8	RX_CNTX_SEL_3	R/W	Receive context select along lane 3 of the PHY specifies the context that needs to be restored for the RX block on lane 3. This field drives the output port 'xpcs_rx3_cntx_sel_o[7:0]'. **Value After Reset:** RX_CNTX_SEL **Exists:** Always
7:0	RX_CNTX_SEL_2	R/W	Receive context select along lane 2 of the PHY specifies the context that needs to be restored for the RX block on lane 0. This field drives the output port 'xpcs_rx2_cntx_sel_o[7:0]'. **Value After Reset:** RX_CNTX_SEL **Exists:** Always

5.2.3.140 VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD2

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G PHY VCO Calibration Load 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0074 : (APB3_16) ? (0x0074*0x2) : (0x0074*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37)

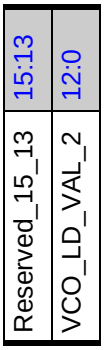


Table 5-270 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD2

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:0	VCO_LD_VAL_2	* Varies	Rx VCO calibration load value on lane 2 of the PHY This field is used to load internal calibration registers in the PHY to perform Rx VCO calibration on lane 2. This field drives the output port 'xpcs_rx2_vco_ld_val_o[5:0]'. **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0 : (RPCS_PRESENT&&REF_CLK_78) ? ((E25||SNPS_RSVDPARAM_33) ? 0x5AC : SNPS_RSVDPARAM_39 ? 0x420 : 0x56A) : (RPCS_PRESENT) ? ((E25||SNPS_RSVDPARAM_33) ? 0x56A : SNPS_RSVDPARAM_39 ? 0x3FF : 0x549) : SNPS_RSVDPARAM_39 ? 0x3E8 :0x550" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" :\"read-only\"""

5.2.3.141 VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD3

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY VCO Calibration Load 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0075 : (APB3_16) ? (0x0075*0x2) : (0x0075*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37

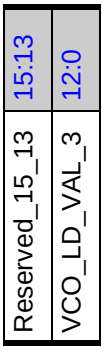


Table 5-271 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD3

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:0	VCO_LD_VAL_3	* Varies	Rx VCO calibration load value on lane 3 of the PHY This field is used to load internal calibration registers in the PHY to perform Rx VCO calibration on lane 3. This field drives the output port 'xpcs_rx3_vco_ld_val_o[5:0]' **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0 : (RPCS_PRESENT&&REF_CLK_78) ? ((E25||SNPS_RSVDPARAM_33) ? 0x5AC : SNPS_RSVDPARAM_39 ? 0x420 : 0x56A) : (RPCS_PRESENT) ? ((E25||SNPS_RSVDPARAM_33) ? 0x56A : SNPS_RSVDPARAM_39 ? 0x3FF : 0x549) : SNPS_RSVDPARAM_39 ? 0x3E8 : 0x550" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" :\"read-only\""

5.2.3.142 VR_XS_PMA_MP_32G_REF_CLK_CTRL0

- **Description:** VR XS or PMA Synopsys Multi-protocol 32G Reference clock control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0075 : (APB3_16) ? (0x0075*0x2) : (0x0075*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&SNPS_XAUI_PHY&&E32

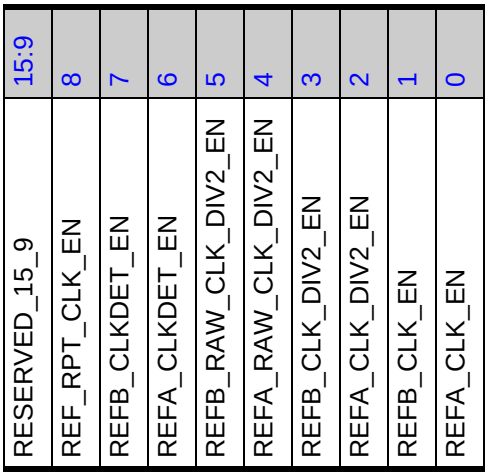


Table 5-272 Fields for Register: VR_XS_PMA_MP_32G_REF_CLK_CTRL0

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	REF_RPT_CLK_EN	R/W	Ref Repeat clock enable Selects the input reference clock for CML output clocks. This field drives the output port 'xpcs_ref_repeat_clock_en_o' **Value After Reset:** 0x0 **Exists:** Always
7	REFB_CLKDET_EN	R/W	Refb Clock detect enable Enables the reference clock detection for refb **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
6	REFA_CLKDET_EN	R/W	Refa Clock detect enable Enables the reference clock detection for refa **Value After Reset:** 0x0 **Exists:** Always
5	REFB_RAW_CLK_DIV2_EN	R/W	Reference clock divider for refb in RAW PCS. When asserted the refb clock in RAW PCS gets divided by 2. This field drives the output port 'xpcs_refb_raw_clk_div2_en_o' **Value After Reset:** "(REF_CLK_78||REF_CLK_125) ? 0x0 : 0x1" **Exists:** Always
4	REFA_RAW_CLK_DIV2_EN	R/W	Reference clock divider for refa in RAW PCS. When asserted the refa clock in RAW PCS gets divided by 2. This field drives the output port 'xpcs_refa_raw_clk_div2_en_o' **Value After Reset:** "(REF_CLK_78||REF_CLK_125) ? 0x0 : 0x1" **Exists:** Always
3	REFB_CLK_DIV2_EN	R/W	Reference clock divider control for refb. When asserted the reference clock refa is divided by 2. This field drives the output port 'xpcs_refb_clk_div2_en_o'. **Value After Reset:** "REF_CLK_312 ? 0x1 : 0x0" **Exists:** Always
2	REFA_CLK_DIV2_EN	R/W	Reference clock divider control for refa. When asserted the reference clock refa is divided by 2. This field drives the output port 'xpcs_refa_clk_div2_en_o'. **Value After Reset:** "REF_CLK_312 ? 0x1 : 0x0" **Exists:** Always
1	REFB_CLK_EN	R/W	Reference clock enable for refa Enables the reference 'b' clock for the PHY This field drives the output port xpcs_refb_clk_en_o. **Value After Reset:** 0x0 **Exists:** Always
0	REFA_CLK_EN	R/W	Reference clock enable for refa Enables the reference 'a' clock for the PHY This field drives the output port xpcs_refa_clk_en_o. **Value After Reset:** 0x1 **Exists:** Always

5.2.3.143 VR_XS_PMA_MP_12G_VCO_CAL_REF0

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G PHY VCO Calibration Reference 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0076 : (APB3_16) ? (0x0076*0x2) : (0x0076*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&E12)

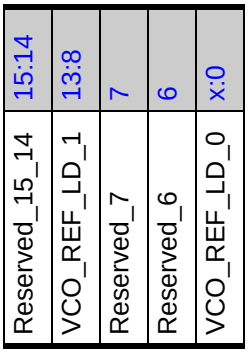


Table 5-273 Fields for Register: VR_XS_PMA_MP_12G_VCO_CAL_REF0

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:8	VCO_REF_LD_1	* Varies	**For Multi-lane configurations:** Rx VCO calibration reference load value on lane 1 of the PHY This field controls the internal calibration registers of the PHY and is used to perform Rx VCO calibration on lane 1. This field drives the output port 'xpcs_rx1_ref_ld_val_o[5:0]'. **For single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0 : (RPCS_PRESENT&&REF_CLK_78) ? (SNPS_RSVDPARAM_39 ? 0x10 : SNPS_RSVDPARAM_33 ? 0xB: 0x15) : (RPCS_PRESENT) ? (SNPS_RSVDPARAM_39 ? 0x1F : SNPS_RSVDPARAM_33 ? 0x15: 0x29) : REF_CLK_78 ? (SNPS_RSVDPARAM_39 ? 0x19 : 0x11) : 0x22" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" :\"read-only\""

Bits	Name	Memory Access	Description
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6	Reserved_6	R	Reserved **Value After Reset:** 0x0 **Exists:** !SNPS_RSVDPARAM_22
x:0	VCO_REF_LD_0	R/W	Rx VCO calibration reference load value -lane 0 This field controls the internal calibration registers of the PHY and is used to perform Rx VCO calibration on lane 0. This field drives the output port 'xpcs_rx0_ref_ld_val_o'. **Value After Reset:** "(BASE1000X_ONLY||KX_ONLY)&&!SNPS_RSVDPARAM_35 ? (REF_CLK_78 ? 0x15 : 0x2A): (RPCS_PRESENT&&REF_CLK_78) ? (SNPS_RSVDPARAM_39 ? 0x10 : SNPS_RSVDPARAM_33 ? 0xB : 0x15) : (RPCS_PRESENT) ? (SNPS_RSVDPARAM_39&&SNPS_RSVDPARAM_13 ? 0x14 :(SNPS_RSVDPARAM_39&&REF_CLK_25) ? 0xA : SNPS_RSVDPARAM_33 ? 0x15: 0x29) : (REF_CLK_78||(SNPS_RSVDPARAM_35&&!SNPS_RSVDPARAM_13)) ? SNPS_RSVDPARAM_39 ? 0x19 : 0x11 : SNPS_RSVDPARAM_13 ? 0x1B : 0x22" **Exists:** Always **Range Variable[x]:** "SNPS_RSVDPARAM_22 ? 7 : 6" - 1

5.2.3.144 VR_XS_PMA_MP_16G_25G_VCO_CAL_REF0

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G/25G PHY VCO Calibration Reference 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0076 : (APB3_16) ? (0x0076*0x2) : (0x0076*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||E25)&&!SNPS_RSVDPAR-AM_37

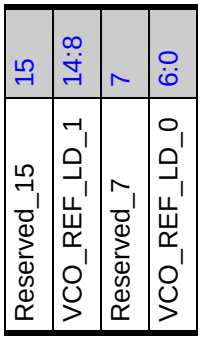


Table 5-274 Fields for Register: VR_XS_PMA_MP_16G_25G_VCO_CAL_REF0

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:8	VCO_REF_LD_1	* Varies	**For Multi-lane configurations:** Rx VCO calibration reference load value on lane 1 of the PHY This field controls the internal calibration registers of the PHY and is used to perform Rx VCO calibration on lane 1. This field drives the output port 'xpcs_rx1_ref_ld_val_o[6:0]'. **For single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0 : (RPCS_PRESENT&&REF_CLK_78) ? (E25 ? 0xB : 0x15) : (RPCS_PRESENT) ? (E25 ? 0x15 : 0x29) : REF_CLK_78 ? 0x11 : 0x22" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" :\"read-only\""

Bits	Name	Memory Access	Description
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:0	VCO_REF_LD_0	R/W	Rx VCO calibration reference load value -lane 0 This field controls the internal calibration registers of the PHY and is used to perform Rx VCO calibration on lane 0. This field drives the output port 'xpcs_rx0_ref_ld_val_o[6:0]'. **Value After Reset:** "(SNPS_RSVDPARAM_36&&KX_ONLY&&SNPS_RSVDPARAM_13) ? 0xE : SNPS_RSVDPARAM_16 ? 0x5 : (BASE1000X_ONLY||KX_ONLY) ? (REF_CLK_78 ? (E25 ? 0xB :0x15) : (E25 ? 0x15 : 0x2A)): (RPCS_PRESENT&&REF_CLK_78) ? (E25 ? 0xB : 0x15) : (RPCS_PRESENT) ? (E25 ? 0x15 : 0x29) : (REF_CLK_78) ? 0x11 : 0x22" **Exists:** Always

5.2.3.145 VR_XS_PMA_MP_32G_REF_CLK_CTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 32G PHY Reference Clock Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0076 : (APB3_16) ? (0x0076*0x2) : (0x0076*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&SNPS_XAUI_PHY&&E32

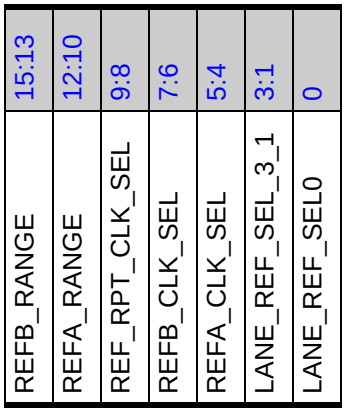


Table 5-275 Fields for Register: VR_XS_PMA_MP_32G_REF_CLK_CTRL1

Bits	Name	Memory Access	Description
15:13	REFB_RANGE	R/W	Reference clock frequency range for refb Specifies the frequency range of the reference 'b' clock. This field drives the output port 'xpcs_refb_range_o'. **Value After Reset:** "REF_CLK_78 ? 0x3 : REF_CLK_125 ? 0x4 : 0x6" **Exists:** Always
12:10	REFA_RANGE	R/W	Reference clock frequency range for refa Specifies the frequency range of the reference 'a' clock. This field drives the output port 'xpcs_refa_range_o'. **Value After Reset:** "REF_CLK_78 ? 0x3 : REF_CLK_125 ? 0x4 : 0x6" **Exists:** Always
9:8	REF_RPT_CLK_SEL	R/W	Repeat reference clock select Selects the reference clock for CML clocks. This field drives the output port 'xpcs_ref_prt_clk_sel_o'. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
7:6	REFB_CLK_SEL	R/W	Reference clock 'b' Source select Select the reference clock source for refa. ■ 2'b00: ref_alt0_clk_(p/m) ■ 2'b01: ref_pad_clk_(p/m) ■ 2'b10: ref_alt1_clk_(p/m) ■ 2'b11: Reserved. This field drives the output port 'xpcs_refb_clk_sel_o' **Value After Reset:** "(REF_CLK_SRC==1) ? 0x0 : (REF_CLK_SRC==0) ? 0x1 : 0x2" **Exists:** Always
5:4	REFA_CLK_SEL	R/W	Reference clock 'a' Source select Select the reference clock source for refa. ■ 2'b00: ref_alt0_clk_(p/m) ■ 2'b01: ref_pad_clk_(p/m) ■ 2'b10: ref_alt1_clk_(p/m) ■ 2'b11: Reserved. This field drives the output port 'xpcs_refa_clk_sel_o' **Value After Reset:** "(REF_CLK_SRC==1) ? 0x0 : (REF_CLK_SRC==0) ? 0x1 : 0x2" **Exists:** Always
3:1	LANE_REF_SEL_3_1	* Varies	**For Multi lane configurations:** Lane 3_1 reference clock selection. Selects the refa or refb reference clock. 0: Selects REFA clock. 1: Selects REFB clocks. This field drives the output port 'xpcs_lane_ref_sel_o[3:1]. For all Ethernet modes, always select REFA clock' **For Single lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\" "
0	LANE_REF_SEL0	R/W	Lane0 reference clock selection. Selects the refa or refb reference clock. ■ 0: Selects REFA clock. ■ 1: Selects REFB clocks. This field drives the output port 'xpcs_lane_ref_sel_o[0]. For all the Ethernet modes, always choose REFA clock.' **Value After Reset:** 0x0 **Exists:** Always

5.2.3.146 VR_XS_PMA_MP_12G_VCO_CAL_REF1

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G PHY VCO Calibration Reference 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0077 : (APB3_16) ? (0x0077*0x2) : (0x0077*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD | |XGXS_PCS)&&SNPS_XAUI_PHY&&E12&&!SINGLE_LANE_CFG)

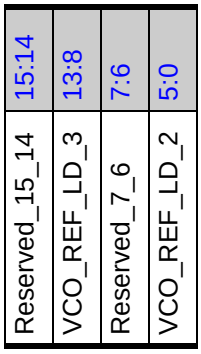


Table 5-276 Fields for Register: VR_XS_PMA_MP_12G_VCO_CAL_REF1

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:8	VCO_REF_LD_3	* Varies	Rx VCO calibration reference load value on lane 3 of the PHY This field controls the internal calibration registers of the PHY and is used to perform Rx VCO calibration on lane 3. This field drives the output port 'xpcs_rx3_ref_ld_val_o[5:0]'. **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0 : (RPCS_PRESENT&&REF_CLK_78) ? (SNPS_RSVDPARAM_39 ? 0x10 : SNPS_RSVDPARAM_33 ? 0xB: 0x15) : (RPCS_PRESENT) ? (SNPS_RSVDPARAM_39 ? 0x1F : SNPS_RSVDPARAM_33 ? 0x15 : 0x29) : REF_CLK_78 ? (SNPS_RSVDPARAM_39 ? 0x19 : 0x11) : 0x22" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" :\"read-only\""
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
5:0	VCO_REF_LD_2	* Varies	Rx VCO calibration reference load value -lane 2 This field controls the internal calibration registers of the PHY and is used to perform Rx VCO calibration on lane 2. This field drives the output port 'xpcs_rx2_ref_ld_val_o[5:0]' **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0 : (RPCS_PRESENT&&REF_CLK_78) ? (SNPS_RSVDPARAM_39 ? 0x10 : SNPS_RSVDPARAM_33 ?0xB : 0x15) : (RPCS_PRESENT) ? (SNPS_RSVDPARAM_39 ? 0x1F : SNPS_RSVDPARAM_33 ? 0x15: 0x29) : REF_CLK_78 ? (SNPS_RSVDPARAM_39 ? 0x19 : 0x11) : 0x22" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" :\"read-only\""

5.2.3.147 VR_XS_PMA_MP_16G_25G_VCO_CAL_REF1

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G/25G PHY VCO Calibration Reference 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0077 : (APB3_16) ? (0x0077*0x2) : (0x0077*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||E25)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37

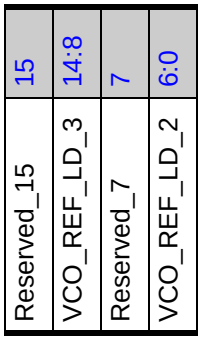


Table 5-277 Fields for Register: VR_XS_PMA_MP_16G_25G_VCO_CAL_REF1

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:8	VCO_REF_LD_3	* Varies	Rx VCO calibration reference load value on lane 3 of the PHY This field controls the internal calibration registers of the PHY and is used to perform Rx VCO calibration on lane 3. This field drives the output port 'xpcs_rx3_ref_ld_val_o[6:0]'. **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0 : (RPCS_PRESENT&&REF_CLK_78) ? (E25 ? 0xB : 0x15) : (RPCS_PRESENT) ? (E25 ? 0x15 : 0x29) : REF_CLK_78 ? 0x11 : 0x22" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" :\"read-only\"""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
6:0	VCO_REF_LD_2	* Varies	Rx VCO calibration reference load value -lane 2 This field controls the internal calibration registers of the PHY and is used to perform Rx VCO calibration on lane 2. This field drives the output port 'xpcs_rx2_ref_ld_val_o[6:0]' **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0 : (RPCS_PRESENT&&REF_CLK_78) ? (E25 ? 0xB : 0x15) : (RPCS_PRESENT) ? (E25 ? 0x15 : 0x29) : REF_CLK_78 ? 0x11 : 0x22" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" :\"read-only\""

5.2.3.148 VR_XS_PMA_MP_32G_LN_LINK_CTRL

- **Description:** VR XS or PMA Synopsys Multi protocol E32 PHY Lane Link Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0077 : (APB3_16) ? (0x0077*0x2) : (0x0077*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&SNPS_XAUI_PHY&&(SNPS_RSVDPARAM_37 | |E32)

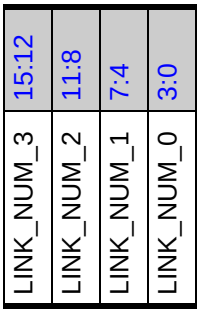


Table 5-278 Fields for Register: VR_XS_PMA_MP_32G_LN_LINK_CTRL

Bits	Name	Memory Access	Description
15:12	LINK_NUM_3	* Varies	**For Multi-lane configurations:** Link number along lane 3. It determines which link the current lane belongs to. This field drives the output port 'xpcs_lane3_link_num_o'. **For single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
11:8	LINK_NUM_2	* Varies	**For Multi-lane configurations:** Link number along lane 2. It determines which link the current lane belongs to. This field drives the output port 'xpcs_lane2_link_num_o'. **For single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
7:4	LINK_NUM_1	* Varies	**For Multi-lane configurations:** Link number along lane 1. It determines which link the current lane belongs to. This field drives the output port 'xpcs_lane1_link_num_o'. **For single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
3:0	LINK_NUM_0	R/W	Link number along lane 0. It determines which link the current lane belongs to. This field drives the output port 'xpcs_lane0_link_num_o'. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.149 VR_XS_PMA_MP_6G_LVL_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 6G PHY Level Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0078 : (APB3_16) ? (0x0078*0x2) : (0x0078*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&SSP)

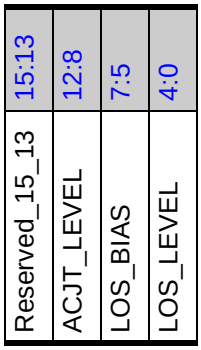


Table 5-279 Fields for Register: VR_XS_PMA_MP_6G_LVL_CTRL

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:8	ACJT_LEVEL	R/W	Receiver Sensitivity Level Control This field controls the sensitivity level for the receiver. For more information, see Synopsys DesignWare Cores PCIe2 PHY For TSMC 65nm GP (Flip-Chip) Databook. **Value After Reset:** 0x0 **Exists:** Always
7:5	LOS_BIAS	R/W	LOS Detector threshold ctrl This field sets the LOS detection threshold level. A positive binary bit setting change results in a +15 mVp incremental change in the LOS threshold. A negative binary setting change results in a -15 mVp incremental change in the LOS threshold. The setting 3'b000 is reserved and must not be used. **Value After Reset:** 0x1 **Exists:** Always

Bits	Name	Memory Access	Description
4:0	LOS_LEVEL	R/W	Loss-of-Signal Detector Sensitivity Control This field sets the sensitivity level for the loss-of-signal detector. For more information, see Synopsys DesignWare Cores PCIe2 PHY For TSMC 65nm GP (Flip-Chip) Databook. **Value After Reset:** 0x9 **Exists:** Always

5.2.3.150 VR_XS_PMA_MP_12G_16G_25G_MISC_STS

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Miscellaneous Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0078 : (APB3_16) ? (0x0078*0x2) : (0x0078*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25||E32)

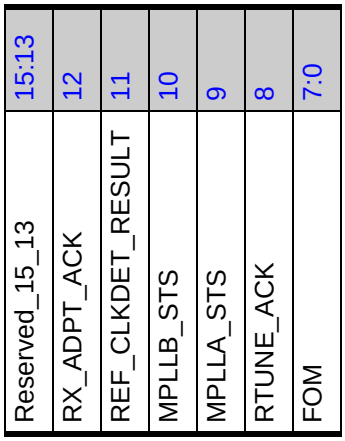


Table 5-280 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_MISC_STS

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	RX_ADPT_ACK	R	**For Configurations with CL72_EN=Disabled:** Receive Adaptation Acknowledgment from PHY.This field reflects the value of rpcs_krx_adapt_ack_i port. **For configurations with CL72_EN = Enabled:** This field is reserved. The receive adaptation acknowledge status is available in VR_PMA_PHY_RX_EQ_CEU register. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
11	REF_CLKDET_RESULT	R	**For Multi-protocol 16G/25G PHY or Multi-protocol 10G Gen2 configurations:**Reference Clock Detecton Result This field is driven by the input xpcs_ref_clkdet_result_i. **For Other configurations:**Reserved **Value After Reset:** 0x0 **Exists:** Always
10	MPLLB_STS	R	**For E32 Configurations :** Reserved **For E12/E16/E25 Configurations :** Status of MPLLB from PHY This bit denotes the value of xpcs_mpllb_state_i input. This valid is not valid for Multi-protocol 8G PHY. **Value After Reset:** 0x0 **Exists:** Always
9	MPLLA_STS	R	**For E32 Configurations :** Reserved **For E12/E16/E25 Configurations :** Status of MPLLA from PHY. This bit denotes the value of xpcs_mplla_state_i input **Value After Reset:** "E32 ? 0x0 :0x1" **Exists:** Always
8	RTUNE_ACK	R	Acknowledgment for Resistor Tune Request This bit denotes the value of 'xgxs_rtune_ack_i' input. **Value After Reset:** 0x0 **Exists:** Always
7:0	FOM	R	Figure of merit output from the PHY is captured in this field. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.151 VR_XS_PMA_MP_6G_MISC_CTRL0

- **Description:** VR XS or PMA Synopsys Multi-protocol 6G PHY Miscellaneous Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0079 : (APB3_16) ? (0x0079*0x2) : (0x0079*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD | |XGXS_PCS)&&SNPS_XAUI_PHY&&SSP)

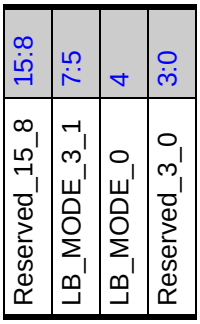


Table 5-281 Fields for Register: VR_XS_PMA_MP_6G_MISC_CTRL0

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	LB_MODE_3_1	* Varies	**For Multi-lane configurations:** Analog Serial Loopback Mode on Lanes[3:1] When set to 1, the PHY enables the data from the Tx predriver to receiver slices on the corresponding lane. When this bit is set to 1, the loss-of-signal detection is not done. ■ LB_MODE_3_1[1]: Loopback mode on Lane 1 ■ LB_MODE_3_1[2]: Loopback mode on Lane 2 ■ LB_MODE_3_1[3]: Loopback mode on Lane 3 **Access Type:** ■ RO in KX_Only or 1000BASEX-Only PCS configurations. ■ RW in other configurations **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
4	LB_MODE_0	R/W	Analog Serial Loopback Mode on Lane 0 When set to 1, the PHY enables the data from the Tx predriver to receiver slices on Lane 0. When this bit is set to 1, the loss-of-signal is not detected. **Value After Reset:** 0x0 **Exists:** Always
3:0	Reserved_3_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.3.152 VR_XS_PMA_MP_12G_16G_25G_MISC_CTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY Miscellaneous Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0079 : (APB3_16) ? (0x0079*0x2) : (0x0079*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25||E32)

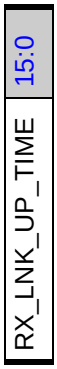


Table 5-282 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_MISC_CTRL1

Bits	Name	Memory Access	Description
15:0	RX_LNK_UP_TIME	R/W	Wait Time before PLL Re-initialization This field determines the number of 'clk_csr_i' clock cycles that can elapse before the Rx link is established (after DWC_xpcs has reached POWER_GOOD state). This field is relevant only if 'PLL_CTRL' bit of VR_XS_PMA_MP_12G_16G_25G_MISC_CTRL0 Register is programmed to 1. If the receive link is not up (or if frame lock does not happen during CL72 Training or if AN page is not received), within this specified interval, DWC_xpcs would program the PHY Rx power state to P1 and would then bring it back to P0 (thus re-initializing the PHY PLL). **Value After Reset:** 0xffff **Exists:** Always

5.2.3.153 VR_XS_PMA_MP_6G_MISC_CTRL1

- **Description:** VR XS or PMA Synopsys Multi-protocol 6G PHY Miscellaneous Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x007A : (APB3_16) ? (0x007A*0x2) : (0x007A*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&SSP)

15:14	13:12	11:10	9:8	7:6	5:4	3:2	1:0
RX_RATE_3	TX_RATE_3	RX_RATE_2	TX_RATE_2	RX_RATE_1	TX_RATE_1	RX_RATE_0	TX_RATE_0

Table 5-283 Fields for Register: VR_XS_PMA_MP_6G_MISC_CTRL1

Bits	Name	Memory Access	Description
15:14	RX_RATE_3	* Varies	**For Multi-lane configurations:** Rx Baud Rate Lane 3 The baud rate selection is as follows: ■ 00: mpll_baud_clk ■ 01: mpll_baud_clk/2 ■ 10: mpll_baud_clk/4 ■ 11: Not supported **For Single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0 : ((RXAUI_DEFAULT) ? 0x0 : 0x1) " **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
13:12	TX_RATE_3	* Varies	**For Multi-lane configurations:** Tx Baud Rate Lane 3 The baud rate selection is as follows: ■ 00: mpll_baud_clk ■ 01: mpll_baud_clk/2 ■ 10: mpll_baud_clk/4 ■ 11: Not supported **For Single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0 : ((RXAUI_DEFAULT) ? 0x0 : 0x1) " **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
11:10	RX_RATE_2	* Varies	**For Multi-lane configurations:** Rx Baud Rate Lane 2 The baud rate selection is as follows: ■ 00: mpll_baud_clk ■ 01: mpll_baud_clk/2 ■ 10: mpll_baud_clk/4 ■ 11: Not supported **For Single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0 : ((RXAUI_DEFAULT) ? 0x0 : 0x1) " **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
9:8	TX_RATE_2	* Varies	**For Multi-lane configurations:** Tx Baud Rate Lane 2 The baud rate selection is as follows: ■ 00: mpll_baud_clk ■ 01: mpll_baud_clk/2 ■ 10: mpll_baud_clk/4 ■ 11: Not supported **For Single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0 : ((RXAUI_DEFAULT) ? 0x0 : 0x1) " **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
7:6	RX_RATE_1	* Varies	**For Multi-lane configurations:** Rx Baud Rate Lane 1 The baud rate selection is as follows: ■ 00: mpll_baud_clk ■ 01: mpll_baud_clk/2 ■ 10: mpll_baud_clk/4 ■ 11: Not supported **For Single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0 : ((RXAUI_DEFAULT) ? 0x0 : 0x1) " **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
5:4	TX_RATE_1	* Varies	**For Multi-lane configurations:** Tx Baud Rate Lane 1 The baud rate selection is as follows: ■ 00: mpll_baud_clk ■ 01: mpll_baud_clk/2 ■ 10: mpll_baud_clk/4 ■ 11: Not supported **For Single-lane configurations:** Reserved **Value After Reset:** "(SINGLE_LANE_CFG) ? 0x0 : ((RXAUI_DEFAULT) ? 0x0 : 0x1) " **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\""
3:2	RX_RATE_0	R/W	Rx Baud Rate Lane 0 The baud rate selection is as follows: ■ 00: mpll_baud_clk ■ 01: mpll_baud_clk/2 ■ 10: mpll_baud_clk/4 ■ 11: Not supported **Value After Reset:** "(RXAUI_DEFAULT) ? 0x0 :((SINGLE_LANE_CFG) ? 0x2 : 0x1)" **Exists:** Always

Bits	Name	Memory Access	Description
1:0	TX_RATE_0	R/W	Tx Baud Rate Lane 0 The baud rate selection is as follows: ■ 00: mpll_baud_clk ■ 01: mpll_baud_clk/2 ■ 10: mpll_baud_clk/4 ■ 11: Not supported **Value After Reset:** "(RXAUI_DEFAULT ? 0x0 :((SINGLE_LANE_CFG) ? 0x2 : 0x1))" **Exists:** Always

5.2.3.154 VR_XS_PMA_MP_12G_16G_25G_EEE_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY EEE Control Register This register is present only in configurations with PCS-R/QSGMII support and with EEE enabled.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x007A : (APB3_16) ? (0x007A*0x2) : (0x007A*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E12||E16||E25||E32)&&(LP_10GBR||(QSGMII_EN&&EEE_EN))

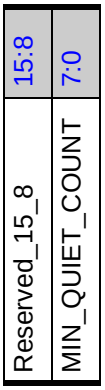


Table 5-284 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_EEE_CTRL

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
7:0	MIN_QUIET_COUNT	R/W	Minimum Rx Quiet Duration This field determines the minimum duration for which Rx LPI FSM remains in the RX_QUIET state.DWC_xpcs implements a vendor-specific 'MIN QUIET counter' in the RX_QUIET state of LPI state machine. This counter (running at clk_rpcs_rx_i clock in PCS-R Mode or running at clk_xgxs_rx0_i clock in QSGMII Mode) starts as soon as RX_QUIET state is reached and it stops when it reaches a value equal to 8 times 'MIN_QUIET_COUNT' field. DWC_xpcs start monitoring for 'energy_detect' (in PCS-R MOde) or 'signal_detect' (in QSGMII Mode) only after this counter reaches its maximum value (8*MIN_QUIET_COUNT). **Note:**In QSGMII Mode, this timer is activated only if SUPRESS_LOS_DET bit of VR_XS_PCS_DEBUG_CTRL Register is set to 1. If this bit is 0,behavior of Rx EEE FSM is as per IEEE specification. **Value After Reset:** 0xa **Exists:** Always

5.2.3.155 VR_XS_PMA_MP_12G_16G_25G_SRAM

- **Description:** VR XS or PMA Synopsys Multi-protocol 12G/16G/25G PHY SRAM Register.
  Note: This is a special register which can be read/written even before DWC_xpcs completes its reset/initialization sequence.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x007B : (APB3_16) ? (0x007B*0x2) : (0x007B*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&(E12||E16||E25||E32))

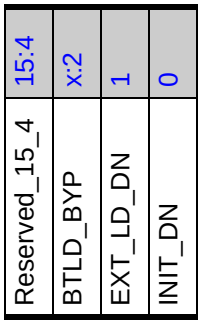


Table 5-285 Fields for Register: VR_XS_PMA_MP_12G_16G_25G_SRAM

Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
x:2	BTLD_BYP	* Varies	SRAM Bootload Bypass. This field drives the xpcs_sram_bootload_byp_o port. **Note:**This field is present only for Multi-protocol 25G/20G/8G/32G PHY configurations. **Value After Reset:** "E32&&!SNPS_RSVDPARAM_3 ? 0x1 : 0x0" **Exists:** Always **Range Variable[x]:** "SNPS_RSVDPARAM_37 ? 2 : 1" + 1 **Memory Access:** "(E25||SNPS_RSVDPARAM_39||E32) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
1	EXT_LD_DN	R/W	SRAM External Loading Done. This field drives the 'xpcs_sram_ext_ld_done_o' output port of DWC_xpcs. Software must set this bit to 1, after PHY has completed its SRAM initialization and after software has completed the (optional) external loading of the SRAM attached to the PHY. The PHY completes its initialization sequence only after this register bit is set to 1. **Note:** ■ After the software sets this bit to 1, it must remain 1 throughout the normal operation. ■ This bit gets cleared during standard soft reset (RST) ■ This bit does not get cleared during vendor soft reset (VR_RST). ■ When the software asserts VR_RST, this bit must be cleared prior to that (in case, software wishes to alter the SRAM contents initialized by the PHY). **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
0	INIT_DN	R	SRAM Initialization Done. This register field reflects the value of the 'xpcs_sram_init_done_i' port of DWC_xpcs. When PHY gets reset, the boot-loader in the Raw PCS loads the code from the internal look-up table to the SRAM. After this SRAM initialization is complete, PHY asserts this signal. After the SRAM initialization is done, software can optionally change the SRAM contents using the CR port interface by addressing the SRAM address space in the register map of the PHY. During power-on-reset or whenever RST or VR_RST register bits are programmed to 1, the PHY also gets reset. Therefore, the software must first poll this register bit until it becomes 1. Thereafter, it must set the 'EXT_LD_DN' bit to 1. Only then the PHY and the DWC_xpcs proceeds to complete its reset sequence. **Note:** Though default value of this register bit is given as 1, this is the value provided by the PHY after the completion of SRAM initialization. **Value After Reset:** 0x1 **Exists:** Always

5.2.3.156 VR_XS_PMA_MP_16G_25G_MISC_CTRL2

- **Description:** VR XS or PMA Synopsys Multi-protocol 16G/25G PHY Miscellaneous Control 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x007C : (APB3_16) ? (0x007C*0x2) : (0x007C*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&(E16||E25||SNPS_RSVDPARAM_39||E32)

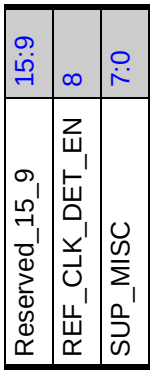


Table 5-286 Fields for Register: VR_XS_PMA_MP_16G_25G_MISC_CTRL2

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	REF_CLK_DET_EN	* Varies	**Non E32 configurations:** Reference Clock Detect enable. This field drives the output port xpcs_ref_clkdet_en_o. **E32 configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!E32 ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
7:0	SUP_MISC	* Varies	Support Miscellaneous Controls.**Non SNPS_RSVDPARAM_37 configurations:** Support Miscellaneous Controls. This field drives the output port xpcs_sup_misc_o[7:0]. **For SNPS_RSVDPARAM_37 configurations:** Reserved **Value After Reset:** "(E25||E32) ? 0x0 : (SNPS_RSVDPARAM_39&&RPCS_PRESENT&&SNPS_RSVDPARAM_13) ? 0x2 : (SNPS_RSVDPARAM_39&&RPCS_PRESENT&&REF_CLK_25) ? 0x1 : SNPS_RSVDPARAM_39 ? 0x2 : 0x1" **Exists:** (!SNPS_RSVDPARAM_37) **Memory Access:** "!SNPS_RSVDPARAM_37 ? \"read-write\" : \"read-only\""

5.2.3.157 VR_XS_PMA_MP_32G_CNTX_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 32G PHY Context Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x007D : (APB3_16) ? (0x007D*0x2) : (0x007D*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&E32

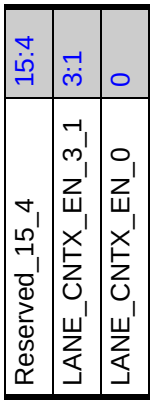


Table 5-287 Fields for Register: VR_XS_PMA_MP_32G_CNTX_CTRL

Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:1	LANE_CNTX_EN_3_1	* Varies	**For multi lane configurations :** Lane Context Enable Lane 3_1. This field drives the output port 'xpcs_lane_cntx_en_o. **For single lane configurations :** Reserved **Value After Reset:** "!SINGLE_LANE_CFG ? 0x7 :0x0" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
0	LANE_CNTX_EN_0	R/W	Lane Context Enable Lane0. </br>This field drives the output port 'xpcs_lane_cntx_en_o' **Value After Reset:** 0x1 **Exists:** Always

5.2.3.158 VR_XS_PMA_MP_32G_MISC_STS1

- **Description:** VR XS or PMA Synopsys Multi-protocol 32G PHY Miscellaneous Status 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x007E : (APB3_16) ? (0x007E*0x2) : (0x007E*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&SNPS_XAUI_PHY&&E32

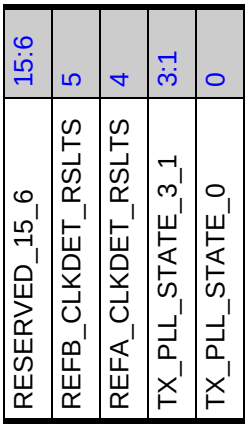


Table 5-288 Fields for Register: VR_XS_PMA_MP_32G_MISC_STS1

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5	REFB_CLKDET_RSLTS	R	Reference clock detection result for refb. Indicates the presence of the reference clock. This field reflects the value of the input port 'xpcs_refb_clkdet_result_i' **Value After Reset:** 0x0 **Exists:** Always
4	REFA_CLKDET_RSLTS	R	Reference clock detection result for refa. Indicates the presence of the reference clock. This field reflects the value of the input port 'xpcs_refa_clkdet_result_i' **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
3:1	TX_PLL_STATE_3_1	R	TX PLL State Indicator. **For Multi lane configurations:** Indicates the state of PLL for the lanes[3:1]. This field reflects the value of the input port 'xpcs_tx_pll_state_i'. **For Single lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always
0	TX_PLL_STATE_0	R	TX PLL State Indicator. Indicates the state of PLL for the lane0. This field reflects the value of the input port 'xpcs_tx_pll_state_i'. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.159 VR_XS_PMA_SNPS_CR_CTRL

- **Description:** VR XS or PMA Synopsys PHY CR Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0080 : (APB3_16) ? (0x0080*0x2) : (0x0080*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&!SNPS_RSVDPARAM_37)

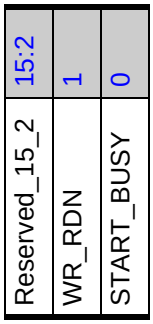


Table 5-289 Fields for Register: VR_XS_PMA_SNPS_CR_CTRL

Bits	Name	Memory Access	Description
15:2	Reserved_15_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1	WR_RDN	R/W	Write or Read Indicator This bit indicates whether a read or write operation is to be performed to the Synopsys PHY registers: ■ 0: Read ■ 1: Write **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	START_BUSY	R/W	Start CR Port Access or Busy Indicator (WS,SC Type) This bit indicates if CR port access is in progress: ■ 0: CR port not busy ■ 1: CR port busy The sequence is: 1. The host sets this bit to start a read or write transfer through the CR port to the Synopsys PHY registers. 2. This bit remains set during the CR port access. 3. The DWC_xpcs clears this bit when the CR port access is complete. **Dependencies:**The host must read this bit as 0 before writing to any of the following registers: ■ VR_XS_PMA_SNPS_CR_CTRL Register (this register) ■ VR_XS_PMA_SNPS_CR_ADDR Register ■ VR_XS_PMA_SNPS_CR_DATA Register During read, the data from the CR port interface is placed into the VR XS or PMA Synopsys PHY CR Data Register. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.160 VR_XS_PMA_SNPS_CR_ADDR

- **Description:** VR XS or PMA Synopsys PHY CR Address Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0081 : (APB3_16) ? (0x0081*0x2) : (0x0081*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&!SNPS_RSVDPARAM_37)

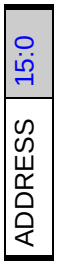


Table 5-290 Fields for Register: VR_XS_PMA_SNPS_CR_ADDR

Bits	Name	Memory Access	Description
15:0	ADDRESS	R/W	CR Port Address This field indicates the address of the register to be accessed through the CR port. **Dependency:** ■ The host must not change this field when Bit 0 of the VR_XS_PMA_SNPS_CR_CTRL Register is set. ■ This field must be written before writing to any of the following registers: ❑ VR_XS_PMA_SNPS_CR_CTRL Register ❑ VR_XS_PMA_SNPS_CR_ADDR Register (this register) ❑ VR_XS_PMA_SNPS_CR_DATA Register **Value After Reset:** 0x0 **Exists:** Always

5.2.3.161 VR_XS_PMA_SNPS_CR_DATA

- **Description:** VR XS or PMA Synopsys CR Data Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0082 : (APB3_16) ? (0x0082*0x2) : (0x0082*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** ((PMA_MMD||XGXS_PCS)&&SNPS_XAUI_PHY&&!SNPS_RSVDPARAM_37)

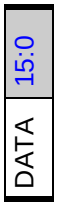


Table 5-291 Fields for Register: VR_XS_PMA_SNPS_CR_DATA

Bits	Name	Memory Access	Description
15:0	DATA	R/W	CR Port Data This field contains the data for CR Port access. ■ During a write operation, the data in this register is written to the Synopsys PHY register pointed by the CR Port Address. ■ During a read operation, the data read from the Synopsys PHY is written to this register. The content becomes valid only when the DWC_xpcs clears Bit 0 of the VR XS or PMA Synopsys PHY CR Control Register. The host must not change this field when Bit 0 of VR_XS_PMA_SNPS_CR_CTRL Register is set. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.162 VR_XS_PMA_MP_25G_RX_WIDTH_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY Rx Width Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0090 : (APB3_16) ? (0x0090*0x2) : (0x0090*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (XGXS_PCS||PMA_MMD)&&((E25&&!SNPS_RSVDPARAM_37)||E32)

15	14:12	11	10:8	7	6:4	3	2:0
Reserved_15	RX3_WIDTH	Reserved_11	RX2_WIDTH	Reserved_7	RX1_WIDTH	Reserved_3	RX0_WIDTH

Table 5-292 Fields for Register: VR_XS_PMA_MP_25G_RX_WIDTH_CTRL

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:12	RX3_WIDTH	* Varies	Rx PHY Width -lane 3. This field drives the xpcs_rx3_width_o output port and controls the width of PHY Rx lane 3. This field is valid only for multi-lane configurations. **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : RPCS_PRESENT ? (R_SRDS_INTF==0 ? 0x2 : R_SRDS_INTF==1 ? 0x4 : R_SRDS_INTF==4 ? 0x3 : 0x5) : RXAUI_DEFAULT ? 0x3 : 0x1" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
10:8	RX2_WIDTH	* Varies	Rx PHY Width -lane 2. This field drives the xpcs_rx2_width_o output port and controls the width of PHY Rx lane 2. This field is valid only for multi-lane configurations. **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : RPCS_PRESENT ? (R_SRDS_INTF==0 ? 0x2 : R_SRDS_INTF==1 ? 0x4 : R_SRDS_INTF==4 ? 0x3 : 0x5) : RXAUI_DEFAULT ? 0x3 : 0x1" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:4	RX1_WIDTH	* Varies	Rx PHY Width -lane 1. This field drives the xpcs_rx1_width_o output port and controls the width of PHY Rx lane 1. This field is valid only for multi-lane configurations. **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : RPCS_PRESENT ? (R_SRDS_INTF==0 ? 0x2 : R_SRDS_INTF==1 ? 0x4 : R_SRDS_INTF==4 ? 0x3 : 0x5) : RXAUI_DEFAULT ? 0x3 : 0x1" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
3	Reserved_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	RX0_WIDTH	R/W	Rx PHY Width -lane 0. This field drives the xpcs_rx0_width_o output port and controls the width of PHY Rx lane 0. **Value After Reset:** "RPCS_PRESENT ? (R_SRDS_INTF==0 ? 0x2 : R_SRDS_INTF==1 ? 0x4 : R_SRDS_INTF==4 ? 0x3 : 0x5) : RXAUI_DEFAULT ? 0x3 : 0x1" **Exists:** Always

5.2.3.163 VR_XS_PMA_MP_25G_RX_AFE_RATE_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY Rx AFE Rate Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0091 : (APB3_16) ? (0x0091*0x2) : (0x0091*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (XGXS_PCS||PMA_MMD)&&E25&&!SNPS_RSVDPARAM_37

15	14:12	11	10:8	7	6:4	3	2:0
Reserved_15	RX3_AFE_RATE	Reserved_11	RX2_AFE_RATE	Reserved_7	RX1_AFE_RATE	Reserved_3	RX0_AFE_RATE

Table 5-293 Fields for Register: VR_XS_PMA_MP_25G_RX_AFE_RATE_CTRL

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:12	RX3_AFE_RATE	* Varies	Rx Equalization AFE Rate -lane 3. This field drives the xpcs_rx3_eq_afe_rate_o output port. This field is valid only for multi-lane configurations. **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : RPCS_PRESENT ? 0x4 : RXAUI_DEFAULT ? 0x6 : 0x7" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
10:8	RX2_AFE_RATE	* Varies	Rx Equalization AFE Rate -lane 2. This field drives the xpcs_rx2_eq_afe_rate_o output port. This field is valid only for multi-lane configurations. **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : RPCS_PRESENT ? 0x4 : RXAUI_DEFAULT ? 0x6 : 0x7" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:4	RX1_AFE_RATE	* Varies	Rx Equalization AFE Rate -lane 1. This field drives the xpcs_rx1_eq_afe_rate_o output port. This field is valid only for multi-lane configurations. **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : RPCS_PRESENT ? 0x4 : RXAUI_DEFAULT ? 0x6 : 0x7" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
3	Reserved_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	RX0_AFE_RATE	R/W	Rx Equalization AFE Rate -lane 0. This field drives the xpcs_rx0_eq_afe_rate_o output port. **Value After Reset:** "RPCS_PRESENT ? 0x4 : RXAUI_DEFAULT ? 0x6 : 0x7" **Exists:** Always

5.2.3.164 VR_XS_PMA_MP_25G_RX_RECAL_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY Rx Re-calibration Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0092 : (APB3_16) ? (0x0092*0x2) : (0x0092*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (XGXS_PCS||PMA_MMD)&&(E25||E32)

15:14	13:12	11:10	9:8	7:5	4	3:1	0
RX3_BNK_SEL	RX2_BNK_SEL	RX1_BNK_SEL	RX0_BNK_SEL	RCAL_SK_EN_3_1	RCAL_SK_EN_0	RCAL_FR_EN_3_1	RCAL_FR_EN_0

Table 5-294 Fields for Register: VR_XS_PMA_MP_25G_RX_RECAL_CTRL

Bits	Name	Memory Access	Description
15:14	RX3_BNK_SEL	* Varies	Rx Recalibration Bank Select-lane 3. This field drives the xpcs_rx1_recal_bank_sel_o output port and selects the bank to store the Rx duty cycle correction and IQ correction calibration values This field is present only for multi-lane configurations. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
13:12	RX2_BNK_SEL	* Varies	Rx Recalibration Bank Select-lane 2. This field drives the xpcs_rx1_recal_bank_sel_o output port and selects the bank to store the Rx duty cycle correction and IQ correction calibration values This field is present only for multi-lane configurations. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
11:10	RX1_BNK_SEL	* Varies	Rx Recalibration Bank Select-lane 1. This field drives the xpcs_rx1_recal_bank_sel_o output port and selects the bank to store the Rx duty cycle correction and IQ correction calibration values This field is present only for multi-lane configurations. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
9:8	RX0_BNK_SEL	R/W	Rx Recalibration Bank Select-lane 0. This field drives the xpcs_rx0_recal_bank_sel_o output port and selects the bank to store the Rx duty cycle correction and IQ correction calibration values **Value After Reset:** 0x0 **Exists:** Always
7:5	RCAL_SK_EN_3_1	* Varies	Rx Skip Re-calibration -lanes 3 to 1 This field drives the xpcs_rx_recal_skip_en_o[3:1] port. This field is valid only for multi-lane configurations. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
4	RCAL_SK_EN_0	R/W	Rx Skip Re-calibration -lane 0 This field drives the xpcs_rx_recal_skip_en_o[0] port. **Value After Reset:** 0x0 **Exists:** Always
3:1	RCAL_FR_EN_3_1	* Varies	Rx Force Re-calibration -lanes 3 to 1 This field drives the xpcs_rx_recal_force_en_o[3:1] port. This field is valid only for multi-lane configurations. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
0	RCAL_FR_EN_0	R/W	Rx Force Re-calibration -lane 0 This field drives the xpcs_rx_recal_force_en_o[0] port. **Value After Reset:** 0x0 **Exists:** Always

5.2.3.165 VR_XS_PMA_MP_25G_RX_DCC_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY Rx DCC Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0093 : (APB3_16) ? (0x0093*0x2) : (0x0093*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (XGXS_PCS||PMA_MMD)&&E25&&!SNPS_RSVDPARAM_37

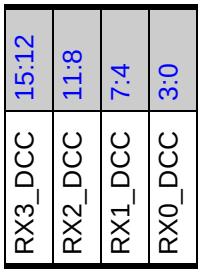


Table 5-295 Fields for Register: VR_XS_PMA_MP_25G_RX_DCC_CTRL

Bits	Name	Memory Access	Description
15:12	RX3_DCC	* Varies	Rx Internal Clock Duty Cycle Correction Control Range for lane 3. **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : (SNPS_RSVDPARAM_16||SNPS_RSVDPARAM_36) ? 0xC : (RPCS_PRESENT) ? 0xB : 0xA" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
11:8	RX2_DCC	* Varies	Rx Internal Clock Duty Cycle Correction Control Range for lane 2. **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : (SNPS_RSVDPARAM_16||SNPS_RSVDPARAM_36) ? 0xC : (RPCS_PRESENT) ? 0xB : 0xA" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
7:4	RX1_DCC	* Varies	Rx Internal Clock Duty Cycle Correction Control Range for lane 1. **Value After Reset:** "SINGLE_LANE_CFG ? 0x0 : (SNPS_RSVDPARAM_16||SNPS_RSVDPARAM_36) ? 0xC : (RPCS_PRESENT) ? 0xB : 0xA" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
3:0	RX0_DCC	R/W	Rx Internal Clock Duty Cycle Correction Control Range for lane 0. **Value After Reset:** "(SNPS_RSVDPARAM_16||SNPS_RSVDPARAM_36) ? 0xC : (RPCS_PRESENT||KX_ONLY) ? 0xB : 0xA" **Exists:** Always

5.2.3.166 VR_XS_PMA_MP_25G_RX_VCO_CFG0

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY Rx CDR VCO Configuration 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0094 : (APB3_16) ? (0x0094*0x2) : (0x0094*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (XGXS_PCS | |PMA_MMD)&&E25&&!SNPS_RSVDPARAM_37

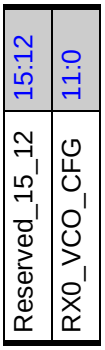


Table 5-296 Fields for Register: VR_XS_PMA_MP_25G_RX_VCO_CFG0

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:0	RX0_VCO_CFG	R/W	Rx CDR VCO configuration Bus - lane 0. This field drives the xpcs_rx0_cdr_vco_cfg_o port and controls the biasing, temperature compensation, DAC settings and frequency range of Rx CDR VCO. **Value After Reset:** "SNPS_RSVDPARAM_16&&!SNPS_RSVDPARAM_36 ? 0x0 : (RPCS_PRESENT||KX_ONLY) ? (SNPS_RSVDPARAM_36 ? 0x300: 0x100) : (SNPS_RSVDPARAM_36 ? 0x303 : 0x103)" **Exists:** Always

5.2.3.167 VR_XS_PMA_MP_25G_RX_VCO_CFG1

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY Rx CDR VCO Configuration 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0095 : (APB3_16) ? (0x0095*0x2) : (0x0095*0x4)" + "(PMA_MMD) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (XGXS_PCS||PMA_MMD)&&E25&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG

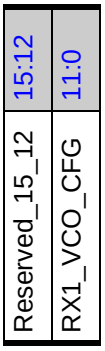


Table 5-297 Fields for Register: VR_XS_PMA_MP_25G_RX_VCO_CFG1

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:0	RX1_VCO_CFG	R/W	Rx CDR VCO configuration Bus -lane 1. This field drives the xpcs_rx1_cdr_vco_cfg_o port and controls the biasing, temperature compensation, DAC settings, and frequency range of Rx CDR VCO. **Value After Reset:** "(RPCS_PRESENT) ? 0x100 : 0x103" **Exists:** Always

5.2.3.168 VR_XS_PMA_MP_25G_RX_VCO_CFG2

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY Rx CDR VCO Configuration 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0096 : (APB3_16) ? (0x0096*0x2) : (0x0096*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (XGXS_PCS | |PMA_MMD)&&E25&&!SINGLE_LANE_CFG

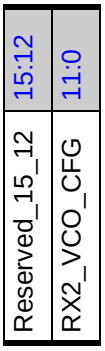


Table 5-298 Fields for Register: VR_XS_PMA_MP_25G_RX_VCO_CFG2

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:0	RX2_VCO_CFG	R/W	Rx CDR VCO configuration Bus -lane 2. This field drives the xpcs_rx2_cdr_vco_cfg_o port and controls the biasing, temperature compensation, DAC settings, and frequency range of Rx CDR VCO. **Value After Reset:** "(RPCS_PRESENT) ? 0x100 : 0x103" **Exists:** Always

5.2.3.169 VR_XS_PMA_MP_25G_RX_VCO_CFG3

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY Rx CDR VCO Configuration 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0097 : (APB3_16) ? (0x0097*0x2) : (0x0097*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (XGXS_PCS | |PMA_MMD)&&E25&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG

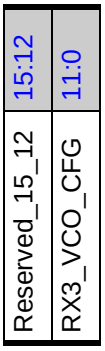


Table 5-299 Fields for Register: VR_XS_PMA_MP_25G_RX_VCO_CFG3

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:0	RX3_VCO_CFG	R/W	Rx CDR VCO configuration Bus -lane 3. This field drives the xpcs_rx3_cdr_vco_cfg_o port and controls the biasing, temperature compensation, DAC settings, and frequency range of Rx CDR VCO. **Value After Reset:** "(RPCS_PRESENT) ? 0x100 : 0x103" **Exists:** Always

5.2.3.170 VR_XS_PMA_MP_25G_RX_LF_TRSHLD

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY RX Low Frequency Threshold Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0098 : (APB3_16) ? (0x0098*0x2) : (0x0098*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&E25&&!SNPS_RSVDPARAM_37

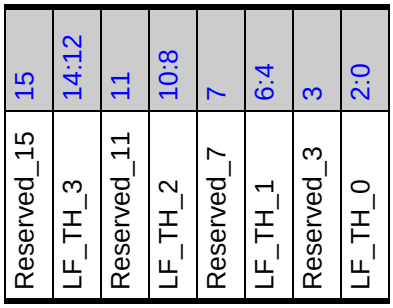


Table 5-300 Fields for Register: VR_XS_PMA_MP_25G_RX_LF_TRSHLD

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:12	LF_TH_3	* Varies	**For Multi-lane configurations:** Rx Low Frequency Signal Detection Threshold on lane 3. This field drives the xpcs_rx3_sigdet_lf_th_o port. **For Single-lane configurations:** Reserved **Value After Reset:** "SNPS_RSVDPARAM_36&&!SINGLE_LANE_CFG ? 0x1 : RXAUI_DEFAULT ? 0x2 : !SINGLE_LANE_CFG ? 0x7 : 0x0" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
10:8	LF_TH_2	* Varies	**For Multi-lane configurations:** Rx Low Frequency Signal Detection Threshold on lane 2. This field drives the xpcs_rx2_sigdet_lf_th_o port. **For Single-lane configurations:** Reserved **Value After Reset:** "SNPS_RSVDPARAM_36&&!SINGLE_LANE_CFG ? 0x1 : RXAUI_DEFAULT ? 0x2 : !SINGLE_LANE_CFG ? 0x7 : 0x0" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:4	LF_TH_1	* Varies	**For Multi-lane configurations:** Rx Low Frequency Signal Detection Threshold on lane 1. This field drives the xpcs_rx1_sigdet_lf_th_o port. **For Single-lane configurations:** Reserved **Value After Reset:** "SNPS_RSVDPARAM_36&&!SINGLE_LANE_CFG ? 0x1 : RXAUI_DEFAULT ? 0x2 : !SINGLE_LANE_CFG ? 0x7 : 0x0" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
3	Reserved_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	LF_TH_0	R/W	Rx Low Frequency Signal Detection Threshold lane 0. This field drives the xpcs_rx0_sigdet_lf_th_o port. **Value After Reset:** "(SNPS_RSVDPARAM_16||SNPS_RSVDPARAM_36) ? 0x1 : (KX_ONLY||RXAUI_DEFAULT) ? 0x2 : 0x7" **Exists:** Always

5.2.3.171 VR_XS_PMA_MP_25G_RX_HF_TRSHLD

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY RX High Frequency Threshold Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0099 : (APB3_16) ? (0x0099*0x2) : (0x0099*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&E25&&!SNPS_RSVDPARAM_37

15	14:12	11	10:8	7	6:4	3	2:0
Reserved_15	HF_TH_3	Reserved_11	HF_TH_2	Reserved_7	HF_TH_1	Reserved_3	HF_TH_0

Table 5-301 Fields for Register: VR_XS_PMA_MP_25G_RX_HF_TRSHLD

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:12	HF_TH_3	* Varies	**For Multi-lane configurations:** Rx High Frequency Signal Detection Threshold on lane 3. This field drives the xpcs_rx3_sigdet_hf_th_o port. **For Single-lane configurations:** Reserved **Value After Reset:** "!SINGLE_LANE_CFG ? 0x2 : 0x0" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
10:8	HF_TH_2	* Varies	**For Multi-lane configurations:** Rx High Frequency Signal Detection Threshold on lane 2. This field drives the xpcs_rx2_sigdet_hf_th_o port. **For Single-lane configurations:** Reserved **Value After Reset:** "!SINGLE_LANE_CFG ? 0x2 : 0x0" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:4	HF_TH_1	* Varies	**For Multi-lane configurations:** Rx High Frequency Signal Detection Threshold on lane 1. This field drives the xpcs_rx1_sigdet_hf_th_o port. **For Single-lane configurations:** Reserved **Value After Reset:** "!SINGLE_LANE_CFG ? 0x2 : 0x0" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
3	Reserved_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	HF_TH_0	R/W	Rx High Frequency Signal Detection Threshold lane 0. This field drives the xpcs_rx0_sigdet_hf_th_o port. **Value After Reset:** 0x2 **Exists:** Always

5.2.3.172 VR_XS_PMA_C16G_RX_EQ_AFE_CTRL0

- **Description:** VR XS or PMA Synopsys Consumer 16G PHY RX Equalization AFE Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x009A : (APB3_16) ? (0x009A*0x2) : (0x009A*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&SNPS_RSVDPARAM_36

15:14	13:12	11:10	9:8	7:6	5:4	3:2	1:0
Reserved_15_14	EQ_AFE_BIAS_3	Reserved_11_10	EQ_AFE_BIAS_2	Reserved_7_6	EQ_AFE_BIAS_1	Reserved_3_2	EQ_AFE_BIAS_0

Table 5-302 Fields for Register: VR_XS_PMA_C16G_RX_EQ_AFE_CTRL0

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:12	EQ_AFE_BIAS_3	* Varies	**For Multi-lane configurations:** RX Equalization AFE input stage bias current control on lane 3. This field drives the xpcs_rx3_eq_afe_bias_o port. **For Single-lane configurations:** Reserved **Value After Reset:** "!SINGLE_LANE_CFG ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
11:10	Reserved_11_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
9:8	EQ_AFE_BIAS_2	* Varies	**For Multi-lane configurations:** RX Equalization AFE input stage bias current control on lane 2. This field drives the xpcs_rx2_eq_afe_bias_o port. **For Single-lane configurations:** Reserved **Value After Reset:** "!SINGLE_LANE_CFG ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	EQ_AFE_BIAS_1	* Varies	**For Multi-lane configurations:** RX Equalization AFE input stage bias current control on lane 1. This field drives the xpcs_rx1_eq_afe_bias_o port. **For Single-lane configurations:** Reserved **Value After Reset:** "!SINGLE_LANE_CFG ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
3:2	Reserved_3_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1:0	EQ_AFE_BIAS_0	R/W	RX Equalization AFE input stage bias current control lane 0. This field drives the xpcs_rx1_eq_afe_bias_o port. **Value After Reset:** 0x1 **Exists:** Always

5.2.3.173 VR_XS_PMA_C16G_RX_EQ_AFE_CTRL1

- **Description:** VR XS or PMA Synopsys Consumer 16G PHY RX Equalization AFE Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x009B : (APB3_16) ? (0x009B*0x2) : (0x009B*0x4)" + "(PMA_MMD) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x018020 : (APB3_16) ? (0x018020*0x2) : (0x018020*0x4)) : ((XGXS_PCS&&!DTE_PHY_SEL) ? ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x058020 : (APB3_16) ? (0x058020*0x2) : (0x058020*0x4)) : ((APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x048020 : (APB3_16) ? (0x048020*0x2) : (0x048020*0x4)))"
- **Exists:** (PMA_MMD | |XGXS_PCS)&&SNPS_RSVDPARAM_36

15	14:12	11	10:8	7	6:4	3	2:0
Reserved_15	EQ_AFE_VC_AD_3	Reserved_11	EQ_AFE_VC_AD_2	Reserved_7	EQ_AFE_VC_AD_1	Reserved_3	EQ_AFE_VC_AD_0

Table 5-303 Fields for Register: VR_XS_PMA_C16G_RX_EQ_AFE_CTRL1

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:12	EQ_AFE_VC_AD_3	* Varies	**For Multi-lane configurations:** RX Equalization Attenuator output common-mode level control on lane 3. This field drives the xpcs_rx3_eq_afe_vcm_adj_o port. **For Single-lane configurations:** Reserved **Value After Reset:** "!SINGLE_LANE_CFG ? 0x3 : 0x0" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
10:8	EQ_AFE_VC_AD_2	* Varies	**For Multi-lane configurations:** RX Equalization Attenuator output common-mode level control on lane 2. This field drives the xpcs_rx2_eq_afe_vcm_adj_o port. **For Single-lane configurations:** Reserved **Value After Reset:** "!SINGLE_LANE_CFG ? 0x3 : 0x0" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:4	EQ_AFE_VC_AD_1	* Varies	**For Multi-lane configurations:** RX Equalization Attenuator output common-mode level control on lane 1. This field drives the xpcs_rx1_eq_afe_vcm_adj_o port. **For Single-lane configurations:** Reserved **Value After Reset:** "!SINGLE_LANE_CFG ? 0x3 : 0x0" **Exists:** Always **Memory Access:** "!SINGLE_LANE_CFG ? \"read-write\" : \"read-only\""
3	Reserved_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	EQ_AFE_VC_AD_0	R/W	RX Equalization Attenuator output common-mode level control lane 0. This field drives the xpcs_rx0_eq_afe_vcm_adj_o port. **Value After Reset:** 0x3 **Exists:** Always

5.2.4 DWC_xpcs_map/AN_MMD Registers

5.2.4.1 SR_AN_CTRL

- **Description:** SR AN MMD Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)

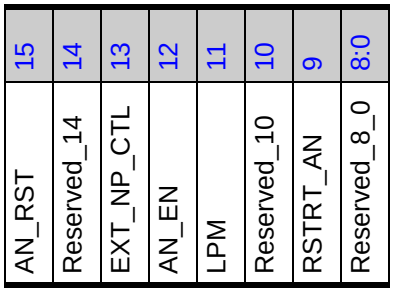


Table 5-304 Fields for Register: SR_AN_CTRL

Bits	Name	Memory Access	Description
15	AN_RST	R/W	Auto-Negotiation Soft Reset (RW,SC Type) When the host sets this bit, the CSR block triggers the software reset process in which all internal blocks, except the Management Interface block, are reset. The registers are reset to their default values. When this bit is set, it also resets the PHY. This bit is self cleared under the following conditions: ■ **For Synopsys PHY:** This bit is self cleared when Bits[4:2] in VR_XS_PCS_DIG_STS Register is equal to 3'b100, that is, Tx/Rx clocks are stable and in Power_Good state. ■ **For Non-Synopsys PHY:** This bit is self cleared after the following: -- 32 clk_csr_i clocks for the MCI interface -- 1 MDC clock period for the MDIO interface**Note:** For description of read or write access to registers during the reset process, see "Special Case Register Access" section in the Databook. **Value After Reset:** 0x0 **Exists:** Always
14	Reserved_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
13	EXT_NP_CTL	R/W	Extended Next Page Control ■ 0: Extended next page is disabled ■ 1: Extended next page is enabled **Value After Reset:** 0x1 **Exists:** Always
12	AN_EN	R/W	Auto-Negotiation Enable ■ 0: Auto-negotiation is disabled. The SR_PMA_CTRL1 Register or SR_XS_PCS_CTRL2 Register determines the link speed. ■ 1: The host determines the link speed based on the outcome of the Clause 73 auto-negotiation. **Value After Reset:** "SNPS_RSVDPARAM_6 ? 0x0 : 0x1" **Exists:** Always **Testable:** untestable
11	LPM	R/W	Low-Power Enable This bit controls the power-down mode of the DWC_xpcs. ■ 0: Normal Operation ■ 1: The DWC_xpcs goes to the power-down mode along with the PHY. For non-Synopsys PHY configurations, DWC_xpcs asserts xpcs_pdown_o port. For Synopsys PHY configurations, DWC_xpcs does the following to trigger the power-down mode: ■ Turns off the PHY Receiver and Transmitter ■ Switches off all the clocks When the host clears this bit, the DWC_xpcs resumes the normal operation. In a Synopsys PHY configuration, after clearing this bit, the host must wait until Bits[4:2] of the VR_XS_PCS_DIG_STS Register indicate that the DWC_xpcs is in the normal state. **Note:** For description of read or write access to registers during the reset process, see "Special Case Register Access" section in the Databook. **Value After Reset:** 0x0 **Exists:** Always
10	Reserved_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
9	RSTRT_AN	R/W	Restart Auto-Negotiation (RW,SC Type) ■ 0: Normal operation ■ 1: The auto-negotiation process is started or restarted, irrespective of the current state of the auto-negotiation process. **Value After Reset:** 0x0 **Exists:** Always
8:0	Reserved_8_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.4.2 SR_AN_STS

- **Description:** SR AN MMD Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1 : (APB3_16) ? (0x1*0x2) : (0x1*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)

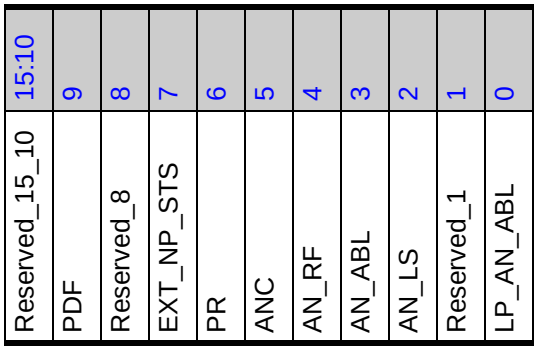


Table 5-305 Fields for Register: SR_AN_STS

Bits	Name	Memory Access	Description
15:10	Reserved_15_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9	PDF	R	Parallel Detection Fault (RO,LH Type) The DWC_xpcs sets this bit when a parallel detection fault is detected in the Clause 73 auto-negotiation ■ 0: No fault detected ■ 1: Parallel detection fault detected A parallel detection fault is declared when the following occurs: 1. The device detects the legacy device by monitoring the Rx Link Up. 2. After Rx Link Up, the Rx Link Down is detected before 50 ms. **Value After Reset:** 0x0 **Exists:** Always
8	Reserved_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
7	EXT_NP_STS	R	Extended Next Page Status ■ 0: Extended next page is not allowed if the local device or the link partner does not support extended next page. ■ 1: Extended next page format is used. It indicates that the local device and the link partner have indicated to use the extended next page support. **Value After Reset:** 0x0 **Exists:** Always
6	PR	R	Page Received (RO, LH Type) ■ 0: No page received ■ 1: Page is received and the corresponding Link Code Word is stored in the following: ❑ SR AN MMD LP Base Page Ability Register 1, 2, or 3 for base page. ❑ SR AN MMD XNP Ability Register 1, 2, or 3 for next page. **Value After Reset:** 0x0 **Exists:** Always
5	ANC	R	Auto-Negotiation Complete ■ 0: Indicates that the auto-negotiation process is not complete. ■ 1: Indicates that the auto-negotiation process is complete. **Value After Reset:** 0x0 **Exists:** Always
4	AN_RF	R	Auto-Negotiation Remote Fault (RO, LH Type) A remote fault occurs when the auto-negotiation state transitions to the TX_DISABLED state from the ACKNOWLEDGE_DETECT or NEXT_PAGE_WAIT state. This happens when a valid DME page reception is not successful. ■ 0: No fault detected ■ 1: Auto-negotiation process detected a remote fault **Value After Reset:** 0x0 **Exists:** Always
3	AN_ABL	R	Auto-Negotiation Ability ■ 0: The local device does not support auto-negotiation. ■ 1: The local device can perform auto-negotiation. This bit is always set to 1. **Value After Reset:** 0x1 **Exists:** Always

Bits	Name	Memory Access	Description
2	AN_LS	R	Auto-negotiation Link Status (RO,LL Type) This bit indicates the link status as determined by the auto-negotiation process. ■ 0: Clause 73 auto-negotiation process is not complete or the process could not determine a valid link. ■ 1: Clause 73 auto-negotiation process is complete and the process successfully determined a valid link. **Value After Reset:** 0x0 **Exists:** Always
1	Reserved_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	LP_AN_ABL	R	Link Partner Auto-Negotiation Ability ■ 0: The link partner is not able to participate in auto-negotiation. ■ 1: The link partner is able to participate in auto-negotiation. **Value After Reset:** 0x0 **Exists:** Always

5.2.4.3 SR_AN_DEV_ID1

- **Description:** SR AN MMD Device Identifier Register 1
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x2 : (APB3_16) ? (0x2*0x2) : (0x2*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)



Table 5-306 Fields for Register: SR_AN_DEV_ID1

Bits	Name	Memory Access	Description
15:0	AN_DEV_OUI_3_18	R	Organizationally Unique Identifier This field indicates the Organizationally Unique Identifier (OUI) of the manufacturer of AN device from bits 3 to 18. You can configure this value. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:0] of SR Control MMD AN Device Identifier Register 1 modifies the content of this register. **Value After Reset:** "(PMA_MMD) ? IEEE_AN_DEV_OUI_3_18 : 0x0" **Exists:** Always

5.2.4.4 SR_AN_DEV_ID2

- **Description:** SR AN MMD Device Identifier Register 2
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x3 : (APB3_16) ? (0x3*0x2) : (0x3*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)

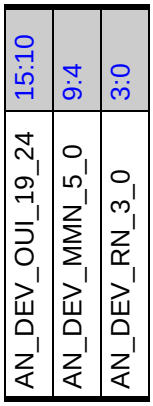


Table 5-307 Fields for Register: SR_AN_DEV_ID2

Bits	Name	Memory Access	Description
15:10	AN_DEV_OUI_19_24	R	Organizationally Unique Identifier This indicates the Organizationally Unique Identifier (OUI) of the AN device manufacturer from bits 19 to 24. You can configure this value. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:10] of SR Control MMD AN Device Identifier Register 2 modifies the content of this register. **Value After Reset:** "(PMA_MMD) ? IEEE_AN_DEV_OUI_19_24 : 0x0" **Exists:** Always
9:4	AN_DEV_MMN_5_0	R	Model Number The 5-bit Model Number of the AN Device. You can configure this value. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[9:4] of SR Control MMD AN Device Identifier Register 2 modifies the content of this register. **Value After Reset:** "(PMA_MMD) ? IEEE_AN_DEV_MODEL_NO_0_5 : 0x0" **Exists:** Always

Bits	Name	Memory Access	Description
3:0	AN_DEV_RN_3_0	R	Revision Number The 4-bit Revision Number of the AN Device. You can configure this value. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[3:0] of SR Control MMD AN Device Identifier Register 2 modifies the content of this register. **Value After Reset:** "(PMA_MMD) ? IEEE_AN_DEV_REV_NO_0_3 : 0x0" **Exists:** Always

5.2.4.5 SR_AN_DEV_PKG1

- **Description:** SR AN MMD Devices in Package Register 1
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x5 : (APB3_16) ? (0x5*0x2) : (0x5*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)

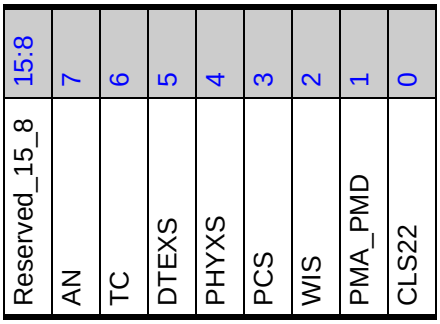


Table 5-308 Fields for Register: SR_AN_DEV_PKG1

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7	AN	R	Auto-Negotiation MMD This bit indicates that an AN MMD is present in the package. **Value After Reset:** "(BACKPLANE_ETHERNET_PCS) ? 0x1 : 0x0" **Exists:** Always
6	TC	R	TC MMD This bit indicates that the device has the TC MMDs in the package. This bit is always 0. **Value After Reset:** 0x0 **Exists:** Always
5	DTEXS	R	DTE XGXS MMD This bit indicates that the device has the DTE XS MMD in the package. **Value After Reset:** "(XGXS_PCS&&(DTE_PHY_SEL==0)) ? 1 : 0" **Exists:** Always

Bits	Name	Memory Access	Description
4	PHYXS	R	PHY XGXS MMD This bit indicates that the device has the PHY XS MMD in the package. **Value After Reset:** "(XGXS_PCS&&(DTE_PHY_SEL==1)) ? 1 : 0" **Exists:** Always
3	PCS	R	PCS MMD This bit indicates that the device has PCS MMD in the package. **Default Value:** If MAIN_MODE = Backplane Ethernet PCS, 10GBASE-X PCS, or 10GBASE-R PCS, this bit returns 1. For all other configurations, this bit returns 0. **Value After Reset:** "(PCS_MMD) ? 0x1 : 0x0" **Exists:** Always
2	WIS	R	WIS MMD This bit indicates that the device has the WIS MMD in the package. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
1	PMA_PMD	R	PMA or PMD MMD This bit indicates that the device has the PMA or PMD MMD in the package. **Default Value:** This bit is set to 1 for the following configurations: ■ Backplane Ethernet PCS ■ 10GBASE-X PCS for Synopsys PHY ■ 10GBASE-R PCS for Synopsys PHY This bit returns 0 for all other configurations. **Value After Reset:** "(PMA_MMD&&!BASE1000X_ONLY) ? 0x1 : 0x0" **Exists:** Always
0	CLS22	R	Clause 22 Register Support This bit indicates that the Clause 22 (1G) functional register sets are implemented in this package. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always

5.2.4.6 SR_AN_DEV_PKG2

- **Description:** SR AN MMD Devices in Package Register 2
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x6 : (APB3_16) ? (0x6*0x2) : (0x6*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)

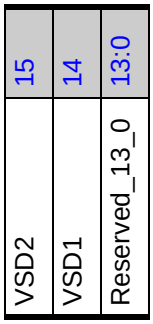


Table 5-309 Fields for Register: SR_AN_DEV_PKG2

Bits	Name	Memory Access	Description
15	VSD2	R	Vendor-Specific Device 2 This bit indicates that a vendor-specific device 2 is present in the package. **Value After Reset:** "(BASE1000X_ONLY||CL37_AN) ? 0x1 : 0x0" **Exists:** Always
14	VSD1	R	Vendor-Specific Device 1 This bit indicates that the vendor-specific device 1 is present in the package. This bit always returns 1. **Value After Reset:** 0x1 **Exists:** Always
13:0	Reserved_13_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.4.7 SR_AN_PKG1

- **Description:** SR AN MMD Package Identifier Register 1
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0xe : (APB3_16) ? (0xe*0x2) : (0xe*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)



Table 5-310 Fields for Register: SR_AN_PKG1

Bits	Name	Memory Access	Description
15:0	AN_PKG_OUI_3_18	R	Organizationally Unique Identifier DWC_xpcs offers 24 configurable bits[24:1], known as OUI, for identifying the device manufacturer. The value is configured within coreConsultant. This read-only field contains Bits[18:3] of 24-bit OUI of device manufacturer. This is a unique identifier for all the MMDs in the device. It is common for all MMDs (PMA, PCS, XS, AN, and MII) present in the package. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:0] of SR Control MMD Package Identifier Register 1 modifies the content of this register. **Value After Reset:** "(!BASE1000X_ONLY) ? IEEE_PKG_OUI_3_18 : 0x0" **Exists:** Always

5.2.4.8 SR_AN_PKG2

- **Description:** SR AN MMD Package Identifier Register 2
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0xf : (APB3_16) ? (0xf*0x2) : (0xf*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)

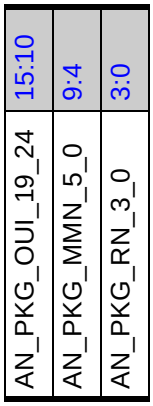


Table 5-311 Fields for Register: SR_AN_PKG2

Bits	Name	Memory Access	Description
15:10	AN_PKG_OUI_19_24	R	Organizationally Unique Identifier DWC_xpcs offers 24 configurable bits[24:1], known as OUI, for identifying the device manufacturer. This value is configured by the host. This read-only field contains Bits[24:19] of 24-bit OUI of device manufacturer. This is a unique identifier for all the MMDs in the device. It is common for all MMDs (PMA, PCS, XS, AN, and MII) present in the package. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:10] of SR Control MMD Package Identifier Register 2 modifies the content of this register. **Value After Reset:** "(!BASE1000X_ONLY) ? IEEE_PKG_OUI_19_24 : 0x0" **Exists:** Always
9:4	AN_PKG_MMN_5_0	R	Model Number This read-only field contains the 5-bit Model Number of the Package in which the various MMDs are installed. This value is configured by the host. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[9:4] of SR Control MMD Package Identifier Register 2 modifies the content of this register. **Value After Reset:** "(!BASE1000X_ONLY) ? IEEE_PKG_MODEL_NO_0_5 : 0x0" **Exists:** Always

Bits	Name	Memory Access	Description
3:0	AN_PKG_RN_3_0	R	Revision Number This read-only field contains the 4-bit Revision Number of the Package. This value is configured by the host. It is common for all the MMDs in the device. It is common for all the MMDs (PMA, PCS, XS, AN, and MII) present in the package. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[3:0] of SR Control MMD Package Identifier Register 2 modifies the content of this register. **Value After Reset:** "(!BASE1000X_ONLY) ? IEEE_PKG_REV_NO_0_3 : 0x0" **Exists:** Always

5.2.4.9 SR_AN_ADV1

- **Description:** SR AN MMD Advertisement Register 1
  The AN Advertisement Register 1, 2, and 3 contain the Base Page Link Code Word for the local device. The host can program these fields. However, the host must program the AN Advertisement Register 1, 2, and 3 within 60 ms after Power On Reset or Soft Reset.
  Note: Before writing to this register, the host must write to the AN Advertisement Register 2 and 3.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x10 : (APB3_16) ? (0x10*0x2) : (0x10*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)

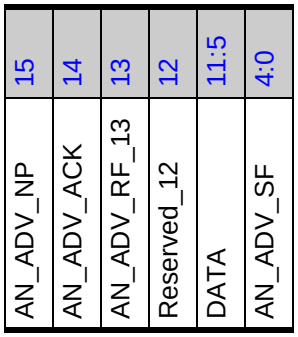


Table 5-312 Fields for Register: SR_AN_ADV1

Bits	Name	Memory Access	Description
15	AN_ADV_NP	R/W	Next Page ■ 0: The management does not intend to exchange the next page after the base page. If the link partner supports Next Page (indicated by Bit 15 of SR AN MMD LP Base Page Ability Register 1 being set to 1), the management performs the next page exchange by programming the SR AN MMD XNP Transmit Register 1, 2, or 3 with a NULL message. ■ 1: The management intends to exchange the next page after the base page. The management performs the next page exchange by programming the AN XNP Transmit Register 1, 2, or 3 with the message it intends to convey to its Link Partner. **Value After Reset:** 0x0 **Exists:** Always
14	AN_ADV_ACK	R	Acknowledgment ■ 0: The device did not receive three consecutive and valid DME pages. ■ 1: The device received three consecutive and valid DME pages. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
13	AN_ADV_RF_13	R/W	Remote Fault The management interface sets this bit to provide the Remote Fault information to its link partner: ■ 0: No fault detected ■ 1: Remote fault detected **Value After Reset:** 0x0 **Exists:** Always
12	Reserved_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:5	DATA	R/W	Link Code Word Encoding DATA[11:5] Link Code Word Encoding as defined in the IEEE Std 802.3, Clause 73, Section 73.6. **Value After Reset:** 0x0 **Exists:** Always
4:0	AN_ADV_SF	R/W	Selector Field This field is set to 5'b0001 to indicate compliance to the IEEE Std 802.3, Annex 28A. **Value After Reset:** 0x1 **Exists:** Always

5.2.4.10 SR_AN_ADV2

- **Description:** SR AN MMD Advertisement Register 2
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x11 : (APB3_16) ? (0x11*0x2) : (0x11*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)

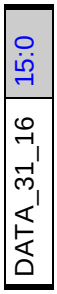


Table 5-313 Fields for Register: SR_AN_ADV2

Bits	Name	Memory Access	Description
15:0	DATA_31_16	R/W	Link Code Word Encoding DATA[31:16] Link Code Word Encoding as defined in the IEEE Std 802.3, Clause 73, Section 73.6. **Value After Reset:** "KX4_ONLY ? 0x40 : KX_ONLY ? 0x20 : KX4_KX ? 0x60 : KR_KX4_KX ? 0xe0 : KR_ONLY ? 0x80 : KR_KX4 ? 0xc0 : 0xa0" **Exists:** Always **Testable:** untestable

5.2.4.11 SR_AN_ADV3

- **Description:** SR AN MMD Advertisement Register 3
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x12 : (APB3_16) ? (0x12*0x2) : (0x12*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)

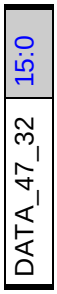


Table 5-314 Fields for Register: SR_AN_ADV3

Bits	Name	Memory Access	Description
15:0	DATA_47_32	R/W	Link Code Word Encoding DATA[47:32] Link Code Word Encoding as defined in the IEEE Std 802.3, Clause 73, Section 73.6. When DATA[46] is set to 1, it indicates that the local device supports FEC. When DATA[47] is set to 1, it indicates that the local device is requesting the link partner to enable FEC. The host must enable the FEC feature by programming the SR_PMA_KR_FEC_CTRL Register only when both devices advertise FEC capability and at the least one device requests to enable FEC. When DATA[32] is set to 1, it indicates that the local device supports 2.5GBASE-KX speed mode. When DATA[33] is set to 1, it indicates that the local device supports 5GBASE-KR speed mode **Value After Reset:** "(ADD_5G&&ADD_2PT5G) ? 0x3 : ADD_5G ? 0x2 : ADD_2PT5G ? 0x1 : 0x0" **Exists:** Always

5.2.4.12 SR_AN_LP_ABL1

- **Description:** SR AN MMD LP Base Page Ability Register 1
  The management uses the SR AN LP Base Page Ability Registers1, 2, and 3 to get the Base Page information received from its link partner. These register contents are valid when bit 6 of the SR AN MMD Status Register is set to 1.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x13 : (APB3_16) ? (0x13*0x2) : (0x13*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)

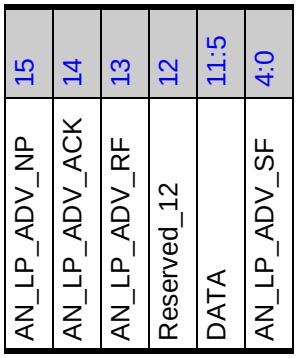


Table 5-315 Fields for Register: SR_AN_LP_ABL1

Bits	Name	Memory Access	Description
15	AN_LP_ADV_NP	R	Next Page ■ 0: The link partner does not want to exchange the Next Page after the Base Page. ■ 1: The link partner wants to exchange the Next Page after the Base Page. **Value After Reset:** 0x0 **Exists:** Always
14	AN_LP_ADV_ACK	R	Acknowledgment ■ 0: The link partner did not receive three consecutive and valid DME pages. ■ 1: The link partner received three consecutive and valid DME pages. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
13	AN_LP_ADV_RF	R	Remote Fault ■ 0: No fault. ■ 1: The link partner wants to convey the Remote Fault information. **Value After Reset:** 0x0 **Exists:** Always
12	Reserved_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:5	DATA	R	Link Code Word Encoding DATA[11:5] Link Code Word Encoding as defined in the IEEE Std 802.3, Clause 73, Section 73.6. **Value After Reset:** 0x0 **Exists:** Always
4:0	AN_LP_ADV_SF	R	Selector Field This field is the selector field of autonegotiation page received from the link partner. A value of 5'h01 indicates compliance to the IEEE Std 802.3, Annex 28A. **Value After Reset:** 0x0 **Exists:** Always

5.2.4.13 SR_AN_LP_ABL2

- **Description:** SR AN MMD LP Base Page Ability Register 2
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x14 : (APB3_16) ? (0x14*0x2) : (0x14*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)

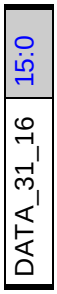


Table 5-316 Fields for Register: SR_AN_LP_ABL2

Bits	Name	Memory Access	Description
15:0	DATA_31_16	R	Link Code Word Encoding DATA[31:16] Link Code Word Encoding as defined in the IEEE Std 802.3, Clause 73, Section 73.6. **Value After Reset:** 0x0 **Exists:** Always

5.2.4.14 SR_AN_LP_ABL3

- **Description:** SR AN MMD LP Base Page Ability Register 3
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x15 : (APB3_16) ? (0x15*0x2) : (0x15*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)

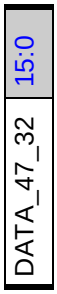


Table 5-317 Fields for Register: SR_AN_LP_ABL3

Bits	Name	Memory Access	Description
15:0	DATA_47_32	R	Link Code Word Encoding DATA[47:32] Link Code Word Encoding as defined in the IEEE Std 802.3, Clause 73, Section 73.6. When DATA[46] is set to 1, it indicates that the link partner supports FEC. When DATA[47] is set to 1, it indicates that the link partner is requesting the local device to enable FEC. The host must enable the FEC feature by programming the SR_PMA_KR_FEC_CTRL register only when both devices advertise FEC capability and at the least one device requests to enable FEC. **Value After Reset:** 0x0 **Exists:** Always

5.2.4.15 SR_AN_XNP_TX1

- **Description:** SR AN MMD XNP Transmit Register 1
  This register holds the next page Link Code Word to be transmitted when extended next page is enabled by the host by using Bit 15 of SR AN MMD Advertisement Register 1. On reset, this register contains the Message page with Message code set to NULL PAGE. The host can update this register with any valid Message page that it intends to transmit.
  Note: Before writing to this register, the host must write to the SR AN XNP Transmit Register 2 and 3.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x16 : (APB3_16) ? (0x16*0x2) : (0x16*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)

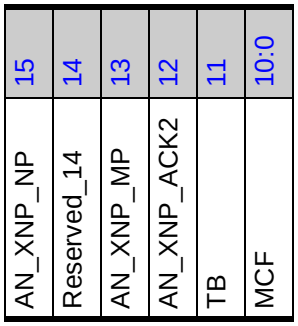


Table 5-318 Fields for Register: SR_AN_XNP_TX1

Bits	Name	Memory Access	Description
15	AN_XNP_NP	R/W	Next Page ■ 0: The management does not intend to exchange the next page after the current page. If the link partner supports the next page (as indicated by Bit 15 of SR AN MMD LP Base Page Ability Register 1 being set to 1), the management performs the next page exchange by programming the SR AN MMD XNP Transmit Register 1, 2, and 3 with a NULL message. ■ 1: The management intends to exchange the next page after the current page. In such a case, the management needs to program the SR AN XNP Transmit Register 1, 2, and 3 with the message to send to its link partner. **Value After Reset:** 0x0 **Exists:** Always
14	Reserved_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
13	AN_XNP_MP	R/W	Message Page This bit indicates that the management contains a formatted message. ■ 0: Unformatted Message Page ■ 1: Formatted Message Page **Value After Reset:** 0x0 **Exists:** Always
12	AN_XNP_ACK2	R/W	Acknowledgment2 ■ 0: Management did not act on the information received from the link partner. ■ 1: Management acted on the information received from the link partner. **Value After Reset:** 0x0 **Exists:** Always
11	TB	R	Toggle Bit The link partner uses this bit to differentiate between two successive pages. The DWC_xpcs toggles this bit after sending any new next page. **Value After Reset:** 0x0 **Exists:** Always
10:0	MCF	R/W	Message Code Field or Unformatted Message This bit depends on Bit 13: ■ AN_XNP_MP = 1: Indicates the Message Code as in IEEE Std 802.3, Annex 28C ■ AN_XNP_MP = 0: Indicates an Unformatted Message which is vendor-specific **Value After Reset:** 0x0 **Exists:** Always

5.2.4.16 SR_AN_XNP_TX2

- **Description:** SR AN MMD XNP Transmit Register 2
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x17 : (APB3_16) ? (0x17*0x2) : (0x17*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)



Table 5-319 Fields for Register: SR_AN_XNP_TX2

Bits	Name	Memory Access	Description
15:0	UMCF1	R/W	Unformatted Message Code Field1 **Value After Reset:** 0x0 **Exists:** Always

5.2.4.17 SR_AN_XNP_TX3

- **Description:** SR AN MMD XNP Transmit Register 3
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x18 : (APB3_16) ? (0x18*0x2) : (0x18*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)

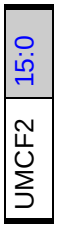


Table 5-320 Fields for Register: SR_AN_XNP_TX3

Bits	Name	Memory Access	Description
15:0	UMCF2	R/W	Unformatted Message Code Field2 **Value After Reset:** 0x0 **Exists:** Always

5.2.4.18 SR_AN_LP_XNP_ABL1

- **Description:** SR AN MMD LP XNP Ability Register 1
  The management uses the AN LP XNP Ability Register 1, 2, or 3 to get the next page information from the link partner. These register contents are valid when Bit 6 of SR AN MMD Status Register is set to 1 during next page exchange.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x19 : (APB3_16) ? (0x19*0x2) : (0x19*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)

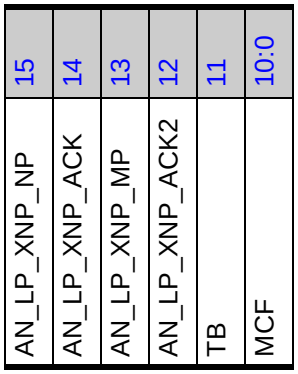


Table 5-321 Fields for Register: SR_AN_LP_XNP_ABL1

Bits	Name	Memory Access	Description
15	AN_LP_XNP_NP	R	Next Page When set, this bit indicates that the link partner wants to exchange the Next Page after the current page. **Value After Reset:** 0x0 **Exists:** Always
14	AN_LP_XNP_ACK	R	Acknowledgment This bit indicates that the link partner is successfully able to receive the Local Device Link Code Word. ■ 0: The link partner not able to receive the Local Device Link Code Word. ■ 1: The link partner is able to receive the Local Device Link Code Word. **Value After Reset:** 0x0 **Exists:** Always
13	AN_LP_XNP_MP	R	Message Page ■ 1: SR AN MMD LP XNP Ability Register 1, 2, and 3 together contain message pages. ■ 0: SR AN MMD LP XNP Ability Register 1, 2, and 3 together contain unformatted message pages. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
12	AN_LP_XNP_ACK2	R	Acknowledgment2 ■ 0: The link partner has not received the page, or it is unable to act on the information received from the local device. ■ 1: The link partner has received the page and it is able to act on the information received from the local device. **Value After Reset:** 0x0 **Exists:** Always
11	TB	R	Toggle Bit This bit is set to make the Management differentiate successive messages: ■ 0: Same Message ■ 1: Next Message **Value After Reset:** 0x0 **Exists:** Always
10:0	MCF	R	Message Code Field or Unformatted Message This field depends on Bit 13 of this register: ■ AN_LP_XNP_MP = 1: This field contains a Message Code described in IEEE Std 802.3, Annexure 28C. ■ AN_LP_XNP_MP = 0: This field contains an unformatted message which is vendor-specific. **Value After Reset:** 0x0 **Exists:** Always

5.2.4.19 SR_AN_LP_XNP_ABL2

- **Description:** SR AN MMD LP XNP Ability Register 2
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1A : (APB3_16) ? (0x1A*0x2) : (0x1A*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)

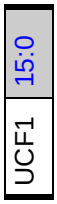


Table 5-322 Fields for Register: SR_AN_LP_XNP_ABL2

Bits	Name	Memory Access	Description
15:0	UCF1	R	Unformatted Message Code Field1 **Value After Reset:** 0x0 **Exists:** Always

5.2.4.20 SR_AN_LP_XNP_ABL3

- **Description:** SR AN MMD LP XNP Ability Register 3
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1B : (APB3_16) ? (0x1B*0x2) : (0x1B*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)

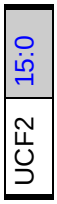
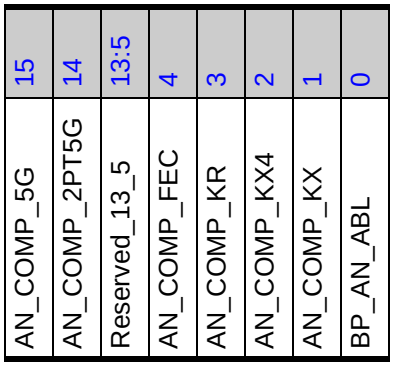


Table 5-323 Fields for Register: SR_AN_LP_XNP_ABL3

Bits	Name	Memory Access	Description
15:0	UCF2	R	Unformatted Message Code Field2 **Value After Reset:** 0x0 **Exists:** Always

5.2.4.21 SR_AN_COMP_STS

- **Description:** SR AN MMD Completion Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x30 : (APB3_16) ? (0x30*0x2) : (0x30*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)



15	14	13:5	4	3	2	1	0
AN_COMP_5G	AN_COMP_2PT5G	Reserved_13_5	AN_COMP_FEC	AN_COMP_KR	AN_COMP_KX4	AN_COMP_KX	BP_AN_ABL

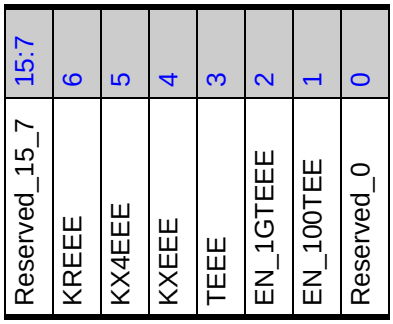
Table 5-324 Fields for Register: SR_AN_COMP_STS

Bits	Name	Memory Access	Description
15	AN_COMP_5G	R	**For Configurations with 5GBASE-KR Support:** 5GBASE-KR Mode Auto-negotiation Completion This bit is valid when the DWC_xpcs is configured for the 5GBASE-KR mode. When this bit is set, it indicates that the auto-negotiation is complete in the 5GBASE-KR mode. **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always
14	AN_COMP_2PT5G	R	**For Configurations with 2.5GBASE-KX Support:** 2.5GBASE-KX Mode Auto-negotiation Completion This bit is valid when the DWC_xpcs is configured for the 2.5GBASE-KX mode. When this bit is set, it indicates that the auto-negotiation is complete in the 2.5GBASE-KX mode. **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always
13:5	Reserved_13_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4	AN_COMP_FEC	R	BASE-R FEC Auto-negotiation Completion This bit is valid when DWC_xpcs is configured to operate in BASE-R FEC mode.When this bit is set, it indicates that the auto-negotiation is complete in the BASE-R FEC mode. **Value After Reset:** 0x0 **Exists:** Always
3	AN_COMP_KR	R	**For KR_Present Configurations:** 10GBASE-KR Mode Auto-negotiation Completion This bit is valid when the DWC_xpcs is configured for the 10GBASE-KR mode. When this bit is set, it indicates that the auto-negotiation is complete in the 10GBASE-KR mode. **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always
2	AN_COMP_KX4	R	**For KX4_Present Configurations:**KX4 Mode Auto-negotiation Completion This bit is valid when the DWC_xpcs is configured for the KX4 mode. When this bit is set, it indicates that the auto-negotiation is complete in the KX4 mode. **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always
1	AN_COMP_KX	R	**For KX_Present Configurations:**KX Mode Auto-negotiation Completion This bit is valid when the DWC_xpcs is configured for the KX mode. When this bit is set, it indicates that the auto-negotiation is complete in the KX mode. **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always
0	BP_AN_ABL	R	Backplane Auto-Negotiation Ability This bit indicates the backplane auto-negotiation ability. This bit always returns 1. **Value After Reset:** 0x1 **Exists:** Always

5.2.4.22 SR_AN_EEE_ABL1

- **Description:** SR AN MMD EEE Capability 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x3C : (APB3_16) ? (0x3C*0x2) : (0x3C*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS&&EEE_EN)



15:7	6	5	4	3	2	1	0
Reserved_15_7	KREEE	KX4EEE	KXEEE	TEEE	EN_1GTEEE	EN_100TEE	Reserved_0

Table 5-325 Fields for Register: SR_AN_EEE_ABL1

Bits	Name	Memory Access	Description
15:7	Reserved_15_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6	KREEE	R/W	10GBASE-KR Mode EEE Support This bit indicates that the local device wants to advertise to the link partner that it supports EEE in the 10GBASE-KR mode. ■ 0: EEE is not supported in the 10GBASE-KR mode ■ 1: EEE is supported in the 10GBASE-KR mode **Value After Reset:** "(LP_10GBR) ? 0x1 : 0x0" **Exists:** Always
5	KX4EEE	R/W	10GBASE-KX4 Mode EEE Support This bit indicates that the local device wants to advertise to the link partner that it supports EEE in the 10GBASE-KX4 mode. ■ 0: EEE is not supported in the 10GBASE-KX4 mode ■ 1: EEE is supported in the 10GBASE-KX4 mode **Value After Reset:** "(LP_10GBX) ? 0x1 : 0x0" **Exists:** Always

Bits	Name	Memory Access	Description
4	KXEEE	R/W	1000BASE-KX Mode EEE Support This bit indicates that the local device wants to advertise to the link partner that it supports EEE in the 1000BASE-KX mode ■ 0: EEE is not supported in the 1000BASE-KX mode ■ 1: EEE is supported in the 1000BASE-KX mode **Value After Reset:** "(LP_1GBX) ? 0x1 : 0x0" **Exists:** Always
3	TEEE	R	10GBASE-T Mode EEE Support This bit indicates the EEE support in the 10GBASE-T mode: ■ 0: EEE is not supported in the 10GBASE-T mode ■ 1: EEE is supported in the 10GBASE-T mode This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
2	EN_1GTEEE	R	1000BASE-T Mode EEE Support This bit indicates the EEE support in the 1000BASE-T mode: ■ 0: EEE is not supported in the 1000BASE-T mode ■ 1: EEE is supported in the 1000BASE-T mode This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
1	EN_100TEE	R	100BASE-T Mode EEE Support This bit indicates the EEE support in the 100BASE-T mode: ■ 0: EEE is not supported in the 100BASE-T mode ■ 1: EEE is supported in the 100BASE-T mode This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always
0	Reserved_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.4.23 SR_AN_EEE_LP_ABL1

- **Description:** SR AN MMD EEE Link Partner Capability Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x3D : (APB3_16) ? (0x3D*0x2) : (0x3D*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS&&EEE_EN)

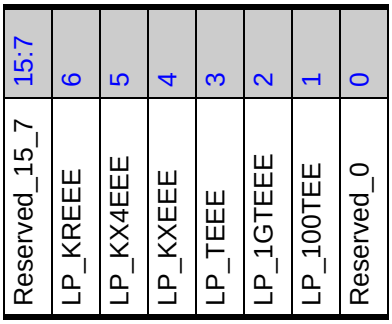


Table 5-326 Fields for Register: SR_AN_EEE_LP_ABL1

Bits	Name	Memory Access	Description
15:7	Reserved_15_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6	LP_KREEE	R	10GBASE-KR Mode EEE Support in Link Partner This bit indicates that the link partner advertised to the local device that it supports EEE in the 10GBASE-KR mode ■ 1: The link partner supports EEE in the 10GBASE-KR mode. ■ 0: The link partner does not support EEE in the 10GBASE-KR mode. **Value After Reset:** 0x0 **Exists:** Always
5	LP_KX4EEE	R	KX4 Mode EEE Support in Link Partner This bit indicates that the link partner advertised to the local device that it supports EEE in the 10GBASE-KX4 mode. ■ 1: The link partner supports EEE in the 10GBASE-KX4 mode. ■ 0: The link partner does not support EEE in the 10GBASE-KX4 mode. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4	LP_KXEEE	R	KX Mode EEE Support in Link Partner This bit indicates that the link partner advertised to the local device that it supports EEE in the 1000BASE-KX mode. ■ 1: The link partner supports EEE in the 1000BASE-KX mode. ■ 0: The link partner does not support EEE in the 1000BASE-KX mode. **Value After Reset:** 0x0 **Exists:** Always
3	LP_TEEE	R	10GBASE-T Mode EEE Support This bit indicates if the link partner supports EEE in the 10GBASE-T mode: ■ 0: EEE is not supported in the 10GBASE-T mode ■ 1: EEE is supported in the 10GBASE-T mode **Value After Reset:** 0x0 **Exists:** Always
2	LP_1GTEEE	R	1000BASE-T Mode EEE Support This bit indicates if the link partner supports EEE in the 1000BASE-T mode: ■ 0: EEE is not supported in the 1000BASE-T mode ■ 1: EEE is supported in the 1000BASE-T mode **Value After Reset:** 0x0 **Exists:** Always
1	LP_100TEE	R	100BASE-T Mode EEE Support This bit indicates if the link partner supports EEE in the 100BASE-T mode: ■ 0: EEE is not supported in the 100BASE-T mode ■ 1: EEE is supported in the 100BASE-T mode **Value After Reset:** 0x0 **Exists:** Always
0	Reserved_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.4.24 SR_AN_EEE_ABL2

- **Description:** SR AN MMD EEE Capability 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x3E : (APB3_16) ? (0x3E*0x2) : (0x3E*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS&&EEE_EN)

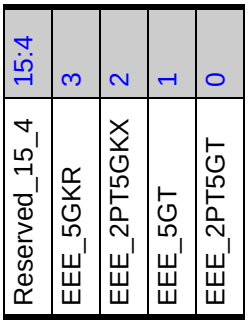


Table 5-327 Fields for Register: SR_AN_EEE_ABL2

Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3	EEE_5GKR	R/W	5GBASE-KR Mode EEE Support This bit indicates that the local device wants to advertise to the link partner that it supports EEE in the 5GBASE-KR mode ■ 0: EEE is not supported in the 5GBASE-KR mode ■ 1: EEE is supported in the 5GBASE-KR mode **Value After Reset:** "(ADD_5G&&EEE_EN) ? 0x1 : 0x0" **Exists:** Always
2	EEE_2PT5GKX	R/W	2.5GBASE-KX Mode EEE Support This bit indicates that the local device wants to advertise to the link partner that it supports EEE in the 2.5GBASE-KX mode ■ 0: EEE is not supported in the 2.5GBASE-KX mode ■ 1: EEE is supported in the 2.5GBASE-KX mode **Value After Reset:** "(ADD_2PT5G&&EEE_EN) ? 0x1 : 0x0" **Exists:** Always
1	EEE_5GT	R	5GBASE-T Mode EEE Support This bit indicates if EEE is supported in the 5GBASE-T mode. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	EEE_2PT5GT	R	2.5GBASE-T Mode EEE Support This bit indicates if EEE is supported in the 2.5GBASE-T mode. This bit always returns 0. **Value After Reset:** 0x0 **Exists:** Always

5.2.4.25 SR_AN_EEE_LP_ABL2

- **Description:** SR AN MMD EEE Link Partner Capability 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x3F : (APB3_16) ? (0x3F*0x2) : (0x3F*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS&&EEE_EN)

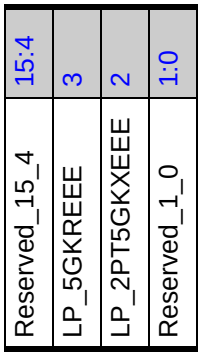


Table 5-328 Fields for Register: SR_AN_EEE_LP_ABL2

Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3	LP_5GKREEE	R	5GBASE-KR Mode EEE Support in Link Partner This bit indicates that the link partner advertised to the local device that it supports EEE in the 5GBASE-KR mode ■ 0: The link partner does not support EEE in the 5GBASE-KR mode ■ 1: The link partner supports EEE in the 5GBASE-KR mode. **Value After Reset:** 0x0 **Exists:** Always
2	LP_2PT5GKXEEE	R	2.5GBASE-KX Mode EEE Support in Link Partner This bit indicates that the link partner advertised to the local device that it supports EEE in the 2.5GBASE-KX mode ■ 0: The link partner does not support EEE in the 2.5GBASE-KX mode ■ 1: The link partner supports EEE in the 2.5GBASE-KX mode. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
1:0	Reserved_1_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.4.26 VR_AN_DIG_CTRL1

- **Description:** VR AN MMD Digital Control1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8000 : (APB3_16) ? (0x8000*0x2) : (0x8000*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)

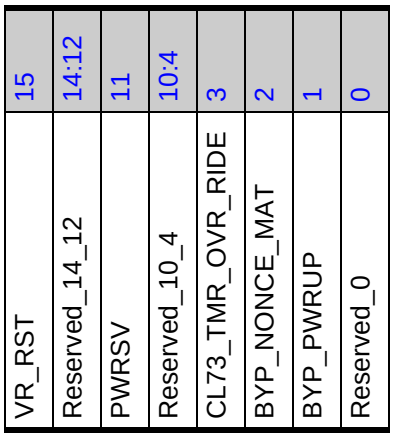


Table 5-329 Fields for Register: VR_AN_DIG_CTRL1

Bits	Name	Memory Access	Description
15	VR_RST	R/W	Vendor-Specific Soft Reset (RW,SC Type) When the host sets this bit, the CSR block triggers the vendor-specific software reset process in which all internal blocks are reset, except the Management Interface block and the CSR block. The registers are reset to their default values. When this bit is set, it also resets the PHY. This bit is self cleared under the following conditions: ■ For Synopsys PHY This bit is self cleared when Bits[4:2] in VR_XS_PCS_DIG_STS Register is equal to 3'b100, that is, Tx/Rx clocks are stable and in Power_Good state. ■ For Non-Synopsys PHY This bit is self cleared after the following: -- 32 clk_csr_i clocks for the MCI interface, -- 1 MDC clock period for the MDIO interface**Note:** For description of read or write access for any register while this bit is set, see "Special Case Register Access" section of the Databook. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
14:12	Reserved_14_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11	PWRSV	* Varies	**For Synopsys PHY configurations:** Power Save This bit is present only when the DWC_xpcs is configured for Synopsys PHY. ■ 0: Normal operation. ■ 1: DWC_xpcs and the PHY enter the power-save mode. When this bit is set, the DWC_xpcs triggers the power-down mode by turning off the PHY Receiver and Transmitter, and without turning off MPLL. For Synopsys multi-protocol 12G/16G PHY, DWC_xpcs asserts xpcs_tx_lpd_o{lane} and xpcs_rx_lpd_o{lane} when this bit is set. When the host clears this bit, DWC_xpcs resumes normal operation. After clearing this bit, the host must wait until Bits[4:2] of the VR_XS_PCS_DIG_STS Register indicate that the DWC_xpcs is in a normal state. **Note:** For description of read or write access for any register while this bit is set, see "Special Case Register Access" section of the Databook. **For non-Synopsys PHY configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_XAUI_PHY) ? \"read-write\" : \"read-only\""
10:4	Reserved_10_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3	CL73_TMR_OVR_RIDE	R/W	Over-Ride Control for Clause 73 Auto-negotiation Timers. This bit can be set to override the default values of Clause 73 timers used by DWC_xpcs for auto-negotiation. If this bit is set, the values programmed in VR_AN_TIMER_CTRL0 Register and VR_AN_TIMER_CTRL1 Register are used to compute the duration of Auto-neg timers. Set this bit only after programming the desired values to the timer control registers **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
2	BYP_NONCE_MAT	R/W	Bypass Nonce-Match Check during Clause 73 auto-negotiation. This bit can be set to 1 if application wishes to perform Clause 73 Auto-negotiation in Loopback mode. As per IEEE specification, auto-negotiation does not work in loopback, as it requires the nonce field of both the devices to be different. When this bit is set, the nonce check is bypassed and the auto-negotiation arbiter FSM can transition from ABILITY_DETECT to ACKNOWLEDGE_DETECT state whenever ability_match is attained (irrespective of nonce_match). This is a debug-mode bit. **Value After Reset:** 0x0 **Exists:** Always
1	BYP_PWRUP	* Varies	**For Synopsys PHY configurations:** Bypass Power-Up Sequence This bit is present only when the DWC_xpcs is configured for the Synopsys PHY. In all other configurations, this bit is reserved. When this bit is set to 1, the DWC_xpcs bypasses the normal flow of the power-up sequence and reaches the Power_Good state to enable transmission or reception. When this bit is set, the DWC_xpcs does not wait for the MPLL and Transmit or Receive PLL status from the Synopsys PHY. You can use this feature in the following scenarios: ■ When the DWC_xpcs is configured to interface with a specific Synopsys PHY. ■ When the data path needs to be interfaced to some other PHY. When this bit is set to 0, the DWC_xpcs waits for the MPLL, Tx, or Rx PLL status from the Synopsys PHY before resuming the normal transmission and reception. **Note:** This bit is shared with Bit 1 of following registers: ■ VR_XS_PCS_DIG_CTRL1 Register ■ VR_AN_DIG_CTRL1 Register ■ VR_MII_DIG_CTRL1 Register **Access Type:** If SNPS_XAUI_PHY= Enabled, this bit is RW. Otherwise, this bit is RO. **For non-Synopsys PHY configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_XAUI_PHY) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
0	Reserved_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.4.27 VR_AN_INTR_MSK

- **Description:** VR AN MMD Interrupt Mask Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8001 : (APB3_16) ? (0x8001*0x2) : (0x8001*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)

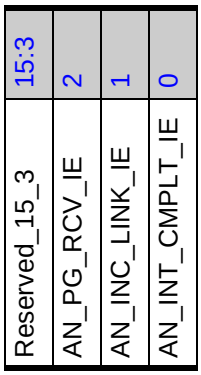


Table 5-330 Fields for Register: VR_AN_INTR_MSK

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2	AN_PG_RCV_IE	R/W	AN Page Received Interrupt Enable ■ 0: The AN_PG_RCV interrupt is blocked. ■ 1: The AN_PG_RCV interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always
1	AN_INC_LINK_IE	R/W	AN Incompatible Link Interrupt Enable ■ 0: The AN_INC_LINK interrupt is blocked. ■ 1: The AN_INC_LINK interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always
0	AN_INT_CMPLT_IE	R/W	AN COMPLETE Interrupt Enable ■ 0: AN_INT_CMPLT Interrupt is blocked. ■ 1: AN_INT_CMPLT Interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always

5.2.4.28 VR_AN_INTR

- **Description:** VR AN MMD Interrupt Register
  Note: The interrupt output port 'sbd_intr_o' goes high when any bit of this register is set along with the corresponding bit of VR_AN_INTR_MSK Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8002 : (APB3_16) ? (0x8002*0x2) : (0x8002*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)

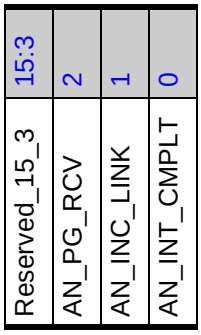


Table 5-331 Fields for Register: VR_AN_INTR

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2	AN_PG_RCV	R/W	AN Page Received (SS, WC Type) The DWC_xpcs sets this bit when it receives a new auto-negotiation page (as defined in IEEE Std 802.3, Clause 73). **Note:** The host must clear this bit, by writing 0 to this bit, after recognizing the interrupt. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

Bits	Name	Memory Access	Description
1	AN_INC_LINK	R/W	AN Incompatible Link (SS,WC Type) This bit indicates that the DWC_xpcs detected an incompatible link during auto-negotiation. ■ 0: No incompatibility detected. ■ 1: Incompatible link. An incompatible link occurs when all of the following conditions are true: ■ The DWC_xpcs starts auto-negotiation. ■ The DWC_xpcs detects that a DME page is not received by the link partner. ■ The PCS link status for the DWC_xpcs remains Not_OK. ■ A time-out for a duration equal to link_fail_inhibit_timer occurs when the DWC_xpcs auto-negotiation arbiter state is in Ability_Detect state. **Note:** The host must clear this bit, by writing 0 to this bit, after recognizing the interrupt. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
0	AN_INT_CMPLT	R/W	AN Complete Interrupt (SS,WC Type) The DWC_xpcs sets this bit after the auto-negotiation is complete (as defined in IEEE Std 802.3, Clause 73). **Note:** The host must clear this bit, by writing 0 to this bit, after recognizing the interrupt. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

5.2.4.29 VR_AN_KR_MODE_CTRL

- **Description:** VR AN MMD KR Mode Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8003 : (APB3_16) ? (0x8003*0x2) : (0x8003*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)



Table 5-332 Fields for Register: VR_AN_KR_MODE_CTRL

Bits	Name	Memory Access	Description
15:1	Reserved_15_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	PDET_EN	* Varies	Parallel Detection Enable When set, this bit enables parallel detection in the KR speed mode during Clause 73 auto-negotiation. Otherwise, parallel detection is not done for the KR speed mode. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(KR_PRESENT) ? \"read-write\" : \"read-only\""

5.2.4.30 VR_AN_TIMER_CTRL0

- **Description:** VR AN MMD Timer Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8004 : (APB3_16) ? (0x8004*0x2) : (0x8004*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)



Table 5-333 Fields for Register: VR_AN_TIMER_CTRL0

Bits	Name	Memory Access	Description
15:0	BRK_LINK_TIME	R/W	Break Link Timer Value for Clause 73 auto-negotiation. This field can be programmed to any desired value if application wishes to change the duration of Break Link Timer used during CL73 Auto negotiation. DWC_xpcs has a timer (27-bit in KR_Present configurations; 25-bit otherwise) that runs at the frequency of clk_xpcs_tx_i clock. The value programmed to this field are used as the upper 16-bit of the value to be loaded to this timer. The lower bits are hardcoded (11'h2EE in KR_Present configurations; 9'hc8 otherwise). Application must program appropriate values to this field based on the frequency of clk_xpcs_tx_i clock as per the current operating mode. After programming this field to the desired value, application must perform either of the following steps for the newly programmed values to take effect: ■ Set CL73_TMR_OVR_RIDE bit of VR_AN_DIG_CTRL1 Register to 1 ■ Set FAST_SIM bit of SR Control MMD Control Register to 1 **Value After Reset:** 0x0 **Exists:** Always

5.2.4.31 VR_AN_TIMER_CTRL1

- **Description:** VR AN MMD Timer Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8005 : (APB3_16) ? (0x8005*0x2) : (0x8005*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)



Table 5-334 Fields for Register: VR_AN_TIMER_CTRL1

Bits	Name	Memory Access	Description
15:0	INHBT_OR_WAIT_TIME	R/W	Link Fail Inhibit/Autoneg Wait Timer Value for Clause 73 auto-negotiation. This field can be programmed to any desired value if application wishes to change the duration of Link Fail Inhibit/Autoneg Wait Timer used during CL73 Autonegotiation. DWC_xpcs uses same timer values for Link Fail Inhibit Timer and Autoneg Wait Timer (as 40-50ms is a common range for both timers as per IEEE spec) DWC_xpcs has a timer (27-bit in KR_Present configurations; 25-bit otherwise) that runs at the frequency of clk_xpcs_tx_i clock. The value programmed to this field is used as the upper 16-bit of the value to be loaded to this timer. The lower bits are hardcoded (11'h2EE in KR_Present configurations; 9'hc8 otherwise). Application must program appropriate values to this field based on the frequency of clk_xpcs_tx_i clock as per the current operating mode. After programming this field to the desired value, application must perform either of the following steps for the newly programmed values to take effect: ■ Set CL73_TMR_OVR_RIDE bit of VR_AN_DIG_CTRL1 Register to 1 ■ Set FAST_SIM bit of SR Control MMD Control Register to 1 **Value After Reset:** 0x0 **Exists:** Always

5.2.4.32 VR_AN_DIG_STS

- **Description:** VR AN MMD Digital Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8010 : (APB3_16) ? (0x8010*0x2) : (0x8010*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070000 : (APB3_16) ? (0x070000*0x2) : (0x070000*0x4)"
- **Exists:** (BACKPLANE_ETHERNET_PCS)

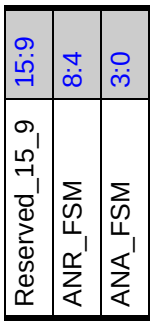


Table 5-335 Fields for Register: VR_AN_DIG_STS

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
8:4	ANR_FSM	R	Auto-negotiation Receive Logic FSM state This FSM is a vendor-specific FSM implementation and encoding is as follows: **5GBASE-KR Mode:** ■ 5'd1: IDLE ■ 5'd2: DELIMETER_DETECT ■ 5'd4: DME_DETECT **Other modes:** ■ 5'd0: RX_IDLE ■ 5'd1: RX_DLD ■ 5'd2: WFC0 ■ 5'd3: WFC1 ■ 5'd4: WFC2 ■ 5'd5: WFC3 ■ 5'd6: WFC4 ■ 5'd7: WFC5 ■ 5'd8: WFC6 ■ 5'd9: WFC7 ■ 5'd10: WFC8 ■ 5'd11: WFPT0(KX_ONLY configuration)/WFC9(Other configurations) ■ 5'd12: WFD0 ■ 5'd13: WFD1 ■ 5'd14: WFD2 ■ 5'd15: WFD3 ■ 5'd16: WFD4 ■ 5'd17: WFD5 ■ 5'd18: WFD6 ■ 5'd19: WFD7 ■ 5'd20: WFD8 ■ 5'd21: WFD9 ■ 5'd22: WFD18 ■ 5'd23: WFD19 ■ 5'd24: WFPT0(Configurations with KX4 but no KR support)/WFC21D17 (Other Configurations) ■ 5'd25: WFC21D18 ■ 5'd26: WFC21D19 ■ 5'd27: WFPT0 (Configurations with KR support) **Note 1:**This field is not reliable when read around the time at which FSM transitions from one state to another. **Note 2:**Though reset value of field is specified as 0, you shall see a non-zero value on read operation as FSM would moved out of default state immediately after coming out of reset. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
3:0	ANA_FSM	R	Auto-negotiation Arbiter FSM state This field indicates the current status of AN arbiter FSM. Encoding is as follows: ■ 4'b0000: AN_ENABLE ■ 4'b0001: TRANSMIT_DISABLE ■ 4'b0010: ABILITY_DETECT ■ 4'b0011: ACKNOWLEDGE_DETECT ■ 4'b0100: COMPLETE_ACKNOWLEDGE ■ 4'b0101: NEXT_PAGE_WAIT ■ 4'b0110: LINK_STATUS_CHECK ■ 4'b0111: PARALLEL_DETECT_FAULT ■ 4'b1000: AN_GOOD_CHECK ■ 4'b1001: AN_GOOD **Note 1:**This field is not reliable when read around the time at which FSM transitions from one state to another. **Note 2:**Though reset value of field is specified as 0, you shall see a non-zero value on read operation as FSM would moved out of default state immediately after coming out of reset. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

5.2.5 DWC_xpcs_map/VS_MII_1_MMD Registers

5.2.5.1 SR_MII_1_CTRL

- **Description:** SR MII_1 MMD Control Register
  The host can use this register to control (enable or disable) some of the features supported by the DWC_xpcs (for Port 1).
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** QSGMII_EN | | MP_USXGMII

15	14	13	12	11	10	9	8	7	6	5	4:0
RST	LBE	SS13	AN_ENABLE	LPM	Reserved_10	RESTART_AN	DUPLEX_MODE	Reserved_7	SS6	SS5	Reserved_4_0

Table 5-336 Fields for Register: SR_MII_1_CTRL

Bits	Name	Memory Access	Description
15	RST	R/W	Soft Reset (RW,SC Type) **For Synopsys PHY configurations:** When the host sets this bit, the CSR block triggers the software reset process in which all internal blocks are reset, except the Management Interface block. The registers are reset to their default values. When this bit is set, it also resets the PHY. This bit is self-cleared when PSEQ_STATE in VR_XS_PCS_DIG_STS/VR_MII_DIG_STS Register is equal to 3'b100 , that is, Tx/Rx clocks are stable and in Power_Good state. **For Non-Synopsys PHY configurations:** When the host sets this bit, the CSR block triggers the software reset process in which all internal blocks are reset, except the Management Interface block. The registers are reset to their default values. When this bit is set, it also resets the PHY. This bit is self-cleared after the following: ■ 32 clk_csr_i clocks for the MCI interface ■ 1 MDC clock period for the MDIO interface **Note:** For description of read or write access for registers during reset process, see "Special Case Register Access" section of the Databook. **Value After Reset:** 0x0 **Exists:** Always
14	LBE	R	Loopback Enable This bit reflects the value of LBE bit of SR_MII_CTRL Register. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
13	SS13	R/W	Speed Selection (LSB) This bit, along with the SS6 bit and SS5 bit of this register, indicates the speed of GMII/XGMII port 1. Speed encoding as follows: **For USXGMII:** ■ When SS5=1, SS6=0 and SS13=1, speed is 5 Gbps ■ When SS5=1, SS6=0 and SS13=0, speed is 2.5 Gbps ■ When SS5=0, SS6=1 and SS13=1, speed is 10 Gbps ■ When SS5=0, SS6=1 and SS13=0, speed is 1000 Mbps ■ When SS5=0, SS6=0 and SS13=1, speed is 100 Mbps ■ When SS5=0, SS6=0 and SS13=0, speed is 10 Mbps **For QSGMII:** ■ When SS6=1 and SS13=0, speed is 1000 Mbps ■ When SS6=0 and SS13=1, speed is 100 Mbps ■ When SS6=0 and SS13=0, speed is 10 Mbps **Value After Reset:** 0x0 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
12	AN_ENABLE	R/W	Enable Auto-Negotiation When set to 1, this bit enables the Clause 37 auto-negotiation process for port 1. **Value After Reset:** 0x0 **Exists:** Always
11	LPM	R/W	Power-Down Mode This bit controls the power-down mode of the DWC_xpcs. ■ 0: Normal operation ■ 1: The DWC_xpcs goes to the power-down mode along with the PHY. For non-Synopsys PHY, the xpcs_pdown_o port is asserted. For Synopsys PHY, the DWC_xpcs does the following to trigger the power-down mode: ■ Turns off the PHY Receiver and Transmitter ■ Switches off all the clocks When the host clears this bit, the DWC_xpcs resumes the normal operation. In a Synopsys PHY configuration, after clearing this bit, the host must wait until Bits[4:2] of VR_XS_PCS_DIG_STS/VR_MII_DIG_STS Register indicate that the DWC_xpcs is in the normal state. **Note:** For information about read or write access to registers during power-down mode, see "Special Case Register Access" section of the Databook. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
10	Reserved_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9	RESTART_AN	R/W	Restart Auto-Negotiation (RW,SC Type) When the host writes this bit, the DWC_xpcs initiates the auto-negotiation process for port 1. This bit is used to restart the auto-negotiation which is already initiated by setting Bit 12. The DWC_xpcs clears this bit after restarting the auto-negotiation. **Value After Reset:** 0x0 **Exists:** Always
8	DUPLEX_MODE	R/W	Duplex Mode This bit specifies the duplex mode of the DWC_xpcs along port 1. ■ 0: Half duplex ■ 1: Full duplex If Bit 12 is set to 0, this bit determines the PHY link duplex mode. If Bit 12 is set to 1, then the PHY link duplex mode is independent of this bit (although the host can write any value) and is determined by the outcome of the Clause 37 auto-negotiation process. **Note:** USXGMII supports only full duplex mode (irrespective of the value programmed to this field). **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6	SS6	R/W	Speed Selection This bit, along with the SS5 and SS13 bits of this register indicates the speed of operation of GMII/XGMII Port 1. For more information, see description of the SS13 bit. **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""

Bits	Name	Memory Access	Description
5	SS5	* Varies	Speed Selection This bit, along with SS6 and SS13 bits, control the speed of operation of XGMII/GMII Port 1. For more information, see description of the SS13 bit. This field is valid only for USXGMII mode. **Value After Reset:** 0x0 **Exists:** Always **Testable:** "(MP_USXGMII) ? \"untestable\" : \"unconstrained\"" **Memory Access:** "MP_USXGMII ? \"read-write\" : \"read-only\""
4:0	Reserved_4_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.5.2 SR_MII_1_STS

- **Description:** SR MII_1 MMD Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1 : (APB3_16) ? (0x1*0x2) : (0x1*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** QSGMII_EN

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ABL100T4	FD100ABL	HD100ABL	FD10ABL	HD10ABL	FD100T	HD100T	EXT_STS_ABL	UN_DIR_ABL	MF_PRE_SUP	AN_CMPL	RF	AN_ABL	LINK_STS	Reserved_1	EXT_REG_CAP

Table 5-337 Fields for Register: SR_MII_1_STS

Bits	Name	Memory Access	Description
15	ABL100T4	R	100BASE-T4 Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always
14	FD100ABL	R	100BASE-X Full-Duplex Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always
13	HD100ABL	R	100BASE-X Half-Duplex Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always
12	FD10ABL	R	10 Mbps Full-Duplex Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
11	HD10ABL	R	10 Mbps Half-Duplex Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always
10	FD100T	R	100BASE-T2 Full-Duplex Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always
9	HD100T	R	100BASE-T2 Half-Duplex Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always
8	EXT_STS_ABL	R	Extended Status Information ■ 0: No Extended Status information is present at register address 16'h000F of this MMD device. ■ 1: Extended Status information is present at register address 16'h000F of this MMD device. DWC_xpcs returns this bit as 1. **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
7	UN_DIR_ABL	R	Unidirectional Ability ■ 1: The DWC_xpcs is able to transmit GMII irrespective of whether device has determined the valid link or not. ■ 0: The DWC_xpcs is able to transmit GMII only when the device has determined the valid link. The DWC_xpcs always returns this bit as 0. **Value After Reset:** 0x0 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""

Bits	Name	Memory Access	Description
6	MF_PRE_SUP	R	MF Preamble Suppression ■ 1: The DWC_xpcs accepts the MDIO frames with preamble suppressed. ■ 0: The DWC_xpcs does not accept the MDIO frames with preamble suppressed. This bit is always set to 0. **Value After Reset:** 0x0 **Exists:** Always
5	AN_CMPL	R	Auto-negotiation Complete for Port 1 ■ 1: The AN process is complete ■ 0: The AN process is not complete This bit returns 0 if AN_ENABLE of SR_MII_1_CTRL Register is set to 0. **Value After Reset:** 0x0 **Exists:** Always
4	RF	R	Remote Fault This register bit is shared with RF bit of SR_MII_STS register. Software must ignore the value returned by this field, as it is not valid in QSGMII/SGMII/USXGMII modes. **Value After Reset:** 0x0 **Exists:** Always
3	AN_ABL	R	Auto-negotiation Ability The DWC_xpcs always returns this bit as 1. ■ 1: The DWC_xpcs is able to perform auto-negotiation. ■ 0: The DWC_xpcs is not able to perform auto-negotiation. **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
2	LINK_STS	R	Link Status This register bit is shared with RLU bit of SR_MII_STS register. **Value After Reset:** 0x0 **Exists:** Always **Testable:** "QSGMII_EN ? \"untestable\" : \"unconstrained\""
1	Reserved_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	EXT_REG_CAP	R	Extended Register Capability ■ 1: Extended Register capability exists. This bit is always set to 1. ■ 0: Extended Register capability does not exist. **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""

5.2.5.3 SR_MII_1_DEV_ID1

- **Description:** SR MII_1 MMD Device Identifier Register 1
  This register returns the configurable Organizationally Unique Identifier (OUI) to the host.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x2 : (APB3_16) ? (0x2*0x2) : (0x2*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** QSGMII_EN||MP_USXGMII



Table 5-338 Fields for Register: SR_MII_1_DEV_ID1

Bits	Name	Memory Access	Description
15:0	VS_MII_DEV_OUI_3_18	R	Organizationally Unique Identifier[3:18] This field contains Bits [18:3] of 24-bit OUI of device manufacturer. The DWC_xpcs offers 24 configurable Bits[24:1] for identifying the device manufacturer. This register reflects the same value as that of SR_MII_DEV_ID1 register. **Value After Reset:** IEEE_DEV_OUI_3_18 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""

5.2.5.4 SR_MII_1_DEV_ID2

- **Description:** SR MII_1 MMD Device Identifier Register 2
  This register returns the configurable Organizationally Unique Identifier (OUI), Model Number, and Revision Number of the device to the host.
  This register returns the same value as that of SR_MII_DEV_ID2 Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x3 : (APB3_16) ? (0x3*0x2) : (0x3*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** QSGMII_EN||MP_USXGMII

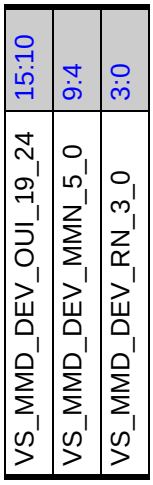


Table 5-339 Fields for Register: SR_MII_1_DEV_ID2

Bits	Name	Memory Access	Description
15:10	VS_MMD_DEV_OUI_19_24	R	Organizationally Unique Identifier [19:24] This field contains Bits[24:19] of 24-bit OUI of device manufacturer. The DWC_xpcs offers 24 configurable Bits[24:1] for identifying the device manufacturer. **Value After Reset:** IEEE_DEV_OUI_19_24 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
9:4	VS_MMD_DEV_MMN_5_0	R	Model Number This field contains the 5-bit Model Number of the device. **Value After Reset:** IEEE_DEV_MODEL_NO_0_5 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""

Bits	Name	Memory Access	Description
3:0	VS_MMD_DEV_RN_3_0	R	Revision Number This field contains the 4-bit Revision Number of the device. **Value After Reset:** IEEE_DEV_REV_NO_0_3 **Exists:** Always

5.2.5.5 SR_MII_1_AN_ADV

- **Description:** SR MII_1 MMD AN Advertisement Register
  Except for bit[5], rest of the fields of this register reflects the corresponding fields of SR_MII_AN_ADV Register.
  Note: This register is valid only when the current operating mode is QSGMII.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x4 : (APB3_16) ? (0x4*0x2) : (0x4*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** QSGMII_EN

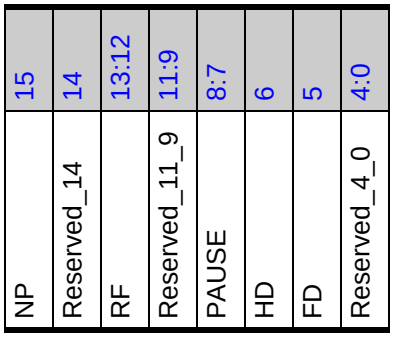


Table 5-340 Fields for Register: SR_MII_1_AN_ADV

Bits	Name	Memory Access	Description
15	NP	R	Next Page This bit reflects the value of NP bit of SR_MII_AN_ADV Register. **Value After Reset:** 0x0 **Exists:** Always
14	Reserved_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:12	RF	R	Remote Fault This bit reflects the value of RF bit of SR_MII_AN_ADV Register. **Value After Reset:** 0x0 **Exists:** Always
11:9	Reserved_11_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
8:7	PAUSE	R	Pause Ability. This bit reflects the value of PAUSE bit of SR_MII_AN_ADV Register. **Value After Reset:** 0x0 **Exists:** Always
6	HD	R	Half Duplex This bit reflects the value of HD bit of SR_MII_AN_ADV Register. **Value After Reset:** 0x0 **Exists:** Always
5	FD	R/W	Full Duplex When this bit is set, it indicates the device (port 1) can operate in full-duplex mode. This bit is used in Clause 37 auto-negotiation (Tx_Config_Reg) done by port 1, when DWC_xpcs is configured as PHY-side QSGMII. **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
4:0	Reserved_4_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.5.6 SR_MII_1_LP_BABL

- **Description:** SR MII_1 MMD AN Link Partner Base Ability Register
  This register merely reflects the value of SR_MII_LP_BABL Register and is not applicable when working in QSGMII/USXGMII Mode.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x5 : (APB3_16) ? (0x5*0x2) : (0x5*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** QSGMII_EN

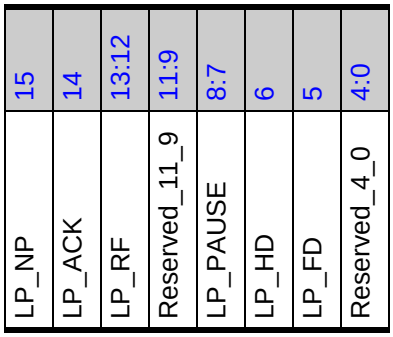


Table 5-341 Fields for Register: SR_MII_1_LP_BABL

Bits	Name	Memory Access	Description
15	LP_NP	R	Next Page **Value After Reset:** 0x0 **Exists:** Always
14	LP_ACK	R	ACK bit from the Link Partner **Value After Reset:** 0x0 **Exists:** Always
13:12	LP_RF	R	Remote Fault **Value After Reset:** 0x0 **Exists:** Always
11:9	Reserved_11_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8:7	LP_PAUSE	R	Pause Ability **Value After Reset:** 0x0 **Exists:** Always
6	LP_HD	R	Half Duplex **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
5	LP_FD	R	Full Duplex **Value After Reset:** 0x0 **Exists:** Always
4:0	Reserved_4_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.5.7 SR_MII_1_AN_EXPN

- **Description:** SR MII_1 MMD AN Expansion Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x6 : (APB3_16) ? (0x6*0x2) : (0x6*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** QSGMII_EN||MP_USXGMII

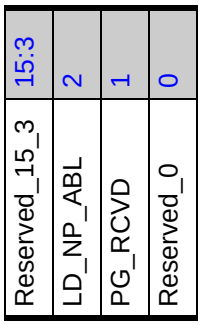


Table 5-342 Fields for Register: SR_MII_1_AN_EXPN

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2	LD_NP_ABL	R	Local Device NP Able ■ 1: The local device has the next page ability ■ 0: The local device does not have the next page ability The DWC_xpcs always returns this bit as 0 because it does not support Next Page. **Value After Reset:** 0x0 **Exists:** Always
1	PG_RCVD	R	Page Received (RO, LH Type) This bit indicates that the local device received a page from the link partner (during port 1 auto-negotiation). ■ 1: The local device received a new page ■ 0: The local device did not receive a new page **Value After Reset:** 0x0 **Exists:** Always
0	Reserved_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.5.8 SR_MII_1_EXT_STS

- **Description:** SR MII_1 MMD Extended Status Register
  This register reflects the same value as that of SR_MII_EX_STS register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0F : (APB3_16) ? (0x0F*0x2) : (0x0F*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** QSGMII_EN

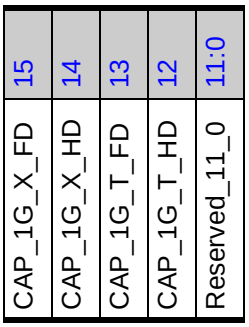


Table 5-343 Fields for Register: SR_MII_1_EXT_STS

Bits	Name	Memory Access	Description
15	CAP_1G_X_FD	R	1000BASE-X Full-Duplex Capable **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
14	CAP_1G_X_HD	R	1000BASE-X Half-Duplex Capable **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
13	CAP_1G_T_FD	R	1000BASE-T Full-Duplex Capable **Value After Reset:** 0x0 **Exists:** Always
12	CAP_1G_T_HD	R	1000BASE-T Half-Duplex Capable **Value After Reset:** 0x0 **Exists:** Always
11:0	Reserved_11_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.5.9 SR_MII_1_TIME_SYNC_ABL

- **Description:** SR MII 1 MMD Time Sync Capability Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0708 : (APB3_16) ? (0x0708*0x2) : (0x0708*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** (QSGMII_EN||MP_USXGMII)

15:14	13	12	11	10	9:4	3	2	1	0
Reserved_15_14	TREF_SFD	TREF_FS	MLS	TXRX_NUC	Reserved_9_4	FINE_TX_DLY	FINE_RX_DLY	PCS_TX_DLY_ABL	PCS_RX_DLY_ABL

Table 5-344 Fields for Register: SR_MII_1_TIME_SYNC_ABL

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13	TREF_SFD	R	Timestamp reference at SFD - 0: DWC_xpcs does not support data delay measurement point at the beginning of SFD - 1: DWC_xpcs supports data delay measurement point at the beginning of SFD **Value After Reset:** "SNPS_RSVDPARAM_11 ? 0x1 : 0x0" **Exists:** Always
12	TREF_FS	R	Timestamp reference at first symbol after SFD - 0: DWC_xpcs does not support data delay measurement point at the beginning of the first symbol after SFD - 1: DWC_xpcs supports data delay measurement point at the beginning of the first symbol after SFD **Value After Reset:** "SNPS_RSVDPARAM_11 ? 0x1 : 0x0" **Exists:** Always

Bits	Name	Memory Access	Description
11	MLS	R	Multi-lane Support. ■ 0: DWC_xpcs does not support reporting of multi-PCS lane path data delay ■ 1: DWC_xpcs supports reporting of multi-PCS lane path data delay **Value After Reset:** "SNPS_RSVDPARAM_11&&!SINGLE_LANE_CFG ? 0x1 : 0x0" **Exists:** Always
10	TXRX_NUC	R	Transmit or Receive Num_Unit_Change Support Available for Port 1. If this bit is set, it indicates that DWC_xpcs supports Transmit/Receive Num_Unit_Change indication capability **Value After Reset:** "SNPS_RSVDPARAM_11 ? 0x1 : 0x0" **Exists:** Always
9:4	Reserved_9_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3	FINE_TX_DLY	R	Fine Resolution Transmit Path Data Delay Available for Port 1. If this bit is set, it indicates that DWC_xpcs provides information on transmit data path delay with sub-ns resolution in SR_MII_1_TIME_SYNC_TX_MAX_FINE_DLY and SR_MII_1_TIME_SYNC_TX_MIN_FINE_DLY registers. **Value After Reset:** "SNPS_RSVDPARAM_11 ? 0x1 : 0x0" **Exists:** Always
2	FINE_RX_DLY	R	Fine Resolution Receive Path Data Delay Available for Port 1. If this bit is set, it indicates that DWC_xpcs provides information on receive data path delay with sub-ns resolution in SR_MII_1_TIME_SYNC_RX_MAX_FINE_DLY and SR_MII_1_TIME_SYNC_RX_MIN_FINE_DLY registers. **Value After Reset:** "SNPS_RSVDPARAM_11 ? 0x1 : 0x0" **Exists:** Always
1	PCS_TX_DLY_ABL	R	PCS Transmit Path Data Delay Information Available for Port 1. If this bit is set, it indicates that the information regarding the maximum and minimum transmit data delay of PCS is available in ns-resolution in MII_1 MMD Tx Time Delay Registers. **Value After Reset:** 0x1 **Exists:** Always

Bits	Name	Memory Access	Description
0	PCS_RX_DLY_ABL	R	PCS Receive Path Data Delay Information Available for Port 1. If this bit is set, it indicates that the information regarding the maximum and minimum receive data delay of PCS is available in ns-resolution in MII_1 MMD Rx Time Delay Registers. **Value After Reset:** 0x1 **Exists:** Always

5.2.5.10 SR_MII_1_TIME_SYNC_TX_MAX_DLY_LWR

- **Description:** SR MII 1 MMD Time Sync Tx Max Delay Lower Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x0709 : (APB3_16) ? (0x0709*0x2) : (0x0709*0x4)" + "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** (QSGMII_EN | | MP_USXGMII)

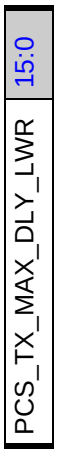


Table 5-345 Fields for Register: SR_MII_1_TIME_SYNC_TX_MAX_DLY_LWR

Bits	Name	Memory Access	Description
15:0	PCS_TX_MAX_DLY_LWR	R	PCS Transmit Max Delay Lower along Port 1 This field indicates the lower 16-bit of the 32-bit value which indicates integer nano-seconds portion of the Maximum Data Delay in the Transmit Path of DWC_xpcs along Port 1. The value returned by this field depends on the current operating mode. If TS_MODE bit is 0, this register indicates the end-to-end (MAC interface to PCS interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the transmit path from the DDMP point until the final Tx data output port along port 1. **Value After Reset:** 0x0 **Exists:** Always

5.2.5.11 SR_MII_1_TIME_SYNC_TX_MAX_DLY_UPR

- **Description:** SR MII 1 MMD Time Sync Tx Max Delay Upper Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070A : (APB3_16) ? (0x070A*0x2) : (0x070A*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** (QSGMII_EN||MP_USXGMII)



Table 5-346 Fields for Register: SR_MII_1_TIME_SYNC_TX_MAX_DLY_UPR

Bits	Name	Memory Access	Description
15:0	PCS_TX_MAX_DLY_UPR	R	PCS Transmit Max Delay Upper along Port 1 This field indicates the upper 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Maximum Data Delay in the Transmit Path of DWC_xpcs along Port 1. If TS_MODE bit is 0, this register indicates the end-to-end (MAC interface to PCS interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the transmit path from the DDMP point until the final Tx data output port along port 1 **Value After Reset:** 0x0 **Exists:** Always

5.2.5.12 SR_MII_1_TIME_SYNC_TX_MIN_DLY_LWR

- **Description:** SR MII 1 MMD Time Sync Tx Min Delay Lower Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x070B : (APB3_16) ? (0x070B*0x2) : (0x070B*0x4)" + "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** (QSGMII_EN | | MP_USXGMII)



Table 5-347 Fields for Register: SR_MII_1_TIME_SYNC_TX_MIN_DLY_LWR

Bits	Name	Memory Access	Description
15:0	PCS_TX_MIN_DLY_LWR	R	PCS Transmit Min Delay Lower along Port 1 This field indicates the lower 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Minimum Data Delay in the Transmit Path of DWC_xpcs along Port 1. The value returned by this register field depends on the current operating mode. If TS_MODE bit is 0, this register indicates the end-to-end (MAC interface to PCS interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the transmit path from the DDMP point until the final Tx data output port along port 1. **Value After Reset:** 0x0 **Exists:** Always

5.2.5.13 SR_MII_1_TIME_SYNC_TX_MIN_DLY_UPR

- **Description:** SR MII 1 MMD Time Sync Tx Min Delay Upper Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x070C : (APB3_16) ? (0x070C*0x2) : (0x070C*0x4)" + "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** (QSGMII_EN | | MP_USXGMII)



Table 5-348 Fields for Register: SR_MII_1_TIME_SYNC_TX_MIN_DLY_UPR

Bits	Name	Memory Access	Description
15:0	PCS_TX_MIN_DLY_UPR	R	PCS Transmit Min Delay Upper along Port 1 This field indicates the upper 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Minimum Data Delay in the Transmit Path of DWC_xpcs along Port 1. If TS_MODE bit is 0, this register indicates the end-to-end (MAC interface to PCS interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the transmit path from the DDMP point until the final Tx data output port along Port 1 **Value After Reset:** 0x0 **Exists:** Always

5.2.5.14 SR_MII_1_TIME_SYNC_RX_MAX_DLY_LWR

- **Description:** SR MII 1 MMD Time Sync Rx Max Delay Lower Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF| |MCI_INTF| |MDIO_INTF) ? 0x070D : (APB3_16) ? (0x070D*0x2) : (0x070D*0x4)" + "(APB3_INTF| |MCI_INTF| |MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** (QSGMII_EN| |MP_USXGMII)

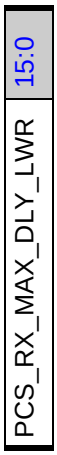


Table 5-349 Fields for Register: SR_MII_1_TIME_SYNC_RX_MAX_DLY_LWR

Bits	Name	Memory Access	Description
15:0	PCS_RX_MAX_DLY_LWR	R	PCS Receive Max Delay Lower along Port 1 This field indicates the lower 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Maximum Data Delay in the Receive Path of DWC_xpcs.The value returned by this register field depends on the current operating mode and is valid after Rx link-up is attained. If TS_MODE bit is 0, this register indicates the end-to-end (PHY interface to MAC interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the receive path from Rx data input port until the DDMP point along port 1. **Value After Reset:** 0x0 **Exists:** Always

5.2.5.15 SR_MII_1_TIME_SYNC_RX_MAX_DLY_UPR

- **Description:** SR MII 1 MMD Time Sync Rx Max Delay Upper Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x070E : (APB3_16) ? (0x070E*0x2) : (0x070E*0x4)" + "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** (QSGMII_EN | | MP_USXGMII)



Table 5-350 Fields for Register: SR_MII_1_TIME_SYNC_RX_MAX_DLY_UPR

Bits	Name	Memory Access	Description
15:0	PCS_RX_MAX_DLY_UPR	R	PCS Receive Max Delay Upper along Port 1 This field indicates the upper 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Maximum Data Delay in the Receive Path of DWC_xpcs along Port 1. If TS_MODE bit is 0, this register indicates the end-to-end (PHY interface to MAC interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the receive path from Rx data input port until the DDMP point along Port 1. **Value After Reset:** 0x0 **Exists:** Always

5.2.5.16 SR_MII_1_TIME_SYNC_RX_MIN_DLY_LWR

- **Description:** SR MII 1 MMD Time Sync Rx Min Delay Lower Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070F : (APB3_16) ? (0x070F*0x2) : (0x070F*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** (QSGMII_EN||MP_USXGMII)



Table 5-351 Fields for Register: SR_MII_1_TIME_SYNC_RX_MIN_DLY_LWR

Bits	Name	Memory Access	Description
15:0	PCS_RX_MIN_DLY_LWR	R	PCS Receive Min Delay Lower along Port 1 This field indicates the lower 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Minimum Data Delay in the Receive Path of DWC_xpcs.The value returned by this register field depends on the current operating mode and is valid after Rx link-up is attained. If TS_MODE bit is 0, this register indicates the end-to-end (PHY interface to MAC interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the receive path from Rx data input port until the DDMP point along Port 1. **Value After Reset:** 0x0 **Exists:** Always

5.2.5.17 SR_MII_1_TIME_SYNC_RX_MIN_DLY_UPR

- **Description:** SR MII 1 MMD Time Sync Rx Min Delay Upper Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0710 : (APB3_16) ? (0x0710*0x2) : (0x0710*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** (QSGMII_EN||MP_USXGMII)



Table 5-352 Fields for Register: SR_MII_1_TIME_SYNC_RX_MIN_DLY_UPR

Bits	Name	Memory Access	Description
15:0	PCS_RX_MIN_DLY_UPR	R	PCS Receive Min Delay Upper along Port 1 This field indicates the upper 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Minimum Data Delay in the Receive Path of DWC_xpcs. If TS_MODE bit is 0, this register indicates the end-to-end (PHY interface to MAC interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the receive path from Rx data input port until the DDMP point along Port 1. **Value After Reset:** 0x0 **Exists:** Always

5.2.5.18 SR_MII_1_TIME_SYNC_TX_MAX_FINE_DLY

- **Description:** SR MII 1 MMD Time Sync Tx Max Fine Delay Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0711 : (APB3_16) ? (0x0711*0x2) : (0x0711*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** ((QSGMII_EN||MP_USXGMII)&&SNPS_RSVDPARAM_11)

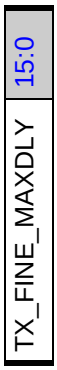


Table 5-353 Fields for Register: SR_MII_1_TIME_SYNC_TX_MAX_FINE_DLY

Bits	Name	Memory Access	Description
15:0	TX_FINE_MAXDLY	R	Transmit Maximum Fine Delay along Port 1 This register indicates the sub-nano-seconds portion of the maximum delay inside the DWC_xpcs transmit path from the DDMP point until the final transmit output port. The delay is indicated in units of 2^-16 ns. Contents of this register is valid only when TS_MODE bit is programmed to 1. Default value of this register is 0, as reset value of TS_MODE bit is 0 **Value After Reset:** 0x0 **Exists:** Always

5.2.5.19 SR_MII_1_TIME_SYNC_TX_MIN_FINE_DLY

- **Description:** SR MII 1 MMD Time Sync Tx Min Fine Delay Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0712 : (APB3_16) ? (0x0712*0x2) : (0x0712*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** ((QSGMII_EN||MP_USXGMII)&&SNPS_RSVDPARAM_11)



Table 5-354 Fields for Register: SR_MII_1_TIME_SYNC_TX_MIN_FINE_DLY

Bits	Name	Memory Access	Description
15:0	TX_FINE_MINDLY	R	Transmit Minimum Fine Delay along Port 1 This register indicates the sub-nano-seconds portion of the minimum delay inside the DWC_xpcs transmit path from the DDMP point until the final transmit output port. The delay is indicated in units of 2^-16 ns. Contents of this register is valid only when TS_MODE bit is programmed to 1. Default value of this register is 0, as reset value of TS_MODE bit is 0 **Value After Reset:** 0x0 **Exists:** Always

5.2.5.20 SR_MII_1_TIME_SYNC_RX_MAX_FINE_DLY

- **Description:** SR MII 1 MMD Time Sync Rx Max Fine Delay Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0713 : (APB3_16) ? (0x0713*0x2) : (0x0713*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** ((QSGMII_EN||MP_USXGMII)&&SNPS_RSVDPARAM_11)

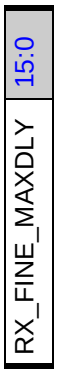


Table 5-355 Fields for Register: SR_MII_1_TIME_SYNC_RX_MAX_FINE_DLY

Bits	Name	Memory Access	Description
15:0	RX_FINE_MAXDLY	R	Receive Maximum Fine Delay along Port 1 This register indicates the sub-nano-seconds portion of the maximum delay inside the DWC_xpcs receive path from Rx data input port until the DDMP point. The delay is indicated in units of 2^-16 ns. Contents of this register is valid only when TS_MODE bit is programmed to 1. Default value of this register is 0, as reset value of TS_MODE bit is 0. **Value After Reset:** 0x0 **Exists:** Always

5.2.5.21 SR_MII_1_TIME_SYNC_RX_MIN_FINE_DLY

- **Description:** SR MII 1 MMD Time Sync Rx Min Fine Delay Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0714 : (APB3_16) ? (0x0714*0x2) : (0x0714*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** ((QSGMII_EN||MP_USXGMII)&&SNPS_RSVDPARAM_11)



Table 5-356 Fields for Register: SR_MII_1_TIME_SYNC_RX_MIN_FINE_DLY

Bits	Name	Memory Access	Description
15:0	RX_FINE_MINDLY	R	Receive Minimum Fine Delay along Port 1 This register indicates the sub-nano-seconds portion of the minimum delay inside the DWC_xpcs receive path from Rx data input port until the DDMP point. The delay is indicated in units of 2^-16 ns. Contents of this register is valid only when TS_MODE bit is programmed to 1. Default value of this register is 0, as reset value of TS_MODE bit is 0. **Value After Reset:** 0x0 **Exists:** Always

5.2.5.22 SR_MII_1_TIME_SYNC_CFG

- **Description:** SR MII 1 MMD Time Sync Configuration Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0715 : (APB3_16) ? (0x0715*0x2) : (0x0715*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** ((QSGMII_EN||MP_USXGMII)&&SNPS_RSVDPARAM_11)

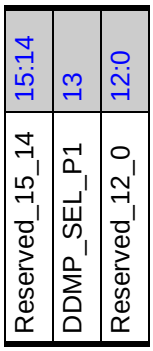


Table 5-357 Fields for Register: SR_MII_1_TIME_SYNC_CFG

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13	DDMP_SEL_P1	R/W	Data Delay Measurement Point Selection along port 1 ■ 0: DWC_xpcs is configured to use the data delay measurement point at the beginning of SFD ■ 1: DWC_xpcs is configured to use the data delay measurement point at the beginning of the first symbol after SFD **Value After Reset:** 0x0 **Exists:** Always
12:0	Reserved_12_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.5.23 VR_MII_1_DIG_CTRL1

- **Description:** VR MII_1 MMD Digital Control1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8000 : (APB3_16) ? (0x8000*0x2) : (0x8000*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** QSGMII_EN||MP_USXGMII

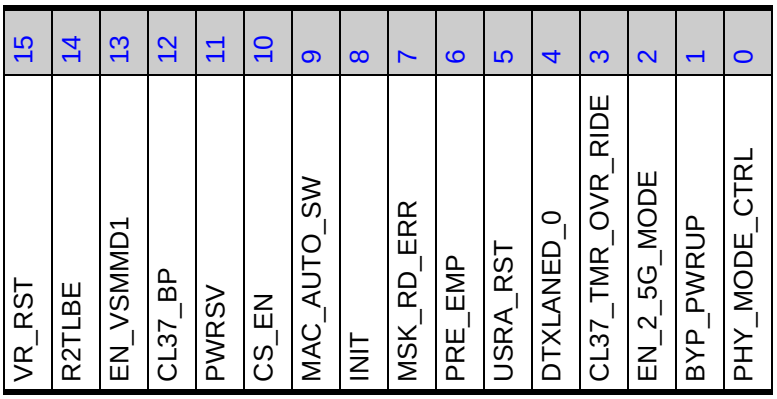


Table 5-358 Fields for Register: VR_MII_1_DIG_CTRL1

Bits	Name	Memory Access	Description
15	VR_RST	R/W	Vendor-Specific Soft Reset (RW,SC Type) When the host sets this bit, the CSR block triggers the vendor-specific software reset process in which all internal blocks, except the Management Interface block and CSR block, are reset. When this bit is set, it also resets the PHY. This bit is self cleared under the following conditions: **For Synopsys PHY:** This bit is self cleared when Bits[4:2] in VR_MII_DIG_STS Register are equal to 3'b100, that is, Tx/Rx clocks are stable and in Power_Good state. **For Non-Synopsys PHY:**This bit is self cleared after the following: ■ 32 clk_csr_i clocks for the MCI interface ■ 1 MDC clock period for the MDIO interface **Note:** For information about the read or write access for any register during the reset process, see "Special Case Register Access" section. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
14	R2TLBE	R/W	Rx to Tx Loopback Enable This bit controls the loopback path from the GMII/XGMII Rx to the GMII/XGMII Tx at the GMII/XGMII interface (port 1). ■ 0: Loopback path is disabled ■ 1: Loopback path is enabled **Value After Reset:** 0x0 **Exists:** Always
13	EN_VSMMD1	R	Enable Vendor-Specific MMD1 This is a read-only bit which reflects the value of EN_VSMMD1 bit of VR_MII_DIG_CTRL1 Register. **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
12	CL37_BP	R	Enable Clause 37 AN in Backplane Configuration This is a read-only bit which reflects the value of CL37_BP bit of VR_MII_DIG_CTRL1 Register. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
11	PWRSV	* Varies	**For Synopsys PHY configurations:** Power Save This bit is present only when the DWC_xpcs is configured for Synopsys PHY. ■ 0: Normal operation. ■ 1: DWC_xpcs and the PHY enter the power-save mode. When this bit is set, the DWC_xpcs triggers the power-down mode by turning off the PHY Receiver and Transmitter, and without turning off MPLL. For Synopsys Multi-protocol 12G/16G PHY, DWC_xpcs asserts xpcs_tx_lpd_o{lane} and xpcs_rx_lpd_o{lane} when this bit is set. When the host clears this bit, DWC_xpcs resumes normal operation. After clearing this bit, the host must wait until Bits[4:2] of the VR_MII_DIG_STS Status Register indicate that the DWC_xpcs is in a normal state. **Note:** For a description of read or write access for any register while this bit is set, see "Special Case Register Access" section of the Databook. **For non-Synopsys PHY configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_XAUI_PHY) ? \"read-write\" : \"read-only\""
10	CS_EN	* Varies	Clock Stop Enable ■ 1: The PHY might stop the clock during LPI mode. ■ 0: The clock cannot be stopped during LPI mode. You must program this bit based on the capability of the MAC (connected to GMII Port 1) during Rx LPI mode. **Note:** This field is valid only in QSGMII mode. **Value After Reset:** "EEE_EN&&QSGMII_EN ? \"0x1\" : \"0x0\"" **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\"" **Memory Access:** "EEE_EN&&QSGMII_EN ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
9	MAC_AUTO_SW	R/W	Automatic Speed Mode Change after CL37 AN (for port 1) If this field is set to 1, DWC_xpcs automatically switches to the negotiated USXGMII/QSGMII(port1) speed, after the completion of Clause 37 auto-negotiation. This mode is valid only when DWC_xpcs is configured as MAC-side USXGMII/QSGMII and must be set only when Auto-negotiation is enabled (AN_ENABLE bit is set to 1). If this bit is set to 0, DWC_xpcs operates at the speed/duplex mode as per the values programmed to SR_MII_1_CTRL Register. In that case, after the completion of CL37 AN, application has to read the negotiated Speed/Duplex Mode from VR_MII_1_AN_INTR_STS Register and then program SR_MII_1_CTRL Register appropriately. If this bit is set to 1 in QSGMII mode, xpcs_qsgmii_link_sts_p1_o, xpcs_link_speed_p1_o and xpcs_qsgmii_full_duplex_p1_o outputs reflect the auto-negotiated values, that is, values from CL37_ANSGM_STS field of VR_MII_1_AN_INTR_STS Register. For USXGMII mode, if clk_xgmii_tx_p1_i and clk_xgmii_rx_p1_i do not stabilize at the new operating frequency (based on the selected speed) immediately after the completion of auto-negotiation, then software might need to program 'USRA_RST' bit prior to starting packet transfer in the new speed mode. **Note:** This bit must be set only when DWC_xpcs is configured as USXGMII/QSGMII MAC, that is, TX_CONFIG = 0. **Note:** Whenever this bit is set, the values programmed to SS5,SS6 and SS13 bits are ignored. Moreover, the values of these bits do not change based on auto-negotiation outcome. **Value After Reset:** 0x0 **Exists:** Always
8	INIT	R	INIT This bit reflects the value of INIT bit of VR_MII_DIG_CTRL1 Register. **Value After Reset:** 0x0 **Exists:** Always
7	MSK_RD_ERR	R	Mask Running Disparity Error. This bit reflects the value of MSK_RD_ERR bit of VR_MII_DIG_CTRL1 Register. **Value After Reset:** 0x0 **Exists:** Always

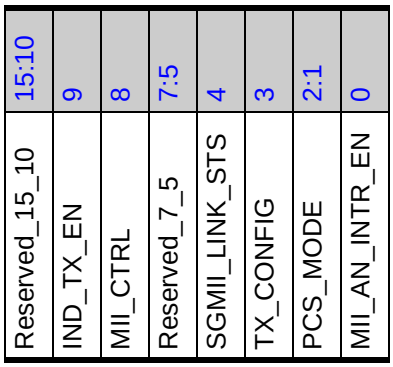
Bits	Name	Memory Access	Description
6	PRE_EMP	* Varies	Pre-emption Packet Enable. Set this bit to 1 to allow the DWC_xpcs to properly receive/transmit preemption packets along GMII port 1, when operating in 10M/100M Modes. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "QSGMII_EN ? \"read-write\" : \"read-only\""
5	USRA_RST	* Varies	USXGMII Rate Adapter Reset (Port 1) This bit can be set to 1 to reset/initialize the USXGMII Rate Adapter Logic (for port 1) inside DWC_xpcs. The Rate Adapter Logic maps GMII data to XGMII (and vice-versa in Rx path) data and perform data replication for lower USXGMII speeds. This is a self-clear bit (RW,SC) and clears itself after the reset of Tx and Rx Rate Adapter Logic is complete. After setting this bit to 1, application must poll this bit till it self-clears. **Note:** If CLKCOMP = Enabled, this bit also initializes the clock-compensation FIFO (corresponding to port 1). When this bit is programmed to 1, RXFIFO_OVF/RXFIFO_UNF bits of VR_MII_1_DIG_STS Register might get set incorrectly. Therefore, read these register bits (RXFIFO_OVF and RXFIFO_UNF) so that they get cleared. Thereafter, RXFIOF_UNF and RXFIFO_UNF bits are reliable. **Value After Reset:** 0x0 **Exists:** Always **Testable:** "(MP_USXGMII) ? \"untestable\" : \"unconstrained\"" **Memory Access:** "MP_USXGMII ? \"read-write\" : \"read-only\""
4	DTXLANED_0	R	Tx Lane 0 Disable This bit reflects the value of DTXLANED_0 bit of VR_MII_DIG_CTRL1 Register. **Value After Reset:** 0x0 **Exists:** Always
3	CL37_TMR_OVR_RIDE	R	Over-Ride Control for CL37 Link Timer. This bit reflects the value of CL37_TMR_OVR_RIDE bit of VR_MII_DIG_CTRL1 Register. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
2	EN_2_5G_MODE	R	Enable 2.5G GMII Mode (GMII interface over-clocked 2.5 times). This bit reflects the value of EN_2_5G_MODE bit of VR_MII_DIG_CTRL1 Register. **Value After Reset:** 0x0 **Exists:** Always
1	BYP_PWRUP	* Varies	**For Synopsys PHY configurations:** Bypass Power-Up Sequence When this bit is set to 1, the DWC_xpcs bypasses the normal flow of the power-up sequence and reaches the Power_Good state to enable transmission or reception. When this bit is set, the DWC_xpcs does not wait for the MPLL and Transmit or Receive PLL status from the Synopsys PHY. You can use this feature in the following scenarios: ■ When the DWC_xpcs is configured to interface with a specific Synopsys PHY. ■ When the data path needs to be interfaced to some other PHY. When this bit is set to 0, the DWC_xpcs waits for the MPLL, Tx, or Rx PLL status from the Synopsys PHY before resuming the normal transmission and reception. **Note:** This bit is shared with Bit 1 of following registers: ■ VR_XS_PCS_DIG_CTRL1 Register ■ VR_AN_DIG_CTRL1 Register ■ VR_PMA_DIG_CTRL1 Register ■ VR_MII_1_DIG_CTRL1 Register ■ VR_MII_2_DIG_CTRL1 Register ■ VR_MII_3_DIG_CTRL1 Register **For non-Synopsys PHY configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_XAUI_PHY) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
0	PHY_MODE_CTRL	* Varies	QSGMII/USXGMII PHY mode control on Port 1 This bit controls the Clause 37 auto-negotiation when operating in QSGMII/USXGMII (Port 1) PHY mode. **QSGMI**: When this bit is set to 1, DWC_xpcs advertises the values of input ports xpcs_qsgmii_link_sts_p1_i, xpcs_qsgmii_link_speed_p1_i and xpcs_qsgmii_full_duplex_p1_i during QSGMII(Port 1) auto-negotiation. When this bit is set to 0, QSGMII(Port 1) auto-negotiation advertises the values programmed to: ■ bit 4 (SGMII_LINK_STS) of VR_MII_1_AN_CTRL Register ■ bit 13 (ss13) and 6 (ss6) of SR_MII_1_CTRL Register and ■ bit 5 (FD) of SR_MII_1_AN_ADV Register **USXGMII**: When this bit is set to 1, DWC_xpcs advertises the values of input ports xpcs_usxg_link_sts_p1_i, xpcs_usxg_link_speed_p1_i,xpcs_usxg_full_duplex_p1_i,xpcs_usxg_eee_abl_p1_i and xpcs_usxg_ck_stp_abl_p1_i during USXGMII (port 1) auto-negotiation. When this bit is set to 0, USXGMI (Port 1) auto-negotiation advertises the following : ■ bit 4 (SGMII_LINK_STS) of VR_MII_1_AN_CTRL Register ■ bit 13 (ss13) and 6 (ss6) and 5 (ss5) of SR_MII_1_CTRL Register and ■ Value of 1 for Duplex-Mode ■ Value of 1 for EEE-capability, if EEE = Enabled; Value of 0 if EEE_EN = Disabled ■ Value of 1 for EEE clock-stop capability,if EEE = Enabled; Value of 0 if EEE_EN = Disabled **Note:** This field is present only for the configurations satisfying either of the following conditions: ■ QSGMII_PHY_AN_AUTO_RESTART= Enabled ■ Multi-port USXGMII configurations with USXG_-PHY_AN_AUTO_RESTART **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(QSGMII_PHY_AN_AUTO_RESTART||(USXG_PHY_AN_AUTO_RESTART&&MP_USXGMII)) ? \"read-write\" : \"read-only\""

5.2.5.24 VR_MII_1_AN_CTRL

- **Description:** VR MII_1 MMD AN Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x8001 : (APB3_16) ? (0x8001*0x2) : (0x8001*0x4)" + "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** QSGMII_EN | | MP_USXGMII



15:10	9	8	7:5	4	3	2:1	0
Reserved_15_10	IND_TX_EN	MII_CTRL	Reserved_7_5	SGMII_LINK_STS	TX_CONFIG	PCS_MODE	MII_AN_INTR_EN

Table 5-359 Fields for Register: VR_MII_1_AN_CTRL

Bits	Name	Memory Access	Description
15:10	Reserved_15_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9	IND_TX_EN	* Varies	Independent Transmit Enable on port 1 during QSGMII mode. If this bit is set to 1, DWC_xpcs is able to transmit the port 1 GMII Tx data, irrespective of its receive link status during QSGMII mode (provided auto-negotiation is not enabled). If this bit is set to 0, DWC_xpcs sends IDLE (for port 1) till its receiver has attained synchronization. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "QSGMII_EN ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
8	MII_CTRL	R/W	MII Control This bit controls the width of the MAC interface (connected to GMII port 1) when operating at 10 Mbps or 100 Mbps ■ 0: 4-bit MII ■ 1: 8-bit MII This bit also controls the xpcs_mii_ctrl_p1_o signal which is used for external clock multiplexing of the clk_mii_tx_p1_i/clk_xgmii_tx_p1_i and clk_mii_rx_p1_i/clk_xgmii_rx_p1_i signals. **Value After Reset:** 0x0 **Exists:** Always
7:5	Reserved_7_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	SGMII_LINK_STS	R/W	QSGMII/USXGMII Link Status (port 1) This bit is used in Bit 15 of the Tx_Config_Reg during Clause 37 auto-negotiation (along port 1) when DWC_xpcs is programmed as QSGMII-PHY (or as USXGMII-PHY) and when PHY_MODE_CTRL bit of VR_MII_1_DIG_CTRL1 Register is 0. ■ 0: Link Down ■ 1: Link Up **Value After Reset:** 0x0 **Exists:** Always
3	TX_CONFIG	R	Transmit Configuration This field reflects the value of TX_CONFIG of VR_MII_AN_CTRL Register. **Value After Reset:** 0x0 **Exists:** Always
2:1	PCS_MODE	R	PCS Mode This field reflects the value of PCS_MODE of VR_MII_AN_CTRL Register for configurations with QSGMII support. For other configurations, this field returns 2'b00. **Value After Reset:** 0x0 **Exists:** Always
0	MII_AN_INTR_EN	R/W	Clause 37 AN Complete Interrupt Enable (port 1) When set to 1, this bit enables the generation of Clause 37 auto-negotiation complete (on port 1) interrupt output. When set to 0, it disables the generation of Clause 37 auto-negotiation complete interrupt. **Value After Reset:** 0x0 **Exists:** Always

5.2.5.25 VR_MII_1_AN_INTR_STS

- **Description:** VR MII_1 MMD AN Interrupt and Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8002 : (APB3_16) ? (0x8002*0x2) : (0x8002*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** QSGMII_EN | |MP_USXGMII

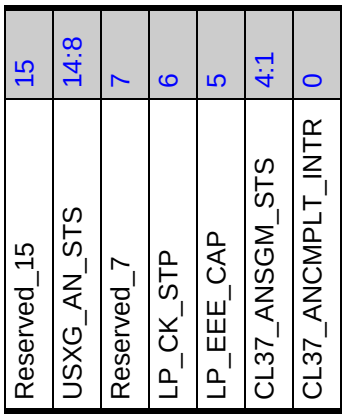


Table 5-360 Fields for Register: VR_MII_1_AN_INTR_STS

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
14:8	USXG_AN_STS	R	USXGMII Clause 37 AN Status (along Port 1) This field is valid only when clause 37 auto-negotiation is complete in USXGMII Mode along Port 1. It indicates the status received from remote link after the USXGMII auto-negotiation is complete. USXG_AN_STS[6] ■ 0: Link is Down ■ 1: Link is Up USXG_AN_STS[5] ■ 0: Half Duplex ■ 1: Full Duplex USXG_AN_STS[4:2] ■ 000: 10 Mbps speed link ■ 001: 100 Mbps speed link ■ 010: 1000 Mbps speed link ■ 011: 10 Gbps speed link ■ 100: 2.5 Gbps speed link ■ 101: 5 Gbps speed link USXG_AN_STS[1] ■ 1: EEE supported ■ 0: EEE not supported USXG_AN_STS[0] ■ 1: EEE clock-stop supported ■ 0: EEE clock-stop not supported **Note:** This field is present only in configurations with Multi-port USXGMII support. **Value After Reset:** 0x0 **Exists:** Always
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
6	LP_CK_STP	R	Link Partner EEE Clock Stop Capability This field indicates the EEE clock stop capability (or enable in case far-end is acting as QSGMII MAC) advertised by the far-end device. This field is valid only when PCS_MODE[1:0] is set to the QSGMII mode and the auto-negotiation is complete along port 1. **Note:** This field is present only in configurations with QSGMII. **Value After Reset:** 0x0 **Exists:** Always
5	LP_EEE_CAP	R	Link Partner EEE Capability This field indicates the EEE capability advertised by the far-end device (Port 1 QSGMII PHY). This field is valid only when PCS_MODE[1:0] is set to the QSGMII mode and the auto-negotiation is complete along port 1. **Note:** This field is present only in configurations with QSGMII. **Value After Reset:** 0x0 **Exists:** Always
4:1	CL37_ANSGM_STS	R	Clause 37 AN QSGMII Status (port 1) This field is valid only when the PCS_MODE[1:0] is set to the QSGMII mode and the auto-negotiation is complete along port 1. It indicates the status received from remote link partner. CL37_ANSGM_STS[0] ■ 0: Half Duplex ■ 1: Full Duplex CL37_ANSGM_STS[2:1] ■ 00: 10 Mbps speed link ■ 01: 100 Mbps speed link ■ 10: 1000 Mbps speed link CL37_ANSGM_STS[3] ■ 0: Link is Down ■ 1: Link is Up **Note:** This field is present only in configurations with QSGMII. **Value After Reset:** 0x5 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
0	CL37_ANCMPLT_INTR	R/W	Clause 37 AN Complete Interrupt (SS,WC Type) The DWC_xpcs sets this bit when Clause 37 auto-negotiation is complete for port 1. The host must clear this bit by writing 0 to it. **Value After Reset:** 0x0 **Exists:** Always

5.2.5.26 VR_MII_1_TC

- **Description:** VR MII_1 MMD Test Control Register.
  Note: This register is present only in 1000BaseX-Only PCS configurations with QSGMII_EN = Enabled.
  This register merely reflects the value of VR_MII_TC Register and is not applicable when working in QSGMII/USXGMII Mode.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8003 : (APB3_16) ? (0x8003*0x2) : (0x8003*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** BASE1000X_ONLY&&QSGMII_EN

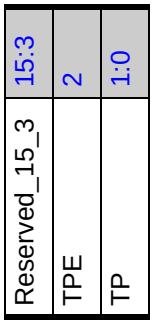


Table 5-361 Fields for Register: VR_MII_1_TC

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2	TPE	R	Test Pattern Enable Lanes **Value After Reset:** 0x0 **Exists:** Always
1:0	TP	R	Test Pattern Select **Value After Reset:** 0x0 **Exists:** Always

5.2.5.27 VR_MII_1_DIG_CTRL3

- **Description:** VR MII_1 MMD Digital Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8004 : (APB3_16) ? (0x8004*0x2) : (0x8004*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** MP_USXGMII

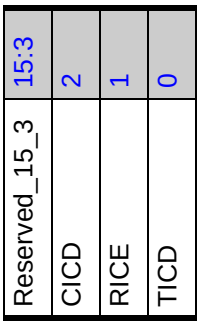


Table 5-362 Fields for Register: VR_MII_1_DIG_CTRL3

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2	CICD	R/W	Clock-compensation IPG check Deviation on Port 1 During USXGMII mode, when operating at GMII speed modes (1G/100M/10M), by default, clock-compensation logic deletes IDLE (to compensate for ppm difference) only if incoming IPG >= 12B. This ensures that output IPG is >= 8B provided input IPG is also greater than or equal to 12B However, if this bit is 1, clock-compensation IPG check deviates from the standard behavior and deletes IDLE even if incoming IPG is only >= 9B. This ensures that output IPG is >= 5B provided input IPG is also greater than or equal to 5B ■ 0: output IPG >= 8B ■ 1: output IPG >= 5B This mode is useful when incoming Rx data is an already clock-compensated data and does not have enough number of IPGs with >=12B of IDLE. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
1	RICE	R/W	Receive IPG Check Enable during XGMII-to-GMII conversion on Port 1. Whenever operating in 1G and lower speeds, DWC_xpcs receiver converts the received data from XGMII format to GMII/MII format before driving it out on GMII/MII Rx interface of DWC_xpcs. This logic blindly carries-out this conversion process without performing any check for number of IPG bytes present in this received data. RICE bit can be set to 1 to enable a IPG check mechanism during XGMII-to-GMII conversion process so as to ensure minimum IPG of 8 bytes at GMII/MII Rx interface. Whenever IPG in XGMII data is less than 8 bytes, DWC_xpcs introduces additional IPG bytes so as to ensure 8 bytes minimum IPG at GMII/MII Rx interface. Addition of IPG is compensated later by deleting IPG bytes from other packets which have larger IPG. If packets with IPG less than 8 bytes appear quite frequently in the incoming data stream, then the buffer can overflow and can cause packet corruption. This feature can work reliably only if not more than one packet with IPG less than 8 bytes occur within a span of 4 packets. This feature is useful in scenarios where link-partner device can send packets with IPG as low as 5 bytes. This bit shall not be set to 1, if CICD is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	TICD	R/W	Transmit IPG Check Deviation on Port 1. When operating in multi-port USXGMII modes, USXGMII Tx Rate Module compensates for AM insertion by deleting IDLEs.This deletion operation involves a IPG check logic which allows deletion only if 8 bytes of IPG can be ensured at transmitter output when operating in GMII/MII modes (1G and lower speed). However, this IPG check logic introduces a limit on the packet size that can be transmitted. If jumbo packets (packets of size 2K bytes to 9K bytes) are transmitted continuously, DWC_xpcs might not find an opportunity to delete sufficient IDLE bytes before the next AM insertion. Therefore, this can result in a gradual fill-up of Tx RAL FIFO and can result in eventual packet corruption. This bit can be set to 1 to disable the standard IPG check and perform a modified IPG check so as to support jumbo packets. Whenever this bit is set to 1, DWC_xpcs perform IPG deletion in such a manner so as to ensure minimum of 5 bytes of IPG at transmitter output when operating in GMII/MII modes (1G and lower speed). This mode is useful if the link-partner device is capable of accepting packets with 5 bytes IPG when operating in GMII/MII modes. When operating in XGMII mode (2.5G and above speeds), irrespective of the value of this bit, DWC_xpcs ensures minimum of 5 bytes IPG at the transmitter output. **Value After Reset:** 0x0 **Exists:** Always

5.2.5.28 VR_MII_1_DBG_CTRL

- **Description:** VR MII_1 MMD Debug Control Register
  The contents of this register is valid only in QSGMII Mode.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8005 : (APB3_16) ? (0x8005*0x2) : (0x8005*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** QSGMII_EN

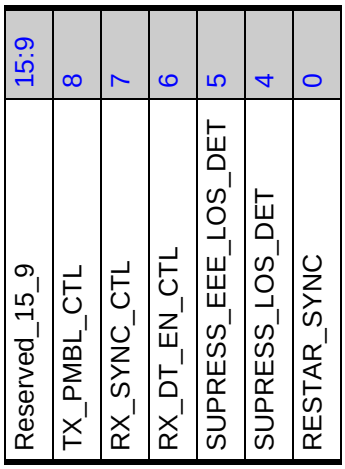


Table 5-363 Fields for Register: VR_MII_1_DBG_CTRL

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	TX_PMBL_CTL	R	Transmit Preamble Control This bit merely reflects the value of TX_PMBL_CTL field of VR_MII_DBG_CTRL Register (for 1000BaseX-Only PCS configurations) or VR_XS_PCS_DEBUG_CTRL Register (for other configurations). **Value After Reset:** 0x0 **Exists:** Always
7	RX_SYNC_CTL	R	Receive Synchronization Control . This bit merely reflects the value of RX_SYNC_CTL field of VR_MII_DBG_CTRL Register (for 1000BaseX-Only PCS configurations) or VR_XS_PCS_DEBUG_CTRL Register (for other configurations). **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
6	RX_DT_EN_CTL	R	Rx Data Enable Control. This bit merely reflects the value of RX_DT_EN_CTL field of VR_MII_DBG_CTRL Register (for 1000BaseX-Only PCS configurations) or VR_XS_PCS_DEBUG_CTRL Register (for other configurations). **Value After Reset:** 0x0 **Exists:** Always
5	SUPRESS_EEE_LOS_DET	R	Suppress EEE Loss of Signal Detection. This bit merely reflects the value of SUPRESS_EEE_LOS_DET field of VR_MII_DBG_CTRL Register (for 1000BaseX-Only PCS configurations) or VR_XS_PCS_DEBUG_CTRL Register (for other configurations). **Value After Reset:** 0x0 **Exists:** Always
4	SUPRESS_LOS_DET	R	Suppress Loss of Signal Detection This bit merely reflects the value of SUPRESS_LOS_DET field of VR_MII_DBG_CTRL Register (for 1000BaseX-Only PCS configurations) or VR_XS_PCS_DEBUG_CTRL Register (for other configurations). **Value After Reset:** 0x0 **Exists:** Always
0	RESTAR_SYNC	R/W	Restart Synchronization When set to 1, this bit restarts the Rx Synchronization State machine on port 1. The host must clear this bit to 0 before setting it to 1 next time. **Value After Reset:** 0x0 **Exists:** Always

5.2.5.29 VR_MII_1_EEE_MCTRL0

- **Description:** VR MII_1 MMD EEE Mode Control Register
  This register is present only for configurations with MAIN_MODE = 1000BASEX-Only PCS and QSGMII_EN = Enabled and EEE_EN = Enabled.
  This is a read-only register which merely reflects the value of VR_MII_EEE_MCTRL Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8006 : (APB3_16) ? (0x8006*0x2) : (0x8006*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** BASE1000X_ONLY&&QSGMII_EN&&EEE_EN

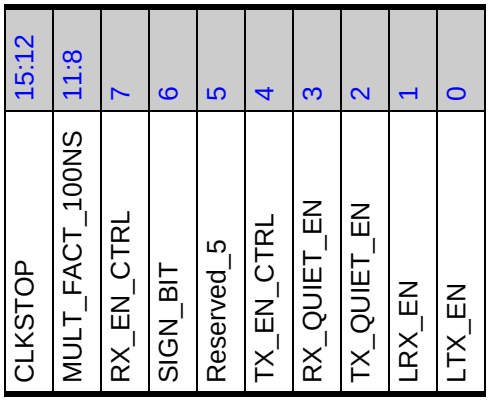


Table 5-364 Fields for Register: VR_MII_1_EEE_MCTRL0

Bits	Name	Memory Access	Description
15:12	CLKSTOP	R	Clock Stop **Value After Reset:** 0x8 **Exists:** Always
11:8	MULT_FACT_100NS	R	100 ns Clock Tic Multiplying Factor **Value After Reset:** 0x9 **Exists:** Always
7	RX_EN_CTRL	R	Rx Control Enable **Value After Reset:** 0x1 **Exists:** Always
6	SIGN_BIT	R	Effective 100 ns Tic Value **Value After Reset:** 0x0 **Exists:** Always
5	Reserved_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4	TX_EN_CTRL	R	Tx Control Enable **Value After Reset:** 0x1 **Exists:** Always
3	RX_QUIET_EN	R	Rx Quiet Enable **Value After Reset:** 0x1 **Exists:** Always
2	TX_QUIET_EN	R	Tx Quiet Enable **Value After Reset:** 0x1 **Exists:** Always
1	LRX_EN	R	LPI Rx Enable **Value After Reset:** 0x0 **Exists:** Always
0	LTX_EN	R	LPI Tx Enable **Value After Reset:** 0x0 **Exists:** Always

5.2.5.30 VR_MII_1_EEE_TXTIMER

- **Description:** VR MII_1 MMD EEE Tx Timer Register
  This register is present only for configurations with MAIN_MODE = 1000BASEX-Only PCS and QSGMII_EN = Enabled and EEE_EN = Enabled.
  This is a read-only register which merely reflects the value of VR_MII_EEE_TXTIMER Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8008 : (APB3_16) ? (0x8008*0x2) : (0x8008*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** BASE1000X_ONLY&&QSGMII_EN&&EEE_EN

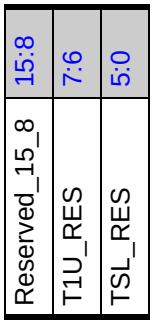


Table 5-365 Fields for Register: VR_MII_1_EEE_TXTIMER

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
7:6	T1U_RES	R	T1u Resolution This field stores the resolution value for the T1u timer. When the generated 100 ns tic timer is not exactly 100 ns, this field is programmed with the required value. The programmed value is equivalent to the number of more (SIGN_BIT is low) or less (SIGN_BIT is high) counts than the ideal count to meet the range of the T1u timer. The DWC_xpcs maintains the default value of the T1u timer as 11, assuming clk_eee_i_time_period * (MULT_FACT_100NS + 1) is equal to 100 ns to produce 1100 ns (1.1 us) as per the T1u requirement in the IEEE standard. Example: For 10GBASE-R or KR_Present configuration, if clk_eee_i_time_period * (MULT_FACT_100NS + 1)is 90, the default T1u timer produces 990 ns (0.99 us) which is lesser than the standard 1.1 us. To make it 1.1 us, you must program this register such that (11 +/- T1U_RES) * ((MULT_FACT_100NS + 1) * clk_eee_i_time_period) is greater than 1.1 us and less than 1.3 us (max limit), that is, (11+ T1U_RES)*90 = 1.1 us. T1U_RES = 2 meets the requirement which produces 1.17 us. Program the SIGN_BIT as 1. **Value After Reset:** 0x0 **Exists:** Always
5:0	TSL_RES	R	TSL Resolution **Value After Reset:** 0x0 **Exists:** Always

5.2.5.31 VR_MII_1_EEE_RXTIMER

- **Description:** VR MII_1 MMD EEE Rx Timer Register
  This register is present only for configurations with MAIN_MODE = 1000BASEX-Only PCS and QSGMII_EN = Enabled and EEE_EN = Enabled.
  This is a read-only register which merely reflects the value of VR_MII_EEE_RXTIMER Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8009 : (APB3_16) ? (0x8009*0x2) : (0x8009*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** BASE1000X_ONLY&&QSGMII_EN&&EEE_EN

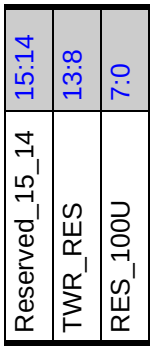


Table 5-366 Fields for Register: VR_MII_1_EEE_RXTIMER

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:8	TWR_RES	R	TWR Resolution **Value After Reset:** 0x0 **Exists:** Always
7:0	RES_100U	R	100 us Resolution **Value After Reset:** 0x0 **Exists:** Always

5.2.5.32 VR_MII_1_LINK_TIMER_CTRL

- **Description:** VR MII_1 MMD Link Timer Control Register
  This register merely reflects the value of VR_MII_LINK_TIMER_CTRL Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x800A : (APB3_16) ? (0x800A*0x2) : (0x800A*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** QSGMII_EN||MP_USXGMII



Table 5-367 Fields for Register: VR_MII_1_LINK_TIMER_CTRL

Bits	Name	Memory Access	Description
15:0	CL37_LINK_TIME	R	Programmable Link Timer Value for Clause 37 auto-negotiation. **Value After Reset:** 0x0 **Exists:** Always

5.2.5.33 VR_MII_1_EEE_MCTRL1

- **Description:** VR MII_1 MMD EEE Mode Control 1 Register
  This register is present only for configurations with MAIN_MODE = 1000BASEX-Only PCS and QSGMII_EN = Enabled and EEE_EN = Enabled.
  This is a read-only register which merely reflects the value of VR_MII_EEE_MCTRL1 Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x800B : (APB3_16) ? (0x800B*0x2) : (0x800B*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** BASE1000X_ONLY&&QSGMII_EN&&EEE_EN

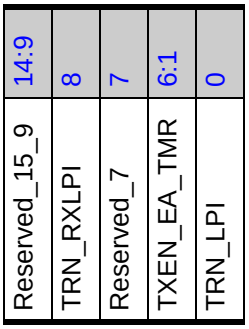
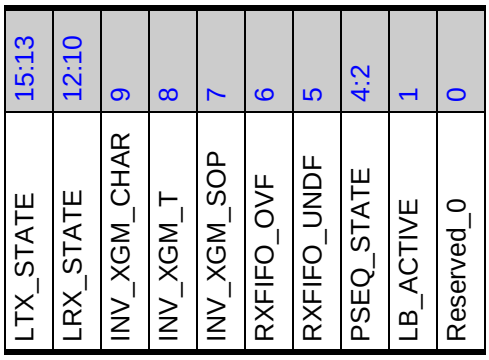


Table 5-368 Fields for Register: VR_MII_1_EEE_MCTRL1

Bits	Name	Memory Access	Description
14:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	TRN_RXLPI	R	Transparent Rx LPI Mode Enable. **Value After Reset:** 0x0 **Exists:** Always
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:1	TXEN_EA_TMR	R	Transmit Enable Early Assert Time Duration **Value After Reset:** 0x0 **Exists:** Always
0	TRN_LPI	R	Transparent Tx LPI Mode Enable **Value After Reset:** 0x0 **Exists:** Always

5.2.5.34 VR_MII_1_DIG_STS

- **Description:** VR MII_1 MMD Digital Status Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x8010 : (APB3_16) ? (0x8010*0x2) : (0x8010*0x4)" + "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** QSGMII_EN | | MP_USXGMII



15:13	12:10	9	8	7	6	5	4:2	1	0
LTX_STATE	LRX_STATE	INV_XGM_CHAR	INV_XGM_T	INV_XGM_SOP	RXFIFO_OVF	RXFIFO_UNDF	PSEQ_STATE	LB_ACTIVE	Reserved_0

Table 5-369 Fields for Register: VR_MII_1_DIG_STS

Bits	Name	Memory Access	Description
15:13	LTX_STATE	R	LPI Transmit State. This field reflects the value of LTX_STATE field of VR_MII_DIG_STS Register (for 1000BaseX-Only PCS configurations) or VR_XS_PCS_DIG_STS Register (for other configurations). **Value After Reset:** 0x0 **Exists:** Always
12:10	LRX_STATE	R	LPI Receive State This field reflects the value of LRX_STATE field of VR_MII_DIG_STS Register (for 1000BaseX-Only PCS configurations) or VR_XS_PCS_DIG_STS Register (for other configurations). **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
9	INV_XGM_CHAR	R	Invalid XGMII Character (RO,LH Type) -Port 1 This bit indicates an invalid XGMII character on the Transmit path (port 1). ■ 0: No Invalid XGMII character ■ 1: Invalid character This bit is set when one of the following conditions is true on the XGMII interface: ■ |Q| is detected on a lane other than Lane 0. ■ |Q| is detected on Lane 0 and also any of the following conditions is true: ❑ No 8'h00 data in Lane 1. ❑ No 8'h00 data in Lane 2. ❑ No 8'h01 or 8'h02 data on Lane 3. ■ |E| or |T| is encountered in non-data phase and |Q| is encountered in data phase. ■ |D| is detected during the Idle phase (after Terminate and before valid SOF). ■ |I| is detected during the data phase (after SOF and before Terminate). **Note:** This field is present only for Multi-port USXGMII configurations. **Value After Reset:** 0x0 **Exists:** Always
8	INV_XGM_T	R	Invalid XGMII T Character (RO,LH Type) -Port 1 This bit indicates that the transmit data (port 1) received on XGMII has an invalid Terminate character. ■ 0: Normal operation ■ 1: Invalid Terminate character The following can cause an invalid Terminate character: ■ In Terminate column, before lane corresponds to the |T| character, there is a |I| or |Q| character. ■ In Terminate column, after lane corresponds to the |T| character, there is a non-|I| character **Note:** This field is present only for Multi-port USXGMII configurations. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
7	INV_XGM_SOP	R	Invalid XGMII Start Character (RO,LH Type) -Port 1 This bit indicates that the XGMII transmit (Port 1) frame contains invalid SOP character. ■ 0: Normal operation ■ 1: Invalid SOP character An invalid SOP event occurs when a |S| character is detected in a lane other than Lane 0. **Note:** This field is present only for Multi-port USXGMII configurations. **Value After Reset:** 0x0 **Exists:** Always
6	RXFIFO_OVF	R	Rx FIFO Overflow (RO,LH Type) This bit indicates the clock rate compensation FIFO overflow for port 1. ■ 0: Normal operation ■ 1: FIFO overflow **Value After Reset:** 0x0 **Exists:** Always
5	RXFIFO_UNDF	R	Rx FIFO Underflow (RO,LH Type) This bit indicates the clock rate compensation FIFO underflow for port 1. ■ 0: Normal operation ■ 1: FIFO underflow **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4:2	PSEQ_STATE	R	**For Synopsys PHY configurations:** Power Up Sequence State This field indicates the state variable value of the power-up sequence module. For Synopsys Multi-protocol 6G PHY: ■ 3'b000: Wait for MPLL ON ■ 3'b001: Wait for TX up ■ 3'b010: Wait for RX up ■ 3'b011/3'b100: Tx/Rx Stable (Power_Good state) ■ 3'b101: Wait for Rx down (MPLL still ON) ■ 3'b110: MPLL OFF For Synopsys Multi-protocol 12G/16G/25G/32G PHY ■ 3'b000: Wait for ACK High 0 ■ 3'b001: Wait for ACK Low 0 ■ 3'b010: Wait for ACK High 1 ■ 3'b011: Wait for ACK Low 1 ■ 3'b100: Tx/Rx stable (Power_Good state) ■ 3'b101: Power Save state ■ 3'b110: Power Down state ■ 3'b111: Other intermediate states (valid only for 32G PHY configurations) **For non-Synopsys PHY configurations:** Reserved **Value After Reset:** 0x4 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
1	LB_ACTIVE	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	Reserved_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.5.35 VR_MII_1_ICG_ERRCNT1

- **Description:** VR MII_1 MMD Invalid Code Group Error Count1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8011 : (APB3_16) ? (0x8011*0x2) : (0x8011*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** QSGMII_EN

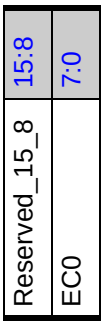


Table 5-370 Fields for Register: VR_MII_1_ICG_ERRCNT1

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:0	EC0	R	Invalid Code Group Count -Port 1 (RO,LH Type) This field gives the invalid code group count in port 1 (of QSGMII) when Bit 4 of VR MII_1 MMD Digital Error Count Select Register is set to 1. **Value After Reset:** 0x0 **Exists:** Always

5.2.5.36 VR_MII_1_GPIO

- **Description:** VR MII_2 MMD GPIO Register
  This register is present only if the DWC_xpcs is configured to have the GPIO interface and QSGMII_EN = Enabled and USXGMI_EN = Disabled.
  This register merely reflects the value of VR_MII_GPIO Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8015 : (APB3_16) ? (0x8015*0x2) : (0x8015*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** QSGMII_EN&&GPIO_EN&&!USXGMII_EN

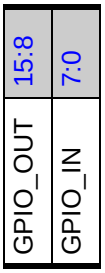


Table 5-371 Fields for Register: VR_MII_1_GPIO

Bits	Name	Memory Access	Description
15:8	GPIO_OUT	R	GPIO Output This field reflects the value of GPIO_OUT field in VR_MII_GPIO register. **Value After Reset:** 0x0 **Exists:** Always
7:0	GPIO_IN	R	GPIO Input This field indicates the content of the xpcs_gpo_i[7:0] port. **Value After Reset:** 0x0 **Exists:** Always

5.2.5.37 VR_MII_1_DIG_CTRL2

- **Description:** VR MII_1 MMD Digital Control2 Register
  Note: This register is present only in 1000BaseX-Only PCS configurations with QSGMII_EN = Enabled.
  This register merely reflects the value of VR_MII_DIG_CTRL2 Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80E1 : (APB3_16) ? (0x80E1*0x2) : (0x80E1*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** QSGMII_EN&&BASE1000X_ONLY

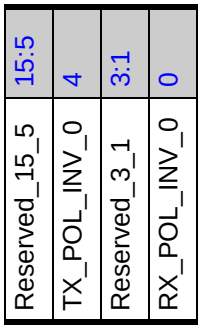


Table 5-372 Fields for Register: VR_MII_1_DIG_CTRL2

Bits	Name	Memory Access	Description
15:5	Reserved_15_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_POL_INV_0	R	Tx Polarity Invert on Lane 0. **Value After Reset:** 0x0 **Exists:** Always
3:1	Reserved_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	RX_POL_INV_0	R	Rx Polarity Invert on Lane 0. **Value After Reset:** 0x0 **Exists:** Always

5.2.5.38 VR_MII_1_DIG_ERRCNT_SEL

- **Description:** VR MII_1 MMD Digital Error Count Select Register
  This register is applicable only in QSGMII mode.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80E2 : (APB3_16) ? (0x80E2*0x2) : (0x80E2*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1A0000 : (APB3_16) ? (0x1A0000*0x2) : (0x1A0000*0x4)"
- **Exists:** QSGMII_EN

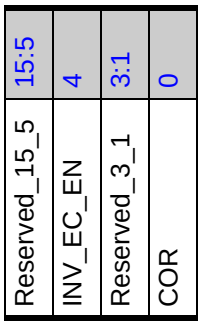


Table 5-373 Fields for Register: VR_MII_1_DIG_ERRCNT_SEL

Bits	Name	Memory Access	Description
15:5	Reserved_15_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	INV_EC_EN	R/W	Invalid Code Group Error Counter Enable When this bit is set, the counting of invalid code group errors (on port 1) is enabled. ■ 0: The counting of errors is disabled. ■ 1: The counting of errors is enabled. For information about the fields containing the number of errors counted, see VR_MII_1_ICG_ERRCNT1 register. **Value After Reset:** 0x0 **Exists:** Always
3:1	Reserved_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	COR	R/W	Clear on Read When this bit is set and the host reads port 1 error counter (VR_MII_1_ICG_ERRCNT1 Register), that counter is cleared after the read cycle. ■ 0: Normal operation ■ 1: Clear error counter that is read **Value After Reset:** 0x0 **Exists:** Always

5.2.6 DWC_xpcs_map/VS_MII_2_MMD Registers

5.2.6.1 SR_MII_2_CTRL

- **Description:** SR MII_2 MMD Control Register
  The host can use this register to control (enable or disable) some of the features supported by the DWC_xpcs (for Port 2).
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** QSGMII_EN | | USXG_QPORT

15	14	13	12	11	10	9	8	7	6	5	4:0
RST	LBE	SS13	AN_ENABLE	LPM	Reserved_10	RESTART_AN	DUPLEX_MODE	Reserved_7	SS6	SS5	Reserved_4_0

Table 5-374 Fields for Register: SR_MII_2_CTRL

Bits	Name	Memory Access	Description
15	RST	R/W	Soft Reset (RW,SC Type) **For Synopsys PHY configurations:** When the host sets this bit, the CSR block triggers the software reset process in which all internal blocks are reset, except the Management Interface block. The registers are reset to their default values. When this bit is set, it also resets the PHY. This bit is self-cleared when PSEQ_STATE in VR_XS_PCS_DIG_STS/VR_MII_DIG_STS Register is equal to 3'b100, that is, Tx/Rx clocks are stable and in Power_Good state. **For Non-Synopsys PHY configurations:** When the host sets this bit, the CSR block triggers the software reset process in which all internal blocks are reset, except the Management Interface block. The registers are reset to their default values. When this bit is set, it also resets the PHY. This bit is self-cleared after the following: ■ 32 clk_csr_i clocks for the MCI interface ■ 1 MDC clock period for the MDIO interface **Note:** For description of read or write access for registers during reset process, see "Special Case Register Access" section of the Databook. **Value After Reset:** 0x0 **Exists:** Always
14	LBE	R	Loopback Enable This bit reflects the value of LBE bit of SR_MII_CTRL Register. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
13	SS13	R/W	Speed Selection (LSB) This bit, along with the SS6 bit and SS5 bit of this register, controls the speed of GMII/XGMII port 2. The speed encoding is as follows: **For USXGMII:** ■ When SS5=1, SS6=0 and SS13=1, speed is 5 Gbps ■ When SS5=1, SS6=0 and SS13=0, speed is 2.5 Gbps ■ When SS5=0, SS6=1 and SS13=1, speed is 10 Gbps ■ When SS5=0, SS6=1 and SS13=0, speed is 1000 Mbps ■ When SS5=0, SS6=0 and SS13=1, speed is 100 Mbps ■ When SS5=0, SS6=0 and SS13=0, speed is 10 Mbps **For QSGMII:** ■ When SS6=1 and SS13=0, speed is 1000 Mbps ■ When SS6=0 and SS13=1, speed is 100 Mbps ■ When SS6=0 and SS13=0, speed is 10 Mbps **Value After Reset:** 0x0 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\"""
12	AN_ENABLE	R/W	Enable Auto-Negotiation When set to 1, this bit enables the Clause 37 auto-negotiation process for port 2. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
11	LPM	R/W	Power-Down Mode This bit controls the power-down mode of the DWC_xpcs. ■ 0: Normal operation ■ 1: The DWC_xpcs goes to the power-down mode along with the PHY. For non-Synopsys PHY, the xpcs_pdown_o port is asserted. For Synopsys PHY, the DWC_xpcs does the following to trigger the power-down mode: ■ Turns off the PHY Receiver and Transmitter ■ Switches off all the clocks When the host clears this bit, the DWC_xpcs resumes the normal operation. In a Synopsys PHY configuration, after clearing this bit, the host must wait until Bits[4:2] of VR_XS_PCS_DIG_STS/VR_MII_DIG_STS Register indicate that the DWC_xpcs is in the normal state. **Note:** For information about read or write access to registers during power-down mode, see "Special Case Register Access" section of the Databook. **Value After Reset:** 0x0 **Exists:** Always
10	Reserved_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9	RESTART_AN	R/W	Restart Auto-Negotiation (RW,SC Type) When the host writes this bit, the DWC_xpcs initiates the auto-negotiation process for port 2. This bit is used to restart the auto-negotiation which is already initiated by setting Bit 12. The DWC_xpcs clears this bit after restarting the auto-negotiation. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
8	DUPLEX_MODE	R/W	Duplex Mode This bit specifies the duplex mode of the DWC_xpcs along GMII port 2. ■ 0: Half duplex ■ 1: Full duplex If Bit 12 is set to 0, this bit determines the PHY link duplex mode. If Bit 12 is set to 1, then the PHY link duplex mode is independent of this bit (although the host can write any value) and is determined by the outcome of the Clause 37 auto-negotiation process. **Note:** USXGMII supports only full duplex mode (irrespective of the value programmed to this field) **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6	SS6	R/W	Speed Selection This bit, along with the SS5 and SS13 bits of this register indicates the speed of operation of GMII/XGMII Port 2. For more information, see description of the SS13 bit. **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
5	SS5	* Varies	Speed Selection This bit, along with SS6 and SS13 bits, control the speed of operation of XGMII/GMII Port 2. For more information, see description of the SS13 bit. This field is valid only in QXGMII (4-port USXGMII) mode. **Value After Reset:** 0x0 **Exists:** Always **Testable:** "(MP_USXGMII) ? \"untestable\" : \"unconstrained\"" **Memory Access:** "USXG_QPORT ? \"read-write\" : \"read-only\""
4:0	Reserved_4_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.6.2 SR_MII_2_STS

- **Description:** SR MII_2 MMD Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1 : (APB3_16) ? (0x1*0x2) : (0x1*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** QSGMII_EN

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ABL100T4	FD100ABL	HD100ABL	FD10ABL	HD10ABL	FD100T	HD100T	EXT_STS_ABL	UN_DIR_ABL	MF_PRE_SUP	AN_CMPL	RF	AN_ABL	LINK_STS	Reserved_1	EXT_REG_CAP

Table 5-375 Fields for Register: SR_MII_2_STS

Bits	Name	Memory Access	Description
15	ABL100T4	R	100BASE-T4 Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always
14	FD100ABL	R	100BASE-X Full-Duplex Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always
13	HD100ABL	R	100BASE-X Half-Duplex Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always
12	FD10ABL	R	10 Mbps Full-Duplex Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
11	HD10ABL	R	10 Mbps Half-Duplex Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always
10	FD100T	R	100BASE-T2 Full-Duplex Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always
9	HD100T	R	100BASE-T2 Half-Duplex Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always
8	EXT_STS_ABL	R	Extended Status Information ■ 0: No Extended Status information is present at register address 16'h000F of this MMD device. ■ 1: Extended Status information is present at register address 16'h000F of this MMD device. DWC_xpcs returns this bit as 1. **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
7	UN_DIR_ABL	R	Unidirectional Ability ■ 1: The DWC_xpcs is able to transmit GMII irrespective of whether device has determined the valid link or not. ■ 0: The DWC_xpcs is able to transmit GMII only when the device has determined the valid link. The DWC_xpcs always returns this bit as 0. **Value After Reset:** 0x0 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""

Bits	Name	Memory Access	Description
6	MF_PRE_SUP	R	MF Preamble Suppression ■ 1: The DWC_xpcs accepts the MDIO frames with preamble suppressed. ■ 0: The DWC_xpcs does not accept the MDIO frames with preamble suppressed. This bit is always set to 0. **Value After Reset:** 0x0 **Exists:** Always
5	AN_CMPL	R	Auto-negotiation Complete for Port 2 ■ 1: The AN process is complete ■ 0: The AN process is not complete This bit returns 0 if AN_ENABLE of SR_MII_1_CTRL Register is set to 0. **Value After Reset:** 0x0 **Exists:** Always
4	RF	R	Remote Fault This register bit is shared with RF bit of SR_MII_STS register. Software must ignore the value returned by this field, as it is not valid in QSGMII/SGMII/USXGMII Modes. **Value After Reset:** 0x0 **Exists:** Always
3	AN_ABL	R	Auto-negotiation Ability The DWC_xpcs always returns this bit as 1. ■ 1: The DWC_xpcs is able to perform auto-negotiation. ■ 0: The DWC_xpcs is not able to perform auto-negotiation. **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
2	LINK_STS	R	Link Status This register bit is shared with RLU bit of SR_MII_STS register. **Value After Reset:** 0x0 **Exists:** Always **Testable:** "QSGMII_EN ? \"untestable\" : \"unconstrained\""
1	Reserved_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	EXT_REG_CAP	R	Extended Register Capability ■ 1: Extended Register capability exists. This bit is always set to 1. ■ 0: Extended Register capability does not exist. **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""

5.2.6.3 SR_MII_2_DEV_ID1

- **Description:** SR MII_2 MMD Device Identifier Register 1
  This register returns the configurable Organizationally Unique Identifier (OUI) to the host.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x2 : (APB3_16) ? (0x2*0x2) : (0x2*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** QSGMII_EN||USXG_QPORT



Table 5-376 Fields for Register: SR_MII_2_DEV_ID1

Bits	Name	Memory Access	Description
15:0	VS_MII_DEV_OUI_3_18	R	Organizationally Unique Identifier[3:18] This field contains Bits [18:3] of 24-bit OUI of device manufacturer. The DWC_xpcs offers 24 configurable Bits[24:1] for identifying the device manufacturer. This register reflects the same value as that of SR_MII_DEV_ID1 register. **Value After Reset:** IEEE_DEV_OUI_3_18 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""

5.2.6.4 SR_MII_2_DEV_ID2

- **Description:** SR MII_2 MMD Device Identifier Register 2
  This register returns the configurable Organizationally Unique Identifier (OUI), Model Number, and Revision Number of the device to the host.
  This register returns the same value as that of SR_MII_DEV_ID2 Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x3 : (APB3_16) ? (0x3*0x2) : (0x3*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** QSGMII_EN | |USXG_QPORT

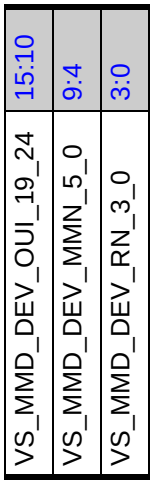


Table 5-377 Fields for Register: SR_MII_2_DEV_ID2

Bits	Name	Memory Access	Description
15:10	VS_MMD_DEV_OUI_19_24	R	Organizationally Unique Identifier [19:24] This field contains Bits[24:19] of 24-bit OUI of device manufacturer. The DWC_xpcs offers 24 configurable Bits[24:1] for identifying the device manufacturer. **Value After Reset:** IEEE_DEV_OUI_19_24 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
9:4	VS_MMD_DEV_MMN_5_0	R	Model Number This field contains the 5-bit Model Number of the device. **Value After Reset:** IEEE_DEV_MODEL_NO_0_5 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""

Bits	Name	Memory Access	Description
3:0	VS_MMD_DEV_RN_3_0	R	Revision Number This field contains the 4-bit Revision Number of the device. **Value After Reset:** IEEE_DEV_REV_NO_0_3 **Exists:** Always

5.2.6.5 SR_MII_2_AN_ADV

- **Description:** SR MII_2 MMD AN Advertisement Register
  Except for bit[5], rest of the fields of this register reflects the corresponding fields of SR_MII_AN_ADV Register.
  Note: This register is valid only when the current operating mode is QSGMII.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x4 : (APB3_16) ? (0x4*0x2) : (0x4*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** QSGMII_EN

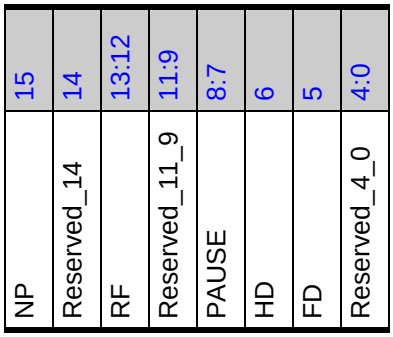


Table 5-378 Fields for Register: SR_MII_2_AN_ADV

Bits	Name	Memory Access	Description
15	NP	R	Next Page This bit reflects the value of NP bit of SR_MII_AN_ADV Register. **Value After Reset:** 0x0 **Exists:** Always
14	Reserved_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:12	RF	R	Remote Fault This bit reflects the value of RF bit of SR_MII_AN_ADV Register. **Value After Reset:** 0x0 **Exists:** Always
11:9	Reserved_11_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
8:7	PAUSE	R	Pause Ability This bit reflects the value of PAUSE bit of SR_MII_AN_ADV Register. **Value After Reset:** 0x0 **Exists:** Always
6	HD	R	Half Duplex. This bit reflects the value of NP bit of SR_MII_AN_ADV Register. **Value After Reset:** 0x0 **Exists:** Always
5	FD	R/W	Full Duplex When this bit is set, it indicates the device (port 2) can operate in full-duplex mode. This bit is used in Clause 37 auto-negotiation (Tx_Config_Reg) done by port 2, when DWC_xpcs is configured as PHY-side QSGMII. **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
4:0	Reserved_4_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.6.6 SR_MII_2_LP_BABL

- **Description:** SR MII_2 MMD AN Link Partner Base Ability Register
  This register merely reflects the value of SR_MII_LP_BABL Register and is not applicable when working in QSGMII/USXGMII Mode.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x5 : (APB3_16) ? (0x5*0x2) : (0x5*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** QSGMII_EN

15	14	13:12	11:9	8:7	6	5	4:0
LP_NP	LP_ACK	LP_RF	Reserved_11_9	LP_PAUSE	LP_HD	LP_FD	Reserved_4_0

Table 5-379 Fields for Register: SR_MII_2_LP_BABL

Bits	Name	Memory Access	Description
15	LP_NP	R	Next Page **Value After Reset:** 0x0 **Exists:** Always
14	LP_ACK	R	ACK bit from the Link Partner **Value After Reset:** 0x0 **Exists:** Always
13:12	LP_RF	R	Remote Fault **Value After Reset:** 0x0 **Exists:** Always
11:9	Reserved_11_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8:7	LP_PAUSE	R	Pause Ability **Value After Reset:** 0x0 **Exists:** Always
6	LP_HD	R	Half Duplex **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
5	LP_FD	R	Full Duplex **Value After Reset:** 0x0 **Exists:** Always
4:0	Reserved_4_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.6.7 SR_MII_2_AN_EXPN

- **Description:** SR MII_2 MMD AN Expansion Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x6 : (APB3_16) ? (0x6*0x2) : (0x6*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** QSGMII_EN||USXG_QPORT

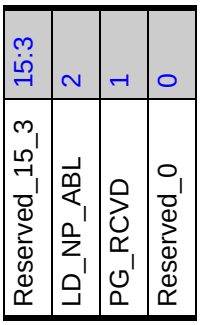


Table 5-380 Fields for Register: SR_MII_2_AN_EXPN

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2	LD_NP_ABL	R	Local Device NP Able ■ 1: The local device has the next page ability ■ 0: The local device does not have the next page ability The DWC_xpcs always returns this bit as 0 because it does not support Next Page. **Value After Reset:** 0x0 **Exists:** Always
1	PG_RCVD	R	Page Received (RO, LH Type) This bit indicates that the local device received a page from the link partner (during port 2 auto-negotiation). ■ 1: The local device received a new page ■ 0: The local device did not receive a new page **Value After Reset:** 0x0 **Exists:** Always
0	Reserved_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.6.8 SR_MII_2_EXT_STS

- **Description:** SR MII_2 MMD Extended Status Register
  This register reflects the same value as that of SR_MII_EX_STS register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0F : (APB3_16) ? (0x0F*0x2) : (0x0F*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** QSGMII_EN

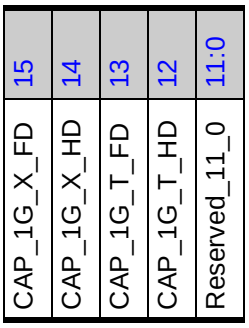


Table 5-381 Fields for Register: SR_MII_2_EXT_STS

Bits	Name	Memory Access	Description
15	CAP_1G_X_FD	R	1000BASE-X Full-Duplex Capable **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
14	CAP_1G_X_HD	R	1000BASE-X Half-Duplex Capable **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
13	CAP_1G_T_FD	R	1000BASE-T Full-Duplex Capable **Value After Reset:** 0x0 **Exists:** Always
12	CAP_1G_T_HD	R	1000BASE-T Half-Duplex Capable **Value After Reset:** 0x0 **Exists:** Always
11:0	Reserved_11_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.6.9 SR_MII_2_TIME_SYNC_PCS_ABL

- **Description:** SR MII 2 MMD Time Sync Capability Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0708 : (APB3_16) ? (0x0708*0x2) : (0x0708*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** (QSGMII_EN||USXG_QPORT)

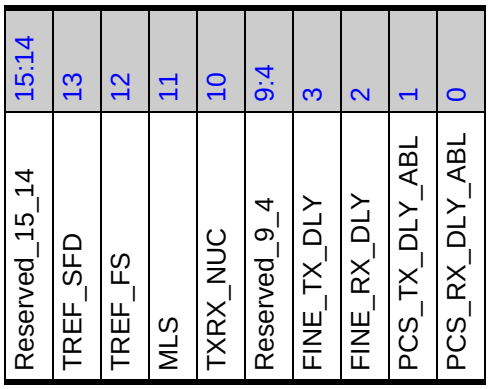


Table 5-382 Fields for Register: SR_MII_2_TIME_SYNC_PCS_ABL

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13	TREF_SFD	R	Timestamp reference at SFD - 0: DWC_xpcs does not support data delay measurement point at the beginning of SFD - 1: DWC_xpcs supports data delay measurement point at the beginning of SFD **Value After Reset:** "SNPS_RSVDPARAM_11 ? 0x1 : 0x0" **Exists:** Always
12	TREF_FS	R	Timestamp reference at first symbol after SFD - 0: DWC_xpcs does not support data delay measurement point at the beginning of the first symbol after SFD - 1: DWC_xpcs supports data delay measurement point at the beginning of the first symbol after SFD **Value After Reset:** "SNPS_RSVDPARAM_11 ? 0x1 : 0x0" **Exists:** Always

Bits	Name	Memory Access	Description
11	MLS	R	Multi-lane Support. ■ 0: DWC_xpcs does not support reporting of multi-PCS lane path data delay ■ 1: DWC_xpcs supports reporting of multi-PCS lane path data delay **Value After Reset:** "SNPS_RSVDPARAM_11&&!SINGLE_LANE_CFG ? 0x1 : 0x0" **Exists:** Always
10	TXRX_NUC	R	Transmit or Receive Num_Unit_Change Support Available for Port 2. If this bit is set, it indicates that DWC_xpcs supports Transmit/Receive Num_Unit_Change indication capability **Value After Reset:** "SNPS_RSVDPARAM_11 ? 0x1 : 0x0" **Exists:** Always
9:4	Reserved_9_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3	FINE_TX_DLY	R	Fine Resolution Transmit Path Data Delay Available for Port 2. If this bit is set, it indicates that DWC_xpcs provides information on transmit data path delay with sub-ns resolution in SR_MII_2_TIME_SYNC_TX_MAX_FINE_DLY and SR_MII_2_TIME_SYNC_TX_MIN_FINE_DLY registers. **Value After Reset:** "SNPS_RSVDPARAM_11 ? 0x1 : 0x0" **Exists:** Always
2	FINE_RX_DLY	R	Fine Resolution Receive Path Data Delay Available for Port 2. If this bit is set, it indicates that DWC_xpcs provides information on receive data path delay with sub-ns resolution in SR_MII_2_TIME_SYNC_RX_MAX_FINE_DLY and SR_MII_2_TIME_SYNC_RX_MIN_FINE_DLY registers. **Value After Reset:** "SNPS_RSVDPARAM_11 ? 0x1 : 0x0" **Exists:** Always
1	PCS_TX_DLY_ABL	R	PCS Transmit Path Data Delay Information Available for Port 2. If this bit is set, it indicates that the information regarding the maximum and minimum transmit data delay of PCS is available in ns-resolution in MII_2 MMD Rx Time Delay Registers. **Value After Reset:** 0x1 **Exists:** Always

Bits	Name	Memory Access	Description
0	PCS_RX_DLY_ABL	R	PCS Receive Path Data Delay Information Available for Port 2. If this bit is set, it indicates that the information regarding the maximum and minimum receive data delay of PCS is available in ns-resolution in MII_2 MMD Rx Time Delay Registers. **Value After Reset:** 0x1 **Exists:** Always

5.2.6.10 SR_MII_2_TIME_SYNC_TX_MAX_DLY_LWR

- **Description:** SR MII 2 MMD Time Sync Tx Max Delay Lower Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x0709 : (APB3_16) ? (0x0709*0x2) : (0x0709*0x4)" + "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** (QSGMII_EN | | USXG_QPORT)

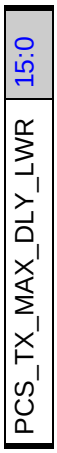


Table 5-383 Fields for Register: SR_MII_2_TIME_SYNC_TX_MAX_DLY_LWR

Bits	Name	Memory Access	Description
15:0	PCS_TX_MAX_DLY_LWR	R	PCS Transmit Max Delay Lower along Port 2 This field indicates the lower 16-bit of the 32-bit value which indicates integer nano-seconds portion of the Maximum Data Delay in the Transmit Path of DWC_xpcs along Port 2. The value returned by this field depends on the current operating mode. If TS_MODE bit is 0, this register indicates the end-to-end (MAC interface to PCS interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the transmit path from the DDMP point until the final Tx data output port. **Value After Reset:** 0x0 **Exists:** Always

5.2.6.11 SR_MII_2_TIME_SYNC_TX_MAX_DLY_UPR

- **Description:** SR MII 2 MMD Time Sync Tx Max Delay Upper Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070A : (APB3_16) ? (0x070A*0x2) : (0x070A*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** (QSGMII_EN||USXG_QPORT)



Table 5-384 Fields for Register: SR_MII_2_TIME_SYNC_TX_MAX_DLY_UPR

Bits	Name	Memory Access	Description
15:0	PCS_TX_MAX_DLY_UPR	R	PCS Transmit Max Delay Upper along Port 2 This field indicates the upper 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Maximum Data Delay in the Transmit Path of DWC_xpcs along Port 2. If TS_MODE bit is 0, this register indicates the end-to-end (MAC interface to PCS interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the transmit path from the DDMP point until the final Tx data output port **Value After Reset:** 0x0 **Exists:** Always

5.2.6.12 SR_MII_2_TIME_SYNC_TX_MIN_DLY_LWR

- **Description:** SR MII 2 MMD Time Sync Tx Min Delay Lower Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070B : (APB3_16) ? (0x070B*0x2) : (0x070B*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** (QSGMII_EN||USXG_QPORT)



Table 5-385 Fields for Register: SR_MII_2_TIME_SYNC_TX_MIN_DLY_LWR

Bits	Name	Memory Access	Description
15:0	PCS_TX_MIN_DLY_LWR	R	PCS Transmit Min Delay Lower along Port 2 This field indicates the lower 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Minimum Data Delay in the Transmit Path of DWC_xpcs along Port 2. The value returned by this register field depends on the current operating mode. If TS_MODE bit is 0, this register indicates the end-to-end (MAC interface to PCS interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the transmit path from the DDMP point until the final Tx data output port. **Value After Reset:** 0x0 **Exists:** Always

5.2.6.13 SR_MII_2_TIME_SYNC_TX_MIN_DLY_UPR

- **Description:** SR MII 2 MMD Time Sync Tx Min Delay Upper Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070C : (APB3_16) ? (0x070C*0x2) : (0x070C*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** (QSGMII_EN||USXG_QPORT)



Table 5-386 Fields for Register: SR_MII_2_TIME_SYNC_TX_MIN_DLY_UPR

Bits	Name	Memory Access	Description
15:0	PCS_TX_MIN_DLY_UPR	R	PCS Transmit Min Delay Upper along Port 2 This field indicates the upper 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Minimum Data Delay in the Transmit Path of DWC_xpcs along Port 2. If TS_MODE bit is 0, this register indicates the end-to-end (MAC interface to PCS interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the transmit path from the DDMP point until the final Tx data output port **Value After Reset:** 0x0 **Exists:** Always

5.2.6.14 SR_MII_2_TIME_SYNC_RX_MAX_DLY_LWR

- **Description:** SR MII 2 MMD Time Sync Rx Max Delay Lower Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070D : (APB3_16) ? (0x070D*0x2) : (0x070D*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** (QSGMII_EN||USXG_QPORT)

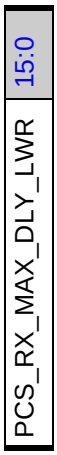


Table 5-387 Fields for Register: SR_MII_2_TIME_SYNC_RX_MAX_DLY_LWR

Bits	Name	Memory Access	Description
15:0	PCS_RX_MAX_DLY_LWR	R	PCS Receive Max Delay Lower along Port 2 This field indicates the lower 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Maximum Data Delay in the Receive Path of DWC_xpcs.The value returned by this register field depends on the current operating mode and is valid after Rx link-up is attained. If TS_MODE bit is 0, this register indicates the end-to-end (PHY interface to MAC interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the receive path from Rx data input port until the DDMP point **Value After Reset:** 0x0 **Exists:** Always

5.2.6.15 SR_MII_2_TIME_SYNC_RX_MAX_DLY_UPR

- **Description:** SR MII 2 MMD Time Sync Rx Max Delay Upper Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070E : (APB3_16) ? (0x070E*0x2) : (0x070E*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** (QSGMII_EN||USXG_QPORT)



Table 5-388 Fields for Register: SR_MII_2_TIME_SYNC_RX_MAX_DLY_UPR

Bits	Name	Memory Access	Description
15:0	PCS_RX_MAX_DLY_UPR	R	PCS Receive Max Delay Upper along Port 2 This field indicates the upper 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Maximum Data Delay in the Receive Path of DWC_xpcs along Port 2. If TS_MODE bit is 0, this register indicates the end-to-end (PHY interface to MAC interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the receive path from Rx data input port until the DDMP point **Value After Reset:** 0x0 **Exists:** Always

5.2.6.16 SR_MII_2_TIME_SYNC_RX_MIN_DLY_LWR

- **Description:** SR MII 2 MMD Time Sync Rx Min Delay Lower Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070F : (APB3_16) ? (0x070F*0x2) : (0x070F*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** (QSGMII_EN||USXG_QPORT)



Table 5-389 Fields for Register: SR_MII_2_TIME_SYNC_RX_MIN_DLY_LWR

Bits	Name	Memory Access	Description
15:0	PCS_RX_MIN_DLY_LWR	R	PCS Receive Min Delay Lower along Port 2 This field indicates the lower 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Minimum Data Delay in the Receive Path of DWC_xpcs.The value returned by this register field depends on the current operating mode and is valid after Rx link-up is attained. If TS_MODE bit is 0, this register indicates the end-to-end (PHY interface to MAC interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the receive path from Rx data input port until the DDMP point. **Value After Reset:** 0x0 **Exists:** Always

5.2.6.17 SR_MII_2_TIME_SYNC_RX_MIN_DLY_UPR

- **Description:** SR MII 2 MMD Time Sync Rx Min Delay Upper Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0710 : (APB3_16) ? (0x0710*0x2) : (0x0710*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** (QSGMII_EN||USXG_QPORT)



Table 5-390 Fields for Register: SR_MII_2_TIME_SYNC_RX_MIN_DLY_UPR

Bits	Name	Memory Access	Description
15:0	PCS_RX_MIN_DLY_UPR	R	PCS Receive Min Delay Upper along Port 2 This field indicates the upper 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Minimum Data Delay in the Receive Path of DWC_xpcs. If TS_MODE bit is 0, this register indicates the end-to-end (PHY interface to MAC interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the receive path from Rx data input port until the DDMP point **Value After Reset:** 0x0 **Exists:** Always

5.2.6.18 SR_MII_2_TIME_SYNC_TX_MAX_FINE_DLY

- **Description:** SR MII 2 MMD Time Sync Tx Max Fine Delay Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0711 : (APB3_16) ? (0x0711*0x2) : (0x0711*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** ((QSGMII_EN||USXG_QPORT)&&SNPS_RSVDPARAM_11)

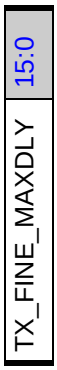


Table 5-391 Fields for Register: SR_MII_2_TIME_SYNC_TX_MAX_FINE_DLY

Bits	Name	Memory Access	Description
15:0	TX_FINE_MAXDLY	R	Transmit Maximum Fine Delay along Port 2 This register indicates the sub-nano-seconds portion of the maximum delay inside the DWC_xpcs transmit path from the DDMP point until the final transmit output port. The delay is indicated in units of 2^-16 ns. Contents of this register is valid only when TS_MODE bit is programmed to 1. Default value of this register is 0, as reset value of TS_MODE bit is 0 **Value After Reset:** 0x0 **Exists:** Always

5.2.6.19 SR_MII_2_TIME_SYNC_TX_MIN_FINE_DLY

- **Description:** SR MII 2 MMD Time Sync Tx Min Fine Delay Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0712 : (APB3_16) ? (0x0712*0x2) : (0x0712*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** ((QSGMII_EN||USXG_QPORT)&&SNPS_RSVDPARAM_11)



Table 5-392 Fields for Register: SR_MII_2_TIME_SYNC_TX_MIN_FINE_DLY

Bits	Name	Memory Access	Description
15:0	TX_FINE_MINDLY	R	Transmit Minimum Fine Delay along Port 2 This register indicates the sub-nano-seconds portion of the minimum delay inside the DWC_xpcs transmit path from the DDMP point until the final transmit output port. The delay is indicated in units of 2^-16 ns. Contents of this register is valid only when TS_MODE bit is programmed to 1. Default value of this register is 0, as reset value of TS_MODE bit is 0 **Value After Reset:** 0x0 **Exists:** Always

5.2.6.20 SR_MII_2_TIME_SYNC_RX_MAX_FINE_DLY

- **Description:** SR MII 2 MMD Time Sync Rx Max Fine Delay Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0713 : (APB3_16) ? (0x0713*0x2) : (0x0713*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** ((QSGMII_EN||USXG_QPORT)&&SNPS_RSVDPARAM_11)

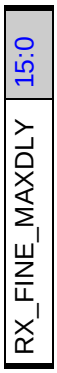


Table 5-393 Fields for Register: SR_MII_2_TIME_SYNC_RX_MAX_FINE_DLY

Bits	Name	Memory Access	Description
15:0	RX_FINE_MAXDLY	R	Receive Maximum Fine Delay along Port 2 This register indicates the sub-nano-seconds portion of the maximum delay inside the DWC_xpcs receive path from Rx data input port until the DDMP point. The delay is indicated in units of 2^-16 ns. Contents of this register is valid only when TS_MODE bit is programmed to 1. Default value of this register is 0, as reset value of TS_MODE bit is 0. **Value After Reset:** 0x0 **Exists:** Always

5.2.6.21 SR_MII_2_TIME_SYNC_RX_MIN_FINE_DLY

- **Description:** SR MII 2 MMD Time Sync Rx Min Fine Delay Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0714 : (APB3_16) ? (0x0714*0x2) : (0x0714*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** ((QSGMII_EN||USXG_QPORT)&&SNPS_RSVDPARAM_11)



Table 5-394 Fields for Register: SR_MII_2_TIME_SYNC_RX_MIN_FINE_DLY

Bits	Name	Memory Access	Description
15:0	RX_FINE_MINDLY	R	Receive Minimum Fine Delay along Port 2 This register indicates the sub-nano-seconds portion of the minimum delay inside the DWC_xpcs receive path from Rx data input port until the DDMP point. The delay is indicated in units of 2^-16 ns. Contents of this register is valid only when TS_MODE bit is programmed to 1. Default value of this register is 0, as reset value of TS_MODE bit is 0. **Value After Reset:** 0x0 **Exists:** Always

5.2.6.22 SR_MII_2_TIME_SYNC_CFG

- **Description:** SR MII 2 MMD Time Sync Configuration Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0715 : (APB3_16) ? (0x0715*0x2) : (0x0715*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** ((QSGMII_EN&&USXG_QPORT)&&SNPS_RSVDPARAM_11)

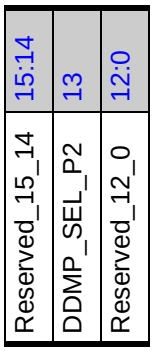


Table 5-395 Fields for Register: SR_MII_2_TIME_SYNC_CFG

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13	DDMP_SEL_P2	R/W	Data Delay Measurement Point Selection on Port 2 ■ 0: DWC_xpcs is configured to use the data delay measurement point at the beginning of SFD ■ 1: DWC_xpcs is configured to use the data delay measurement point at the beginning of the first symbol after SFD **Value After Reset:** 0x0 **Exists:** Always
12:0	Reserved_12_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.6.23 VR_MII_2_DIG_CTRL1

- **Description:** VR MII_2 MMD Digital Control1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8000 : (APB3_16) ? (0x8000*0x2) : (0x8000*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** QSGMII_EN||USXG_QPORT

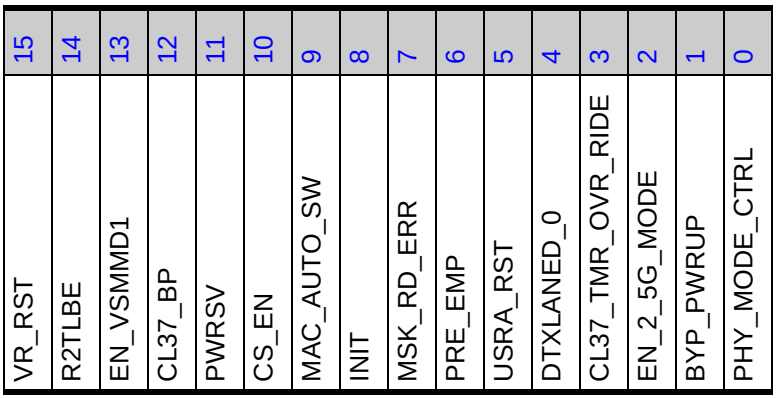


Table 5-396 Fields for Register: VR_MII_2_DIG_CTRL1

Bits	Name	Memory Access	Description
15	VR_RST	R/W	Vendor-Specific Soft Reset (RW,SC Type) When the host sets this bit, the CSR block triggers the vendor-specific software reset process in which all internal blocks, except the Management Interface block and CSR block, are reset. When this bit is set, it also resets the PHY. This bit is self cleared under the following conditions: **For Synopsys PHY:** This bit is self cleared when Bits[4:2] in VR_XS_PCS_DIG_STS Register are equal to 3'b100, that is, Tx/Rx clocks are stable and in Power_Good state. **For Non-Synopsys PHY:**This bit is self cleared after the following: ■ 32 clk_csr_i clocks for the MCI interface ■ 1 MDC clock period for the MDIO interface **Note:** For information about the read or write access for any register during the reset process, see "Special Case Register Access" section. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
14	R2TLBE	R/W	Rx to Tx Loopback Enable This bit controls the loopback path from the GMII/XGMII Rx to the GMII/XGMII Tx at the GMII/XGMII interface (port 2). ■ 0: Loopback path is disabled ■ 1: Loopback path is enabled **Value After Reset:** 0x0 **Exists:** Always
13	EN_VSMMD1	R	Enable Vendor-Specific MMD1 This is a read-only bit which reflects the value of EN_VSMMD1 bit of VR_MII_DIG_CTRL1 Register. **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
12	CL37_BP	R	Enable Clause 37 AN in Backplane Configuration This is a read-only bit which reflects the value of CL37_BP bit of VR_MII_DIG_CTRL1 Register. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
11	PWRSV	* Varies	**For Synopsys PHY configurations:** Power Save This bit is present only when the DWC_xpcs is configured for Synopsys PHY. ■ 0: Normal operation. ■ 1: DWC_xpcs and the PHY enter the power-save mode. When this bit is set, the DWC_xpcs triggers the power-down mode by turning off the PHY Receiver and Transmitter, and without turning off MPLL. For Synopsys Multi-protocol 12G/16G PHY, DWC_xpcs asserts xpcs_tx_lpd_o{lane} and xpcs_rx_lpd_o{lane} when this bit is set. When the host clears this bit, DWC_xpcs resumes normal operation. After clearing this bit, the host must wait until Bits[4:2] of the VR_XS_PCS_DIG_STS/VR_MII_DIG_STS Register indicate that the DWC_xpcs is in a normal state. **Note:** For a description of read or write access for any register while this bit is set, see "Special Case Register Access" section of the Databook. **For non-Synopsys PHY configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_XAUI_PHY) ? \"read-write\" : \"read-only\""
10	CS_EN	* Varies	Clock Stop Enable ■ 1: The PHY might stop the clock during LPI mode ■ 0: The clock cannot be stopped during LPI mode You must program this bit based on the capability of the MAC (connected to GMII Port 2) during Rx LPI mode. **Note:** This field is applicable only in QSGMII mode. **Value After Reset:** "EEE_EN&&QSGMII_EN ? \"0x1\" : \"0x0\"" **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\"" **Memory Access:** "EEE_EN&&QSGMII_EN ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
9	MAC_AUTO_SW	R/W	Automatic Speed Mode Change after CL37 AN (for port 2) If this field is set to 1, DWC_xpcs automatically switches to the negotiated USXGMII/QSGMII(port2) speed, after the completion of Clause 37 auto-negotiation. This mode is valid only when DWC_xpcs is configured as MAC-side USXGMII/QSGMII and must be set only when Auto-negotiation is enabled (AN_ENABLE bit is set to 1). If this bit is set to 0, DWC_xpcs operates at the speed/duplex mode as per the values programmed to SR_MII_2_CTRL Register. In that case, after the completion of CL37 AN, application has to read the negotiated Speed/Duplex Mode from VR_MII_2_AN_INTR_STS Register and then program SR_MII_2_CTRL Register appropriately. If this bit is set to 1 in QSGMII mode, xpcs_qsgmii_link_sts_p2_o, xpcs_link_speed_p2_o and xpcs_qsgmii_full_duplex_p2_o outputs reflect the auto-negotiated values, that is, values from CL37_ANSGM_STS field of VR_MII_2_AN_INTR_STS Register. For USXGMII mode, if clk_xgmii_tx_p2_i and clk_xgmii_rx_p2_i do not stabilize at the new operating frequency (based on the selected speed) immediately after the completion of auto-negotiation, then software might need to program 'USRA_RST' bit prior to starting packet transfer in the new speed mode. **Note:** Set this bit only when DWC_xpcs is configured as USXGMII/QSGMII MAC, that is, TX_CONFIG = 0 **Note:** Whenever this bit is set, the values programmed to SS5,SS6, and SS13 bits are ignored. Moreover, the values of these bits do not change based on auto-negotiation outcome. **Value After Reset:** 0x0 **Exists:** Always
8	INIT	R	INIT This bit reflects the value of INIT bit of VR_MII_DIG_CTRL1 Register. **Value After Reset:** 0x0 **Exists:** Always
7	MSK_RD_ERR	R	Mask Running Disparity Error. This bit reflects the value of MSK_RD_ERR bit of VR_MII_DIG_CTRL1 Register. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
6	PRE_EMP	* Varies	Pre-emption Packet Enable. Set this bit to 1, to allow the DWC_xpcs to properly receive/transmit preemption packets along GMII port 2, when operating in 10M/100M Modes. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "QSGMII_EN ? \"read-write\" : \"read-only\""
5	USRA_RST	* Varies	USXGMII Rate Adapter Reset (Port 2) This bit can be set to 1 to reset/initialize the USXGMII Rate Adapter Logic (for port 2) inside DWC_xpcs. The Rate Adapter Logic maps GMII data to XGMII (and vice-versa in Rx path) data and perform data replication for lower USXGMII speeds. This is a self-clear bit (RW,SC) and clears itself after the reset of Tx and Rx Rate Adapter Logic is complete. After setting this bit to 1, application must poll this bit till it self-clears. **Note:** If CLKCOMP = Enabled, this bit also initializes the clock-compensation FIFO (corresponding to port 2). When this bit is programmed to 1, RXFIFO_OVF/RXFIFO_UNF bits of VR_MII_2_DIG_STS Register might get set incorrectly. Therefore, read these register bits (RXFIFO_OVF and RXFIFO_UNF) so that they get cleared. Thereafter, RXFIOF_UNF and RXFIFO_UNF bits are reliable. **Value After Reset:** 0x0 **Exists:** Always **Testable:** "(MP_USXGMII) ? \"untestable\" : \"unconstrained\"" **Memory Access:** "USXG_QPORT ? \"read-write\" : \"read-only\""
4	DTXLANED_0	R	Tx Lane 0 Disable This bit reflects the value of DTXLANED_0 bit of VR_MII_DIG_CTRL1 Register. **Value After Reset:** 0x0 **Exists:** Always
3	CL37_TMR_OVR_RIDE	R	Over-Ride Control for CL37 Link Timer. This bit reflects the value of CL37_TMR_OVR_RIDE bit of VR_MII_DIG_CTRL1 Register. **Value After Reset:** 0x0 **Exists:** Always
2	EN_2_5G_MODE	R	Enable 2.5G GMII Mode. This bit reflects the value of EN_2_5G_MODE bit of VR_MII_DIG_CTRL1 Register. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
1	BYP_PWRUP	* Varies	**For Synopsys PHY configurations:** Bypass Power-Up Sequence When this bit is set to 1, the DWC_xpcs bypasses the normal flow of the power-up sequence and reaches the Power_Good state to enable transmission or reception. When this bit is set, the DWC_xpcs does not wait for the MPLL and Transmit or Receive PLL status from the Synopsys PHY. You can use this feature in the following scenarios: ■ When the DWC_xpcs is configured to interface with a specific Synopsys PHY. ■ When the data path needs to be interfaced to some other PHY. When this bit is set to 0, the DWC_xpcs waits for the MPLL, Tx, or Rx PLL status from the Synopsys PHY before resuming the normal transmission and reception. **Note:** This bit is shared with Bit 1 of following registers: ■ VR_XS_PCS_DIG_CTRL1 Register ■ VR_AN_DIG_CTRL1 Register ■ VR_PMA_DIG_CTRL1 Register ■ VR MII/MII_2/MII_3_DIG_CTRL1 Register **Access Type:** If SNPS_XAUI_PHY= Enabled, this bit is RW. Otherwise, this bit is RO. **For non-Synopsys PHY configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_XAUI_PHY) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
0	PHY_MODE_CTRL	* Varies	QSGMII/USXGMII PHY mode control on Port 2 This bit controls the Clause 37 auto-negotiation when operating in QSGMII/USXGMII (Port 2) PHY mode. **QSGMI**: When this bit is set to 1, DWC_xpcs advertises the values of input ports xpcs_qsgmii_link_sts_p2_i, xpcs_qsgmii_link_speed_p2_i and xpcs_qsgmii_full_duplex_p2_i during QSGMII(Port 2) auto-negotiation. When this bit is set to 0, QSGMII(Port 2) auto-negotiation advertises the values programmed to: ■ bit 4 (SGMII_LINK_STS) of VR_MII_2_AN_CTRL Register ■ bit 13 (ss13) and 6 (ss6) of SR_MII_2_CTRL Register and ■ bit 5 (FD) of SR_MII_2_AN_ADV Register **USXGMII**: When this bit is set to 1, DWC_xpcs advertises the values of input ports xpcs_usxg_link_sts_p2_i, xpcs_usxg_link_speed_p2_i,xpcs_usxg_full_duplex_p2_i,xpcs_usxg_eee_abl_p2_i and xpcs_usxg_ck_stp_abl_p2_i during USXGMII (port 2) auto-negotiation. When this bit is set to 0, USXGMI (Port 2) auto-negotiation advertises the following : ■ bit 4 (SGMII_LINK_STS) of VR_MII_2_AN_CTRL Register ■ bit 13 (ss13) and 6 (ss6) and 5 (ss5) of SR_MII_2_CTRL Register and ■ Value of 1 for Duplex-Mode ■ Value of 1 for EEE-capability, if EEE = Enabled; Value of 0 if EEE_EN = Disabled ■ Value of 1 for EEE clock-stop capability,if EEE = Enabled; Value of 0 if EEE_EN = Disabled **Note:** This field is present only for the configurations satisfying either of the following conditions: ■ QSGMII_PHY_AN_AUTO_RESTART= Enabled ■ Quad-port USXGMII configurations with USXG_-PHY_AN_AUTO_RESTART = Enabled **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(QSGMII_PHY_AN_AUTO_RESTART||(USXG_QPORT&&USXG_PHY_AN_AUTO_RESTART)) ? \"read-write\" : \"read-only\""

5.2.6.24 VR_MII_2_AN_CTRL

- **Description:** VR MII_2 MMD AN Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8001 : (APB3_16) ? (0x8001*0x2) : (0x8001*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** QSGMII_EN||USXG_QPORT

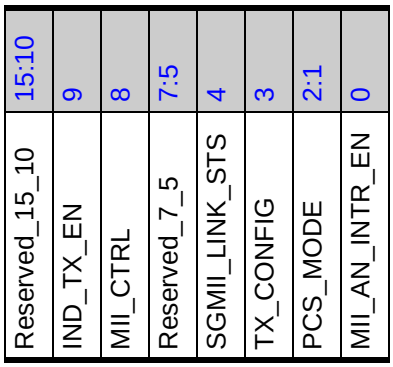


Table 5-397 Fields for Register: VR_MII_2_AN_CTRL

Bits	Name	Memory Access	Description
15:10	Reserved_15_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9	IND_TX_EN	* Varies	Independent Transmit Enable on port 2 during QSGMII mode. If this bit is set to 1, DWC_xpcs is able to transmit the port 2 GMII Tx data, irrespective of its receive link status during QSGMII mode (provided auto-negotiation is not enabled). If this bit is set to 0, DWC_xpcs sends IDLE (for port 2) till its receiver has attained synchronization. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "QSGMII_EN ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
8	MII_CTRL	R/W	MII Control This bit controls the width of the MAC interface (connected to GMII port 2) when operating at 10 Mbps or 100 Mbps ■ 0: 4-bit MII ■ 1: 8-bit MII This bit also controls the xpcs_mii_ctrl_p2_o signal which is used for external clock multiplexing of the clk_mii_tx_p2_i/clk_xgmii_tx_p2_i and clk_mii_rx_p2_i/clk_xgmii_rx_p2_i signals. **Value After Reset:** 0x0 **Exists:** Always
7:5	Reserved_7_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	SGMII_LINK_STS	R/W	QSGMII/USXGMII Link Status (port 2) This bit is used in Bit 15 of the Tx_Config_Reg during Clause 37 auto-negotiation (along port 2) when DWC_xpcs is programmed as QSGMII-PHY (or USXGMII-PHY) and when PHY_MODE_CTRL bit of VR_MII_2_DIG_CTRL1 Register is 0 . ■ 0: Link Down ■ 1: Link Up **Value After Reset:** 0x0 **Exists:** Always
3	TX_CONFIG	R	Transmit Configuration This field reflects the value of TX_CONFIG of VR_MII_AN_CTRL Register. **Value After Reset:** 0x0 **Exists:** Always
2:1	PCS_MODE	R	PCS Mode This field reflects the value of PCS_MODE of VR_MII_AN_CTRL Register for configurations with QSGMII support. For other configurations, this field returns 2'b00.. **Value After Reset:** 0x0 **Exists:** Always
0	MII_AN_INTR_EN	R/W	Clause 37 AN Complete Interrupt Enable (port 2) When set to 1, this bit enables the generation of Clause 37 auto-negotiation complete (on port 2) interrupt output. When set to 0, it disables the generation of Clause 37 auto-negotiation complete interrupt. **Value After Reset:** 0x0 **Exists:** Always

5.2.6.25 VR_MII_2_AN_INTR_STS

- **Description:** VR MII_2 MMD AN Interrupt and Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8002 : (APB3_16) ? (0x8002*0x2) : (0x8002*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** QSGMII_EN||USXG_QPORT

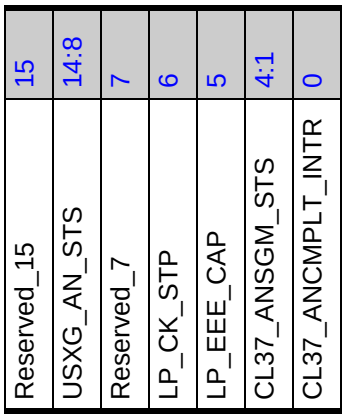


Table 5-398 Fields for Register: VR_MII_2_AN_INTR_STS

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
14:8	USXG_AN_STS	R	USXGMII Clause 37 AN Status (along Port 2) This field is valid only when clause 37 auto-negotiation is complete in USXGMII Mode along Port 2. It indicates the status received from remote link after the USXGMII auto-negotiation is complete. USXG_AN_STS[6] ■ 0: Link is Down ■ 1: Link is Up USXG_AN_STS[5] ■ 0: Half Duplex ■ 1: Full Duplex USXG_AN_STS[4:2] ■ 000: 10 Mbps speed link ■ 001: 100 Mbps speed link ■ 010: 1000 Mbps speed link ■ 011:10 Gbps speed link ■ 100:2.5 Gbps speed link ■ 101: 5 Gbps speed link USXG_AN_STS[1] ■ 1: EEE supported ■ 0: EEE not supported USXG_AN_STS[0] ■ 1: EEE clock-stop supported ■ 0: EEE clock-stop not supported **Note:** This field is present only in configurations with QXGMII (4-port USXGMII) support. **Value After Reset:** 0x0 **Exists:** Always
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6	LP_CK_STP	R	Link Partner EEE Clock Stop Capability This field indicates the EEE clock stop capability (or clock-stop enable - in case far-end is acting as QSGMII MAC) advertised by the far-end device. This field is valid only when PCS_MODE[1:0] is set to the QSGMII mode and the auto-negotiation is complete along port 2. **Note:** This field is present only in configurations with QSGMII. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
5	LP_EEE_CAP	R	Link Partner EEE Capability This field indicates the EEE capability advertised by the far-end device (Port 2 QSGMII PHY). This field is valid only when PCS_MODE[1:0] is set to the QSGMII mode and the auto-negotiation is complete along port 2. **Note:** This field is present only in configurations with QSGMII. **Value After Reset:** 0x0 **Exists:** Always
4:1	CL37_ANSGM_STS	R	Clause 37 AN QSGMII Status (port 2) This field is valid only when the PCS_MODE[1:0] is set to the QSGMII mode and the auto-negotiation is complete along port 2. It indicates the status received from remote link partner. CL37_ANSGM_STS[0] ■ 0: Half Duplex ■ 1: Full Duplex CL37_ANSGM_STS[2:1] ■ 00: 10 Mbps speed link ■ 01: 100 Mbps speed link ■ 10: 1000 Mbps speed link CL37_ANSGM_STS[3] ■ 0: Link is Down ■ 1: Link is Up **Note:** This field is present only in configurations with QSGMII. **Value After Reset:** 0x5 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
0	CL37_ANCMPLT_INTR	R/W	Clause 37 AN Complete Interrupt (SS,WC Type) The DWC_xpcs sets this bit when Clause 37 auto-negotiation is complete for port 2. The host must clear this bit by writing 0 to it. **Value After Reset:** 0x0 **Exists:** Always

5.2.6.26 VR_MII_2_TC

- **Description:** VR MII_2 MMD Test Control Register.
  Note: This register is present only in 1000BaseX-Only PCS configurations with QSGMII_EN = Enabled.
  This register merely reflects the value of VR_MII_TC Register and is not applicable when working in QSGMII/USXGMII Mode.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8003 : (APB3_16) ? (0x8003*0x2) : (0x8003*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** BASE1000X_ONLY&&QSGMII_EN

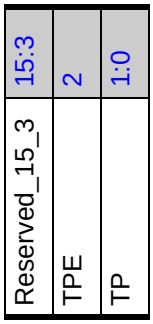


Table 5-399 Fields for Register: VR_MII_2_TC

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2	TPE	R	Test Pattern Enable Lanes **Value After Reset:** 0x0 **Exists:** Always
1:0	TP	R	Test Pattern Select **Value After Reset:** 0x0 **Exists:** Always

5.2.6.27 VR_MII_2_DIG_CTRL3

- **Description:** VR MII_2 MMD Digital Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8004 : (APB3_16) ? (0x8004*0x2) : (0x8004*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** USXG_QPORT

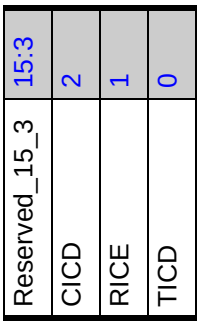


Table 5-400 Fields for Register: VR_MII_2_DIG_CTRL3

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2	CICD	R/W	Clock-compensation IPG check Deviation on Port 2 During USXGMII mode, when operating at GMII speed modes (1G/100M/10M), by default, clock-compensation logic deletes IDLE (to compensate for ppm difference) only if incoming IPG >= 12B. This ensures that output IPG is >= 8B provided input IPG is also greater than or equal to 12B However, if this bit is 1, clock-compensation IPG check deviates from the standard behavior and deletes IDLE even if incoming IPG is only >= 9B. This ensures that output IPG is >= 5B provided input IPG is also greater than or equal to 5B. ■ 0: output IPG >= 8B ■ 1: output IPG >= 5B This mode is useful when incoming Rx data is an already clock-compensated data and does not have enough number of IPGs with >= 12B of IDLE. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
1	RICE	R/W	Receive IPG Check Enable during XGMII-to-GMII conversion on Port 2. Whenever operating in 1G and lower speeds, DWC_xpcs receiver converts the received data from XGMII format to GMII/MII format before driving it out on GMII/MII Rx interface of DWC_xpcs. This logic blindly carries-out this conversion process without performing any check for number of IPG bytes present in this received data. RICE bit can be set to 1 to enable a IPG check mechanism during XGMII-to-GMII conversion process so as to ensure minimum IPG of 8 bytes at GMII/MII Rx interface. Whenever IPG in XGMII data is less than 8 bytes, DWC_xpcs introduces additional IPG bytes so as to ensure 8 bytes minimum IPG at GMII/MII Rx interface. Addition of IPG is compensated later by deleting IPG bytes from other packets which have larger IPG. If packets with IPG less than 8 bytes appear quite frequently in the incoming data stream, then the buffer can overflow and can cause packet corruption. This feature can work reliably only if not more than one packet with IPG less than 8 bytes occur within a span of 4 packets. This feature is useful in scenarios where link-partner device can send packets with IPG as low as 5 bytes. This bit shall not be set to 1, if CICD is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	TICD	R/W	Transmit IPG Check Deviation on Port 2 When operating in multi-port USXGMII modes, USXGMII Tx Rate Module compensates for AM insertion by deleting IDLEs.This deletion operation involves a IPG check logic which allows deletion only if 8 bytes of IPG can be ensured at transmitter output when operating in GMII/MII modes (1G and lower speed). However, this IPG check logic introduces a limit on the packet size that can be transmitted. If jumbo packets (packets of size 2K bytes to 9K bytes) are transmitted continuously, DWC_xpcs might not find an opportunity to delete sufficient IDLE bytes before the next AM insertion. Therefore, this can result in a gradual fill-up of Tx RAL FIFO and can result in eventual packet corruption. This bit can be set to 1 to disable the standard IPG check and perform a modified IPG check so as to support jumbo packets. Whenever this bit is set to 1, DWC_xpcs perform IPG deletion in such a manner so as to ensure minimum of 5 bytes of IPG at transmitter output when operating in GMII/MII modes (1G and lower speed). This mode is useful if the link-partner device is capable of accepting packets with 5 bytes IPG when operating in GMII/MII modes. When operating in XGMII mode (2.5G and above speeds), irrespective of the value of this bit, DWC_xpcs ensures minimum of 5 bytes IPG at the transmitter output. **Value After Reset:** 0x0 **Exists:** Always

5.2.6.28 VR_MII_2_DBG_CTRL

- **Description:** VR MII_2 MMD Debug Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8005 : (APB3_16) ? (0x8005*0x2) : (0x8005*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** QSGMII_EN

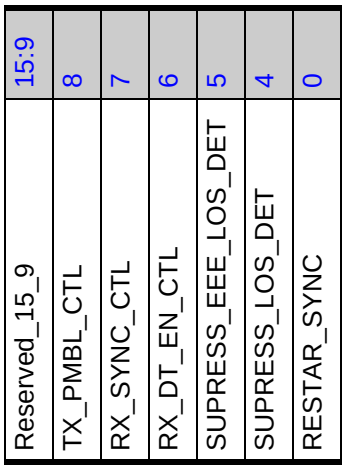


Table 5-401 Fields for Register: VR_MII_2_DBG_CTRL

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	TX_PMBL_CTL	R	Transmit Preamble Control This bit merely reflects the value of TX_PMBL_CTL field of VR_MII_DBG_CTRL Register (for 1000BaseX-Only PCS configurations) or VR_XS_PCS_DEBUG_CTRL Register (for other configurations). **Value After Reset:** 0x0 **Exists:** Always
7	RX_SYNC_CTL	R	Receive Synchronization Control . This bit merely reflects the value of RX_SYNC_CTL field of VR_MII_DBG_CTRL Register (for 1000BaseX-Only PCS configurations) or VR_XS_PCS_DEBUG_CTRL Register (for other configurations). **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
6	RX_DT_EN_CTL	R	Rx Data Enable Control. This bit merely reflects the value of RX_DT_EN_CTL field of VR_MII_DBG_CTRL Register (for 1000BaseX-Only PCS configurations) or VR_XS_PCS_DEBUG_CTRL Register (for other configurations). **Value After Reset:** 0x0 **Exists:** Always
5	SUPRESS_EEE_LOS_DET	R	Suppress EEE Loss of Signal Detection. This bit merely reflects the value of SUPRESS_EEE_LOS_DET field of VR_MII_DBG_CTRL Register (for 1000BaseX-Only PCS configurations) or VR_XS_PCS_DEBUG_CTRL Register (for other configurations). **Value After Reset:** 0x0 **Exists:** Always
4	SUPRESS_LOS_DET	R	Suppress Loss of Signal Detection This bit merely reflects the value of SUPRESS_LOS_DET field of VR_MII_DBG_CTRL Register (for 1000BaseX-Only PCS configurations) or VR_XS_PCS_DEBUG_CTRL Register (for other configurations). **Value After Reset:** 0x0 **Exists:** Always
0	RESTAR_SYNC	R/W	Restart Synchronization When set to 1, this bit restarts the Rx Synchronization State machine on port 2. The host must clear this bit to 0 before setting it to 1 next time. **Value After Reset:** 0x0 **Exists:** Always

5.2.6.29 VR_MII_2_EEE_MCTRL0

- **Description:** VR MII_2 MMD EEE Mode Control Register
  This register is present only for configurations with MAIN_MODE = 1000BASEX-Only PCS and QSGMII_EN = Enabled and EEE_EN = Enabled.
  This is a read-only register which merely reflects the value of VR_MII_EEE_MCTRL Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8006 : (APB3_16) ? (0x8006*0x2) : (0x8006*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** BASE1000X_ONLY&&QSGMII_EN&&EEE_EN

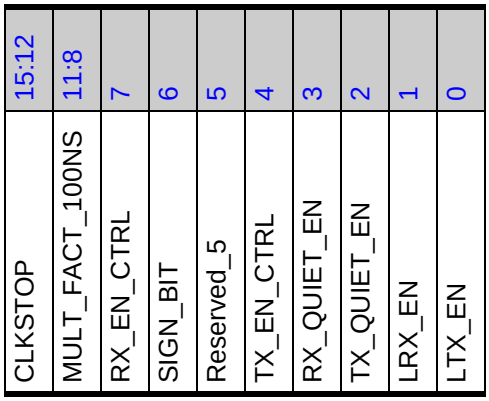


Table 5-402 Fields for Register: VR_MII_2_EEE_MCTRL0

Bits	Name	Memory Access	Description
15:12	CLKSTOP	R	Clock Stop **Value After Reset:** 0x8 **Exists:** Always
11:8	MULT_FACT_100NS	R	100 ns Clock Tic Multiplying Factor **Value After Reset:** 0x9 **Exists:** Always
7	RX_EN_CTRL	R	Rx Control Enable **Value After Reset:** 0x1 **Exists:** Always
6	SIGN_BIT	R	Effective 100 ns Tic Value **Value After Reset:** 0x0 **Exists:** Always
5	Reserved_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4	TX_EN_CTRL	R	Tx Control Enable **Value After Reset:** 0x1 **Exists:** Always
3	RX_QUIET_EN	R	Rx Quiet Enable **Value After Reset:** 0x1 **Exists:** Always
2	TX_QUIET_EN	R	Tx Quiet Enable **Value After Reset:** 0x1 **Exists:** Always
1	LRX_EN	R	LPI Rx Enable **Value After Reset:** 0x0 **Exists:** Always
0	LTX_EN	R	LPI Tx Enable **Value After Reset:** 0x0 **Exists:** Always

5.2.6.30 VR_MII_2_EEE_TXTIMER

- **Description:** VR MII_2 MMD EEE Tx Timer Register
  This register is present only for configurations with MAIN_MODE = 1000BASEX-Only PCS and QSGMII_EN = Enabled and EEE_EN = Enabled.
  This is a read-only register which merely reflects the value of VR_MII_EEE_TXTIMER Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8008 : (APB3_16) ? (0x8008*0x2) : (0x8008*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** BASE1000X_ONLY&&QSGMII_EN&&EEE_EN

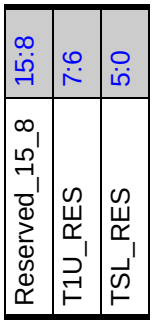


Table 5-403 Fields for Register: VR_MII_2_EEE_TXTIMER

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
7:6	T1U_RES	R	T1u Resolution This field stores the resolution value for the T1u timer. When the generated 100 ns tic timer is not exactly 100 ns, this field is programmed with the required value. The programmed value is equivalent to the number of more (SIGN_BIT is low) or less (SIGN_BIT is high) counts than the ideal count to meet the range of the T1u timer. The DWC_xpcs maintains the default value of the T1u timer as 11, assuming clk_eee_i_time_period * (MULT_FACT_100NS + 1) is equal to 100 ns to produce 1100 ns (1.1 us) as per the T1u requirement in the IEEE standard. Example: For 10GBASE-R or KR_Present configuration, if clk_eee_i_time_period * (MULT_FACT_100NS + 1)is 90, the default T1u timer produces 990 ns (0.99 us) which is lesser than the standard 1.1 us. To make it 1.1 us, you must program this register such that (11 +/- T1U_RES) * ((MULT_FACT_100NS + 1) * clk_eee_i_time_period) is greater than 1.1 us and less than 1.3 us (max limit), that is, (11+ T1U_RES)*90 = 1.1 us. T1U_RES = 2 meets the requirement which produces 1.17 us. Program the SIGN_BIT as 1. **Value After Reset:** 0x0 **Exists:** Always
5:0	TSL_RES	R	TSL Resolution **Value After Reset:** 0x0 **Exists:** Always

5.2.6.31 VR_MII_2_EEE_RXTIMER

- **Description:** VR MII_2 MMD EEE Rx Timer Register
  This register is present only for configurations with MAIN_MODE = 1000BASEX-Only PCS and QSGMII_EN = Enabled and EEE_EN = Enabled.
  This is a read-only register which merely reflects the value of VR_MII_EEE_RXTIMER Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8009 : (APB3_16) ? (0x8009*0x2) : (0x8009*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** BASE1000X_ONLY&&QSGMII_EN&&EEE_EN

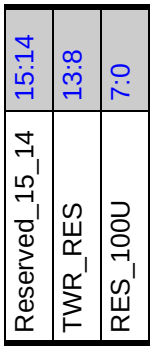


Table 5-404 Fields for Register: VR_MII_2_EEE_RXTIMER

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:8	TWR_RES	R	TWR Resolution **Value After Reset:** 0x0 **Exists:** Always
7:0	RES_100U	R	100 us Resolution **Value After Reset:** 0x0 **Exists:** Always

5.2.6.32 VR_MII_2_LINK_TIMER_CTRL

- **Description:** VR MII_2 MMD Link Timer Control Register
  This register merely reflects the value of VR_MII_LINK_TIMER_CTRL Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x800A : (APB3_16) ? (0x800A*0x2) : (0x800A*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** QSGMII_EN | |USXG_QPORT



Table 5-405 Fields for Register: VR_MII_2_LINK_TIMER_CTRL

Bits	Name	Memory Access	Description
15:0	CL37_LINK_TIME	R	Programmable Link Timer Value for Clause 37 auto-negotiation. **Value After Reset:** 0x0 **Exists:** Always

5.2.6.33 VR_MII_2_EEE_MCTRL1

- **Description:** VR MII_2 MMD EEE Mode Control 1 Register
  This register is present only for configurations with MAIN_MODE = 1000BASEX-Only PCS and QSGMII_EN = Enabled and EEE_EN = Enabled.
  This is a read-only register which merely reflects the value of VR_MII_EEE_MCTRL1 Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x800B : (APB3_16) ? (0x800B*0x2) : (0x800B*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** BASE1000X_ONLY&&QSGMII_EN&&EEE_EN

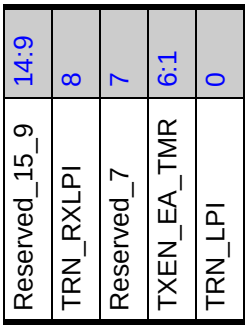
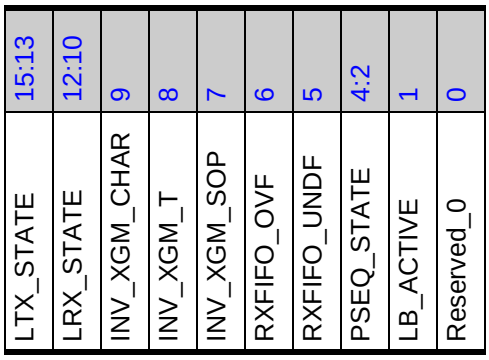


Table 5-406 Fields for Register: VR_MII_2_EEE_MCTRL1

Bits	Name	Memory Access	Description
14:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	TRN_RXLPI	R	Transparent Rx LPI Mode Enable. **Value After Reset:** 0x0 **Exists:** Always
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:1	TXEN_EA_TMR	R	Transmit Enable Early Assert Time Duration **Value After Reset:** 0x0 **Exists:** Always
0	TRN_LPI	R	Transparent Tx LPI Mode Enable **Value After Reset:** 0x0 **Exists:** Always

5.2.6.34 VR_MII_2_DIG_STS

- **Description:** VR MII_2 MMD Digital Status Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8010 : (APB3_16) ? (0x8010*0x2) : (0x8010*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** QSGMII_EN | |USXG_QPORT



15:13	12:10	9	8	7	6	5	4:2	1	0
LTX_STATE	LRX_STATE	INV_XGM_CHAR	INV_XGM_T	INV_XGM_SOP	RXFIFO_OVF	RXFIFO_UNDF	PSEQ_STATE	LB_ACTIVE	Reserved_0

Table 5-407 Fields for Register: VR_MII_2_DIG_STS

Bits	Name	Memory Access	Description
15:13	LTX_STATE	R	LPI Transmit State. This field reflects the value of LTX_STATE field of VR_MII_DIG_STS Register (for 1000BaseX-Only PCS configurations) or VR_XS_PCS_DIG_STS Register (for other configurations). **Value After Reset:** 0x0 **Exists:** Always
12:10	LRX_STATE	R	LPI Receive State This field reflects the value of LRX_STATE field of VR_MII_DIG_STS Register (for 1000BaseX-Only PCS configurations) or VR_XS_PCS_DIG_STS Register (for other configurations). **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
9	INV_XGM_CHAR	R	Invalid XGMII Character (RO,LH Type) -Port 2 This bit indicates an invalid XGMII character on the Transmit path (port 2). ■ 0: No Invalid XGMII character ■ 1: Invalid character This bit is set when one of the following conditions is true on the XGMII interface: ■ |Q| is detected on a lane other than Lane 0. ■ |Q| is detected on Lane 0 and also any of the following conditions is true: ❑ No 8'h00 data in Lane 1. ❑ No 8'h00 data in Lane 2. ❑ No 8'h01 or 8'h02 data on Lane 3. ■ |E| or |T| is encountered in non-data phase and |Q| is encountered in data phase. ■ |D| is detected during the Idle phase (after Terminate and before valid SOF). ■ |I| is detected during the data phase (after SOF and before Terminate). **Note:** This field is present only for Multi-port USXGMII configurations. **Value After Reset:** 0x0 **Exists:** Always
8	INV_XGM_T	R	Invalid XGMII T Character (RO,LH Type) -Port 2 This bit indicates that the transmit data (port 2) received on XGMII has an invalid Terminate character. ■ 0: Normal operation ■ 1: Invalid Terminate character The following can cause an invalid Terminate character: ■ In Terminate column, before lane corresponds to the |T| character, there is a |I| or |Q| character. ■ In Terminate column, after lane corresponds to the |T| character, there is a non-|I| character **Note:** This field is present only for Multi-port USXGMII configurations. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
7	INV_XGM_SOP	R	Invalid XGMII Start Character (RO,LH Type) -Port 2 This bit indicates that the XGMII transmit (Port 2) frame contains invalid SOP character. ■ 0: Normal operation ■ 1: Invalid SOP character An invalid SOP event occurs when a |S| character is detected in a lane other than Lane 0. **Note:** This field is present only for Multi-port USXGMII configurations. **Value After Reset:** 0x0 **Exists:** Always
6	RXFIFO_OVF	R	Rx FIFO Overflow (RO,LH Type) This bit indicates the clock rate compensation FIFO overflow for port 2. ■ 0: Normal operation ■ 1: FIFO overflow **Value After Reset:** 0x0 **Exists:** Always
5	RXFIFO_UNDF	R	Rx FIFO Underflow (RO,LH Type) This bit indicates the clock rate compensation FIFO underflow for port 2. ■ 0: Normal operation ■ 1: FIFO underflow **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4:2	PSEQ_STATE	R	**For Synopsys PHY configurations:** Power Up Sequence State This field indicates the state variable value of the power-up sequence module. For Synopsys Multi-protocol 6G PHY: ■ 3'b000: Wait for MPLL ON ■ 3'b001: Wait for TX up ■ 3'b010: Wait for RX up ■ 3'b011/3'b100: Tx/Rx Stable (Power_Good state) ■ 3'b101: Wait for Rx down (MPLL still ON) ■ 3'b110: MPLL OFF For Synopsys Multi-protocol 12G/16G/25G/32G PHY ■ 3'b000: Wait for ACK High 0 ■ 3'b001: Wait for ACK Low 0 ■ 3'b010: Wait for ACK High 1 ■ 3'b011: Wait for ACK Low 1 ■ 3'b100: Tx/Rx stable (Power_Good state) ■ 3'b101: Power Save state ■ 3'b110: Power Down state ■ 3'b111: Other intermediate states (valid only for 32G PHY configurations) **For non-Synopsys PHY configurations:** Reserved **Value After Reset:** 0x4 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
1	LB_ACTIVE	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	Reserved_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.6.35 VR_MII_2_ICG_ERRCNT1

- **Description:** VR MII_2 MMD Invalid Code Group Error Count1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8011 : (APB3_16) ? (0x8011*0x2) : (0x8011*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** QSGMII_EN

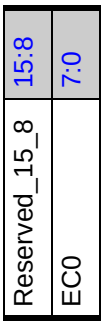


Table 5-408 Fields for Register: VR_MII_2_ICG_ERRCNT1

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:0	EC0	R	Invalid Code Group Count -Port 2 (RO,LH Type) This field gives the invalid code group count in port 2 (of QSGMII) when Bit 4 of VR MII_2 MMD Digital Error Count Select Register is set to 1. **Value After Reset:** 0x0 **Exists:** Always

5.2.6.36 VR_MII_2_GPIO

- **Description:** VR MII_2 MMD GPIO Register
  This register is present only if the DWC_xpcs is configured to have the GPIO interface and QSGMII_EN = Enabled and USXGMII_EN = Disabled.
  This register merely reflects the value of VR_MII_GPIO Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8015 : (APB3_16) ? (0x8015*0x2) : (0x8015*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** QSGMII_EN&&GPIO_EN&&!USXGMII_EN

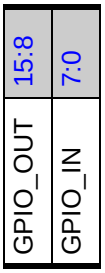


Table 5-409 Fields for Register: VR_MII_2_GPIO

Bits	Name	Memory Access	Description
15:8	GPIO_OUT	R	GPIO Output This field reflects the value of GPIO_OUT field in VR_MII_GPIO register. **Value After Reset:** 0x0 **Exists:** Always
7:0	GPIO_IN	R	GPIO Input This field indicates the content of the xpcs_gpo_i[7:0] port. **Value After Reset:** 0x0 **Exists:** Always

5.2.6.37 VR_MII_2_DIG_CTRL2

- **Description:** VR MII_2 MMD Digital Control2 Register
  Note: This register is present only in 1000BaseX-Only PCS configurations with QSGMII_EN = Enabled.
  This register merely reflects the value of VR_MII_DIG_CTRL2 Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80E1 : (APB3_16) ? (0x80E1*0x2) : (0x80E1*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** QSGMII_EN&&BASE1000X_ONLY

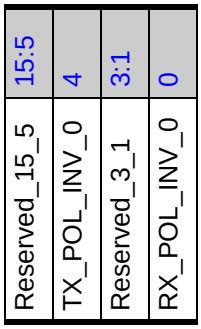


Table 5-410 Fields for Register: VR_MII_2_DIG_CTRL2

Bits	Name	Memory Access	Description
15:5	Reserved_15_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_POL_INV_0	R	Tx Polarity Invert on Lane 0 **Value After Reset:** 0x0 **Exists:** Always
3:1	Reserved_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	RX_POL_INV_0	R	Rx Polarity Invert on Lane 0 **Value After Reset:** 0x0 **Exists:** Always

5.2.6.38 VR_MII_2_DIG_ERRCNT_SEL

- **Description:** VR MII_2 MMD Digital Error Count Select Register
  This register is applicable only in QSGMII mode.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80E2 : (APB3_16) ? (0x80E2*0x2) : (0x80E2*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1B0000 : (APB3_16) ? (0x1B0000*0x2) : (0x1B0000*0x4)"
- **Exists:** QSGMII_EN

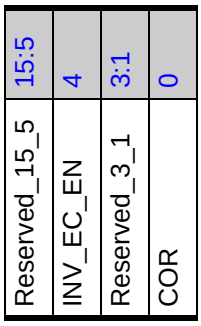


Table 5-411 Fields for Register: VR_MII_2_DIG_ERRCNT_SEL

Bits	Name	Memory Access	Description
15:5	Reserved_15_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	INV_EC_EN	R/W	Invalid Code Group Error Counter Enable When this bit is set, the counting of invalid code group errors (on port 2) is enabled. ■ 0: The counting of errors is disabled ■ 1: The counting of errors is enabled For information about the fields containing the number of errors counted, see VR_MII_2_ICG_ERRCNT1 Register. **Value After Reset:** 0x0 **Exists:** Always
3:1	Reserved_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	COR	R/W	Clear on Read When this bit is set and the host reads port 2 error counter (VR_MII_2_ICG_ERRCNT1 Register), that counter is cleared after the read cycle. ■ 0: Normal operation ■ 1: Clear error counter that is read **Value After Reset:** 0x0 **Exists:** Always

5.2.7 DWC_xpcs_map/VS_MII_3_MMD Registers

5.2.7.1 SR_MII_3_CTRL

- **Description:** SR MII_3 MMD Control Register
  The host can use this register to control (enable or disable) some of the features supported by the DWC_xpcs (for Port 3).
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** QSGMII_EN | | USXG_QPORT

15	14	13	12	11	10	9	8	7	6	5	4:0
RST	LBE	SS13	AN_ENABLE	LPM	Reserved_10	RESTART_AN	DUPLEX_MODE	Reserved_7	SS6	SS5	Reserved_4_0

Table 5-412 Fields for Register: SR_MII_3_CTRL

Bits	Name	Memory Access	Description
15	RST	R/W	Soft Reset (RW, SC Type) **For Synopsys PHY configurations:** When the host sets this bit, the CSR block triggers the software reset process in which all internal blocks are reset, except the Management Interface block. The registers are reset to their default values. When this bit is set, it also resets the PHY. This bit is self-cleared when PSEQ_STATE in VR_XS_PCS_DIG_STS/VR_MII_DIG_STS Register is equal to 3'b100, that is, Tx/Rx clocks are stable and in Power_Good state. **For Non-Synopsys PHY configurations:** When the host sets this bit, the CSR block triggers the software reset process in which all internal blocks are reset, except the Management Interface block. The registers are reset to their default values. When this bit is set, it also resets the PHY. This bit is self-cleared after the following: ■ 32 clk_csr_i clocks for the MCI interface ■ 1 MDC clock period for the MDIO interface **Note:** For description of read or write access for registers during reset process, see "Special Case Register Access" section of the Databook. **Value After Reset:** 0x0 **Exists:** Always
14	LBE	R	Loopback Enable This bit reflects the value of LBE bit of SR_MII_CTRL Register. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
13	SS13	R/W	Speed Selection (LSB) This bit, along with the SS6 bit and SS5 bit of this register, indicates the speed of port 3. The speed encoding is as follows: **For USXGMII:** ■ When SS5=1, SS6=0 and SS13=1, speed is 5 Gbps ■ When SS5=1, SS6=0 and SS13=0, speed is 2.5 Gbps ■ When SS5=0, SS6=1 and SS13=1, speed is 10 Gbps ■ When SS5=0, SS6=1 and SS13=0, speed is 1000 Mbps ■ When SS5=0, SS6=0 and SS13=1, speed is 100 Mbps ■ When SS5=0, SS6=0 and SS13=0, speed is 10 Mbps **For QSGMII:** ■ When SS6=1 and SS13=0, speed is 1000 Mbps ■ When SS6=0 and SS13=1, speed is 100 Mbps ■ When SS6=0 and SS13=0, speed is 10 Mbps **Value After Reset:** 0x0 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
12	AN_ENABLE	R/W	Enable Auto-Negotiation When set to 1, this bit enables the Clause 37 auto-negotiation process for port 2. **Value After Reset:** 0x0 **Exists:** Always
11	LPM	R/W	Power-Down Mode This bit controls the power-down mode of the DWC_xpcs. ■ 0: Normal operation ■ 1: The DWC_xpcs goes to the power-down mode along with the PHY. For non-Synopsys PHY, the xpcs_pdown_o port is asserted. For Synopsys PHY, the DWC_xpcs does the following to trigger the power-down mode: ■ Turns off the PHY Receiver and Transmitter ■ Switches off all the clocks When the host clears this bit, the DWC_xpcs resumes the normal operation. In a Synopsys PHY configuration, after clearing this bit, the host must wait until Bits[4:2] of VR_XS_PCS_DIG_STS/VR_MII_DIG_STS Register indicate that the DWC_xpcs is in the normal state. **Note:** For information about read or write access to registers during power-down mode, see "Special Case Register Access" section of the Databook. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
10	Reserved_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9	RESTART_AN	R/W	Restart Auto-Negotiation (RW,SC Type) When the host writes this bit, the DWC_xpcs initiates the auto-negotiation process for port 3. This bit is used to restart the auto-negotiation which is already initiated by setting Bit 12. The DWC_xpcs clears this bit after restarting the auto-negotiation. **Value After Reset:** 0x0 **Exists:** Always
8	DUPLEX_MODE	R/W	Duplex Mode This bit specifies the duplex mode of the DWC_xpcs along GMII port 2. ■ 0: Half duplex ■ 1: Full duplex If Bit 12 is set to 0, this bit determines the PHY link duplex mode. If Bit 12 is set to 1, then the PHY link duplex mode is independent of this bit (although the host can write any value) and is determined by the outcome of the Clause 37 auto-negotiation process. **Note:**USXGMII supports only full duplex mode (irrespective of the value programmed to this field). **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6	SS6	R/W	Speed Selection This bit, along with the SS13 bit of this register indicates the speed of operation of Port 3. For more information, see description of the SS13 bit. **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""

Bits	Name	Memory Access	Description
5	SS5	* Varies	Speed Selection This bit, along with SS6 and SS13 bits, control the speed of operation of XGMII/GMII Port 2. For more information, see description of the SS13 bit. This field is valid only in QXGMII (4-port USXGMII) mode. **Value After Reset:** 0x0 **Exists:** Always **Testable:** "(MP_USXGMII) ? \"untestable\" : \"unconstrained\"" **Memory Access:** "USXG_QPORT ? \"read-write\" : \"read-only\""
4:0	Reserved_4_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.7.2 SR_MII_3_STS

- **Description:** SR MII_3 MMD Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1 : (APB3_16) ? (0x1*0x2) : (0x1*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** QSGMII_EN

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ABL100T4	FD100ABL	HD100ABL	FD10ABL	HD10ABL	FD100T	HD100T	EXT_STS_ABL	UN_DIR_ABL	MF_PRE_SUP	AN_CMPL	RF	AN_ABL	LINK_STS	Reserved_1	EXT_REG_CAP

Table 5-413 Fields for Register: SR_MII_3_STS

Bits	Name	Memory Access	Description
15	ABL100T4	R	100BASE-T4 Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always
14	FD100ABL	R	100BASE-X Full-Duplex Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always
13	HD100ABL	R	100BASE-X Half-Duplex Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always
12	FD10ABL	R	10 Mbps Full-Duplex Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
11	HD10ABL	R	10 Mbps Half-Duplex Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always
10	FD100T	R	100BASE-T2 Full-Duplex Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always
9	HD100T	R	100BASE-T2 Half-Duplex Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always
8	EXT_STS_ABL	R	Extended Status Information ■ 0: No Extended Status information is present at register address 16'h000F of this MMD device. ■ 1: Extended Status information is present at register address 16'h000F of this MMD device. DWC_xpcs returns this bit as 1. **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
7	UN_DIR_ABL	R	Unidirectional Ability ■ 1: The DWC_xpcs is able to transmit GMII irrespective of whether device has determined the valid link or not. ■ 0: The DWC_xpcs is able to transmit GMII only when the device has determined the valid link. The DWC_xpcs always returns this bit as 0. **Value After Reset:** 0x0 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""

Bits	Name	Memory Access	Description
6	MF_PRE_SUP	R	MF Preamble Suppression ■ 1: The DWC_xpcs accepts the MDIO frames with preamble suppressed. ■ 0: The DWC_xpcs does not accept the MDIO frames with preamble suppressed. This bit is always set to 0. **Value After Reset:** 0x0 **Exists:** Always
5	AN_CMPL	R	Auto-negotiation Complete for Port 2 ■ 1: The AN process is complete ■ 0: The AN process is not complete This bit returns 0 if AN_ENABLE of SR_MII_1_CTRL Register is set to 0. **Value After Reset:** 0x0 **Exists:** Always
4	RF	R	Remote Fault This register bit is shared with RF bit of SR_MII_STS register. Software must ignore the value returned by this field as it is not valid in QSGMII/SGMII/USXGMII modes. **Value After Reset:** 0x0 **Exists:** Always
3	AN_ABL	R	Auto-negotiation Ability The DWC_xpcs always returns this bit as 1. ■ 1: The DWC_xpcs is able to perform auto-negotiation. ■ 0: The DWC_xpcs is not able to perform auto-negotiation. **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
2	LINK_STS	R	Link Status This register bit is shared with RLU bit of SR_MII_STS register. **Value After Reset:** 0x0 **Exists:** Always **Testable:** "QSGMII_EN ? \"untestable\" : \"unconstrained\""
1	Reserved_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	EXT_REG_CAP	R	Extended Register Capability ■ 1: Extended Register capability exists. This bit is always set to 1. ■ 0: Extended Register capability does not exist. **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""

5.2.7.3 SR_MII_3_DEV_ID1

- **Description:** SR MII_3 MMD Device Identifier Register 1
  This register returns the configurable Organizationally Unique Identifier (OUI) to the host.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x2 : (APB3_16) ? (0x2*0x2) : (0x2*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** QSGMII_EN||USXG_QPORT



Table 5-414 Fields for Register: SR_MII_3_DEV_ID1

Bits	Name	Memory Access	Description
15:0	VS_MII_DEV_OUI_3_18	R	Organizationally Unique Identifier[3:18] This field contains Bits [18:3] of 24-bit OUI of device manufacturer. The DWC_xpcs offers 24 configurable Bits[24:1] for identifying the device manufacturer. This register reflects the same value as that of SR_MII_DEV_ID1 register. **Value After Reset:** IEEE_DEV_OUI_3_18 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""

5.2.7.4 SR_MII_3_DEV_ID2

- **Description:** SR MII_3 MMD Device Identifier Register 2
  This register returns the configurable Organizationally Unique Identifier (OUI), Model Number, and Revision Number of the device to the host.
  This register returns the same value as that of SR_MII_DEV_ID2 Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x3 : (APB3_16) ? (0x3*0x2) : (0x3*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** QSGMII_EN | |USXG_QPORT

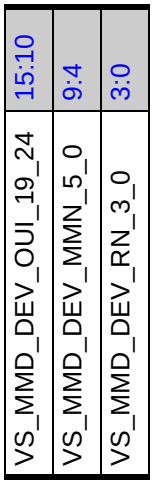


Table 5-415 Fields for Register: SR_MII_3_DEV_ID2

Bits	Name	Memory Access	Description
15:10	VS_MMD_DEV_OUI_19_24	R	Organizationally Unique Identifier [19:24] This field contains Bits[24:19] of 24-bit OUI of device manufacturer. The DWC_xpcs offers 24 configurable Bits[24:1] for identifying the device manufacturer. **Value After Reset:** IEEE_DEV_OUI_19_24 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
9:4	VS_MMD_DEV_MMN_5_0	R	Model Number This field contains the 5-bit Model Number of the device. **Value After Reset:** IEEE_DEV_MODEL_NO_0_5 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""

Bits	Name	Memory Access	Description
3:0	VS_MMD_DEV_RN_3_0	R	Revision Number This field contains the 4-bit Revision Number of the device. **Value After Reset:** IEEE_DEV_REV_NO_0_3 **Exists:** Always

5.2.7.5 SR_MII_3_AN_ADV

- **Description:** SR MII_3 MMD AN Advertisement Register
  Except for bit[5], rest of the fields of this register reflects the corresponding fields of SR_MII_AN_ADV Register.
  Note: This register is valid only when the current operating mode is QSGMII.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x4 : (APB3_16) ? (0x4*0x2) : (0x4*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** QSGMII_EN

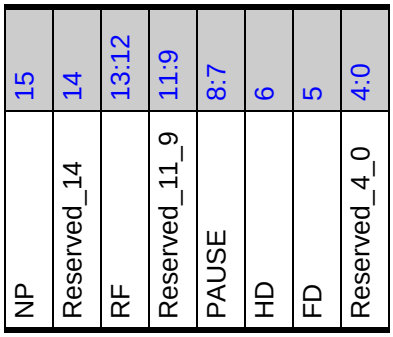


Table 5-416 Fields for Register: SR_MII_3_AN_ADV

Bits	Name	Memory Access	Description
15	NP	R	Next Page. This bit reflects the value of NP bit of SR_MII_AN_ADV Register. **Value After Reset:** 0x0 **Exists:** Always
14	Reserved_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:12	RF	R	Remote Fault. This bit reflects the value of RF bit of SR_MII_AN_ADV Register **Value After Reset:** 0x0 **Exists:** Always
11:9	Reserved_11_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
8:7	PAUSE	R	Pause Ability. This bit reflects the value of PAUSE bit of SR_MII_AN_ADV Register **Value After Reset:** 0x0 **Exists:** Always
6	HD	R	Half Duplex. This bit reflects the value of HD bit of SR_MII_AN_ADV Register **Value After Reset:** 0x0 **Exists:** Always
5	FD	R/W	Full Duplex. When this bit is set, it indicates the device (port 3) can operate in full-duplex mode. This bit is used in Clause 37 auto-negotiation (Tx_Config_Reg) done by port 3, when DWC_xpcs is configured as PHY-side QSGMII. **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
4:0	Reserved_4_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.7.6 SR_MII_3_LP_BABL

- **Description:** SR MII_3 MMD AN Link Partner Base Ability Register
  This register merely reflects the value of SR_MII_LP_BABL Register and is not applicable when working in QSGMII/USXGMII Mode.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x5 : (APB3_16) ? (0x5*0x2) : (0x5*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** QSGMII_EN

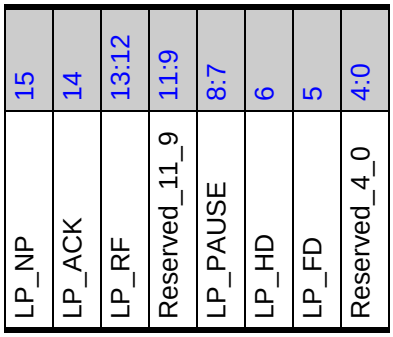


Table 5-417 Fields for Register: SR_MII_3_LP_BABL

Bits	Name	Memory Access	Description
15	LP_NP	R	Next Page **Value After Reset:** 0x0 **Exists:** Always
14	LP_ACK	R	ACK bit from the Link Partner **Value After Reset:** 0x0 **Exists:** Always
13:12	LP_RF	R	Remote Fault **Value After Reset:** 0x0 **Exists:** Always
11:9	Reserved_11_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8:7	LP_PAUSE	R	Pause Ability **Value After Reset:** 0x0 **Exists:** Always
6	LP_HD	R	Half Duplex **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
5	LP_FD	R	Full Duplex **Value After Reset:** 0x0 **Exists:** Always
4:0	Reserved_4_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.7.7 SR_MII_3_AN_EXPN

- **Description:** SR MII_3 MMD AN Expansion Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x6 : (APB3_16) ? (0x6*0x2) : (0x6*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** QSGMII_EN||USXG_QPORT

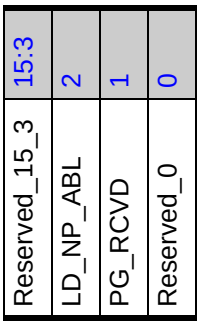


Table 5-418 Fields for Register: SR_MII_3_AN_EXPN

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2	LD_NP_ABL	R	Local Device NP Able ■ 1: The local device has the next page ability ■ 0: The local device does not have the next page ability The DWC_xpcs always returns this bit as 0 because it does not support Next Page. **Value After Reset:** 0x0 **Exists:** Always
1	PG_RCVD	R	Page Received (RO, LH Type) This bit indicates that the local device received a page from the link partner (during port 3 auto-negotiation). ■ 1: The local device received a new page ■ 0: The local device did not receive a new page **Value After Reset:** 0x0 **Exists:** Always
0	Reserved_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.7.8 SR_MII_3_EXT_STS

- **Description:** SR MII_3 MMD Extended Status Register
  This register reflects the same value as that of SR_MII_EX_STS register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0F : (APB3_16) ? (0x0F*0x2) : (0x0F*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** QSGMII_EN

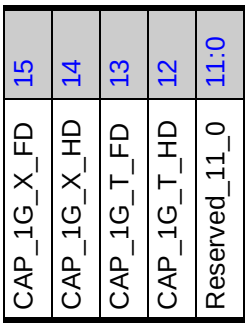


Table 5-419 Fields for Register: SR_MII_3_EXT_STS

Bits	Name	Memory Access	Description
15	CAP_1G_X_FD	R	1000BASE-X Full-Duplex Capable **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
14	CAP_1G_X_HD	R	1000BASE-X Half-Duplex Capable **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
13	CAP_1G_T_FD	R	1000BASE-T Full-Duplex Capable **Value After Reset:** 0x0 **Exists:** Always
12	CAP_1G_T_HD	R	1000BASE-T Half-Duplex Capable **Value After Reset:** 0x0 **Exists:** Always
11:0	Reserved_11_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.7.9 SR_MII_3_TIME_SYNC_PCS_ABL

- **Description:** SR MII 3 MMD Time Sync Capability Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0708 : (APB3_16) ? (0x0708*0x2) : (0x0708*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** (QSGMII_EN||USXG_QPORT)

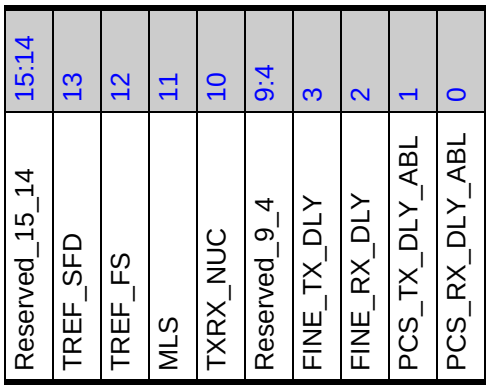


Table 5-420 Fields for Register: SR_MII_3_TIME_SYNC_PCS_ABL

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13	TREF_SFD	R	Timestamp reference at SFD - 0: DWC_xpcs does not support data delay measurement point at the beginning of SFD - 1: DWC_xpcs supports data delay measurement point at the beginning of SFD **Value After Reset:** "SNPS_RSVDPARAM_11 ? 0x1 : 0x0" **Exists:** Always
12	TREF_FS	R	Timestamp reference at first symbol after SFD - 0: DWC_xpcs does not support data delay measurement point at the beginning of the first symbol after SFD - 1: DWC_xpcs supports data delay measurement point at the beginning of the first symbol after SFD **Value After Reset:** "SNPS_RSVDPARAM_11 ? 0x1 : 0x0" **Exists:** Always

Bits	Name	Memory Access	Description
11	MLS	R	Multi-lane Support. ■ 0: DWC_xpcs does not support reporting of multi-PCS lane path data delay ■ 1: DWC_xpcs supports reporting of multi-PCS lane path data delay **Value After Reset:** "SNPS_RSVDPARAM_11&&!SINGLE_LANE_CFG ? 0x1 : 0x0" **Exists:** Always
10	TXRX_NUC	R	Transmit or Receive Num_Unit_Change Support Available for Port 3. If this bit is set, it indicates that DWC_xpcs supports Transmit/Receive Num_Unit_Change indication capability **Value After Reset:** "SNPS_RSVDPARAM_11 ? 0x1 : 0x0" **Exists:** Always
9:4	Reserved_9_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3	FINE_TX_DLY	R	Fine Resolution Transmit Path Data Delay Available for Port 3. If this bit is set, it indicates that DWC_xpcs provides information on transmit data path delay with sub-ns resolution in SR_MII_3_TIME_SYNC_TX_MAX_FINE_DLY and SR_MII_3_TIME_SYNC_TX_MIN_FINE_DLY registers. **Value After Reset:** "SNPS_RSVDPARAM_11 ? 0x1 : 0x0" **Exists:** Always
2	FINE_RX_DLY	R	Fine Resolution Receive Path Data Delay Available for Port 3. If this bit is set, it indicates that DWC_xpcs provides information on receive data path delay with sub-ns resolution in SR_MII_3_TIME_SYNC_RX_MAX_FINE_DLY and SR_MII_3_TIME_SYNC_RX_MIN_FINE_DLY registers. **Value After Reset:** "SNPS_RSVDPARAM_11 ? 0x1 : 0x0" **Exists:** Always
1	PCS_TX_DLY_ABL	R	PCS Transmit Path Data Delay Information Available for Port 3. If this bit is set, it indicates that the information regarding the maximum and minimum transmit data delay of PCS is available in ns-resolution in MII_3 MMD Rx Time Delay Registers. **Value After Reset:** 0x1 **Exists:** Always

Bits	Name	Memory Access	Description
0	PCS_RX_DLY_ABL	R	PCS Receive Path Data Delay Information Available for Port 3. If this bit is set, it indicates that the information regarding the maximum and minimum receive data delay of PCS is available in ns-resolution in MII_3 MMD Rx Time Delay Registers. **Value After Reset:** 0x1 **Exists:** Always

5.2.7.10 SR_MII_3_TIME_SYNC_TX_MAX_DLY_LWR

- **Description:** SR MII 3 MMD Time Sync Tx Max Delay Lower Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x0709 : (APB3_16) ? (0x0709*0x2) : (0x0709*0x4)" + "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** (QSGMII_EN | | USXG_QPORT)

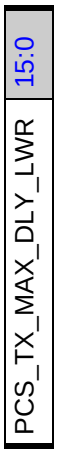


Table 5-421 Fields for Register: SR_MII_3_TIME_SYNC_TX_MAX_DLY_LWR

Bits	Name	Memory Access	Description
15:0	PCS_TX_MAX_DLY_LWR	R	PCS Transmit Max Delay Lower along Port 3 This field indicates the lower 16-bit of the 32-bit value which indicates integer nano-seconds portion of the Maximum Data Delay in the Transmit Path of DWC_xpcs along Port 3. The value returned by this field depends on the current operating mode. If TS_MODE bit is 0, this register indicates the end-to-end (MAC interface to PCS interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the transmit path from the DDMP point until the final Tx data output port along port 3. **Value After Reset:** 0x0 **Exists:** Always

5.2.7.11 SR_MII_3_TIME_SYNC_TX_MAX_DLY_UPR

- **Description:** SR MII 3 MMD Time Sync Tx Max Delay Upper Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x070A : (APB3_16) ? (0x070A*0x2) : (0x070A*0x4)" + "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** (QSGMII_EN | | USXG_QPORT)



Table 5-422 Fields for Register: SR_MII_3_TIME_SYNC_TX_MAX_DLY_UPR

Bits	Name	Memory Access	Description
15:0	PCS_TX_MAX_DLY_UPR	R	PCS Transmit Max Delay Upper along Port 3 This field indicates the upper 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Maximum Data Delay in the Transmit Path of DWC_xpcs along Port 3. If TS_MODE bit is 0, this register indicates the end-to-end (MAC interface to PCS interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the transmit path from the DDMP point until the final Tx data output port along port 3 **Value After Reset:** 0x0 **Exists:** Always

5.2.7.12 SR_MII_3_TIME_SYNC_TX_MIN_DLY_LWR

- **Description:** SR MII 3 MMD Time Sync Tx Min Delay Lower Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x070B : (APB3_16) ? (0x070B*0x2) : (0x070B*0x4)" + "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** (QSGMII_EN | | USXG_QPORT)



Table 5-423 Fields for Register: SR_MII_3_TIME_SYNC_TX_MIN_DLY_LWR

Bits	Name	Memory Access	Description
15:0	PCS_TX_MIN_DLY_LWR	R	PCS Transmit Min Delay Lower along Port 3 This field indicates the lower 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Minimum Data Delay in the Transmit Path of DWC_xpcs along Port 3. The value returned by this register field depends on the current operating mode. If TS_MODE bit is 0, this register indicates the end-to-end (MAC interface to PCS interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the transmit path from the DDMP point until the final Tx data output port along port 3. **Value After Reset:** 0x0 **Exists:** Always

5.2.7.13 SR_MII_3_TIME_SYNC_TX_MIN_DLY_UPR

- **Description:** SR MII 3 MMD Time Sync Tx Min Delay Upper Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x070C : (APB3_16) ? (0x070C*0x2) : (0x070C*0x4)" + "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** (QSGMII_EN | | USXG_QPORT)



Table 5-424 Fields for Register: SR_MII_3_TIME_SYNC_TX_MIN_DLY_UPR

Bits	Name	Memory Access	Description
15:0	PCS_TX_MIN_DLY_UPR	R	PCS Transmit Min Delay Upper along Port 3 This field indicates the upper 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Minimum Data Delay in the Transmit Path of DWC_xpcs along Port 3. If TS_MODE bit is 0, this register indicates the end-to-end (MAC interface to PCS interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the transmit path from the DDMP point until the final Tx data output port along Port 3 **Value After Reset:** 0x0 **Exists:** Always

5.2.7.14 SR_MII_3_TIME_SYNC_RX_MAX_DLY_LWR

- **Description:** SR MII 3 MMD Time Sync Rx Max Delay Lower Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x070D : (APB3_16) ? (0x070D*0x2) : (0x070D*0x4)" + "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** (QSGMII_EN | | USXG_QPORT)

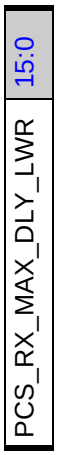


Table 5-425 Fields for Register: SR_MII_3_TIME_SYNC_RX_MAX_DLY_LWR

Bits	Name	Memory Access	Description
15:0	PCS_RX_MAX_DLY_LWR	R	PCS Receive Max Delay Lower along Port 3 This field indicates the lower 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Maximum Data Delay in the Receive Path of DWC_xpcs.The value returned by this register field depends on the current operating mode and is valid after Rx link-up is attained. If TS_MODE bit is 0, this register indicates the end-to-end (PHY interface to MAC interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the receive path from Rx data input port until the DDMP point along port 3. **Value After Reset:** 0x0 **Exists:** Always

5.2.7.15 SR_MII_3_TIME_SYNC_RX_MAX_DLY_UPR

- **Description:** SR MII 3 MMD Time Sync Rx Max Delay Upper Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070E : (APB3_16) ? (0x070E*0x2) : (0x070E*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** (QSGMII_EN||USXG_QPORT)



Table 5-426 Fields for Register: SR_MII_3_TIME_SYNC_RX_MAX_DLY_UPR

Bits	Name	Memory Access	Description
15:0	PCS_RX_MAX_DLY_UPR	R	PCS Receive Max Delay Upper along Port 3 This field indicates the upper 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Maximum Data Delay in the Receive Path of DWC_xpcs along Port 3. If TS_MODE bit is 0, this register indicates the end-to-end (PHY interface to MAC interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the receive path from Rx data input port until the DDMP point along Port 3. **Value After Reset:** 0x0 **Exists:** Always

5.2.7.16 SR_MII_3_TIME_SYNC_RX_MIN_DLY_LWR

- **Description:** SR MII 3 MMD Time Sync Rx Min Delay Lower Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070F : (APB3_16) ? (0x070F*0x2) : (0x070F*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** (QSGMII_EN||USXG_QPORT)



Table 5-427 Fields for Register: SR_MII_3_TIME_SYNC_RX_MIN_DLY_LWR

Bits	Name	Memory Access	Description
15:0	PCS_RX_MIN_DLY_LWR	R	PCS Receive Min Delay Lower along Port 3 This field indicates the lower 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Minimum Data Delay in the Receive Path of DWC_xpcs.The value returned by this register field depends on the current operating mode and is valid after Rx link-up is attained. If TS_MODE bit is 0, this register indicates the end-to-end (PHY interface to MAC interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the receive path from Rx data input port until the DDMP point along Port 3. **Value After Reset:** 0x0 **Exists:** Always

5.2.7.17 SR_MII_3_TIME_SYNC_RX_MIN_DLY_UPR

- **Description:** SR MII 3 MMD Time Sync Rx Min Delay Upper Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0710 : (APB3_16) ? (0x0710*0x2) : (0x0710*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** (QSGMII_EN||USXG_QPORT)



Table 5-428 Fields for Register: SR_MII_3_TIME_SYNC_RX_MIN_DLY_UPR

Bits	Name	Memory Access	Description
15:0	PCS_RX_MIN_DLY_UPR	R	PCS Receive Min Delay Upper along Port 3 This field indicates the upper 16-bit of the 32-bit value which indicates the integer nano-seconds portion of the Minimum Data Delay in the Receive Path of DWC_xpcs. If TS_MODE bit is 0, this register indicates the end-to-end (PHY interface to MAC interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the receive path from Rx data input port until the DDMP point along Port 3. **Value After Reset:** 0x0 **Exists:** Always

5.2.7.18 SR_MII_3_TIME_SYNC_TX_MAX_FINE_DLY

- **Description:** SR MII 3 MMD Time Sync Tx Max Fine Delay Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0711 : (APB3_16) ? (0x0711*0x2) : (0x0711*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** ((QSGMII_EN||USXG_QPORT)&&SNPS_RSVDPARAM_11)

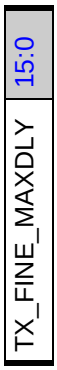


Table 5-429 Fields for Register: SR_MII_3_TIME_SYNC_TX_MAX_FINE_DLY

Bits	Name	Memory Access	Description
15:0	TX_FINE_MAXDLY	R	Transmit Maximum Fine Delay along Port 3 This register indicates the sub-nano-seconds portion of the maximum delay inside the DWC_xpcs transmit path from the DDMP point until the final transmit output port. The delay is indicated in units of 2^-16 ns. Contents of this register is valid only when TS_MODE bit is programmed to 1. Default value of this register is 0, as reset value of TS_MODE bit is 0 **Value After Reset:** 0x0 **Exists:** Always

5.2.7.19 SR_MII_3_TIME_SYNC_TX_MIN_FINE_DLY

- **Description:** SR MII 3 MMD Time Sync Tx Min Fine Delay Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0712 : (APB3_16) ? (0x0712*0x2) : (0x0712*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** ((QSGMII_EN||USXG_QPORT)&&SNPS_RSVDPARAM_11)



Table 5-430 Fields for Register: SR_MII_3_TIME_SYNC_TX_MIN_FINE_DLY

Bits	Name	Memory Access	Description
15:0	TX_FINE_MINDLY	R	Transmit Minimum Fine Delay along Port 3 This register indicates the sub-nano-seconds portion of the minimum delay inside the DWC_xpcs transmit path from the DDMP point until the final transmit output port. The delay is indicated in units of 2^-16 ns. Contents of this register is valid only when TS_MODE bit is programmed to 1. Default value of this register is 0, as reset value of TS_MODE bit is 0 **Value After Reset:** 0x0 **Exists:** Always

5.2.7.20 SR_MII_3_TIME_SYNC_RX_MAX_FINE_DLY

- **Description:** SR MII 3 MMD Time Sync Rx Max Fine Delay Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0713 : (APB3_16) ? (0x0713*0x2) : (0x0713*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** ((QSGMII_EN||USXG_QPORT)&&SNPS_RSVDPARAM_11)

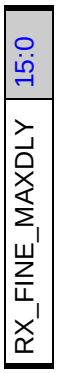


Table 5-431 Fields for Register: SR_MII_3_TIME_SYNC_RX_MAX_FINE_DLY

Bits	Name	Memory Access	Description
15:0	RX_FINE_MAXDLY	R	Receive Maximum Fine Delay along Port 3 This register indicates the sub-nano-seconds portion of the maximum delay inside the DWC_xpcs receive path from Rx data input port until the DDMP point. The delay is indicated in units of 2^-16 ns. Contents of this register is valid only when TS_MODE bit is programmed to 1. Default value of this register is 0, as reset value of TS_MODE bit is 0. **Value After Reset:** 0x0 **Exists:** Always

5.2.7.21 SR_MII_3_TIME_SYNC_RX_MIN_FINE_DLY

- **Description:** SR MII 3 MMD Time Sync Rx Min Fine Delay Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0714 : (APB3_16) ? (0x0714*0x2) : (0x0714*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** ((QSGMII_EN||USXG_QPORT)&&SNPS_RSVDPARAM_11)



Table 5-432 Fields for Register: SR_MII_3_TIME_SYNC_RX_MIN_FINE_DLY

Bits	Name	Memory Access	Description
15:0	RX_FINE_MINDLY	R	Receive Minimum Fine Delay along Port 3 This register indicates the sub-nano-seconds portion of the minimum delay inside the DWC_xpcs receive path from Rx data input port until the DDMP point. The delay is indicated in units of 2^-16 ns. Contents of this register is valid only when TS_MODE bit is programmed to 1. Default value of this register is 0, as reset value of TS_MODE bit is 0. **Value After Reset:** 0x0 **Exists:** Always

5.2.7.22 SR_MII_3_TIME_SYNC_CFG

- **Description:** SR MII 3 MMD Time Sync Configuration Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0715 : (APB3_16) ? (0x0715*0x2) : (0x0715*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** ((QSGMII_EN&&USXG_QPORT)&&SNPS_RSVDPARAM_11)

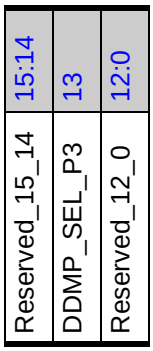


Table 5-433 Fields for Register: SR_MII_3_TIME_SYNC_CFG

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13	DDMP_SEL_P3	R/W	Data Delay Measurement Point Selection on Port 3 ■ 0: DWC_xpcs is configured to use the data delay measurement point at the beginning of SFD ■ 1: DWC_xpcs is configured to use the data delay measurement point at the beginning of the first symbol after SFD **Value After Reset:** 0x0 **Exists:** Always
12:0	Reserved_12_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.7.23 VR_MII_3_DIG_CTRL1

- **Description:** VR MII_3 MMD Digital Control1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8000 : (APB3_16) ? (0x8000*0x2) : (0x8000*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** QSGMII_EN||USXG_QPORT

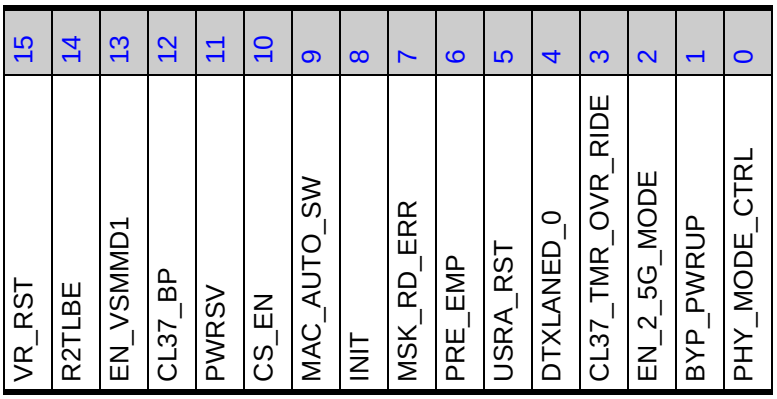


Table 5-434 Fields for Register: VR_MII_3_DIG_CTRL1

Bits	Name	Memory Access	Description
15	VR_RST	R/W	Vendor-Specific Soft Reset (RW, SC Type) When the host sets this bit, the CSR block triggers the vendor-specific software reset process in which all internal blocks, except the Management Interface block and CSR block, are reset. When this bit is set, it also resets the PHY. This bit is self cleared under the following conditions: **For Synopsys PHY:**This bit is self cleared when Bits[4:2] in VR_XS_PCS_DIG_STS/VR_MII_DIG_STS Register are equal to 3'b100, that is, Tx/Rx clocks are stable and in Power_Good state. **For Non-Synopsys PHY:**This bit is self cleared after the following: ■ 32 clk_csr_i clocks for the MCI interface ■ 1 MDC clock period for the MDIO interface **Note:** For information about the read or write access for any register during the reset process, see "Specail Case Register Access" section. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
14	R2TLBE	R/W	Rx to Tx Loopback Enable This bit controls the loopback path from the GMII/XGMII Rx to the GMII/XGMII Tx at the GMII/XGMII interface (port 3). ■ 0: Loopback path is disabled ■ 1: Loopback path is enabled **Value After Reset:** 0x0 **Exists:** Always
13	EN_VSMMD1	R	Enable Vendor-Specific MMD1 This is a read-only bit which reflects the value of EN_VSMMD1 bit of VR_MII_DIG_CTRL1 Register. **Value After Reset:** 0x1 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
12	CL37_BP	R	Enable Clause 37 AN in Backplane Configuration This is a read-only bit which reflects the value of CL37_BP bit of VR_MII_DIG_CTRL1 Register. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
11	PWRSV	* Varies	**For Synopsys PHY configurations:** Power Save This bit is present only when the DWC_xpcs is configured for Synopsys PHY. ■ 0: Normal operation. ■ 1: DWC_xpcs and the PHY enter the power-save mode. When this bit is set, the DWC_xpcs triggers the power-down mode by turning off the PHY Receiver and Transmitter, and without turning off MPLL. For Synopsys Multi-protocol 12G/16G PHY, DWC_xpcs asserts xpcs_tx_lpd_o{lane} and xpcs_rx_lpd_o{lane} when this bit is set. When the host clears this bit, DWC_xpcs resumes normal operation. After clearing this bit, the host must wait until Bits[4:2] of the VR_XS_PCS_DIG_STS/VR_MII_DIG_STS Status Register indicate that the DWC_xpcs is in a normal state. **Note:** For a description of read or write access for any register while this bit is set, see "Special Case Register Access" section of the Databook. **For non-Synopsys PHY configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_XAUI_PHY) ? \"read-write\" : \"read-only\""
10	CS_EN	* Varies	Clock Stop Enable ■ 1: The PHY might stop the clock during LPI mode ■ 0: The clock cannot be stopped during LPI mode You must program this bit based on the capability of the MAC (connected to GMII Port 3) during Rx LPI mode. **Note:** This field is applicable only in QSGMII mode. **Value After Reset:** "EEE_EN&&QSGMII_EN ? \"0x1\" : \"0x0\"" **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\"" **Memory Access:** "EEE_EN&&QSGMII_EN ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
9	MAC_AUTO_SW	R/W	Automatic Speed Mode Change after CL37 AN (for port 3) If this field is set to 1, DWC_xpcs automatically switches to the negotiated USXGMII/QSGMII(port3) speed, after the completion of Clause 37 auto-negotiation. This mode is valid only when DWC_xpcs is configured as MAC-side USXGMII/QSGMII and must be set only when Auto-negotiation is enabled (AN_ENABLE bit is set to 1). If this bit is set to 0, DWC_xpcs operates at the speed/duplex mode as per the values programmed to SR_MII_3_CTRL Register. In that case, after the completion of CL37 AN, application has to read the negotiated Speed/Duplex Mode from VR_MII_3_AN_INTR_STS Register and then program SR_MII_3_CTRL Register appropriately. If this bit is set to 1 in QSGMII mode, xpcs_qsgmii_link_sts_p3_o, xpcs_link_speed_p3_o and xpcs_qsgmii_full_duplex_p3_o outputs reflect the auto-negotiated values, that is, values from CL37_ANSGM_STS field of VR_MII_3_AN_INTR_STS Register. For USXGMII mode, if clk_xgmii_tx_p3_i and clk_xgmii_rx_p3_i do not stabilize at the new operating frequency (based on the selected speed) immediately after the completion of auto-negotiation, then software might need to program 'USRA_RST' bit prior to starting packet transfer in the new speed mode. **Note:** Set this bit only when DWC_xpcs is configured as USXGMII/QSGMII MAC, that is, TX_CONFIG = 0 **Note:** Whenever this bit is set, the values programmed to SS5, SS6, and SS13 bits are ignored. Moreover, the values of these bits do not change based on auto-negotiation outcome. **Value After Reset:** 0x0 **Exists:** Always
8	INIT	R	INIT This bit reflects the value of INIT bit of VR_MII_DIG_CTRL1 Register. **Value After Reset:** 0x0 **Exists:** Always
7	MSK_RD_ERR	R	Mask Running Disparity Error. This bit reflects the value of MSK_RD_ERR bit of VR_MII_DIG_CTRL1 Register. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
6	PRE_EMP	* Varies	Pre-emption Packet Enable. Set this bit to 1 to allow the DWC_xpcs to properly receive/transmit preemption packets along GMII port 3, when operating in 10M/100M Modes. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "QSGMII_EN ? \"read-write\" : \"read-only\""
5	USRA_RST	* Varies	USXGMII Rate Adapter Reset (Port 3) This bit can be set to 1 to reset/initialize the USXGMII Rate Adapter Logic (for port 3) inside DWC_xpcs. The Rate Adapter Logic maps GMII data to XGMII (and vice-versa in Rx path) data and perform data replication for lower USXGMII speeds. This is a self-clear bit (RW,SC) and clears itself after the reset of Tx and Rx Rate Adapter Logic is complete. After setting this bit to 1, application must poll this bit till it self-clears. **Note:** If CLKCOMP = Enabled, this bit also initializes the clock-compensation FIFO (corresponding to port 3). When this bit is programmed to 1, RXFIFO_OVF/RXFIFO_UNF bits of VR_MII_3_DIG_STS Register might get set incorrectly. Therefore, read these register bits (RXFIFO_OVF and RXFIFO_UNF) so that they get cleared. Thereafter, RXFIOF_UNF and RXFIFO_UNF bits are reliable. **Value After Reset:** 0x0 **Exists:** Always **Testable:** "(MP_USXGMII) ? \"untestable\" : \"unconstrained\"" **Memory Access:** "USXG_QPORT ? \"read-write\" : \"read-only\""
4	DTXLANED_0	R	Tx Lane 0 Disable This bit reflects the value of DTXLANED_0 bit of VR_MII_DIG_CTRL1 Register. **Value After Reset:** 0x0 **Exists:** Always
3	CL37_TMR_OVR_RIDE	R	Over-Ride Control for CL37 Link Timer. This bit reflects the value of CL37_TMR_OVR_RIDE bit of VR_MII_DIG_CTRL1 Register. **Value After Reset:** 0x0 **Exists:** Always
2	EN_2_5G_MODE	R	Enable 2.5G GMII Mode. This bit reflects the value of EN_2_5G_MODE bit of VR_MII_DIG_CTRL1 Register. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
1	BYP_PWRUP	* Varies	**For Synopsys PHY configurations:** Bypass Power-Up Sequence When this bit is set to 1, the DWC_xpcs bypasses the normal flow of the power-up sequence and reaches the Power_Good state to enable transmission or reception. When this bit is set, the DWC_xpcs does not wait for the MPLL and Transmit or Receive PLL status from the Synopsys PHY. You can use this feature in the following scenarios: ■ When the DWC_xpcs is configured to interface with a specific Synopsys PHY. ■ When the data path needs to be interfaced to some other PHY. When this bit is set to 0, the DWC_xpcs waits for the MPLL, Tx, or Rx PLL status from the Synopsys PHY before resuming the normal transmission and reception. **Note:** This bit is shared with Bit 1 of following registers: ■ VR_XS_PCS_DIG_CTRL1 Register ■ VR_AN_DIG_CTRL1 Register ■ VR_PMA_DIG_CTRL1 Register ■ VR MII/MII_1/MII_2_DIG_CTRL1 Register **Access Type:** If SNPS_XAUI_PHY= Enabled, this bit is RW. Otherwise, this bit is RO. **For non-Synopsys PHY configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_XAUI_PHY) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
0	PHY_MODE_CTRL	* Varies	QSGMII/USXGMII PHY mode control on Port 3 This bit controls the Clause 37 auto-negotiation when operating in QSGMII/USXGMII (Port 3) PHY mode. **QSGMI**: When this bit is set to 1, DWC_xpcs advertises the values of input ports xpcs_qsgmii_link_sts_p3_i, xpcs_qsgmii_link_speed_p3_i and xpcs_qsgmii_full_duplex_p3_i during QSGMII(Port 3) auto-negotiation. When this bit is set to 0, QSGMII(Port 3) auto-negotiation advertises the values programmed to: ■ bit 4 (SGMII_LINK_STS) of VR_MII_3_AN_CTRL Register ■ bit 13 (ss13) and 6 (ss6) of SR_MII_3_CTRL Register and ■ bit 5 (FD) of SR_MII_3_AN_ADV Register **USXGMII**: When this bit is set to 1, DWC_xpcs advertises the values of input ports xpcs_usxg_link_sts_p3_i, xpcs_usxg_link_speed_p3_i,xpcs_usxg_full_duplex_p3_i,xpcs_usxg_eee_abl_p3_i and xpcs_usxg_ck_stp_abl_p3_i during USXGMII (port 2) auto-negotiation. When this bit is set to 0, USXGMI (Port 3) auto-negotiation advertises the following : ■ bit 4 (SGMII_LINK_STS) of VR_MII_3_AN_CTRL Register ■ bit 13 (ss13) and 6 (ss6) and 5 (ss5) of SR_MII_3_CTRL Register and ■ Value of 1 for Duplex-Mode ■ Value of 1 for EEE-capability, if EEE = Enabled; Value of 0 if EEE_EN = Disabled ■ Value of 1 for EEE clock-stop capability,if EEE = Enabled; Value of 0 if EEE_EN = Disabled **Note:** This field is present only for the configurations satisfying either of the following conditions: ■ QSGMII_PHY_AN_AUTO_RESTART= Enabled ■ Quad-port USXGMII configurations with USXG_-PHY_AN_AUTO_RESTART = Enabled **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(QSGMII_PHY_AN_AUTO_RESTART||(USXG_QPORT&&USXG_PHY_AN_AUTO_RESTART)) ? \"read-write\" : \"read-only\""

5.2.7.24 VR_MII_3_AN_CTRL

- **Description:** VR MII_3 MMD AN Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x8001 : (APB3_16) ? (0x8001*0x2) : (0x8001*0x4)" + "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** QSGMII_EN | | USXG_QPORT

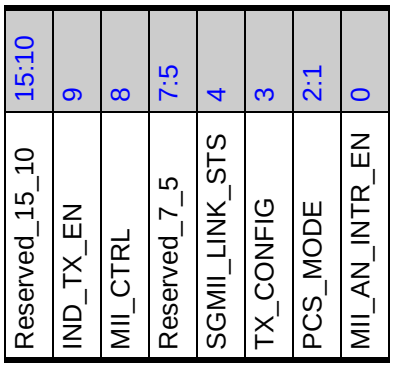


Table 5-435 Fields for Register: VR_MII_3_AN_CTRL

Bits	Name	Memory Access	Description
15:10	Reserved_15_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9	IND_TX_EN	* Varies	Independent Transmit Enable on port 3 during QSGMII mode. If this bit is set to 1, DWC_xpcs is able to transmit the port 3 GMII Tx data, irrespective of its receive link status during QSGMII mode (provided auto-negotiation is not enabled). If this bit is set to 0, DWC_xpcs sends IDLE (for port 3) till its receiver has attained synchronization. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "QSGMII_EN ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
8	MII_CTRL	R/W	MII Control This bit controls the width of the MAC interface (connected to GMII port 3) when operating at 10 Mbps or 100 Mbps ■ 0: 4-bit MII ■ 1: 8-bit MII This bit also controls the xpcs_mii_ctrl_p3_o signal which is used for external clock multiplexing of the clk_mii_tx_p3_i/clk_xgmii_tx_p3_i and clk_mii_rx_p3_i/clk_xgmii_rx_p3_i signals. **Value After Reset:** 0x0 **Exists:** Always
7:5	Reserved_7_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	SGMII_LINK_STS	R/W	QSGMII/USXGMII Link Status (port 3) This bit is used in Bit 15 of the Tx_Config_Reg during Clause 37 auto-negotiation (along port 3) when DWC_xpcs is programmed as QSGMII-PHY (or USXGMII-PHY) and when PHY_MODE_CTRL bit of VR_MII_3_DIG_CTRL1 Register is 0 . ■ 0: Link Down ■ 1: Link Up **Value After Reset:** 0x0 **Exists:** Always
3	TX_CONFIG	R	Transmit Configuration This field reflects the value of TX_CONFIG of VR_MII_AN_CTRL Register. **Value After Reset:** 0x0 **Exists:** Always
2:1	PCS_MODE	R	PCS Mode This field reflects the value of PCS_MODE of VR_MII_AN_CTRL Register for configurations with QSGMII support. For other configurations, this field returns 2'b00. **Value After Reset:** 0x0 **Exists:** Always
0	MII_AN_INTR_EN	R/W	Clause 37 AN Complete Interrupt Enable (port 3) When set to 1, this bit enables the generation of Clause 37 auto-negotiation complete (on port 3) interrupt output. When set to 0, it disables the generation of Clause 37 auto-negotiation complete interrupt. **Value After Reset:** 0x0 **Exists:** Always

5.2.7.25 VR_MII_3_AN_INTR_STS

- **Description:** VR MII_3 MMD AN Interrupt and Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x8002 : (APB3_16) ? (0x8002*0x2) : (0x8002*0x4)" + "(APB3_INTF | | MCI_INTF | | MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** QSGMII_EN | | USXG_QPORT

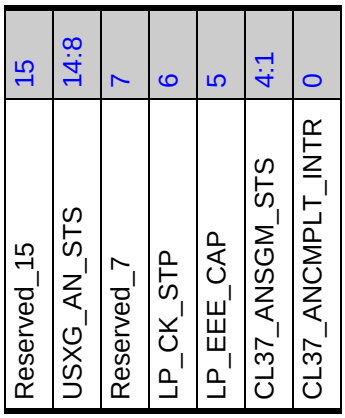


Table 5-436 Fields for Register: VR_MII_3_AN_INTR_STS

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
14:8	USXG_AN_STS	R	USXGMII Clause 37 AN Status (along Port 3) This field is valid only when clause 37 auto-negotiation is complete in USXGMII Mode along Port 3. It indicates the status received from remote link after the USXGMII auto-negotiation is complete. USXG_AN_STS[6] ■ 0: Link is Down ■ 1: Link is Up USXG_AN_STS[5] ■ 0: Half Duplex ■ 1: Full Duplex USXG_AN_STS[4:2] ■ 000: 10 Mbps speed link ■ 001: 100 Mbps speed link ■ 010: 1000 Mbps speed link ■ 011: 10 Gbps speed link ■ 100: 2.5 Gbps speed link ■ 101: 5 Gbps speed link USXG_AN_STS[1] ■ 1: EEE supported ■ 0: EEE not supported USXG_AN_STS[0] ■ 1: EEE clock-stop supported ■ 0: EEE clock-stop not supported **Note:** This field is present only in configurations with QXGMII (4-port USXGMII) support. **Value After Reset:** 0x0 **Exists:** Always
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
6	LP_CK_STP	R	Link Partner EEE Clock Stop Capability This field indicates the EEE clock stop capability (or clock-stop enable - in case far-end is acting as QSGMII MAC) advertised by the far-end device. This field is valid only when PCS_MODE[1:0] is set to the QSGMII mode and the auto-negotiation is complete along port 3. **Note:** This field is present only in configurations with QSGMII. **Value After Reset:** 0x0 **Exists:** Always
5	LP_EEE_CAP	R	Link Partner EEE Capability This field indicates the EEE capability advertised by the far-end device (Port 3 QSGMII PHY). This field is valid only when PCS_MODE[1:0] is set to the QSGMII mode and the auto-negotiation is complete along port 3. **Note:** This field is present only in configurations with QSGMII. **Value After Reset:** 0x0 **Exists:** Always
4:1	CL37_ANSGM_STS	R	Clause 37 AN QSGMII Status (port 3) This field is valid only when the PCS_MODE[1:0] is set to the QSGMII mode and the auto-negotiation is complete along port 3. It indicates the status received from remote link partner. CL37_ANSGM_STS[0] ■ 0: Half Duplex ■ 1: Full Duplex CL37_ANSGM_STS[2:1] ■ 00: 10 Mbps speed link ■ 01: 100 Mbps speed link ■ 10: 1000 Mbps speed link CL37_ANSGM_STS[3] ■ 0: Link is Down ■ 1: Link is Up **Note:** This field is present only in configurations with QSGMII. **Value After Reset:** "QSGMII_EN ? 0x5 : 0x0" **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""

Bits	Name	Memory Access	Description
0	CL37_ANCMPLT_INTR	R/W	Clause 37 AN Complete Interrupt (SS,WC Type) The DWC_xpcs sets this bit when Clause 37 auto-negotiation is complete for port 3. The host must clear this bit by writing 0 to it. **Value After Reset:** 0x0 **Exists:** Always

5.2.7.26 VR_MII_3_TC

- **Description:** VR MII_3 MMD Test Control Register.
  Note: This register is present only in 1000BaseX-Only PCS configurations with QSGMII_EN = Enabled.
  This register merely reflects the value of VR_MII_TC Register and is not applicable when working in QSGMII/USXGMII Mode.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8003 : (APB3_16) ? (0x8003*0x2) : (0x8003*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** BASE1000X_ONLY&&QSGMII_EN

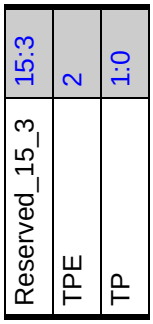


Table 5-437 Fields for Register: VR_MII_3_TC

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2	TPE	R	Test Pattern Enable Lanes **Value After Reset:** 0x0 **Exists:** Always
1:0	TP	R	Test Pattern Select **Value After Reset:** 0x0 **Exists:** Always

5.2.7.27 VR_MII_3_DIG_CTRL3

- **Description:** VR MII_3 MMD Digital Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8004 : (APB3_16) ? (0x8004*0x2) : (0x8004*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** USXG_QPORT

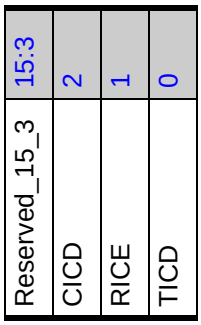


Table 5-438 Fields for Register: VR_MII_3_DIG_CTRL3

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2	CICD	R/W	Clock-compensation IPG check Deviation on Port 3 During USXGMII mode, when operating at GMII speed modes (1G/100M/10M), by default, clock-compensation logic deletes IDLE (to compensate for ppm difference) only if incoming IPG >= 12B. This ensures that output IPG is >= 8B provided input IPG is also greater than or equal to 12B However, if this bit is 1, clock-compensation IPG check deviates from the standard behavior and deletes IDLE even if incoming IPG is only >= 9B. This ensures that output IPG is >= 5B provided input IPG is also greater than or equal to 5B ■ 0: output IPG >= 8B ■ 1: output IPG >= 5B This mode is useful when incoming Rx data is an already clock-compensated data and does not have enough number of IPGs with >= 12B of IDLE. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
1	RICE	R/W	Receive IPG Check Enable during XGMII-to-GMII conversion on Port 3. Whenever operating in 1G and lower speeds, DWC_xpcs receiver converts the received data from XGMII format to GMII/MII format before driving it out on GMII/MII Rx interface of DWC_xpcs. This logic blindly carries-out this conversion process without performing any check for number of IPG bytes present in this received data. RICE bit can be set to 1 to enable a IPG check mechanism during XGMII-to-GMII conversion process so as to ensure minimum IPG of 8 bytes at GMII/MII Rx interface. Whenever IPG in XGMII data is less than 8 bytes, DWC_xpcs introduces additional IPG bytes so as to ensure 8 bytes minimum IPG at GMII/MII Rx interface. Addition of IPG is compensated later by deleting IPG bytes from other packets which have larger IPG. If packets with IPG less than 8 bytes appear quite frequently in the incoming data stream, then the buffer can overflow and can cause packet corruption. This feature can work reliably only if not more than one packet with IPG less than 8 bytes occur within a span of 4 packets. This feature is useful in scenarios where link-partner device can send packets with IPG as low as 5 bytes. This bit shall not be set to 1, if CICD is programmed to 1. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	TICD	R/W	Transmit IPG Check Deviation on Port 3 When operating in multi-port USXGMII modes, USXGMII Tx Rate Module compensates for AM insertion by deleting IDLEs.This deletion operation involves a IPG check logic which allows deletion only if 8 bytes of IPG can be ensured at transmitter output when operating in GMII/MII modes (1G and lower speed). However, this IPG check logic introduces a limit on the packet size that can be transmitted. If jumbo packets (packets of size 2K bytes to 9K bytes) are transmitted continuously, DWC_xpcs might not find an opportunity to delete sufficient IDLE bytes before the next AM insertion. Therefore, this can result in a gradual fill-up of Tx RAL FIFO and can result in eventual packet corruption. This bit can be set to 1 to disable the standard IPG check and perform a modified IPG check so as to support jumbo packets. Whenever this bit is set to 1, DWC_xpcs perform IPG deletion in such a manner so as to ensure minimum of 5 bytes of IPG at transmitter output when operating in GMII/MII modes (1G and lower speed). This mode is useful if the link-partner device is capable of accepting packets with 5 bytes IPG when operating in GMII/MII modes. When operating in XGMII mode (2.5G and above speeds), irrespective of the value of this bit, DWC_xpcs ensures minimum of 5 bytes IPG at the transmitter output. **Value After Reset:** 0x0 **Exists:** Always

5.2.7.28 VR_MII_3_DBG_CTRL

- **Description:** VR MII_3 MMD Debug Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8005 : (APB3_16) ? (0x8005*0x2) : (0x8005*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** QSGMII_EN

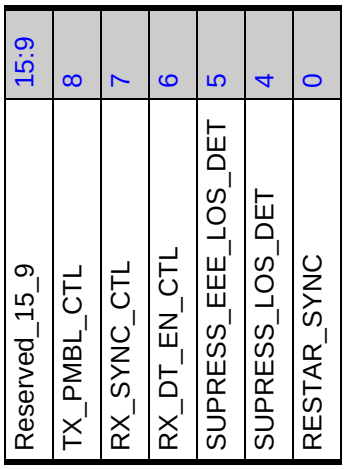


Table 5-439 Fields for Register: VR_MII_3_DBG_CTRL

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	TX_PMBL_CTL	R	Transmit Preamble Control This bit merely reflects the value of TX_PMBL_CTL field of VR_MII_DBG_CTRL Register (for 1000BaseX-Only PCS configurations) or VR_XS_PCS_DEBUG_CTRL Register (for other configurations). **Value After Reset:** 0x0 **Exists:** Always
7	RX_SYNC_CTL	R	Receive Synchronization Control . This bit merely reflects the value of RX_SYNC_CTL field of VR_MII_DBG_CTRL Register (for 1000BaseX-Only PCS configurations) or VR_XS_PCS_DEBUG_CTRL Register (for other configurations). **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
6	RX_DT_EN_CTL	R	Rx Data Enable Control. This bit merely reflects the value of RX_DT_EN_CTL field of VR_MII_DBG_CTRL Register (for 1000BaseX-Only PCS configurations) or VR_XS_PCS_DEBUG_CTRL Register (for other configurations). **Value After Reset:** 0x0 **Exists:** Always
5	SUPRESS_EEE_LOS_DET	R	Suppress EEE Loss of Signal Detection. This bit merely reflects the value of SUPRESS_EEE_LOS_DET field of VR_MII_DBG_CTRL Register (for 1000BaseX-Only PCS configurations) or VR_XS_PCS_DEBUG_CTRL Register (for other configurations). **Value After Reset:** 0x0 **Exists:** Always
4	SUPRESS_LOS_DET	R	Suppress Loss of Signal Detection This bit merely reflects the value of SUPRESS_LOS_DET field of VR_MII_DBG_CTRL Register (for 1000BaseX-Only PCS configurations) or VR_XS_PCS_DEBUG_CTRL Register (for other configurations). **Value After Reset:** 0x0 **Exists:** Always
0	RESTAR_SYNC	R/W	Restart Synchronization When set to 1, this bit restarts the Rx Synchronization State machine on port 3. The host must clear this bit to 0 before setting it to 1 next time. **Value After Reset:** 0x0 **Exists:** Always

5.2.7.29 VR_MII_3_EEE_MCTRL0

- **Description:** VR MII_3 MMD EEE Mode Control Register
  This register is present only for configurations with MAIN_MODE = 1000BASEX-Only PCS and QSGMII_EN = Enabled and EEE_EN = Enabled.
  This is a read-only register which merely reflects the value of VR_MII_EEE_MCTRL Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8006 : (APB3_16) ? (0x8006*0x2) : (0x8006*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** BASE1000X_ONLY&&QSGMII_EN&&EEE_EN

15:12	11:8	7	6	5	4	3	2	1	0
CLKSTOP	MULT_FACT_100NS	RX_EN_CTRL	SIGN_BIT	Reserved_5	TX_EN_CTRL	RX_QUIET_EN	TX_QUIET_EN	LRX_EN	LTX_EN

Table 5-440 Fields for Register: VR_MII_3_EEE_MCTRL0

Bits	Name	Memory Access	Description
15:12	CLKSTOP	R	Clock Stop **Value After Reset:** 0x8 **Exists:** Always
11:8	MULT_FACT_100NS	R	100 ns Clock Tic Multiplying Factor **Value After Reset:** 0x9 **Exists:** Always
7	RX_EN_CTRL	R	Rx Control Enable **Value After Reset:** 0x1 **Exists:** Always
6	SIGN_BIT	R	Effective 100 ns Tic Value **Value After Reset:** 0x0 **Exists:** Always
5	Reserved_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4	TX_EN_CTRL	R	Tx Control Enable **Value After Reset:** 0x1 **Exists:** Always
3	RX_QUIET_EN	R	Rx Quiet Enable **Value After Reset:** 0x1 **Exists:** Always
2	TX_QUIET_EN	R	Tx Quiet Enable **Value After Reset:** 0x1 **Exists:** Always
1	LRX_EN	R	LPI Rx Enable **Value After Reset:** 0x0 **Exists:** Always
0	LTX_EN	R	LPI Tx Enable **Value After Reset:** 0x0 **Exists:** Always

5.2.7.30 VR_MII_3_EEE_TXTIMER

- **Description:** VR MII_3 MMD EEE Tx Timer Register
  This register is present only for configurations with MAIN_MODE = 1000BASEX-Only PCS and QSGMII_EN = Enabled and EEE_EN = Enabled.
  This is a read-only register which merely reflects the value of VR_MII_EEE_TXTIMER Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8008 : (APB3_16) ? (0x8008*0x2) : (0x8008*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&QSGMII_EN&&EEE_EN)

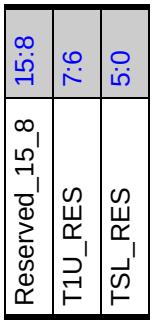


Table 5-441 Fields for Register: VR_MII_3_EEE_TXTIMER

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
7:6	T1U_RES	R	T1u Resolution This field stores the resolution value for the T1u timer. When the generated 100 ns tic timer is not exactly 100 ns, this field is programmed with the required value. The programmed value is equivalent to the number of more (SIGN_BIT is low) or less (SIGN_BIT is high) counts than the ideal count to meet the range of the T1u timer. The DWC_xpcs maintains the default value of the T1u timer as 11, assuming clk_eee_i_time_period * (MULT_FACT_100NS + 1) is equal to 100 ns to produce 1100 ns (1.1 us) as per the T1u requirement in the IEEE standard. Example: For 10GBASE-R or KR_Present configuration, if clk_eee_i_time_period * (MULT_FACT_100NS + 1)is 90, the default T1u timer produces 990 ns (0.99 us) which is lesser than the standard 1.1 us. To make it 1.1 us, you must program this register such that (11 +/- T1U_RES) * ((MULT_FACT_100NS + 1) * clk_eee_i_time_period) is greater than 1.1 us and less than 1.3 us (max limit), that is, (11+ T1U_RES)*90 = 1.1 us. T1U_RES = 2 meets the requirement which produces 1.17 us. Program the SIGN_BIT as 1. **Value After Reset:** 0x0 **Exists:** Always
5:0	TSL_RES	R	TSL Resolution **Value After Reset:** 0x0 **Exists:** Always

5.2.7.31 VR_MII_3_EEE_RXTIMER

- **Description:** VR MII_3 MMD EEE Rx Timer Register
  This register is present only for configurations with MAIN_MODE = 1000BASEX-Only PCS and QSGMII_EN = Enabled and EEE_EN = Enabled.
  This is a read-only register which merely reflects the value of VR_MII_EEE_RXTIMER Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8009 : (APB3_16) ? (0x8009*0x2) : (0x8009*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** BASE1000X_ONLY&&QSGMII_EN&&EEE_EN

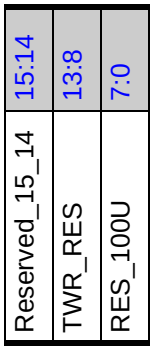


Table 5-442 Fields for Register: VR_MII_3_EEE_RXTIMER

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:8	TWR_RES	R	TWR Resolution **Value After Reset:** 0x0 **Exists:** Always
7:0	RES_100U	R	100 us Resolution **Value After Reset:** 0x0 **Exists:** Always

5.2.7.32 VR_MII_3_LINK_TIMER_CTRL

- **Description:** VR MII_3 MMD Link Timer Control Register
  This register merely reflects the value of VR_MII_LINK_TIMER_CTRL Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x800A : (APB3_16) ? (0x800A*0x2) : (0x800A*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** QSGMII_EN



Table 5-443 Fields for Register: VR_MII_3_LINK_TIMER_CTRL

Bits	Name	Memory Access	Description
15:0	CL37_LINK_TIME	R	Programmable Link Timer Value for Clause 37 auto-negotiation. **Value After Reset:** 0x0 **Exists:** Always

5.2.7.33 VR_MII_3_EEE_MCTRL1

- **Description:** VR MII_3 MMD EEE Mode Control 1 Register
  This register is present only for configurations with MAIN_MODE = 1000BASEX-Only PCS and QSGMII_EN = Enabled and EEE_EN = Enabled.
  This is a read-only register which merely reflects the value of VR_MII_EEE_MCTRL1 Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x800B : (APB3_16) ? (0x800B*0x2) : (0x800B*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** BASE1000X_ONLY&&QSGMII_EN&&EEE_EN

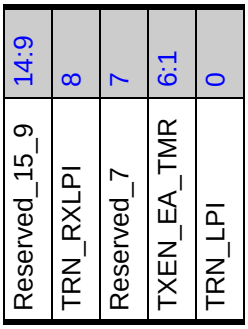


Table 5-444 Fields for Register: VR_MII_3_EEE_MCTRL1

Bits	Name	Memory Access	Description
14:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	TRN_RXLPI	R	Transparent Rx LPI Mode Enable. **Value After Reset:** 0x0 **Exists:** Always
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:1	TXEN_EA_TMR	R	Transmit Enable Early Assert Time Duration **Value After Reset:** 0x0 **Exists:** Always
0	TRN_LPI	R	Transparent Tx LPI Mode Enable **Value After Reset:** 0x0 **Exists:** Always

5.2.7.34 VR_MII_3_DIG_STS

- **Description:** VR MII_3 MMD Digital Status Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8010 : (APB3_16) ? (0x8010*0x2) : (0x8010*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** QSGMII_EN||USXG_QPORT

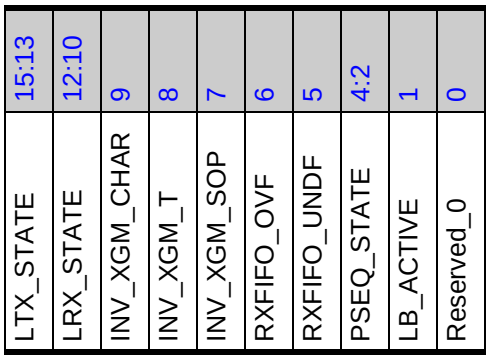


Table 5-445 Fields for Register: VR_MII_3_DIG_STS

Bits	Name	Memory Access	Description
15:13	LTX_STATE	R	LPI Transmit State. This field reflects the value of LTX_STATE field of VR_MII_DIG_STS Register (for 1000BaseX-Only PCS configurations) or VR_XS_PCS_DIG_STS Register (for other configurations). **Value After Reset:** 0x0 **Exists:** Always
12:10	LRX_STATE	R	LPI Receive State This field reflects the value of LRX_STATE field of VR_MII_DIG_STS Register (for 1000BaseX-Only PCS configurations) or VR_XS_PCS_DIG_STS Register (for other configurations). **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
9	INV_XGM_CHAR	R	Invalid XGMII Character (RO,LH Type) -Port 3 This bit indicates an invalid XGMII character on the Transmit path (port 3). ■ 0: No Invalid XGMII character ■ 1: Invalid character This bit is set when one of the following conditions is true on the XGMII interface: ■ |Q| is detected on a lane other than Lane 0. ■ |Q| is detected on Lane 0 and also any of the following conditions is true: ❑ No 8'h00 data in Lane 1. ❑ No 8'h00 data in Lane 2. ❑ No 8'h01 or 8'h02 data on Lane 3. ■ |E| or |T| is encountered in non-data phase and |Q| is encountered in data phase. ■ |D| is detected during the Idle phase (after Terminate and before valid SOF). ■ |I| is detected during the data phase (after SOF and before Terminate). **Note:** This field is present only for Multi-port USXGMII configurations. **Value After Reset:** 0x0 **Exists:** Always
8	INV_XGM_T	R	Invalid XGMII T Character (RO,LH Type) -Port 3 This bit indicates that the transmit data (port 3) received on XGMII has an invalid Terminate character. ■ 0: Normal operation ■ 1: Invalid Terminate character The following can cause an invalid Terminate character: ■ In Terminate column, before lane corresponds to the |T| character, there is a |I| or |Q| character. ■ In Terminate column, after lane corresponds to the |T| character, there is a non-|I| character **Note:** This field is present only for Multi-port USXGMII configurations. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
7	INV_XGM_SOP	R	Invalid XGMII Start Character (RO,LH Type) -Port 3 This bit indicates that the XGMII transmit (Port 3) frame contains invalid SOP character. ■ 0: Normal operation ■ 1: Invalid SOP character An invalid SOP event occurs when a |S| character is detected in a lane other than Lane 0. **Note:** This field is present only for Multi-port USXGMII configurations. **Value After Reset:** 0x0 **Exists:** Always
6	RXFIFO_OVF	R	Rx FIFO Overflow (RO,LH Type) This bit indicates the clock rate compensation FIFO overflow for port 3. ■ 0: Normal operation ■ 1: FIFO overflow **Value After Reset:** 0x0 **Exists:** Always
5	RXFIFO_UNDF	R	Rx FIFO Underflow (RO,LH Type) This bit indicates the clock rate compensation FIFO underflow for port 3. ■ 0: Normal operation ■ 1: FIFO underflow **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4:2	PSEQ_STATE	R	**For Synopsys PHY configurations:** Power Up Sequence State This field indicates the state variable value of the power-up sequence module. For Synopsys Multi-protocol 6G PHY: ■ 3'b000: Wait for MPLL ON ■ 3'b001: Wait for TX up ■ 3'b010: Wait for RX up ■ 3'b011/3'b100: Tx/Rx Stable (Power_Good state) ■ 3'b101: Wait for Rx down (MPLL still ON) ■ 3'b110: MPLL OFF For Synopsys Multi-protocol 12G/16G/25G/32G PHY ■ 3'b000: Wait for ACK High 0 ■ 3'b001: Wait for ACK Low 0 ■ 3'b010: Wait for ACK High 1 ■ 3'b011: Wait for ACK Low 1 ■ 3'b100: Tx/Rx stable (Power_Good state) ■ 3'b101: Power Save state ■ 3'b110: Power Down state ■ 3'b111: Other intermediate states (valid only for 32G PHY configurations) **For non-Synopsys PHY configurations:** Reserved **Value After Reset:** 0x4 **Exists:** Always **Testable:** "(!(BASE1000X_ONLY||KX_ONLY)&&QSGMII_EN) ? \"untestable\" : \"unconstrained\""
1	LB_ACTIVE	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	Reserved_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.7.35 VR_MII_3_ICG_ERRCNT1

- **Description:** VR MII_3 MMD Invalid Code Group Error Count1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8011 : (APB3_16) ? (0x8011*0x2) : (0x8011*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** QSGMII_EN

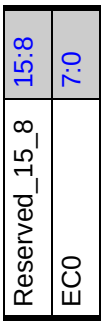


Table 5-446 Fields for Register: VR_MII_3_ICG_ERRCNT1

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:0	EC0	R	Invalid Code Group Count in Port 2 (RO, LH Type) This field gives the invalid code group count in port 2 (of QSGMII) when Bit 4 of VR MII_2 MMD Digital Error Count Select Register is set to 1. **Value After Reset:** 0x0 **Exists:** Always

5.2.7.36 VR_MII_3_GPIO

- **Description:** VR MII_3 MMD GPIO Register
  This register is present only if the DWC_xpcs is configured to have the GPIO interface and QSGMII_EN = Enabled and USXGMII_EN = Disabled.
  This register merely reflects the value of VR_MII_GPIO Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8015 : (APB3_16) ? (0x8015*0x2) : (0x8015*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** QSGMII_EN&&GPIO_EN&&!USXGMII_EN

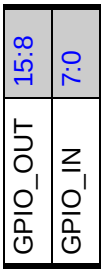


Table 5-447 Fields for Register: VR_MII_3_GPIO

Bits	Name	Memory Access	Description
15:8	GPIO_OUT	R	GPIO Output This field reflects the value of GPIO_OUT field in VR_MII_GPIO register. **Value After Reset:** 0x0 **Exists:** Always
7:0	GPIO_IN	R	GPIO Input This field indicates the content of the xpcs_gpo_i[7:0] port. **Value After Reset:** 0x0 **Exists:** Always

5.2.7.37 VR_MII_3_DIG_CTRL2

- **Description:** VR MII_2 MMD Digital Control2 Register
  Note: This register is present only in 1000BaseX-Only PCS configurations with QSGMII_EN = Enabled.
  This register merely reflects the value of VR_MII_DIG_CTRL2 Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80E1 : (APB3_16) ? (0x80E1*0x2) : (0x80E1*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** QSGMII_EN&&BASE1000X_ONLY

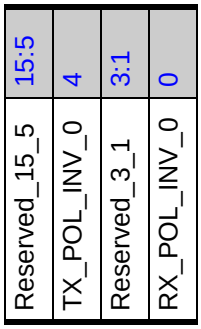


Table 5-448 Fields for Register: VR_MII_3_DIG_CTRL2

Bits	Name	Memory Access	Description
15:5	Reserved_15_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_POL_INV_0	R	Tx Polarity Invert on Lane 0 **Value After Reset:** 0x0 **Exists:** Always
3:1	Reserved_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	RX_POL_INV_0	R	Rx Polarity Invert on Lane 0 **Value After Reset:** 0x0 **Exists:** Always

5.2.7.38 VR_MII_3_DIG_ERRCNT_SEL

- **Description:** VR MII_3 MMD Digital Error Count Select Register
  This register is applicable only in QSGMII mode.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x80E2 : (APB3_16) ? (0x80E2*0x2) : (0x80E2*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1C0000 : (APB3_16) ? (0x1C0000*0x2) : (0x1C0000*0x4)"
- **Exists:** QSGMII_EN

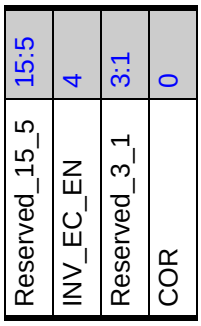


Table 5-449 Fields for Register: VR_MII_3_DIG_ERRCNT_SEL

Bits	Name	Memory Access	Description
15:5	Reserved_15_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	INV_EC_EN	R/W	Invalid Code Group Error Counter Enable When this bit is set, the counting of invalid code group errors (on port 3) is enabled. ■ 0: The counting of errors is disabled ■ 1: The counting of errors is enabled For information about the fields containing the number of errors counted, see VR_MII_3_ICG_ERRCNT1 register. **Value After Reset:** 0x0 **Exists:** Always
3:1	Reserved_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	COR	R/W	Clear on Read When this bit is set and the host reads port 3 error counter (VR_MII_3_ICG_ERRCNT1 Register), that counter is cleared after the read cycle. ■ 0: Normal operation ■ 1: Clear error counter that is read **Value After Reset:** 0x0 **Exists:** Always

5.2.8 DWC_xpcs_map/VS_MMD1 Registers

5.2.8.1 SR_VSMMD_PMA_ID1

- **Description:** SR Control MMD PMA Device Identifier Register 1
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1E0000 : (APB3_16) ? (0x1E0000*0x2) : (0x1E0000*0x4)"
- **Exists:** Always



Table 5-450 Fields for Register: SR_VSMMD_PMA_ID1

Bits	Name	Memory Access	Description
15:0	PMADOUI_3_18	* Varies	**For configurations with PMA MMD:** Organizationally Unique Identifier[3:18] for PMA MMD This field contains Bits[18:3] of 24-bit OUI of device manufacturer. The DWC_xpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. The default value of this register is similar to the default value of SR_PMA_DEV_ID_1 Register. The value written in this register is reflected in SR_PMA_DEV_ID_1 Register. **Access Type:** ■ RW: For PMA MMD configurations with IEEE_REG_WR_-SUPPORT = Enabled ■ RO: For all other configurations **For other configurations:** Reserved **Value After Reset:** "(PMA_MMD) ? IEEE_PMA_DEV_OUI_3_18 : 0x0" **Exists:** Always **Memory Access:** "(PMA_MMD&&IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""

5.2.8.2 SR_VSMMD_PMA_ID2

- **Description:** SR Control MMD PMA Device Identifier Register 2
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1 : (APB3_16) ? (0x1*0x2) : (0x1*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1E0000 : (APB3_16) ? (0x1E0000*0x2) : (0x1E0000*0x4)"
- **Exists:** Always

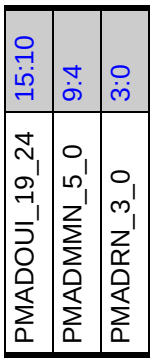


Table 5-451 Fields for Register: SR_VSMMD_PMA_ID2

Bits	Name	Memory Access	Description
15:10	PMADOUI_19_24	* Varies	**For configurations with PMA MMD:** Organizationally Unique Identifier[19:24] for PMA MMD The DWC_xpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer of PMA MMD. This field contains Bits[24:19] of 24-bit OUI of device manufacturer. The default value of this field is similar to the default value of the corresponding field in SR_PMA_DEV_ID_2 Register. The value written in this field is reflected in the corresponding field of SR_PMA_DEV_ID_2 Register. **Access Type:** ■ RW: For PMA MMD configurations with IEEE_REG_WR_-SUPPORT = Enabled. ■ RO: For all other configurations. **For other configurations:** Reserved **Value After Reset:** "(PMA_MMD) ? IEEE_PMA_DEV_OUI_19_24 : 0x0" **Exists:** Always **Memory Access:** "(PMA_MMD&&IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
9:4	PMADMMN_5_0	* Varies	**For configurations with PMA MMD:** Model Number for PMA MMD This field contains the 6-bit Model number of the PMA MMD. The default value of this field is similar to the default value of the corresponding field in SR_PMA_DEV_ID_2 Register. The value written in this field is reflected in the corresponding field of SR_PMA_DEV_ID_2 Register. **Access Type:** ■ RW: For PMA MMD configurations with IEEE_REG_WR_-SUPPORT = Enabled. ■ RO: For all other configurations. **For other configurations:** Reserved **Value After Reset:** "(PMA_MMD) ? IEEE_PMA_DEV_MODEL_NO_0_5 : 0x0" **Exists:** Always **Memory Access:** "(PMA_MMD&&IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""
3:0	PMADRN_3_0	* Varies	**For configurations with PMA MMD:** Revision Number for PMA MMD This field contains the 4-bit Revision number of the PMA MMD. The default value of this field is similar to the default value of the corresponding field in SR_PMA_DEV_ID_2 Register. The value written in this field is reflected in the corresponding field of SR_PMA_DEV_ID_2 Register. **Access Type:** ■ RW: For PMA MMD configurations with IEEE_REG_WR_-SUPPORT = Enabled. ■ RO: For all other configurations. **For other configurations:** Reserved **Value After Reset:** "(PMA_MMD) ? IEEE_PMA_DEV_REV_NO_0_3 : 0x0" **Exists:** Always **Memory Access:** "(PMA_MMD&&IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""

5.2.8.3 SR_VSMMD_DEV_ID1

- **Description:** SR Control MMD Device Identifier Register 1
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x2 : (APB3_16) ? (0x2*0x2) : (0x2*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1E0000 : (APB3_16) ? (0x1E0000*0x2) : (0x1E0000*0x4)"
- **Exists:** Always

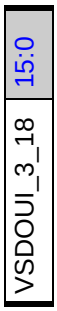


Table 5-452 Fields for Register: SR_VSMMD_DEV_ID1

Bits	Name	Memory Access	Description
15:0	VSDOUI_3_18	* Varies	Organizationally Unique Identifier[3:18] for Vendor-Specific MMD1 This field contains Bits[18:3] of 24-bit OUI of device manufacturer. The DWC_xpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. You can configure the value of each MMD using coreConsultant. **Access Type:** ■ RW: For configurations with IEEE_REG_WR_SUPPORT = Enabled. ■ RO: For all other configurations. **Value After Reset:** IEEE_VS_DEV_OUI_3_18 **Exists:** Always **Memory Access:** "(IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""

5.2.8.4 SR_VSMMD_DEV_ID2

- **Description:** SR Control MMD Device Identifier Register 2
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x3 : (APB3_16) ? (0x3*0x2) : (0x3*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1E0000 : (APB3_16) ? (0x1E0000*0x2) : (0x1E0000*0x4)"
- **Exists:** Always

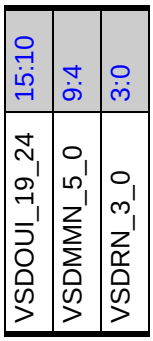


Table 5-453 Fields for Register: SR_VSMMD_DEV_ID2

Bits	Name	Memory Access	Description
15:10	VSDOUI_19_24	* Varies	Organizationally Unique Identifier[19:24] for Vendor-Specific MMD1 This field contains Bits[24:19] of the device manufacturer's 24-bit OUI. The DWC_xpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. You can configure the value of each MMD using coreConsultant. **Access Type:** ■ RW: For configurations with IEEE_REG_WR_SUPPORT = Enabled. ■ RO: For all other configurations. **Value After Reset:** IEEE_VS_DEV_OUI_19_24 **Exists:** Always **Memory Access:** "(IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\"""

Bits	Name	Memory Access	Description
9:4	VSDMMN_5_0	* Varies	Model Number for Vendor-Specific MMD1 This field contains the 6-bit Model number of the vendor-specific MMD1.You can configure the default value through coreConsultant. **Access Type:** ■ RW: For configurations with IEEE_REG_WR_SUPPORT = Enabled. ■ RO: For all other configurations. **Value After Reset:** IEEE_VS_DEV_MODEL_NO_0_5 **Exists:** Always **Memory Access:** "(IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""
3:0	VSDRN_3_0	* Varies	Revision Number for Vendor-Specific MMD1 This field contains the 4-bit Revision number of the vendor-specific MMD1. You can configure the default value through coreConsultant. **Access Type:** ■ RW: For configurations with IEEE_REG_WR_SUPPORT = Enabled. ■ RO: For all other configurations. **Value After Reset:** IEEE_VS_DEV_REV_NO_0_3 **Exists:** Always **Memory Access:** "(IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""

5.2.8.5 SR_VSMMD_PCS_ID1

- **Description:** SR Control MMD PCS Device Identifier Register 1
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x4 : (APB3_16) ? (0x4*0x2) : (0x4*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1E0000 : (APB3_16) ? (0x1E0000*0x2) : (0x1E0000*0x4)"
- **Exists:** Always



Table 5-454 Fields for Register: SR_VSMMD_PCS_ID1

Bits	Name	Memory Access	Description
15:0	PCSDOUI_3_18	* Varies	Organizationally Unique Identifier[3:18] for PCS MMD This field contains Bits[18:3] of 24-bit OUI of device manufacturer. The DWC_xpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. The default value of this register is similar to the default value of SR XS or PCS MMD Device Identifier Register 1. The value written in this register is reflected in SR XS or PCS MMD Device Identifier Register 1. **Access Type:** ■ RW: For configurations with IEEE_REG_WR_SUPPORT = Enabled. ■ RO: For all other configurations. **Value After Reset:** IEEE_DEV_OUI_3_18 **Exists:** Always **Memory Access:** "(IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""

5.2.8.6 SR_VSMMD_PCS_ID2

- **Description:** SR Control MMD PCS Device Identifier Register 2
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x5 : (APB3_16) ? (0x5*0x2) : (0x5*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1E0000 : (APB3_16) ? (0x1E0000*0x2) : (0x1E0000*0x4)"
- **Exists:** Always

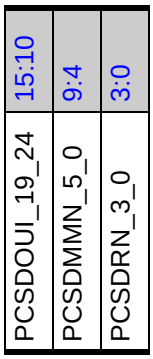


Table 5-455 Fields for Register: SR_VSMMD_PCS_ID2

Bits	Name	Memory Access	Description
15:10	PCSDOUI_19_24	* Varies	Organizationally Unique Identifier[19:24] for PCS MMD This field contains Bits[24:19] of 24-bit OUI of device manufacturer. The DWC_xpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. The default value of this field is similar to the default value of the corresponding field in SR XS or PCS MMD Device Identifier Register 2. The value written in this field is reflected in the corresponding field of SR XS or PCS MMD Device Identifier Register 2. **Access Type:** ■ RW: For configurations with IEEE_REG_WR_SUPPORT = Enabled. ■ RO: For all other configurations. **Value After Reset:** IEEE_DEV_OUI_19_24 **Exists:** Always **Memory Access:** "(IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\"""

Bits	Name	Memory Access	Description
9:4	PCSDMMN_5_0	* Varies	Model Number for XS or PCS MMD This field contains the 6-bit Model number of vendor-specific XS or PCS MMD. The default value of this field is similar to the default value of the corresponding field in SR XS or PCS MMD Device Identifier Register 2. The value written in this field is reflected in the corresponding field of SR XS or PCS MMD Device Identifier Register 2. **Access Type:** ■ RW: For configurations with IEEE_REG_WR_SUPPORT = Enabled ■ RO: For all other configurations **Value After Reset:** IEEE_DEV_MODEL_NO_0_5 **Exists:** Always **Memory Access:** "(IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\"""
3:0	PCSDRN_3_0	* Varies	Revision Number for XS or PCS MMD This field contains the 4-bit Revision number of vendor-specific XS or PCS MMD. The default value of this field is similar to the default value of the corresponding field in SR XS or PCS MMD Device Identifier Register 2. The value written in this field is reflected in the corresponding field of SR XS or PCS MMD Device Identifier Register 2. **Access Type:** ■ RW: For configurations with IEEE_REG_WR_SUPPORT = Enabled ■ RO: For all other configurations **Value After Reset:** IEEE_DEV_REV_NO_0_3 **Exists:** Always **Memory Access:** "(IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\"""

5.2.8.7 SR_VSMMD_AN_ID1

- **Description:** SR Control MMD AN Device Identifier Register 1
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x6 : (APB3_16) ? (0x6*0x2) : (0x6*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1E0000 : (APB3_16) ? (0x1E0000*0x2) : (0x1E0000*0x4)"
- **Exists:** Always

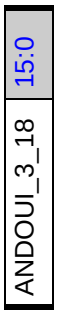


Table 5-456 Fields for Register: SR_VSMMD_AN_ID1

Bits	Name	Memory Access	Description
15:0	ANDOUI_3_18	* Varies	**For Backplane Ethernet Configurations:** Organizationally Unique Identifier[3:18] for AN MMD This field contains Bits[18:3] of the device manufacturer's 24-bit OUI. The DWC_xpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. The default value of this register is similar to the default value of SR AN MMD Device Identifier Register 1. The value written in this register is reflected in SR AN MMD Device Identifier Register 1. **Default Value:** Configurable only when MAIN_MODE = Backplane Ethernet PCS; Otherwise, the default value is 0. **For other configurations:** Reserved **Value After Reset:** "(BACKPLANE_ETHERNET_PCS) ? IEEE_AN_DEV_OUI_3_18 : 0x0" **Exists:** Always **Memory Access:** "(BACKPLANE_ETHERNET_PCS&&IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""

5.2.8.8 SR_VSMMD_AN_ID2

- **Description:** SR Control MMD AN Device Identifier Register 2
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x7 : (APB3_16) ? (0x7*0x2) : (0x7*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1E0000 : (APB3_16) ? (0x1E0000*0x2) : (0x1E0000*0x4)"
- **Exists:** Always

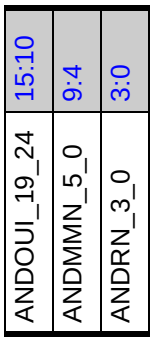


Table 5-457 Fields for Register: SR_VSMMD_AN_ID2

Bits	Name	Memory Access	Description
15:10	ANDOUI_19_24	* Varies	**For Backplane Ethernet Configurations:** Organizationally Unique Identifier[19:24] for AN MMD The DWC_xpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. This field contains Bits[24:19] of the 24-bit OUI of device manufacturer. The default value of this field is similar to the default value of the corresponding field in SR AN MMD Device Identifier Register 2. The value written in this field is reflected in the corresponding field of SR AN MMD Device Identifier Register 2. **Default Value:** Configurable only when MAIN_MODE = Backplane Ethernet PCS; Otherwise, the default value is 0. **For other configurations:** Reserved **Value After Reset:** "(BACKPLANE_ETHERNET_PCS) ? IEEE_AN_DEV_OUI_19_24 : 0x0" **Exists:** Always **Memory Access:** "(BACKPLANE_ETHERNET_PCS&&IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
9:4	ANDMMN_5_0	* Varies	**For Backplane Ethernet Configurations:**Model Number This field contains the 6-bit Model number of AN MMD. The default value of this field is similar to the default value of the corresponding field in SR AN MMD Device Identifier Register 2. The value written in this field is reflected in the corresponding field of SR AN MMD Device Identifier Register 2. **Default Value:** Configurable only when MAIN_MODE = Backplane Ethernet PCS; Otherwise, the default value is 0. **For other configurations:** Reserved **Value After Reset:** "(BACKPLANE_ETHERNET_PCS) ? IEEE_AN_DEV_MODEL_NO_0_5 : 0x0" **Exists:** Always **Memory Access:** "(BACKPLANE_ETHERNET_PCS&&IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\"""
3:0	ANDRN_3_0	* Varies	**For Backplane Ethernet Configurations:** Revision Number This field contains the 4-bit Revision number of AN MMD. The default value of this field is similar to the default value of the corresponding field in SR AN MMD Device Identifier Register 2. The value written in this field is reflected in the corresponding field of SR AN MMD Device Identifier Register 2. **Default Value:**Configurable only when MAIN_MODE = Backplane Ethernet PCS; Otherwise, the default value is 0. **For other configurations:** Reserved **Value After Reset:** "(BACKPLANE_ETHERNET_PCS) ? IEEE_AN_DEV_REV_NO_0_3 : 0x0" **Exists:** Always **Memory Access:** "(BACKPLANE_ETHERNET_PCS&&IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\"""

5.2.8.9 SR_VSMMD_STS

- **Description:** SR Control MMD Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8 : (APB3_16) ? (0x8*0x2) : (0x8*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1E0000 : (APB3_16) ? (0x1E0000*0x2) : (0x1E0000*0x4)"
- **Exists:** Always

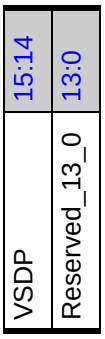


Table 5-458 Fields for Register: SR_VSMMD_STS

Bits	Name	Memory Access	Description
15:14	VSDP	R	Control MMD Device Present This field indicates if the control MMD device is present and responding to this address: ■ 10: Device responding at this address ■ 11, 01, or 00: No device responding at this address **Value After Reset:** 0x2 **Exists:** Always
13:0	Reserved_13_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.8.10 SR_VSMMD_CTRL

- **Description:** SR Control MMD Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x9 : (APB3_16) ? (0x9*0x2) : (0x9*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1E0000 : (APB3_16) ? (0x1E0000*0x2) : (0x1E0000*0x4)"
- **Exists:** Always

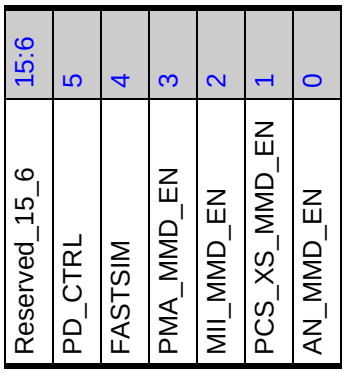


Table 5-459 Fields for Register: SR_VSMMD_CTRL

Bits	Name	Memory Access	Description
15:6	Reserved_15_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5	PD_CTRL	* Varies	**For Non-Synopsys PHY configurations:** Power Down Control This bit is used to control the output port 'xpcs_pdown_o'. If this bit is set, xpcs_pdown_o port is NOT asserted when LPM (Low Power Enable) bit is programmed to 1. If this bit is low, xpcs_down_o port is asserted when LPM (Low Power Enable) bit is programmed to 1. **For Synopsys PHY configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!SNPS_XAUI_PHY) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
4	FASTSIM	R/W	Fast Simulation Enable When set, this bit indicates that the Fast simulation is enabled. When this bit is set to 1, all IEEE Std 802.3 defined long timers, implemented in the DWC_xpcs are reduced to shorter time period to reduce the simulation time. The long timers are implemented in Clause 73 and Clause 37 auto-negotiation modules and also EEE Tx and Rx modules. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
3	PMA_MMD_EN	* Varies	**For configurations with PMA MMD:** PMA MMD Enable When set, this bit indicates that the PMA MMD device is accessible. When reset, this bit indicates that the PMA MMD device is not accessible. **For configurations without PMA MMD:**Reserved **Value After Reset:** "(PMA_MMD) ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(PMA_MMD) ? \"read-write\" : \"read-only\""
2	MII_MMD_EN	* Varies	For configurations with CL37_AN = Enabled: VS MMD Enable When set, this bit indicates that the vendor-specific MMD1 device is accessible. When reset, this bit indicates that the vendor-specific MMD2 (MII MMD) device is not accessible. **For configurations with CL37_AN=Disabled:**Reserved **Value After Reset:** "(CL37_AN) ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(CL37_AN) ? \"read-write\" : \"read-only\""
1	PCS_XS_MMD_EN	* Varies	**For Configurations with PCS/XS MMD:** PCS MMD Enable When set, this bit indicates that the PCS MMD device is accessible. When reset, this bit indicates that the PCS MMD device is not accessible. **For other configurations:** Reserved **Value After Reset:** "(PCS_MMD||XGXS_PCS) ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(PCS_MMD||XGXS_PCS) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
0	AN_MMD_EN	* Varies	**For Backplane Ethernet configurations:**AN MMD Enable When set, this bit indicates that the AN MMD device is accessible. When reset, this bit indicates that the AN MMD device is not accessible. **For other configurations:** Reserved **Value After Reset:** "(BACKPLANE_ETHERNET_PCS) ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(BACKPLANE_ETHERNET_PCS) ? \"read-write\" : \"read-only\"""

5.2.8.11 SR_VSMMD_PKGID1

- **Description:** SR Control MMD Package Identifier Register 1
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF| |MCI_INTF| |MDIO_INTF) ? 0xE : (APB3_16) ? (0xE*0x2) : (0xE*0x4)" + "(APB3_INTF| |MCI_INTF| |MDIO_INTF) ? 0x1E0000 : (APB3_16) ? (0x1E0000*0x2) : (0x1E0000*0x4)"
- **Exists:** Always



Table 5-460 Fields for Register: SR_VSMMD_PKGID1

Bits	Name	Memory Access	Description
15:0	MMDPOUI_3_18	* Varies	**For 1000BASEX-Only Configurations:**Reserved**For other configurations:** Organizationally Unique Identifier The DWC_xpcs offers 24 configurable bits[24:1], known as OUI, for identifying the device manufacturer. This field contains Bits[18:3] of 24-bit OUI of device manufacturer. This is a unique identifier for all MMDs present in the DWC_xpcs. The value written in this register is reflected in the following registers: ■ SR_PMA_PKG1 Register ■ SR_XS_PCS_PKG1 Register ■ SR_AN_PKG1 Register **Default Value:** Configurable only when MAIN_MODE != 10GBASE-X PCS. Otherwise, the default value is 0. **Value After Reset:** "(!BASE1000X_ONLY) ? IEEE_PKG_OUI_3_18 : 0x0" **Exists:** Always **Memory Access:** "(!BASE1000X_ONLY&&IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""

5.2.8.12 SR_VSMMD_PKGID2

- **Description:** SR Control MMD Package Identifier Register 2
  Note: The values written in the fields of this register are reflected in the corresponding fields of the following registers:
  - SR_PMA_PKG2 Register
  - SR_XS_PCS_PKG2 Register
  - SR_AN_PKG2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0xF : (APB3_16) ? (0xF*0x2) : (0xF*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1E0000 : (APB3_16) ? (0x1E0000*0x2) : (0x1E0000*0x4)"
- **Exists:** Always

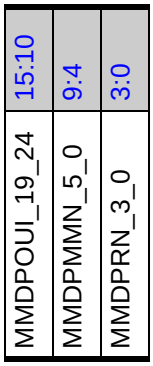


Table 5-461 Fields for Register: SR_VSMMD_PKGID2

Bits	Name	Memory Access	Description
15:10	MMDPOUI_19_24	* Varies	**For 1000BASEX-Only Configurations:**Reserved**For other configurations:** Organizationally Unique Identifier The DWC_xpcs offers 24 configurable bits [24:1], known as OUI, for identifying the device manufacturer. This field contains Bits[24:19] of 24-bit OUI of device manufacturer. This is a unique identifier for all MMDs present in the DWC_xpcs. **Default Value:** Configurable only when MAIN_MODE != 10GBaseX-Only PCS; Otherwise, the default value is 0. **Value After Reset:** "(!BASE1000X_ONLY) ? IEEE_PKG_OUI_19_24 : 0x0" **Exists:** Always **Memory Access:** "(!BASE1000X_ONLY&&IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
9:4	MMDPMMN_5_0	* Varies	**For 1000BASEX-Only Configurations:**Reserved **For other configurations:** Model Number This field contains the 5-bit Model Number of the package in which the MMDs are present in DWC_xpcs. **Default Value:** Configurable only when MAIN_MODE != 1000BaseX-Only PCS; Otherwise, the default value is 0. **Value After Reset:** "(!BASE1000X_ONLY) ? IEEE_PKG_MODEL_NO_0_5 : 0x0" **Exists:** Always **Memory Access:** "(!BASE1000X_ONLY&&IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\"""
3:0	MMDPRN_3_0	* Varies	**For 1000BASEX-Only Configurations:**Reserved**For other configurations:** Revision Number This field contains the 4-bit revision number of the package in which the MMDs are present in DWC_xpcs. **Default Value:** Configurable only when MAIN_MODE != 1000BASEX-Only PCS; Otherwise, the default value is 0. **Value After Reset:** "(!BASE1000X_ONLY) ? IEEE_PKG_REV_NO_0_3 : 0x0" **Exists:** Always **Memory Access:** "(!BASE1000X_ONLY&&IEEE_REG_WR_SUPPORT) ? \"read-write\" : \"read-only\"""

5.2.9 DWC_xpcs_map/VS_MII_MMD Registers

5.2.9.1 SR_MII_CTRL

- **Description:** SR MII MMD Control Register
  The host can use this register to control (enable or disable) some of the features supported by the DWC_xpcs with 1G (1000BASE-X) or USXGMII support) and Clause 37 auto-negotiation support.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (CL37_AN||BASE1000X_ONLY)

15	14	13	12	11	10	9	8	7	6	5	4:0
RST	LBE	SS13	AN_ENABLE	LPM	Reserved_10	RESTART_AN	DUPLEX_MODE	Reserved_7	SS6	SS5	Reserved_4_0

Table 5-462 Fields for Register: SR_MII_CTRL

Bits	Name	Memory Access	Description
15	RST	R/W	Soft Reset (RW,SC Type) **For Synopsys PHY configurations:** When the host sets this bit, the CSR block triggers the software reset process in which all internal blocks are reset, except the Management Interface block. The registers are reset to their default values. When this bit is set, it also resets the PHY. This bit is self-cleared when PSEQ_STATE in VR_XS_PCS_DIG_STS Register is equal to 3'b100, that is, Tx/Rx clocks are stable and in Power_Good state. **For Non-Synopsys PHY configurations:** When the host sets this bit, the CSR block triggers the software reset process in which all internal blocks are reset, except the Management Interface block. The registers are reset to their default values. When this bit is set, it also resets the PHY. This bit is self-cleared after the following: ■ 32 clk_csr_i clocks for the MCI interface ■ 1 MDC clock period for the MDIO interface **Note:** For description of read or write access for registers during reset process, see "Special Case Register Access" section of the Databook. **Value After Reset:** 0x0 **Exists:** Always
14	LBE	R/W	Loopback Enable When set, this bit loops the PHY Tx lanes back to the PHY Rx lanes. ■ 0: The DWC_xpcs de-asserts xpcs_loopback_en_o signal. ■ 1: The DWC_xpcs asserts xpcs_loopback_en_o signal. **Value After Reset:** 0x0 **Exists:** Always

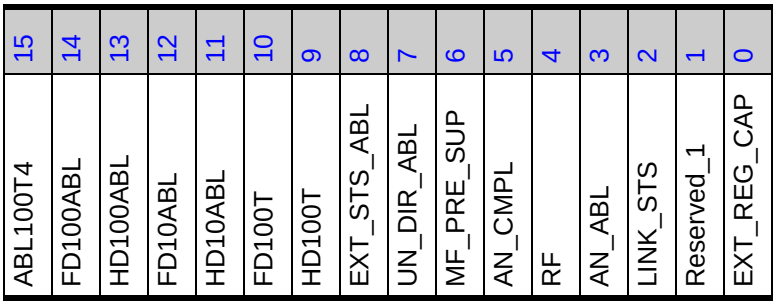
Bits	Name	Memory Access	Description
13	SS13	* Varies	Speed Selection (LSB) This bit, along with the SS6 bit and SS5 bit of this register, indicates the speed. **For USXGMII configurations:** ■ When SS5=1, SS6=0 and SS13=1, speed is 5 Gbps ■ When SS5=1, SS6=0 and SS13=0, speed is 2.5 Gbps ■ When SS5=0, SS6=1 and SS13=1, speed is 10 Gbps ■ When SS5=0, SS6=1 and SS13=0, speed is 1000 Mbps ■ When SS5=0, SS6=0 and SS13=1, speed is 100 Mbps ■ When SS5=0, SS6=0 and SS13=0, speed is 10 Mbps **For SGMII configurations:** ■ When SS6=1 and SS13=0, speed is 1000 Mbps ■ When SS6=0 and SS13=1, speed is 100 Mbps ■ When SS6=0 and SS13=0, speed is 10 Mbps This bit is reserved when SGMII_EN = Disabled and QSGMII_EN=Disabled and USXGMII_EN = Disabled. **Value After Reset:** "USXGMII_EN ? 0x1 : 0x0" **Exists:** Always **Testable:** "USXGMII_EN ? \"untestable\" : \"writeAsRead\"" **Memory Access:** "(SGMII_EN||QSGMII_EN||USXGMII_EN) ? \"read-write\" : \"read-only\""
12	AN_ENABLE	R/W	Enable Auto-Negotiation When set to 1, this bit enables the Clause 37 auto-negotiation process. **Value After Reset:** "(USXGMII_EN||BACKPLANE_ETHERNET_PCS) ? \"0x0\" : \"0x1\"" **Exists:** Always **Testable:** untestable

Bits	Name	Memory Access	Description
11	LPM	R/W	Power-Down Mode This bit controls the power-down mode of the DWC_xpcs. ■ 0: Normal operation ■ 1: The DWC_xpcs goes to the power-down mode along with the PHY. For non-Synopsys PHY, the xpcs_pdown_o port is asserted. For Synopsys PHY, the DWC_xpcs does the following to trigger the power-down mode: ■ Turns off the PHY Receiver and Transmitter ■ Switches off all the clocks When the host clears this bit, the DWC_xpcs resumes the normal operation. In a Synopsys PHY configuration, after clearing this bit, the host must wait until Bits[4:2] of VR_XS_PCS_DIG_STS Register indicate that the DWC_xpcs is in the normal state. **Note:** For information about read or write access to registers during power-down mode, see "Special Case Register Access" section of the Databook. **Value After Reset:** 0x0 **Exists:** Always
10	Reserved_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9	RESTART_AN	R/W	Restart Auto-Negotiation (RW,SC Type) When the host writes this bit, the DWC_xpcs initiates the auto-negotiation process. This bit is used to restart the auto-negotiation which is already initiated by setting Bit 12. The DWC_xpcs clears this bit after restarting the auto-negotiation. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
8	DUPLEX_MODE	R/W	Duplex Mode This bit specifies the duplex mode of the DWC_xpcs. ■ 0: Half duplex ■ 1: Full duplex If Bit 12 is set to 0, this bit determines the PHY link duplex mode. If Bit 12 is set to 1, then the PHY link duplex mode is independent of this bit (although the host can write any value) and is determined by the outcome of the Clause 37 auto-negotiation process. **Note:** USXGMII mode supports only FULL duplex mode, irrespective of the value programmed to this bit. **Value After Reset:** 0x1 **Exists:** Always **Testable:** untestable
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6	SS6	* Varies	Speed Selection This bit, along with the SS13 bit (and SS5 bit, in case of USXGMII configuration) of this register indicates the speed. For more information, see description of the SS13 bit. This bit is reserved when SGMII_EN = Disabled and QSGMII_EN =Disabled and USXGMII_EN=Disabled. **Value After Reset:** 0x1 **Exists:** Always **Testable:** untestable **Memory Access:** "(SGMII_EN||QSGMII_EN||USXGMII_EN) ? \"read-write\" : \"read-only\""
5	SS5	* Varies	**For USXGMII configurations:** Speed Selection bit This bit along with SS13 and SS6 bit of this register indicates the speed, For more information, see description of the SS13 bit. **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** "(USXGMII_EN) ? \"untestable\" : \"unconstrained\"" **Memory Access:** "USXGMII_EN ? \"read-write\" : \"read-only\""
4:0	Reserved_4_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.9.2 SR_MII_STS

- **Description:** SR MII MMD Status Register
  The host uses this register to know the features supported by the DWC_xpcs in the 1000BASE-X mode.
  Note: This register is present if the selected configuration has 1G/KX support.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1 : (APB3_16) ? (0x1*0x2) : (0x1*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** ((CL37_AN&&!RPCS_ONLY)||BASE1000X_ONLY)



15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ABL100T4	FD100ABL	HD100ABL	FD10ABL	HD10ABL	FD100T	HD100T	EXT_STS_ABL	UN_DIR_ABL	MF_PRE_SUP	AN_CMPL	RF	AN_ABL	LINK_STS	Reserved_1	EXT_REG_CAP

Table 5-463 Fields for Register: SR_MII_STS

Bits	Name	Memory Access	Description
15	ABL100T4	R	100BASE-T4 Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always
14	FD100ABL	R	100BASE-X Full-Duplex Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always
13	HD100ABL	R	100BASE-X Half-Duplex Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always
12	FD10ABL	R	10 Mbps Full-Duplex Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
11	HD10ABL	R	10 Mbps Half-Duplex Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always
10	FD100T	R	100BASE-T2 Full-Duplex Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always
9	HD100T	R	100BASE-T2 Half-Duplex Ability The DWC_xpcs always returns 0 because it does not support this functionality. **Value After Reset:** 0x0 **Exists:** Always
8	EXT_STS_ABL	R	Extended Status Information ■ 0: No Extended Status information is present at register address 16'h000F of this MMD device. ■ 1: Extended Status information is present at register address 16'h000F of this MMD device. DWC_xpcs returns this bit as 1. **Value After Reset:** 0x1 **Exists:** Always **Testable:** untestable
7	UN_DIR_ABL	R	Unidirectional Ability ■ 1: The DWC_xpcs is able to transmit GMII irrespective of whether device has determined the valid link or not. ■ 0: The DWC_xpcs is able to transmit GMII only when the device has determined the valid link. The DWC_xpcs always returns this bit as 0. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
6	MF_PRE_SUP	R	MF Preamble Suppression ■ 1: The DWC_xpcs accepts the MDIO frames with preamble suppressed. ■ 0: The DWC_xpcs does not accept the MDIO frames with preamble suppressed. This bit is always set to 0. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
5	AN_CMPL	R	Auto-negotiation Complete ■ 1: The AN process is complete ■ 0: The AN process is not complete When this bit is set to 1, the contents of the AN MMD Advertisement, AN MMD Link partner Ability, and AN MMD Expansion registers are valid. This bit returns 0 if AN_ENABLE is set to 0. **Value After Reset:** 0x0 **Exists:** Always
4	RF	R	Remote Fault (RO, LH Type) When set to 1, this bit indicates that the receive link of the link partner is down. This bit is set based on the auto-negotiated (1000BASE-X auto-negotiation) information from the link partner. ■ 1: The DWC_xpcs detected a remote fault. ■ 0: The DWC_xpcs did not detect a remote fault. This bit is not valid in SGMII/QSGMII/USXGMII Modes. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
3	AN_ABL	R	Auto-negotiation Ability The DWC_xpcs always returns this bit as 1. ■ 1: The DWC_xpcs is able to perform auto-negotiation. ■ 0: The DWC_xpcs is not able to perform auto-negotiation. **Value After Reset:** 0x1 **Exists:** Always **Testable:** untestable
2	LINK_STS	R	Link Status (RO, LL Type) When the DWC_xpcs sets this bit to 1, it indicates that the Rx link is up. If the link goes down, it is latched until the host perform the read operation to this register. ■ 1: Link Up ■ 0: Link Down **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
1	Reserved_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	EXT_REG_CAP	R	Extended Register Capability ■ 1: Extended Register capability exists. This bit is always set to 1. ■ 0: Extended Register capability does not exist. **Value After Reset:** 0x1 **Exists:** Always **Testable:** untestable

5.2.9.3 SR_MII_DEV_ID1

- **Description:** SR MII MMD Device Identifier Register 1
  This register returns the configurable Organizationally Unique Identifier (OUI) to the host.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x2 : (APB3_16) ? (0x2*0x2) : (0x2*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (CL37_AN | |BASE1000X_ONLY)



Table 5-464 Fields for Register: SR_MII_DEV_ID1

Bits	Name	Memory Access	Description
15:0	VS_MII_DEV_OUI_3_18	R	Organizationally Unique Identifier[3:18] This field contains Bits [18:3] of 24-bit OUI of device manufacturer. The DWC_xpcs offers 24 configurable Bits[24:1] for identifying the device manufacturer. The default value of this register is similar to the value of SR XS or PCS MMD Device Identifier Register 1. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:0] of SR XS or PCS MMD Device Identifier Register 1 modifies the content of this register. **Value After Reset:** IEEE_DEV_OUI_3_18 **Exists:** Always **Testable:** untestable

5.2.9.4 SR_MII_DEV_ID2

- **Description:** SR MII MMD Device Identifier Register 2
  This register returns the configurable Organizationally Unique Identifier (OUI), Model Number, and Revision Number of the device to the host.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x3 : (APB3_16) ? (0x3*0x2) : (0x3*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (CL37_AN||BASE1000X_ONLY)

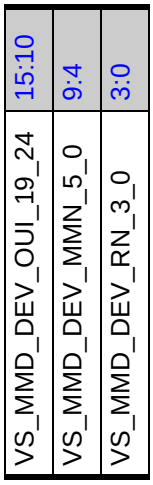


Table 5-465 Fields for Register: SR_MII_DEV_ID2

Bits	Name	Memory Access	Description
15:10	VS_MMD_DEV_OUI_19_24	R	Organizationally Unique Identifier [19:24] This field contains Bits[24:19] of 24-bit OUI of device manufacturer. The DWC_xpcs offers 24 configurable Bits[24:1] for identifying the device manufacturer. The default value of this register is similar to the value of SR XS or PCS MMD Device Identifier Register 2. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[15:10] of SR XS or PCS MMD Device Identifier Register 2 modifies the content of this register. **Value After Reset:** IEEE_DEV_OUI_19_24 **Exists:** Always **Testable:** untestable

Bits	Name	Memory Access	Description
9:4	VS_MMD_DEV_MMN_5_0	R	Model Number This field contains the 5-bit Model Number of the device. The default value of this register is similar to the value of SR XS or PCS MMD Device Identifier Register 2. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[9:4] of SR XS or PCS MMD Device Identifier Register 2 modifies the content of this register. **Value After Reset:** IEEE_DEV_MODEL_NO_0_5 **Exists:** Always **Testable:** untestable
3:0	VS_MMD_DEV_RN_3_0	R	Revision Number This field contains the 4-bit Revision Number of the device. The default value of this register is similar to the value of SR XS or PCS MMD Device Identifier Register 2. If IEEE_REG_WR_SUPPORT = Enabled, writing to Bits[3:0] of SR XS or PCS MMD Device Identifier Register 2 modifies the content of this register. **Value After Reset:** IEEE_DEV_REV_NO_0_3 **Exists:** Always

5.2.9.5 SR_MII_AN_ADV

- **Description:** SR MII MMD AN Advertisement Register
  The host uses this register to advertise the abilities and status of the local device to its link partner through Clause 37 auto-negotiation protocol.
  Note: This register is present only for configurations with 1G/KX support.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x4 : (APB3_16) ? (0x4*0x2) : (0x4*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** ((CL37_AN&&!RPCS_ONLY)||BASE1000X_ONLY)

15	14	13:12	11:9	8:7	6	5	4:0
NP	Reserved_14	RF	Reserved_11_9	PAUSE	HD	FD	Reserved_4_0

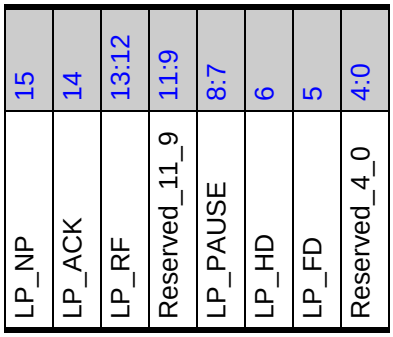
Table 5-466 Fields for Register: SR_MII_AN_ADV

Bits	Name	Memory Access	Description
15	NP	R	Next Page The DWC_xpcs always returns this bit as 0 because it does not support the Next Page feature. **Value After Reset:** 0x0 **Exists:** Always
14	Reserved_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:12	RF	R/W	Remote Fault This field indicates the fault signaling of the local device to be communicated to the link partner. Bits[12:13] are encoded as: ■ 00: No Error ■ 01: Offline ■ 10: Link Failure ■ 11: Auto-negotiation Error **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
11:9	Reserved_11_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8:7	PAUSE	R/W	Pause Ability This field indicates the Pause ability of the device. Bits[7:8] are encoded as: ■ 00: No Pause ■ 01: Asymmetric Pause towards the link partner ■ 10: Symmetric Pause ■ 11: Symmetric Pause and Asymmetric Pause towards the local device. Software can program suitable values based on the capability of the MAC. **Value After Reset:** 0x0 **Exists:** Always
6	HD	R/W	Half Duplex When this bit is set, it indicates that the device can operate in the half-duplex mode. **Value After Reset:** 0x0 **Exists:** Always
5	FD	R/W	Full Duplex When this bit is set, it indicates that the device can operate in the full-duplex mode. **Value After Reset:** 0x1 **Exists:** Always **Testable:** untestable
4:0	Reserved_4_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.9.6 SR_MII_LP_BABL

- **Description:** SR MII MMD AN Link Partner Base Ability Register
  The host uses this page to know the link partner's ability when the base page is received through Clause 37 auto-negotiation. The content of this register is valid only for 1000BASE-X auto-negotiation. It is not valid in SGMII/QSGMII/USXGMII auto-negotiation.
  Note: This register is present only for configurations with 1G/KX support.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x5 : (APB3_16) ? (0x5*0x2) : (0x5*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** ((CL37_AN&&!RPCS_ONLY)||BASE1000X_ONLY)



15	14	13:12	11:9	8:7	6	5	4:0
LP_NP	LP_ACK	LP_RF	Reserved_11_9	LP_PAUSE	LP_HD	LP_FD	Reserved_4_0

Table 5-467 Fields for Register: SR_MII_LP_BABL

Bits	Name	Memory Access	Description
15	LP_NP	R	Next Page This bit indicates that the link partner can handle Next Page. **Note:** To exchange information through Next Page, both devices (local and remote) must have the capability to handle Next Page. The DWC_xpcs does not support Next Page. Therefore, the Next Page exchange does not happen. **Value After Reset:** 0x0 **Exists:** Always
14	LP_ACK	R	ACK bit from the Link Partner This bit indicates that the link partner has successfully received the page sent by the local device. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
13:12	LP_RF	R	Remote Fault This field indicates the fault signaling of the link partner. Bits[12:13] are encoded as: ■ 00: No Error ■ 01: Offline ■ 10: Link Failure ■ 11: Auto-negotiation Error **Value After Reset:** 0x0 **Exists:** Always
11:9	Reserved_11_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8:7	LP_PAUSE	R	Pause Ability This field indicates the Pause ability of the link partner. Bits[7:8] are encoded as: ■ 00: No Pause ■ 01: Asymmetric Pause towards the link partner ■ 10: Symmetric Pause ■ 11: Both Symmetric Pause and Asymmetric Pause towards the local device **Value After Reset:** 0x0 **Exists:** Always
6	LP_HD	R	Half Duplex When this bit is set, it indicates that the link partner is capable of operating in the half-duplex mode. **Value After Reset:** 0x0 **Exists:** Always
5	LP_FD	R	Full Duplex When this bit is set, it indicates that the link partner is capable of operating in the full-duplex mode. **Value After Reset:** 0x0 **Exists:** Always
4:0	Reserved_4_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.9.7 SR_MII_AN_EXPN

- **Description:** SR MII MMD AN Expansion Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x6 : (APB3_16) ? (0x6*0x2) : (0x6*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (CL37_AN||BASE1000X_ONLY)

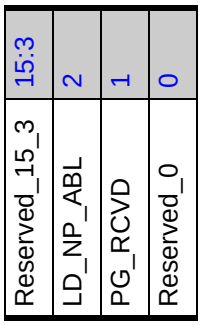


Table 5-468 Fields for Register: SR_MII_AN_EXPN

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2	LD_NP_ABL	R	Local Device NP Able ■ 1: The local device has the next page ability ■ 0: The local device does not have the next page ability The DWC_xpcs always returns this bit as 0 because it does not support Next Page. **Value After Reset:** 0x0 **Exists:** Always
1	PG_RCVD	R	Page Received (RO, LH Type) This bit indicates that the local device received a page from the link partner. ■ 1: The local device received a new page ■ 0: The local device did not receive a new page **Value After Reset:** 0x0 **Exists:** Always
0	Reserved_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.9.8 SR_MII_EXT_STS

- **Description:** SR MII MMD Extended Status Register
  This register is present only for configurations with 1G/KX support.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0F : (APB3_16) ? (0x0F*0x2) : (0x0F*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** ((CL37_AN&&!RPCS_ONLY)||BASE1000X_ONLY)

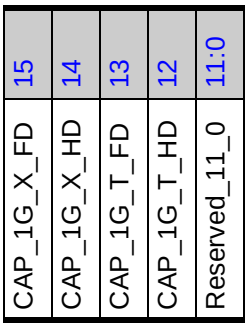


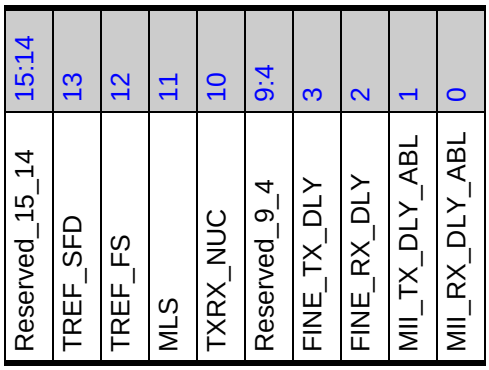
Table 5-469 Fields for Register: SR_MII_EXT_STS

Bits	Name	Memory Access	Description
15	CAP_1G_X_FD	R	1000BASE-X Full-Duplex Capable ■ 1: 1000BASE-X full-duplex capable ■ 0: Not 1000BASE-X full-duplex capable The DWC_xpcs always returns 1 because it supports this feature. **Value After Reset:** 0x1 **Exists:** Always **Testable:** untestable
14	CAP_1G_X_HD	R	1000BASE-X Half-Duplex Capable ■ 1: 1000BASE-X half-duplex capable ■ 0: Not 1000BASE-X half-duplex capable The DWC_xpcs always returns 1 because it supports this feature. **Value After Reset:** 0x1 **Exists:** Always **Testable:** untestable

Bits	Name	Memory Access	Description
13	CAP_1G_T_FD	R	1000BASE-T Full-Duplex Capable ■ 1: 1000BASE-T full-duplex capable ■ 0: Not 1000BASE-T full-duplex capable The DWC_xpcs always returns 0 because it does not support this feature. **Value After Reset:** 0x0 **Exists:** Always
12	CAP_1G_T_HD	R	1000BASE-T Half-Duplex Capable ■ 1: 1000BASE-T half-duplex capable ■ 0: Not 1000BASE-T half-duplex capable The DWC_xpcs always returns 0 because it does not support this feature. **Value After Reset:** 0x0 **Exists:** Always
11:0	Reserved_11_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.9.9 SR_MII_TIME_SYNC_ABL

- **Description:** SR MII MMD Time Sync Capability Register.
  This register is present only in 1000BaseX-Only PCS configurations
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0708 : (APB3_16) ? (0x0708*0x2) : (0x0708*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY



15:14	13	12	11	10	9:4	3	2	1	0
Reserved_15_14	TREF_SFD	TREF_FS	MLS	TXRX_NUC	Reserved_9_4	FINE_TX_DLY	FINE_RX_DLY	MII_TX_DLY_ABL	MII_RX_DLY_ABL

Table 5-470 Fields for Register: SR_MII_TIME_SYNC_ABL

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13	TREF_SFD	R	Timestamp reference at SFD - 0: DWC_xpcs does not support data delay measurement point at the beginning of SFD - 1: DWC_xpcs supports data delay measurement point at the beginning of SFD **Value After Reset:** "SNPS_RSVDPARAM_11 ? 0x1 : 0x0" **Exists:** Always
12	TREF_FS	R	Timestamp reference at first symbol after SFD - 0: DWC_xpcs does not support data delay measurement point at the beginning of the first symbol after SFD - 1: DWC_xpcs supports data delay measurement point at the beginning of the first symbol after SFD **Value After Reset:** "SNPS_RSVDPARAM_11 ? 0x1 : 0x0" **Exists:** Always

Bits	Name	Memory Access	Description
11	MLS	R	Multi-lane Support. ■ 0: DWC_xpcs does not support reporting of multi-PCS lane path data delay ■ 1: DWC_xpcs supports reporting of multi-PCS lane path data delay In this mode(1000BASEX_ONLY), this bit is always 0. **Value After Reset:** 0x0 **Exists:** Always
10	TXRX_NUC	R	Transmit or Receive Num_Unit_Change Support Available. If this bit is set, it indicates that DWC_xpcs supports Transmit/Receive Num_Unit_Change indication capability **Value After Reset:** "SNPS_RSVDPARAM_11 ? 0x1 : 0x0" **Exists:** Always
9:4	Reserved_9_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3	FINE_TX_DLY	R	Fine Resolution Transmit Path Data Delay Available. If this bit is set, it indicates that DWC_xpcs provides information on transmit data path delay with sub-ns resolution in SR_MII_TIME_SYNC_TX_MAX_FINE_DLY and SR_MII_TIME_SYNC_TX_MIN_FINE_DLY registers. **Value After Reset:** "SNPS_RSVDPARAM_11 ? 0x1 : 0x0" **Exists:** Always
2	FINE_RX_DLY	R	Fine Resolution Receive Path Data Delay Available. If this bit is set, it indicates that DWC_xpcs provides information on receive data path delay with sub-ns resolution in SR_MII_TIME_SYNC_RX_MAX_FINE_DLY and SR_MII_TIME_SYNC_RX_MIN_FINE_DLY registers. **Value After Reset:** "SNPS_RSVDPARAM_11 ? 0x1 : 0x0" **Exists:** Always
1	MII_TX_DLY_ABL	R	DWC_xpcs Transmit Path Data Delay Information Available. If this bit is set, it indicates that the information regarding the maximum and minimum transmit data delay of DWC_xpcs is available in MII MMD Tx Time Delay Registers **Value After Reset:** 0x1 **Exists:** Always

Bits	Name	Memory Access	Description
0	MII_RX_DLY_ABL	R	DWC_xpcs Receive Path Data Delay Information Available. If this bit is set, it indicates that the information regarding the maximum and minimum receive data delay of DWC_xpcs is available in MII MMD Rx Time Delay Registers **Value After Reset:** 0x1 **Exists:** Always

5.2.9.10 SR_MII_TIME_SYNC_TX_MAX_DLY_LWR

- **Description:** SR MII MMD Time Sync Tx Max Delay Lower Register.
  This register is present only in 1000BaseX-Only PCS configurations.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0709 : (APB3_16) ? (0x0709*0x2) : (0x0709*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY

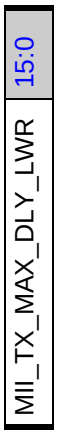


Table 5-471 Fields for Register: SR_MII_TIME_SYNC_TX_MAX_DLY_LWR

Bits	Name	Memory Access	Description
15:0	MII_TX_MAX_DLY_LWR	R	This field indicates the lower 16-bit of the 32-bit value which indicates the Maximum Data Delay in the Transmit Path (in nanoseconds) of DWC_xpcs. If TS_MODE bit is 0, this register indicates the end-to-end (MAC interface to PCS interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the transmit path from the DDMP point until the final Tx data output port. **Value After Reset:** 0x38 **Exists:** Always

5.2.9.11 SR_MII_TIME_SYNC_TX_MAX_DLY_UPR

- **Description:** SR MII MMD Time Sync Tx Max Delay Upper Register.
  This register is present only in 1000BaseX-Only PCS configurations.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070A : (APB3_16) ? (0x070A*0x2) : (0x070A*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY



Table 5-472 Fields for Register: SR_MII_TIME_SYNC_TX_MAX_DLY_UPR

Bits	Name	Memory Access	Description
15:0	MII_TX_MAX_DLY_UPR	R	This field indicates the upper 16-bit of the 32-bit value which indicates the Maximum Data Delay in the Transmit Path (in nanoseconds) of DWC_xpcs. If TS_MODE bit is 0, this register indicates the end-to-end (MAC interface to PCS interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the transmit path from the DDMP point until the final Tx data output port. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.12 SR_MII_TIME_SYNC_TX_MIN_DLY_LWR

- **Description:** SR MII MMD Time Sync Tx Min Delay Lower Register.
  This register is present only in 1000BaseX-Only PCS configurations.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070B : (APB3_16) ? (0x070B*0x2) : (0x070B*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY



Table 5-473 Fields for Register: SR_MII_TIME_SYNC_TX_MIN_DLY_LWR

Bits	Name	Memory Access	Description
15:0	MII_TX_MIN_DLY_LWR	R	This field indicates the lower 16-bit of the 32-bit value which indicates the Minimum Data Delay in the Transmit Path (in nanoseconds) of DWC_xpcs. If TS_MODE bit is 0, this register indicates the end-to-end (MAC interface to PCS interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the transmit path from the DDMP point until the final Tx data output port **Value After Reset:** 0x38 **Exists:** Always

5.2.9.13 SR_MII_TIME_SYNC_TX_MIN_DLY_UPR

- **Description:** SR MII MMD Time Sync Tx Min Delay Upper Register.
  This register is present only in 1000BaseX-Only PCS configurations.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070C : (APB3_16) ? (0x070C*0x2) : (0x070C*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY



Table 5-474 Fields for Register: SR_MII_TIME_SYNC_TX_MIN_DLY_UPR

Bits	Name	Memory Access	Description
15:0	MII_TX_MIN_DLY_UPR	R	This field indicates the upper 16-bit of the 32-bit value which indicates the Minimum Data Delay in the Transmit Path (in nanoseconds) of DWC_xpcs. If TS_MODE bit is 0, this register indicates the end-to-end (MAC interface to PCS interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the transmit path from the DDMP point until the final Tx data output port. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.14 SR_MII_TIME_SYNC_RX_MAX_DLY_LWR

- **Description:** SR MII MMD Time Sync Rx Max Delay Lower Register.
  This register is present only in 1000BaseX-Only PCS configurations.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070D : (APB3_16) ? (0x070D*0x2) : (0x070D*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY



Table 5-475 Fields for Register: SR_MII_TIME_SYNC_RX_MAX_DLY_LWR

Bits	Name	Memory Access	Description
15:0	MII_RX_MAX_DLY_LWR	R	This field indicates the lower 16-bit of the 32-bit value which indicates the Maximum Data Delay in the Receive Path (in nanoseconds) of DWC_xpcs. If TS_MODE bit is 0, this register indicates the end-to-end (PHY interface to MAC interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the receive path from Rx data input port until the DDMP point **Value After Reset:** "(SGMII_EN||QSGMII_EN)&&CLKCOMP ? \"0xB8\" : (SGMII_EN||QSGMII_EN)&&!CLKCOMP ? \"0x58\" : !(SGMII_EN||QSGMII_EN)&&CLKCOMP ? \"0xB0\" :\"0x50\"" **Exists:** Always

5.2.9.15 SR_MII_TIME_SYNC_RX_MAX_DLY_UPR

- **Description:** SR MII MMD Time Sync Rx Max Delay Upper Register.
  This register is present only in 1000BaseX-Only PCS configurations.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x070E : (APB3_16) ? (0x070E*0x2) : (0x070E*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY



Table 5-476 Fields for Register: SR_MII_TIME_SYNC_RX_MAX_DLY_UPR

Bits	Name	Memory Access	Description
15:0	MII_RX_MAX_DLY_UPR	R	This field indicates the upper 16-bit of the 32-bit value which indicates the Maximum Data Delay in the Receive Path (in nanoseconds) of DWC_xpcs. If TS_MODE bit is 0, this register indicates the end-to-end (PHY interface to MAC interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the receive path from Rx data input port until the DDMP point. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.16 SR_MII_TIME_SYNC_RX_MIN_DLY_LWR

- **Description:** SR MII MMD Time Sync Rx Min Delay Lower Register.
  This register is present only in 1000BaseX-Only PCS configurations.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x070F : (APB3_16) ? (0x070F*0x2) : (0x070F*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY



Table 5-477 Fields for Register: SR_MII_TIME_SYNC_RX_MIN_DLY_LWR

Bits	Name	Memory Access	Description
15:0	MII_RX_MIN_DLY_LWR	R	This field indicates the lower 16-bit of the 32-bit value which indicates the Minimum Data Delay in the Receive Path (in nanoseconds) of PCS. If TS_MODE bit is 0, this register indicates the end-to-end (PHY interface to MAC interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the receive path from Rx data input port until the DDMP point. **Value After Reset:** "(SGMII_EN||QSGMII_EN)&&CLKCOMP ? \"0x88\" : (SGMII_EN||QSGMII_EN)&&!CLKCOMP ? \"0x50\" : !(SGMII_EN||QSGMII_EN)&&CLKCOMP ? \"0x80\" :\"0x48\"" **Exists:** Always

5.2.9.17 SR_MII_TIME_SYNC_RX_MIN_DLY_UPR

- **Description:** SR MII MMD Time Sync Rx Min Delay Upper Register
  This register is present only in 1000BaseX-Only PCS configurations.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x0710 : (APB3_16) ? (0x0710*0x2) : (0x0710*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY



Table 5-478 Fields for Register: SR_MII_TIME_SYNC_RX_MIN_DLY_UPR

Bits	Name	Memory Access	Description
15:0	MII_RX_MIN_DLY_UPR	R	This field indicates the upper 16-bit of the 32-bit value which indicates the Minimum Data Delay in the Receive Path (in nanoseconds) of PCS. If TS_MODE bit is 0, this register indicates the end-to-end (PHY interface to MAC interface) delay and if TS_MODE bit is 1, this register indicates the delay inside DWC_xpcs in the receive path from Rx data input port until the DDMP point. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.18 SR_MII_TIME_SYNC_CFG

- **Description:** SR MII MMD Time Sync Configuration Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x0715 : (APB3_16) ? (0x0715*0x2) : (0x0715*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_RSVDPARAM_11)

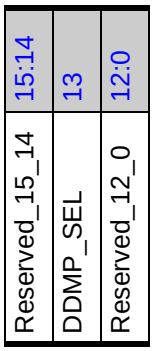


Table 5-479 Fields for Register: SR_MII_TIME_SYNC_CFG

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13	DDMP_SEL	R/W	Data Delay Measurement Point Selection ■ 0: DWC_xpcs is configured to use the data delay measurement point at the beginning of SFD ■ 1: DWC_xpcs is configured to use the data delay measurement point at the beginning of the first symbol after SFD **Value After Reset:** 0x0 **Exists:** Always
12:0	Reserved_12_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.9.19 VR_MII_DIG_CTRL1

- **Description:** VR MII MMD Digital Control1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8000 : (APB3_16) ? (0x8000*0x2) : (0x8000*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (CL37_AN | |BASE1000X_ONLY)

15	14	13	12	11	10	9	8	7	6	4	3	2	1	0
VR_RST	R2TLBE	EN_VSMMD1	CL37_BP	PWRSV	CS_EN	MAC_AUTO_SW	INIT	MSK_RD_ERR	PRE_EMP	DTXLANED_0	CL37_TMR_OVR_RIDE	EN_2_5G_MODE	BYP_PWRUP	PHY_MODE_CTRL

Table 5-480 Fields for Register: VR_MII_DIG_CTRL1

Bits	Name	Memory Access	Description
15	VR_RST	R/W	Vendor-Specific Soft Reset (RW, SC Type) When the host sets this bit, the CSR block triggers the vendor-specific software reset process in which all internal blocks, except the Management Interface block and CSR block, are reset. When this bit is set, it also resets the PHY. This bit is self cleared under the following conditions: **For Synopsys PHY:** This bit is self cleared when Bits[4:2] in VR_XS_PCS_DIG_STS Register are equal to 3'b100, that is, Tx/Rx clocks are stable and in Power_Good state. **For Non-Synopsys PHY:** This bit is self cleared after the following: ■ 32 clk_csr_i clocks for the MCI interface ■ 1 MDC clock period for the MDIO interface **Note:** For information about the read or write access for any register during the reset process, see "Special Case Register Access" section. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
14	R2TLBE	* Varies	Rx to Tx Loopback Enable This bit controls the loopback path from the GMII Rx to the GMII Tx at the GMII interface. ■ 0: Loopback path is disabled ■ 1: Loopback path is enabled In configurations with CL37_AN = Enabled and MAIN_MODE!=1000BaseX-Only PCS, though this bit is read-only, it reflects the value programmed to R2TLBE bit of VR_XS_PCS_DIG_CTRL1 Register **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "BASE1000X_ONLY ? \"read-write\" : \"read-only\""
13	EN_VSMMD1	* Varies	Enable Vendor-Specific MMD1 When this bit is set to 1, the vendor-specific MMD1 (VSMMD1) is enabled. When this bit is set to zero, VSMMD1 is disabled. **Value After Reset:** "BASE1000X_ONLY ? 0x1 : 0x0" **Exists:** Always **Testable:** "BASE1000X_ONLY ? \"untestable\" : \"unconstrained\"" **Memory Access:** "BASE1000X_ONLY ? \"read-write\" : \"read-only\""
12	CL37_BP	R	Enable Clause 37 AN in Backplane Configuration This bit is present when MAIN_MODE = Backplane Ethernet PCS and CL37_AN = Enabled. This is a read-only bit and it reflects the value programmed to CL37_BP bit of VR PCS MMD Digital Control 1 Register. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
11	PWRSV	* Varies	**For Synopsys PHY configurations:** Power Save This bit is present only when the DWC_xpcs is configured for Synopsys PHY. ■ 0: Normal operation. ■ 1: DWC_xpcs and the PHY enter the power-save mode. When this bit is set, the DWC_xpcs triggers the power-down mode by turning off the PHY Receiver and Transmitter, and without turning off MPLL. For Synopsys multi-protocol 12G/16G PHY, DWC_xpcs asserts xpcs_tx_lpd_o{lane} and xpcs_rx_lpd_o{lane} when this bit is set. When the host clears this bit, DWC_xpcs resumes normal operation. After clearing this bit, the host must wait until Bits[4:2] of the VR_XS_PCS_DIG_STS Register indicate that the DWC_xpcs is in a normal state. **Note:** For a description of read or write access for any register while this bit is set, see "Special Case Register Access" section of Databook. **For non-Synopsys PHY configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_XAUI_PHY) ? \"read-write\" : \"read-only\""
10	CS_EN	* Varies	**For 1000BASEX_Only PCS configurations with EEE:** Clock Stop Enable ■ 1: The PHY might stop the clock during LPI mode ■ 0: The clock cannot be stopped during LPI mode You must program this bit based on the capability of the MAC during Rx LPI mode. **Access Type:** ■ For configurations with MAIN_MODE = 1000BASEX-Only PCS and EEE_EN = Enabled, this bit is RW. ■ For all other configurations, this bit is reserved and RO. **For other configurations:** Reserved **Value After Reset:** "(BASE1000X_ONLY&&EEE_EN) ? \"0x1\" : \"0x0\"" **Exists:** Always **Memory Access:** "(BASE1000X_ONLY&&EEE_EN) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
9	MAC_AUTO_SW	* Varies	Automatic Speed Mode Change after CL37 AN **When SGMII_EN = Enabled or QSGMII_EN = Enabled or USXGMII_EN = Enabled:** ■ 1: DWC_xpcs automatically switches to the negotiated SGMII/USXGMII/QSGMII(port0) speed, after the completion of Clause 37 auto-negotiation. This mode is valid only when DWC_xpcs is configured as MAC-side SGMII/USXGMII/QSGMII and must be set only when auto-negotiation is enabled (AN_ENABLE bit is set to 1). ■ 0: DWC_xpcs operates at the speed or in duplex mode as per the values programmed in the SR_MII_CTRL register. After the completion of CL37 AN, application must read the negotiated speed or the duplex mode from the VR_MII_AN_INTR_STS register and then program the SR_MII_CTRL register appropriately. When this bit is set to 1 in SGMII mode, xpcs_sgmii_link_sts_o, xpcs_link_speed_o and xpcs_sgmii_full_duplex_o outputs reflect the auto-negotiated values that is, values from CL37_ANSGM_STS field of VR_MII_AN_INTR_STS register. For USXGMII mode, if clk_xgmii_tx_i and clk_xgmii_rx_i do not stabilize at the new operating frequency (based on the selected speed) immediately after the completion of auto-negotiation, then software might need to program 'USRA_RST' bit prior to starting packet transfer in the new speed mode. **Note:** Set this bit only when DWC_xpcs is configured as SGMII/USXGMII/QSGMII MAC, that is, TX_CONFIG = 0. **Note:** When this bit is set, the values programmed to SS5,SS6 and SS13 bits are ignored. Moreover, the values of these bits do not change based on auto-negotiation outcome. **For other configurations:** This is a read-only reserved field and returns 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SGMII_EN||QSGMII_EN||USXGMII_EN) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
8	INIT	* Varies	**For 1000BASEX_Only PCS configurations:**Datapath Initialization Control This bit can be set to flush/initialize the various FIFOs implemented inside DWC_xpcs. This is a Self-Clear bit. After writing 1 to this bit, software must continuously poll this bit. Software must proceed to any other operation, only after reading this bit as 0. When this bit is programmed to 1, RXFIFO_OVF/RXFIFO_UNF bits of VR_MII_DIG_STS Register might get set incorrectly. Therefore, read these register bits (RXFIFO_OVF and RXFIFO_UNF) so that they get cleared. Thereafter, RXFIOF_UNF and RXFIFO_UNF bits are reliable. **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "BASE1000X_ONLY ? \"read-write\" : \"read-only\""
7	MSK_RD_ERR	* Varies	Mask Running Disparity Error. In QSGMII mode of operation, an error in the data received from a particular port of a far-end device can result in unprecedented running disparity errors for other ports (even if data is received correctly). To resolve this issue, DWC_xpcs provides an option to mask the running disparity errors. If this register bit is set, then running disparity errors are ignored by DWC_xpcs receiver in evaluating the validity of received code-groups. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "QSGMII_EN ? \"read-write\" : \"read-only\""
6	PRE_EMP	* Varies	Pre-emption Packet Enable. Set this bit to 1 to allow the DWC_xpcs to properly receive or transmit preemption packets in SGMII 10M/100M Modes. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SGMII_EN||QSGMII_EN ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
4	DTXLANED_0	* Varies	Tx Lane 0 Disable When this bit is set, the DWC_xpcs disables the Tx Lane 0 of the PHY. When reset, the DWC_xpcs enables the Tx Lane 0 of the PHY. This bit is Read-Write only when MAIN_MODE= 1000BASEX-Only PCS. **Note:** This bit is shared with the following bits: ■ Bit 4 of VR_XS_PCS_DIG_CTRL1 Register and VR_MII-ID_DIG_CTRL1 Register ■ Bit 1 of SR_PMA_TX_DIS Register ■ Bit 0 of SR_PMA_KX_CTRL Register **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "BASE1000X_ONLY ? \"read-write\" : \"read-only\""
3	CL37_TMR_OVR_RIDE	R/W	Over-Ride Control for CL37 Link Timer. This bit can be set to override the default value of Clause 37 link_timer used by DWC_xpcs for auto-negotiation. If this bit is set, the value programmed to VR_MII_LINK_TIMER_CTRL Register is used to compute the duration of Link Timer. Set this bit, only after programming the appropriate value to VR_MII_LINK_TIMER_CTRL Register. **Value After Reset:** 0x0 **Exists:** Always
2	EN_2_5G_MODE	* Varies	Enable 2.5G GMII Mode. Set this bit to 1 to enable 2.5G GMII Mode of operation.This bit drives the output port 'xpcs_2pt5g_mode_o'. This bit is shared with bit[2] of VR_XS_PCS_DIG_CTR1 Register. For configurations without 1G (1000BASE-X/KX) support , this bit is reserved and read-only. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SUPPORT_1G||BASE1000X_ONLY||KX_PRESENT)? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
1	BYP_PWRUP	* Varies	**For Synopsys PHY configurations:** Bypass Power-Up Sequence When this bit is set to 1, the DWC_xpcs bypasses the normal flow of the power-up sequence and reaches the Power_Good state to enable transmission or reception. When this bit is set, the DWC_xpcs does not wait for the MPLL and Transmit or Receive PLL status from the Synopsys PHY. You can use this feature in the following scenarios: ■ When the DWC_xpcs is configured to interface with a specific Synopsys PHY. ■ When the data path needs to be interfaced to some other PHY. When this bit is set to 0, the DWC_xpcs waits for the MPLL, Tx, or Rx PLL status from the Synopsys PHY before resuming the normal transmission and reception. **Note:** This bit is shared with Bit 1 of following registers: ■ VR_XS_PCS_DIG_CTRL1 Register ■ VR_AN_DIG_CTRL1 Register ■ VR_PMA_DIG_CTRL1 Register **Access Type:** If SNPS_XAUI_PHY= Enabled, this bit is RW. Otherwise, this bit is RO. **For non-Synopsys PHY configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_XAUI_PHY) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
0	PHY_MODE_CTRL	* Varies	**When SGMII_PHY_AN_AUTO_RESTART = Enabled or QSGMII_PHY_AN_AUTO_RESTART = Enabled or USXG_PHY_AN_AUTO_RESTART = Enabled:**PHY mode control This bit controls the Clause 37 auto-negotiation when operating in SGMII/QSGMII/USXGMII (Port 0) PHY mode. **SGMII/QSGMI**: When this bit is set to 1, DWC_xpcs advertises the values of input ports xpcs_sgmii_link_sts_i, xpcs_sgmii_link_speed_i and xpcs_sgmii_full_duplex_i during SGMII/(Port0)QSGMII auto-negotiation. When this bit is set to 0, SGMII/QSGMII(Port0) auto-negotiation advertises the values programmed to: ■ bit 4 (SGMII_LINK_STS) of VR_MII_AN_CTRL Register ■ bit 13 (ss13) and 6 (ss6) of SR_MII_CTRL Register and ■ bit 5 (FD) of SR_MII_AN_ADV Register **USXGMII**: When this bit is set to 1, DWC_xpcs advertises the values of input ports xpcs_usxg_link_sts_i, xpcs_usxg_link_speed_i,xpcs_usxg_full_duplex_i,xpcs_usxg_eee_abl_i and xpcs_usxg_ck_stp_abl_i during USXGMII (port 0) auto-negotiation. When this bit is set to 0, USXGMI (Port0) auto-negotiation advertises the following : ■ bit 4 (SGMII_LINK_STS) of VR_MII_AN_CTRL Register ■ bit 13 (ss13) and 6 (ss6) and 5 (ss5) of SR_MII_CTRL Register and ■ Value of 1 for Duplex-Mode ■ Value of 1 for EEE-capability, if EEE = Enabled; Value of 0 if EEE_EN = Disabled ■ Value of 1 for EEE clock-stop capability, if EEE = Enabled; Value of 0 if EEE_EN = Disabled **Note:** Set this bit only when DWC_xpcs is configured as SGMII/QSGMII/USXGMII PHY, that is, TX_CONFIG = 1. In other configurations, this field is reserved and read-only. **Value After Reset:** "(SGMII_PHY_AN_AUTO_RESTART&&SGMII_PHY_DFLT)?\"0x1\" : \"0x0\"" **Exists:** Always **Testable:** "(SGMII_PHY_DFLT&&SGMII_PHY_AN_AUTO_RESTART)?\"untestable\" : \"unconstrained\"" **Memory Access:** "(SGMII_PHY_AN_AUTO_RESTART||QSGMII_PHY_AN_AUTO_RESTART||USXG_PHY_AN_AUTO_RESTART) ?\"read-write\" : \"read-only\""

5.2.9.20 VR_MII_AN_CTRL

- **Description:** VR MII MMD AN Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8001 : (APB3_16) ? (0x8001*0x2) : (0x8001*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (CL37_AN||BASE1000X_ONLY)

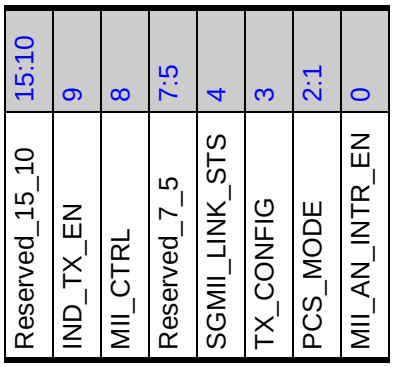


Table 5-481 Fields for Register: VR_MII_AN_CTRL

Bits	Name	Memory Access	Description
15:10	Reserved_15_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9	IND_TX_EN	* Varies	Independent Transmit Enable in 1000BASE-X/KX mode. If this bit is set to 1, DWC_xpcs transmits the GMII Tx data, irrespective of its receive link status (provided auto-negotiation is not enabled). If this bit is set to 0, DWC_xpcs sends IDLE till its receiver has attained synchronization. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(BASE1000X_ONLY||KX_PRESENT||SUPPORT_1G) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
8	MII_CTRL	* Varies	MII Control **When SGMII_EN = Enabled or QSGMII_EN = Enabled or USXGMII_EN = Enabled :** This bit controls the width of the MAC interface when operating at SGMII/QSGMII/USXGMII speed modes of 10 Mbps or 100 Mbps ■ 0: 4-bit MII ■ 1: 8-bit MII This bit also controls the xpcs_mii_ctrl_o signal which is used for external clock multiplexing of the clk_mii_tx_i and clk_mii_rx_i signals. **For other configurations:** This is a read-only reserved field and returns 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SGMII_EN||QSGMII_EN||USXGMII_EN) ? \"read-write\" : \"read-only\""
7:5	Reserved_7_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	SGMII_LINK_STS	* Varies	**When SGMII_EN = Enabled or QSGMII_EN = Enabled or USXGMII Link Status:**SGMII Link Status/ USXGMII Link Status /QSGMII Port0 Link Status This bit is used in Bit 15 of the Config_Reg during Clause 37 auto-negotiation when the TX_CONFIG bit of this register is set to 1 in the SGMII/QSGMII/USXGMII mode and when PHY_MODE_CTRL bit of VR_MII_DIG_CTRL1 Register is 0. ■ 0: Link Down ■ 1: Link Up **>For other configurations:** This is a read-only reserved field and returns 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SGMII_EN||QSGMII_EN||USXGMII_EN) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
3	TX_CONFIG	* Varies	**When SGMII_EN = Enabled or QSGMII_EN = Enabled or USXMII_EN = Enabled:**Transmit Configuration This bit controls the Config_Reg value to be used during the Clause 37 auto-negotiation in the SGMII/QSGMII/USXGMII mode. ■ 1: Configures the DWC_xpcs as the PHY side SGMII/QSGMII/USXGMII. ■ 0: Configures the DWC_xpcs as the MAC side SGMII/QSGMII/USXGMII. **For other configurations:** This is a read-only reserved field and returns 0. **Value After Reset:** "SGMII_PHY_DFLT ? \"0x1\" : \"0x0\"" **Exists:** Always **Testable:** "SGMII_PHY_DFLT ? \"untestable\" : \"unconstrained\"" **Memory Access:** "(SGMII_EN||QSGMII_EN||USXGMII_EN) ? \"read-write\" : \"read-only\""
2:1	PCS_MODE	* Varies	**When SGMII_EN = Enabled or QSGMII_EN = Enabled:**PCS Mode This field controls the auto-negotiation (and operating) mode. ■ 00: 1000BASE-X mode (clause 37 auto-negotiation is as per 1000BaseX). ■ 10: SGMII mode (clause 37 auto-negotiation is as per SGMII). ■ 11: QSGMII mode (clause 37 auto-negotiation conforms to QSGMII) If this field is set as 2'b11, then DWC_xpcs asserts the 'xgxs_qsgmii_mode_o' port. **For other configurations:** This is a read-only reserved field and returns 0. **Value After Reset:** "SGMII_PHY_DFLT ? \"0x2\" : \"0x0\"" **Exists:** Always **Testable:** "SGMII_PHY_DFLT ? \"untestable\" : \"unconstrained\"" **Memory Access:** "(SGMII_EN||QSGMII_EN) ? \"read-write\" : \"read-only\""
0	MII_AN_INTR_EN	R/W	Clause 37 AN Complete Interrupt Enable When set to 1, this bit enables the generation of Clause 37 auto-negotiation complete interrupt output. When set to 0, it disables the generation of Clause 37 auto-negotiation complete interrupt. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.21 VR_MII_AN_INTR_STS

- **Description:** VR MII MMD AN Interrupt and Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8002 : (APB3_16) ? (0x8002*0x2) : (0x8002*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (CL37_AN||BASE1000X_ONLY)

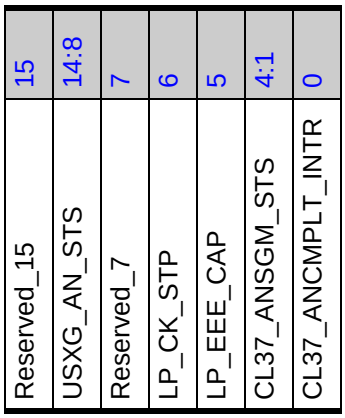


Table 5-482 Fields for Register: VR_MII_AN_INTR_STS

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
14:8	USXG_AN_STS	R	**When USXGMII_EN = Enabled:** USXGMII Clause 37 AN Status This field is valid only when clause 37 auto-negotiation is complete in USXGMII Mode. It indicates the status received from remote link after the USXGMII auto-negotiation is complete. USXG_AN_STS[6] ■ 0: Link is Down ■ 1: Link is Up USXG_AN_STS[5] ■ 0: Half Duplex ■ 1: Full Duplex USXG_AN_STS[4:2] ■ 000: 10 Mbps speed link ■ 001: 100 Mbps speed link ■ 010: 1000 Mbps speed link ■ 011 :10 Gbps speed link ■ 100 :2.5 Gbps speed link ■ 101 : 5 Gbps speed link USXG_AN_STS[1] ■ 1 : EEE supported ■ 0 : EEE not supported USXG_AN_STS[0] ■ 1 : EEE clock-stop supported ■ 0 : EEE clock-stop not supported **When USXGMII_EN=Disabled:** Reserved **Value After Reset:** 0x0 **Exists:** Always
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6	LP_CK_STP	R	Link Partner EEE Clock Stop Capability This field indicates the EEE clock stop capability (or clock-stop enable - in case far-end is acting as QSGMII MAC) advertised by the far-end device. This field is valid only when PCS_MODE[1:0] is set to the QSGMII mode and the auto-negotiation is complete along port 0. **Note:** This field is present only in configurations with QSGMII_EN = Enabled **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
5	LP_EEE_CAP	R	Link Partner EEE Capability This field indicates the EEE capability advertised by the far-end device (Port 0 QSGMII PHY). This field is valid only when PCS_MODE[1:0] is set to the QSGMII mode and the auto-negotiation is complete along port 0. **Note:** This field is present only in configurations with QSGMII_EN = Enabled **Value After Reset:** 0x0 **Exists:** Always
4:1	CL37_ANSGM_STS	R	**When SGMII_EN = Enabled or QSGMII_EN = Enabled:**Clause 37 AN SGMII Status/QSGMII Port 0 Status This field is valid only when the PCS_MODE[1:0] is set to the SGMII/QSGMII mode and the auto-negotiation is complete. It indicates the status received from remote link after the SGMII/QSGMII (Port 0) auto-negotiation is complete. CL37_ANSGM_STS[0] ■ 0: Half Duplex ■ 1: Full Duplex CL37_ANSGM_STS[2:1] ■ 00: 10 Mbps speed link ■ 01: 100 Mbps speed link ■ 10: 1000 Mbps speed link CL37_ANSGM_STS[3] ■ 0: Link is Down ■ 1: Link is Up **For other configurations:** This is a read-only reserved field and returns 0. **Value After Reset:** "(SGMII_EN||QSGMII_EN) ? 0x5 : 0x0" **Exists:** Always **Testable:** "(!BASE1000X_ONLY) ? \"untestable\" : \"unconstrained\"""
0	CL37_ANCMPLT_INTR	R/W	Clause 37 AN Complete Interrupt (SS,WC Type) The DWC_xpcs sets this bit when Clause 37 auto-negotiation is complete. The host must clear this bit by writing 0 to it. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.22 VR_MII_TC

- **Description:** VR MII MMD Test Control Register.
  Note: This register is present only in 1000BaseX-Only PCS configurations.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8003 : (APB3_16) ? (0x8003*0x2) : (0x8003*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY)

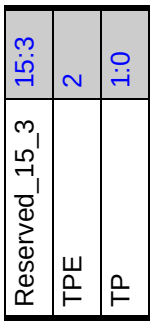


Table 5-483 Fields for Register: VR_MII_TC

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2	TPE	R/W	Test Pattern Enable Lanes This bit indicates that a test pattern can be enabled in the Tx path after the current normal frame transmission is complete. ■ 0: Test pattern disabled ■ 1: Test pattern enabled **Dependencies:** The specific test pattern that is generated is based on Bits[1:0] of this register. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
1:0	TP	R/W	Test Pattern Select This field indicates the pattern type that is enabled with Bit 2 of this register. The following are the supported test patterns: ■ 2'b00: High Frequency Test Pattern ■ 2'b01: Low Frequency Test Pattern ■ 2'b10: Mixed Frequency Test Pattern ■ 2'b11: Reserved The definition of these test patterns is specified in IEEE Std 802.3ae, Annex 48A. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.23 VR_MII_DIG_CTRL3

- **Description:** VR MII MMD Digital Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8004 : (APB3_16) ? (0x8004*0x2) : (0x8004*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** USXGMII_EN

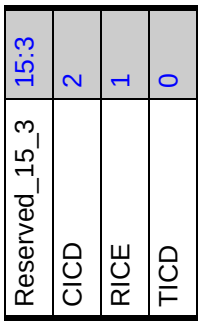


Table 5-484 Fields for Register: VR_MII_DIG_CTRL3

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2	CICD	R/W	Clock-compensation IPG check Deviation on Port 0 During USXGMII mode, when operating at GMII speed modes (1G/100M/10M), by default, clock-compensation logic deletes IDLE (to compensate for ppm difference) only if incoming IPG >=12B. This ensures that output IPG is >= 8B provided input IPG is also greater than or equal to 12B However, if this bit is 1, clock-compensation IPG check deviates from the standard behavior and deletes IDLE even if incoming IPG is only >= 9B. This ensures that output IPG is >=5B provided input IPG is also greater than or equal to 5B ■ 0: output IPG >= 8B ■ 1: output IPG >= 5B This mode is useful when incoming Rx data is an already clock-compensated data and does not have enough number of IPGs with >=12B of IDLE. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
1	RICE	* Varies	Receive IPG Check Enable during XGMII-to-GMII conversion on Port 0. Whenever operating in 1G and lower speeds, DWC_xpcs receiver converts the received data from XGMII format to GMII/MII format before driving it out on GMII/MII Rx interface of DWC_xpcs. This logic blindly carries-out this conversion process without performing any check for number of IPG bytes present in this received data. RICE bit can be set to 1 to enable a IPG check mechanism during XGMII-to-GMII conversion process so as to ensure minimum IPG of 8 bytes at GMII/MII Rx interface. Whenever IPG in XGMII data is less than 8 bytes, DWC_xpcs introduces additional IPG bytes so as to ensure 8 bytes minimum IPG at GMII/MII Rx interface. Addition of IPG is compensated later by deleting IPG bytes from other packets which have larger IPG. If packets with IPG less than 8 bytes appear quite frequently in the incoming data stream, then the buffer can overflow and can cause packet corruption. This feature can work reliably only if not more than one packet with IPG less than 8 bytes occur within a span of 4 packets. This feature is useful in scenarios where link-partner device can send packets with IPG as low as 5 bytes. This bit shall not be set to 1, if CICD is programmed to 1. This feature is valid in both single-port and multi-port USXGMII modes. **Note:** This bit is present only for multi-port USXGMII configurations. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "MP_USXGMII ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
0	TICD	* Varies	Transmit IPG Check Deviation on Port 0. When operating in multi-port USXGMII modes, USXGMII Tx Rate Module compensates for AM insertion by deleting IDLEs.This deletion operation involves a IPG check logic which allows deletion only if 8 bytes of IPG can be ensured at transmitter output when operating in GMII/MII modes (1G and lower speed). However, this IPG check logic introduces a limit on the packet size that can be transmitted. If jumbo packets (packets of size 2K bytes to 9K bytes) are transmitted continuously, DWC_xpcs might not find an opportunity to delete sufficient IDLE bytes before the next AM insertion. Therefore, this can result in a gradual fill-up of Tx RAL FIFO and can result in eventual packet corruption. This bit can be set to 1 to disable the standard IPG check and perform a modified IPG check so as to support jumbo packets. Whenever this bit is set to 1, DWC_xpcs perform IPG deletion in such a manner so as to ensure minimum of 5 bytes of IPG at transmitter output when operating in GMII/MII modes (1G and lower speed). This mode is useful if the link-partner device is capable of accepting packets with 5 bytes IPG when operating in GMII/MII modes. ■ 0: output IPG >= 8B ■ 1: output IPG >= 5B When operating in XGMII mode (2.5G and above speeds), irrespective of the value of this bit, DWC_xpcs ensures minimum of 5 bytes IPG at the transmitter output. This bit has no effect when operating in single-port USXGMII modes. **Note:** This bit is present only for multi-port USXGMII configurations. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "MP_USXGMII ? \"read-write\" : \"read-only\""

5.2.9.24 VR_MII_DBG_CTRL

- **Description:** VR MII MMD Debug Control Register
  Note: This register is present only in 1000BaseX-Only PCS configurations.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8005 : (APB3_16) ? (0x8005*0x2) : (0x8005*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY)

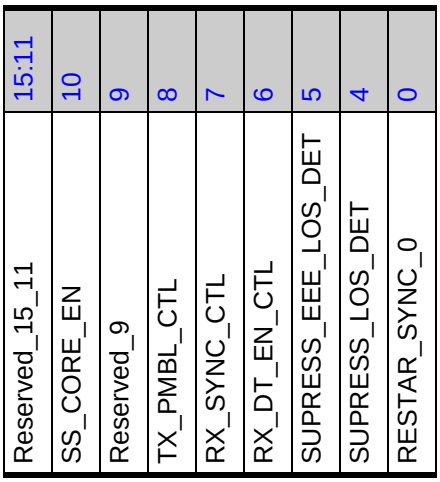


Table 5-485 Fields for Register: VR_MII_DBG_CTRL

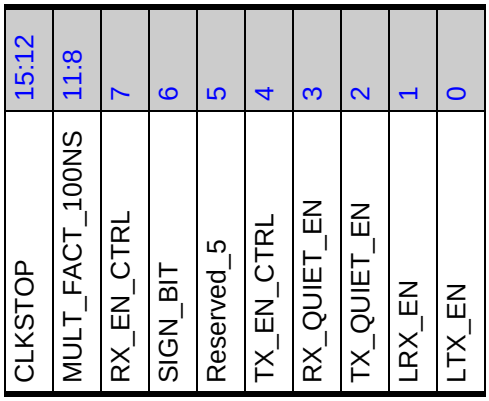
Bits	Name	Memory Access	Description
15:11	Reserved_15_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
10	SS_CORE_EN	* Varies	**When Safety Subordinate feature is enabled** Safety Subordinate Core enable To enable the safety subordinate to monitor errors in DWC_xpcs set this bit to 1. **Other Configurations**Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "DWC_XPCS_SS ? \"read-write\" : \"read-only\""
9	Reserved_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
8	TX_PMBL_CTL	R/W	Transmit Preamble Control This bit can be set to 1 to prevent possible preamble truncation in the DWC_xpcs transmitter when operating in 1000BASE-X (or 2.5G Mode over GMII) Mode. As per IEEE spec, it is permissible for a compliant 1000BASE-X PCS transmit process to truncate the first byte of preamble to align the start of the packet on the EVEN boundary. If this bit is set to 1, DWC_xpcs does not delete any preamble bytes (received from GMII Tx interface) and passes on the same number of preamble bytes to its output. DWC_xpcs carries out this operation by adjusting the IPG. This mode is a deviation from Clause 36 of IEEE802.3 specification. But it can prove useful when operating at 2.5G Mode (over clocked 1000BASE-X Mode) and the far-end device is compliant to IEEE802.3bz/cb specification. **Value After Reset:** 0x0 **Exists:** Always
7	RX_SYNC_CTL	* Varies	Receive Synchronization Control DWC_xpcs, by default, does not monitor the 'xgxs_rx_valid_i' input signal to determine Receive Synchronization Status. Receive Synchronization (sync_status) is attained by merely monitoring and detecting COMMA pattern in the input data, irrespective of the value of 'xgxs_rx_valid_i' signal. If this bit is set to 1, DWC_xpcs starts looking for COMMA in the input data stream only after PHY has asserted 'xgxs_rx_valid_i'. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SNPS_XAUI_PHY ? \"read-write\" : \"read-only\""
6	RX_DT_EN_CTL	* Varies	**For multi-protocol 12G/16G PHY configurations:** Rx Data Enable Control. This bit controls the behavior of xpcs_rx_data_en_o[0] output from DWC_xpcs. During normal operation DWC_xpcs de-asserts xpcs_rx_data_en_o[0] on detecting of Loss-of-Signal. However, if this bit is set, DWC_xpcs does not de-assert xpcs_rx_data_en_o[0] on detection of Loss-of-Signal.Instead, the value of RX_DT_EN_0 bit from VR_MII_MP_12G_16G_25G_RX_GENCTRL0 Register is driven on xpcs_rx_data_en_o[0] output port. **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "E12||E16||E25||E32 ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
5	SUPRESS_EEE_LOS_DET	* Varies	**For PCS-X configurations with EEE:** Suppress EEE Loss of Signal Detection. When this field is set to 1, Loss of Signal indicated by the PHY (based on input port xpcs_los_i) is ignored by DWC_xpcs while evaluating the Receive link when operating in EEE mode. Receive link is evaluated based the on data received by DWC_xpcs from PHY. When this field is set to 0, Loss of signal indicated by the PHY is considered by the DWC_xpcs while evaluating the Receive link status in EEE mode. **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(LP_1GBX||LP_10GBX) ? \"read-write\" : \"read-only\""
4	SUPRESS_LOS_DET	R/W	Suppress Loss of Signal Detection When this field is set to 1, Loss of Signal indicated by the PHY (based on input port xpcs_los_i) is ignored by DWC_xpcs while evaluating the Receive link. Receive link is evaluated based the Rx data received on the 'xgxs_rx_data{lane}_i' port of DWC_xpcs. When this field is set to 0, Loss of signal indicated by the PHY is considered by the DWC_xpcs while evaluating the Receive link status. **Note:** This register bit has an impact on the EEE Rx behavior of DWC_xpcs when operating in QSGMII Mode for multi-protocol 12G/16G/25G PHY configurations. if this register bit is set to 1, then entry to RX_QUIET state is based on 'code_sync_status' (generated by PCS Synchronization State Diagram) instead of 'signal_detect'. **Value After Reset:** 0x0 **Exists:** Always
0	RESTAR_SYNC_0	R/W	Restart Synchronization When set to 1, this bit restarts the Rx Synchronization State machine on Lane 0. The host must clear this bit to 0 before setting it to 1 next time. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.25 VR_MII_EEE_MCTRL0

- **Description:** VR MII MMD EEE Mode Control Register
  This register enables or disables the Energy Efficient Ethernet (EEE) support. This register is present only for configurations with MAIN_MODE = 1000BASEX-Only PCS and EEE_EN = Enabled.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8006 : (APB3_16) ? (0x8006*0x2) : (0x8006*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&EEE_EN)



15:12	11:8	7	6	5	4	3	2	1	0
CLKSTOP	MULT_FACT_100NS	RX_EN_CTRL	SIGN_BIT	Reserved_5	TX_EN_CTRL	RX_QUIET_EN	TX_QUIET_EN	LRX_EN	LTX_EN

Table 5-486 Fields for Register: VR_MII_EEE_MCTRL0

Bits	Name	Memory Access	Description
15:12	CLKSTOP	* Varies	Clock Stop This field holds the count value after which the Rx clock to the MAC can be stopped. The default value is 8. The host must program this value depending on the capability of the MAC. This field is used to control the assertion of the xpcs_lrx_clk_gat_o signal (to gate the Rx clock of the MAC) By default, the xpcs_lrx_clk_gat_o signal is asserted after the PCS Rx module detects the LPI pattern followed by the following: Number of clk_xgxs_rx0_i or clk_rpcs_rx_i clocks (specified in this field) + clk_eee_i synchronization delay + FIFO initialization delay **Value After Reset:** 0x8 **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
11:8	MULT_FACT_100NS	* Varies	100 ns Clock Tic Multiplying Factor This bit is the multiplying factor to the clk_eee_i clock time period to make it closer to 100 ns. For example, if the clk_eee_i clock time period (clk_eee_i_time_period) is 10 ns, the value of this field is 9, that is, 1 less than 10. Program a value such that the clk_eee_i_time_period * (MULT_FACT_100NS + 1) is within 80 ns to 120 ns. The default value of this register is 9, assuming the default clk_eee_i time period to be 10 ns. **Value After Reset:** 0x9 **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\""
7	RX_EN_CTRL	* Varies	Rx Control Enable This bit controls the generation of the xpcs_rx_en_o{lane} signal. When this bit is set to 1, the xpcs_rx_en_o{lane} signal is de-asserted when the EEE receive controller reaches the Quiet state. When this bit set to 0, the xpcs_rx_en_o{lane} signal is not de-asserted when the EEE transmit controller reaches the Quiet state. **Value After Reset:** "(EEE_EN) ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\""
6	SIGN_BIT	* Varies	Effective 100 ns Tic Value The host must use this bit to fine tune the EEE timing requirement. The host must set this bit to 0 when the clk_eee_i_time_period * (MULT_FACT_100NS + 1) value is more than 100 ns. The host must set this bit to 1 when the clk_eee_i clock period * (MULT_FACT_100NS + 1) value is less than or equal to 100 ns. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\""
5	Reserved_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4	TX_EN_CTRL	* Varies	Tx Control Enable This bit controls the generation of the following signals: ■ xgxs_tx{lane}_en_o (for Synopsys PHY) ■ xpcs_tx_en_o{lane} (for non-Synopsys PHY) When this bit is set to 1, the xgxs_tx{lane}_en_o or xpcs_tx_en_o{lane} signal is de-asserted when the EEE transmit controller reaches the Quiet state. When this bit set to 0, the xgxs_tx{lane}_en_o or xpcs_tx_en_o{lane} signal is not de-asserted when the EEE transmit controller reaches the Quiet state. **Value After Reset:** "(EEE_EN) ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\"""
3	RX_QUIET_EN	* Varies	Rx Quiet Enable This bit controls the generation of the xpcs_lpitx_quiet_o output. When this bit is set to 1, the xpcs_lpitx_quiet_o output is set to 1 when the EEE receive controller reaches the Quiet state. When this bit set to 0, the xpcs_lpitx_quiet_o output is not set to 1. **Value After Reset:** "(EEE_EN) ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\"""
2	TX_QUIET_EN	* Varies	Tx Quiet Enable This bit controls the generation of the xpcs_lpitx_quiet_o output. When this bit is set to 1, the xpcs_lpitx_quiet_o output is set to 1 when the EEE transmit controller reaches the Quiet state. When this bit set to 0, the xpcs_lpitx_quiet_o output is not set to 1. **Value After Reset:** "(EEE_EN) ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\"""
1	LRX_EN	* Varies	LPI Rx Enable When set to 1, this bit enables the Energy Efficient Ethernet support in the DWC_xpcs receive path. When set to 0, it disables the support for Energy Efficient Ethernet in the DWC_xpcs receive path. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\"""

Bits	Name	Memory Access	Description
0	LTX_EN	* Varies	LPI Tx Enable When set to 1, this bit enables the Energy Efficient Ethernet support in the DWC_xpcs Transmit path. When set to 0, it disables the support for Energy Efficient Ethernet in the DWC_xpcs Transmit path. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\""

5.2.9.26 VR_MII_EEE_TXTIMER

- **Description:** VR MII MMD EEE Tx Timer Register
  This register is present only for configurations with MAIN_MODE = 1000BASEX-Only PCS and EEE_EN = Enabled.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8008 : (APB3_16) ? (0x8008*0x2) : (0x8008*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&EEE_EN)

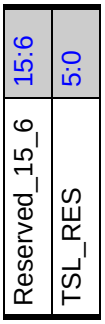


Table 5-487 Fields for Register: VR_MII_EEE_TXTIMER

Bits	Name	Memory Access	Description
15:6	Reserved_15_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
5:0	TSL_RES	* Varies	TSL Resolution This field stores the resolution value for the TSL timer. When the generated 100 ns tic timer is not exactly 100 ns, this field is programmed with the required value. The programmed value is equivalent to the number of more (SIGN_BIT is low) or less (SIGN_BIT is high) counts than the ideal count to meet the range of the TSL timer. The DWC_xpcs maintains the default value of the TSL timer as 199, assuming clk_eee_i_time_period * (MULT_FACT_100NS + 1) is equal to 100 ns to produce 19900 ns (19.9 us) as per the TSL requirement in the IEEE standard. If clk_eee_i_time_period * (MULT_FACT_100NS + 1) is 90, the default TSL timer produces 17910 ns (17.91 us) which is lesser than the standard 19.9 us. To make it 19.9 us, you must program this register such that (199 +/- TSL_RES) * ((MULT_FACT_100NS + 1) * clk_eee_i_time_period) is greater than 19.9 us and less than 20.1 us (max limit), that is, (199 + TSL_RES)*90 = 19.9 us. TWL_RES = 23 meets the requirement which produces 19.98 us. Program the SIGN_BIT with a value 1. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\""

5.2.9.27 VR_MII_EEE_RXTIMER

- **Description:** VR MII MMD EEE Rx Timer Register
  This register is present only for configurations with MAIN_MODE = 1000BASEX-Only PCS and EEE_EN = Enabled.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8009 : (APB3_16) ? (0x8009*0x2) : (0x8009*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&EEE_EN)

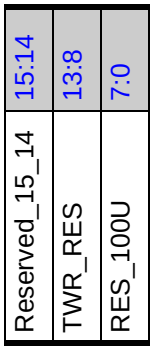


Table 5-488 Fields for Register: VR_MII_EEE_RXTIMER

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
13:8	TWR_RES	* Varies	TWR Resolution This field stores the resolution value for the TWR timer. When the generated 100 ns tic timer is not exactly 100 ns, this field is programmed with the required value. The programmed value is equivalent to the number of more (SIGN_BIT is low) or less (SIGN_BIT is high) counts than the ideal count to meet the range of the TWR timer. The default value is 11 us for 1000BASEX-Only PCS mode. This value is as per the requirements specified in the standard assuming that clk_eee_i_time_period * (MULT_FACT_100NS + 1) produces 100 ns tick. Example: For KX mode, the time requirement is 11 us. The DWC_xpcs maintains the default value of the TWR timer as 110, assuming clk_eee_i_time_period * (MULT_FACT_100NS + 1) is equal to 100 ns to produce 11000 ns (11 us). If clk_eee_i_time_period * (MULT_FACT_100NS + 1) is 110, the default TWR timer produces 12100 ns (12.1 us) which is more than 11 us. To make it 11 us, you must program this register such that (110 +/- TWR_RES) * ((MULT_FACT_100NS + 1) * clk_eee_i_time_period) = 11 us. Therefore, use the following equation: (110 - TWR_RES) * 110 = 11000 ns TWL_RES = 10 meets the requirement which produces 11 us. Program the SIGN_BIT as 0. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
7:0	RES_100U	* Varies	100 us Resolution This field stores the resolution value for the 100 us timer. If the generated 100 ns tic timer is not exactly 100 ns, this field is programmed with the required value. The programmed value is equivalent to the number of more (SIGN_BIT is low) or less (SIGN_BIT is high) counts than the ideal count to meet the range of the 100 us timer. This field is used to control the generation of 100 us time tick using the clk_eee_i clock and MULT_FACT_100NS field. By default, the 100 us timer is generated assuming clk_eee_i_time_period * (MULT_FACT_100NS + 1) is equal to 100 ns. Therefore, the DWC_xpcs maintains the default value of the 100 us timer as 1000 to achieve 100000 ns (100 us). You must use this field if clk_eee_i_time_period * (MULT_FACT_100NS + 1) is not equal to 100 ns. To program this field, use the following equation: (1000 +/- RES_100US)* ((MULT_FACT_100NS + 1) * clk_eee_i_time_period) = 100 us Program the SIGN_BIT as 1 for (+) and 0 for (-). **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\""

5.2.9.28 VR_MII_LINK_TIMER_CTRL

- **Description:** VR MII MMD Link Timer Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x800A : (APB3_16) ? (0x800A*0x2) : (0x800A*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY||CL37_AN)



Table 5-489 Fields for Register: VR_MII_LINK_TIMER_CTRL

Bits	Name	Memory Access	Description
15:0	CL37_LINK_TIME	R/W	Programmable Link Timer Value for Clause 37 auto-negotiation. This field can be programmed to any desired value if application wishes to override the standard specified values for Link Timer used during Clause 37 Auto negotiation. Link timer is implemented in DWC_xpcs using a 24-bit timer.When operating in USXGMII mode, link timer runs at 156.25 MHz. When operating in 1000BaseX/SGMII mode, this timer operates at 125 MHz. **For USXGMII configurations:**This field forms the upper 16-bit of the 24-bit value that gets loaded to the link timer.The lower 8-bits are hard-coded as zero. For example, if CL37_LINK_TIME = 1, the value that is loaded to the timer is 24'h100, which corresponds to a duration of 1638 ns (256*6.4ns) in USXGMII mode or 2048 ns (256*8ns) in 1000BaseX/SGMII mode.**For configurations without USXGMII:**This field forms the upper 16-bit of the 24-bit value that gets loaded to the link timer.The lower 8-bits are hardcoded as 8'h7D. For example, if CL37_LINK_TIME = 1, the value that is loaded to the timer is 24'h17D, which corresponds to a duration of 3048 ns (381*8ns). After programming this register, application must perform either of the following steps, so that the new values takes effect: ■ Set CL37_TMR_OVR_RIDE bit (bit[3]) of VR_MII_DIG_CTRL1 Register to 1 ■ Clear the FAST_SIM bit of SR Control MMD Control register (if already set) and then set it back to 1. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.29 VR_MII_EEE_MCTRL1

- **Description:** VR MII MMD EEE Mode Control 1 Register
  This register also controls the Energy Efficient Ethernet (EEE) related settings of DWC_xpcs.This register is present only for configurations with MAIN_MODE = 1000BASEX-Only PCS and EEE_EN = Enabled.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x800B : (APB3_16) ? (0x800B*0x2) : (0x800B*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&EEE_EN

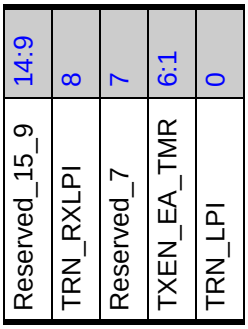


Table 5-490 Fields for Register: VR_MII_EEE_MCTRL1

Bits	Name	Memory Access	Description
14:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	TRN_RXLPI	R/W	Transparent Rx LPI Mode Enable. When set to 1 (along with LRX_EN=1), transparent LPI mode gets activated in the DWC_xpcs Receive path. On receiving Lower-Power Idles from link-partner, DWC_xpcs transparently decodes it and passes the LPI pattern to MAC GMII Rx interface without initiating any low power actions (like switching PHY to low power state or gating-off clocks to DWC_xpcs)) The Receive LPI FSM goes to LRX_SLEEP/LRX_LPI_K state on receiving LPI pattern from link-partner.Thereafter, FSM transitions back to RX_ACTIVE state on receiving normal IDLE patterns from link-partner. To prevent DWC_xpcs from asserting xpcs_lrx_clk_gat_o during the transparent Rx LPI mode, it is recommended to program CS_EN bit of VR_MII_DIG_CTRL1/VR_MII_1/2/3_DIG_CTRL1 register to 0. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:1	TXEN_EA_TMR	R/W	Transmit Enable Early Assert Time Duration During normal Tx EEE operation, DWC_xpcs drives xpcs_tx_en_o/xgxs_tx_data_en_o low for the entire duration for which Tx EEE FSM remains in TX_QUIET state. On exiting TX_QUIET state, it asserts back xpcs_tx_en_o/xgxs_tx_data_en_o to 1. This field can be set to a non-zero value (ranging from 1 to 63), to allow early assertion of xpcs_tx_en_o/xgxs_tx_data_en_o signal prior to Tx FSM exiting TX_QUIET state. This field controls how many micro-seconds earlier to the exit of TX_QUIET state, should DWC_xpcs be driving xpcs_tx_en_o/xgxs_tx_data_en_o back to 1. This feature can be used to enable back the PHY transmitter, a little earlier than desired, so that it can be fully ready to transmit the desired patterns when Tx FSM reaches TX_WAKE state. If the clk_eee_i period is other than 10ns, then program "MULT_FACT_100NS", "SIGN_BIT" of VR_MII_EEE_MCTRL0 register, & "T1U_RES" of VR_MII_EEE_TXTIMER appropriately to generate the 1us tic. If the value of 0 is programmed, DWC_xpcs retains the original legacy behavior, that is, no early exit. For Synopsys-PHY configurations, DWC_xpcs additionally de-asserts xpcs_lpitx_quiet_o signal earlier if a non-zero value is programmed. This is done to ensure that Tx clocks are available to DWC_xpcs and therefore xgxs_tx_data_en_o can be asserted synchronously(to Tx clock) as required by the Synopsys PHY. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	TRN_LPI	R/W	Transparent Tx LPI Mode Enable. When set to 1 (along with LTX_EN=1), transparent LPI mode gets activated in the DWC_xpcs Transmit path. In this mode, the transmit LPI state-machine does not move to TX_QUIET state. On detecting Lower-Power Idle on GMII Tx interface, DWC_xpcs goes to the TX_SLEEP state and remains in this state till MAC stops sending LPI. In this mode, DWC_xpcs merely sends the encoded LPI pattern to the serdes. This mode does not involve gating-off of any clocks to DWC_xpcs. In addition, the serdes transmitter is not disabled (xpcs_tx_data_en_o signal remains high). MAC must ensure that it does not gate-off XGMII/GMII Tx clock to DWC_xpcs during this mode of operation. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.30 VR_MII_DIG_STS

- **Description:** VR MII MMD Digital Status Register.
  Note: This register is present only in 1000BaseX-Only PCS configurations.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8010 : (APB3_16) ? (0x8010*0x2) : (0x8010*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY)

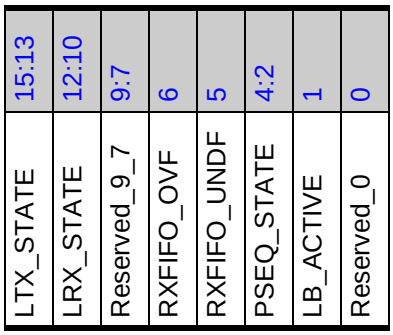


Table 5-491 Fields for Register: VR_MII_DIG_STS

Bits	Name	Memory Access	Description
15:13	LTX_STATE	R	**For EEE configurations:**LPI Transmit State. This field indicates the current state of the LPI Transmit state Machine. This is a 2-bit field if the selected configuration does not have RPCS.Otherwise, it is a 3-bit field. ■ 000: LTX_ACTIVE ■ 001: LTX_SLEEP ■ 010: LTX_QUIET ■ 011: LTX_REF_WAKE ■ 100: LTX_ALERT (valid only if RPCS is present) ■ 101: LTX_SCR_BYP (valid only if RPCS is present) **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
12:10	LRX_STATE	R	**For EEE configurations:**LPI Receive State This field indicates the current state of the LPI Receive State Machine: ■ 000: LRX_ACTIVE ■ 001: LRX_SLEEP ■ 010: LRX_QUIET ■ 011: LRX_WAKE ■ 100: LRX_WTF ■ 101: LRX_LINK_FAIL ■ 110: LRX_LPI_K **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always
9:7	Reserved_9_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6	RXFIFO_OVF	R	Rx FIFO Overflow (RO, LH Type) This bit indicates the clock rate compensation FIFO overflow. ■ 0: Normal operation ■ 1: FIFO overflow **Value After Reset:** 0x0 **Exists:** Always
5	RXFIFO_UNDF	R	Rx FIFO Underflow (RO, LH Type) This bit indicates the clock rate compensation FIFO underflow. ■ 0: Normal operation ■ 1: FIFO underflow **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4:2	PSEQ_STATE	R	**For Synopsys PHY configurations:** Power Up Sequence State This field indicates the state variable value of the power-up sequence module. For Synopsys multi-protocol 6G PHY: ■ 3'b000: Wait for MPLL ON ■ 3'b001: Wait for TX up ■ 3'b010: Wait for RX up ■ 3'b011/3'b100: Tx/Rx Stable (Power_Good state) ■ 3'b101: Wait for Rx down (MPLL still ON) ■ 3'b110: MPLL OFF For Other Synopsys multi-protocol PHYs ■ 3'b000: Wait for ACK High 0 ■ 3'b001: Wait for ACK Low 0 ■ 3'b010: Wait for ACK High 1 ■ 3'b011: Wait for ACK Low 1 ■ 3'b100: Tx/Rx stable (Power_Good state) ■ 3'b101: Power Save state ■ 3'b110: Power Down state ■ 3'b111: Other intermediate states (valid only for 32G PHY configurations) **For non-Synopsys PHY configurations:** Reserved **Value After Reset:** 0x4 **Exists:** Always
1	LB_ACTIVE	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	Reserved_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.9.31 VR_MII_ICG_ERRCNT1

- **Description:** VR MII MMD Invalid Code Group Error Count1 Register.
  Note: This register is present only for 1000BaseX-Only PCS configurations
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8011 : (APB3_16) ? (0x8011*0x2) : (0x8011*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY

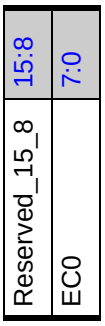


Table 5-492 Fields for Register: VR_MII_ICG_ERRCNT1

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:0	EC0	R	Invalid Code Group Count Lane 0 (RO, LH Type) This field gives the invalid code group count in Lane 0 when Bit 4 of VR_MII_DIG_ERRCNT_SEL Register is set to 1. It is recommended to wait at the least for a time duration equal to 5 clocks of clk_xgxs_rx0_i before initiating back-to-back register access. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.32 VR_MII_GPIO

- **Description:** VR MII MMD GPIO Register
  This register is present only if the DWC_xpcs is configured to have the GPIO interface.
  Note: This register is not present if USXGMII_EN = Enabled
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8015 : (APB3_16) ? (0x8015*0x2) : (0x8015*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** ((BASE1000X_ONLY||(CL37_AN&&!USXGMII_EN))&&GPIO_EN)

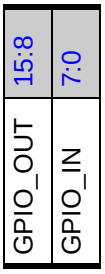


Table 5-493 Fields for Register: VR_MII_GPIO

Bits	Name	Memory Access	Description
15:8	GPIO_OUT	R/W	GPIO Output The content written on this field is driven to the xpcs_gpo_o[7:0] output port. Dependency: This field is valid only when GPIO_EN = Enabled. **Value After Reset:** 0x0 **Exists:** Always
7:0	GPIO_IN	R	GPIO Input This field indicates the content of the xpcs_gpo_i[7:0] port. Dependency: This field is valid only when GPIO_EN = Enabled. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.33 VR_MII_MISC_STS

- **Description:** VR MII MMD Miscellaneous Status Register.
  This register is present only when the DWC_xpcs is configured in 1000BASEX-Only PCS mode.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8018 : (APB3_16) ? (0x8018*0x2) : (0x8018*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY

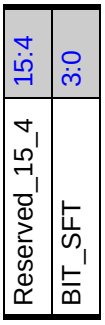


Table 5-494 Fields for Register: VR_MII_MISC_STS

Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:0	BIT_SFT	R	Bit Shift This field indicates the number of bit-shifts carried-out by comma-detect logic so as to align the incoming 10-bit XGXS Rx data **Default Value:**The default value of this field can be any value, depending on the status of comma-detect logic. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

5.2.9.34 VR_MII_NSNPS_EEE_CTRL

- **Description:** VR MII MMD Non-Synopsys PHY EEE Control Register
  This register is present only for configurations with MAIN_MODE = 1000BASEX-Only PCS, QSGMII_EN = Enabled and EEE_EN = Enabled.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x801F : (APB3_16) ? (0x801F*0x2) : (0x801F*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&!SNPS_XAUI_PHY&&QSGMII_EN&&EEE_EN

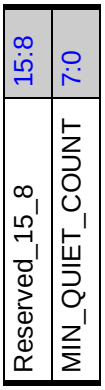


Table 5-495 Fields for Register: VR_MII_NSNPS_EEE_CTRL

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:0	MIN_QUIET_COUNT	R/W	Minimum Rx Quiet Duration This field determines the minimum duration for which Rx LPI FSM remains in the RX_QUIET state.DWC_xpcs implements a vendor-specific 'MIN QUIET counter' in the RX_QUIET state of LPI state machine. This counter (running at 125 MHz in QSGMII mode) starts as soon as RX_QUIET state is reached and it stops when it reaches a value equal to 8 times the value mentioned as 'MIN QUIET COUNT' field. DWC_xpcs start monitoring for 'signal_detect' (in QSGMII Mode) only after this count has reached its maximum value (8 times MIN_QUIET_COUNT). **Value After Reset:** 0xa **Exists:** Always

5.2.9.35 VR_MII_RX_LSTS

- **Description:** VR MII PHY Rx Lane Status Register
  In KX_Only, KR_Only, KR_KX, 10GBASE-R PCS, and 1000BASEX-Only PCS configurations, only Lane 0 control and status information is present. In configurations with both 1G (KX) and 10G (XGXS PCS, 10GBASE-X PCS, or KX4) modes, only Lane 0 control and status information is used when the DWC_xpcs is operating in the 1G mode.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8020 : (APB3_16) ? (0x8020*0x2) : (0x8020*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY

15:13	12	11:9	8	7:5	4	3:0
RX_VALID_3_1	RX_VALID_0	RX_PLL_STATE_3_1	RX_PLL_STATE_0	SIG_DET_3_1	SIG_DET_0	Reserved_3_0

Table 5-496 Fields for Register: VR_MII_RX_LSTS

Bits	Name	Memory Access	Description
15:13	RX_VALID_3_1	R	DPLL Lock Status for Lanes[3:1] This field indicates that the DPLL in the PHY is locked on the serial data in the corresponding lane. ■ 3'b**1: Lane 1 DPLL bit locked ■ 3'b*1*: Lane 2 DPLL bit locked ■ 3'b1**: Lane 3 DPLL bit locked Dependency: This field is present only the following configurations: ■ Synopsys PHY configurations ■ Backplane Ethernet PCS configurations with non-Synopsys PHY **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

Bits	Name	Memory Access	Description
12	RX_VALID_0	R	DPLL Lock Status for Lane 0 This field indicates that the DPLL in the PHY is locked on the serial data in Lane 0. The value 1'b1 indicates that Lane 0 DPLL bit is locked. Dependency: This field is present only the following configurations: ■ Synopsys PHY configurations ■ Backplane Ethernet PCS configurations with non-Synopsys PHY **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
11:9	RX_PLL_STATE_3_1	R	**For Multi-lane Synopsys Multi-protocol 6G PHY configurations:** DPLL State for Lanes[3:1] This field indicates that the Rx has successfully reached the PLL Power_Good during the power up sequence on the corresponding lane. ■ 3'b**1: Lane 1 Rx successful ■ 3'b*1*: Lane 2 Rx successful ■ 3'b1**: Lane 3 Rx successful **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
8	RX_PLL_STATE_0	R	**For Synopsys Multi-protocol 6G PHY configurations:**Rx DPLL State for Lane 0 This field indicates that the Rx has successfully reached the PLL Power_Good state during the power up sequence on Lane 0. The value 1'b1 indicates that Lane 0 Rx is successful. **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

Bits	Name	Memory Access	Description
7:5	SIG_DET_3_1	R	**For Multi-lane configurations:** Rx Signal Detect for Lanes[3:1] This field indicates that the Rx has detected the signal on the corresponding lanes. The bits in this field are the complement value of the signals input from PHY (xpcs_los_i{lane}) on the corresponding lane. ■ 3'b**1: Lane 1 signal detected ■ 3'b*1*: Lane 2 signal detected ■ 3'b1**: Lane 3 signal detected **For Single-lane configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
4	SIG_DET_0	R	Rx Signal Detect for Lane 0 This bit indicates that the Rx detected the signal on Lane 0. This bit is the complement value of the signals input from PHY (xpcs_los_i[0]) on Lane 0. The value 1'b1 indicates that Lane 0 signal is detected. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
3:0	Reserved_3_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.9.36 VR_MII_MP_12G_16G_25G_TX_GENCTRL0

- **Description:** VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Tx General Register 0
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8030 : (APB3_16) ? (0x8030*0x2) : (0x8030*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16||E25||E32))

15:13	12	11:9	8	7:5	4	3:1	0
TX_DT_EN_3_1	TX_DT_EN_0	TX_RST_3_1	TX_RST_0	TX_INV_3_1	TX_INV_0	TXBCN_EN_3_1	TXBCN_EN_0

Table 5-497 Fields for Register: VR_MII_MP_12G_16G_25G_TX_GENCTRL0

Bits	Name	Memory Access	Description
15:13	TX_DT_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	TX_DT_EN_0	R/W	Tx Data Enable on lane 0 of PHY When this bit is set, Transmit Output Driver in the PHY is enabled. This bit controls the output port 'xgxs_tx_data_en_o[0]'. Clear this field when PHY Tx is not in P0 power state. **Value After Reset:** 0x1 **Exists:** Always
11:9	TX_RST_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	TX_RST_0	R/W	Tx Reset on lane 0 of PHY When this bit is set, PHY transmitter is reset, including common-mode adjustment and receiver detection state machines. This signal drives the output port 'xgxs_tx_reset_o[0]' when DWC_xpcs is in POWER_GOOD state. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
7:5	TX_INV_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INV_0	R/W	Tx Invert on lane 0 of PHY When this bit is set, the data on PHY Tx serial lines are logically inverted. This signal drives the output port 'xgxs_tx_invert_o[0]'. **Value After Reset:** 0x0 **Exists:** Always
3:1	TXBCN_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	TXBCN_EN_0	R/W	Tx Beaconing Enable on lane 0 of PHY When this bit is set, PHY enables transmitter beaconing (LFPS). The period for transmit pulses is between 20-50ns. This field drives the output port 'xgxs_tx_beacon_en_o[0]'. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.37 VR_MII_MP_12G_16G_25G_TX_GENCTRL1

- **Description:** VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Tx General Register 1
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8031 : (APB3_16) ? (0x8031*0x2) : (0x8031*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16||E25||E32))

15:13	12	11	10:8	7:5	4	3:1	0
TX_CLK_RDY_3_1	TX_CLK_RDY_0	Reserved_11	VBOOST_LVL	VBOOST_EN_3_1	VBOOST_EN_0	DET_RX_REQ_3_1	DET_RX_REQ_0

Table 5-498 Fields for Register: VR_MII_MP_12G_16G_25G_TX_GENCTRL1

Bits	Name	Memory Access	Description
15:13	TX_CLK_RDY_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	TX_CLK_RDY_0	R/W	Transmitter Input clock ready on lane 0 This field drives the output xpcs_tx_clk_rdy_o[0]. This field must remain high, as long as PHY Tx clock is active **Value After Reset:** 0x1 **Exists:** Always
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
10:8	VBOOST_LVL	R/W	Tx Voltage Boost Maximum Level This field controls the maximum achievable Tx swing in the PHY This field drives the output port 'xpcs_tx_vboost_lvl_o[2:0]'. **Value After Reset:** "(SNPS_RSVDPARAM_39) ? 0x4 : (E16&&SNPS_RSVDPARAM_15) ? 0x5 : (SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_22) ? 0x3 : 0x5" **Exists:** !(SNPS_RSVDPARAM_37||E32)

Bits	Name	Memory Access	Description
7:5	VBOOST_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	VBOOST_EN_0	* Varies	Tx voltage Boost Enable on lane 0 of PHY When this bit is set, the current mode Tx Swing boost in the PHY is enabled. This bit drives the output port 'xpcs_tx_vboost_en_o[0]'. **Value After Reset:** "(E25||SNPS_RSVDPARAM_39) ? 0x0 : 0x1" **Exists:** !(SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_37||E32) **Memory Access:** "SNPS_RSVDPARAM_22 ? \"read-only\" : \"read-write\""
3:1	DET_RX_REQ_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	DET_RX_REQ_0	R/W	Transmitter Rx-Detection request on lane 0 of PHY. This field drives the output port 'xpcs_tx_detrx_req_o[0]'. Whenever this bit is set, a receiver detection request is made towards the PHY on lane 0. When this bit is set, it must remain high till 'TX_ACK_0' bit is read as high. When 'TX_ACK_0' is read as high, the result of the Rx Detection operation is available at field 'DETRX_RSLT_0'. After obtaining the result of 'Rx-Detect operation', clear this bit. Rx Detection operation is valid only when transmitter is in P1 power state **Value After Reset:** 0x0 **Exists:** Always

5.2.9.38 VR_MII_MP_12G_16G_TX_GENCTRL2

- **Description:** VR MII Synopsys Multi-protocol 12G/16G PHY Tx General Register 2
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8032 : (APB3_16) ? (0x8032*0x2) : (0x8032*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16))

15:14	13:12	11:10	9:8	7:5	4	3:1	0
TX3_WIDTH	TX2_WIDTH	TX1_WIDTH	TX0_WIDTH	TX_LPD_3_1	TX_LPD_0	TX_REQ_3_1	TX_REQ_0

Table 5-499 Fields for Register: VR_MII_MP_12G_16G_TX_GENCTRL2

Bits	Name	Memory Access	Description
15:14	TX3_WIDTH	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:12	TX2_WIDTH	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:10	TX1_WIDTH	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9:8	TX0_WIDTH	R/W	Tx Datapath Width on lane 0 of the PHY This field controls the width of input transmit data on lane 0. The encoding of the width is as follows: ■ 2'b00: 8 bit ■ 2'b01: 10 bit ■ 2'b10: 16 bit ■ 2'b11: 20 bit This field drives the output port xpcs_tx0_width_o[1:0]. **Value After Reset:** 0x1 **Exists:** Always

Bits	Name	Memory Access	Description
7:5	TX_LPD_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_LPD_0	R/W	Transmitter Lane Power Down on lane 0 of PHY This field drives the output 'xpcs_tx_lpd_o[3:1]'. This field can be asserted to put the phy transmitter to a power state equivalent to that of P1. **Value After Reset:** 0x0 **Exists:** Always
3:1	TX_REQ_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	TX_REQ_0	R/W	Transmitter operation request on lane 0 of PHY (RW,SC Type) This bit drives the output port 'xpcs_tx_req_o[0]. This bit can be set to initiate a new transmitter setting request. Set this bit only if TX_ACK_0 field is low. When this bit is set, application must monitor TX_ACK_0 field till it becomes 1'b1. This implies that the transmitter request operation has been successfully completed. Then application must clear this bit. Assert this bit to make sure that the PHY accepts any change to the following signals: ■ xpcs_tx0_pstate_o ■ xpcs_tx0_lpd_o ■ xpcs_tx0_rate_o ■ xpcs_tx0_width_o ■ xpcs_tx0_mpllb_sel **Value After Reset:** 0x0 **Exists:** Always

5.2.9.39 VR_MII_MP_25G_TX_GENCTRL2

- **Description:** VR MII Synopsys Multi-Protocol 25G PHY Tx General 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8032 : (APB3_16) ? (0x8032*0x2) : (0x8032*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&(E25||E32)

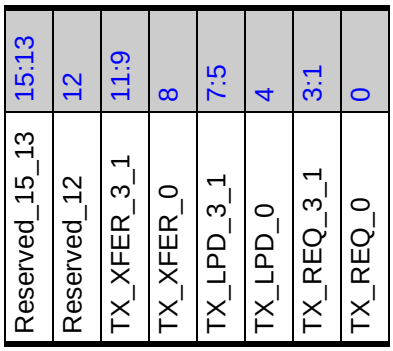


Table 5-500 Fields for Register: VR_MII_MP_25G_TX_GENCTRL2

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	Reserved_12	R	Reserved **Value After Reset:** 0x0 **Exists:** !SNPS_RSVDPARAM_37
11:9	TX_XFER_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	TX_XFER_0	* Varies	Transmitter Alignment Wide Transfer Enable on lane 0 When this field is set, the internal data transfer (in PHY) from txX_clk domain to the internal clock domain is performed using a width of 32 bits. This field drives the xpcs_tx_align_wide_xfer_en_o[0] port. **Value After Reset:** 0x0 **Exists:** (!(SNPS_RSVDPARAM_37||E32)) **Memory Access:** "(!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37&&!E32) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
7:5	TX_LPD_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_LPD_0	R/W	Transmitter Lane Power Down on lane 0 of PHY This field drives the output 'xpcs_tx_lpd_o[3:1]'. This field can be asserted to put the PHY transmitter to a power state equivalent to that of P1. **Value After Reset:** 0x0 **Exists:** Always
3:1	TX_REQ_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	TX_REQ_0	R/W	Transmitter operation request on lane 0 of PHY (RW,SC Type) This bit drives the output port 'xpcs_tx_req_o[0]. This bit can be set to initiate a new transmitter setting request. Set this bit only if TX_ACK_0 field is low. This bit gets self-cleared after the transmitter request operation has been successfully completed. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.40 VR_MII_MP_12G_16G_25G_TX_BOOST_CTRL

- **Description:** VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Tx Boost Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8033 : (APB3_16) ? (0x8033*0x2) : (0x8033*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16||E25)&&!(SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_37))

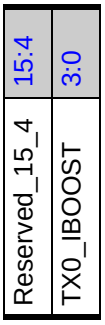


Table 5-501 Fields for Register: VR_MII_MP_12G_16G_25G_TX_BOOST_CTRL

Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:0	TX0_IBOOST	R/W	Tx current boost level on lane 0 of the PHY This bit drives the output port xpcs_tx0_iboost_lvl_o[3:0]. For details on how to set this field, see the PHY Databook. **Value After Reset:** 0xf **Exists:** Always

5.2.9.41 VR_MII_MP_32G_TX_GEN_CTRL3

- **Description:** VR MII Synopsys Multi-Protocol 32G PHY Tx General Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8033 : (APB3_16) ? (0x80333*0x2) : (0x8033*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** ((BASE1000X_ONLY)&&SNPS_XAUI_PHY&&(E32))

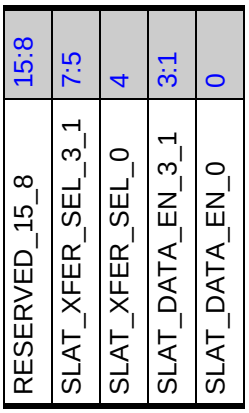


Table 5-502 Fields for Register: VR_MII_MP_32G_TX_GEN_CTRL3

Bits	Name	Memory Access	Description
15:8	RESERVED_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	SLAT_XFER_SEL_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	SLAT_XFER_SEL_0	R/W	Transmitter Short Latency Transfer Select - for Lane0. This field drives the output port xpcs_tx_slat_xfer_sel_o. **Value After Reset:** 0x0 **Exists:** Always
3:1	SLAT_DATA_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	SLAT_DATA_EN_0	R/W	Transmitter Short Latency Data Enable - for Lane0. This field drives the output port xpcs_tx_slat_data_en_o. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.42 VR_MII_MP_12G_16G_25G_TX_RATE_CTRL

- **Description:** VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Tx Rate Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8034 : (APB3_16) ? (0x8034*0x2) : (0x8034*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16||E25)&&!SNPS_RSVDPARAM_37)

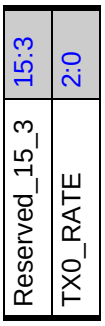


Table 5-503 Fields for Register: VR_MII_MP_12G_16G_25G_TX_RATE_CTRL

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	TX0_RATE	R/W	Tx date rate on lane 0 of the PHY Data rate encoding is as follows: ■ 3'b000: baud ■ 3'b001: baud/2 ■ 3'b010: baud/4 ■ 3'b011: baud/8 ■ 3'b111: baud/10 ■ 3'b100-3'b110: Not supported **Value After Reset:** "(E25||SNPS_RSVDPARAM_39) ? 0x2 : SNPS_RSVDPARAM_22 ? 0x0: SNPS_RSVDPARAM_35 ? 0x2 : (E16&&SNPS_RSVDPARAM_15) ? 0x6 : 0x3" **Exists:** Always

5.2.9.43 VR_MII_MP_32G_TX_GEN_CTRL4

- **Description:** VR XS or PMA Synopsys Multi-Protocol 32G PHY Tx General Control 4 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8034 : (APB3_16) ? (0x8034*0x2) : (0x8034*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** ((BASE1000X_ONLY)&&SNPS_XAUI_PHY&&(E32))

15:14	13:12	11:10	9:8	7:5	4	3:1	0
TX_RCL_BANK_SEL_3	TX_RCL_BANK_SEL_2	TX_RCL_BANK_SEL_1	TX_RCL_BANK_SEL_0	TX_RCL_SKIP_EN_3_1	TX_RCL_SKIP_EN_0	TX_RCL_FRC_EN_3_1	TX_RCL_FRC_EN_0

Table 5-504 Fields for Register: VR_MII_MP_32G_TX_GEN_CTRL4

Bits	Name	Memory Access	Description
15:14	TX_RCL_BANK_SEL_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:12	TX_RCL_BANK_SEL_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:10	TX_RCL_BANK_SEL_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9:8	TX_RCL_BANK_SEL_0	R/W	Tx Recalibration Bank select for Lane 0 Selects the bank to store the Tx Duty cycle correction (DCC) values for Lane 0. **Value After Reset:** 0x0 **Exists:** Always
7:5	TX_RCL_SKIP_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4	TX_RCL_SKIP_EN_0	R/W	Tx Recalibration Skip Enable on Lane 0 Enables the skipping of Tx recalibration on lane0. This field drives the output port 'xpcs_tx_recal_skip_en_o'. **Value After Reset:** 0x0 **Exists:** Always
3:1	TX_RCL_FRC_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	TX_RCL_FRC_EN_0	R/W	Tx Force Recalibration Enable on Lane 0 Forces Recalibration on lane 0. This field drives the output port 'xpcs_tx_recal_force_en_o' **Value After Reset:** 0x0 **Exists:** Always

5.2.9.44 VR_MII_MP_12G_16G_25G_TX_POWER_STATE_CTRL

- **Description:** VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Tx Power State Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8035 : (APB3_16) ? (0x8035*0x2) : (0x8035*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16||E25||E32))

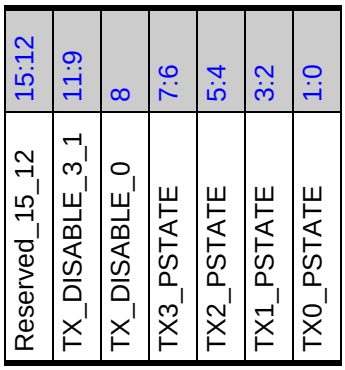


Table 5-505 Fields for Register: VR_MII_MP_12G_16G_25G_TX_POWER_STATE_CTRL

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:9	TX_DISABLE_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	TX_DISABLE_0	R/W	Transmitter Disable on lane 0 This field drives the output port 'xpcs_tx_disable_o[0]'. **Value After Reset:** 0x0 **Exists:** Always
7:6	TX3_PSTATE	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	TX2_PSTATE	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:2	TX1_PSTATE	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
1:0	TX0_PSTATE	R/W	Tx power state control for lane 0 of PHY Power state encoding is as follows: ■ 2'b00: P0 ■ 2'b01: P0s ■ 2'b10: P1 ■ 2'b11: P2 **Value After Reset:** 0x0 **Exists:** Always

5.2.9.45 VR_MII_MP_12G_16G_25G_TX_EQ_CTRL0

- **Description:** VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Tx Equalization Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8036 : (APB3_16) ? (0x8036*0x2) : (0x8036*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16||E25||E32))

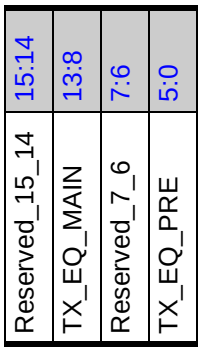


Table 5-506 Fields for Register: VR_MII_MP_12G_16G_25G_TX_EQ_CTRL0

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:8	TX_EQ_MAIN	R/W	Control for setting Tx driver output amplitude This field drives the output port 'rpcs_ktx_main_o' if 'TX_EQ_OVR_RIDE' bit of VR_MII_MP_12G_16G_TX_EQ_CTRL1 Register is set or in configurations with CL72_EN = Disabled. **Value After Reset:** "SNPS_RSVDPARAM_37 ? 0x30: SNPS_RSVDPARAM_22 ? 0x3E : SNPS_RSVDPARAM_35 ? 0x18 : (E25||SNPS_RSVDPARAM_39||E32) ? 0x18 : 0x28" **Exists:** Always
7:6	Reserved_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
5:0	TX_EQ_PRE	R/W	Tx Pre-Emphasis level adjustment Control This field controls the transmitter driver output pre-emphasis (pre-shoot coefficient).This field drives the output port 'rpcs_ktx_pre_o' if 'TX_EQ_OVR_RIDE' bit of VR_MII_MP_12G_16G_25G_TX_EQ_CTRL1 Register is set or in configurations with CL72_EN = Disabled. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.46 VR_MII_MP_12G_16G_25G_TX_EQ_CTRL1

- **Description:** VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Tx Equalization Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8037 : (APB3_16) ? (0x8037*0x2) : (0x8037*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16||E25||E32))

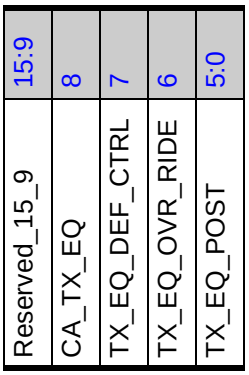


Table 5-507 Fields for Register: VR_MII_MP_12G_16G_25G_TX_EQ_CTRL1

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	CA_TX_EQ	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7	TX_EQ_DEF_CTRL	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6	TX_EQ_OVR_RIDE	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:0	TX_EQ_POST	R/W	Tx Post-Emphasis level adjustment Control This field controls the transmitter driver output pre-emphasis (pre-shoot coefficient). This field drives the output port 'rpcs_ktx_post_o' if 'TX_EQ_OVR_RIDE' bit is set or in configurations with CL72_EN = Disabled. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.47 VR_MII_MP_6G_TX_BSTCTRL0

- **Description:** VR MII Synopsys Multi-protocol 6G Tx Boost Control Register 0
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8038 : (APB3_16) ? (0x8038*0x2) : (0x8038*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&SSP

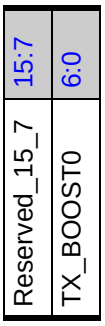


Table 5-508 Fields for Register: VR_MII_MP_6G_TX_BSTCTRL0

Bits	Name	Memory Access	Description
15:7	Reserved_15_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:0	TX_BOOST0	R/W	Transmit Pre-Emphasis Level Adjustment Control for Lane 0 This field is similar to the TX_BOOST1 field. It sets the amplitude of the transmitter driver output on Lane 0. This is specified in decibels (dB) and is equal to the following: (TX_BOOST0[6:0] + 1) / 4 **Value After Reset:** 0xa **Exists:** Always

5.2.9.48 VR_MII_MP_6G_TX_LVLCTRL0

- **Description:** VR MII Synopsys Multi-protocol 6G Tx Level Control Register 0
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x803A : (APB3_16) ? (0x803A*0x2) : (0x803A*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&SSP

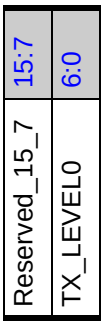


Table 5-509 Fields for Register: VR_MII_MP_6G_TX_LVLCTRL0

Bits	Name	Memory Access	Description
15:7	Reserved_15_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:0	TX_LEVEL0	R/W	Transmit Amplitude Adjustment Control for Lane 0 This is the control field for adjusting the amplitude of the transmit driver output. **Value After Reset:** 0x7f **Exists:** Always

5.2.9.49 VR_MII_MP_6G_TXGENCTRL0

- **Description:** VR MII Synopsys Multi-protocol 6G Tx General Control Register 0
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x803C : (APB3_16) ? (0x803C*0x2) : (0x803C*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&SSP

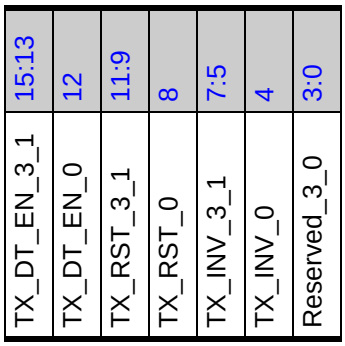


Table 5-510 Fields for Register: VR_MII_MP_6G_TXGENCTRL0

Bits	Name	Memory Access	Description
15:13	TX_DT_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	TX_DT_EN_0	R/W	Output Driver Enable on Lane 0 When set to 1, this bit enables the TX_EN[] and TX_CM_EN[] signals inside the PHY (if not already set to 1 in its IO). This bit enables the output driver on Lane 0. **Value After Reset:** 0x1 **Exists:** Always
11:9	TX_RST_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	TX_RST_0	R/W	Transmit Reset on Lane 0 This field disables the corresponding TX_EN[] and TX_DATA_EN[] signals. It overrides the IO settings and resets all transmit state machines, including the common-mode settings. This bit resets transmit on Lane 0. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
7:5	TX_INV_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_INV_0	R/W	Transmit Invert on Lane 0 This bit inverts the logic levels of tx0_p and tx0_m of the PHY output. **Value After Reset:** 0x0 **Exists:** Always
3:0	Reserved_3_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.9.50 VR_MII_MP_32G_TX_CM_CNTX_SEL0

- **Description:** VR XS or PMA Synopsys Multi-Protocol 32G Tx Common Context Select 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x803C : (APB3_16) ? (0x803C*0x2) : (0x803C*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY)&&SNPS_XAUI_PHY&&E32

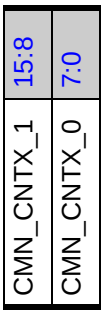


Table 5-511 Fields for Register: VR_MII_MP_32G_TX_CM_CNTX_SEL0

Bits	Name	Memory Access	Description
15:8	CMN_CNTX_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:0	CMN_CNTX_0	R/W	TX Common Context Select Lane0. This field drives the output port 'xpcs_tx_cm0_cntx_sel_o'. **Value After Reset:** CM_CNTX_SEL **Exists:** Always

5.2.9.51 VR_MII_MP_16G_25G_TX_GENCTRL3

- **Description:** VR MII Synopsys Multi-Protocol 16G/25G PHY TX General Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x803C : (APB3_16) ? (0x803C*0x2) : (0x803C*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37

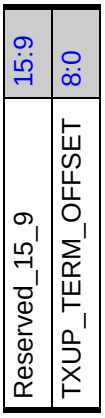
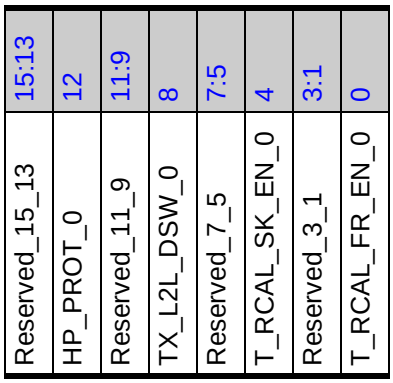


Table 5-512 Fields for Register: VR_MII_MP_16G_25G_TX_GENCTRL3

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8:0	TXUP_TERM_OFFSET	R/W	Offset for TX Up Termination. This field drives the output port xpcs_txup_term_offset_o[8:0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.52 VR_MII_Consumer_20G_TX_GENCTRL3

- **Description:** VR MII Synopsys Consumer 20G PHY TX General Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x803C : (APB3_16) ? (0x803C*0x2) : (0x803C*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&SNPS_RSVDPARAM_37



15:13	12	11:9	8	7:5	4	3:1	0
Reserved_15_13	HP_PROT_0	Reserved_11_9	TX_L2L_DSW_0	Reserved_7_5	T_RCAL_SK_EN_0	Reserved_3_1	T_RCAL_FR_EN_0

Table 5-513 Fields for Register: VR_MII_Consumer_20G_TX_GENCTRL3

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	HP_PROT_0	R/W	Hot-Plug Protection Enable for lane 0 This field drives the xpcs_tx_hp_prot_en_o[0] port. **Value After Reset:** 0x0 **Exists:** Always
11:9	Reserved_11_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	TX_L2L_DSW_0	R/W	Enable lane-to-lane clock deskewing for lane 0 This field drives the xpcs_tx_lane2lane_dskw_en_o[0] port. **Value After Reset:** 0x0 **Exists:** Always
7:5	Reserved_7_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4	T_RCAL_SK_EN_0	R/W	Tx Skip Re-calibration -lane 0 This field drives the xpcs_tx_recal_skip_en_o[0] port. **Value After Reset:** 0x0 **Exists:** Always
3:1	Reserved_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	T_RCAL_FR_EN_0	R/W	Tx Force Re-calibration -lane 0 This field drives the xpcs_tx_recal_force_en_o[0] port. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.53 VR_MII_MP_6G_TXGENCTRL1

- **Description:** VR MII Synopsys Multi-protocol 6G Tx General Control Register 1
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x803D : (APB3_16) ? (0x803D*0x2) : (0x803D*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&SSP)

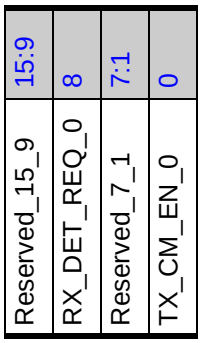


Table 5-514 Fields for Register: VR_MII_MP_6G_TXGENCTRL1

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	RX_DET_REQ_0	R/W	Receiver Detection Request on Lane 0 When asserted, this bit initiates the receiver detection operation for the Synopsys Multi-protocol 6G PHY on Lane 0. Set this bit to 1 only when the corresponding bit (RX_DET_ACK_ 0) in the VR_MII_MP_6G_TX_STS register is set to 0. When this bit is 1, do not change it to 0 until the RX_DET_ACK_0 is set to 1. **Value After Reset:** 0x0 **Exists:** Always
7:1	Reserved_7_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	TX_CM_EN_0	R/W	Transmit Common Mode Control Enable on Lane 0 This field enables the transmit common mode to be held at the half value of the vptx on Lane 0. Otherwise, the common mode is weakly held to ground through a high impedance connection. **Value After Reset:** 0x1 **Exists:** Always

5.2.9.54 VR_MII_MP_16G_25G_TX_GENCTRL4

- **Description:** VR MII Synopsys Multi-Protocol 16G/25G PHY TX General Control 4 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x803D : (APB3_16) ? (0x803D*0x2) : (0x803D*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37

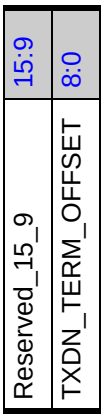


Table 5-515 Fields for Register: VR_MII_MP_16G_25G_TX_GENCTRL4

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8:0	TXDN_TERM_OFFSET	R/W	Offset for TX Down Termination. This field drives the output port xpcs_txdn_term_offset_o[8:0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.55 VR_MII_MP_6G_TXGENCTRL2

- **Description:** VR MII Synopsys Multi-protocol 6G PHY Tx General Control Register 2
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x803E : (APB3_16) ? (0x803E*0x2) : (0x803E*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&SSP)

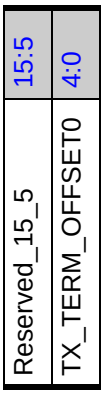


Table 5-516 Fields for Register: VR_MII_MP_6G_TXGENCTRL2

Bits	Name	Memory Access	Description
15:5	Reserved_15_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4:0	TX_TERM_OFFSET0	R/W	Tx Termination Offset Lane 0 This field applies an offset to the resistor calibration value on lane 0. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.56 VR_MII_MP_16G_25G_TX_MISC_CTRL0

- **Description:** VR MII Synopsys Multi-Protocol 16G/25G PHY TX Miscellaneous Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x803E : (APB3_16) ? (0x803E*0x2) : (0x803E*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37

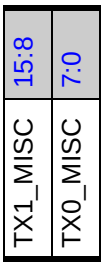


Table 5-517 Fields for Register: VR_MII_MP_16G_25G_TX_MISC_CTRL0

Bits	Name	Memory Access	Description
15:8	TX1_MISC	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:0	TX0_MISC	R/W	TX Miscellaneous control for lane0. This field drives the output port xpcs_tx0_misc_o[7:0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.57 VR_MII_MP_32G_TX_CNTX_CTRL0

- **Description:** VR MII Synopsys Multi Protocol E32 PHY TX Context Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x803E : (APB3_16) ? (0x803E*0x2) : (0x803E*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(SNPS_RSVDPARAM_37||E32))



Table 5-518 Fields for Register: VR_MII_MP_32G_TX_CNTX_CTRL0

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:0	TX_CNTX_SEL_0	R/W	Transmit context select along lane 0 of the PHY specifies the context that needs to be restored for the TX block on lane 0. This field drives the output port 'xpcs_tx0_cntx_sel_o[7:0]'. **Value After Reset:** TX_CNTX_SEL **Exists:** Always

5.2.9.58 VR_MII_MP_12G_16G_25G_TX_STS

- **Description:** VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Tx Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8040 : (APB3_16) ? (0x8040*0x2) : (0x8040*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16||E25||E32))

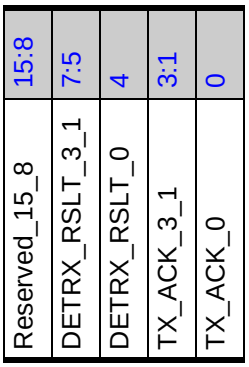


Table 5-519 Fields for Register: VR_MII_MP_12G_16G_25G_TX_STS

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	DETRX_RSLT_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	DETRX_RSLT_0	R	Receiver Detection Result on lane 0 This field captures the value of the input port 'xpcs_tx_detrx_result_i[0]'. The value of this field is valid when 'TX_ACK_0' is high. ■ 1'b0: Receiver not detected ■ 1'b1: Receiver detected **Value After Reset:** 0x0 **Exists:** Always
3:1	TX_ACK_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	TX_ACK_0	R	Tx Acknowledge on lane 0 of PHY This bit captures the value of the input port 'xpcs_tx_ack_i[0]'. Whenever this bit is read as high, it indicates that the requested transmitter setting is complete or the requested RX-detection operation is complete. After this bit is read high, it remains high till bit TX_REQ_0 or DET_RX_REQ_0 is cleared **Value After Reset:** 0x0 **Exists:** Always

5.2.9.59 VR_MII_MP_25G_TX_TERM_CTRL

- **Description:** VR MII Synopsys Multi-Protocol 25G PHY Transmit Termination Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8044 : (APB3_16) ? (0x8044*0x2) : (0x8044*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&E25&&!SNPS_RSVDPARAM_37

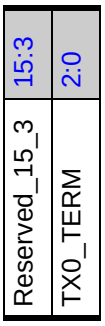


Table 5-520 Fields for Register: VR_MII_MP_25G_TX_TERM_CTRL

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	TX0_TERM	R/W	Transmit Termination Control for lane 0 **Value After Reset:** "(SNPS_RSVDPARAM_36) ? 0x7 : 0x2" **Exists:** Always

5.2.9.60 VR_MII_MP_25G_TX_DCC_CTRL

- **Description:** VR MII Synopsys Multi-Protocol 25G PHY Tx DCC Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8045 : (APB3_16) ? (0x8045*0x2) : (0x8045*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&E25&&!SNPS_RSVDPARAM_37

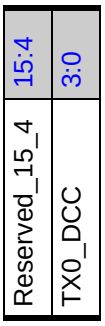


Table 5-521 Fields for Register: VR_MII_MP_25G_TX_DCC_CTRL

Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:0	TX0_DCC	R/W	Tx Internal Clock Duty Cycle Correction Control Range for lane 0. **Value After Reset:** "SNPS_RSVDPARAM_36 ? 0xC : 0xA" **Exists:** Always

5.2.9.61 VR_MII_MP_25G_TX_WIDTH_CTRL

- **Description:** VR MII Synopsys Multi-Protocol 25G PHY Tx Width Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8046 : (APB3_16) ? (0x8046*0x2) : (0x8046*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&((E25&&!SNPS_RSVDPARAM_37)||E32)

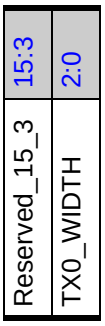
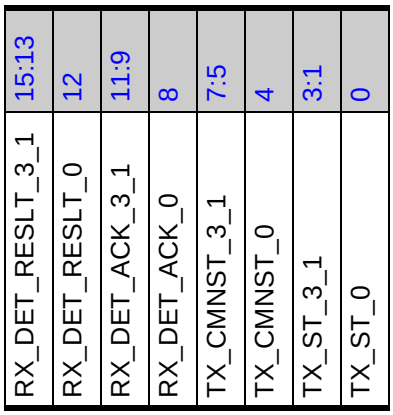


Table 5-522 Fields for Register: VR_MII_MP_25G_TX_WIDTH_CTRL

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	TX0_WIDTH	R/W	Tx PHY Width -lane 0. This field drives the xpcs_tx0_width_o output port and controls the width of PHY Tx lane 0. **Value After Reset:** 0x1 **Exists:** Always

5.2.9.62 VR_MII_MP_6G_TX_STS

- **Description:** VR MII Synopsys Multi-protocol 6G PHY Tx Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8048 : (APB3_16) ? (0x8048*0x2) : (0x8048*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&SSP)



15:13	12	11:9	8	7:5	4	3:1	0
RX_DET_RESLT_3_1	RX_DET_RESLT_0	RX_DET_ACK_3_1	RX_DET_ACK_0	TX_CMNST_3_1	TX_CMNST_0	TX_ST_3_1	TX_ST_0

Table 5-523 Fields for Register: VR_MII_MP_6G_TX_STS

Bits	Name	Memory Access	Description
15:13	RX_DET_RESLT_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	RX_DET_RESLT_0	R	Transmit Receiver Detect Result on Lane 0 When set to 1, this field indicates that a receiver is detected on Lane 0. This signal is valid when the corresponding TX_DET_RX_ACK[] signal is read as 1. **Value After Reset:** 0x0 **Exists:** Always
11:9	RX_DET_ACK_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	RX_DET_ACK_0	R	Transmit Receiver Detect Acknowledgment Signal When set to 1, this field indicates that the receiver detection operation, requested by setting the TX_DET_RX_REQ[] signal, is complete on Lane 0. This is de-asserted only when the corresponding TX_DET_RX_REQ[] signal is set to 0. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
7:5	TX_CMNST_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_CMNST_0	R	Transmit Common Mode Control State on Lane 0 When set to 1, this field indicates that the Tx common mode has reached the intended level on Lane 0. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
3:1	TX_ST_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	TX_ST_0	R	Transmit Active Status of the PHY on Lane 0 When set to 1, this field indicates that the PHY Tx has reached the intended level on Lane 0 and is ready to transmit the data. **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable

5.2.9.63 VR_MII_MP_12G_16G_25G_RX_GENCTRL0

- **Description:** VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Rx General Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8050 : (APB3_16) ? (0x8050*0x2) : (0x8050*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16||E25||E32)

15:13	12	11:9	8	7:5	4	3:1	0
RX_CLKSFT_3_1	RX_CLKSFT_0	RX_DT_EN_3_1	RX_DT_EN_0	RX_ALIGN_EN_3_1	RX_ALIGN_EN_0	RX_TERM_EN_3_1	RX_TERM_EN_0

Table 5-524 Fields for Register: VR_MII_MP_12G_16G_25G_RX_GENCTRL0

Bits	Name	Memory Access	Description
15:13	RX_CLKSFT_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	RX_CLKSFT_0	* Varies	Rx clock shift on lane 0 of Multi-protocol 12G PHY When this bit is set, a 1-bit shift of receive data happens relate to receive clock. This operation works only if alignment enable is disabled. This bit drives the output port 'xpcs_rx_clk_shift_o[0]'. This field is reserved for multi-protocol 16G/25G PHY configurations. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "E12 ? \"read-write\" : \"read-only\""
11:9	RX_DT_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
8	RX_DT_EN_0	R/W	Rx Data Enable on lane 0 of PHY Set this bit to enable the PHY receiver data output on lane 0. This bit drives the output port 'xgxs_rx_data_en_o[0]'. **Value After Reset:** 0x1 **Exists:** Always
7:5	RX_ALIGN_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	RX_ALIGN_EN_0	* Varies	Rx Data Alignment Enable on lane 0 of Multi-Protocol 12G PHY This bit can be set to enable word alignment (based on k28.5 character) in the PHY. This field drives the output port 'xgxs_rx_align_en_o[0]'. This field is reserved for multi-protocol 16G/25G PHY configurations. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "E12 ? \"read-write\" : \"read-only\""
3:1	RX_TERM_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	RX_TERM_EN_0	R/W	Rx Termination Enable on lane 0 of PHY When this bit is set, PHY Rx is terminated with a nominal 50 ohm resistance. Otherwise, the termination is in high impedance. This field drives the output port 'xpcs_rx_term_en_o[0]'. **Value After Reset:** 0x1 **Exists:** Always

5.2.9.64 VR_MII_MP_12G_16G_25G_RX_GENCTRL1

- **Description:** VR MII Synopsys Multi-protocol 12G/16G/25G PHY Rx General Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8051 : (APB3_16) ? (0x8051*0x2) : (0x8051*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16||E25||E32)

15:13	12	11:9	8	7:5	4	3:1	0
RX_DIV16P5_CLK_EN_3_1	RX_DIV16P5_CLK_EN_0	RX_TERM_ACDC_3_1	RX_TERM_ACDC_0	RX_RST_3_1	RX_RST_0	RX_INV_3_1	RX_INV_0

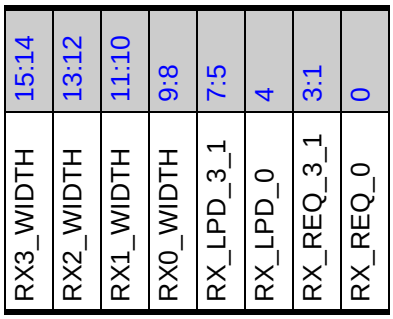
Table 5-525 Fields for Register: VR_MII_MP_12G_16G_25G_RX_GENCTRL1

Bits	Name	Memory Access	Description
15:13	RX_DIV16P5_CLK_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	RX_DIV16P5_CLK_EN_0	* Varies	**For Multi-protocol 12G,16G,25G configurations:** Receiver Divide by 16.5 output clock on lane 0 When this bit is set, it enables the divide by 16.5 Rx VCO recovered clock. This clock is available at rx0_div16p5_clk output port of PHY. This bit must be set when operating in PCS-R mode. **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!(SNPS_RSVDPARAM_35||(SNPS_RSVDPARAM_22&&SNPS_RSVDPARAM_2)||SNPS_RSVDPARAM_37||E32) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
11:9	RX_TERM_ACDC_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	RX_TERM_ACDC_0	* Varies	**For E32 Configurations:** Reserved **For non E32 Configurations:** Rx Termination control on lane 0 of PHY. ■ 0: DC Termination (Floating Rx) ■ 1: AC Termination (Grounded Rx) This field drives the output port xpcs_rx_term_acdc_o[0]. **Value After Reset:** "E32 ? 0x0 : 0x1" **Exists:** Always **Memory Access:** "E32 ? \"read-only\" : \"read-write\"""
7:5	RX_RST_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	RX_RST_0	R/W	Rx reset on lane 0 of PHY When this bit is set, RX data path, all the receiver settings and state machines of the PHY are reset This field drives the output port xgxs_rx_reset_o[0] when DWC_xpcs is in POWER_GOOD state. **Value After Reset:** 0x0 **Exists:** Always
3:1	RX_INV_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	RX_INV_0	R/W	Rx Data Invert on lane 0 of PHY When this bit is set, the data on PHY Rx serial lines are logically inverted. This signal drives the output port xgxs_rx_invert_o[0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.65 VR_MII_MP_12G_16G_RX_GENCTRL2

- **Description:** VR MII Synopsys Multi-protocol 12G/16G PHY Rx General Control 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8052 : (APB3_16) ? (0x8052*0x2) : (0x8052*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16))



15:14	13:12	11:10	9:8	7:5	4	3:1	0
RX3_WIDTH	RX2_WIDTH	RX1_WIDTH	RX0_WIDTH	RX_LPD_3_1	RX_LPD_0	RX_REQ_3_1	RX_REQ_0

Table 5-526 Fields for Register: VR_MII_MP_12G_16G_RX_GENCTRL2

Bits	Name	Memory Access	Description
15:14	RX3_WIDTH	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:12	RX2_WIDTH	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:10	RX1_WIDTH	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9:8	RX0_WIDTH	R/W	Rx Datapath Width on lane 0 of the PHY This field controls the width of output receive data from PHY on lane 3. The encoding of the width is as follows: ■ 2'b00: 8-bit ■ 2'b01: 10-bit ■ 2'b10: 16-bit ■ 2'b11: 20-bit This field drives the output port xpcs_rx0_width[1:0]. **Value After Reset:** 0x1 **Exists:** Always

Bits	Name	Memory Access	Description
7:5	RX_LPD_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	RX_LPD_0	R/W	Receiver Lane Power Down on lane 0 of PHY This bit can be set to power down the receiver to a power state equivalent to that of P1. This bit drives the output port 'xpcs_rx_lpd_o[0]'. **Value After Reset:** 0x0 **Exists:** Always
3:1	RX_REQ_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	RX_REQ_0	R/W	Receiver operation request on lane 0 of PHY (RW,SC Type). This bit can be set to 1 by the application. This bit is self-cleared when 'xpcs_tx_ack_i[0]' is asserted. When this bit is set, a new receiver setting request is made towards the PHY.This bit drives the output port xpcs_rx_req_o[0]. When this bit is set, PHY captures its following input signals: ■ rx0_pstate[1:0] ■ rx0_lpd ■ rx0_rate[1:0] ■ rx0_width[1:0] ■ rx0_data_en ■ rx0_adapt_afe_en ■ rx0_adapt_dfe_en ■ rx0_eq_att_lvl[2:0] ■ rx0_eq_vga1_gain[3:0] ■ rx0_eq_vga2_gain[3:0] ■ rx0_eq_ctle_pole[2:0] ■ rx0_eq_ctle_boost[4:0] ■ rx0_eq_dfe_tap1[7:0] A successful completion of receiver request operation on lane 0 is indicated by bit RX_ACK_0. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.66 VR_MII_MP_25G_RX_GENCTRL2

- **Description:** VR XS or PMA Synopsys Multi-protocol 25G PHY Rx General Control 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8052 : (APB3_16) ? (0x8052*0x2) : (0x8052*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&(E25||E32)

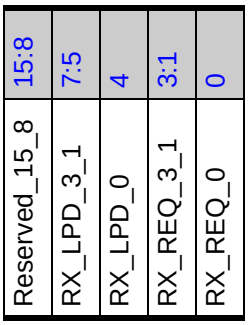


Table 5-527 Fields for Register: VR_MII_MP_25G_RX_GENCTRL2

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	RX_LPD_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	RX_LPD_0	R/W	Receiver Lane Power Down on lane 0 of PHY This bit can be set to power down the receiver to a power state equivalent to that of P1. This bit drives the output port 'xpcs_rx_lpd_o[0]'. **Value After Reset:** 0x0 **Exists:** Always
3:1	RX_REQ_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	RX_REQ_0	R/W	Receiver operation request on lane 0 of PHY (RW,SC Type). When this bit is set, a new receiver setting request is made towards the PHY.This bit drives the output port xpcs_rx_req_o[0]. A successful completion of receiver request operation on lane 0 is indicated by bit RX_ACK_0. This bit is self-cleared when 'xpcs_tx_ack_i[0]' is asserted. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.67 VR_MII_MP_12G_16G_RX_GENCTRL3

- **Description:** VR MII Synopsys Multi-protocol 12G/16G PHY Rx General Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8053 : (APB3_16) ? (0x8053*0x2) : (0x8053*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16)&&!SNPS_RSVDPARAM_22)

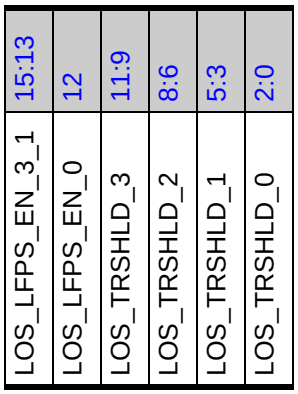


Table 5-528 Fields for Register: VR_MII_MP_12G_16G_RX_GENCTRL3

Bits	Name	Memory Access	Description
15:13	LOS_LFPS_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	LOS_LFPS_EN_0	R/W	Rx LOS LFPS Enable on lane 0 of the PHY This field drives the output port xpcs_rx_los_lfps_en_o[0] to enable the LFPS filter on lane 0 of the PHY. **Value After Reset:** 0x0 **Exists:** Always
11:9	LOS_TRSHLD_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8:6	LOS_TRSHLD_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:3	LOS_TRSHLD_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
2:0	LOS_TRSHLD_0	R/W	Loss of signal threshold on lane 0 of PHY. This field drives the output port xpcs_rx0_los_threshold_o[2:0]. Threshold voltages for various values are as follows: ■ 3'b000: Reserved ■ 3'b001: 90 mVpp ■ 3'b010: 120 mVpp ■ 3'b011: 150 mVpp ■ 3'b100: 180 mVpp ■ 3'b101: 210 mVpp ■ 3'b110: 240 mVpp ■ 3'b111: 270 mVpp **Value After Reset:** "(E16&&SNPS_RSVDPARAM_15) ? 0x2 : E16 ? 0x1 : SNPS_RSVDPARAM_39 ? 0x5 : 0x3 " **Exists:** Always

5.2.9.68 VR_MII_MP_25G_RX_GENCTRL3

- **Description:** VR MII Synopsys Multi-protocol 25G PHY Rx General Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8053 : (APB3_16) ? (0x8053*0x2) : (0x8053*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&E25&&!SNPS_RSVDPARAM_37

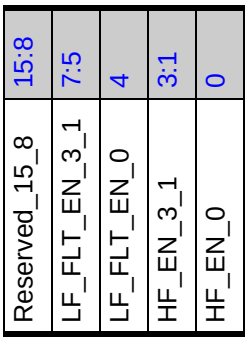


Table 5-529 Fields for Register: VR_MII_MP_25G_RX_GENCTRL3

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	LF_FLT_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	LF_FLT_EN_0	R/W	Receiver LFPS Signal Detection Filter Enable on lane 0 This field drives the xpcs_rx_sigdet_lf_flt_en_o[0] port. **Value After Reset:** 0x0 **Exists:** Always
3:1	HF_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	HF_EN_0	R/W	High frequency Signal Detect Enable on lane 0 This field drives the xpcs_rx_sigdet_hf_en_o[0] port. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.69 VR_MII_Consumer_10G_RX_GENCTRL3

- **Description:** VR MII Synopsys Consumer 10G PHY Rx General Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8053 : (APB3_16) ? (0x8053*0x2) : (0x8053*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** ((BASE1000X_ONLY)&&SNPS_XAUI_PHY&&(SNPS_RSVDPARAM_22))

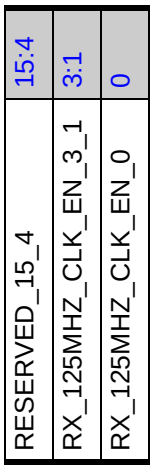


Table 5-530 Fields for Register: VR_MII_Consumer_10G_RX_GENCTRL3

Bits	Name	Memory Access	Description
15:4	RESERVED_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:1	RX_125MHZ_CLK_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	RX_125MHZ_CLK_EN_0	* Varies	**For SNPS_RSVDPARAM_35 Gen2 PHY without USXGMII support** Reserved **For SNPS_RSVDPARAM_35 Gen 2 PHY with USXGMII support** Rx 125Mhz Clock Generation Enable on Lane 1. When this bit is set 125Mhz recovered clock is generated by PHY.This bit drives the output port xpcs_rx_125mhz_clk_en_o. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!SNPS_RSVDPARAM_2 ? \"read-write\" : \"read-only\""

5.2.9.70 VR_MII_MP_12G_16G_25G_RX_RATE_CTRL

- **Description:** VR MII Synopsys Multi-protocol 12G/16G/25G PHY Rx Rate Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8054 : (APB3_16) ? (0x8054*0x2) : (0x8054*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16||E25)&&!SNPS_RSVDPARAM_37)

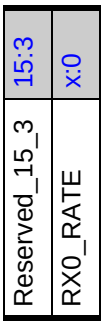


Table 5-531 Fields for Register: VR_MII_MP_12G_16G_25G_RX_RATE_CTRL

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
x:0	RX0_RATE	R/W	Rx date rate on lane 0 of the PHY Data Rate Encoding for Multi-protocol 25G PHY is as follows: ■ 3'b000: baud ■ 3'b001: baud/2 ■ 3'b010: baud/4 ■ 3'b011: baud/8 ■ 3'b100: baud/16 Data Rate Encoding for Multi-protocol 12G/16G PHY is as follows: ■ 2'b00: baud ■ 2'b01: baud/2 ■ 2'b10: baud/4 ■ 2'b11: baud/8 **Value After Reset:** "E25 ? 0x4 : SNPS_RSVDPARAM_39 ? 0x2 : 0x3" **Exists:** Always **Range Variable[x]:** "E25 ? 3 : 2" - 1

5.2.9.71 VR_MII_MP_32G_RX_MRGN_CTRL4

- **Description:** VR MII Synopsys Multi protocol E32 PHY Rx Margin Control 4 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8054 : (APB3_16) ? (0x8054*0x2) : (0x8054*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(SNPS_RSVDPARAM_37||E32))

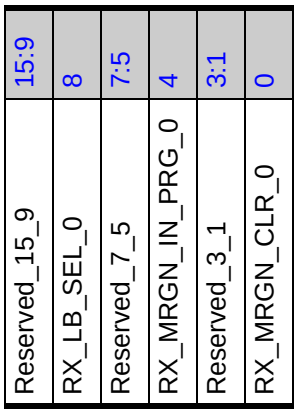


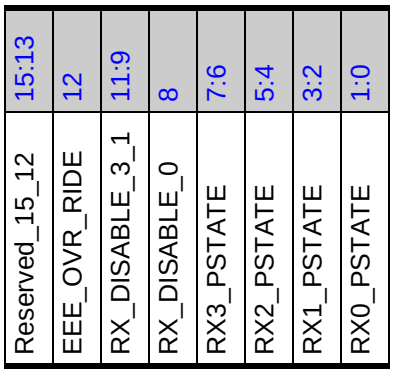
Table 5-532 Fields for Register: VR_MII_MP_32G_RX_MRGN_CTRL4

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	RX_LB_SEL_0	* Varies	**SNPS_RSVDPARAM_37 configurations:** RX Loopback path select on lane 0 of the PHY This signal is connected to xpcs_rx_lb_sel_o[0] output port. **E32 configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SNPS_RSVDPARAM_37 ? \"read-write\" : \"read-only\""
7:5	Reserved_7_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4	RX_MRGN_IN_PRG_0	R/W	Receiver margining in progress on lane 0 of the PHY This signal is connected to xpcs_rx_margin_in_prg_o[0] output port. **Value After Reset:** 0x0 **Exists:** Always
3:1	Reserved_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	RX_MRGN_CLR_0	R/W	Clear signal for RX margining on lane 0 of the PHY This signal is connected to xpcs_rx_margin_clr_o[0] output port. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.72 VR_MII_MP_12G_16G_25G_RX_POWER_STATE_CTRL

- **Description:** VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Rx Power State Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8055 : (APB3_16) ? (0x8055*0x2) : (0x8055*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16||E25||E32))



15:13	12	11:9	8	7:6	5:4	3:2	1:0
Reserved_15_12	EEE_OVR_RIDE	RX_DISABLE_3_1	RX_DISABLE_0	RX3_PSTATE	RX2_PSTATE	RX1_PSTATE	RX0_PSTATE

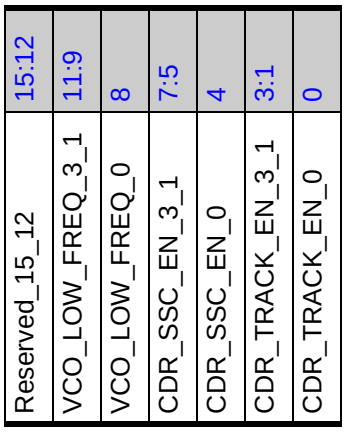
Table 5-533 Fields for Register: VR_MII_MP_12G_16G_25G_RX_POWER_STATE_CTRL

Bits	Name	Memory Access	Description
15:13	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	EEE_OVR_RIDE	* Varies	For configurations with EEE: Rx Power state override control during EEE DWC_xpcs switches Rx power state to P0s by itself during Rx QUIET state of EEE to save power in the PHY. Set this bit to 1, to override this behavior. When this bit is set, Rx power state remains unchanged during Rx EEE and retains the values as programmed in the 'RX{lane}_PSTATE' fields of this register. For configurations without EEE: Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(EEE_EN) ? \"read-write\" : \"read-only\" "
11:9	RX_DISABLE_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
8	RX_DISABLE_0	R/W	Receiver Disable on lane 0 This bit can be set in P1 power state to put the receiver in a low-power mode. This field drives the output port 'xpcs_rx_disable_o[0]'. **Value After Reset:** 0x0 **Exists:** Always
7:6	RX3_PSTATE	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5:4	RX2_PSTATE	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:2	RX1_PSTATE	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1:0	RX0_PSTATE	R/W	Rx power state control for lane 0 of PHY Power state encoding is as follows: ■ 2'b00: P0 ■ 2'b01: P0s ■ 2'b10: P1 ■ 2'b11: P2 **Value After Reset:** 0x0 **Exists:** Always

5.2.9.73 VR_MII_MP_12G_16G_25G_RX_CDR_CTRL

- **Description:** VR MII Synopsys Multi-protocol 12G/16G PHY/25G Rx CDR Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8056 : (APB3_16) ? (0x8056*0x2) : (0x8056*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16||E25||E32))



15:12	11:9	8	7:5	4	3:1	0
Reserved_15_12	VCO_LOW_FREQ_3_1	VCO_LOW_FREQ_0	CDR_SSC_EN_3_1	CDR_SSC_EN_0	CDR_TRACK_EN_3_1	CDR_TRACK_EN_0

Table 5-534 Fields for Register: VR_MII_MP_12G_16G_25G_RX_CDR_CTRL

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:9	VCO_LOW_FREQ_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	VCO_LOW_FREQ_0	* Varies	**For Multi-Protocol 12G PHY configurations:** Rx VCO lower frequency band mode on lane 0 of the PHY This field controls the frequency of the Rx VCO to a lower-frequency operating band. This field drives the output port xpcs_rx_cdr_vco_lowfreq_o[0]. **For other configurations:** Reserved **Value After Reset:** "(E16||SNPS_RSVDPARAM_35||E25||SNPS_RSVDPARAM_39||SNPS_RSVDPARAM_22||E32) ? 0x0 : 0x1" **Exists:** Always **Memory Access:** "E12 ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
7:5	CDR_SSC_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	CDR_SSC_EN_0	R/W	Rx CDR SSC Mode Enable on lane 0 of the PHY This field controls the CDR tracking gains and duration. Set this bit to 1 when receive data has a spread spectrum clock. Clear this bit if receive data does not have SSC. This bit drives the output port 'xpcs_rx_cdr_ssc_en_o[0]'. **Value After Reset:** 0x0 **Exists:** Always
3:1	CDR_TRACK_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	CDR_TRACK_EN_0	* Varies	**For Multi-Protocol 12G PHY configurations:** Rx CDR Tracking Enable on lane 0 of the PHY Set this bit to enable CDR tracking of receive data on lane 0 of the PHY. This bit drives the output port 'xpcs_rx_cdr_track_en_o[0]'. **For other configurations:** Reserved **Value After Reset:** "E12 ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "E12 ? \"read-write\" : \"read-only\""

5.2.9.74 VR_MII_MP_12G_16G_25G_RX_ATTN_CTRL

- **Description:** VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Rx Attenuation Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8057 : (APB3_16) ? (0x8057*0x2) : (0x8057*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16||E25)&&!SNPS_RSVDPARAM_37

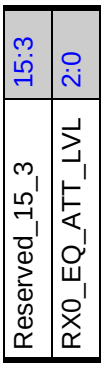


Table 5-535 Fields for Register: VR_MII_MP_12G_16G_25G_RX_ATTN_CTRL

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	RX0_EQ_ATT_LVL	R/W	Rx Equalization Attenuation level for lane 0 of the PHY This field drives the output port xpcs_rx0_eq_att_lvl_o[2:0]. This field controls the AFE attenuation level of the PHY. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.75 VR_MII_MP_6G_RXGENCTRL0

- **Description:** VR MII Synopsys Multi-protocol 6G PHY Rx General Control Register 0
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8058 : (APB3_16) ? (0x8058*0x2) : (0x8058*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&SSP)

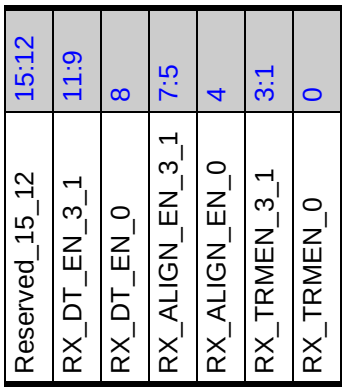


Table 5-536 Fields for Register: VR_MII_MP_6G_RXGENCTRL0

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:9	RX_DT_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	RX_DT_EN_0	R/W	Receiver Clock and Data Output Enable on Lane 0 This bit enables receiver clock and data output on Lane 0. **Value After Reset:** 0x1 **Exists:** Always
7:5	RX_ALIGN_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	RX_ALIGN_EN_0	R/W	Receiver Alignment Enable on Lane 0 When set to 1, this field enables the 10-bit word alignment on Lane 0 based on the COMMA character. You can set this field only when the corresponding RX_DATA_EN[] signal is 0. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
3:1	RX_TRMEN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	RX_TRMEN_0	R/W	Receiver Termination Enable on Lane 0 When set to 1, this bit enables the 100 Ohm termination on rx0_p and rx0_m. Change this field only when the corresponding RX_DATA_EN[] signal is 0. **Value After Reset:** 0x1 **Exists:** Always

5.2.9.76 VR_MII_MP_12G_RX_EQ_CTRL0

- **Description:** VR MII Synopsys Multi-Protocol 12G PHY Rx Equalization Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8058 : (APB3_16) ? (0x8058*0x2) : (0x8058*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&E12)

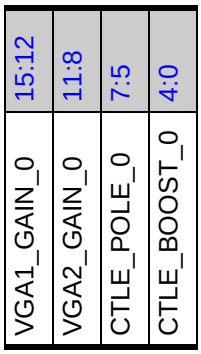


Table 5-537 Fields for Register: VR_MII_MP_12G_RX_EQ_CTRL0

Bits	Name	Memory Access	Description
15:12	VGA1_GAIN_0	R/W	Rx Equalization VGA1 Gain on lane 0 of the PHY This field drives the output port xpcs_rx0_eq_vga1_gain_o[3:0]. This field controls the AFE first stage Variable Gain amplifier gain. **Value After Reset:** "SNPS_RSVDPARAM_39 ? 0x3 : 0x7" **Exists:** !(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_22)
11:8	VGA2_GAIN_0	R/W	Rx Equalization VGA2 Gain on lane 0 of the PHY This field drives the output port xpcs_rx0_eq_vga2_gain_o[3:0].This field controls the AFE second stage Variable Gain amplifier gain. **Value After Reset:** "SNPS_RSVDPARAM_39 ? 0x3 : 0x7" **Exists:** !(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_22)
7:5	CTLE_POLE_0	R/W	Rx Equalization CTLE Pole Value on lane 0 of the PHY This field drives the output port xpcs_rx0_eq_ctle_pole_o[2:0].This field controls the continuous time linear equalizer boost pole location of the PHY. **Value After Reset:** "(SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_39) ? 0x0 : 0x5" **Exists:** !(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_22)

Bits	Name	Memory Access	Description
4:0	CTLE_BOOST_0	R/W	Rx Equalization CTLE Boost value on lane 0 of the PHY This field drives the output port xpcs_rx0_eq_ctle_boost_o[4:0].This field controls the CTLE boost level. **Value After Reset:** "(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_22) ? 0x7 : 0x6" **Exists:** Always

5.2.9.77 VR_MII_MP_16G_25G_RX_EQ_CTRL0

- **Description:** VR MII Synopsys Multi-Protocol 16G/25G PHY Rx Equalization Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8058 : (APB3_16) ? (0x8058*0x2) : (0x8058*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E16||E25)&&!SNPS_RSVDPARAM_37

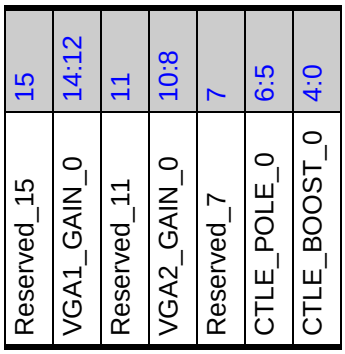


Table 5-538 Fields for Register: VR_MII_MP_16G_25G_RX_EQ_CTRL0

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:12	VGA1_GAIN_0	R/W	Rx Equalization VGA1 Gain on lane 0 of the PHY This field drives the output port xpcs_rx0_eq_vga1_gain_o[3:0]. This field controls the AFE first stage Variable Gain amplifier gain. **Value After Reset:** 0x4 **Exists:** Always
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
10:8	VGA2_GAIN_0	R/W	Rx Equalization VGA2 Gain on lane 0 of the PHY This field drives the output port xpcs_rx0_eq_vga2_gain_o[3:0].This field controls the AFE second stage Variable Gain amplifier gain. **Value After Reset:** 0x4 **Exists:** Always
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
6:5	CTLE_POLE_0	R/W	Rx Equalization CTLE Pole Value on lane 0 of the PHY This field drives the output port xpcs_rx0_eq_ctle_pole_o[2:0].This field controls the continuous time linear equalizer boost pole location of the PHY. **Value After Reset:** 0x0 **Exists:** Always
4:0	CTLE_BOOST_0	R/W	Rx Equalization CTLE Boost value on lane 0 of the PHY This field drives the output port xpcs_rx0_eq_ctle_boost_o[4:0].This field controls the CTLE boost level. **Value After Reset:** "SNPS_RSVDPARAM_36 ? 0xE : (E25||E16&&SNPS_RSVDPARAM_15) ? 0x6 : 0x12" **Exists:** Always

5.2.9.78 VR_MII_MP_32G_RX_MRGN_CTRL0

- **Description:** VR MII Synopsys Multi-Protocol E32 PHY Rx Margin Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8058 : (APB3_16) ? (0x8058*0x2) : (0x8058*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&(SNPS_RSVDPARAM_37||E32)

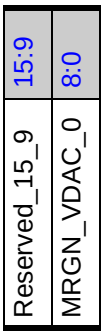
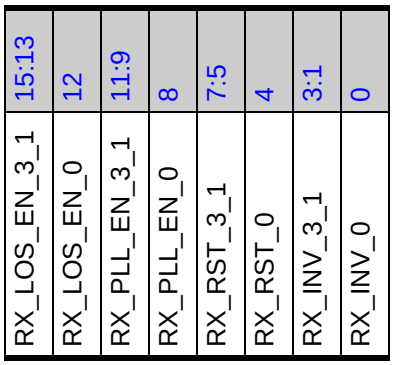


Table 5-539 Fields for Register: VR_MII_MP_32G_RX_MRGN_CTRL0

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8:0	MRGN_VDAC_0	R/W	Value for RX voltage margining along lane 0 of the PHY This field drives the output port xpcs_rx0_margin_vdac_o[8:0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.79 VR_MII_MP_6G_RXGENCTRL1

- **Description:** VR MII Synopsys Multi-protocol 6G Rx General Control Register 1
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8059 : (APB3_16) ? (0x8059*0x2) : (0x8059*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&SSP)



15:13	12	11:9	8	7:5	4	3:1	0
RX_LOS_EN_3_1	RX_LOS_EN_0	RX_PLL_EN_3_1	RX_PLL_EN_0	RX_RST_3_1	RX_RST_0	RX_INV_3_1	RX_INV_0

Table 5-540 Fields for Register: VR_MII_MP_6G_RXGENCTRL1

Bits	Name	Memory Access	Description
15:13	RX_LOS_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	RX_LOS_EN_0	R/W	Receiver Loss-of-Signal Enable on Lane 0 When set to 1, this bit enables the receiver loss-of-signal detection on Lane 0. **Value After Reset:** 0x1 **Exists:** Always
11:9	RX_PLL_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	RX_PLL_EN_0	R/W	Receiver DPLL Enable on Lane 0 When set to 1, this field enables the receiver DPLL on Lane 0. Otherwise, it disables the receiver DPLL. **Value After Reset:** 0x1 **Exists:** Always
7:5	RX_RST_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4	RX_RST_0	R/W	Receiver Reset on Lane 0 When set to 1, this field resets the PHY receiver functionality including RX_DATA_EN[] and RX_PLL_EN[] signals inside the PHY. **Value After Reset:** 0x0 **Exists:** Always
3:1	RX_INV_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	RX_INV_0	R/W	Receiver Invert on Lane 0 When set to 1, this bit inverts the logic levels of rx0_p and rx0_m. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.80 VR_MII_MP_6G_RXEQ_CTRL

- **Description:** VR MII Synopsys Multi-Protocol 6G PHY Rx Equalization Control Register
  All four Rx Equalization fields use the following equation:
  Internal equalizer boost = ~(rx_eq_val+1)*0.5 dB
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x805B : (APB3_16) ? (0x805B*0x2) : (0x805B*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&SSP)

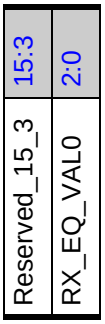


Table 5-541 Fields for Register: VR_MII_MP_6G_RXEQ_CTRL

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	RX_EQ_VAL0	R/W	Rx Equalization Control Lane 0 This field controls the internal linear equalizer boost on Lane 0. **Value After Reset:** 0x2 **Exists:** Always

5.2.9.81 VR_MII_MP_12G_16G_25G_RX_EQ_CTRL4

- **Description:** VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Rx Equalization Control 4 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x805C : (APB3_16) ? (0x805C*0x2) : (0x805C*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16||E25)&&!SNPS_RSVDPARAM_37)

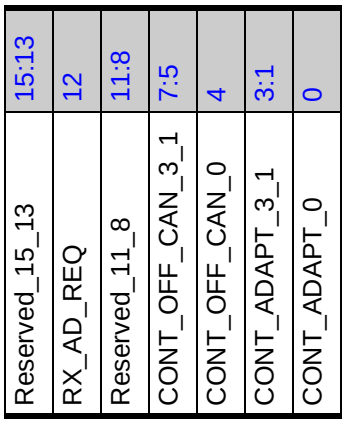


Table 5-542 Fields for Register: VR_MII_MP_12G_16G_25G_RX_EQ_CTRL4

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	RX_AD_REQ	R/W	Receive Adaptation Request This bit drives the 'rpcs_krx_adapt_req_o' port.Software can set this bit to make an Receiver Adaptation Request to the PHY.The acknowledgment from the PHY (for the adaptation request made) is reflected in bit[12] (RX_ADPT_ACK) of VR_MII_MP_12G_16G_25G_MISC_STS Register.After receiving the acknowledgment, software must clear this register field. **Value After Reset:** 0x0 **Exists:** Always
11:8	Reserved_11_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	CONT_OFF_CAN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4	CONT_OFF_CAN_0	R/W	Receiver offset cancellation continuous operation on lane 0 This bit can be set if continuous receiver offset cancellation is required. If this bit is 0, offset cancellation runs when receiver exits P2 power state. This bit drives the output port 'xpcs_rx_offcan_cont_o[0]'. **Value After Reset:** 0x1 **Exists:** Always
3:1	CONT_ADAPT_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	CONT_ADAPT_0	R/W	Receiver Adaptation Continuous Operation on lane 0 This bit can be set to enable continuous receiver adaptation in the PHY. This bit drives the output port 'xpcs_rx_offcan_cont_o'. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.82 VR_MII_MP_32G_RX_EQ_CTRL4

- **Description:** VR MII Synopsys Multi-Protocol E32 PHY Rx Equalization Control 4 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x805C : (APB3_16) ? (0x805C*0x2) : (0x805C*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(SNPS_RSVDPARAM_37||E32))

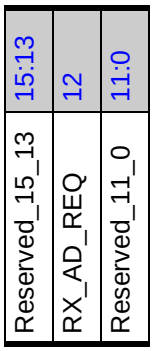


Table 5-543 Fields for Register: VR_MII_MP_32G_RX_EQ_CTRL4

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	RX_AD_REQ	R/W	Receive Adaptation Request This bit drives the 'rpcs_krx_adapt_req_o' port.Software can set this bit to make an Receiver Adaptation Request to the PHY.The acknowledgment from the PHY (for the adaptation request made) is reflected in bit[12] (RX_ADPT_ACK) of VR_MII_MP_12G_16G_25G_MISC_STS Register.After receiving the acknowledgment, software must clear this register field. **Value After Reset:** 0x0 **Exists:** Always
11:0	Reserved_11_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.9.83 VR_MII_MP_6G_RXLOS_CTRL0

- **Description:** VR MII Synopsys Multi-Protocol 6G PHY Rx Loss-of-Signal Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x805D : (APB3_16) ? (0x805D*0x2) : (0x805D*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&SSP)

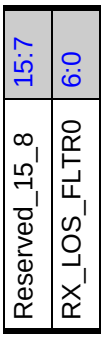


Table 5-544 Fields for Register: VR_MII_MP_6G_RXLOS_CTRL0

Bits	Name	Memory Access	Description
15:7	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:0	RX_LOS_FLTR0	R/W	Loss-of-Signal Filter Value for Lane 0 This field controls the loss-of-signal filter mode on Lane 0 by driving the output port 'xgxs_los0_ctl_o' of DWC_xpcs. **Value After Reset:** 0xe **Exists:** Always

5.2.9.84 VR_MII_MP_12G_AFE_DFE_EN_CTRL

- **Description:** VR MII Synopsys Multi-Protocol 12G PHY AFE-DFE Enable Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x805D : (APB3_16) ? (0x805D*0x2) : (0x805D*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&E12)

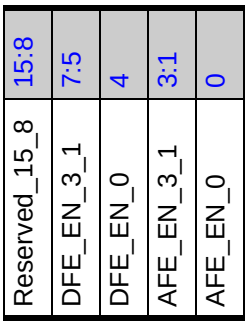


Table 5-545 Fields for Register: VR_MII_MP_12G_AFE_DFE_EN_CTRL

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	DFE_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	DFE_EN_0	R/W	Rx DFE Enable on lane 0 of the PHY This bit drives the output port xpcs_rx_adapt_dfe_en_o[0]. This bit can be set to enable Rx adaption and decision feedback equalization (DFE) circuitry and applies the input setting of DFE Tap1: -rx0_eq_dfe_tap1[7:0]. If this bit is not set, the change on the input DFE settings can be applied using a 'RX_REQ_0/RX_ACK_0' handshake. If this bit is set, DFE settings are initialized to the inputs and RX DFE adaption routines are executed to achieve optimal DFE settings. **Value After Reset:** 0x0 **Exists:** Always
3:1	AFE_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	AFE_EN_0	R/W	Rx Adaptation AFE Enable on lane 0 of the PHY This bit drives the output port xpcs_rx_adapt_afe_en_o[0]. This bit can be set to enable Rx adaption circuitry and applies the following input receiver equalization settings to the PHY: - rx0_eq_att_lvl[2:0] - rx0_eq_vga1_gain[3:0] - rx0_eq_vga2_gain[3:0] - rx0_eq_ctle_pole[2:0] - rx0_eq_ctle_boost[4:0] If this bit is not set, change on input AFE settings can be applied using RX_REQ_0/RX_ACK_0 handshake. If this bit is set, AFE equalization settings are initialized to the PHY inputs, but any further change on these inputs are ignored by PHY and RX AFE adaption routines are executed to achieve optimal AFE settings. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.85 VR_MII_MP_16G_25G_RX_EQ_CTRL5

- **Description:** VR MII Synopsys Multi-Protocol 16G/25G PHY Receive Equalization Control 5 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x805D : (APB3_16) ? (0x805D*0x2) : (0x805D*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E16||E25)&&!SNPS_RSVDPARAM_37

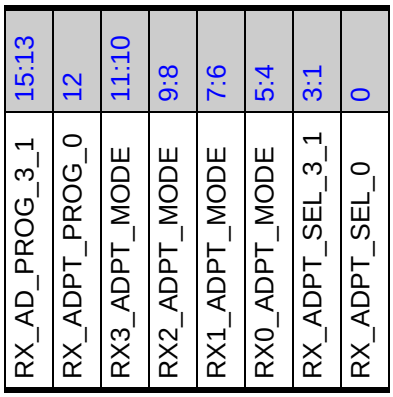


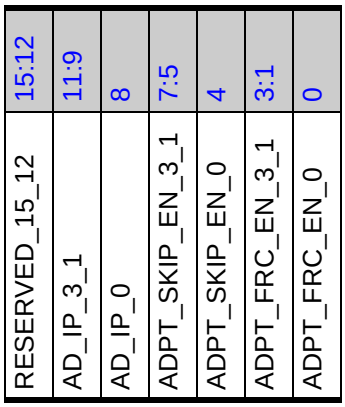
Table 5-546 Fields for Register: VR_MII_MP_16G_25G_RX_EQ_CTRL5

Bits	Name	Memory Access	Description
15:13	RX_AD_PROG_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	RX_ADPT_PROG_0	R/W	Receiver Adaptation in Progress -lane 0 This field drives the output port xpcs_rx_adpt_in_prog_o[0]. **Value After Reset:** 0x0 **Exists:** Always
11:10	RX3_ADPT_MODE	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9:8	RX2_ADPT_MODE	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:6	RX1_ADPT_MODE	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
5:4	RX0_ADPT_MODE	R/W	RX Adaptation for lane0 This field drives the output port xpcs_rx0_adpt_mode_o[1:0]. **Value After Reset:** 0x0 **Exists:** Always
3:1	RX_ADPT_SEL_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	RX_ADPT_SEL_0	R/W	Select Storage Bank for RX adaptation on Lane0 This field drives the output port xpcs_rx_adpt_sel_o[0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.86 VR_MII_MP_32G_RX_ADPT_CTRL

- **Description:** VR XS PMA Synopsys Multi-Protocol 32G PHY Rx Adapt Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x805D : (APB3_16) ? (0x805D*0x2) : (0x805D*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY)&&SNPS_XAUI_PHY&&(E32)



15:12	11:9	8	7:5	4	3:1	0
RESERVED_15_12	AD_IP_3_1	AD_IP_0	ADPT_SKIP_EN_3_1	ADPT_SKIP_EN_0	ADPT_FRC_EN_3_1	ADPT_FRC_EN_0

Table 5-547 Fields for Register: VR_MII_MP_32G_RX_ADPT_CTRL

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:9	AD_IP_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	AD_IP_0	R/W	Rx Adaptation in Progress on lane 0 This field drives the output port xpcs_rx_adpt_in_prog_o[0]. **Value After Reset:** 0x0 **Exists:** Always
7:5	ADPT_SKIP_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	ADPT_SKIP_EN_0	R/W	Rx Coarse adaptation skip enable on lane 0 Enables the skipping of Rx coarse adaptation on lane 0. This field drives the output port xpcs_rx_crse_adpt_skip_en_o. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
3:1	ADPT_FRC_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	ADPT_FRC_EN_0	R/W	Rx Force Coarse adaptation enable on lane 0 Forces Coarse adaptation on lane0. This field drives the output port xpcs_rx_crse_adpt_frc_en_o. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.87 VR_MII_MP_12G_16G_25G_DFE_TAP_CTRL0

- **Description:** VR MII Synopsys Multi-Protocol 12G/16G/25G PHY DFE Tap Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x805E : (APB3_16) ? (0x805E*0x2) : (0x805E*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16||E25)&&!SNPS_RSVDPARAM_37)

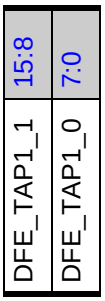
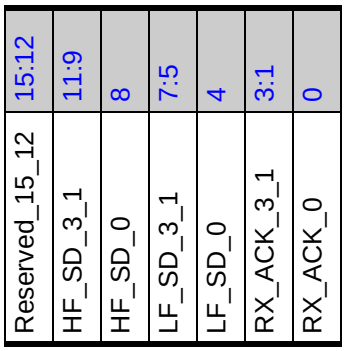


Table 5-548 Fields for Register: VR_MII_MP_12G_16G_25G_DFE_TAP_CTRL0

Bits	Name	Memory Access	Description
15:8	DFE_TAP1_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:0	DFE_TAP1_0	R/W	Rx Equalization DFE Tap1 value on lane 0 of the PHY This field drives the output port xpcs_rx0_eq_dfe_tap1_o[7:0] **Value After Reset:** 0x0 **Exists:** Always

5.2.9.88 VR_MII_MP_12G_16G_25G_RX_STS

- **Description:** VR MII Synopsys Multi-Protocol 12G/16G/25G PHY Rx Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8060 : (APB3_16) ? (0x8060*0x2) : (0x8060*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16||E25||E32))



15:12	11:9	8	7:5	4	3:1	0
Reserved_15_12	HF_SD_3_1	HF_SD_0	LF_SD_3_1	LF_SD_0	RX_ACK_3_1	RX_ACK_0

Table 5-549 Fields for Register: VR_MII_MP_12G_16G_25G_RX_STS

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:9	HF_SD_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	HF_SD_0	R	**Multi-protocol 25G PHY or Multi-protocol 32G PHY or for multi-protocol 10G Gen2 PHY configurations:** High-frequency Signal Detect on lane 0. This field captures the value of the input port 'xpcs_rx_sigdet_hf_i[0]'. >**For Other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	LF_SD_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4	LF_SD_0	R	**For multi-protocol 25G PHY or multi-protocol 32G PHY or Multi-protocol 10G Gen2 PHY configurations:** Low-frequency Signal Detect on lane 0. This field captures the value of the input port 'xpcs_rx_sigdet_lf_i[0]'. **For Other configurations:** Reserved **Value After Reset:** "(SNPS_RSVDPARAM_22||E25||E32) ? 0x1 : 0x0" **Exists:** Always
3:1	RX_ACK_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	RX_ACK_0	R	Rx Acknowledge on lane 0 of PHY This bit captures the value of the input port xpcs_rx_ack_i[0]. If this bit is set, it indicates that the requested receiver setting is complete. This bit forms a hand-shake with 'RX_REQ_0' bit. When this bit is set, RX_REQ_0 bit is self-cleared. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.89 VR_MII_MP_16G_25G_RX_PPM_STS0

- **Description:** VR MII Synopsys Multi-Protocol 16G/25G PHY Rx PPM Status 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8061 : (APB3_16) ? (0x8061*0x2) : (0x8061*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E16||E25||SNPS_RSVDPARAM_39||E32)

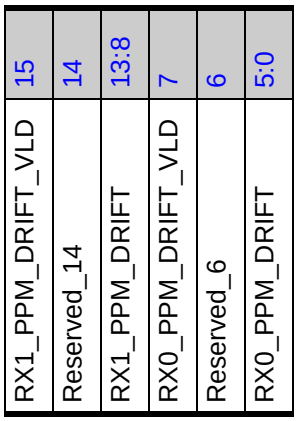


Table 5-550 Fields for Register: VR_MII_MP_16G_25G_RX_PPM_STS0

Bits	Name	Memory Access	Description
15	RX1_PPM_DRIFT_VLD	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14	Reserved_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:8	RX1_PPM_DRIFT	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7	RX0_PPM_DRIFT_VLD	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6	Reserved_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
5:0	RX0_PPM_DRIFT	R	RX CDR PPM Drift on lane0 This field is driven by the input port xpcs_rx0_ppm_drift_i[5:0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.90 VR_MII_MP_25G_RX_TERM_CTRL

- **Description:** VR MII Synopsys Multi-Protocol 25G PHY Receive Termination Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8064 : (APB3_16) ? (0x8064*0x2) : (0x8064*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&E25&&!SNPS_RSVDPARAM_37

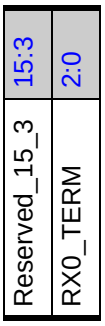
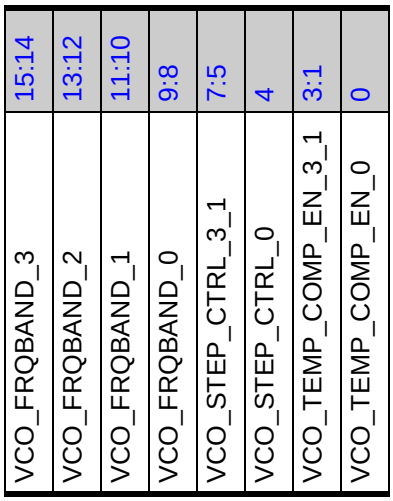


Table 5-551 Fields for Register: VR_MII_MP_25G_RX_TERM_CTRL

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	RX0_TERM	R/W	Receive Termination Control for lane 0 **Value After Reset:** 0x2 **Exists:** Always

5.2.9.91 VR_MII_MP_16G_RX_CDR_CTRL1

- **Description:** VR MII Synopsys Multi-Protocol 16G PHY Rx CDR Control1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8064 : (APB3_16) ? (0x8064*0x2) : (0x8064*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E16||SNPS_RSVDPARAM_39))



15:14	13:12	11:10	9:8	7:5	4	3:1	0
VCO_FRQBAND_3	VCO_FRQBAND_2	VCO_FRQBAND_1	VCO_FRQBAND_0	VCO_STEP_CTRL_3_1	VCO_STEP_CTRL_0	VCO_TEMP_COMP_EN_3_1	VCO_TEMP_COMP_EN_0

Table 5-552 Fields for Register: VR_MII_MP_16G_RX_CDR_CTRL1

Bits	Name	Memory Access	Description
15:14	VCO_FRQBAND_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:12	VCO_FRQBAND_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:10	VCO_FRQBAND_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9:8	VCO_FRQBAND_0	R/W	RX CDR VCO Frequency Band lane0 This field drives the output port xpcs_vco_freqband0_o[1:0]. **Value After Reset:** "E16 ? 0x1 : 0x0" **Exists:** Always
7:5	VCO_STEP_CTRL_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4	VCO_STEP_CTRL_0	* Varies	RX_CDR VCO Step Control Lane 0 This field drives the output port xpcs_vco_step_ctrl_o[0]. **Value After Reset:** "E16 ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(E16) ? \"read-write\" : \"read-only\""
3:1	VCO_TEMP_COMP_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	VCO_TEMP_COMP_EN_0	* Varies	RX_CDR VCO Temperature Compensation Enable Lane 0 This field drives the output port xpcs_vco_temp_comp_en_o[0]. **Value After Reset:** "(E16&&!SNPS_RSVDPARAM_15) ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "(E16) ? \"read-write\" : \"read-only\""

5.2.9.92 VR_MII_MP_16G_25G_RX_PPM_CTRL0

- **Description:** VR MII Synopsys Multi-Protocol 16G/25G PHY Rx PPM Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8065 : (APB3_16) ? (0x8065*0x2) : (0x8065*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37

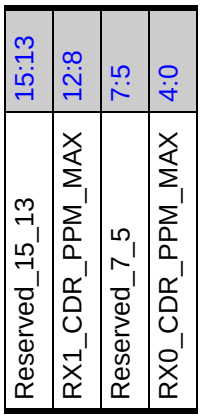


Table 5-553 Fields for Register: VR_MII_MP_16G_25G_RX_PPM_CTRL0

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:8	RX1_CDR_PPM_MAX	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	Reserved_7_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4:0	RX0_CDR_PPM_MAX	R/W	Maximum Allowed PPM on the RX0 CDR Clock This field drives the output port xpcs_rx0_cdr_ppm_max_o[4:0]. **Value After Reset:** "(E16&&SNPS_RSVDPARAM_15) ? 0x13 : SNPS_RSVDPARAM_39 ? 0xD : ((E25&&REF_CLK_78)||(SNPS_RSVDPARAM_36&&SNPS_RSVDPARAM_13)) ? 0x13 : 0x12" **Exists:** Always

5.2.9.93 VR_MII_MP_16G_25G_RX_GENCTRL4

- **Description:** VR MII Synopsys Multi-Protocol 16G/25G PHY Rx General Control 4 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8068 : (APB3_16) ? (0x8068*0x2) : (0x8068*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37

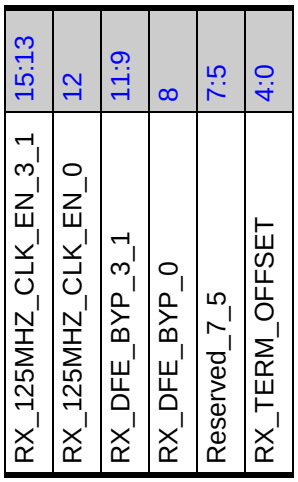


Table 5-554 Fields for Register: VR_MII_MP_16G_25G_RX_GENCTRL4

Bits	Name	Memory Access	Description
15:13	RX_125MHZ_CLK_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	RX_125MHZ_CLK_EN_0	* Varies	RX 125MHz clock generation Enable on Lane 0 This field drives the output port xpcs_rx_125mhz_clk_en_o[0]. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_RSVDPARAM_39) ? \"read-write\" : \"read-only\""
11:9	RX_DFE_BYP_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
8	RX_DFE_BYP_0	* Varies	RX DFE Bypass Enable on Lane o This field drives the output port xpcs_rx_dfe_bypass_o[0]. **Value After Reset:** "SNPS_RSVDPARAM_39 ? 0x0 : 0x1" **Exists:** Always **Memory Access:** "SNPS_RSVDPARAM_39 ? \"read-only\" : \"read-write\""
7:5	Reserved_7_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4:0	RX_TERM_OFFSET	R/W	Offset for RX Termination This field drives the output port xpcs_rx_term_offset_o[4:0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.94 VR_MII_MP_16G_25G_RX_MISC_CTRL0

- **Description:** VR MII Synopsys Multi-Protocol 16G/25G PHY RX Miscellaneous Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8069 : (APB3_16) ? (0x8069*0x2) : (0x8069*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37

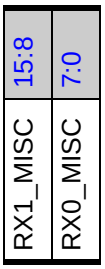


Table 5-555 Fields for Register: VR_MII_MP_16G_25G_RX_MISC_CTRL0

Bits	Name	Memory Access	Description
15:8	RX1_MISC	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:0	RX0_MISC	R/W	RX Miscellaneous control for lane0. This field drives the output port xpcs_rx0_misc_o[7:0]. **Value After Reset:** "(E25) ? 0x0 : (SNPS_RSVDPARAM_39) ? 0x1 : (E16&&SNPS_RSVDPARAM_15) ? 0x47 : 0x16" **Exists:** Always

5.2.9.95 VR_MII_MP_32G_RX_MRGN_STS0

- **Description:** VR MII Synopsys Multi-Protocol E32 PHY RX Margin Status 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8069 : (APB3_16) ? (0x8069*0x2) : (0x8069*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&(SNPS_RSVDPARAM_37||E32)

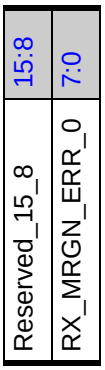


Table 5-556 Fields for Register: VR_MII_MP_32G_RX_MRGN_STS0

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:0	RX_MRGN_ERR_0	R	Error counter for RX margining along Lane 0. This field is driven by xpcs_rx0_margin_err_i[7:0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.96 VR_MII_MP_16G_25G_RX_IQ_CTRL0

- **Description:** VR MII Synopsys Multi-Protocol 16G/25G PHY RX IQ Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x806B : (APB3_16) ? (0x806B*0x2) : (0x806B*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E16||E25||SNPS_RSVDPARAM_39||E32)

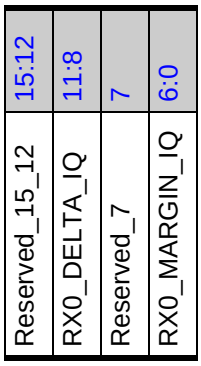


Table 5-557 Fields for Register: VR_MII_MP_16G_25G_RX_IQ_CTRL0

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:8	RX0_DELTA_IQ	* Varies	**Non SNPS_RSVDPARAM_37 and Non E32 configurations:** RX IQ Offset Value for lane0. This field drives the output port xpcs_rx0_delta_iq_o[3:0]. **For SNPS_RSVDPARAM_37/E32 configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(!(SNPS_RSVDPARAM_37||E32)) ? \"read-write\" :\"read-only\""
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
6:0	RX0_MARGIN_IQ	* Varies	Value of RX IQ Margining on lane0. This field drives the output port xpcs_rx0_margin_iq_o[6:0]. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!SNPS_RSVDPARAM_39 ? \"read-write\" : \"read-only\""

5.2.9.97 VR_MII_MP_12G_16G_25G_MPLL_CMN_CTRL

- **Description:** VR MII Synopsys Multi-Protocol 12G/16G/25G PHY MPLL Common Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8070 : (APB3_16) ? (0x8070*0x2) : (0x8070*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16||E25))

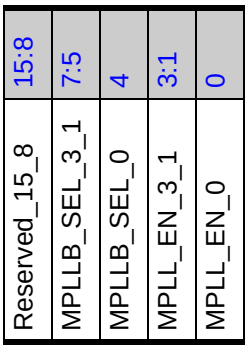


Table 5-558 Fields for Register: VR_MII_MP_12G_16G_25G_MPLL_CMN_CTRL

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:5	MPLLB_SEL_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	MPLLB_SEL_0	* Varies	Tx MPLLB Select-lane 0 When this bit is set, PHY selects MPLLB to generate Tx analog clocks on lane 0 **Value After Reset:** "(E16&&SNPS_RSVDPARAM_15) ? 0x1 : 0x0" **Exists:** (!SNPS_RSVDPARAM_37) **Memory Access:** "SNPS_RSVDPARAM_39||SNPS_RSVDPARAM_37 ? \"read-only\" : \"read-write\"""
3:1	MPLL_EN_3_1	* Varies	Reserved **Value After Reset:** "(!SINGLE_LANE_CFG) ? 0x7 : 0x0" **Exists:** Always **Memory Access:** "(!SINGLE_LANE_CFG) ? \"read-write\" : \"read-only\"""

Bits	Name	Memory Access	Description
0	MPLL_EN_0	R/W	Tx MPLL Enable-lane 0 Set this bit to power-up the MPLL. This bit must be 1, for normal operation. **Value After Reset:** 0x1 **Exists:** Always

5.2.9.98 VR_MII_MP_12G_16G_MPLLA_CTRL0

- **Description:** VR MII Synopsys Multi-Protocol 12G/16G PHY MPLLA Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8071 : (APB3_16) ? (0x8071*0x2) : (0x8071*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16))

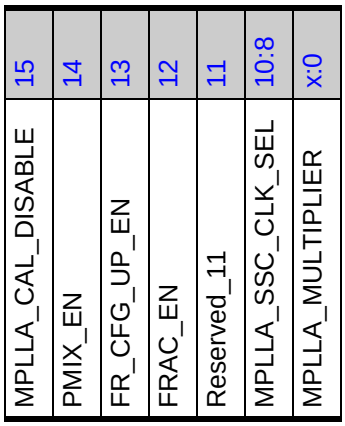


Table 5-559 Fields for Register: VR_MII_MP_12G_16G_MPLLA_CTRL0

Bits	Name	Memory Access	Description
15	MPLLA_CAL_DISABLE	R/W	MPLLA Calibration Disable This field can be programmed to 1, to disable calibration of MPLLA by PHY firmware. **Value After Reset:** 0x0 **Exists:** Always
14	PMIX_EN	* Varies	**For Multi-protocol 10G Gen2 configurations:** MPLLA PMIX Enabled This field drives the output port 'xpcs_mplla_pmix_en_o'. **For Other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SNPS_RSVDPARAM_22 ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
13	FR_CFG_UP_EN	* Varies	**For Multi-protocol 10G Gen2 configurations:** MPLLA Fractional Configuration Latch Signal This field drives the xpcs_mplla_fracn_cfg_updt_en_o port. **For Other configurations:** Reserved **Value After Reset:** "SNPS_RSVDPARAM_22 ? 0x1 : 0x0" **Exists:** Always **Memory Access:** "SNPS_RSVDPARAM_22 ? \"read-write\" : \"read-only\""
12	FRAC_EN	* Varies	**For Multi-protocol 10G Gen2 configurations:** MPLLA Fractional Mode Enable This field drives the xpcs_mplla_fracn_en_o port. **For Other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SNPS_RSVDPARAM_22 ? \"read-write\" : \"read-only\""
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** !SNPS_RSVDPARAM_22
10:8	MPLLA_SSC_CLK_SEL	* Varies	This field controls MPLLA spread-spectrum clock select of Multi-protocol 10G PHY. For other configurations, this is a reserved field. **Value After Reset:** 0x0 **Exists:** !SNPS_RSVDPARAM_22 **Memory Access:** "SNPS_RSVDPARAM_35 ? \"read-write\" : \"read-only\""
x:0	MPLLA_MULTIPLIER	R/W	MPLLA frequency Multiplier Control This field controls the multiplication of reference clock to a frequency suitable for operating speed Any change in this field must be followed by a vendor-specific Soft Reset to ensure that PHY is properly powered-up in the desired mode. **Value After Reset:** "(E16&&SNPS_RSVDPARAM_15) ? 0xc1 : !SNPS_RSVDPARAM_35 ? (SNPS_RSVDPARAM_39&&SNPS_RSVDPARAM_13 ? 0x64 : SNPS_RSVDPARAM_39&&REF_CLK_20 ? 0x7D : 0x20) : SNPS_RSVDPARAM_35&&!SNPS_RSVDPARAM_13 ? 0x50 : SNPS_RSVDPARAM_35&&SNPS_RSVDPARAM_13 ? 0x19 : (RPCS_PRESENT) ? 0x21 : 0x28" **Exists:** Always **Range Variable[x]:** "SNPS_RSVDPARAM_22 ? 12 : 8" - 1

5.2.9.99 VR_MII_MP_25G_MPLLA_CTRL0

- **Description:** VR MII Synopsys Multi-protocol 25G PHY MPLLA Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8071 : (APB3_16) ? (0x8071*0x2) : (0x8071*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&E25&&!SNPS_RSVDPARAM_37

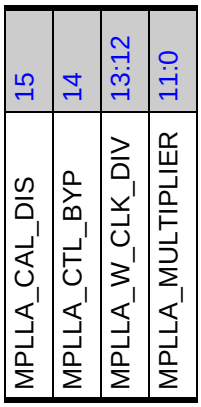


Table 5-560 Fields for Register: VR_MII_MP_25G_MPLLA_CTRL0

Bits	Name	Memory Access	Description
15	MPLLA_CAL_DIS	R/W	MPLLA Calibration Disable This field drives the xpcs_mplla_init_cal_disable_o port. **Value After Reset:** 0x0 **Exists:** Always
14	MPLLA_CTL_BYP	R/W	MPLLA Control Buffer Bypass This field drives the xpcs_mplla_ctl_buf_byp_o port. **Value After Reset:** 0x0 **Exists:** Always
13:12	MPLLA_W_CLK_DIV	R/W	MPLLA Word Clock Divider This field controls the MPLLA word clock dividers for generating mplla_word_clk.This field drives the xpcs_mplla_word_clk_div_o port. **Value After Reset:** "SNPS_RSVDPARAM_36 ? 0x3 : 0x1" **Exists:** Always

Bits	Name	Memory Access	Description
11:0	MPLLA_MULTIPLIER	R/W	MPLLA frequency Multiplier Control This field controls the multiplication of reference clock to a frequency suitable for operating speed. Any change in this field must be followed by a vendor-specific Soft Reset to ensure that PHY is properly powered-up in the desired mode. **Value After Reset:** "SNPS_RSVDPARAM_36 ? 0X64 : 0x80" **Exists:** Always

5.2.9.100 VR_MII_MP_12G_MPLLA_CTRL1

- **Description:** VR MII Synopsys Multi-protocol 12G PHY MPLLA Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8072 : (APB3_16) ? (0x8072*0x2) : (0x8072*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&E12&&!SNPS_RSVDPARAM_39&&!SNPS_RSVDPARAM_22)

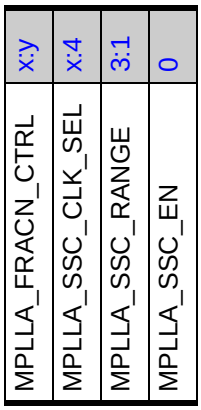


Table 5-561 Fields for Register: VR_MII_MP_12G_MPLLA_CTRL1

Bits	Name	Memory Access	Description
x:y	MPLLA_FRACN_CTRL	R/W	MPLLA Fractional Control This field drives the output port xpcs_mplla_fracn_ctrl_o. **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "SNPS_RSVDPARAM_35 ? 11 : 9" + "SNPS_RSVDPARAM_35 ? 5 : 7" - 1 **Range Variable[y]:** "SNPS_RSVDPARAM_35 ? 5 : 7"
x:4	MPLLA_SSC_CLK_SEL	* Varies	**For Multi-protocol 10G Gen1 configurations:** Reserved **For other configurations:** MPLLA SSC Clock Select This field drives the output port xpcs_mplla_ssc_clk_sel_o. **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "!SNPS_RSVDPARAM_35 ? 3 : 1" + 3 **Memory Access:** "!SNPS_RSVDPARAM_35 ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
3:1	MPLLA_SSC_RANGE	R/W	MPLLA SSC Range This field selects the modulation to insert. This field applies a fixed PPM offset to phase accumulator as follows: ■ 3'b000: -4980 ■ 3'b001: -4492 ■ 3'b010: -4003 ■ 3'b011: -3027 ■ 3'b100: 4980 ■ 3'b101: 4492 ■ 3'b110: 4003 ■ 3'b111: 3027 **Value After Reset:** 0x0 **Exists:** Always
0	MPLLA_SSC_EN	R/W	MPLLA SSC Enable This field can be set to enable spread-spectrum generation on mplla_div_clk output. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.101 VR_MII_MP_16G_MPLLA_CTRL1

- **Description:** VR MII Synopsys Multi-protocol 16G PHY MPLLA Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8072 : (APB3_16) ? (0x8072*0x2) : (0x8072*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E16||SNPS_RSVDPARAM_39))

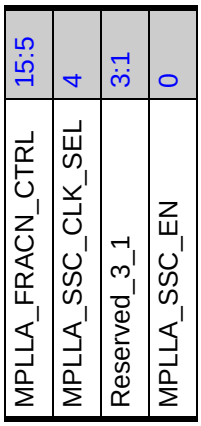
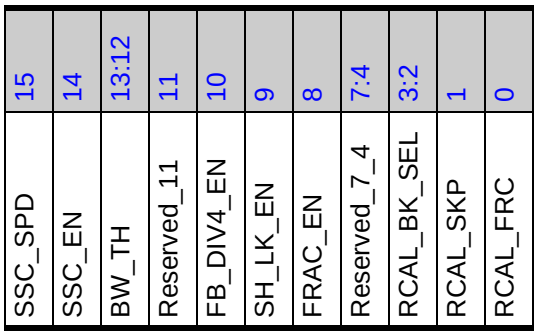


Table 5-562 Fields for Register: VR_MII_MP_16G_MPLLA_CTRL1

Bits	Name	Memory Access	Description
15:5	MPLLA_FRACN_CTRL	R/W	MPLLA Fractional Control This field drives the output port xpcs_mplla_fracn_ctrl_o. **Value After Reset:** "SNPS_RSVDPARAM_15 ? 0x188 : 0x0" **Exists:** Always
4	MPLLA_SSC_CLK_SEL	R/W	MPLLA SSC Clock Select This field drives the output port xpcs_mplla_ssc_clk_sel_o. **Value After Reset:** 0x0 **Exists:** Always
3:1	Reserved_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	MPLLA_SSC_EN	R/W	MPLLA SSC Enable This field can be set to enable spread-spectrum generation on mplla_div_clk output. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.102 VR_MII_MP_25G_MPLLA_CTRL1

- **Description:** VR MII Synopsys Multi-protocol 25G PHY MPLLA Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8072 : (APB3_16) ? (0x8072*0x2) : (0x8072*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&E25&&!SNPS_RSVDPARAM_37



15	14	13:12	11	10	9	8	7:4	3:2	1	0
SSC_SPD	SSC_EN	BW_TH	Reserved_11	FB_DIV4_EN	SH_LK_EN	FRAC_EN	Reserved_7_4	RCAL_BK_SEL	RCAL_SKP	RCAL_FRC

Table 5-563 Fields for Register: VR_MII_MP_25G_MPLLA_CTRL1

Bits	Name	Memory Access	Description
15	SSC_SPD	R/W	MPLLA Up Spread Enable This field drives the xpcs_mplla_ssc_spd_en_o port. **Value After Reset:** 0x0 **Exists:** Always
14	SSC_EN	R/W	MPLLA Spread Spectrum Enable This field drives the xpcs_mplla_ssc_en_o port. **Value After Reset:** 0x0 **Exists:** Always
13:12	BW_TH	R/W	MPLLA Bandwidth Threshold This field drives the xpcs_mplla_bw_th_o port. **Value After Reset:** 0x3 **Exists:** Always
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
10	FB_DIV4_EN	R/W	MPLLA DIV clock output frequency divider control This field drives the xpcs_mplla_fb_clk_div4_en_o port. **Value After Reset:** 0x1 **Exists:** Always

Bits	Name	Memory Access	Description
9	SH_LK_EN	R/W	MPLLA Short Lock Enable This field drives the xpcs_mplla_short_lock_en_o port. **Value After Reset:** 0x0 **Exists:** Always
8	FRAC_EN	R/W	MPLLA Fractional Enable This field drives the xpcs_mplla_fracn_en_o port. **Value After Reset:** 0x0 **Exists:** Always
7:4	Reserved_7_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:2	RCAL_BK_SEL	R/W	MPLLA Recalibration Bank Select This field drives the xpcs_mplla_recal_bank_sel_o port. **Value After Reset:** 0x0 **Exists:** Always
1	RCAL_SKP	R/W	MPLLA Recalibration Skip Enable This field drives the xpcs_mplla_recal_skip_en_o port. **Value After Reset:** 0x0 **Exists:** Always
0	RCAL_FRC	R/W	MPLLA Recalibration Force Enable This drives the xpcs_mplla_recal_force_en_o port. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.103 VR_MII_MP_32G_MPLLA_CTRL1

- **Description:** VR MII Synopsys Multi Protocol 32G PHY MPLLA Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8072 : (APB3_16) ? (0x8072*0x2) : (0x8072*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&(SNPS_RSVDPARAM_37||E32)

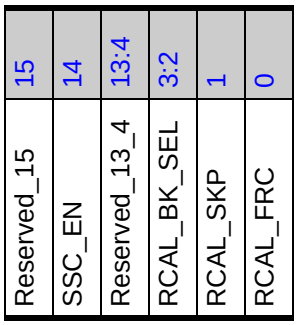


Table 5-564 Fields for Register: VR_MII_MP_32G_MPLLA_CTRL1

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14	SSC_EN	R/W	MPLLA Spread Spectrum Enable This field drives the xpcs_mplla_ssc_en_o port. **Value After Reset:** 0x0 **Exists:** Always
13:4	Reserved_13_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:2	RCAL_BK_SEL	R/W	MPLLA Recalibration Bank Select This field drives the xpcs_mplla_recal_bank_sel_o port. **Value After Reset:** 0x0 **Exists:** Always
1	RCAL_SKP	R/W	MPLLA Recalibration Skip Enable This field drives the xpcs_mplla_recal_skip_en_o port. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	RCAL_FRC	R/W	MPLLA Recalibration Force Enable This drives the xpcs_mplla_recal_force_en_o port. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.104 VR_MII_MP_12G_16G_MPLLA_CTRL2

- **Description:** VR MII Synopsys Multi-protocol 12G/16G PHY MPLLA Control 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8073 : (APB3_16) ? (0x8073*0x2) : (0x8073*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16)&&!SNPS_RSVDPARAM_22)

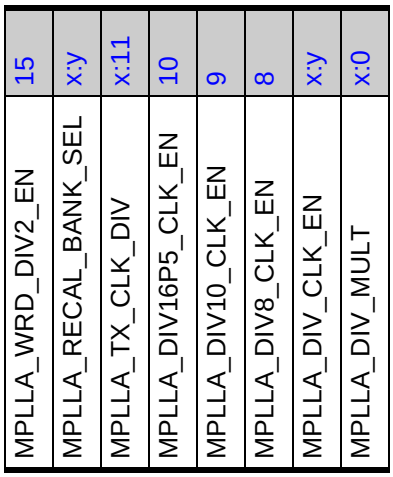


Table 5-565 Fields for Register: VR_MII_MP_12G_16G_MPLLA_CTRL2

Bits	Name	Memory Access	Description
15	MPLLA_WRD_DIV2_EN	* Varies	MPLLA Word Clock Divide by 2 This field controls the xpcs_mplla_word_div2_en_o port. **Note:** This field is valid only for Multi-protocol 8G PHY configurations. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SNPS_RSVDPARAM_39 ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
x:y	MPLLA_RECAL_BANK_SEL	* Varies	**For Multi-protocol 16G configurations:** MPLLB Re-calibration Bank Select This field drives the output port 'xpcs_mplla_recal_bank_sel_o[1:0]'. **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "!(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_39) ? 2 : 1" + "!(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_39) ? 13 : 14" - 1 **Range Variable[y]:** "!(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_39) ? 13 : 14" **Memory Access:** "(E16) ? \"read-write\" :\"read-only\""
x:11	MPLLA_TX_CLK_DIV	* Varies	MPLLA Tx Clock Divider This field drives the output port 'xpcs_mplla_tx_clk_div_o[1:0]'. This field is present only for Multi-protocol 16G configurations and Multi-protocol 10G PHY configurations. **Value After Reset:** "(SNPS_RSVDPARAM_35) ? 0x1 : 0x0" **Exists:** Always **Range Variable[x]:** "SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_39 ? 3 : 2" + 10 **Memory Access:** "(E16||SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_39) ? \"read-write\" :\"read-only\""
10	MPLLA_DIV16P5_CLK_EN	R/W	MPLLA Divide by 16.5 Enable This bit can be set to enable output clocks derived from MPLLA based on 16.5,33 and 66 division ratios. **Value After Reset:** 0x0 **Exists:** !SNPS_RSVDPARAM_35
9	MPLLA_DIV10_CLK_EN	R/W	MPLLA Divide by 10 Enable When this bit is set, the frequency of the mplla_word_clk output clock from PHY is MPLLA frequency divided by 10. **Value After Reset:** 0x1 **Exists:** Always
8	MPLLA_DIV8_CLK_EN	R/W	MPLLA Divide by 8 Enable When this bit is set, the frequency of the mplla_word_clk output clock from PHY is MPLLA frequency divided by 8. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
x:y	MPLLA_DIV_CLK_EN	R/W	Enable mplla_div_clk from PHY. When asserted, the frequency of mplla_div_clk from PHY is the MPLLA frequency divided by 'mplla_div_multiplier' **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "SNPS_RSVDPARAM_35 ? 10 **Range Variable[y]:** 7"
x:0	MPLLA_DIV_MULT	R/W	MPLLA Output Frequency Multiplier Control This field controls the frequency multiplication factor used to generate MPLLA clock output from the reference clock input as seen by the MPLL. **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "SNPS_RSVDPARAM_35 ? 8 : 7" - 1

5.2.9.105 VR_MII_MP_25G_MPLLA_CTRL2

- **Description:** VR MII Synopsys Multi-protocol 25G PHY MPLLA Control 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8073 : (APB3_16) ? (0x8073*0x2) : (0x8073*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&E25&&!SNPS_RSVDPARAM_37

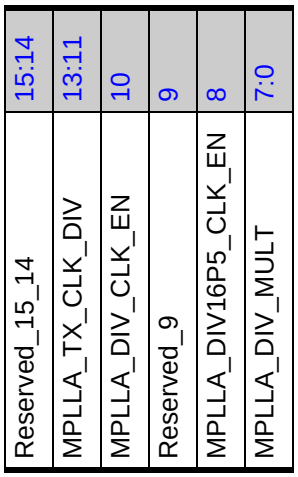


Table 5-566 Fields for Register: VR_MII_MP_25G_MPLLA_CTRL2

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:11	MPLLA_TX_CLK_DIV	R/W	MPLLA Tx Clock Divider This field drives the output port 'xpcs_mplla_tx_clk_div_o[1:0]'. **Value After Reset:** 0x4 **Exists:** Always
10	MPLLA_DIV_CLK_EN	R/W	Enable mplla_div_clk from PHY. When asserted, the frequency of mplla_div_clk from PHY is the MPLLA frequency divided by 'mplla_div_multiplier' **Value After Reset:** 0x0 **Exists:** Always
9	Reserved_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
8	MPLLA_DIV16P5_CLK_EN	R/W	MPLLA Divide by 16.5 Enable This bit can be set to enable output clocks derived from MPLLA based on 16.5,33 and 66 division ratios. **Value After Reset:** 0x0 **Exists:** Always
7:0	MPLLA_DIV_MULT	R/W	MPLLA Output Frequency Multiplier Control This field controls the frequency multiplication factor used to generate MPLLA clock output from the reference clock input as seen by the MPLL.This field drives the xpcs_mplla_div_mult_o port. **Value After Reset:** "(SNPS_RSVDPARAM_36) ? 0x28 : 0x0" **Exists:** Always

5.2.9.106 VR_MII_MP_12G_16G_MPLLB_CTRL0

- **Description:** VR MII Synopsys Multi-protocol 12G/16G PHY MPLLB Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8074 : (APB3_16) ? (0x8074*0x2) : (0x8074*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16)&&!SNPS_RSVDPARAM_39)

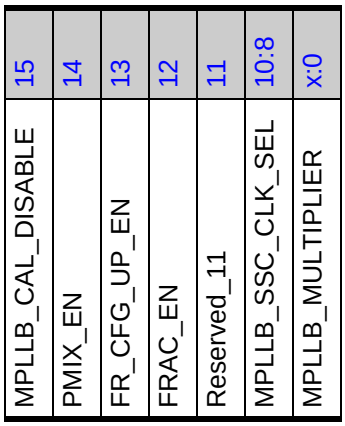


Table 5-567 Fields for Register: VR_MII_MP_12G_16G_MPLLB_CTRL0

Bits	Name	Memory Access	Description
15	MPLLB_CAL_DISABLE	R/W	MPLLB Calibration Disable This field can be programmed to 1, to disable calibration of MPLLB by PHY firmware. **Value After Reset:** "(E16&&SNPS_RSVDPARAM_15) ? 0x0 : 0x1" **Exists:** Always
14	PMIX_EN	* Varies	**For Multi-protocol 10G Gen2 configurations:** MPLLA PMIX Enabled This field drives the output port 'xpcs_mpllb_pmix_en_o'. **For Other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SNPS_RSVDPARAM_22 ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
13	FR_CFG_UP_EN	* Varies	**For Multi-protocol 10G Gen2 configurations:** MPLLB Fractional Configuration Latch Signal This field drives the xpcs_mpllb_fracn_cfg_updt_en_o port. **For Other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SNPS_RSVDPARAM_22 ? \"read-write\" : \"read-only\""
12	FRAC_EN	* Varies	**For Multi-protocol 10G Gen2 configurations:** MPLLB Fractional Mode Enable This field drives the xpcs_mpllb_fracn_en_o port. **For Other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SNPS_RSVDPARAM_22 ? \"read-write\" : \"read-only\""
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** !SNPS_RSVDPARAM_22
10:8	MPLLB_SSC_CLK_SEL	* Varies	This field controls MPLLB spread-spectrum clock select of Multi-protocol 10G PHY. For other configurations, this is a reserved field. **Value After Reset:** 0x0 **Exists:** !SNPS_RSVDPARAM_22 **Memory Access:** "SNPS_RSVDPARAM_35 ? \"read-write\" : \"read-only\""
x:0	MPLLB_MULTIPLIER	R/W	MPLLB frequency Multiplier Control This field controls the multiplication of reference clock to a frequency suitable for operating speed Any change in this field must be followed by a vendor-specific Soft Reset to ensure that PHY is properly powered-up in the desired mode. **Value After Reset:** "(E16&&SNPS_RSVDPARAM_15) ? 0xce : SNPS_RSVDPARAM_35&&!SNPS_RSVDPARAM_13 ? 0x7d : SNPS_RSVDPARAM_35 ? 0x9c : 0x0" **Exists:** Always **Range Variable[x]:** "SNPS_RSVDPARAM_22 ? 12 : 8" - 1

5.2.9.107 VR_MII_MP_25G_MPLLB_CTRL0

- **Description:** VR MII Synopsys Multi-protocol 25G PHY MPLLB Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8074 : (APB3_16) ? (0x8074*0x2) : (0x8074*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&E25&&!SNPS_RSVDPARAM_37

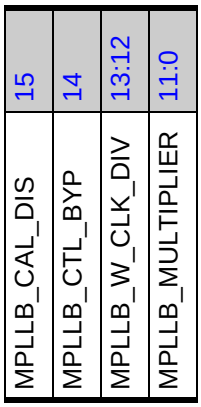


Table 5-568 Fields for Register: VR_MII_MP_25G_MPLLB_CTRL0

Bits	Name	Memory Access	Description
15	MPLLB_CAL_DIS	R/W	MPLLB Calibration Disable This field drives the xpcs_mpllb_init_cal_disable_o port. **Value After Reset:** 0x1 **Exists:** Always
14	MPLLB_CTL_BYP	* Varies	MPLLB Control Buffer Bypass This field drives the xpcs_mpllb_ctl_buf_byp_o port. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_RSVDPARAM_36) ? \"read-only\" :\"read-write\""
13:12	MPLLB_W_CLK_DIV	R/W	MPLLB Word Clock Divider This field controls the MPLLB word clock dividers for generating mpllb_word_clk.This field drives the xpcs_mpllb_word_clk_div_o port. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
11:0	MPLLB_MULTIPLIER	R/W	MPLLB frequency Multiplier Control This field controls the multiplication of reference clock to a frequency suitable for operating speed. Any change in this field must be followed by a vendor-specific Soft Reset to ensure that PHY is properly powered-up in the desired mode. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.108 VR_MII_MP_12G_MPLLB_CTRL1

- **Description:** VR MII Synopsys Multi-protocol 12G PHY MPLLB Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8075 : (APB3_16) ? (0x8075*0x2) : (0x8075*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&E12&&!SNPS_RSVDPARAM_39&&!SNPS_RSVDPARAM_22)

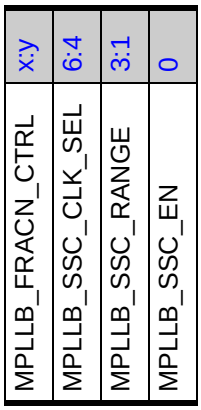


Table 5-569 Fields for Register: VR_MII_MP_12G_MPLLB_CTRL1

Bits	Name	Memory Access	Description
x:y	MPLLB_FRACN_CTRL	R/W	MPLLB Fractional Control This field drives the output port 'xpcs_mpllb_fracn_ctrl_o'. **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "SNPS_RSVDPARAM_35 ? 11 : 9" + "SNPS_RSVDPARAM_35 ? 5 : 7" - 1 **Range Variable[y]:** "SNPS_RSVDPARAM_35 ? 5 : 7"
6:4	MPLLB_SSC_CLK_SEL	R/W	MPLLB Spread Spectrum Clock Select This field drives the output port 'xpcs_mpllb_ssc_clk_sel_o' **Value After Reset:** 0x0 **Exists:** !SNPS_RSVDPARAM_35

Bits	Name	Memory Access	Description
3:1	MPLLB_SSC_RANGE	R/W	MPLLB Spread Spectrum Range This field selects the modulation to insert. This field applies a fixed PPM offset to phase accumulator as follows: ■ 3'b000: -4980 ■ 3'b001: -4492 ■ 3'b010: -4003 ■ 3'b011: -3027 ■ 3'b100: 4980 ■ 3'b101: 4492 ■ 3'b110: 4003 ■ 3'b111: 3027 drives the output port 'xpcs_mpllb_ssc_range_o'. **Value After Reset:** 0x0 **Exists:** Always
0	MPLLB_SSC_EN	R/W	MPLLB Spread Spectrum Enable This field can be set to enable spread-spectrum generation on mplla_div_clk output.This bit drives the output port xpcs_mpllb_ssc_en_o. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.109 VR_MII_MP_16G_MPLLB_CTRL1

- **Description:** VR MII Synopsys Multi-protocol 16G PHY MPLLB Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8075 : (APB3_16) ? (0x8075*0x2) : (0x8075*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&E16

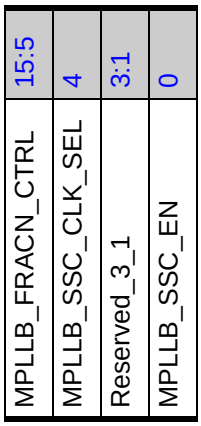


Table 5-570 Fields for Register: VR_MII_MP_16G_MPLLB_CTRL1

Bits	Name	Memory Access	Description
15:5	MPLLB_FRACN_CTRL	R/W	MPLLB Fractional Control This field drives the output port 'xpcs_mpllb_fracn_ctrl_o'. **Value After Reset:** "SNPS_RSVDPARAM_15 ? 0x188 : 0x0" **Exists:** Always
4	MPLLB_SSC_CLK_SEL	R/W	MPLLB Spread Spectrum Clock Select This field drives the output port 'xpcs_mpllb_ssc_clk_sel_o' **Value After Reset:** 0x0 **Exists:** Always
3:1	Reserved_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	MPLLB_SSC_EN	R/W	MPLLB Spread Spectrum Enable This field can be set to enable spread-spectrum generation on mplla_div_clk output.This bit drives the output port xpcs_mpllb_ssc_en_o. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.110 VR_MII_MP_25G_MPLLB_CTRL1

- **Description:** VR MII Synopsys Multi-protocol 25G PHY MPLLB Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8075 : (APB3_16) ? (0x8075*0x2) : (0x8075*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&E25&&!SNPS_RSVDPARAM_37

15	14	13:12	11	10	9	8	7:6	5:4	3:2	1	0
SSC_SPD	SSC_EN	BW_TH	Reserved_11	FB_DIV4_EN	SH_LK_EN	FRAC_EN	MPLLB_V2I	MPLLB_FREQ_VCO	RCAL_BK_SEL	RCAL_SKP	RCAL_FRC

Table 5-571 Fields for Register: VR_MII_MP_25G_MPLLB_CTRL1

Bits	Name	Memory Access	Description
15	SSC_SPD	R/W	MPLLB Up Spread Enable This field drives the xpcs_mpllb_ssc_spd_en_o port. **Value After Reset:** 0x0 **Exists:** Always
14	SSC_EN	R/W	MPLLB Spread Spectrum Enable This field drives the xpcs_mpllb_ssc_en_o port. **Value After Reset:** 0x0 **Exists:** Always
13:12	BW_TH	* Varies	MPLLB Bandwidth Threshold This field drives the xpcs_mpllb_bw_th_o port. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_RSVDPARAM_36) ? \"read-only\" :\"read-write\""
11	Reserved_11	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
10	FB_DIV4_EN	R/W	MPLLB DIV clock output frequency divider control This field drives the xpcs_mpllb_fb_clk_div4_en_o port. **Value After Reset:** 0x0 **Exists:** Always
9	SH_LK_EN	* Varies	MPLLB Short Lock Enable This field drives the xpcs_mpllb_short_lock_en_o port. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_RSVDPARAM_36) ? \"read-only\" :\"read-write\""
8	FRAC_EN	R/W	MPLLB Fractional Enable This field drives the xpcs_mpllb_fracn_en_o port. **Value After Reset:** 0x0 **Exists:** Always
7:6	MPLLB_V2I	* Varies	**For configurations with SNPS_RSVDPARAM_36:** MPLLB V2I Operating range This field drives the xpcs_mpllb_v2i_o port and allows the configuration of V2I operating range. **For non SNPS_RSVDPARAM_36 configurations:** Reserved For configurations without SNPS_RSVDPARAM_36 , this bit is Reserved **Value After Reset:** "SNPS_RSVDPARAM_36 ? 0x2 : 0x0" **Exists:** Always **Memory Access:** "(SNPS_RSVDPARAM_36) ? \"read-write\" :\"read-only\""
5:4	MPLLB_FREQ_VCO	* Varies	**For configurations with SNPS_RSVDPARAM_36:** MPLLB VCO Range Control This field drives the xpcs_mpllb_freq_vco_o port and allows to control the desired operating VCO range. **For non SNPS_RSVDPARAM_36 configurations:** Reserved For configurations without SNPS_RSVDPARAM_36 , this bit is Reserved **Value After Reset:** "SNPS_RSVDPARAM_36 ? 0x3 : 0x0" **Exists:** Always **Memory Access:** "(SNPS_RSVDPARAM_36) ? \"read-write\" :\"read-only\""

Bits	Name	Memory Access	Description
3:2	RCAL_BK_SEL	R/W	MPLLB Recalibration Bank Select This field drives the xpcs_mpllb_recal_bank_sel_o port. **Value After Reset:** 0x0 **Exists:** Always
1	RCAL_SKP	R/W	MPLLB Recalibration Skip Enable This field drives the xpcs_mpllb_recal_skip_en_o port. **Value After Reset:** 0x0 **Exists:** Always
0	RCAL_FRC	R/W	MPLLB Recalibration Force Enable This drives the xpcs_mpllb_recal_force_en_o port. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.111 VR_MII_MP_32G_MPLLB_CTRL1

- **Description:** VR MII Synopsys Multi-protocol E32 PHY MPLLB Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8075 : (APB3_16) ? (0x8075*0x2) : (0x8075*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&(SNPS_RSVDPARAM_37||E32)

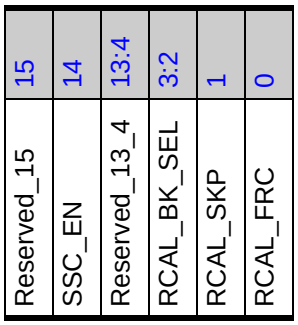


Table 5-572 Fields for Register: VR_MII_MP_32G_MPLLB_CTRL1

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14	SSC_EN	R/W	MPLLB Spread Spectrum Enable This field drives the xpcs_mplla_ssc_en_o port. **Value After Reset:** 0x0 **Exists:** Always
13:4	Reserved_13_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:2	RCAL_BK_SEL	R/W	MPLLB Recalibration Bank Select This field drives the xpcs_mplla_recal_bank_sel_o port. **Value After Reset:** 0x0 **Exists:** Always
1	RCAL_SKP	R/W	MPLLB Recalibration Skip Enable This field drives the xpcs_mplla_recal_skip_en_o port. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	RCAL_FRC	R/W	MPLLB Recalibration Force Enable This drives the xpcs_mplla_recal_force_en_o port. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.112 VR_MII_MP_12G_16G_MPLLB_CTRL2

- **Description:** VR MII Synopsys Multi-protocol 12G/16G PHY MPLLB Control 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8076 : (APB3_16) ? (0x8076*0x2) : (0x8076*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16)&&!SNPS_RSVDPARAM_22&&!SNPS_RSVDPARAM_39)

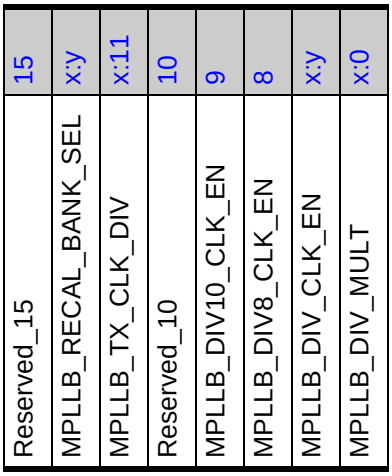


Table 5-573 Fields for Register: VR_MII_MP_12G_16G_MPLLB_CTRL2

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
x:y	MPLLB_RECAL_BANK_SEL	* Varies	**For Multi-protocol 16G configurations:** MPLLB Re-calibration Bank Select This field drives the output port 'xpcs_mpllb_recal_bank_sel_o[1:0]'. **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "!(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_39) ? 2 : 1" + "!(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_39) ? 13 : 14" - 1 **Range Variable[y]:** "!(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_39) ? 13 : 14" **Memory Access:** "(E16) ? \"read-write\" :\"read-only\""
x:11	MPLLB_TX_CLK_DIV	* Varies	MPLLB Tx Clock Divider This field drives the output port 'xpcs_mpllb_tx_clk_div_o[1:0]'. This field is present only for Multi-protocol 16G/Multi-protocol 10G PHY configurations. **Value After Reset:** "SNPS_RSVDPARAM_35 ? 0x2 : 0x0" **Exists:** Always **Range Variable[x]:** "SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_39 ? 3 : 2" + 10 **Memory Access:** "(E16||SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_39) ? \"read-write\" :\"read-only\""
10	Reserved_10	R	Reserved **Value After Reset:** 0x0 **Exists:** !SNPS_RSVDPARAM_35
9	MPLLB_DIV10_CLK_EN	R/W	MPLLB Divide by 10 Enable When this bit is set, the frequency of the mpllb_word_clk output clock from PHY is MPLLB frequency divided by 10. **Value After Reset:** "(E16&&SNPS_RSVDPARAM_15) ? 0x1 : 0x0" **Exists:** Always
8	MPLLB_DIV8_CLK_EN	R/W	MPLLB Divide by 8 Enable When this bit is set, the frequency of the mpllb_word_clk output clock from PHY is MPLLB frequency divided by 8. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
x:y	MPLLB_DIV_CLK_EN	R/W	Enable mpllb_div_clk from PHY When asserted, the frequency of mpllb_div_clk output from PHY is MPLLB frequency divided by 'mpllb_div_multiplier' **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "SNPS_RSVDPARAM_35 ? 10 **Range Variable[y]:** 7"
x:0	MPLLB_DIV_MULT	R/W	MPLLB Output Frequency Multiplier Control This field controls the frequency multiplication factor used to generate MPLLB clock output from the reference clock input as seen by the MPLL. **Value After Reset:** 0x0 **Exists:** Always **Range Variable[x]:** "SNPS_RSVDPARAM_35 ? 8 : 7" - 1

5.2.9.113 VR_MII_MP_25G_MPLLB_CTRL2

- **Description:** VR MII Synopsys Multi-protocol 25G PHY MPLLB Control 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8076 : (APB3_16) ? (0x8076*0x2) : (0x8076*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&E25&&!SNPS_RSVDPARAM_37

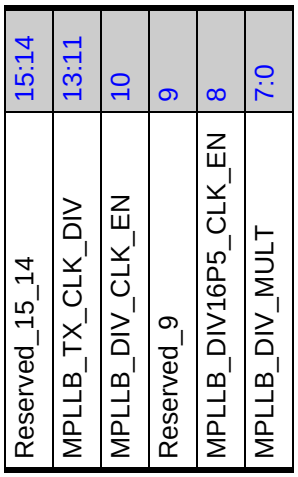


Table 5-574 Fields for Register: VR_MII_MP_25G_MPLLB_CTRL2

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:11	MPLLB_TX_CLK_DIV	R/W	MPLLB Tx Clock Divider This field drives the output port 'xpcs_mpllb_tx_clk_div_o[1:0]'. **Value After Reset:** 0x0 **Exists:** Always
10	MPLLB_DIV_CLK_EN	R/W	Enable mpllb_div_clk from PHY. When asserted, the frequency of mpllb_div_clk from PHY is the MPLLB frequency divided by 'mpllb_div_multiplier' **Value After Reset:** 0x0 **Exists:** Always
9	Reserved_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
8	MPLLB_DIV16P5_CLK_EN	* Varies	MPLLB Divide by 16.5 Enable This bit can be set to enable output clocks derived from MPLLB based on 16.5,33 and 66 division ratios. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_RSVDPARAM_36) ? \"read-only\" :\"read-write\""
7:0	MPLLB_DIV_MULT	R/W	MPLLB Output Frequency Multiplier Control This field controls the frequency multiplication factor used to generate MPLLB clock output from the reference clock input as seen by the MPLL.This field drives the xpcs_mpllb_div_mult_o port. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.114 VR_MII_MP_12G_MPLLA_CTRL3

- **Description:** VR MII Synopsys Multi-protocol 12G PHY MPLLA Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8077 : (APB3_16) ? (0x8077*0x2) : (0x8077*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&E12&&!SNPS_RSVDPARAM_39&&!SNPS_RSVDPARAM_22)

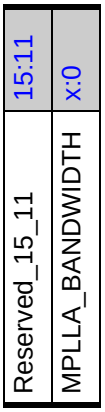


Table 5-575 Fields for Register: VR_MII_MP_12G_MPLLA_CTRL3

Bits	Name	Memory Access	Description
15:11	Reserved_15_11	R	Reserved **Value After Reset:** 0x0 **Exists:** !SNPS_RSVDPARAM_35
x:0	MPLLA_BANDWIDTH	R/W	MPLLA Bandwidth Control This field controls the bandwidth of MPLLA present in Multi-protocol 12G PHY. This field drives the output port 'xpcs_mplla_bandwidth_o'. **Value After Reset:** "(SNPS_RSVDPARAM_35&&SNPS_RSVDPARAM_13) ? 0x165 : SNPS_RSVDPARAM_35 ? 0x2B : 0x798" **Exists:** Always **Range Variable[x]:** "SNPS_RSVDPARAM_35 ? 16 : 11" - 1

5.2.9.115 VR_MII_MP_16G_MPLLA_CTRL3

- **Description:** VR MII Synopsys Multi-protocol 16G PHY MPLLA Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8077 : (APB3_16) ? (0x8077*0x2) : (0x8077*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&E16)



Table 5-576 Fields for Register: VR_MII_MP_16G_MPLLA_CTRL3

Bits	Name	Memory Access	Description
15:0	MPLLA_BANDWIDTH	R/W	MPLLA Bandwidth Control This field controls the bandwidth of MPLLA present in Multi-protocol 16G PHY. This field drives the output port 'xpcs_mplla_bandwidth_o'. **Value After Reset:** "(E16&&SNPS_RSVDPARAM_15) ? 0xf01c : 0xA035" **Exists:** Always

5.2.9.116 VR_MII_MP_25G_MPLLA_CTRL3

- **Description:** VR MII Synopsys Multi-protocol 25G PHY MPLLA Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8077 : (APB3_16) ? (0x8077*0x2) : (0x8077*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&E25&&!SNPS_RSVDPARAM_37



Table 5-577 Fields for Register: VR_MII_MP_25G_MPLLA_CTRL3

Bits	Name	Memory Access	Description
15:0	MPLLA_FRACN_QUOT	R/W	MPLLA fractional Quotient Control This field drives the xpcs_mplla_fracn_quot_o port. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.117 VR_MII_MP_6G_MPLL_CTRL0

- **Description:** VR MII Synopsys Multi-protocol 6G MPLL Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8078 : (APB3_16) ? (0x8078*0x2) : (0x8078*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&SSP)

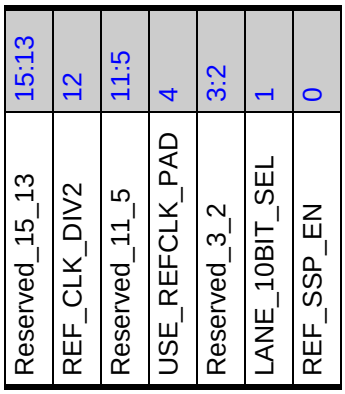


Table 5-578 Fields for Register: VR_MII_MP_6G_MPLL_CTRL0

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	REF_CLK_DIV2	R/W	Divides the reference clock by 2 before passing it to MPLL **Default Value:** ■ For 156.25 MHz, 125 MHz, 50 MHz Reference Clock Frequency: 1'b1 ■ For 62.5 MHz and 25 MHz Reference Clock Frequency: 1'b0 **Value After Reset:** "((REF_CLK_62==1||REF_CLK_25==1) ? 0x0 : 0x1)" **Exists:** Always
11:5	Reserved_11_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4	USE_REFCLK_PAD	R/W	Use Normal Reference Clock ■ When set to 1, PHY selects the clock source connected to ref_pad_clk_p/m as its reference clock. ■ When this bit is set to 0, clock source connected to ref_alt_-clk_p/m is selected. **Value After Reset:** "(REF_CLK_SRC==0) ? 0x1 : 0x0" **Exists:** Always
3:2	Reserved_3_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1	LANE_10BIT_SEL	R/W	Lane Width Control When set to 1, the lane width is 10 bit and when set to 0, the lane width is 20 bit **Value After Reset:** "(RXAUI_DEFAULT)? 0x0 : 0x1" **Exists:** Always
0	REF_SSP_EN	R/W	Enables Reference Clock When set to 1, the reference clock to the PHY is enabled. **Value After Reset:** 0x1 **Exists:** Always

5.2.9.118 VR_MII_MP_12G_MPLLB_CTRL3

- **Description:** VR MII Synopsys Multi-protocol 12G PHY MPLLB Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8078 : (APB3_16) ? (0x8078*0x2) : (0x8078*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&E12&&!SNPS_RSVDPARAM_39&&!SNPS_RSVDPARAM_22)

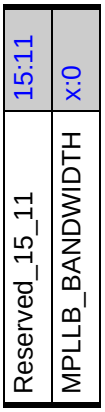


Table 5-579 Fields for Register: VR_MII_MP_12G_MPLLB_CTRL3

Bits	Name	Memory Access	Description
15:11	Reserved_15_11	R	Reserved **Value After Reset:** 0x0 **Exists:** !SNPS_RSVDPARAM_35
x:0	MPLLB_BANDWIDTH	R/W	MPLLB Bandwidth Control This field controls the bandwidth of MPLLB present in Multi-protocol 12G PHY. This field drives the output port 'xpcs_mpllb_bandwidth_o'. **Value After Reset:** "SNPS_RSVDPARAM_35 ? 0x44 : 0x0" **Exists:** Always **Range Variable[x]:** "SNPS_RSVDPARAM_35 ? 16 : 11" - 1

5.2.9.119 VR_MII_MP_16G_MPLLB_CTRL3

- **Description:** VR MII Synopsys Multi-protocol 16G PHY MPLLB Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8078 : (APB3_16) ? (0x8078*0x2) : (0x8078*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&E16



Table 5-580 Fields for Register: VR_MII_MP_16G_MPLLB_CTRL3

Bits	Name	Memory Access	Description
15:0	MPLLB_BANDWIDTH	R/W	MPLLB Bandwidth Control This field controls the bandwidth of MPLLB present in Multi-protocol 16G PHY. This field drives the output port 'xpcs_mpllb_bandwidth_o'. **Value After Reset:** "(E16&&SNPS_RSVDPARAM_15) ? 0xa02c : SNPS_RSVDPARAM_39 ? 0x502A : 0x0" **Exists:** Always

5.2.9.120 VR_MII_MP_25G_MPLLB_CTRL3

- **Description:** VR MII Synopsys Multi-protocol 25G PHY MPLLB Control 3 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8078 : (APB3_16) ? (0x8078*0x2) : (0x8078*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&E25&&!SNPS_RSVDPARAM_37



Table 5-581 Fields for Register: VR_MII_MP_25G_MPLLB_CTRL3

Bits	Name	Memory Access	Description
15:0	MPLLB_FRACN_QUOT	R/W	MPLLB fractional Quotient Control This field drives the xpcs_mpllb_fracn_quot_o port. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.121 VR_MII_MP_6G_MPLL_CTRL1

- **Description:** VR MII Synopsys Multi-protocol 6G MPLL Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8079 : (APB3_16) ? (0x8079*0x2) : (0x8079*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&SSP)

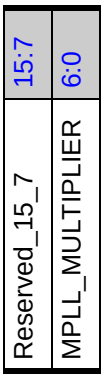


Table 5-582 Fields for Register: VR_MII_MP_6G_MPLL_CTRL1

Bits	Name	Memory Access	Description
15:7	Reserved_15_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
6:0	MPLL_MULTIPLIER	R/W	MPLL Frequency Multiplier Control**Default Value:** The default value of this bit depends on the reference clock frequency and the configuration: 156.25 MHz Reference Clock Frequency ■ 7'h20 for KX_Only or 1000BASEX-Only PCS configurations ■ 7'h28 for all other configurations. 125 MHz and 62.5 MHz Reference Clock Frequency ■ 7'h28 for KX_Only or 1000BASEX-Only PCS configurations ■ 7'h32 for all other configurations 50 MHz and 25 MHz Reference Clock Frequency ■ 7'h64 KX_Only or 1000BASEX-Only PCS configurations ■ 7'h7D for all other configurations **Value After Reset:** "(REF_CLK_156==1) ? ((SINGLE_LANE_CFG) ? 0x20 :0x28): ((REF_CLK_125==1||REF_CLK_62==1) ? ((SINGLE_LANE_CFG) ? 0x28 :0x32) : ((SINGLE_LANE_CFG) ? 0x64 :0x7d))" **Exists:** Always

5.2.9.122 VR_MII_MP_32G_PLL_CMN_CTRL

- **Description:** VR MII Synopsys Multi-protocol 32G PLL Common Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8079 : (APB3_16) ? (0x8079*0x2) : (0x8079*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** ((BASE1000X_ONLY)&&SNPS_XAUI_PHY&&E32)

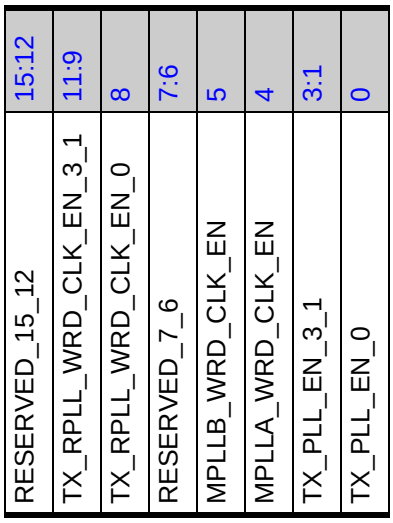


Table 5-583 Fields for Register: VR_MII_MP_32G_PLL_CMN_CTRL

Bits	Name	Memory Access	Description
15:12	RESERVED_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:9	TX_RPLL_WRD_CLK_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	TX_RPLL_WRD_CLK_EN_0	R/W	Tx ROPLL Word clock output enable Lane 0 The field drives the output port 'xpcs_tx_ropll_wrd_clk_en_o'. **Value After Reset:** 0x1 **Exists:** Always
7:6	RESERVED_7_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
5	MPLLB_WRD_CLK_EN	R/W	MPLLB word clock output enable. This field drives the output port 'xpcs_mpllb_wrd_clk_en_o'. **Value After Reset:** 0x0 **Exists:** Always
4	MPLLA_WRD_CLK_EN	R/W	MPLLA word clock output enable. This field drives the output port 'xpcs_mplla_wrd_clk_en_o'. **Value After Reset:** 0x0 **Exists:** Always
3:1	TX_PLL_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	TX_PLL_EN_0	R/W	TX PLL Enable Lane0. Powers up or powers down the ROPLL or MPLLA or MPLLB. This field drives the output port 'xpcs_tx_pll_en_o'. **Value After Reset:** 0x1 **Exists:** Always

5.2.9.123 VR_MII_MP_16G_MPLLA_CTRL4

- **Description:** VR MII Synopsys Multi-protocol 16G PHY MPLLA Control 4 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8079 : (APB3_16) ? (0x8079*0x2) : (0x8079*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E16||SNPS_RSVDPARAM_39))



Table 5-584 Fields for Register: VR_MII_MP_16G_MPLLA_CTRL4

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:0	MPLLA_SSC_FRQ_CNT_INT	R/W	MPLLA SSC Frequency Counter Initialization. This field drives the output port xpcs_mplla_ssc_freq_cnt_init_o[11:0]. **Value After Reset:** "SNPS_RSVDPARAM_39&&(BASE1000X_ONLY) ? 0x4D :0x0" **Exists:** Always

5.2.9.124 VR_MII_MP_25G_MPLLA_CTRL4

- **Description:** VR MII Synopsys Multi-protocol 25G PHY MPLLA Control 4 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8079 : (APB3_16) ? (0x8079*0x2) : (0x8079*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&E25&&!SNPS_RSVDPARAM_37



Table 5-585 Fields for Register: VR_MII_MP_25G_MPLLA_CTRL4

Bits	Name	Memory Access	Description
15:0	MPLLA_FRACN_REM	R/W	MPLLA fractional Remainder Control This field drives the xpcs_mplla_fracn_rem_o port. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.125 VR_MII_MP_25G_MPLLB_CTRL4

- **Description:** VR MII Synopsys Multi-protocol 25G PHY MPLLB Control 4 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x807A : (APB3_16) ? (0x807A*0x2) : (0x807A*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&E25&&!SNPS_RSVDPARAM_37



Table 5-586 Fields for Register: VR_MII_MP_25G_MPLLB_CTRL4

Bits	Name	Memory Access	Description
15:0	MPLLB_FRACN_REM	R/W	MPLLB fractional Remainder Control This field drives the xpcs_mpllb_fracn_rem_o port. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.126 VR_MII_MP_16G_MPLLA_CTRL5

- **Description:** VR MII Synopsys Multi-protocol 16G PHY MPLLA Control 5 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x807A : (APB3_16) ? (0x807A*0x2) : (0x807A*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E16||SNPS_RSVDPARAM_39))

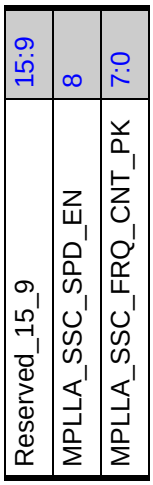


Table 5-587 Fields for Register: VR_MII_MP_16G_MPLLA_CTRL5

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	MPLLA_SSC_SPD_EN	R/W	MPLLA SSC Up Spread Enable. This field drives the output port xpcs_mplla_ssc_spd_en_o. **Value After Reset:** 0x0 **Exists:** Always
7:0	MPLLA_SSC_FRQ_CNT_PK	R/W	MPLLA SSC Frequency Counter Peak. This field drives the output port xpcs_mplla_ssc_freq_cnt_pk_o[7:0]. **Value After Reset:** "SNPS_RSVDPARAM_39&&BASE1000X_ONLY ? 0x33 : 0x0" **Exists:** Always

5.2.9.127 VR_MII_MP_16G_MPLLB_CTRL4

- **Description:** VR MII Synopsys Multi-protocol 16G PHY MPLLB Control 4 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x807B : (APB3_16) ? (0x807B*0x2) : (0x807B*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&E16



Table 5-588 Fields for Register: VR_MII_MP_16G_MPLLB_CTRL4

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:0	MPLLB_SSC_FRQ_CNT_INT	R/W	MPLLB SSC Frequency Counter Initialization. This field drives the output port xpcs_mpllb_ssc_freq_cnt_init_o[11:0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.128 VR_MII_MP_25G_MPLLA_CTRL5

- **Description:** VR MII Synopsys Multi-protocol 25G PHY MPLLA Control 5 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x807B : (APB3_16) ? (0x807B*0x2) : (0x807B*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&E25&&!SNPS_RSVDPARAM_37



Table 5-589 Fields for Register: VR_MII_MP_25G_MPLLA_CTRL5

Bits	Name	Memory Access	Description
15:0	MPLLA_FRACN_DEN	R/W	MPLLA fractional Denominator Control This field drives the xpcs_mplla_fracn_den_o port. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.129 VR_MII_MP_16G_MPLLB_CTRL5

- **Description:** VR MII Synopsys Multi-protocol 16G PHY MPLLB Control 5 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x807C : (APB3_16) ? (0x807C*0x2) : (0x807C*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&E16

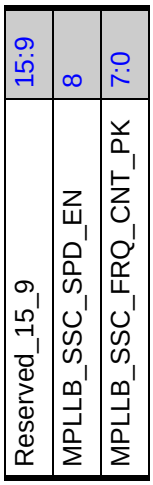


Table 5-590 Fields for Register: VR_MII_MP_16G_MPLLB_CTRL5

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	MPLLB_SSC_SPD_EN	R/W	MPLLB SSC Up Spread Enable. This field drives the output port xpcs_mpllb_ssc_spd_en_o. **Value After Reset:** 0x0 **Exists:** Always
7:0	MPLLB_SSC_FRQ_CNT_PK	R/W	MPLLB SSC Frequency Counter Peak. This field drives the output port xpcs_mpllb_ssc_freq_cnt_pk_o[7:0]. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.130 VR_MII_MP_25G_MPLLB_CTRL5

- **Description:** VR MII Synopsys Multi-protocol 25G PHY MPLLB Control 5 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x807C : (APB3_16) ? (0x807C*0x2) : (0x807C*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&E25&&!SNPS_RSVDPARAM_37



Table 5-591 Fields for Register: VR_MII_MP_25G_MPLLB_CTRL5

Bits	Name	Memory Access	Description
15:0	MPLLB_FRACN_DEN	R/W	MPLLB fractional Denominator Control This field drives the xpcs_mpllb_fracn_den_o port. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.131 VR_MII_MP_25G_MPLLA_BW_HI_CTRL

- **Description:** VR MII Synopsys Multi-protocol 25G PHY MPLLA Bandwidth High Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x807D : (APB3_16) ? (0x807D*0x2) : (0x807D*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&E25&&!SNPS_RSVDPARAM_37

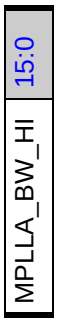


Table 5-592 Fields for Register: VR_MII_MP_25G_MPLLA_BW_HI_CTRL

Bits	Name	Memory Access	Description
15:0	MPLLA_BW_HI	R/W	MPLLA Bandwidth Control -High This field controls the bandwidth of MPLLA present in Multi-protocol 25G PHY. This field drives the output port 'xpcs_mplla_bw_high_o'. **Value After Reset:** "SNPS_RSVDPARAM_36 ? 0x26 : 0x7" **Exists:** Always

5.2.9.132 VR_MII_MP_25G_MPLLA_BW_LO_CTRL

- **Description:** VR MII Synopsys Multi-protocol 25G PHY MPLLA Bandwidth Low Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x807E : (APB3_16) ? (0x807E*0x2) : (0x807E*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&E25&&!SNPS_RSVDPARAM_37



Table 5-593 Fields for Register: VR_MII_MP_25G_MPLLA_BW_LO_CTRL

Bits	Name	Memory Access	Description
15:0	MPLLA_BW_LO	R/W	MPLLA Bandwidth Control -Low This field controls the bandwidth of MPLLA present in Multi-protocol 25G PHY. This field drives the output port 'xpcs_mplla_bw_low_o'. **Value After Reset:** "SNPS_RSVDPARAM_36 ? 0x26 : 0x7" **Exists:** Always

5.2.9.133 VR_MII_MP_25G_MPLLB_BW_HI_CTRL

- **Description:** VR MII Synopsys Multi-protocol 25G PHY MPLLB Bandwidth High Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x807F : (APB3_16) ? (0x807F*0x2) : (0x807F*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&E25&&!SNPS_RSVDPARAM_36&&!SNPS_RSVDPARAM_37

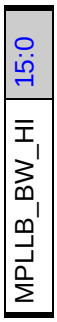


Table 5-594 Fields for Register: VR_MII_MP_25G_MPLLB_BW_HI_CTRL

Bits	Name	Memory Access	Description
15:0	MPLLB_BW_HI	R/W	MPLLB Bandwidth Control -High This field controls the bandwidth of MPLLB present in Multi-protocol 25G PHY. This field drives the output port 'xpcs_mpllb_bw_high_o'. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.134 VR_MII_MP_25G_MPLLB_BW_LO_CTRL

- **Description:** VR MII Synopsys Multi-protocol 25G PHY MPLLB Bandwidth Low Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8081 : (APB3_16) ? (0x8081*0x2) : (0x8081*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&E25&&!SNPS_RSVDPARAM_36&&!SNPS_RSVDPARAM_37



Table 5-595 Fields for Register: VR_MII_MP_25G_MPLLB_BW_LO_CTRL

Bits	Name	Memory Access	Description
15:0	MPLLB_BW_LO	R/W	MPLLB Bandwidth Control -Low This field controls the bandwidth of MPLLB present in Multi-protocol 25G PHY. This field drives the output port 'xpcs_mpllb_bw_low_o'. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.135 VR_MII_MP_6G_MPLL_STS

- **Description:** VR MII Synopsys Multi-protocol 6G PHY MPLL Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8088 : (APB3_16) ? (0x8088*0x2) : (0x8088*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&SSP)

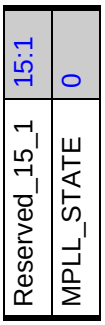


Table 5-596 Fields for Register: VR_MII_MP_6G_MPLL_STS

Bits	Name	Memory Access	Description
15:1	Reserved_15_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	MPLL_STATE	R	MPLL State Indicator This bit indicates the state of MPLL. When set to 1, it indicates that the MPLL is powered up and phase-locked. **Value After Reset:** 0x1 **Exists:** Always

5.2.9.136 VR_MII_MP_12G_16G_25G_MISC_CTRL0

- **Description:** VR MII Synopsys Multi-protocol 12G/16G/25G PHY Miscellaneous Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8090 : (APB3_16) ? (0x8090*0x2) : (0x8090*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16||E25||E32))

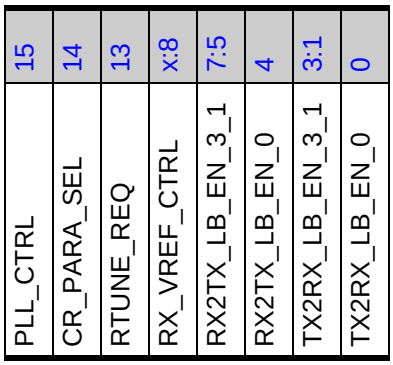


Table 5-597 Fields for Register: VR_MII_MP_12G_16G_25G_MISC_CTRL0

Bits	Name	Memory Access	Description
15	PLL_CTRL	R/W	PLL Re-initialization Control If the PHY PLL does not lock for a long time, DWC_xpcs initiates a Rx power-state change from P0 to P1 and then back to P0, if this bit is set. This process is done to re-initialize the PLL. The time duration to wait before reinitializing the PLL is determined by VR_MII_MP_12G_16G_25G_MISC_CTRL1 Register **Value After Reset:** 0x0 **Exists:** Always
14	CR_PARA_SEL	R/W	Select CR Para/APB Port to access PHY registers This bit select the interface for accessing PHY registers: ■ 0: JTAG ■ 1: CR parallel port/APB interface Change this bit only after disabling 'jtag_tck' to PHY. **Value After Reset:** 0x1 **Exists:** Always
13	RTUNE_REQ	R/W	Resistor Tuning Request This bit can be set to trigger a resistor tune request to the PHY. This bit controls the 'xgxs_rtune_req_o' output port. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
x:8	RX_VREF_CTRL	* Varies	**Non SNPS_RSVDPARAM_37/Non E32 configurations:** Rx Biasing Current Control This field drives the output port 'xpcs_rx_vref_ctrl_o[4:0]'. This field sets the Rx biasing current for Rx analog front end. **For SNPS_RSVDPARAM_37/E32 configurations:** Reserved **Value After Reset:** "SNPS_RSVDPARAM_37||E32 ? 0x0 : ((SNPS_RSVDPARAM_22||E25) ? 0x5 : 0x11)" **Exists:** Always **Range Variable[x]:** "SNPS_RSVDPARAM_22 ? 3 : 5" + 7 **Memory Access:** "(!(SNPS_RSVDPARAM_37||E32)) ? \"read-write\" :\"read-only\"""
7:5	RX2TX_LB_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	RX2TX_LB_EN_0	R/W	Enable Parallel Rx-to-Tx Loopback on lane 0 When this bit is set, recovered parallel data from PHY receiver is looped back to the transmit serializer. This loop-back takes place internal to the PHY (not within DWC_xpcs). **Value After Reset:** 0x0 **Exists:** Always
3:1	TX2RX_LB_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	TX2RX_LB_EN_0	R/W	Enable Analog Tx-to-Rx Serial Loopback on lane 0 This bit can be set to enable serial loopback in the PHY from Tx pre-driver to Rx analog front-end. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.137 VR_MII_MP_12G_16G_25G_REF_CLK_CTRL

- **Description:** VR MII Synopsys Multi-protocol 12G/16G/25G PHY Reference Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8091 : (APB3_16) ? (0x8091*0x2) : (0x8091*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16||E25)&&!SNPS_RSVDPARAM_37

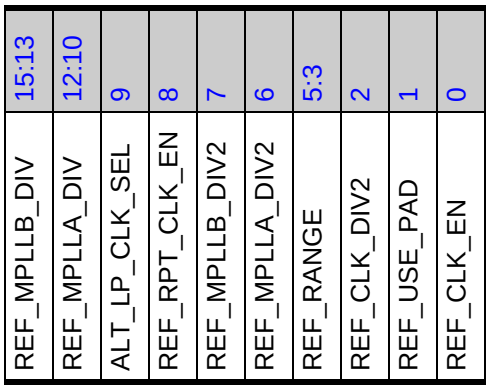


Table 5-598 Fields for Register: VR_MII_MP_12G_16G_25G_REF_CLK_CTRL

Bits	Name	Memory Access	Description
15:13	REF_MPLLB_DIV	* Varies	**For Multi-protocol 25G PHY or Multi-protocol 10G Gen2 PHY configurations:** This field drives the xpcs_ref_clk_mpllb_div_o port. **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "(SNPS_RSVDPARAM_22||E25) ? \"read-write\" : \"read-only\""
12:10	REF_MPLLA_DIV	* Varies	**For Multi-protocol 25G PHY or Multi-protocol 10G Gen2 PHY configurations:** This field drives the xpcs_ref_clk_mplla_div_o port. **For other configurations:** Reserved **Value After Reset:** "SNPS_RSVDPARAM_22 ? 0x1 : E25 ? (REF_CLK_312 ? 0x2 : SNPS_RSVDPARAM_13 ? 0x0 : REF_CLK_156 ? 0x1 : 0x0) : 0x0" **Exists:** Always **Memory Access:** "(SNPS_RSVDPARAM_22||E25) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
9	ALT_LP_CLK_SEL	* Varies	**For Multi-protocol 10G Gen2 PHY configurations:** Alternate Low Precision Clock Select. This field drives the xpcs_alt_clk_lp_sel_o port. **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SNPS_RSVDPARAM_22 ? \"read-write\" : \"read-only\""
8	REF_RPT_CLK_EN	R/W	Repeat Reference Clock Enable If this bit is set, ref_repeat_clk_{p,m} clock from PHY is enabled. **Value After Reset:** 0x0 **Exists:** Always
7	REF_MPLLB_DIV2	R/W	MPLLB Reference Clock Divider Control The reference clock used for MPLLB calibration and locking can be divided by 2 by setting this bit to 1. This division is applied after ref_clk_div2_en. Therefore, the total division ratio (from the input reference clock) can be 1, 2 or 4 **Value After Reset:** 0x0 **Exists:** !SNPS_RSVDPARAM_22&&!E25&&!SNPS_RSVDPARAM_39
6	REF_MPLLA_DIV2	R/W	MPLLA Reference Clock Divider Control The reference clock used for MPLLA calibration and locking can be divided by 2 by setting this bit to 1. This division is applied after ref_clk_div2_en. Therefore, the total division ratio (from the input reference clock) can be 1, 2 or 4. **Value After Reset:** "(E16&&SNPS_RSVDPARAM_15) ? 0x0 : (REF_CLK_FREQ==2)||(SNPS_RSVDPARAM_35&&SNPS_RSVDPARAM_13) ? 0x0 : (SNPS_RSVDPARAM_39&&SNPS_RSVDPARAM_13) ? 0x1 : (SNPS_RSVDPARAM_39&&REF_CLK_20) ? 0x0 : 0x1" **Exists:** !SNPS_RSVDPARAM_22&&!E25

Bits	Name	Memory Access	Description
5:3	REF_RANGE	R/W	Input Reference Clock Range This field specifies the frequency range of the input reference clock (post ref_clk_div2_en division if any). The code mapping of REF_RANGE is as follows: ■ 3'b000: 20 - 26 MHz ■ 3'b001: 26.1 - 52 MHz ■ 3'b010: 52.1 - 78 MHz ■ 3'b011: 78.1 - 104 MHz ■ 3'b100: 104.1 - 130 MHz ■ 3'b101: 130.1 - 156 MHz ■ 3'b110: 156.1 - 182 MHz ■ 3'b111: 182.1 - 200 MHz Set this field to appropriate values based on your reference clock frequency. **Value After Reset:** "(E16&&SNPS_RSVDPARAM_15) ? 0x0 : (SNPS_RSVDPARAM_39&&SNPS_RSVDPARAM_13) ? 0x1 : (SNPS_RSVDPARAM_39&&REF_CLK_20) ? 0x0 : SNPS_RSVDPARAM_22 ? (SNPS_RSVDPARAM_2 ? 0x5 : 0x6) : ((SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_36)&&SNPS_RSVDPARAM_13) ? 0x3 : SNPS_RSVDPARAM_35 ? 0x2 : (REF_CLK_FREQ==2) ? 0x3 : 0x6" **Exists:** Always
2	REF_CLK_DIV2	R/W	Reference Clock divide by 2 If this bit is set, reference clock provided to PHY gets divided by 2 internally in the PHY. **Value After Reset:** "((SNPS_RSVDPARAM_15&&E16)||(SNPS_RSVDPARAM_36&&SNPS_RSVDPARAM_13)) ? 0x0 : (REF_CLK_FREQ==6)||(SNPS_RSVDPARAM_35&&!SNPS_RSVDPARAM_13)||(SNPS_RSVDPARAM_39&&SNPS_RSVDPARAM_13) ? 0x1 : 0x0" **Exists:** !SNPS_RSVDPARAM_22
1	REF_USE_PAD	R/W	Use Pad Clk as Reference Clock If this bit is set, PHY selects the clock connected to ref_pad_clk_p/m as its reference clock; otherwise it selects the clock connected to ref_alt_clk_p/m. **Value After Reset:** "(REF_CLK_SRC==0) ? 0x1 : 0x0" **Exists:** Always

Bits	Name	Memory Access	Description
0	REF_CLK_EN	R/W	Reference Clock Enable Set this bit to enable reference clock to the PHY. This bit controls the xpcs_ref_clk_en_o output port. The value programmed to this bit is driven on xpcs_ref_clk_en_o when LPM (power-down) bit is programmed to 1. When the LPM bit is programmed to 0, xpcs_ref_clk_en_o remains high. **Value After Reset:** 0x1 **Exists:** Always

5.2.9.138 VR_MII_MP_6G_EEE_CTRL

- **Description:** VR MII Synopsys Multi-Protocol 6G PHY EEE Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8092 : (APB3_16) ? (0x8092*0x2) : (0x8092*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SSP&&QSGMII_EN&&EEE_EN

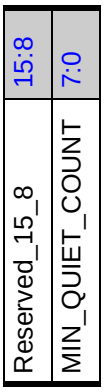


Table 5-599 Fields for Register: VR_MII_MP_6G_EEE_CTRL

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:0	MIN_QUIET_COUNT	R/W	This field allows application to determine the maximum value of MIN QUIET COUNTER. MIN QUIET counter is implemented in the RX_QUIET state of LPI state machine. This counter starts when DWC_xpcs reaches RX_QUIET state (if SUPRESS_LOS_DET bit is set) and it stops when it reaches a value equal to 8 times the value indicated by this field. DWC_xpcs starts monitoring for 'signal_detect' in RX_QUIET state only after this timer has reached its maximum value (8 times MIN_QUIET_COUNT). **Note:** This is a vendor-specific feature applicable only in QSGMII Mode.It is useful if xpcs_los_i input is unreliable at QSGMII rate. **Value After Reset:** 0xa **Exists:** Always

5.2.9.139 VR_MII_MP_12G_16G_25G_VCO_CAL_LD0

- **Description:** VR MII Synopsys Multi-protocol 12G/16G/25G PHY VCO Calibration Load 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8092 : (APB3_16) ? (0x8092*0x2) : (0x8092*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16||E25)&&!SNPS_RSVDPARAM_37)

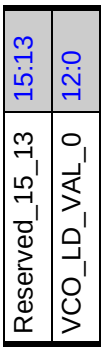


Table 5-600 Fields for Register: VR_MII_MP_12G_16G_25G_VCO_CAL_LD0

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:0	VCO_LD_VAL_0	R/W	Rx VCO calibration load value on lane 0 of the PHY This field is used to load internal calibration registers in the PHY to perform Rx VCO calibration on lane 0. This field drives the output port 'xpcs_rx0_vco_ld_val_o[5:0]'. **Value After Reset:** "(SNPS_RSVDPARAM_36&&SNPS_RSVDPARAM_13) ? 0x578 : (E16&&SNPS_RSVDPARAM_15) ? 0x5b2 : (E25&&REF_CLK_78) ? 0x580 : SNPS_RSVDPARAM_39 ? 0x3E8 : SNPS_RSVDPARAM_35 ? (SNPS_RSVDPARAM_13 ? 0x546 : 0x550) : 0x540 " **Exists:** Always

5.2.9.140 VR_MII_MP_32G_RX_CNTX_CTRL0

- **Description:** VR MII Synopsys Multi protocol E32 PHY RX Contex Control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8092 : (APB3_16) ? (0x8092*0x2) : (0x8092*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(SNPS_RSVDPARAM_37||E32))



Table 5-601 Fields for Register: VR_MII_MP_32G_RX_CNTX_CTRL0

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:0	RX_CNTX_SEL_0	R/W	Receive context select along lane 0 of the PHY specifies the context that needs to be restored for the RX block on lane 0. This field drives the output port 'xpcs_rx0_cntx_sel_o[7:0]'. **Value After Reset:** RX_CNTX_SEL **Exists:** Always

5.2.9.141 VR_MII_MP_6G_MISC_CTRL2

- **Description:** VR MII Synopsys Multi-Protocol 6G PHY Miscellaneous Control 2 Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8093 : (APB3_16) ? (0x8093*0x2) : (0x8093*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&SSP)

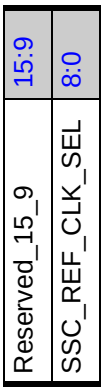


Table 5-602 Fields for Register: VR_MII_MP_6G_MISC_CTRL2

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8:0	SSC_REF_CLK_SEL	R/W	Spread Spectrum Reference Clock Shifting Control. This register field drives the output port 'xgxs_ssc_ref_clk_sel_o' of DWC_xpcs, which in turn is connected to 'ssc_ref_clk_sel' input port of the PHY. This helps in enabling non-standard oscillator frequencies to generate targeted MPLL output rates. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.142 VR_MII_MP_32G_REF_CLK_CTRL0

- **Description:** VR MII Synopsys Multi-protocol 32G Reference clock control 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8095 : (APB3_16) ? (0x8095*0x2) : (0x8095*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY)&&SNPS_XAUI_PHY&&E32

15:9	8	7	6	5	4	3	2	1	0
RESERVED_15_9	REF_RPT_CLK_EN	REFB_CLKDET_EN	REFA_CLKDET_EN	REFB_RAW_CLK_DIV2_EN	REFA_RAW_CLK_DIV2_EN	REFB_CLK_DIV2_EN	REFA_CLK_DIV2_EN	REFB_CLK_EN	REFA_CLK_EN

Table 5-603 Fields for Register: VR_MII_MP_32G_REF_CLK_CTRL0

Bits	Name	Memory Access	Description
15:9	RESERVED_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	REF_RPT_CLK_EN	R/W	Ref Repeat clock enable Selects the input reference clock for CML outputclocks. This field drives the output port 'xpcs_ref_repeat_clock_en_o' **Value After Reset:** 0x0 **Exists:** Always
7	REFB_CLKDET_EN	R/W	Refb Clock detect enable Enables the reference clock detection for refb **Value After Reset:** 0x0 **Exists:** Always
6	REFA_CLKDET_EN	R/W	Refa Clock detect enable Enables the reference clock detection for refa **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
5	REFB_RAW_CLK_DIV2_EN	R/W	Reference clock divider for refb in RAW PCS. When asserted the refb clock in RAW PCS gets divided by 2. This field drives the output port 'xpcs_refb_raw_clk_div2_en_o' **Value After Reset:** "(REF_CLK_78||REF_CLK_125) ? 0x0 : 0x1" **Exists:** Always
4	REFA_RAW_CLK_DIV2_EN	R/W	Reference clock divider for refa in RAW PCS. When asserted the refa clock in RAW PCS gets divided by 2. This field drives the output port 'xpcs_refa_raw_clk_div2_en_o' **Value After Reset:** "(REF_CLK_78||REF_CLK_125) ? 0x0 : 0x1" **Exists:** Always
3	REFB_CLK_DIV2_EN	R/W	Reference clock divider control for refb. When asserted the reference clock refa is divided by 2. This field drives the output port 'xpcs_refb_clk_div2_en_o'. **Value After Reset:** "REF_CLK_312 ? 0x1 : 0x0" **Exists:** Always
2	REFA_CLK_DIV2_EN	R/W	Reference clock divider control for refa. When asserted the reference clock refa is divided by 2. This field drives the output port 'xpcs_refa_clk_div2_en_o'. **Value After Reset:** "REF_CLK_312 ? 0x1 : 0x0" **Exists:** Always
1	REFB_CLK_EN	R/W	Reference clock enable for refa Enables the reference 'b' clock for the PHY This field drives the output port xpcs_refb_clk_en_o. **Value After Reset:** 0x0 **Exists:** Always
0	REFA_CLK_EN	R/W	Reference clock enable for refa Enables the reference 'a' clock for the PHY This field drives the output port xpcs_refa_clk_en_o. **Value After Reset:** 0x1 **Exists:** Always

5.2.9.143 VR_MII_MP_32G_REF_CLK_CTRL1

- **Description:** VR MII Synopsys Multi-Protocol 32G PHY Reference Clock Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8096 : (APB3_16) ? (0x8096*0x2) : (0x8096*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY)&&SNPS_XAUI_PHY&&E32

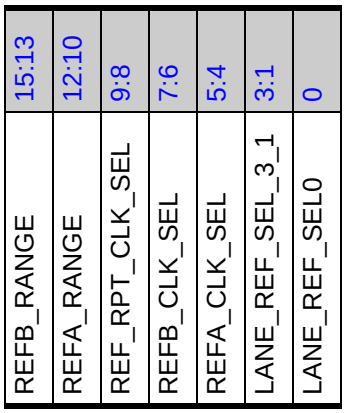


Table 5-604 Fields for Register: VR_MII_MP_32G_REF_CLK_CTRL1

Bits	Name	Memory Access	Description
15:13	REFB_RANGE	R/W	Reference clock frequency range for refb. Specifies the frequency range of the reference 'b' clock. This field drives the output port 'xpcs_refb_range_o'. **Value After Reset:** "REF_CLK_78 ? 0x3 :REF_CLK_125 ? 0x4 : 0x6" **Exists:** Always
12:10	REFA_RANGE	R/W	Reference clock frequency range for refa. Specifies the frequency range of the reference 'a' clock. This field drives the output port 'xpcs_refa_range_o'. **Value After Reset:** "REF_CLK_78 ? 0x3 :REF_CLK_125 ? 0x4 : 0x6" **Exists:** Always
9:8	REF_RPT_CLK_SEL	R/W	Repeat reference clock select. Selects the reference clock for CML clocks. This field drives the output port 'xpcs_ref_prt_clk_sel_o'. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
7:6	REFB_CLK_SEL	R/W	Reference clock 'b' Source select. Select the reference clock source for refa. ■ 2'b00: ref_alt0_clk_(p/m) ■ 2'b01: ref_pad_clk_(p/m) ■ 2'b10: ref_alt1_clk_(p/m) ■ 2'b11: Reserved. This field drives the output port 'xpcs_refb_clk_sel_o' **Value After Reset:** "(REF_CLK_SRC==1) ? 0x0 : (REF_CLK_SRC==0) ? 0x1 : 0x2" **Exists:** Always
5:4	REFA_CLK_SEL	R/W	Reference clock 'a' Source select Select the reference clock source for refa. ■ 2'b00 : ref_alt0_clk_(p/m) ■ 2'b01 : ref_pad_clk_(p/m) ■ 2'b10 : ref_alt1_clk_(p/m) ■ 2'b11 : Reserved. This field drives the output port 'xpcs_refa_clk_sel_o' **Value After Reset:** "(REF_CLK_SRC==1) ? 0x0 : (REF_CLK_SRC==0) ? 0x1 : 0x2" **Exists:** Always
3:1	LANE_REF_SEL_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	LANE_REF_SEL0	R/W	Lane0 reference clock selection. Selects the refa or refb reference clock. ■ 0: Selects REFA clock. ■ 1: Selects REFB clocks. This field drives the output port 'xpcs_lane_ref_sel_o. For all Ethernet modes, always select REFA clock' **Value After Reset:** 0x0 **Exists:** Always

5.2.9.144 VR_MII_MP_12G_VCO_CAL_REF0

- **Description:** VR MII Synopsys Multi-protocol 12G PHY VCO Calibration Reference Register 0
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x8096 : (APB3_16) ? (0x8096*0x2) : (0x8096*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&E12)

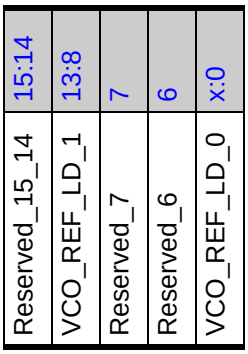


Table 5-605 Fields for Register: VR_MII_MP_12G_VCO_CAL_REF0

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13:8	VCO_REF_LD_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6	Reserved_6	R	Reserved **Value After Reset:** 0x0 **Exists:** !SNPS_RSVDPARAM_22

Bits	Name	Memory Access	Description
x:0	VCO_REF_LD_0	R/W	Rx VCO calibration reference load value -lane 0 This field controls the internal calibration registers of the PHY and is used to perform Rx VCO calibration on lane 0. This field drives the output port 'xpcs_rx0_ref_ld_val_o[5:0]'. **Value After Reset:** "SNPS_RSVDPARAM_39&&SNPS_RSVDPARAM_13 ? 0x14 : SNPS_RSVDPARAM_39&&REF_CLK_20 ? 0x8 : !SNPS_RSVDPARAM_35 ? (REF_CLK_78 ? 0x15 : 0x2A): (REF_CLK_78||(SNPS_RSVDPARAM_35&&!SNPS_RSVDPARAM_13)) ? 0x11 : SNPS_RSVDPARAM_13 ? 0x1B : 0x22" **Exists:** Always **Range Variable[x]:** "SNPS_RSVDPARAM_22 ? 7 : 6" - 1

5.2.9.145 VR_MII_MP_16G_25G_VCO_CAL_REF0

- **Description:** VR MII Synopsys Multi-protocol 16G/25G PHY VCO Calibration Reference Register 0
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8096 : (APB3_16) ? (0x8096*0x2) : (0x8096*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E16||E25)&&!SNPS_RSVDPARAM_37

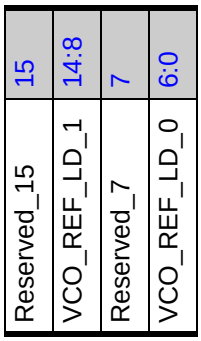


Table 5-606 Fields for Register: VR_MII_MP_16G_25G_VCO_CAL_REF0

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:8	VCO_REF_LD_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7	Reserved_7	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
6:0	VCO_REF_LD_0	R/W	Rx VCO calibration reference load value -lane 0 This field controls the internal calibration registers of the PHY and is used to perform Rx VCO calibration on lane 0. This field drives the output port 'xpcs_rx0_ref_ld_val_o[6:0]'. **Value After Reset:** "(SNPS_RSVDPARAM_36&&KX_ONLY&&SNPS_RSVDPARAM_13) ? 0xE :(E16&&SNPS_RSVDPARAM_15) ? 0x7 : REF_CLK_78 ? (E25 ? 0xB :0x15) : (E25 ? 0x15 : 0x2A)" **Exists:** Always

5.2.9.146 VR_MII_MP_32G_LN_LINK_CTRL

- **Description:** VR MII Synopsys Multi protocol E32 PHY Lane Link Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8097 : (APB3_16) ? (0x8097*0x2) : (0x8097*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&(SNPS_RSVDPARAM_37||E32)

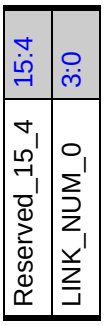


Table 5-607 Fields for Register: VR_MII_MP_32G_LN_LINK_CTRL

Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:0	LINK_NUM_0	R/W	Link number along lane 0. It determines which link the current lane belongs to. This field drives the output port 'xpcs_lane0_link_num_o'. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.147 VR_MII_MP_6G_LVL_CTRL

- **Description:** VR MII Synopsys Multi-protocol 6G Level Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8098 : (APB3_16) ? (0x8098*0x2) : (0x8098*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&SSP)

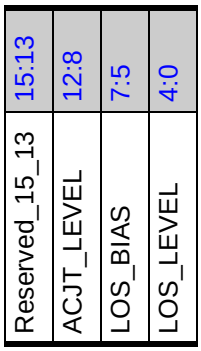


Table 5-608 Fields for Register: VR_MII_MP_6G_LVL_CTRL

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12:8	ACJT_LEVEL	R/W	Receiver Sensitivity Level Control. This field controls the sensitivity level for the receiver. For more information, see Synopsys DesignWare Cores PCIe2 PHY For TSMC 65nm GP (Flip-Chip) Databook. **Value After Reset:** 0x0 **Exists:** Always
7:5	LOS_BIAS	R/W	LOS threshold control. This field sets the LOS detection threshold level. A positive binary bit setting change results in a +15 mVp incremental change in the LOS threshold. A negative binary setting change results in a -15 mVp incremental change in the LOS threshold. The setting 3'b000 is reserved and must not be used. **Value After Reset:** 0x1 **Exists:** Always

Bits	Name	Memory Access	Description
4:0	LOS_LEVEL	R/W	Loss-of-Signal Detector Sensitivity Control This field sets the sensitivity level for the loss-of-signal detector. For more information, see Synopsys DesignWare Cores PCIe2 PHY For TSMC 65nm GP (Flip-Chip) Databook. **Value After Reset:** 0x9 **Exists:** Always

5.2.9.148 VR_MII_MP_12G_16G_25G_MISC_STS

- **Description:** VR MII Synopsys Multi-protocol 12G/16G/25G PHY Miscellaneous Status Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8098 : (APB3_16) ? (0x8098*0x2) : (0x8098*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16||E25||E32)

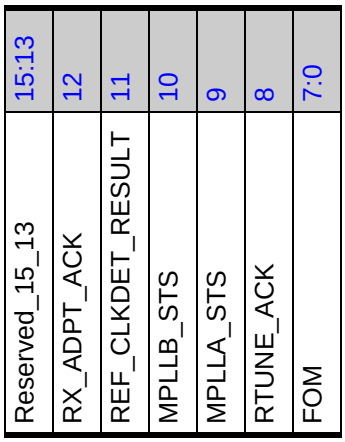


Table 5-609 Fields for Register: VR_MII_MP_12G_16G_25G_MISC_STS

Bits	Name	Memory Access	Description
15:13	Reserved_15_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	RX_ADPT_ACK	R	Receive Adaptation Acknowledgment from PHY. This field reflects the value of rpcs_krx_adapt_ack_i port. **Value After Reset:** 0x0 **Exists:** Always
11	REF_CLKDET_RESULT	R	**For Multi-protocol 16G/25G PHY configurations:** Reference Clock Detect enable This field is driven by the input xpcs_ref_clkdet_result_i. **For other configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
10	MPLLB_STS	R	**For E32 Configurations:** Reserved **For E12/E16/E25 Configurations :** Status of MPLLB from PHY This bit denotes the value of xpcs_mpllb_state_i input. This valid is not valid for Multi-protocol 8G PHY. **Value After Reset:** "E16&&SNPS_RSVDPARAM_15 ? 0x1 : 0x0" **Exists:** Always
9	MPLLA_STS	R	**For E32 Configurations:** Reserved **For E12/E16/E25 Configurations:** Status of MPLLA from PHY. This bit denotes the value of xpcs_mplla_state_i input **Value After Reset:** "(E16&&SNPS_RSVDPARAM_15)||E32 ? 0x0 : 0x1" **Exists:** Always
8	RTUNE_ACK	R	Acknowledgment for Resistor Tune Request This bit denotes the value of 'xgxs_rtune_ack_i' input. **Value After Reset:** 0x0 **Exists:** Always
7:0	FOM	R	Figure of Merit from the PHY. This field reflects the value of rpcs_krx_fom_i port. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.149 VR_MII_MP_6G_MISC_CTRL0

- **Description:** VR MII Synopsys Multi-Protocol 6G Miscellaneous Control 0 Register.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8099 : (APB3_16) ? (0x8099*0x2) : (0x8099*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&SSP)

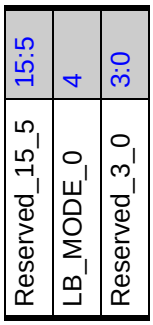


Table 5-610 Fields for Register: VR_MII_MP_6G_MISC_CTRL0

Bits	Name	Memory Access	Description
15:5	Reserved_15_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	LB_MODE_0	R/W	Analog Serial Loopback Mode on Lane 0. ■ 1: The PHY enables the data from the Tx predriver to receiver slices on Lane 0. When this bit is 1, the loss-of-signal is not detected. **Value After Reset:** 0x0 **Exists:** Always
3:0	Reserved_3_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.9.150 VR_MII_MP_12G_16G_25G_MISC_CTRL1

- **Description:** VR MII Synopsys Multi-protocol 12G/16G/25G PHY Miscellaneous Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x8099 : (APB3_16) ? (0x8099*0x2) : (0x8099*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16||E25||E32)

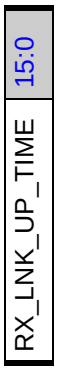


Table 5-611 Fields for Register: VR_MII_MP_12G_16G_25G_MISC_CTRL1

Bits	Name	Memory Access	Description
15:0	RX_LNK_UP_TIME	R/W	Wait Time before PLL Re-initialization This field determines the number of 'clk_csr_i' clock cycles that can elapse before the Rx link is established (after DWC_xpcs has reached POWER_GOOD state). This field is relevant only if 'PLL_CTRL' bit of VR_MII_MP_12G_16G_25G_MISC_CTRL0 Register is programmed to 1. If the receive link is not up (or if frame lock does not happen during CL72 Training or if AN page is not received), within this specified interval, DWC_xpcs programs the PHY Rx power state to P1 and then bring it back to P0 (thus re-initializing the PHY PLL). **Value After Reset:** 0xffff **Exists:** Always

5.2.9.151 VR_MII_MP_6G_MISC_CTRL1

- **Description:** VR MII Synopsys Multi-protocol 6G Miscellaneous Control 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x809A : (APB3_16) ? (0x809A*0x2) : (0x809A*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&SSP)

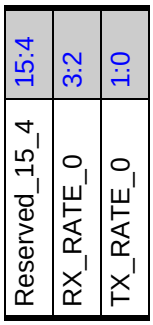


Table 5-612 Fields for Register: VR_MII_MP_6G_MISC_CTRL1

Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:2	RX_RATE_0	R/W	Rx Baud Rate Lane 0. The baud rate selection is as follows: ■ 00: mpll_baud_clk ■ 01: mpll_baud_clk/2 ■ 10: mpll_baud_clk/4 ■ 11: Not supported **Value After Reset:** "((RXAUI_DEFAULT) ? 0x0 :((SINGLE_LANE_CFG) ? 0x2 : 0x1))" **Exists:** Always
1:0	TX_RATE_0	R/W	Tx Baud Rate Lane 0. The baud rate selection is as follows: ■ 00: mpll_baud_clk ■ 01: mpll_baud_clk/2 ■ 10: mpll_baud_clk/4 ■ 11: Not supported **Value After Reset:** "((RXAUI_DEFAULT) ? 0x0 :((SINGLE_LANE_CFG) ? 0x2 : 0x1))" **Exists:** Always

5.2.9.152 VR_MII_MP_12G_16G_25G_EEE_CTRL

- **Description:** VR MII Synopsys Multi-protocol 12G/16G/25G PHY EEE Control Register
  This register is present only in configurations with QSGMII_EN and EEE enabled.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x809A : (APB3_16) ? (0x809A*0x2) : (0x809A*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:**
  (BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E12||E16||E25||E32)&&QSGMII_EN&&EEE_EN)

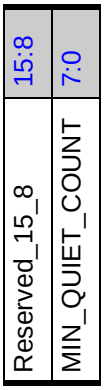


Table 5-613 Fields for Register: VR_MII_MP_12G_16G_25G_EEE_CTRL

Bits	Name	Memory Access	Description
15:8	Reserved_15_8	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
7:0	MIN_QUIET_COUNT	R/W	Minimum Rx Quiet Duration This field determines the minimum duration for which Rx LPI FSM remains in the RX_QUIET state.DWC_xpcs implements a vendor-specific 'MIN QUIET counter' in the RX_QUIET state of LPI state machine. This counter (running at 125 MHz in QSGMII Mode) starts as soon as RX_QUIET state is reached and it stops when it reaches a value equal to 8 times the value mentioned in this field. DWC_xpcs start monitoring for 'signal_detect' (in QSGMII Mode) only after this counter reaches its maximum value (8 times MIN_QUIET_COUNT). **Note:** In the QSGMII mode, this counter is activated only if SUPRESS_LOS_DET bit of VR_MII_DBG_CTRL register is set to 1. If this bit is 0, behavior of Rx EEE FSM is as per IEEE specification. **Value After Reset:** 0xa **Exists:** Always

5.2.9.153 VR_MII_MP_12G_16G_25G_SRAM

- **Description:** VR MII Synopsys Multi-protocol 12G/16G/25G PHY SRAM Register.
  Note: This is a special register which can be read/written even before DWC_xpcs completes its reset/initialization sequence.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x809B : (APB3_16) ? (0x809B*0x2) : (0x809B*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (E12||E16||E25||E32)&&SNPS_XAUI_PHY&&BASE1000X_ONLY

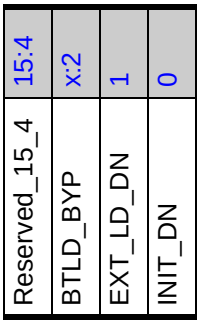


Table 5-614 Fields for Register: VR_MII_MP_12G_16G_25G_SRAM

Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
x:2	BTLD_BYP	* Varies	SRAM Bootload Bypass. This field drives the xpcs_sram_bootload_byp_o port. **Note:** This field is present only for Multi-protocol 25G/8G/32G PHY configurations. **Value After Reset:** "E32&&!SNPS_RSVDPARAM_3 ? 0x1 : 0x0" **Exists:** Always **Range Variable[x]:** "SNPS_RSVDPARAM_37 ? 2 : 1" + 1 **Memory Access:** "(E25||SNPS_RSVDPARAM_39||E32) ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
1	EXT_LD_DN	R/W	SRAM External Loading Done. This field drives the output port 'xpcs_sram_ext_ld_done_o' of DWC_xpcs. Software must set this bit to 1, after PHY has completed its SRAM initialization and after software has completed the (optional) external loading of the SRAM attached to the PHY. The PHY completes its initialization sequence only after this register bit is set to 1. **Note:** ■ After the software sets this bit as 1, it must remain 1 throughout normal operation. ■ This bit is cleared during standard soft reset (RST) ■ This bit does not get cleared during vendor soft reset (VR_RST). ■ When software asserts VR_RST, this bit must be cleared prior to that (in case, software wishes to alter the SRAM contents initialized by the PHY). **Value After Reset:** 0x0 **Exists:** Always **Testable:** untestable
0	INIT_DN	R	SRAM Initialization Done. This register field reflects the value of the 'xpcs_sram_init_done_i' port of DWC_xpcs. Whenever PHY gets reset, the boot-loader in the Raw PCS loads the code from the internal look-up table to the SRAM. After this SRAM initialization is complete, PHY asserts this signal. After the SRAM initialization is done, software can optionally change the SRAM contents using the CR port interface by addressing the SRAM address space in the register map of the PHY. During power-on-reset or whenever RST or VR_RST register bits are programmed to 1, the PHY also gets reset. Therefore, the software must first poll this register bit to become 1. Thereafter, it must set the 'EXT_LD_DN' bit to 1. Only then the PHY and the DWC_xpcs proceeds to complete its reset sequence. **Note:** Though default value of this register bit is given as 1, this is the value provided by the PHY after the completion of SRAM initialization. **Value After Reset:** 0x1 **Exists:** Always

5.2.9.154 VR_MII_MP_16G_25G_MISC_CTRL2

- **Description:** VR MII Synopsys Multi-protocol 16G/25G PHY Miscellaneous Control 2 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x809C : (APB3_16) ? (0x809C*0x2) : (0x809C*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_XAUI_PHY&&(E16||E25||SNPS_RSVDPARAM_39||E32)

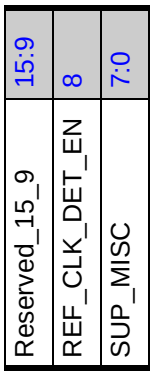


Table 5-615 Fields for Register: VR_MII_MP_16G_25G_MISC_CTRL2

Bits	Name	Memory Access	Description
15:9	Reserved_15_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
8	REF_CLK_DET_EN	* Varies	**Non E32 configurations:** Reference Clock Detect enable. This field drives the output port xpcs_ref_clkdet_en_o. **E32 configurations:** Reserved **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "!E32 ? \"read-write\" : \"read-only\" "

Bits	Name	Memory Access	Description
7:0	SUP_MISC	* Varies	**Non SNPS_RSVDPARAM_37 configurations:** Support Miscellaneous Controls. This field drives the output port xpcs_sup_misc_o[7:0]. **For SNPS_RSVDPARAM_37 configurations:** Reserved **Value After Reset:** "(E25||E32) ? 0x0 : SNPS_RSVDPARAM_39&&SNPS_RSVDPARAM_13 ? 0x2 : 0x1" **Exists:** (!SNPS_RSVDPARAM_37) **Memory Access:** "!SNPS_RSVDPARAM_37 ? \"read-write\" : \"read-only\""

5.2.9.155 VR_MII_MP_32G_CNTX_CTRL

- **Description:** VR XS or PMA Synopsys Multi-protocol 32G PHY Context Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x809D : (APB3_16) ? (0x809D*0x2) : (0x809D*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY)&&SNPS_XAUI_PHY&&E32

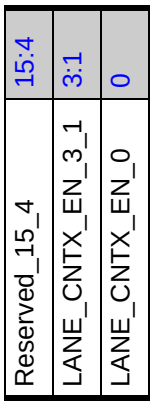


Table 5-616 Fields for Register: VR_MII_MP_32G_CNTX_CTRL

Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:1	LANE_CNTX_EN_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	LANE_CNTX_EN_0	R/W	Lane Context Enable Lane0. This field drives the output port 'xpcs_lane_cntx_en_o'. **Value After Reset:** 0x1 **Exists:** Always

5.2.9.156 VR_MII_MP_32G_MISC_STS1

- **Description:** VR XS or PMA Synopsys Multi-protocol 32G PHY Miscellaneous Status 1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x809E : (APB3_16) ? (0x809E*0x2) : (0x809E*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY)&&SNPS_XAUI_PHY&&E32

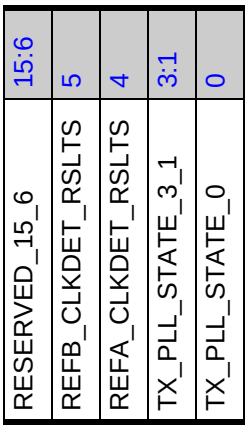


Table 5-617 Fields for Register: VR_MII_MP_32G_MISC_STS1

Bits	Name	Memory Access	Description
15:6	RESERVED_15_6	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
5	REFB_CLKDET_RSLTS	R	Reference clock detection result for refb. Indicates the presence of the reference clock. This field reflects the value of the input port 'xpcs_refb_clkdet_result_i' **Value After Reset:** 0x0 **Exists:** Always
4	REFA_CLKDET_RSLTS	R	Reference clock detection result for refa. Indicates the presence of the reference clock. This field reflects the value of the input port 'xpcs_refa_clkdet_result_i' **Value After Reset:** 0x0 **Exists:** Always
3:1	TX_PLL_STATE_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	TX_PLL_STATE_0	R	TX PLL State Indicator. Indicates the state of PLL for the lane0. This field reflects the value of the input port 'xpcs_tx_pll_state_i'. **Value After Reset:** 0x1 **Exists:** Always

5.2.9.157 VR_MII_SNPS_CR_CTRL

- **Description:** VR MII Synopsys PHY CR Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80A0 : (APB3_16) ? (0x80A0*0x2) : (0x80A0*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&!SNPS_RSVDPARAM_37)

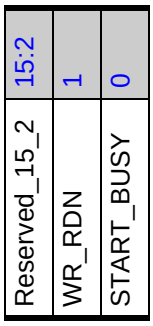


Table 5-618 Fields for Register: VR_MII_SNPS_CR_CTRL

Bits	Name	Memory Access	Description
15:2	Reserved_15_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1	WR_RDN	R/W	Write or Read Indicator This bit indicates whether a read or write operation is to be performed to the Synopsys PHY registers: ■ 0: Read ■ 1: Write **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	START_BUSY	R/W	Start CR Port Access or Busy Indicator (WS,SC Type) This bit indicates if CR port access is in progress: ■ 0: CR port not busy ■ 1: CR port busy The sequence is: 1. The host sets this bit to start a read or write transfer through the CR port to the Synopsys PHY registers. 2. This bit remains set during the CR port access. 3. The DWC_xpcs clears this bit when the CR port access is complete. Dependencies: The host must read this bit as 0 before writing to any of the following registers: ■ VR_MII_SNPS_CTRL Register (this register) ■ VR_MII_SNPS_CR_ADDR Register ■ VR_MII_SNPS_CR_DATA Register During read, the data from the CR port interface is placed into the VR_MII_SNPS_CR_DATA Register. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.158 VR_MII_SNPS_CR_ADDR

- **Description:** VR MII Synopsys PHY CR Address Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80A1 : (APB3_16) ? (0x80A1*0x2) : (0x80A1*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&!SNPS_RSVDPARAM_37)

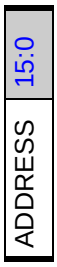


Table 5-619 Fields for Register: VR_MII_SNPS_CR_ADDR

Bits	Name	Memory Access	Description
15:0	ADDRESS	R/W	CR Port Address This field indicates the address of the register to be accessed through the CR port. Dependency: * The host must not change this field when Bit 0 of the VR_MII_SNPS_CR_CTRL Register is set. * This field must be written before writing to any of the following registers: ■ VR_MII_SNPS_CR_CTRL Register ■ VR_MII_SNPS_CR_ADDR Register (this register) ■ VR_MII_SNPS_CR_DATA Register **Value After Reset:** 0x0 **Exists:** Always

5.2.9.159 VR_MII_SNPS_CR_DATA

- **Description:** VR MII Synopsys CR Data Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80A2 : (APB3_16) ? (0x80A2*0x2) : (0x80A2*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY&&SNPS_XAUI_PHY&&!SNPS_RSVDPARAM_37)

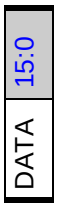


Table 5-620 Fields for Register: VR_MII_SNPS_CR_DATA

Bits	Name	Memory Access	Description
15:0	DATA	R/W	CR Port Data This field contains the data for CR Port access. ■ During a write operation, the data in this register is written to the Synopsys PHY register pointed by the CR Port Address. ■ During a read operation, the data read from the Synopsys PHY is written to this register. The content becomes valid only when the DWC_xpcs clears Bit 0 of the VR_MII_SNPS_CR_CTRL Register. The host must not change this field when Bit 0 of VR_MII_SNPS_CR_CTRL Register is set. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.160 VR_MII_MP_25G_RX_WIDTH_CTRL

- **Description:** VR MII Synopsys Multi-protocol 25G PHY Rx Width Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80B0 : (APB3_16) ? (0x80B0*0x2) : (0x80B0*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&((E25&&!SNPS_RSVDPARAM_37)||E32)

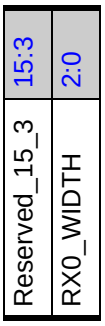


Table 5-621 Fields for Register: VR_MII_MP_25G_RX_WIDTH_CTRL

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	RX0_WIDTH	R/W	Rx PHY Width -lane 0. This field drives the xpcs_rx0_width_o output port and controls the width of PHY Rx lane 0. **Value After Reset:** 0x1 **Exists:** Always

5.2.9.161 VR_MII_MP_25G_RX_AFE_RATE_CTRL

- **Description:** VR MII Synopsys Multi-protocol 25G PHY Rx AFE Rate Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80B1 : (APB3_16) ? (0x80B1*0x2) : (0x80B1*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&E25&&!SNPS_RSVDPARAM_37

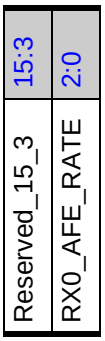
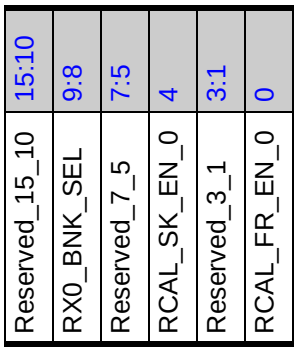


Table 5-622 Fields for Register: VR_MII_MP_25G_RX_AFE_RATE_CTRL

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	RX0_AFE_RATE	R/W	Rx Equalization AFE Rate -lane 0. This field drives the xpcs_rx0_eq_afe_rate_o output port. **Value After Reset:** 0x7 **Exists:** Always

5.2.9.162 VR_MII_MP_25G_RX_RECAL_CTRL

- **Description:** VR MII Synopsys Multi-protocol 25G PHY Rx Re-calibration Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80B2 : (APB3_16) ? (0x80B2*0x2) : (0x80B2*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&(E25||E32)



15:10	9:8	7:5	4	3:1	0
Reserved_15_10	RX0_BNK_SEL	Reserved_7_5	RCAL_SK_EN_0	Reserved_3_1	RCAL_FR_EN_0

Table 5-623 Fields for Register: VR_MII_MP_25G_RX_RECAL_CTRL

Bits	Name	Memory Access	Description
15:10	Reserved_15_10	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
9:8	RX0_BNK_SEL	R/W	Rx Recalibration Bank Select-lane 0. This field drives the xpcs_rx0_recal_bank_sel_o output port and selects the bank to store the Rx duty cycle correction and IQ correction calibration values **Value After Reset:** 0x0 **Exists:** Always
7:5	Reserved_7_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	RCAL_SK_EN_0	R/W	Rx Skip Re-calibration -lane 0 This field drives the xpcs_rx_recal_skip_en_o[0] port. **Value After Reset:** 0x0 **Exists:** Always
3:1	Reserved_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	RCAL_FR_EN_0	R/W	Rx Force Re-calibration -lane 0 This field drives the xpcs_rx_recal_force_en_o[0] port. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.163 VR_MII_MP_25G_RX_DCC_CTRL

- **Description:** VR MII Synopsys Multi-protocol 25G PHY Rx DCC Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80B3 : (APB3_16) ? (0x80B3*0x2) : (0x80B3*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&E25&&!SNPS_RSVDPARAM_37



Table 5-624 Fields for Register: VR_MII_MP_25G_RX_DCC_CTRL

Bits	Name	Memory Access	Description
15:4	Reserved_15_4	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
3:0	RX0_DCC	R/W	Rx Internal Clock Duty Cycle Correction Control Range for lane 0. **Value After Reset:** "(SNPS_RSVDPARAM_36) ? 0xC : 0xB" **Exists:** Always

5.2.9.164 VR_MII_MP_25G_RX_VCO_CFG0

- **Description:** VR MII Synopsys Multi-protocol 25G PHY Rx CDR VCO Configuration 0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80B4 : (APB3_16) ? (0x80B4*0x2) : (0x80B4*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&E25&&!SNPS_RSVDPARAM_37

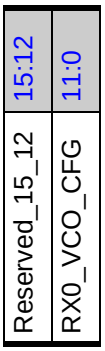


Table 5-625 Fields for Register: VR_MII_MP_25G_RX_VCO_CFG0

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:0	RX0_VCO_CFG	R/W	Rx CDR VCO configuration Bus -lane 0. This field drives the xpcs_rx0_cdr_vco_cfg_o port and controls the biasing, temperature compensation, DAC settings and frequency range of Rx CDR VCO. **Value After Reset:** "SNPS_RSVDPARAM_36 ? 0x300 : 0x100" **Exists:** Always

5.2.9.165 VR_MII_MP_25G_RX_LF_TRSHLD

- **Description:** VR MII Synopsys Multi-protocol 25G PHY RX Low Frequency Threshold Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80B8 : (APB3_16) ? (0x80B8*0x2) : (0x80B8*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&E25&&!SNPS_RSVDPARAM_37

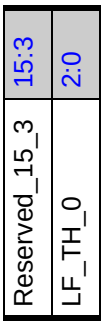


Table 5-626 Fields for Register: VR_MII_MP_25G_RX_LF_TRSHLD

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	LF_TH_0	R/W	Rx Low Frequency Signal Detection Threshold lane 0. This field drives the xpcs_rx0_sigdet_lf_th_o port. **Value After Reset:** "(SNPS_RSVDPARAM_36) ? 0x1 : 0x2" **Exists:** Always

5.2.9.166 VR_MII_MP_25G_RX_HF_TRSHLD

- **Description:** VR MII Synopsys Multi-protocol 25G PHY RX High Frequency Threshold Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80B9 : (APB3_16) ? (0x80B9*0x2) : (0x80B9*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&E25&&!SNPS_RSVDPARAM_37

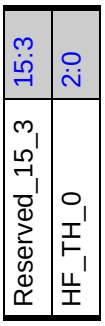


Table 5-627 Fields for Register: VR_MII_MP_25G_RX_HF_TRSHLD

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	HF_TH_0	R/W	Rx High Frequency Signal Detection Threshold lane 0. This field drives the xpcs_rx0_sigdet_hf_th_o port. **Value After Reset:** 0x2 **Exists:** Always

5.2.9.167 VR_MII_C16G_RX_EQ_AFE_CTRL0

- **Description:** VR MII Synopsys Consumer 16G PHY RX Equalization AFE Control0 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x80BA : (APB3_16) ? (0x80BA*0x2) : (0x80BA*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_RSVDPARAM_36

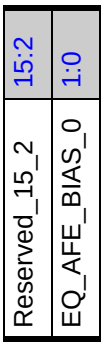


Table 5-628 Fields for Register: VR_MII_C16G_RX_EQ_AFE_CTRL0

Bits	Name	Memory Access	Description
15:2	Reserved_15_2	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
1:0	EQ_AFE_BIAS_0	R/W	RX Equalization AFE input stage bias current control lane 0. This field drives the xpcs_rx0_eq_afe_bias_o port. **Value After Reset:** 0x1 **Exists:** Always

5.2.9.168 VR_MII_C16G_RX_EQ_AFE_CTRL1

- **Description:** VR MII Synopsys Consumer 16G PHY RX Equalization AFE Control1 Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80BB : (APB3_16) ? (0x80BB*0x2) : (0x80BB*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&SNPS_RSVDPARAM_36

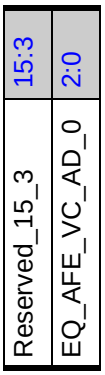


Table 5-629 Fields for Register: VR_MII_C16G_RX_EQ_AFE_CTRL1

Bits	Name	Memory Access	Description
15:3	Reserved_15_3	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
2:0	EQ_AFE_VC_AD_0	R/W	RX Equalization Attenuator output common-mode level control lane 0. This field drives the xpcs_rx0_eq_afe_vcm_adj_o port. **Value After Reset:** 0x3 **Exists:** Always

5.2.9.169 VR_MII_DIG_CTRL2

- **Description:** VR MII MMD Digital Control2 Register
  This register is present only when the DWC_xpcs is configured in 1000BASEX-Only PCS mode.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80E1 : (APB3_16) ? (0x80E1*0x2) : (0x80E1*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY)

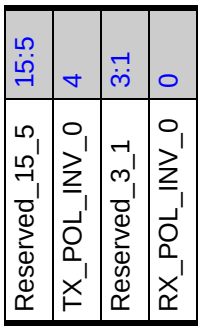


Table 5-630 Fields for Register: VR_MII_DIG_CTRL2

Bits	Name	Memory Access	Description
15:5	Reserved_15_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	TX_POL_INV_0	R/W	Tx Polarity Invert on Lane 0 When set to 1, this bit reverses the data polarity on the Tx differential lines of Lane 0. **Value After Reset:** 0x0 **Exists:** Always
3:1	Reserved_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	RX_POL_INV_0	R/W	Rx Polarity Invert on Lane 0 ■ 1: The bits within this field indicate that the data received on Rx serial line is inverted on Lane 0. This reverses the polarity on the data received from the PHY core. The value 1 indicates that Rx data on Lane 0 is inverted. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.170 VR_MII_DIG_ERRCNT_SEL

- **Description:** VR MII MMD Digital Error Count Select Register
  This register is present only when the DWC_xpcs is configured in 1000BASEX-Only PCS mode.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80E2 : (APB3_16) ? (0x80E2*0x2) : (0x80E2*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** (BASE1000X_ONLY)

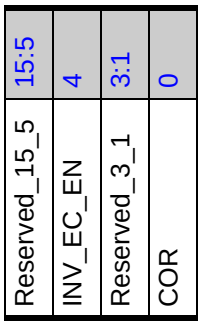


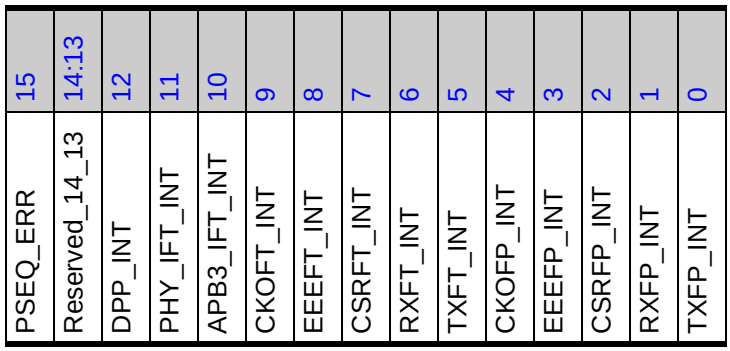
Table 5-631 Fields for Register: VR_MII_DIG_ERRCNT_SEL

Bits	Name	Memory Access	Description
15:5	Reserved_15_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
4	INV_EC_EN	R/W	Invalid Code Group Error Counter Enable When this bit is set, the counting of invalid code group errors is enabled. ■ 0: The counting of errors is disabled ■ 1: The counting of errors is enabled For information about the fields containing the number of errors counted, see VR_MII_ICG_ERRCNT1 Register. **Value After Reset:** 0x0 **Exists:** Always
3:1	Reserved_3_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	COR	R/W	Clear on Read When this bit is set and the host reads any error counter, that counter is cleared after the read cycle. ■ 0: Normal operation ■ 1: Clear any error counter that is read **Value After Reset:** 0x0 **Exists:** Always

5.2.9.171 VR_MII_SFTY_UE_INTR0

- **Description:** VR MII MMD Uncorrectable Safety Error Interrupt Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80F0 : (APB3_16) ? (0x80F0*0x2) : (0x80F0*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&DWC_XPCS_ASP



15	14:13	12	11	10	9	8	7	6	5	4	3	2	1	0
PSEQ_ERR	Reserved_14_13	DPP_INT	PHY_IFT_INT	APB3_IFT_INT	CKOFT_INT	EEEFT_INT	CSRFT_INT	RXFT_INT	TXFT_INT	CKOFP_INT	EEEFP_INT	CSRFP_INT	RXFP_INT	TXFP_INT

Table 5-632 Fields for Register: VR_MII_SFTY_UE_INTR0

Bits	Name	Memory Access	Description
15	PSEQ_ERR	R	Error in power-up sequence FSM This bit is read as 1 when the power-up sequence FSM is stuck in a state other than 'POWER_GOOD'. This field is present only for Synopsys PHY configurations. **Note:** Whenever, power-up sequence FSM is any state other than 'POWER_GOOD', read to any register returns bit[15] as 1 and bits[4:2] indicate power-up sequence FSM state (PSEQ_STAE field of VR_MII_DIG_STS register). **Value After Reset:** 0x0 **Exists:** Always
14:13	Reserved_14_13	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
12	DPP_INT	R/W	Datapath Parity Error Interrupt This bit gets set automatically when the safety feature in DWC_xpcs detects a parity error in the receive data-path. When this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always

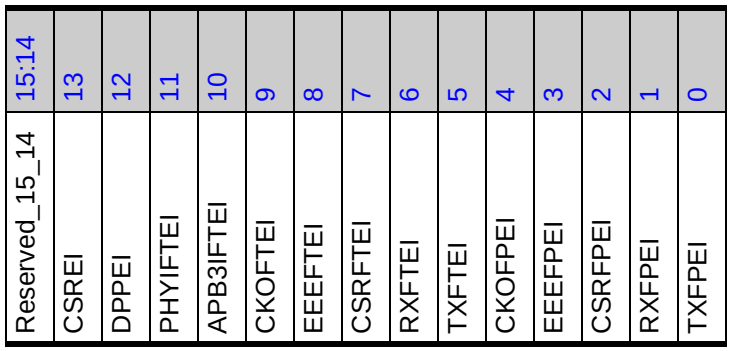
Bits	Name	Memory Access	Description
11	PHY_IFT_INT	* Varies	PHY Interface Time-out Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects a time-out Error at the PHY interface. When this bit is set, it remains high till software clears it by writing 0. This bit is valid only for configurations with Synopsys PHYs. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SNPS_XAUI_PHY ? \"read-write\" : \"read-only\""
10	APB3_IFT_INT	* Varies	APB3/MCI Interface Time-out Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects a time-out Error at the APB3/MCI interface. When this bit is set, it remains high till software clears it by writing 0. This bit is not valid for MDIO configurations. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "MDIO_INTF ? \"read-only\" :\"read-write\""
9	CKOFT_INT	* Varies	CKO Alive FSM Time-out Error Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects a parity error in FSMs operating in clk_cko_alive_i domain. When this bit is set, it remains high till software clears it by writing 0. This bit is valid only for configurations with Synopsys multi-protocol 6G PHY. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SSP ? \"read-write\" : \"read-only\""
8	EEEFT_INT	* Varies	EEE FSM Time-out Error Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects a time-out error in FSMs operating in clk_eee_i domain. When this bit is set, it remains high till software clears it by writing 0. This bit is valid only for configurations with EEE_EN = Enabled. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "EEE_EN ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
7	CSRFT_INT	R/W	CSR FSM Time-out Error Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects a parity Error in FSMs operating in CSR clock domain. When this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always
6	RXFT_INT	R/W	Rx FSM Time-out Error Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects a time-out Error in FSMs operating in Rx clock domain. When this bit is set, it remains high till software clears it by writing 0 **Value After Reset:** 0x0 **Exists:** Always
5	TXFT_INT	R/W	Tx FSM Time-out Error Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects a Time-out Error in FSMs operating in Tx clock domain. When this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always
4	CKOFP_INT	* Varies	CKO Alive FSM Parity Error Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects a parity Error in FSMs operating in clk_cko_alive_i domain. When this bit is set, it remains high till software clears it by writing 0. When this bit is cleared by software, it remains 0 even if the parity error continues to exist. This bit can get set again only if a new parity error is seen after the current parity error subsides. This bit is valid only for configurations with Synopsys multi-protocol 6G PHY. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SSP ? \"read-write\" : \"read-only\""

Bits	Name	Memory Access	Description
3	EEEFP_INT	* Varies	EEE FSM Parity Error Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects a parity Error in FSMs operating in clk_eee_i domain. When this bit is set, it remains high till software clears it by writing 0. When this bit is cleared by software, it remains 0 even if the parity error continues to exist. This bit can get set again only if a new parity error is seen after the current parity error subsides. This bit is valid only for configurations with EEE_EN = Enabled. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "EEE_EN ? \"read-write\" : \"read-only\""
2	CSRFP_INT	R/W	CSR FSM Parity Error Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects a parity Error in FSMs operating in CSR clock domain. When this bit is set, it remains high till software clears it by writing 0. When this bit is cleared by software, it remains 0 even if the parity error continues to exist. This bit can get set again only if a new parity error is seen after the current parity error subsides. **Value After Reset:** 0x0 **Exists:** Always
1	RXFP_INT	R/W	Rx FSM Parity Error Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects a parity Error in FSMs operating in Rx clock domain. When this bit is set, it remains high till software clears it by writing 0. When this bit is cleared by software, it remains 0 even if the parity error continues to exist. This bit can get set again only if a new parity error is seen after the current parity error subsides. **Value After Reset:** 0x0 **Exists:** Always
0	TXFP_INT	R/W	Tx FSM Parity Error Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects a parity Error in FSMs operating in Tx clock domain. When this bit is set, it remains high till software clears it by writing 0. When this bit is cleared by the software, it remains 0 even if the parity error continues to exist. This bit can get set again only if a new parity error is seen after the current parity error subsides. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.172 VR_MII_SFTY_ERR_INJ

- **Description:** VR MII MMD Safety Error Injection Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x80F1 : (APB3_16) ? (0x80F1*0x2) : (0x80F1*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&DWC_XPCS_ASP



15:14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved_15_14	CSREI	DPPEI	PHYIFTEI	APB3IFTEI	CKOFTEI	EEEFTEI	CSRFTEI	RXFTEI	TXFTEI	CKOFPEI	EEEFPEI	CSRFPEI	RXFPEI	TXFPEI

Table 5-633 Fields for Register: VR_MII_SFTY_ERR_INJ

Bits	Name	Memory Access	Description
15:14	Reserved_15_14	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
13	CSREI	R/W	CSR Error Injection Whenever this bit is set to 1, DWC_xpcs injects error into the CSR protection feature. Consequent CSR_INT interrupt gets set. **Note:** This bit self-clears after the error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always
12	DPPEI	R/W	Datapath Parity Error Inject ■ 1: DWC_xpcs injects parity error in its data-path. **Note:** This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
11	PHYIFTEI	* Varies	PHY Interface Time-out Error Injection Enable This bit can set to 1 to inject time-out error at PHY (CR port) interface. This bit is valid only for configurations with Synopsys PHYs. **Note:** This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SNPS_XAUI_PHY ? \"read-write\" : \"read-only\""
10	APB3IFTEI	* Varies	APB3/MCI Interface Time-out Error Injection Enable ■ 1: Inject time-out error at APB3/MCI interface. This bit is not valid for MDIO configurations. **Note:** This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "MDIO_INTF ? \"read-only\" :\"read-write\""
9	CKOFTEI	* Varies	CKO Alive FSM Time-out Error Injection Enable This bit can be set to 1 to inject time-out error in all the FSMs operating with clk_cko_alive_i. This bit is valid only for configurations with Synopsys multi-protocol 6G PHY. **Note:** This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SSP ? \"read-write\" : \"read-only\""
8	EEEFTEI	* Varies	EEE FSM Time-out Error Injection Enable This bit can be set to 1 to inject time-out error in all the FSMs operating with clk_eee_i. This bit is valid only for configurations with EEE_EN = Enabled. **Note:** This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "EEE_EN ? \"read-write\" : \"read-only\""
7	CSRFTEI	R/W	CSR FSM Time-out Error Injection Enable This bit can be set to 1 to inject time-out error in all the FSMs operating with clk_csr_i. **Note:** This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
6	RXFTEI	R/W	Rx FSM Time-out Error Injection Enable This bit can be set to 1 to inject time-out error in all the FSMs operating in receive clock domain(s). **Note:** This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always
5	TXFTEI	R/W	Tx FSM Time-out Error Injection Enable This bit can be set to 1 to inject time-out error in all the FSMs operating in transmit clock domain(s). **Note:** This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always
4	CKOFPEI	* Varies	CKO Alive FSM Parity Error Injection Enable This bit can be set to 1 to inject parity error in all the FSMs operating with clk_cko_alive_i. This bit is valid only for configurations with Synopsys multi-protocol 6G PHY. **Note:** This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "SSP ? \"read-write\" : \"read-only\""
3	EEEFPEI	* Varies	EEE FSM Parity Error Injection Enable This bit can be set to 1 to inject parity error in all the FSMs operating with clk_eee_i. This bit is valid only for configurations with EEE_EN = Enabled. **Note:** This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always **Memory Access:** "EEE_EN ? \"read-write\" : \"read-only\""
2	CSRFPEI	R/W	CSR FSM Parity Error Injection Enable This bit can be set to 1 to inject parity error in all the FSMs operating with clk_csr_i. **Note:** This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always
1	RXFPEI	R/W	Rx FSM Parity Error Injection Enable This bit can be set to 1 to inject parity error in all the FSMs operating in receive clock domain(s). **Note:** This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
0	TXFPEI	R/W	Tx FSM Parity Error Injection Enable This bit can be set to 1 to inject parity error in all the FSMs operating in transmit clock domain(s). **Note:** This bit self-clears after error interrupt is generated. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.173 VR_MII_SFTY_DBG_CTRL

- **Description:** VR MII MMD Safety Debug Control Register
  This register provides the flexibility to control the safety error injection mode. Program the fields of this register to desired values prior to enabling the error injection enable bits in VR_MII_SFTY_ERR_INJ register. If the fields of this register are changed after enabling the error injection enable bits, it can lead to ambiguous results.
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80F3 : (APB3_16) ? (0x80F3*0x2) : (0x80F3*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&DWC_XPCS_ASP

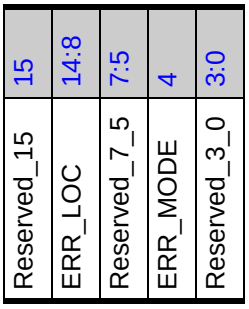


Table 5-634 Fields for Register: VR_MII_SFTY_DBG_CTRL

Bits	Name	Memory Access	Description
15	Reserved_15	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
14:8	ERR_LOC	R/W	Error Location This field controls the bit-location at which Data Path Parity (DPP) error is introduced. Software must program judicious values to this field depending on the width of the data and parity bus applicable to the feature selected and also depending on the current operating mode. If an invalid value is programmed to this field, then error injection behavior is undefined. The legal values for this field are as follows: ■ 7'd0 to 7'd7: Introduce error in data bits (when ERR_MODE=0) ■ 7'd0: Introduce error in parity bit (when ERR_MODE=1) **Value After Reset:** 0x0 **Exists:** Always
7:5	Reserved_7_5	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
4	ERR_MODE	R/W	Error Mode This bit determines whether DDP error must be introduced in data bits or parity bits. Encoding of this field is as follows: ■ 0: Introduce error in data bits ■ 1: Introduce error in parity bits **Value After Reset:** 0x0 **Exists:** Always
3:0	Reserved_3_0	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

5.2.9.174 VR_MII_SFTY_DISABLE

- **Description:** VR MII MMD Safety Disable Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x80F4 : (APB3_16) ? (0x80F4*0x2) : (0x80F4*0x4)" + "(APB3_INTF||MCI_INTF||MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&DWC_XPCS_ASP

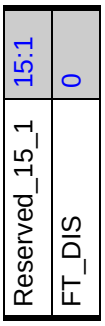


Table 5-635 Fields for Register: VR_MII_SFTY_DISABLE

Bits	Name	Memory Access	Description
15:1	Reserved_15_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	FT_DIS	R/W	FSM Time-out Disable. When this bit is set to 1, the FSM time-out safety mechanism inside DWC_xpcs gets disabled. **Value After Reset:** 0x0 **Exists:** Always

5.2.9.175 VR_MII_SFTY_TMR_CTRL

- **Description:** VR MII MMD Safety Timer Control Register
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x80F5 : (APB3_16) ? (0x80F5*0x2) : (0x80F5*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&DWC_XPCS_ASP

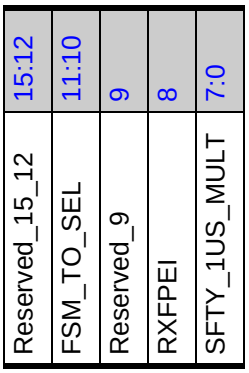


Table 5-636 Fields for Register: VR_MII_SFTY_TMR_CTRL

Bits	Name	Memory Access	Description
15:12	Reserved_15_12	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
11:10	FSM_TO_SEL	R/W	FSM Time-out Select DWC_xpcs allocates appropriate time-out values to various FSMs inside it. These pre-allocated time-out values range from 10 micro-seconds to 1 second. This field provides an option to scale the time-out values if required. ■ 2'b00: Timers get scaled by 0.1 ■ 2'b01: Timers run for standard duration ■ 2'b10: Timers get scaled by 10 ■ 2'b11: Reserved This field has no effect if FASTSIM bit is set to 1. **Value After Reset:** 0x1 **Exists:** Always
9	Reserved_9	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
8	RXFPEI	R/W	Interface Time-out Select. This field can set to appropriate value to configure the time-out value of Interface time-outs. ■ 0: Interface time-out is 10 us ■ 1: Interface time-out is 50 us **Value After Reset:** 0x0 **Exists:** Always
7:0	SFTY_1US_MULT	R/W	Safety 1us Clock Tic Multiplying Factor This bit is the multiplying factor to the clk_csr_i clock time period to make it equal to 1us. Program this bit with a value such that the (clk_csr_i time_period * 2 * SFTY_1US_MULT) is 1 us or as close as possible to 1 us. For example, if the clk_csr_i clock time period is 10 ns, the value of this field needs to be set to 50. The default value of this register is set as 50 corresponding to a 100 MHz clk_csr_i clock. **Value After Reset:** 0x32 **Exists:** Always

5.2.9.176 VR_MII_SFTY_UE_INTR1

- **Description:** VR MII MMD Uncorrectable Safety Error Interrupt Register 1
- **Size:** "(APB3_32) ? 32 : 16" bits
- **Offset:** "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x80F8 : (APB3_16) ? (0x80F8*0x2) : (0x80F8*0x4)" + "(APB3_INTF | |MCI_INTF | |MDIO_INTF) ? 0x1F0000 : (APB3_16) ? (0x1F0000*0x2) : (0x1F0000*0x4)"
- **Exists:** BASE1000X_ONLY&&DWC_XPCS_ASP

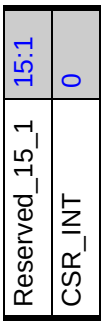


Table 5-637 Fields for Register: VR_MII_SFTY_UE_INTR1

Bits	Name	Memory Access	Description
15:1	Reserved_15_1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
0	CSR_INT	R/W	Control & Status register Error Interrupt (SS,WC) This bit gets set automatically when the safety feature in DWC_xpcs detects error in Control or Status register. When this bit is set, it remains high till software clears it by writing 0. **Value After Reset:** 0x0 **Exists:** Always

5.3 DWC_xpcs_safetyslave_map Memory Map Registers

Register definitions for each component memory map.

Table 5-638 Registers for the DWC_xpcs_safetyslave_map Memory Map

Register	Offset	Description
This register block defines the Safety Subordinate register set DWC_xpcs_safetyslave_block Exists: Always		
"SS_STRT_COREOFF_REG" on page 1379	0x0	Core Register Start Offset Register
"SS_CNTRL_REG" on page 1380	0x4	Safety Subordinate Control Register
"SS_RESET_REG" on page 1384	0x8	Safety Subordinate Reset Register
"SS_ERRSRC_REG" on page 1385	0xc	Safety Error Source Register
"SS_ERRFORCE_REG" on page 1387	0x10	Safety Error Force Register
"SS_DIAG0_RERRSTS_REG" on page 1388	0x14	Safety Subordinate Recoverable Error Status Register
"SS_DIAG1_RERRSTS_REG" on page 1389	0x18	Safety Subordinate Recoverable Error Status Register
"SS_CORE_VERNUM_REG" on page 1390	0x20	Safety Subordinate Core Version Number Register
"SS_CORE_VERTYPE_REG" on page 1391	0x24	Safety Subordinate Core Version Type Register
"SS_CORE_URERRSTS_REG" on page 1392	0x28	Safety Subordinate Core Unrecoverable Error Status Register
"SS_CORE_RERRSTS_REG" on page 1393	0x2c	Safety Subordinate Core Recoverable Error Status Register
"SS_CORE_DIAG_REG0" on page 1394	0x30	Core Diagnostic Register0
"SS_CORE_DIAG_REG1" on page 1395	0x34	Core Diagnostic Register1
"SS_CORE_DIAG_REG2" on page 1396	0x38	Core Diagnostic Register2
"SS_CORE_DIAG_REG3" on page 1397	0x3c	Core Diagnostic Register3
"SS_CORE_DIAG_REG4" on page 1398	0x40	Core Diagnostic Register4
"SS_CORE_DIAG_REG5" on page 1399	0x44	Core Diagnostic Register5

5.3.1 DWC_xpcs_safetyslave_map/DWC_xpcs_safetyslave_block Registers

5.3.1.1 SS_STRT_COREOFF_REG

- **Name:** Core Register Start Offset Register
- **Description:** This register is used to indicate the size of the core registers and start address of the core register map.
- **Size:** 32 bits
- **Offset:** 0x0
- **Exists:** Always

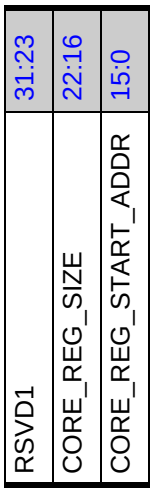


Table 5-639 Fields for Register: SS_STRT_COREOFF_REG

Bits	Name	Memory Access	Description
31:23	RSVD1	R	Reserved **Value After Reset:** 0x0 **Exists:** Always
22:16	CORE_REG_SIZE	R	Core Register Size, this field returns the configured value of the core register depth. The SS ignores all writes to this register **Value After Reset:** SS_CORE_REG_DEPTH **Exists:** Always **Volatile:** true
15:0	CORE_REG_START_ADDR	R	Core Register Start Pointer, this field returns the configured value of the core register start offset. The SS ignores all writes to this register **Value After Reset:** SS_CORE_REG_START_OFFSET **Exists:** Always **Volatile:** true

5.3.1.2 SS_CNTRL_REG

- **Name:** Safety Subordinate Control Register
- **Description:** This register is the safety subordinate control register. It has control signals to enable error checks, and controls for injecting different errors.
- **Size:** 32 bits
- **Offset:** 0x4
- **Exists:** Always

31:30	29:14	13:12	11:10	9:8	7:6	5:4	3:2	1:0
SS_INT_RERR_SET	RSVD	SS_DIAG1_RERR_SET	SS_DIAG0_RERR_SET	CORE_URERR_SET	CORE_RERR_SET	CORE_URMON_EN	CORE_RMON_EN	SS_EMON_EN

Table 5-640 Fields for Register: SS_CNTRL_REG

Bits	Name	Memory Access	Description
31:30	SS_INT_RERR_SET	R/W	Inject SS Internal Error When enabled by the application, this will facilitate in injecting ss recoverable error to test the functionality ■ 2'b10: Disabled ■ 2'b01: Enabled ■ 2'b00, 2'b11: Reserved On setting this field with either 2'b00 or 2'b11 no write to the register will happen, but the transfer will be completed on APB side. When this field flips to any value from the enabled state or disabled state (when the application is not writing), then 'safety error' will get generated, this error will get reflected in the SS_ERRSRC_R[SS_INT_ERR_SRC] bit. **Value After Reset:** 0x2 **Exists:** Always **Volatile:** true
29:14	RSVD	R	Reserved **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
13:12	SS_DIAG1_RERR_SET	R/W	Inject SS Diagnostic1 Recoverable Error When enabled by the application, this will facilitate in injecting ss recoverable diagnostic1 error status (SS_DIAG1_RERRSTS_REG) to test the functionality ■ 2'b10: Disabled ■ 2'b01: Enabled ■ 2'b00, 2'b11: Reserved On setting this field with either 2'b00 or 2'b11 no write to the register will happen, but the transfer will be completed on APB side. When this field flips to any value from the enabled state or disabled state (when the application is not writing), then 'safety error' will get generated, this error will get reflected in the SS_ERRSRC_R[SS_INT_ERR_SRC] bit. **Value After Reset:** 0x2 **Exists:** Always **Volatile:** true
11:10	SS_DIAG0_RERR_SET	R/W	Inject SS Diagnostic0 Recoverable Error When enabled by the application, this will facilitate in injecting ss recoverable diagnostic0 error status (SS_DIAG0_RERRSTS_REG) to test the functionality ■ 2'b10: Disabled ■ 2'b01: Enabled ■ 2'b00, 2'b11: Reserved On setting this field with either 2'b00 or 2'b11 no write to the register will happen, but the transfer will be completed on APB side. When this field flips to any value from the enabled state or disabled state (when the application is not writing), then 'safety error' will get generated, this error will get reflected in the SS_ERRSRC_R[SS_INT_ERR_SRC] bit. **Value After Reset:** 0x2 **Exists:** Always **Volatile:** true
9:8	CORE_URERR_SET	R/W	Inject Unrecoverable Error When enabled by the application, this will facilitate in injecting unrecoverable error to test the functionality ■ 2'b10: Disabled ■ 2'b01: Enabled ■ 2'b00, 2'b11: Reserved When this field flips to any value from the enabled state or disabled state (when the application is not writing), then 'safety error' will get generated. **Value After Reset:** 0x2 **Exists:** Always **Volatile:** true

Bits	Name	Memory Access	Description
7:6	CORE_RERR_SET	R/W	Inject Recoverable Error When enabled by the application, this will facilitate in injecting recoverable error to test the functionality ■ 2'b10 : Disabled ■ 2'b01 : Enabled ■ 2'b00, 2'b11 : Reserved When this field flips to any value from the enabled state or disabled state (when the application is not writing), then 'safety error' will get generated. **Value After Reset:** 0x2 **Exists:** Always **Volatile:** true
5:4	CORE_URMON_EN	R/W	Unrecoverable Error Monitor Enable When enabled by the application, SS monitors for unrecoverable errors generated by the core. Current design of the Safety Subordinate does not generate unrecoverable errors. ■ 2'b10 : Disabled ■ 2'b01 : Enabled ■ 2'b00, 2'b11 : Reserved When this field flips to any value from the enabled state or disabled state (when the application is not writing), then 'safety error' will get generated. **Value After Reset:** 0x2 **Exists:** Always **Volatile:** true
3:2	CORE_RMON_EN	R/W	Recoverable Error Monitor Enable When enabled by the application, SS monitors for recoverable errors generated both by the Safety Subordinate and the core. ■ 2'b10 : Disabled ■ 2'b01 : Enabled ■ 2'b00, 2'b11 : Reserved When this field flips to any value from the enabled state or disabled state (when the application is not writing), then 'safety error' will get generated. **Value After Reset:** 0x2 **Exists:** Always **Volatile:** true

Bits	Name	Memory Access	Description
1:0	SS_EMON_EN	R/W	Safety Subordinate Error Monitor Enable When enabled by the safety manager, the safety subordinate starts monitoring for errors in the SS_CNTRL_REG register and the following port inputs signals. ■ core_swreset ■ core_enable ■ core_version_num_reg ■ core_version_type_reg ■ core_diagnostic_reg0-63 ■ 2'b10: Disabled ■ 2'b01: Enabled ■ 2'b00, 2'b11: Reserved Writing 2'b00 or 2'b11 from SSM bus will be ignored. When this field flips to any value from the enabled state or disabled state due to any internal fault conditions, then 'ss_safety_error' will get generated. **Value After Reset:** 0x2 **Exists:** Always **Volatile:** true

5.3.1.3 SS_RESET_REG

- **Name:** Safety Subordinate Reset Register
- **Description:** This register is the safety subordinate reset register. It has control signals to clear the error status registers.
- **Size:** 32 bits
- **Offset:** 0x8
- **Exists:** Always

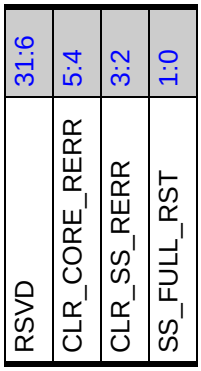


Table 5-641 Fields for Register: SS_RESET_REG

Bits	Name	Memory Access	Description
31:6	RSVD	R	These bits of the register are reserved. They always return 0. **Value After Reset:** 0x0 **Exists:** Always
5:4	CLR_CORE_RERR	R/W	Clear Core Recoverable Error Clear the core recoverable error set in the SS Error Status register **Value After Reset:** 0x2 **Exists:** Always **Volatile:** true
3:2	CLR_SS_RERR	R/W	Clear Safety Subordinate Recoverable Error Clear the SS recoverable error set in the SS Error Status register **Value After Reset:** 0x2 **Exists:** Always **Volatile:** true
1:0	SS_FULL_RST	R/W	Safety Subordinate Full Reset When enabled the Safety Subordinate is reset. **Value After Reset:** 0x2 **Exists:** Always **Volatile:** true

5.3.1.4 SS_ERRSRC_REG

- **Name:** Safety Error Source Register
- **Description:** This register is the safety subordinate error source register. It shows the source of the safety subordinate error.
- **Size:** 32 bits
- **Offset:** 0xc
- **Exists:** Always

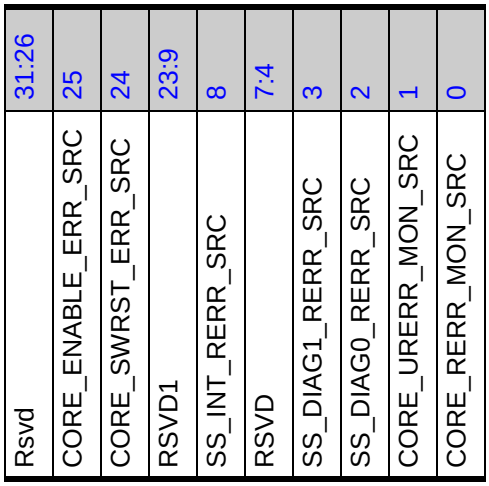


Table 5-642 Fields for Register: SS_ERRSRC_REG

Bits	Name	Memory Access	Description
31:26			**Reserved Field:** Yes
25	CORE_ENABLE_ERR_SRC	R	Core Enable Port Error This bit will get set when the SS encounters an error in the core_enable port input. **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true
24	CORE_SWRST_ERR_SRC	R	Core SW Reset Port Error This bit will get set when the SS encounters an error in the core_swreset port input. **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true
23:9	RSVD1	R	These bits (9-15) of the register are reserved. They always return 0. **Value After Reset:** 0x0 **Exists:** Always

Bits	Name	Memory Access	Description
8	SS_INT_RERR_SRC	R	Safety Subordinate Internal Recoverable Error This bit will get set when the SS encounters an error in any one of the following register or port input signals. 1. core_version_num_reg 2. core_version_type_reg 3. SS_CNTRL_REG **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true
7:4	RSVD	R	These bits of the register are reserved. They always return 0. **Value After Reset:** 0x0 **Exists:** Always
3	SS_DIAG1_RERR_SRC	R	SS Recoverable DIagnostic1 Error Source This bit is set when the SS encounters ss_safety_err due to SS recoverable error i.e. parity mismatch on diagnostic registers 32-64 **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true
2	SS_DIAG0_RERR_SRC	R	SS Recoverable DIagnostic0 Error Source This bit is set when the SS encounters ss_safety_err due to SS recoverable error i.e. parity mismatch on diagnostic registers 0-31 **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true
1	CORE_URERR_MON_SRC	R	Core Unrecoverable Monitor Error Source This bit will get set when the SS encounters an error in the core_urerr_mon port input. **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true
0	CORE_RERR_MON_SRC	R	Core Recoverable Monitor Error Source This bit will get set when the SS encounters an error in the core_rerr_mon port input. **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

5.3.1.5 SS_ERRFORCE_REG

- **Name:** Safety Error Force Register
- **Description:** This register is the safety Subordinate error force register.
- **Size:** 32 bits
- **Offset:** 0x10
- **Exists:** Always



Table 5-643 Fields for Register: SS_ERRFORCE_REG

Bits	Name	Memory Access	Description
31:0	SS_ERR_FORCE_VAL	W	Safety Subordinate Error Force Write the register with 1 (multiple 1s are allowed) for the supported recoverable error monitor size (SS_CORE_RERR_MON_SIZE) when register field SS_CNTRL_REG[CORE_RERR_SET] is to `SAFETY_HI for forcing Core Recoverable Error Or Write the register with 1 (multiple 1s are allowed) for the supported unrecoverable error monitor size (SS_CORE_URERR_MON_SIZE) when the register field SS_CNTRL_REG[CORE_URERR_SET] to `SAFETY_HI for forcing Core Unrecoverable Error Or Write the register SS_ERRFORCE_REG with 0x00000001 when the register field SS_CNTRL_REG[SS_RERR_SET] to `SAFETY_HI for forcing Safety subordinate Recoverable Error This register always returns 0 when read. **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

5.3.1.6 SS_DIAG0_RERRSTS_REG

- **Name:** Safety Subordinate Recoverable Error Status Register
- **Description:** This register is the safety subordinate recoverable error status register. It shows the status of parity error on Diagnostic register 0-31
- **Size:** 32 bits
- **Offset:** 0x14
- **Exists:** Always



Table 5-644 Fields for Register: SS_DIAG0_RERRSTS_REG

Bits	Name	Memory Access	Description
31:0	SS_CORE_DIAG0_ERR_STS	R	Core Diagnostic0 Error Status If the SS encounters any parity error (if configured for) in the diagnostic register port input set from core_diagnostic_reg0 to core_diagnostic_reg31, then this register captures the error status. For example if any parity error is encountered in core_diagnostic_reg15, then bit15 of this register will get set. **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

5.3.1.7 SS_DIAG1_RERRSTS_REG

- **Name:** Safety Subordinate Recoverable Error Status Register
- **Description:** This register is the safety subordinate recoverable error status register. It shows the status of parity error on Diagnostic register 32-63
- **Size:** 32 bits
- **Offset:** 0x18
- **Exists:** Always



Table 5-645 Fields for Register: SS_DIAG1_RERRSTS_REG

Bits	Name	Memory Access	Description
31:0	SS_CORE_DIAG1_ERR_STS	R	Core Diagnostic1 Error Status If the SS encounters any parity error (if configured for) in the diagnostic register port input set from core_diagnostic_reg32 to core_diagnostic_reg63, then this register captures the error status. For example if any parity error is encountered in core_diagnostic_reg47, then bit15 of this register will get set. **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

5.3.1.8 SS_CORE_VERNUM_REG

- **Name:** Safety Subordinate Core Version Number Register
- **Description:** This register is the core version number register.
- **Size:** 32 bits
- **Offset:** 0x20
- **Exists:** Always



Table 5-646 Fields for Register: SS_CORE_VERNUM_REG

Bits	Name	Memory Access	Description
31:0	SS_CORE_VERNUM	R	Core i/p core_version_num_reg is captured in this register **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

5.3.1.9 SS_CORE_VERTYPE_REG

- **Name:** Safety Subordinate Core Version Type Register
- **Description:** This register is the core version type register.
- **Size:** 32 bits
- **Offset:** 0x24
- **Exists:** Always



Table 5-647 Fields for Register: SS_CORE_VERTYPE_REG

Bits	Name	Memory Access	Description
31:0	SS_CORE_VERTYPE	R	Core i/p core_version_type_reg is captured in this register **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

5.3.1.10 SS_CORE_URERRSTS_REG

- **Name:** Safety Subordinate Core Unrecoverable Error Status Register
- **Description:** This register reflects the Unrecoverable error status
- **Size:** 32 bits
- **Offset:** 0x28
- **Exists:** Always



Table 5-648 Fields for Register: SS_CORE_URERRSTS_REG

Bits	Name	Memory Access	Description
31:0	SS_CORE_URERR_STS	R	If the SS encounters either `SAFETY_HI or any invalid value on any one of the anti-valent pair of the core_urerr_mon (only up to configured value SS_CORE_URERR_MON_SIZE) input port, then it sets corresponding bit of the SS_CORE_URERRSTS_REG register. **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

5.3.1.11 SS_CORE_RERRSTS_REG

- **Name:** Safety Subordinate Core Recoverable Error Status Register
- **Description:** This register reflects the Recoverable error status
- **Size:** 32 bits
- **Offset:** 0x2c
- **Exists:** Always

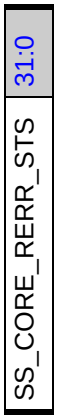


Table 5-649 Fields for Register: SS_CORE_RERRSTS_REG

Bits	Name	Memory Access	Description
31:0	SS_CORE_RERR_STS	R	If the SS encounters either `SAFETY_HI or any invalid value on any one of the anti-valent pair of the core_rerr_mon (only up to configured value SS_CORE_RERR_MON_SIZE) input port, then it sets corresponding bit of the SS_CORE_RERRSTS_REG register. **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

5.3.1.12 SS_CORE_DIAG_REG0

- **Name:** Core Diagnostic Register0
- **Description:** This is Core Diagnostic Register0
- **Size:** 32 bits
- **Offset:** 0x30
- **Exists:** Always

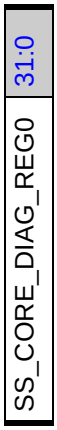


Table 5-650 Fields for Register: SS_CORE_DIAG_REG0

Bits	Name	Memory Access	Description
31:0	SS_CORE_DIAG_REG0	R	This register stores the diagnostic register0 from the core **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

5.3.1.13 SS_CORE_DIAG_REG1

- **Name:** Core Diagnostic Register1
- **Description:** This is Core Diagnostic Register1
- **Size:** 32 bits
- **Offset:** 0x34
- **Exists:** Always

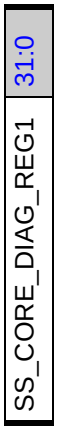


Table 5-651 Fields for Register: SS_CORE_DIAG_REG1

Bits	Name	Memory Access	Description
31:0	SS_CORE_DIAG_REG1	R	This register stores the diagnostic register1 from the core **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

5.3.1.14 SS_CORE_DIAG_REG2

- **Name:** Core Diagnostic Register2
- **Description:** This is Core Diagnostic Register2
- **Size:** 32 bits
- **Offset:** 0x38
- **Exists:** Always

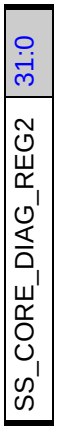


Table 5-652 Fields for Register: SS_CORE_DIAG_REG2

Bits	Name	Memory Access	Description
31:0	SS_CORE_DIAG_REG2	R	This register stores the diagnostic register2 from the core **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

5.3.1.15 SS_CORE_DIAG_REG3

- **Name:** Core Diagnostic Register3
- **Description:** This is Core Diagnostic Register3
- **Size:** 32 bits
- **Offset:** 0x3c
- **Exists:** Always

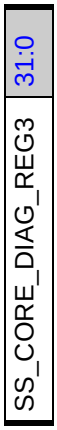


Table 5-653 Fields for Register: SS_CORE_DIAG_REG3

Bits	Name	Memory Access	Description
31:0	SS_CORE_DIAG_REG3	R	This register stores the diagnostic register3 from the core **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

5.3.1.16 SS_CORE_DIAG_REG4

- **Name:** Core Diagnostic Register4
- **Description:** This is Core Diagnostic Register4
- **Size:** 32 bits
- **Offset:** 0x40
- **Exists:** Always

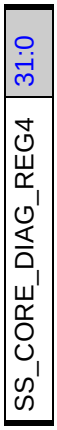


Table 5-654 Fields for Register: SS_CORE_DIAG_REG4

Bits	Name	Memory Access	Description
31:0	SS_CORE_DIAG_REG4	R	This register stores the diagnostic register4 from the core **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

5.3.1.17 SS_CORE_DIAG_REG5

- **Name:** Core Diagnostic Register5
- **Description:** This is Core Diagnostic Register5
- **Size:** 32 bits
- **Offset:** 0x44
- **Exists:** Always

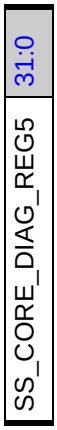


Table 5-655 Fields for Register: SS_CORE_DIAG_REG5

Bits	Name	Memory Access	Description
31:0	SS_CORE_DIAG_REG5	R	This register stores the diagnostic register5 from the core **Value After Reset:** 0x0 **Exists:** Always **Volatile:** true

6

Signal Descriptions

This chapter details all possible I/O signals in the IP. For configurable IP titles, your actual configuration might not contain all of these signals.

Inputs are on the left of the signal diagrams; outputs are on the right.

Attention: For configurable IP titles, do not use this document to determine the exact I/O footprint of the controller. It is for reference purposes only.

When you configure the controller in coreConsultant, you must access the I/O signals for your actual configuration at workspace/report/IO.html or workspace/report/IO.xml after you have completed the report creation activity. That report comes from the exact same source as this chapter but removes all the I/O signals that are not in your actual configuration. This does not apply to non-configurable IP titles. In addition, all parameter expressions are evaluated to actual values. Therefore, the widths might change depending on your actual configuration.

Some expressions might refer to TCL functions or procedures (sometimes identified as **<functionof>**) that coreConsultant uses to make calculations. The exact formula used by these TCL functions is not provided in this chapter. However, when you configure the controller in coreConsultant, all TCL functions and parameters are evaluated completely; and the resulting values are displayed where appropriate in the coreConsultant GUI reports.

In addition to describing the function of each signal, the signal descriptions in this chapter include the following information:

- **Active State:** Indicates whether the signal is active high or active low. When a signal is not intended to be used in a particular application, then this signal needs to be tied or driven to the inactive state (opposite of the active state).
- **Registered:** Indicates whether or not the signal is registered directly inside the IP boundary without intervening logic (excluding simple buffers). A value of *No* does not imply that the signal is not synchronous, only that there is some combinatorial logic between the signal's origin or destination register and the boundary of the controller. A value of N/A indicates that this information is not provided for this IP title.
- **Synchronous to:** Indicates which clocks in the IP sample this input (drive for an output). This clock might not be the same as the clock that your application logic should use to clock (sample/drive) this pin. For more details, consult the clock section in the databook. The presence of the postfix SuperList indicates a list of all possible clocks over all possible configs. Consult coreConsultant report for which clock applies to your specific configuration.
- **Exists:** Name of configuration parameter that populates this signal in your configuration.

- **Power Domain:** Name of power/voltage domain that this signal is part of when power/voltage islands are used. The SINGLE_DOMAIN value indicates that there are no islands.

The I/O signals are grouped as follows:

- "System Interface Signals" on page 1404
- "MDIO Management Interface Signals" on page 1406
- "XGMII Transmit Interface Signals" on page 1409
- "XGMII Receive Interface Signals" on page 1418
- "GMII Transmit Interface Signals" on page 1427
- "GMII Receive Interface Signals" on page 1433
- "Micro Controller Interface Signals" on page 1442
- "APB3 Management Interface Signals" on page 1445
- "Transmit Interface to PCS-X PHY Signals" on page 1447
- "Energy Efficient Ethernet PHY Interface Signals" on page 1450
- "Receive Interface from PCS-X PHY Signals" on page 1454
- "Miscellaneous Control/Status Interface Signals" on page 1457
- "Synopsys Multi-protocol 6G PHY Transmit Status and Control Interface Signals" on page 1463
- "Synopsys Multi-protocol 12G/16G/25G PHY Receive Status and Control Interface Signals" on page 1470
- "Synopsys Multi-protocol 6G PHY Receive Status and Control Interface Signals" on page 1487
- "Synopsys Multi-protocol 6G/12G PHY Receive Status and Control Interface Signals" on page 1491
- "Synopsys PHY Receive Status and Control Interface Signals" on page 1492
- "Synopsys PHY CR Port Interface Signals" on page 1495
- "Synopsys Multi-protocol 12G/16G/25G PHY Transmit Status and Control Interface Signals" on page 1497
- "Clause 72 KR Training Signals for Synopsys PHY" on page 1504
- "Synopsys Multi-protocol 16G/25G PHY Transmit Status and Control Interface Signals" on page 1507
- "Synopsys Multi-protocol 12G/16G PHY Receive Status and Control Interface Signals" on page 1510
- "Synopsys Multi-protocol 12G PHY Receive Status and Control Interface Signals" on page 1512
- "Synopsys Multi-protocol 25G PHY Receive Status and Control Interface Signals" on page 1514
- "Synopsys Multi-protocol 12G/16G/25G PHY MPLL Control and Status Interface Signals" on page 1519
- "Synopsys Multi-protocol 12G PHY MPLL Control and Status Interface Signals" on page 1527
- "Synopsys Multi-protocol 12G/16G PHY MPLL Control and Status Interface Signals" on page 1528
- "Synopsys Multi-protocol 25G PHY MPLL Control and Status Interface Signals" on page 1535
- "Synopsys Multi-protocol 16G/25G PHY Miscellaneous Interface Signals" on page 1541
- "Synopsys Multi-protocol 12G/16G/25G PHY Miscellaneous Interface Signals" on page 1542
- "Synopsys Multi-protocol 12G/16G PHY Miscellaneous Interface Signals" on page 1545

- "Synopsys Multi-protocol 16G/25G PHY Transmit Control and Status Interface Signals" on page 1546
- "Synopsys Multi-protocol 16G/25G PHY Receive Control and Status Interface Signals" on page 1549
- "Synopsys Multi-protocol 16G/25G PHY Receive MPLL Control and Status Interface Signals" on page 1557
- "Synopsys Multi-protocol 25G PHY Receive Control and Status Interface Signals" on page 1558
- "Synopsys Multi-protocol 32G PHY Rx Control and Status Interface Signals" on page 1560
- "Synopsys Multi-protocol 32G PHY Tx Control and Status Interface Signals" on page 1564
- "Synopsys Multi-protocol 32G PHY Common Control and Status Interface Signals" on page 1570
- "Synopsys Multi-protocol 16G PHY Receive Control and Status Interface Signals" on page 1575
- "Synopsys Multi-protocol 16G/25G PHY Recieve Control and Status Interface Signals" on page 1578
- "Synopsys Multi-protocol 25G PHY Transmit Status and Control Interface Signals" on page 1579
- "GPIO Interface Signals" on page 1582
- "SGMII/QSGMII Control and Clock Interface Signals" on page 1583
- "SGMII/QSGMII/USXGMII Control Interface Signals" on page 1588
- "QSGMII Control and Clock Interface Signals" on page 1589
- "QSGMII/USXGMII Control Interface Signals" on page 1599
- "USXGMII Control and Status Interface Signals" on page 1601
- "SGMII/QSGMII/USXGMII Control and Clock Interface Signals" on page 1609
- "Synopsys PHY Transmit Status and Control Interface Signals" on page 1617
- "Synopsys Multi-protocol 6G PHY Miscellaneous Interface Signals" on page 1620
- "Synopsys Multi-protocol 6G PHY MPLL Control Interface Signals" on page 1621
- "Synopsys PHY Resistor Tune Interface Signals" on page 1623
- "Synopsys PHY Clock Control Interface Signals" on page 1624
- "PCS-R PHY Interface Signals" on page 1625
- "PCS-X and PCS-R Common Interface Signals" on page 1628
- "FEC Error Correction Memory Interface Signals" on page 1629
- "FEC Error Forward Memory Interface Signals" on page 1631
- "Clause 72 KR Training Signals for Non-Synopsys PHY" on page 1633
- "Non-Synopsys PHY Interface Signals" on page 1644
- "SSM Bus Interface Signals" on page 1645

6.1 System Interface Signals



Table 6-1 System Interface Signals

Port Name	I/O	Description
pwr_on_rst_n	I	Power On Reset. Power on reset input. **Exists:** Always **Synchronous To:** None **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** Low
scan_mode	I	Scan Mode. This signal is used to get the good Automatic Test Pattern Generator (ATPG) test vector coverage. When asserted, this signal bypasses the internal reset generation and directly connects the pwr_on_rst_n signal to all internally-generated resets. **Exists:** Always **Synchronous To:** System Test Clock **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_link_status_o	O	PCS Receive Link Status. This signal indicates whether the receive link of the PCS is up or down. **Exists:** Always **Synchronous To:** {RPCS_ONLY ? "clk_rpcs_rx_i" : !RPCS_EN ? "clk_xgxs_rx0_i" : "clk_xpcs_rx_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
sbd_intr_o	O	Interrupt Output (Optional). This signal indicates the following events of Clause 73 and Clause 37 auto-negotiation. ■ Page Received (for Clause 73 auto-negotiation) ■ Auto-negotiation is completed (for Clause 73 and Clause 37 auto-negotiation) ■ Incompatible link partner (for Clause 73 auto-negotiation) This interrupt generation is disabled by default. The Management must enable interrupt generation by writing to the Interrupt Mask register. **Exists:** BACKPLANE_ETHERNET_PCS||CL37_AN **Synchronous To:** None **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_sfty_ue_intr_o	O	Uncorrectable Error Safety Interrupt **Exists:** DWC_XPCS_ASP **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_sfty_ce_intr_o	O	Correctable Error Safety Interrupt **Exists:** DWC_XPCS_ASP&&FEC_EN **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.2 MDIO Management Interface Signals

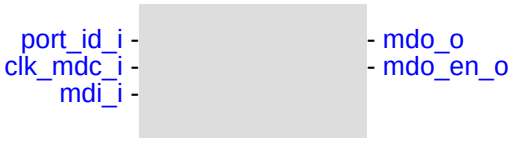


Table 6-2 MDIO Management Interface Signals

Port Name	I/O	Description
port_id_i[4:0]	I	ID of the Port. This is the Port Address input which indicates the PHY address to which DWC_xpcs is expected to respond. **Exists:** MDIO_INTF **Synchronous To:** clk_mdc_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
clk_mdc_i	I	Management Clock. This clock input is a management clock to do the following: ■ To register management data mdi_i ■ To drive data mdo_o The IEEE Std 802.3, Clause 45 defines the clk_mdc_i as 2.5 MHz. However, the DWC_xpcs can work at higher clk_mdc_i frequencies for non-Synopsys PHY configurations. The maximum frequency of the clk_mdc_i clock depends the current speed mode: ■ 1000BASE-X, KX, or 1G mode: 25 MHz ■ 10GBASE-R PCS or KR mode: 31.25 MHz ■ XAUI, RXAUI or 10GBASE-X PCS mode: 62.5 MHz **Exists:** MDIO_INTF **Synchronous To:** None **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
mdi_i	I	Management Data Input. This signal is a serial bit input to program internal device registers. **Exists:** MDIO_INTF **Synchronous To:** clk_mdc_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
mdo_o	O	Management Data Output. This output is the serialized version of the register that is being read. This output is valid only when the mdo_en_o signal is asserted. **Exists:** MDIO_INTF **Synchronous To:** clk_mdc_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
mdo_en_o	O	Management Data IO Enable. This signal enables selection of input or output mode because the management data pin is a bidirectional pin. *Values:* ■ 0: The DWC_xpcs is sampling the data mdi_i. ■ 1: The DWC_xpcs is driving valid data mdo_o. **Exists:** MDIO_INTF **Synchronous To:** clk_mdc_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.3 XGMII Transmit Interface Signals

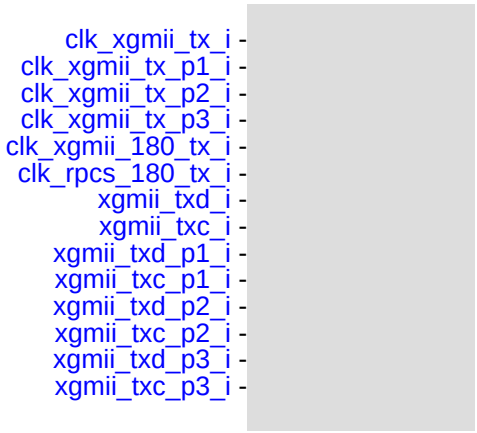


Table 6-3 XGMII Transmit Interface Signals

Port Name	I/O	Description
clk_xgmii_tx_i	I	XGMII or GMII Transmit Clock. If XGMII = DDR or DDW in the PCS-X mode or XGMII = SDR in the PCS-R mode, this signal registers the following XGMII input signals: ■ xgmii_txd_i[{xdw}:0] ■ xgmii_txc_i[{xcw}:0] If ADD_GMII = Enabled, SGMII_EN = Disabled, and the current speed mode is 1G or when USXGMII_EN=Enabled, this signal also registers the following input signals: ■ gmii_txd_i[7:0] ■ gmii_tx_en_i ■ gmii_tx_er_i ***Dependencies:*** *PCS-X Only Configurations (XGXS PCS, 10GBASE-X PCS, KX4_KX, KX4_Only):* ■ If XGMII = SDR, this clock is not present. ■ If XGMII = DDR or DDW, for 10G mode, connect this clock to a clock which has exactly half frequency of the clk_xgxs_tx_i clock. For 1G mode, connect this clock to a clock which has exactly same frequency as the clk_xgxs_tx_i clock. *PCS-R Only Configurations (10GBASE-R PCS, KR_Only):* ■ If XGMII = DDR or DDW, this clock is not present. ■ If XGMII = SDR, connect this clock to a clock which has exactly double frequency than the clk_rpcs_tx_i clock. *PCS-X and PCS-R Configurations (KR_KX4_KX, KR_KX4, KR_KX):* ■ PCS-X mode when XGMII = DDR or DDW: In 10G PCS-X mode, connect this clock to a clock which has exactly half frequency of the clk_xgxs_tx_i clock. In 1G PCS-X mode, connect this clock to a clock which has exactly same frequency as the clk_xgxs_tx_i clock. ■ PCS-X mode when XGMII = SDR: Connect this clock to a clock which has the same frequency as the clk_xgxs_tx_i clock. ■ PCS-R mode when XGMII = DDR or DDW: Connect this clock to a clock which has the same frequency as the clk_rpcs_tx_i clock. ■ PCS-R mode when XGMII = SDR: Connect this clock to a clock which has exactly double frequency of the clk_rpcs_tx_i clock. This clock is not present in the following configurations: ■ Configurations with XGMII = SDR and one of the following modes: XGXS PCS, 10GBASE-X PCS, KX4_KX, or KX4_Only ■ Configurations with XGMII = DDW or DDR and one of the following modes: 10GBASE-R PCS, KR_Only, or KR_KX ■ Configurations with 1000BASEX-Only PCS or KX_Only mode **Exists:**

Port Name	I/O	Description
clk_xgmii_tx_i...(cont.)	I.	(!XGMII_SDR&&!RPCS_EN)||(XGMII_SDR&&RPCS_ONLY)||(!RPCS_ONLY&&RPCS_EN&&(!KRKX_OR_RX||XGMII_SDR))||ADD_2PT5G||SNPS_RSVDPARAM_14||USXGMII_EN **Synchronous To:** None **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
clk_xgmii_tx_p1_i	I	XGMII or GMII Transmit Clock for Port 1 (for multi-port USXGMII). This clock registers the following signals: ■ xgmii_txd_p1_i[{xdw}:0] ■ xgmii_txc_p1_i[{xcw}:0] If the current operating mode is USXGMII, this clock additionally registers the following input signals: ■ gmii_txd_p1_i[7:0] ■ gmii_tx_en_p1_i ■ gmii_tx_er_p1_i ***Dependencies:*** This clock is present only in multi-port USXGMII configurations. **Exists:** MP_USXGMII **Synchronous To:** None **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
clk_xgmii_tx_p2_i	I	XGMII or GMII Transmit Clock for Port 2 (for multi-port USXGMII). This clock registers the following signals: ■ xgmii_txd_p2_i[{xdw}:0] ■ xgmii_txc_p2_i[{xcw}:0] If the current operating mode is USXGMII, this clock additionally registers the following input signals: ■ gmii_txd_p2_i[7:0] ■ gmii_tx_en_p2_i ■ gmii_tx_er_p2_i ***Dependencies:*** This clock is present only in Quad-Port USXGMII configurations. **Exists:** USXG_QPORT **Synchronous To:** None **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
clk_xgmii_tx_p3_i	I	XGMII or GMII Transmit Clock for Port 3 (for multi-port USXGMII). This clock registers the following signals: ■ xgmii_txd_p3_i[{xdw}:0] ■ xgmii_txc_p3_i[{xcw}:0] If the current operating mode is USXGMII, this clock additionally registers the following input signals: ■ gmii_txd_p3_i[7:0] ■ gmii_tx_en_p3_i ■ gmii_tx_er_p3_i ***Dependencies:*** This clock is present only in Quad-Port USXGMII configurations. **Exists:** USXG_QPORT **Synchronous To:** None **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
clk_xgmii_180_tx_i	I	Inverted XGMII Transmission Clock. This clock is an inverted version of the clk_xgmii_tx_i clock (180 degrees phase-shifted). This clock is used to register the following XGMII Transmit signals: ■ xgmii_txd_i[{xdw}:0] ■ xgmii_txc_i[{xcw}:0] ***Dependencies:*** ■ This clock is present but not used if the DWC_xpcs is operating in the 1G mode. ■ This clock is present in the following configurations when XGMII = DDR: PCS-X Only Configurations (XGXS PCS, 10GBASE-X PCS, KX4_KX, KX4_Only) and PCS-X and PCS-R Configurations (KR_KX4_KX, KR_KX4) In these configurations, connect this clock to inverted version of the clk_xgmii_tx_i clock. ■ This clock is not present in the following configurations: PCS-R Only (10GBASE-R PCS, KR_Only) or KR_KX Configurations. The clk_rpcs_180_tx_i clock is used to sample the xgmii_txd_i and xgmii_txc_i signals in the DDR mode. KX_Only or 1000BASEX-Only PCS configurations **Exists:** XGMII_DDR&&!RPCS_ONLY&&!KRKX_OR_RX **Synchronous To:** clk_xgmii_tx_i **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
clk_rpcs_180_tx_i	I	PCS-R XGMII Interface Clock in the DDR Mode. This clock is inverted version of the clk_rpcs_tx_i clock. This clock samples the following signals in the 10GBASE-R PCS, KR_Only, or KR_KX XGMII DDR configurations: ■ xgmii_txd_i[{xdw}:0] ■ xgmii_txc_i[{xcw}:0] ***Dependencies:*** This clock is present only in the following configurations: ■ Configurations with XGMII = DDR and MAIN_MODE = 10GBASE-R PCS ■ Configurations with XGMII = DDR, MAIN_MODE = Backplane Ethernet PCS, and BACKPLANE_ETHER_CFG = KR_Only or KR_KX **Exists:** XGMII_DDR&&(RPCS_ONLY||KRKX_OR_RX) **Synchronous To:** clk_rpcs_tx_i **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xgmii_txd_i[(XTI_WR_DATA_WIDTH-1):0]	I	XGMII Transmit Data Lanes. This port is the data lane of XGMII, indicating the data or control character to be transmitted. The data width depends the XGMII interface: ■ If XGMII = DDW, then xgmii_txd_i[63:0] ■ If XGMII = SDR or DDR, then xgmii_txd_i[31:0] ***Dependencies:*** ■ In 1G mode, this port is used only when ADD_GMII = Disabled. ■ In 1G mode, only xgmii_txd_i[7:0] is sampled and it is used as gmii_txd_i[7:0]. ■ This port is not present in the configurations with MAIN_MODE = 1000BASEX-Only PCS or configurations with BACKPLANE_ETH_-CONFIG = KX_Only **Exists:** (!(KX_ONLY||BASE1000X_ONLY))||SNPS_RSVDPARAM_14 **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** **PCS-X Only Configurations (XGXS PCS, 10GBASE-X PCS, KX4_KX, KX4_Only):** ■ If XGMII = DDW, synchronous to clk_xgmii_tx_i ■ If XGMII = DDR, synchronous to clk_xgmii_tx_i and clk_xg-mii_180_tx_i ■ If XGMII = SDR, synchronous to clk_xgxs_tx_i **PCS-R Only (10GBASE-R PCS, KR_Only) or KR_KX Configurations:** ■ If XGMII = DDW, synchronous to clk_rpcs_tx_i ■ If XGMII = DDR, synchronous to clk_rpcs_tx_i and clk_rpcs_180_tx_i ■ If XGMII = SDR, synchronous to clk_xgmii_tx_i **PCS-X and PCS-R Configurations (KR_KX4_KX, KR_KX4):** ■ If XGMII = DDW or SDR, synchronous to clk_xgmii_tx_i ■ If XGMII = DDR, synchronous to clk_xgmii_tx_i and clk_xg-mii_180_tx_i

Port Name	I/O	Description
xgmii_txc_i[(XTI_WR_CTRL_WIDTH-1):0]	I	XGMII Transmit Control. These are the control lines of the XGMII interface to indicate whether xgmii_txd_i[{xdw}:0] is a data or control character. ■ xgmii_txc_i[0]: ❑ 0: xgmii_txd_i[7:0] is a data character. ❑ 1: xgmii_txd_i[7:0] is a control character. ■ xgmii_txc_i[1]: ❑ 0: xgmii_txd_i[15:8] is a data character. ❑ 1: xgmii_txd_i[15:8] is a control character. ... ■ xgmii_txc_i[7]: ❑ 0: xgmii_txd_i[63:56] is a data character. ❑ 1: xgmii_txd_i[63:56] is a control character. The width of this signal depends the XGMII mode. ■ If XGMII = DDW, then xgmii_txc_i[7:0] ■ If XGMII = SDR or DDR, then xgmii_txc_i[3:0] ***Dependencies:*** ■ In 1G mode, this port is used only when ADD_GMII = Disabled. ■ In 1G mode, only xgmii_txc_i[1:0] is sampled. xgmii_txc_i[0] is used as gmii_tx_en and xgmii_txc_i[1] is used as gmii_tx_er. ■ This port is not present in configurations with MAIN_MODE = 1000BASEX-Only PCS or configurations with BACKPLANE_ETH_-CONFIG = KX_Only **Exists:** (!(KX_ONLY||BASE1000X_ONLY))||SNPS_RSVDPARAM_14 **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** **PCS-X Only Configurations (XGXS PCS, 10GBASE-X PCS, KX4_KX, KX4_Only):** ■ If XGMII = DDW, synchronous to clk_xgmii_tx_i ■ If XGMII = DDR, synchronous to clk_xgmii_tx_i and clk_xg-mii_180_tx_i ■ If XGMII = SDR, synchronous to clk_xgxs_tx_i **PCS-R Only (10GBASE-R PCS, KR_Only) or KR_KX Configurations:** ■ If XGMII = DDW, synchronous to clk_rpcs_tx_i ■ If XGMII = DDR, synchronous to clk_rpcs_tx_i and clk_rpcs_180_tx_i ■ If XGMII = SDR, synchronous to clk_xgmii_tx_i **PCS-X and PCS-R Configurations (KR_KX4_KX, KR_KX4):** ■ If XGMII = DDW or SDR, synchronous to clk_xgmii_tx_i ■ If XGMII = DDR, synchronous to clk_xgmii_tx_i and clk_xg-mii_180_tx_i

Port Name	I/O	Description
xgmii_txd_p1_i[(XTI_WR_DATA_WIDTH-1):0]	I	XGMII Transmit Data Lanes for Port 1. ***Dependencies:*** This port is present only in multi-port USXGMII configurations. This port is sampled only when the current operating mode is multi-port USXGMII. **Exists:** MP_USXGMII **Synchronous To:** clk_xgmii_tx_p1_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgmii_txc_p1_i[(XTI_WR_CTRL_WIDTH-1):0]	I	XGMII Transmit Control for Port 1 These are the control lines of the XGMII interface to indicate whether xgmii_txd_p1_i[{xdw}:0] is a data or control character. ***Dependencies:***This port is present only in multi-port USXGMII configurations. This port is sampled only when the current operating mode is multi-port USXGMII. **Exists:** MP_USXGMII **Synchronous To:** clk_xgmii_tx_p1_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgmii_txd_p2_i[(XTI_WR_DATA_WIDTH-1):0]	I	XGMII Transmit Data Lanes for Port 2. ***Dependencies:*** This port is present only in Quad-port USXGMII configurations. This port is sampled only when the current operating mode is Quad-port USXGMII. **Exists:** USXG_QPORT **Synchronous To:** clk_xgmii_tx_p2_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgmii_txc_p2_i[(XTI_WR_CTRL_WIDTH-1):0]	I	XGMII Transmit Control for Port 2 These are the control lines of the XGMII interface to indicate whether xgmii_txd_p2_i[{xdw}:0] is a data or control character. ***Dependencies:*** This port is present only in Quad-port USXGMII configurations. This port is sampled only when the current operating mode is Quad-port USXGMII. **Exists:** USXG_QPORT **Synchronous To:** clk_xgmii_tx_p2_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xgmii_txd_p3_i[(XTI_WR_DATA_WIDTH-1):0]	I	XGMII Transmit Data Lanes for Port 3. ***Dependencies:*** This port is present only in Quad-port USXGMII configurations. This port is sampled only when the current operating mode is Quad-port USXGMII. **Exists:** USXG_QPORT **Synchronous To:** clk_xgmii_tx_p3_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgmii_txc_p3_i[(XTI_WR_CTRL_WIDTH-1):0]	I	XGMII Transmit Control for Port 3 These are the control lines of the XGMII interface to indicate whether xgmii_txd_p3_i[{xdw}:0] is a data or control character. ***Dependencies:*** This port is present only in Quad-port USXGMII configurations. This port is sampled only when the current operating mode is Quad-port USXGMII. **Exists:** USXG_QPORT **Synchronous To:** clk_xgmii_tx_p3_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.4 XGMII Receive Interface Signals

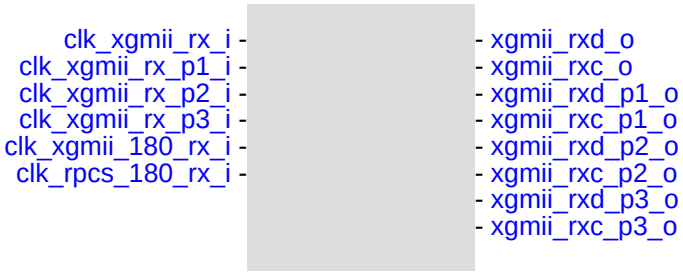


Table 6-4 XGMII Receive Interface Signals

Port Name	I/O	Description
clk_xgmii_rx_i	I	XGMII Receive Clock. This clock input is used to drive the following XGMII signals when XGMII = DDR or DDW: ■ xgmii_rxd_o[{xdw}:0] ■ xgmii_rxc_o[{xcw}:0] This is also used to drive the following signals when ADD_GMII = Enabled and the current speed mode is 1G or when USXGMII_EN=Enabled: ■ gmii_rxd_o[7:0] ■ gmii_rx_dv_o ■ gmii_rx_er_o ***Dependencies:*** ■ **PCS-X Only Configurations (XGXS PCS, 10GBASE-X PCS, KX4_KX, KX4_Only):** ❑ If XGMII = DDR or DDW and CLKCOMP = Enabled, connect this clock to the clk_xgmii_tx_i clock. ❑ If XGMII = DDR or DDW and CLKCOMP = Disabled, in 10G mode, connect this clock to a clock which has exact half frequency of the clk_xgxs_rx{lane}_i clock. In 1G mode, connect this clock to a clock which has exact same frequency of the clk_xgxs_rx{lane}_i clock. ❑ If XGMII = SDR, this clock is not present. ■ **PCS-R Only Configurations (10GBASE-R PCS, KR_Only):** ❑ If XGMII = DDR or DDW, this clock is not present. ❑ If XGMII = SDR and CLKCOMP = Enabled, connect this clock to the clk_xgmii_tx_i clock. ❑ If XGMII = SDR and CLKCOMP = Disabled, connect this clock to a clock which has double frequency of the clk_rpcs_rx{lane}_i clock. ■ **PCS-X and PCS-R Configurations (KR_KX4_KX, KR_KX4, KR_KX):**If CLKCOMP = Enabled, connect this clock to the clk_xgmii_tx_i clock. If CLKCOMP = Disabled, follow these guidelines: ❑ **PCS-X mode when XGMII = DDR or DDW:**In 10G PCS-X mode, connect this clock to a clock which has exactly half frequency of the clk_xgxs_rx0_i clock. In 1G PCS-X mode, connect this clock to a clock which has the same frequency as the clk_xgxs_rx0_i clock. ❑ **PCS-X mode when XGMII = SDR:**Connect this clock to a clock which has the same frequency as the clk_xgxs_rx0_i clock. ❑ **PCS-R mode when XGMII = DDR or DDW:**Connect this clock to a clock which has the same frequency as the clk_rpcs_rx{lane}_i clock. ❑ **PCS-R mode when XGMII = SDR:**Connect this clock to a clock which has exactly double frequency than the clk_rpcs_rx_i clock. This clock is not present in the following configurations: ■ Configurations with XGMII = SDR and one of the following modes: XGXS PCS, 10GBASE-X PCS, KX4_KX, or KX4_Only

Port Name	I/O	Description
clk_xgmii_rx_i...(cont.)	I.	■ Configurations with XGMII = DDW or DDR and one of the following modes: 10GBASE-R PCS, KR_Only, or KR_KX ■ Configurations with 1000BASEX-Only PCS or KX_Only mode **Exists:** (!XGMII_SDR&&!RPCS_EN)||(XGMII_SDR&&RPCS_ONLY)||(!RPCS_ONLY&&RPCS_EN&&(!KRKX_OR_RX||XGMII_SDR))||ADD_2PT5G||USXGMII_EN **Synchronous To:** None **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
clk_xgmii_rx_p1_i	I	XGMII or GMII Receive Clock for Port 1 (for multi-port USXGMII). This clock input is used to drive the following XGMII signals: ■ xgmii_rxd_p1_o[{xdw}:0] ■ xgmii_rxc_p1_o[{xcw}:0] This is also used to drive the following signals when current operating mode is multi-port USXGMII. ■ gmii_rxd_p1_o[7:0] ■ gmii_rx_dv_p1_o ■ gmii_rx_er_p1_o ***Dependencies:*** ■ This clock is present only in configurations with multi-port USXGMII. ■ If CLKCOMP = Enabled, connect this clock to the clk_xgmii_tx_p1_i clock. **Exists:** MP_USXGMII **Synchronous To:** None **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
clk_xgmii_rx_p2_i	I	XGMII or GMII Receive Clock for Port 2 (for Quad-port USXGMII). This clock input is used to drive the following XGMII signals: ■ xgmii_rxd_p2_o[{xdw}:0] ■ xgmii_rxc_p2_o[{xcw}:0] This is also used to drive the following signals when current operating mode is Quad-Port USXGMII. ■ gmii_rxd_p2_o[7:0] ■ gmii_rx_dv_p2_o ■ gmii_rx_er_p2_o ***Dependencies:*** ■ This clock is present only in configurations with quad-port USXGMII. ■ If CLKCOMP = Enabled, connect this clock to the clk_xgmii_tx_p2_i clock. **Exists:** USXG_QPORT **Synchronous To:** None **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
clk_xgmii_rx_p3_i	I	XGMII or GMII Receive Clock for Port 3 (for Quad-port USXGMII). This clock input is used to drive the following XGMII signals: ■ xgmii_rxd_p3_o[{xdw}:0] ■ xgmii_rxc_p3_o[{xcw}:0] This is also used to drive the following signals when current operating mode is Quad-Port USXGMII. ■ gmii_rxd_p3_o[7:0] ■ gmii_rx_dv_p3_o ■ gmii_rx_er_p3_o ***Dependencies:*** ■ This clock is present only in configurations with quad-port USXGMII. ■ If CLKCOMP = Enabled, connect this clock to the clk_xgmii_tx_p3_i clock. **Exists:** USXG_QPORT **Synchronous To:** None **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
clk_xgmii_180_rx_i	I	Inverted XGMII Receive Clock. This clock is an inverted version of the clk_xgmii_rx_i (180 degrees phase-shifted) clock. If XGMII = DDR, this clock is used to drive the following XGMII Receive signals: ■ xgmii_rxd_o[{xdw}:0] ■ xgmii_rxc_o[{xcw}:0] ***Dependencies:*** ■ This clock is present but not used in 1G mode. ■ This clock is present if XGMII = DDR in the following configurations: ❑ PCS-X Only Configurations (XGXS PCS, 10GBASE-X PCS, KX4_KX, KX4_Only) ❑ PCS-X and PCS-R Configurations (KR_KX4_KX, KR_KX4) In these configurations, connect this clock to inverted version of the clk_xgmii_rx_i clock. ■ This clock is not present in the following configurations: ❑ PCS-R Only (10GBASE-R PCS, KR_Only) or KR_KX Configurations: The clk_rpcs_180_rx_i clock is used to drive the xgmii_rxd_o and xgmii_rxc_o signals. ❑ KX_Only or 1000BASEX-Only PCS configurations **Exists:** XGMII_DDR&&!RPCS_ONLY&&!KRKX_OR_RX **Synchronous To:** clk_xgmii_rx_i **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
clk_rpcs_180_rx_i	I	PCS-R XGMII Rx Interface Clock in the DDR Mode. This clock is inverted version of the clk_rpcs_rx_i clock (when CLKCOMP=Disabled) or inverted version of clk_rpcs_tx_i (when CLKCOMP=Enabled). This clock is used to drive the following signals in the 10GBASE-R PCS, KR_Only, or KR_KX XGMII DDR configurations: ■ xgmii_rxd_o[{xdw}:0] ■ xgmii_rxc_o[{xcw}:0] ***Dependencies:*** This clock is present only in the following configurations: ■ Configurations with MAIN_MODE = 10GBASE-R PCS and XGMII = DDR ■ Configurations with BACKPLANE_ETH_CONFIG = KR_Only or KR_KX and XGMII = DDR **Exists:** XGMII_DDR&&(RPCS_ONLY||KRKX_OR_RX) **Synchronous To:** clk_rpcs_rx_i **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xgmii_rxd_o[(XRI_RD_DATA_WIDTH-1):0]	O	XGMII Receive Data Lanes. This port is the data lane of XGMII, indicating received data. The width of this port depends the XGMII mode: ■ If XGMII = DDW, then xgmii_rxd_o[63:0] ■ If XGMII = SDR or DDR, then xgmii_rxd_o[31:0] ***Dependencies:*** ■ In 1G mode, this port is used only if ADD_GMII = Disabled. Only xgmii_rxd_o[7:0] is driven with the received data in 1G mode and its functionality is similar to gmii_rxd_o[7:0]. If ADD_GMII = Enabled, this port is driven with 32'h0100009C when operating in 1G mode (BASE-X). For USXGMII configurations, when current operating mode is USXGMII (USXG_EN bit=1) and if the current operating speed is 10M/100M/1G/(2.5G via GMII), then IDLE pattern (32'h07070707) is driven on this output port. ■ This port is not present in the following configurations: ❑ Configurations with MAIN_MODE = 1000BASEX-Only PCS ❑ Configurations with BACKPLANE_ETH_CONFIG = KX_Only **Exists:** (!(KX_ONLY||BASE1000X_ONLY)) **Synchronous To:** None **Registered:** (XGMII!=1) ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** **10GBASE-R PCS, KR_Only, and KR_KX Configurations:** ■ If XGMII = DDW and CLKCOMP = Disabled, synchronous to clk_rpcs_rx{lane}_i ■ If XGMII = DDW and CLKCOMP = Enabled, synchronous to clk_rpcs_tx_i ■ If XGMII = DDR and CLKCOMP = Disabled, synchronous to clk_rpcs_rx{lane}_i and clk_rpcs_180_rx_i ■ If XGMII = DDR and CLKCOMP = Enabled, synchronous to clk_rpcs_tx_i and clk_rpcs_180_tx_i ■ If XGMII = SDR, synchronous to clk_xgmii_rx_i **XGXS PCS, 10GBASE-X PCS, KX4_KX, and KX4_Only Configurations:** ■ If XGMII = DDW, synchronous to clk_xgmii_rx_i ■ If XGMII = DDR, synchronous to clk_xgmii_rx_i and clk_xgmii_180_rx_i ■ If XGMII = SDR and CLKCOMP = Disabled, synchronous to clk_xgxs_rx0_i. ■ If XGMII = SDR and CLKCOMP = Enabled, synchronous to clk_xgxs_tx_i. **KR_KX4_KX and KR_KX4 Configurations:** ■ If XGMII = DDW or SDR, synchronous to clk_xgmii_rx_i ■ If XGMII = DDR, synchronous to clk_xgmii_rx_i and clk_xgmii_180_rx_i

Port Name	I/O	Description
xgmii_rxc_o[(XRI_RD_CTRL_WIDTH-1):0]	O	XGMII Receive Control Lines. These are the control lines of the XGMII interface to indicate whether xgmii_rxd_o[{xdw}:0] is a data or control character each lane. Any bit going low this bus indicates that corresponding data byte xgmii_rxd_o[{xdw}:0] is a data character. ■ xgmii_rxc_o[0]: ❑ 0: xgmii_rxd_o[7:0] is a data character. ❑ 1: xgmii_rxd_o[7:0] is a control character. ■ xgmii_rxc_o[1]: ❑ 0: xgmii_rxd_o[15:8] is a data character. ❑ 1: xgmii_rxd_o[15:8] is a control character. ... ■ xgmii_rxc_o[7]: ❑ 0: xgmii_rxd_o[63:56] is a data character. ❑ 1: xgmii_rxd_o[63:56] is a control character. The width of this signal depends the XGMII Interface: ■ If XGMII = DDW, then xgmii_rxc_o[7:0] ■ If XGMII = SDR or DDR, then xgmii_rxc_o[3:0] ***Dependencies:*** ■ This port is not present in the following configurations: ❑ Configurations with MAIN_MODE = 1000BASEX-Only PCS ❑ Configurations with BACKPLANE_ETH_CONFIG = KX_Only ■ In 1G mode, if ADD_GMII = Disabled, this port is driven as follows: ❑ xgmii_rxc_o[0]: The functionality is similar to gmii_rx_dv_o. ❑ xgmii_rxc_o[1]: The functionality is similar to gmii_rx_er_o. ❑ xgmii_rxc_o[2]: The functionality is reserved. ❑ xgmii_rxc_o[3]: The functionality is reserved. ■ In 1G mode, if ADD_GMII = Enabled, this port is driven with 4'h01(which corresponds to the LOCAL_FAULT signaling -for configurations without USXGMII) or 4'hF (which corresponds to IDLE signaling -for configurations with USXGMII). **Exists:** (!(KX_ONLY||BASE1000X_ONLY)) **Synchronous To:** None **Registered:** (XGMII!=1) ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** / **10GBASE-R PCS, KR_Only, and KR_KX Configurations:** ■ If XGMII = DDW and CLKCOMP = Disabled, it is synchronous to clk_rpcs_rx{lane}_i ■ If XGMII = DDW and CLKCOMP = Enabled, it is synchronous to clk_rpcs_tx_i ■ If XGMII = DDR and CLKCOMP = Disabled, synchronous to clk_rpcs_rx{lane}_i and clk_rpcs_180_rx_i ■ If XGMII = DDR and CLKCOMP = Enabled, synchronous to clk_rpcs_tx_i and clk_rpcs_180_tx_i ■ If XGMII = SDR, synchronous to clk_xgmii_rx_i

Port Name	I/O	Description
xgmii_rxc_o[(XRI_RD_CTRL_WIDTH-1):0]...(cont.)	O.	**XGXS PCS, 10GBASE-X PCS, KX4_KX, and KX4_Only Configurations:** ■ If XGMII = DDW, synchronous to clk_xgmii_rx_i ■ If XGMII = DDR, synchronous to clk_xgmii_rx_i and clk_xgmii_180_rx_i ■ If XGMII = SDR and CLKCOMP = Disabled, synchronous to clk_xgxs_rx0_i ■ If XGMII = SDR and CLKCOMP = Enabled, synchronous to clk_xgxs_tx_i **KR_KX4_KX and KR_KX4 Configurations:** ■ If XGMII = DDW or SDR, synchronous to clk_xgmii_rx_i ■ If XGMII = DDR, synchronous to clk_xgmii_rx_i and clk_xgmii_180_rx_i
xgmii_rxd_p1_o[(XRI_RD_DATA_WIDTH-1):0]	O	XGMII Receive Data Lanes for Port 1. ***Dependencies:*** This port is present only in multi-port USXGMII configurations. This port is used only when the current operating mode is multi-port USXGMII. **Exists:** MP_USXGMII **Synchronous To:** clk_xgmii_rx_p1_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgmii_rxc_p1_o[(XRI_RD_CTRL_WIDTH-1):0]	O	XGMII Receive Control Lines for Port 1. These are the control lines of the XGMII interface to indicate whether xgmii_rxd_p1_o[{xdw}:0] is a data or control character in each lane. Any bit going low in this bus indicates that corresponding data byte xgmii_rxd_p1_o[{xdw}:0] is a data character. ***Dependencies:*** This port is present only in multi-port USXGMII configurations. This port is used only when the current operating mode is multi-port USXGMII. **Exists:** MP_USXGMII **Synchronous To:** clk_xgmii_rx_p1_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xgmii_rxd_p2_o[(XRI_RD_DATA_WIDTH-1):0]	O	XGMII Receive Data Lanes for Port 2. ***Dependency:*** This port is present only in quad-port USXGMII configurations. This port is used only when the current operating mode is quad-port USXGMII. **Exists:** USXG_QPORT **Synchronous To:** clk_xgmii_rx_p2_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgmii_rxc_p2_o[(XRI_RD_CTRL_WIDTH-1):0]	O	XGMII Receive Control Lines for Port 2. These are the control lines of the XGMII interface to indicate whether xgmii_rxd_p2_o[{xdw}:0] is a data or control character in each lane. Any bit going low in this bus indicates that corresponding data byte xgmii_rxd_p2_o[{xdw}:0] is a data character. ***Dependencies:*** This port is present only in quad-port USXGMII configurations. This port is used only when the current operating mode is quad-port USXGMII. **Exists:** USXG_QPORT **Synchronous To:** clk_xgmii_rx_p2_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgmii_rxd_p3_o[(XRI_RD_DATA_WIDTH-1):0]	O	XGMII Receive Data Lanes for Port 3. ***Dependency:*** This port is present only in quad-port USXGMII configurations. This port is used only when the current operating mode is quad-port USXGMII. **Exists:** USXG_QPORT **Synchronous To:** clk_xgmii_rx_p3_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xgmii_rxc_p3_o[(XRI_RD_CTRL_WIDTH-1):0]	O	XGMII Receive Control Lines for Port 3. These are the control lines of the XGMII interface to indicate whether xgmii_rxd_p3_o[{xdw}:0] is a data or control character in each lane. Any bit going low in this bus indicates that corresponding data byte xgmii_rxd_p3_o[{xdw}:0] is a data character. ***Dependencies:*** This port is present only in multi-port USXGMII configurations. This port is used only when the current operating mode is multi-port USXGMII. **Exists:** USXG_QPORT **Synchronous To:** clk_xgmii_rx_p3_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.5 GMII Transmit Interface Signals

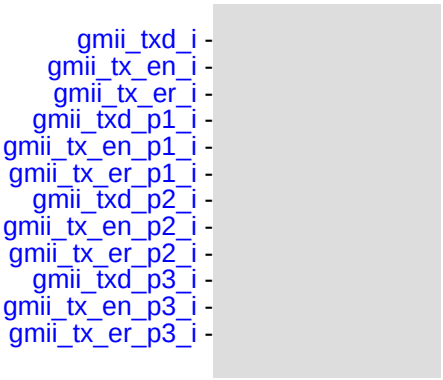


Table 6-5 GMII Transmit Interface Signals

Port Name	I/O	Description
gmii_txd_i[7:0]	I	GMII Transmit Data. This bus contains the Transmit data or Control character when the gmii_tx_en_i or gmii_tx_er_i signal is high. ■ gmii_tx_en_i = 0, gmii_tx_er_i = 0: Normal Inter Frame Gap ■ gmii_tx_en_i = 0, gmii_tx_er_i = 1: gmii_txd_i[7:0] = 8'h0F, Carrier Extend gmii_txd_i[7:0] = 8'h1F, Carrier Extend Error gmii_txd_i[7:0] = 8'h01, EEE LPI ■ gmii_tx_en_i = 1, gmii_tx_er_i = 0: Normal frame data (gmii_txd_i[7:0] indicates the data) ■ gmii_tx_en_i = 1, gmii_tx_er_i = 1: Error Indication **Exists:** GMII_PORT **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** Configuration dependent: ■ clk_mii_tx_i: For configurations with SGMII_EN = Enabled or QSGMII_EN= Enabled ■ clk_xgxs_tx_i: ❑ For KR_KX configurations or (10GBASE-R PCS + 1G support) configurations with SGMII_EN = Disabled and QSGMII_EN=Disabled and USXGMII_EN=Disabled ❑ For configurations with XGMII = SDR and SGMII_EN = Disabled and QSGMII_EN=Disabled and USXGMII_EN=Disabled ■ clk_xgmii_tx_i: ❑ For configurations with XGMII = DDR or DDW and SGMII_EN = Disabled and QSGMII_EN=Disabled ❑ For configurations with USXGMII_EN=Enabled or when current operating mode is USXGMII

Port Name	I/O	Description
gmii_tx_en_i	I	GMII Data Enable. When this signal is asserted and the gmii_tx_er_i signal is de-asserted, this signal indicates that the frame data is available the gmii_txd_i[7:0] bus. For more information, see description of the gmii_txd_i[7:0] signal. **Exists:** GMII_PORT **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** Configuration dependent: ■ clk_mii_tx_i: For configurations with SGMII_EN = Enabled or QSGMII_EN= Enabled ■ clk_xgxs_tx_i: ❑ For KR_KX configurations or (10GBASE-R PCS + 1G support) configurations with SGMII_EN = Disabled and QSGMII_EN=Disabled and USXGMII_EN=Disabled ❑ For configurations with XGMII = SDR and SGMII_EN = Disabled and QSGMII_EN=Disabled and USXGMII_EN=Disabled ■ clk_xgmii_tx_i: ❑ For configurations with XGMII = DDR or DDW and SGMII_EN = Disabled and QSGMII_EN=Disabled ❑ For configurations with USXGMII_EN=Enabled or when current operating mode is USXGMII
gmii_tx_er_i	I	GMII Error indicator. When high, this signal indicates the Error or Control character to be sent. For more information, see description of the gmii_txd_i[7:0] signal. **Exists:** GMII_PORT **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** Configuration dependent: ■ clk_mii_tx_i: For configurations with SGMII_EN = Enabled or QSGMII_EN= Enabled ■ clk_xgxs_tx_i: ❑ For KR_KX configurations or (10GBASE-R PCS + 1G support) configurations with SGMII_EN = Disabled and QSGMII_EN=Disabled and USXGMII_EN=Disabled ❑ For configurations with XGMII = SDR and SGMII_EN = Disabled and QSGMII_EN=Disabled and USXGMII_EN=Disabled ■ clk_xgmii_tx_i: ❑ For configurations with XGMII = DDR or DDW and SGMII_EN = Disabled and QSGMII_EN=Disabled ❑ For configurations with USXGMII_EN=Enabled or when current operating mode is USXGMII

Port Name	I/O	Description
gmii_txd_p1_i[7:0]	I	GMII Transmit Data along Port1. This bus contains the Transmit data or Control character when the gmii_tx_en_p1_i or gmii_tx_er_p1_i signal is high. ■ gmii_tx_en_p1_i = 0, gmii_tx_er_p1_i = 0: Normal Inter Frame Gap ■ gmii_tx_en_p1_i = 0, gmii_tx_er_p1_i = 1: gmii_txd_p1_i[7:0] = 8'h0F, Carrier Extend gmii_txd_p1_i[7:0] = 8'h1F, Carrier Extend Error gmii_txd_p1_i[7:0] = 8'h01, EEE LPI ■ gmii_tx_en_p1_i = 1, gmii_tx_er_p1_i = 0: Normal frame data (gmii_txd_p1_i[7:0] indicates the data) ■ gmii_tx_en_p1_i = 1, gmii_tx_er_p1_i = 1: Error Indication **Exists:** QSGMII_EN||MP_USXGMII **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** ■ clk_mii_tx_p1_i: When operating in QSGMII mode ■ clk_xgmii_tx_p1_i: When operating in USXGMII mode
gmii_tx_en_p1_i	I	GMII Data Enable along Port1. When this signal is asserted and the gmii_tx_er_p1_i signal is de-asserted, this signal indicates that the frame data is available the gmii_txd_p1_i[7:0] bus. For more information, see description of the gmii_txd_p1_i[7:0] signal. **Exists:** QSGMII_EN||MP_USXGMII **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** ■ clk_mii_tx_p1_i: When operating in QSGMII mode ■ clk_xgmii_tx_p1_i: When operating in USXGMII mode
gmii_tx_er_p1_i	I	GMII Error indicator along Port1. When high, this signal indicates the Error or Control character to be sent. For more information, see description of the gmii_txd_p1_i[7:0] signal. **Exists:** QSGMII_EN||MP_USXGMII **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** ■ clk_mii_tx_p1_i: When operating in QSGMII mode ■ clk_xgmii_tx_p1_i: When operating in USXGMII mode

Port Name	I/O	Description
gmii_txd_p2_i[7:0]	I	GMII Transmit Data along Port2 in QSGMII mode. This bus contains the Transmit data or Control character when the gmii_tx_en_p2_i or gmii_tx_er_p2_i signal is high. ■ gmii_tx_en_p2_i = 0, gmii_tx_er_p2_i = 0: Normal Inter Frame Gap ■ gmii_tx_en_p2_i = 0, gmii_tx_er_p2_i = 1: gmii_txd_p2_i[7:0] = 8'h0F, Carrier Extend gmii_txd_p2_i[7:0] = 8'h1F, Carrier Extend Error gmii_txd_p2_i[7:0] = 8'h01, EEE LPI ■ gmii_tx_en_p2_i = 1, gmii_tx_er_p2_i = 0: Normal frame data (gmii_txd_p2_i[7:0] indicates the data) ■ gmii_tx_en_p2_i = 1, gmii_tx_er_p2_i = 1: Error Indication **Exists:** QSGMII_EN||USXG_QPORT **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** ■ clk_mii_tx_p2_i: When operating in QSGMII mode ■ clk_xgmii_tx_p2_i: When operating in USXGMII mode
gmii_tx_en_p2_i	I	GMII Data Enable along Port2. When this signal is asserted and the gmii_tx_er_p2_i signal is de-asserted, this signal indicates that the frame data is available the gmii_txd_p2_i[7:0] bus. For more information, see description of the gmii_txd_p2_i[7:0] signal. **Exists:** QSGMII_EN||USXG_QPORT **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** ■ clk_mii_tx_p2_i: When operating in QSGMII mode ■ clk_xgmii_tx_p2_i: When operating in USXGMII mode
gmii_tx_er_p2_i	I	GMII Error indicator along Port2. When high, this signal indicates the Error or Control character to be sent. For more information, see description of the gmii_txd_p2_i[7:0] signal. **Exists:** QSGMII_EN||USXG_QPORT **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** ■ clk_mii_tx_p2_i: When operating in QSGMII mode ■ clk_xgmii_tx_p2_i: When operating in USXGMII mode

Port Name	I/O	Description
gmii_txd_p3_i[7:0]	I	GMII Transmit Data along Port3 in QSGMII mode. This bus contains the Transmit data or Control character when the gmii_tx_en_p3_i or gmii_tx_er_p3_i signal is high. ■ gmii_tx_en_p3_i = 0, gmii_tx_er_p3_i = 0: Normal Inter Frame Gap ■ gmii_tx_en_p3_i = 0, gmii_tx_er_p3_i = 1: gmii_txd_p3_i[7:0] = 8'h0F, Carrier Extend gmii_txd_p3_i[7:0] = 8'h1F, Carrier Extend Error gmii_txd_p3_i[7:0] = 8'h01, EEE LPI ■ gmii_tx_en_p3_i = 1, gmii_tx_er_p3_i = 0: Normal frame data (gmii_txd_p3_i[7:0] indicates the data) ■ gmii_tx_en_p3_i = 1, gmii_tx_er_p3_i = 1: Error Indication **Exists:** QSGMII_EN||USXG_QPORT **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** ■ clk_mii_tx_p3_i: When operating in QSGMII mode ■ clk_xgmii_tx_p3_i: When operating in USXGMII mode
gmii_tx_en_p3_i	I	GMII Data Enable along Port3. When this signal is asserted and the gmii_tx_er_p3_i signal is de-asserted, this signal indicates that the frame data is available the gmii_txd_p3_i[7:0] bus. For more information, see description of the gmii_txd_p3_i[7:0] signal. **Exists:** QSGMII_EN||USXG_QPORT **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** ■ clk_mii_tx_p3_i: When operating in QSGMII mode ■ clk_xgmii_tx_p3_i: When operating in USXGMII mode
gmii_tx_er_p3_i	I	GMII Error indicator along Port3. When high, this signal indicates the Error or Control character to be sent. For more information, see description of the gmii_txd_p3_i[7:0] signal. **Exists:** QSGMII_EN||USXG_QPORT **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** ■ clk_mii_tx_p3_i: When operating in QSGMII mode ■ clk_xgmii_tx_p3_i: When operating in USXGMII mode

6.6 GMII Receive Interface Signals

- gmii_rxd_o
- gmii_rx_dv_o
- gmii_rx_er_o
- gmii_crs_o
- gmii_col_o
- gmii_rxd_p1_o
- gmii_rx_dv_p1_o
- gmii_rx_er_p1_o
- gmii_crs_p1_o
- gmii_col_p1_o
- gmii_rxd_p2_o
- gmii_rx_dv_p2_o
- gmii_rx_er_p2_o
- gmii_crs_p2_o
- gmii_col_p2_o
- gmii_rxd_p3_o
- gmii_rx_dv_p3_o
- gmii_rx_er_p3_o
- gmii_crs_p3_o
- gmii_col_p3_o

Table 6-6 GMII Receive Interface Signals

Port Name	I/O	Description
gmii_rxd_o[7:0]	O	GMII Receive Data. This bus contains the Receive data or Control character when the gmii_rx_dv_o or gmii_rx_er_o signal is high. ■ gmii_rx_dv_o = 0, gmii_rx_er_o = 0: Normal Inter Frame Gap ■ gmii_rx_dv_o = 0, gmii_rx_er_o = 1: gmii_rxd_o[7:0] = 8'h00, Normal Inter Frame Gap gmii_rxd_o = 8'h01, EEE LPI gmii_rxd_o[7:0] = 8'h0E, False Carrier gmii_rxd_o[7:0] = 8'h0F, Carrier Extend gmii_rxd_o[7:0] = 8'h1F, Carrier Extend Error ■ gmii_rx_dv_o = 1, gmii_rx_er_o = 0: Normal frame data (gmii_rxd_o[7:0] indicates the data) ■ gmii_rx_dv_o = 1, gmii_rx_er_o = 1: Error Indication **Exists:** GMII_PORT **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** Configuration dependent: ■ clk_mii_rx_i: For configurations with SGMII_EN = Enabled or QSGMII_EN=Enabled ■ clk_xgmii_rx_i: ❑ For configurations with XGMII = DDR or DDW and SGMII_EN = Disabled and QSGMII_EN=Enabled ❑ For Configurations with USXGMII_EN=Enabled or when current operating mode is USXGMII ■ clk_xgxs_rx0_i: For KR_KX configurations or (10GBASE-R PCS+1G Support) configurations or configurations with XGMII = SDR, CLKCOMP = Disabled, and SGMII_EN = Disabled and QSGMII_EN=Disabled ■ clk_xgxs_tx_i: For KR_KX configurations or (10GBASE-R PCS+1G Support) configurations or configurations with XGMII = SDR, CLKCOMP = Enabled, and SGMII_EN = Disabled and QSGMII_EN=Disabled.

Port Name	I/O	Description
gmii_rx_dv_o	O	GMII Rx Valid. When this signal is asserted and the gmii_rx_er_o signal is de-asserted, this signal indicates that the valid frame data is available the gmii_rxd_o[7:0] bus. For more information, see description of the gmii_rxd_o[7:0] signal. **Exists:** GMII_PORT **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** Configuration dependent: ■ clk_mii_rx_i: For configurations with SGMII_EN = Enabled or QSGMII_EN=Enabled ■ clk_xgmii_rx_i: ❑ For configurations with XGMII = DDR or DDW and SGMII_EN = Disabled and QSGMII_EN=Enabled ❑ For Configurations with USXGMII_EN=Enabled or when current operating mode is USXGMII ■ clk_xgxs_rx0_i: For KR_KX configurations or (10GBASE-R PCS+1G Support) configurations or configurations with XGMII = SDR, CLKCOMP = Disabled, and SGMII_EN = Disabled and QSGMII_EN=Disabled and USXGMII_EN=Disabled ■ clk_xgxs_tx_i: For KR_KX configurations or (10GBASE-R PCS+1G Support) configurations or configurations with XGMII = SDR, CLKCOMP = Enabled, and SGMII_EN = Disabled and QSGMII_EN=Disabled.

Port Name	I/O	Description
gmii_rx_er_o	O	GMII Rx Error Indicator. When high, this signal indicates the Rx Error or Control character. For more information, see description of the gmii_rxd_o[7:0] signal. **Exists:** GMII_PORT **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** Configuration dependent: ■ clk_mii_rx_i: For configurations with SGMII_EN = Enabled or QSGMII_EN=Enabled ■ clk_xgmii_rx_i: ❑ For configurations with XGMII = DDR or DDW and SGMII_EN = Disabled and QSGMII_EN=Enabled ❑ For Configurations with USXGMII_EN=Enabled ■ clk_xgxs_rx0_i: For KR_KX configurations or (10GBASE-R PCS+1G Support) configurations or configurations with XGMII = SDR, CLKCOMP = Disabled, and SGMII_EN = Disabled and QSGMII_EN=Disabled ■ clk_xgxs_tx_i: For KR_KX configurations or (10GBASE-R PCS+1G Support) configurations or configurations with XGMII = SDR, CLKCOMP = Enabled, and SGMII_EN = Disabled and QSGMII_EN=Disabled.
gmii_crs_o	O	GMII Carrier Sense. This signal indicates that the DWC_xpcs is either transmitting or receiving valid data. **Exists:** (KX_PRESENT||BASE1000X_ONLY||SUPPORT_1G)&&!SNPS_RSVD PARAM_14 **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
gmii_col_o	O	GMII Collision. This signal indicates that a collision occurred. **Exists:** (KX_PRESENT||BASE1000X_ONLY||SUPPORT_1G)&&!SNPS_RSVD PARAM_14 **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
gmii_rxd_p1_o[7:0]	O	GMII Receive Data along Port1. This bus contains the Receive data or Control character when the gmii_rx_dv_p1_o or gmii_rx_er_p1_o signal is high. ■ gmii_rx_dv_p1_o = 0, gmii_rx_er_p1_o = 0: Normal Inter Frame Gap ■ gmii_rx_dv_p1_o = 0, gmii_rx_er_p1_o = 1: gmii_rxd_p1_o[7:0] = 8'h00, Normal Inter Frame Gap gmii_rxd_p1_o = 8'h01, EEE LPI gmii_rxd_p1_o[7:0] = 8'h0E, False Carrier gmii_rxd_p1_o[7:0] = 8'h0F, Carrier Extend gmii_rxd_p1_o[7:0] = 8'h1F, Carrier Extend Error ■ gmii_rx_dv_p1_o = 1, gmii_rx_er_p1_o = 0: Normal frame data (gmii_rxd_p1_o[7:0] indicates the data) ■ gmii_rx_dv_p1_o = 1, gmii_rx_er_p1_o = 1: Error Indication **Exists:** QSGMII_EN||MP_USXGMII **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** ■ clk_mii_rx_p1_i: When operating in QSGMII mode ■ clk_xgmii_rx_p1_i: When operating in USXGMII mode
gmii_rx_dv_p1_o	O	GMII Rx Valid along Port1. When this signal is asserted and the gmii_rx_er_p1_o signal is de-asserted, this signal indicates that the valid frame data is available the gmii_rxd_p1_o[7:0] bus. For more information, see description of the gmii_rxd_p1_o[7:0] signal. **Exists:** QSGMII_EN||MP_USXGMII **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** ■ clk_mii_rx_p1_i: When operating in QSGMII mode ■ clk_xgmii_rx_p1_i: When operating in USXGMII mode

Port Name	I/O	Description
gmii_rx_er_p1_o	O	GMII Rx Error Indicator along Port1. When high, this signal indicates the Rx Error or Control character. For more information, see description of the gmii_rxd_p1_o[7:0] signal. **Exists:** QSGMII_EN||MP_USXGMII **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** ■ clk_mii_rx_p1_i: When operating in QSGMII mode ■ clk_xgmii_rx_p1_i: When operating in USXGMII mode
gmii_crs_p1_o	O	GMII Carrier Sense along Port1. This signal indicates that the DWC_xpcs is either transmitting or receiving valid data. **Exists:** QSGMII_EN **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
gmii_col_p1_o	O	GMII Collision along Port1. This signal indicates that a collision occurred. **Exists:** QSGMII_EN **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
gmii_rxd_p2_o[7:0]	O	GMII Receive Data along Port2. This bus contains the Receive data or Control character when the gmii_rx_dv_p2_o or gmii_rx_er_p2_o signal is high. ■ gmii_rx_dv_p2_o = 0, gmii_rx_er_p2_o = 0: Normal Inter Frame Gap ■ gmii_rx_dv_p2_o = 0, gmii_rx_er_p2_o = 1: gmii_rxd_p2_o[7:0] = 8'h00, Normal Inter Frame Gap gmii_rxd_p2_o = 8'h01, EEE LPI gmii_rxd_p2_o[7:0] = 8'h0E, False Carrier gmii_rxd_p2_o[7:0] = 8'h0F, Carrier Extend gmii_rxd_p2_o[7:0] = 8'h1F, Carrier Extend Error ■ gmii_rx_dv_p2_o = 1, gmii_rx_er_p2_o = 0: Normal frame data (gmii_rxd_p2_o[7:0] indicates the data) ■ gmii_rx_dv_p2_o = 1, gmii_rx_er_p2_o = 1: Error Indication **Exists:** QSGMII_EN||USXG_QPORT **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** ■ clk_mii_rx_p2_i: When operating in QSGMII mode ■ clk_xgmii_rx_p2_i: When operating in USXGMII mode
gmii_rx_dv_p2_o	O	GMII Rx Valid along Port2. When this signal is asserted and the gmii_rx_er_p1_o signal is de-asserted, this signal indicates that the valid frame data is available the gmii_rxd_p2_o[7:0] bus. For more information, see description of the gmii_rxd_p2_o[7:0] signal. **Exists:** QSGMII_EN||USXG_QPORT **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** ■ clk_mii_rx_p2_i: When operating in QSGMII mode ■ clk_xgmii_rx_p2_i: When operating in USXGMII mode

Port Name	I/O	Description
gmii_rx_er_p2_o	O	GMII Rx Error Indicator along Port2. When high, this signal indicates the Rx Error or Control character. For more information, see description of the gmii_rxd_p2_o[7:0] signal. **Exists:** QSGMII_EN||USXG_QPORT **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** ■ clk_mii_rx_p2_i: When operating in QSGMII mode ■ clk_xgmii_rx_p2_i: When operating in USXGMII mode
gmii_crs_p2_o	O	GMII Carrier Sense along Port2. This signal indicates that the DWC_xpcs is either transmitting or receiving valid data. **Exists:** QSGMII_EN **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
gmii_col_p2_o	O	GMII Collision along Port2. This signal indicates that a collision occurred. **Exists:** QSGMII_EN **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
gmii_rxd_p3_o[7:0]	O	GMII Receive Data along Port3. This bus contains the Receive data or Control character when the gmii_rx_dv_p3_o or gmii_rx_er_p3_o signal is high. ■ gmii_rx_dv_p3_o = 0, gmii_rx_er_p3_o = 0: Normal Inter Frame Gap ■ gmii_rx_dv_p3_o = 0, gmii_rx_er_p3_o = 1: gmii_rxd_p3_o[7:0] = 8'h00, Normal Inter Frame Gap gmii_rxd_p3_o = 8'h01, EEE LPI gmii_rxd_p3_o[7:0] = 8'h0E, False Carrier gmii_rxd_p3_o[7:0] = 8'h0F, Carrier Extend gmii_rxd_p3_o[7:0] = 8'h1F, Carrier Extend Error ■ gmii_rx_dv_p3_o = 1, gmii_rx_er_p3_o = 0: Normal frame data (gmii_rxd_p3_o[7:0] indicates the data) ■ gmii_rx_dv_p3_o = 1, gmii_rx_er_p3_o = 1: Error Indication **Exists:** QSGMII_EN||USXG_QPORT **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** ■ clk_mii_rx_p3_i: When operating in QSGMII mode ■ clk_xgmii_rx_p3_i: When operating in USXGMII mode
gmii_rx_dv_p3_o	O	GMII Rx Valid along Port3. When this signal is asserted and the gmii_rx_er_p3_o signal is de-asserted, this signal indicates that the valid frame data is available the gmii_rxd_p3_o[7:0] bus. For more information, see description of the gmii_rxd_p3_o[7:0] signal. **Exists:** QSGMII_EN||USXG_QPORT **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** ■ clk_mii_rx_p3_i: When operating in QSGMII mode ■ clk_xgmii_rx_p3_i: When operating in USXGMII mode

Port Name	I/O	Description
gmii_rx_er_p3_o	O	GMII Rx Error Indicator along Port3. When high, this signal indicates the Rx Error or Control character. For more information, see description of the gmii_rxd_p3_o[7:0] signal. **Exists:** QSGMII_EN||USXG_QPORT **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** ■ clk_mii_rx_p3_i: When operating in QSGMII mode ■ clk_xgmii_rx_p3_i: When operating in USXGMII mode
gmii_crs_p3_o	O	GMII Carrier Sense along Port3. This signal indicates that the DWC_xpcs is either transmitting or receiving valid data. **Exists:** QSGMII_EN **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
gmii_col_p3_o	O	GMII Collision along Port3. This signal indicates that a collision occurred. **Exists:** QSGMII_EN **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.7 Micro Controller Interface Signals

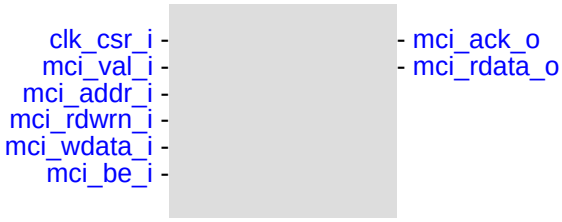


Table 6-7 Micro Controller Interface Signals

Port Name	I/O	Description
clk_csr_i	I	CSR Clock. This clock is used for the CSR block. This can be of any suitable frequency.However,for Synopsys PHY configurations, you need to ensure that frequency of clk_csr_i is less than 5.3 times clock frequency of clk_cko_alive_i. Also note that, DWC_xpcs is tested in simulation only for 31.25 MHz-178.5 MHz range. ***Dependencies:*** ■ If CSR_INTERFACE = MDIO and SNPS_XAUI_PHY = Enabled, this clock is connected to the clk_cko_alive_i clock. ■ If CSR_INTERFACE = MDIO and SNPS_XAUI_PHY = Disabled, this clock is not present. ■ If CSR_INTERFACE = MCI, this clock is the host clock for the MCI interface. ■ If CSR_INTERFACE = APB3, this clock is the host clock for the APB3 interface. **Exists:** SNPS_XAUI_PHY||!MDIO_INTF **Synchronous To:** None **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
mci_val_i	I	MCI Access Valid. This signal is asserted when the host accesses the DWC_xpcs CSR registers, indicating to the DWC_xpcs that the following signals are valid: ■ mci_addr_i[20:0 or 7:0] ■ mci_rdwrn_i ■ mci_be_i[1:0] ■ mci_wdata_i[15:0] (for Write operation) The assertion of this signal indicates the start of the transaction. **Exists:** MCI_INTF **Synchronous To:** clk_csr_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
mci_addr_i[(ADDRS_WIDTH-1):0]	I	MCI Address. This signal carries the device or register address of the CSR module from the host. This bus is valid only when the mci_val_i signal is asserted. ■ If INDIRECT_ACCESS = Disabled: ❑ The mci_addr_i[20:16] signal indicates the MMD Device Address ❑ The mci_addr_i[15:0] signal indicates the register address of the addressed MMD device ■ If INDIRECT_ACCESS = Enabled, the value of the mci_addr_i signal is decoded as follows: ❑ 0xFF: Access to the Base address register ❑ Any other value: Access to the register (decoded by the mci_addr_i[7:0] signal) within the group, indicated by the value in Base address register **Exists:** MCI_INTF **Synchronous To:** clk_csr_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
mci_rdwrn_i	I	MCI Transfer Type. This signal indicates to the DWC_xpcs whether to perform read or write transfer. **Values:** ■ 0: Write transfer ■ 1: Read transfer **Exists:** MCI_INTF **Synchronous To:** clk_csr_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
mci_wdata_i[15:0]	I	MCI Write Data. This bus carries the Write data from the host to the Control registers of the CSR module. The validity of each byte in this bus is qualified with the byte enables the mci_be_i[1:0] port. **Exists:** MCI_INTF **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
mci_be_i[1:0]	I	MCI Byte Enable. This port indicates valid bytes the mci_wdata_i[15:0] bus. This signal is valid only when mci_val_i is asserted. **Exists:** MCI_INTF **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
mci_ack_o	O	MCI Acknowledge. The DWC_xpcs acknowledges all data transfers through this signal. The assertion of this signal indicates the end of a transaction. **Exists:** MCI_INTF **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
mci_rdata_o[15:0]	O	MCI Read Data. This port carries the read data from the DWC_xpcs CSRs to the host. The DWC_xpcs supplies all the bytes, irrespective of the byte enable signal. **Exists:** MCI_INTF **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.8 APB3 Management Interface Signals

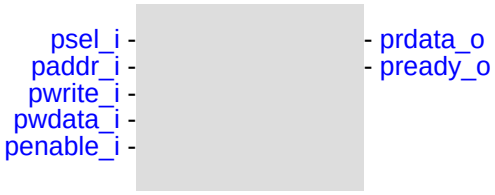


Table 6-8 APB3 Management Interface Signals

Port Name	I/O	Description
psel_i	I	APB Slave Select. When this signal is asserted, it indicates the start of transaction and validity of the paddr_i[] and pwrite_i signals. In addition, during a write operation, this signal indicates validity of the pwdata_i[] write data. **Exists:** APB3_INTF||APB3_16||APB3_32 **Synchronous To:** clk_csr_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
paddr_i[(ADDRS_WIDTH-1):0]	I	APB Address. This signal carries the register address of the CSR module. The address is valid only when the psel_i signal is asserted. ■ If INDIRECT_ACCESS = Disabled: ❑ The paddr_i[20:16] signal indicates the MMD Device Address ❑ The paddr_i[15:0] signal indicates the register address of the addressed MMD device ■ If INDIRECT_ACCESS = Enabled, the value of the paddr_i signal is decoded as follows: ❑ 0xFF (APB3_INTF) or 0x1FE (APB3_16) or 0x3FC (APB3_32): Access to the Base address register ❑ Any other value: Access to the register (decoded by paddr_i[7/8/9:0]) within the group, indicated by the value in Base address register **Exists:** APB3_INTF||APB3_16||APB3_32 **Synchronous To:** clk_csr_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
pwrite_i	I	APB Read or Write. When high, this signal indicates a write operation. When low, it indicates a read operation. **Exists:** APB3_INTF||APB3_16||APB3_32 **Synchronous To:** clk_csr_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
pwdata_i[(APB_DATA_W-1):0]	I	Write Data. This signal carries the write data from the host for the CSR module Control registers. **Exists:** APB3_INTF||APB3_16||APB3_32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
penable_i	I	APB Enable. This signal is a part of the standard APB 3 interface. This signal is not used in the DWC_xpcs. **Exists:** APB3_INTF||APB3_16||APB3_32 **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
prdata_o[(APB_DATA_W-1):0]	O	Read Data. This port carries the read data from the DWC_xpcs CSR to the host. This data is valid only when the pready_o signal is asserted. **Exists:** APB3_INTF||APB3_16||APB3_32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
pready_o	O	Read Data Ready. When high, this signal indicates the following: ■ Data the prdata_o[15:0] signal is valid. The host can accept the data. ■ End of write transfer **Exists:** APB3_INTF||APB3_16||APB3_32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.9 Transmit Interface to PCS-X PHY Signals

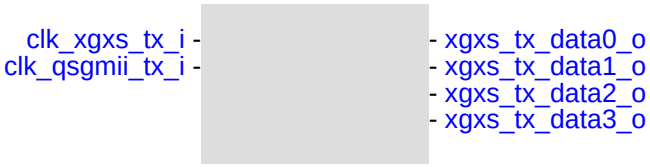


Table 6-9 Transmit Interface to PCS-X PHY Signals

Port Name	I/O	Description
clk_xgxs_tx_i	I	PHY Transmission Clock. This is the clock interface from the PHY used for data transmit logic. The following is the frequency of this clock: ■ 312.5 MHz in 10G mode ■ 125 MHz in 1G mode **Exists:** (!RPCS_ONLY) **Synchronous To:** None **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
clk_qsgmii_tx_i	I	PHY Transmit Clock in QSGMII mode. This is the clock interface from the PHY used for data transmit logic. The following is the frequency of this clock: ■ 250 MHz in QSGMII enabled mode ■ same as clk_xgxs_tx_i in other modes **Exists:** QSGMII_EN **Synchronous To:** None **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xgxs_tx_data0_o[9:0]	O	PHY Transmit Data Lane 0. This port transmits 10 bits of data (per lane) to the PHY. *Note:* In RXAUI mode, combined {xgxs_tx_data0_o, xgxs_tx_data1_o} form the 20-bit data for Lane 0 and combined {xgxs_tx_data2_o, xgxs_tx_data3_o} form the 20-bit data for Lane 1. In QSGMII mode, combined {xgxs_tx_data0_o, xgxs_tx_data1_o} form the 20-bit data for Lane 0 **Exists:** (!RPCS_ONLY) **Synchronous To:** None **Registered:** (!RXAUI&&!QSGMII_EN) ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** Configuration/Operating Mode dependent: ■ clk_qsgmii_tx_i: When operating in QSGMII mode ■ clk_xgxs_tx_i: In all other configurations/modes.
xgxs_tx_data1_o[9:0]	O	PHY Transmit Data Lane 1. This port transmits 10 bits of data (per lane) to the PHY. *Note:* In RXAUI mode, combined {xgxs_tx_data0_o, xgxs_tx_data1_o} form the 20-bit data for Lane 0 and combined {xgxs_tx_data2_o, xgxs_tx_data3_o} form the 20-bit data for Lane 1. In QSGMII mode, combined {xgxs_tx_data0_o, xgxs_tx_data1_o} form the 20-bit data for Lane 0 **Exists:** (!(KX_ONLY||BASE1000X_ONLY||RPCS_ONLY||KRKX_OR_RX))||QSGMII_EN **Synchronous To:** None **Registered:** (!RXAUI&&!QSGMII_EN) ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** Configuration/Operating Mode dependent: ■ clk_qsgmii_tx_i: When operating in QSGMII mode ■ clk_xgxs_tx_i: In all other configurations/modes.
xgxs_tx_data2_o[9:0]	O	PHY Transmit Data Lane 2. This port transmits 10 bits of data (per lane) to the PHY. *Note:* In RXAUI mode, combined {xgxs_tx_data0_o, xgxs_tx_data1_o} form the 20-bit data for Lane 0 and combined {xgxs_tx_data2_o, xgxs_tx_data3_o} form the 20-bit data for Lane 1. **Exists:** (!(KX_ONLY||BASE1000X_ONLY||RPCS_ONLY||KRKX_OR_RX)) **Synchronous To:** clk_xgxs_tx_i **Registered:** (!RXAUI) ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xgxs_tx_data3_o[9:0]	O	PHY Transmit Data Lane 3. This port transmits 10 bits of data (per lane) to the PHY. *Note:* In RXAUI mode, combined {xgxs_tx_data0_o, xgxs_tx_data1_o} form the 20-bit data for Lane 0 and combined {xgxs_tx_data2_o, xgxs_tx_data3_o} form the 20-bit data for Lane 1. **Exists:** (!(KX_ONLY||BASE1000X_ONLY||RPCS_ONLY||KRKX_OR_RX)) **Synchronous To:** clk_xgxs_tx_i **Registered:** (!RXAUI) ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.10 Energy Efficient Ethernet PHY Interface Signals



Table 6-10 Energy Efficient Ethernet PHY Interface Signals

Port Name	I/O	Description
clk_xgxs_rtx_i	I	XGXS Clock Compensation Read Clock. This clock is synchronous to the clk_xgxs_tx_i clock. It is used for clock compensation in the Rx path. **Exists:** (!RPCS_ONLY)&&EEE_EN&&CLKCOMP **Synchronous To:** clk_xgxs_tx_i **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
clk_rpcs_rtx_i	I	PCS-R Clock Compensation Read Clock. This clock is synchronous to the clk_rpcs_tx_i clock. It is used for clock compensation in the Rx path in the KR or PCS-R speed mode. This signal is present when EEE_EN = Enabled in 10GBASE-R PCS or KR_Only configurations. **Exists:** RPCS_EN&&EEE_EN&&CLKCOMP **Synchronous To:** clk_rpcs_tx_i **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
clk_eee_i	I	Free Running Clock. Clock for the Energy Efficient Ethernet block. This is a free running clock, irrespective of current state of the Transmit and Receive. The frequency of this clock can be between 10 MHz to 200 MHz. However, a 100MHz clock is optimal, as it eliminates the need to program EEE Timer related registers inside DWC_xpcs. **Exists:** EEE_EN **Synchronous To:** None **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
rpcs_energy_det_i	I	PCS-R (10GBASE-R or 10GBASE-KR) Energy Detect Signal from PMD. This signal is asserted when the PCS-R PMD detects energy its Rx link. **Exists:** LP_10GBR&&!SNPS_XAUI_PHY **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_lpitx_alert_o	O	This signal when asserted indicates that the PCS controller is sending EEE Alert Pattern. This signal is present only when EEE is selected when DWC_xpcs is configured for 10GBASE-R or KR_KX4_KX or KR_KX4 or KR_KX or KR_Only. **Exists:** LP_10GBR **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_lpitx_quiet_o	O	Transmit Path in Quiet State. This signal is asserted when the EEE controller in Tx path is in the Quiet (power-save) state. When this signal is asserted, you can disable the Tx data path of the PHY or the PMD. In addition, you can use this signal to stop the Transmit clocks to the DWC_xpcs and the PHY. **Exists:** EEE_EN **Synchronous To:** clk_eee_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_lrx_active_o	O	Receive Path in EEE Mode. This signal is asserted when the DWC_xpcs receives LPI Idle the Rx path. It is deactivated when it receives the normal Idle or Data. This signal provides an additional status to the PHY or the PMD. **Exists:** EEE_EN **Synchronous To:** clk_xpcs_rx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_lpirx_quiet_o	O	Receive Path in Quiet State. This signal is asserted when the EEE controller in Rx path is in the Quiet (power-save) state. When this signal is asserted, you can disable the Rx path of the PHY or the PMD. In addition, you can use this signal to stop the Receive clocks to the DWC_xpcs. **Exists:** EEE_EN **Synchronous To:** clk_eee_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_lrx_clk_gat_o	O	MAC Clock Gating Enable Control. This signal can be used for gating the XGMII or GMII Rx Clock towards the MAC. This signal is asserted when the DWC_xpcs receives the LPI Idle Rx path for a period equal to or greater than the number of clk_xpcs_rx_i cycles as programmed in Bits[15:12] of VR XS or PCS MMD EEE Mode Control Register and Bit 10 of SR XS or PCS MMD Control1 Register or VR MII MMD Digital Control1 Register is set to 1. **Exists:** EEE_EN **Synchronous To:** clk_eee_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_lrx_clk_gat_p1_o	O	Port1 MAC Clock Gating Enable Control. This signal can be used for gating the GMII Rx Clock towards the Port1 MAC. This signal is asserted when the DWC_xpcs receives the LPI Idle Rx path for a period equal to or greater than the number of clk_xpcs_rx_i cycles as programmed in Bits[15:12] of VR XS or PCS MMD EEE Mode Control Register and Bit 10 of SR XS or PCS MMD Control1 Register or VR MII MMD_1 Digital Control1 Register is set to 1. **Exists:** EEE_EN&&QSGMII_EN **Synchronous To:** clk_eee_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_lrx_clk_gat_p2_o	O	Port2 MAC Clock Gating Enable Control. This signal can be used for gating the GMII Rx Clock towards the Port2 MAC. This signal is asserted when the DWC_xpcs receives the LPI Idle Rx path for a period equal to or greater than the number of clk_xpcs_rx_i cycles as programmed in Bits[15:12] of VR XS or PCS MMD EEE Mode Control Register and Bit 10 of SR XS or PCS MMD Control1 Register or VR MII MMD_2 Digital Control1 Register is set to 1. **Exists:** EEE_EN&&QSGMII_EN **Synchronous To:** clk_eee_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_lrx_clk_gat_p3_o	O	Port3 MAC Clock Gating Enable Control. This signal can be used for gating the GMII Rx Clock towards the Port3 MAC. This signal is asserted when the DWC_xpcs receives the LPI Idle Rx path for a period equal to or greater than the number of clk_xpcs_rx_i cycles as programmed in Bits[15:12] of VR XS or PCS MMD EEE Mode Control Register and Bit 10 of SR XS or PCS MMD Control1 Register or VR MII MMD_3 Digital Control1 Register is set to 1. **Exists:** EEE_EN&&QSGMII_EN **Synchronous To:** clk_eee_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.11 Receive Interface from PCS-X PHY Signals

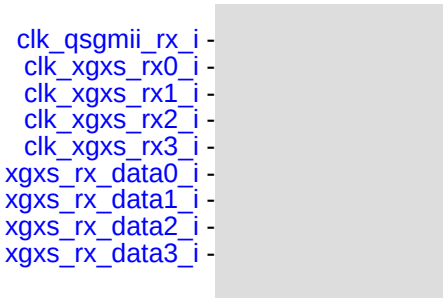


Table 6-11 Receive Interface from PCS-X PHY Signals

Port Name	I/O	Description
clk_qsgmii_rx_i	I	PHY Receive Clock in QSGMII mode. This is the recovered clock from the PHY used for data receive logic. The following is the frequency of this clock: ■ 250 MHz in QSGMII enabled mode ■ same as clk_xgxs_rx0_i in other modes. **Exists:** QSGMII_EN **Synchronous To:** None **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
clk_xgxs_rx0_i	I	PHY Receive Clock for Lane 0. This signal is the recovered clock (per lane) from the PHY. In 10G mode, the frequency of this clock is 312.5 MHz. In 1G mode, the frequency of this clock is 125 MHz. **Exists:** (!RPCS_ONLY) **Synchronous To:** None **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
clk_xgxs_rx1_i	I	PHY Receive Clock for Lane 1. This signal is the recovered clock (per lane) from the PHY. In 10G mode, the frequency of this clock is 312.5 MHz. In 1G mode, the frequency of this clock is 125 MHz. **Exists:** (!(KX_ONLY||BASE1000X_ONLY||RPCS_ONLY||KRKX_OR_RX)) **Synchronous To:** None **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
clk_xgxs_rx2_i	I	PHY Receive Clock for Lane 2. This signal is the recovered clock (per lane) from the PHY. In 10G mode, the frequency of this clock is 312.5 MHz. In 1G mode, the frequency of this clock is 125 MHz. **Exists:** (!(KX_ONLY||BASE1000X_ONLY||RPCS_ONLY||KRKX_OR_RX)) **Synchronous To:** None **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
clk_xgxs_rx3_i	I	PHY Receive Clock for Lane 3. This signal is the recovered clock (per lane) from the PHY. In 10G mode, the frequency of this clock is 312.5 MHz. In 1G mode, the frequency of this clock is 125 MHz. **Exists:** (!(KX_ONLY||BASE1000X_ONLY||RPCS_ONLY||KRKX_OR_RX)) **Synchronous To:** None **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_rx_data0_i[9:0]	I	PHY Receive Data Lane 0. This port receives the 10-bit recovered deserialized data (per lane) from the PHY. *Note:* In RXAUI mode, combined {xgxs_rx_data0_o, xgxs_rx_data1_o} form the 20-bit data for Lane 0 and combined{xgxs_rx_data2_o, xgxs_rx_data3_o} form the 20-bit data for Lane 1. **Exists:** (!RPCS_ONLY) **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** Configuration/Operating Mode dependent: ■ clk_qsgmii_rx_i: When operating in QSGMII Mode ■ clk_xgxs_rx{lane}_i: In all other configurations/modes

Port Name	I/O	Description
xgxs_rx_data1_i[9:0]	I	PHY Receive Data Lane 1. This port receives the 10-bit recovered deserialized data (per lane) from the PHY. *Note:* In RXAUI mode, combined {xgxs_rx_data0_o, xgxs_rx_data1_o} form the 20-bit data for Lane 0 and combined{xgxs_rx_data2_o, xgxs_rx_data3_o} form the 20-bit data for Lane 1. **Exists:** (!(KX_ONLY||BASE1000X_ONLY||RPCS_ONLY||KRKX_OR_RX))||QSGMII_EN **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** Configuration/Operating Mode dependent: ■ clk_qsgmii_rx_i: When operating in QSGMII Mode ■ clk_xgxs_rx{lane}_i: In all other configurations/modes
xgxs_rx_data2_i[9:0]	I	PHY Receive Data Lane 2. This port receives the 10-bit recovered deserialized data (per lane) from the PHY. *Note:* In RXAUI mode, combined {xgxs_rx_data0_o, xgxs_rx_data1_o} form the 20-bit data for Lane 0 and combined{xgxs_rx_data2_o, xgxs_rx_data3_o} form the 20-bit data for Lane 1. **Exists:** (!(KX_ONLY||BASE1000X_ONLY||RPCS_ONLY||KRKX_OR_RX)) **Synchronous To:** clk_xgxs_rx{lane}_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_rx_data3_i[9:0]	I	PHY Receive Data Lane 3. This port receives the 10-bit recovered deserialized data (per lane) from the PHY. *Note:*In RXAUI mode, combined {xgxs_rx_data0_o, xgxs_rx_data1_o} form the 20-bit data for Lane 0 and combined{xgxs_rx_data2_o, xgxs_rx_data3_o} form the 20-bit data for Lane 1. **Exists:** (!(KX_ONLY||BASE1000X_ONLY||RPCS_ONLY||KRKX_OR_RX)) **Synchronous To:** clk_xgxs_rx{lane}_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.12 Miscellaneous Control/Status Interface Signals

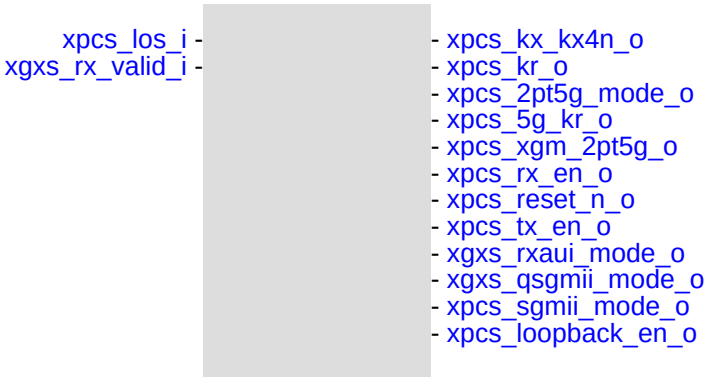


Table 6-12 Miscellaneous Control/Status Interface Signals

Port Name	I/O	Description
xpcs_los_i[(NUM_OF_LANES-1):0]	I	Loss of Signal. This signal is the inverse of the signal-detect signal (defined in IEEE Std 802.3) from the PMA or PMD Layer. When asserted, this signal indicates that the PMA or PMD layer has detected an error condition the physical connectivity. Otherwise, it indicates the normal condition. In the KR_KX4_KX, KR_KX4, or KR_KX mode, when the xpcs_kr_o is asserted, this signal indicates the loss of signal from the PCS-R PHY. Otherwise, it indicates the loss of signal from the PCS-X PHY. *Values:* ■ 0: Link restored. ■ 1: Receive side is unable to receive data, indicating loss of signal. The synchronization state in DWC_xpcs resets to LOSS_OF_SYNC. *Note:* This signal is not used in the RXAUI mode. **Exists:** !KR_PHY **Synchronous To:** Asynchronous **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High **Registered:** Yes, with: ■ clk_csr_i ■ clk_xgxs_rx{lane}_i ■ clk_eee_i

Port Name	I/O	Description
xpcs_kx_kx4n_o	O	KX or KX4 Selection Control. This signal controls the PHY speed. Values: ■ 0: 10G mode ■ 1: 1G mode **Exists:** KX4_KX_PRESENT||SUPPORT_10GX_1GX **Synchronous To:** (!SNPS_XAUI_PHY&&MDIO_INTF) ? "clk_mdc_i" : "clk_csr_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_kr_o	O	PHY Speed Control. When asserted, this signal indicates that the current speed mode is 10GBASE-R or 10GBASE-KR (10.3125 Gbps link speed). If 'Add 5GBASE-KR Support' option is enabled, this signal indicates that the operating mode is BASE-R. The operating speed can be 10G or 5G, based on the value of xpcs_5g_kr_o signal. **Exists:** RPCS_EN&&!RPCS_ONLY **Synchronous To:** (!SNPS_XAUI_PHY&&MDIO_INTF) ? "clk_mdc_i" : "clk_csr_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_2pt5g_mode_o	O	2.5G GMII Mode Control. This signal is used to switch between 1G and 2.5G GMII Modes. When this signal is set, it indicates that current mode of operation is 2.5G. This signal can be used for clock multiplexing to enable switching between 1G and 2.5G Modes. **Exists:** (SUPPORT_1G||BASE1000X_ONLY||KX_PRESENT) **Synchronous To:** (!SNPS_XAUI_PHY&&MDIO_INTF) ? "clk_mdc_i" : "clk_csr_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_5g_kr_o	O	5G KR Mode Control. When this signal is high, it indicates that the current operating speed is 5Gbps (as per IEEE clause 129). In addition, xpcs_kr_o signal is high, to indicate that the operating mode is BASE-R (64/66b encoding). **Exists:** ADD_5G **Synchronous To:** {(!SNPS_XAUI_PHY&&MDIO_INTF) ? "clk_mdc_i" : "clk_csr_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_xgm_2pt5g_o	O	2.5GBASE-KX (through XGMII) Mode Control. When this signal is high, it indicates that the current operating mode is 2.5GBASE-KX (as per IEEE clause 127). During this mode, data rate at XGMII interface is 2.5Gbps and the serial data rate is 3.125Gbps (due to 8b/10b encoding). **Exists:** ADD_2PT5G **Synchronous To:** {(!SNPS_XAUI_PHY&&MDIO_INTF) ? "clk_mdc_i" : "clk_csr_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx_en_o[(NUM_OF_LANES-1):0]	O	For Synopsys PHY:. Enable Receive Data This signal enables the Receiver Data Path of the PHY. **Exists:** (SNPS_XAUI_PHY||(BACKPLANE_ETHERNET_PCS||EEE_EN||USXGMII_EN))&&!KR_PHY&&!E12&&!E16&&!E25&&!E32 **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xgxs_rx_valid_i[(NUM_OF_LANES-1):0]	I	Receiver Valid. This signal indicates if DPLL is locked. *Values:* ■ 0: Unlocked ■ 1: Locked This signal also validates data the xgxs_rx_-data{lane}_i[9:0]/rpcs_rx_data_i signal. This signal is present only if SNPS_XAUI_PHY = Enabled or MAIN_MODE = Backplane Ethernet PCS. **Exists:** (SNPS_XAUI_PHY||PMA_MMD)&&!KR_PHY **Synchronous To:** Asynchronous **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High **Registered:** Yes, with: ■ clk_csr_i ■ clk_cko_alive_i
xpcs_reset_n_o	O	PHY Reset. Active Low Reset signal to PHY. **Exists:** Always **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** Low
xpcs_tx_en_o[(NUM_OF_LANES-1):0]	O	Transmit Lane Enable or Disable Control. This signal controls the Transmit data path of the PHY. *Values:* ■ 0: Corresponding Transmit lane is disabled. ■ 1: Corresponding Transmit lane is enabled (default). **Exists:** !SNPS_XAUI_PHY&&(BACKPLANE_ETHERNET_PCS||EEE_EN||USXGMII_EN) **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xgxs_rxaui_mode_o	O	RXAUI Mode Control Output. This output is used to switch between the XAUI and RXAUI mode. When set to 1, the PHY switches to the RXAUI interface (20-bit interface working at 312.5 MHz with a serial baud rate of 6.25 Gbps). When set to 0, the PHY switches to the XAUI mode (10-bit interface working at 312.5 MHz with serial baud rate of 3.125 Gbps). *Note:* The external data and clock multiplexing is required between the PHY and the DWC_xpcs. For more information, see DesignWare Cores Ethernet PCS User Guide. **Exists:** RXAUI **Synchronous To:** {(!SNPS_XAUI_PHY&&MDIO_INTF) ? "None" : "clk_csr_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xgxs_qsgmii_mode_o	O	QSGMII Mode Control Output. When set to 1, the PHY switches to the QSGMII interface (Lane0 width changed to 20-bit interface working at 250MHz with a serial baud rate of 5 Gbps). *Note:* The external data and clock multiplexing is required between the PHY and the DWC_xpcs. For more information, see DesignWare Cores Ethernet PCS User Guide. **Exists:** QSGMII_EN **Synchronous To:** {(!SNPS_XAUI_PHY&&MDIO_INTF) ? "clk_mdc_i" : "clk_csr_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_sgmii_mode_o	O	SGMII Mode Indication Status. This signal will be high when DWC_xpcs is configured to SGMII mode. ie, PCS_MODE field of VR_MII_AN_CTRL is programmed to 2'b10 **Exists:** SGMII_EN||QSGMII_EN **Synchronous To:** {(!SNPS_XAUI_PHY&&MDIO_INTF) ? "clk_mdc_i" : "clk_csr_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_loopback_en_o[(LOOPBACK_W-1):0]	O	Enable Loopback (Tx-to-Rx) in the PHY This signal indicates that software wants to enable loopback mode in the PHY. **Exists:** (!SNPS_XAUI_PHY&&!PCS_10G)||UP5||KR_PHY **Synchronous To:** None **Registered:** (!CL37_AN) ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** Configuration dependent: ■ clk_mdc_i: For configurations with CSR_INTERFACE = MDIO and SNPS_XAUI_PHY=Disabled ■ clk_csr_i: For all other configurations

6.13 Synopsys Multi-protocol 6G PHY Transmit Status and Control Interface Signals

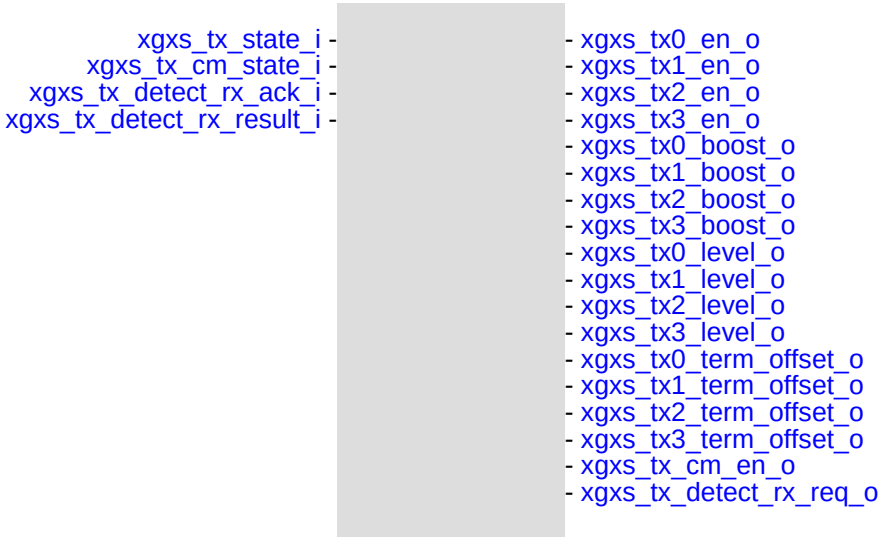


Table 6-13 Synopsys Multi-protocol 6G PHY Transmit Status and Control Interface Signals

Port Name	I/O	Description
xgxs_tx0_en_o	O	Transmitter Enable Lane 0. **Exists:** SNPS_XAUI_PHY&&(UP3||UP5) **Synchronous To:** clk_cko_alive_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_tx1_en_o	O	Transmitter Enable Lane 1. **Exists:** SNPS_XAUI_PHY&&(UP3||UP5)&&(!(KX_ONLY||BASE1000X_ONLY)) **Synchronous To:** clk_cko_alive_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_tx2_en_o	O	Transmitter Enable Lane 2. **Exists:** SNPS_XAUI_PHY&&(UP3||UP5)&&(!(KX_ONLY||BASE1000X_ONLY)) **Synchronous To:** clk_cko_alive_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xgxs_tx3_en_o	O	Transmitter Enable Lane 3. **Exists:** SNPS_XAUI_PHY&&(UP3||UP5)&&(!(KX_ONLY||BASE1000X_ONLY)) **Synchronous To:** clk_cko_alive_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_tx0_boost_o[6:0]	O	Transmitter Pre-Emphasis Level Adjustment Control of the PHY. This port sets the amplitude of the transmitter driver output. **Exists:** SNPS_XAUI_PHY&&(UP3||UP5) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_tx1_boost_o[6:0]	O	Transmitter Pre-Emphasis Level Adjustment Control of the PHY lane 1. This port sets the amplitude of the transmitter driver output. **Exists:** SNPS_XAUI_PHY&&(UP3||UP5)&&(!(KX_ONLY||BASE1000X_ONLY)) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_tx2_boost_o[6:0]	O	Transmitter Pre-Emphasis Level Adjustment Control of the PHY lane 2. This port sets the amplitude of the transmitter driver output. **Exists:** SNPS_XAUI_PHY&&(UP3||UP5)&&(!(KX_ONLY||BASE1000X_ONLY)) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_tx3_boost_o[6:0]	O	Transmitter Pre-Emphasis Level Adjustment Control of the PHY lane 3. This port sets the amplitude of the transmitter driver output. **Exists:** SNPS_XAUI_PHY&&(UP3||UP5)&&(!(KX_ONLY||BASE1000X_ONLY)) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xgxs_tx0_level_o[(LVL_W-1):0]	O	Transmitter Amplitude Adjustment Control of the PHY for Lane 0. This signal controls the adjustment of the output amplitude of the Transmit driver. **Exists:** UP5 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_tx1_level_o[(LVL_W-1):0]	O	Transmitter Amplitude Adjustment Control of the PHY for Lane 1. This signal controls the adjustment of the output amplitude of the transmit driver. **Exists:** UP5&&(!(KX_ONLY||BASE1000X_ONLY)) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_tx2_level_o[(LVL_W-1):0]	O	Transmitter Amplitude Adjustment Control of the PHY for Lane 2. This signal controls the adjustment of the output amplitude of the transmit driver. **Exists:** UP5&&(!(KX_ONLY||BASE1000X_ONLY)) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_tx3_level_o[(LVL_W-1):0]	O	Transmitter Amplitude Adjustment Control of the PHY for Lane 3. This signal controls the adjustment of the output amplitude of the transmit driver. **Exists:** UP5&&(!(KX_ONLY||BASE1000X_ONLY)) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_tx0_term_offset_o[4:0]	O	Controls Transmitter termination offset in Lane 0. This output can be controlled by programming VR XS,PMA or MII Synopsys Multi-protocol 6G PHY Tx General Control Register 2 and VR XS,PMA or MII Synopsys Multi-protocol 6G PHY Tx General Control Register 3 **Exists:** UP5&&SSP **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xgxs_tx1_term_offset_o[4:0]	O	Controls Transmitter termination offset in Lane 1. This output can be controlled by programming VR XS,PMA or MII Synopsys Multi-protocol 6G PHY Tx General Control Register 2 and VR XS,PMA or MII Synopsys Multi-protocol 6G PHY Tx General Control Register 3 **Exists:** UP5&&SSP&&(!(KX_ONLY||BASE1000X_ONLY)) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_tx2_term_offset_o[4:0]	O	Controls Transmitter termination offset in Lane 2. This output can be controlled by programming VR XS,PMA or MII Synopsys Multi-protocol 6G PHY Tx General Control Register 2 and VR XS,PMA or MII Synopsys Multi-protocol 6G PHY Tx General Control Register 3 **Exists:** UP5&&SSP&&(!(KX_ONLY||BASE1000X_ONLY)) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_tx3_term_offset_o[4:0]	O	Controls Transmitter termination offset in Lane 3. This output can be controlled by programming VR XS,PMA or MII Synopsys Multi-protocol 6G PHY Tx General Control Register 2 and VR XS,PMA or MII Synopsys Multi-protocol 6G PHY Tx General Control Register 3 **Exists:** UP5&&SSP&&(!(KX_ONLY||BASE1000X_ONLY)) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_tx_cm_en_o[(NUM_OF_LANES-1):0]	O	PHY Transmit Common-Mode Control Enable. When asserted, this signal enables the Transmit common mode to be held at the half value of vptx. Otherwise, the common mode is weakly held to ground through a high-impedance connection. **Exists:** UP5 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xgxs_tx_state_i[(NUM_OF_LANES-1):0]	I	Transmit Active State Machine Status. The PHY asserts this signal when the transmitter has reached the intended level, and it is ready to Transmit data. **Exists:** UP5 **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_tx_cm_state_i[(NUM_OF_LANES-1):0]	I	Transmit Common Mode Control State Machine Status. The PHY asserts this signal when the Transmit common mode has reached the intended level. During a receiver detection or recovery period, this signal is set to 0. **Exists:** UP5 **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xgxs_tx_detect_rx_req_o[(NUM_OF_LANES-1):0]	O	PHY Receiver Detection Request. When asserted, this signal initiates the receiver detection operation in the PHY. In addition, this signal internally enables the xgxs_tx_cm_en_o[] and xpcs_tx_en_o[] signals if they are not already enabled. **Exists:** UP5 **Synchronous To:** clk_xgxs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xgxs_tx_detect_rx_ack_i[(NUM_OF_LANES-1):0]	I	Receiver Detect Acknowledge Status. When asserted, this signal indicates that the receiver detection operation is complete. When the xgxs_tx_detect_rx_req_o[] signal is set to 1'b0, this signal is de-asserted. **Exists:** UP5 **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xgxs_tx_detect_rx_result_i[(NUM_OF_LANES-1):0]	I	Receiver Detect Result Status of the PHY. When asserted, this signal indicates that a receiver is detected. This signal is valid only while the xgxs_tx_detect_rx_ack_i signal is set to 1'b1. **Exists:** UP5 **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.14 Synopsys Multi-protocol 12G/16G/25G PHY Receive Status and Control Interface Signals

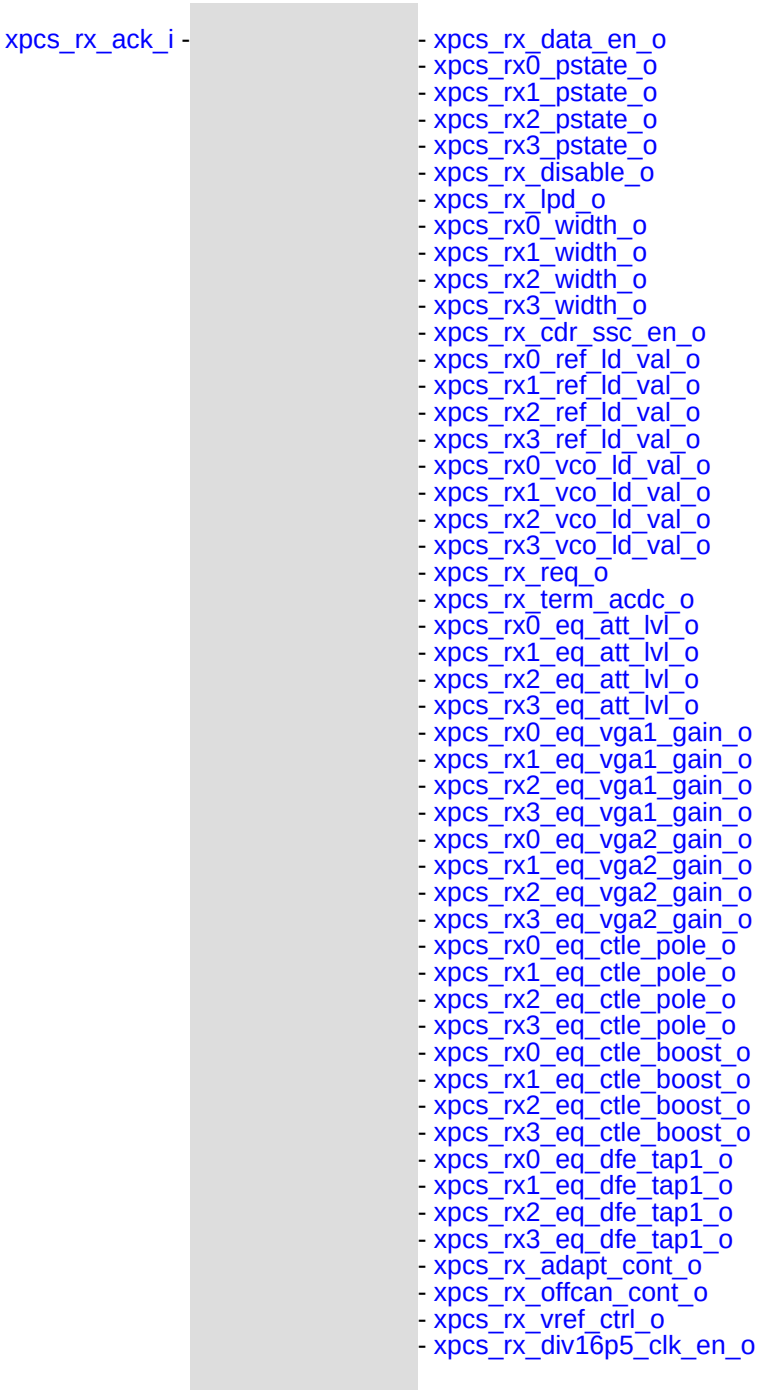


Table 6-14 Synopsys Multi-protocol 12G/16G/25G PHY Receive Status and Control Interface Signals

Port Name	I/O	Description
xpcs_rx_data_en_o[(NUM_OF_LANES-1):0]	O	Receiver Data Enable. This signal enables the receiver data output from PHY. **Exists:** E12||E16||E25||E32 **Synchronous To:** {EEE_EN ? "None" : "clk_csr_i"} **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx0_pstate_o[1:0]	O	Receiver Power state -Lane 0. This signal controls the power state of the receiver. This signal should be connected to rx0_pstate[1:0] input port of the PHY. **Exists:** KR_PHY||E12||E16||E25||E32 **Synchronous To:** {EEE_EN ? "None" : "clk_csr_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx1_pstate_o[1:0]	O	Receiver Power state -Lane 1. This signal controls the power state of the receiver. **Exists:** (KR_PHY||E12||E16||E25||E32)&&!SINGLE_LANE_CFG **Synchronous To:** {EEE_EN ? "None" : "clk_csr_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx2_pstate_o[1:0]	O	Receiver Power state -Lane 2. This signal controls the power state of the receiver. **Exists:** (KR_PHY||E12||E16||E25||E32)&&!SINGLE_LANE_CFG **Synchronous To:** {EEE_EN ? "None" : "clk_csr_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx3_pstate_o[1:0]	O	Receiver Power state -Lane 3. This signal controls the power state of the receiver. **Exists:** (KR_PHY||E12||E16||E25||E32)&&!SINGLE_LANE_CFG **Synchronous To:** {EEE_EN ? "None" : "clk_csr_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_rx_disable_o[(NUM_OF_LANES-1):0]	O	Receiver Disable. This signal is driven from RX_DISABLE fields of VR XS,PMA or MII Synopsys Multi-protocol 12G/16G PHY Rx Power State Control Register. This signal should be connected to rxX_disable input port of the PHY. **Exists:** E12||E16||E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx_lpd_o[(NUM_OF_LANES-1):0]	O	Receive Lane Power Down. This signal is driven from RX_LPD_3_1 and RX_LPD_0 fields of VR_XS_PMA_MP_12G_16G_RX_GENCTRL2/VR_XS_PMA_MP_25G_RX_GENCTRL2/VR_MII_MP_12G_16G_RX_GENCTRL2/VR_MII_MP_25G_RX_GENCTRL2 Register.Also this signal is controlled by PWRSV bit[11] of VR_XS_PCS_DIG_CTRL1 Register This signal powers down the PHY receiver into a power state equivalent of P1. This signal should be connected to rxX_lpd input pin of the PHY. **Exists:** E12||E16||E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx0_width_o[(RX_W-1):0]	O	Rx Data Width -Lane 0. This signal is driven from RX0_WIDTH field of VR_XS_PMA_MP_12G_16G_RX_GENCTRL2/VR_MII_MP_12G_16G_RX_GENCTRL2/VR_XS_PMA_MP_25G_RX_WIDTH_CTRL/VR_MII_MP_25G_RX_WIDTH_CTRL Register. This signal selects the width of the receive data output from the PHY. This signal should be connected to rx0_width input pin of the PHY. **Exists:** (E12||E16||E25||E32)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx1_width_o[(RX_W-1):0]	O	Rx Data Width -Lane 1. **Exists:** (E12||E16||E25||E32)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_rx2_width_o[(RX_W-1):0]	O	Rx Data Width -Lane 2. **Exists:** (E12||E16||E25||E32)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx3_width_o[(RX_W-1):0]	O	Rx Data Width -Lane 3. **Exists:** (E12||E16||E25||E32)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx_cdr_ssc_en_o[(NUM_OF_LANES-1):0]	O	Rx CDR SSC-Mode Enable. This signal is driven from CDR_SSC_EN_0/CDR_SSC_EN_3_1 fields of VR_XS_PMA_MP_12G_16G_25G_RX_CDR_CTRL/VR_MII_MP_12G_16G_25G_RX_CDR_CTRL Register. This signal controls the CDR tracking gains and duration of the PHY. This signal must be asserted when Rx data has SS Clock. This signal must be connected to rxX_cdr_ssc_en input pin of the PHY. **Exists:** E12||E16||E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx0_ref_ld_val_o[(RF_LD_W-1):0]	O	Rx VCO calibration reference load value -Lane 0. This signal is driven from VCO_REF_LD_0 field of VR_XS_PMA_MP_12G_VCO_CAL_REF0/VR_XS_PMA_MP_16G_25G_VCO_CAL_REF0 Register. This signal is used to load internal calibration registers (in the PHY) to perform Rx VCO calibration. This signal must be connected to rx0_ref_ld_val input port of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_rx1_ref_ld_val_o[(RF_LD_W-1):0]	O	Rx VCO calibration reference load value -Lane 1. This signal is driven from VCO_REF_LD_1 field of VR_XS_PMA_MP_12G_VCO_CAL_REF0/VR_XS_PMA_MP_16G_25G_VCO_CAL_REF0 Register. This signal is used to load internal calibration registers (in the PHY) to perform Rx VCO calibration. This signal must be connected to rx1_ref_ld_val input port of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx2_ref_ld_val_o[(RF_LD_W-1):0]	O	Rx VCO calibration reference load value -Lane 2. This signal is driven from VCO_REF_LD_2 field of VR_XS_PMA_MP_12G_VCO_CAL_REF1/VR_XS_PMA_MP_16G_25G_VCO_CAL_REF1 Register. This signal is used to load internal calibration registers (in the PHY) to perform Rx VCO calibration. This signal must be connected to rx2_ref_ld_val input port of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx3_ref_ld_val_o[(RF_LD_W-1):0]	O	Rx VCO calibration reference load value -Lane 3. This signal is driven from VCO_REF_LD_3 field of VR_XS_PMA_MP_12G_VCO_CAL_REF1/VR_XS_PMA_MP_16G_25G_VCO_CAL_REF1 Register. This signal is used to load internal calibration registers (in the PHY) to perform Rx VCO calibration. This signal must be connected to rx3_ref_ld_val input port of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_rx0_vco_ld_val_o[12:0]	O	Rx VCO calibration load value -Lane 0. This signal is driven from VCO_LD_VAL_0 fields of VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD0/VR_MII_MP_12G_16G_25G_VCO_CAL_LD0 Register. This signal is used to load internal calibration registers (in the PHY) to perform Rx VCO calibration. This signal must be connected to rx0_vco_ld_val input port of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx1_vco_ld_val_o[12:0]	O	Rx VCO calibration load value -Lane 1. This signal is driven from VCO_LD_VAL_1 fields of VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD1 Register. This signal is used to load internal calibration registers (in the PHY) to perform Rx VCO calibration. This signal must be connected to rx1_vco_ld_val input port of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx2_vco_ld_val_o[12:0]	O	Rx VCO calibration load value -Lane 2. This signal is driven from VCO_LD_VAL_2 fields of VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD2 Register. This signal is used to load internal calibration registers (in the PHY) to perform Rx VCO calibration. This signal must be connected to rx2_vco_ld_val input port of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_rx3_vco_ld_val_o[12:0]	O	Rx VCO calibration load value -Lane 3. This signal is driven from VCO_LD_VAL_3 fields of VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD3 Register. This signal is used to load internal calibration registers (in the PHY) to perform Rx VCO calibration. This signal must be connected to rx3_vco_ld_val input port of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx_req_o[(NUM_OF_LANES-1):0]	O	Receiver Request. This signal is driven from RX_REQ_0/RX_REQ_3_1 fields of VR_XS_PMA_MP_12G_16G_RX_GENCTRL2/VR_XS_PMA_MP_25G_RX_GENCTRL2/VR_MII_MP_12G_16G_RX_GENCTRL2/VR_MII_MP_25G_RX_GENCTRL2 Register. This signal indicates a new receive setting request. This signal must be connected to rxX_req input pin of the PHY. **Exists:** E12||E16||E25||E32 **Synchronous To:** {EEE_EN ? "None" : "clk_csr_i"} **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx_ack_i[(NUM_OF_LANES-1):0]	I	Receive Acknowledge. The value of this signal can be read from RX_ACK_0/RX_ACK_3_1 fields of VR_XS_PMA_MP_12G_16G_25G_RX_STS/VR_MII_MP_12G_16G_25G_RX_STS register. Assertion of this signal indicates that the request receiver setting is complete. This signal must be connected to rxX_ack output pin of the PHY. **Exists:** E12||E16||E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_rx_term_acdc_o[(NUM_OF_LANES-1):0]	O	Rx termination control. This signal is driven from RX_TERM_ACDC_0/RX_TERM_ACDC_3_1 fields of VR_XS_PMA_MP_12G_16G_25G_RX_GENCTRL1/VR_MII_MP_12G_16G_25G_RX_GENCTRL1 Register. This controls the PHY receiver termination. This signal must be connected to rxX_term_acdc. **Exists:** E12||E16||E25 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx0_eq_att_lvl_o[2:0]	O	Rx Equalization Attenuation Level -Lane 0. This signal is driven from RX0_EQ_ATT_LVL field of VR_XS_PMA_MP_12G_16G_25G_RX_ATTN_CTRL/VR_MII_MP_12G_16G_25G_RX_ATTN_CTRL Register. This signal controls the AFE attenuation level. This signal should be connected to rx0_eq_att_lvl input port of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx1_eq_att_lvl_o[2:0]	O	Rx Equalization Attenuation Level -Lane 1. This signal is driven from RX1_EQ_ATT_LVL field of VR_XS_PMA_MP_12G_16G_25G_RX_ATTN_CTRL Register. This signal controls the AFE attenuation level. This signal should be connected to rx1_eq_att_lvl input port of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx2_eq_att_lvl_o[2:0]	O	Rx Equalization Attenuation Level -Lane 2. This signal is driven from RX2_EQ_ATT_LVL field of VR_XS_PMA_MP_12G_16G_25G_RX_ATTN_CTRL Register. This signal controls the AFE attenuation level. This signal should be connected to rx2_eq_att_lvl input port of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_rx3_eq_att_lvl_o[2:0]	O	Rx Equalization Attenuation Level -Lane 3. This signal is driven from RX3_EQ_ATT_LVL field of VR_XS_PMA_MP_12G_16G_25G_RX_ATTN_CTRL Register. This signal controls the AFE attenuation level. This signal should be connected to rx3_eq_att_lvl input port of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx0_eq_vga1_gain_o[(VGA_W-1):0]	O	Rx Equalization VGA Gain1 -Lane 0. This signal is driven from VGA1_GAIN_0 field of VR_XS_PMA_MP_12G_RX_EQ_CTRL0/VR_MII_MP_12G_RX_EQ_CTRL0/VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL0/VR_MII_MP_16G_25G_RX_EQ_CTRL0 Register. This signal controls the AFE first stage Variable Gain Amplifier Gain of the PHY. This signal should be connected to rx0_eq_vga1_gain input port of the PHY. **Exists:** (E12||E16||E25)&&!(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx1_eq_vga1_gain_o[(VGA_W-1):0]	O	Rx Equalization VGA Gain1 -Lane 1. This signal is driven from VGA1_GAIN_1 field of VR_XS_PMA_MP_12G_RX_EQ_CTRL1/VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL1 Register. This signal controls the AFE first stage Variable Gain Amplifier Gain of the PHY. This signal should be connected to rx1_eq_vga1_gain input port of the PHY. **Exists:** (E12||E16||E25)&&!SINGLE_LANE_CFG&&!(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_rx2_eq_vga1_gain_o[(VGA_W-1):0]	O	Rx Equalization VGA Gain1 -Lane 2. This signal is driven from VGA1_GAIN_2 field of VR_XS_PMA_MP_12G_RX_EQ_CTRL2/VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL2 Register. This signal controls the AFE first stage Variable Gain Amplifier Gain of the PHY. This signal should be connected to rx2_eq_vga1_gain input port of the PHY. **Exists:** (E12||E16||E25)&&!SINGLE_LANE_CFG&&!(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx3_eq_vga1_gain_o[(VGA_W-1):0]	O	Rx Equalization VGA Gain1 -Lane 3. This signal is driven from VGA1_GAIN_0 field of VR_XS_PMA_MP_12G_RX_EQ_CTRL3/VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL3 Register. This signal controls the AFE first stage Variable Gain Amplifier Gain of the PHY. This signal should be connected to rx3_eq_vga1_gain input port of the PHY. **Exists:** (E12||E16||E25)&&!SINGLE_LANE_CFG&&!(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_rx0_eq_vga2_gain_o[(VGA_W-1):0]	O	Rx Equalization VGA Gain2 -Lane 0. This signal is driven from VGA2_GAIN_0 fields of VR_XS_PMA_MP_12G_RX_EQ_CTRL0/VR_MII_MP_12G_RX_EQ_CTRL0/VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL0/VR_MII_MP_16G_25G_RX_EQ_CTRL0 Register. This signal controls the AFE second stage Variable Gain Amplifier Gain of the PHY. This signal should be connected to rx0_eq_vga2_gain input port of the PHY. **Exists:** (E12||E16||E25)&&!(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx1_eq_vga2_gain_o[(VGA_W-1):0]	O	Rx Equalization VGA Gain2 -Lane 1. This signal is driven from VGA2_GAIN_1 fields of VR_XS_PMA_MP_12G_RX_EQ_CTRL1/VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL1 Register. This signal controls the AFE second stage Variable Gain Amplifier Gain of the PHY. This signal should be connected to rx1_eq_vga2_gain input port of the PHY. **Exists:** (E12||E16||E25)&&!SINGLE_LANE_CFG&&!(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_rx2_eq_vga2_gain_o[(VGA_W-1):0]	O	Rx Equalization VGA Gain2 -Lane 2. This signal is driven from VGA2_GAIN_2 fields of VR_XS_PMA_MP_12G_RX_EQ_CTRL2/VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL2 Register. This signal controls the AFE second stage Variable Gain Amplifier Gain of the PHY. This signal should be connected to rx2_eq_vga2_gain input port of the PHY. **Exists:** (E12||E16||E25)&&!SINGLE_LANE_CFG&&!(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx3_eq_vga2_gain_o[(VGA_W-1):0]	O	Rx Equalization VGA Gain2 -Lane 3. This signal is driven from VGA2_GAIN_3 fields of VR_XS_PMA_MP_12G_RX_EQ_CTRL3/VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL3 Register. This signal controls the AFE second stage Variable Gain Amplifier Gain of the PHY. This signal should be connected to rx3_eq_vga2_gain input port of the PHY. **Exists:** (E12||E16||E25)&&!SINGLE_LANE_CFG&&!(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_rx0_eq_ctle_pole_o[(POL_W-1):0]	O	Rx Equalization CTLE Pole -Lane 0. This signal is driven from CTLE_POLE_0 field of VR_XS_PMA_MP_12G_RX_EQ_CTRL0/VR_MII_MP_12G_RX_EQ_CTRL0/VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL0/VR_MII_MP_16G_25G_RX_EQ_CTRL0 Register. This signal controls the continuous time linear equalization (CTLE) boost pole location of the PHY. This signal should be connected to rx0_eq_ctle_pole input port of the PHY. **Exists:** (E12||E16||E25)&&!(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx1_eq_ctle_pole_o[(POL_W-1):0]	O	Rx Equalization CTLE Pole -Lane 1. This signal is driven from CTLE_POLE_1 field of VR_XS_PMA_MP_12G_RX_EQ_CTRL1/VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL1 Register. This signal controls the continuous time linear equalization (CTLE) boost pole location of the PHY. This signal should be connected to rx1_eq_ctle_pole input port of the PHY. **Exists:** (E12||E16||E25)&&!SINGLE_LANE_CFG&&!(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_rx2_eq_ctle_pole_o[(POL_W-1):0]	O	Rx Equalization CTLE Pole -Lane 2. This signal is driven from CTLE_POLE_2 field of VR_XS_PMA_MP_12G_RX_EQ_CTRL2/VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL2 Register. This signal controls the continuous time linear equalization (CTLE) boost pole location of the PHY. This signal should be connected to rx2_eq_ctle_pole input port of the PHY. **Exists:** (E12||E16||E25)&&!SINGLE_LANE_CFG&&!(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx3_eq_ctle_pole_o[(POL_W-1):0]	O	Rx Equalization CTLE Pole -Lane 3. This signal is driven from CTLE_POLE_3 field of VR_XS_PMA_MP_12G_RX_EQ_CTRL3/VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL3 Register. This signal controls the continuous time linear equalization (CTLE) boost pole location of the PHY. This signal should be connected to rx3_eq_ctle_pole input port of the PHY. **Exists:** (E12||E16||E25)&&!SINGLE_LANE_CFG&&!(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx0_eq_ctle_boost_o[4:0]	O	Rx Equalization CTLE Boost -Lane 0. This signal is driven from CTLE_BOOST_0 field of VR_XS_PMA_MP_12G_RX_EQ_CTRL0/VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL0/VR_MII_MP_12G_RX_EQ_CTRL0/VR_MII_MP_16G_25G_RX_EQ_CTRL0 Register. This signal controls the CTLE Boost Level of the PHY. This signal should be connected to rx0_eq_ctle_boost input port of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_rx1_eq_ctle_boost_o[4:0]	O	Rx Equalization CTLE Boost -Lane 1. This signal is driven from CTLE_BOOST_1 field of VR_XS_PMA_MP_12G_RX_EQ_CTRL1/VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL1 Register. This signal controls the CTLE Boost Level of the PHY. This signal should be connected to rx1_eq_ctle_boost input port of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx2_eq_ctle_boost_o[4:0]	O	Rx Equalization CTLE Boost -Lane 2. This signal is driven from CTLE_BOOST_2 field of VR_XS_PMA_MP_12G_RX_EQ_CTRL2/VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL2 Register. This signal controls the CTLE Boost Level of the PHY. This signal should be connected to rx2_eq_ctle_boost input port of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx3_eq_ctle_boost_o[4:0]	O	Rx Equalization CTLE Boost -Lane 3. This signal is driven from CTLE_BOOST_3 field of VR_XS_PMA_MP_12G_RX_EQ_CTRL3/VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL3 Register. This signal controls the CTLE Boost Level of the PHY. This signal should be connected to rx3_eq_ctle_boost input port of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_rx0_eq_dfe_tap1_o[7:0]	O	Rx Equalization DFE Tap1 -Lane 0. This signal is driven from DFE_TAP1_0 field of VR_XS_PMA_MP_12G_16G_25G_DFE_TAP_CTRL0/VR_MII_MP_12G_16G_25G_DFE_TAP_CTRL0 Register. This signal is used to control the value of DFE data Tap 1 of the PHY. This signal should be connected to rx0_eq_dfe_tap1 input port of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx1_eq_dfe_tap1_o[7:0]	O	Rx Equalization DFE Tap1 -Lane 1. This signal is driven from DFE_TAP1_1 field of VR_XS_PMA_MP_12G_16G_25G_DFE_TAP_CTRL0 Register. This signal is used to control the value of DFE data Tap 1 of the PHY. This signal should be connected to rx1_eq_dfe_tap1 input port of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx2_eq_dfe_tap1_o[7:0]	O	Rx Equalization DFE Tap1 -Lane 2. This signal is driven from DFE_TAP1_2 field of VR_XS_PMA_MP_12G_16G_25G_DFE_TAP_CTRL1 Register. This signal is used to control the value of DFE data Tap 1 of the PHY. This signal should be connected to rx2_eq_dfe_tap1 input port of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_rx3_eq_dfe_tap1_o[7:0]	O	Rx Equalization DFE Tap1 -Lane 3. This signal is driven from DFE_TAP1_3 field of VR_XS_PMA_MP_12G_16G_25G_DFE_TAP_CTRL1 Register. This signal is used to control the value of DFE data Tap 1 of the PHY. This signal should be connected to rx3_eq_dfe_tap1 input port of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx_adapt_cont_o[(NUM_OF_LANES-1):0]	O	Receiver Adaption Continuous Operation. If this signal is high, PHY receiver adapts continuously. This signal should be connected to rxX_adapt_cont input pin of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx_offcan_cont_o[(NUM_OF_LANES-1):0]	O	Receiver offset cancellation continuous operation. If this signal is high, continuous receiver offset cancellation in PHY is carried out. Otherwise, offset cancellation runs only when receiver exits the P2 power state. This signal should be connected to rxX_offcan_cont input port of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx_vref_ctrl_o[(VREF_W-1):0]	O	Rx Biasing Current Control. This signal is driven from RX_VREF_CTRL field of VR_XS_PMA_MP_12G_16G_25G_MISC_CTRL0/VR_MII_MP_12G_16G_25G_MISC_CTRL0 Register. This signal should be connected to rx_vref_ctrl input pin of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_rx_div16p5_clk_en_o[(NUM_OF_LANES-1):0]	O	Rx Divide 16.5 output clock Enable. This signal is driven from RX_DIV16P5_CLK_EN_0 fields of VR_XS_PMA_MP_12G_16G_25G_RX_GENCTRL1/VR_MII_MP_12G_16G_25G_RX_GENCTRL1 Register. This signal should be connected to rxX_div16p5_clk_en input port of the PHY. **Exists:** (E12&&!(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_22))||E16||(E25&&!SNPS_RSVDPARAM_37)||(SNPS_RSVDPARAM_22&&!SNPS_RSVDPARAM_2) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.15 Synopsys Multi-protocol 6G PHY Receive Status and Control Interface Signals



Table 6-15 Synopsys Multi-protocol 6G PHY Receive Status and Control Interface Signals

Port Name	I/O	Description
xgxs_rx_pll_state_i[(NUM_OF_LANES-1):0]	I	Receive PLL State. This signal indicates the current state of the Receive PLL. This signal matches the xgxs_rx_pll_pwron_o signal when the PLL reaches the requested state. The PLL changes states as long as the MPLL is powered on, regardless of the state of the incoming data lines. *Note:* This signal does not indicate that the link is ready. When the xpcs_rx_en_o{lane} signal is asserted, the xgxs_rx_valid_i{lane} signal must be used for the link ready status. **Exists:** SNPS_XAUI_PHY&&(UP3||UP5) **Synchronous To:** Asynchronous **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High **Registered:** Yes, with: ■ clk_csr_i ■ clk_cko_alive_i
xgxs_rx0_eq_val_o[2:0]	O	Receiver Equalization Control in lane 0. This signal controls the Rx equalization settings of the PHY. **Exists:** (SNPS_XAUI_PHY&&UP3) || (SNPS_XAUI_PHY&&UP5&&SSP) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xgxs_rx1_eq_val_o[2:0]	O	Receiver Equalization Control in lane 1. **Exists:** ((SNPS_XAUI_PHY&&UP3) || (SNPS_XAUI_PHY&&UP5&&SSP))&&(!(KX_ONLY||BASE1000X_ONLY)) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_rx2_eq_val_o[2:0]	O	Receiver Equalization Control in lane 2. **Exists:** ((SNPS_XAUI_PHY&&UP3) || (SNPS_XAUI_PHY&&UP5&&SSP))&&(!(KX_ONLY||BASE1000X_ONLY)) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_rx3_eq_val_o[2:0]	O	Receiver Equalization Control in lane 3. **Exists:** ((SNPS_XAUI_PHY&&UP3) || (SNPS_XAUI_PHY&&UP5&&SSP))&&(!(KX_ONLY||BASE1000X_ONLY)) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_rx_pll_en_o[(NUM_OF_LANES-1):0]	O	Receive PLL Enable. This signal enables or disables the receiver DPLL. **Exists:** UP5 **Synchronous To:** clk_cko_alive_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xgxs_rx_los_en_o[(NUM_OF_LANES-1):0]	O	PHY Loss-of-Signal Detection Enable. This signal enables the loss-of-signal detector. **Exists:** UP5 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xgxs_los_lvl_o[4:0]	O	Loss-of-Signal Threshold. This signal controls the setting of the LOS detection threshold level. For more information, see the following Databooks: ■ Synopsys DesignWare Cores XAUI PHY for TSMC 40nm LP 1.1/2.5V Databook *Note:* <BS>This signal is valid only for Synopsys Consumer PHYs. **Exists:** SNPS_XAUI_PHY&&(UP3||UP5) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_los0_ctl_o[(LOS_W-1):0]	O	Loss-of-Signal Control for Lane 0. This signal enables the LOS detector and/or filtering of raw xpcs_los_i[0]. **Exists:** ((SNPS_XAUI_PHY&&(UP3||UP5)&&!SSP)||SSP_V42) **Synchronous To:** clk_xgxs_rx0_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_los1_ctl_o[(LOS_W-1):0]	O	Loss-of-Signal Control for Lane 1. This signal enables the LOS detector and/or filtering of raw xpcs_los_i[1]. **Exists:** ((SNPS_XAUI_PHY&&(UP3||UP5)&&!SSP)||SSP_V42)&&(!(KX_ONLY||BASE1000X_ONLY)) **Synchronous To:** clk_xgxs_rx1_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_los2_ctl_o[(LOS_W-1):0]	O	Loss-of-Signal Control for Lane 2. This signal enables the LOS detector and/or filtering of raw xpcs_los_i[2] **Exists:** ((SNPS_XAUI_PHY&&(UP3||UP5)&&!SSP)||SSP_V42)&&(!(KX_ONLY||BASE1000X_ONLY)) **Synchronous To:** clk_xgxs_rx2_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xgxs_los3_ctl_o[(LOS_W-1):0]	O	Loss-of-Signal Control for Lane 3. This signal enables the LOS detector and/or filtering of raw xpcs_los_i[3]. **Exists:** ((SNPS_XAUI_PHY&&(UP3||UP5)&&!SSP)||SSP_V42)&&(!(KX_ONLY||BASE1000X_ONLY)) **Synchronous To:** clk_xgxs_rx3_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_los_bias_o[2:0]	O	Loss Of Signal Bias. Sets the LOS detection threshold level **Exists:** SNPS_XAUI_PHY&&UP5&&SSP **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_acjt_lvl_o[4:0]	O	ACJT Level. This signal controls the receiver comparator level. **Exists:** SNPS_XAUI_PHY&&(UP3||UP5) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.16 Synopsys Multi-protocol 6G/12G PHY Receive Status and Control Interface Signals

Table 6-16 Synopsys Multi-protocol 6G/12G PHY Receive Status and Control Interface Signals

Port Name	I/O	Description
xgxs_rx_align_en_o[(NUM_OF_LANES-1):0]	O	Receiver Data Alignment Enable. This signal configures the Synopsys PHY to enable comma-detection and word-alignment. This signal should be connected to rxX_align_en input port of the PHY. **Exists:** SNPS_XAUI_PHY&&(UP3||UP5||E12) **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** Configuration dependent: ■ clk_xgxs_rx{lane}_i: For Synopsys Multi-protocol 6G PHY configurations ■ clk_csr_i: For Synopsys Multi-protocol 12G PHY configurations

6.17 Synopsys PHY Receive Status and Control Interface Signals

Table 6-17 Synopsys PHY Receive Status and Control Interface Signals

Port Name	I/O	Description
xgxs_rx_term_en_o[(NUM_OF_LANES-1):0]	O	Receiver Termination Enable. If this signal is high, PHY Rx serial line is terminated with nominal 50 ohm resistance, else high-impedance. **Exists:** SNPS_XAUI_PHY&&(UP3||UP5||E12||E16||E25||E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xgxs_rx_invert_o[(NUM_OF_LANES-1):0]	O	Invert logical sense of PHY Rx serial inputs. This signal has the effect of inverting the received input serial data stream. **Exists:** UP5||E12||E16||E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xgxs_rx_reset_o[(NUM_OF_LANES-1):0]	O	Receiver Reset. This signal resets the PHY receiver. This signal should be connected to rxX_reset input port of the PHY. **Exists:** UP5||E12||E16||E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xgxs_rx0_rate_o[(RX_RT_W-1):0]	O	Receiver Data Rate -Lane 0. This signal selects the baud-rate for the receive path. For Multi-protocol 12G/16G PHY configurations, this signal is driven from RX0_RATE field of VR_XS_PMA_MP_12G_16G_25G_RX_RATE_CTRL/VR_MII_MP_12G_16G_25G_RX_RATE_CTRL Register. For Multi-protocol 6G PHY configurations, this signal is controlled by programming RX_RATE_0 field of VR_XS_PMA_MP_6G_MISC_CTRL1/VR_MII_MP_6G_MISC_CTRL1 Register. **Exists:** ((UP5&&SSP)||E12||E16||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_rx1_rate_o[(RX_RT_W-1):0]	O	Receiver Data Rate -Lane 1. This signal selects the baud-rate for the receive path. For configurations with Multi-protocol 12G/16G PHY, this signal is driven RX1_RATE field of VR_XS_PMA_MP_12G_16G_25G_RX_RATE_CTRL Register. For configurations with Multi-protocol 6G PHY, this signal is controlled by programming RX_RATE_1 field of VR_XS_PMA_MP_6G_MISC_CTRL1 Register **Exists:** ((UP5&&SSP)||E12||E16||E25)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_rx2_rate_o[(RX_RT_W-1):0]	O	Receiver Data Rate -Lane 2. This signal selects the baud-rate for the receive path. For configurations with Multi-protocol 12G/16G PHY, this signal is driven RX2_RATE field of VR_XS_PMA_MP_12G_16G_25G_RX_RATE_CTRL Register. For configurations with Multi-protocol 6G PHY, this signal is controlled by programming RX_RATE_2 field of VR_XS_PMA_MP_6G_MISC_CTRL1 Register **Exists:** ((UP5&&SSP)||E12||E16||E25)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xgxs_rx3_rate_o[(RX_RT_W-1):0]	O	Receiver Data Rate -Lane 3. This signal selects the baud-rate for the receive path. For configurations with Multi-protocol 12G/16G PHY, this signal is driven RX3_RATE field of VR_XS_PMA_MP_12G_16G_25G_RX_RATE_CTRL Register. For configurations with Multi-protocol 6G PHY, this signal is controlled by programming RX_RATE_3 field of VR_XS_PMA_MP_6G_MISC_CTRL1 Register **Exists:** ((UP5&&SSP)||E12||E16||E25)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.18 Synopsys PHY CR Port Interface Signals

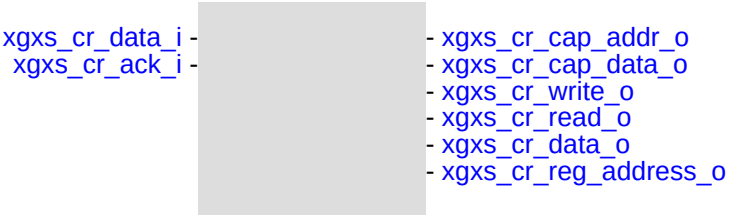


Table 6-18 Synopsys PHY CR Port Interface Signals

Port Name	I/O	Description
xgxs_cr_data_i[15:0]	I	CR Data. This signal transmits the data read from the Synopsys PHY. **Exists:** SNPS_XAUI_PHY&&!SNPS_RSVDPARAM_37 **Synchronous To:** Asynchronous **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_cr_ack_i	I	CR Acknowledge. This signal acknowledges the assertion of read or write operation for Synopsys Multi-protocol 12G/16G/25G PHY. For multi-protocol 6G PHY, it acknowledges the assertion of any of the following controls: ■ xgxs_cr_cap_addr_o ■ xgxs_cr_cap_data_o ■ xgxs_cr_write_o ■ xgxs_cr_read_o **Exists:** SNPS_XAUI_PHY&&!SNPS_RSVDPARAM_37 **Synchronous To:** Asynchronous **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High **Registered:** Yes, ■ with clk_csr_i when SNPS_PHY_TYPE = Multi-protocol 6G. ■ with clk_cko_alive_i for other configurations
xgxs_cr_cap_addr_o	O	CR Address Capture. This signal is used to capture the xgxs_cr_data_o[15:0] signal into the address register of the Synopsys PHY. **Exists:** SNPS_XAUI_PHY&&(UP3||UP5) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xgxs_cr_cap_data_o	O	CR Data Capture. This signal is used to capture the xgxs_cr_data_o[15:0] signal into the write data register of the Synopsys PHY. **Exists:** SNPS_XAUI_PHY&&(UP3||UP5) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xgxs_cr_write_o	O	CR Write. This signal is used to write the content of the xgxs_cr_data_o[15:0] signal into the dereferenced address register of the Synopsys PHY. **Exists:** SNPS_XAUI_PHY&&!SNPS_RSVDPARAM_37 **Synchronous To:** {(E12||E16||E25) ? "clk_cko_alive_i" : "clk_csr_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xgxs_cr_read_o	O	CR Read. This signal is asserted to read the data from dereferenced address register of the Synopsys PHY the xgxs_cr_data_i signal. **Exists:** SNPS_XAUI_PHY&&!SNPS_RSVDPARAM_37 **Synchronous To:** {(E12||E16||E25) ? "clk_cko_alive_i" : "clk_csr_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xgxs_cr_data_o[15:0]	O	CR Data. This bus is used to send the address (only for Multi-protocol 6G PHY) and the write data (for all PHYs) to the Synopsys PHY. **Exists:** SNPS_XAUI_PHY&&!SNPS_RSVDPARAM_37 **Synchronous To:** {(E12||E16||E25) ? "clk_cko_alive_i" : "clk_csr_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xgxs_cr_reg_address_o[15:0]	O	Register Address. Specifies the address of the PHY register that is being accessed. **Exists:** (KR_PHY||E12||E16||E25||E32)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_cko_alive_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.19 Synopsys Multi-protocol 12G/16G/25G PHY Transmit Status and Control Interface Signals

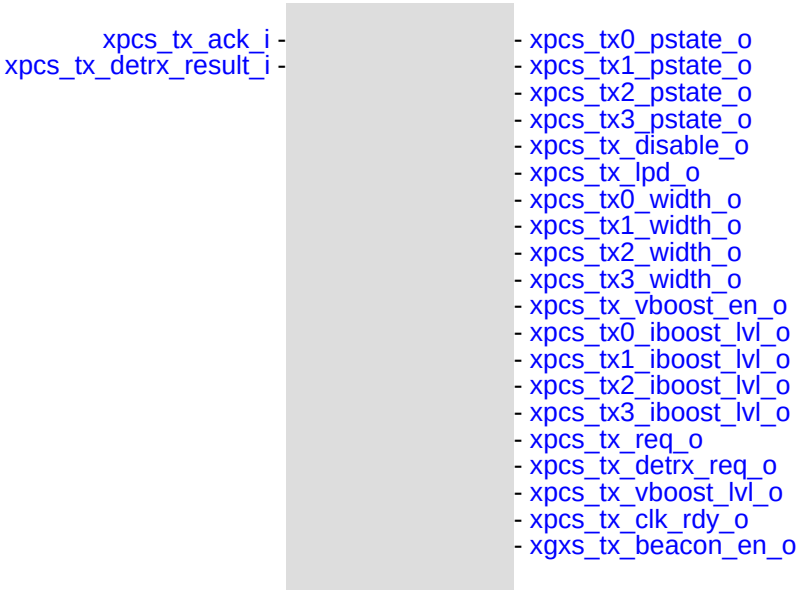


Table 6-19 Synopsys Multi-protocol 12G/16G/25G PHY Transmit Status and Control Interface Signals

Port Name	I/O	Description
xpcs_tx0_pstate_o[1:0]	O	Transmit Power state -Lane 0. This signal should be connected to tx0_pstate[1:0] input port of the PHY. **Exists:** KR_PHY||E12||E16||E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx1_pstate_o[1:0]	O	Transmit Power state -Lane 1. This signal controls the power-state of PHY Transmitter. **Exists:** (KR_PHY||E12||E16||E25||E32)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_tx2_pstate_o[1:0]	O	Transmit Power state -Lane 2. This signal controls the power-state of PHY Transmitter. **Exists:** (KR_PHY||E12||E16||E25||E32)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx3_pstate_o[1:0]	O	Transmit Power state -Lane 3. This signal controls the power-state of PHY Transmitter. **Exists:** (KR_PHY||E12||E16||E25||E32)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx_disable_o[(NUM_OF_LANES-1):0]	O	Transmitter Disable. This signal is driven from the TX_DISABLE fields of VR XS,PMA or MII Synopsys Multi-protocol 12G/16G PHY Tx Power State Control Register. This signal must be connected to the txX_disable input signal of the PHY. **Exists:** E12||E16||E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_tx_lpd_o[(NUM_OF_LANES-1):0]	O	Transmitter Lane Power Down. This signal is driven from the TX_LPD_3_1 and TX_LPD_0 fields of VR_XS_PMA_MP_12G_16G_TX_GENCTRL2/VR_XS_PMA_MP_25G_TX_GENCTRL2/VR_MII_MP_12G_16G_TX_GENCTRL2/VR_MII_MP_25G_TX_GENCTRL2 Register.Also, this signal is controlled by PWRSV bit[11] of VR_XS_PCS_DIG_CTRL1 Register. This signal powers down the PHY transmitter to a power state equivalent to P1. This signal should be connected to txX_lpd input signal of the PHY. **Exists:** E12||E16||E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_tx0_width_o[(TX_W-1):0]	O	Tx Data Width control lane 0. This signal is driven from the TX0_WIDTH field of VR_XS_PMA_MP_12G_16G_TX_GENCTRL2/VR_MII_MP_12G_16G_TX_GENCTRL2/VR_XS_PMA_MP_25G_TX_WIDTH_CTRL/VR_MII_MP_25G_TX_WIDTH_CTRL Register. This signal selects the width of the input transmit data to the PHY lane 0. This signal must be connected to the tx0_width input signal of the PHY. **Exists:** (E12||E16||E25||E32)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx1_width_o[(TX_W-1):0]	O	Tx Data Width control lane 1. **Exists:** (E12||E16||E25||E32)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx2_width_o[(TX_W-1):0]	O	Tx Data Width Control Lane 2. **Exists:** (E12||E16||E25||E32)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx3_width_o[(TX_W-1):0]	O	Tx Data Width Control Lane 3. **Exists:** (E12||E16||E25||E32)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_tx_vboost_en_o[(NUM_OF_LANES-1):0]	O	Tx Voltage Boost Enable. This signal is driven from the VBOOST_EN_0/VBOOST_EN_3_1 field of VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1/VR_MII_MP_12G_16G_25G_TX_GENCTRL1 Register. This signal enables the current mode Tx Swing boost. This signal must be connected to the txX_vboost_en input signal of the PHY. **Exists:** (E12&&!SNPS_RSVDPARAM_22)||E16||(E25&&!SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx0_iboost_lvl_o[3:0]	O	Tx Current Boost Level -Lane 0. This signal is driven from VR_XS_PMA_MP_12G_16G_25G_TX_BOOST_CTRL/VR_MII_MP_12G_16G_25G_TX_BOOST_CTRL Register. This signal controls the Tx swing level Lane 0. This signal must be connected to the txX_iboost_lvl input signal of the PHY. **Exists:** (E12&&!SNPS_RSVDPARAM_22)||E16||(E25&&!SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx1_iboost_lvl_o[3:0]	O	Tx Current Boost Level -Lane 1. This signal is driven from VR_XS_PMA_MP_12G_16G_25G_TX_BOOST_CTRL Register. This signal controls the Tx swing level Lane 1. This signal must be connected to the txX_iboost_lvl input signal of the PHY. Active State: N/A **Exists:** ((E12&&!SNPS_RSVDPARAM_22)||E16||E25)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_tx2_iboost_lvl_o[3:0]	O	Tx Current Boost Level -Lane 2. This signal is driven from VR_XS_PMA_MP_12G_16G_25G_TX_BOOST_CTRL Register. This signal controls the Tx swing level Lane 2. This signal must be connected to the txX_iboost_lvl input signal of the PHY. Active State: N/A **Exists:** ((E12&&!SNPS_RSVDPARAM_22)||E16||E25)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx3_iboost_lvl_o[3:0]	O	Tx Current Boost Level -Lane 3. This signal is driven from VR_XS_PMA_MP_12G_16G_25G_TX_BOOST_CTRL Register. This signal controls the Tx swing level Lane 3. This signal must be connected to the txX_iboost_lvl input signal of the PHY. Active State: N/A **Exists:** ((E12&&!SNPS_RSVDPARAM_22)||E16||E25)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx_req_o[(NUM_OF_LANES-1):0]	O	Transmitter Request. This signal is driven from the TX_REQ_0/TX_REQ_3_1 fields of VR_XS_PMA_MP_12G_16G_TX_GENCTRL2/VR_XS_PMA_MP_25G_TX_GENCTRL2/VR_MII_MP_12G_16G_TX_GENCTRL2/VR_MII_MP_25G_TX_GENCTRL2 Register. This signal indicates a new transmitter setting request to the PHY. This signal must be connected to the txX_req input signal of the PHY. **Exists:** E12||E16||E25||E32 **Synchronous To:** clk_csr_i **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_tx_ack_i[(NUM_OF_LANES-1):0]	I	Transmitter Acknowledge. The value of this signal can be read from the TX_ACK_0/TX_ACK_3_1 fields of VR_XS_PMA_MP_12G_16G_25G_TX_STS/VR_MII_MP_12G_16G_25G_TX_STS Register. When this signal is asserted, it indicates that the request transmitter setting is complete or the requested Rx detection operation is complete. This signal must be connected to the txX_ack output signal of the PHY. **Exists:** E12||E16||E25||E32 **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_tx_detrx_req_o[(NUM_OF_LANES-1):0]	O	Receiver Detection Request. This signal is asserted to make a receiver detection request to the PHY. This signal is driven from DET_RX_REQ_0/DET_RX_REQ_3_1 fields of VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1/VR_MII_MP_12G_16G_25G_TX_GENCTRL1 Register. **Exists:** E12||E16||E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_tx_detrx_result_i[(NUM_OF_LANES-1):0]	I	Receiver Detection Result. This signal indicates the status of receiver detection operation. If this signal is high, it implies that a remote-end receiver is successfully detected. This signal must be connected to tx{lane}_detrx_result input port of the PHY. **Exists:** E12||E16||E25||E32 **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_tx_vboost_lvl_o[2:0]	O	Tx Voltage Boost Maximum Level. This signal is driven from the VBOOST_LVL field of VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1/VR_MII_MP_12G_16G_25G_TX_GENCTRL1 Register. This signal must be connected to the tx_vboost_lvl[2:0] input signal of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx_clk_rdy_o[(NUM_OF_LANES-1):0]	O	Transmitter Input Clock Ready. This signal is driven from TX_CLK_RDY_0/TX_CLK_RDY_3_1 fields of VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1/VR_MII_MP_12G_16G_25G_TX_GENCTRL1 Register. This signal is used to synchronize Tx lanes of PHY in a multi-lane link. This signal should be connected to txX_clk_rdy input port of the PHY. **Exists:** E12||E16||E25||E32 **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** Configuration/Operating Mode dependent: ■ clk_xsbi_tx_i when operating in KR mode ■ clk_qsgmii_tx_i when operating in QSGMII mode ■ clk_xgxs_tx_i when operating in other modes ■ clk_csr_i in case of Multi-protocol 32G PHY
xgxs_tx_beacon_en_o[(NUM_OF_LANES-1):0]	O	Transmitter Beaconing/LFPS Enable. This signal is driven from the TXBCN_EN fields of the VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL0/VR_MII_MP_12G_16G_25G_TX_GENCTRL0 Register. This signal enables and disables transmitter beaconing. The period of transmit pulses is between 20-50ns. This signal must be connected to the txX_beacon_en input signal of the PHY. **Exists:** (UP5&&!SSP)||E12||E16||E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.20 Clause 72 KR Training Signals for Synopsys PHY

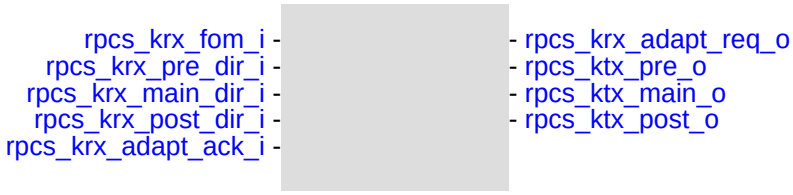


Table 6-20 Clause 72 KR Training Signals for Synopsys PHY

Port Name	I/O	Description
rpcs_krx_fom_i[7:0]	I	Receiver Figure of Merit. This signal indicates the quality of the received data eye. The value of this signal is available at FOM field of VR_XS_PMA_MP_12G_16G_25G_MISC_STS Register. This signal should be connected to rx0_adapt_fom port of the PHY. **Exists:** (KR_PHY&&CL72_EN)||E12||E16||E25||E32 **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_pre_dir_i[1:0]	I	Direction Change for far-end Tx Coefficient C(-1). This signal provides the recommended direction change for far-end Tx equalization settings. This signal should be connected to rx0_txpre_dir port of the PHY. **Exists:** (KR_PHY||E12||E16||E25||E32)&&CL72_EN **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_main_dir_i[1:0]	I	Direction Change for far-end Transmitter Coefficient C(0). This signal provides the recommended direction change for far-end Tx equalization settings. The encoding is as follows: ■ 2'b00: IDLE ■ 2'b01: INC ■ 2'b10: DEC ■ 2'b11: HOLD This signal should be connected to rx0_txmain_dir port of the PHY. **Exists:** (KR_PHY||E12||E16||E25||E32)&&CL72_EN **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
rpcs_krx_post_dir_i[1:0]	I	Direction Change for far-end Tx Coefficient C(1). This signal provides the recommended direction change for far-end Tx equalization settings. This signal should be connected to rx0_txpost_dir port of the PHY **Exists:** (KR_PHY||E12||E16||E25||E32)&&CL72_EN **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_adapt_req_o	O	Receiver Adaptation Request. This signal indicates a request for the receiver to adapt to incoming data. This signal should be connected to rx0_adapt_req port of the PHY. **Exists:** (E12||E16||E25||E32) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_krx_adapt_ack_i	I	Receiver adaptation acknowledge. The assertion of this signal indicates the completion of receiver adaptation sequence and the values rxX_txmain_dir[1:0], rxX_txpre_dir[1:0] and rxX_txpost_dir[1:0] are valid. This signal should be connected to rx0_adapt_ack port of the PHY. **Exists:** (E12||E16||E25||E32) **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktx_pre_o[5:0]	O	Transmitter Equalization filter coefficient C(-1). The value of this signal is computed based KR Training algorithm when CL72_EN=Enabled. Otherwise, this signal gets the value of TX_EQ_PRE field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL0/VR_MII_MP_12G_16G_25G_TX_EQ_CTRL0 Register. This signal should be connected to tx0_eq_pre port of the PHY. **Exists:** KR_PHY||E12||E16||E25||E32 **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** ■ clk_xsbi_tx_i when operating in KR mode ■ clk_qsgmii_tx_i when operating in QSGMII mode ■ clk_xgxs_tx_i when operating in other modes.

Port Name	I/O	Description
rpcs_ktx_main_o[5:0]	O	Transmitter Equalization filter coefficient C(0). The value of this signal is computed based KR Training algorithm when CL72_EN=Enabled. Otherwise, this signal gets the value of TX_EQ_MAIN field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL0/VR_MII_MP_12G_16G_25G_TX_EQ_CTRL0 Register. This signal should be connected to tx0_eq_main port of the PHY. **Exists:** E12||E16||E25||E32 **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** Configuration dependent: ■ clk_xsbi_tx_i when operating in KR mode ■ clk_qsgmii_tx_i when operating in QSGMII mode ■ clk_xgxs_tx_i when operating in other modes
rpcs_ktx_post_o[5:0]	O	Transmitter Equalization filter coefficient C(1). The value of this signal is computed based KR Training algorithm when CL72_EN=Enabled. Otherwise, this signal gets the value of TX_EQ_POST field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL1/VR_MII_MP_12G_16G_25G_TX_EQ_CTRL1 Register. This signal should be connected to tx0_eq_post port of the PHY. **Exists:** KR_PHY||E12||E16||E25||E32 **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** ■ clk_xsbi_tx_i when operating in KR mode ■ clk_qsgmii_tx_i when operating in QSGMII mode ■ clk_xgxs_tx_i when operating in other modes

6.21 Synopsys Multi-protocol 16G/25G PHY Transmit Status and Control Interface Signals

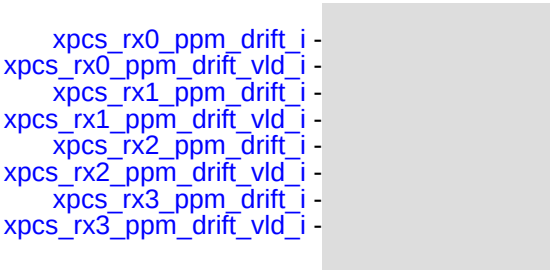


Table 6-21 Synopsys Multi-protocol 16G/25G PHY Transmit Status and Control Interface Signals

Port Name	I/O	Description
xpcs_rx0_ppm_drift_i[5:0]	I	RX CDR PPM Drift (Along lane0). This value represents the amount of ppm on the rx0_clk(of PHY) with respect to the ideal desired frequency. This input is valid when xpcs_rx0_ppm_drift_vld_i is asserted. This signal is driven by rx0_ppm_drift output port of the PHY. **Exists:** E16||E25||SNPS_RSVDPARAM_39||E32 **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx0_ppm_drift_vld_i	I	RX CDR PPM Drift Valid(Along lane0). Indicates when the xpcs_rx0_ppm_drift_i input is valid and can be sampled. This signal is driven by rx0_ppm_drift_vld output port of the PHY. **Exists:** E16||E25||SNPS_RSVDPARAM_39||E32 **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx1_ppm_drift_i[5:0]	I	RX CDR PPM Drift (Along lane1). This value represents the amount of ppm on the rx1_clk(of PHY) with respect to the ideal desired frequency. This input is valid when xpcs_rx1_ppm_drift_vld_i is asserted. This signal is driven by rx1_ppm_drift output port of the PHY. **Exists:** (E16||E25||SNPS_RSVDPARAM_39||E32)&&!SINGLE_LANE_CFG **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_rx1_ppm_drift_vld_i	I	RX CDR PPM Drift Valid(Along lane1). Indicates when the xpcs_rx1_ppm_drift_i input is valid and can be sampled. This signal is driven by rx1_ppm_drift_vld output port of the PHY. **Exists:** (E16||E25||SNPS_RSVDPARAM_39||E32)&&!SINGLE_LANE_CFG **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx2_ppm_drift_i[5:0]	I	RX CDR PPM Drift (Along lane2). This value represents the amount of ppm on the rx2_clk(of PHY) with respect to the ideal desired frequency. This input is valid when xpcs_rx2_ppm_drift_vld_i is asserted. This signal is driven by rx2_ppm_drift output port of the PHY. **Exists:** (E16||E25||SNPS_RSVDPARAM_39||E32)&&!SINGLE_LANE_CFG **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx2_ppm_drift_vld_i	I	RX CDR PPM Drift Valid(Along lane2). Indicates when the xpcs_rx2_ppm_drift_i input is valid and can be sampled. This signal is driven by rx2_ppm_drift_vld output port of the PHY. **Exists:** (E16||E25||SNPS_RSVDPARAM_39||E32)&&!SINGLE_LANE_CFG **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx3_ppm_drift_i[5:0]	I	RX CDR PPM Drift (Along lane3). This value represents the amount of ppm on the rx3_clk(of PHY) with respect to the ideal desired frequency. This input is valid when xpcs_rx3_ppm_drift_vld_i is asserted. This signal is driven by rx3_ppm_drift output port of the PHY. **Exists:** (E16||E25||SNPS_RSVDPARAM_39||E32)&&!SINGLE_LANE_CFG **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_rx3_ppm_drift_vld_i	I	RX CDR PPM Drift Valid(Along lane3). Indicates when the xpcs_rx3_ppm_drift_i input is valid and can be sampled. This signal is driven by rx3_ppm_drift_vld output port of the PHY. **Exists:** (E16||E25||SNPS_RSVDPARAM_39||E32)&&!SINGLE_LANE_CFG **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.22 Synopsys Multi-protocol 12G/16G PHY Receive Status and Control Interface Signals

Table 6-22 Synopsys Multi-protocol 12G/16G PHY Receive Status and Control Interface Signals

Port Name	I/O	Description
xpcs_rx_los_lfps_en_o[(NUM_OF_LANES -1):0]	O	Rx LOS LFPS Enable. This signal is driven from LOS_LFPS_EN_0/LOS_LFPS_EN_3_1 fields of VR_XS_PMA_MP_12G_16G_RX_GENCTRL3/VR_MII_MP_12G_16G_RX_GENCTRL3 Register. This signal enables the Low Frequency Periodic Signal (LFPS) filter in the PHY. This signal must be connected to rxX_los_lfps_en input pin of the PHY. **Exists:** (E12&&!SNPS_RSVDPARAM_22)||E16 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx0_los_threshold_o[2:0]	O	Rx Loss-of-Signal Threshold Lane 0. This signal is driven from LOS_TRSHLD_0 field of VR_XS_PMA_MP_12G_16G_RX_GENCTRL3/VR_MII_MP_12G_16G_RX_GENCTRL3 Register. This signal sets the LOS threshold trip point of the PHY receiver. This signal must be connected to rx0_los_threshold input port of the PHY. **Exists:** (E12&&!SNPS_RSVDPARAM_22)||E16 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_rx1_los_threshold_o[2:0]	O	Rx Loss-of-Signal Threshold Lane 1. This signal is driven from LOS_TRSHLD_1 field of VR_XS_PMA_MP_12G_16G_RX_GENCTRL3 Register. This signal sets the LOS threshold trip point of the PHY receiver. This signal must be connected to rx1_los_threshold input port of the PHY. **Exists:** ((E12&&!SNPS_RSVDPARAM_22)||E16)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx2_los_threshold_o[2:0]	O	Rx Loss-of-Signal Threshold Lane 2. This signal is driven from LOS_TRSHLD_2 field of VR_XS_PMA_MP_12G_16G_RX_GENCTRL3 Register. This signal sets the LOS threshold trip point of the PHY receiver. This signal must be connected to rx2_los_threshold input port of the PHY. **Exists:** ((E12&&!SNPS_RSVDPARAM_22)||E16)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx3_los_threshold_o[2:0]	O	Rx Loss-of-Signal Threshold Lane 3. This signal is driven from LOS_TRSHLD_3 field of VR_XS_PMA_MP_12G_16G_RX_GENCTRL3 Register. This signal sets the LOS threshold trip point of the PHY receiver. This signal must be connected to rx3_los_threshold input port of the PHY. **Exists:** ((E12&&!SNPS_RSVDPARAM_22)||E16)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.23 Synopsys Multi-protocol 12G PHY Receive Status and Control Interface Signals

Table 6-23 Synopsys Multi-protocol 12G PHY Receive Status and Control Interface Signals

Port Name	I/O	Description
xpcs_rx_cdr_track_en_o[(NUM_OF_LANES-1):0]	O	Rx CDR Tracking Enable. This signal is driven from CDR_TRACK_EN_0/CDR_TRACK_EN_3_1 fields of VR_XS_PMA_MP_12G_16G_25G_RX_CDR_CTRL/VR_MII_MP_12G_16G_25G_RX_CDR_CTRL Register. This signal is used to enable CDR tracking of receive data in the PHY. This signal must be connected to rxX_cdr_track_en input pin of the PHY. **Exists:** E12 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx_cdr_vco_lowfreq_o[(NUM_OF_LANES-1):0]	O	Rx VCO lower frequency band mode. This signal is driven from VCO_LOW_FREQ_0/VCO_LOW_FREQ_3_1 fields of VR_XS_PMA_MP_12G_16G_25G_RX_CDR_CTRL/VR_MII_MP_12G_16G_25G_RX_CDR_CTRL Register. This signal controls the frequency of the Rx VCO to a lower frequency operating band. This signal must be connected to rxX_cdr_vco_lowfreq input pin of the PHY. **Exists:** E12 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_rx_adapt_afe_en_o[(NUM_OF_LANES-1):0]	O	Rx AFE Enable. This signal is driven from AFE_EN_0/AFE_EN_3_1 fields of VR_XS_PMA_MP_12G_AFE_DFE_EN_CTRL/VR_MII_MP_12G_AFE_DFE_EN_CTRL Register. This signal enables the Rx adaption circuitry and applies the input receiver AFE equalization settings. This signal must be connected to rxX_adapt_afe_en input pin of the PHY. **Exists:** E12 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx_adapt_dfe_en_o[(NUM_OF_LANES-1):0]	O	Rx DFE Enable. This signal is driven from DFE_EN_0/DFE_EN_3_1 fields of VR_XS_PMA_MP_12G_AFE_DFE_EN_CTRL/VR_MII_MP_12G_AFE_DFE_EN_CTRL Register. This signal enables the Rx adaption and decision feedback equalization (DFE) circuitry and applies the input setting for DFE Tap1. This signal must be connected to irxX_adapt_dfe_en input pin of the PHY. **Exists:** E12 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xgxs_rx_clk_shift_o[(NUM_OF_LANES-1):0]	O	Receiver Clock shift. This signal causes a 1-bit shift on the Receive Data(xgxs_rx_data_i/rpcs_rx_data_i) relative to Receive clock (clk_xgxs_rx{lane}_i/clk_xsbi_rxi). **Exists:** (UP5&&!SSP)||E12 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.24 Synopsys Multi-protocol 25G PHY Receive Status and Control Interface Signals

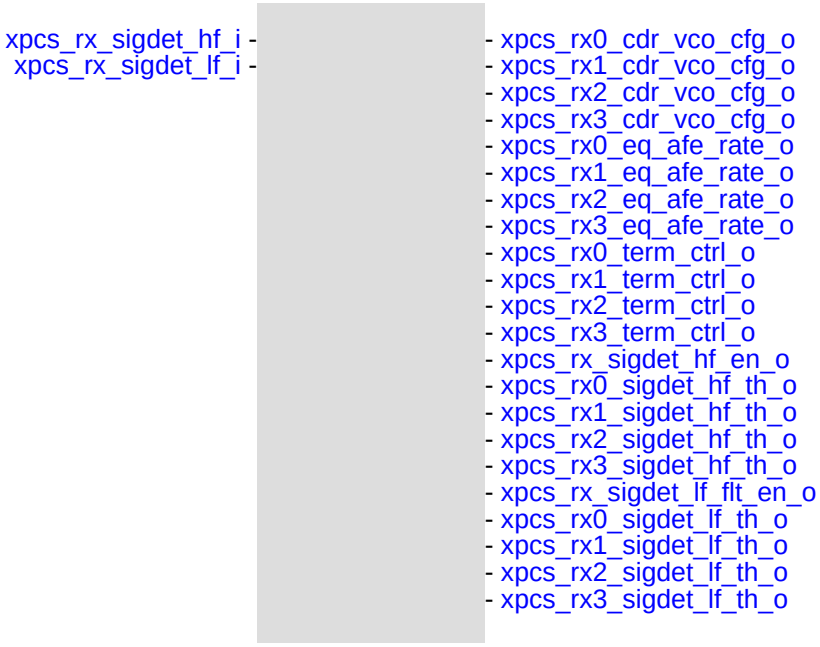


Table 6-24 Synopsys Multi-protocol 25G PHY Receive Status and Control Interface Signals

Port Name	I/O	Description
xpcs_rx0_cdr_vco_cfg_o[11:0]	O	Receive CDR VCO configuration Bias -lane 0 This signal controls the biasing, temperature compensation,DAC settings and frequency rate of the Rx CDR VCO. **Exists:** E25&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx1_cdr_vco_cfg_o[11:0]	O	Receive CDR VCO configuration Bias -lane 1 **Exists:** E25&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx2_cdr_vco_cfg_o[11:0]	O	Receive CDR VCO configuration Bias -lane 2 **Exists:** E25&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_rx3_cdr_vco_cfg_o[11:0]	O	Receive CDR VCO configuration Bias -lane 3 **Exists:** E25&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx0_eq_afe_rate_o[2:0]	O	Receive Equalization AFE Rate -lane 0 **Exists:** E25&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx1_eq_afe_rate_o[2:0]	O	Receive Equalization AFE Rate -lane 1 **Exists:** E25&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx2_eq_afe_rate_o[2:0]	O	Receive Equalization AFE Rate -lane 2 **Exists:** E25&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx3_eq_afe_rate_o[2:0]	O	Receive Equalization AFE Rate -lane 3 **Exists:** E25&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx0_term_ctrl_o[2:0]	O	Receive Termination Control -lane 0 **Exists:** (SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_rx1_term_ctrl_o[2:0]	O	Receive Termination Control -lane 1 **Exists:** (SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||E25)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx2_term_ctrl_o[2:0]	O	Receive Termination Control -lane 2 **Exists:** (SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||E25)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx3_term_ctrl_o[2:0]	O	Receive Termination Control -lane 3 **Exists:** (SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||E25)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx_sigdet_hf_en_o[(NUM_OF_LANES-1):0]	O	High frequency Signal detection enable **Exists:** SNPS_RSVDPARAM_22||E25 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx_sigdet_hf_i[(NUM_OF_LANES-1):0]	I	High frequency signal detection result **Exists:** SNPS_RSVDPARAM_22||E25||E32 **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx0_sigdet_hf_th_o[2:0]	O	Threshold for High frequency signal detection-lane 0 **Exists:** (SNPS_RSVDPARAM_22||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_rx1_sigdet_hf_th_o[2:0]	O	Threshold for High frequency signal detection-lane 1 **Exists:** (SNPS_RSVDPARAM_22||E25)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx2_sigdet_hf_th_o[2:0]	O	Threshold for High frequency signal detection-lane 2 **Exists:** (SNPS_RSVDPARAM_22||E25)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx3_sigdet_hf_th_o[2:0]	O	Threshold for High frequency signal detection-lane 3 **Exists:** (SNPS_RSVDPARAM_22||E25)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx_sigdet_lf_flt_en_o[(NUM_OF_LANES-1):0]	O	Low frequency Signal detection filter enable **Exists:** (SNPS_RSVDPARAM_22||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx_sigdet_lf_i[(NUM_OF_LANES-1):0]	I	Low frequency signal detection result **Exists:** SNPS_RSVDPARAM_22||E25||E32 **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx0_sigdet_lf_th_o[2:0]	O	Threshold for Low frequency signal detection-lane 0 **Exists:** (SNPS_RSVDPARAM_22||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_rx1_sigdet_lf_th_o[2:0]	O	Threshold for Low frequency signal detection-lane 1 **Exists:** (SNPS_RSVDPARAM_22||E25)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx2_sigdet_lf_th_o[2:0]	O	Threshold for Low frequency signal detection-lane 2 **Exists:** (SNPS_RSVDPARAM_22||E25)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx3_sigdet_lf_th_o[2:0]	O	Threshold for Low frequency signal detection-lane 3 **Exists:** (SNPS_RSVDPARAM_22||E25)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.25 Synopsys Multi-protocol 12G/16G/25G PHY MPLL Control and Status Interface Signals

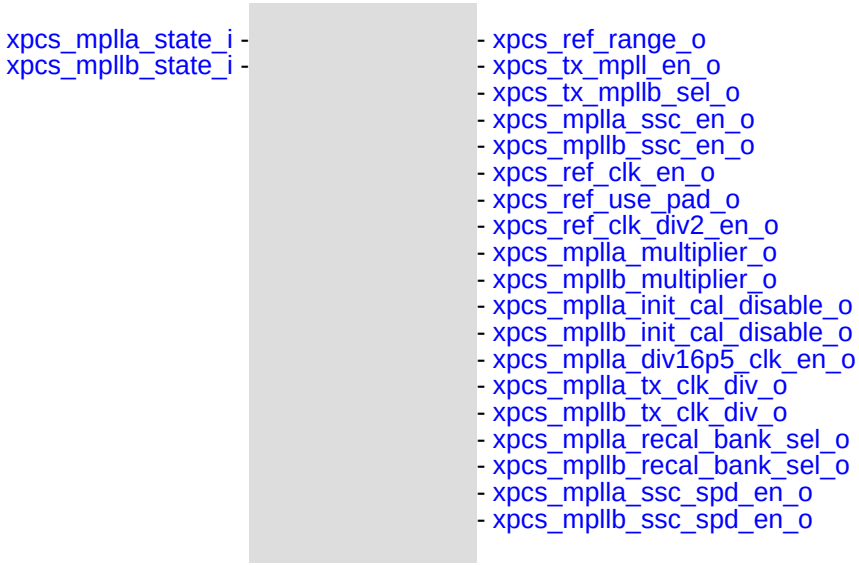


Table 6-25 Synopsys Multi-protocol 12G/16G/25G PHY MPLL Control and Status Interface Signals

Port Name	I/O	Description
xpcs_ref_range_o[(REF_R_W-1):0]	O	Input Reference Clock Range. This signal is used to specify the range of input reference clock supplied to the PHY. This signal should be connected to ref_range input port of the PHY. **Exists:** E12||E16|E25 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx_mpll_en_o[(NUM_OF_LANES-1):0]	O	Tx MPLL Enable. This signal is driven from MPLL_EN_0/MPLL_EN_3_1 fields of VR_XS_PMA_MP_12G_16G_25G_MPLL_CMN_CTRL/VR_MII_MP_12G_16G_25G_MPLL_CMN_CTRL Register. This signal should be connected to txX_mpll_en input port of the PHY. **Exists:** E12||E16||E25 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_tx_mpllb_sel_o[(NUM_OF_LANES-1):0]	O	Tx MPLLB select. This signal is driven from MPLLB_SEL_0/MPLLB_SEL_3_1 fields of VR_XS_PMA_MP_12G_16G_25G_MPLL_CMN_CTRL/VR_MII_MP_12G_16G_25G_MPLL_CMN_CTRL Register. This signal should be connected to txX_mpllb_sel input port of the PHY. **Exists:** (E12&&!SNPS_RSVDPARAM_39)||E16||(E25&&!SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mplla_ssc_en_o	O	MPLLA Spread Spectrum Enable. This signal is used to enable spread-spectrum clock generation the mplla_div_clk output of the PHY. This signal should be connected to mplla_ssc_en input of the PHY. **Exists:** E12||E16||E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mpllb_ssc_en_o	O	MPLLB Spread Spectrum Enable.This signal is used to enable spread-spectrum clock generation the 'mpllb_div_clk' output of the PHY. This signal should be connected to mpllb_ssc_en input of the PHY. **Exists:** (E12&&!SNPS_RSVDPARAM_39)||E16||E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_ref_clk_en_o	O	Reference Clock Enable. This signal is driven from REF_CLK_EN field of VR_XS_PMA_MP_12G_16G_25G_REF_CLK_CTRL/VR_MII_MP_12G_16G_25G_REF_CLK_CTRL Register. This signal is used to enable reference clock to the PHY. This signal should be connected to ref_clk_en input pin of the PHY. **Exists:** E12||E16||E25 **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_ref_use_pad_o	O	Select Pad Reference Clock. This signal is driven from REF_USE_PAD field of VR_XS_PMA_MP_12G_16G_25G_REF_CLK_CTRL/VR_MII_MP_12G_16G_25G_REF_CLK_CTRL Register. This signal is used to select the reference clock connected to ref_pad_clk_p/m input pins of the PHY. This signal should be connected to ref_use_pad input pin of the PHY. **Exists:** E12||E16||E25 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_ref_clk_div2_en_o	O	PHY Reference clock divider control. This signal is driven form REF_CLK_DIV2 field of VR_XS_PMA_MP_12G_16G_25G_REF_CLK_CTRL/VR_MII_MP_12G_16G_25G_REF_CLK_CTRL Register. When this signal is asserted, input reference clock to the PHY is divided by 2. This signal should be connected to ref_clk_div2_en input pin of the PHY. **Exists:** (E12&&!SNPS_RSVDPARAM_22)||E16||(E25&&!SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mplla_multiplier_o[(MULT_W-1):0]	O	MPLLA Frequency Multiplier Control. This signal is driven from MPLLA_MULTIPLIER field of VR_XS_PMA_MP_12G_16G_MPLLA_CTRL0/VR_MII_MP_12G_16G_MPLLA_CTRL0 Register. This signal is used to multiply the reference clock to a frequency suitable for intended operating speed. This signal should be connected to mplla_multiplier input port of the PHY. **Exists:** E12||E16||(E25&&!SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_mpllb_multiplier_o[(MULT_W-1):0]	O	MPLLB Frequency Multiplier Control. This signal is used to multiply the reference clock to a frequency suitable for intended operating speed. This signal should be connected to mpllb_multiplier input port of the PHY. **Exists:** (E12&&!SNPS_RSVDPARAM_39)||E16||(E25&&!SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_mplla_init_cal_disable_o	O	MPLLA Calibration Disable. This signal is driven from MPLLA_CAL_DISABLE field of VR_XS_PMA_MP_12G_16G_MPLLA_CTRL0/VR_MII_MP_12G_16G_MPLLA_CTRL0 Register. This signal is used to disable the calibration operation of MPLLA, that is performed by the PHY firmware. This signal should be connected to the mplla_init_cal_disable input port of the PHY. **Exists:** E12||E16||E25 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mpllb_init_cal_disable_o	O	MPLLA Calibration Disable. This signal is driven from MPLLB_CAL_DISABLE field of VR_XS_PMA_MP_12G_16G_MPLLB_CTRL0/VR_MII_MP_12G_16G_MPLLB_CTRL0 Register. This signal is used to disable the calibration operation of MPLLB, that is performed by the PHY firmware. This signal should be connected to the mpllb_init_cal_disable input port of the PHY. **Exists:** (E12&&!SNPS_RSVDPARAM_39)||E16||E25 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_mplla_div16p5_clk_en_o	O	MPLLA Divided by 16.5 clock. This signal is driven from MPLLA_DIV16P5_CLK_EN field of VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2/VR_MII_MP_12G_16G_MPLLA_CTRL2 Register. This signal is used to enable the output clocks derived from MPLLA based 16.5,33 and 66 division ratios. This signal should be connected to mplla_div16p5_clk_en input pin of the PHY. **Exists:** (E12&&!(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_22))||E16||(E25&&!SNPS_RSVDPARAM_37)||(SNPS_RSVDPARAM_22&&!SNPS_RSVDPARAM_2) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mplla_tx_clk_div_o[(TX_DIV_W-1):0]	O	MPLLA TX Clock Divider. Sets the divide ratio on the MPLLA clock sent to the TX as follows: ■ 2'b00: div1 ■ 2'b01: div2 ■ 2'b10: div4 ■ 2'b11: div1 (duplicate state) ***Note:*** This signal can only be changed when the TX clocks are disabled within the analog. This corresponds to the power states, P1 and P2 in PHY(E16). This signal is driven from MPLLA_TX_CLK_DIV field of VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2/VR_MII_MP_12G_16G_MPLLA_CTRL2/VR_XS_PMA_MP_25G_MPLLA_CTRL2/VR_MII_MP_25G_MPLLA_CTRL2 Register. This signal should be connected to mplla_tx_clk_div input pin of the PHY. **Exists:** E16||SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||(E25&&!SNPS_RSVDPARAM_37)||SNPS_RSVDPARAM_39 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_mpllb_tx_clk_div_o[(TX_DIV_W-1):0]	O	MPLLB TX Clock Divider. Sets the divide ratio on the MPLLB clock sent to the TX as follows: ■ 2'b00: div1 ■ 2'b01: div2 ■ 2'b10: div4 ■ 2'b11: div1 (duplicate state) ***Note:*** This signal can only be changed when the TX clocks are disabled within the analog. This corresponds to the power states, P1 and P2 in PHY(E16). This signal is driven from MPLLB_TX_CLK_DIV field of VR_XS_PMA_MP_12G_16G_MPLLB_CTRL2/VR_MII_MP_12G_16G_MPLLB_CTRL2/VR_XS_PMA_MP_12G_16G_MPLLB_CTRL2/VR_MII_MP_12G_16G_MPLLB_CTRL2 Register. This signal should be connected to mpllb_tx_clk_div input pin of the PHY. **Exists:** E16||SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||(E25&&!SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mplla_recal_bank_sel_o[1:0]	O	MPLLA Re-calibration Bank Select. Selects the bank to store the MPLL calibration value. ***Note:*** This signal can only be changed when the TX clocks are disabled within the analog. This corresponds to the power states, P1 and P2 in PHY. This signal is driven from MPLLA_RECAL_BANK_SEL fields of VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2/VR_MII_MP_12G_16G_MPLLA_CTRL2 Register. This signal should be connected to mplla_recal_bank_sel input port of the PHY. **Exists:** E16||E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_mpllb_recal_bank_sel_o[1:0]	O	MPLLB Re-calibration Bank Select. Selects the bank to store the MPLL calibration value. ***Note:*** This signal can only be changed when the TX clocks are disabled within the analog. This corresponds to the power states, P1 and P2 in PHY. This signal is driven from MPLLB_RECAL_BANK_SEL fields of VR_XS_PMA_MP_12G_16G_MPLLB_CTRL2/VR_MII_MP_12G_16G_MPLLB_CTRL2 Register. This signal should be connected to mpllb_recal_bank_sel input port of the PHY. **Exists:** E16||E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mplla_state_i	I	MPLLA state. This signal indicates the status of MPLLA. The value of this signal is available in MPLLA_STS field of VR_XS_PMA_MP_12G_16G_25G_MISC_STS/VR_MII_MP_12G_16G_25G_MISC_STS Register. This input port of DWC_xpcs should be connected to mplla_state output of the PHY. **Exists:** E12||E16||E25 **Synchronous To:** Asynchronous **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mpllb_state_i	I	MPLLB state. This signal indicates the status of MPLLB. The value of this signal is available in MPLLB_STS field of VR_XS_PMA_MP_12G_16G_25G_MISC_STS/VR_MII_MP_12G_16G_25G_MISC_STS Register. This input port of DWC_xpcs should be connected to mpllb_state output of the PHY. **Exists:** (E12&&!SNPS_RSVDPARAM_39)||E16||E25 **Synchronous To:** Asynchronous **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_mplla_ssc_spd_en_o	O	MPLLA SSC Up Spread Enable. When asserted, the SSC is applied in upwards direction (positive ppm) and when de-asserted the SSC is applied downwards (negative ppm). This signal should be connected to mplla_ssc_up_spread input port of the PHY. **Exists:** (E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mpllb_ssc_spd_en_o	O	MPLLB SSC Up Spread Enable. When asserted, the SSC is applied in upwards direction (positive ppm) and when de-asserted the SSC is applied downwards (negative ppm). This signal should be connected to mpllb_ssc_up_spread input port of the PHY. **Exists:** (E16||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.26 Synopsys Multi-protocol 12G PHY MPLL Control and Status Interface Signals

Table 6-26 Synopsys Multi-protocol 12G PHY MPLL Control and Status Interface Signals

Port Name	I/O	Description
xpcs_mplla_ssc_range_o[2:0]	O	MPLLA Spread Spectrum Clock Range. This signal is used to control the fixed offset applied to the phase accumulator in the PHY. This signal should be connected to mplla_ssc_range input of the PHY. **Exists:** E12&&!(SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_39) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_mpllb_ssc_range_o[2:0]	O	MPLLB Spread Spectrum Clock Range. This signal is used to control the fixed offset applied to the phase accumulator in the PHY. This signal should be connected to mpllb_ssc_range input of the PHY. **Exists:** E12&&!(SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_39) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.27 Synopsys Multi-protocol 12G/16G PHY MPLL Control and Status Interface Signals

- xpcs_mplla_fracn_ctrl_o
- xpcs_mpllb_fracn_ctrl_o
- xpcs_mplla_ssc_clk_sel_o
- xpcs_mpllb_ssc_clk_sel_o
- xpcs_mplla_div_mult_o
- xpcs_mpllb_div_mult_o
- xpcs_mplla_div_clk_en_o
- xpcs_mpllb_div_clk_en_o
- xpcs_mplla_bandwidth_o
- xpcs_mpllb_bandwidth_o
- xpcs_mplla_div8_clk_en_o
- xpcs_mpllb_div8_clk_en_o
- xpcs_mplla_div10_clk_en_o
- xpcs_mpllb_div10_clk_en_o
- xpcs_ref_clk_mplla_div2_en_o
- xpcs_ref_clk_mpllb_div2_en_o
- xpcs_mplla_ssc_freq_cnt_init_o
- xpcs_mplla_ssc_freq_cnt_pk_o
- xpcs_mpllb_ssc_freq_cnt_init_o
- xpcs_mpllb_ssc_freq_cnt_pk_o

Table 6-27 Synopsys Multi-protocol 12G/16G PHY MPLL Control and Status Interface Signals

Port Name	I/O	Description
xpcs_mplla_fracn_ctrl_o[(FR_W-1):0]	O	MPLLA Fractional Control. This signal should be connected to mplla_fracn_ctrl input port of the PHY. **Exists:** (E12&&!SNPS_RSVDPARAM_22)||E16 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_mpllb_fracn_ctrl_o[(FR_W-1):0]	O	MPLLB Fractional Control. This signal should be connected to mpllb_fracn_ctrl input port of the PHY. **Exists:** (E12&&!SNPS_RSVDPARAM_22&&!SNPS_RSVDPARAM_39)||E16 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_mplla_ssc_clk_sel_o[(SSC_W-1):0]	O	MPLLA Spread Spectrum Clock Select. This signal selects the clock for generating spread spectrum generation in the PHY. This signal should be connected to mplla_ssc_clk_sel input port of the PHY. **Exists:** (E12&&!SNPS_RSVDPARAM_22)||E16 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_mpllb_ssc_clk_sel_o[(SSC_W-1):0]	O	MPLLB Spread Spectrum Clock Select. This signal selects the clock for generating spread spectrum generation in the PHY. This signal should be connected to mpllb_ssc_clk_sel input port of the PHY. **Exists:** (E12&&!SNPS_RSVDPARAM_22&&!SNPS_RSVDPARAM_39)||E16 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_mplla_div_mult_o[(DIV_M_W-1):0]	O	MPLLA Output Frequency Multiplier Control. This signal should be connected to mplla_div_multiplier input port of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_mpllb_div_mult_o[(DIV_M_W-1):0]	O	MPLLB Output Frequency Multiplier Control. This frequency multiplication factor is used to control MPLL clock generation from the reference clock input. This signal should be connected to mpllb_div_multiplier input port of the PHY. **Exists:** (E12&&!SNPS_RSVDPARAM_39)||E16||(E25&&!SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_mplla_div_clk_en_o	O	MPLLA Divide Clock Enable. If this signal is asserted, frequency of mplla_div_clk output from PHY is MPLLA frequency divided by mplla_div_multiplier. This signal should be connected to mplla_div_clk_en input pin of the PHY. **Exists:** (E12||E16||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mpllb_div_clk_en_o	O	MPLLB Divide Clock Enable. If this signal is asserted, frequency of mpllb_div_clk output from PHY is MPLLB frequency divided by mpllb_div_multiplier. This signal should be connected to mpllb_div_clk_en input pin of the PHY. **Exists:** (E12&&!SNPS_RSVDPARAM_39)||E16||(E25&&!SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mplla_bandwidth_o[(BW_W-1):0]	O	MPLLA Bandwidth Control. This signal is driven from MPLLA_BANDWIDTH field of VR_XS_PMA_MP_12G_MPLLA_CTRL0/VR_XS_PMA_MP_16G_MPLLA_CTRL3 Register. This signal is set appropriately to control the bandwidth of MPLLA. This signal should be connected to mplla_bandwidth input port of the PHY. **Exists:** (E12&&!SNPS_RSVDPARAM_22&&!SNPS_RSVDPARAM_39)||E16 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_mpllb_bandwidth_o[(BW_W-1):0]	O	MPLLB Bandwidth Control. This signal is driven from MPLLB_BANDWIDTH field of VR_XS_PMA_MP_12G_MPLLB_CTRL0/VR_XS_PMA_MP_16G_MPLLB_CTRL0 Register. This signal is set appropriately to control the bandwidth of MPLLB. This signal should be connected to mpllb_bandwidth input port of the PHY. **Exists:** (E12&&!SNPS_RSVDPARAM_22&&!SNPS_RSVDPARAM_39)||E16 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_mplla_div8_clk_en_o	O	MPLLA Divide by 8 Enable. This signal is driven from MPLLA_DIV8_CLK_EN field of VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2/VR_MII_MP_12G_16G_MPLLA_CTRL2 Register. When asserted, the frequency of mplla_word_clk output from PHY is MPLLA frequency divided by 8. This signal should be connected to mplla_div8_clk_en input port of the PHY. **Exists:** (E12&&!SNPS_RSVDPARAM_22)||E16||(SNPS_RSVDPARAM_22&&!SNPS_RSVDPARAM_2) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mpllb_div8_clk_en_o	O	MPLLB Divide by 8 Enable. This signal is driven from MPLLB_DIV8_CLK_EN field of VR_XS_PMA_MP_12G_16G_MPLLB_CTRL2/VR_MII_MP_12G_16G_MPLLB_CTRL2 Register. When asserted, the frequency of mpllb_word_clk output from PHY is MPLLB frequency divided by 8. This signal should be connected to mpllb_div8_clk_en input port of the PHY. **Exists:** (E12&&!SNPS_RSVDPARAM_22&&!SNPS_RSVDPARAM_39)||E16 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_mplla_div10_clk_en_o	O	MPLLA Divide by 10 Enable. This signal is driven from MPLLA_DIV10_CLK_EN field of VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2/VR_MII_MP_12G_16G_MPLLA_CTRL2 Register. When asserted, the frequency of mplla_word_clk output from PHY is MPLLA frequency divided by 10. This signal should be connected to mplla_div10_clk_en input port of the PHY. **Exists:** (E12&&!SNPS_RSVDPARAM_22)||E16||(SNPS_RSVDPARAM_22&&!SNPS_RSVDPARAM_2) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mpllb_div10_clk_en_o	O	MPLLB Divide by 10 Enable. This signal is driven from MPLLB_DIV8_CLK_EN field of VR_XS_PMA_MP_12G_16G_MPLLB_CTRL2/VR_MII_MP_12G_16G_MPLLB_CTRL2 Register. When asserted, the frequency of mpllb_word_clk output from PHY is MPLLB frequency divided by 10. This signal should be connected to mpllb_div10_clk_en input port of the PHY. **Exists:** (E12&&!SNPS_RSVDPARAM_22&&!SNPS_RSVDPARAM_39)||E16 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_ref_clk_mplla_div2_en_o	O	Divide reference clock to MPLLA by 2. This signal is driven from REF_MPLLA_DIV2 field of VR_XS_PMA_MP_12G_16G_REF_CLK_CTRL/VR_MII_MP_12G_16G_REF_CLK_CTRL Register. This signal should be connected to ref_clk_mplla_div2_en input pin of the PHY. **Exists:** (E12&&!SNPS_RSVDPARAM_22)||E16 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_ref_clk_mpllb_div2_en_o	O	Divide reference clock to MPLLB by 2. This signal is driven from REF_MPLLB_DIV2 field of VR_XS_PMA_MP_12G_16G_REF_CLK_CTRL/VR_MII_MP_12G_16G_REF_CLK_CTRL Register. This signal should be connected to ref_clk_mpllb_div2_en input pin of the PHY. **Exists:** (E12&&!SNPS_RSVDPARAM_22&&!SNPS_RSVDPARAM_39)||E16 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mplla_ssc_freq_cnt_init_o[11:0]	O	MPLLA SSC Frequency Counter Initialization. Sets the initialization value for the SSC frequency counter. This signal is driven from MPLLA_SSC_FREQ_CNT_INT fields of VR_XS_PMA_MP_16G_MPLLA_CTRL4/VR_MII_MP_16G_MPLLA_CTRL4 Register. This signal should be connected to mplla_ssc_freq_cnt_init input port of the PHY. **Exists:** E16||SNPS_RSVDPARAM_39 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mplla_ssc_freq_cnt_pk_o[7:0]	O	MPLLA SSC Frequency Counter Peak. Sets the peak value for the SSC frequency counter. This signal is driven from MPLLA_SSC_FRQ_CNT_PK fields of VR_XS_PMA_MP_16G_MPLLA_CTRL5/VR_MII_MP_16G_MPLLA_CTRL5 Register. This signal should be connected to mplla_ssc_frq_cnt_peak[7:0] input port of the PHY. **Exists:** E16||SNPS_RSVDPARAM_39 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_mpllb_ssc_freq_cnt_init_o[11:0]	O	MPLLB SSC Frequency Counter Initialization. Sets the initialization value for the SSC frequency counter. This signal is driven from MPLLB_SSC_FREQ_CNT_INT fields of VR_XS_PMA_MP_16G_MPLLB_CTRL4/VR_MII_MP_16G_MPLLB_CTRL4 Register. This signal should be connected to mpllb_ssc_freq_cnt_init input port of the PHY. **Exists:** E16 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mpllb_ssc_freq_cnt_pk_o[7:0]	O	MPLLB SSC Frequency Counter Peak. Sets the peak value for the SSC frequency counter. This signal is driven from MPLLB_SSC_FRQ_CNT_PK fields of VR_XS_PMA_MP_16G_MPLLB_CTRL5/VR_MII_MP_16G_MPLLB_CTRL5 Register. This signal should be connected to mpllb_ssc_frq_cnt_peak[7:0] input port of the PHY. **Exists:** E16 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.28 Synopsys Multi-protocol 25G PHY MPLL Control and Status Interface Signals

- xpcs_mplla_fracn_den_o
- xpcs_mpllb_fracn_den_o
- xpcs_mplla_fracn_en_o
- xpcs_mpllb_fracn_en_o
- xpcs_mplla_fracn_quot_o
- xpcs_mpllb_fracn_quot_o
- xpcs_mplla_fracn_rem_o
- xpcs_mpllb_fracn_rem_o
- xpcs_mplla_bw_high_o
- xpcs_mpllb_bw_high_o
- xpcs_mplla_bw_low_o
- xpcs_mpllb_bw_low_o
- xpcs_mplla_bw_th_o
- xpcs_mpllb_bw_th_o
- xpcs_mplla_ctl_buf_byp_o
- xpcs_mpllb_ctl_buf_byp_o
- xpcs_mplla_short_lock_en_o
- xpcs_mpllb_short_lock_en_o
- xpcs_mplla_fb_clk_div4_en_o
- xpcs_mpllb_fb_clk_div4_en_o
- xpcs_mpllb_div16p5_clk_en_o
- xpcs_ref_clk_mplla_div_o
- xpcs_ref_clk_mpllb_div_o
- xpcs_mplla_recal_force_en_o
- xpcs_mpllb_recal_force_en_o
- xpcs_mplla_recal_skip_en_o
- xpcs_mpllb_recal_skip_en_o
- xpcs_mplla_word_clk_div_o
- xpcs_mpllb_word_clk_div_o

Table 6-28 Synopsys Multi-protocol 25G PHY MPLL Control and Status Interface Signals

Port Name	I/O	Description
xpcs_mplla_fracn_den_o[15:0]	O	MPLLA fractional denominator control This signal allows to control the PLL modulated frequency fractional ratio **Exists:** (SNPS_RSVDPARAM_22||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_mpllb_fracn_den_o[15:0]	O	MPLLB fractional denominator control This signal allows to control the PLL modulated frequency fractional ratio. **Exists:** (SNPS_RSVDPARAM_22||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_mplla_fracn_en_o	O	MPLLA fractional mode enable **Exists:** (SNPS_RSVDPARAM_22||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mpllb_fracn_en_o	O	MPLLB fractional mode enable **Exists:** (SNPS_RSVDPARAM_22||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mplla_fracn_quot_o[15:0]	O	MPLLA fractional quotient control This signal allows to control the PLL modulated frequency fractional ratio. **Exists:** (SNPS_RSVDPARAM_22||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_mpllb_fracn_quot_o[15:0]	O	MPLLB fractional quotient control This signal allows to control the PLL modulated frequency fractional ratio. **Exists:** (SNPS_RSVDPARAM_22||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_mplla_fracn_rem_o[15:0]	O	MPLLA fractional remainder control This signal allows to control the PLL modulated frequency fractional ratio. **Exists:** (SNPS_RSVDPARAM_22||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_mpllb_fracn_rem_o[15:0]	O	MPLLB fractional remainder control This signal allows to control the PLL modulated frequency fractional ratio. **Exists:** (SNPS_RSVDPARAM_22||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_mplla_bw_high_o[15:0]	O	MPLLA Bandwidth Control-High **Exists:** E25&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_mpllb_bw_high_o[15:0]	O	MPLLB Bandwidth Control-High **Exists:** E25&&!(SNPS_RSVDPARAM_36||SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_mplla_bw_low_o[15:0]	O	MPLLA Bandwidth Control-Low **Exists:** E25&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_mpllb_bw_low_o[15:0]	O	MPLLB Bandwidth Control-Low **Exists:** E25&&!(SNPS_RSVDPARAM_36||SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_mplla_bw_th_o[1:0]	O	MPLLA Bandwidth Threshold This signal controls the threshold that the PHY uses to select between xpcs_mplla_bw_low_o and xpcs_mplla_bw_high_o values. **Exists:** E25&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_mpllb_bw_th_o[1:0]	O	MPLLB Bandwidth Threshold This signal controls the threshold that the PHY uses to select between xpcs_mpllb_bw_low_o and xpcs_mpllb_bw_high_o values. **Exists:** E25&&!(SNPS_RSVDPARAM_36||SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_mplla_ctl_buf_byp_o	O	MPLLA Control Buffer Bypass Enable **Exists:** E25&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mpllb_ctl_buf_byp_o	O	MPLLB Control Buffer Bypass Enable **Exists:** E25&&!(SNPS_RSVDPARAM_36||SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mplla_short_lock_en_o	O	MPLLA Short Lock Enable When this signal is asserted, the MPLLA locking time gets reduced. **Exists:** E25&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mpllb_short_lock_en_o	O	MPLLB Short Lock Enable When this signal is asserted, the MPLLB locking time gets reduced. **Exists:** E25&&!(SNPS_RSVDPARAM_36||SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_mplla_fb_clk_div4_en_o	O	mplla_div_clk Output frequency Divider Control **Exists:** E25&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mpllb_fb_clk_div4_en_o	O	mpllb_div_clk Output frequency Divider Control **Exists:** E25&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mpllb_div16p5_clk_en_o	O	MPLLB Divided by 16.5 clock. **Exists:** E25&&!(SNPS_RSVDPARAM_36||SNPS_RSVDPARAM_37) **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_ref_clk_mplla_div_o[2:0]	O	MPLLA reference clock divider control **Exists:** (SNPS_RSVDPARAM_22||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_ref_clk_mpllb_div_o[2:0]	O	MPLLB reference clock divider control **Exists:** (SNPS_RSVDPARAM_22||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mplla_recal_force_en_o	O	MPLLA Force Re-calibration. This signal forces recalibration of MPLLA and TX DCC in the selected bank. **Exists:** E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_mpllb_recal_force_en_o	O	MPLLB Force Re-calibration. This signal forces recalibration of MPLLB and TX DCC in the selected bank. **Exists:** E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mplla_recal_skip_en_o	O	MPLLA Skip Re-calibration This signal skips re-calibration of MPLLA and TX DCC in the selected bank **Exists:** E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mpllb_recal_skip_en_o	O	MPLLB Skip Re-calibration This signal skips re-calibration of MPLLB and TX DCC in the selected bank **Exists:** E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mplla_word_clk_div_o[1:0]	O	MPLLA Word Clock Divider Control This signal controls the MPLLA Word Clock dividers for generating the mplla_word_clk **Exists:** E25&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_mpllb_word_clk_div_o[1:0]	O	MPLLB Word Clock Divider Control This signal controls the MPLLB Word Clock dividers for generating the mpllb_word_clk **Exists:** E25&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.29 Synopsys Multi-protocol 16G/25G PHY Miscellaneous Interface Signals

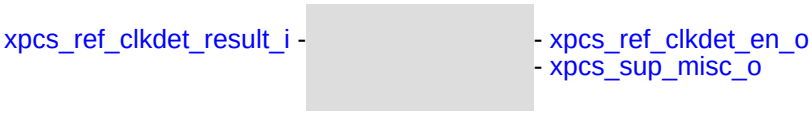


Table 6-29 Synopsys Multi-protocol 16G/25G PHY Miscellaneous Interface Signals

Port Name	I/O	Description
xpcs_ref_clkdet_result_i	I	Reference Clock Detection result. Indicates the presence of the reference clock on the pads input of PHY. The value of this signal is available in REF_CLKDET_RESULT field of VR_MII_MP_12G_16G_MISC_STS Register. This signal is driven by ref_clkdet_result output port of PHY. **Exists:** E16||SNPS_RSVDPARAM_22||E25||SNPS_RSVDPARAM_39 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_ref_clkdet_en_o	O	Enable Reference Clock Detection. Enables detection of the reference clock on the pads input. This signal is driven from REF_CLK_DET_EN fields of VR_XS_PMA_MP_16G_25G_MISC_CTRL2/VR_MII_MP_16G_25G_MISC_CTRL2 Register. This signal should be connected to ref_clkdet_en input port of the PHY. **Exists:** E16||SNPS_RSVDPARAM_22||E25||SNPS_RSVDPARAM_39 **Synchronous To:** None **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_sup_misc_o[7:0]	O	Support Miscellaneous Controls. This input controls miscellaneous settings in the Support block of PHY. This signal is driven from SUP_MISC fields of VR_XS_PMA_MP_16G_25G_MISC_CTRL2/VR_MII_MP_16G_25G_MISC_CTRL2 Register. This signal should be connected to sup_misc input port of the PHY. **Exists:** (E16||E25||SNPS_RSVDPARAM_39||E32)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.30 Synopsys Multi-protocol 12G/16G/25G PHY Miscellaneous Interface Signals



Table 6-30 Synopsys Multi-protocol 12G/16G/25G PHY Miscellaneous Interface Signals

Port Name	I/O	Description
xpcs_ref_repeat_clk_en_o	O	Enable Repeat Reference Clock from PHY. This signal is driven from REF_RPT_CLK_EN field of VR_XS_PMA_MP_12G_16G_REF_CLK_CTRL/VR_MII_MP_12G_16G_REF_CLK_CTRL Register. This signal should be connected to ref_repeat_clk_en input of the PHY. **Exists:** E12||E16||E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx2tx_par_lb_en_o[(NUM_OF_LANES-1):0]	O	Parallel (Rx to Tx) loopback enable. This signal is driven from RX2TX_LB_EN_0/RX2TX_LB_EN_3_1 fields of VR_XS_PMA_MP_12G_16G_25G_MISC_CTRL0/VR_MII_MP_12G_16G_25G_MISC_CTRL0 Register. When this signal is asserted, recovered parallel data from the PHY receiver is looped back to the PHY transmit serializer. This signal should be connected to laneX_rx2tx_par_lb_en input port of the PHY. **Exists:** E12||E16||E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_tx2rx_ser_lb_en_o[(NUM_OF_LANES-1):0]	O	Analog serial (Tx to Rx) Loopback Enable. This signal is driven from TX2RX_LB_EN_0/TX2RX_LB_EN_3_1 fields of VR_XS_PMA_MP_12G_16G_25G_MISC_CTRL0/VR_MII_MP_12G_16G_25G_MISC_CTRL0 Register. When this signal is asserted, data from Tx pre-driver in PHY is looped back to Rx analog front end. This signal should be connected to laneX_tx2rx_ser_lb_en input port of the PHY. **Exists:** E12||E16||E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_sram_bootload_byp_o[(SRAM_BLBYP_W-1):0]	O	SRAM Boot-Load Bypass. This signal is driven from BTLD_BYP field of VR_XS_PMA_MP_12G_16G_25G_SRAM/VR_MII_MP_12G_16G_25G_SRAM Register. This signal can be connected to 'sram_bootload_bypass' input pin of the PHY (if available). **Exists:** E25||SNPS_RSVDPARAM_39||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_sram_ext_ld_done_o	O	Control to indicate that external loading of SRAM is Done. This signal is driven from EXT_LD_DN field of VR_XS_PMA_MP_12G_16G_25G_SRAM/VR_MII_MP_12G_16G_25G_SRAM Register. Software should program this register field after it has completed the (optional) external loading of SRAM connected to Multi-protocol 12G/16G PHY. Only after this step, the PHY proceeds to complete its intialization sequence. This signal should be connected to 'sram_ext_ld_done' input pin of the PHY. **Exists:** E12||E16||E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_sram_init_done_i	I	Status from PHY indication that SRAM initialization is Complete. Value of this signal is reflected in INIT_DN field of VR_XS_PMA_MP_12G_16G_25G_SRAM/VR_MII_MP_12G_16G_25G_SRAM Register. Whenever power-on-reset/soft-reset is asserted, software should first poll for this signal to go high. Thereafter, it should program 'EXT_LD_DN' bit (after completing the optional SRAM loading operation). Only after these steps are completed, the reset sequence of PHY (and in turn DWC_xpcs) gets over. This signal should be connected to 'sram_init_done' output pin of the PHY. In case, the PHY does not have SRAM interface, tie this input pin to zero. **Exists:** E12||E16||E25||E32 **Synchronous To:** Asynchronous **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.31 Synopsys Multi-protocol 12G/16G PHY Miscellaneous Interface Signals

- xpcs_cr_para_sel_o

Table 6-31 Synopsys Multi-protocol 12G/16G PHY Miscellaneous Interface Signals

Port Name	I/O	Description
xpcs_cr_para_sel_o	O	CR parallel port Select. This signal is driven from CR_PARA_SEL field of VR_XS_PMA_MP_12G_16G_25G_MISC_CTRL0/VR_MII_MP_12G_16G_25G_MISC_CTRL0 Register. This signal should be connected to cr_para_sel input pin of the PHY. **Exists:** (E12||E16||E25||E32)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.32 Synopsys Multi-protocol 16G/25G PHY Transmit Control and Status Interface Signals

Table 6-32 Synopsys Multi-protocol 16G/25G PHY Transmit Control and Status Interface Signals

Port Name	I/O	Description
xpcs_tx0_misc_o[7:0]	O	TX Miscellaneous Controls along lane0. It controls miscellaneous settings in the TX analog block of PHY for lane0 Tx path. This signal is driven from TX0_MISC fields of VR_XS_PMA_MP_16G_25G_TX_MISC_CTRL0/VR_MII_MP_16G_25G_TX_MISC_CTRL0 Register. This signal should be connected to tx0_misc input port of the PHY. **Exists:** (E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_tx1_misc_o[7:0]	O	TX Miscellaneous Controls along lane1. It controls miscellaneous settings in the TX analog block of PHY for lane1 Tx path. This signal is driven from TX1_MISC fields of VR_XS_PMA_MP_16G_25G_TX_MISC_CTRL0 Register. This signal should be connected to tx1_misc input port of the PHY. **Exists:** (E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_tx2_misc_o[7:0]	O	TX Miscellaneous Controls along lane2. It controls miscellaneous settings in the TX analog block of PHY for lane2 Tx path. This signal is driven from TX2_MISC fields of VR_XS_PMA_MP_16G_25G_TX_MISC_CTRL1 Register. This signal should be connected to tx2_misc input port of the PHY. **Exists:** (E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_tx3_misc_o[7:0]	O	TX Miscellaneous Controls along lane3. It controls miscellaneous settings in the TX analog block of PHY for lane3 Tx path. This signal is driven from TX3_MISC fields of VR_XS_PMA_MP_16G_25G_TX_MISC_CTRL1 Register. This signal should be connected to tx3_misc input port of the PHY. **Exists:** (E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_txup_term_offset_o[8:0]	O	Offset for TX Up Termination. Specifies an additional fixed offset to calibrated TX up termination value. This signal is driven from TXUP_TERM_OFFSET fields of VR_XS_PMA_MP_16G_25G_TX_GENCTRL3/VR_MII_MP_16G_25G_TX_GENCTRL3 Register. This signal should be connected to txup_term_offset input port of the PHY. **Exists:** (E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_txdn_term_offset_o[8:0]	O	Offset for TX Down Termination. Specifies an additional fixed offset to calibrated TX down termination value. This signal is driven from TXDN_TERM_OFFSET fields of VR_XS_PMA_MP_16G_25G_TX_GENCTRL4/VR_MII_MP_16G_25G_TX_GENCTRL4 Register. This signal should be connected to txdn_term_offset input port of the PHY. **Exists:** (E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.33 Synopsys Multi-protocol 16G/25G PHY Receive Control and Status Interface Signals

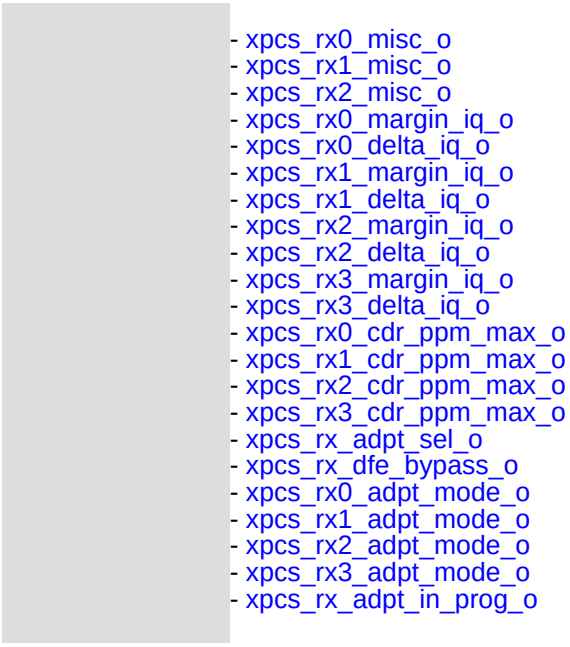


Table 6-33 Synopsys Multi-protocol 16G/25G PHY Receive Control and Status Interface Signals

Port Name	I/O	Description
xpcs_rx0_misc_o[7:0]	O	RX Miscellaneous Controls along lane0. It controls miscellaneous settings in the RX analog block of PHY for lane0 Rx path. This signal is driven from RX0_MISC fields of VR_XS_PMA_MP_16G_25G_RX_MISC_CTRL0/VR_MII_MP_16G_25G_RX_MISC_CTRL0 Register. This signal should be connected to rx0_misc input port of the PHY. **Exists:** (E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_rx1_misc_o[7:0]	O	TX Miscellaneous Controls along lane1. It controls miscellaneous settings in the RX analog block of PHY for lane1 Rx path. This signal is driven from RX1_MISC fields of VR_XS_PMA_MP_16G_25G_RX_MISC_CTRL0 Register. This signal should be connected to rx1_misc input port of the PHY. **Exists:** (E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx2_misc_o[7:0]	O	RX Miscellaneous Controls along lane2. It controls miscellaneous settings in the RX analog block of PHY for lane2 Rx path. This signal is driven from RX2_MISC fields of VR_XS_PMA_MP_16G_25G_RX_MISC_CTRL1 Register. This signal should be connected to rx2_misc input port of the PHY. **Exists:** (E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx0_margin_iq_o[6:0]	O	Value for RX IQ margining (Along Rx Lane0). This is a signed value for skewing the IQ separation for the purpose of RX margining. This signal is driven from RX0_MARGIN_IQ fields of VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL0/VR_MII_MP_16G_25G_RX_IQ_CTRL0 Register. This signal should be connected to rx0_margin_iq[6:0] input port of the PHY. **Exists:** E16||E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_rx0_delta_iq_o[3:0]	O	IQ offset value (Along Rx Lane0). Indicates the amount of additional offset to apply to the In-phase/Quadrature (IQ) separation. This signal is driven from RX0_DELTA_IQ fields of VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL0/VR_MII_MP_16G_25G_RX_IQ_CTRL0 Register. This signal should be connected to rx0_delta_iq[3:0] input port of the PHY. **Exists:** E16||SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||(E25&&!SNPS_RSVDPARAM_37)||SNPS_RSVDPARAM_39 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx1_margin_iq_o[6:0]	O	Value for RX IQ margining (Along Rx Lane1). This is a signed value for skewing the IQ separation for the purpose of RX margining. This signal is driven from RX1_MARGIN_IQ fields of VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL1 Register. This signal should be connected to rx1_margin_iq[6:0] input port of the PHY. **Exists:** (E16||E25||E32)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx1_delta_iq_o[3:0]	O	IQ offset value (Along Rx Lane1). Indicates the amount of additional offset to apply to the In-phase/Quadrature (IQ) separation. This signal is driven from RX1_DELTA_IQ fields of VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL1 Register. This signal should be connected to rx1_delta_iq[3:0] input port of the PHY. **Exists:** (E16||SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||E25||SNPS_RSVDPARAM_39)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_rx2_margin_iq_o[6:0]	O	Value for RX IQ margining (Along Rx Lane2). This is a signed value for skewing the IQ separation for the purpose of RX margining. This signal is driven from RX2_MARGIN_IQ fields of VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL2 Register. This signal should be connected to rx2_margin_iq[6:0] input port of the PHY. **Exists:** (E16||E25||E32)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx2_delta_iq_o[3:0]	O	IQ offset value (Along Rx Lane2). Indicates the amount of additional offset to apply to the In-phase/Quadrature (IQ) separation. This signal is driven from RX2_DELTA_IQ fields of VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL2 Register. This signal should be connected to rx2_delta_iq[3:0] input port of the PHY. **Exists:** (E16||SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||E25||SNPS_RSVDPARAM_39)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx3_margin_iq_o[6:0]	O	Value for RX IQ margining (Along Rx Lane3). This is a signed value for skewing the IQ separation for the purpose of RX margining. This signal is driven from RX3_MARGIN_IQ fields of VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL3 Register. This signal should be connected to rx3_margin_iq[6:0] input port of the PHY. **Exists:** (E16||E25||E32)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_rx3_delta_iq_o[3:0]	O	IQ offset value (Along Rx Lane3). Indicates the amount of additional offset to apply to the In-phase/Quadrature (IQ) separation. This signal is driven from RX3_DELTA_IQ fields of VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL3 Register. This signal should be connected to rx3_delta_iq[3:0] input port of the PHY. **Exists:** (E16||SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||E25||SNPS_RSVDPARAM_39)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx0_cdr_ppm_max_o[4:0]	O	Maximum Allowed PPM on the RX CDR Clock(Along Lane0). This signal specifies the maximum PPM drift to be allowed on the lane0 RX clock (relative to ideal desired frequency) before the CDR recovery circuit is engaged. This signal is driven from RX0_CDR_PPM_MAX fields of VR_XS_PMA_MP_16G_25G_RX_PPM_CTRL0/VR_MII_MP_16G_25G_RX_PPM_CTRL0 Register. This signal should be connected to rx0_cdr_ppm_max input port of the PHY. **Exists:** (E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx1_cdr_ppm_max_o[4:0]	O	Maximum Allowed PPM on the RX CDR Clock(Along Lane1). This signal specifies the maximum PPM drift to be allowed on the lane1 RX clock (relative to ideal desired frequency) before the CDR recovery circuit is engaged. This signal is driven from RX1_CDR_PPM_MAX fields of VR_XS_PMA_MP_16G_25G_RX_PPM_CTRL0 Register. This signal should be connected to rx1_cdr_ppm_max input port of the PHY. **Exists:** (E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_rx2_cdr_ppm_max_o[4:0]	O	Maximum Allowed PPM on the RX CDR Clock(Along Lane2). This signal specifies the maximum PPM drift to be allowed on the lane2 RX clock (relative to ideal desired frequency) before the CDR recovery circuit is engaged. This signal is driven from RX2_CDR_PPM_MAX fields of VR_XS_PMA_MP_16G_25G_RX_PPM_CTRL1 Register. This signal should be connected to rx2_cdr_ppm_max input port of the PHY. **Exists:** (E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx3_cdr_ppm_max_o[4:0]	O	Maximum Allowed PPM on the RX CDR Clock(Along Lane3). This signal specifies the maximum PPM drift to be allowed on the lane3 RX clock (relative to ideal desired frequency) before the CDR recovery circuit is engaged. This signal is driven from RX3_CDR_PPM_MAX fields of VR_XS_PMA_MP_16G_25G_RX_PPM_CTRL1 Register. This signal should be connected to rx3_cdr_ppm_max input port of the PHY. **Exists:** (E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx_adpt_sel_o[(NUM_OF_LANES-1):0]	O	Select Storage Bank for RX Adaptation. The PHY supports two internal register storage banks for saving adaptation results-this signal specifies which one is selected. This signal is driven from RX_ADPT_SEL_0/RX_ADPT_SEL_3_1 fields of VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL5/VR_MII_MP_16G_25G_RX_EQ_CTRL5 Register. This signal should be connected to rxX_adapt_sel input port of the PHY. **Exists:** (E16||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_rx_dfe_bypass_o[(NUM_OF_LANES -1):0]	O	RX DFE By-pass Enable. Indicates by-passing of the decision feedback equalization (DFE) circuitry of PHY. This signal is driven from RX_DFE_BYP_0/RX_DFE_BYP_3_1 fields of VR_XS_PMA_MP_16G_25G_RX_GENCTRL4/VR_MII_MP_16G_25G _RX_GENCTRL4 Register. This signal should be connected to rxX_dfe_bypass input port of the PHY. **Exists:** (E16||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx0_adpt_mode_o[1:0]	O	RX Adaptation mode select (Along Rx Lane0). Selects what RX adaptation mode is performed along Rx Lane0 of PHY. This signal is driven from RX0_ADPT_MODE fields of VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL5/VR_MII_MP_16G_25G _RX_EQ_CTRL5 Register. This signal should be connected to rx0_adapt_mode[1:0] input port of the PHY. **Exists:** (E16||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx1_adpt_mode_o[1:0]	O	RX Adaptation mode select (Along Rx Lane1). Selects what RX adaptation mode is performed along Rx Lane1 of PHY. This signal is driven from RX1_ADPT_MODE fields of VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL5 Register. This signal should be connected to rx1_adapt_mode[1:0] input port of the PHY. **Exists:** (E16||E25)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_rx2_adpt_mode_o[1:0]	O	RX Adaptation mode select (Along Rx Lane2). Selects what RX adaptation mode is performed along Rx Lane2 of PHY. This signal is driven from RX2_ADPT_MODE fields of VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL5 Register. This signal should be connected to rx2_adapt_mode[1:0] input port of the PHY. **Exists:** (E16||E25)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx3_adpt_mode_o[1:0]	O	RX Adaptation mode select (Along Rx Lane3). Selects what RX adaptation mode is performed along Rx Lane3 of PHY. This signal is driven from RX3_ADPT_MODE fields of VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL5 Register. This signal should be connected to rx3_adapt_mode[1:0] input port of the PHY. **Exists:** (E16||E25)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx_adpt_in_prog_o[(NUM_OF_LANES-1):0]	O	Receive Adaptation in Progress. **Exists:** E16||E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.34 Synopsys Multi-protocol 16G/25G PHY Receive MPLL Control and Status Interface Signals

Table 6-34 Synopsys Multi-protocol 16G/25G PHY Receive MPLL Control and Status Interface Signals

Port Name	I/O	Description
xpcs_rx3_misc_o[7:0]	O	RX Miscellaneous Controls along lane3. It controls miscellaneous settings in the RX analog block of PHY for lane3 Rx path. This signal is driven from RX3_MISC fields of VR_XS_PMA_MP_16G_25G_RX_MISC_CTRL1 Register. This signal should be connected to rx3_misc input port of the PHY. **Exists:** (E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.35 Synopsys Multi-protocol 25G PHY Receive Control and Status Interface Signals

- xpcs_rx_recal_force_en_o
- xpcs_rx_recal_skip_en_o
- xpcs_rx0_recal_bank_sel_o
- xpcs_rx1_recal_bank_sel_o
- xpcs_rx2_recal_bank_sel_o
- xpcs_rx3_recal_bank_sel_o

Table 6-35 Synopsys Multi-protocol 25G PHY Receive Control and Status Interface Signals

Port Name	I/O	Description
xpcs_rx_recal_force_en_o[(NUM_OF_LANES-1):0]	O	Receive Force Re-calibration **Exists:** E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx_recal_skip_en_o[(NUM_OF_LANES-1):0]	O	Receive Skip Re-Calibration **Exists:** E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx0_recal_bank_sel_o[1:0]	O	Receive Re-calibration Bank Select -lane 0 **Exists:** E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx1_recal_bank_sel_o[1:0]	O	Receive Re-calibration Bank Select -lane 1 **Exists:** (E25||E32)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx2_recal_bank_sel_o[1:0]	O	Receive Re-calibration Bank Select -lane 2 **Exists:** (E25||E32)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_rx3_recal_bank_sel_o[1:0]	O	Receive Re-calibration Bank Select -lane 3 **Exists:** (E25||E32)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.36 Synopsys Multi-protocol 32G PHY Rx Control and Status Interface Signals

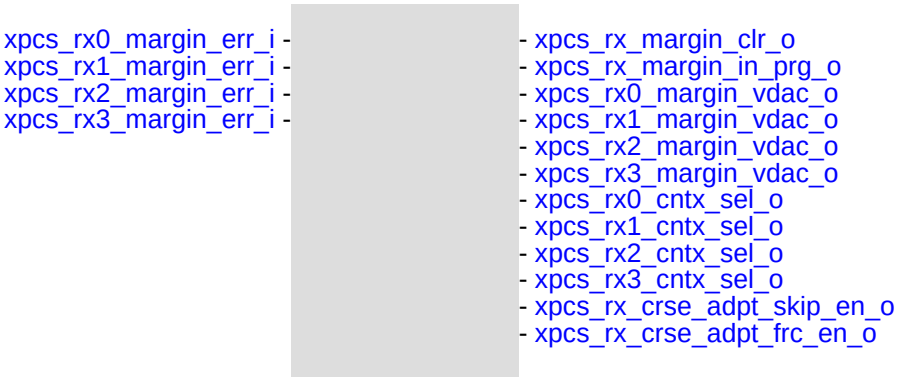


Table 6-36 Synopsys Multi-protocol 32G PHY Rx Control and Status Interface Signals

Port Name	I/O	Description
xpcs_rx0_margin_err_i[5:0]	I	Error counter for RX margining along Lane 0 This signal is used to indicate the number of bit-errors during RX margining procedure. This output is continuously updated when rx0_margin_in_prog is high except when rx0_ack is asserted. **Exists:** SNPS_RSVDPARAM_37||E32 **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx1_margin_err_i[5:0]	I	Error counter for RX margining along Lane 1 This signal is used to indicate the number of bit-errors during RX margining procedure. This output is continuously updated when rx0_margin_in_prog is high except when rx0_ack is asserted. **Exists:** (SNPS_RSVDPARAM_37||E32)&&!SINGLE_LANE_CFG **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx2_margin_err_i[5:0]	I	Error counter for RX margining along Lane 2 This signal is used to indicate the number of bit-errors during RX margining procedure. This output is continuously updated when rx0_margin_in_prog is high except when rx0_ack is asserted. **Exists:** (SNPS_RSVDPARAM_37||E32)&&!SINGLE_LANE_CFG **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_rx3_margin_err_i[5:0]	I	Error counter for RX margining along Lane 3 This signal is used to indicate the number of bit-errors during RX margining procedure. This output is continuously updated when rx0_margin_in_prog is high except when rx0_ack is asserted. **Exists:** (SNPS_RSVDPARAM_37||E32)&&!SINGLE_LANE_CFG **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx_margin_clr_o[(NUM_OF_LANES-1):0]	O	Clear signal for RX margining This signal clears the rxX_margin_error counter. **Exists:** SNPS_RSVDPARAM_37||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx_margin_in_prg_o[(NUM_OF_LANES-1):0]	O	Receiver margining in progress This input indicates that RX margining is in progress. Assert this signal at the same time or prior to any change in rxX_margin_vdac or rxX_margin_iq. It should stay asserted during the entire RX margining sequence and should be de-asserted to complete the RX margining sequence. **Exists:** SNPS_RSVDPARAM_37||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx0_margin_vdac_o[8:0]	O	Value for RX voltage margining along lane 0 This is a signed value. It covers a range from -50 mV to +50 mV. **Exists:** SNPS_RSVDPARAM_37||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx1_margin_vdac_o[8:0]	O	Value for RX voltage margining along lane 1 This is a signed value. It covers a range from -50 mV to +50 mV. **Exists:** (SNPS_RSVDPARAM_37||E32)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_rx2_margin_vdac_o[8:0]	O	Value for RX voltage margining along lane 2 This is a signed value. It covers a range from -50 mV to +50 mV. **Exists:** (SNPS_RSVDPARAM_37||E32)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx3_margin_vdac_o[8:0]	O	Value for RX voltage margining along lane 3 This is a signed value. It covers a range from -50 mV to +50 mV. **Exists:** (SNPS_RSVDPARAM_37||E32)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx0_cntx_sel_o[7:0]	O	Receive context select along Lane 0 This input specifies the context that needs to be restored for the RX block. If there is any change on this input when rx0_req is asserted, the context restore feature detects that the RX block context restore should be performed. **Note:** The rx_cntx_sel of all the lanes in the same link must be stable before rx_reset of any lane in the link gets de-asserted or before rx0_req is asserted. **Exists:** SNPS_RSVDPARAM_37||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx1_cntx_sel_o[7:0]	O	Receive context select along Lane 1 This input specifies the context that needs to be restored for the RX block. If there is any change on this input when rx1_req is asserted, the context restore feature detects that the RX block context restore should be performed. **Note:** The rx_cntx_sel of all the lanes in the same link must be stable before rx_reset of any lane in the link gets de-asserted or before rx1_req is asserted. **Exists:** (SNPS_RSVDPARAM_37||E32)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_rx2_cntx_sel_o[7:0]	O	Receive context select along Lane 2 This input specifies the context that needs to be restored for the RX block. If there is any change on this input when rx2_req is asserted, the context restore feature detects that the RX block context restore should be performed. **Note:** The rx_cntx_sel of all the lanes in the same link must be stable before rx_reset of any lane in the link gets de-asserted or before rx2_req is asserted. **Exists:** (SNPS_RSVDPARAM_37||E32)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx3_cntx_sel_o[7:0]	O	Receive context select along Lane 3 This input specifies the context that needs to be restored for the RX block. If there is any change on this input when rx3_req is asserted, the context restore feature detects that the RX block context restore should be performed. **Note:** The rx_cntx_sel of all the lanes in the same link must be stable before rx_reset of any lane in the link gets de-asserted or before rx3_req is asserted. **Exists:** (SNPS_RSVDPARAM_37||E32)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx_crse_adpt_skip_en_o[(NUM_OF_LANES-1):0]	O	Skips coarse adaptation in the selected bank and mode regardless of the state of the corresponding done signal **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_rx_crse_adpt_frc_en_o[(NUM_OF_LANES-1):0]	O	Forces a coarse adaptation in the selected bank and mode regardless of the state of the corresponding done signal **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.37 Synopsys Multi-protocol 32G PHY Tx Control and Status Interface Signals

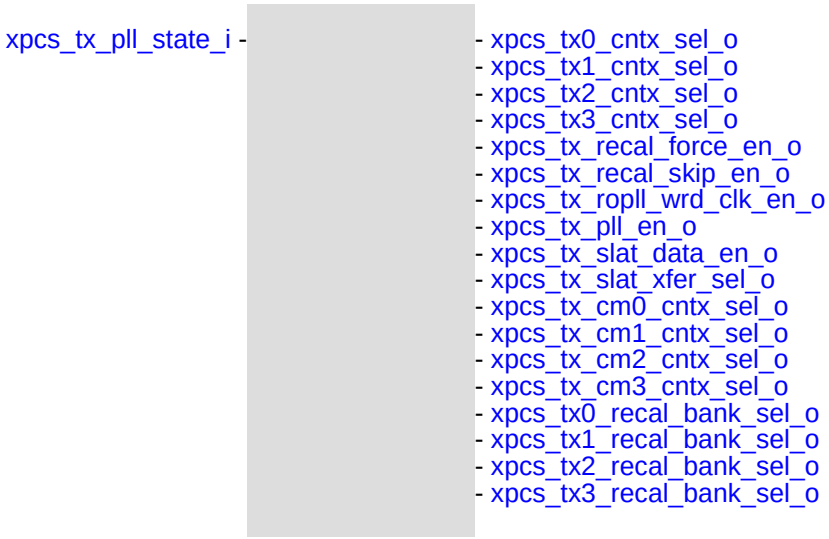


Table 6-37 Synopsys Multi-protocol 32G PHY Tx Control and Status Interface Signals

Port Name	I/O	Description
xpcs_tx0_cntx_sel_o[7:0]	O	Transmit context select along Lane 0 This input specifies the context that needs to be restored for the TX block. If there is any change on this input when tx0_req is asserted, the context // restore feature detects that the TX block context restore should be performed. **Note:** The tx_cntx_sel of all the lanes in the same link must be stable before tx_reset of any lane in the link gets de-asserted or before tx0_req is asserted. **Exists:** SNPS_RSVDPARAM_37||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_tx1_cntx_sel_o[7:0]	O	Transmit context select along Lane 1 This input specifies the context that needs to be restored for the TX block. If there is any change on this input when tx1_req is asserted, the context // restore feature detects that the TX block context restore should be performed. **Note:** The tx_cntx_sel of all the lanes in the same link must be stable before tx_reset of any lane in the link gets de-asserted or before tx1_req is asserted. **Exists:** (SNPS_RSVDPARAM_37||E32)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_tx2_cntx_sel_o[7:0]	O	Transmit context select along Lane 2 This input specifies the context that needs to be restored for the TX block. If there is any change on this input when tx2_req is asserted, the context // restore feature detects that the TX block context restore should be performed. **Note:** The tx_cntx_sel of all the lanes in the same link must be stable before tx_reset of any lane in the link gets de-asserted or before tx2_req is asserted. **Exists:** (SNPS_RSVDPARAM_37||E32)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_tx3_cntx_sel_o[7:0]	O	Transmit context select along Lane 3 This input specifies the context that needs to be restored for the TX block. If there is any change on this input when tx3_req is asserted, the context // restore feature detects that the TX block context restore should be performed. **Note:** The tx_cntx_sel of all the lanes in the same link must be stable before tx_reset of any lane in the link gets de-asserted or before tx3_req is asserted. **Exists:** (SNPS_RSVDPARAM_37||E32)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_tx_recal_force_en_o[(NUM_OF_LANES-1):0]	O	TX Force Re-Calibration Forces a re-calibration of the TX DCC in the selected bank regardless of the state of the corresponding done signal. **Note:** This input is sampled by the PHY with the assertion of txX_req. This input has a higher priority than the txX_recal_skip_en signal. **Exists:** SNPS_RSVDPARAM_37||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_tx_recal_skip_en_o[(NUM_OF_LANES-1):0]	O	TX Skip Re-Calibration Skips re-calibration of the TX DCC in the selected bank regardless of the state of the corresponding done signal. **Note:** This input is sampled by the PHY with the assertion of txX_req. This input has a lower priority than the txX_recal_force_en signal. **Exists:** SNPS_RSVDPARAM_37||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_tx_ropll_wrd_clk_en_o[(NUM_OF_LANES-1):0]	O	TX ROPLL word clock output enable **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_tx_pll_state_i[(NUM_OF_LANES-1):0]	I	Indicates the state of PLL for the lane. This signal is asserted when lane PLL is powered up and phase-locked. **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_tx_pll_en_o[(NUM_OF_LANES-1):0]	O	TX PLL Enable Powers up or powers down the ROPLL or MPLLA or MPLLB. For normal operation, this output is set to 1 when txX_pstate[1:0] is in P0 or P0s state. **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_tx_slat_data_en_o[(NUM_OF_LANES-1):0]	O	Per-lane Transmitter data enable for short latency mode When asserted and the TX lane is in P0, then the TX output driver of PHY is enabled. If de-asserted or the TX lane is not in P0, then the TX output driver is disabled. **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx_slat_xfer_sel_o[(NUM_OF_LANES-1):0]	O	Per-lane TX short latency transfer enable. Static input signal that selects between short latency Tx mode and normal latency Tx mode. txX_reset or phy_reset is required after changing this port of PHY. Tie low if unused. **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx_cm0_cntx_sel_o[7:0]	O	Per-lane TX common context select lane0. A change in tx0_cmn_cntx_sel along with tx0_req triggers changes in MPLLA/B, RTUNE etc. So to avoid conflict, shared MPLL and ext RTUNE are disabled when this changes. **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx_cm1_cntx_sel_o[7:0]	O	Per-lane TX common context select lane1. A change in tx1_cmn_cntx_sel along with tx1_req triggers changes in MPLLA/B, RTUNE etc. So to avoid conflict, shared MPLL and ext RTUNE are disabled when this changes. **Exists:** E32&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_tx_cm2_cntx_sel_o[7:0]	O	Per-lane TX common context select lane2. A change in tx2_cmn_cntx_sel along with tx2_req triggers changes in MPLLA/B, RTUNE etc. So to avoid conflict, shared MPLL and ext RTUNE are disabled when this changes. **Exists:** E32&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx_cm3_cntx_sel_o[7:0]	O	Per-lane TX common context select lane3. A change in tx3_cmn_cntx_sel along with tx2_req triggers changes in MPLLA/B, RTUNE etc. So to avoid conflict, shared MPLL and ext RTUNE are disabled when this changes. **Exists:** E32&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx0_recal_bank_sel_o[1:0]	O	Selects the bank to store the RX duty cycle correction (DCC) values for Lane0 **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx1_recal_bank_sel_o[1:0]	O	Selects the bank to store the RX duty cycle correction (DCC) values for Lane1 **Exists:** E32&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx2_recal_bank_sel_o[1:0]	O	Selects the bank to store the RX duty cycle correction (DCC) values for Lane2 **Exists:** E32&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_tx3_recal_bank_sel_o[1:0]	O	Selects the bank to store the RX duty cycle correction (DCC) values for Lane3 **Exists:** E32&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.38 Synopsys Multi-protocol 32G PHY Common Control and Status Interface Signals

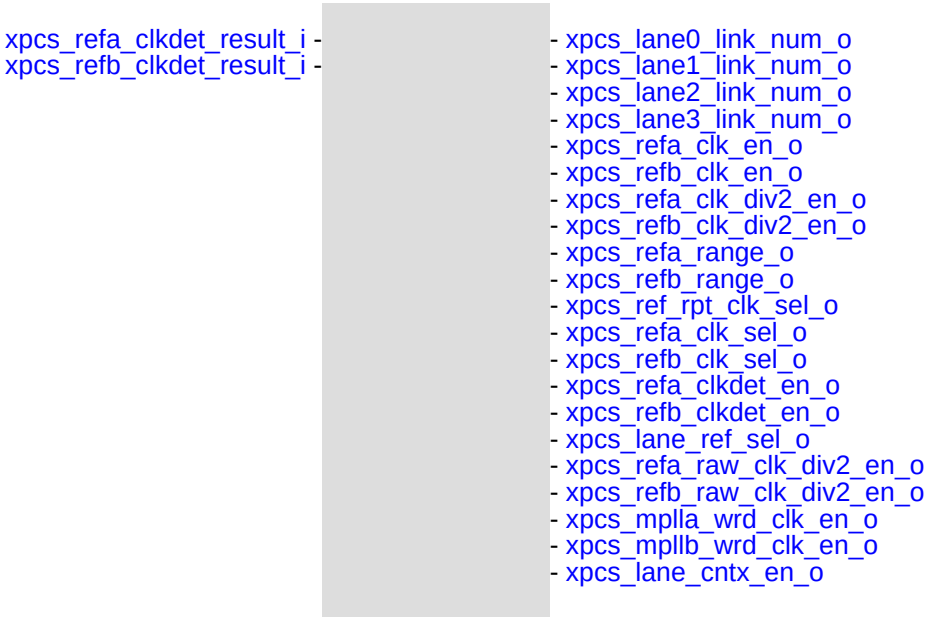


Table 6-38 Synopsys Multi-protocol 32G PHY Common Control and Status Interface Signals

Port Name	I/O	Description
xpcs_lane0_link_num_o[3:0]	O	Link number along lane 0 This input determines which link the current lane belongs to. **Exists:** SNPS_RSVDPARAM_37||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_lane1_link_num_o[3:0]	O	Link number along lane 1 This input determines which link the current lane belongs to. **Exists:** (SNPS_RSVDPARAM_37||E32)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_lane2_link_num_o[3:0]	O	Link number along lane 2 This input determines which link the current lane belongs to. **Exists:** (SNPS_RSVDPARAM_37||E32)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_lane3_link_num_o[3:0]	O	Link number along lane 3 This input determines which link the current lane belongs to. **Exists:** (SNPS_RSVDPARAM_37||E32)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_refa_clk_en_o	O	Reference clock enable for refa Enables the reference clock for the PHY. The xpcs_refa_clk_en_o signal must remain de-asserted until the reference clock is running at the appropriate frequency **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_refb_clk_en_o	O	Reference clock enable for refb Enables the reference clock for the PHY. The xpcs_refb_clk_en_o signal must remain de-asserted until the reference clock is running at the appropriate frequency **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_refa_clk_div2_en_o	O	Output reference clock divider control for refa When asserted, the reference clock is divided by 2 **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_refb_clk_div2_en_o	O	Output reference clock divider control for refb When asserted, the reference clock is divided by 2 **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_refa_range_o[2:0]	O	Output reference clock frequency range for refa Specifies the frequency range of the PHY's input reference clock. **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_refb_range_o[2:0]	O	Output reference clock frequency range for refb Specifies the frequency range of the PHY's input reference clock. **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_ref_rpt_clk_sel_o[1:0]	O	Selects the input reference clock for the CML output clocks xpcs_ref_repeat_clk_(p,m). **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_refa_clk_sel_o[1:0]	O	xpcs_refa_clk_sel - Select reference clock source from ref_pad_clk_p/m or ref_alt0/1_clk_p/m **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_refb_clk_sel_o[1:0]	O	xpcs_refb_clk_sel - Select reference clock source from ref_pad_clk_p/m or ref_alt0/1_clk_p/m **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_refa_clkdet_en_o	O	Enable reference clock detection for refa **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_refb_clkdet_en_o	O	Enable reference clock detection for refb **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_lane_ref_sel_o[(NUM_OF_LANES-1):0]	O	Selects the refa or refb reference clock - 0: Select refa clock - 1: Select refb clock **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low
xpcs_refa_clkdet_result_i	I	Reference clock detection result for refa **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_refb_clkdet_result_i	I	Reference clock detection result for refb **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_refa_raw_clk_div2_en_o	O	Reference clock divider control for refa clock in RAW PCS. Asserted to divide refa clock by 2 in RAW PCS. phy_reset should be asserted only after this signal is changed **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_refb_raw_clk_div2_en_o	O	Reference clock divider control for refb clock in RAW PCS. Asserted to divide refb clock by 2 in RAW PCS. phy_reset should be asserted only after this signal is changed **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_mplla_wrd_clk_en_o	O	MPLLA word clock output enable Synopsys UPCS uses xpcs_txX_ropll_word clock instead of mpll(a,b)_word_clk. So, this is set to 0 in functional mode. In scan mode, it is set to 1. **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_mpllb_wrd_clk_en_o	O	MPLLB word clock output enable Synopsys UPCS uses xpcs_txX_ropll_word clock instead of mpll(a,b)_word_clk. So, this is set to 0 in functional mode. In scan mode, it is set to 1. **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_lane_cntx_en_o[(NUM_OF_LANES-1):0]	O	Per-lane context enable. This signal is changed only during phy_reset. Here, tied off to low using programmable register. **Exists:** E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.39 Synopsys Multi-protocol 16G PHY Receive Control and Status Interface Signals

- xpcs_vco_temp_comp_en_o
- xpcs_vco_step_ctrl_o
- xpcs_vco_fqband0_o
- xpcs_vco_fqband1_o
- xpcs_vco_fqband2_o
- xpcs_vco_fqband3_o

Table 6-39 Synopsys Multi-protocol 16G PHY Receive Control and Status Interface Signals

Port Name	I/O	Description
xpcs_vco_temp_comp_en_o[(NUM_OF_LANES-1):0]	O	Value for RX CDR Temperature Compensation Enable. Control of CDR VCO temperature compensation circuit based on data rate. This signal is driven from CDR_VCO_TEMP_CTRL_EN_0/CDR_VCO_TEMP_CTRL_EN_3_1 fields of VR_XS_PMA_MP_16G_RX_CDR_CTRL1/VR_MII_MP_16G_RX_CDR_CTRL1 Register. This signal should be connected to rxX_cdr_vco_temp_comp_en input ports of the PHY. **Exists:** E16 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_vco_step_ctrl_o[(NUM_OF_LANES-1):0]	O	RX CDR VCO Step Control . Controls CDR VCO biasing based on data rate. This signal is driven from CDR_VCO_STEP_CTRL0/CDR_VCO_STEP_CTRL_3_1 fields of VR_XS_PMA_MP_16G_RX_CDR_CTRL1/VR_MII_MP_16G_RX_CDR_CTRL1 Register. This signal should be connected to rxX_cdr_vco_step_ctrl input ports of the PHY. **Exists:** E16 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_vco_fqband0_o[1:0]	O	RX CDR VCO Frequency Band (Along Lane0 Rx Path). Controls Lane0 coarse-DAC step-size based on data rate and VPH level. This signal is driven from CDR_VCO_FREQBAND_0 fields of VR_XS_PMA_MP_16G_RX_CDR_CTRL1/VR_MII_MP_16G_RX_CDR_CTRL1 Register. This signal should be connected to rx0_cdr_vco_freqband input ports of the PHY. **Exists:** E16||SNPS_RSVDPARAM_39 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_vco_fqband1_o[1:0]	O	RX CDR VCO Frequency Band (Along Lane1 Rx Path). Controls Lane1 coarse-DAC step-size based on data rate and VPH level. This signal is driven from CDR_VCO_FREQBAND_1 fields of VR_XS_PMA_MP_16G_RX_CDR_CTRL1 Register. This signal should be connected to rx1_cdr_vco_freqband input ports of the PHY. **Exists:** (E16||SNPS_RSVDPARAM_39)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_vco_fqband2_o[1:0]	O	RX CDR VCO Frequency Band (Along Lane2 Rx Path). Controls Lane2 coarse-DAC step-size based on data rate and VPH level. This signal is driven from CDR_VCO_FREQBAND_2 fields of VR_XS_PMA_MP_16G_RX_CDR_CTRL1 Register. This signal should be connected to rx2_cdr_vco_freqband input ports of the PHY. **Exists:** (E16||SNPS_RSVDPARAM_39)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_vco_fqband3_o[1:0]	O	RX CDR VCO Frequency Band (Along Lane3 Rx Path). Controls Lane3 coarse-DAC step-size based on data rate and VPH level. This signal is driven from CDR_VCO_FREQBAND_3 fields of VR_XS_PMA_MP_16G_RX_CDR_CTRL1 Register. This signal should be connected to rx3_cdr_vco_freqband input ports of the PHY. **Exists:** (E16||SNPS_RSVDPARAM_39)&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.40 Synopsys Multi-protocol 16G/25G PHY Recieve Control and Status Interface Signals

- xpcs_rx_term_offset_o

Table 6-40 Synopsys Multi-protocol 16G/25G PHY Recieve Control and Status Interface Signals

Port Name	I/O	Description
xpcs_rx_term_offset_o[4:0]	O	Offset for RX Termination. Specifies an additional fixed offset to calibrated RX termination value. This signal is driven from RX_TERM_OFFSET field of VR_XS_PMA_MP_16G_25G_RX_GENCTRL4/VR_MII_MP_16G_25G_RX_GENCTRL4 Register. This signal should be connected to rx_term_offset input port of the PHY. **Exists:** (E16||E25||SNPS_RSVDPARAM_39)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.41 Synopsys Multi-protocol 25G PHY Transmit Status and Control Interface Signals

Table 6-41 Synopsys Multi-protocol 25G PHY Transmit Status and Control Interface Signals

Port Name	I/O	Description
xpcs_tx0_term_ctrl_o[2:0]	O	Transmit Termination Control -lane 0 **Exists:** (SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx1_term_ctrl_o[2:0]	O	Transmit Termination Control -lane 1 **Exists:** (SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||E25)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx2_term_ctrl_o[2:0]	O	Transmit Termination Control -lane 2 **Exists:** (SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||E25)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_tx3_term_ctrl_o[2:0]	O	Transmit Termination Control -lane 3 **Exists:** (SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||E25)&&!SINGLE_LANE_CFG&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx0_dcc_ctrl_range_o[3:0]	O	Transmit Internal Clock Duty Cycle Control Range -lane 0 **Exists:** E25&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx1_dcc_ctrl_range_o[3:0]	O	Transmit Internal Clock Duty Cycle Control Range -lane 1 **Exists:** E25&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx2_dcc_ctrl_range_o[3:0]	O	Transmit Internal Clock Duty Cycle Control Range -lane 2 **Exists:** E25&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx3_dcc_ctrl_range_o[3:0]	O	Transmit Internal Clock Duty Cycle Control Range -lane 3 **Exists:** E25&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx0_dcc_ctrl_range_o[3:0]	O	Receive Internal Clock Duty Cycle Control Range -lane 0 **Exists:** E25&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_rx1_dcc_ctrl_range_o[3:0]	O	Receive Internal Clock Duty Cycle Control Range -lane 1 **Exists:** E25&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx2_dcc_ctrl_range_o[3:0]	O	Receive Internal Clock Duty Cycle Control Range -lane 2 **Exists:** E25&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_rx3_dcc_ctrl_range_o[3:0]	O	Receive Internal Clock Duty Cycle Control Range -lane 3 **Exists:** E25&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_tx_align_wide_xfer_en_o[(NUM_OF_LANES-1):0]	O	Transmitter Alignment Wide Transfer Enable **Exists:** E25&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.42 GPIO Interface Signals



Table 6-42 GPIO Interface Signals

Port Name	I/O	Description
xpcs_gpo_o[7:0]	O	General Purpose Output. This output indicates the content of Bits[15:8] of VR XS or PCS MMD GPIO Register. **Exists:** GPIO_EN **Synchronous To:** {(!SNPS_XAUI_PHY&&MDIO_INTF) ? "clk_mdc_i" : "clk_csr_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_gpo_i[7:0]	I	General Purpose Input. The contents of this input can be read by reading Bits[7:0] of VR XS or PCS MMD GPIO Register. **Exists:** GPIO_EN **Synchronous To:** {(!SNPS_XAUI_PHY&&MDIO_INTF) ? "clk_mdc_i" : "clk_csr_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.43 SGMII/QSGMII Control and Clock Interface Signals

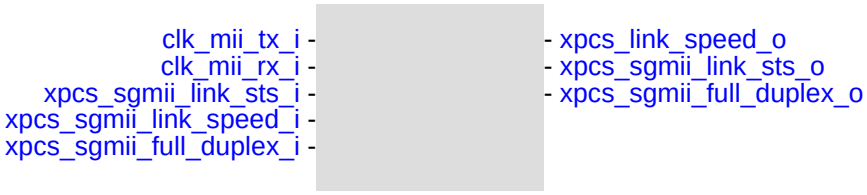


Table 6-43 SGMII/QSGMII Control and Clock Interface Signals

Port Name	I/O	Description
clk_mii_tx_i	I	SGMII Transmit MAC Interface Clock. This clock is used to sample the following signals when SGMII_EN = Enabled or when QSGMII_EN = Enabled and the current speed mode is 1G or less: ■ gmii_txd_i[7:0] ■ gmii_tx_en_i ■ gmii_tx_er_i ***Note:*** This clock must be synchronous to the clk_xgxs_tx_i clock. The frequency of this clock depends the SGMII speed mode which is determined based the xpcs_link_speed_o and xpcs_mii_ctrl_o signals as described in the following list: ■ 2.5 MHz: If xpcs_link_speed_o is 00 and xpcs_mii_ctrl_o is 0 ■ 25 MHz: If xpcs_link_speed_o is 01 and xpcs_mii_ctrl_o is 0 ■ 1.25 MHz: If xpcs_link_speed_o is 00 and xpcs_mii_ctrl_o is 1 ■ 12.5 MHz: If xpcs_link_speed_o is 01 and xpcs_mii_ctrl_o is 1 ■ 125 MHz: If xpcs_link_speed_o is 10 and xpcs_mii_ctrl_o is X (any value) **Exists:** SGMII_EN||QSGMII_EN **Synchronous To:** clk_xgxs_tx_i **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
clk_mii_rx_i	I	SGMII Receive MAC Interface Clock. This clock is used to drive the following signals if SGMII_EN = Enabled or when QSGMII_EN = Enabled and the current speed mode is 1G or less. ■ gmii_rxd_o[7:0] ■ gmii_rx_dv_o ■ gmii_rx_er_o ***Note:*** If CLKCOMP = Enabled, this clock input must be externally connected to the clk_mii_tx_i clock. If CLKCOMP = Disabled, this clock must be synchronous to the clk_xgxs_rx0_i clock. The frequency of this clock depends the SGMII speed mode which is determined based the xpcs_link_speed_o and xpcs_mii_ctrl_o signals as described in the following list: ■ 2.5 MHz: If xpcs_link_speed_o is 00 and xpcs_mii_ctrl_o is 0 ■ 25 MHz: If xpcs_link_speed_o is 01 and xpcs_mii_ctrl_o is 0 ■ 1.25 MHz: If xpcs_link_speed_o is 00 and xpcs_mii_ctrl_o is 1 ■ 12.5 MHz: If xpcs_link_speed_o is 01 and xpcs_mii_ctrl_o is 1 ■ 125 MHz: If xpcs_link_speed_o is 10 and xpcs_mii_ctrl_o can be any value **Exists:** SGMII_EN||QSGMII_EN **Synchronous To:** (CLKCOMP) ? "clk_xgxs_tx_i" : "clk_xgxs_rx0_i" **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_link_speed_o[1:0]	O	SGMII MAC Speed Control. This signal indicates the current operating speed of the DWC_xpcs in the SGMII mode. This signal is used for selecting the frequency of the clk_mii_tx_i and clk_mii_rx_i signals. This signal is derived from the Speed Select bits (Bit 13 and Bit 6) of the SR MII MMD Control Register. The following is the encoded speed: ■ 2'b00: 10 Mbps ■ 2'b01: 100 Mbps ■ 2'b10: 1000 Mbps **Exists:** SGMII_EN||QSGMII_EN **Synchronous To:** None **Registered:** (!SGMII_PHY_AN_AUTO_RESTART) ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A **Synchronous to:** Configuration dependent: ■ Asynchronous: For configurations with SGMII_PHY_AN_AUTO_RESTART=Enabled ■ clk_mdc_i: For configurations with SNPS_XAUI_PHY = Disabled and CSR_INTERFACE=MDIO and SGMII_PHY_AN_AUTO_RESTART=Disabled ■ clk_csr_i: For all other configurations
xpcs_sgmii_link_sts_o	O	SGMII Link Status. This signal indicates ■ Link status as programmed in the register bit[4] of VR MII MMD AN Control register ■ Link status of the auto-negotiation output bit[4] of VR MII MMD AN Interrupt and Status Register ■ Link status as indicated in the input port xpcs_sgmii_link_sts_i For more details, see bit[0] and bit[9] of VR MII MMD Digital Control 1 Register which controls the behavior of this output. **Exists:** SGMII_EN||QSGMII_EN **Synchronous To:** (!SNPS_XAUI_PHY&&MDIO_INTF)? "Asynchronous" : "clk_csr_i" **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_sgmii_full_duplex_o	O	SGMII full duplex. This signal indicates the whether SGMII supports full duplex or half duplex. This signal is used during CL37_AN when operating in SGMII PHY mode. ■ Duplex mode as programmed in the register bit[8] of SR MII MMD Control Register ■ Duplex mode of the auto-negotiation output come bit[1] of VR MII MMD AN Interrupt and Status Register ■ Duplex mode as indicated in the input port xpcs_sgmii_full_duplex_i For more details, see bit[0] and bit[9] of VR MII MMD Digital Control 1 Register which controls the behavior of this output. **Exists:** SGMII_EN||QSGMII_EN **Synchronous To:** (!SNPS_XAUI_PHY&&MDIO_INTF)? "Asynchronous" : "clk_csr_i" **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_sgmii_link_sts_i	I	SGMII Link Status. This signal indicates the SGMII Link status. This signal is used during CL37_AN when operating in SGMII PHY mode. In the PHY mode this signal indicates that the underlying physical PHY's Link Status. When this bit is set then the underlying physical PHY Link is Up. When this bit is reset then the underlying physical PHY Link is Down. Also, whenever there is any change in this input, restarts the SGMII auto-negotiation when the PCS is configured in SGMII PHY mode. **Exists:** SGMII_PHY_AN_AUTO_RESTART||QSGMII_PHY_AN_AUTO_RESTART **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_sgmii_link_speed_i[1:0]	I	SGMII Link Speed. This signal indicates the SGMII Link Speed determined by SGMII PHY. This signal is used during CL37_AN when operating in SGMII PHY mode. The encoding of the speed is as follows: ■ 2'b00: 10 Mbps ■ 2'b01: 100 Mbps ■ 2'b10: 1000 Mbps Any change to this input restarts SGMII auto-negotiation when DWC_xpcs is configured in SGMII PHY mode. **Exists:** SGMII_PHY_AN_AUTO_RESTART||QSGMII_PHY_AN_AUTO_RESTART **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_sgmii_full_duplex_i	I	SGMII full duplex. This signal indicates the whether SGMII supports full duplex or half duplex. This signal is used during CL37_AN when operating in SGMII PHY mode. **Exists:** SGMII_PHY_AN_AUTO_RESTART||QSGMII_PHY_AN_AUTO_RESTART **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.44 SGMII/QSGMII/USXGMII Control Interface Signals

Table 6-44 SGMII/QSGMII/USXGMII Control Interface Signals

Port Name	I/O	Description
xpcs_mii_ctrl_o	O	MII Data Width Control. This signal controls the data width of the MII interface. This signal is applicable only if the MAC is working at 10 Mbps or 100 Mbps speed. The following is the encoded data width: ■ 0: 4-bit MII ■ 1: 8-bit MII **Exists:** SGMII_EN||QSGMII_EN||USXGMII_EN **Synchronous To:** (!SNPS_XAUI_PHY&&MDIO_INTF)? "clk_mdc_i" : "clk_csr_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.45 QSGMII Control and Clock Interface Signals

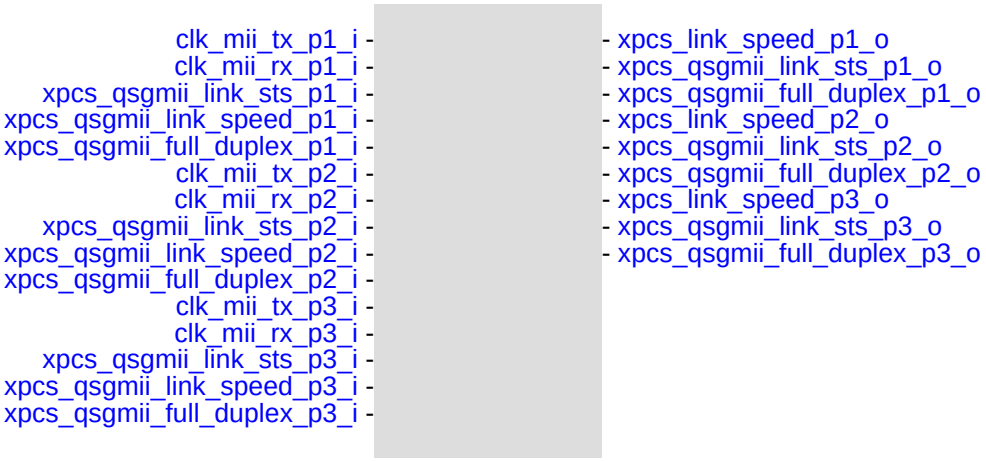


Table 6-45 QSGMII Control and Clock Interface Signals

Port Name	I/O	Description
clk_mii_tx_p1_i	I	QSGMII Transmit MAC Interface Clock along Port1. This clock is used to sample the following signals in QSGMII mode and the current speed mode is 1G or less: ■ gmii_txd_p1_i[7:0] ■ gmii_tx_en_p1_i ■ gmii_tx_er_p1_i ***Note:*** This clock must be synchronous to the clk_xgxs_tx_i clock. The frequency of this clock depends the QSGMII speed mode which is determined based the xpcs_link_speed_p1_o and xpcs_mii_ctrl_p1_o signals as described in the following list: ■ 2.5 MHz: If xpcs_link_speed_p1_o is 00 and xpcs_mii_ctrl_p1_o is 0 ■ 25 MHz: If xpcs_link_speed_p1_o is 01 and xpcs_mii_ctrl_p1_o is 0 ■ 1.25 MHz: If xpcs_link_speed_p1_o is 00 and xpcs_mii_ctrl_p1_o is 1 ■ 12.5 MHz: If xpcs_link_speed_p1_o is 01 and xpcs_mii_ctrl_p1_o is 1 ■ 125 MHz: If xpcs_link_speed_p1_o is 10 and xpcs_mii_ctrl_p1_o is X (any value) **Exists:** QSGMII_EN **Synchronous To:** clk_xgxs_tx_i **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
clk_mii_rx_p1_i	I	QSGMII Receive MAC Interface Clock along Port1. This clock is used to drive the following signals in QSGMII mode and the current speed mode is 1G or less. ■ gmii_rxd_p1_o[7:0] ■ gmii_rx_dv_p1_o ■ gmii_rx_er_p1_o ***Note:*** If CLKCOMP = Enabled, this clock input must be externally connected to the clk_mii_tx_p1_i clock. If CLKCOMP = Disabled, this clock must be synchronous to the clk_xgxs_rx0_i clock. The frequency of this clock depends the QSGMII speed mode which is determined based the xpcs_link_speed_p1_o and xpcs_mii_ctrl_p1_o signals as described in the following list: ■ 2.5 MHz: If xpcs_link_speed_p1_o is 00 and xpcs_mii_ctrl_p1_o is 0 ■ 25 MHz: If xpcs_link_speed_p1_o is 01 and xpcs_mii_ctrl_p1_o is 0 ■ 1.25 MHz: If xpcs_link_speed_p1_o is 00 and xpcs_mii_ctrl_p1_o is 1 ■ 12.5 MHz: If xpcs_link_speed_p1_o is 01 and xpcs_mii_ctrl_p1_o is 1 ■ 125 MHz: If xpcs_link_speed_p1_o is 10 and xpcs_mii_ctrl_p1_o can be any value **Exists:** QSGMII_EN **Synchronous To:** {CLKCOMP ? "clk_xgxs_tx_i" : "clk_xgxs_rx0_i"} **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_link_speed_p1_o[1:0]	O	QSGMII MAC Speed Control along Port1. This signal indicates the current operating speed of the DWC_xpcs in the QSGMII mode. This signal is used for selecting the frequency of the clk_mii_tx_p1_i and clk_mii_rx_p1_i signals. This signal is derived from the Speed Select bits (Bit 13 and Bit 6) of the SR MII MMD P1 Control Register. The following is the encoded speed: ■ 2'b00: 10 Mbps ■ 2'b01: 100 Mbps ■ 2'b10: 1000 Mbps **Exists:** QSGMII_EN **Synchronous To:** {(!SNPS_XAUI_PHY&&MDIO_INTF) ? "None" : "clk_csr_i"} **Registered:** (!QSGMII_PHY_AN_AUTO_RESTART) ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_qsgmii_link_sts_p1_o	O	QSGMII Link Status along Port1. This signal indicates ■ Link status as programmed in the register bit[4] of VR MII MMD AN P1 Control register ■ Link status of the auto-negotiation output bit[4] of VR MII MMD AN P1 Interrupt and Status Register ■ Link status as indicated in the input port xpcs_qsgmii_link_sts_p1_i For more details, see bit[0] and bit[9] of VR MII MMD P1 Digital Control 1 Register which controls the behavior of this output. **Exists:** QSGMII_EN **Synchronous To:** {(!SNPS_XAUI_PHY&&MDIO_INTF) ? "None" : "clk_csr_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_qsgmii_full_duplex_p1_o	O	QSGMII full duplex along Port1. This signal indicates the whether QSGMII supports full duplex or half duplex. This signal is used during CL37_AN when operating in QSGMII PHY mode. ■ Duplex mode as programmed in the register bit[8] of SR MII MMD P1 Control Register ■ Duplex mode of the auto-negotiation output come bit[1] of VR MII MMD AN P1 Interrupt and Status Register ■ Duplex mode as indicated in the input port xpcs_qsgmii_full_duplex-_p1_i For more details, see bit[0] and bit[9] of VR MII MMD P1 Digital Control 1 Register which controls the behavior of this output. **Exists:** QSGMII_EN **Synchronous To:** {(!SNPS_XAUI_PHY&&MDIO_INTF) ? "None" : "clk_csr_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_qsgmii_link_sts_p1_i	I	QSGMII Link Status along Port1. This signal indicates the QSGMII link status. This signal is used during CL37_AN when operating in QSGMII PHY mode. In the PHY mode this signal indicates that the underlying physical PHY's Link Status. When this bit is set then the underlying physical PHY Link is Up. When this bit is reset then the underlying physical PHY Link is Down. Also, whenever there is any change in this input restarts the QSGMII auto-negotiation when the PCS is configured in QSGMII PHY mode. **Exists:** QSGMII_PHY_AN_AUTO_RESTART **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_qsgmii_link_speed_p1_i[1:0]	I	QSGMII Link Speed along Port1. This signal indicates the QSGMII Link Speed along Port1. This signal is used during CL37_AN when operating in QSGMII PHY mode. The encoding of the speed is as follows: ■ 2'b00: 10 Mbps ■ 2'b01: 100 Mbps ■ 2'b10: 1000 Mbps **Exists:** QSGMII_PHY_AN_AUTO_RESTART **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_qsgmii_full_duplex_p1_i	I	QSGMII full duplex along Port1. This signal indicates the whether QSGMII supports full duplex or half duplex. This signal is used during CL37_AN when operating in QSGMII PHY mode. **Exists:** QSGMII_PHY_AN_AUTO_RESTART **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
clk_mii_tx_p2_i	I	QSGMII Transmit MAC Interface Clock along Port2. This clock is used to sample the following signals in QSGMII mode and the current speed mode is 1G or less: ■ gmii_txd_p2_i[7:0] ■ gmii_tx_en_p2_i ■ gmii_tx_er_p2_i ***Note:*** This clock must be synchronous to the clk_xgxs_tx_i clock. The frequency of this clock depends the QSGMII speed mode which is determined based the xpcs_link_speed_p2_o and xpcs_mii_ctrl_p2_o signals as described in the following list: ■ 2.5 MHz: If xpcs_link_speed_p2_o is 00 and xpcs_mii_ctrl_p2_o is 0 ■ 25 MHz: If xpcs_link_speed_p2_o is 01 and xpcs_mii_ctrl_p2_o is 0 ■ 1.25 MHz: If xpcs_link_speed_p2_o is 00 and xpcs_mii_ctrl_p2_o is 1 ■ 12.5 MHz: If xpcs_link_speed_p2_o is 01 and xpcs_mii_ctrl_p2_o is 1 ■ 125 MHz: If xpcs_link_speed_p2_o is 10 and xpcs_mii_ctrl_p2_o is X (any value) **Exists:** QSGMII_EN **Synchronous To:** clk_xgxs_tx_i **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
clk_mii_rx_p2_i	I	QSGMII Receive MAC Interface Clock along Port2. This clock is used to drive the following signals in QSGMII mode and the current speed mode is 1G or less. ■ gmii_rxd_p2_o[7:0] ■ gmii_rx_dv_p2_o ■ gmii_rx_er_p2_o ***Note:*** If CLKCOMP = Enabled, this clock input must be externally connected to the clk_mii_tx_p2_i clock. If CLKCOMP = Disabled, this clock must be synchronous to the clk_xgxs_rx0_i clock. The frequency of this clock depends the QSGMII speed mode which is determined based the xpcs_link_speed_p2_o and xpcs_mii_ctrl_p2_o signals as described in the following list: ■ 2.5 MHz: If xpcs_link_speed_p2_o is 00 and xpcs_mii_ctrl_p2_o is 0 ■ 25 MHz: If xpcs_link_speed_p2_o is 01 and xpcs_mii_ctrl_p2_o is 0 ■ 1.25 MHz: If xpcs_link_speed_p2_o is 00 and xpcs_mii_ctrl_p2_o is 1 ■ 12.5 MHz: If xpcs_link_speed_p2_o is 01 and xpcs_mii_ctrl_p2_o is 1 ■ 125 MHz: If xpcs_link_speed_p2_o is 10 and xpcs_mii_ctrl_p2_o can be any value **Exists:** QSGMII_EN **Synchronous To:** {CLKCOMP ? "clk_xgxs_tx_i" : "clk_xgxs_rx0_i"} **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_link_speed_p2_o[1:0]	O	QSGMII MAC Speed Control along Port1. This signal indicates the current operating speed of the DWC_xpcs in the QSGMII mode. This signal is used for selecting the frequency of the clk_mii_tx_p2_i and clk_mii_rx_p2_i signals. This signal is derived from the Speed Select bits (Bit 13 and Bit 6) of the SR MII MMD P2 Control Register. The following is the encoded speed: ■ 2'b00: 10 Mbps ■ 2'b01: 100 Mbps ■ 2'b10: 1000 Mbps **Exists:** QSGMII_EN **Synchronous To:** {(!SNPS_XAUI_PHY&&MDIO_INTF) ? "None" : "clk_csr_i"} **Registered:** (!QSGMII_PHY_AN_AUTO_RESTART) ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_qsgmii_link_sts_p2_o	O	QSGMII Link Status along Port2. This signal indicates ■ Link status as programmed in the register bit[4] of VR MII MMD AN P2 Control register ■ Link status of the auto-negotiation output bit[4] of VR MII MMD AN P2 Interrupt and Status Register ■ Link status as indicated in the input port xpcs_qsgmii_link_sts_p2_i For more details, see bit[0] and bit[9] of VR MII MMD P2 Digital Control 1 Register which controls the behavior of this output. **Exists:** QSGMII_EN **Synchronous To:** {(!SNPS_XAUI_PHY&&MDIO_INTF) ? "None" : "clk_csr_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_qsgmii_full_duplex_p2_o	O	QSGMII full duplex along Port2. This signal indicates the whether QSGMII supports full duplex or half duplex. This signal is used during CL37_AN when operating in QSGMII PHY mode. ■ Duplex mode as programmed in the register bit[8] of SR MII MMD P2 Control Register ■ Duplex mode of the auto-negotiation output come bit[1] of VR MII MMD AN P2 Interrupt and Status Register ■ Duplex mode as indicated in the input port xpcs_qsgmii_full_duplex-_p2_i For more details, see bit[0] and bit[9] of VR MII MMD P2 Digital Control 1 Register which controls the behavior of this output. **Exists:** QSGMII_EN **Synchronous To:** {(!SNPS_XAUI_PHY&&MDIO_INTF) ? "None" : "clk_csr_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_qsgmii_link_sts_p2_i	I	QSGMII Link Status along Port2. This signal indicates the QSGMII Link status. This signal is used during CL37_AN when operating in QSGMII PHY mode. In the PHY mode this signal indicates that the underlying physical PHY's Link Status. When this bit is set then the underlying physical PHY Link is Up. When this bit is reset then the underlying physical PHY Link is Down. Also, whenever there is any change in this input restarts the QSGMII auto-negotiation when the PCS is configured in QSGMII PHY mode. **Exists:** QSGMII_PHY_AN_AUTO_RESTART **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_qsgmii_link_speed_p2_i[1:0]	I	QSGMII Link Speed along Port2. This signal indicates the QSGMII Link Speed along Port2. This signal is used during CL37_AN when operating in QSGMII PHY mode. The encoding of the speed is as follows: ■ 2'b00: 10 Mbps ■ 2'b01: 100 Mbps ■ 2'b10: 1000 Mbps **Exists:** QSGMII_PHY_AN_AUTO_RESTART **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_qsgmii_full_duplex_p2_i	I	QSGMII full duplex along Port2. This signal indicates the whether QSGMII supports full duplex or half duplex. This signal is used during CL37_AN when operating in QSGMII PHY mode. **Exists:** QSGMII_PHY_AN_AUTO_RESTART **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
clk_mii_tx_p3_i	I	QSGMII Transmit MAC Interface Clock along Port3. This clock is used to sample the following signals in QSGMII mode and the current speed mode is 1G or less: ■ gmii_txd_p3_i[7:0] ■ gmii_tx_en_p3_i ■ gmii_tx_er_p3_i ***Note:*** This clock must be synchronous to the clk_xgxs_tx_i clock. The frequency of this clock depends the QSGMII speed mode which is determined based the xpcs_link_speed_p3_o and xpcs_mii_ctrl_p3_o signals as described in the following list: ■ 2.5 MHz: If xpcs_link_speed_p3_o is 00 and xpcs_mii_ctrl_p3_o is 0 ■ 25 MHz: If xpcs_link_speed_p3_o is 01 and xpcs_mii_ctrl_p3_o is 0 ■ 1.25 MHz: If xpcs_link_speed_p3_o is 00 and xpcs_mii_ctrl_p3_o is 1 ■ 12.5 MHz: If xpcs_link_speed_p3_o is 01 and xpcs_mii_ctrl_p3_o is 1 ■ 125 MHz: If xpcs_link_speed_p3_o is 10 and xpcs_mii_ctrl_p3_o is X (any value) **Exists:** QSGMII_EN **Synchronous To:** clk_xgxs_tx_i **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
clk_mii_rx_p3_i	I	QSGMII Receive MAC Interface Clock along Port3. This clock is used to drive the following signals in QSGMII mode and the current speed mode is 1G or less. ■ gmii_rxd_p3_o[7:0] ■ gmii_rx_dv_p3_o ■ gmii_rx_er_p3_o ***Note:*** If CLKCOMP = Enabled, this clock input must be externally connected to the clk_mii_tx_p3_i clock. If CLKCOMP = Disabled, this clock must be synchronous to the clk_xgxs_rx0_i clock. The frequency of this clock depends the QSGMII speed mode which is determined based the xpcs_link_speed_p3_o and xpcs_mii_ctrl_p3_o signals as described in the following list: ■ 2.5 MHz: If xpcs_link_speed_p3_o is 00 and xpcs_mii_ctrl_p3_o is 0 ■ 25 MHz: If xpcs_link_speed_p3_o is 01 and xpcs_mii_ctrl_p3_o is 0 ■ 1.25 MHz: If xpcs_link_speed_p3_o is 00 and xpcs_mii_ctrl_p3_o is 1 ■ 12.5 MHz: If xpcs_link_speed_p3_o is 01 and xpcs_mii_ctrl_p3_o is 1 ■ 125 MHz: If xpcs_link_speed_p3_o is 10 and xpcs_mii_ctrl_p3_o can be any value **Exists:** QSGMII_EN **Synchronous To:** {CLKCOMP ? "clk_xgxs_tx_i" : "clk_xgxs_rx0_i"} **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_link_speed_p3_o[1:0]	O	QSGMII MAC Speed Control along Port3. This signal indicates the current operating speed of the DWC_xpcs in the QSGMII mode. This signal is used for selecting the frequency of the clk_mii_tx_p3_i and clk_mii_rx_p3_i signals. This signal is derived from the Speed Select bits (Bit 13 and Bit 6) of the SR MII MMD P3 Control Register. The following is the encoded speed: ■ 2'b00: 10 Mbps ■ 2'b01: 100 Mbps ■ 2'b10: 1000 Mbps **Exists:** QSGMII_EN **Synchronous To:** {(!SNPS_XAUI_PHY&&MDIO_INTF) ? "None" : "clk_csr_i"} **Registered:** (!QSGMII_PHY_AN_AUTO_RESTART) ? "Yes" : "No" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_qsgmii_link_sts_p3_o	O	QSGMII Link Status along Port3. This signal indicates ■ Link status as programmed in the register bit[4] of VR MII MMD AN P3 Control register ■ Link status of the auto-negotiation output bit[4] of VR MII MMD AN P3 Interrupt and Status Register ■ Link status as indicated in the input port xpcs_qsgmii_link_sts_p3_i For more details, see bit[0] and bit[9] of VR MII MMD P3 Digital Control 1 Register which controls the behavior of this output. **Exists:** QSGMII_EN **Synchronous To:** {(!SNPS_XAUI_PHY&&MDIO_INTF) ? "None" : "clk_csr_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_qsgmii_full_duplex_p3_o	O	QSGMII full duplex along Port3. This signal indicates the whether QSGMII supports full duplex or half duplex. This signal is used during CL37_AN when operating in QSGMII PHY mode. ■ Duplex mode as programmed in the register bit[8] of SR MII MMD P3 Control Register ■ Duplex mode of the auto-negotiation output come bit[1] of VR MII MMD AN P2 Interrupt and Status Register ■ Duplex mode as indicated in the input port xpcs_qsgmii_full_duplex-_p3_i For more details, see bit[0] and bit[9] of VR MII MMD P3 Digital Control 1 Register which controls the behavior of this output. **Exists:** QSGMII_EN **Synchronous To:** {(!SNPS_XAUI_PHY&&MDIO_INTF) ? "None" : "clk_csr_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_qsgmii_link_sts_p3_i	I	QSGMII Link Status along Port3. This signal indicates the QSGMII Link status. This signal is used during CL37_AN when operating in QSGMII PHY mode. In the PHY mode this signal indicates that the underlying physical PHY's Link Status. When this bit is set then the underlying physical PHY Link is Up. When this bit is reset then the underlying physical PHY Link is Down. Also, whenever there is any change in this input restarts the QSGMII auto-negotiation when the PCS is configured in QSGMII PHY mode. **Exists:** QSGMII_PHY_AN_AUTO_RESTART **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_qsgmii_link_speed_p3_i[1:0]	I	QSGMII Link Speed along Port3. This signal indicates the QSGMII Link Speed along Port3. This signal is used during CL37_AN when operating in QSGMII PHY mode. The encoding of the speed is as follows: ■ 2'b00: 10 Mbps ■ 2'b01: 100 Mbps ■ 2'b10: 1000 Mbps **Exists:** QSGMII_PHY_AN_AUTO_RESTART **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_qsgmii_full_duplex_p3_i	I	QSGMII full duplex along Port3. This signal indicates the whether QSGMII supports full duplex or half duplex. This signal is used during CL37_AN when operating in QSGMII PHY mode. **Exists:** QSGMII_PHY_AN_AUTO_RESTART **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.46 QSGMII/USXGMII Control Interface Signals

Table 6-46 QSGMII/USXGMII Control Interface Signals

Port Name	I/O	Description
xpcs_mii_ctrl_p1_o	O	MII Data Width Control along Port1. This signal controls the data width of the MII interface when operating in 10Mbps/100Mbps. The following is the encoded data width: ■ 0: 4-bit MII ■ 1: 8-bit MII **Exists:** QSGMII_EN||MP_USXGMII **Synchronous To:** {(!SNPS_XAUI_PHY&&MDIO_INTF) ? "clk_mdc_i" : "clk_csr_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_mii_ctrl_p2_o	O	MII Data Width Control along Port2. This signal controls the data width of the MII interface when operating in 10Mbps/100Mbps. The following is the encoded data width: ■ 0: 4-bit MII ■ 1: 8-bit MII **Exists:** QSGMII_EN||USXG_QPORT **Synchronous To:** {(!SNPS_XAUI_PHY&&MDIO_INTF) ? "None" : "clk_csr_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_mii_ctrl_p3_o	O	MII Data Width Control along Port3. This signal controls the data width of the MII interface when operating in 10Mbps/100Mbps. The following is the encoded data width: ■ 0: 4-bit MII ■ 1: 8-bit MII **Exists:** QSGMII_EN||USXG_QPORT **Synchronous To:** {(!SNPS_XAUI_PHY&&MDIO_INTF) ? "clk_mdc_i" : "clk_csr_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.47 USXGMII Control and Status Interface Signals

- xpcs_usxg_speed_o
- xpcs_usxg_speed_p1_o
- xpcs_usxg_speed_p2_o
- xpcs_usxg_speed_p3_o
- xpcs_usxg_lp_full_duplex_o
- xpcs_usxg_lp_full_duplex_p1_o
- xpcs_usxg_lp_full_duplex_p2_o
- xpcs_usxg_lp_full_duplex_p3_o
- xpcs_usxg_lp_link_sts_o
- xpcs_usxg_lp_link_sts_p1_o
- xpcs_usxg_lp_link_sts_p2_o
- xpcs_usxg_lp_link_sts_p3_o
- xpcs_usxg_lp_eee_abl_o
- xpcs_usxg_lp_eee_abl_p1_o
- xpcs_usxg_lp_eee_abl_p2_o
- xpcs_usxg_lp_eee_abl_p3_o
- xpcs_usxg_lp_ck_stp_abl_o
- xpcs_usxg_lp_ck_stp_abl_p1_o
- xpcs_usxg_lp_ck_stp_abl_p2_o
- xpcs_usxg_lp_ck_stp_abl_p3_o
- xpcs_usxg_mode_o

Table 6-47 USXGMII Control and Status Interface Signals

Port Name	I/O	Description
xpcs_usxg_speed_o[2:0]	O	USXGMII Speed Mode -Port 0. This signal indicates the current operating speed of Port 0 when operating in USXGMII mode. This signal can be used as mux-select to appropriately select the clocks to be connected to clk_xgmii_tx_i and clk_xgmii_rx_i ports. The speed encoding is as follows: ■ 3'b000: 10Mbps ■ 3'b001: 100Mbps ■ 3'b010: 1Gbps ■ 3'b011: 10Gbps (default) ■ 3'b100: 2.5Gbps ■ 3'b101: 5Gbps ■ 3'b110/3'b111: Reserved This field is driven from {SS5, SS6, SS13} bits of SR_MII_CTRL Register if auto-negotiation is turned-off or if bit [9] (MAC_AUTO_SW) of VR_MII_DIG_CTRL1 Register is 0. If auto-negotiation is enabled and if bit [1] (MAC_AUTO_SW) of VR_MII_DIG_CTRL1 Register is set to 1, after the completion of auto-negotiation, DWC_xpcs automatically switches to the speed mode indicated (in VR MII MMD AN Interrupt and Status Register) by the link partner. Hence, xpcs_usxg_speed_o signal indicates the value of bit [12:10] of this register. If DWC_xpcs is configured in PHY mode and PHY_MODE_CTRL bit of VR_MII_DIG_CTRL1 Register is set, this port reflects the value of xpcs_usxg_link_speed_i input port. **Exists:** USXGMII_EN **Synchronous To:** {(RPCS_ONLY) ? "clk_rpcs_rx_i" : "clk_xpcs_rx_i"} **Registered:** (!SNPS_XAUI_PHY&&MDIO_INTF) ? "No" : "Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_usxg_speed_p1_o[2:0]	O	USXGMII Speed Mode -Port 1. This signal indicates the current operating speed of Port 1 when operating in Dual-port/Quad-port USXGMII mode. This signal can be used as mux-select to appropriately select the clocks to be connected to clk_xgmii_tx_p1_i and clk_xgmii_rx_p1_i ports. The speed encoding is same as that of xpcs_usxg_speed_o port. This field is driven from {SS5, SS6, SS13} bits of SR_MII_1_CTRL Register if auto-negotiation is turned-off or if bit [9] (MAC_AUTO_SW) of VR_MII_1_DIG_CTRL1 Register is 0. If auto-negotiation is enabled and if bit [1] (MAC_AUTO_SW) of VR_MII_1_DIG_CTRL1 Register is set to 1, after the completion of auto-negotiation, DWC_xpcs automatically switches to the speed mode indicated (in VR MII MMD AN Interrupt and Status Register) by the link partner. Hence, xpcs_usxg_speed_o signal indicates the value of bit [12:10] of this register. If DWC_xpcs is configured in PHY mode and PHY_MODE_CTRL bit of VR_MII_1_DIG_CTRL1 Register is set, this port reflects the value of xpcs_usxg_link_speed_p1_i input port. **Exists:** MP_USXGMII **Synchronous To:** {(RPCS_ONLY) ? "clk_rpcs_rx_i" : "clk_xpcs_rx_i"} **Registered:** (!SNPS_XAUI_PHY&&MDIO_INTF) ? "No" : "Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_usxg_speed_p2_o[2:0]	O	USXGMII Speed Mode -Port 2. This signal indicates the current operating speed of Port 2 when operating in Quad-port USXGMII mode. This signal can be used as mux-select to appropriately select the clocks to be connected to clk_xgmii_tx_p2_i and clk_xgmii_rx_p2_i ports. The speed encoding is same as that of xpcs_usxg_speed_o port. This field is driven from {SS5, SS6, SS13} bits of SR_MII_2_CTRL Register if auto-negotiation is turned-off or if bit [9] (MAC_AUTO_SW) of VR_MII_2_DIG_CTRL1 Register is 0. If auto-negotiation is enabled and if bit [1] (MAC_AUTO_SW) of VR_MII_2_DIG_CTRL1 Register is set to 1, after the completion of auto-negotiation, DWC_xpcs automatically switches to the speed mode indicated (in VR MII MMD AN Interrupt and Status Register) by the link partner. Hence, xpcs_usxg_speed_o signal indicates the value of bit [12:10] of this register. If DWC_xpcs is configured in PHY mode and PHY_MODE_CTRL bit of VR_MII_2_DIG_CTRL1 Register is set, this port reflects the value of xpcs_usxg_link_speed_p2_i input port. **Exists:** USXG_QPORT **Synchronous To:** {(RPCS_ONLY) ? "clk_rpcs_rx_i" : "clk_xpcs_rx_i"} **Registered:** (!SNPS_XAUI_PHY&&MDIO_INTF) ? "No" : "Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_usxg_speed_p3_o[2:0]	O	USXGMII Speed Mode -Port 3. This signal indicates the current operating speed of Port 3 when operating in Quad-port USXGMII mode. This signal can be used as mux-select to appropriately select the clocks to be connected to clk_xgmii_tx_p3_i and clk_xgmii_rx_p3_i ports. The speed encoding is same as that of xpcs_usxg_speed_o port. This field is driven from {SS5, SS6, SS13} bits of SR_MII_3_CTRL Register if auto-negotiation is turned-off or if bit [9] (MAC_AUTO_SW) of VR_MII_3_DIG_CTRL1 Register is 0. If auto-negotiation is enabled and if bit [1] (MAC_AUTO_SW) of VR MII_3 MMD Digital Control 1 Register is set to 1, after the completion of auto-negotiation, DWC_xpcs automatically switches to the speed mode indicated (in VR MII MMD AN Interrupt and Status Register) by the link partner. Hence, xpcs_usxg_speed_o signal indicates the value of bit [12:10] of this register. If DWC_xpcs is configured in PHY mode and PHY_MODE_CTRL bit of VR_MII_3_DIG_CTRL1 Register is set, this port reflects the value of xpcs_usxg_link_speed_p3_i input port. **Exists:** USXG_QPORT **Synchronous To:** {(RPCS_ONLY) ? "clk_rpcs_rx_i" : "clk_xpcs_rx_i"} **Registered:** (!SNPS_XAUI_PHY&&MDIO_INTF) ? "No" : "Yes" **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_usxg_lp_full_duplex_o	O	USXGMII Auto-negotiation Duplex Mode (Port 0). This signal indicates the duplex mode indicated by the (Port 0) link partner during auto-negotiation. ■ 0: Half Duplex ■ 1: Full Duplex If clause 37 auto-negotiation is enabled, this signal indicates the duplex mode information received from link-partner through Clause 37 auto-negotiation. Therefore, this signal reflects the value of bit[13] of VR MII MMD AN Interrupt and Status Register. If clause 37 auto-negotiation is turned-off, this signal is driven 1. DWC_xpcs always operates in full-duplex mode during USXGMII operation, irrespective of the duplex-mode indicated by the link partner. **Exists:** USXGMII_EN **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_usxg_lp_full_duplex_p1_o	O	USXGMII Auto-negotiation Duplex Mode (Port 1). This signal indicates the duplex mode indicated by the (Port 1) link partner during auto-negotiation. If clause 37 auto-negotiation is enabled, this signal indicates the duplex mode information received from link-partner through Clause 37 auto-negotiation. Hence, this signal reflects the value of bit[13] of VR MII_1 MMD AN Interrupt and Status Register. If clause 37 auto-negotiation is turned-off, this signal is driven 1. DWC_xpcs always operates in full-duplex mode during USXGMII operation, irrespective of the duplex-mode indicated by the link partner. **Exists:** MP_USXGMII **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_usxg_lp_full_duplex_p2_o	O	USXGMII Auto-negotiation Duplex Mode (Port 2). This signal indicates the duplex mode indicated by the (Port 2) link partner during auto-negotiation. If clause 37 auto-negotiation is enabled, this signal indicates the duplex mode information received from link-partner through Clause 37 auto-negotiation. Hence, this signal reflects the value of bit[13] of VR MII_2 MMD AN Interrupt and Status Register. If clause 37 auto-negotiation is turned-off, this signal is driven 1. DWC_xpcs always operates in full-duplex mode during USXGMII operation, irrespective of the duplex-mode indicated by the link partner. **Exists:** USXG_QPORT **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_usxg_lp_full_duplex_p3_o	O	USXGMII Auto-negotiation Duplex Mode (Port 3). This signal indicates the duplex mode indicated by the (Port 3) link partner during auto-negotiation. If clause 37 auto-negotiation is enabled, this signal indicates the duplex mode information received from link-partner through Clause 37 auto-negotiation. Hence, this signal reflects the value of bit[13] of VR MII_3 MMD AN Interrupt and Status Register. If clause 37 auto-negotiation is turned-off, this signal is driven 1. DWC_xpcs always operates in full-duplex mode during USXGMII operation, irrespective of the duplex-mode indicated by the link partner. **Exists:** USXG_QPORT **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_usxg_lp_link_sts_o	O	USXGMII Auto-Negotiation Link Status (Port 0). This signal indicates the link status (status of network link) information sent from the link-partner (PHY) to MAC, for Port 0, through USXGMII Clause 37 auto-negotiation. This signal is relevant when DWC_xpcs is configured as MAC-side USXGMII. This signal reflects the value of bit [14] of VR MII MMD AN Interrupt and Status Register **Exists:** USXGMII_EN **Synchronous To:** {(RPCS_ONLY) ? "clk_rpcs_rx_i" : "clk_xpcs_rx_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_usxg_lp_link_sts_p1_o	O	USXGMII Auto-Negotiation Link Status (Port 1). This signal indicates the link status (status of network link) information sent from the link-partner (PHY) to MAC, for Port 1, through USXGMII Clause 37 auto-negotiation. This signal is relevant when DWC_xpcs is configured as MAC-side USXGMII. This signal reflects the value of bit [14] of VR MII_1 MMD AN Interrupt and Status Register **Exists:** MP_USXGMII **Synchronous To:** {(RPCS_ONLY) ? "clk_rpcs_rx_i" : "clk_xpcs_rx_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_usxg_lp_link_sts_p2_o	O	USXGMII Auto-Negotiation Link Status (Port 2). This signal indicates the link status (status of network link) information sent from the link-partner (PHY) to MAC, for Port 2, through USXGMII Clause 37 auto-negotiation. This signal is relevant when DWC_xpcs is configured as MAC-side USXGMII. This signal reflects the value of bit [14] of VR MII_2 MMD AN Interrupt and Status Register **Exists:** USXG_QPORT **Synchronous To:** {(RPCS_ONLY) ? "clk_rpcs_rx_i" : "clk_xpcs_rx_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_usxg_lp_link_sts_p3_o	O	USXGMII Auto-Negotiation Link Status (Port 3). This signal indicates the link status (status of network link) information sent from the link-partner (PHY) to MAC, for Port 3, through USXGMII Clause 37 auto-negotiation. This signal is relevant when DWC_xpcs is configured as MAC-side USXGMII. This signal reflects the value of bit [14] of VR MII_3 MMD AN Interrupt and Status Register **Exists:** USXG_QPORT **Synchronous To:** {(RPCS_ONLY) ? "clk_rpcs_rx_i" : "clk_xpcs_rx_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_usxg_lp_eee_abl_o	O	USXGMII EEE Capability of the Link Partner (Port 0). This signal indicates the EEE capability status information received from the link partner (port 0) through USXGMII Clause 37 auto-negotiation. This signal reflects the value of bit [9] of VR MII MMD AN Interrupt and Status Register. **Exists:** USXGMII_EN **Synchronous To:** {(RPCS_ONLY) ? "clk_rpcs_rx_i" : "clk_xpcs_rx_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_usxg_lp_eee_abl_p1_o	O	USXGMII EEE Capability of the Link Partner (Port 1). This signal indicates the EEE capability status information received from the link partner (port 1) through USXGMII Clause 37 auto-negotiation. This signal reflects the value of bit [9] of VR MII_1 MMD AN Interrupt and Status Register. **Exists:** MP_USXGMII **Synchronous To:** {(RPCS_ONLY) ? "clk_rpcs_rx_i" : "clk_xpcs_rx_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_usxg_lp_eee_abl_p2_o	O	USXGMII EEE Capability of the Link Partner (Port 2). This signal indicates the EEE capability status information received from the link partner (port 2) through USXGMII Clause 37 auto-negotiation. This signal reflects the value of bit [9] of VR MII_2 MMD AN Interrupt and Status Register. **Exists:** USXG_QPORT **Synchronous To:** {(RPCS_ONLY) ? "clk_rpcs_rx_i" : "clk_xpcs_rx_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_usxg_lp_eee_abl_p3_o	O	USXGMII EEE Capability of the Link Partner (Port 3). This signal indicates the EEE capability status information received from the link partner (port 3) through USXGMII Clause 37 auto-negotiation. This signal reflects the value of bit [9] of VR MII_3 MMD AN Interrupt and Status Register. **Exists:** USXG_QPORT **Synchronous To:** {(RPCS_ONLY) ? "clk_rpcs_rx_i" : "clk_xpcs_rx_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_usxg_lp_ck_stp_abl_o	O	USXGMII EEE Clock Stop Capability of the Link Partner (Port 0). This signal indicates the EEE Clock-Stop capability status information received from the link partner (port 0) through USXGMII Clause 37 auto-negotiation. This signal reflects the value of bit [8] of VR MII MMD AN Interrupt and Status Register. **Exists:** USXGMII_EN **Synchronous To:** {(RPCS_ONLY) ? "clk_rpcs_rx_i" : "clk_xpcs_rx_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_usxg_lp_ck_stp_abl_p1_o	O	USXGMII EEE Clock Stop Capability of the Link Partner (Port 1). This signal indicates the EEE Clock-Stop capability status information received from the link partner (port 1) through USXGMII Clause 37 auto-negotiation. This signal reflects the value of bit [8] of VR MII_1 MMD AN Interrupt and Status Register. **Exists:** MP_USXGMII **Synchronous To:** {(RPCS_ONLY) ? "clk_rpcs_rx_i" : "clk_xpcs_rx_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_usxg_lp_ck_stp_abl_p2_o	O	USXGMII EEE Clock Stop Capability of the Link Partner (Port 2). This signal indicates the EEE Clock-Stop capability status information received from the link partner (port 2) through USXGMII Clause 37 auto-negotiation. This signal reflects the value of bit [8] of VR MII_2 MMD AN Interrupt and Status Register. **Exists:** USXG_QPORT **Synchronous To:** {(RPCS_ONLY) ? "clk_rpcs_rx_i" : "clk_xpcs_rx_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_usxg_lp_ck_stp_abl_p3_o	O	USXGMII EEE Clock Stop Capability of the Link Partner (Port 3). This signal indicates the EEE Clock-Stop capability status information received from the link partner (port 3) through USXGMII Clause 37 auto-negotiation. This signal reflects the value of bit [8] of VR MII_3 MMD AN Interrupt and Status Register. **Exists:** USXG_QPORT **Synchronous To:** {(RPCS_ONLY) ? "clk_rpcs_rx_i" : "clk_xpcs_rx_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_usxg_mode_o[2:0]	O	USXGMII Mode Indicator. This signal indicates the current USXGMII mode in which DWC_xpcs operates. The mode encoding is as follows: ■ 3'b000: USXGMII is disabled ■ 3'b001: 10G-SXGMII ■ 3'b010: 5G-SXGMII ■ 3'b011: 2.5G-SXGMII ■ 3'b100: 10G-DXGMII ■ 3'b101: 5G-DXGMII ■ 3'b110: 10G-QXGMII ■ 3'b111: Reserved **Exists:** USXGMII_EN **Synchronous To:** {(!SNPS_XAUI_PHY&&MDIO_INTF) ? "clk_mdc_i" : "clk_csr_i"} **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.48 SGMII/QSGMII/USXGMII Control and Clock Interface Signals

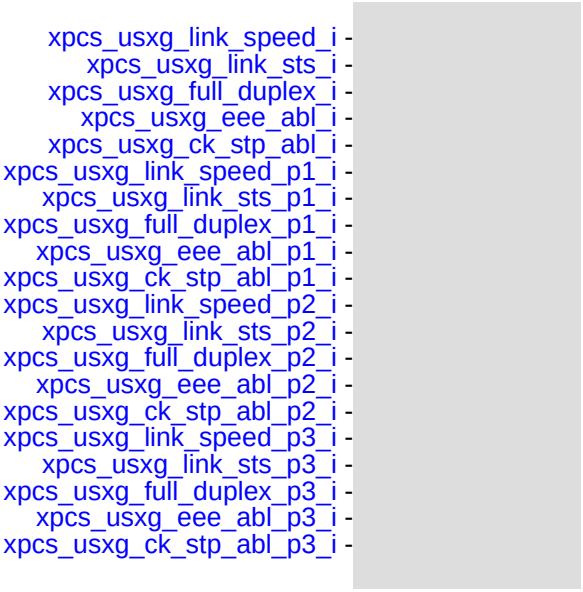


Table 6-48 SGMII/QSGMII/USXGMII Control and Clock Interface Signals

Port Name	I/O	Description
xpcs_usxg_link_speed_i[2:0]	I	USXGMII PHY Mode Link Speed Input - Port 0. This signal can be used to pass-on the USXGMII Link Speed (determined by USXGMII PHY) information to DWC_xpcs.This input is valid only when DWC_xpcs is configured in PHY mode and PHY_MODE_CTRL bit of VR_MII_DIG_CTRL1 register is set. The value of this signal is advertised to the Link Partner during Clause 37 Auto-negotiation and any change to this input restarts USXGMII auto-negotiation (along Port 0) The encoding of the speed is as follows: ■ 3'b000: 10Mbps ■ 3'b001: 100Mbps ■ 3'b010: 1Gbps ■ 3'b011: 10Gbps ■ 3'b100: 2.5Gbps ■ 3'b101: 5Gbps ■ 3'b110/3'b111: Reserved **Exists:** USXG_PHY_AN_AUTO_RESTART **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_usxg_link_sts_i	I	USXGMII PHY Mode Link Status Input - Port 0. This signal can be used to pass-on the USXGMII Link Status (determined by USXGMII PHY) information to DWC_xpcs.This input is valid only when DWC_xpcs is configured in PHY mode and PHY_MODE_CTRL bit of VR_MII_DIG_CTRL1 register is set. The value of this signal is advertised to the Link Partner during Clause 37 Auto-negotiation and any change to this input results in restart of USXGMII auto-negotiation (along Port 0) **Exists:** USXG_PHY_AN_AUTO_RESTART **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_usxg_full_duplex_i	I	USXGMII PHY Duplex Mode Input - Port 0. This signal can be used to pass-on the USXGMII Duplex Mode (determined by USXGMII PHY) information to DWC_xpcs.This input is valid only when DWC_xpcs is configured in PHY mode and PHY_MODE_CTRL bit of VR_MII_DIG_CTRL1 register is set. The value of this signal is advertised to the Link Partner during Clause 37 Auto-negotiation and any change to this input results in restart of USXGMII auto-negotiation (along Port 0) **Note:** Though USXGMII defines the duplex-mode field in auto-negotiation page, USXGMII always supports full-duplex only. **Exists:** USXG_PHY_AN_AUTO_RESTART **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_usxg_eee_abl_i	I	USXGMII PHY Mode EEE Capability Input - Port 0. This signal can be used to pass-on the USXGMII EEE capability (determined by USXGMII PHY) information to DWC_xpcs.This input is valid only when DWC_xpcs is configured in PHY mode and PHY_MODE_CTRL bit of VR_MII_DIG_CTRL1 register is set. The value of this signal is advertised to the Link Partner during Clause 37 Auto-negotiation and any change to this input results in restart of USXGMII auto-negotiation (along Port 0) **Exists:** USXG_PHY_AN_AUTO_RESTART **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_usxg_ck_stp_abl_i	I	USXGMII PHY Mode EEE Clock Stop Capability Input - Port 0. This signal can be used to pass-on the USXGMII EEE Clock Stop capability (determined by USXGMII PHY) information to DWC_xpcs.This input is valid only when DWC_xpcs is configured in PHY mode and PHY_MODE_CTRL bit of VR_MII_DIG_CTRL1 register is set. The value of this signal is advertised to the Link Partner during Clause 37 Auto-negotiation and any change to this input results in restart of USXGMII auto-negotiation (along Port 0) **Exists:** USXG_PHY_AN_AUTO_RESTART **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_usxg_link_speed_p1_i[2:0]	I	USXGMII PHY Mode Link Speed Input - Port 1. This signal can be used to pass-on the USXGMII Link Speed (determined by USXGMII PHY) information to DWC_xpcs.This input is valid only when DWC_xpcs is configured in PHY mode and PHY_MODE_CTRL bit of VR_MII_1_DIG_CTRL1 register is set. The value of this signal is advertised to the Link Partner during Clause 37 Auto-negotiation and any change to this input results in restart of USXGMII auto-negotiation (along Port 1) **Exists:** USXG_PHY_AN_AUTO_RESTART&&MP_USXGMII **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_usxg_link_sts_p1_i	I	USXGMII PHY Mode Link Status Input - Port 1. This signal can be used to pass-on the USXGMII Link Status (determined by USXGMII PHY) information to DWC_xpcs.This input is valid only when DWC_xpcs is configured in PHY mode and PHY_MODE_CTRL bit of VR_MII_1_DIG_CTRL1 register is set. The value of this signal is advertised to the Link Partner during Clause 37 Auto-negotiation and any change to this input results in restart of USXGMII auto-negotiation (along Port 1) **Exists:** USXG_PHY_AN_AUTO_RESTART&&MP_USXGMII **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_usxg_full_duplex_p1_i	I	USXGMII PHY Duplex Mode Input - Port 1. This signal can be used to pass-on the USXGMII Duplex Mode (determined by USXGMII PHY) information to DWC_xpcs.This input is valid only when DWC_xpcs is configured in PHY mode and PHY_MODE_CTRL bit of VR_MII_1_DIG_CTRL1 register is set. The value of this signal is advertised to the Link Partner during Clause 37 Auto-negotiation and any change to this input results in restart of USXGMII auto-negotiation (along Port 1) **Note:** Though USXGMII defines the duplex-mode field in auto-negotiation page, USXGMII always supports full-duplex only. **Exists:** USXG_PHY_AN_AUTO_RESTART&&MP_USXGMII **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_usxg_eee_abl_p1_i	I	USXGMII PHY Mode EEE Capability Input - Port 1. This signal can be used to pass-on the USXGMII EEE capability (determined by USXGMII PHY) information to DWC_xpcs.This input is valid only when DWC_xpcs is configured in PHY mode and PHY_MODE_CTRL bit of VR_MII_1_DIG_CTRL1 register is set. The value of this signal is advertised to the Link Partner during Clause 37 Auto-negotiation and any change to this input results in restart of USXGMII auto-negotiation (along Port 1) **Exists:** USXG_PHY_AN_AUTO_RESTART&&MP_USXGMII **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_usxg_ck_stp_abl_p1_i	I	USXGMII PHY Mode EEE Clock-stop Capability Input - Port 1. This signal can be used to pass-on the USXGMII EEE Clock-stop capability (determined by USXGMII PHY) information to DWC_xpcs.This input is valid only when DWC_xpcs is configured in PHY mode and PHY_MODE_CTRL bit of VR_MII_1_DIG_CTRL1 register is set. The value of this signal is advertised to the Link Partner during Clause 37 Auto-negotiation and any change to this input results in restart of USXGMII auto-negotiation (along Port 1) **Exists:** USXG_PHY_AN_AUTO_RESTART&&MP_USXGMII **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_usxg_link_speed_p2_i[2:0]	I	USXGMII PHY Mode Link Speed Input - Port 2. This signal can be used to pass-on the USXGMII Link Speed (determined by USXGMII PHY) information to DWC_xpcs.This input is valid only when DWC_xpcs is configured in PHY mode and PHY_MODE_CTRL bit of VR_MII_2_DIG_CTRL1 register is set. The value of this signal is advertised to the Link Partner during Clause 37 Auto-negotiation and any change to this input results in restart of USXGMII auto-negotiation (along Port 2) **Exists:** USXG_PHY_AN_AUTO_RESTART&&USXG_QPORT **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xpcs_usxg_link_sts_p2_i	I	USXGMII PHY Mode Link Status Input - Port 2. This signal can be used to pass-on the USXGMII Link Status (determined by USXGMII PHY) information to DWC_xpcs.This input is valid only when DWC_xpcs is configured in PHY mode and PHY_MODE_CTRL bit of VR_MII_2_DIG_CTRL1 register is set. The value of this signal is advertised to the Link Partner during Clause 37 Auto-negotiation and any change to this input results in restart of USXGMII auto-negotiation (along Port 2). **Exists:** USXG_PHY_AN_AUTO_RESTART&&USXG_QPORT **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_usxg_full_duplex_p2_i	I	USXGMII PHY Duplex Mode Input - Port 2. This signal can be used to pass-on the USXGMII Duplex Mode (determined by USXGMII PHY) information to DWC_xpcs.This input is valid only when DWC_xpcs is configured in PHY mode and PHY_MODE_CTRL bit of VR_MII_2_DIG_CTRL1 register is set. The value of this signal is advertised to the Link Partner during Clause 37 Auto-negotiation and any change to this input results in restart of USXGMII auto-negotiation (along Port 2) **Note:** Though USXGMII defines the duplex-mode field in auto-negotiation page, USXGMII always supports full-duplex only. **Exists:** USXG_PHY_AN_AUTO_RESTART&&USXG_QPORT **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_usxg_eee_abl_p2_i	I	USXGMII PHY Mode EEE Capability Input - Port 2. This signal can be used to pass-on the USXGMII EEE capability (determined by USXGMII PHY) information to DWC_xpcs.This input is valid only when DWC_xpcs is configured in PHY mode and PHY_MODE_CTRL bit of VR_MII_2_DIG_CTRL1 register is set. The value of this signal is advertised to the Link Partner during Clause 37 Auto-negotiation and any change to this input results in restart of USXGMII auto-negotiation (along Port 2) **Exists:** USXG_PHY_AN_AUTO_RESTART&&USXG_QPORT **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_usxg_ck_stp_abl_p2_i	I	USXGMII PHY Mode EEE Clock-stop Capability Input - Port 2. This signal can be used to pass-on the USXGMII EEE Clock-stop capability (determined by USXGMII PHY) information to DWC_xpcs.This input is valid only when DWC_xpcs is configured in PHY mode and PHY_MODE_CTRL bit of VR_MII_2_DIG_CTRL1 register is set. The value of this signal is advertised to the Link Partner during Clause 37 Auto-negotiation and any change to this input results in restart of USXGMII auto-negotiation (along Port 2) **Exists:** USXG_PHY_AN_AUTO_RESTART&&USXG_QPORT **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_usxg_link_speed_p3_i[2:0]	I	USXGMII PHY Mode Link Speed Input - Port 3. This signal can be used to pass-on the USXGMII Link Speed (determined by USXGMII PHY) information to DWC_xpcs.This input is valid only when DWC_xpcs is configured in PHY mode and PHY_MODE_CTRL bit of VR_MII_3_DIG_CTRL1 register is set. The value of this signal is advertised to the Link Partner during Clause 37 Auto-negotiation and any change to this input results in restart of USXGMII auto-negotiation (along Port 3). **Exists:** USXG_PHY_AN_AUTO_RESTART&&USXG_QPORT **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xpcs_usxg_link_sts_p3_i	I	USXGMII PHY Mode Link Status Input - Port 3. This signal can be used to pass-on the USXGMII Link Status (determined by USXGMII PHY) information to DWC_xpcs.This input is valid only when DWC_xpcs is configured in PHY mode and PHY_MODE_CTRL bit of VR_MII_3_DIG_CTRL1 register is set. The value of this signal is advertised to the Link Partner during Clause 37 Auto-negotiation and any change to this input results in restart of USXGMII auto-negotiation (along Port 3) **Exists:** USXG_PHY_AN_AUTO_RESTART&&USXG_QPORT **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_usxg_full_duplex_p3_i	I	USXGMII PHY Duplex Mode Input - Port 3. This signal can be used to pass-on the USXGMII Duplex Mode (determined by USXGMII PHY) information to DWC_xpcs.This input is valid only when DWC_xpcs is configured in PHY mode and PHY_MODE_CTRL bit of VR_MII_3_DIG_CTRL1 register is set. The value of this signal is advertised to the Link Partner during Clause 37 Auto-negotiation and any change to this input results in restart of USXGMII auto-negotiation (along Port 3) **Note:** Though USXGMII defines the duplex-mode field in auto-negotiation page, USXGMII always supports full-duplex only. **Exists:** USXG_PHY_AN_AUTO_RESTART&&USXG_QPORT **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_usxg_eee_abl_p3_i	I	USXGMII PHY Mode EEE Capability Input - Port 3. This signal can be used to pass-on the USXGMII EEE capability (determined by USXGMII PHY) information to DWC_xpcs.This input is valid only when DWC_xpcs is configured in PHY mode and PHY_MODE_CTRL bit of VR_MII_3_DIG_CTRL1 register is set. The value of this signal is advertised to the Link Partner during Clause 37 Auto-negotiation and any change to this input results in restart of USXGMII auto-negotiation (along Port 3) **Exists:** USXG_PHY_AN_AUTO_RESTART&&USXG_QPORT **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_usxg_ck_stp_abl_p3_i	I	USXGMII PHY Mode EEE Clock-stop Capability Input - Port 3. This signal can be used to pass-on the USXGMII EEE Clock-stop capability (determined by USXGMII PHY) information to DWC_xpcs.This input is valid only when DWC_xpcs is configured in PHY mode and PHY_MODE_CTRL bit of VR_MII_3_DIG_CTRL1 register is set. The value of this signal is advertised to the Link Partner during Clause 37 Auto-negotiation and any change to this input results in restart of USXGMII auto-negotiation (along Port 3) **Exists:** USXG_PHY_AN_AUTO_RESTART&&USXG_QPORT **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.49 Synopsys PHY Transmit Status and Control Interface Signals

Table 6-49 Synopsys PHY Transmit Status and Control Interface Signals

Port Name	I/O	Description
xgxs_tx_invert_o[(NUM_OF_LANES-1):0]	O	Invert logical sense of Tx serial outputs of the PHY. This signal has the effect of inverting the serial output data stream of the transmitter **Exists:** UP5||E12||E16||E25||E32 **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** Configuration/Operating Mode dependent: ■ clk_xsbi_tx_i when operating in KR mode ■ clk_qsgmii_tx_i when operating in QSGMII Mode ■ clk_xgxs_tx_i when operating in other modes
xgxs_tx_reset_o[(NUM_OF_LANES-1):0]	O	PHY Transmitter reset. This signal internally resets the Tx data-path, all PHY Tx settings and state machines. This signal must be connected to the txX_reset signal of the PHY. **Exists:** UP5||E12||E16||E25||E32 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xgxs_tx_data_en_o[(NUM_OF_LANES-1):0]	O	Transmitter Data Enable. When this signal is asserted, the Tx output driver in the PHY gets enabled. **Exists:** UP5||E12||E16||E25||E32 **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High **Synchronous to:** Configuration/Operating Mode dependent: ■ clk_xsbi_tx_i when operating in KR mode ■ clk_qsgmii_tx_i when operating in QSGMII Mode ■ clk_xgxs_tx_i when operating in other modes
xgxs_tx0_rate_o[(TX_RT_W-1):0]	O	Transmit data rate - Lane 0. This signal controls the data rate PHY transmit serial lines. For configurations with Multi-protocol 12G/16G PHY, this signal is driven TX0_RATE field of VR_XS_PMA_MP_12G_16G_25G_TX_RATE_CTRL/VR_MII_MP_12G_16G_25G_TX_RATE_CTRL Register. For configurations with Multi-protocol 6G PHY, this signal is controlled by programming TX_RATE_0 field of VR_XS_PMA_MP_6G_MISC_CTRL1/VR_MII_MP_6G_MISC_CTRL1 Register **Exists:** ((UP5&&SSP)||E12||E16||E25)&&!SNPS_RSVDPARAM_37 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_tx1_rate_o[(TX_RT_W-1):0]	O	Transmit data rate - Lane 1. For configurations with Multi-protocol 12G/16G PHY, this signal is driven TX1_RATE field of VR_XS_PMA_MP_12G_16G_25G_TX_RATE_CTRL Register. For configurations with Multi-protocol 6G PHY, this signal is controlled by programming TX_RATE_1 field of VR_XS_PMA_MP_6G_MISC_CTRL1 Register. **Exists:** ((UP5&&SSP)||E12||E16||E25)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
xgxs_tx2_rate_o[(TX_RT_W-1):0]	O	Transmit data rate - Lane 2. For configurations with Multi-protocol 12G/16G PHY, this signal is driven TX2_RATE field of VR_XS_PMA_MP_12G_16G_25G_TX_RATE_CTRL Register. For configurations with Multi-protocol 6G PHY, this signal is controlled by programming TX_RATE_2 field of VR_XS_PMA_MP_6G_MISC_CTRL1 Register. **Exists:** ((UP5&&SSP)||E12||E16||E25)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_tx3_rate_o[(TX_RT_W-1):0]	O	Transmit data rate - Lane 3. For configurations with Multi-protocol 12G/16G PHY, this signal is driven TX3_RATE field of VR_XS_PMA_MP_12G_16G_25G_TX_RATE_CTRL Register. For configurations with Multi-protocol 6G PHY, this signal is controlled by programming TX_RATE_3 field of VR_XS_PMA_MP_6G_MISC_CTRL1 Register. **Exists:** ((UP5&&SSP)||E12||E16||E25)&&!SNPS_RSVDPARAM_37&&!SINGLE_LANE_CFG **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.50 Synopsys Multi-protocol 6G PHY Miscellaneous Interface Signals

- xgxs_ref_ssp_en_o
- xgxs_lane_10bit_sel_o
- xgxs_ssc_ref_clk_sel_o

Table 6-50 Synopsys Multi-protocol 6G PHY Miscellaneous Interface Signals

Port Name	I/O	Description
xgxs_ref_ssp_en_o	O	Enables reference clock for PHY. This is normally set to 1. When the controller is powered down, this output reflects the REF_SSP_EN bit of VR XS, PMA, or MII Synopsys Consumer Gen2 PHY or Synopsys Multi-protocol 6G PHY MPLL Control 0 Register. **Exists:** SNPS_XAUI_PHY&&UP5&&SSP **Synchronous To:** None **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
xgxs_lane_10bit_sel_o	O	Controls the width of the lanes of Synopsys Multi-protocol 6G PHY. This output is controlled by programming the LANE_10BIT_SEL bit of VR XS, PMA, or MII Synopsys Consumer Gen2 PHY or Synopsys Multi-protocol 6G PHY MPLL Control 0 Register. *Values:* ■ 0: 20 bit lane ■ 1: 10 bit lane **Exists:** UP5&&SSP **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
xgxs_ssc_ref_clk_sel_o[8:0]	O	Spread Spectrum reference clock control. This signal is driven by SSC_REF_CLK_SEL field of VR XS, PMA, or MII Synopsys Multi-protocol 6G PHY Miscellaneous Control 2 Register. This field can be programmed to control the MPLL frequency of the PHY. **Exists:** UP5&&SSP&&SSP_V42 **Synchronous To:** clk_csr_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.51 Synopsys Multi-protocol 6G PHY MPLL Control Interface Signals



Table 6-51 Synopsys Multi-protocol 6G PHY MPLL Control Interface Signals

Port Name	I/O	Description
xgxs_mpll_pwr_on_o	O	MPLL Clock ON. This signal causes the MPLL to power-on. *Values:* ■ 0: During the power-down sequence. ■ 1: During the power-up sequence. **Exists:** SNPS_XAUI_PHY&&(UP3||UP5) **Synchronous To:** clk_cko_alive_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xgxs_use_refclk_pad_o	O	Additional Reference Clock Control. This signal, when asserted, selects the external ref_pad_clk_{p,m} inputs of the Synopsys Multi-protocol 6G PHY as the reference clock. **Exists:** UP5 **Synchronous To:** clk_cko_alive_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xgxs_ref_clk_div2_o	O	Divides the reference clock by two. This output is controlled by programming the REF_CLK_DIV2 bit of the VR XS, PMA, or MII Synopsys Consumer Gen2 PHY or Synopsys Multi-protocol 6G PHY MPLL Control 0 Register. **Exists:** (UP5&&SSP) **Synchronous To:** clk_cko_alive_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xgxs_mpll_state_i	I	PHY MPLL State. This signal indicates the state of the MPLL. This signal is asserted when the MPLL is powered up and phase-locked. **Exists:** UP5 **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xgxs_mpll_multiplier_o[6:0]	O	MPLL Frequency Multiplier Control of the PHY. This signal multiplies the reference clock to a frequency suitable for required operating speed. **Exists:** UP5 **Synchronous To:** clk_cko_alive_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.52 Synopsys PHY Resistor Tune Interface Signals



Table 6-52 Synopsys PHY Resistor Tune Interface Signals

Port Name	I/O	Description
xgxs_rtune_req_o	O	Resistor Tune Request. This signal, when asserted, triggers a resistor tune request to the PHY (if one request is already not in progress). This signal is valid for Consumer Gen2 PHY and Multi-protocol 12G PHY configurations. **Exists:** (UP5&&!SSP)||E12||E16||E25||E32 **Synchronous To:** E12 ? "clk_cko_alive" : "clk_csr_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xgxs_rtune_ack_i	I	Resistor Tune acknowledge. This signal indicates that resistor tune has completed. This signal is valid for Consumer Gen2 PHY and Multi-protocol 12G PHY configurations. **Exists:** (UP5&&!SSP)||E12||E16||E25||E32 **Synchronous To:** Asynchronous **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High

6.53 Synopsys PHY Clock Control Interface Signals

clk_cko_alive_i -

Table 6-53 Synopsys PHY Clock Control Interface Signals

Port Name	I/O	Description
clk_cko_alive_i	I	Synopsys PHY Reference Output Clock. For Synopsys Multi-protocol 6G PHY, you must connect this input to the ref_output_clk clock of the PHY. For Synopsys Multi-protocol 12G/16G PHYs, you must connect this input to 'ref_dig_fr_clk' clock of the PHY (after suitable division by 2 or 4, if needed). **Exists:** SNPS_XAUI_PHY **Synchronous To:** None **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.54 PCS-R PHY Interface Signals

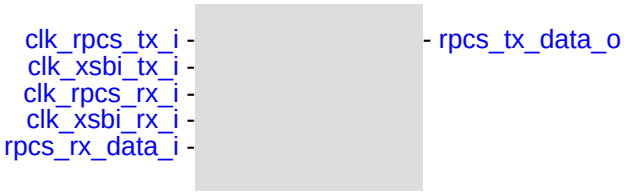


Table 6-54 PCS-R PHY Interface Signals

Port Name	I/O	Description
clk_rpcs_tx_i	I	PCS-R Transmit Data Path Clock. This clock is used in the Transmit data path of the PCS-R block. The clock frequency is 156.25 MHz. If R_SRDS_INTF = 66 x 156.25 MHz, this clock is also used to drive the rpcs_tx_data_o[PHY_DW-1:0] data. Note: In configurations with USXGMII support, the frequency of this clock may be scaled down by a factor of 2 or 4 depending upon the current USXGMII mode. **Exists:** RPCS_EN **Synchronous To:** None **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
clk_xsbi_tx_i	I	XSBI Transmit Clock. If R_SRDS_INTF != 66 x 156.25 MHz, this clock drives the rpcs_tx_data_o[PHY_DW-1:0] data. The frequency of this clock depends the PCS-R PHY Interface Type: ■ If R_SRDS_INTF = 16 x 644.53 MHz, Frequency = 644.53 MHz ■ If R_SRDS_INTF = 20 x 515.625 MHz, Frequency = 515.625 MHz ■ If R_SRDS_INTF = 32 x 322.265 MHz, Frequency = 322.265 MHz ■ If R_SRDS_INTF = 33 x 312.5 MHz, Frequency = 312.5 MHz ■ If R_SRDS_INTF = 40 x 257.8125 MHz, Frequency = 257.8125 MHz ■ If R_SRDS_INTF = 64 x 161.13 MHz, Frequency = 161.13 MHz Note: In configurations with USXGMII support, the frequency of this clock may be scaled down by a factor of 2 or 4 depending upon the current USXGMII mode. **Exists:** (R_SRDS_INTF!=3&&RPCS_EN) **Synchronous To:** None **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
rpcs_tx_data_o[(PHY_DW-1):0]	O	XSBI Transmit data. This is the Transmit data the PCS-R PHY interface. The width of this signal (PHY_DW) depends the PCS-R PHY Interface Type: ■ If R_SRDS_INTF = 16 x 644.53 MHz, Width = 16 ■ If R_SRDS_INTF = 20 x 515.625 MHz, Width = 20 ■ If R_SRDS_INTF = 32 x 322.265 MHz, Width = 32 ■ If R_SRDS_INTF = 33 x 312.5 MHz, Width = 33 ■ If R_SRDS_INTF = 40 x 257.8125 MHz, Width = 40 ■ If R_SRDS_INTF = 64 x 161.13 MHz, Width = 64 ■ If R_SRDS_INTF = 66 x 156.25 MHz, Width = 66 **Exists:** RPCS_EN **Synchronous To:** (R_SRDS_INTF==3) ? "clk_rpcs_rx_i" : "clk_xsbi_rx_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
clk_rpcs_rx_i	I	PCS-R Receive Data Path Clock. This clock is used in the Receive data path of the PCS-R block. The clock frequency is 156.25 MHz. The rpcs_rx_data_i[PHY_DW-1:0] signal is synchronous to this clock when R_SRDS_INTF = 66 x 156.25 MHz. **Note:** In configurations with USXGMII support, the frequency of this clock may be scaled down by a factor of 2 or 4 depending upon the current USXGMII mode. **Exists:** RPCS_EN **Synchronous To:** None **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
clk_xsbi_rx_i	I	XSBI Receive Clock. This clock samples the rpcs_rx_data_i[PHY_DW-1:0] data for all PCS-R PHY Interface types except 66 x 156.25 MHz. The frequency of this clock depends the PCS-R PHY Interface Type: ■ R_SRDS_INTF = 16 x 644.53 MHz, Frequency = 644.53 MHz ■ R_SRDS_INTF = 20 x 515.625 MHz, Frequency = 515.625 MHz ■ R_SRDS_INTF = 32 x 322.265 MHz, Frequency = 322.265 MHz ■ R_SRDS_INTF = 33 x 312.5 MHz, Frequency = 312.5 MHz ■ R_SRDS_INTF = 40 x 257.8125 MHz, Frequency = 257.8125 MHz ■ R_SRDS_INTF = 64 x 161.13 MHz, Frequency = 161.13 MHz **Note:** In configurations with USXGMII support, the frequency of this clock may be scaled down by a factor of 2 or 4 depending upon the current USXGMII mode. **Exists:** (R_SRDS_INTF!=3&&RPCS_EN) **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_rx_data_i[(PHY_DW-1):0]	I	PCS-R Receive Data. This is the Receive data the PCS-R PHY interface. The width of this signal (rpcs_phw) depends the PCS-R PHY Interface Type: ■ R_SRDS_INTF = 16 x 644.53 MHz, Width = 16 ■ R_SRDS_INTF = 20 x 515.625 MHz, Width = 20 ■ R_SRDS_INTF = 32 x 322.265 MHz, Width = 32 ■ R_SRDS_INTF = 33 x 312.5 MHz, Width = 33 ■ R_SRDS_INTF = 40 x 257.8125 MHz, Width = 40 ■ R_SRDS_INTF = 64 x 161.13 MHz, Width = 64 ■ R_SRDS_INTF = 66 x 156.25 MHz, Width = 66 **Exists:** RPCS_EN **Synchronous To:** (R_SRDS_INTF==3) ? "clk_rpcs_rx_i" : "clk_xsbi_rx_i" **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.55 PCS-X and PCS-R Common Interface Signals

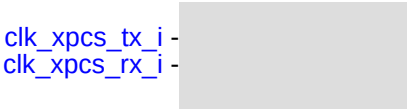


Table 6-55 PCS-X and PCS-R Common Interface Signals

Port Name	I/O	Description
clk_xpcs_tx_i	I	Common Transmit Data Path Clock for PCS-R and PCS-X. This clock is the multiplexed version of the clk_rpcs_tx_i or clk_xgxs_tx_i clock depending the current operating mode. In the PCS-R mode (xpcs_kr_o is asserted), this clock is synchronous to the clk_rpcs_tx_i clock. Similarly, in the PCS-X mode (xpcs_kr_o is de-asserted), this clock is synchronous to the clk_xgxs_tx_i clock. **Exists:** RPCS_EN&&!RPCS_ONLY **Synchronous To:** clk_xgxs_tx_i or clk_rpcs_tx_i **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
clk_xpcs_rx_i	I	Common Receive Data Path Clock for PCS-R and PCS-X. This clock is the multiplexed version of the clk_rpcs_rx_i and clk_xgxs_rx0_i clock depending the current operating mode. In the PCS-R mode (xpcs_kr_o is asserted), this clock is synchronous to the clk_rpcs_rx_i clock. Similarly, in the PCS-X mode (xpcs_kr_o is de-asserted), this clock is synchronous to the clk_xgxs_tx_i clock. **Exists:** RPCS_EN&&!RPCS_ONLY **Synchronous To:** clk_xgxs_rx0_i or clk_rpcs_rx_i **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.56 FEC Error Correction Memory Interface Signals

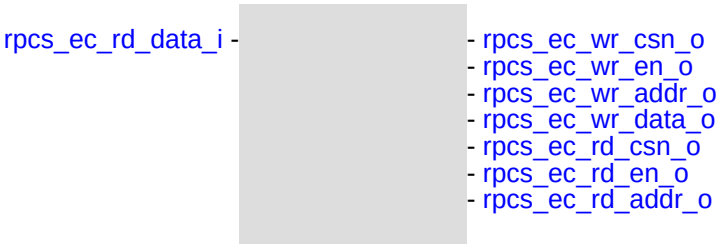


Table 6-56 FEC Error Correction Memory Interface Signals

Port Name	I/O	Description
rpcs_ec_wr_csn_o	O	FEC Error Correction Memory Chip Select Signal of the Write Port. This signal qualifies the rpcs_ec_wr_addr_o[4:0] signal. **Exists:** FEC_EN **Synchronous To:** clk_rpcs_rx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low
rpcs_ec_wr_en_o	O	FEC Error Correction Memory Enable Signal of the Write Port. When high, this signal indicates that the rpcs_ec_wr_data_o[65:0] data is valid and it is written to the memory. **Exists:** FEC_EN **Synchronous To:** clk_rpcs_rx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ec_wr_addr_o[4:0]	O	FEC Error Correction Memory Address Bus of the Write Port. This signal is the memory address bus of the write port. It is valid when the rpcs_ec_wr_csn_o signal is asserted. **Exists:** FEC_EN **Synchronous To:** clk_rpcs_rx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ec_wr_data_o[(FEC_DW-1):0]	O	FEC Error Correction Memory Data Bus of the Write Port. This signal carries the data to be written into the memory. It is valid only when the rpcs_ec_wr_en_o signal is asserted. **Exists:** FEC_EN **Synchronous To:** clk_rpcs_rx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
rpcs_ec_rd_csn_o	O	FEC Error Forward Memory Chip Select Signal of the Read Port. This signal qualifies the rpcs_ec_rd_addr_o[4:0] signal. **Exists:** FEC_EN **Synchronous To:** clk_rpcs_rx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low
rpcs_ec_rd_en_o	O	FEC Error Forward Memory Enable Signal of the Read Port. When high, this signal indicates that the rpcs_ec_rd_data_i[65:0] signal contains valid data, and it sampled by the DWC_xpcs. **Exists:** FEC_EN **Synchronous To:** clk_rpcs_rx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ec_rd_addr_o[4:0]	O	FEC Error Forward Memory Address Bus of the Read Port. This signal is the memory address bus of the read port. It is valid when the rpcs_ec_rd_csn_o signal is asserted. **Exists:** FEC_EN **Synchronous To:** clk_rpcs_rx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ec_rd_data_i[(FEC_DW-1):0]	I	FEC Error Forward Memory Data Bus of the Read Port. This signal carries the read data from the FEC error correction memory. It is valid when the rpcs_ec_rd_en_o signal is asserted. **Exists:** FEC_EN **Synchronous To:** clk_rpcs_rx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.57 FEC Error Forward Memory Interface Signals

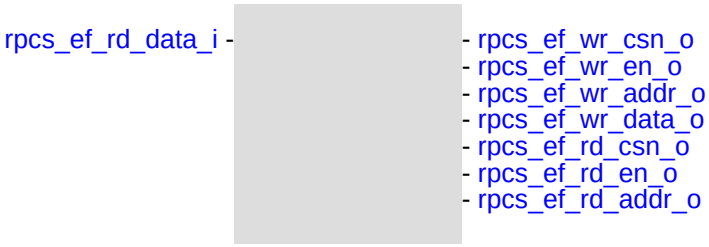


Table 6-57 FEC Error Forward Memory Interface Signals

Port Name	I/O	Description
rpcs_ef_wr_csn_o	O	FEC Error Forward Memory Chip Select Signal of the Write Port. This signal qualifies the rpcs_ef_wr_addr_o[4:0] signal. **Exists:** FEC_ERROR_FWD **Synchronous To:** clk_rpcs_rx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low
rpcs_ef_wr_en_o	O	FEC Error Forward Memory Enable Signal of the Write Port. When high, this signal indicates that the rpcs_ef_wr_data_o[65:0] data is valid and it is written in the memory. **Exists:** FEC_ERROR_FWD **Synchronous To:** clk_rpcs_rx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ef_wr_addr_o[4:0]	O	FEC Error Forward Memory Address Bus of the Write Port. This signal is the memory address bus of the write port. It is valid when the rpcs_ef_wr_csn_o signal is asserted. **Exists:** FEC_ERROR_FWD **Synchronous To:** clk_rpcs_rx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ef_wr_data_o[(FEC_DW-1):0]	O	FEC Error Forward Memory Data Bus of the Write Port. This signal carries the data to be written into the memory. It is valid only when the rpcs_ef_wr_en_o signal is asserted. **Exists:** FEC_ERROR_FWD **Synchronous To:** clk_rpcs_rx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
rpcs_ef_rd_csn_o	O	FEC Error Forward Memory Chip Select Signal of the Read Port. This signal qualifies the rpcs_ef_rd_addr_o[4:0] signal. **Exists:** FEC_ERROR_FWD **Synchronous To:** clk_rpcs_rx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** Low
rpcs_ef_rd_en_o	O	FEC Error Forward Memory Enable Signal of the Read Port. When high, this signal indicates that the rpcs_ef_rd_data_i[65:0] signal contains valid data, and it is sampled by the DWC_xpcs. **Exists:** FEC_ERROR_FWD **Synchronous To:** clk_rpcs_rx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ef_rd_addr_o[4:0]	O	FEC Error Forward Memory Address Bus of the Read Port. This signal is the memory address bus of the read port. It is valid when the rpcs_ef_rd_csn_o signal is asserted. **Exists:** FEC_ERROR_FWD **Synchronous To:** clk_rpcs_rx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ef_rd_data_i[(FEC_DW-1):0]	I	FEC Error Forward Memory Data Bus of the Read Port. This signal carries the read data from the FEC error forward memory. It is valid when the rpcs_ef_rd_en_o signal is asserted. **Exists:** FEC_ERROR_FWD **Synchronous To:** clk_rpcs_rx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.58 Clause 72 KR Training Signals for Non-Synopsys PHY

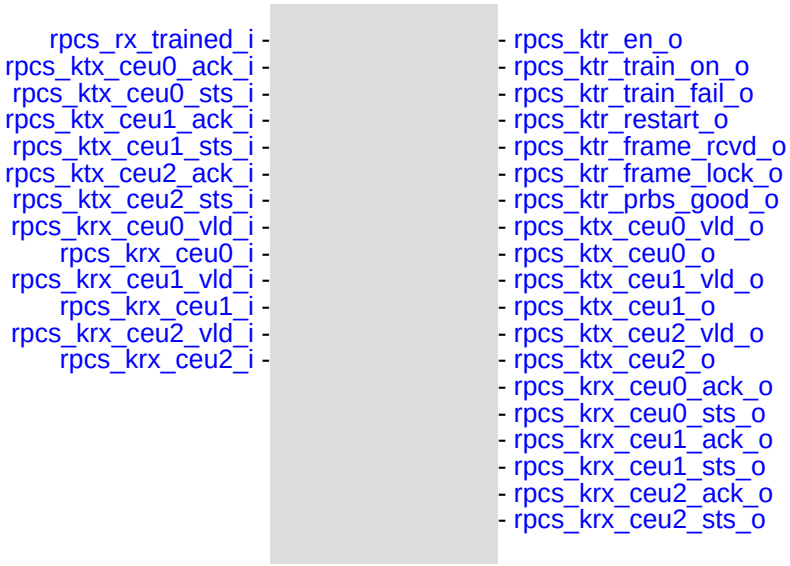


Table 6-58 Clause 72 KR Training Signals for Non-Synopsys PHY

Port Name	I/O	Description
rpcs_rx_trained_i	I	Local Receiver is Trained. When asserted, this signal indicates that the local KR adaptation process has determined that the local receiver and remote transmit equalization have been optimized to operate at 10GBASE-KR speed. **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktr_en_o	O	10GBASE-KR Training Enable. The DWC_xpcs asserts this signal when the host enables the Clause 72 KR Training by programming Bit 1 of SR PMA MMD 10GBASE-KR PMD Control Register. This signal can be used in the KR PMA receiver adaptation process. **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
rpcs_ktr_train_on_o	O	Training is ON. This signal is asserted when the host has enabled the Clause 72 Training process and the training process is not yet complete. This signal is de-asserted when the Clause 72 KR training is successfully completed. This signal can be used in the KR receiver adaptation process. **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktr_train_fail_o	O	Training Fail Status. This signal is asserted when the Clause 72 training process fails. The Clause 72 training process fails when both local and remote adaptation processes fail to establish reliable KR link through Transmit filter coefficient settings within 500 ms time. This signal is asserted until the management restarts the KR training process. **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktr_restart_o	O	Restart Training. The DWC_xpcs asserts this signal when the host restarts the Clause 72 KR Training by programming Bit 0 of SR PMA MMD 10GBASE-KR PMD Control Register. *Note:*This signal is a pulse of 1 clock period of the clk_rpcs_tx_i clock. **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktr_frame_rcvd_o	O	Training Frame Received. This signal indicates that the new training frame is received. This (pulse) signal is asserted for 1 clock period of the clk_rpcs_tx_i clock. **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
rpcs_ktr_frame_lock_o	O	Training Frame Synchronization Status. This signal is asserted when the Received Training frame synchronization is successful. Otherwise, this signal is de-asserted. This signal can be used in the KR receiver adaptation process. **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktr_prbs_good_o	O	PRBS Good Indication. This signal indicates the PRBS data decoding status of the currently received training frame when the rpcs_ktr_frame_rcvd_o signal is asserted. ■ 0: Decoding is not successful ■ 1: Decoding is successful **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktx_ceu0_vld_o	O	Local Transmitter Coefficient Update Valid of Corresponding Filter Tap c(-1). This signal is asserted to request the local KR PMA Transmitter to update its filter coefficient the corresponding filter tap. The request is placed the rpcs_ktx_ceu0_o signal. When this signal is asserted, it continues to be asserted till the rpcs_ktx_ceu0_ack_i signal is asserted from the Local KR PMA. **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
rpcs_ktx_ceu0_o[1:0]	O	Local Device Filter Coefficient Update Request of Corresponding Filter Tap c(-1). This signal contains the content of the received training sequence from the link partner to set the filter coefficient value of the local transmitter. *Values:* ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_ceu0_ack_i	I	Acknowledgment for rpcs_ktx_ceu0_vld_o from the KR Adaptation Process. This signal is asserted in response to the rpcs_ktx_ceu0_vld_o signal. This signal can be de-asserted only after the rpcs_ktx_ceu0_vld_o is de-asserted. When this signal is asserted, the rpcs_ktx_ceu0_sts_i signal is valid. **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktx_ceu0_sts_i[1:0]	I	Status for rpcs_ktx_ceu0_vld_o Request. This signal indicates the status of the rpcs_ktx_ceu0_vld_o request. The status is valid only when the rpcs_ktx_ceu0_ack_i is asserted. This information is used in the Status Update field of the transmitted training frame to the link partner. *Values:* ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
rpcs_ktx_ceu1_vld_o	O	Local Transmitter Coefficient Update Valid of transmit filter value for c(0). This signal is asserted to request the local KR PMA Transmitter to update its filter coefficient the corresponding filter tap. The request is placed the rpcs_ktx_ceu1_o signal. When this signal is asserted, it continues to be asserted till the rpcs_ktx_ceu1_ack_i signal is asserted from the Local KR PMA. **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktx_ceu1_o[1:0]	O	Local Device Filter Coefficient Update Request of Corresponding Filter Tap c(0). This signal contains the content of the received training sequence from the link partner to set the filter coefficient value of the local transmitter. *Values:* ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_ceu1_ack_i	I	Acknowledgment for rpcs_ktx_ceu1_vld_o from the KR Adaptation Process. This signal is asserted in response to the rpcs_ktx_ceu1_vld_o signal. This signal can be de-asserted only after the rpcs_ktx_ceu1_vld_o is de-asserted. When this signal is asserted, the rpcs_ktx_ceu1_sts_i signal is valid. **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
rpcs_ktx_ceu1_sts_i[1:0]	I	Status for rpcs_ktx_ceu1_vld_o Request. This signal indicates the status of the rpcs_ktx_ceu1_vld_o request. The status is valid only when the rpcs_ktx_ceu1_ack_i is asserted. This information is used in the Status Update field of the transmitted training frame to the link partner. *Values:* ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_ktx_ceu2_vld_o	O	Local Transmitter Coefficient Update Valid of Corresponding Filter Tap c(1). This signal is asserted to request the local KR PMA Transmitter to update its filter coefficient the corresponding filter tap. The request is placed the rpcs_ktx_ceu2_o signal. When this signal is asserted, it continues to be asserted till the rpcs_ktx_ceu2_ack_i signal is asserted from the Local KR PMA. **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktx_ceu2_o[1:0]	O	Local Device Filter Coefficient Update Request of Corresponding Filter Tap c(1). This signal contains the content of the received training sequence from the link partner to set the filter coefficient value of the local transmitter. *Values:* ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
rpcs_ktx_ceu2_ack_i	I	Acknowledgment for rpcs_ktx_ceu2_vld_o from the KR Adaptation Process. This signal is asserted in response to the rpcs_ktx_ceu2_vld_o signal. This signal can be de-asserted only after the rpcs_ktx_ceu2_vld_o is de-asserted. When this signal is asserted, the rpcs_ktx_ceu2_sts_i signal is valid. **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_ktx_ceu2_sts_i[1:0]	I	Status for rpcs_ktx_ceu2_vld_o Request. This signal indicates the status of the rpcs_ktx_ceu2_vld_o request. The status is valid only when the rpcs_ktx_ceu2_ack_i is asserted. This information is used in the Status Update field of the transmitted training frame to the link partner. *Values:* ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not Applicable **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu0_vld_i	I	Request from the Local KR Adaptation Process to the Link Partner Transmitter. When this signal is asserted, the rpcs_krx_ceu0_i[1:0] is valid. When this signal is asserted, the KR adaptation process should not de-assert it until the rpcs_krx_ceu0_ack_o signal is asserted. **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
rpcs_krx_ceu0_i[1:0]	I	Link Partner Coefficient Update of Corresponding Filter Tap c(-1). The KR adaptation process provides this input based the received training frame PRBS Status from the link partner. This information is used in the Coefficient Update field of the transmitted training frame. *Values:* ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu0_ack_o	O	Acknowledgement for the rpcs_krx_ceu0_vld_i Request. When this signal is asserted, the rpcs_krx_ceu0_sts_o[1:0] status is valid. This signal is de-asserted after the rpcs_krx_ceu0_vld_i signal is de-asserted. **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_krx_ceu0_sts_o[1:0]	O	Status for the rpcs_krx_ceu0_i[1:0] Request. This signal is valid when the rpcs_krx_ceu0_ack_o signal is asserted. This signal indicates the status of the rpcs_krx_ceu0_i[1:0] request. This information is generated based the Receive training from the link partner. *Values:* ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not-Updated **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
rpcs_krx_ceu1_vld_i	I	Request from the Local KR Adaptation Process to the Link Partner Transmitter. When this signal is asserted, the rpcs_krx_ceu1_i[1:0] is valid. When this signal is asserted, the KR adaptation process should not de-assert it until the rpcs_krx_ceu1_ack_o signal is asserted. **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu1_i[1:0]	I	Link Partner Coefficient Update of Corresponding Filter Tap c(0). The KR adaptation process provides this input based the received training frame PRBS Status from the link partner. This information is used in the Coefficient Update field of the transmitted training frame. *Values:* ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu1_ack_o	O	Acknowledgement for the rpcs_krx_ceu1_vld_i Request. When this signal is asserted, the rpcs_krx_ceu1_sts_o[1:0] status is valid. This signal is de-asserted after the rpcs_krx_ceu1_vld_i signal is de-asserted. **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
rpcs_krx_ceu1_sts_o[1:0]	O	Status for the rpcs_krx_ceu1_i[1:0] Request. This signal is valid when the rpcs_krx_ceu1_ack_o signal is asserted. This signal indicates the status of the rpcs_krx_ceu1_i[1:0] request. This information is generated based the Receive training from the link partner. *Values:* ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not-Updated **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A
rpcs_krx_ceu2_vld_i	I	Request from the Local KR Adaptation Process to the Link Partner Transmitter. When this signal is asserted, the rpcs_krx_ceu2_i[1:0] is valid. When this signal is asserted, the KR adaptation process should not de-assert it until the rpcs_krx_ceu2_ack_o signal is asserted. **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_krx_ceu2_i[1:0]	I	Link Partner Coefficient Update of Corresponding Filter Tap c(1). The KR adaptation process provides this input based the received training frame PRBS Status from the link partner. This information is used in the Coefficient Update field of the transmitted training frame. *Values:* ■ 11: Initialize ■ 01: Increment ■ 10: Decrement ■ 00: Preset **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** Asynchronous **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

Port Name	I/O	Description
rpcs_krx_ceu2_ack_o	O	Acknowledgement for the rpcs_krx_ceu2_vld_i Request. When this signal is asserted, the rpcs_krx_ceu0_sts_o[1:0] status is valid. This signal is de-asserted after the rpcs_krx_ceu2_vld_i signal is de-asserted. **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
rpcs_krx_ceu2_sts_o[1:0]	O	Status for the rpcs_krx_ceu2_i[1:0] Request. This signal is valid when the rpcs_krx_ceu2_ack_o signal is asserted. This signal indicates the status of the rpcs_krx_ceu2_i[1:0] request. This information is generated based the Receive training from the link partner. *Values:* ■ 10: Minimum ■ 11: Maximum ■ 01: Updated ■ 00: Not-Updated **Exists:** CL72_EN&&!SNPS_XAUI_PHY **Synchronous To:** clk_rpcs_tx_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** N/A

6.59 Non-Synopsys PHY Interface Signals



Table 6-59 Non-Synopsys PHY Interface Signals

Port Name	I/O	Description
xpcs_pdown_o	O	Power-Down Enable. This signal indicates that the host programmed the device for the power-down mode. **Exists:** !SNPS_XAUI_PHY **Synchronous To:** {(!SNPS_XAUI_PHY&&MDIO_INTF) ? "None" : "clk_csr_i"} **Registered:** No **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_phy_rdy_i	I	PHY Ready Indication. This signal should be set to High when the PHY is ready and all the clocks of the PHY are stable. When this input is low the internal data path of the DWC_xpcs is in reset state. **Exists:** !SNPS_XAUI_PHY&&ADD_PHY_RDY **Synchronous To:** Asynchronous **Registered:** N/A **Power Domain:** SINGLE_DOMAIN **Active State:** High **Registered:** ■ No, in MDIO configurations ■ Yes, with clk_csr_i in APB3/MCI configurations

6.60 SSM Bus Interface Signals

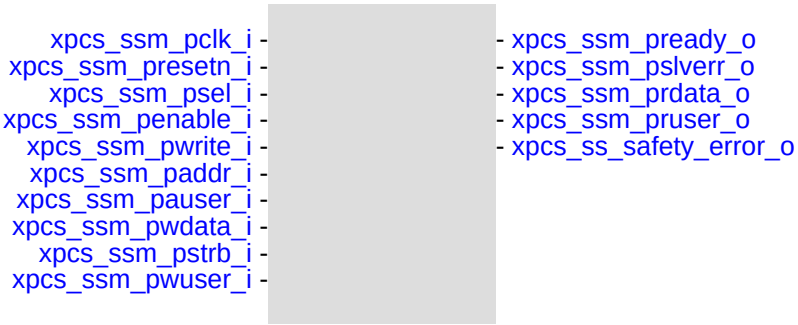


Table 6-60 SSM Bus Interface Signals

Port Name	I/O	Description
xpcs_ssm_pclk_i	I	SSM clock. The clock may be enabled only when accessing the Safety Subordinate. **Exists:** DWC_XPCS_SS **Synchronous To:** None **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_ssm_presetn_i	I	Safety Subordinate Reset. This signal must be asserted along with pwr_on_rst_n and deasserted synchronously with ssm_pclk after a minimum of 10 csr clocks has elapsed after pwr_on_rst_n goes inactive. **Exists:** DWC_XPCS_SS **Synchronous To:** xpcs_ssm_pclk_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_ssm_psel_i	I	Safety Subordinate Select. When this signal is asserted, it indicates the start of transaction and validity of the xpcs_ssm_paddr_i[] and xpcs_ssm_pwrite_i signals. In addition, during a write operation, this signal indicates validity of the xpcs_ssm_pwdata_i[] write data. **Exists:** DWC_XPCS_SS **Synchronous To:** xpcs_ssm_pclk_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_ssm_penable_i	I	Safety Subordinate Enable **Exists:** DWC_XPCS_SS **Synchronous To:** xpcs_ssm_pclk_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_ssm_pwrite_i	I	Safety Subordinate Write/Read Indicator. When high, this signal indicates a write operation. When low, it indicates a read operation. Safety Subordinate Address (configurable to 8 or 16 bits) **Exists:** DWC_XPCS_SS **Synchronous To:** xpcs_ssm_pclk_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_ssm_paddr_i[15:0]	I	Safety Subordinate Address. This signal carries the register address of the Control or Status Register of the safety_slave module. The address is valid only when the xpcs_ssm_psel_i signal is asserted. **Exists:** DWC_XPCS_SS **Synchronous To:** xpcs_ssm_pclk_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_ssm_pauser_i[5:0]	I	User defined address bus. This consists of the parity bit for the address and Antivalent bits of the control signals pwrite, psel and penable. {ssm_paddr_parity, 1'b0, ssm_pwrite_av, ssm_penable_av, ssm_psel_av} **Exists:** DWC_XPCS_SS **Synchronous To:** xpcs_ssm_pclk_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_ssm_pwdata_i[31:0]	I	Write Data. This signal carries the write data from the host (Safety Manager Application) for the Control registers inside the safety_slave. **Exists:** DWC_XPCS_SS **Synchronous To:** xpcs_ssm_pclk_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_ssm_pstrb_i[3:0]	I	Write Data Strobe. **Exists:** DWC_XPCS_SS **Synchronous To:** xpcs_ssm_pclk_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_ssm_pwuser_i[4:0]	I	User defined write data bus. It contains the parity for write data and anti-valent bit write_strb. {ssm_pwdata_parity, ssm_pstrb_parity} **Exists:** DWC_XPCS_SS **Synchronous To:** xpcs_ssm_pclk_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_ssm_pready_o	O	Safety Subordinate ready. When high, this signal indicates the following: - Data on the xpcs_ssm_prdata_o[15:0] signal is valid. The host can accept the data. - End of write transfer Parity error detected if enabled on xpcs_ssm_paddr_i or xpcs_ssm_pwdata_i or xpcs_ssm_pstrb_i. **Exists:** DWC_XPCS_SS **Synchronous To:** xpcs_ssm_pclk_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_ssm_pslverr_o	O	Safety Subordinate Error. Parity error detected if enabled on xpcs_ssm_paddr_i or xpcs_ssm_pwdata_i or xpcs_ssm_pstrb_i. **Exists:** DWC_XPCS_SS **Synchronous To:** xpcs_ssm_pclk_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_ssm_prdata_o[31:0]	O	SSM Read Data. This port carries the read data from the Safety Subordinate Registers to the host (Safety Manager Application). This data is valid only when the xpcs_ssm_ pready_o signal is asserted. **Exists:** DWC_XPCS_SS **Synchronous To:** xpcs_ssm_pclk_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

Port Name	I/O	Description
xpcs_ssm_pruser_o[5:0]	O	User defined read data bus. It contains the parity for read data and anti -valent bit for xpcs_ssm_pslverr_o, xpcs_ssm_pready_o. {ssm_prdata_parity, ssm_pslverr_av, ssm_pready_av}. It contains the parity for read data and read parity. {ssm_prdata_parity, ssm_pslverr_av, ssm_pready_av}. **Exists:** DWC_XPCS_SS **Synchronous To:** xpcs_ssm_pclk_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High
xpcs_ss_safety_error_o[1:0]	O	Safety Error. Sticky until reset. No Error = 2'b10 Error = 2'b01 **Exists:** DWC_XPCS_SS **Synchronous To:** xpcs_ssm_pclk_i **Registered:** Yes **Power Domain:** SINGLE_DOMAIN **Active State:** High

7

Programming Guidelines

This chapter provides the programming guidelines for switching between various modes and enabling auto-negotiation and Energy Efficient Ethernet features. It contains the following sections:

- "Initializing the DWC_xpcs Controller" on page 1650
- "Switching to 10G XAUI Mode or KX4 Mode" on page 1652
- "Switching to 10G RXAUI Mode" on page 1661
- "Switching to 1G or KX Mode and 10 or 100 Mbps SGMII Speed" on page 1669
- "Switching to 10G/5G KR Speed Mode" on page 1676
- "Switching to USXGMII Mode" on page 1682
- "Programming Guidelines for QSGMII Mode" on page 1684
- "Programming Guidelines for 2.5G GMII Mode" on page 1692
- "Programming Guidelines for 2.5GBASE-KX Mode (over XGMII)" on page 1696
- "Bifurcation Mode Usage of Synopsys Multi-Protocol PHY" on page 1698
- "Programming Guidelines for CR Port Read or Write Operation" on page 1702
- "Programming Guidelines for Clause 37 Auto-Negotiation" on page 1704
- "Programming Guidelines for Clause 72 Start-Up Protocol" on page 1708
- "Programming Guidelines for Clause 73 Auto-Negotiation" on page 1712
- "Programming Guidelines for Energy Efficient Ethernet" on page 1719
- "Programming Guidelines for ECC Protection for Memories" on page 1721
- "Programming Guidelines for On-Chip Datapath Parity Protection" on page 1722
- "Programming Guidelines for FSM Parity Protection" on page 1723
- "Programming Guidelines for FSM Timeout Protection" on page 1724
- "Programming Guidelines for CSR Protection" on page 1725
- "Programming Guidelines for Interface Timeout Protection" on page 1726

Note If your configuration does not have XS or PCS or PMA MMD and if the programming guidelines mentioned in this chapter make reference to a register in XS/PCS/PMA MMD, you can program the corresponding register in MII MMD.

7.1 Initializing the DWC_xpcs Controller

Complete the following steps to initialize the DWC_xpcs controller:

1. Switch on the power supply. Reset is asserted (pwr_on_rst_n = 0).
2. Wait for the required amount of time depending on the PHY requirements.
3. De-assert reset (pwr_on_rst_n = 1).
4. Follow these steps:
   - For configurations with Synopsys multi-protocol 12G/16G/25G/32G PHY:

     Poll bit [0], INIT_DN field of the following register until it becomes 1.

     VR_MII_MP_12G_16G_25G_SRAM register for 1000BASE-X Only PCS configurations.

     VR_XS_PMA_MP_12G_16G_25G_SRAM register for other configurations.
   - For Synopsys multi-protocol 12G/16G/25G configurations:

     After waiting for at least 10 clock cycles of cr_para_clk, software can optionally write to CR registers inside DWC_xpcs to modify the firmware that has boot-loaded to the SRAM connected to the PHY.
   - For Synopsys Multi-protocol 32G PHY configurations:

     After waiting for at least 10 clock cycles of cr_para_clk, load the SRAM with 'firmware and context restore' content by initiating the CR write operation through DWC_xpcs. For more details, see “Programming Guidelines for CR Port Read or Write Operation” on page 1702.

   After loading the SRAM, program bit[1], EXT_LD_DN field to 1, in the following register:
   - VR_MII_MP_12G_16G_SRAM register for 1000BASE-X Only PCS configurations.
   - VR_XS_PMA_MP_25G_16G_SRAM register for other configurations.
5. Read the following registers and wait till the 15th bit of these registers is read as 0:
   - SR_MII_CTRL, when configured in Base1000-X Only configuration.
   - SR_XS_PCS_CTRL1, for configurations other than Base1000-X Only configuration.



Note

- Completion of this step indicates that both the PHY and DWC_xpcs have successfully completed the reset sequence in the default speed mode. Default speed of DWC_xpcs is the highest speed mode selected in the configuration.
- For Synopsys multi-protocol 32G PHY configurations, this step completes, only if you have correctly set the values of TX_CNTX_SEL, RX_CNTX_SEL and CM_CNTX_SEL parameters during RTL configuration.

6. For Synopsys PHY configurations with 10BASE-(K)R mode support, program the following bits to 1 in the VR_XS_PCS_DEBUG_CTRL register:
   - bit[4], SUPRESS_LOS_DET
   - bit[6], RX_DT_EN_CTL
7. For configurations with Backplane Ethernet PCS as the mode of operation, follow these steps:

- To perform Clause 73 auto-negotiation, to negotiate the link capabilities with link-partner, see "Programming Guidelines for Clause 73 Auto-Negotiation" on page 1712.

  If you have selected the Add Clause 72 KR Training Support parameter while RTL configuration and the auto-negotiated speed is 10G-KR, DWC_xpcs performs the Clause 72 Training after the exchange of all the AN pages.
- Program bit[12], AN_EN field of the SR_AN_CTRL register to 0, if you do not require DWC_xpcs to perform Clause 73 auto-negotiation. To switch DWC_xpcs and the PHY to the required speed mode, follow the steps in sections 7.2 through 7.9.

8. If you select "Enable Clause 37 Auto-Negotiation" in the configuration, then follow the programming flow outlined in "Programming Guidelines for Clause 37 Auto-Negotiation" on page 1704, to perform Clause 37 auto-negotiation operation, which is required for 1000BASE-X, SGMII, QSGMII, and USXGMII links.

   Program AN_EN field of the SR_MII_CTRL register to 0 if you do not require DWC_xpcs to perform Clause 37 auto-negotiation.
9. Wait for the following register bit to be set to 1:
   - RLU, bit[2] of the SR_XS_PCS_STS1 register, for configurations other than 1000BASE-X Only PCS.
   - LINK_STS, bit[2] of the SR_MII_STS register, for 1000BASE-X Only PCS configurations.



This step completes successfully only when you send appropriate data to the DWC_xpcs receiver as required by the current operating speed mode.

The DWC_xpcs controller is now ready to transmit or receive data.

7.2 Switching to 10G XAUI Mode or KX4 Mode

In 10G XAUI mode, the DWC_xpcs to the PHY interface works at 10-bit parallel data with the clock rate of 312.5 MHz. All four lanes are functional. Each serial lane works at the baud rate of 3.125 Gbps. The effective data transfer rate, considering the overhead of 8B/10B encoding or decoding, is 2.5 Gbps in each lane (10G data transfer rate). To switch between 1G and 10G modes, program Bit 13 of the following registers:

- SR_PMA_CTRL1 in Backplane Ethernet PCS configurations
- SR_XS_PCS_CTRL1 in all other configurations

When this bit is set to 1, the DWC_xpcs assumes the 10GBASE-X PCS (10G) mode of operation.

7.2.1 Programming Sequence for Synopsys Multi-Protocol 6G PHY

To switch to the 10G XAUI Mode or KX4 Mode when using Synopsys Multi-Protocol 6G PHY:

1. (Only for RXAUI configurations) Set Bit 0 of VR_XS_PCS_XAUI_CTRL Register to 0.
2. Set Bit 13 of SR_PMA_CTRL1 Register (only for Backplane Ethernet PCS configurations) or SR_XS_PCS_CTRL1 Register to 1.
3. Program the register bits shown in Table 7-1.

Table 7-1 Synopsys Multi-Protocol 6G PHY: 10G XAUI Mode Switching

Field	Values
MPLL_MULTIPLIER	Based on the frequency of the reference clock, set the value of this field to the following in VR_XS_PMA_MP_6G_MPLL_CTRL1or VR_MII_MP_6G_MPLL_CTRL1 ■ 7'h28 for 156.25 MHz reference clock frequency ■ 7'h32 for 125 MHz and 62.5 MHz reference clock frequency ■ 7'h7D for 50 MHz and 25 MHz reference clock frequency
LANE_10BIT_SEL	Set the value of this field to 1'b1 in VR_XS_PMA_MP_6G_MPLL_CTRL0 or VR_MII_MP_6G_MPLL_CTRL0.
TX_RATE_* and RX_RATE_*	Set the values of these fields for all the lanes to 2'b01 (16'h5555) in VR_XS_PMA_MP_6G_MISC_CTRL1or VR_MII_MP_6G_MISC_CTRL1.
REF_CLK_DIV2	Based on the frequency of the reference clock, set the value of this field in VR_XS_PMA_MP_6G_MPLL_CTRL0 or VR_MII_MP_6G_MPLL_CTRL0. to: ■ 1'b1 for 156.25 MHz, 125 MHz and 50 MHz reference clock frequency ■ 1'b0 for 62.5 MHz and 25 MHz reference clock frequency

Note To enable non-standard XAUI link to have (2.5G or 5.0G or 7.5G speed modes using one or two or three lanes respectively) program bit[11:9] and bit[15:13] of VR_XS_PCS_DIG_CTRL2.

4. Set Bit 11 of the SR_PMA_CTRL1 Register (only for Backplane Ethernet PCS configurations) or SR_XS_PCS_CTRL1 Register to 1 to power down the DWC_xpcs.
5. Wait till Bits[4:2] of VR_XS_PCS_DIG_STS are set to 3'b110.

6. Clear Bit 11 of SR_PMA_CTRL1 Register (only for Backplane Ethernet PCS configurations) or SR_X-S_PCS_CTRL1 to 0 to power up the DWC_xpcs.
7. Wait till Bits[4:2] of VR_XS_PCS_DIG_STS Register are set to 3'b100.



If you have enabled non-standard XAUI link to have (2.5G or 5.0G or 7.5G speed modes using one or two or three lanes respectively) you can also disable the unused transmit lanes by programming bit[7:5] of VR_XS_PCS_DIG_CTRL1 and disable the unused receive lanes by programming bit[7:5] of VR_XS_PMA_MP_6G_RXGENCTRL1 or VR_MII_MP_6G_RXGENCTRL1.

7.2.2 Programming Sequence for Synopsys Multi-Protocol 12G/16G PHY

To switch to the 10G XAUI mode or KX4 mode, follow these steps:

1. (Only for RXAUI configurations) Set Bit 0 of VR_XS_PCS_XAUI_CTRL Register to 0.
2. Write 4'b0001 to Bits[3:0] of SR_XS_PCS_CTRL2 Register.
3. Set Bit 13 of SR_PMA_CTRL1 Register (only for Backplane Ethernet PCS configurations) or SR_X-S_PCS_CTRL1 Register to 1.
4. Program the register bits shown in Table 7-2 for 12G PHY and Table 7-3 for 16G PHY.

Table 7-2 Synopsys Multi-Protocol 12G PHY: 10G XAUI Mode Switching

Field	Values
MPLLA_MULTIPLIER	Set the values of this field in the register VR_XS_PMA_MP_12G_16G_MPLLA_CTRL0 Register to 8’d40
MPPLA_BANDWIDTH	Set the values of this field in the register VR_XS_PMA_MP_12G_MPLLA_CTRL3 Register based on the PHY model used: ■ GF 14 LPP: 11’d1923 ■ TSMC 16FFLL: 11’d83 ■ TSMC 28HPC: 11'd86 Note: These values might not be accurate. See the latest PHY data book corresponding to your version of the PHY and program suitably.
VCO_LD_VAL_{lane}	Set the value of these fields for all the lanes in VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD {lane} Register to 13’d1360
VCO_REF_LD_{lane}	Set the value of these fields for all the lanes in VR_XS_PMA_MP_12G_VCO_CAL_REF 0/1 Registers as follows: ■ 6'd17 for 78.125 MHz Reference Clock ■ 6'd37 for 156.25 MHz/ 312.5 MHz Reference Clock
AFE_EN_0, AFE_EN_3_1, DFE_EN_0, DFE_EN_3_1	Set the value of these fields in VR_XS_PMA_MP_12G_AFE_DFE_EN_CTRL Register to 0.
CONT_ADAPT_0, CONT_ADAPT_3_1	Set the value of these fields in VR_XS_PMA_MP_12G_16G_25G_RX_EQ_CTRL4 Register to 0.

Field	Values
TX{lane}_RATE	Set the value of these fields for all the lanes to 3'b010 in VR_XS_PMA_MP_12G_16G_25G_TX_RATE_CTRL Register
RX{lane}_RATE	Set the value of these fields for all the lanes to 2'b10 in VR_XS_PMA_MP_12G_16G_25G_RX_RATE_CTRL Register
TX{lane}_WIDTH	Set the value of these fields for all the lanes to 2'b01 in VR_XS_PMA_MP_12G_16G_TX_GENCTRL2 Register
RX{lane}_WIDTH	Set the value of these fields for all the lanes to 2'b01 in VR_XS_PMA_MP_12G_16G_RX_GENCTRL2 Register
MPLLA_DIV16P5_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 Register to 0
MPLLA_DIV10_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 Register to 1
MPLLA_DIV8_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 Register to 0
TX_CLK_RDY_{lane}	Set the value of this field om VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 0

Table 7-3 Synopsys Multi-Protocol 16G PHY: 10G XAUI Mode Switching

Field	Values
MPLLA_MULTIPLIER	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL0 register to 8'd40
MPLLA_BANDWIDTH	Set the value of this field in VR_XS_PMA_MP_16G_MPLLA_CTRL3 register to 16'd41022
VCO_LD_VAL_{lane}	Set the values of these fields in the following registers to 13'd1360 ■ VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD0 ■ VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD1 ■ VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD2 ■ VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD3
VCO_REF_LD_{lane}	Set the values of these fields in the following registers to 7'd17: For 78.125 MHz Ref Clock or 7'd34: For 156.25 MHz/312.5 MHz Ref Clock ■ VR_XS_PMA_MP_16G_25G_VCO_CAL_REF0 ■ VR_XS_PMA_MP_16G_25G_VCO_CAL_REF1
CONT_ADAPT_0, CONT_ADAPT_3_1	Set the value of this field in VR_MII_MP_12G_16G_25G_RX_EQ_CTRL4 register to 0
TX{lane}_RATE	Set the values of these fields in VR_XS_PMA_MP_12G_16G_25G_TX_RATE_CTRL register to 2
RX{lane}_RATE	Set the values of these fields in VR_XS_PMA_MP_12G_16G_25G_RX_RATE_CTRL register to 2

Field	Values
TX{lane}_WIDTH	Set the values of these fields in VR_XS_PMA_MP_12G_16G_TX_GENCTRL2 register to 1
RX{lane}_WIDTH	Set the values of these fields in VR_XS_PMA_MP_12G_16G_RX_GENCTRL2 register to 1
MPLLA_DIV16P5_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 register to 0
MPLLA_DIV10_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 register to 1
MPLLA_DIV8_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 register to 0
TX_CLK_RDY_{lane}	Set the value of this field in VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 0
VBOOST_EN_0	Set the value of this field in VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 1
VBOOST_EN_3_1	Set the value of this field in VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 7
CTLE_BOOST_{lane}	Set the values of these fields in the following registers to 6. ■ VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL0 ■ VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL1 ■ VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL2 ■ VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL3
VCO_STEP_CTRL_0, VCO_STEP_CTRL_3_1	Set the values of these fields in VR_XS_PMA_MP_16G_RX_CDR_CTRL1 register to 0
VCO_TEMP_COMP_EN_0, VCO_TEMP_COMP_EN_3_1	Set the values of these fields in VR_XS_PMA_MP_16G_RX_CDR_CTRL1 register to 0
RX{lane}_MISC	Set the value of these fields in the following registers to 7 ■ VR_XS_PMA_MP_16G_25G_RX_MISC_CTRL0 ■ VR_XS_PMA_MP_16G_25G_RX_MISC_CTRL1
RX_DFE_BYP_0	Set the value of this field in VR_XS_PMA_MP_16G_25G_RX_GENCTRL4 register to 1
RX_DFE_BYP_3_1	Set the value of this field in VR_XS_PMA_MP_16G_25G_RX_GENCTRL4 register to 7
VCO_FRQBAND_{lane}	Set the values of these fields in VR_XS_PMA_MP_16G_25G_RX_CDR_CTRL1 register to 2
RX{lane}_DELTA_IQ	Set the values of these fields in the following register to 0 ■ VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL0 ■ VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL1 ■ VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL2 ■ VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL3

Field	Values
RX{lane}_ADAPT_MODE	Set the values of these fields in VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL5 register to 0

Note If the values provided in these tables contradict with the corresponding values in the PHY data book, the values provided in PHY databook take precedence.

5. Initiate the vendor-specific Software Reset by writing 1'b1 to VR_RST bit[15] of VR_XS_PCS_DIG_CTRL1 Register.
6. (Optional) Execute step 4 from the section, "Initializing the DWC_xpcs Controller" on page 1650.
7. Wait for the bit[15] (VR_RST) of the VR_XS_PCS_DIG_CTRL1
8. Program TX_CLK_RDY{lane} field of VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 1
9. Program TX_EQ_PRE field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL0 register to 0
10. Program TX_EQ_MAIN field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL0 register to 6'd40
11. Program TX_EQ_POST field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL1 register to 0
12. Program TX_EQ_OVR_RIDE field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL1 register to 1 (only for configurations with CL72 support)
13. Program bit [4] (SUPRESS_LOS_DET) and bit [6] (RX_DT_EN_CTL) of the VR_XS_PCS_DEBUG_CTRL register to 0.

7.2.3 Programming Sequence for Synopsys Multi-Protocol 25G PHY

To switch to 10G XAUI or KX4 mode, follow these steps:

1. Program bits [3:0] of SR_XS_PCS_CTRL2 register to 4'b0001
2. Program bit[0] of VR_XS_PCS_XAUI_CTRL register to 0 (only for RXAUI configurations)
3. Set bit [13] of SR_PMA_CTRL1 register (for Backplane Ethernet configurations) or SR_XS_PCS_CTRL1 (for other configurations) to 1.
4. Program MPLLA_MULTIPLIER field of VR_XS_PMA_SNPS_MP_25G_MPLLA_CTRL0 register to 12'd120
5. Program MPLLA_TX_CLK_DIV field of VR_XS_PMA_SNPS_MP_25G_MPLLA_CTRL2 register to 3
6. Program MPLLA_W_CLK_DIV field of VR_XS_PMA_SNPS_MP_25G_MPLLA_CTRL0 register to 1
7. Program BW_TH field of VR_XS_PMA_SNPS_MP_25G_MPLLA_CTRL1 register to 2
8. Program VCO_REF_LD_{lane} field of VR_XS_PMA_SNPS_MP_16G_25G_VCO_CAL_REF0/1 register to 7'd34
9. Program VCO_LD_VAL_{lane} field of VR_XS_PMA_SNPS_MP_12G_16G_25G _VCO_CAL_LD{lane} register to 13'd1360
10. Program TX{lane}_DCC fields of VR_XS_PMA_SNPS_MP_25G_TX_DCC_CTRL register to 4'd10
11. Program TX{lane}_WIDTH field of VR_XS_PMA_SNPS_MP_25G_TX_WIDTH_CTRL register to 1
12. Program TX{lane}_RATE field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_RATE_CTRL register to 1

13. Program VBOOST_EN_0 field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_GENCTRL1 register to 1
14. Program VBOOST_EN_3_1 field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_GENCTRL1 register to 3'd7
15. Program RX{lane}_VCO_CFG field of VR_XS_PMA_SNPS_MP_25G_RX_VCO_CFG{lane} register to 12'd259
16. Program LF_TH_{lane} field of VR_XS_PMA_SNPS_MP_25G_RX_LF_TRSHLD register to 7
17. Program RX{lane}_WIDTH field of VR_XS_PMA_SNPS_MP_25G_RX_WIDTH_CTRL register to 1
18. Program RX{lane}_RATE field of VR_XS_PMA_SNPS_MP_12G_16G_25G_RX_RATE_CTRL register to 2
19. Program RX_DIV16P5_CLK_EN_0 field of VR_XS_PMA_SNPS_MP_12G_16G_25G_RX_GENCTRL1 to 0
20. Program RX{lane}_ADPT_MODE field of VR_XS_PMA_SNPS_MP_16G_25G_RX_EQ_CTRL5 register to 0
21. Program RX_DFE_BYP_0 field of VR_XS_PMA_SNPS_MP_16G_25G_RX_EQ_CTRL4 to 1
22. Program RX_DFE_TAP1_{lane} field of VR_XS_PMA_SNPS_MP_12G_16G_25G_DFE_TAP_CTRL0 to 10
23. Program VGA1_GAIN_{lane} field of VR_XS_PMA_SNPS_MP_16G_25G_RX_EQ_CTRL0 to 4
24. Program VGA2_GAIN_{lane} field of VR_XS_PMA_SNPS_MP_16G_25G_RX_EQ_CTRL0 to 4
25. Program RX{lane}_AFE_RATE field of VR_XS_PMA_SNPS_MP_25G_RX_AFE_RATE_CTRL to 7
26. Program CTLE_BOOST_{lane} field of VR_XS_PMA_SNPS_MP_16G_25G_RX_EQ_CTRL0 register to 6
27. Program RX{lane}_CDR_PPM_MAX field of VR_XS_PMA_SNPS_MP_16G_25G_RX_PPM_CTRL0 register to 5'd15
28. Program TX_CLK_RDY_0/ TX_CLK_RDY_3_1 field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_-GENCTRL1 register to 0
29. Initiate a Vendor-specific Software Reset by writing 1 to bit [15] of VR_XS_PCS_DIG_CTRL1 register.
30. (Optional) Execute step 4 from the section, “Initializing the DWC_xpcs Controller” on page 1650.
31. Wait for bit [15] of VR_XS_PCS_DIG_CTRL1 register to get cleared.
32. Program TX_CLK_RDY_0 field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_GENCTRL1 register to 1
33. Program TX_CLK_RDY_3_1 field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_GENCTRL1 register to 3'd7
34. Program TX_EQ_PRE field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_EQ_CTRL0 register to 0
35. Program TX_EQ_MAIN field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_EQ_CTRL0 register to 6'd24
36. Program TX_EQ_POST field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_EQ_CTRL1 register to 0
37. Program TX_EQ_OVR_RIDE field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_EQ_CTRL1 register to 1
38. Program bit [4], SUPRESS_LOS_DET field and bit [6], RX_DT_EN_CTL field of the VR_XS_PCS_DE-BUG_CTRL register to 0.

7.2.4 Programming Sequence for Synopsys Multi-Protocol 32G PHY for 10GBASE-KX4/XAUI Mode

To switch 32G PHY to 10GBASE-KX4/XAUI mode, follow these steps:

1. Program 4'b0001 to bits[3:0] of the SR_XS_PCS_CTRL2 register.
2. Program 0 to bit[0] of the VR_XS_PCS_XAUI_CTRL register. This step is required only for RXAUI configurations.
3. Program 1 in the bit[13] of the
   - SR_PMA_CTRL1 register for Backplane Ethernet PCS configurations.
   - SR_XS_PCS_CTRL1 register for other configurations.
4. Program 0 to bit[2] (EN_2_5G_MODE) of the VR_XS_PCS_DIG_CTRL1 register.
5. Program the register bits as shown in Table 7-4, to switch the PHY to KX4/XAUI mode.

Table 7-4 Settings to Configures Multi-Protocol 32G PHY for 10GBASE-KX4/XAUI Mode

Register	Field	Values
VR_XS_PMA_MP_32G_RX_CNTX_CTRL0	RX_CNTX_SEL_0 RX_CNTX_SEL_1	Value indicated in the context_sel_ID.txt file of PHY package for 10GBASE-KX4/XAUI mode
VR_XS_PMA_MP_32G_RX_CNTX_CTRL0	RX_CNTX_SEL_2 RX_CNTX_SEL_3	
VR_XS_PMA_MP_32G_TX_CNTX_CTRL0	TX_CNTX_SEL_0 TX_CNTX_SEL_1	
VR_XS_PMA_MP_32G_TX_CNTRX_CTRL1	TX_CNTX_SEL_2 TX_CNTX_SEL_3	
VR_XS_PMA_MP_32G_TX_CM_CNTX_SEL0	CMN_CNTX_0 CMN_CNTX_1	
VR_XS_PMA_MP_32G_TX_CM_CNTX_SEL1	CMN_CNTX_2 CMN_CNTX_3	
VR_XS_PMA_MP_25G_RX_WIDTH_CTRL	RX0_WIDTH RX1_WIDTH RX2_WIDTH RX3_WIDTH	3’b001 (10-bit)
VR_XS_PMA_MP_25G_TX_WIDTH_CTRL	TX0_WIDTH TX1_WIDTH TX2_WIDTH TX3_WIDTH	
VR_XS_PMA_MP_32G_LN_LINK_CTRL	LINK_NUM_0 LINK_NUM_1 LINK_NUM_2 LINK_NUM_3	Program all the 4 fields to any identical and unique value. If bifurcation mode is not used, you can leave all these fields at the default value of 0.

6. Program the LPM bit of the SR_XS_PCS_CTRL1 register to 1.
7. Wait for the PSEQ_STATE of the VR_XS_PCS_DIG_STS register to indicate ‘Power Down’, a value of 3’b110.
8. Program the LPM bit of the SR_XS_PCS_CTRL1 register to 0. Program the other bits of this register to values present prior to step 6.

   Note: To avoid assertion of soft-reset do not perform read-modify-write operation.
9. Wait for the PSEQ_STATE of the VR_XS_PCS_DIG_STS register to indicate ‘Power Good’, a value of 3’b100.
10. Program bit[4] (SUPRESS_LOS_DET) and bit[6] (RX_DT_EN_CTL) of the VR_XS_PCS_DEBUG_CTRL register to 1.

7.2.5 Programming Sequence for Non-Synopsys PHY

In configurations with non-Synopsys PHY, the PHY should switch among the speed modes based on the xpcs_kr_o, xpcs_kx_kx4n_o, and xpcs_rxaui_mode_o signals. Table 7-5 describes the speed modes and signal dependencies.

Table 7-5 SGMII Speed Modes and Signals Dependencies

Speed mode	xpcs_kr_o[a]	xpcs_kx_kx4n_o[b]	xpcs_rxaui_mode_o[c]	xpcs_link_speed_o[1:0][d]
10G (10GBASE-R PCS or 10GBASE-KR)	1	X	X	XX
10G (XAUI, 10GBASE-X PCS, or 10GBASE-KX4)	0	0	0	XX
10G RXAUI	0	0	1	XX
1G (1000BASE-X, 1000BASE-KX, or SGMII)	0	1	0	10
100 Mbps SGMII	0	1	0	01
10 Mbps SGMII	0	1	0	00

a. xpcs_kr_o is present only in the KR_KX4_KX, KR_KX4, and KR_KX configurations.
b. xpcs_kx_kx4n is present only in the KR_KX4_KX, KR_KX4, and KX4_KX configurations or if SUPPORT_1G = Enabled.
c. xpcs_rxaui_mode_o is present only if RXAUI mode is selected while configuring the DWC_xpcs.
d. xpcs_link_speed_o[1:0] is present only when the SGMII_EN = Enabled.

To switch to 10G XAUI or KX4 mode when using non-Synopsys PHY, complete the following steps:

1. Write 4'b0001 to Bits[3:0] of SR_XS_PCS_CTRL2 Register to set the xpcs_kr_o output to 0.
2. Clear Bit 0 of VR_XS_PCS_XAUI_CTRL Register to 0. This bit controls the xpcs_rxaui_mode_o output.
3. Set Bit 13 of SR_PMA_CTRL1 Register (only for Backplane Ethernet PCS configurations) or SR_X-S_PCS_CTRL1 Register to 1. This bit controls the xpcs_kx_kx4n_o. This bit is inverted to drive the xpcs_kx_kx4n_o output.
4. Set Bit 11 of SR_XS_PCS_CTRL1 Register to 1 to power down DWC_xpcs.

Note	To enable non-standard XAUI link to have (2.5G or 5.0G or 7.5G speed modes using one or two or three lanes respectively) program bit[11:9] and bit[15:13] of VR_XS_PCS_DIG_CTRL2.

5. Wait till Bits[4:2] of VR_XS_PCS_DIG_STS are set to any value other than 3'b100.
6. Clear Bit 11 of SR_XS_PCS_CTRL1 register to 0 to power up DWC_xpcs.

7.3 Switching to 10G RXAUI Mode

In 10G RXAUI mode, the DWC_xpcs to PHY interface works at 20-bit parallel data with the clock rate of 312.5 MHz. Only Lane 0 and Lane 1 are functional. Each serial line works at the baud rate of 6.25 Gbps. The effective data transfer rate, considering the overhead of 8B/10B encoding or decoding, is 5 Gbps in each lane.

You should program the PHY to work at 10G RXAUI mode and also set Bit 0 (XAUI_MODE) of the VR_XS_PCS_XAUI_CTRL Register to 1. When this bit is set to 1, the xpcs_rxaui_mode_o output is asserted and the Synopsys PHY switches the speed or interface to the RXAUI mode.

7.3.1 Programming Sequence for Synopsys Multi-Protocol 6G PHY or Non-Synopsys PHY

To switch to the 10G RXAUI mode when using Synopsys Multi-Protocol 6G PHY or non-Synopsys PHY, complete the following steps:

1. Write 4'b0001 to Bits[3:0] of SR_XS_PCS_CTRL2 Register to set the xpcs_kr_o output to 0.
2. Set Bit 0 of VR_XS_PCS_XAUI_CTRL Register to 1.
3. Set Bit 13 of SR_PMA_CTRL1 register (only for Backplane Ethernet PCS configurations) or SR_X-S_PCS_CTRL1 Register to 1.
4. (Only for Synopsys Multi-Protocol 6G) Program the register bits shown in Table 7-6.

Table 7-6 Synopsys Multi-Protocol 6G PHY: 10G RXAUI Mode Switching

Field	Values
MPLL_MULTIPLIER	Based on the frequency of the reference clock, set the value of this field to the following in VR_XS_PMA_MP_6G_MPLL_CTRL1: ■ 7'h28 for 156.25 MHz reference clock frequency ■ 7'h32 for 125 MHz and 62.5 MHz reference clock frequency ■ 7'h7D for 50 MHz and 25 MHz reference clock frequency
LANE_10BIT_SEL	Set the value of this field to 1'b0 in VR_XS_PMA_MP_6G_MPLL_CTRL0.
TX_RATE_* and RX_RATE_*	Set the values of these fields on all the lanes to be 2'b00 (16'h0000) in VR_XS_PMA_MP_6G_MISC_CTRL1.
REF_CLK_DIV2	Based on the frequency of the reference clock, set the value of this field in VR_XS_PMA_MP_6G_MPLL_CTRL0 to: ■ 1'b1 for 156.25 MHz, 125 MHz and 50 MHz reference clock frequency ■ 1'b0 for 62.5 MHz and 25 MHz reference clock frequency

5. Set Bit 11 of the SR_PMA_CTRL1 register (only for Backplane Ethernet PCS configurations) or SR_X-S_PCS_CTRL1 Register to 1 to power down DWC_xpcs.
6. Wait till Bits[4:2] of VR_XS_PCS_DIG_STS register are set to 3'b110.
7. Clear Bit 11 of SR_PMA_CTRL1 Register (only for Backplane Ethernet PCS configurations) or SR_X-S_PCS_CTRL1 register to 0 to power up DWC_xpcs.
8. Wait till Bits[4:2] of VR_XS_PCS_DIG_STS Register are set to 3'b100.

7.3.2 Programming Sequence for Synopsys Multi-Protocol 12G/16G PHY

To switch to the 10G RXAUI mode, complete the following steps:

1. Set Bit 0 of VR_XS_PCS_XAUI_CTRL register to 1.
2. Write 4'b0001 to Bits[3:0] of SR_XS_PCS_CTRL2 register to set xpcs_kr_o output to 0.
3. Set Bit 13 of SR_PMA_CTRL1 register (only for Backplane Ethernet PCS configurations) or SR_X-S_PCS_CTRL1 Register to 1.
4. Program the register bits shown in Table 7-7 for 12G PHY and Table 7-8 for 16G PHY.

Table 7-7 Synopsys Multi-Protocol 12G PHY: 10G RXAUI Mode Switching

Field	Values
MPLLA_MULTIPLIER	Set the values of this field in the register VR_XS_PMA_MP_12G_16G_MPLLA_CTRL0 to 8’d40
MPPLA_BANDWIDTH	Set the values of this field in the register VR_XS_PMA_MP_12G_MPLLA_CTRL3 based on the PHY Model used: ■ GF 14 LPP: 11'd1923 ■ TSMC 16 FFLL: 11'd83 ■ TSMC 28HPC: 11'd86 Note: These values might not be accurate. See the latest PHY databook corresponding to your version of the PHY and program suitably.
VCO_LD_VAL_{lane}	Set the value of these fields for all the lanes in VR_XS_PMA_MP_12G_16G_25G_VCO_LD {lane} Register 13’d1360
VCO_REF_LD_{lane}	Set the value of these fields for all the lanes in VR_XS_PMA_MP_12G_VCO_CAL_REF 0/1 Registers as follows: ■ 6'd17 for 78.125 MHz Reference Clock ■ 6'd37 for 156.25 MHz/ 312.5 MHz Reference Clock
AFE_EN_0, AFE_EN_3_1, DFE_EN_0, DFE_EN_3_1	Set the value of these fields in VR_XS_PMA_MP_12G_AFE_DFE_EN_CTRL to 0.
CONT_ADAPT_0, CONT_ADAPT_3_1	Set the value of these fields in VR_XS_PMA_MP_12G_16G_25G_RX_EQ_CTRL4 to 0.
TX{lane}_RATE	Set the value of these fields for lanes 0 and 1 to 3'b001 in VR_XS_PMA_MP_12G_16G_25G_TX_RATE_CTRL.
RX{lane}_RATE	Set the value of these fields for all the lanes to 2'b01 in VR_XS_PMA_MP_12G_16G_25G_RX_RATE_CTRL.
TX{lane}_WIDTH	Set the value of these fields for all the lanes to 2'b11 in VR_XS_PMA_MP_12G_16G_TX_GENCTRL2
RX{lane}_WIDTH	Set the value of these fields for all the lanes to 2'b11 in VR_XS_PMA_MP_12G_16G_RX_GENCTRL2

Field	Values
MPLLA_DIV16P5_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 to 0
MPLLA_DIV10_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 to 1
MPLLA_DIV8_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 to 0
TX_CLK_RDY_{lane}	Set the value of this field in VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 0

Table 7-8 Synopsys Multi-Protocol 16G PHY: 10G RXAUI Mode Switching

Field	Values
MPLLA_MULTIPLIER	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL0 register to 8'd40
MPLLA_BANDWIDTH	Set the value of this field in VR_MII_MP_16G_MPLLA_CTRL3 register to 16'd41022
VCO_LD_VAL_{lane}	Set the values of these fields in the following register to 13'd1360 ■ VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD0 ■ VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD1 ■ VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD2 ■ VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD3
VCO_REF_LD_{lane}	Set the values of these fields in the following register as 7'd17: For 78.125 MHz Ref Clock or 7'd34: For 156.25 MHz/312.5 MHz Ref Clock: ■ VR_XS_PMA_MP_16G_25G_VCO_CAL_REF0 ■ VR_XS_PMA_MP_16G_25G_VCO_CAL_REF1
CONT_ADAPT_0, CONT_ADAPT_3_1	Set the value of this field in VR_XS_PMA_MP_12G_16G_25G_RX_EQ_CTRL4 register to 0
TX{lane}_RATE	Set the values of these fields in VR_XS_PMA_MP_12G_16G_25G_TX_RATE_CTRL register to 1
RX{lane}_RATE	Set the values of these fields in VR_XS_PMA_MP_12G_16G_25G_RX_RATE_CTRL register to 1
TX{lane}_WIDTH	Set the values of these fields in VR_XS_PMA_MP_12G_16G_TX_GENCTRL2 register to 3
RX{lane}_WIDTH	Set the values of these fields in VR_XS_PMA_MP_12G_16G_RX_GENCTRL2 register to 3
MPLLA_DIV16P5_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 register to 0
MPLLA_DIV10_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 register to 1
MPLLA_DIV8_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 register to 0

Field	Values
TX_CLK_RDY_0	Set the value of this field in VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 0
VBOOST_EN_0	Set the value of this field in VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 1
VBOOST_EN_3_1	Set the value of this field in VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 7
CTLE_BOOST_{lane}	Set the values of these fields in the following registers to 6 VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL0 VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL1 VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL2 VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL3
VCO_STEP_CTRL_0, VCO_STEP_CTRL_3_1	Set the values of these fields in VR_XS_PMA_MP_16G_RX_CDR_CTRL1 register to 0
VCO_TEMP_COMP_EN_0, VCO_TEMP_COMP_EN_3_1	Set the values of these fields in VR_XS_PMA_MP_16G_RX_CDR_CTRL1 register to 0
RX{lane}_MISC	Set the value of these fields in VR_XS_PMA_MP_16G_25G_RX_MISC_CTRL0 and VR_XS_PMA_MP_16G_25G_RX_MISC_CTRL1 registers to 5
RX_DFE_BYP_0, RX_DFE_BYP_3_1	Set the values of these fields in VR_XS_PMA_MP_16G_RX_GEN_CTRL4 register to 0
VCO_FRQBAND_{lane}	Set the values of these fields in VR_XS_PMA_MP_16G_RX_CDR_CTRL1 register to 2
RX{lane}_DELTA_IQ	Set the values of these fields in the following register to 0 ■ VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL0 ■ VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL1 ■ VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL2 ■ VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL3
RX{lane}_ADAPT_MODE	Set the values of these fields in VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL5 register to 3

Note If the values provided in these tables contradict with the corresponding the values in the PHY data book, the values provided in PHY Databook take precedence.

5. Initiate the Vendor specific software reset by writing 1'b1 to VR_RST bit[15] of VR_XS_PCS_DIG_CTRL1.
6. (Optional) Execute step 4 from the section, "Initializing the DWC_xpcs Controller" on page 1650.
7. Wait for the bit[15] (VR_RST) of the VR_XS_PCS_DIG_CTRL1 to get cleared.

8. Program TX_CLK_RDY {lane} field of VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 1
9. Program TX_EQ_PRE field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL0 register to 0.
10. Program TX_EQ_MAIN field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL0 register to 6'd30.
11. Program TX_EQ_POST field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL1 register to 6'd40.
12. Program TX_EQ_OVR_RIDE field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL1 register to 1 (only for configurations with CL72 support).

7.3.3 Programming Sequence for Synopsys Multi-Protocol 25G PHY

To switch to the RXAUI mode, complete the following steps:

1. Program bit[0] of VR_XS_PCS_XAUI_CTRL register to 1.
2. Write 4'b0001 to bits[3:0] of SR_XS_PCS_CTRL2 register to set xpcs_kr_o output signal to 0.
3. Program bit[13] of the SR_PMA_CTRL1 register (only for Backplane Ethernet PCS configurations) or SR_XS_PCS_CTRL1 register to 1.
4. Program the register bits shown in Table 7-9 for 25G PHY.

Table 7-9 Synopsys Multi-Protocol 25G PHY: RXAUI Mode Switching

Field	Value
MPLLA_MULTIPLIER	Set the value of this field in the VR_XS_PMA_SNPS_MP_25G_MPLLA_CTRL0Register to12'd120
MPLLA_TX_CLK_DIV	Set the value of this field in the VR_XS_PMA_SNPS_MP_25G_MPLLA_CTRL2register to 3
MPLLA_W_CLK_DIV	Set the value of this field in the VR_XS_PMA_SNPS_MP_25G_MPLLA_CTRL0register to1
BW_TH	Set the value of this field in the VR_XS_PMA_SNPS_MP_25G_MPLLA_CTRL1register to 2
VCO_REF_LD_0,1	Set the value of this field in the VR_XS_PMA_SNPS_MP_16G_25G_VCO_CAL_REF0 registerto7'd17if REF_CLK_78 Else 7'd34
VCO_REF_LD_2,3	Set the value of this field in the VR_XS_PMA_SNPS_MP_16G_25G_VCO_CAL_REF1 registerto7'd17if REF_CLK_78. Else 7'd34
VCO_LD_VAL_{lane}	Set the value of this field in the VR_XS_PMA_SNPS_MP_12G_16G_25G_VCO_CAL_LD{lane}registerto13'd1360
TX{lane}_DCC	Set the value of this field in the VR_XS_PMA_SNPS_MP_25G_TX_DCC_CTRLregister to4'd10
TX{lane}_WIDTH	Set the value of this field in the VR_XS_PMA_SNPS_MP_25G_TX_WIDTH_CTRL register to 3
TX{lane}_RATE	Set the value of this field in the VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_RATE_CTRLregister to 0
VBOOST_EN_0	Set the value of this field in the VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_GENCTRL1 register to 1

Field	Value
VBOOST_EN_3_1	Set the value of this field in the VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_GENCTRL1 register to 3'd7
RX{lane}_VCO_CFG	Set the value of this field in the VR_XS_PMA_SNPS_MP_25G_RX_VCO_CFG{lane} register to 12'd259
LF_TH_{lane}	Set the value of this field in the VR_XS_PMA_SNPS_MP_25G_RX_LF_TRSHLD register to 2
RX{lane}_WIDTH	Set the value of this field in the VR_XS_PMA_SNPS_MP_25G_RX_WIDTH_CTRL register to 3
RX{lane}_RATE	Set the value of this field in the VR_XS_PMA_SNPS_MP_12G_16G_25G_RX_RATE_CTRL register to 1
RX_DIV16P5_CLK_EN_0	Set the value of this field in the VR_XS_PMA_SNPS_MP_12G_16G_25G_RX_GENCTRL1 to 0
RX{lane}_ADPT_MODE	Set the value of this field in the VR_XS_PMA_SNPS_MP_16G_25G_RX_EQ_CTRL5 register to 3
RX{lane}_DCC	Set the value of this field in the VR_XS_PMA_SNPS_MP_25G_RX_DCC_CTRL register to 4'd11
RX_DFE_BYP_{lane}	Set the value of this field in the VR_XS_PMA_SNPS_MP_16G_25G_RX_EQ_CTRL4 to 1
VGA1_GAIN_{lane}	Set the value of this field in the VR_XS_PMA_SNPS_MP_16G_25G_RX_EQ_CTRL{lane} to 4
VGA2_GAIN_{lane}	Set the value of this field in the VR_XS_PMA_SNPS_MP_16G_25G_RX_EQ_CTRL{lane} to 4
RX{lane}_ AFE_RATE	Set the value of this field in the VR_XS_PMA_SNPS_MP_25G_RX_AFE_RATE_CTRL to 6
CTLE_BOOST_{lane}	Set the value of this field in the VR_XS_PMA_SNPS_MP_16G_25G_RX_EQ_CTRL{lane} register to 8
RX{lane}_CDR_PPM_MAX	Set the value of this field in the VR_XS_PMA_SNPS_MP_16G_25G_RX_PPM_CTRL0 register to 5'd18

5. Program TX_CLK_RDY_0/TX_CLK_RDY_3_1 field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_-GENCTRL1 register to 0
6. Initiate a Vendor-specific Software Reset by writing 1 to bit [15] of VR_XS_PCS_DIG_CTRL1 register. If you enable SRAM support in the PHY, optionally, execute step 4 from the section, "Initializing the DWC_xpcs Controller" on page 1650.
7. Wait for bit [15] of VR_XS_PCS_DIG_CTRL1 register to get cleared.
8. Program TX_CLK_RDY_0 field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_GENCTRL1 register to 1
9. Program TX_CLK_RDY_3_1 field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_GENCTRL1 register to 3'd7
10. 10.Program TX_EQ_PRE field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_EQ_CTRL0 register to 0

11. 11.Program TX_EQ_MAIN field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_EQ_CTRL0 register to 6'd18
12. Program TX_EQ_POST field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_EQ_CTRL1 register to 24
13. Program TX_EQ_OVR_RIDE field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_EQ_CTRL1 register to 1

7.3.4 Programming Sequence for Synopsys Multi-Protocol 32G PHY

To switch 32G PHY to RXAUI mode, follow these steps:

1. Program 4'b0001 to bits[3:0] of the SR_XS_PCS_CTRL2 register.
2. Program 1 to bit[0] of the VR_XS_PCS_XAUI_CTRL register.
3. Program 1 in the bit[13] of the
   - SR_PMA_CTRL1 register for Backplane Ethernet PCS configurations.
   - SR_XS_PCS_CTRL1 register for other configurations.
4. Program 0 to bit[2] (EN_2_5G_MODE) of the VR_XS_PCS_DIG_CTRL1 register.
5. Program the register bits as shown in Table 7-10, to switch the PHY to RXAUI mode.

Table 7-10 Settings to Configures Multi-Protocol 32G PHY for RXAUI Mode

Register	Field	Values
VR_XS_PMA_MP_32G_RX_CNTX_CTRL0	RX_CNTX_SEL_0 RX_CNTX_SEL_1	Value indicated in the context_sel_ID.txt file of PHY package for RXAUI mode
VR_XS_PMA_MP_32G_TX_CNTX_CTRL0	TX_CNTX_SEL_0 TX_CNTX_SEL_1	
VR_XS_PMA_MP_32G_TX_CM_CNTX_SEL0	CMN_CNTX_0 CMN_CNTX_1	
VR_XS_PMA_MP_25G_RX_WIDTH_CTRL	RX0_WIDTH RX1_WIDTH	3’b011 (20-bit)
VR_XS_PMA_MP_25G_TX_WIDTH_CTRL	TX0_WIDTH TX1_WIDTH	
VR_XS_PMA_MP_32G_LN_LINK_CTRL	LINK_NUM_0 LINK_NUM_1	Program both the fields to any identical and unique value. If bifurcation mode is not used, you can leave all these fields at default value of 0.

6. Program the LPM bit of the SR_XS_PCS_CTRL1 register to 1.
7. Wait for the PSEQ_STATE of the VR_XS_PCS_DIG_STS register to indicate ‘Power Down’, a value of 3’b110.
8. Program the LPM bit of the SR_XS_PCS_CTRL1 register to 0. Program the other bits of this register to values present prior to step 6.

   Note: To avoid assertion of soft-reset do not perform read-modify-write operation.

9. Wait for the PSEQ_STATE of the VR_XS_PCS_DIG_STS register to indicate ‘Power Good’, a value of 3’b100.
10. Program bit[4] (SUPRESS_LOS_DET) and bit[6] (RX_DT_EN_CTL) of the VR_XS_PCS_DEBUG_CTRL register to 1.

7.4 Switching to 1G or KX Mode and 10 or 100 Mbps SGMII Speed

In 1G mode, the DWC_xpcs-PHY interface works at 10-bit parallel data with the clock rate of 125 MHz. Only Lane 0 is functional. The serial lane works at the baud rate of 1.25 Gbps. The effective data transfer rate, considering the overhead of 8B/10B encoding or decoding, is 1 Gbps.

To switch between the 1G and 10G modes, program Bit 13 (SS13) of SR_PMA_CTRL1and SR_XS_PCS_CTRL1 Registers. When this bit is cleared to 1, DWC_xpcs operates in the 1000BASE-X (1G) mode. The output signal xpcs_kx_kx4n_o is asserted when the DWC_xpcs works in this mode.

To switch from SGMII and 1000BASE-X mode and vice-versa, program PCS_MODE field of VR_MII_AN_CTRL register to 2'b10 or 2'b00 respectively.his

7.4.1 Programming Sequence for Synopsys Multi-Protocol 6G PHY

To switch to the 1G or KX mode when using Synopsys Multi-Protocol 6G PHY, complete the following steps:

1. Set Bit 0 of the VR_XS_PCS_XAUI_CTRL Register to 0.
2. Clear Bit 13 of SR_PMA_CTRL1 Register (only for Backplane Ethernet PCS configurations) or SR_X-S_PCS_CTRL1 Register to 0.
3. (Optional) If the SGMII mode is enabled, you can program the Bit 13 and Bit 6 of SR_MII_CTRL Register to set the SGMII speed as shown in the following table:

Table 7-11 SGMII Speed Modes

Speed mode	SS6	SS13
1000 Mbps SGMII	1	0
100 Mbps SGMII	0	1
10 Mbps SGMII	0	0

4. Program the register bits shown in Table 7-12.

Table 7-12 Synopsys Multi-Protocol 6G: 1G or KX Mode Switching

Field	Values
MPLL_MULTIPLIER	Based on the frequency of the reference clock, set the value of this field in VR_XS_PMA_MP_6G_MPLL_CTRL1 or VR_MII_MP_6G_MPLL_CTRL1 to the following: ■ 7'h20 for 156.25 MHz reference clock frequency ■ 7'h28 for 125 MHz and 62.5 MHz reference clock frequency ■ 7'h64 for 50 MHz and 25 MHz reference clock frequency
LANE_10BIT_SEL	Set the value of this field to 1’b1 in VR_XS_PMA_MP_6G_MPLL_CTRL0 or VR_MII_Gen2_Gen4_MPLL_CTRL0
TX_RATE_* and RX_RATE_*	Set the values of these fields on all the lanes to be 2'b10. (16'hAAAA) in VR_XS_PMA_MP_6G_MISC_CTRL1 or VR_MII_MP_6G_MISC_CTRL1.

Field	Values
REF_CLK_DIV2	Based on the frequency of the reference clock, set the value of this field in VR_XS_PMA_MP_6G_MPLL_CTRL0 or VR_MII_MP_6G_MPLL_CTRL0 to: ■ 1'b1 for 156.25 MHz, 125 MHz and 50 MHz reference clock frequency ■ 1'b0 for 62.5 MHz and 25 MHz reference clock frequency

5. Set Bit 11 of SR_PMA_CTRL1 Register (only for Backplane Ethernet PCS configurations) or SR_X-S_PCS_CTRL1 Register to 1 to power down the DWC_xpcs.
6. Wait till Bits[4:2] of VR_XS_PCS_DIG_STS Register are set to 3'b110.
7. Clear Bit 11 of SR_PMA_CTRL1 Register (only for Backplane Ethernet PCS configurations) or SR_X-S_PCS_CTRL1 Register to 0 to power up the DWC_xpcs.
8. Wait till Bits [4:2] of VR_XS_PCS_DIG_STS are set to 3'b100

Note You can also disable the unused transmit lanes by programming bit[7:5] of VR_XS_PCS_DIG_CTRL1 Register and disable the unused receive lanes by programming bit[7:5] of VR_XS_PMA_MP_6G_RXGENCTRL1 or VR_MII_MP_6G_RXGENCTRL1.

7.4.2 Programming Sequence to Switch to KX Mode for Synopsys Multi-Protocol 12G/16G PHY

To switch to 1G or KX mode, complete the following steps:

1. (Only for RXAUI configurations) set Bit 0 of VR_XS_PCS_XAUI_CTRL to 0.
2. Write 4'b0001 to Bits[3:0] of SR_XS_PCS_CTRL2
3. Clear Bit 13 of SR_PMA_CTRL1 (only for Backplane Ethernet PCS configurations) or SR_XS_PCS_C-TRL1 to 0.
4. (Optional) If SGMII mode is enabled, you can program bits 13 and 6 of SR_MII_CTRL Register to set the SGMII speed as shown in Table 7-11.
5. Program the register bits shown in Table 7-13 for 12G PHY or Table 7-14 for 16G PHY.

Table 7-13 Synopsys Multi-Protocol 12G PHY: 1G or KX Mode Switching

Field	Values
MPLLA_MULTIPLIER	Set the value of this field in the register VR_XS_PMA_MP_12G_16G_MPLLA_CTRL0 to 8'd32
MPPLA_BANDWIDTH	Set the value of this field in the register VR_XS_PMA_MP_12G_MPLLA_CTRL3, based on the PHY Model used: ■ GF 14 LPP: 11'd1944 ■ TSMC 16 FFLL: 11'd65 ■ TSMC28HPC: 11'd70 Note: These values might not be accurate. See the latest PHY databook corresponding to your version of the PHY and program suitably.
VCO_LD_VAL_0	Set the value of this field in VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD0 to 13'd1344

Field	Values
VCO_REF_LD_0	Set the value of this field in VR_XS_PMA_MP_12G_VCO_CAL_REF0 as follows: ■ 6'd21 for 78.125 MHz Reference Clock ■ 6'd42 for 156.25 MHz/ 312.5 MHz Reference Clock
AFE_EN_0, DFE_EN_0	Set the value of these fields in VR_XS_PMA_MP_12G_AFE_DFE_EN_CTRL to 0.
CONT_ADAPT_0	Set the value of these fields in VR_XS_PMA_MP_12G_16G_25G_RX_EQ_CTRL4 to 0.
TX0_RATE	Set the value of this field to 3'b011 in VR_XS_PMA_MP_12G_16G_25G_TX_RATE_CTRL
RX0_RATE	Set the value of this field to 2'b11 in VR_XS_PMA_MP_12G_16G_25G_RX_RATE_CTRL
TX0_WIDTH	Set the value of this field in VR_XS_PMA_MP_12G_16G_TX_GENCTRL2 to 2’b01
RX0_WIDTH	Set the value of this field in VR_XS_PMA_MP_12G_16G_RX_GENCTRL2 to 2’b01
MPLLA_DIV16P5_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 to 0
MPLLA_DIV10_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 to 1
MPLLA_DIV8_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 to 0
TX_CLK_RDY_0	Set the value of this field in VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 to 0

Table 7-14 Synopsys Multi-Protocol 16G PHY: 1G or KX Mode Switching

Field	Values
MPLLA_MULTIPLIER	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL0 register to 8'd32
MPLLA_BANDWIDTH	Set the value of this field in VR_XS_PMA_MP_16G_MPLLA_CTRL3 register to 16'd41022
VCO_LD_VAL_0	Set the values of this field in VR_XS_PMA_MP_16G_25G_VCO_CAL_LD0 register to 13'd1344
VCO_REF_LD_0	Set the values of this field as 7'd21: For 78.125 MHz Ref Clock or 7'd42: For 156.25/312.5 MHz Ref Clock in the following register: ■ VR_XS_PMA_MP_16G__25G_VCO_CAL_REF0
CONT_ADAPT_0	Set the value of this field in VR_XS_PMA_MP_12G_16G_25G_RX_EQ_CTRL4 register to 0
TX0_RATE	Set the value of this field in VR_XS_PMA_MP_12G_16G_25G_TX_RATE_CTRL register to 3
RX0_RATE	Set the value of this field in VR_XS_PMA_MP_12G_16G_25G_RX_RATE_CTRL register to 3
TX0_WIDTH	Set the values of this field in VR_XS_PMA_MP_12G_16G_TX_GENCTRL2 register to 1
RX0_WIDTH	Set the values of this field in VR_XS_PMA_MP_12G_16G_RX_GENCTRL2 register to 1

Field	Values
MPLLA_DIV16P5_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 register to 0
MPLLA_DIV10_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 register to 1
MPLLA_DIV8_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 register to 0
VBOOST_EN_0	Set the value of this field in VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 0
CTLE_BOOST_0	Set the values of this field in VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL0 registers to 5'd6
VCO_STEP_CTRL_0	Set the values of this field in VR_XS_PMA_MP_16G_RX_CDR_CTRL1 register to 1
VCO_TEMP_COMP_EN_0	Set the values of this field in VR_XS_PMA_MP_16G_RX_CDR_CTRL1 register to 1
RX0_MISC	Set the value of this field in VR_XS_PMA_MP_16G_25G_RX_MISC_CTRL0 register to 6
RX_DFE_BYP_0	Set the value of this field in VR_XS_PMA_MP_16G_25G_RX_GENCTRL4 register to 1
VCO_FRQBAND_0	Set the value of this field in VR_XS_PMA_MP_16G_RX_CDR_CTRL1 register to 1
RX0_DELTA_IQ	Set the value of this field in VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL0 register to 0
RX0_ADAPT_MODE	Set the value of this field in VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL5 register to 0
TX_CLK_RDY_0	Set the value of this filed in VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 0

Note In 1000BaseX-Only PCS configuration, program the corresponding registers in MII MMD (instead of XS_PMA MMD)

Note If the values provided in these tables contradict with the corresponding values in the PHY data book, the values provided in PHY data-book take precedence.

6. Initiate the Vendor specific software reset by writing 1'b1 to VR_RST bit[15] of VR_XS_PCS_DIG_CTRL1.
7. If you enable SRAM support in the PHY, optionally, execute step 4 from the section, "Initializing the DWC_xpcs Controller" on page 1650.
8. Wait for the bit[15] (VR_RST) of the VR_XS_PCS_DIG_CTRL1 to get cleared.
9. Program TX_CLK_RDY {lane} field of the VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 1.
10. Program TX_EQ_PRE field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL0register to 0.
11. Program TX_EQ_MAIN field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL0 register to 6'd40.
12. Program TX_EQ_POST field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL1 register to 0.

13. Program TX_EQ_OVR_RIDE field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL1 register to 1 (only for configurations with CL72 support)
14. Program bit[4], SUPRESS_LOS_DET field and bit[6], RX_DT_EN_CTL field of the VR_XS_PCS_DEBUG_CTRL register to 0.

7.4.3 Programming Sequence for Multi-Protocol 25G PHY

To switch to 1G or KX mode, complete the following steps:

1. Program 4'b0001 to bits [3:0] of SR_XS_PCS_CTRL2 register.
2. Program bit [0] of VR_XS_PCS_XAUI_CTRL register to 0 (needed only for RXAUI configurations).
3. Program bit [2] (EN_2_5G_MODE) of VR_XS_PCS_DIG_CTRL1 to 0.
4. If SGMII Mode is enabled, you can program SS6 (bit [6]) and SS13 (bit [13]) bits of SR_MII_CTRL register to set the SGMII speed, as shown in Table 7-11.
5. Program MPLLA_MULTIPLIER field of VR_XS_PMA_MP_25G_MPLLA_CTRL0 register to 12'd128
6. Program MPLLA_TX_CLK_DIV field of VR_XS_PMA_MP_25G_MPLLA_CTRL2 register to 4
7. Program MPLLA_W_CLK_DIV field of VR_XS_PMA_MP_25G_MPLLA_CTRL0 register to 1
8. Program BW_TH field of VR_XS_PMA_MP_25G_MPLLA_CTRL1 register to 3
9. Program VCO_REF_LD_0 field of VR_XS_PMA_MP_16G_25G_VCO_CAL_REF0 register to 7'd21
10. Program VCO_LD_VAL_0 field of VR_XS_PMA_MP_12G_16G_25G _VCO_CAL_LD0 register to 13'd1344
11. Program TX0_DCC field of VR_XS_PMA_MP_25G_TX_DCC_CTRL register to 4'd10
12. Program TX0_WIDTH field of VR_XS_PMA_MP_25G_TX_WIDTH_CTRL register to 1
13. Program TX0_RATE field of VR_XS_PMA_MP_12G_16G_25G_TX_RATE_CTRL register to 2
14. Program VBOOST_EN_0 field of VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 0
15. Program RX0_VCO_CFG field of VR_XS_PMA_MP_25G_RX_VCO_CFG0 register to 12'd256
16. Program LF_TH_0 field of VR_XS_PMA_MP_25G_RX_LF_TRSHLD register to 2
17. Program RX0_WIDTH field of VR_XS_PMA_MP_25G_RX_WIDTH_CTRL register to 1
18. Program RX0_RATE field of VR_XS_PMA_MP_12G_16G_25G_RX_RATE_CTRL register to 4
19. Program RX_DIV16P5_CLK_EN_0 field of VR_XS_PMA_MP_12G_16G_25G_RX_GENCTRL1 register to 0
20. Program RX0_ADPT_MODE field of VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL5 register to 0
21. Program RX_DFE_BYP_0 field of VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL4 register to 1
22. Program RX_DFE_TAP1_0 field of VR_XS_PMA_MP_12G_16G_25G_DFE_TAP_CTRL0 register to 0
23. Program VGA1_GAIN_0 field of VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL0 register to 4
24. Program VGA2_GAIN_0 field of VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL0 register to 4
25. Program RX0_AFE_RATE field of VR_XS_PMA_MP_25G_RX_AFE_RATE_CTRL register to 7
26. Program CTLE_BOOST_0 of VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL0 register to 6
27. Program RX0_CDR_PPM_MAX field of VR_XS_PMA_MP_16G_25G_RX_PPM_CTRL0 register to 5'd18
28. Program TX_CLK_RDY_0 field of VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 0

29. Initiate a Vendor-specific Software Reset by writing 1 to bit [15] of VR_XS_PCS_DIG_CTRL1 register.
30. If SRAM support is enabled in the PHY, you can optionally execute step 4 from "Initializing the DWC_xpcs Controller" on page 1650.
31. Wait for bit [15] of VR_XS_PCS_DIG_CTRL1 register to get cleared.
32. Program TX_CLK_RDY_0 field of VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 1.
33. Program TX_EQ_PRE field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL0 register to 0.
34. Program TX_EQ_MAIN field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL0 register to 6'd24.
35. Program TX_EQ_POST field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL1 register to 0.
36. Program TX_EQ_OVR_RIDE field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL1 register to 1.
37. Program bit [4] (SUPRESS_LOS_DET) and bit [6] (RX_DT_EN_CTL) of VR_XS_PCS_DEBUG_CTRL register to 0.

7.4.4 Programming Sequence for Multi-Protocol 32G PHY

Follow these steps:

1. Write 4'b0001 to bits[3:0] of SR_XS_PCS_CTRL2 register.
2. For RXAUI configurations, program bit[0] of the VR_XS_PCS_XAUI_CTRL register to 0.
3. Program bit[2], EN_2_5G_MODE field of theVR_XS_PCS_DIG_CTRL1 to 0.
4. (Optional) If you enable SGMII mode, you can program SS6 (bit[6]) and SS13 (bit[13]) bits of the SR_MII_CTRL register to specify the SGMII speed shown in Table 7-11.
5. Configure the PHY to 1.25Gbps by programming various registers as shown in Table 7-15.

Table 7-15 Synopsys Multi-Protocol 32G PHY for 1.25Gbps Rate

Register	Field	Values
VR_XS_PMA_MP_32G_RX_CNTX_CTRL0	RX_CNTX_SEL_0	Value indicated in context_sel_ID.txt file of PHY package for 1000BASE-KX (10-bit) mode
VR_XS_PMA_MP_32G_TX_CNTX_CTRL0	TX_CNTX_SEL_0	Value indicated in context_sel_ID.txt file of PHY package for 1000BASE-KX (10-bit) mode
VR_XS_PMA_MP_32G_TX_CM_CNTX_SEL0	CM_CNTX_SEL_0	Value indicated in context_sel_ID.txt file of PHY package for 1000BASE-KX (10-bit) mode
VR_XS_PMA_MP_25G_RX_WIDTH_CTRL	RX0_WIDTH	3'b001 (10-bit)
VR_XS_PMA_MP_25G_TX_WIDTH_CTRL	TX0_WIDTH	3'b001 (10-bit)
VR_XS_PMA_MP_32G_LN_LINK_CTRL	LINK_NUM_0	When operating in the bifurcation mode, program to a unique value for each instance of the IP.

6. Program the LPM bit of the SR_XS_PCS_CTRL1 register to 1.
7. Wait for the PSEQ_STATE of the VR_XS_PCS_DIG_STS register to indicate 'Power Down' (3'b110).
8. Program the LPM bit of SR_XS_PCS_CTRL1 register to 0. Program the other bits of this register to values that was available in this register prior to step 6.

   Note: Do not perform read-modify-write to avoid assertion of soft-reset.

9. Wait for the PSEQ_STATE of the VR_XS_PCS_DIG_STS register to indicate 'Power Good' (3'b100).
10. Program bit[4], SUPRESS_LOS_DET field and bit[6], RX_DT_EN_CTL field of the VR_XS_PCS_DEBUG_CTRL register to 0.

7.4.5 Programming Sequence for Non-Synopsys PHY

To switch to 1G or KX mode when using non-Synopsys PHY, complete the following steps:

1. Write 4'b0001 to Bits[3:0] of SR_XS_PCS_CTRL2
2. Clear Bit 0 of VR_XS_PCS_XAUI_CTRL to 0. This bit controls the xpcs_rxaui_mode_o output.
3. Set Bit 13 of SR_PMA_CTRL1 (only for Backplane Ethernet PCS configurations) or SR_XS_PCS_CTRL1 to 0. This bit controls the xpcs_kx_kx4n_o output. This bit is inverted to drive the xpcs_kx_kx4n_o output.
4. (Optional) If the SGMII mode is enabled, you can program Bit 13 and Bit 6 of the SR_MII_CTRL to set the SGMII speed as shown in Table 7-11.
5. Set Bit 11 of SR_XS_PCS_CTRL1 to 1 to power down DWC_xpcs.
6. Wait till Bits[4:2] of VR_XS_PCS_DIG_STS are set to any value other than 3'b100.
7. Clear Bit 11 of the SR_XS_PCS_CTRL1 to 0 to power up DWC_xpcs.

7.5 Switching to 10G/5G KR Speed Mode

When operating in BASE-R/KR mode, DWC_xpcs can operate either at 10G or at 5G speed. Because of the 64B/66B encoding or decoding, the serial baud-rates are 10.3125Gbps or 5.15625Gbps respectively. To configure DWC_xpcs, program Bits[3:0] of the SR_XS_PCS_CTRL2 register as follows:

- To operate at 10G BASE-R/KR mode, write 4'b0000
- To operate at 5G BASE-R/KR mode, write 4'b1111

To configure the PHY for the required speed mode, program the PHY related registers accordingly.

7.5.1 Programming Sequence for Synopsys Multi-Protocol 12G/16G PHY

To switch to the 10G/5G (K)R mode when using Synopsys multi-protocol 5 12G/16G PHY, complete the following steps:

1. Write 4'b0000 (for 10G) or 4'b1111 (for 5G) to Bits[3:0] of SR_XS_PCS_CTRL2 register.
2. For USXGMII configurations, program bit[9], USXG_EN field of the VR_XS_PCS_DIG_CTRL1 register to 0.
3. Program the register bits shown in Table 7-16 for 12G PHY or Table 7-17 for 16G PHY.

Table 7-16 Synopsys Multi-Protocol 12G PHY: 10G/5G (K)R Mode Speed Switching

Field	Values
MPLLA_MULTIPLIER	Set the values of this field in the register VR_XS_PMA_MP_12G_M16G_PLLA_CTRL0 to 8’d33
MPLLA_BANDWIDTH	Set the values of this field in the register VR_XS_PMA_MP_12G_MPLLA_CTRL3 based on the PHY Model used: ■ GF 14 LPP: 11’d1936 ■ TSMC 16 FFLL: 11’d65 ■ TSMC 28HPC: 11'd4 Note: These values might not be accurate. See the latest PHY data book corresponding to your version of the PHY and program suitably.
VCO_LD_VAL_0	Set the value of this field in VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD0 as follows: ■ 13’d1386 for 78.125 MHz Reference Clock ■ 13'd1353 for 156.25 MHz/ 312.5 MHz Reference Clock
VCO_REF_LD_0	Set the value of these fields in VR_XS_PMA_MP_12G_VCO_CAL_REF0 Register as follows: ■ 6'd21 for 78.125 MHz Reference Clock ■ 6'd41 for 156.25 MHz/ 312.5 MHz Reference Clock
AFE_EN_0, DFE_EN_0	Set the value of these fields in VR_XS_PMA_MP_12G_AFE_DFE_EN_CTRL Register to 1.
CONT_ADAPT_0	Set the value of these fields in VR_XS_PMA_MP_12G_16G_25G_RX_EQ_CTRL4 to 1.

Field	Values
TX0_RATE	Set the value of this field to 3'b000 (for 10G) or 3'b001 (for 5G) or 3'b010 (for 2.5G SXGMII mode) in VR_XS_PMA_MP_12G_16G_25G_TX_RATE_CTRL
RX0_RATE	Set the value of this field to 2'b00 (for 10G) or 2'b01 (for 5G) or 2'b10 (for 2.5G) in VR_XS_PMA_MP_12G_16G_25G_RX_RATE_CTRL
TX0_WIDTH	Set the value of this field in VR_XS_PMA_MP_12G_16G_TX_GENCTRL2 as follows: ■ 2'b11 for 20-bit Gear Box Interface ■ 2'b10 for 16-bit Gear Box Interface
RX0_WIDTH	Set the value of this field in VR_XS_PMA_MP_12G_16G_RX_GENCTRL2 as follows: ■ 2'b11 for 20-bit Gear Box Interface ■ 2'b10 for 16-bit Gear Box Interface
MPLLA_DIV16P5_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 to 1
MPLLA_DIV10_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 as follows: ■ 1 for 20-bit Gear Box Interface ■ 0 for 16-bit Gear Box Interface
MPLLA_DIV8_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 as follows: ■ 0 for 20-bit Gear Box Interface ■ 1 for 16-bit Gear Box Interface
TX_CLK_RDY_0	Program this field of VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 0.

Table 7-17 Synopsys Multi-Protocol 16G PHY: 10G/5G (K)R Mode Switching

Field	Values
MPLLA_MULTIPLIER	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL0 register to 8'd33
MPLLA_BANDWIDTH	Set the value of this field in VR_XS_PMA_16G_MPLLA_CTRL3 register to 16'd41022
VCO_LD_VAL_0	Set the values of this field in VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD0 register as follows: ■ 13'd1386: For 78.125 MHz Ref Clock ■ 13'd1353: For 156.25/312.5 MHz Ref Clock
VCO_REF_LD_0	Set the values of this field in VR_XS_PMA_MP_16G_25G_VCO_CAL_REF0 register as follows: ■ 7'd21: For 78.125 MHz Ref Clock ■ 7'd41: For 156.25/312.5 MHz Ref Clock
CONT_ADAPT_0	Set the value of this field in VR_XS_PMA_12G_16G_25G_RX_EQ_CTRL4 register to 1
TX0_RATE	Set the values of this field in VR_XS_PMA_MP_12G_16G_25G_TX_RATE_CTRL register to 0 (for 10G) or 1 (for 5G) or 2 (for 2.5G)
RX0_RATE	Set the values of these fields in VR_XS_PMA_MP_12G_16G_25G_RX_RATE_CTRL register to 0 (for 10G) or 1 (for 5G) or 2 (for 2.5G)

Field	Values
TX0_WIDTH	Set the values of this field in VR_XS_PMA_MP_12G_16G_TX_GENCTRL2 register as follows: ■ 2'b10 for 16-bit PHY interface ■ 2'b11 for 20-bit PHY interface
RX0_WIDTH	Set the values of this field in VR_XS_PMA_MP_12G_16G_RX_GENCTRL2 register as follows: ■ 2'b10 for 16-bit PHY interface ■ 2'b11 for 20-bit PHY interface
MPLLA_DIV16P5_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 register to 1
MPLLA_DIV10_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 register as follows: ■ 1 for 20-bit PHY interface ■ 0 for 16-bit PHY interface
MPLLA_DIV8_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 register as follows: ■ 0 for 20-bit PHY interface ■ 1 for 16-bit PHY interface
VBOOST_EN_0	Set the value of this field in VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 1
CTLE_BOOST_0	Set the values of this field in VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL0 registers to 5'd10
VCO_STEP_CTRL_0	Set the values of this field in VR_XS_PMA_MP_16G_RX_CDR_CTRL1 register to 1
VCO_TEMP_COMP_EN_0	Set the values of this field in VR_XS_PMA_MP_16G_RX_CDR_CTRL1 register to 1
RX0_MISC	Set the value of this field in VR_XS_PMA_MP_16G_25G_RX_MISC_CTRL0 register to 2
RX_DFE_BYP_0	Set the value of this field in VR_XS_PMA_MP_16G_25G_RX_GENCTRL4 register to 0
VCO_FRQBAND_0	Set the value of this field in VR_XS_PMA_MP_16G_RX_CDR_CTRL1 register to 1
RX0_DELTA_IQ	Set the value of this field in VR_XS_PMA_MP_16G_25G_RX_IQ_CTRL0 register to 3
RX0_ADAPT_MODE	Set the value of this field in VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL5 register to 3
TX_CLK_RDY_0	Program this field of VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1registerto0.



If the values in these tables contradict with the corresponding values in the PHY data book, the values provided in PHY data-book takes precedence.

4. Initiate the Vendor specific software reset by writing 1'b1 to VR_RST bit[15] of VR_XS_PCS_DIG_CTRL1
5. If you enable SRAM support in the PHY, optionally execute step 4 from the section, "Initializing the DWC_xpcs Controller" on page 1650.
6. Wait for the bit[15] (VR_RST) of the VR_XS_PCS_DIG_CTRL1 to get cleared.
7. Program TX_CLK_RDY_0 field of VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 1
8. Program bit[4](SUPRESS_LOS_DET) and bit[6] (RX_DT_EN_CTL) of VR_XS_PCS_DEBUG_CTRL to 1.
9. Wait for RX_VALID_0 bit of VR_XS_PMA_RX_LSTS register to indicate 1.
10. Program RX_AD_REQ bit of VR_XS_PMA_MP_12G_16G_25G_RX_EQ_CTRL4 register to 1.
11. Poll for bit[8] of VR_PMA_PHY_RX_EQ_CEU register (for configurations with CL72_EN=Enabled) or bit [12] of VR_XS_PMA_MP_12G_16G_25G_MISC_STS register (for configurations with CL72_EN=Disabled) to indicate 1.
12. Program the RX_AD_REQ bit of VR_XS_PMA_MP_12G_16G_25G_RX_EQ_CTRL4 register to 0.

7.5.2 Programming Sequence for Synopsys Multi-Protocol 25G PHY

To switch to 10G/5G (K)R mode complete the following steps:

1. Program 4'b0000 to bits [3:0] of SR_XS_PCS_CTRL2 register.
2. For USXGMII configurations, program bit[9], USXG_EN field of the VR_XS_PCS_DIG_CTRL1 register to 0.
3. Program MPLLA_MULTIPLIER field of VR_XS_PMA_MP_25G_MPLLA_CTRL0 register to 12'd132.
4. Program MPLLA_DIV16P5_EN field of VR_XS_PMA_MP_25G_MPLLA_CTRL2 register to 1.
5. Program MPLLA_TX_CLK_DIV field of VR_XS_PMA_MP_25G_MPLLA_CTRL2 register to 1.
6. Program MPLLA_W_CLK_DIV field of VR_XS_PMA_MP_25G_MPLLA_CTRL0 register to 2.
7. Program BW_TH field of VR_XS_PMA_MP_25G_MPLLA_CTRL1 to 3.
8. Program VCO_REF_LD_0 field of VR_XS_PMA_MP_16G_25G_VCO_CAL_REF0 register to 7'd21.
9. Program VCO_LD_VAL_0 field of VR_XS_PMA_MP_12G_16G_25G _VCO_CAL_LD0 register to 13'd1386.
10. Program TX0_DCC field of VR_XS_PMA_MP_25G_TX_DCC_CTRL register to 4'd11.
11. Program TX0_WIDTH of VR_XS_PMA_MP_25G_TX_WIDTH_CTRL register to 4 (32-bit interface).
12. Program TX0_RATE field of VR_XS_PMA_MP_12G_16G_25G_TX_RATE_CTRL register to 1 (for 10.3125Gbps) or 2 (for 5.15625Gbps) or 3 (for 2.578125Gbps).
13. Program VBOOST_EN_0 field of VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 1.
14. Program RX0_VCO_CFG field of VR_XS_PMA_MP_25G_RX_VCO_CFG0 to 12'd256.
15. Program LF_TH_0 field of VR_XS_PMA_MP_25G_RX_LF_TRSHLD register to 7.
16. Program RX0_WIDTH field of VR_XS_PMA_MP_25G_RX_WIDTH_CTRL register to 4 (32-bit interface).
17. Program RX0_RATE field of VR_XS_PMA_MP_12G_16G_25G_RX_RATE_CTRL register to 1 (for 10.3125Gbps) or 2 (for 5.15625Gbps) or 3 (for 2.578125Gbps).
18. Program RX_DIV16P5_CLK_EN_0 field of VR_XS_PMA_MP_12G_16G_25G_RX_GENCTRL1 to 1.
19. Program RX0_ADPT_MODE field of VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL5 register to 3.

20. Program RX_DFE_BYP_0 field of VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL4 register to 0.
21. Program RX_DFE_TAP1_0 field of VR_XS_PMA_MP_12G_16G_25G_DFE_TAP_CTRL0 register to 8'd10
22. Program VGA1_GAIN_0 field of VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL0 register to 5
23. Program VGA2_GAIN_0 field of VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL0 register to 5
24. Program RX0_AFE_RATE field of VR_XS_PMA_MP_25G_RX_AFE_RATE_CTRL register to 4
25. Program CTLE_BOOST_0 field of VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL0 register to 5'd16
26. Program RX0_CDR_PPM_MAX field of VR_XS_PMA_MP_16G_25G_RX_PPM_CTRL0 register to 5'd18
27. Program TX_CLK_RDY_0 field of VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 0
28. Initiate a Vendor-specific Software Reset by writing 1 to bit [15] of VR_XS_PCS_DIG_CTRL1 register.
29. If you enable SRAM support in the PHY, optionally perform step 4 of "Initializing the DWC_xpcs Controller" on page 1650.
30. Wait for bit [15] of VR_XS_PCS_DIG_CTRL1 register to get cleared.
31. Program TX_CLK_RDY_0 field of VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 1
32. Program bit[4](SUPRESS_LOS_DET) and bit[6] (RX_DT_EN_CTL) of VR_XS_PCS_DEBUG_CTRL to 1.
33. Wait for RX_VALID_0 bit of VR_XS_PMA_RX_LSTS register to indicate 1.
34. Program RX_AD_REQ bit of VR_XS_PMA_MP_12G_16G_25G_RX_EQ_CTRL4 register to 1.
35. Poll for bit[8] of VR_PMA_PHY_RX_EQ_CEU register (for configurations with CL72_EN=Enabled) or bit [12] of VR_XS_PMA_MP_12G_16G_25G_MISC_STS register (for configurations with CL72_EN=Disabled) to indicate 1.
36. Program RX_AD_REQ bit of VR_XS_PMA_MP_12G_16G_25G_RX_EQ_CTRL4 register to 0.

7.5.3 Programming Sequence for Synopsys Multi-Protocol 32G PHY

Follow these steps:

1. Program bits[3:0] of SR_XS_PCS_CTRL2 register as follows:
   - 4'b0000 for 10G
   - 4'b1111 for 5G
2. For USXGMII configurations, program bit[9], USXG_EN field of the VR_XS_PCS_DIG_CTRL1 register to 0.
3. Program bit[2], EN_2_5G_MODE of the VR_XS_PCS_CTRL1 register to 0.
4. Configure the PHY to operate at 10GBASE-(K)R mode by programming the registers as shown in Table 7-18.

Table 7-18 Configuring PHY for 10GBASE-(K)R Mode

Register	Field	Values
VR_XS_PMA_MP_32G_RX_CNTX_CTRL0	RX_CNTX_SEL_0	Value indicated in context_sel_ID.txt file of PHY package for 10GBASE-(K)R (32-bit)/ 5GBASE-(K)R (32-bit) mode.

Register	Field	Values
VR_XS_PMA_MP_32G_TX_CNTX_CTRL0	TX_CNTX_SEL_0	Value indicated in context_sel_ID.txt file of PHY package for 10GBASE-(K)R (32-bit)/5GBASE-(K)R (32-bit) mode.
VR_XS_PMA_MP_32G_TX_CM_CNTX_SEL0	CM_CNTX_SEL_0	Value indicated in context_sel_ID.txt file of PHY package for 10GBASE-(K)R (32-bit)/5GBASE-(K)R (32-bit) mode.
VR_XS_PMA_MP_25G_RX_WIDTH_CTRL	RX0_WIDTH	3'b100 (32-bit)
VR_XS_PMA_MP_25G_TX_WIDTH_CTRL	TX0_WIDTH	3'b100 (32-bit)
VR_XS_PMA_MP_32G_LN_LINK_CTRL	LINK_NUM_0	When operating in the bifurcation mode, program to a unique value for each instance of the IP.

5. Program LPM bit of the SR_XS_PCS_CTRL1 register to 1.
6. Wait for the PSEQ_STATE of the VR_XS_PCS_DIG_STS register to indicate 'Power Down' (3'b110).
7. Program the LPM bit of the SR_XS_PCS_CTRL1 register to 0. Program the other bits of this register to values that was available in this register prior to step 5.

   Note: To avoid assertion of soft-reset, do not perform read-modify-write.
8. Wait for the PSEQ_STATE of the VR_XS_PCS_DIG_STS register to indicate 'Power Good' (3'b100).
9. Program bit[4], SUPRESS_LOS_DET field and bit[6], RX_DT_EN_CTL field of the VR_XS_PCS_DEBUG_CTRL register to 1.

7.5.4 Programming Sequence When Using Non-Synopsys PHY

To switch to the 10G/5G (K)R mode when using non-Synopsys PHY, complete the following steps:

1. Write 4'b0000 (for 10G) or 4'b1111 (for 5G) to Bits[3:0] of SR_XS_PCS_CTRL2 register.
2. For USXGMII configurations, program bit[9], USXG_EN field of the VR_XS_PCS_DIG_CTRL1 register to 0.
3. Set bit 11 of SR_XS_PCS_CTRL1 to power down DWC_xpcs.
4. After arbitrary delay, clear bit 11 of the SR_XS_PCS_CTRL1 to 0 to power up DWC_xpcs.

7.6 Switching to USXGMII Mode

USXGMII Mode is based on BASE-R/KR PCS. The serial baud rate can be 10.3125Gbps or 5.15625 Gbps or 2.578125Gbps. Therefore, to operate in USXGMII mode, follow these steps:

1. Write 4'b0000 to bits[3:0] of SR_XS_PCS_CTRL2 register to switch DWC_xpcs to BASE-R mode.
2. Program bit[9] (USXG_EN) of VR_XS_PCS_DIG_CTRL1 register to 1 to enable USXGMII Mode inside DWC_xpcs.
3. Select the appropriate USXGMII mode by programming USXG_MODE field of VR_XS_PCS_KR_CTRL register to suitable values.
4. Program the PHY to operate at 10.3125Gbps/5.15625Gbps/2.578125Gbps.

   For Synopsys multi-protocol 12G PHY, see Table 7-16 on page 1676. For Synopsys multi-protocol 16G PHY, see Table 7-17. For Synopsys multi-protocol 25G PHY, execute steps 3 to 27 of "Programming Sequence for Synopsys Multi-Protocol 25G PHY" on page 1679. For multi-protocol 32G PHY, program the registers mentioned in Table 7-19.

Table 7-19 Settings to Configure Multi-Protocol 32G PHY for USXGMII Mode

Register	Field	Values
VR_XS_PMA_MP_32G_RX_CNTX_CTRL0	RX_CNTX_SEL_0	Value indicated in context_sel_ID.txt file of PHY package for 10G-SXGMII/5G-SXGMII.
VR_XS_PMA_MP_32G_TX_CNTX_CTRL0	TX_CNTX_SEL_0	
VR_XS_PMA_MP_32G_TX_CM_CNTX_SEL0	CM_CNTX_SEL_0	
VR_XS_PMA_MP_25G_RX_WIDTH_CTRL	RX0_WIDTH	3'b100 (32-bit)
VR_XS_PMA_25G_TX_WIDTH_CTRL	TX0_WIDTH	
VR_XS_PMA_MP_32G_LN_LINK_CTRL	LINK_NUM_0	Program to a unique value for each instance of the IP when operating in bifurcation mode.

5. Based on the PHY, follow these steps:
   - Multi-protocol 12G/16G PHY:

     Execute steps 4 to 12 in "Programming Sequence for Synopsys Multi-Protocol 12G/16G PHY" on page 1676.
   - Multi-protocol 25G PHY:

     Execute steps 28 to 36 in "Programming Sequence for Synopsys Multi-Protocol 25G PHY" on page 1679.
   - Multi-protocol 32G PHY:

     Execute steps 5 to 9 in "Programming Sequence for Synopsys Multi-Protocol 32G PHY" on page 1680.
   - Non-Synopsys PHY:

     Execute steps 3 and 4 in "Programming Sequence When Using Non-Synopsys PHY" on page 1681.
6. In Backplane Ethernet PCS configurations, program bit[12] (AN_EN) of SR_AN_CTRL Register to 0 and bit[12] (CL37_BP) of VR_XS_PCS_DIG_CTRL1 Register to 1.
7. Program various bits of VR_MII_AN_CTRL Register as follows:

- MII_AN_INTR_EN to 1, to enable auto-negotiation complete interrupt (optional step).
- MII_CTRL to 0 or 1, based on your MAC capability.
- TX_CONFIG to 1 (PHY side USXGMII) or 0 (MAC side USXGMII) based on your requirement.
- If TX_CONFIG is set to 1, program SGMII_LINK_STS bit to a suitable value to indicate the link status to the MAC side of USXGMII link.

8. (Optional step) Duration of link timer can be changed (default setting corresponds to 1.6ms) by programming VR_MII_LINK_TIMER_CTRL Register suitably and by setting bit [3] of VR_MII_DIG_CTRL1 Register to 1.
9. If DWC_xpcs is configured as PHY-side USXGMII, program SS5, SS6 and SS13 bits of SR_MII_CTRL Register to the desired speed mode. The value programmed to this register gets advertised to the MAC during auto-negotiation.
10. Enable Clause 37 auto-negotiation by programming bit [12] of SR_MII_CTRL Register to 1.
11. (If interrupt has been enabled) After the completion of auto-negotiation, DWC_xpcs generates an interrupt on sbd_intr_o port.
12. Read the auto-negotiation status register (VR_MII_AN_INTR_STS Register). Bit [0] is set to indicate that auto-negotiation is complete. Bits [14:8] indicate the link-speed, duplex mode, EEE capability and EEE clock-stop capability indicated by the link partner (PHY chip).
13. Clear the Interrupt by writing 0 to bit [0] of VR_MII_AN_INTR_STS Register.
14. Program SS13, SS6 and SS5 bits of SR_MII_CTRL Register to configure DWC_xpcs to the USXGMII speed mode (for port 0) indicated by PHY in step 12. The values programmed to these bits will reflect in the output port xpcs_usxg_speed_o. This step is required only if DWC_xpcs is configured as MAC-side USXGMII.
15. Wait for some time, 1 micro second, so that XGMII clocks (clk_xgmii_tx_i/clk_xgmii_rx_i/clk_xgmii_tx_p{1,2,3}_i/clk_xgmii_rx_p{1,2,3}_i) get stabilized at the desired frequencies.
16. Program USRA_RST (bit [10]) of VR_XS_PCS_DIG_CTRL1 Register to 1 and wait for it to get self-cleared. For multi-port configurations, program the following bits too (and wait for it to get self-cleared).
    - USRA_RST bit of VR_MII_1_DIG_CTRL1 (for Port 1)
    - USRA_RST bit of VR_MII_2_DIG_CTRL1 (for Port 2)
    - USRA_RST bit of VR_MII_3_DIG_CTRL1 (for Port 3)

DWC_xpcs is now ready to operate in the desired USXGMII speed mode.

7.7 Programming Guidelines for QSGMII Mode

7.7.1 Programming Sequence with Synopsys Multi-Protocol 12G/16G PHY

To switch to QSGMII mode, perform the following steps:

1. Write 4'b0001 to Bits [3:0] of SR_XS_PCS_CTRL2 Register (Optional- Required only if the configuration has KR/10GBASE-R support).
2. Clear Bit 13 of SR_PMA_CTRL1 Register (only for Backplane Ethernet PCS configurations) or SR_X-S_PCS_CTRL1 to 0.
3. Program bits [2:1] of VR_MII_AN_CTRL Register to 2'b11.
4. Program the register bits as shown in Table 7-20 for 12G PHY or Table 7-21 for 16G PHY, to switch the PHY to QSGMII mode.

Table 7-20 Synopsys Multi-Protocol 12G PHY: QSGMII Mode Switching

Field	Values
MPLLA_MULTIPLIER	Set the value of this field in the register VR_XS_PMA_MP_12G_16G_MPLLA_CTRL0 or VR_MII_MP_12G_16G_MPLLA_CTRL0 Register to 8'd32.
MPLLA_BANDWIDTH	Set the value of this field in the register VR_XS_PMA_MP_12G_MPLLA_CTRL3 or VR_MII_MP_12G_MPLLA_CTRL3 Register to a suitable value after referring to PHY Databook.
VCO_LD_VAL_0	Set the value of this field in VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD0 or VR_MII_MP_12G_16G_25G_VCO_CAL_LD0 Register to a suitable value after referring to PHY Databook.
VCO_REF_LD_0	Set the value of this field in VR_XS_PMA_MP_12G_VCO_CAL_REF0 or VR_MII_MP_12G_VCO_CAL_REF0 Register to a suitable value after referring to PHY Databook.
AFE_EN_0, DFE_EN_0	Set the value of these fields in VR_XS_PMA_MP_12G_AFE_DFE_EN_CTRL orVR_MII_MP_12G_AFE_DFE_EN_CTRL Register to 1.
CONT_ADAPT_0	Set the value of these fields in VR_XS_PMA_MP_12G_16G_25G_RX_EQ_CTRL4 or VR_MII_MP_12G_16G_25G_RX_EQ_CTRL4 Register to 0.
TX0_RATE	Set the value of this field to 3'b001 in VR_XS_PMA_MP_12G_16G_25G_TX_RATE_CTRL or VR_MII_MP_12G_16G_25G_TX_RATE_CTRL Register.
RX0_RATE	Set the value of this field to 2'b01 in VR_XS_PMA_MP_12G_16G_25G_RX_RATE_CTRL or VR_MII_MP_12G_16G_25G_RX_RATE_CTRL Register.
TX0_WIDTH	Set the value of this field in VR_XS_PMA_MP_12G_16G_TX_GENCTRL2 or VR_MII_MP_12G_16G_TX_GENCTRL2 to 2'b11.
RX0_WIDTH	Set the value of this field in VR_XS_PMA_MP_12G_16G_RX_GENCTRL2 or VR_MII_MP_12G_16G_RX_GENCTRL2 to 2'b11.

Field	Values
MPLLA_DIV16P5_CLK_ EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 or VR_MII_MP_12G_16G_MPLLA_CTRL2 to 0.
MPLLA_DIV10_CLK_ EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 or VR_MII_MP_12G_16G_MPLLA_CTRL2 Register to 1.
MPLLA_DIV8_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 or VR_MII_MP_12G_16G_MPLLA_CTRL2 Register to 0.
TX_CLK_RDY_0	Program this field of VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 0.

Table 7-21 Synopsys Multi-Protocol 16G PHY: QSGMII Mode Switching

Field	Values
MPLLA_MULTIPLIER	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL0 register to 8'd32
MPLLA_BANDWIDTH	Set the value of this field in VR_MII_MP_16G_MPLLA_CTRL3 register to 16'd41022
VCO_LD_VAL_0	Set the values of this field in VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD0 register to 13'd1344
VCO_REF_LD_0	Set the values of this field in VR_XS_PMA_MP_16G_25G_VCO_CAL_REF0 register as follows: ■ 7'd21: For 78.125 MHz Ref Clock ■ 7'd42: For 156.25/312.5 MHz Ref Clock
CONT_ADAPT_0	Set the value of this field in VR_XS_PMA_MP_12G_16G_25G_RX_EQ_CTRL4 register to 0
TX0_RATE	Set the values of this field in VR_XS_PMA_MP_12G_16G_25G_TX_RATE_CTRL register to 1
RX0_RATE	Set the values of these fields in VR_XS_PMA_MP_12G_16G_25G_RX_RATE_CTRL register to 1
TX0_WIDTH	Set the values of this field in VR_XS_PMA_MP_12G_16G_TX_GENCTRL2 register to 3
RX0_WIDTH	Set the values of this field in VR_XS_PMA_MP_12G_16G_RX_GENCTRL2 register to 3
MPLLA_DIV16P5_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 register to 0
MPLLA_DIV10_CLK_ EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 register to 1
MPLLA_DIV8_CLK_EN	Set the value of this field in VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 register to 0
VBOOST_EN_0	Set the value of this field in VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 0

Field	Values
CTLE_BOOST_0	Set the values of this field in VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL0 registers to 6
VCO_STEP_CTRL_0	Set the values of this field in VR_XS_PMA_MP_16G_RX_CDR_CTRL1 register to 1
VCO_TEMP_COMP_EN_0	Set the values of this field in VR_XS_PMA_MP_16G_RX_CDR_CTRL1 register to 1
RX0_MISC	Set the value of this field in VR_XS_PMA_MP_16G_RX_MISC_CTRL0 register to 4
RX_DFE_BYP_0	Set the value of this field in VR_XS_PMA_MP_16G_RX_GEN_CTRL4 register to 1
VCO_FRQBAND_0	Set the value of this field in VR_XS_PMA_MP_16G_RX_CDR_CTRL1 register to 1
RX0_DELTA_IQ	Set the value of this field in VR_XS_PMA_MP_16G_RX_IQ_CTRL0 register to 0
RX0_ADPT_MODE	Set the value of this field in VR_XS_PMA_MP_16G_25G_RX_EQ_CTRL5 register to 0
TX_CLK_RDY_0	Program the value of this field in VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 0.

Note If your configuration is 1000BaseX-Only PCS configuration, program the corresponding registers in MII MMD (instead of XS_PMA MMD).

5. Initiate the Vendor specific software reset by writing 1'b1 to VR_RST bit [15] of VR_XS_PCS_DIG_CTRL1 Register.
6. If you enable SRAM support in the PHY, optionally perform step 4 of "Initializing the DWC_xpcs Controller" on page 1650.
7. Wait for the bit [15] (VR_RST) of the VR_XS_PCS_DIG_CTRL1 Register to get cleared.
8. Program TX_CLK_RDY{lane} filed of VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 1.
9. Program TX_EQ_PRE field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL0 register to appropriate value.
10. Program TX_EQ_MAIN field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL0 register to appropriate value.
11. Program TX_EQ_POST field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL1 register to appropriate value
12. Program TX_EQ_OVR_RIDE field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL1 register to 1

DWC_xpcs is now ready to operate in QSGMII mode.

Note

1. The speed mode (1000Mbps, 100Mbps, 10Mbps) of the 4 ports can controlled by programming Bit [13] (SS13) & Bit [6] (SS6) of SR MII MMD/ SR MII_1 MMD/ SR MII_2 MMD/ SR MII_3 MMD Control Register to the required value.
2. When operating at 100M/10M speed mode, width of the GMII ports is determined by bit [0] of VR MII MMD/ VR MII_1 MMD/ VR MII_2 MMD/ VR MII_3 MMD AN Control Register.
3. The rxX_los signal (used by DWC_xpcs to infer loss/presence of signal) from Enterprise Gen5 12G/16G PHY might be unreliable in QSGMII Mode. In such cases, program bit[4](SUPRESS_LOS_DET) and bit[6] (RX_DT_EN_CTL) of VR_XS_PCS_DEBUG_CTRL Register to 1 to ensure proper operation of DWC_xpcs.

7.7.2 Programming Sequence when Using Non-Synopsys PHY

To switch to QSGMII mode, perform the following steps:

1. Write 4'b0001 to Bits [3:0] of SR_XS_PCS_CTRL2 Register (Optional- Required only if the configuration has KR/10GBASE-R support).
2. Clear Bit [13] of SR_PMA_CTRL1 Register (for Backplane Ethernet PCS configurations) or SR_XS_PCS_CTRL1 Register (for other configurations) to 0.
3. Program bits [2:1] of VR_MII_AN_CTRL Register to 2'b11. This asserts 'xgxs_qsgmii_mode_o' port of DWC_xpcs.
4. Set Bit [11] of SR_XS_PCS_CTRL1 Register (or SR_MII_CTRL Register for 1000BaseX-Only PCS configurations) to 1 to power down DWC_xpcs.
5. Wait till Bits[4:2] of VR_XS_PCS_DIG_STS Register (or VR_MII_DIG_STS Register for 1000BaseX-Only PCS configurations) are set to any value other than 3'b100.
6. Clear Bit [11] of the SR_XS_PCS_CTRL1 Register (or SR_MII_CTRL Register for 1000BaseX-Only PCS configurations) to 0 to power up DWC_xpcs.

7.7.3 Programming Sequence for Synopsys Multi-Protocol 6G PHY

To switch to QSGMII mode, perform the following steps:

1. (Only for RXAUI configurations) Set Bit 0 of VR_XS_PCS_XAUI_CTRL Register to 0).
2. Clear Bit [13] of SR_PMA_CTRL1 Register (for Backplane Ethernet PCS configurations) or SR_XS_PCS_CTRL1 Register to 0.
3. Program bits [2:1] of VR_MII_AN_CTRL Register to 2'b11.
4. Program the register bits as shown in the following table to switch the PHY to QSGMII mode.

Table 7-22 Synopsys Multi-Protocol 6G PHY: QSGMII Mode Switching

Field	Values
MPLL_MULTIPLIER	Based on the frequency of the reference clock, set the value of this field to the following in VR_XS_PMA_MP_6G_MPLL_CTRL1or VR_MII_MP_6G_MPLL_CTRL1 to 8'd32. ■ 7'h20 for 156.25 MHz reference clock frequency ■ 7'h28 for 125 MHz reference clock frequency
LANE_10BIT_SEL	Set the value of this field to 1'b0 in VR_XS_PMA_MP_6G_MPLL_CTRL0or VR_MII_MP_6G_MPLL_CTRL0.

Field	Values
TX_RATE_* and RX_RATE_*	Set the value of these fields for all the lanes to 2'b00 in VR_XS_PMA_MP_6G_MISC_CTRL1or VR_MII_MP_6G_MISC_CTRL1.
REF_CLK_DIV2	Set the value of this field in VR_XS_PMA_Gen2_Gen4_MPLL_CTRL0 or VR_MII_MP_6G_MPLL_CTRL0 register to 1.

5. Set Bit 11 of the SR_PMA_CTRL1 Register (only for Backplane Ethernet PCS configurations) or SR_X-S_PCS_CTRL1 Register to 1 to power down the DWC_xpcs.
6. Wait till Bits[4:2] of VR_XS_PCS_DIG_STS are set to 3′b110.
7. Clear Bit 11 of the SR_PMA_CTRL1 Register (only for Backplane Ethernet PCS configurations) or SR_XS_PCS_CTRL1 Register to 0 to power up DWC_xpcs.
8. Wait till Bits [4:2] of VR_XS_PCS_DIG_STS Register are set to 3′b100.

7.7.4 Programming Sequence for Synopsys Multi-Protocol 25G PHY

To switch to QSGMII mode, perform the following steps:

1. Write 4'b0001 to Bits [3:0] of SR_XS_PCS_CTRL2 Register (Optional- Required only if the configuration has KR/10GBASE-R support).
2. Clear Bit [13] of SR_PMA_CTRL1 Register (for Backplane Ethernet PCS configurations) or SR_X-S_PCS_CTRL1 Register to 0.
3. Program bits [2:1] of VR_MII_AN_CTRL Register to 2′b11.
4. Program the register bits as shown in the following table to switch the PHY to QSGMII mode.

Table 7-23 Synopsys Multi-Protocol 25G PHY: QSGMII Mode Switching

Field	Values
MPLLA_MULTIPLIER	Set the value of this field in the VR_XS_PMA_MP_12G_16G_MPLLA_CTRL0or VR_XS_PMA_SNPS_MP_25G_MPLLA_CTRL0Registerto 12'd128
MPLLA_TX_CLK_DIV	Set the value of this field in the VR_MII_MP_25G_MPLLA_CTRL2or VR_XS_PMA_SNPS_MP_25G_MPLLA_CTRL2registerto 4
MPLLA_W_CLK_DIV	Set the value of this field in the VR_MII_MP_25G_MPLLA_CTRL0or VR_XS_PMA_SNPS_MP_25G_MPLLA_CTRL0registerto 1
BW_TH	Set the value of this field in the VR_MII_MP_25G_MPLLA_CTRL1or VR_XS_PMA_SNPS_MP_25G_MPLLA_CTRL1registerto 2
VCO_REF_LD_0	Set the value of this field in the VR_MII_MP_16G_25G_VCO_CAL_REF0or VR_XS_PMA_SNPS_MP_16G_25G_VCO_CAL_REF0registerto 7'd11 if REF_CLK_78 Else7'd21
VCO_LD_VAL_0	Set the value of this field in the VR_MII_MP_12G_16G_25G_VCO_CAL_LD0or VR_XS_PMA_SNPS_MP_12G_16G_25G_VCO_CAL_LD0register to 13'd1408if REF_CLK_78 or 13'd1344
TX0_DCC	Set the value of this field in the VR_MII_MP_25G_TX_DCC_CTRLor VR_XS_PMA_SNPS_MP_25G_TX_DCC_CTRLregisterto 4'd10

Field	Values
TX0_WIDTH	Set the value of this field in the VR_MII_MP_25G_TX_WIDTH_CTRLor VR_XS_PMA_SNPS_MP_25G_TX_WIDTH_CTRLregisterto 3
TX0_RATE	Set the value of this field in the VR_MII_MP_12G_16G_25G_TX_RATE_CTRLor VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_RATE_CTRLregisterto 0
VBOOST_EN_0	Set the value of this field in the VR_MII_MP_12G_16G_25G_TX_GENCTRL1or VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_GENCTRL1registerto 1
RX0_VCO_CFG	Set the value of this field in the VR_MII_MP_25G_RX_VCO_CFG0or VR_XS_PMA_SNPS_MP_25G_RX_VCO_CFG0registerto 12'd256
LF_TH_0	Set the value of this field in the VR_MII_MP_25G_RX_LF_TRSHLDor VR_XS_PMA_SNPS_MP_25G_RX_LF_TRSHLDregisterto 2
RX0_WIDTH	Set the value of this field in the VR_MII_MP_25G_RX_WIDTH_CTRor VR_XS_PMA_SNPS_MP_25G_RX_WIDTH_CTRLregisterto 3
RX0_RATE	Set the value of this field in the VR_MII_MP_12G_16G_25G_RX_RATE_CTRLor VR_XS_PMA_SNPS_MP_12G_16G_25G_RX_RATE_CTRLregisterto 2
RX_DIV16P5_CLK_EN_0	Set the value of this field in the VR_MII_MP_12G_16G_25G_RX_GENCTRL1or VR_XS_PMA_SNPS_MP_12G_16G_25G_RX_GENCTRL1to0
RX0_ADPT_MODE	Set the value of this field in the VR_MII_MP_16G_25G_RX_EQ_CTRL5or VR_XS_PMA_SNPS_MP_16G_25G_RX_EQ_CTRL5registerto 0
RX_DFE_BYP_0	Set the value of this field in the VR_MII_MP_16G_25G_RX_GENCTRL4or VR_XS_PMA_SNPS_MP_16G_25G_RX_EQ_CTRL4to1
VGA1_GAIN_0	Set the value of this field in the VR_MII_MP_16G_25G_RX_EQ_CTRL0or VR_XS_PMA_SNPS_MP_16G_25G_RX_EQ_CTRL0to4
VGA2_GAIN_0	Set the value of this field in the VR_MII_MP_16G_25G_RX_EQ_CTRL0or VR_XS_PMA_SNPS_MP_16G_25G_RX_EQ_CTRL0to4
RX0_ AFE_RATE	Set the value of this field in the VR_MII_MP_25G_RX_AFE_RATE_CTRLor VR_XS_PMA_SNPS_MP_25G_RX_AFE_RATE_CTRLto6
CTLE_BOOST_0	Set the value of this field in the VR_MII_MP_16G_25G_RX_EQ_CTRL0or VR_XS_PMA_SNPS_MP_16G_25G_RX_EQ_CTRL0registerto 0
RX0_CDR_PPM_MAX	Set the value of this field in the VR_MII_MP_16G_25G_RX_PPM_CTRL0or VR_XS_PMA_SNPS_MP_16G_25G_RX_PPM_CTRL0registerto 5'd19 REF_CLK_78 or5'd18

Note If your configuration is 1000BaseX-Only PCS configuration, program the corresponding registers in MII MMD (instead of XS_PMA MMD).

5. Program TX_CLK_RDY_0/ TX_CLK_RDY_3_1 field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_-GENCTRL1 register to 0.

6. Initiate a Vendor-specific Software Reset by writing 1 to bit [15] of VR_XS_PCS_DIG_CTRL1 register.
7. If SRAM support is enabled in the PHY, perform step 4 of "Initializing the DWC_xpcs Controller" on page 1650.
8. Wait for bit [15] of VR_XS_PCS_DIG_CTRL1 register to get cleared.
9. Program TX_CLK_RDY_0 field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_GENCTRL1 register to 1.
10. Program TX_CLK_RDY_3_1 field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_GENCTRL1 register to 3'd7
11. Program TX_EQ_PRE field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_EQ_CTRL0 register to 0
12. Program TX_EQ_MAIN field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_EQ_CTRL0 register to 6'd24
13. Program TX_EQ_POST field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_EQ_CTRL1 register to 0
14. Program TX_EQ_OVR_RIDE field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_EQ_CTRL1 register to 1

7.7.5 Programming Sequence for Synopsys Multi-Protocol 32G PHY

To switch 32G PHY to QSGMII mode, follow these steps:

1. Write 4'b0001 to bits[3:0] of the SR_XS_PCS_CTRL2 register.
2. Write 0 to bit[13] of the
   - SR_PMA_CTRL1 register for Backplane Ethernet PCS configurations.
   - SR_XS_PCS_CTRL1 register for other configurations.
3. Write 0 to bit[2] (EN_2_5G_MODE) of the VR_XS_PCS_DIG_CTRL1 register.
4. Write 2'b11 to bits[2:1] of the VR_MII_AN_CTRL register.
5. Program the register bits as shown in Table 7-24, to switch the PHY to QSGMII mode.

Table 7-24 Settings to Configures Multi-Protocol 32G PHY for QSGMII Mode

Register	Field	Values
VR_XS_PMA_MP_32G_RX_CNTX_CTRL0	RX_CNTX_SEL_0	Value indicated in the context_sel_ID.txt file of PHY package for QSGMII mode
VR_XS_PMA_MP_32G_TX_CNTX_CTRL0	TX_CNTX_SEL_0	
VR_XS_PMA_MP_32G_TX_CM_CNTX_SEL0	CMN_CNTX_0	
VR_XS_PMA_MP_25G_RX_WIDTH_CTRL	RX0_WIDTH	3'b011 (20-bit)
VR_XS_PMA_MP_25G_TX_WIDTH_CTRL	TX0_WIDTH	
VR_XS_PMA_MP_32G_LN_LINK_CTRL	LINK_NUM_0	Program to a unique value

6. Program the LPM bit of the SR_XS_PCS_CTRL1 register to 1.
7. Wait for the PSEQ_STATE of the VR_XS_PCS_DIG_STS register to indicate 'Power Down', a value of 3'b110.

8. Program the LPM bit of the SR_XS_PCS_CTRL1 register to 0. Program the other bits of this register to values present prior to step 6.

   Note: To avoid assertion of soft-reset do not perform read-modify-write operation.

9. Wait for the PSEQ_STATE of the VR_XS_PCS_DIG_STS register to indicate 'Power Good', a value of 3'b100.

10. Program bit[4] (SUPRESS_LOS_DET) and bit[6] (RX_DT_EN_CTL) of the VR_XS_PCS_DEBUG_CTRL register to 1.

7.8 Programming Guidelines for 2.5G GMII Mode

To operate at 2.5G Speed Mode across GMII Interface, the programming should be such that DWC_xpcs is configured to operate at 1000BASE-X (1G) Mode and PHY is configured to operate at XAUI/KX4 Mode.

This ensures that PHY generates 312.5 MHz clock (instead of 125 MHz) and it can operate at a serial baud rate of 3.125Gpbs (instead of 1.25Gbps). DWC_xpcs operates in 1000BASE-X Mode (but at a higher clock-rate).

7.8.1 Programming Sequence While Using Synopsys Multi-Protocol 12G/16G PHYs

To operate at 2.5G GMII Mode, follow these steps:

1. Write 4'b0001 to bits [3:0] of SR_XS_PCS_CTRL2 (to disable KR Mode).
2. Clear bit [0] of VR_XS_PCS_XAUI_CTRL to 0 (to disable RXAUI Mode).
3. Clear Bit [13] of SR_PMA_CTRL1 (for Backplane Ethernet PCS configurations) or SR_XS_PCS_CTRL1 to 0.
4. Program bit [2] (EN_2_5G_MODE) of VR_MII_DIG_CTRL1 (in 1000BASEX-Only PCS configuration) or VR_XS_PCS_DIG_CTRL1 (in all other configurations) to 1.
5. If SGMII Mode is enabled, program SS6 (bit [6]) and SS13 (bit [13]) bits of SR_MII_CTRL to set speed as 1000Mbps (which corresponds to 2.5Gbps due to higher clock-rate).
6. Program the register bits as shown in Table 7-2 (for 12G PHY) or Table 7-3 (for 16G PHY).
7. Initiate a Vendor-specific Software Reset by writing 1 to bit [15] of VR_XS_PCS_DIG_CTRL1.
8. If you enable SRAM support in the PHY, optionally execute step 4 from the section, “Initializing the DWC_xpcs Controller” on page 1650.
9. Wait for bit [15] of VR_XS_PCS_DIG_CTRL1 to get cleared.
10. Program TX_CLK_RDY{lane} field of VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 1
11. Program TX_EQ_PRE field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL0 register to 0
12. Program TX_EQ_MAIN field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL0 register to 6'd40
13. Program TX_EQ_POST field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL1 register to 0
14. Program TX_EQ_OVR_RIDE field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL1 register to 1 (only for configurations with CL72 support).
15. Program bit[4], SUPRESS_LOS_DET field and bit[6], RX_DT_EN_CTL field of the VR_XS_PCS_DEBUG_CTRL register to 0.
16. (Optional) To perform clause 37 Auto-negotiation in 2.5G SGMII mode, program VR_MII_LINK_TIMER_CTRL to 16'h07A1 (for 1.6 ms in SGMII Mode) or 16'h2FAF (for 10ms in 1000BASE-X mode) to ensure that the link timer runs for the duration prescribed by the standard. Then, program bit[3] (CL37_TMR_OVR_RIDE) of VR_MII_DIG_CTRL1 to 1.

Note

Depending on your PHY variant, use either MPLLA or MPLLB. If you use MPLLB, the following additional steps should be performed prior to step 7 (Issuing of Vendor Soft Reset) in the sequence mentioned:

- Program 'MPLLB_CAL_DISABLE' bit of any one of the following registers to 0.
  - VR_MII_MP_12G_16G_MPLLB_CTRL0 (for 1000BaseX-Only PCS configurations)
  - VR_XS_PMA_MP_12G_16G_MPLLB_CTRL0 Register (for other configurations)
- Program 'MPLLB_SEL_0' bit of any one of the following registers to 1.
  - VR_MII_MP_12G_16G_25G_MPLL_CMN_CTRL Register (for 1000BaseX-Only PCS configurations)
  - VR_XS_PMA_MP_12G_16G_25G_MPLL_CMN_CTRL Register (for other configurations)

7.8.2 Programming Guidelines for Multi-Protocol 25G PHY

To operate in the 2.5G GMII mode, follow these steps:

1. Program 4'b0001 to bits [3:0] of SR_XS_PCS_CTRL2 register
2. Program bit [0] of VR_XS_PCS_XAUI_CTRL register to 0 (only for RXAUI configurations
3. Program bit [13] of SR_PMA_CTRL1 register (for Backplane Ethernet configurations) or SR_XS_PCS_CTRL1 (for other configurations) to 0.
4. Program bit [2] (EN_2_5G_MODE) of VR_MII_DIG_CTRL1 (for 1000BASE-X Only PCS configurations) or VR_XS_PCS_DIG_CTRL1 (in all other configurations) to 1.
5. Program MPLLA_MULTIPLIER field of VR_XS_PMA_SNPS_MP_25G_MPLLA_CTRL0 register to 12'd120
6. Program MPLLA_TX_CLK_DIV field of VR_XS_PMA_SNPS_MP_25G_MPLLA_CTRL2 register to 3
7. Program MPLLA_W_CLK_DIV field of VR_XS_PMA_SNPS_MP_25G_MPLLA_CTRL0 register to 1
8. Program BW_TH field of VR_XS_PMA_SNPS_MP_25G_MPLLA_CTRL1 register to 2
9. Program VCO_REF_LD_0 field of VR_XS_PMA_SNPS_MP_16G_25G_VCO_CAL_REF0 register to 7'd34
10. Program VCO_LD_VAL_0 field of VR_XS_PMA_SNPS_MP_12G_16G_25G _VCO_CAL_LD0 register to 13'd1360
11. Program TX0_DCC fields of VR_XS_PMA_SNPS_MP_25G_TX_DCC_CTRL register to 4'd10
12. Program TX0_WIDTH field of VR_XS_PMA_SNPS_MP_25G_TX_WIDTH_CTRL register to 1
13. Program TX0_RATE field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_RATE_CTRL register to 1
14. Program VBOOST_EN_0 field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_GENCTRL1 register to 1
15. Program RX0_VCO_CFG field of VR_XS_PMA_SNPS_MP_25G_RX_VCO_CFG0 register to 12'd259
16. Program LF_TH_0 field of VR_XS_PMA_SNPS_MP_25G_RX_LF_TRSHLD register to 7
17. Program RX0_WIDTH field of VR_XS_PMA_SNPS_MP_25G_RX_WIDTH_CTRL register to 1
18. Program RX0_RATE field of VR_XS_PMA_SNPS_MP_12G_16G_25G_RX_RATE_CTRL register to 2
19. Program RX_DIV16P5_CLK_EN_0 field of VR_XS_PMA_SNPS_MP_12G_16G_25G_RX_GENCTRL1 to 0

20. Program RX0_ADPT_MODE field of VR_XS_PMA_SNPS_MP_16G_25G_RX_EQ_CTRL5 register to 0
21. Program RX_DFE_BYP_0 field of VR_XS_PMA_SNPS_MP_16G_25G_RX_EQ_CTRL4 to 1
22. Program RX_DFE_TAP1_0 field of VR_XS_PMA_SNPS_MP_12G_16G_25G_DFE_TAP_CTRL0 to 10
23. Program VGA1_GAIN_0 field of VR_XS_PMA_SNPS_MP_16G_25G_RX_EQ_CTRL0 to 4
24. Program VGA2_GAIN_0 field of VR_XS_PMA_SNPS_MP_16G_25G_RX_EQ_CTRL0 to 4
25. Program RX0_AFE_RATE field of VR_XS_PMA_SNPS_MP_25G_RX_AFE_RATE_CTRL to 7
26. Program CTLE_BOOST_0 field of VR_XS_PMA_SNPS_MP_16G_25G_RX_EQ_CTRL0 register to 6
27. Program RX0_CDR_PPM_MAX field of VR_XS_PMA_SNPS_MP_16G_25G_RX_PPM_CTRL0 register to 5'd15
28. Program TX_CLK_RDY_0 field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_GENCTRL1 register to 0
29. Initiate a Vendor-specific Software Reset by writing 1 to bit [15] of VR_XS_PCS_DIG_CTRL1 register
30. If you enable SRAM support in the PHY, optionally execute step 4 from the section, “Initializing the DWC_xpcs Controller” on page 1650.
31. Wait for bit [15] of VR_XS_PCS_DIG_CTRL1 register to get cleared.
32. Program TX_CLK_RDY_0 field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_GENCTRL1 register to 1
33. Program TX_EQ_PRE field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_EQ_CTRL0 register to 0
34. Program TX_EQ_MAIN field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_EQ_CTRL0 register to 6'd24
35. Program TX_EQ_POST field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_EQ_CTRL1 register to 0
36. Program TX_EQ_OVR_RIDE field of VR_XS_PMA_SNPS_MP_12G_16G_25G_TX_EQ_CTRL1 register to 1
37. Program TX_EQ_OVR_RIDE field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL1 register to 1 (only for configurations with CL72 support)
38. (Optional) To perform clause 37 Auto-negotiation in 2.5G SGMII mode, program VR_MII_LINK_TIM-ER_CTRL to 16'h07A1 (for 1.6 ms in SGMII Mode) or 16'h2FAF (for 10ms in 1000BASE-X mode) to ensure that the link timer runs for the duration prescribed by the standard. Then, program bit [3] (CL37_TMR_OVR_RIDE) of VR_MII_DIG_CTRL1 to 1

7.8.3 Programming Sequence While Using Synopsys Multi-Protocol 6G PHY

To operate at 2.5G GMII mode using the Synopsys multi-protocol 6G PHYs, follow these steps:

1. Clear bit [0] of VR_XS_PCS_XAUI_CTRL Register to 0 (to disable RXAUI Mode)
2. Clear Bit [13] of SR_PMA_CTRL1 Register (for Backplane Ethernet PCS configurations) or SR_X-S_PCS_CTRL1 Register (for other configurations) to 0.
3. Program bit [2] (EN_2_5G_MODE) of VR_MII_DIG_CTRL1 Register (in 1000BASEX-Only PCS configuration) or SR_XS_PCS_CTRL1 Register (in all other configurations) to 1.
4. If SGMII Mode is enabled, program SS6 (bit [6]) and SS13 (bit [13]) bits of SR_MII_CTRL Register to set speed as 1000Mbps.
5. Program register bits as shown in Table 7-1.
6. Set bit [11] of SR_PMA_CTRL1 Register (only for Backplane Ethernet PCS configurations) or SR_X-S_PCS_CTRL1 Register to power down DWC_xpcs.

7. Wait until bits [4:2] of VR_XS_PCS_DIG_STS Register are set to 3'b110.
8. Clear bit [11] of SR_PMA_CTRL1 Register (only for Backplane Ethernet PCS configurations) or SR_X-S_PCS_CTRL1 Register (for other configurations) to 0 to power up DWC_xpcs.
9. Wait until bits [4:2] of VR_XS_PCS_DIG_STS Register are set to 3'b100.

7.8.4 Programming Sequence for Multi-Protocol 32G PHY

Follow these steps:

1. Program 4'b0001 in bits[3:0] of SR_XS_PCS_CTRL2 register.
2. Program bit[2], EN_2_5G_MODE field of the VR_XS_PCS_DIG_CTRL1 register to 1.
3. Program bit[9], USXG_EN field of the VR_XS_PCS_DIG_CTRL1 register to 0.
4. Configure the PHY to operate at 3.125Gbps rate by programming registers as shown in Table 7-25.

Table 7-25 Configuring PHY for 3.125 Gbps Rate

Register	Field	Values
VR_XS_PMA_MP_32G_RX_CNTX_CTRL0	RX_CNTX_SEL_0	Value indicated in context_sel_ID.txt file of PHY package for 2.5G mode
VVR_XS_PMA_MP_32G_TX_CNTX_CTRL0	TX_CNTX_SEL_0	Value indicated in context_sel_ID.txt file of PHY package for 2.5G mode
VR_XS_PMA_MP_32G_TX_CM_CNTX_SEL0	CM_CNTX_SEL_0	Value indicated in context_sel_ID.txt file of PHY package for 2.5G mode
VR_XS_PMA_MP_25G_RX_WIDTH_CTRL	RX0_WIDTH	3'b001 (10-bit)
VR_XS_PMA_MP_25G_TX_WIDTH_CTRL	TX0_WIDTH	3'b001 (10-bit)
VR_XS_PMA_MP_32G_LN_LINK_CTRL	LINK_NUM_0	In bifurcation mode, program to a unique value for each instance of the IP.

5. Program LPM bit of SR_XS_PCS_CTRL1 register to 1.
6. Wait for PSEQ_STATE of the VR_XS_PCS_DIG_STS register to indicate 'Power Down' (3'b110).
7. Program LPM bit of the SR_XS_PCS_CTRL1 register to 0. Program other bits of this register to values that was available in this register prior to step 5.

   Note: To avoid assertion of soft-reset, do not perform read-modify-write operation.
8. Wait for the PSEQ_STATE of the VR_XS_PCS_DIG_STS register to indicate 'Power Good' (3'b100).
9. Program bit[4], SUPRESS_LOS_DET field and bit[6], RX_DT_EN_CTL field of the VR_XS_PCS_DE-BUG_CTRL register to 0.

7.9 Programming Guidelines for 2.5GBASE-KX Mode (over XGMII)

In the 2.5GBASE-(K)X mode, DWC_xpcs interfaces with the MAC through the XGMII interface. It interfaces with the PHY through a parallel 10-bit interface operating at 312.5 MHz clock.

When operating in this mode, the serial baud-rate is 3.125Gbps. To configure DWC_xpcs to this mode, write 4'b1110 to the bits[3:0] of the SR_XS_PCS_CTRL2 register.

To configure the PHY to operate at 3.125Gbps, program the PHY related registers accordingly.

7.9.1 Programming Sequence While Using Synopsys Multi-Protocol 12G/16G PHY

Follow these steps:

1. Write 4'b1110 to bits [3:0] of SR_XS_PCS_CTRL2 register.
2. Clear Bit [13] of SR_PMA_CTRL1 to 0.
3. Program register bits as shown in Table 7-2 (for 12G PHY) or Table 7-3 (for 16G PHY) .
4. Initiate a Vendor-specific Software Reset by writing 1 to bit[15]ofVR_XS_PCS_DIG_CTRL1.
5. If you enable SRAM support in the PHY, optionally execute step 4 from the section, "Initializing the DWC_xpcs Controller" on page 1650.
6. Wait for bit [15] of VR_XS_PCS_DIG_CTRL1 to get cleared.
7. Program TX_CLK_RDY{lane} field of VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 register to 1
8. Program TX_EQ_PRE field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL0 register to 0
9. Program TX_EQ_MAIN field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL0 register to 6'd40
10. Program TX_EQ_POST field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL1 register to 0
11. Program TX_EQ_OVR_RIDE field of VR_XS_PMA_MP_12G_16G_25G_TX_EQ_CTRL1 register to 1 (only for configurations with CL72 support)

7.9.2 Programming Sequence While Using Multi-Protocol 25G PHY

Follow these steps:

1. Write 4'b1110 to bits [3:0] of SR_XS_PCS_CTRL2 Register.
2. Clear bit [13] of SR_PMA_CTRL1 register to 0.
3. Follow step 5 onwards in "Programming Guidelines for Multi-Protocol 25G PHY" on page 1693.

7.9.3 Programming Sequence While Using Synopsys Multi-Protocol 6G PHY or Non-Synopsys PHY

Follow these steps:

1. Write 4'b1110 to bits [3:0] of SR_XS_PCS_CTRL2 register.
2. Clear Bit [13] of SR_PMA_CTRL1 to 0.
3. Program register bits as shown in Table 6-4 (for Synopsys Multi-Protocol 6G PHY). For Non-Synopsys PHY, configure the PHY to operate at a baud-rate of 3.125Gpbs.
4. Set Bit [11] of the SR_PMA_CTRL1 Register to 1 to power down the DWC_xpcs.
5. Wait till Bits [4:2] of VR_XS_PCS_DIG_STS are set to 3'b110 (for Synopsys Multi-Protocol 6G PHY only).
6. Clear Bit [11] of SR_PMA_CTRL1 Register to 0 to power up the DWC_xpcs.

7. Wait till Bits [4:2] of VR_XS_PCS_DIG_STS Register are set to 3'b100.

7.9.4 Programming Sequence for Multi-Protocol 32G PHY

Follow these steps:

1. Program 4’b1110 in bits[3:0] of SR_XS_PCS_CTRL2 register.
2. Program bit[2], EN_2_5G_MODE field of the VR_XS_PCS_DIG_CTRL1 register to 0.
3. Program bit[9], (USXG_EN) of VR_XS_PCS_DIG_CTRL1 register to 0.
4. Configure the PHY to operate at 3.125Gbps rate by programming registers as shown in Table 7-25.
5. Program LPM bit of SR_XS_PCS_CTRL1 register to 1.
6. Wait for PSEQ_STATE of VR_XS_PCS_DIG_STS register to indicate 'Power Down' (3'b110)
7. Program the LPM bit of the SR_XS_PCS_CTRL1 register to 0. Program the other bits of this register to values that was available in this register prior to step 5.

   Note: To avoid assertion of soft-reset, do not perform read-modify-write.
8. Wait for the PSEQ_STATE field of the VR_XS_PCS_DIG_STS register to indicate 'Power Good' (3'b100).
9. Program bit[4], SUPRESS_LOS_DET field and bit[6], RX_DT_EN_CTL field of the VR_XS_PCS_DEBUG_CTRL register to 0.

7.10 Bifurcation Mode Usage of Synopsys Multi-Protocol PHY

Synopsys PHY supports bifurcation mode operation. Different lanes of the PHY can operate at different Ethernet protocols at the same-time. The presence of two MPLLs -MPLLA and MPLLB is the key factor that allows this bifurcation mode.

Note

This section describes programming steps specifically for multi-protocol12G PHY. Similar programming steps can be used for Synopsys multi-protocol 16G and 25G PHY configurations as well.

Synopsys multi-protocol 32G PHY configurations do not need any special programming sequence. The programming steps described in sections 7.2 to 7.9 can be used in both bifurcation and non-bifurcation modes.

When operating in bifurcation mode, switching of a particular lane from one mode to another can be done without issuing a VR_RST (or phy_reset). A particular lane can be made to operate from MPLLA or MPLLB by controlling "txX_mpllb_sel" input of the PHY (which is controlled by MPLLB_SEL _* fields of VR_XS_PMA_MP_12G_16G_25G_MPLL_CMN_CTRL or VR_MII_MP_12G_16G_25G_MPLL_CMN_CTRL Register).

To enable operation in bifurcation mode, both MPLL A and B should be programmed and calibrated upfront. This can be done by programming appropriate values (provided in PHY databook) to various fields of the following registers:

- VR_XS_PMA_MP_12G_16G_MPLLA_CTRL0 or VR_MII_MP_12G_16G_MPLLA_CTRL0
- VR_XS_PMA_MP_12G_MPLLA_CTRL1 or VR_MII_MP_12G_MPLLA_CTRL1
- VR_XS_PMA_MP_12G_16G_MPLLA_CTRL2 or VR_MII_MP_12G_16G_MPLLA_CTRL2
- VR_XS_PMA_MP_12G_MPLLA_CTRL3 or VR_MII_MP_12G_MPLLA_CTRL3
- VR_XS_PMA_MP_12G_16G_MPLLB_CTRL0 or VR_MII_MP_12G_16G_MPLLB_CTRL0
- VR_XS_PMA_MP_12G_MPLLB_CTRL1 or VR_MII_MP_12G_MPLLB_CTRL1
- VR_XS_PMA_MP_12G_16G_MPLLB_CTRL2 or VR_MII_MP_12G_16G_MPLLB_CTRL2
- VR_XS_PMA_MP_12G_MPLLB_CTRL3 or VR_MII_MP_12G_MPLLB_CTRL3

and then issuing a VR_RST (Vendor Soft Reset).

Note

'MPLLB_CAL_DISABLE' bit of VR_XS_PMA_Gen5_12G_MPLLB_CTRL0 or VR_MII_MP_12G_16G_MPLLB_CTRL0 should be programmed to 0.

Thereafter, a particular lane of the PHY can be made to operate at KR/2.5G/KX mode by selecting txX_mpllb_sel input appropriately. Transmit and Receive settings of a particular lane should be programmed based on the desired operating mode.

For multi-protocol 12G PHYs, typically, when operating in

- KX mode: MPLLB is used.
- KR/2.5G mode: MPLLA is used.

With this assumption, the following steps illustrate the main operations involved in switching a lane (say lane 0) from KR (or 2.5G) to KX mode:

1. Program MPLLA registers for KR (or 2.5G) mode and MPLLB registers for KX mode (Refer to PHY databook for details).
2. Program 'VR_RST' bit (bit [15]) of VR_XS_PCS_DIG_CTRL1 Register to 1.
3. Wait for 'VR_RST' bit to self-clear.



Note

These 3 steps are required only once after start-up. They are not required for any further speed mode changes. These steps ensure that both MPLLA and MPLLB are calibrated suitably to operate at the desired speed mode.

4. Switch DWC_xpcs to KX mode by programming:
   - bit[13] of SR_PMA_CTRL1 Register to 0
   - bits[1:0] of SR_XS_PCS_CTRL2 Register to 2'b01
5. Set Transmit and Receive controls of the PHY to KX mode (Refer to PHY databook for exact settings in bifurcation mode) by programming appropriate registers in DWC_xpcs. The registers fields to be programmed are:
   - VCO_LD_VAL_0 of VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD0 or VR_MII_MP_12G_16G_25G_VCO_CAL_LD0
   - VCO_REF_LD_VAL_0 of VR_XS_PMA_MP_12G_VCO_CAL_REF0 or VR_MII_MP_12G_VCO_-CAL_REF0
   - AFE_EN_0, DFE_EN_0 of VR_XS_PMA_MP_12G_AFE_DFE_EN_CTRL or VR_MII_MP_12G_AFE_DFE_EN_CTRL
   - CONT_ADAPT_0 of VR_XS_PMA_MP_12G_16G_25G_RX_EQ_CTRL4 or VR_MII_MP_12G_16G_25G_RX_EQ_CTRL4
   - TX0_RATE of VR_XS_PMA_MP_12G_16G_25G_TX_RATE_CTRL or VR_MII_MP_12G_16G_25G_TX_RATE_CTRL
   - RX0_RATE of VR_XS_PMA_MP_12G_16G_25G_RX_RATE_CTRL or VR_MII_MP_12G_16G_25G_RX_RATE_CTRL
   - TX0_WIDTH of VR_XS_PMA_MP_12G_16G_TX_GENCTRL2 or VR_MII_MP_12G_16G_TX_-GENCTRL2
   - RX0_WIDTH of VR_XS_PMA_MP_12G_16G_RX_GENCTRL2 or VR_MII_MP_12G_16G_RX_-GENCTRL2
6. For configurations other than 32G PHY, program TX_CLK_RDY_0 field (bit [12]) of VR_XS_P-MA_MP_12G_16G_25G_TX_GENCTRL1 or VR_MII_MP_12G_16G_25G_TX_GENCTRL1to 0.
7. Program MPLLB_SEL_0 field (bit [4]) of VR_XS_PMA_MP_12G_16G_25G_MPLL_CMN_CTRL or VR_MII_MP_12G_16G_25G_MPLL_CMN_CTRLto 1.
8. Program TX_REQ_0 field (bit [0]) of VR_XS_PMA_MP_12G_16G_TX_GENCTRL2 or VR_MII_MP_12G_16G_TX_GENCTRL2 to 1.
9. Wait for 'TX_REQ_0' bit to self-clear.
10. Program 'RX_REQ_0' (bit [0]) of VR_XS_PMA_MP_12G_16G_RX_GENCTRL2 or VR_MII_MP_12G_16G_RX_GENCTRL2 to 1.

11. Wait for 'RX_REQ_0' bit to self-clear.
12. For configurations other than 32G PHY, program 'TX_CLK_RDY_0' (bit [12]) of VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 or VR_MII_MP_12G_16G_25G_TX_GENCTRL1 to 1.
13. Program 'INIT' (bit [8]) of VR_XS_PCS_DIG_CTRL1 Register to 1.
14. Wait for 'INIT' bit to self-clear.

Now DWC_xpcs and PHY are ready to operate in KX mode.

Note For multi-port USXGMII configurations, instead of programming 'INIT' (steps 10 and 11), use the LPM bit of the SR_XS_PCS_CTRL1 register; program the LPM bit to 1 and after some time program it to 0.

A similar sequence can be carried-out to switch a particular lane from KX to KR (or 2.5G) mode. Following steps illustrate how to switch lane 0 from KX to KR (or 2.5G) mode:

1. Switch DWC_xpcs to KR mode by programming bits[1:0] of SR_XS_PCS_CTRL2 Register to 2'b00 OR switch XPCS to 2.5G mode by programming:
   - bit[13] of SR_PMA_CTRL1 Register to 0
   - bits[1:0] of SR_XS_PCS_CTRL2 Register to 2'b01
   - bit[2] of VR_XS_PCS_DIG_CTRL1 Register to 1
2. Set Transmit and Receive controls of the PHY to KR (or 2.5G) mode (See the PHY databook for exact settings in bifurcation mode) by programming appropriate registers in DWC_xpcs. The register fields to be programmed are:
   - VCO_LD_VAL_0 of VR_XS_PMA_MP_12G_16G_25G_VCO_CAL_LD0 or VR_MII_MP_12G_16G_25G_VCO_CAL_LD0
   - VCO_REF_LD_VAL_0 of VR_XS_PMA_MP_12G_VCO_CAL_REF0 or VR_MII_MP_12G_VCO_CAL_REF0
   - AFE_EN_0, DFE_EN_0 of VR_XS_PMA_MP_12G_AFE_DFE_EN_CTRL or VR_MII_MP_12G_AFE_DFE_EN_CTRL
   - CONT_ADAPT_0 of VR_XS_PMA_MP_12G_16G_25G_RX_EQ_CTRL4 or VR_MII_MP_12G_16G_25G_RX_EQ_CTRL4
   - TX0_RATE of VR_XS_PMA_MP_12G_16G_25G_TX_RATE_CTRL or VR_MII_MP_12G_16G_25G_TX_RATE_CTRL
   - RX0_RATE of VR_XS_PMA_MP_12G_16G_25G_RX_RATE_CTRL or VR_MII_MP_12G_16G_25G_RX_RATE_CTRL
   - TX0_WIDTH of VR_XS_PMA_MP_12G_16G_TX_GENCTRL2 or VR_MII_Gen5_12G_16G_TX_GENCTRL2
   - RX0_WIDTH of VR_XS_PMA_MP_12G_16G_RX_GENCTRL2 or VR_MII_MP_12G_16G_RX_GENCTRL2
3. For configurations other than 32G PHY, program 'TX_CLK_RDY_0' (bit [12]) of VR_XS_PMA_12G_16G_25G_TX_GENCTRL1 or VR_MII_12G_16G_25G_TX_GENCTRL1 to 0
4. Program 'MPLLB_SEL_0' (bit [4]) of VR_XS_PMA_12G_16G_25G_MPLL_CMN_CTRL or VR_MII_12G_16G_25G_MPLL_CMN_CTRL to 0.

5. Program 'TX_REQ_0' (bit [0]) of VR_XS_PMA_MP_12G_16G_TX_GENCTRL2 or VR_MII_MP_12G_16G_TX_GENCTRL2 to 1.
6. Wait for 'TX_REQ_0' bit to self-clear.
7. Program 'RX_REQ_0' (bit [0]) of VR_XS_PMA_MP_12G_16G_RX_GENCTRL2 or VR_MII_MP_12G_16G_RX_GENCTRL2 to 1.
8. Wait for 'RX_REQ_0' bit to self-clear.
9. For configurations other than 32G PHY, program 'TX_CLK_RDY_0' field (bit [12]) of VR_XS_PMA_MP_12G_16G_25G_TX_GENCTRL1 or VR_MII_MP_12G_16G_25G_TX_GENCTRL1 to 1.
10. Program INIT field (bit [8]) of VR_XS_PCS_DIG_CTRL1 Register to 1.
11. Wait for INIT bit to self-clear.

DWC_xpcs and PHY are now ready to operate in KR (or 2.5G) mode.



For multi-port USXGMII configurations, instead of programming 'INIT' (steps 10 and 11), use LPM bit of SR_XS_PCS_CTRL1 register; program the LPM bit to 1 and after some time, program it to 0.

When operating in bifurcation KX mode, connect the mpllb_div_clk (instead of mpll{a,b}_qword_clk) from PHY to clk_xgxs_tx_i and clk_xgxs_rx_i ports of DWC_xpcs and also use the mpllb_div_clk as 'mac_tx_clock'.

7.11 Programming Guidelines for CR Port Read or Write Operation

The following lists provide the programming steps for a read or write operation on the CR port:

7.11.1 Write Operation

To write any register in Synopsys PHY, complete the following:

1. Read the VR_XS_PMA_SNPS_CR_CTRL or VR_MII_SNPS_CR_CTRL and wait till Bit 0 of this register reads as 0.
2. Write the Synopsys PHY register address to VR_XS_PMA_SNPS_CR_ADDR or VR_MII_SNPS_CR_ADDR. These addresses are defined in the Synopsys PHY Databooks.
3. Write the data to be written to the VR_XS_PMA_SNPS_CR_DATA or VR_MII_SNPS_CR_DATA.
4. Write 1 to Bit 1 of the VR_XS_PMA_SNPS_CR_CTRL or VR_MII_SNPS_CR_CTRL Register.
5. Write 1 to Bit 0 of the VR_XS_PMA_SNPS_CR_CTRL or VR_MII_SNPS_CR_CTRL Register.
6. Read the VR_XS_PMA_SNPS_CR_CTRL or VR_MII_SNPS_CR_CTRL and wait till Bit 0 of this register reads as 0.

7.11.2 Read Operation

To read any register in Synopsys PHY, complete the following:

1. Read the VR_XS_PMA_SNPS_CR_CTRL or VR_MII_SNPS_CR_CTRL and wait till Bit 0 of this register reads as 0.
2. Write the Synopsys PHY register address to VR_XS_PMA_SNPS_CR_ADDR or VR_MII_SNPS_CR_ADDR Register. These addresses are defined in the Synopsys PHY Data Books.
3. Write 0 to Bit 1 of the VR_XS_PMA_SNPS_CR_CTRL or VR_MII_SNPS_CR_CTRL Register. In addition, if Synopsys Enterprise Gen5 10G PHY is used, you must select the PHY MMD address by programming bits[6:2] and select PHY LANE_SEL by programming bits[11:8] of this register.
4. Write 1 to Bit 0 of the VR_XS_PMA_SNPS_CR_CTRL or VR_MII_SNPS_CR_CTRL Register.
5. Read the VR_XS_PMA_SNPS_CR_CTRL or VR_MII_SNPS_CR_CTRL Register and wait till Bit 0 of this register reads as 0.
6. Read the VR_XS_PMA_SNPS_CR_DATA or VR_MII_SNPS_CR_DATA Register. The register read in this step contains the Read Data

7.12 Firmware Loading for Synopsys Multi-Protocol 32G PHY Configurations

The data and corresponding addresses for the CR write operation is available in the sram_firmware file, dwc_e32mp_phy_x4_ns_pcs_raw_mem_sram_cr_para.fw, provided with the PHY package. You must regenerate the firmware file by suitably selecting the target operating modes and executing the dwc_e32mp_phy_x4_ns_image_gen.pl perl script, which is also provided with the PHY package. Figure 7-1 shows the snippet of a sample SRAM firmware file.

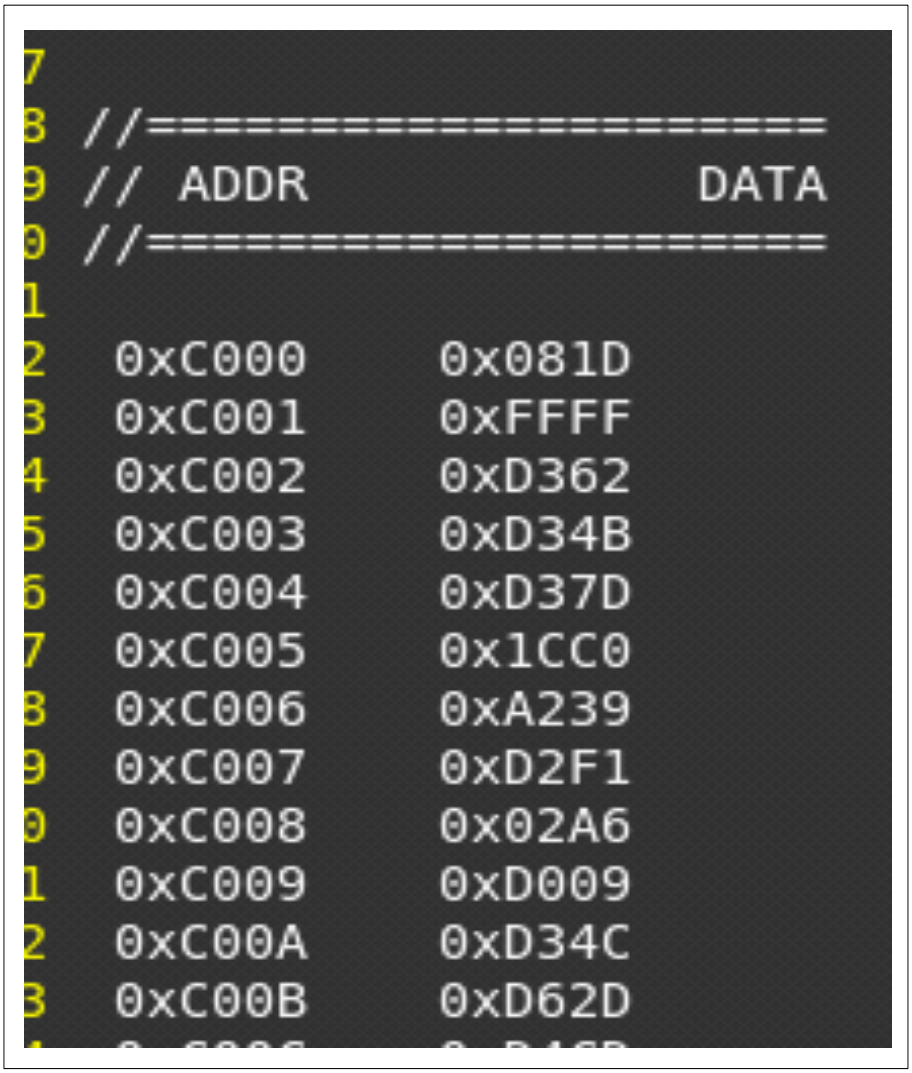
First column indicates the CR address to be written to the

- VR_MII_SNPS_CR_DATA register for 1000BASE-X Only PCS configurations
- VR_PMA_SNPS _CR_ADDR register for other configurations

Second column indicates the content, CR write data to be written to the

- VR_MII_SNPS_CR_DATA register for 1000BASE-X Only PCS configurations
- VR_PMA_SNPS_ CR_DATA register for other configurations

Figure 7-1 Snippet of the Sample SRAM Firmware File



7.13 Programming Guidelines for Clause 37 Auto-Negotiation

The Clause 37 auto-negotiation is enabled only when Bit 12 of the SR_MII_CTRL Register is set. For Backplane Ethernet or USXGMII configurations, the default value of this bit is 0. For all other configurations, the default value of this bit is 1. When you enable this bit, the Clause 37 auto-negotiation is initiated on the following events:

- Power on Reset
- Soft Reset

The DWC_xpcs initiates the auto-negotiation on the following events:

- When the link partner requests auto-negotiation by transmitting configuration code groups.
- When the receive data path loses code group synchronization for more than
  - 10 ms in 1000BASE-X mode
  - 1.6 ms in SGMII, QSGMII, or USXGMII mode
- When an error condition is detected while receiving the /C/ or /I/ order sets.
- When the host or software requests auto-negotiation by setting Bit 9 in the SR_MII_CTRL Register.

The following list explains the auto-negotiation process:

1. The DWC_xpcs starts auto-negotiation by first transmitting the configuration words with all zeroes for
   - 10 ms in 1000BASE-X mode
   - 1.6 ms in SGMII, QSGMII, or USXGMII mode
2. In the 1000BASE-X mode, the content of the SR_MII_AN_ADV register is transmitted in the configuration word.

   In the SGMII mode, the values given in Table 2-25 on page 145 are transmitted in the configuration word.

   In the USXGMII mode, the values in Table 2-11 on page 88 are transmitted in the configuration word.

   In the QSGMII mode, the values in Table 2-28 on page 153 are transmitted in the configuration word.

   The auto-negotiation is complete when both link partners have exchanged their base pages.
3. DWC_xpcs generates an interrupt on completion (sbd_intr_o) of auto-negotiation when bit[0] of VR_MII_AN_CTRL Register is set to 1.
4. The auto-negotiation completion is indicated in the VR_MII_AN_INTR_STS Register.

Note

- In configurations with MDIO interface, you must read the VR_MII_AN_INTR_STS Register after the auto-negotiation is complete.
- Auto-negotiation Status reads incorrect value when Status is not read in the previous Auto-negotiation session and MDC clock is not active when the Management is not accessing the PHY/PCS registers.

Because of this limitation, management might get incorrect information of Link partner and this can cause link failure.

However, this limitation is not visible if the programming guidelines are followed as mentioned in the databook. This is because the minimum time between two successive auto-negotiations has at least 3.2ms and Management or Host is expected to read the current status of the Auto-negotiation before the next auto-negotiation is completed.

5. In the MAC attached to the DWC_xpcs, the Transmit and Receive Flow Control mode is determined based on the capabilities of the partner (given in SR_MII_LP_BABL) and the half-duplex or full-duplex operating mode.

   In SGMII, QSXGMII, or USXGMII auto-negotiation, the received link status such as speed, duplex mode is also given in the VR_MII_AN_INTR_STS register.

   When Clause 37 auto-negotiation is complete, DWC_xpcs automatically resolves the duplex mode by selecting the duplex mode of its link partner. If auto-negotiation is disabled, the DWC_xpcs selects the duplex mode defined in Bit 8 of SR_MII_CTRL register.

Note

- DWC_xpcs supports only 1000 Mbps link speed in the 1000BASE-X mode. It does not support the 10 or 100 Mbps link speed. In SGMII mode, the DWC_xpcs supports 10 or 100 Mbps link speed.
- DWC_xpcs does not support the Next Page exchange.
- DWC_xpcs support only full-duplex in USXGMII mode.

7.13.1 SGMII Auto-Negotiation

Clause 37 auto-negotiation can be performed in the SGMII mode by programming various registers as follows:

1. In configurations with both 10G and 1G mode speed mode, switch DWC_xpcs to 1G speed mode by following the steps described in "Switching to 1G or KX Mode and 10 or 100 Mbps SGMII Speed" on page 1669. If the target operating mode is 2.5G SGMII, instead of the standard 1G SGMII, switch to 2.5G mode by following the steps mentioned in "Programming Guidelines for 2.5G GMII Mode" on page 1692.
2. In Backplane Ethernet PCS configurations, program bit[12] (AN_EN) of SR_AN_CTRL Register to 0 and bit[12] (CL37_BP) of VR_XS_PCS_DIG_CTRL1 Register to 1.
3. Disable Clause 37 auto-negotiation by programming bit [12] (AN_ENABLE) of SR_MII_CTRL Register to 0 (in case it is already enabled).
4. Program various fields of VR_MII_AN_CTRL Register appropriately as follows:
   - Program PCS_MODE to 2'b10
   - Program TX_CONFIG to 1 (PHY side SGMII) or 0 (MAC side SGMII) based on your requirement
   - Program MII_AN_INTR_EN to 1, to enable auto-negotiation complete interrupt
   - If TX_CONFIG is set to 1 and bit[0] of VR_MII_DIG_CTRL1 Register is set to 0, program SGMII_LINK_STS to indicate the link status to the MAC side SGMII.
   - Program MII_CTRL to 0 or 1, as per your requirement.
5. If DWC_xpcs is configured as PHY-side SGMII in the previous step, you can program bit [0] of VR_MII_DIG_CTRL1 Register to 1, if you wish to use the values of the xpcs_sgmii_link_sts_i input ports. xpcs_sgmii_full_duplex_i and xpcs_sgmii_link_speed_i as the transmitted configuration word.
6. If DWC_xpcs is configured as PHY-side SGMII and if bit[0] of VR_MII_DIG_CTRL1 Register is set to 0,
   - Program SS13 and SS6 bits of SR_MII_CTRL Register to the required SGMII Speed
   - Program bit [5] (FD) of SR_MII_AN_ADV to the desired mode. This step is mandatory even if you wish to leave the FD register bit to its default value.

7. If DWC_xpcs is configured as MAC-side SGMII in step 4, program bit[9] of VR_MII_DIG_CTRL1 Register to 1, for DWC_xpcs to automatically switch to the negotiated link-speed, after the completion of auto-negotiation.
8. Only in case of 2.5G SGMII mode,
   - Program the VR_MII_LINK_TIMER_CTRL register to 16'h07A1 so that link timer runs for 1.6 ms.
   - Program bit[3], CL37_TMR_OVR_RIDE, of the VR_MII_DIG_CTRL1 register to 1.
9. Enable CL37 Auto-negotiation, by programming bit[12] of the SR_MII_CTRL Register to 1.

Completion of the SGMII auto-negotiation is indicated by bit[0] of the VR_MII_AN_INTR_STS register. The negotiated capabilities are indicated in the CL37_SGM_STS field of this register.

7.13.2 QSGMII Auto-Negotiation

Clause 37 auto-negotiation can be independently performed for each of the 4 ports, when operating in QSGMII mode. DWC_xpcs has 4 separate MMDs (MII, MII_1, MII_2 and MII_3) to control (and to reflect the status) the auto-negotiation operation.

- MII MMD -> Port 0
- MII_1 MMD -> Port 1
- MII_2 MMD -> Port 2
- MII_3 MMD -> Port 3

After switching to QSGMII mode (as described in "Programming Guidelines for QSGMII Mode" on page 1684), follow these steps to enable auto-negotiation on a particular port (say port 2).

1. In Backplane Ethernet PCS configurations, program bit [12] (AN_EN) of SR AN MMD Control Register to 0 and bit [12] (CL37_BP) of VR XS or PCS MMD Digital Control1 Register to 1.
2. Disable Clause 37 auto-negotiation by programming bit [12] (AN_ENABLE) of SR MII_2 MMD Control Register to 0 (required only in case it is already enabled).
3. Program various fields of VR MII_2 MMD AN Control Register appropriately as follows:
   - Program TX_CONFIG to 1 (PHY side QSGMII) or 0 (MAC side QSGMII) based on your requirement
   - Program MII_AN_INTR_EN to 1, to enable auto-negotiation complete interrupt (optional step)
   - Program QSGMII_LINK_STS suitably to indicate the link status to the MAC side QSGMII, if TX_CONFIG is set to 1 (QSGMII-PHY) and bit [0] of VR MII_2 MMD Digital Control 1 Register is set to 0.
   - Program MII_CTRL to 0 or 1, as per your requirement.
4. If DWC_xpcs is configured as PHY-side QSGMII in the previous step, you can program bit [0] of VR MII_2 MMD Digital Control 1 Register to 1, if you wish to use the values of the xpcs_qsgmii_link_sts_p2_i, xpcs_qsgmii_full_duplex_p2_i and xpcs_qsgmii_link_speed_p2_i input ports as the transmitted configuration word.
5. If DWC_xpcs is configured as PHY-side QSGMII and if bit [0] of VR MII_2 MMD Digital Control 1 Register is set to 0,
   - Program SS13 and SS6 bits of SR MII_2 MMD Control Register to the required Speed Mode.
   - Program bit [5] (FD) of SR MII_2 MMD AN Advertisement Register to the desired mode

   These values get advertised in the configuration word transmitted to the link partner.

6. If DWC_xpcs is configured as MAC-side QSGMII in step 3, program bit [9] of VR MII_2 MMD Digital Control 1 Register to 1, so that DWC_xpcs automatically switches to the negotiated link-speed, after the completion of auto-negotiation. (optional step)
7. Enable CL37 Auto-negotiation, by programming bit [12] of the SR MII_2 MMD Control Register to 1.

Completion of auto-negotiation is indicated by bit [0] of VR MII_2 MMD AN Interrupt Status Register. If step 6 has not been performed, program SS13 and SS6 bits of SR MII_2 MMD Control Register suitably, based on the outcome of auto-negotiation (CL37_ANSGM_STS field of VR MII_2 MMD AN Interrupt Status Register).



Note These steps correspond to auto-negotiation on port 2. Identical steps can be performed for all the other ports by programming the respective MMD.

7.13.3 USXGMII Auto-Negotiation

Clause 37 auto-negotiation can also be performed in USXGMII Mode. For more details, see "Switching to USXGMII Mode" on page 1682.

7.14 Programming Guidelines for Clause 72 Start-Up Protocol

The Clause 72 start-up protocol should be initiated when all pages are exchanged during Clause 73 auto-negotiation and when the auto-negotiation process is waiting for link status to be UP for 500 ms after exchanging all the pages.

For information about programming guidelines for Clause 73 auto-negotiation, see "Programming Guidelines for Clause 73 Auto-Negotiation" on page 1712.

The following list explains the start-up protocol process:

1. After power on reset, soft reset, or speed switching, the DWC_xpcs initiates the Clause 73 Auto-negotiation by default. The host can also restart the Clause 73 auto-negotiation at any time by setting Bit 9 of SR_AN_CTRL Register.
2. The host must enable the Clause 72 start-up protocol by setting bits[1:0] of the SR_PMA_KR_PMD_CTRL register.
3. After Clause 73 auto-negotiation is initiated, the host must wait until all the pages are exchanged with the link partner.

   For more information, see Figure 7-2 on page 1714.
4. The host should restart the Clause 72 start-up protocol by setting Bit 0 of SR_PMA_KR_PMD_CTRL Register.

   At this stage, the host can do any of the following optional steps:

   - Get the status of the start-up protocol (frame lock, local receiver ready, start-up protocol in progress, training failure) by reading the SR_PMA_KR_PMD_CTRL register.
   - Get the latest received coefficient update or status information from the link partner by reading the following registers:
     - SR_PMA_KR_LP_CEU
     - SR_PMA_KR_LP_CESTS
   - Monitor the coefficient update or status information sent from the local device to the link partner by reading the following registers:
     - SR_PMA_KR_LD_CEU
     - SR_PMA_KR_LD_CESTS
5. The DWC_xpcs performs the start-up KR training as described in "Start-Up Protocol Flow" on page 125. If the training is successfully complete (when the local and the far-end receivers give the Receiver Ready status), the Clause 73 auto-negotiation is resumed (to look for link-okay). The KR training is successful when the DWC_xpcs gets the auto-negotiation complete interrupt as indicated by Bit 0 of VR_AN_INTR.

   If the KR training is unsuccessful (500 msec time out), the link-fault is declared and the Clause 73 AN is automatically restarted.
6. Clear the auto-negotiation complete interrupt Bit 0 of VR_AN_INTR.

If the Clause 72 start-up protocol is disabled (Bit 1 of SR_PMA_KR_PMD_CTRL is cleared), the host can directly tune the local transmitter filter coefficient. The host can do this by writing to the SR PMA MMD 10GBASE-KR LP Coefficient Update Register. The host can monitor the update status by reading the SR

PMA MMD 10GBASE-KR LD Coefficient Status before subsequent updates. The following list explains the flow for c(-1) filter which is also applicable for other filters:

1. Read Bits[1:0] of the SR_PMA_KR_LP_CESTS Register and wait till it reads as 2'b00 (NOT_UPDATED status).



This flow is applicable only for increment and decrement requests from the host. For preset and initialize requests, the host should wait till all the Tx filter status indicates as NOT_UPDATED before issuing the preset or initialize request.

2. Write 2'b01 (to increment) or 2'b10 (to decrement) to Bit[1:0] of the SR_PMA_KR_LP_CEU Register.
3. Read Bits[1:0] of the register SR_PMA_KR_LP_CESTS Register and wait till it read as 2'b11 (max), 2'b10 (min) or 2'b01 (updated) as a status.
4. Repeat step 1 through step 3 to further tune the local transmit filters.

7.14.1 Software control during Receive Equalization

DWC_xpcs allows the software (host) to control the coefficient update commands sent to the link-partner (over-riding the commands provided by the local PHY receiver) during KR training. Also, host can control the 'status' information that is returned to the local PHY receiver (over-riding the status sent by the link partner). To use this software mode, follow these steps:

1. Set bit [15] of VR_PMA_KRTR_RX_EQ_CTRL Register to 1.
2. Set bit [1] of SR_PMA_KR_PMD_CTRL Register to 1 to enable KR training.
3. Set bit [0] of SR_PMA_KR_PMD_CTRL Register to 1 to restart KR training.
4. To make an adaptation request to the PHY, program the RX_AD_REQ bit of VR_XS_PMA_MP_12G_16G_25G_RX_EQ_CTRL4 Register to 1.
5. Poll till bits [8], [9], and [10] of the VR_PMA_PHY_RX_EQ_CEU register indicate 1. CFF_UPDTM1, CFF_UPDT0 and CFF_UPDT1 fields of this register indicate the coefficient update commands given by the local PHY.
6. Issue INC/DEC/PRESET/INIT coefficient update requests to link partner by programming any (one or more) of the fields (PRST, INIT, CFF_UPDT1, CFF_UPDT0, CFF_UPDTM1) of VR_PMA_KRTR_RX_EQ_CTRL Register.
7. Poll SR_PMA_KR_LP_CESTS Register till it indicates UPDATED/MAX/MIN status for the corresponding coefficients for which an update request was made.
8. Software can write any desired status to VR_PMA_KRTR_RX_EQ_STS_CTRL Register (for non-Synopsys PHY configurations), which gets driven out to the local PHY.
9. Issue a HOLD request to the link-partner by writing 2'b00 to the same fields that is programmed in step 6.
10. Poll SR_PMA_KR_LP_CESTS register till it indicates NOT_UPDATED status for that particular coefficients.
11. Repeat step 4 through step 10, for further tuning of the (same or other) filter coefficients.
12. Program RR_RDY field of VR_PMA_KRTR_RX_EQ_CTRL Register to 1 to indicate to the link-partner that the local receiver is trained.


Note When performing Clause 72 KR training in conjunction with Clause 73 auto-negotiation, initiate step 3 only after all AN page exchanges are complete.

7.14.2 Software control during Transmit Equalization

DWC_xpcs allows the software (host) to control the coefficient update commands driven to the local PHY transmitter (over-riding the commands received from link partner) during KR training. Also, host can control the 'status' information that is returned to the link partner (over-riding the status sent by local PHY). To use this software mode, follow these steps:

1. Program bit [15] of VR_PMA_KRTR_TX_EQ_STS_CTRL Register to 1.
2. Set bit [1] of SR_PMA_KR_PMD_CTRL Register to 1 to enable KR training.
3. Set bit [0] of SR_PMA_KR_PMD_CTRL Register to 1 to restart KR training.
4. Poll for INC/DEC/PRESET/INIT commands in LP_CFF_UPDTM1/LP_CFF_UPDT0/ LP_CFF_UPDTM1/LP_INIT/PRST fields of SR_PMA_KR_LP_CEU Register (which indicate the coefficient update commands received from the link partner).
5. Issue INC/DEC/PRESET/INIT coefficient update requests to the local PHY transmitter by programming any fields (CFF_PRST, CFF_INIT, CFF_UPDT1, CFF_UPDT0, CFF_UPDTM1) of VR_PMA_KRTR_TX_EQ_CFF_CTRL Register to appropriate values.
6. Poll STSM1_VLD/STS0_VLD/STS1_VLD bits of VR_PMA_PHY_TX_EQ_STS Register till it indicate 1 (for whichever coefficient a command is issued in step 5). The EQ_STSM1, EQ_STS0 and EQ_STS1 fields of this register indicate the 'Status' provided by the local PHY.
7. Write UPDATED/MIN/MAX Status to CFF_STSM1/CFF_STS0/CFF_STS1 fields of VR_PMA_KRTR_TX_EQ_STS_CTRL Register (which gets transmitted to the link partner via training frames).
8. Poll for HOLD commands in SR_PMA_KR_LP_CEU Register.
9. Write HOLD commands to VR_PMA_KRTR_TX_EQ_CFF_CTRL Register.
10. Poll till STSM1_VLD/STS0_VLD/STS1_VLD bits of VR_PMA_PHY_TX_EQ_STS Register indicate 0.
11. Write NOT_UPDATED status to VR_PMA_TX_EQ_STS_CTRL Register.
12. Repeat step 4 to step 11, for further tuning the same or other filter coefficients.

Note

When INIT/PRESET command is received in SR_PMA_KR_LP_CEU Register, software should return the UPDATED/MIN/MAX status for all the three coefficients.

Whenever INIT or PRESET commands are programed to VR_PMA_KRTR_TX_EQ_CFF_CTRL Register, the Status values are returned (by the local PHY) for all the three coefficients. As a result, software should subsequently write HOLD commands for all the three coefficients

For Synopsys PHY configurations, if more than one coefficients are simultaneously set to INC/DEC in VR_PMA_KRTR_TX_EQ_CFF_CTRL register, DWC_xpcs processes them one by one, based on the priority order defined by the PHY. Software should check which valid field (STS1_VLD/STS0_VLD/STSM1_VLD) is getting asserted first. Then, it should program HOLD for that particular coefficient. DWC_xpcs proceeds to serve the INC/DEC command for the other coefficients only after that. To simplify the software sequence in such cases, it is recommended that software does not program all three coefficients (to INC/DEC) at the same time. Instead, only one coefficient is programmed at a time.

7.15 Programming Guidelines for Clause 73 Auto-Negotiation

This section describes the hardware-software interaction required for the Clause 73 auto-negotiation process.

7.15.1 Definitions

Base Page: First Page exchange between local device and link partner after the auto-negotiation started or restarted

Next Page: The pages that are exchanged between the local device and the link partner after the Base page

7.15.2 Clause 73 Auto-Negotiation Process

The clause 73 auto-negotiation consists of the following:

1. Auto-negotiation initiation

   The DWC_xpcs automatically starts auto-negotiation when you power on, soft reset, or power cycle (that is, power down and then power up) the DWC_xpcs. The host can restart the auto-negotiation by setting Bit 9 of the SR_AN_CTRL Register to 1.

   The host can disable the auto-negotiation by clearing Bit 12 of SR_AN_CTRL Register. When auto-negotiation is disabled, link speed is determined by the speed at which DWC_xpcs was powered up.

Note

- The host can change the default values of the Base Page register by writing to the following registers before starting or restarting the Clause 73 auto-negotiation, in the following order:
  - Write to SR_AN_ADV3 Register.
  - Write to SR_AN_ADV2 Register.
  - Write to SR_AN_ADV1 Register.
- If required, the Next page registers should be written in the following order:
  - Write to SR_AN_XNP_TX3 Register.
  - Write to SR_AN_XNP_TX2 Register.
  - Write to SR_AN_XNP_TX1 Register.

This sequence is required because the DWC_xpcs considers that the 48-bit Base page and Next Page is ready for transmission once the host writes into SR AN Advertisement Register 1 or SR AN XNP Transmit Register 1, respectively.

2. Speed Mode Dependency

   The DWC_xpcs supports only one of the configured speed modes at a time. If the DWC_xpcs is configured to have multiple speed modes (KR_KX4_KX, KR_KX4, KR_KX, and KX4_KX), speed switching is implemented as explained in "Switching to 10G XAUI Mode or KX4 Mode" on page 1652, "Switching to 1G or KX Mode and 10 or 100 Mbps SGMII Speed" on page 1669, "Switching to 10G/5G KR Speed Mode" on page 1676, and"Programming Guidelines for 2.5GBASE-KX Mode (over XGMII)" on page 1696to switch to KX4, 1000BASE-KX, KR, 2.5GBASE-KX mode, respectively.

Note

In configurations with MDIO interface, you must read the VR_AN_INTR Register after the auto-negotiation is complete.

Figure 7-2 describes the auto-negotiation flow when the current speed mode is 10G KR. When you select Add Clause 72 KR Training Support in your configuration, follow these steps before you proceed with the instructions in Figure 7-2.

1. Disable auto-negotiation by programming the AN_EN bit of the SR_AN_CTRL register to 0.
2. (For Synopsys PHY configurations only) program RX_DT_EN_0 field of VR_XS_PMA_MP_12G_16G_25G_RX_GENCTRL0 register to 0.
3. (For Synopsys PHY configurations only) program RX_EQ_MM field of VR_PMA_KRTR_RX_EQ_CTRL register to 1.
4. To enable KR training, program the SR_PMA_KR_PMD_CTRL register to 2'b11.
5. Enable auto-negotiation by programming AN_EN bit of SR_AN_CTRL register to 1.

Figure 7-2 Clause 73 Auto-negotiation in the 10G KR Speed Mode

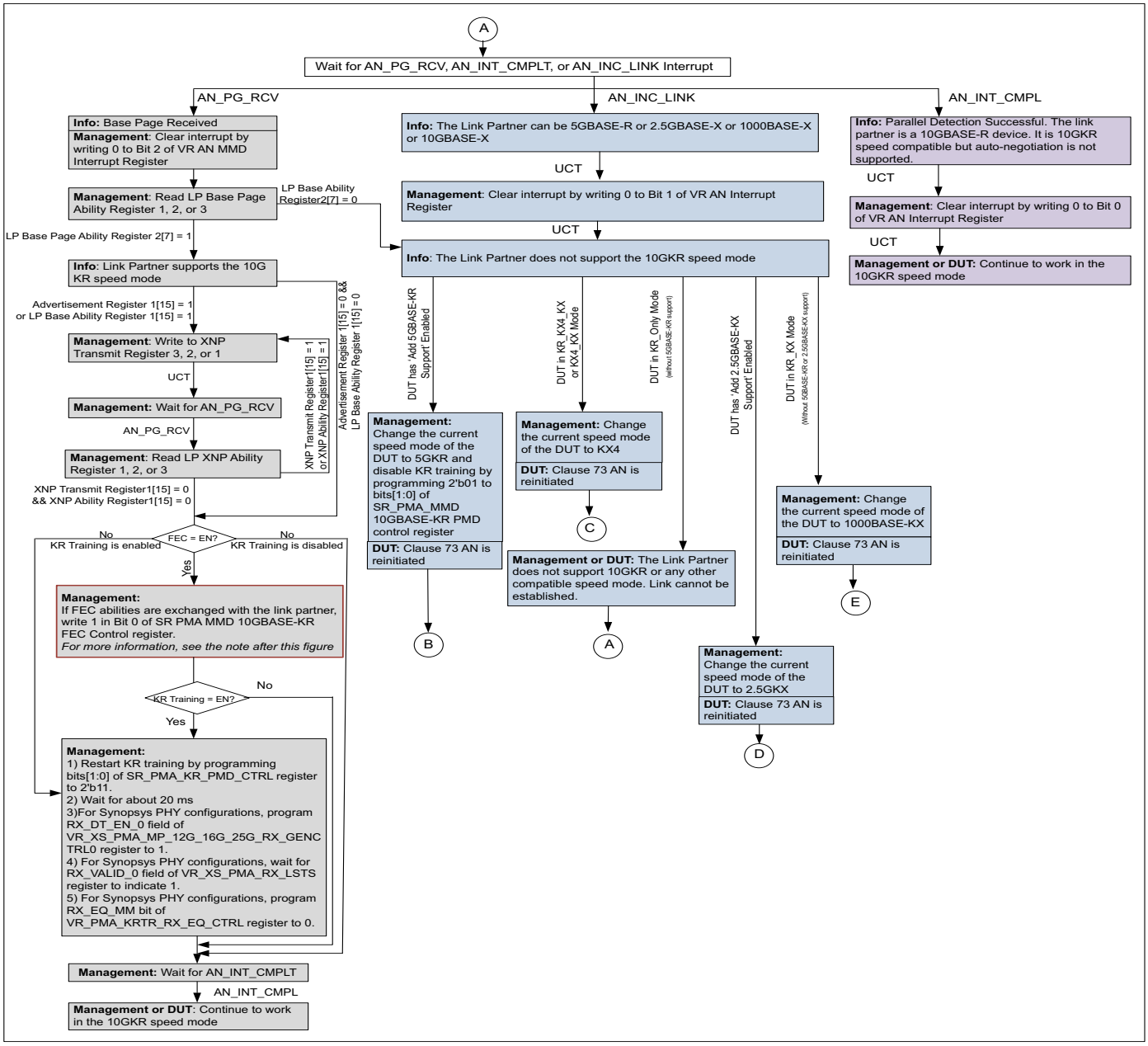


Figure 7-3, Figure 7-4, Figure 7-5, and Figure 7-6 describe the auto-negotiation flow when the current speed mode is 5GKR, KX4, 2.5GKX, or 1000BASE-KX, respectively.

Figure 7-3 Clause 73 Auto-negotiation in 5GKR Speed Mode

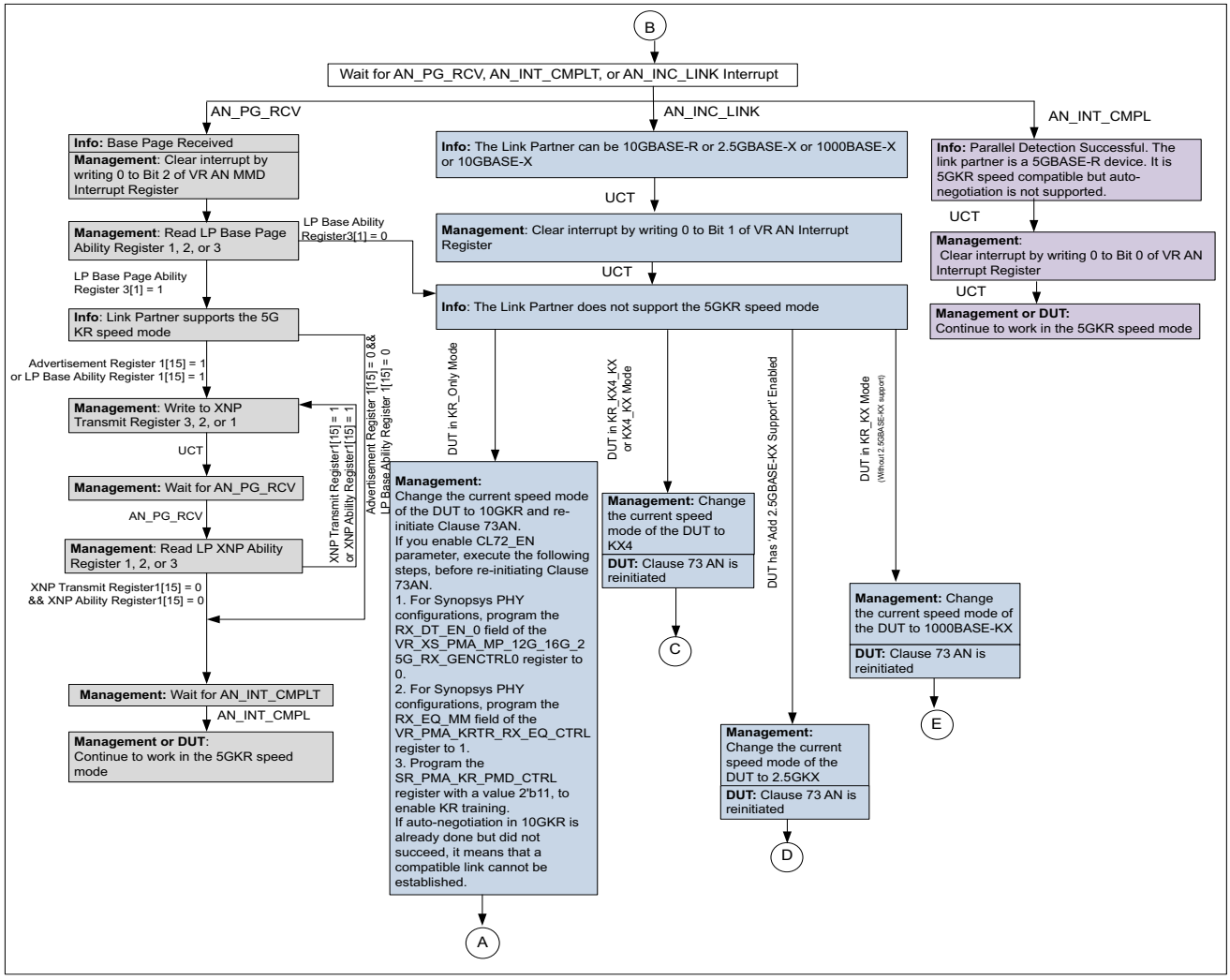


Figure 7-4 Clause 73 Auto-negotiation in the KX4 Speed Mode

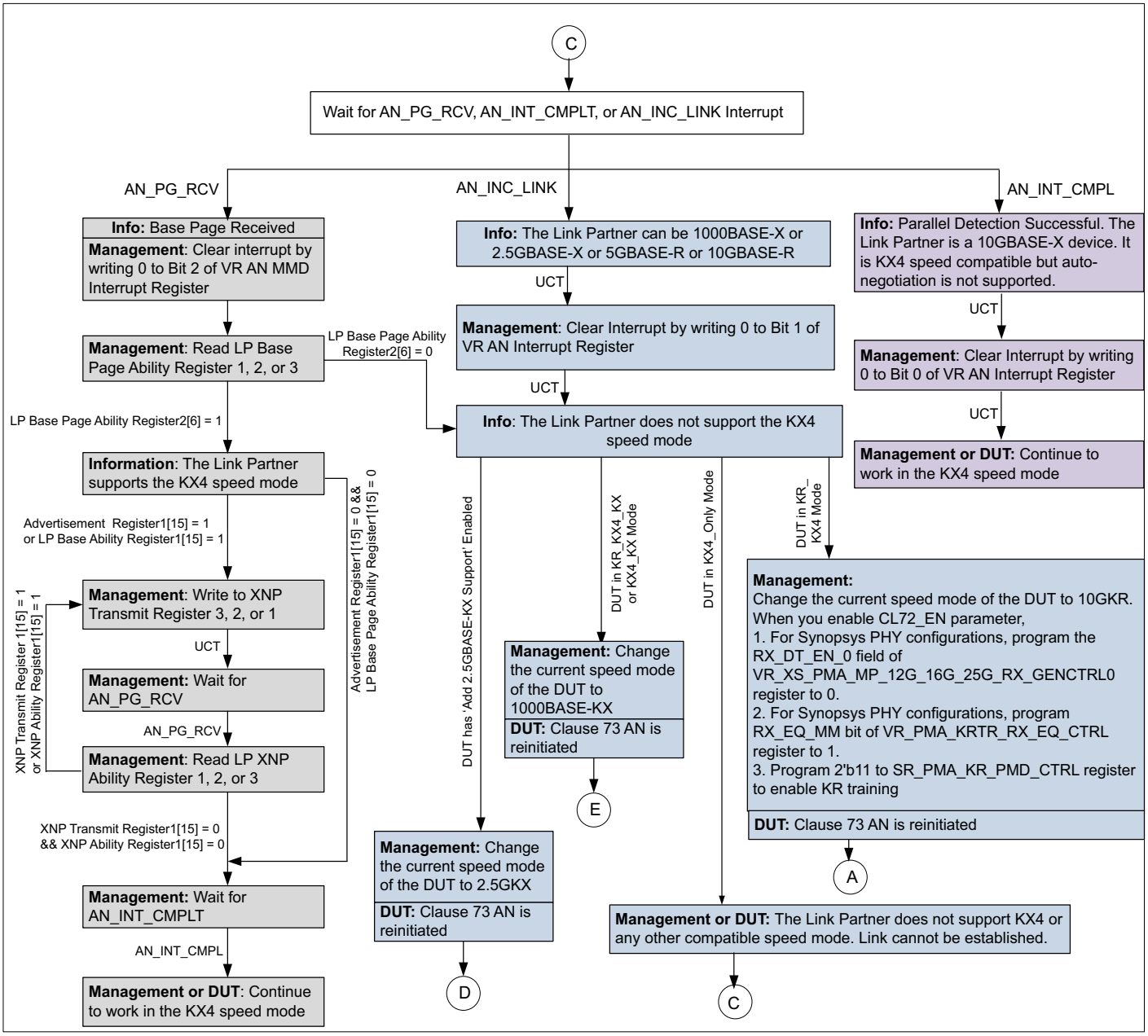


Figure 7-5 Clause 73 Auto-negotiation in 2.5GBASE-KX Mode

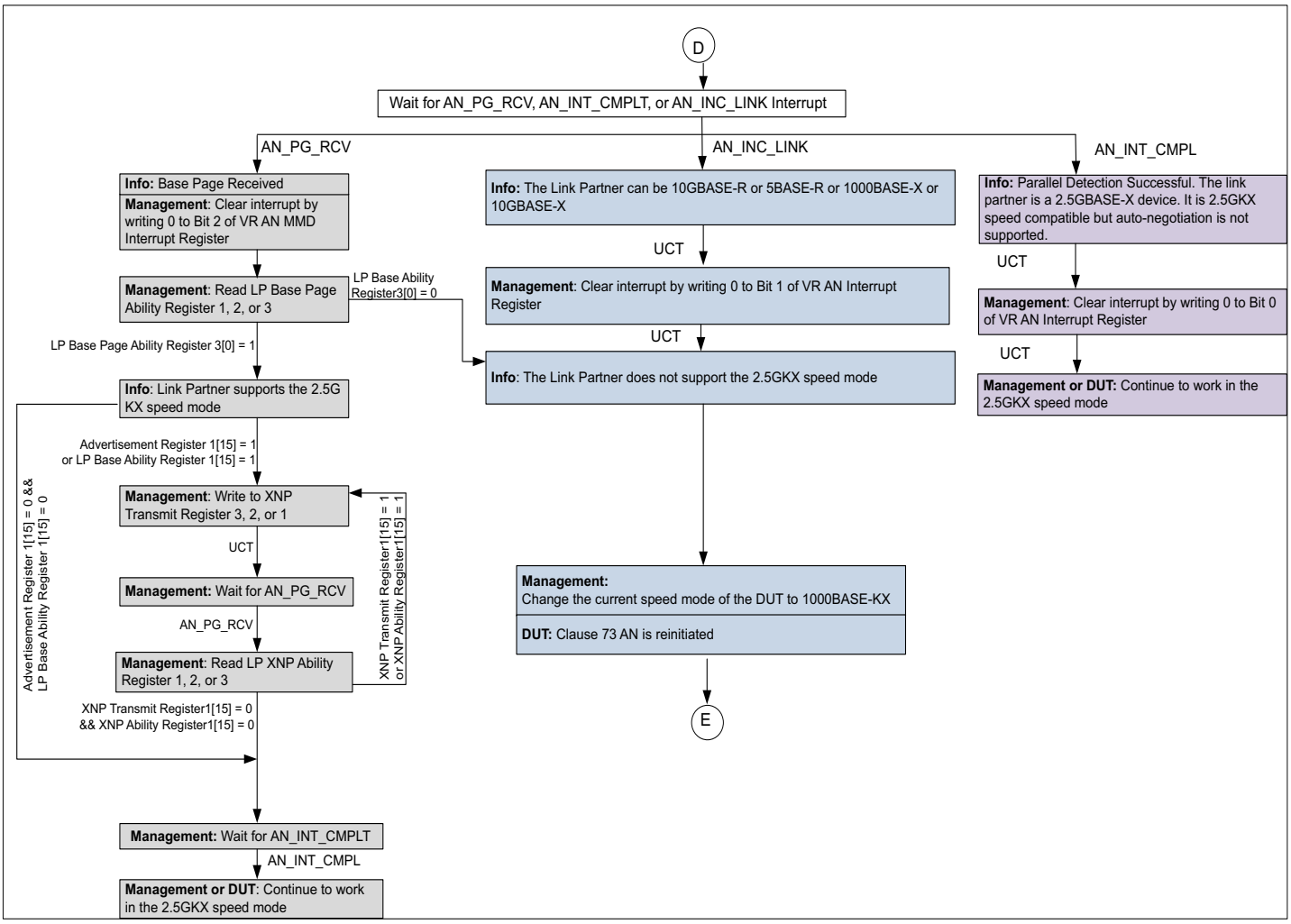
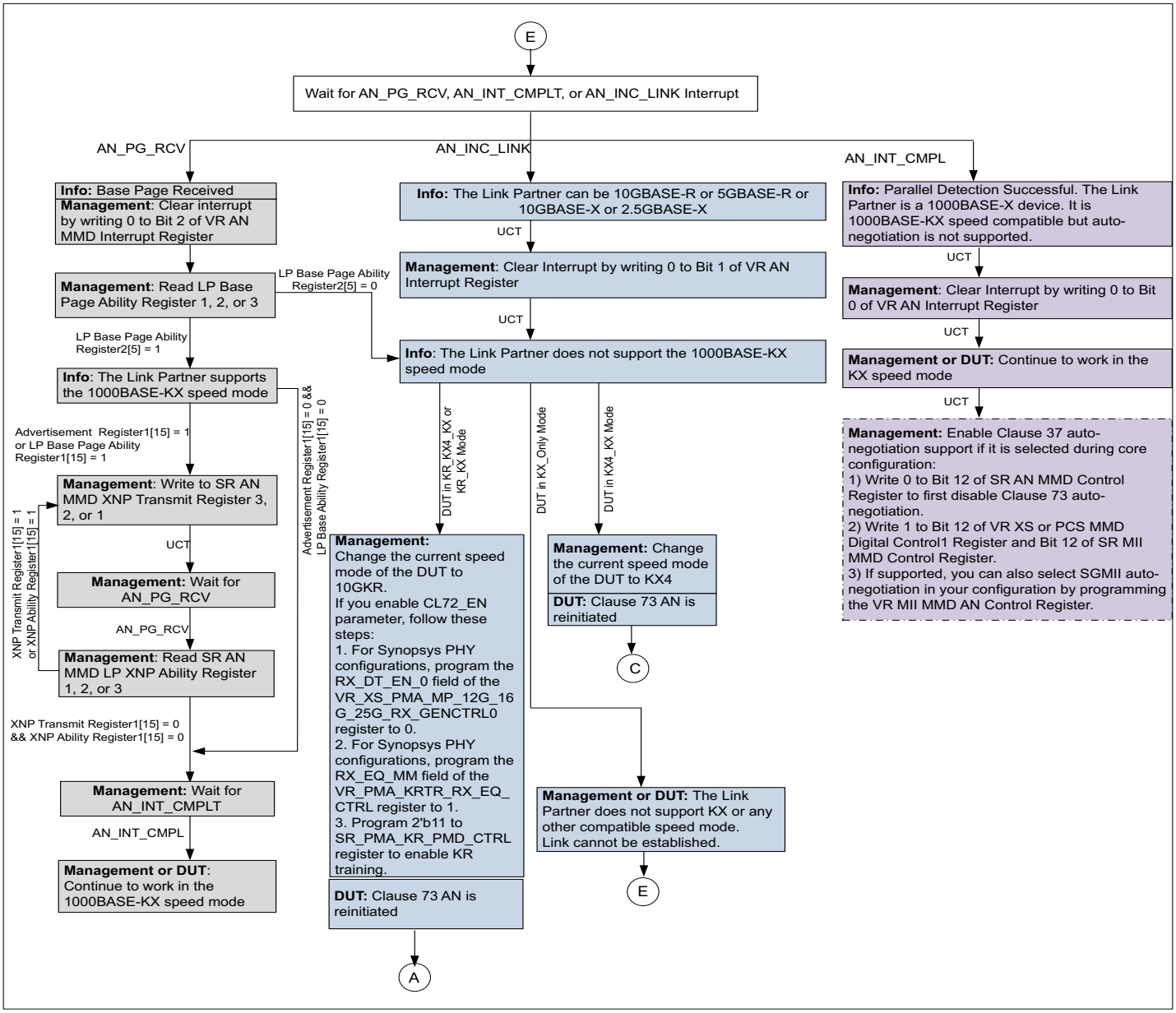


Figure 7-6 Clause 73 Auto-negotiation in the 1000BASE-KX Speed Mode



Note

The IEEE Std 802.3ap does not define the relationship between Clause 37 and Clause 73 auto-negotiation flows and when to switch between these modes. You should enable the Clause 37 auto-negotiation mode only when the far-end device also supports the Clause 37 auto-negotiation, and it has enabled the Clause 37 auto-negotiation (and disabled Clause 73 auto-negotiation). Otherwise, the auto-negotiation does not complete successfully and the link fault may be declared. Therefore, you should be aware of and careful about the implications of enabling Clause 37 auto-negotiation between Backplane Ethernet capable devices.

One use of this feature is when the far-end device is a legacy device (not capable of Backplane Ethernet). In such a case, the Clause 73 auto-negotiation in local device completes with parallel-detect link-okay status in the KX mode. In such a scenario, the Clause 37 auto-negotiation (with SGMII) can be enabled to check the capabilities of the far-end link.

7.16 Programming Guidelines for Energy Efficient Ethernet

To enable the Energy Efficient Ethernet (EEE) feature, the host should complete the following:

1. Check if DWC_xpcs supports the EEE feature by reading the SR_XS_PCS_EEE_ABL register. If Bit 4 of this register is set, it indicates that DWC_xpcs supports EEE in1000BASE-X or KX mode. Similarly, Bit 5 indicates that DWC_xpcs supports EEE in 10GBASE-X PCS or 10GBASE-KX4 modes, and Bit 6 indicates that DWC_xpcs supports EEE in 10GBASE-R PCS or 10GBASE-KR mode.
2. Program various timers used in the EEE mode depending on the clk_eee_i clock frequency:
   - For 1000BASEX-Only PCS configurations, program the following registers:
     - VR_MII_EEE_MCTRL0
     - VR_MII_EEE_TXTIMER
     - VR_MII_EEE_RXTIMER
   - For all other configurations, program the following registers:
     - VR_XS_PCS_EEE_MCTRL0
     - VR_XS_PCS_EEE_TXTIMER
     - VR_XS_PCS_EEE_RXTIMER



Note

The clk_eee_i clock can be independent of Tx and Rx path clocks. It should be free running. The frequency of this clock can be between 10 MHz to 200 MHz (excluding the frequency range of 12.5 MHz to 16.67 MHz).

A TCL script named `eee_reg_program_guide.tcl` is provided in the `<workspace>/custom/resources/eee` directory along with a `README_eee_reg_prog.txt` file. This tcl file provides programming assistance for EEE-related registers, depending upon the clk_eee_i time period and the DWC_xpcs mode of operation. For more information about how to operate the TCL script, see the README file.

3. If FEC is enabled in the KR mode, set Bit 5 of VR_XS_PCS_EEE_MCTRL0 Register to 1 to enable the FEC fast synchronization.
4. Enable the EEE feature on the Tx path by setting Bit 0 of VR_XS_PCS_EEE_MCTRL0 Register. Similarly, enable the EEE feature on the Rx path by setting Bit 1 of VR_XS_PCS_EEE_MCTRL0 Register.

   When these bits are set, DWC_xpcs enables the encoding or decoding of the LPI code groups and also enables the EEE Tx and Rx controllers.



Note

DWC_xpcs has recovery time after it comes out of the LPI mode and before it is ready to send the frames. The worst-case recovery time is equal to the following:

`4 Clocks of clk_eee_i period + 3 clocks of clk_xgmii_tx_i, clk_xgxs_i, or clk_rpcs_tx_i` (whichever is slower among these three clocks).

Note

DWC_xpcs supports 'Transparent LPI Mode' in the Tx and Rx data paths. In this mode, DWC_xpcs does not support clock-gating. You can enable this mode in the

- Tx path by programming TRN_LPI bit of VR XS or PCS MMD EEE Mode Control 1 Register or VR MII MMD EEE Mode Control 1 Register.
- Rx path by programming TRN_RXLPI field of the VR XS or PCS MMD EEE Mode Control 1 Register or VR MII MMD EEE Mode Control 1 Register to 1.

7.17 Programming Guidelines for ECC Protection for Memories

Programming guidelines for the ECC protection for memories:

- In DWC_xpcs, when you program the FEC_EN field of the SR_PMA_KR_FEC_CTRL register, ECC protection feature is enabled.
- A correctable error interrupt (xpcs_sfty_ce_intr_o) is generated to indicate to the application if any correctable/uncorrectable/address errors are detected. Appropriate status is indicated in the VR_XS_PCS_SFTY_UE_INTR0 or VR_XS_PCS_SFTY_CE_INTR register.
- Debug information is available in VR_XS_PCS_EC_MEM_ECC_STS and VR_XS_PCS_EF_MEM_ECC_STS registers.

7.17.1 Programming Guidelines for ECC Hardware Error Injection

Follow these steps for ECC Hardware Error Injection.

- Program the following fields of the VR_XS_PCS_SFTY_DBG_CTRL register to suitable values to control the error injection mode for 'Error Correction' or 'Error forwarding' memories:
  - EC_EI_MOD/EF_EI_MODE
  - ERR_MODE
  - ERR_LOC
- Program EC_ECCEI bit of VR_XS_PCS_SFTY_ERR_INJ register to 1 to enable error injection in 'Error Correction' memory.
- Program EF_ECCEI bit of VR_XS_PCS_SFTY_ERR_INJ register to 1 to enable error injection in 'Error Forwarding' memory.
- On injecting error, an uncorrectable error interrupt is asserted. Interrupts can be cleared by writing 0 to appropriate bits of VR_XS_PCS_SFTY_UE_INTR0 register.

7.18 Programming Guidelines for On-Chip Datapath Parity Protection

- Datapath parity protection is enabled by default
- An uncorrectable error interrupt (xpcs_sfty_ue_intr_o) is generated to indicate to the application if any parity mismatch is detected. In addition, DPP_INT bit of VR_XS_PCS_SFTY_UE_INTR0 register gets set.

7.18.1 Programming Guidelines for On-chip Datapath Parity Error Injection

- Program the following fields of VR_XS_PCS_SFTY_DBG_CTRL register to suitable values to control the error injection mode:
  - ERR_MODE
  - ERR_LOC
- Program DPEI field of VR_XS_PCS_SFTY_ERR_INJ register to inject DPP error.
- An uncorrectable error interrupt (xpcs_sfty_ue_intr_o) is generated on injecting errors. In addition, DPP_INT interrupt bit of VR_XS_PCS_SFTY_UE_INTR0 gets set.
- Clear the interrupt bit by writing 0 to resume normal operation.



Note Datapath Parity checkers are not applicable/present for certain configurations or speed modes (depending on presence/absence of active FIFOs in receive datapath) Therefore, error interrupt is not generated in such cases even if DPEI bit is set. For more details, see “Automotive Safety Features” on page 1727.

7.19 Programming Guidelines for FSM Parity Protection

Follow these steps to program the FSM parity and timeout

- FSM parity protection is enabled by default
- An uncorrectable error interrupt (xpcs_sfty_ue_intr_o) is generated when an FSM parity error (FSM state with even number of 1s) is detected.Variousbitsof VR_XS_PCS_SFTY_UE_INTR0 register indicates the clock domain of the FSM which encountered parity error.

7.19.1 Programming Guidelines for FSM Parity Error Injection

- Program the following bits of VR_XS_PCS_SFTY_ERR_INJ register to inject parity error to FSMs operating in various clock domains:
  - TXFPEI
  - RXFPEI
  - CSRFPEI
  - EEEFPEI
  - CKOFPEI
- An uncorrectable error interrupt (xpcs_sfty_ue_intr_o) is generated on injecting errors. Various bits of VR_XS_PCS_SFTY_UE_INTR0 register indicates the clock domain of the FSM which encountered parity error.
- Clear the interrupt bits of VR_XS_PCS_SFTY_UE_INTR0 register by writing 0 to resume normal operation

7.20 Programming Guidelines for FSM Timeout Protection

- FSM time-out protection is enabled by default. You can disable this feature by programming the FT_DIS of VR_XS_PCS_SFTY_DISABLE register to 1.
- Program SFTY_1US_MULT field of VR_XS_PCS_SFTY_TMR_CTRL register with appropriate number of CSR clock cycles to generate 1 us tic.
- Time-out values can be modified (scaled) by programming FSM_TO_SEL field of VR_XS_PCS_SFTY_TMR_CTRL register.
- An uncorrectable error interrupt (xpcs_sfty_ue_intr_o) is generated when an FSM time-out error is detected. Various bits of VR_XS_PCS_SFTY_UE_INTR0 register indicates the clock domain of the FSM which encountered time-out error.



Note To prevent false time-out interrupts during speed mode switching sequence, it is recommended that you program FT_DIS bit to 1 before the speed-switch sequence and after completing the switching sequence, program it back to 0.

7.20.1 Programming Guidelines for FSM Timeout Error Injection

- Program the following bits of VR_XS_PCS_SFTY_ERR_INJ register to inject timeout error to FSMs operating in various clock domains:
  - TXFPEI
  - RXFPEI
  - CSRFPEI
  - EEEFPEI
  - CKOFPEI
- An uncorrectable error interrupt (xpcs_sfty_ue_intr_o) is generated on injecting errors, after the lapse of the time-out interval. Various bits of VR_XS_PCS_SFTY_UE_INTR0 register indicates the clock domain of the FSM which encountered timeout error.
- Clear the interrupt bits of VR_XS_PCS_SFTY_UE_INTR0 register by writing 0 to resume normal operation

7.21 Programming Guidelines for CSR Protection

- CSR protection feature is enabled by default.
- An uncorrectable error interrupt (xpcs_sfty_ue_intr_o) is generated to indicate to the application if any CSR bit has flipped. In addition, CSR_INT bit of VR_XS_PCS_SFTY_UE_INTR1 register gets set.

7.21.1 Programming Guidelines for CSR Error Injection

- Errors can be inserted in CSR protection feature by programming CSREI bit of VR_XS_PCS_SFTY_ERR_INJ register to 1
- An uncorrectable error interrupt (xpcs_sfty_ue_intr_o) is generated on injecting errors. In addition, CSR_INT bit of VR_XS_PCS_SFTY_UE_INTR1 register gets set
- Clear the interrupt bits of VR_XS_PCS_SFTY_UE_INTR1 register by writing 0 to resume normal operation.

7.22 Programming Guidelines for Interface Timeout Protection

- Interface timeout protection feature is enabled by default.
- Program SFTY_1US_MULT field of VR_XS_PCS_SFTY_TMR_CTRL register with appropriate number of CSR clock cycles to generate 1 us tic.
- Interface timeout values can be selected by programming IFT_SEL field of VR_XS_PCS_SFTY_TMR_CTRL register
- An uncorrectable error interrupt (xpcs_sfty_ue_intr_o) is generated when an interface time-out error is detected. Various bits of VR_XS_PCS_SFTY_UE_INTR0 register indicate which interface encountered a time-out error

7.22.1 Programming Guidelines for Interface Timeout Error Injection

- Program APB3IFTEI or PHYIFTEI fields of VR_XS_PCS_SFTY_ERR_INJ register to 1 to inject interface timeout errors
- An uncorrectable error interrupt (xpcs_sfty_ue_intr_o) is generated on injecting errors after the lapse of the time-out interval. In addition, PHY_IFT_INT or APB3_IFT_INT interrupt bit of VR_XS_PCS_SFTY_UE_INTR0 register is set.
- Clear the interrupt bits of VR_XS_PCS_SFTY_UE_INTR1 register by writing 0 to resume normal operation.



Note After enabling the error injection bits (APB3IFTEI/PHYIFTEI), application must not perform interface access operations until the interrupt is generated by DWC_xpcs.

8

Automotive Safety Features

This chapter describes the automotive safety features supported by DWC_xpcs and enabled by the DWC_XPCS_ASP parameter when you have the corresponding license.

This chapter has the following sections:

- "Error Correction Code (ECC) Protection for Memories" on page 1728
- "On-Chip Data Path Parity Protection" on page 1731
- "FSM Parity and Timeout Protection" on page 1736
- "Interface Timeout Protection" on page 1739
- "CSR Protection" on page 1741

8.1 Error Correction Code (ECC) Protection for Memories

8.1.1 Overview

The Error Correction Code (ECC) block can correct single-bit error and detect double-bit error.

8.1.2 Description of ECC Protection for Memories

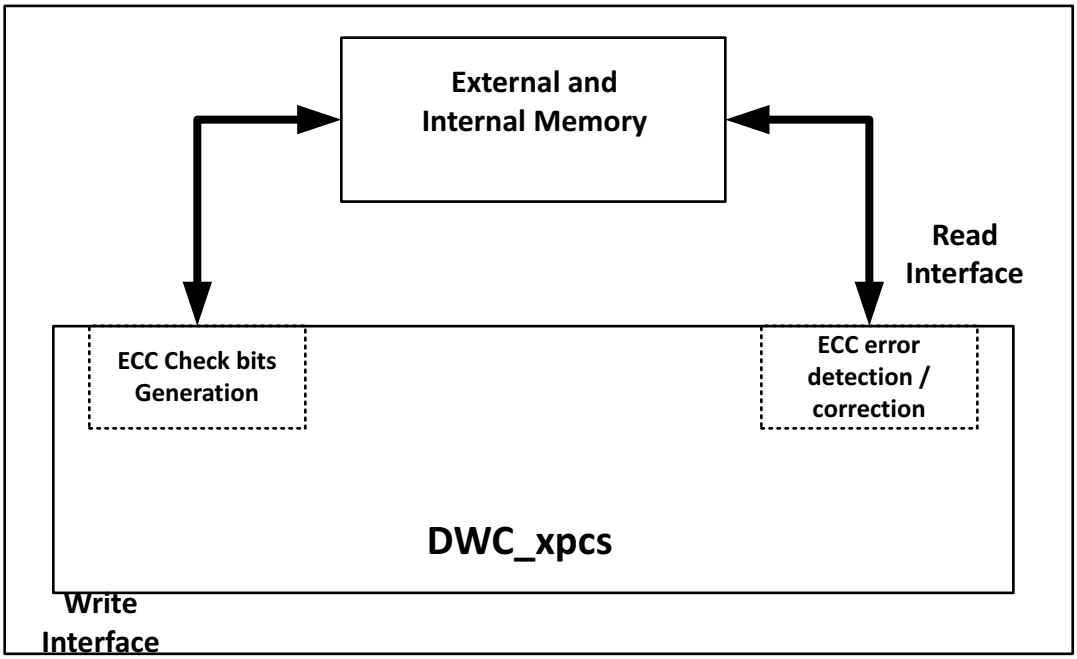
The Error Correction Code (ECC) block can correct single-bit error and detect double-bit error.

At the write interface, ECC checkbits are generated by computing ECC on the contents of the data bus and respective address is appended with the data that is written to the memory.

At the read interface, ECC checkbits are recomputed on the content of the read data and the respective address is compared with the received checkbits in the memory.

Figure 8-1 shows the block diagram of ECC protection for memories

Figure 8-1 Block Diagram of ECC Protection for Memories



The number of ECC-checkbits generated for each data block depends on the size of the data bus and address bus. Table 8-1 describes the memory bus widths, ECC code size and the generated checkbits size.

Table 8-1 Memory Bus Width, ECC Code Size, and Checkbits Size

Memory Block	Data Width (Without ECC)	Address Width	ECC Code Reference Range	Required Checkbits	Data Width (with ECC)
FEC Error Correction Memory (when FEC_EN =Enabled)	66 bits	5 bits	Data width + Address width <= 120 bits	8 bits	74 bits
FEC Error Forwarding Memory (when FEC_ERROR_FWD= Enabled)	66 bits	5 bits	Data width + Address width <= 120 bits	8 bits	74 bits

8.1.2.1 Handling the Address Mismatch

ECC is calculated over the combination of memory data and memory address. However only data is written in the memory along with the ECC. (address is excluded).

On the read interface, the ECC is checked over the combination of memory DATA and address. When an ECC correctable error is detected in the address bit position, it is treated as an uncorrectable error. This is because data might have been read from a different location.

8.1.2.2 Diagnostic Support for the Error Management

Following statistics are provided for monitoring the error behavior on each of the memory blocks.

- Error status provided to management
  - Uncorrectable error status is reported in the VR_XS_PCS_SFTY_UE_INTR0 register. Separate status bits are available for EC and EF memories.
  - Correctable error status is reported in VR_XS_PCS_SFTY_CE_INTR register. Separate status bits are available for EC and EF memories
  - The most recent address location at which correctable error is detected is reported in the VR_XS_PCS_EC_MEM_STS/VR_XS_PCS_EF_MEM_STS register. In addition, these registers indicate the number of ECC code-words that are corrected.
- Interrupts provided to management
  - Separate interrupts are generated for correctable and uncorrectable errors.
  - xpcs_sfty_ue_intr_o interrupt is generated when uncorrectable errors are detected and these errors cannot be masked. Application should reset DWC_xpcs IP on detecting uncorrectable errors.
  - xpcs_sfty_ce_intr_o interrupt is generated when correctable errors are detected.

For more details, see the "*Register Descriptions*" chapter.

8.1.2.3 ECC Error Injection Capabilities

ECC error protection in DWC_xpcs supports a hardware debug mode where the hardware injects error at the memory interface and reports it to the application through a safety interrupt. Failure to report Error/Safety Interrupt indicates an Error in the ECC Safety Mechanism.

To enable ECC Error Injection program the EC_ECCEI or EF_ECCEI fields of VR_XS_PCS_SFTY_ERR_INJ register.

DWC_xpcs supports the following types of error injection:

- 1-bit data error
- 2-bit data error

This can be controlled by programming the EC_EI_MODE/EF_EI_MODE fields of the VR_XS_PCS_MII_SFTY_DBG_CTRL register.

Error can be introduced either in the data bits or the parity bits feeding the EC checker. Error is introduced only in the logic path feeding the checker. This error injection mode does not impact the normal datapath. To control the error position (bit position) program the ERR_LOC field of the VR_XS_PCS_SFTY_DBG_CTRL register. For the ECC parity bit calculation, both data bits (66-bits) and

address bits (5-bits) are considered as 'data' field, address errors can be introduced by programming appropriate values in the ERR_LOC field. For more details, see ERR_LOC field in the Register Description Chapter.

8.1.3 Enabling ECC Protection for Memories

To enable the ECC support during the Specify Configuration activity in CoreConsultant, select the Add Safety Support and Add Forward Error Correction Support, in the System Configurations Settings window.

For details about these options, see Automotive Safety Features Parameters in the "*Parameter Descriptions*" chapter.



See the "Enabled" field of the parameter descriptions to understand the dependencies.

8.1.4 Signals Related to ECC Protection for Memories

- xpcs_sfty_ue_intr_o
- xpcs_sfty_ce_intr_o

For a description of signals related to ECC Protection for Memories, see the "*Signal Descriptions*" chapter.

8.1.5 Registers Related to ECC Protection for Memories

- `VR_XS_PCS_SFTY_UE_INTR0`
- `VR_XS_PCS_EC_MEM_ECC_STS`
- `VR_XS_PCS_EF_MEM_ECC_STS`
- `VR_XS_PCS_SFTY_DBG_CTRL`
- `VR_XS_PCS_SFTY_ERR_INJ`

For more details, see the "*Register Descriptions*" chapter.

8.1.6 Programming Guidelines for ECC Protection for Memories

For more details about programming guidelines for ECC protection for memories, see "Programming Guidelines for ECC Protection for Memories" on page 1721.

8.2 On-Chip Data Path Parity Protection

8.2.1 Overview

On-Chip Data Path Parity Protection supports protection for Transmit and Receive datapaths, where feasible. For every byte of data, one bit parity is generated. Only limited datapath parity protection is implemented in DWC_xpcs as data gets manipulated inside various sub-modules of DWC_xpcs (due to encoding, decoding operations and so on). There is no parity protection at all in the transmit path. Error introduced in the DWC_xpcs Tx datapath is detected by the far-end MAC. Similarly, any error in the DWC_xpcs receiver is detected by the local end MAC.

Note DWC_xpcs implements an odd parity.

The following sections describe the parity generation and checking scheme implemented in various operating modes of DWC_xpcs.

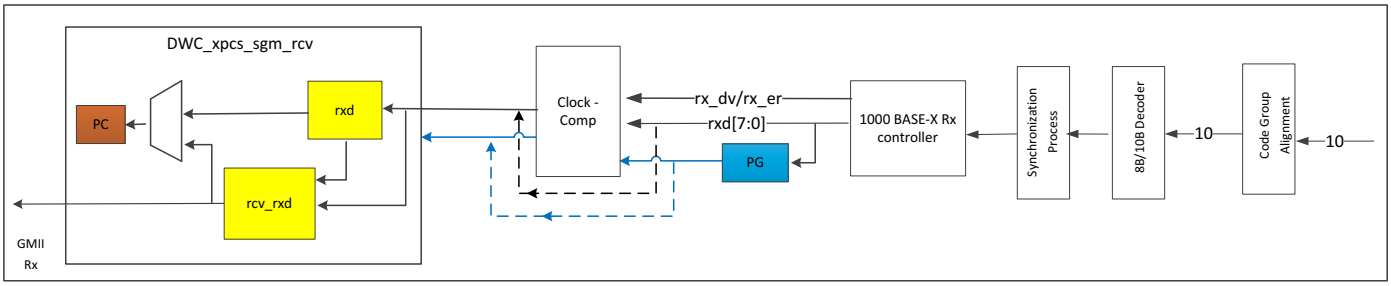
8.2.2 PCS-X - 1G Mode

In the PCS-X 1G mode of operation, datapath is 8 bits wide. Therefore, 1-bit parity is added at the input of the clock compensation module.

8.2.2.1 Configurations With SGMII_EN = Enabled

When operating at 1G mode or 8-bit 10M/100M mode, parity check is done on the final register stage that drives the gmii_rxd_o[7:0] port. When operating in 4-bit mode, parity check is done on a 8-bit internal data, before the final 8-bit to 4-bit conversion, as shown in Figure 8-2.

Figure 8-2 Parity Check When SGMII_EN = Enabled



Note Parity bit is added per byte. So, no checkisperformedontherx_er or rx_dv signals.

8.2.2.2 Configurations With SGMII_EN = Disabled

For configurations without SGMII support, parity check is performed at the following locations based on the configuration:

- Output of DWC_xpcs_xdr_rgif module for multi-lane configurations with XGMII interface as DDW
- Output of clock-compensation module for configurations with XGMII interface as SDR and CLKCOMP=Enabled

For configurations that do not have any downstream logic between '1000BASE-X Rx controller' and GMII Rx interface, parity check is not performed.

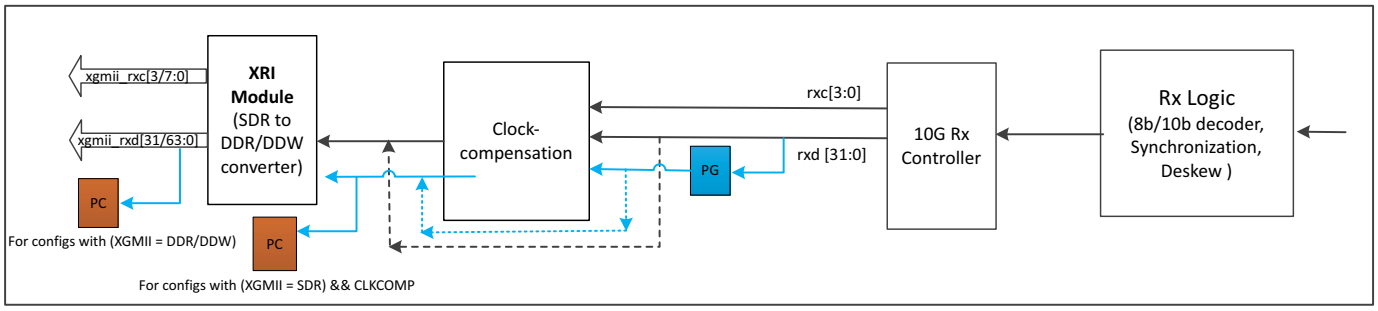
8.2.3 PCS-X - QSGMII Mode

QSGMII configurations have four parity generators and checkers. Each of the four parity generators generates parity bits for the respective data outputs from the four 1000 BASE-X receive controllers. The parity bits then pass through the corresponding clock compensation module (for configurations with CLKCOMP=Enabled) and finally reach the corresponding SGMII/QSGMII Rate adapter module (DWC_xpcs_sgm_rcv). The parity checker inside the rate-adapter module (4 instances) performs parity check on the data (similar to 1G mode) and reports when there is a parity error.

8.2.4 PCS-X - 10GBASE-X/KX4 Mode

The 4-bit parity (1-bit per byte) is added at the 32-bit data output from '10G Rx controller'. Parity check is performed at the XGMII Rx interface for DDW configurations. For SDR configurations (as XRI module is absent), parity check is performed at the output of the clock compensation module (for configurations with CLKCOMP=Enabled).

Figure 8-3 Parity Check for 10GBASE-X/KX4 Mode



For configurations that do not have downstream logic between 10G Rx controller and XGMII Rx interface, there is no parity check being performed.

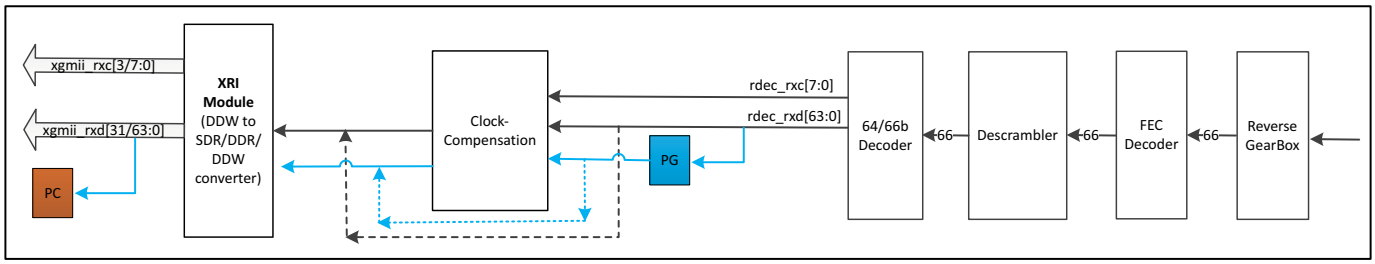
8.2.5 PCS-X - 2.5GBASE-X Mode

In 2.5GBASE-X mode, the output of clock-compensation module (if CLKCOMP=Enabled) or 1000BASE-X controller (if CLKCOMP=Disabled) is input to DWC_xpcs_xdr_2pt5g module. Octet to word conversion and word-decode functions are performed in this module. Because the data is manipulated in this module, parity bits are not passed on to this module. So, parity check is not performed at the output of this module. However, for configurations with ADD_2PT5G=Enabled and CLKCOMP=Enabled, a parity check is performed at the output of the clock-compensation module.

8.2.6 PCS-R - Configurations without USXGMII

In BASE-R mode, parity bits are computed on the output of 64/66b decoder. The data and parity bits then pass through various blocks and finally reach the XGMII interface. A parity check is performed on the final data that is driven out on XGMII Rx port.

Figure 8-4 Parity Check for Configurations Without USXGMII



8.2.7 PCS-R - Configurations with USXGMII

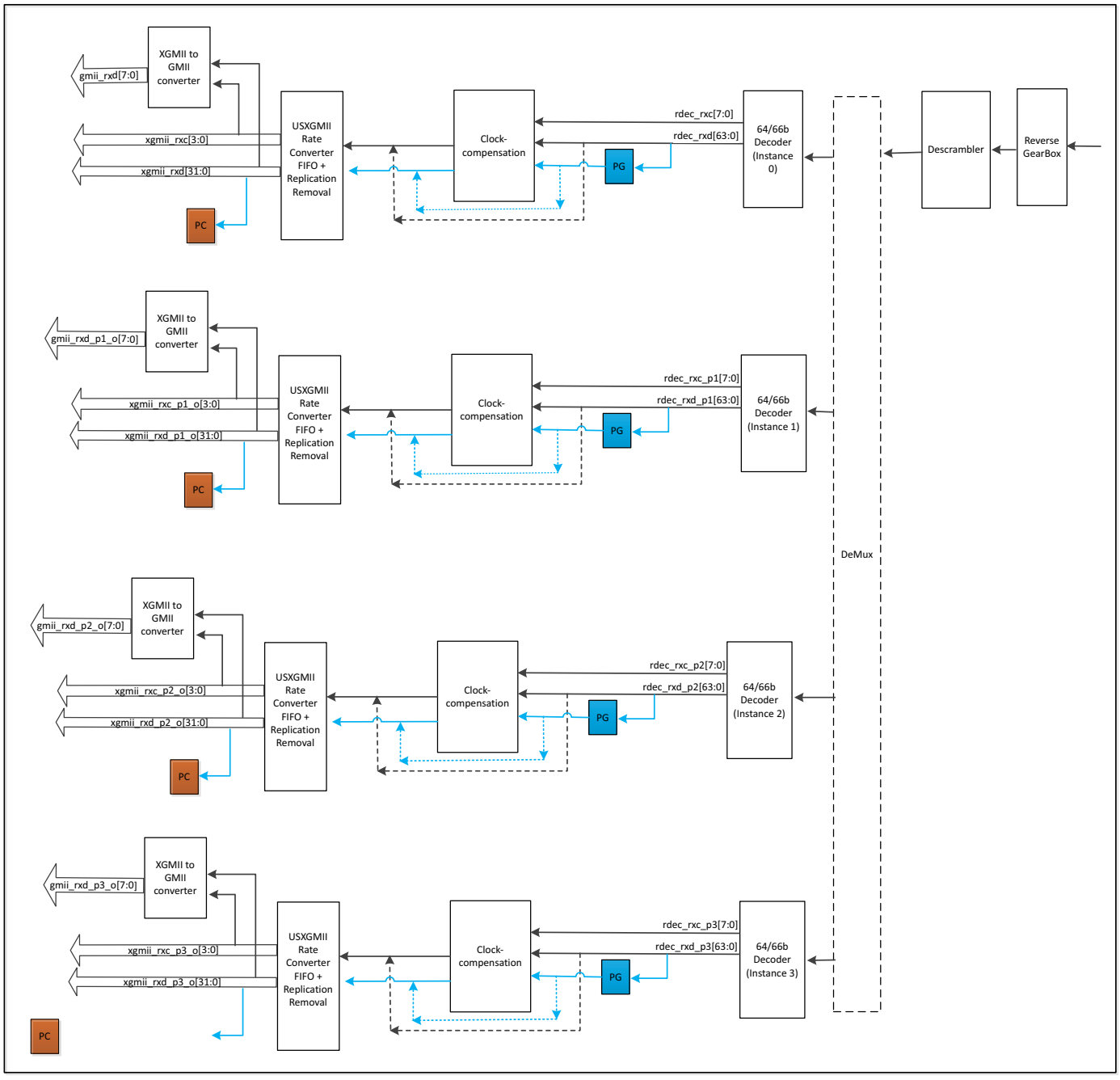
Parity generation and checks are performed for configurations with USXGMII, as shown in Figure 8-5.

Note

Figure 8-5 considers quad-port USXGMII configuration.

If the configuration has only single-port or dual-port support, the number of parity generators and checkers is one and two respectively(insteadof 4).

Figure 8-5 Parity Check for Configuration with USXGMII



8.2.8 Parity Error Injection

The datapath parity protection feature supports a hardware debug mode where the hardware injects an error and upon detection of an error, reports it to the application through a safety interrupt. Failure to report error/safety interrupt indicates an error in the datapath parity safety mechanism.

8.2.9 Interrupt for Parity Error

DWC_xpcs generates an uncorrectable interrupt, xpcs_sfty_ue_intr_o to indicate to the application if any parity error is detected. In addition, it sets the DPP_INT field of VR_XS_PCS_SFTY_UE_INTR0 or VR_MII_SFTY_UE_INTR0 register.

Note A parity error is uncorrectable, and in most cases the operation of the DWC_xpcs is not guaranteed after the parity error. A software reset or a hardware reset is required to safely recover from the parity error reported by the assertion of xpcs_sfty_ue_intr_o signal.

8.2.10 Enabling On-Chip Data Path Parity Protection

To enable this feature, set the DWC_XPCS_ASP parameter during configuration.

For details about these options, see Automotive Safety Features Parameters in the "*Parameter Descriptions*" chapter.

Note See the "Enabled" field of the parameter descriptions to understand the dependencies.

8.2.11 Signals Related to On-Chip Data Path Parity Protection

- xpcs_sfty_ue_intr_o

For a description of signals related to On-Chip Data Path Protection, see the "*Signal Descriptions*" chapter.

8.2.12 Registers Related to On-Chip Data Path Parity Protection

- `VR_XS_PCS_SFTY_UE_INTR0/VR_MII_SFTY_UE_INTR0`
- `VR_XS_PCS_SFTY_ERR_INJ/VR_MII_SFTY_ERR_INJ`
- `VR_XS_PCS_SFTY_DBG_CTRL/VR_MII_SFTY_DBG_CTRL`

For more details, see the "*Register Descriptions*" chapter.

8.2.13 Programming Guidelines for On-Chip Data Path Parity Protection

For details about programming guidelines for on-chip data path parity protection, see "Programming Guidelines for On-Chip Datapath Parity Protection" on page 1722.

8.3 FSM Parity and Timeout Protection

8.3.1 Overview

The FSM protection feature supports FSM parity and time out protection.

8.3.1.1 Description of FSM State Parity Protection

DWC_xpcs implements one-hot encoding scheme for its various internal FSMs. A one-hot encoding check is implemented on all the FSM state register bits. FSMs are monitored for every clock, after the reset is de-asserted. When a bit flips due to transient errors or permanent faults, the erroneous FSM is set to its default state and an uncorrectable error is indicated.

FSM state parity protection is enabled by default. When more than one bit in an FSM state variable becomes high, a parity error is detected and DWC_xpcs asserts the xpcs_sfty_ue_intr_o interrupt port.

FSM parity error information is provided as interrupt-status bits in the VR_XS_PCS_SFTY_UE_INTR0/VR_MII_SFTY_UE_INTR0 register. Individual FSM status information is not provided; instead only status information is provided ‘per clock-domain’. FSM parity error status bits are provided for the following clock-domains:

- Transmit clock domain
- Receive clock domain
- CSR clock domain
- EEE clock domain
- CKO ALIVE clock domain

Though this protection feature checks for one-hot encoding, for simplicity, this is referred to as a parity check.

Error Injection mode is also supported for FSM parity error check. An additional register bit is added to each FSM state register bus. When error injection mode is enabled, this additional register bit (called as parity bit) gets corrupted. Program the Error Injection enable for the respective clock domain denoted by [4:0] bits of `VR_XS_PCS_SFTY_ERR_INJ` or VR_MII_SFTY_ERR_INJ Register.



The FSM Parity error is classified as non-recoverable error and software intervention (IP reset) is required to recover from the error state.

8.3.1.2 Description of FSM Timeout Protection

FSM timeout protection feature supports a mechanism to ensure that all the FSMs complete a transaction and returns to a known completion state within a programmed time-out value. If an FSM is hung in a state for a duration longer than expected, it indicates to the application layer by asserting xpcs_sfty_ue_intr_o interrupt.

To reduce area overhead, individual FSM timers are not implemented; instead generic 'tic' timers are implemented. Application should program SFTY_1US_MULT field of VR_XS_PCS_SFTY_TMR_CTRL/VR_MII_SFTY_TMR_CTRL register with a value that indicates the number of CSR clock cycles required to generate a 1 microsecond tic. This microsecond tic is internally used

to generate various time-out values. DWC_xpcs allocates appropriate time-out values for various FSMs. The time-out values are judiciously determined based on the design intent and based on operating speed mode (contact Synopsys Support in case you require exact details of time-out values associated with various FSMs). However, DWC_xpcs provides an option to scale the timers by a factor of 0.1 or 10 if desired. This can be performed by programming suitable values to FSM_TO_SEL field of VR_XS_PCS_SFTY_TMR_CTRL/VR_MII_SFTY_TMR_CTRL register.

At every timer tick, all the FSMs are monitored for being in active state (non-IDLE state, which indicates FSM is active processing the data or handshakes) and an FSM active flag bit is set. The active flag bit is implemented for every FSM. Whenever the FSM reaches a safe states or safe conditions (for FSMs which does not have any safe state), the active flag for that FSM is reset. At the subsequent timer tick, the active flag bit is monitored and if it still set, it is treated as a time-out error. This process of setting the flag and checking at subsequent timer tick is repeated for every tic.

Contact Synopsys Support for details of the safe states or safe conditions associated with each FSM.

FSM parity error information is provided as interrupt-status bits in the VR_XS_PCS_SFTY_UE_INTR0/VR_MII_SFTY_UE_INTR0 register. Individual FSM status information is not provided; instead only status information is provided 'per clock-domain'. FSM parity error status bits are provided for the following clock-domains:

- Transmit clock domain
- Receive clock domain
- CSR clock domain
- EEE clock domain
- CKO ALIVE clock domain

Timeout error injection feature is also supported in DWC_xpcs to verify the sanity of time-out feature. Per clock-domain time-out errors can be introduced by programming bits [9:5] of the VR_XS_PCS_SFTY_ERR_INJ/VR_MII_SFTY_ERR_INJ.

Note

- Based on the time at which the FSM enters the active state relative to the timer ticks the timeout period could be any value between the programmed timeout and 2x programmed timeout.
- FSM time-out error is classified as non-recoverable error and software intervention (DWC_xpcs reset) is required to recover from the error state.

8.3.1.3 Enabling FSM Parity and Timeout Protection

To enable FSM protection, set the DWC_XPCS_ASP parameter to 1 during the configuration.

By default, FSM time-out feature is enabled. There is an additional register bit which controls FSM time-out feature. However, it can be disabled by programming FT_DIS bit of VR_XS_PCS_SFTY_DISABLE/VR_MII_SFTY_DISABLE register to 1.

For details about these options, see Automotive Safety Features Parameters in the "*Parameter Descriptions*" chapter.



See the “Enabled” field of the parameter descriptions to understand the dependencies.

8.3.1.4 Signals Related to FSM Parity and Timeout Protection

- xpcs_stfy_ue_intr_o

For a description of signals related to FSM Parity and Timeout Protection, see the “*Signal Descriptions*” chapter.

8.3.1.5 Registers Related to FSM Parity and Timeout Protection

- `VR_XS_PCS_SFTY_UE_INTR0/VR_MII_SFTY_UE_INTR0`
- `VR_XS_PCS_SFTY_ERR_INJ/VR_MII_SFTY_ERR_INJ`
- `VR_XS_PCS_SFTY_TMR_CTRL/VR_MII_SFTY_TMR_CTRL`
- `VR_XS_PCS_SFTY_DISABLE/VR_MII_SFTY_DISABLE_`

For more details, see the “*Register Descriptions*” chapter.

8.3.1.6 Programming Guidelines for FSM Parity and Timeout Protection

For details about programming guidelines for FSM protection, see “Programming Guidelines for FSM Parity Protection” on page 1723.

8.4 Interface Timeout Protection

8.4.1 Overview

This feature provides timeout protection to the various Interfaces of DWC_xpcs.

8.4.2 Description of Interface Timeout Protection

All the interfaces which has the handshake mechanism in DWC_xpcs are monitored for potential hangs due to the lack of response for initiated requests. After the request is initiated the response arrival interval is monitored. If the response does not arrive within a programmed time (controlled by IFT_SEL bit of VR_XS_PCS_SFTY_TMR_CTRL) the timeout is triggered, using similar trigger generation as explained in the Section FSM protection.

The hardware does not attempt to recover from the hang and relies on software to take appropriate corrective action, by resetting DWC_xpcs.

The timeout status is provided in the [11:10] field of the `VR_XS_PCS_SFTY_UE_INTR0` register. The safety interrupt, xpcs_sfty_ue_intr_o is asserted when timeout status is set.

8.4.3 PHY (CR) Master Interface

Synopsys PHYs provide a CR port interface to allow the application to access its internal registers. To access a PHY registers, DWC_xpcs drives the address/data on the corresponding buses and asserts the read/write enable signals. The corresponds responds with a ‘cr_ack’ to indicate a successful completion of the read/write operation. Normally the ‘cr_ack’ is asserted in a few clock cycles. If the acknowledgment from the PHY, is not generated within the programmed time-out, DWC_xpcs generates a time-out interrupt.This feature is available only for Synopsys PHY configurations.The Timeout status is reflected in PHY_IF_INT field of VR_XS_PCS_SFTY_UE_INTR0 register.

Note This error is an uncorrectable error. Application must issue a hard-reset or soft-reset to facilitate a recovery.

8.4.4 APB3/MCI Interface

DWC_xpcs supports a APB3/MCI interface to allow access of its internal registers. When the application makes a write or read requests, it completes the transaction and responds with an acknowledgment within a few clock cycles.

If DWC_xpcs fails to generate the 'ack' within the programmed interval, an interface time-out error is flagged.

Note

- APB3/MCI interface error is an uncorrectable error. To recover from this application, software should issue a hard reset to DWC_xpcs. In addition, reset the APB3/MCI master device to facilitate its recovery.
- Based on the time at which the transfer is initiated relative to the timer ticks, the timeout period can be any value between the programmed timeout and 2x times the programmed-timeout.

8.4.5 Enabling Interface Timeout Protection

To enable interface time-out feature, set the DWC_XPCS_ASP parameter to 1 during the configuration.

8.4.6 Signals Related to Interface Timeout Protection

Following is the main signal related to interface timeout protection feature:

- xpcs_sfty_ue_intr_o

For a description of signals related to Interface Timeout Protection, see the "*Signal Descriptions*" chapter.

8.4.7 Programming Guidelines for Interface Timeout Protection

See "Programming Guidelines for Interface Timeout Protection" on page 1726 for details.

8.5 CSR Protection

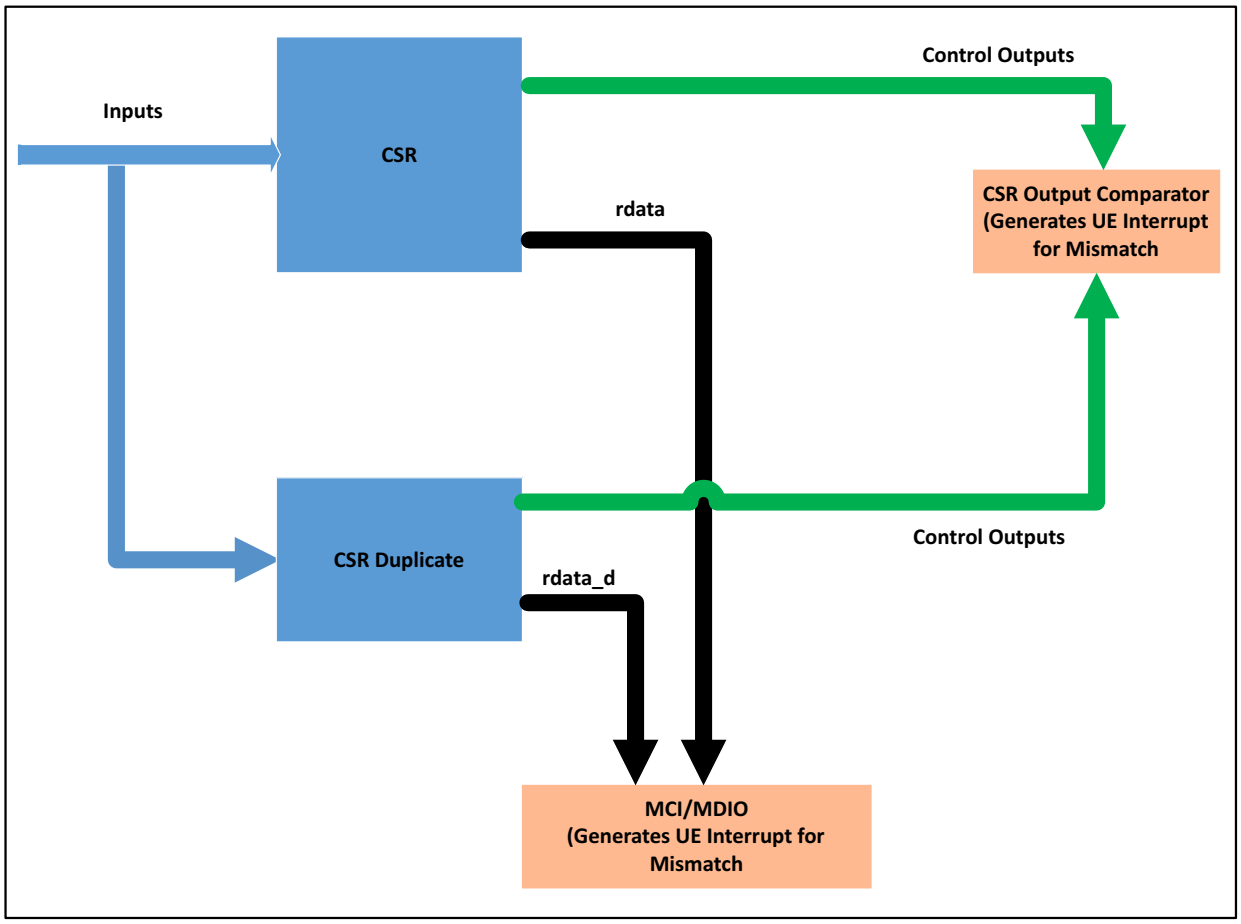
8.5.1 Overview

DWC_xpcs_csr defines several registers which control the operating mode of DWC_xpcs. The status information from various modules are provided in status registers.

8.5.2 Description of CSR Protection

Figure 8-6 shows the block diagram of CSR protection.

Figure 8-6 CSR Protection Block Diagram



8.5.2.1 Replication of Registers

CSR registers control the transmitter and receiver operations of DWC_xpcs. Any accidental bit-flip to the configuration register bit can alter the operating mode of the IP. To prevent such mishaps, DWC_xpcs replicates the CSR module. A comparator continuously checks the output ports of both, CSR module and its replica. DWC_xpcs asserts the xpcs_sfty_ue_intr_o interrupt if any discrepancy is detected.

For status register, when the application tries to read the register, it should get the correct value. To ensure correct value is read by the application, when DWC_xpcs receives a read request from the application, the final output of DWC_xpcs (mci_rdata_o) and the data read from CSR Duplicate (rdata_d) are checked.

8.5.3 Enabling CSR Protection

To enable interface CSR protection, set the DWC_XPCS_ASP parameter to 1 during the configuration. For more details, see the "*Parameter Descriptions*" chapter.

8.5.4 Signals Related to CSR Protection

Following is one of the main signal related to CSR protection.

- xpcs_sfty_ue_intr_o

For a description of signals related to CSR Protection, see the "*Signal Descriptions*" chapter.

8.5.5 Programming Guidelines for CSR Protection

See "Programming Guidelines for CSR Protection" on page 1725 for details.

8.6 Safety Subordinate

When you enable the DWC_XPCS_SS coreConsultant parameter, DWC_xpcs instantiates a safety subordinate module inside it and provides a safety bus interface. For more details, see Appendix “Safety Subordinate Feature” on page 1761.

A

Power, Area, and Latency

This appendix provides power consumption, gate count, and latency information. It contains the following sections:

- "Power" on page 1746
- "Gate Count Summary" on page 1750
- "Latency" on page 1754

A.1 Power

Power dissipation estimates are shown in Table A-1. The power estimates are for simulations with transmission and reception of 15 packets of 64-byte length.

Table A-1 Power by Configuration

Setting	Configuration					
	XGXS PCS	**Backplane Ethernet PCS_KR_KX4_KX**	**1000BASEX-Only PCS**	**10GBASE-R PCS**	**Backplane Ethernet PCS KR_KX**	**10GBASE-R with USXGMII**
Design Compiler	2020.09-SP4					
Technology Synthesized for	Industry Standard 7nm	Industry Standard 7nm	Industry Standard 7nm	Industry Standard 7nm	Industry Standard 7nm	Industry Standard 7nm
Gate Size	NAND2X1					

	Setting	Configuration					
		XGXS PCS	Backplane Ethernet PCS_KR_KX4_KX	1000BASEX-Only PCS	10GBASE-R PCS	Backplane Ethernet PCS KR_KX	10GBASE-R with USXGMII
Configuration Parameter	MAIN_MODE	XGXS PCS	Backplane Ethernet PCS	1000BASEX-Only PCS	10GBASE-R PCS	Backplane Ethernet PCS	10GBASE-R PCS
	Management Clock	2.5 MHz	200 MHz	2.5 MHz	200 MHz	2.5 MHz	200 MHz
	CSR_INTERFACE	MDIO	MCI	MDIO	MCI	MDIO	MCI
	INDIRECT_ACCESS	No	No	No	Yes	No	Yes
	XGMII Clock	156.25 MHz	156.25 MHz	125 MHz	156.25 MHz	312.5 MHz	312.5 MHz
	XGMII	DDR	DDW	-	DDR	SDR	SDR
	BACKPLANE_ETH_CONFIG	-	KR_KX4_KX	-	-	KR_KX	-
	DTE_PHY_SEL	PHY	-	-	-	-	-
	USXG_SPORT	-	-	-	Disabled	-	Enabled
	SUPPORT_1G	Enabled	-	-	Disabled	-	-
	CL37_AN	Enabled	Enabled	-	-	Disabled	Enabled
	SGM_OR_QSGM_SEL	SGMII Only	SGMII Only	SGMII Only	-	None	-
	SNPS_XAUI_PHY	Enabled	Disabled	Enabled	Disabled	Disabled	-
	SNPS_PHY_TYPE	Multi-protocol 6G	-	Multi-protocol 6G	-	-	-

	Setting	Configuration					
		XGXS PCS	Backplane Ethernet PCS_KR_KX4_KX	1000BASEX-Only PCS	10GBASE-R PCS	Backplane Ethernet PCS KR_KX	10GBASE-R with USXGMII
Configuration Parameter	REF_CLK_FREQ	156.25 MHz	N/A	125 MHz	N/A	N/A	N/A
	R_SRDS_INTF (PCS-R PHY Interface)	N/A	16 * 644.53 MHz	N/A	32*322.26 MHz	66*156.25 MHz	32*322.26 MHz
	GPIO_EN	Enabled	Enabled	Disabled	Disabled	Disabled	Disabled
	TX_LSB_FIRST	Disabled	Enabled	N/A	Disabled	Enabled	Disabled
	RX_LSB_FIRST	Disabled	Enabled	N/A	Enabled	Disabled	Disabled
	LANE_CONTROL	Enabled	Enabled	N/A	N/A	N/A	N/A
	EXT_DSKW	Disabled	Enabled	N/A	N/A	N/A	N/A
	RXAUI	Enabled	Enabled	N/A	N/A	N/A	N/A
	RXAUI_DEFAULT	Enabled	Disabled	N/A	N/A	N/A	N/A
	EEE_EN	Enabled	Enabled	Enabled	Disabled	Enabled	Disabled
	FEC_EN	N/A	Enabled	N/A	N/A	Disabled	N/A
	FEC_ERROR_FWD	N/A	Enabled	N/A	N/A	Disabled	N/A
	CL72_EN	N/A	Enabled	N/A	N/A	Disabled	N/A
	CLKCOMP	Enabled	Enabled	Enabled	Disabled	Enabled	Disabled
	ADD_GMII	Enabled	N/A	N/A	Disabled	N/A	N/A
Power	Gate Count	50037	135778	17130	37333	56963	75148
	Comb/Seq Count	17407/32630	61556/74222	5963/11167	17574/19759	24426/32537	34815/40333
	Internal Power (mW)	0.0036	2.05	0.407	0.86	0.059	0.0038
	Switching Power (µW)	0.0005	25.6	6.92	65.4	1.95	0.0117

Setting	Configuration					
	XGXS PCS	Backplane Ethernet PCS_KR_KX4_KX	1000BASEX-Only PCS	10GBASE-R PCS	Backplane Ethernet PCS KR_KX	10GBASE-R with USXGMII
Leakage Power (µW)	0.141	0.375	0.049	0.106	0.164	0.203

A.2 Gate Count Summary

Table A-2 summarizes the area in terms of the gate count numbers (without DFT or clock gating) for various modules and configurations listed in the "*Parameter Descriptions*" chapter. Gate count is calculated (with TSMC 7nm library) with the following parameters:

Table A-2 Area by Operating Mode

Configuration	Gate Count
MAIN_MODE = XGXS PCS or 10GBASE-X PCS with minimum configuration (XGMII = SDR, CSR_INTERFACE = MCI, SNPS_XAUI_PHY = Disabled, and LANE_CONTROL = Disabled)	~27K
MAIN_MODE = Backplane Ethernet PCS with KX_Only minimum configuration (XGMII = SDR, CSR_INTERFACE = MCI, and SNPS_XAUI_PHY = Disabled)	~16K
MAIN_MODE = Backplane Ethernet PCS with KX4_Only minimum configuration (XGMII = SDR, CSR_INTERFACE = MCI, SNPS_XAUI_PHY = Disabled, and LANE_CONTROL = Disabled)	~36K
MAIN_MODE = Backplane Ethernet PCS with KX4_KX minimum configuration (XGMII = SDR, CSR_INTERFACE = MCI, SNPS_XAUI_PHY = Disabled, and LANE_CONTROL = Disabled)	~39K
MAIN_MODE = Backplane Ethernet PCS with KR_Only minimum configuration (XGMII = DDW, CSR_INTERFACE = MCI, R_SRDS_INTF = 66 x 156.25 MHz, and SNPS_XAUI_PHY = Disabled)	~37K
MAIN_MODE = Backplane Ethernet PCS with KR_KX minimum configuration (XGMII = DDW, CSR_INTERFACE = MCI, R_SRDS_INTF = 66 * 156.25 MHz, and SNPS_XAUI_PHY = Disabled)	~46K
MAIN_MODE = Backplane Ethernet PCS with KR_KX4_KX minimum configuration (XGMII = DDW, CSR_INTERFACE = MCI, R_SRDS_INTF = 66 x 156.25 MHz, LANE_CONTROL = Disabled, and SNPS_XAUI_PHY = Disabled)	~73K
MAIN_MODE = 1000BASE-X PCS with minimum configuration (CSR_INTERFACE = MCI, and SNPS_XAUI_PHY = Disabled)	~9K
MAIN_MODE = 10GBASE-R PCS with minimum configuration (XGMII = DDW, CSR_INTERFACE = MCI, R_SRDS_INTF = 66 x 156.25 MHz, and SNPS_XAUI_PHY = Disabled)	~26K
Additional Features	
CSR_INTERFACE = MDIO	~200
XGMII = DDR or DDW Note: This indicates additional gate counts in XGXS PCS, 10GBASE-X PCS, KX4_ONLY, or KX4_KX configuration.	~6K (DDR) ~7K (DDW)

Configuration	Gate Count
XGMII = SDR or DDR Note ■ This indicates additional gate counts in 10GBASE-R PCS, KR_ONLY, or KR_KX configuration without USXGMII. ■ For USXGMII configurations, area due to XGMII=SDR is included in the area for USXG_SPORT parameter.	~6K (SDR) ~1K (DDR)
SNPS_XAUI_PHY = Synopsys multi-protocol 6G PHY For multi-lane configurations.	~5K
SNPS_XAUI_PHY = Synopsys multi-protocol 12G PHY For multi-lane configurations. Note: This area does not include area of gearboxes that are automatically added for BASE-R 12G PHY configurations.	~8K
SNPS_XAUI_PHY = Synopsys multi-protocol 16G PHY For multi-lane configurations. Note: This area does not include area of gearboxes that are automatically added for BASE-R 16G PHY configurations.	~10K
SNPS_XAUI_PHY = Synopsys multi-protocol 25G PHY For multi-lane configurations. Note: This area does not include area of gearboxes that are automatically added for BASE-R 25G PHY configurations.	~12K
SNPS_XAUI_PHY = Synopsys multi-protocol 32G PHY Note: These areas do not include area of gearboxes that are automatically added for BASE-R 32G PHY configurations in ■ Single-lane configurations ■ Multi-lane configurations	~4K ~7K
EXT_DSKW = Enabled	~4K
RXAUI = Enabled	~1K
EEE_EN = Enabled: ■ 1000BASEX-Only PCS configuration ■ XGXS PCS/10GBASE-X PCS configuration ■ 10GBASE-R PCS configuration ■ KR_KX configuration ■ KR_KX4_KX configuration	■ ~4K ■ ~4K ■ ~5K ■ ~6K ■ ~6K
SUPPORT_1G = Enabled	~2K
CL37_AN = Enabled For non-USXGMII configurations	~2K
SGM_OR_QSGM_SEL = SGMII Only Note: Does not include area of CL37_AN parameter	~2K

Configuration	Gate Count
SGM_OR_QSGM_SEL = SGMII and QSGMII Note: Does not include area of CL37_AN parameter	~28K
USXG_SPORT = Enabled Note: This includes area of Clause 37 auto-negotiation logic for USXGMII	~42K
USXG_DPORT = Enabled Note: ■ This is additional area over single port area ■ This includes area of Clause 37 auto-negotiation logic for USXGMII	~76K
USXG_QPORT = Enabled Note: ■ This is additional area over area due to single-port and dual-port ■ This includes area of Clause 37 auto-negotiation logic for USXGMII	~125K
CLKCOMP = Enabled	~300
ADD_2PT5G = Enabled ■ XGMII = SDR ■ XGMII = DDW	~5K ~10K
ADD_5G = Enabled ■ Backplane ethernet configurations ■ "Non-backplane ethernet configurations	4K Negligible
Gear box Enabled ■ 16-bit ■ 20-bit ■ 32-bit ■ 33-bit ■ 40-bit ■ 64-bit	 ~11K ~10K ~11K ~5K ~14K ~11K
FEC_EN = Enabled (without external memory)	~20K
FEC_ERROR_FWD = Enabled (without external memory)	~700
CL72_EN = Enabled ■ For Non-Synopsys PHY configurations ■ For Synopsys PHY configurations	 ~13K ~17K
IEEE_REG_WR_SUPPORT = Enabled	~1K
DWC_XPCS_ASP = Enabled (Note: for maximum configuration)	~47K
For complete parameter settings for these modes, see the *"Parameter Descriptions"* chapter.	

Table A-3 shows the area differences because of scan ready and clock gating for the two configurations.

Table A-3 Area Differences Because of Scan Ready and Clock Gating

	Option	Configuration1	Configuration2
Configuration Parameter	MAIN_MODE	XGXS PCS	Backplane Ethernet PCS in KR_KX4_KX configuration
	CSR_INTERFACE	MDIO	MCI
	XGMII	SDR	SDR
	CL37_AN	Disabled	Enabled
	CL72_EN	N/A	Enabled
	SGMII_EN	Disabled	Enabled
	CLKCOMP	Disabled	Enabled
	ADD_GMII	Disabled	Enabled
	Technology	ts28nmhssmc_ss0p81v125c.db	
	Library	28nm	
	Wire Load Model	Zero	
	Gate Size	NAND 2X1	
	Area of 1 NAND Gate	0.50625	

Table A-4 Gate Count Because of Scan Ready and Clock Gating

Configuration	Clock Gating	Scan Ready	Test Coverage (%)	Gate Count	Clock Gating (%)
Configuration 1	No	No	N/A	27013	N/A
		Yes	97.63	32888	N/A
	Yes	No	N/A	25034	45.57
		Yes	97.27	31559	45.57
Configuration 2	No	No	N/A	101060	N/A
		Yes	98.71	120510	N/A
	Yes	No	N/A	91874	67.00
		Yes	98.30	112230	67.23

A.3 Latency

Table A-5 summarizes the latency numbers (nominal values) for various configurations. The latency is calculated based on the configuration in the following way:

- 10GBASE-X PCS Configurations: Latency is calculated with respect to the 312.5 MHz Transmit (clk_xgxs_tx_i) and Receive (clk_xgxs_rx0_i) clocks.
- 1000BASEX-Only PCS Configurations: Latency is calculated with respect to the 125 MHz Transmit (clk_xgxs_tx_i) and Receive (clk_xgxs_rx0_i) clocks.
- 10GBASE-R PCS Configurations: Latency is calculated with respect to the 156.25 MHz Transmit (clk_rpcs_tx_i) and Receive (clk_rpcs_rx_i) clocks.
- 5GBASE-R PCS Configurations: Latency is calculated with respect to the 78.125 MHz Transmit (clk_rpcs_tx_i) and Receive (clk_rpcs_rx_i) clocks.
- 2.5GBASE-X Configurations: Latency is calculated with respect to the 312.5 MHz Transmit (clk_xgxs_tx_i) and Receive (clk_xgxs_rx0_i) clocks.

Table A-5 Latency

Configuration		10GBASE-X PCS		1000BASEX-Only PCS		10GBASE-R PCS / 5GBASE-RPCS (without USXGMII)		2.5GBASE-X PCS	
XGMII Interface	CLKCOMP	Tx Latency	Rx Latency	Tx Latency	Rx Latency	Tx Latency	Rx Latency	Tx Latency	Rx Latency
SDR	Disabled	6	13	7	10	7	9	20	31
	Enabled	6	21	7	19	7	14	20	40
DDR	Disabled	12	19	11	13	4	5	NA	NA
	Enabled	12	27	11	25	4	9	NA	NA
DDW	Disabled	12	19	11	13	4	5	28	47
	Enabled	12	27	11	25	4	9	28	56

In Table A-5, the latency for 1000BASEX-Only PCS (1G) mode is calculated when SGM_OR_QSGM parameter has a value of 0. In DDR or DDW configurations with SGMII_OR_QSGM_SEL is 1 or 2, the 1G mode latency is similar to the SDR mode latency.

In Table A-5, the latency for 10GBASE-R Mode is for minimum configuration, that is, R_SRDS_INTF = 66 x 156.25 MHz (without Gear box and Reverse Gear box), FEC_EN = Disabled, FEC_ERROR_FWD = Disabled and USXGMI_EN = Disabled.

Table A-6 shows the latency for configurations with USXGMII_EN = Enabled.

Table A-6 Latency for USXGMII Configuration (without Gearbox)

USXGMII Mode		10G-SXGMII		5G-SXGMII		2.5G-SXGMII	
Speed	CLKCOMP	Tx Latency	Rx Latency	Tx Latency	Rx Latency	Tx Latency	Rx Latency
10G	Disabled	8	10	NA	NA	NA	NA
	Enabled	8	15	NA	NA	NA	NA
5G	Disabled	8	16	8	10	NA	NA
	Enabled	8	11	8	15	NA	NA
2.5G	Disabled	12	18	8	11	8	10
	Enabled	12	23	8	16	8	15
1G	Disabled	25	22	15	14	10	12
	Enabled	25	34	15	19	10	17

Table A-7 summarizes the additional latency numbers for 10GBASE-R PCS or KR mode configurations when Gear box or FEC is enabled in the configuration.

Table A-7 Additional Latency Information for 10GBASE-R PCS or KR Mode

R_SRDS_INTF	FEC_EN	FEC_ERROR_FWD	Tx Latency	Rx Latency
Gear box not enabled (Bit Width = 66)	Enabled	Disabled	2	39
	Enabled	Enabled	2	69
GearBox Enabled (Bit width = 16/20/32/40/64)	Disabled	Disabled	2	1
	Enabled	Disabled	4	40
	Enabled	Enabled	4	70
GearBox Enabled (Bit Width = 33)	Disabled	Disabled	3	2
	Enabled	Disabled	5	41
	Enabled	Enabled	5	71

Note The actual latencies might be slightly different from the values mentioned here due to the uncertainties associated with various FIFO (specifically, clock-compensation FIFO) inside DWC_xpcs.

B

DWC_xpcs Automotive Safety

This appendix provides pointers to the Safety Features supported in DWC_xpcs and the Automotive Safety Package available. You must have the corresponding license to use these features.

This appendix has the following sections:

- "Overview of Automotive Safety Features" on page 1758
- "Automotive Safety Package Documents" on page 1759
- "Configuring the Automotive Safety Features" on page 1760

B.1 Overview of Automotive Safety Features

To make an IP better suited for applications in the automotive space, functional safety needs to be considered throughout the IP development process. ISO 26262 addresses functional safety and requires the hardware components to be analyzed with respect to their ability to detect and/or control the effects of random hardware faults. Random hardware faults can occur at any point in time during the life-cycle of the hardware component. They can take the form of permanent faults, or transient faults, and require safety mechanisms to be in place to ensure that severe consequences do not occur as a result of those faults. The DWC_xpcs includes the internal safety mechanisms which assist in detecting and handling random hardware faults. These complement the mechanisms already defined in the DWC_xpcs protocol.

Reference Configuration

As the DWC_xpcs is a highly configurable IP with several optional features, a reference configuration suited for automotive domain usage is used for the Safety Analysis. This configuration has all the major features and includes all the fault detection and reporting related safety logic in the controller. The details of the reference configuration parameters are provided in the Safety Manual.The internal safety mechanisms monitor the IP logic and report any detected errors to the application. The application should take an action to handle the error conditions and ensure safe operation of the hardware component.

FMEDA

A Failure Mode, Effects, and Diagnostic Analysis (FMEDA) has also been performed for this reference configuration. The analysis relies on a per-transistor failure rate derived in accordance to IEC TR 62380. This, in conjunction with area information from design synthesis trials, allows to determine the failure rate for the IP which can then be distributed between the modules of the design according to their relative area.

The list of failure modes and their effects is derived for each module and each of the failure modes is assigned a portion of the failure rate of the module they belong to. If the failure mode has the potential to violate a top level safety requirement (error at a top level output pin) there is a search for a safety mechanism that can prevent that violation, totally or partially. If a safety mechanism exists, for example, ECC protection for RAMs, a diagnostic coverage (DC) value is assigned representing the percentage of the failure rate associated with the specific failure mode that is prevented from violating a top level safety requirement. The process is repeated for all failure modes and finally results are accumulated and IP safety metrics are collected in the FMEDA.

DFMEA

DFMEA document analyses the various failure modes, its effects and root-causes.For each item in the list, the potential failure mode is identified with the corresponding effect of the failure, the severity of the failure (S), the probability of occurrence (O), the detection capability (D), the risk priority number (RPN=S*O*D) and the recommended actions to mitigate the risks of the failure mode.

B.2 Automotive Safety Package Documents

Table B-1 describes the documents included in the DWC_xpcs AC safety package. These documents are available at $DESIGNWARE_HOME/iip/DWC_ac_xpcs/latest directory.

Table B-1 Description of Documents in Automotive Safety AC Package

File Name	Description	Contents
WKPR41_Safety_Manual_DWC_xpcs.pdf	Safety Manual	Describes the Safety goals, safety mechanisms, assumptions of use etc. Summarizes the results of various safety analyses.
SG_Quality_Manual.pdf	Quality Manual	Describes the internal Quality Management System and how its specific activities are related to the Synopsys quality process documents listed in the Safety Manual
WKPR22_FMEDA_DWC_xpcs.xlsx	FMEDA	Calculates FIT rates for each module; calculates SPFM and LFM; enumerates diagnostic capabilities for failure modes.
WKPR17_DFMEA_DWC_xpcs.xlsx	DFMEA	DFMEA analysis
WKPR14_DFA_DWC_xpcs.xls	DFA	Dependent Failure analysis including common cause failures and cascading failures
WKPR40_Safety_Case_Report_DWC_xpcs.pdf	Safety Case Report	Captures details of the Safety Life cycle documentation supporting all functional safety claims
Assessment_Report_ASIL-B_Compliant_DWC_xpcs.pdf	Assessment Report	Assessment Report on the Functional Safety achieved by the IP

B.3 Configuring the Automotive Safety Features

When you have a valid automotive safety package license (DWC-ETHERNET-XPCS-ASP or DWC-AC-ETHERNET-XPCS), you can select the safety features supported by DWC_xpcs, from the Automotive Safety Package pane in the Specify Configuration activity dialog box, while configuring the controller in the coreConsultant GUI.

For details of the various automotive safety features supported by DWC_xpcs, see "Automotive Safety Features" on page 1727.

C

Safety Subordinate Feature

This appendix describes the safety subordinate feature supported by DWC_xpcs and enabled by the DWC_XPCS_SS parameter when you have the automotive AP or AC license.

This chapter has the following sections:

- "Introduction to the Safety Subordinate Feature" on page 1762
- "Safety Error Handling Procedure" on page 1764
- "Details of Safety Subordinate" on page 1765
- "Safety Diagnostic Registers" on page 1768
- "Safety Subordinate Architecture" on page 1769
- "Recoverable and Unrecoverable Errors" on page 1770
- "Safety Error Recovery Flow" on page 1773
- "Forcing Safety Error" on page 1774
- "Forcing Safety Subordinate Internal Recoverable Error" on page 1776

C.1 Introduction to the Safety Subordinate Feature

A safety enabled device, that is, the monitored controller implements the monitor function to look for random corruption of single or multi-bit data within its functional and memory blocks. In case of data corruption leading to a fatal error, the system goes to a safe state as per the functional safety standards.

This chapter describes the function of the safety subordinate block which you can integrate to safety enabled IP. This chapter also provides a common and standard interface to

1. Report the functional safety errors of the IP to a central monitor or safety manager.
2. Allow the central monitor to access the diagnosis registers of the IP through a dedicated safety bus.

The safety subordinate block also supports monitoring and reporting random faults that might occur within its functional blocks. The safety bus

- Has a dedicated bus that connects the safety enabled IPs for error reporting and diagnosis to a centralized safety manager block.
- Can access the existing diagnostic registers of the safety enabled IPs in an SOC.
- Is implemented as a simple, yet modified APB bus, operating at a very low frequency.
- Has all its bus interface signals parity protected or anti-valent protected.
- Segregates the safety diagnosis functions from the functional bus and is independent of the mission mode of operations.
- Works in isolation and does not depend on the primary functional buses. This is because the primary functional bases can be the source of the fault.
- Avoids the need for arbitration as it is a dedicated block (single master).
- Can operate at low frequencies only when required, that is for fault diagnosis, as it is independent of the traffic path. When not in use, you can stop the clock of the safety bus.
- Can be used for reading the diagnostic registers of the monitored controller.

Following are the benefits of using safety subordinate block:

- Supports a central entity (Safety Manager) that has the information on fatal errors across all IPs in the SOC. This provides a single source of information for the higher layer and the data required for safety actions at the SOC level compared to a distributed safety handling mechanism.
- If a fault occurs in the functional or configuration bus that prevents the fatal error being communicated, safety bus is an alternate channel that is available, a form of redundancy.
- A standard and uniform method across IPs to communicate the fatal errors occurring within the IP. You can avoid each IP having a different method of its own to communicate the fatal errors.
- Possible to have a periodic safety function or monitor diagnostic (heartbeat) check across safety monitored IP blocks within an SOC, without interfering in its intended functions and performance.
- Optionally, you can use the safety bus for on-chip debugging for functional errors and performance measurements.



The functionality of the safety manager is not described in this chapter.

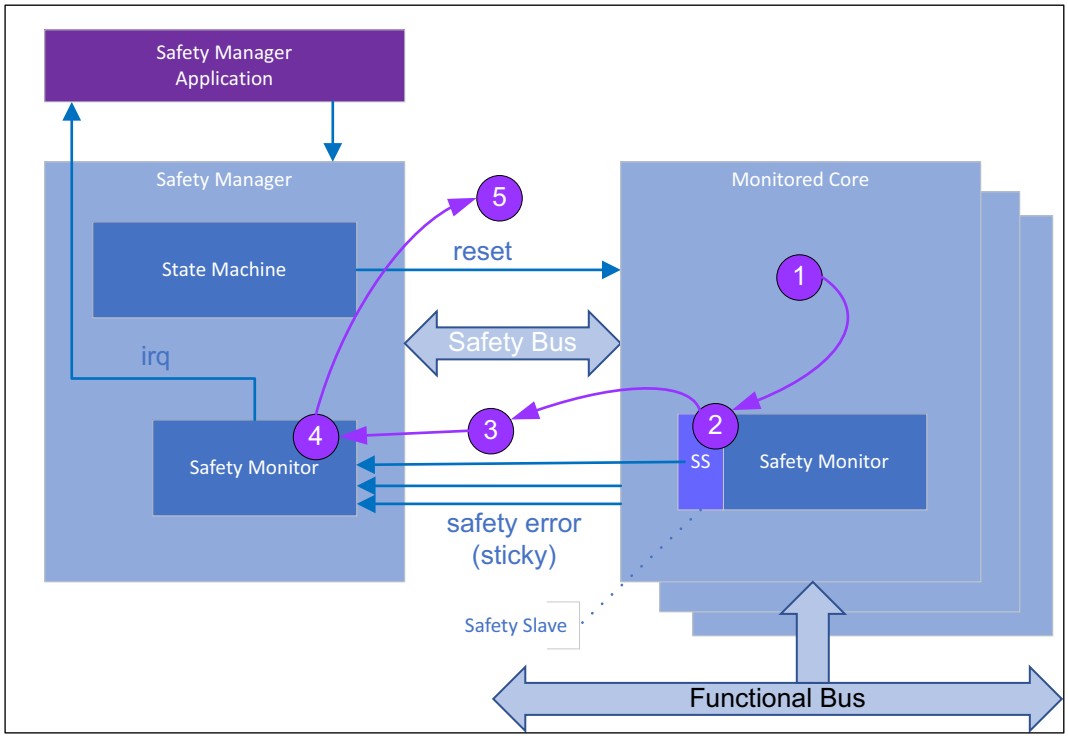
C.1.1 Definitions/Acronyms Used

Term	Definition
SSM	Synopsys Safety Monitor Bus
SS	Safety Subordinate
SM	Safety Manager
ECC	Error Correcting Code
EDC	Error Detecting Code

C.2 Safety Error Handling Procedure

Figure C.1 shows the procedure of how an internal safety error in the controller is detected and communicated to an external safety manager which converts the safety error to an irq.

Figure C-1 Safety Error Procedure



1. An internal safety error, recoverable or unrecoverable (EDC, Parity, lockstep mismatch) is detected in the controller.
2. The safety monitor in the controller converts each internal error to a dual rail signal (anti-valent) and passes on to the safety subordinate module.
3. The safety error state is set to 2'b01 (sticky) from its default state of 2'b10 and held until reset. The state changes in synchronous to core clock of the IP block.
4. The safety monitor in the safety manager module detects the error from each safety monitored IP and asserts an irq to its application.
5. The IRQ handler then identifies the IP source of the error by reading the index of the captured "safety error" vector. Then the IRQ handler may choose to either use the safety bus to diagnose the error or/and issue a software reset to clear the "safety error" by resetting a part of the monitored core or the entire IP/controller if required.

Assertion of the IRQ implies an unrecoverable safety error in the monitored cores. The ISR can perform a diagnosis of the error source of the associated IP block and uses the safety bus to communicate with the monitored devices. The IRQ will be cleared when the device asserting the error is reset.

C.3 Details of Safety Subordinate

C.3.1 Key Features

Following are the key features of safety subordinate:

1. A single dedicated safety monitor bus to connect the safety monitored IPs.
2. The safety monitor bus is the modified APB.
3. Only safety core diagnostic register (of the core) reads are allowed through SSM bus. Core register writes are not allowed and not supported. Write transfers are allowed only to programming the control registers of SS.
4. The Address, Data and control signals of the SSM bus are protected with even parity.
5. The frequency of the SSM bus can be very low and gets enabled only when the safety manager needs an access.

C.3.2 Functional Description

The Safety Subordinate (SS) is a standalone module which provide the necessary hooks for connecting the safety enabled functional IP block/Controller to an external safety manager module through a dedicated safety bus (SSM). The following functions are supported in the safety subordinate module.

1. SSM subordinate state machine with parity and anti-valence bit support for address/data/control signals. Detection and reporting of any SSM bus errors.
2. Address decoder to access core diagnostic registers.
3. Synchronizers for CDC signals (SSM and core clocks)
4. Safety Subordinate reset mechanism with Core reset (hw and sw) and SSM bus resets.
5. Generation of “safety error” state to central Safety Manager.

The SSM Address and control signals are protected with even/odd parity. One bit of parity is generated for every 8 bits.

Similarly, the SSM data bus is also protected with even/odd parity (4 bits parity for the 32-bit data bus).

The generated parity bits are sent through the user signals of address, write data and read data signals on the SSM bus.

C.3.3 Safety Bus Parity

The SS supports an optional parity check for the safety bus signals. The Safety master generates either an odd or an even parity for each byte of ssm_pwdata, ssm_pstrb (1 per nibble) and ssm_paddr. The generated parity bits are passed on to the SS through the respective p*user signals. SS recomputes the parity on ssm_pwdata, ssm_pstrb and ssm_paddr signals and checks for validity before accepting the write transfer. If invalid, SS ignores the writes and registers the error and reports through pslverr.

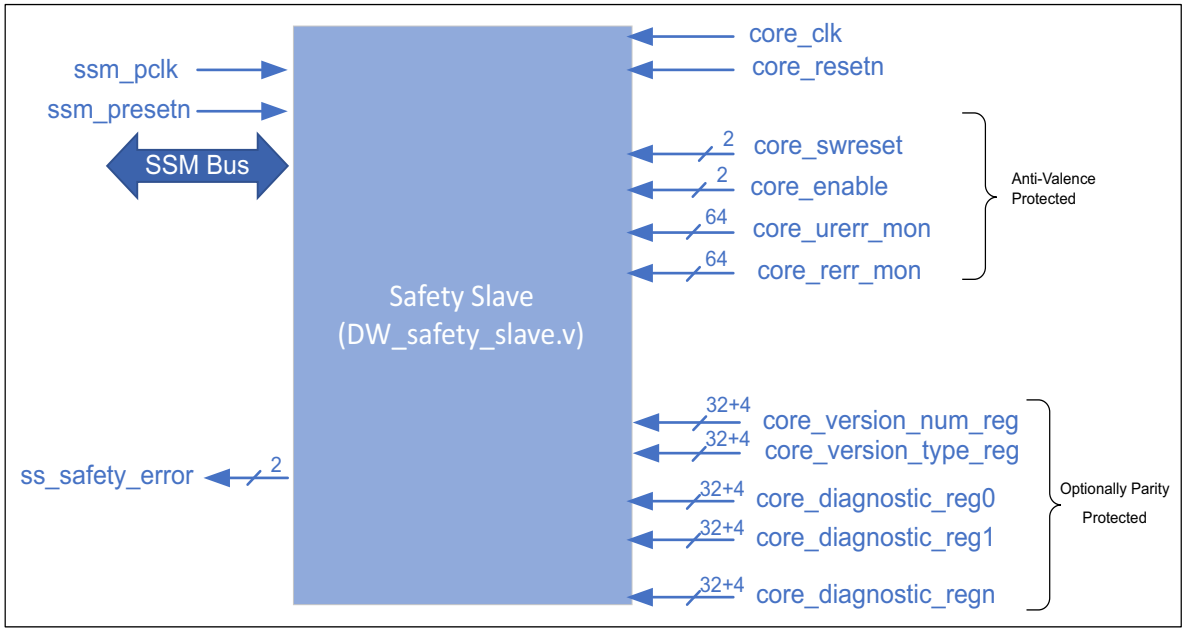
C.3.4 Safety Bus Wait State

The core registers (monitor and diagnostic) are maintained in core clock domain. Any core register read transfer initiated by the SM will get synchronized to the core clock domain before the SS responds with the read data. The SS puts a wait state on the safety bus until the read data is ready.

C.3.5 Safety Subordinate Block Diagram

Figure C-2 shows the top level pin diagram of the DW_safety_slave.v. The core interface signals are shown on the right-hand side of the block and the SSM bus interface signals are shown on the left side of the block.

Figure C-2 Safety Subordinate Block Diagram



C.3.6 Safety Subordinate Timing Diagram

Figure C-3 SSM Bus Write Transfer with Wait State and Error

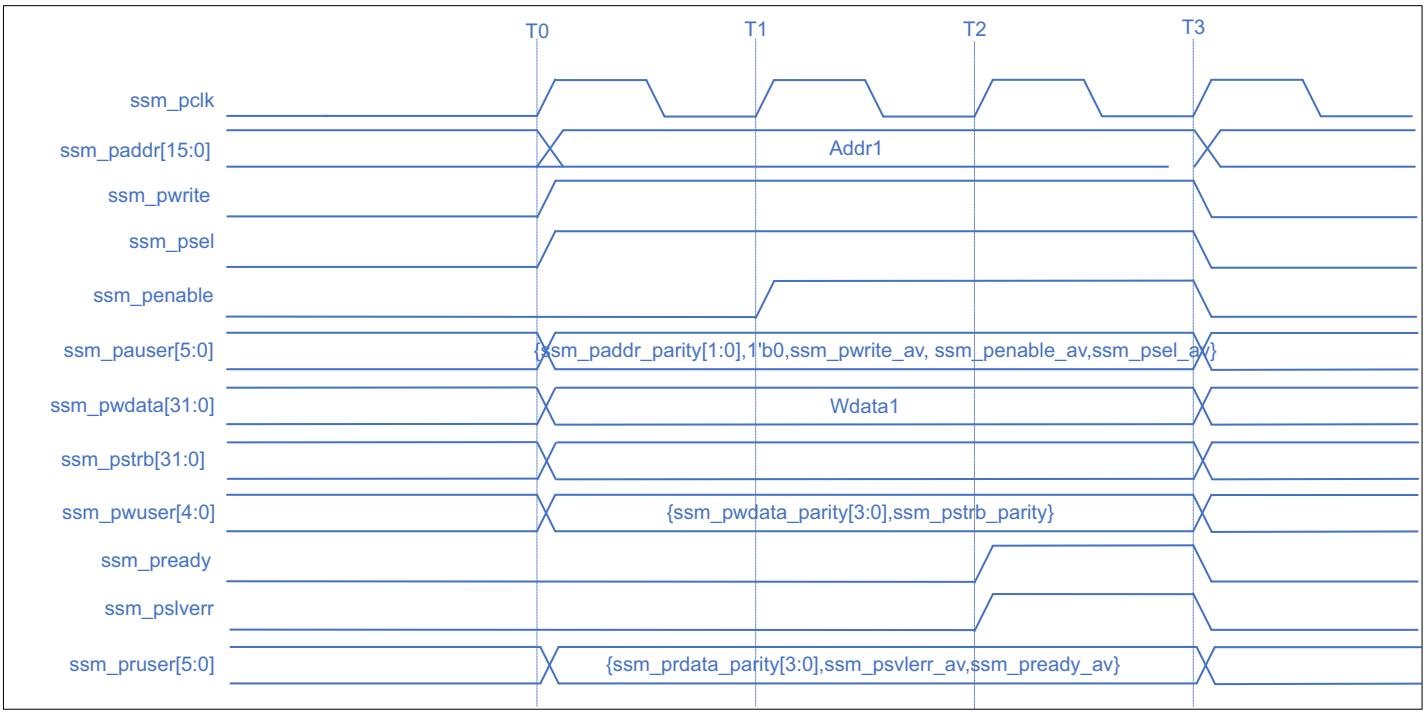
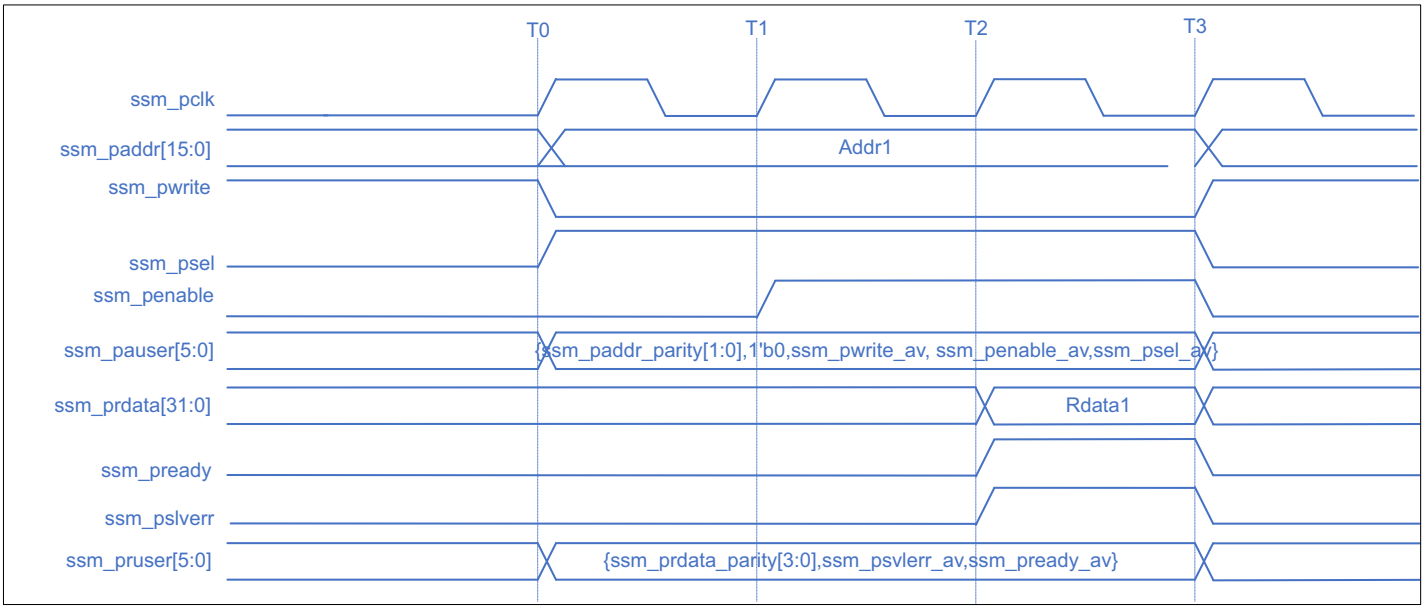


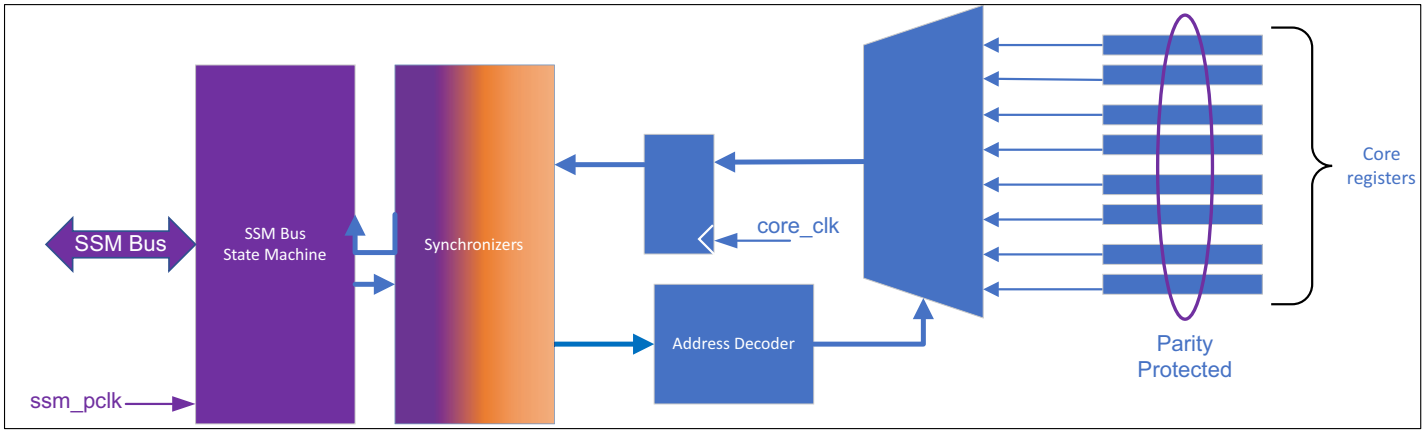
Figure C-4 SSM Bus Read Transfer with Wait State and Error



C.4 Safety Diagnostic Registers

The safety subordinate consists of local (SS) and core registers. In addition to registering the fatal (unrecoverable) and non-fatal (recoverable) errors from the respective monitoring inputs, the Safety Subordinate allows the existing diagnostic registers of the monitored IP to be accessed through the SSM bus. The offsets of these diagnostic registers can be selected during configuration.

Figure C-5 Safety Diagnostic Registers

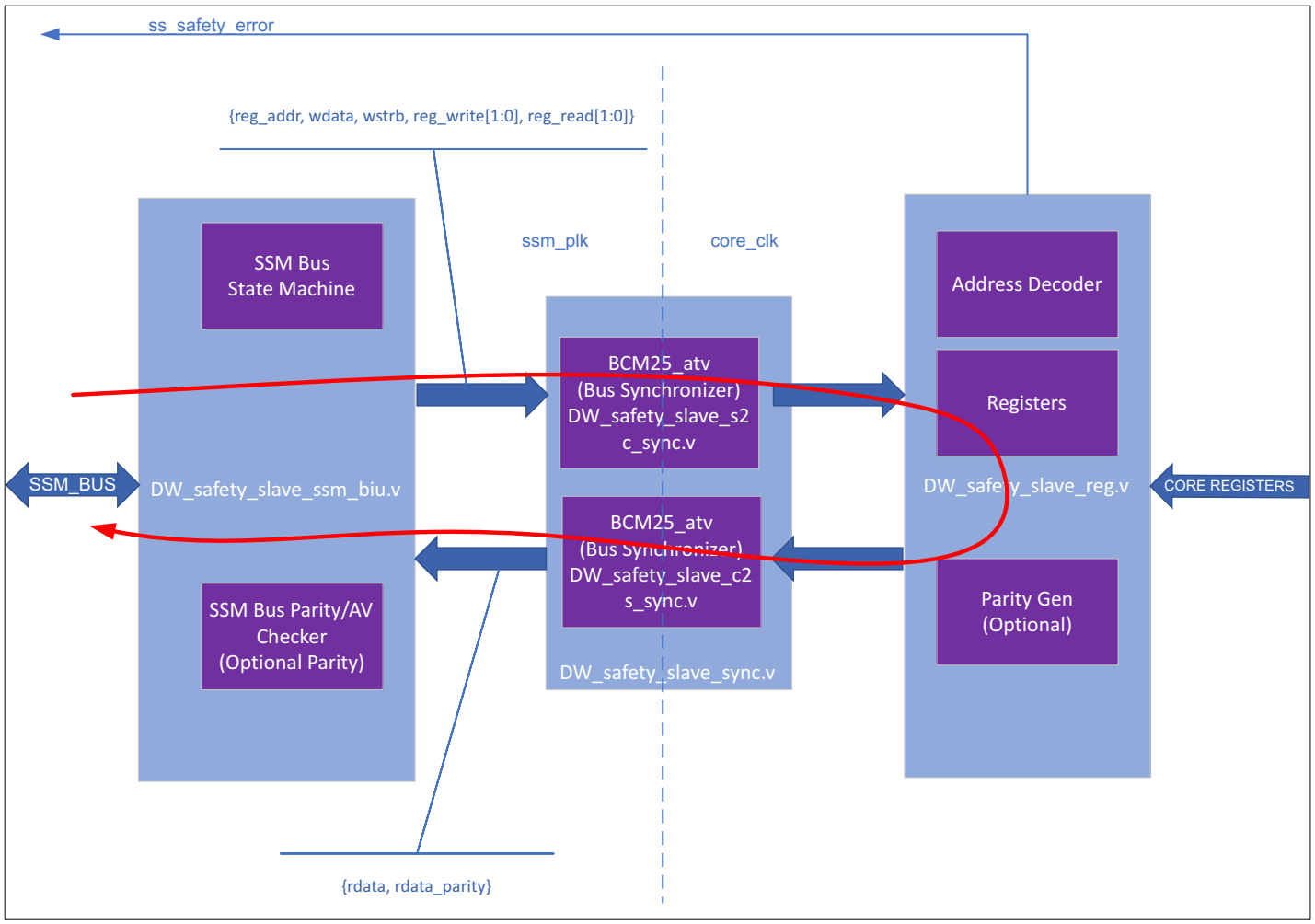


C.5 Safety Subordinate Architecture

Figure C-6 shows the internal blocks of the DW_safety_slave.v module. The red line shows the direction of the data flow. For a register read operation, the register data flows through the data synchronizer (DW_safety_slave_c2s_sync.v) before the SS responds with pready. Similarly, for a register write operation, the SS responds with pready only after detecting a confirmation of register write through the synchronizer (DW_safety_slave_c2s_sync.v).

If the safety subordinate detects an error (parity or antivalent) in the write or read transfer, then it drives the pslverr along with the pready after discarding the current transfer. The host can retry the same transfer on detecting a pslverr.

Figure C-6 Internal Blocks of DWC_safety_slave.v



C.6 Recoverable and Unrecoverable Errors

The Safety Subordinate module supports the monitoring function of the following error types.

1. Safety Subordinate Unrecoverable Error
2. Safety Subordinate Recoverable Error
3. Core Unrecoverable Error
4. Core Recoverable Error

When an error is detected due to a fault either on the core interface port signals or in a critical safety subordinate register, then the safety subordinate generates the “ss_safety_error” error signal (sticky) to inform the host (safety manager). After identifying the root cause of the ss_safety_error, the safety manager can clear all the recoverable error status by writing the SS_RESET_REG register. All unrecoverable error status can be cleared only by either hardware reset or core_swreset port input.

The SS module starts monitoring for the above errors only after the core is enabled (core_enable port input = `SAFETY_HI) in addition to the error enable controls (SS_CNTRL_REG).

C.6.1 Safety Subordinate Unrecoverable Errors

The Safety Subordinate module monitors the following core port signals and reports a ss unrecoverable error when an antivalent mismatch detected.

1. core_swreset
2. core_enable

Enable Condition: SS_CNTRL_REG[SS_EMON_EN] = `SAFETY_HI

Error Status:

SS_ERRSRC_REG[CORE_SWRST_ERR_SRC] = 1’b1 for core_swreset error

SS_ERRSRC_REG[CORE_ENABLE_ERR_SRC] = 1’b1 for core_enable error

Error Clear Condition: core_resetn or core_swreset

C.6.2 Safety Subordinate Recoverable Errors

The Safety Subordinate module monitors the following core port signals for any parity mismatch (if enabled through configuration) and monitors the SS_CNTRL_REG register control flags for any antivalent mismatch and reports a ss recoverable error.

1. core_version_num_reg
2. core_version_type_reg
3. core_diagnostic_reg0-63

Enable Condition: SS_CNTRL_REG[SS_EMON_EN] = `SAFETY_HI

Error Status:

- SS_ERRSRC_REG[SS_INT_RERR_SRC] = 1’b1 for
  - core_version_num_reg error or
  - core_version_type_reg error or
  - SS_CNTRL_REG error

- SS_ERRSRC_REG[SS_DIAG0_RERR_SRC] = 1'b1 for
  - core_diagnostic_reg0-31 errors and
  - SS_CORE_DIAG0_ERR_STS = Error Source Status
- SS_ERRSRC_REG[SS_DIAG1_RERR_SRC] = 1'b1 for
  - core_diagnostic_reg31-63 errors and
  - SS_CORE_DIAG1_ERR_STS = Error Source Status

Error Clear Condition: SS_RESET_REG[CLR_SS_RERR] or SS_RESET_REG[SS_FULL_RST] or core_resetn or core_swreset

Note When you the configure safety subordinate for parity, on the core diagnostic register interface, the SS implements the parity check on each diagnostic register port inputs. The core version register port inputs and monitors for parity mismatch on every rising edge of the core_clk.

C.6.3 Core Unrecoverable Errors

The safety subordinate module monitors the following core port signals and reports a core unrecoverable error when an antivalent mismatch detected.

core_urerr_mon

Enable Condition: SS_CNTRL_REG[CORE_URMON_EN] = `SAFETY_HI

Error Status:

- SS_ERRSRC_REG[CORE_URERR_MON_SRC] = 1'b1
- SS_CORE_URERRSTS_REG = Error Source Status

Error Clear Condition: core_resetn or core_swreset

Note When you enable the core clock, the safety subordinate implements the anti-valent bit error check on each pair of the core_urerr_mon input port on every rising edge of the core clock. For example, core_urerr_mon[1:0], core_urerr_mon[3:2] and so on.

If the safety subordinate encounters either `SAFETY_HI or any invalid value on any one of the anti-valent pair of the core_urerr_mon (only up to the configured value, SS_CORE_URERR_MON_SIZE) input port, it sets the corresponding bit of the SS_CORE_URERRSTS_REG.

C.6.4 Core Recoverable Errors

The Safety Subordinate module monitors the following core port signals and reports a core recoverable error when an antivalent mismatch detected.

core_rerr_mon

Enable Condition: SS_CNTRL_REG[CORE_RMON_EN] = `SAFETY_HI

Error Status:

- SS_ERRSRC_REG[CORE_RERR_MON_SRC] = 1'b1

- SS_CORE_RERRSTS_REG = Error Source Status

Error Clear Condition: SS_RESET_REG[CLR_CORE_RERR] or SS_RESET_REG[SS_FULL_RST] or core_resetn or core_swreset

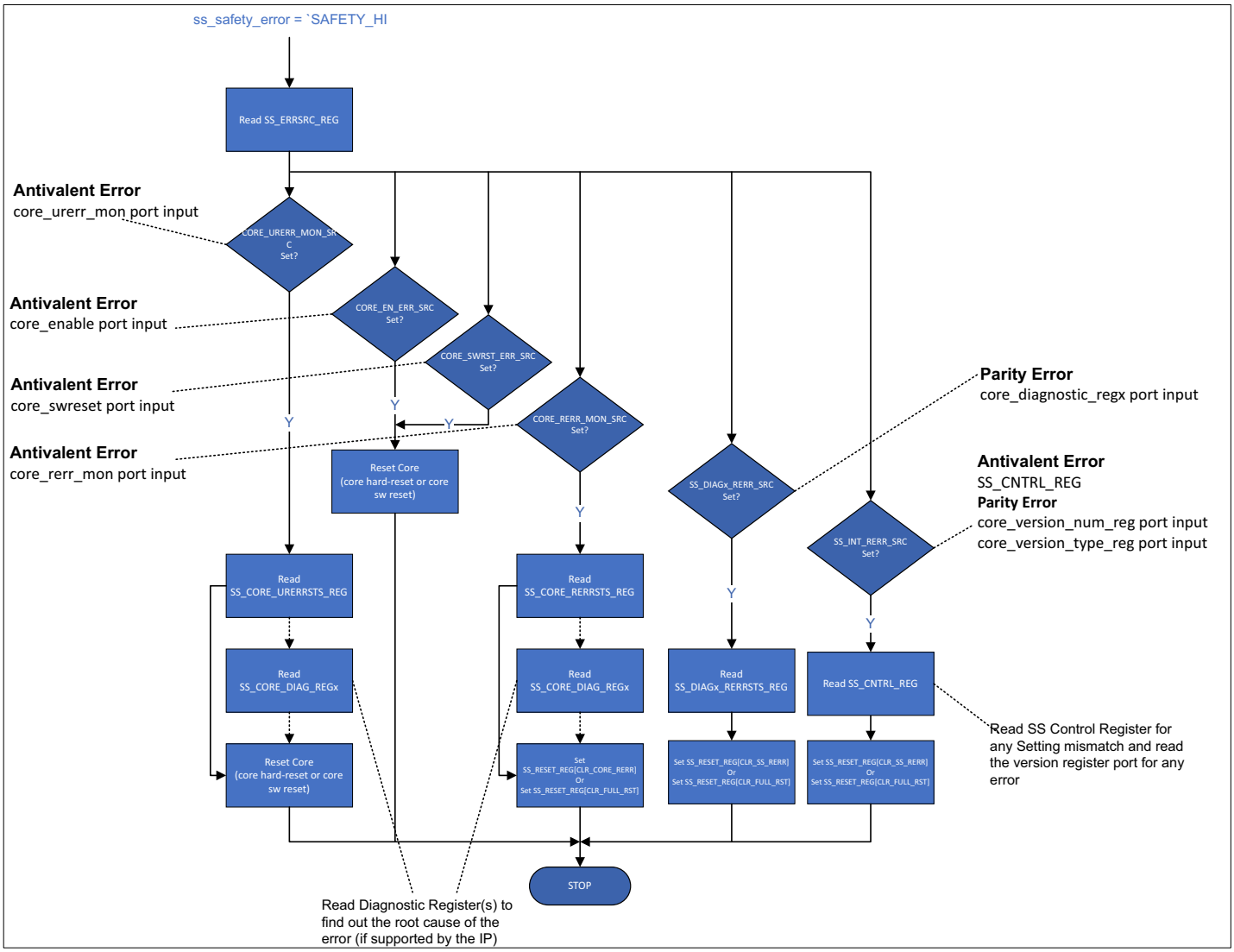
Note

When you enable the core clock, the SS implements the anti-valent bit error check on each pair of the core_rerr_mon input port on every rising edge of core clock. For example, core_rerr_mon[1:0], core_rerr_mon[3:2], and so on. If the SS encounters, `SAFETY_HI or any invalid value on any one of the anti-valent pair of the core_rerr_mon (only up to configured value, SS_CORE_RERR_MON_SIZE) input port, then it sets corresponding bit of the SS_CORE_RERRSTS_REG.

C.7 Safety Error Recovery Flow

Figure C-7 shows the typical error recovery steps taken by the safety master on detecting the ss_safety_error. The actual flow might vary based on the dependencies of other IP blocks in an SOC.

Figure C-7 Safety Subordinate Error Recovery Flow



C.8 Forcing Safety Error

Force the following safety errors at any time to check the error flow functionality without any real error.

1. Core Recoverable Error
2. Core Unrecoverable Error
3. Core Diagnostic Recoverable Error
4. Safety Subordinate Internal Recoverable Error

To force any unrecoverable or recoverable error, set the SS to disabled state for error monitoring. So, set the following registers to `SAFETY_LO to enable error forcing.

- SS_CNTRL_REG[CORE_URMON_EN]
- SS_CNTRL_REG[CORE_RMON_EN]
- SS_CNTRL_REG[SS_EMON_EN]

The core_enable port signal can be in any state (need not be in enabled state) for forcing an error.

Note

The safety master must set only one of the following force controls to `SAFETY_HI at any given time. Setting of more than one force control flag to `SAFETY_HI in a single write results in unpredicted behavior.

- SS_CNTRL_REG[SS_INT_RERR_SET]
- SS_CNTRL_REG[SS_DIAG1_RERR_SET]
- SS_CNTRL_REG[SS_DIAG0_RERR_SET]
- SS_CNTRL_REG[SS_DIAG0_RERR_SET]
- SS_CNTRL_REG[CORE_URERR_SET]
- SS_CNTRL_REG[CORE_RERR_SET]

C.8.1 Forcing Core Recoverable Error

Follow this sequence of operation to force supported diagnostic recoverable error in the Safety Subordinate.

1. Set the register field SS_CNTRL_REG[CORE_RERR_SET] to `SAFETY_HI
2. Write the register SS_ERRFORCE_VAL_REG with the non-zero value that needs to be written into the SS_CORE_RERRSTS_REG register. Any bit value with “1” of the write data to the SS_ERR-FORCE_VAL_REG register will set the corresponding bit in the SS_CORE_RERRSTS_REG register. The number of bits that can be set to “1” from the RHS of the SS_CORE_RERRSTS_REG register is based on the configured monitor size (SS_CORE_RERR_MON_SIZE).
3. After writing a non-zero value to the SS_CORE_RERRSTS_REG register, the SS sets the SS_ERRSRC_REG[CORE_RERR_MON_SRC] and drives the ss_safety_error port signal to `SAFE-TY_HI.
4. The Error clearing mechanism is the same as explained in the section “Safety Error Recovery Flow”.

Rules: Register access other than Write SS_ERRFORCE_VAL_REG in step 2 cancels the “error force” operation. For example, if the application issues a read to any register after step-1 cancels the “error force” operation. The application must continue from step-1 again to force a recoverable error.

C.8.2 Forcing Core Unrecoverable Error

Follow this sequence of operation to force supported diagnostic unrecoverable error in the Safety Subordinate.

1. Set the register field SS_CNTRL_REG[CORE_URERR_SET] to `SAFETY_HI
2. Write the register SS_ERRFORCE_VAL_REG with the non-zero value that needs to be written into the SS_CORE_URERRSTS_REG register. Any bit value with "1" of the write data to the SS_ERRFORCE_VAL_REG register will set the corresponding bit in the SS_CORE_URERRSTS_REG register. The number of bits that can be set to "1" from the RHS of the SS_CORE_URERRSTS_REG register is based on the configured monitor size (SS_CORE_URERR_MON_SIZE).
3. After writing a non-zero value to the SS_CORE_URERRSTS_REG register, the SS sets the SS_ERRSRC_REG[CORE_URERR_MON_SRC] and drives the ss_safety_error port signal to `SAFETY_HI.
4. The Error clearing mechanism is explained in "Safety Error Recovery Flow" on page 1773.

Rules: Register access other than Write SS_ERRFORCE_VAL_REG in step 2 cancels the "error force" operation. For example, if the application issues a read to any register after step-1 cancels the "error force" operation. The application must continue from step-1 again to force a recoverable error.

C.8.3 Forcing Core Diagnostic Recoverable Error

Follow this sequence of operation to force supported diagnostic recoverable error in the Safety Subordinate.

1. Set the SS_CNTRL_REG[SS_DIAGx_RERR_SET register field to `SAFETY_HI
2. Write the register SS_ERRFORCE_VAL_REG with the non-zero value that needs to be written into the SS_DIAGx_RERRSTS_REG register. Any bit value with "1" of the write data to the SS_ERRFORCE_VAL_REG register will set the corresponding bit in the SS_DIAGx_RERRSTS_REG register. The number of bits that can be set to "1" from the RHS of the SS_DIAGx_RERRSTS_REG register is based on the configured diagnostic register size.
3. After writing a non-zero value to the SS_DIAGx_RERRSTS_REG register, the SS sets the SS_ERRSRC_REG[SS_DIAGx_RERR_SRC] and drives the ss_safety_error port signal to `SAFETY_HI.
4. The error clearing mechanism is explained in "Safety Error Recovery Flow" on page 1773.

Rules: Register access other than Write SS_ERRFORCE_VAL_REG in step 2 cancels the "error force" operation. For example, if the application issues a read to any register after step-1 cancels the "error force" operation. The application must continue from step-1 again to force a recoverable error.

C.9 Forcing Safety Subordinate Internal Recoverable Error

Follow this sequence of operation to force supported diagnostic recoverable error in the Safety Subordinate.

1. Set the register field SS_CNTRL_REG[SS_INT_RERR_SET] to `SAFETY_HI
2. Write the register SS_ERRFORCE_VAL_REG with 0x00000001 to set the SS_ERRSRC_REG[SS_INT_RERR_SRC] register.
3. After writing a value of 1 to the SS_ERRSRC_REG[SS_INT_RERR_SRC] register, the SS sets the SS_ERRSRC_REG[SS_INT_RERR_SRC] and drives the ss_safety_error port signal to `SAFETY_HI.
4. The Error clearing mechanism is the same as explained in “Safety Error Recovery Flow” on page 1773.

Register access other than Write SS_ERRFORCE_VAL_REG in step 2 cancels the “error force” operation. For example, if the application issues a read to any register after step-1 cancels the “error force” operation, the application must continue from step-1 again to force a recoverable error.

D

Internal Parameter Descriptions

Provides a description of the internal parameters that might be indirectly referenced in expressions in the Signals, Parameters, or Registers chapters. These parameters are not visible in the coreConsultant GUI and most of them are derived automatically from visible parameters.**You must not set any of these parameters directly.**

Some expressions might refer to TCL functions or procedures (sometimes identified as **function_of**) that coreConsultant uses to make calculations. The exact formula used by these TCL functions is not provided in this chapter. However, when you configure the core in coreConsultant, all TCL functions and parameters are evaluated completely; and the resulting values are displayed where appropriate in the coreConsultant GUI reports.

Table D-1 Internal Parameters

Parameter Name	Equals To
ADDRS_WIDTH	APB3_32 ? (INDIRECT_ACCESS ? 10: 23) : APB3_16 ? (INDIRECT_ACCESS ? 9: 22) : (INDIRECT_ACCESS ? 8: 21)
APB3_16	CSR_INTERFACE == 3
APB3_32	CSR_INTERFACE == 4
APB3_INTF	CSR_INTERFACE == 2
APB_DATA_W	APB3_32 ? 32 : 16
BACKPLANE_ETHERNET_PCS	MAIN_MODE == 2
BASE1000X_ONLY	MAIN_MODE == 3
BW_W	(E16||SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_39) ? 16 : 11
DIV_M_W	((E12||E16)&& !(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22)) ? 7 : 8
E12	(SNPS_PHY_TYPE==1)

Parameter Name	Equals To
E16	(SNPS_PHY_TYPE==2)
E25	(SNPS_PHY_TYPE==3)
E32	(SNPS_PHY_TYPE==4)
FEC_DW	SNPS_RSVDPARAM_29 ? 74 : 66
FR_W	(E12&& !(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_39)) ? 9 : 11
GMII_PORT	!SNPS_RSVDPARAM_14&& (KX_ONLY||BASE1000X_ONLY||ADD_GMII||KRKX_OR_RX)
IEEE_AN_DEV_MODEL_NO_0_5	[function_of: IEEE_AN_DEV_MODEL_NO 0 5]
IEEE_AN_DEV_OUI_19_24	[function_of: IEEE_AN_DEV_OUI 19 24]
IEEE_AN_DEV_OUI_3_18	[function_of: IEEE_AN_DEV_OUI 3 18]
IEEE_AN_DEV_REV_NO_0_3	[function_of: IEEE_AN_DEV_REV_NO 0 3]
IEEE_DEV_MODEL_NO_0_5	[function_of: IEEE_DEV_MODEL_NO 0 5]
IEEE_DEV_OUI_19_24	[function_of: IEEE_DEV_OUI 19 24]
IEEE_DEV_OUI_3_18	[function_of: IEEE_DEV_OUI 3 18]
IEEE_DEV_REV_NO_0_3	[function_of: IEEE_DEV_REV_NO 0 3]
IEEE_PKG_MODEL_NO_0_5	[function_of: IEEE_PKG_MODEL_NO 0 5]
IEEE_PKG_OUI_19_24	[function_of: IEEE_PKG_OUI 19 24]
IEEE_PKG_OUI_3_18	[function_of: IEEE_PKG_OUI 3 18]
IEEE_PKG_REV_NO_0_3	[function_of: IEEE_PKG_REV_NO 0 3]
IEEE_PMA_DEV_MODEL_NO_0_5	[function_of: IEEE_PMA_DEV_MODEL_NO 0 5]
IEEE_PMA_DEV_OUI_19_24	[function_of: IEEE_PMA_DEV_OUI 19 24]
IEEE_PMA_DEV_OUI_3_18	[function_of: IEEE_PMA_DEV_OUI 3 18]
IEEE_PMA_DEV_REV_NO_0_3	[function_of: IEEE_PMA_DEV_REV_NO 0 3]
IEEE_VS_DEV_MODEL_NO_0_5	[function_of: IEEE_VS_DEV_MODEL_NO 0 5]
IEEE_VS_DEV_OUI_19_24	[function_of: IEEE_VS_DEV_OUI 19 24]
IEEE_VS_DEV_OUI_3_18	[function_of: IEEE_VS_DEV_OUI 3 18]
IEEE_VS_DEV_REV_NO_0_3	[function_of: IEEE_VS_DEV_REV_NO 0 3]
KR_KX	(MAIN_MODE == 2)&& (BACKPLANE_ETH_CONFIG == 5)

Parameter Name	Equals To
KR_KX4	(MAIN_MODE == 2)&& (BACKPLANE_ETH_CONFIG == 4)
KR_KX4_KX	(MAIN_MODE == 2)&& (BACKPLANE_ETH_CONFIG == 3)
KRKX_OR_RX	KR_KX||SUPPORT_10GR_1GX
KR_ONLY	(MAIN_MODE == 2)&& (BACKPLANE_ETH_CONFIG == 6)
KR_PHY	0
KR_PRESENT	(KR_ONLY||KR_KX4||KR_KX||KR_KX4_KX)
KX4_KX	(MAIN_MODE == 2)&& (BACKPLANE_ETH_CONFIG == 0)
KX4_KX_PRESENT	(KX4_PRESENT&& KX_PRESENT)
KX4_ONLY	(MAIN_MODE == 2)&& (BACKPLANE_ETH_CONFIG == 1)
KX4_PRESENT	(KX4_ONLY||KX4_KX||KR_KX4||KR_KX4_KX)
KX_ONLY	(MAIN_MODE == 2)&& (BACKPLANE_ETH_CONFIG == 2)
KX_PRESENT	(KX_ONLY||KX4_KX||KR_KX||KR_KX4_KX)
LOOPBACK_W	(UP5||KR_PHY) ? NUM_OF_LANES : 1
LOS_W	(SSP_V42) ? 7 : 2
LP_10GBR	RPCS_PRESENT&& EEE_EN
LP_10GBX	(KX4_PRESENT||PCS_10G||XGXS_PCS)&& EEE_EN
LP_1GBX	(BASE1000X_ONLY||SUPPORT_1G||KX_PRESENT)&& EEE_EN
LVL_W	(UP5&& SSP) ? 7 : 6
MCI_INTF	CSR_INTERFACE == 1
MDIO_INTF	CSR_INTERFACE == 0
MP_USXGMII	USXG_DPORT||USXG_QPORT
MULT_W	(SNPS_RSVDPARAM_22||E25) ? 12 : 8
NUM_OF_LANES	(KX_ONLY||BASE1000X_ONLY||(MAIN_MODE==4)||KR_ONLY||KR_KX) ? 1 : 4
ONLY_1G_CFG	KX_ONLY||BASE1000X_ONLY
PCS_10G	MAIN_MODE == 1
PCS_MMD	(BACKPLANE_ETHERNET_PCS||PCS_10G||RPCS_EN)
PCS_XS_MMD	(XS_MMD||PCS_MMD)

Parameter Name	Equals To
PHY_DW	(SNPS_RSVDPARAM_12) ? 80 : (R_SRDS_INTF==0) ? 16 : (R_SRDS_INTF==1) ? 32 : (R_SRDS_INTF==2) ? 33 : (R_SRDS_INTF==3) ? 66 : (R_SRDS_INTF==4) ? 20 : (R_SRDS_INTF==5) ? 64 : 40
PMA_MMD	(BACKPLANE_ETHERNET_PCS||(PCS_10G&& SNPS_XAUI_PHY)||((MAIN_MODE==4)&& SNPS_XAUI_PHY))
POL_W	(E25||E16) ? 2 : 3
QSGMII_EN	(SGM_OR_QSGM_SEL==2)
REF_CLK_125	REF_CLK_FREQ==1
REF_CLK_156	REF_CLK_FREQ==0
REF_CLK_20	SNPS_RSVDPARAM_39&& (KX_ONLY||BASE1000X_ONLY)
REF_CLK_25	REF_CLK_FREQ==5||(SNPS_RSVDPARAM_39&& RPCS_PRESENT)
REF_CLK_312	REF_CLK_FREQ==6
REF_CLK_62	REF_CLK_FREQ==3
REF_CLK_78	REF_CLK_FREQ==2
REF_R_W	(SNPS_RSVDPARAM_37) ? 4 : 3
RF_LD_W	(E12&& !SNPS_RSVDPARAM_22) ? 6 : 7
RPCS_EN	(MAIN_MODE == 4)||(BACKPLANE_ETH_CONFIG==3||BACKPLANE_ETH_CONFIG==4||BACKPLANE_ETH_CONFIG==5||BACKPLANE_ETH_CONFIG==6)
RPCS_ONLY	(MAIN_MODE == 4 && SUPPORT_1G==0)||(BACKPLANE_ETH_CONFIG==6)
RPCS_PRESENT	(KR_ONLY||KR_KX4||KR_KX||KR_KX4_KX||MAIN_MODE==4)
RX_RT_W	E25 ? 3 : 2
RX_W	(E25||E32) ? 3 : 2
SGMII_EN	(SGM_OR_QSGM_SEL==1)
SINGLE_LANE_CFG	KX_ONLY||BASE1000X_ONLY||KR_ONLY||(MAIN_MODE==4)||KR_KX
SNPS_RSVDPARAM_11	0
SNPS_RSVDPARAM_12	0

Parameter Name	Equals To
SNPS_RSVDPARAM_13	0
SNPS_RSVDPARAM_14	0
SNPS_RSVDPARAM_15	0
SNPS_RSVDPARAM_16	0
SNPS_RSVDPARAM_2	0
SNPS_RSVDPARAM_22	0
SNPS_RSVDPARAM_29	DWC_XPCS_ASP&& FEC_EN
SNPS_RSVDPARAM_3	0
SNPS_RSVDPARAM_33	0
SNPS_RSVDPARAM_35	0
SNPS_RSVDPARAM_36	0
SNPS_RSVDPARAM_37	0
SNPS_RSVDPARAM_39	0
SNPS_RSVDPARAM_6	0
SRAM_BLBYP_W	SNPS_RSVDPARAM_37 ? 2 : 1
SSC_W	(E12&& !(SNPS_RSVDPARAM_22||SNPS_RSVDPARAM_39)) ? 3 : 1
SSP	SNPS_PHY_TYPE==0&& SNPS_XAUI_PHY==1
SSP_V42	SNPS_PHY_TYPE==0&& SNPS_XAUI_PHY==1
SUPPORT_10GR_1GX	(MAIN_MODE == 4)&& (SUPPORT_1G == 1)
SUPPORT_10GX_1GX	(MAIN_MODE == 0 || MAIN_MODE ==1)&& (SUPPORT_1G == 1)
TX_DIV_W	(SNPS_RSVDPARAM_35||SNPS_RSVDPARAM_33||SNPS_RSVDPARAM_22||E25||SNPS_RSVDPARAM_39) ? 3 : 2
TX_RT_W	SSP ? 2 : 3
TX_W	(E25||E32) ? 3 : 2
UP3	0
UP5	SNPS_PHY_TYPE==0&& SNPS_XAUI_PHY==1
USXGMII_EN	USXG_SPORT||USXG_DPORT||USXG_QPORT
USXG_SPORT_FULL	USXG_SPORT

Parameter Name	Equals To
VGA_W	(E25||E16) ? 3 : 4
VREF_W	(E12||E16||E25)&& !SNPS_RSVDPARAM_22 ? 5 : 3
XGMII_DDR	XGMII == 1
XGMII_DDW	XGMII == 2
XGMII_SDR	XGMII == 0
XGXS_PCS	MAIN_MODE == 0
XRI_RD_CTRL_WIDTH	(XGMII_DDW) ? 8 : !(KX_ONLY||BASE1000X_ONLY) ? 4 : 2
XRI_RD_DATA_WIDTH	(XGMII_DDW) ? 64 : !(KX_ONLY||BASE1000X_ONLY) ? 32 : 8
XS_MMD	XGXS_PCS
XTI_WR_CTRL_WIDTH	(XGMII_DDW) ? 8 : !(KX_ONLY||BASE1000X_ONLY)||SNPS_RSVDPARAM_14 ? 4 : 2
XTI_WR_DATA_WIDTH	(XGMII_DDW) ? 64 : !(KX_ONLY||BASE1000X_ONLY)||SNPS_RSVDPARAM_14 ? 32 : 8

Glossary

Term	Meaning
||A||	Indicates |A| control character in all lanes
||D||	Indicates |D| control character in all lanes
||I||	Indicates |I| control character in all lanes
||K||	Indicates |K| control character in all lanes
||R||	Indicates |R| control character in all lanes
|A|	Control character to indicate PCS Align pattern
|D|	Control character to indicate one of PCS Data pattern
|E|	Control character to indicate PCS Error pattern
|I|	Control character to indicate PCS Idle pattern
|K|	Control character to indicate PCS Comma pattern
|Q|	Control character to indicate PCS Sequence pattern
|R|	Control character to indicate PCS Skip pattern
|S|	Control character to indicate PCS SOP pattern
|T|	Control character to indicate PCS Terminate pattern
10GbE	10-Gigabit Ethernet
AN	Auto-negotiation
ATPG	Automatic Test Pattern Generator
BERT	Bit Error Ratio Test set
BFM	Bus Functional Model

Term	Meaning
Byte	8 bits
CJTP	Continuous Jitter Test Pattern supported in 10G mode of operation
CR	Control Register
CRTP	Continuous Random Test Pattern supported in 10G mode of operation
CSMA/CD	Carrier-Sensing, Multiple-Access with Collision Detection
CSR	Control and Status Register
DFT	Design for Test
DTE	Data Terminal Equipment
DTE XGXS	XGXS Device Interfacing with DTE
DUT	Device Under Test
DWORD	64 bits
EEE	Energy Efficient Ethernet
EMI	Electro-magnetic Interference
FEC	Forward Error Correction
FSM	Finite State Machine
Halfword	16 bits
HFTP	High Frequency Test Pattern
IFG	Inter Frame Gap
LFTP	Low Frequency Test Pattern
LPI	Low Power Idle
LWORD	128 bits
MAC	Media Access Control. Corresponds to OSI Layer 2
MCI	Micro Controller Interface
MDC	MDIO Interface Clock
MDIO	Management Data Input/Output Interface
MII	Media Independent Interface
MFTP	Mixed Frequency Test Pattern
MMD	MDIO Manageable Device
MV	Manchester Violation

Term	Meaning
MVD	Manchester Violation Delimiter
OUI	Organizationally Unique Identifier
PCS	Physical Coding Sublayer. A type of PHY
PHY	Physical layer of an 802.3-compliant device
PHY XGXS	XGXS device with a PHY interface
PMA	Physical Medium Attachment Sublayer
PMD	Physical Medium Dependent Sublayer, a type of PHY
Port ID	Port ID (PORT_ID) of an 802.3-compliant device
PRBS	Pseudo Random Binary Sequence
RS	Reconciliation Sublayer
RXAUI	Reduced XAUI
SerDes	Serializer-Deserializer
SGMII	Serial Gigabit Media Independent Interface
SMA	Station Management Agent
SOF	Start of Frame
Word	32 bits
XAUI	10-Gigabit Attachment Unit Interface. An interface extender for the XGMII
XGMAC	10-Gigabit Media Access Controller
XGMII	10-Gigabit Media-Independent Interface
XGXS	XGMII-Extended Sublayer